Geophysical fluid mechanics

volume 1

foundations in mathematical physics and classical mechanics

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Preface

Geophysical fluid mechanics (GFM) is a branch of theoretical physics concerned with natural fluid motion on a rotating and gravitating planet or star. The subject makes use of concepts from classical continuum mechanics and thermodynamics, along with the corresponding methods of mathematical physics. The primary inspiration for our study comes from the motion of fluids in the earth's atmosphere and ocean. Geophysical fluids are in near rigid-body motion with the rotating planet or star, thus prompting a description from the rotating (non-inertial) planetary reference frame. Body forces from gravity plus planetary rotation (Coriolis and centrifugal) are fundamental features of the motion, as are contact forces from stresses (pressure and friction). In this book, we limit attention to the motion of a single phase of matter (gas or liquid), with the study of multiphase geophysical fluid mechanics, which is relevant to a moist atmosphere, outside our scope. Electromagnetic forces, important for the study of astrophysical fluid motions, are also ignored. We also ignore chemical reactions (which transform matter from one form to another), and nuclear reactions (which convert between matter and nuclear energy).

Geophysical fluid flows manifest over a huge range of space and time scales, with linear and nonlinear interactions transferring information across these scales. Physical insights into such flows typically result from examining a hierarchy of conceptual models using a variety of methods and perspectives. Some of the models we consider are formulated within the context of a perfect fluid comprising a single material constituent with fundamental processes limited to the reversible and mechanical (i.e., conservative). Other models are posed using a real fluid that is comprised of multiple matter constituents exposed to an irreversible process such as mixing of momentum through viscous friction, mixing of matter through matter diffusion, and/or the mixing of enthalpy through conduction. Some models consider constant density fluids, as commonly considered in classical hydrodynamics. And some models ignore rotation, thus tacitly applying to flows with length scales too short to feel the planetary Coriolis acceleration, whereas others ignore buoyancy to thus focus on the dynamics of a homogeneous fluid in a rotating reference frame.

We develop geophysical fluid mechanics from a mathematical physics perspective, with a grounding in fundamentals offering a robust and versatile framework for exploring a gamut of special cases and approximations. Topics are approached by establishing general principles prior to the examination of case studies. Consistent with this approach, our treatment focuses on developing the mechanics of geophysical fluid motion, with that focus supporting theoretical explorations that often extend beyond that required for phenomenological purposes. Correspondingly, we embrace the opportunity to examine physics through multiple lenses that render a variety of complementary insights. In a nutshell, if a physical system can be formulated and analyzed in more than one way, then we do so if it enhances pedagogy and exposes layers of understanding. As a result, brevity is sacrificed to support exposition and exploration, with this perspective leading to a book with multiple volumes.

The presentation is based on the premise that skills in theoretical physics are optimally taught by nurturing physical reasoning, with physical reasoning supported by mathematical precision coupled to the elucidation of concepts using words and pictures. Correspondingly, the presentation is both deductive and descriptive. The deductive approach supports a precise understanding through the use of elementary physical notions that are expressed mathematically. The descriptive approach builds skills in reasoning along with the ability to articulate physical ideas using words and pictures that complement the maths. Readers are supported by development of salient physical concepts and mathematical methods in the process of building understanding. With sufficient study, the material in this book should be accessible to the advanced undergraduate student or entering graduate student in fields such as applied mathematics, astrophysics, atmospheric physics, engineering, geophysics, ocean physics, planetary physics, and theoretical physics.

We generally offer details to mathematical derivations. Doing so nurtures the mathematical skills required for the budding theorist, with the reader strongly encouraged to work through the various derivations and exercises to fully embrace each detail and concept. Exposing mathematical details also helps to unpack many of the physical concepts encapsulated by equations. It is notable that the concepts encountered in this book generally accord with common experience (we are doing classical physics), thus affording a means to check on the validity of the maths. Even so, it does take time to wrap one's head around the physics of large-scale ocean and atmosphere circulations, so patience and persistence are required. Furthermore, as we are studying physics, all mathematical equations must satisfy dimensional consistency, with this constraint offering the physicist a powerful tool for exposing spurious mathematical statements.

We consider this book to be an intellectual journey taken together by the author and reader, thus motivating use of the first person plural pronouns *we* and *us*. Furthermore, we cultivate the deductive and descriptive approaches by embracing the synergism between physics and maths, whereby physics informs the maths and maths reveals the physics. This synergism is facilitated by a presentation style inspired by *Mermin* (1989), who identified the following characteristics for clear presentations of mathematical physics.

- RULE 1: All displayed equations are given numbers to facilitate cross-referencing. Additionally, any equation supporting another equation or a discussion is itself afforded an equation number.
- RULE 2: Cross-referenced equations are referred to by their equation number as well as descriptive phrases or names (e.g., "as seen by the vector-invariant velocity equation XX.YY" rather than just "as seen by equation XX.YY"). Coupling maths to words supports learning and reduces the need to flip pages to view the cited equation.
- RULE 3: Equations are part of the prose and are thus subject to punctuation.

Concerning the book's title

The study of rotating and stratified geophysical fluid motion largely started in the first half of the 20th century. During recent decades, the study has seen particular evolution through deepening physical foundations, refining mathematical formulations, increasing the intellectual and predictive value of numerical simulations, extending applications across terrestrial and planetary systems, and expanding observational and laboratory measurements and techniques. What has emerged is a recognition that a fruitful study of rotating and stratified fluid flows makes use of ideas that go beyond the traditional notions of geophysical fluid dynamics (GFD). Rather, the contemporary practitioner develops insights by weaving together concepts and tools from mathematics, Newtonian mechanics, analytical mechanics, fluid mechanics, thermodynamics, classical field theory, numerical simulations, laboratory experiments, field measurements, and data science. Acknowledging this broadening of the practice motivates the term *mechanics* in this book's title, rather than the more focused *dynamics*. It is a minor change in verbiage that reflects a broadening of the perspectives and goals pursued here.

Two pillars of theoretical geophysical fluid mechanics

We conceive of two pillars to theoretical geophysical fluid mechanics that are synergistic, thus offering lessons, guidance, and feedback to the other. The elements pillar of geophysical fluid mechanics comprises the physical and mathematical formulation of conceptual models used to garner insight into rotating and stratified fluid motion. This pillar is concerned with setting the stage by deductively and descriptively exposing how physical concepts are mathematically expressed to describe geophysical fluid flows. We provide a thorough treatment of the element pillar given its foundational importance, and since it is commonly offered only a terse treatment in other presentations. We emphasize that the elements pillar is far more than equation manipulation, although one certainly must become adept at that task. Instead, at its core, the elements pillar allows the physicist to reveal the fundamental physical concepts in a precise mathematical manner. Doing so supports understanding while building the foundations for the emergent phenomena pillar. The emergent phenomena pillar of geophysical fluid mechanics studies solutions to equations that describe phenomena, such as waves, instabilities, turbulence, and general circulation, all of which emerge from the fundamental equations. Phenomena can emerge in manners that are far from simple to understand deductively, particularly when considering nonlinear behavior such as turbulence. Our treatment of the emergent pillar is limited to waves and instabilities, whereas turbulence and general circulation are beyond our scope, though we do touch upon these topics where suited to the discussion.[1]

Some themes found in this book

This multi-volume book covers a number of topics in theoretical geophysical fluid mechanics. Throughout, we encounter many themes that appear in various guises, with the following offering a brief survey.

Causation and budgets

A great deal of this book is concerned with deriving and understanding equations that describe the evolution of fluid properties, with such equations (differential or integral) derived from physical principles such as Newton's laws of motion, Hamilton's principle of stationary action, Noether's theorem, thermodynamic laws, mass conservation, and vorticity mechanics. These budget equations form the theoretical foundation of continuum mechanics. As part of this development we often seek information about what *causes* fluid motion, making use of a variety of kinematic and mathematical frameworks. The causality question is nicely posed by Newton's equation of motion, which says that acceleration (motion) arises from a net force (the cause of motion). Even though seemingly a clear decomposition of cause and effect, this fundamental

[1]The further one moves along the axis of nonlinearity, the more Sisyphean the task of connecting fundamental processes to emergent phenomena. This perspective is lucidly discussed by *Anderson* (1972).

statement of Newtonian mechanics offers little more than the definition of a force. We break the self-referential loop, and thus make physical progress, after specifying the nature of the force (e.g., gravitational, frictional), as well as by offering properties of these forces as per Newton's third law (the action/reaction law).[2]

In geophysical fluid mechanics, we sometimes refer to a time evolving budget equation as an evolution equation or, more commonly, a prognostic equation, with each term in the prognostic equation referred to as a time tendency.[3] For prognostic equations, knowledge of the processes contributing to the net time tendency enables a prediction of flow properties. The question arises how to practically determine the tendencies acting in the fluid, particularly when tendencies are generally dependent on the flow itself. This question is often very difficult to answer. Such is the complexity and beauty inherent in nonlinear field theories such as fluid mechanics, where cause and effect are intrinsically coupled.

We can sometimes make progress by turning the problem around, whereby kinematic knowledge of the motion offers inferential knowledge of the dynamical processes contributing to the motion. This situation is exemplified by pressure forces acting within a non-divergent flow. In a non-divergent flow, pressure provides the force that acts, instantaneously and globally, to maintain the constraint that the velocity is non-divergent.[4] We may also make use of constraints that restrict the flow in manners that assist in prediction and understanding.

Constraints

Determining the forces, either directly or indirectly, provides physical insight into the cause of fluid flow and its changes. This approach is sometimes referred to a momentum based viewpoint since it is based on working directly with the momentum equation (i.e., Newton's second law of motion). However, we are commonly unable to deduce the forces due to complexities inherent in nonlinear field theories. Furthermore, there are many occasions when we are simply uninterested in the forces. In these cases, we are motivated to use constraints that can allow us to sidestep forces but still garner insights into the motion.

One example of a constraint concerns the inability of fluid to flow through a solid static material boundary, such as the solid-earth boundary encountered by geophysical flows. To understand how this constraint impacts the macroscopic fluid motion, we do not need to understand details of the atomic forces that underlie the resistance to macroscopic motion. Instead, we simply impose the kinematic boundary condition whereby the component of the velocity that is normal to the boundary vanishes at the boundary. The forces active within the fluid, no matter what flavor they may take, are constrained to respect the kinematic boundary condition. Another example concerns the study of vorticity. A variety of vorticity constraints offer the means to deduce flow properties without determining forces. Indeed, the vorticity based viewpoint often provides a framework that is more versatile in practice than the momentum-based approach, thus prompting the importance of vortex mechanics in the study of geophysical fluid flows.

Associations and balances

Besides seeking causal relations pointing toward the future, many basic questions of fluid mechanics arise either instantaneously, as in the constraints maintaining non-divergent flows,

[2]For more on this perspective of Newton's laws, see Chapter 1 of *Symon* (1971) or Chapter 2 of *Marion and Thornton* (1988).

[3]This language has its origins in weather forecasting.

[4]For non-divergent flow, pressure acts as the Lagrange multiplier enforcing flow non-divergence.

or when the flow is steady, in which case properties at each point in space have no time dependence. In steady flows, the net acceleration, and hence the net force, vanish at each point within the fluid, although the fluid itself can still be moving (steady flows are not necessarily static). For steady flows we are unconcerned with causality since time changes have been removed. In this manner, a steady state equation is a diagnostic equation rather than a prognostic equation. Diagnostic equations thus provide mechanical statements about associations between physical processes that manifest as balances. The geostrophic balance is the canonical association in geophysical fluid mechanics, where the horizontal Coriolis force is balanced by the horizontal pressure gradient force. Another balance concerns the vertical pressure gradient and its near balance with the weight of fluid above a point in the fluid, with this hydrostatic balance approximately maintained at the large scale even for moving geophysical fluids. Further associations arise when studying steady vorticity balances, with the Sverdrup balance a key example that is commonly used in ocean circulation theory.

We summarize the above by saying that diagnostic equations are concerned with the way things are, whereas prognostic equations point to how things will be. So although a predictive theory requires prognostic equations that manifest causal relations, an understanding of how fluid motion appears, and in particular how it is constrained, is revealed by studying diagnostic relations that expose associations through balances.

Mathematical transformations between kinematic perspectives

Geophysical fluid flows are complex. Hence, it proves useful to avail ourselves of a variety of methods and perspectives that support a mechanistic description of the motion. Many methods are associated with distinct kinematic lenses that reveal particular facets of the flow that might be less visible using alternative lenses. Examples include the Eulerian (spatial) and Lagrangian (material) kinematics used throughout fluid mechanics; the dual position space ($\boldsymbol{x}$-space) and wavevector space ($\boldsymbol{k}$-space) used for wave mechanics; the variety of vertical coordinates used for vertically stratified flows; and the analysis of motion in property spaces exemplified by watermass or thermodynamic analysis. We make use of these perspectives throughout this book, and offer the mathematical tools (e.g., tensor methods) needed to transform between them.

Newtonian mechanics and Hamilton's principle

Throughout this book we pursue the maxim

> PURSUE ALL WAYS TO FORMULATE AND TO SOLVE A PROBLEM.

A canonical example concerns the complementary perspectives available from Newtonian mechanics and Hamilton's principle of stationary action. Each offers logically consistent results yet approaches mechanics from fundamentally distinct conceptual and operational perspectives. In a Newtonian approach to fluid mechanics, governing differential equations are formulated using a continuum version of Newton's law of motion, in which forces (causes) and accelerations (effects) are articulated as a means to understand and predict the flow. The alternative approach of Hamilton's principle of stationary action approaches mechanics via a variational formulation involving the action. Hamilton's principle says that the action functional is extremized by the physically realized system. The action is the space-time integral of the difference between kinetic and potential/internal energies, and by extremizing the action we reveal the governing Euler-Lagrange differential equations. The Euler-Lagrange equations are identical to Newton's equations for those cases where Newton's equations are available,[5] and yet the route to deriving these equations is very distinct. It is by pursuing these disinct paths that we uncover new insights and develop distinct tools for analysis.

Hamilton's principle is not typically covered in fluid mechanics books. This absence contrasts to the ubiquity of Hamilton's principle in other areas of physics. We include facets of Hamilton's principle in this book with the hope that doing so partially remedies the disconnect.[6] Furthermore, we include Hamilton's principle since it provides novel perspectives on the fundamental equations of geophysical fluid mechanics, and renders insights and tools for the study of emergent phenomena such as waves and instabilities. The reader interested in a serious pursuit of theoretical mechanics should, at some point, make friends with Hamilton's principle. The effort is nontrivial as it requires brain muscles not exercised when studying Newtonian mechanics. But the conceptual and technical payoff is significant.

Non-dimensionalization and scale analysis

Mathematical symbols describing a physical system generally have physical dimensions. Examining the physical dimensions of an equation supports an understanding of the physical content of the equation, and provides a powerful means to identify errors in mathematical manipulations. It is for this reason that we prefer to expose physical dimensions throughout this book, rather than the alternative approach of working predominantly with non-dimensional equations. Even so, scale analysis, as realized through non-dimensionalization, offers an essential tool for deriving mathematical equations used to describe particular flow regimes.

There are two general types of dimensional scales that we use to non-dimensionalize a mathematical physics equation. The first is the external scale, with examples in this book being the gravitational acceleration, Coriolis parameter, domain size, and specified properties of the background state such as the buoyancy frequency or prescribed flow. External scales are set by the geophysical parameter regime in which the flow occurs, and as such they are under direct control of the theorist or experimentalist. The second is the emergent scale, which emerges from the flow itself. Emergent scales, such as the length scale and velocity scale of the flow, are specified by the subjective interest of the theorist though these scales are not under direct control. That is, we choose to focus on flows with a particular scale for purposes of examining the corresponding equations that describe that flow regime. A key example concerns our study of planetary geostrophy and quasi-geostrophy, where we choose to focus on flows of a particular scale where the Coriolis acceleration is of leading order importance.

We thus consider the operational aspects of scale analysis to be largely subjective in nature. Namely, we approach the analysis with a subjective bias towards the flow regime of interest, which in turn affects choices for non-dimensional parameters that lead to the corresponding asymptotic equations that describe the regime. Hence, scale analysis is deductive while being strongly guided by subjective interests.

[5] Hamilton's principle yields the Maxwell's equations of electromagnetism, and yet Maxwell's equations are distinct from Newton's equations. Indeed, Hamilton's principle is used throughout modern physics in areas far beyond Newtonian mechanics.

[6] There certainly are examples where Hamilton's principle is discussed in fluid mechanics books, with *Salmon* (1998), *Olbers et al.* (2012), and *Badin and Crisciani* (2018) notable examples that have inspired this author. Even so, these books remain the exception rather than the norm. As a result, the broader geophysical fluid mechanics community, even those pursuing theoretical aspects, are largely unaware of the beauty and power of Hamilton's principle. This situation contrasts to nearly every other area of theoretical physics, in which Hamilton's principle is central to both theory and application.

Approximations

There are a variety of approximations encountered in this book. One type of approximation is concerned with those made at a fundamental level, whereby the equations of motion are approximated based on assumptions about space and time scales of motion or other related assumptions. We are mostly concerned with approximations that lead to self-consistent equation sets, as defined by equations that maintain conservation principles that have a direct analog to the unapproximated equations. For example, in making the Boussinesq ocean approximation, the conservation of mass maintained by a fluid element is converted to conservation of volume for the element. Self-consistent equation sets can be derived using Hamilton's principle, thus ensuring proper accounting for conservation of energy and potential vorticity.

Another level of approximation is more mundane, and yet it is encountered quite frequently in practice. Namely, once deriving a diagnostic equation, we commonly find the need to approximate the equation in order to enable practical calculations to compute numbers and to thus compare to observations or to more exact numerical calculations. For example, in studying steric setup in VOLUME 2, we derive an equation for the difference between sea level at the coast and at an off-shore position, with this equation determined by the depth and off-shore integrated density field. However, this expression involves the sea level on both sides of the equation. Hence, the equation must be approximated to enable a practical calculation, with details of the approximation typically specific to the case in-hand.

Geophysical Fluid Mechanics and Climate Science

Fluid mechanics has a history of applications that span science and engineering, from blood flow to the evolution of galaxies. A key 21st century application of geophysical fluid mechanics concerns the questions of earth system science associated with the uncontrolled greenhouse gas experiment pursued by industrialized civilization's carbon centered energy use. Leading order science questions about climate warming have been sufficiently addressed to recognize that the Earth's biosphere is under great stress. Even so, mechanistic answers to a number of questions remain at the cutting edge of research. What will happen to the atmospheric jet stream and storm tracks in a world without summer Arctic sea ice? Will tropical storms be more powerful in a warmer world? What are the patterns for coastal sea level rise and their connections to large-scale ocean circulation? What are the key processes acting to bring relatively warm ocean waters to the base of high latitude ice shelves? How stable are the large-scale overturning circulations in the ocean and atmosphere, along with their associated heat transport? Are there feasible and sustainable climate intervention options that equitably reduce the negative impacts of climate warming without introducing new problems? These questions, and countless others, constitute key intellectual challenges of climate science in particular and Earth system science more generally.

Numerical circulation models, observational field campaigns (both *in situ* and remote), and laboratory experiments, are core platforms for Earth system science. Many of these platforms have reached a level of maturity allowing them to vividly reveal details of the complex and multi-scaled nature of planetary fluid flow. Geophysical fluid mechanics is key to the design of observational field campaigns and novel laboratory and numerical experiments, and it provides the intellectual framework for developing mechanistic analyses and robust interpretations of measurements and simulations. In this way, geophysical fluid mechanics furthers predictive capability for weather and climate forecast systems and it enhances confidence in projections for future climate. In a world of increasingly large volumes of simulated and measured data, we conjecture that the marriage of fundamental physical theory to data science tools will enable the significant science and engineering advances needed to address key questions of Earth system science.

About the cover

I took the cover photo of an iceberg, ocean, clouds, and sea bird (can you find the bird?) in the Orkney Passage region of the Southern Ocean during a research cruise from March-May 2017 aboard the British ship James Clark Ross. I am grateful to Alberto Naveira Garabato, the chief scientist on this cruise, for taking me to this amazing part of the planet. Although I largely pursue theoretical research, experiences with seagoing field research have greatly enhanced my scientific viewpoint and profoundly deepened a connection to the natural forces and phenomena that are in part described by geophysical fluid mechanics.

Gratitudes

This book greatly benefited from interactions with students in the Princeton University Atmospheric and Oceanic Sciences Program. In particular, parts of this book served as the basis for my teaching, over many years, a two-semester graduate course, AOS 571 and AOS 572. It also supported a variety of special topic classes (AOS/GEO 585) and lecture series. Further inspiration was offered by students, postdocs, and fellow researchers and scholars encountered on my path. I also thank those who provided specific suggestions, corrections, and comments on various drafts of this book, whose names are too many to list.

I am grateful for having been part of the unique research and learning environment cultivated by three of the world's best examples of scientific enterprises. First and foremost, I am the product of NOAA's Geophysical Fluid Dynamics Laboratory (GFDL), where I worked as a research physicist from 1996 until 2025. As part of my life as a US federal research scientist, I was fortunate to also be associated with Princeton University's Atmospheric and Oceanic Sciences (AOS) program, where I was a postdoc from 1993-1996 and then a faculty member from 2014-2026. As of 2026, I entered the most recent part of my career journey as a CNRS research scientist at LOCEAN (Laboratoire d'Océanographie et du Climat) in Paris. This position offers an amazing, and humbling, level of intellectual freedom among an energetic group of brilliant scientists and engineers.

The communities at GFDL, Princeton AOS, and LOCEAN provide an ideal setting for those interested in broadening scientific perspectives while diving deep into particular research areas. As part of my research and mentoring in this community, I have encountered thinkers whose style, questions, and insights have taken root in my work. This work has also afforded me the opportunity to travel the world to interact with colleagues whose wisdom and love of the scientific endeavor are infectious and inspiring. Throughout these interactions, I have entered into trusting and non-judgmental spaces where deep learning and understanding arise. Partaking in these spaces, where heart and mind meld, has been among the most fulfilling experiences of my life.

Developing a book of this nature is a not a simple endeavor. It starts modestly, grows over time, and eventually becomes a passion and obsession. I was particularly drawn to writing during the COVID-19 pandemic that kept the world largely sequestered at home, and I am grateful that my life situation allowed for this work to safely flourish during what were otherwise

very difficult times for civilization. Writing this book has been an exercise in rational thought that exemplifies the maxim “to write is to learn”, as articulated by *Zinnser* (1993). It was furthermore fed by spiritual food from meditation, yoga, family, and community. In particular, each step was supported by my wife, Adi, and our son, Francisco. I am deeply grateful for their patience and trust as I satisfied the goal of writing this book. I treasure being part of our family and I dedicate this work to you two amazing human beings.

Caveats and limitations

This book remains a work in progress that is not yet ready for publication. There are many loose threads detailed at the start of various chapters. In addition, here are items targeted for completion prior to release of this book to a publisher.

- Wave mechanics
  - equatorial shallow water waves
  - Rossby wave packets and motion in non-homogeneous background
  - Shallow water waves on a rotating sphere, including Laplace’s tidal equations, Hough functions, and spherical harmonics
  - Ray theory using Hamilton’s principle as in *Tracy et al.* (2014)
- Flow stability
  - Charney problem of baroclinic instability
  - Arnold’s stability theorem
  - Rayleigh-Benard convection
- Application of Hamilton’s principle
  - Referential flow using Hamilton’s principle
  - shallow water and Hamilton’s principle
  - semi-geostrophy and Hamilton’s principle
  - quasi-geostrophy and Hamilton’s principle
  - waves and mean flow interactions
  - Eddy stress via an eddy internal energy
- Mathematical topics
  - Lie derivative following Section F.3 of *Tromp* (2025a)

Disclaimer

Reference has been made to the published and unpublished literature to support the development of this book, as well as consultations with large language models to verify citations, refine exercise solutions, and clarify glossary entries. All material from such sources has been extensively reworked and refined into the author’s voice for purposes of uniformity, pedagogy, and clarity.

GUIDE TO THIS BOOK

No book is an island, with this book generously making use of other books, review articles, research papers, and online tutorials. Many readers find value in studying a subject from a variety of perspectives and voices, thus justifying the proliferation of books with overlapping subject matter. Sometimes it is merely one or two sentences that allow for an idea or concept to click within the reader's brain, whereas other topics require the full gamut of detailed derivations and discussions coming from multiple voices. For these reasons we provide pointers to written and/or video presentations that offer supportive views on material in this book. Many further resources are available through a quick internet search or consulation with artificial intelligence (AI).

There is no pretense that any reader will study all topics in this multi-volume book. This recognition is particularly apparent in a world where research and educational agendas often spread rather than focus attention. Hence, an attempt has been made to facilitate picking up each book at a variety of starting points. For that purpose, each chapter is written in a reasonably self-contained manner and with a brief guide at the start of each chapter listing pre-requisite material. As such, some equations and derivations are reproduced in more than one place, thus obviating the need to back reference. Certainly each chapter cannot be fully self-contained, since this is a book with material building from other chapters across the volumes. We thus make generous use of cross-referencing to point out allied material treated elsewhere. We also make extensive use of the glossary to help define concepts accessed in one volume that might be more thoroughly treated in another volume.

Organization

This book contains multiple volumes, each of which comprises parts with chapters. Parts and chapters start with a brief guide to the material along with pointers to dependencies. The book's end matter includes a glossary of key concepts and terms. Items highlighted within the text identity terms with a glossary entry. The glossary also serves as an annotated index, with page numbers pointing to where the terms and concepts are examined within a particular volume. Indeed, the glossary is an essential means to navigate this multi-volume book, reducing (though not eliminating) the need to have more than one volume open at a time. The glossary is then followed by a list of acronyms[7] and then by a list of symbols. A bibliography follows, with pages listed for where the book or paper is cited. We close the book with an index.

Not all topics are treated equally, with some probed deeply whereas others are given relatively superficial treatment. Indeed, there are many topics omitted that arguably should

[7]We generally try to avoid acronyms, but some are inevitable.

find a home here. Each shortcoming reflects on the author's limited energy and experience rather than a judgement of relative importance.

Cross-referencing to specific sections and equations is provided when pointing to material within the same volume. Cross-referencing material in other volumes is less specific. In many cases, a cross-reference concerns an item in the glossary and/or index, which can be consulted across volumes to help make the connection.

Volume 1

VOLUME 1 establishes foundations in mathematical physics and classical mechanics.

Mathematical physics

We start the book with a suite of mathematical methods chapters. Many readers can skim these chapters without sacrificing too much from later chapters, assuming they have a working knowledge of Cartesian tensors as well as vector differential and integral calculus. Where unfamiliar mathematics topics arise in later chapters, the reader is encouraged to return to this part of the book to help develop the necessary skills.

Clasical mechanics

We here survey salient topics from classical mechanics with a geophysical perspective, and in turn develop concepts and methods that have direct relevance to the continuum physics of geophysical fluid flows. Of particular note, this part develops an nderstanding of physics as viewed from a rotating reference frame. Doing so allows for the sometimes non-intuitive results of rotating physics to be developed within the context of a particle system as a pedagogical preface to later developments for geophysical fluid motion.

Volume 2

VOLUME 2 treats the fundamentals of fluid mechanics with an emphasis on geophysical fluid mechanics.

Kinematics of fluid flow

Mechanics is comprised of kinematics (the study of intrinsic properties of motion) and dynamics (the study of forces and energies causing motion). In the fluid kinematics part of this book, we initiate a study of fluid mechanics by focusing on the kinematics of fluid flow and matter transported by that flow. Our treatment exposes both the Eulerian and Lagrangian viewpoints and emphasizes the variety of kinematic notions and tools key to describing fluid motion. We also encounter facets of material transport as described by the tracer equation. We emphasize that fluid flow, and the transport of matter within that flow, have many features that are fundamentally distinct from point particle and rigid body motion. It takes practice to intellectually digest these differences.

Some kinematic topics can seem esoteric on first encounter, particularly the study of Lagrangian kinematics. However, an incomplete understanding of fluid kinematics can lead to difficulties appreciating facets of fluid dynamics. The reader is thus encouraged to fully study the kinematics chapters, and to revisit the material as the needs arise in later chapters.

Thermodynamics

We study the rudiments of thermodynamics with a focus on topics arising in the study of geophysical fluids. We pay particular attention to the role of gravity in modifying the treatment of thermodynamic equilibrium states, with gravity an essential facet of geophysical fluids and yet a force that is commonly ignored in standard treatments of thermodynamics. We ignore phase transitions, which is a notable limitation of our treatment, thus making this part of the book a mere introduction to the study of a moist atmosphere or an ocean with sea ice.

Dynamics of geophysical fluid flow

In this lengthy part of the book, we study how Newton's laws of mechanics and the principles of thermodynamics are applied to continuum fluid motion on a rotating and gravitating planet. We approach the subject by focusing on how forces that act on fluid elements lead to accelerations and thus to motion. In particular, we examine body forces that act throughout the volume of a fluid element (e.g., planetary gravity, planetary Coriolis, and planetary centrifugal) as well as contact forces that act on the boundary of a fluid element (e.g., pressure and friction).

Volume 3

Shallow water mechanics

A shallow water fluid is comprised of hydrostatically balanced homogeneous fluid layers. The layers are also typically assumed to be immiscible, so that interactions between layers occur only via mechanical forces from pressure acting at the layer interfaces. The shallow water fluid allows us to focus on planetary rotation and vertical stratification without the complexities of vertically continuous stratification and thermodynamics. Many physical insights garnered by studying shallow water fluids extend to more realistic fluids, thus making the shallow water model very popular among theorists and teachers. Indeed, *Zeitlin* (2018) provides an example of just how far one can go in understanding geophysical fluids with shallow water theory.

Vorticity

Vorticity plays a role in the motion of all geophysical fluids since motion on a rotating planet provides a nonzero planetary vorticity even to fluids at rest on the planet. This feature of geophysical fluids contrasts to many other areas of fluid mechanics, where irrotational flows are commonly encountered. Potential vorticity is a strategically chosen component of the vorticity vector that melds mechanics (vorticity) to thermodynamics (stratification). Material conservation properties of potential vorticity are striking and render important constraints on fluid motion. Indeed, perhaps the most practical reason to study vorticity concerns the various constraints imposed on the flow moving on a rotating and gravitating planet. These constraints provide conceptual insights and predictive power.

Nearly geostrophic balanced flows

Balanced models generally remove the horizontally divergent motions associated with gravity waves, thus allowing a focus on the large-scale vortical motions. Balanced models have a rich history among theoretical geophysical fluid studies, providing insights into both laminar oceanic flows through planetary geostrophy, and wave-turbulent atmospheric and oceanic flows through quasi-geostrophy. When studying balanced models, we focus on the shallow water and continuously stratified versions of quasi-geostrophy and planetary geostrophy.

Generalized vertical coordinates

The chapters on generalized vertical coordinates (GVC) dive into the maths, kinematics, dynamics, and applications of such coordinates for the study of geophysical fluid mechanics. This material is central to many current research activities, including subgrid scale parameterizations and the design of numerical atmosphere and ocean models. The mathematics in this part leans heavily on the general tensors studied in VOLUME 1. Even so, many readers can make the most of these chapter without the full gamut of general tensors.

Scalar fields

Many chapters target the mechanics of scalar fields with a focus mostly on the ocean. Here we consider active tracers (temperature and salinity), passive tracers, and buoyancy. Much of this study forms the basis of tracer mechanics, which has proven very important for the ocean since it is generally very difficult to measure vector fields such as velocity and vorticity, whereas tracer distributions are far more readily measured. We also consider facets of sea level analysis in this part of the book.

Volume 4

Hamilton's principle for geophysical flows

We study the analytical mechanics of geophysical flows using Hamilton's principle. This material forms the heart of field theory, both classical and quantum. It requires a different set of techniques than used in the study of Newtonian fluid mechanics used elsewhere in this book. Hence, it offers complementary insights that deepen our understanding of geophysical fluid flows in particular.

Linear wave mechanics

We study a variety of geophysical waves and associated mathematical methods used for their characterization. Notably, we consider waves not commonly included in a book on geophysical fluids, such as sound and capillary waves, with these waves included due to their ubiquity in the natural environment as well as their pedagogical value. Most focus, however, is given to waves arising from the Coriolis acceleration (inertial waves, planetary Rossby waves, topographic Rossby waves) and gravitational acceleration (surface gravity waves, internal gravity waves). Furthermore, we study linear waves and their corresponding wave packets, first studying their behavior in a homogeneous background environment where Fourier methods are available. Thereafter, we introduce methods needed to study linear waves on a gently varying background, including the methods of geometrical optics and wave action, with these methods of use particularly when Fourier methods are not suited.

Flow instabilities

We study instabilities that arise in geophysical fluid motions, distinguishing two classes of fluid instabilities: local instabilities or parcel instabilities versus global instabilities or wave instabilities. Local instabilities are afforded a local necessary and sufficient condition to determine whether the fluid base state is unstable to perturbations. In contrast, global instabilities arise from the constructive interference of waves and so involve the solution of an eigenvalue problem to determine properties of unstable waves. At most, a necessary condition can be derived to determine whether a global instability exists. Our study of fluid instabilities introduces a suite of case studies that foster analysis and conceptual methods to establish a foundation for further study. Geophysical fluid instability analysis remains an active area of research, with insights into the suite of primary and secondary instabilities providing compelling stories for how the ocean and atmosphere work.

Solution manual

A solution manual is available for students (with partial solutions provided) and instructors (with full solutions). Solutions aim to be instructional as well as utilitarian. For this reason, we expose many of the intermediate steps needed to derive a solution, further supporting an in-depth learning of how to independently solve physics problems. Additionally, exposing details helps to identify when an incorrect result follows from a physical/conceptual error (e.g., incorrect setup of the problem solution) or from a mathematical error (e.g., sign error).

Part I

Methods of mathematical physics

CONCERNING THE VALUE OF MATHEMATICAL GEOPHYSICS

When considering the impressive skill of numerical simulations of geophysical flows now common in the 21st century, the analytical training received by a 20th century theoretical scientist and engineer may seem like an unnecessary relic. Why bother to become adept at the huge variety of methods for deriving and/or analytically solving equations of mathematical physics and geophysics?

One answer is that knowledge and skills in mathematical methods provide access to a wealth of physical understanding. The reason is that mathematics is the most precise and powerful language of physics. To sidestep a training in mathematics risks missing a huge opportunity to grok the physics studied in this book. Namely, conceptual insights arise from the use of mathematics to expose the physical and geometrical meaning of physics equations. We thus propose that there is an increasing, not decreasing, need for practitioners to understand the physical principles and the mathematical methods at the heart of the compelling pictures generated by increasingly powerful computers, satellites, and *in situ* measures of geophysical fluid flows. For these reasons, we present two broad categories of mathematical methods in this part of the book. The first concerns methods of use to formulate the equations of geophysical fluid mechanics, with this category making use of tools from differential calculus and tensor analysis. The second category concerns the solution to the equations of geophysical fluid mechanics, with this category focused on linear partial differential equations arising from geophysical flows.

The depth to which one can pursue mathematical geophysics goes far beyond that presented in this book. For example, a modern mathematical treatment of continuum mechanics makes use of differential geometry and differential forms as presented in such books as *Schutz* (1980), *Frankel* (2004), and *Tromp* (2025b). However, to develop familiarity and trust with differential forms requires a nontrivial intellectual investment. At best, we offer glimpses of that approach through forays into tensor analysis in this part of the book.

PHYSICS PROVIDES RELATIONS BETWEEN GEOMETRIC OBJECTS

Mathematical objects of use for the study of fluid mechanics include scalar fields (e.g., temperature, mass density, specific entropy), vector fields (e.g., velocity, vorticity), and second order tensor fields (e.g., diffusion tensor, stress tensor, moment of inertia tensor). These and other fields have a value at each point in the continuous space and time used in continuum mechanics. Furthermore, their existence is independent of the arbitrary coordinates chosen by physicists to formulate a mathematical description. In the study of geophysical fluid mechanics, we use physical principles to develop differential equations relating geometric objects referred to here as tensors. Mathematical tools of analysis are then used to compute the numbers required to compare with experiments and field measurements, and to formulate discrete equations for numerical simulations.

Our perspective can be summarized by "physics as geometry", which forms a foundation to theoretical physics such as that detailed in *Thorne and Blandford* (2017) and *Tromp* (2025a,b). This perspective has both conceptual and practical use for our study throughout this book. In particular, tensor methods provide tools of geometry and analysis that are suited to formulate the equations of theoretical mechanics of particles and continua as arise in the study of geophysical fluid mechanics. Our aim is to develop mathematics to help pack and unpack the physics encapsulated by equations, pictures, and words.

TENSOR ANALYSIS AND GEOPHYSICAL FLUID MECHANICS

There are many occasions where a geophysical fluid system is more physically transparent when using a particular coordinate description or reference frame. However, there is no *a priori* coordinate choice that fits all cases. Thus, being adept at transforming from one mathematical description to another eases the study and pays huge dividends to the practioner. Tensor analysis is the proven means for systematically performing such transformations, thus motivating its use in fluid mechanics and in other areas of continuum mechanics (e.g., *Tromp* (2025a,b). In its more abstract realization via differential (or exterior) forms (not pursued in this book), tensor analysis provides the means to mathematically express physical ideas without any display of coordinate artefacts, thus exposing the underlying physical and mathematical essence.

The following offers an incomplete list of geophysical fluid systems where various coordinate descriptions or reference frames are encountered, and thus where tensor analysis can be put to use. Granted, many of these physical systems can be studied without the formalism of tensor analysis. However, by doing so one often encounters clumsy manipulations that obfuscate the underlying physical concepts. Indeed, imagine the tedium required to write continuous field equations in multiple dimensions prior to vector analysis.[8] That situation is akin to the tedium and awkward nature required to work across multiple coordinate systems and reference frames absent the formalism of tensor analysis. Hence, an adept use of vector analysis and its generalization to tensor analysis reveals how maths can inform the physics and how physics is embodied by the maths.

- APPROXIMATELY SPHERICAL PLANET: Geophysical fluids move on an approximately spherical planet, making spherical coordinates the preferred choice for studying planetary flows.[9] The nearly spherical geometry of a planet is non-Euclidean. However, we we assume that planet is embedded in a background Euclidean space (we ignore general relativistic effects that lead to more complex space-time embeddings). We make use of tensor methods to transform between planetary Cartesian coordinates (origin at the center of the planet) and spherical coordinates. Notably, spherical coordinates are locally orthogonal, so that one can make use of spherical coordinates without the machinery of general tensor analysis. Even so, having two ways to develop the spherical equations provides useful pedagogy when learning tensor analysis since the results are readily checked with prior experience using alternative methods.
- ROTATING TANK: Rotating laboratory fluids move in a circular tank, with cylindrical polar coordinates respecting symmetry of the domain. We make use of tensor methods to transform between Cartesian and cylindrical polar coordinates when considering rotating tank systems. Cylindrical polar coordinates are locally orthogonal, so that the comments above about spherical coordinates also hold here.
- ROTATING PLANETARY REFERENCE FRAME: Geophysical fluids move around a rotating earth and the fluid maintains a nearly rigid-body motion. Terrestrial observers also move in near rigid-body motion with the planet. We are thus motivated to study geophysical fluids from a rotating reference frame. We use rudimentary tensor methods to transform between a fixed inertial frame and the non-inertial rotating reference frame, with this transformation revealing non-inertial accelerations (planetary Coriolis and planetary centrifugal) that impact on the observed fluid flow.
- GALILEAN TRANSFORMATIONS: Geophysical fluids move in a space-time defined by

[8]The original treatment of electrodynamics by Maxwell was before the notations of vector analysis were invented by Gibbs, Heaviside, and others. Hence, reading Maxwell's original maths is, by today's standards, tedious and challenging.

[9]*Staniforth* (2022) provides arguments for moving beyond spherical coordinates for realistic atmospheric and oceanic modeling.

universal Newtonian time (all clocks measure the same time so that past, present, and future are the same everywhere) and Euclidean space, with the product of these two defining Galilean space-time. When studying fluid motion it can be convenient to move from one Galilean reference frame to another, with the movement referred to as a Galilean transformation. Tensor methods are quite useful to develop the mathematical relations between two Galilean reference frames.

- RELATING EULERIAN AND LAGRANGIAN KINEMATICS: There is a duality in fluid kinematics between Eulerian and Lagrangian descriptions of fluid motion. To develop an understanding of this duality we make use of tensor analysis to facilitate the transformation between the two kinematic descriptions. Whereas Cartesian coordinates offer a complete description for Eulerian kinematics, Lagrangian kinematics requires general tensor analysis since fluid particles deform with the fluid motion and thus render a non-orthogonal and time dependent coordinate description.

  For many purposes in continuum mechanics, one can pursue a description of motion without the choice of an origin, thus making use of some general results from differential geometry and tensor analysis. Our pursuit is less ambitious as we always consider fluids on a rotating planet or a laboratory, each of which are placed in a background Euclidean space with universal Newtonian time. Hence, for our puposes we acknowledge that there is a natural origin for the spatial description (e.g., planetary center or fixed location in the laboratory).
- STRATIFIED FLUIDS AND GENERALIZED VERTICAL COORDINATES: Geophysical fluids move in a gravitational field that acts to stratify the fluid according to its local buoyancy (high buoyancy fluid sits above low buoyancy fluid). When there is a one-to-one invertible relation between the geopotential and the buoyancy, then there are physical motivations to describe the vertical position of a fluid element according to its buoyancy rather than its geopotential. In this case we refer to buoyancy as a generalized vertical coordinate (GVC). Generalized vertical coordinates, as formulated for geophysical fluid motion, are non-orthogonal and time dependent coordinates. Developing the mathematics and physics of generalized vertical coordinates is greatly facilitated by the use of general tensor analysis.

PARTIAL DIFFERENTIAL EQUATIONS

Fluid mechanics is a classical field theory whose mathematical description involves partial differential equations. An understanding of partial differential equations provides useful insights into the physics and maths of geophysical fluids. We thus explore the rudiments of partial differential equations in this part of the book, developing salient mathematical properties with an eye towards applications to geophysical systems. As part of this study we develop the fundamentals of the Green's function method, which provides a mathematically and physically satisfying approach to solving differential equations.

SUMMARY OF THE MATHEMATICAL GEOPHYSICS CHAPTERS

Throughout are study of mathematical physics in this part of the book, we focus on concepts and methods of use for geophysical fluid systems. This focus supports our use of the term "mathematical geophysics" in the title to this part. It furthermore distinguishes our treatment from the many other texts that cover similar topics.

- CARTESIAN TENSOR ALGEBRA: Chapter 1 is a synopsis of Cartesian tensor analysis, which provides a systematization of ideas from Cartesian geometry and linear algebra. We push the envelope a bit with this presentation, making use of covariant and contravariant index notion as a preface to that required for general tensors. Material in this chapter is essential for nearly every topic in this book.
- VECTOR CALCULUS: Chapter 2 extends the algebraic ideas from Chapter 1 to differential and integral calculus. This chapter provides a resume of multivariate or vector calculus of use for fluid mechanics. Material in this chapter is essential for nearly every topic in this book.
- GENERAL TENSORS: Chapter 3 extends the Cartesian tensor algebra and calculus from Chapters 1 and 2 to allow for arbitrary, or general, coordinates. This chapter is essential for the mathematics underlying non-Cartesian Eulerian coordinates, such as spherical coordinates. It is also essential for developing the fluid mechanical equations using a Lagrangian reference frame (VOLUME 1) as well as for the generalized vertical coordinates studied in VOLUME 4.
- ORTHOGONAL COORDINATES: Chapter 4 provides a reference or handbook of results for Cartesian coordinates, cylindrical-polar coordinates, and spherical coordinates.
- DIRAC DELTA: In Chapter 5 we work through a suite of identities satisfied by the Dirac delta, which is an indispensible tool for mathematical physics.
- FOURIER ANALYSIS: In Chapter 6 we summarize many salient features of Fourier analysis, with a focus on the needs for studying wave mechanics in VOLUME 4.
- VARIATIONAL METHODS: In Chapter 7 we introduce variational methods, which find particular use in our study of Hamilton's principle in particle mechanics, fluid mechanics, and wave mechanics.
- ELLIPTIC PARTIAL DIFFERENTIAL EQUATIONS: In Chapter 8 we provide a summary of linear partial differential equations, and then focus on elliptic equations along with their Green's function solutions. Elliptic partial differential equations arise throughout our studies of geophysical fluids, with this chapter providing a foundation in their maths to help support an understanding of their physics.
- PARABOLIC PARTIAL DIFFERENTIAL EQUATIONS: In Chapter 9 we study the mathematics of parabolic partial differential equations, which are equations exemplified by the diffusion equation. We study diffusive processes in VOLUME 4, with the mathematics in Chapter 9 forming a suitable foundation. This chapter includes a study of the Green's function method for the diffusion boundary value problem, building on material from the elliptical equations in Chapter 8.
- HYPERBOLIC PARTIAL DIFFERENTIAL EQUATIONS: In Chapter 10 we study hyperbolic partial differential equations, focusing on the wave equation that describes non-dispersive waves. This study of non-dispersive waves bears fruit in VOLUME 4, both for the study of acoustic and shallow water gravity waves, but also for the study of dispersive geophysical fluid waves.

CLOSING REMARKS FROM MORSE AND FESHBACH

We here offer a quote from page 1 of the milestone textbook on methods of theoretical physics by *Morse and Feshbach* (1953), with their perspective providing inspiration for our treatment of mathematical methods for geophysical fluid mechanics.

> Our task in this book is to discuss the mathematical techniques which are useful in the calculation and analysis of the various types of fields occurring in modern physical

theory. Emphasis will be given primarily to the exposition of the interrelation between the equations and physical properties of the fields, and at times details of the mathematical rigor will be sacrified when it might interfere with the clarification of the physical background. Mathematical rigor is important and cannot be neglected, but the theoretical physicist must first gain a thorough understanding of the physical implications of the symbolic tools [they are] using before formal rigor can be of help. Other volumes are available which provide the rigor; this book will have fulfilled its purpose if it provides physical insight into the manifold field equations which occur in modern theory, together with a comprehension of the physical meaning behind the various mathematical techniques employed for their solution.

Chapter 1

Tensor algebra

In this chapter we introduce tensor analysis with a focus on algebraic relations. A tensor is an objective geometric object that we use as part of a mathematical description of physics. We make use of coordinate representations of tensors for analysis (i.e., to compute numbers), while appreciating that tensors are independent of the coordinate system or coordinate origin. In this manner, tensors are ideally suited for describing objective physical quantities such as scalar fields, forces, velocities, accelerations, and stresses. We are concerned in this book with tensors living in three-dimensional Euclidean space as well as tensors restricted to non-Euclidean manifolds that are embedded within Euclidean space (e.g., a sphere or an isopycnal). In this chapter, we restrict attention to Cartesian tensors, which make use of Cartesian coordinates in Euclidean space. The isomorphism between three-dimensional Euclidean space and the space of three-dimensional real numbers allows us to associate each point of Euclidean space, $\mathcal{P} \in \mathbb{E}^3$, with unique Cartesian coordinates, $\boldsymbol{x} \in \mathbb{R}^3$, for that point. This isomorphism renders a connection between geometry and analysis.

CHAPTER GUIDE

The material in this chapter should be accessible to those with a knowledge of linear algebra. Our treatment shares much with the mathematical physics literature, such as Chapter 2 of *Aris* (1962), Chapter 1 of *Segel* (1987), and Chapter 2 of *Lovelock and Rund* (1989). We make use of orthonormal basis vectors, $\hat{\boldsymbol{e}}_a$, in this chapter, with the associated tensor algebra directly transfering to the coordinate basis vectors, $\boldsymbol{e}_a$, considered with general tensors in Chapter 3.

We make use of covariant index notation for the representation of tensors (indices placed downstairs), as well as the contravariant index notation (indices placed upstairs).[a] For Cartesian coordinates in Euclidean space, there is no numerical distinction between covariant and contravariant index placements (see Section 1.3.5). However, the placement is critical for the general tensors studied elsewhere in this book (e.g., Chapter 3, in particular Section 3.3). We also write $\mathbb{g}_{ab}$ for a particular coordinate representation of the metric tensor, which for Cartesian tensors is the Kronecker delta, $\mathbb{g}_{ab} = \delta_{ab}$.

[a]The mnemonic “co-low” helps remember the index placement.

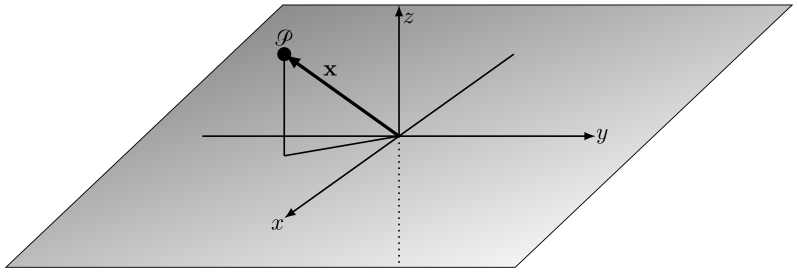


FIGURE 1.1: An arbitrary point in Euclidean space, $\mathcal{P} \in \mathbb{E}^3$, has an objective existence independent of a subjective choice of coordinate system used to describe its position. Its position, $\mathbf{x}$, relative to a chosen origin has an existence independent of the orientation of the coordinate system. The position is here represented by a right-handed Cartesian coordinate system according to $\mathbf{x} = \hat{\boldsymbol{x}}\, x + \hat{\boldsymbol{y}}\, y + \hat{\boldsymbol{z}}\, z$, with the coordinate representation, $\boldsymbol{x} = (x, y, z) \in \mathbb{R}^3$. The Cartesian orthonormal basis directions are given by the normalized triplet, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$. There is a continuous infinity of possible Cartesian coordinate systems that are rotated with respect to the one shown here.

1.1 Points, positions, and notation

Consider a point, $\mathcal{P} \in \mathbb{E}^3$, living in three-dimensional Euclidean space. As a geometric object, a point has an existence that is independent of any coordinate representation and independent of the coordinate system's origin. Now let $\mathbf{x}$ be the position for this point relative to an arbitrary origin. Once an origin is fixed, the position also has an existence that is independent of coordinates, and so we consider $\mathbf{x} \in \mathbb{E}^3$. Namely, the position for a point in Euclidean space is a line segment with an arrow pointing from the origin to the point. In this manner, the point, $\mathcal{P}$, and its position, $\mathbf{x}$, have no concern for coordinate systems. Even so, to make use of geometric objects for analysis requires specifying a set of coordinates, and in this chapter we focus on Cartesian coordinates.

1.1.1 Cartesian representation of the position

The isomorphism between Euclidean space and the real numbers allows us to associate a unique Cartesian coordinate representation, $\boldsymbol{x} \in \mathbb{R}^3$, for the position of every point in Euclidean space. We write this Cartesian coordinate representation as

$$\mathbf{x} = \hat{\boldsymbol{x}}\, x + \hat{\boldsymbol{y}}\, y + \hat{\boldsymbol{z}}\, z \qquad \mathbf{x} \in \mathbb{E}^3 \tag{1.1a}$$

$$\boldsymbol{x} = (x, y, z) \qquad \boldsymbol{x} \in \mathbb{R}^3. \tag{1.1b}$$

These equations state that each position, $\mathbf{x} \in \mathbb{E}^3$, has a unique coordinate representation, $\boldsymbol{x} \in \mathbb{R}^3$, with the coordinate representation here written using three-dimensional Cartesian coordinates. A Cartesian coordinate representation is provided by an ordered triplet of real numbers, (x, y, z). Each element of the triplet measures the coordinate distance (with physical dimensions of length) along the axes defined by their corresponding Cartesian orthonormal unit directions, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$, with these unit directions pointing toward increasing coordinate values. The Cartesian unit directions are a right-handed set of linearly independent directions that form a basis for three dimensional Euclidean space.[1] As such, the position for any point in Euclidean space can be represented in terms of these three basis directions.

[1]The unit directions are sometimes denoted $(\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}})$ in the literature. We do not use that notation in this book.

1.1.2 A soft convention for upright versus slanted notation

We use an upright symbol for a geometric object living in Euclidean space. The position, $\mathbf{x} \in \mathbb{E}^3$, is an example (in LaTeX, the upright is written `\mathbf{x}`), as are the tensors introduced in Section 1.2. The corresponding slanted symbol is used for a coordinate representation of the position, $\boldsymbol{x}$, or of a tensor, with coordinate representations living in the space of real numbers (using LaTeX, the slanted boldface is written `{\bm x}`). Each coordinate representation is realized by specifying a subjectively chosen set of coordinates. For this chapter the coordinates are Cartesian with $\boldsymbol{x} \in \mathbb{R}^3$ and with each coordinate having a physical dimension of length. So in summary, the upright notation is used for the position and for tensors, and the slanted notation is used for their coordinate representations.

The distinction between an upright symbols and slanted symbol is fundamental conceptually, in that it distinguishes geometric objects from their coordinate specific representations. Even so, the distinction is not critical in practice since an equation can be generalized from a specific set of coordinates (e.g., Cartesian) to arbitrary coordinates, so long as the equation respects the tensor rules discussed in Section 3.2.2.

1.1.3 Indices and Einstein's summation convention

Coordinate representations of the position and of tensors require indices to distinguish the various objects that appear in tensor analysis. The practioner of tensor analysis should become quite adept with the rules of index gymnastics, and we nurture the necessary brain-muscle in this chapter. The choice for where to place indices follows a convention that originates from that used for the position. This convention then sets the stage for subsequent usage with tensors.

We choose to write the coordinate representation of the position as

$$\mathbf{x} = \hat{\boldsymbol{x}}\, x^1 + \hat{\boldsymbol{y}}\, x^2 + \hat{\boldsymbol{z}}\, x^3 = \sum_{a=1}^{3} \hat{\boldsymbol{e}}_a\, x^a = \hat{\boldsymbol{e}}_a\, x^a, \tag{1.2}$$

with the superscripts acting as tensor labels rather than as multiplicative powers, and the hats used to denote the unit length of the basis directions. The final equality in equation (1.2) introduced the Einstein summation convention, in which we drop the summation symbol while repeated indices (one upstairs and one downstairs) are summed over their range. The summation convention is central to a variety of tensor manipulations, and so it is important to make friends with it. The second equality in equation (1.2) introduced a generic notation for the orthonormal Cartesian basis vectors

$$\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{x}} \quad \text{and} \quad \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{y}} \quad \text{and} \quad \hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{z}}, \tag{1.3}$$

with this notation allowing us to succinctly write the coordinate representation as $\mathbf{x} = \hat{\boldsymbol{e}}_a\, x^a$.

We emphasize that the indices denote components of a specific coordinate representation, $x^a = (x^1, x^2, x^3)$, as well as members from the set of orthonormal basis vectors, $\hat{\boldsymbol{e}}_a = (\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3)$. These labels are not multiplicative powers nor partial derivative operations.[2] Furthermore, placement of the downstairs tensor indices on the basis vectors pairs with the upstairs indices placed on the coordinate representations of the position. Such index placements establish a convention that then leads to further index placements for other tensors, and for index

[2] In this book we typically eschew the notation where partial derivatives are denoted by a subscript.

gymnastics following the Einstein summation convention.

1.1.4 Alternative Cartesian coordinates

We have thus far written $\boldsymbol{x}$ as the Cartesian coordinate representation of the position, $\mathbf{x}$. Now consider an alternative Cartesian coordinate system in which the position is represented by $\overline{\boldsymbol{x}}$. In Section 1.8 we work through the transformation between the two sets of Cartesian coordinates, $\boldsymbol{x}$ and $\overline{\boldsymbol{x}}$. For now we use the existence of this alternative coordinate system to exemplify the distinction between a geometric object (e.g., the position), versus its coordinate representation. Namely, the position for a particular point in space, $\mathbf{x}$, can be represented in either of the following equivalent manners

$$\mathbf{x} = \sum_{a=1}^{3} \hat{\boldsymbol{e}}_a \, x^a \quad \text{with} \quad \boldsymbol{x} = (x^1, x^2, x^3) \tag{1.4a}$$

$$\mathbf{x} = \sum_{\overline{a}=1}^{3} \hat{\boldsymbol{e}}_{\overline{a}} \, x^{\overline{a}} \quad \text{with} \quad \overline{\boldsymbol{x}} = (x^{\overline{1}}, x^{\overline{2}}, x^{\overline{3}}). \tag{1.4b}$$

That is, the position, $\mathbf{x}$, points from the origin to the point $\mathcal{P}$, and it can be represented using any number of Cartesian coordinates, with two such coordinates here written as $\boldsymbol{x}$ and $\overline{\boldsymbol{x}}$. Since both coordinate sets are Cartesian, their respective basis vectors, $\hat{\boldsymbol{e}}_a$, and $\hat{\boldsymbol{e}}_{\overline{a}}$, are orthonormal.

1.1.5 Concerning the use of orthonormal basis vectors

Much of the tensor analysis used in this book makes us of orthonormal basis vectors, $\hat{\boldsymbol{e}}_a$, to represent tensors. This usage originates from the Cartesian unit vectors, $\hat{\boldsymbol{e}}_a = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$, found in Cartesian tensor analysis, with the hat indicating normalization. Numerically, these orthonormal basis vectors point in three orthogonal directions and so have values

$$(\hat{\boldsymbol{e}}_a)^b = \delta_a{}^b, \tag{1.5}$$

where $\delta_a{}^b$ is the Kronecker tensor, which is unity when $a = b$ and zero otherwise. We extend this usage to orthonormal spherical and cylindrical unit vectors in Chapter 4.

We distinguish $\hat{\boldsymbol{e}}_a$ as a set of non-dimensional orthonormal basis vectors, from the generally unnormalized coordinate basis vectors, $\boldsymbol{e}_a$. The distinction is unimportant for Cartesian tensors. However, the distinction is important in our study of general tensors in Chapter 3, in which coordinate basis vectors, $\boldsymbol{e}_a$, have a geometric interpretation as partial derivative operators ($\boldsymbol{e}_a = \partial_a$), which are not normalized and so generally carry physical dimensions (e.g., inverse length).

1.2 Tensors

We extend the study of points and their positions in Section 1.1 to now consider a tensor as a geometrical object that lives in $\mathbb{E}^3$ or a submanifold. Each tensor has a coordinate representation that allows for its use in analysis, thus enabling its use for physical calculations that result in numbers. Importantly, a tensor has no dependence on either the choice of coordinates used in its representation, nor the choice for coordinate system origin. For Cartesian tensors, we represent tensors using the Cartesian orthonormal basis vectors, $\hat{\boldsymbol{e}}_a$.

1.2.1 Kronecker delta

The Cartesian orthonormal basis vectors have a scalar product[3]

$$\hat{\boldsymbol{x}} \cdot \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}} \cdot \hat{\boldsymbol{y}} = \hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{z}} = 1, \tag{1.6}$$

which can be written more generally as

$$\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b = \delta_{ab}, \tag{1.7}$$

with δ_{ab} the Kronecker delta given numerically

$$\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b = \delta_{ab} = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b. \end{cases} \tag{1.8}$$

As seen in Section 1.3, the Kronecker delta provides the Cartesian coordinate representation of the metric tensor for Euclidean space.

Use of a hat over the basis in equation (1.3) signifies that each component is normalized using the Euclidean norm. It is notable that a normalized basis vector can change only through rotation since by definition it remains of unit norm and so cannot change its magnitude (see Section 2.1.4).

1.2.2 Vectors as first order tensors

Geometrically, a vector is an arrow with a line segment whose length represents the magnitude of the vector. Furthermore, a vector has its origin at an arbitrary point in space. This geometrical construct manifests the objective nature of vectors in that they are independent of any particular coordinates used for their representation, and they are independent of the origin of the coordinate system. We encounter a number of vectors in our studies, such as the velocity, acceleration, and force. We extend this objective property of a vector to all tensors, so that tensors have no dependence on coordinate system nor coordinate origin.

The position, $\mathbf{x}$, identifies the location in space for a point, with the position defined relative to a chosen origin. So although the position is commonly referred to as the position "vector", its dependence on an origin means it is not a vector according to the definition presented in the previous paragraph. In contrast, the velocity is the time derivative of the position that moves through space, and the acceleration is the second time derivative. Taking a time derivative of the position removes the dependence on the coordinate system origin. So although the position is not a vector (since it depends on the arbitrarily chosen origin), its time derivative is a vector (a first order tensor), as is the acceleration. This result hints at the nature of vectors as living in the tangent space to points on a manifold, which is a concept we return to in Section 3.8.2 when studying general tensors.[4]

Following the notation for the position and its coordinate representation, we use an upright bold symbol, $\mathbf{F}$, for a vector and write its coordinate representations with a slanted $\boldsymbol{F}$:

$$\mathbf{F} = F^a \, \hat{\boldsymbol{e}}_a \in \mathbb{E}^3 \quad \text{and} \quad \boldsymbol{F} = (F^1, F^2, F^3) \in \mathbb{R}^3. \tag{1.9}$$

[3]The Cartesian scalar (or dot or inner) product is discussed more formally in Section 1.3.

[4]The flow map or motion field (see kinematics in VOLUME 2), generalize position and trajectory to continuum matter. Like the position, the motion field is not a tensor, though its time derivative, the velocity field, is a tensor.

The coordinate components, (F^1, F^2, F^3) are dependent on the choice for the orthonormal basis, $\hat{\boldsymbol{e}}_a$. We say that the tensor, $\mathbf{F}$, has a tensor order of one since there is one index on the components of the vector. Furthermore, the contravariant representation of a first order tensor, F^a, is referred to as its $(1,0)$ representation.

1.2.3 One-forms are dual to vectors

Equation (1.9) represents the first order tensor, $\mathbf{F}$, in terms of a set of orthonormal basis vectors, $\hat{\boldsymbol{e}}_a$, with the corresponding contravariant coordinate representation, F^a. We define one-forms as the duals to vectors in a manner analogous to how row and column vectors are dual objects in linear algebra.[5] We introduce one-forms to Cartesian tensors by defining an orthonormal one-form basis, $\hat{\boldsymbol{e}}^a$, with the upstairs index chosen to distinguish from the downstairs index used for the orthonormal direction basis, $\hat{\boldsymbol{e}}_a$. Notably, for Cartesian tensors there is absolutely no difference between $\hat{\boldsymbol{e}}^a$ and $\hat{\boldsymbol{e}}_a$. Even so, following Figure 2.8 of *Misner et al.* (1973) and Figure C.6 of *Tromp* (2025a), we offer the following geometric interpretation of the basis of one-forms as a dual to a basis of vectors. Namely, a particular member of the basis $\hat{\boldsymbol{e}}_a$ corresponds geometrically to a unit length line segment with an arrow pointing in the direction of increasing coordinate value. In contrast, a member of the basis, $\hat{\boldsymbol{e}}^a$, has the dual geometric interpretation as a surface or sheet whose outward normal direction is given by the corresponding basis vector.

Equation (1.9) offers a coordinate representation of a first order tensor in terms of the orthonormal vector basis, $\hat{\boldsymbol{e}}_a$. Since a tensor is a geometric object, then we can also represent the same tensor in terms of a set of the dual orthonormal one-form basis, $\hat{\boldsymbol{e}}_a$, in which case

$$\mathbf{F} = F_a \, \hat{\boldsymbol{e}}^a \in \mathbb{E}^3. \tag{1.10}$$

We say that the tensor, $\mathbf{F}$, has a tensor order of one, since it is represented in terms of the basis, $\hat{\boldsymbol{e}}^a$, with the covariant index expansion coefficients, F_a, providing the corresponding $(0,1)$ coordinate representation of the tensor. In this manner, we could refer to $\mathbf{F}$ as either a vector or a one-form, with first order tensor the more generic terminology. Higher order tensors require more indices and thus more basis elements.

1.2.4 Scalar product and the duality condition

Using the language of *Misner et al.* (1973), we say that a one-form is a machine that eats a vector to produce a scalar. Equivalently, a vector is a machine that eats a one-form and also produces the same scalar. We realize this machine analogy by defining the scalar product as follows. Namely, we define a scalar product between a first order tensor represented as a vector, $\mathbf{F} = F^a \, \hat{\boldsymbol{e}}_a$, and a first order tensor represented as a one-form, $\mathbf{G} = G_a \, \hat{\boldsymbol{e}}^a$, as

$$\mathbf{F} \cdot \mathbf{G} = (F^a \, \hat{\boldsymbol{e}}_a) \cdot (G_b \, \hat{\boldsymbol{e}}^b) = F^a \, G_b \, (\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}^b). \tag{1.11}$$

[5] There is no practical motivation to introduce one-forms for Cartesian tensors since there are no numerical distinctions between one-forms and vectors (as shown in Section 1.3.5). So why add this extra bit of formalism? It turns out that one-forms are needed when working with general tensors, with the associated formalism further detailed and motivated in Section 3.8. We introduce the extra baggage in the study of Cartesian tensors to anticipate the treatment with general tensors. It also offers a means to help keep track of the index notation and the associated Einstein summation convenction. The added formalism is not onerous, and it is well worth the price in support of the transition to general tensors.

To proceed, assume the vector basis and one-form basis satisfy the following *duality condition*

$$\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}^b = \delta_a{}^b = \delta^b{}_a. \tag{1.12}$$

We here introduced $\delta_a{}^b = \delta^b{}_a$, which is the representation of the $(1,1)$ Kronecker tensor where one index is downstairs and the other is upstairs. In Cartesian tensors, this representation of the Kronecker tensor is numerically identical to the Kronecker symbols, δ_{ab}, introduced by equation (1.8)

$$\delta_{ab} = \delta_a{}^b = \delta^b{}_a = \begin{cases} 1 & a = b \\ 0 & a \neq b. \end{cases} \tag{1.13}$$

Furthermore, the identity $\delta_a{}^b = \delta^b{}_a$ means that the Kronecker tensor is symmetric; i.e., swapping which index comes first (i.e., rows and columns when expressed as a matrix) does not alter the tensor. Making use of duality condition leads to the equivalent expressions of a scalar product between a vector and one-form

$$\mathbf{F} \cdot \mathbf{G} = (F^a \, \hat{\boldsymbol{e}}_a) \cdot (G_b \, \hat{\boldsymbol{e}}^b) = F^a \, G_b \, \hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}^b = F^a \, G_b \, \delta_a{}^b = F^a \, G_a. \tag{1.14}$$

The result is manifestly a scalar since the indices are fully contracted. Returning to the geometric picture from Section 1.2.3, the duality condition (1.12) says that the scalar product of a vector basis element and one-form basis element equals to unity if the vector basis is the outward normal to the one-form basis, whereas the scalar product is zero otherwise.

1.3 Distance and metric

In defining the Cartesian orthonormal vectors, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$, to have unit magnitudes, we presumed knowledge of how to measure the magnitude of a vector. We here make this notion precise.

1.3.1 Distance between points

Consider two points in Euclidean space, $\mathcal{P}$ and $\mathcal{P} + \mathrm{d}\mathcal{P}$, where the differential, d, symbolizes a small increment in space. These two points are specified by their respective positions, $\mathbf{x}$ and $\mathbf{x} + \mathrm{d}\mathbf{x}$. Representing these positions using a particular set of Cartesian coordinates renders

$$\mathbf{x} = \boldsymbol{x} = x^a \, \hat{\boldsymbol{e}}_a \quad \text{and} \quad \mathbf{x} + \mathrm{d}\mathbf{x} = (x^a + \mathrm{d}x^a) \, \hat{\boldsymbol{e}}_a. \tag{1.15}$$

Euclidean space is afforded a metric whereby the squared distance between two points is measured via Pythagora's theorem

$$\begin{aligned}
[\text{distance}(\mathcal{P}, \mathcal{P} + \mathrm{d}\mathcal{P})]^2 &= (\mathbf{x} + \mathrm{d}\mathbf{x} - \mathbf{x}) \cdot (\mathbf{x} + \mathrm{d}\mathbf{x} - \mathbf{x}) && \text{definition of distance} && (1.16a) \\
&= \mathrm{d}x^a \, \mathrm{d}x^b \, (\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b) && \text{coordinate representation} && (1.16b) \\
&= \mathrm{d}x^a \, \mathrm{d}x^b \, \delta_{ab} && \text{basis orthonormality (1.7)} && (1.16c) \\
&= (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 + (\mathrm{d}x^3)^2 && \text{expanding the sum} && (1.16d) \\
&= (\mathrm{d}x)^2 + (\mathrm{d}y)^2 + (\mathrm{d}z)^2 && \text{familiar } x, y, z \text{ notation.} && (1.16e)
\end{aligned}$$

Evidently, the Kronecker delta, δ_{ab}, provides the means to compute distances in Euclidean space when using Cartesian coordinates. We thus say that the Kronecker delta provides the Cartesian representation of the Euclidean metric tensor.

1.3.2 Use of more general notation for the metric tensor

The Kronecker symbols, δ_{ab}, are but one of the many representations available for the metric tensor applicable to Euclidean space. We encounter a number of other representations when using, for example, spherical coordinates, cylindrical polar coordinates, generalized vertical coordinates, and Lagrangian coordinates. For these other coordinates we need a more general expression for the metric. For this purpose we write the metric tensor representation as g_{ab}, so that

$$\mathrm{g}_{ab} \stackrel{\text{Cartesian}}{=} \delta_{ab} \quad \text{for Euclidean space with Cartesian coordinates.} \tag{1.17}$$

Although we focus on Cartesian tensors in this chapter, we introduce the more general expression for the metric tensor to readily expose those results that hold for general tensors.

1.3.3 Magnitude of a vector

By defining the distance between two points, we now have the means to define the squared magnitude of an arbitrary vector

$$|\mathbf{F}|^2 = \mathbf{F} \cdot \mathbf{F} = F^a \, F^b \, (\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b) = F^a \, F^b \, \mathrm{g}_{ab} \stackrel{\text{Cartesian}}{=} \sum_{a=1}^{3} F^a \, F^a, \tag{1.18}$$

with the final equality holding for Cartesian tensors with $\mathrm{g}_{ab} = \delta_{ab}$. Correspondingly, we have the scalar product between two first order tensors

$$\mathbf{F} \cdot \mathbf{G} = F^a \, G^b \, (\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b) = F^a \, G^b \, \mathrm{g}_{ab} \stackrel{\text{Cartesian}}{=} \sum_{a=1}^{3} F^a \, G^a, \tag{1.19}$$

which we already encountered in Section 1.2.4. Given our expression for the scalar product and the magnitude of vectors, we can introduce a geometrical interpretation by defining the angle between the vectors according to

$$\cos\vartheta \equiv \frac{\mathbf{F} \cdot \mathbf{G}}{|\mathbf{F}| \, |\mathbf{G}|} = \frac{F^a \, G^b \, \mathrm{g}_{ab}}{\sqrt{F^c \, F^d \, \mathrm{g}_{cd}} \, \sqrt{G^e \, G^f \, \mathrm{g}_{ef}}}. \tag{1.20}$$

Though a bit pedantic, the use of distinct indices within each of the terms emphasizes that each of the scalar products is self-contained. We illustrate this equation in Figure 1.2 for Cartesian coordinates. It is useful to verify that this definition is consistent with $-1 \leq \cos\vartheta \leq 1$. Furthermore, note that this definition of an angle between two vectors makes use of the metric tensor, thus emphasizing that angles on a manifold, just like distances, require a metric.[6]

1.3.4 Relating covariant and contravariant tensor representations

When a manifold has a metric tensor, then we can seamlessly transfer between the covariant and contravariant representation of a tensor. In this manner, the metric tensor links the one-forms

[6]This point about the need for a metric tensor might seem pedantic to those having only worked with Euclidean space with Cartesian coordinates, where the formalism of linear algebra and calculus tacitly builds in the Kronecker metric from the start. However, the study of calculus can be generalized to differential manifolds that have no metric. For example, the mathematics of thermodynamics is based on differential manifolds, so there is no sense for distance or angles between two points in thermodynamic space (see *Schutz* (1980) and *Frankel* (2004) for discussions). *Nurser et al.* (2022) offer another example for studies of fluid motion in continuous property spaces rather than Euclidean space.

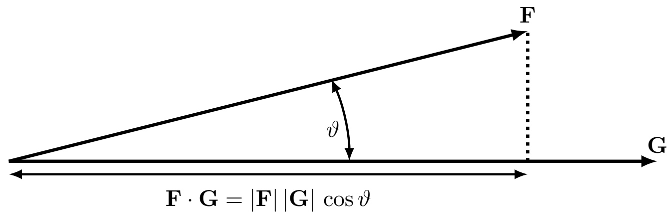


FIGURE 1.2: Illustrating the geometry of the scalar product (also known as the dot product or inner product) between two arbitrary vectors, $\mathbf{F}\cdot\mathbf{G} = |\boldsymbol{F}|\,|\boldsymbol{G}|\,\cos\vartheta$.

and vectors so to make their distinction important conceptually yet unimportant in practice. We here introduce the formalism for this conversion.[7]

The operational means to convert between the covariant to contravariant representations of a tensor arises through *contraction* of a tensor with the metric tensor, where contraction refers to summation over a common index. For Cartesian coordinates on Euclidean space, we saw in Section 1.3 that the metric tensor is represented by the Kronecker symbols. To raise and lower indices on Cartesian tensors, one contracts with the metric tensor and/or its inverse, so that, for example,

$$
\begin{aligned}
\mathbf{F} &= F^{a}\,\hat{\boldsymbol{e}}_{a} && \text{contravariant representation} && (1.21a)\\
&= (\mathbb{g}^{ab}\,F_{b})\,(\mathbb{g}_{ac}\,\hat{\boldsymbol{e}}^{c}) && \text{raise/lower indices with inverse metric, } \mathbb{g}^{ab}\text{, and metric, } \mathbb{g}_{ab} && (1.21b)\\
&= \mathbb{g}^{ab}\,\mathbb{g}_{ac}\,F_{b}\,\hat{\boldsymbol{e}}^{c} && \text{rearrange} && (1.21c)\\
&= F_{a}\,\hat{\boldsymbol{e}}^{a} && \mathbb{g}^{ab}\,\mathbb{g}_{ac} = \delta^{b}{}_{c}. && (1.21d)
\end{aligned}
$$

For the case of Cartesian tensors, with $\mathbb{g}_{ab} \overset{\text{Cartesian}}{=} \delta_{ab}$ and $\mathbb{g}^{ab} \overset{\text{Cartesian}}{=} \delta^{ab}$, these relations are no more than a repackaging of results already presented in earlier subsections. But what we have identified here is the role of the metric tensor in moving an index from upstairs to downstairs, and that is what we mean by converting between the contravariant (upstairs) and covariant (downstairs) representations. This operation holds for any representation of the metric, as indicated here by use of $\mathbb{g}_{ab}$ and its inverse, $\mathbb{g}^{ab}$. In particular, we can trivially move an index up if contracted with another down index, so that the scalar product is given by the equivalent expressions

$$
\mathbf{F}\cdot\mathbf{G} = F^{a}\,G_{a} = F_{a}\,G^{a}. \tag{1.22}
$$

Again, we emphasize that all of these manipulations hold for general coordinates when using the corresponding general coordinate representation of the metric tensor.

1.3.5 Covariant and contravariant are identical for Cartesian tensors

As seen in Section 1.3.1, the Kronecker delta is the metric tensor for Euclidean space using Cartesian coordinates. As a result, there is no distinction for Cartesian tensors between covariant and contravariant representations. For example, consider the identity

$$
F_{a} = \delta_{ab}\,F^{b}, \tag{1.23}
$$

[7]The conversion is superfluous for Cartesian tensors since there is no distinction between covariant and contravariant, which is exhibited in Section 1.3.5. Even so, let us agree to play the game in support of the general tensor case where the distinction is important.

which is how we lower an index to obtain the covariant representation, F_a, from the contravariant representation, F^b. Since δ_{ab} equals unity with $a = b$ and vanishes otherwise, we have, for each index a, the following identity

$$F_a = F^a. \tag{1.24}$$

This identity holds for all Cartesian tensors, so that, for example, the orthonormal basis vectors and orthonormal basis one-forms are identical

$$\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{e}}^1 = \hat{\boldsymbol{x}} \quad \text{and} \quad \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}^2 = \hat{\boldsymbol{y}} \quad \text{and} \quad \hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{e}}^3 = \hat{\boldsymbol{z}}. \tag{1.25}$$

Herein lies the reason that there is no need to distinguish upstairs and downstairs tensor indices when working with Cartesian tensors. Even so, we continue to play the game given the need for a distinction when working with general tensors.

1.4 Tensor fields

To make use of tensor analysis for studying continuum mechanics requires the use of tensor fields, which provides a tensor at each point in space and time. Space and time relations between tensor fields result from physical theories that are expressed by differential and integral equations.

1.4.1 Scalar fields

A physical scalar field, F, provides a number with physical dimensions at each point in space and time. Example physical scalar fields encountered in this book include temperature, mass density, buoyancy, salinity, humidity, and mechanical energy. Consider a point, $\mathcal{P} \in \mathbb{E}^3$, that has a position, $\mathbf{x}$. The scalar field has a value written

$$\mathrm{F}(\mathbf{x}, t) = \text{scalar property, F, evaluated at position } \mathbf{x} \text{ and time instance } t. \tag{1.26}$$

Now choose a particular set of spatial coordinates, $\boldsymbol{x} \in \mathbb{R}^3$. The scalar field, F, as sampled at a point represented by coordinates $\boldsymbol{x}$ and at time t, is specified by evaluating a function

$$F(\boldsymbol{x}, t) = \text{property F measured using coordinates } \boldsymbol{x} \text{ and time } t. \tag{1.27}$$

As per the notational convention introduced in Section 1.1.2, we here use the upright, F, for the scalar field as a zeroth order tensor, whereas the slanted, F, symbolizes the scalar tensor field as represented by a scalar function that depends on the chosen Cartesian space coordinates.

Bringing equations (1.26) and (1.27) together, we say that with $\boldsymbol{x}$ as the Cartesian coordinate expression for the position, $\mathbf{x}$, then

$$F(\boldsymbol{x}, t) = \mathrm{F}(\mathbf{x}, t) \quad \text{with } \boldsymbol{x} \text{ the coordinate representation of } \mathbf{x}. \tag{1.28}$$

If we instead choose another set of coordinates, $\overline{\boldsymbol{x}}$, then the property F sampled at a spatial point represented by coordinates $\overline{\boldsymbol{x}}$ is specified by another function,

$$\overline{F}(\overline{\boldsymbol{x}}, t) = \text{property F representated by coordinates } \overline{\boldsymbol{x}} \text{ and time } t. \tag{1.29}$$

If the two sets of Cartesian coordinates represent the same position, $\mathbf{x}$, then

$$\mathrm{F}(\mathbf{x}, t) = F(\boldsymbol{x}, t) = \overline{F}(\overline{\boldsymbol{x}}, t). \tag{1.30}$$

The distinction between a scalar field, $\mathrm{F}(\mathbf{x}, t)$, and scalar function, $F(\boldsymbol{x}, t)$, is conceptually fundamental. Even so, in practice the distinction is somewhat pedantic, and so it will be ignored when it is safe to do so.

1.4.2 Vector fields

The vector field, $\mathbf{F}(\mathbf{x}, t)$, provides a vector at each point in space and time, indicated by the position, $\mathbf{x}$, and each instance in time, t. Example vector fields considered in this book include velocity, acceleration, and forces. Choosing a particular coordinate representation of a vector field leads to the representation

$$\mathbf{F}(\mathbf{x}, t) = \boldsymbol{F}(\boldsymbol{x}, t). \tag{1.31}$$

The right hand side provides a vector function, $\boldsymbol{F}$, that represents the vector field, $\mathbf{F}$, with the vector function specific to the chosen Cartesian coordinates, $\boldsymbol{x}$. Another set of Cartesian coordinates, $\overline{\boldsymbol{x}}$, generally requires a distinct vector function, $\overline{\boldsymbol{F}}$, that renders an alternative representation of the vector field $\mathbf{F}(\mathbf{x}, t)$,

$$\mathbf{F}(\mathbf{x}, t) = \overline{\boldsymbol{F}}(\overline{\boldsymbol{x}}, t). \tag{1.32}$$

Just as found for scalar fields in equation (1.30), we have the identity for vector fields and their vector function representations

$$\mathbf{F}(\mathbf{x}, t) = \boldsymbol{F}(\boldsymbol{x}, t) = \overline{\boldsymbol{F}}(\overline{\boldsymbol{x}}, t). \tag{1.33}$$

In Section 1.8 we detail the mechanics of how to transform coordinate components of the vector field written using two sets of Cartesian coordinates, and in Chapter 3 we do the same for general coordinates. Each component, F^a, of a vector field, when written in terms of a particular set of coordinates, is itself a function of those coordinates[8]

$$\boldsymbol{F}(\boldsymbol{x}, t) = F^a(\boldsymbol{x}, t)\, \hat{\boldsymbol{e}}_a \quad \text{and} \quad \overline{\boldsymbol{F}}(\overline{\boldsymbol{x}}, t) = F^{\overline{a}}(\overline{\boldsymbol{x}}, t)\, \hat{\boldsymbol{e}}_{\overline{a}}. \tag{1.34}$$

However, each component function, $F^a(\boldsymbol{x}, t)$, cannot be considered a scalar field since the vector components transform according to the coordinate transformation rules detailed in Section 1.8, with these transformation rules distinct from those holding for a scalar field. This point is very basic to tensor analysis, and yet it can be readily overlooked upon first encounter.

1.5 Second order tensors and tensor products

The metric tensor, stress tensor, moment of inertia tensor, and diffusion tensor are examples of second order tensors encountered in fluid mechanics. Second order tensors have a coordinate representation given by any of the equivalent expressions

$$\mathbf{T} = T^{ab}\, \hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}_b = T^a{}_b\, \hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}^b = T_a{}^b\, \hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}_b = T_{ab}\, \hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}^b, \tag{1.35}$$

[8] In Chapter 3, the basis vectors are functions of space and time, whereas in this chapter we keep them constant, as done in equation (1.34).

with T^{ab}, $T^{a}{}_{b}$, $T_{a}{}^{b}$, and T_{ab} the variety of coordinate representations of the second order tensor, $\mathbf{T}$. This expression for a general second order tensor allows us to write the Kronecker or unit tensor in the equivalent forms

$$\mathbf{I} = \delta^{ab}\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}_b = \delta^{a}{}_{b}\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}^b = \delta_{a}{}^{b}\,\hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}_b = \delta_{ab}\,\hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}^b, \tag{1.36}$$

as well as the metric tensor

$$\mathbb{g} = \mathbb{g}^{ab}\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}_b = \mathbb{g}^{a}{}_{b}\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}^b = \mathbb{g}_{a}{}^{b}\,\hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}_b = \mathbb{g}_{ab}\,\hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}^b, \tag{1.37}$$

where the metric and its inverse satisfy the identities

$$\mathbb{g}^{ac}\,\mathbb{g}_{cb} = \mathbb{g}^{a}{}_{b} = \delta^{a}{}_{b} \quad \text{and} \quad \mathbb{g}_{ac}\,\mathbb{g}^{cb} = \mathbb{g}_{a}{}^{b} = \delta_{a}{}^{b}. \tag{1.38}$$

Notably, there is not a scalar product between the basis vectors in equations (1.35) or (1.36). Rather, the $\otimes$ symbol represents the tensor product of the basis vectors and/or basis one-forms. The tensor product generalizes the outer product of linear algebra, with components given by

$$(\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}_b)_{ij} = (\hat{\boldsymbol{e}}_a)_i\,(\hat{\boldsymbol{e}}_b)_j. \tag{1.39}$$

1.5.1 Three representations of a second order tensor

Equation (1.35) provides three distinct representations for the second order tensor, sometimes referred to as the musical nomenclature. We define these representations as

$$\mathbf{T}^{\natural} = T^{a}{}_{b}\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}^b = T_{a}{}^{b}\,\hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}_b \quad \text{natural or } (1,1) \text{ representation} \tag{1.40a}$$

$$\mathbf{T}^{\sharp} = T^{ab}\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}_b \quad \text{sharp or } (2,0) \text{ representation} \tag{1.40b}$$

$$\mathbf{T}^{\flat} = T_{ab}\,\hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}^b \quad \text{flat or } (0,2) \text{ representation.} \tag{1.40c}$$

1.5.2 More on the tensor product

Consider two arbitrary first order tensors, $\mathbf{A}$ and $\mathbf{B}$, and form a second order tensor by taking their tensor product

$$\mathbf{A} \otimes \mathbf{B} = A^a\,B^b\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}_b = A^a\,B_b\,\hat{\boldsymbol{e}}_a \otimes \hat{\boldsymbol{e}}^b = A_a\,B^b\,\hat{\boldsymbol{e}}^a \otimes \hat{\boldsymbol{e}}_b. \tag{1.41}$$

It can prove useful to organize elements of the tensor product for a particular coordinate representation according to the rules of matrix multiplication

$$\boldsymbol{A} \otimes \boldsymbol{B} = \begin{bmatrix} A^1 \\ A^2 \\ A^3 \end{bmatrix} [B^1 \quad B^2 \quad B^3] = \begin{bmatrix} A^1\,B^1 & A^1\,B^2 & A^1\,B^3 \\ A^2\,B^1 & A^2\,B^2 & A^2\,B^3 \\ A^3\,B^1 & A^3\,B^2 & A^3\,B^3 \end{bmatrix}. \tag{1.42}$$

This example makes it clear that the tensor product between two tensors is not commutative,

$$\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}, \tag{1.43}$$

but instead it satisfies

$$(\mathbf{A} \otimes \mathbf{B})^{\mathrm{T}} = \mathbf{B} \otimes \mathbf{A}, \tag{1.44}$$

where the T superscript is the *transpose operation* (swapping index locations). Likewise, considering the horizontal basis vectors for Cartesian coordinates renders the following tensor products

$$\hat{\boldsymbol{e}}_1 \otimes \hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{x}} \otimes \hat{\boldsymbol{x}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \hat{\boldsymbol{e}}_2 \otimes \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{y}} \otimes \hat{\boldsymbol{y}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{1.45a}$$

$$\hat{\boldsymbol{e}}_1 \otimes \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{x}} \otimes \hat{\boldsymbol{y}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \hat{\boldsymbol{e}}_2 \otimes \hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{y}} \otimes \hat{\boldsymbol{x}} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \tag{1.45b}$$

and similar results when including $\hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{z}}$.

1.6 Vector cross product

The vector cross product provides a means to measure the area of a parallelogram defined by two vectors, and furthermore to specify the orientation of that parallelogram. Note that when the vectors have physical dimensions distinct from length, then their vector cross product is not the area dimensioned as a squared length. Even so, the geometric interpretation remains, only with the physical dimensions adjusted accordingly. We here introduce the vector cross product, focusing on the Cartesian expression. We require a bit more nuance for the case of general tensors, with details presented in Section 3.13. Even so, the algebraic relations derived here also hold for general tensors.

1.6.1 Orienting an area via the Levi-Civita tensor

Consider a flat plane defined by any two of the orthonormal Cartesian basis vectors, $\hat{\boldsymbol{e}}_a$ and $\hat{\boldsymbol{e}}_b$ where $a \neq b$. We seek a means to specify what side of the plane is up and what side is down. Doing so allows us to orient objects within space.[9] Notably, there is no objective means for this specification, since "up" and "down" are subject to our chosen orientation. Therefore, we must choose a convention. For that purpose, we follow the right hand rule, in which the out-stretched thumb, index, and middle fingers of the right hand orient the three Cartesian basis vectors.

We algebraically specify the right hand rule for the basis vectors through the relation

$$\hat{\boldsymbol{e}}_a \times \hat{\boldsymbol{e}}_b = \epsilon_{abc}\, \hat{\boldsymbol{e}}^c. \tag{1.46}$$

The left hand side introduces the vector cross product of two basis vectors, with the $\times$ symbol used for the product. The right hand side identifies the vector cross product as a one-form. That is, as seen in Section 1.6.4, the cross product of two $(1,0)$ tensors produces a $(0,1)$ tensor.

The cross product (1.46) introduced the Levi-Civita tensor, whose Cartesian components are given by the totally anti-symmetric permutation symbol

$$\epsilon_{123} = 1 \tag{1.47a}$$

[9]There are surfaces, such as the Möbius strip and Klein bottle, that are not orientable. We only consider orientiable surfaces in this book.

$$\epsilon_{abc} = \begin{cases} 1, \text{ even permutation of } abc \ (123, 312, 231) \\ -1, \text{ odd permutation of } abc \ (321, 132, 213) \\ 0, \text{ all other } abc. \end{cases} \tag{1.47b}$$

Exchanging any two indices is an odd permutation and results in a sign swap

$$\epsilon_{abc} = -\epsilon_{bac} = -\epsilon_{acb} \Longrightarrow \epsilon_{123} = -\epsilon_{213}. \tag{1.48}$$

In contrast, cycling the indices in the following manner is an even permutation and so it preserves the sign

$$\epsilon_{abc} = \epsilon_{cab} = \epsilon_{bca} \Longrightarrow \epsilon_{123} = \epsilon_{312} = \epsilon_{231}. \tag{1.49}$$

Correspondingly, the vector cross product changes sign when its elements are commuted

$$\hat{\boldsymbol{e}}_a \times \hat{\boldsymbol{e}}_b = \epsilon_{abc}\, \hat{\boldsymbol{e}}^c = -\epsilon_{bac}\, \hat{\boldsymbol{e}}^c = -\hat{\boldsymbol{e}}_b \times \hat{\boldsymbol{e}}_a. \tag{1.50}$$

1.6.2 Axial vector

The vector cross product defines a one-form whose coordinate representation changes its sign upon changing from a right handed to left handed orientation of the coordinate system. One often sees the term axial vector used for this one-form. The angular velocity is a prominent axial vector encountered in physics.

1.6.3 Orthogonality relations between cross products

The permutation symbol ensures that $\hat{\boldsymbol{e}}_a \times \hat{\boldsymbol{e}}_b$ is orthogonal to both $\hat{\boldsymbol{e}}_a$ and $\hat{\boldsymbol{e}}_b$. To prove this property we write

$$\begin{aligned} \hat{\boldsymbol{e}}_a \cdot (\hat{\boldsymbol{e}}_a \times \hat{\boldsymbol{e}}_b) &= \hat{\boldsymbol{e}}_a \cdot \epsilon_{abc}\, \hat{\boldsymbol{e}}^c && \text{equation (1.46)} && (1.51a) \\ &= \epsilon_{abc}\, \delta_a{}^c && \text{duality (1.12)} && (1.51b) \\ &= 0 && \epsilon_{abc}\, \delta_a{}^c = 0. && (1.51c) \end{aligned}$$

The final equality holds since $\epsilon_{abc} = -\epsilon_{cba}$ whereas $\delta_a{}^c = \delta^c{}_a$, with a vanishing double contraction of an anti-symmetric tensor and a symmetric tensor.[10] The same procedure shows that $\hat{\boldsymbol{e}}_b \cdot (\hat{\boldsymbol{e}}_a \times \hat{\boldsymbol{e}}_b) = 0$. Hence, the one-form defined by the vector cross product of two vectors is orthogonal to both of the vectors. Evidently, the vector cross product is oriented orthogonal to the plane defined by the two vectors, with a direction determined by the right hand rule.

1.6.4 Vector product of two arbitrary vectors

The expression (1.46) for the vector product of two basis vectors renders the vector product of two arbitrary vectors

$$\begin{aligned} \mathbf{P} \times \mathbf{Q} &= P^a\, \hat{\boldsymbol{e}}_a \times Q^b\, \hat{\boldsymbol{e}}_b && (1.52a) \\ &= P^a\, Q^b\, \hat{\boldsymbol{e}}_a \times \hat{\boldsymbol{e}}_b && (1.52b) \\ &= P^a\, Q^b\, \epsilon_{abc}\, \hat{\boldsymbol{e}}^c && (1.52c) \\ &= (P^2\, Q^3 - P^3\, Q^2)\, \hat{\boldsymbol{e}}^1 + (P^3\, Q^1 - P^1\, Q^3)\, \hat{\boldsymbol{e}}^2 + (P^1\, Q^2 - P^2\, Q^1)\, \hat{\boldsymbol{e}}^3. && (1.52d) \end{aligned}$$

[10] We further illustrate this property for matrices in Exercise 1.2.

For a particular representation, we can write the vector product in the form of a determinant

$$\boldsymbol{P} \times \boldsymbol{Q} = \det \begin{bmatrix} \hat{\boldsymbol{e}}^1 & \hat{\boldsymbol{e}}^2 & \hat{\boldsymbol{e}}^3 \\ P^1 & P^2 & P^3 \\ Q^1 & Q^2 & Q^3 \end{bmatrix}. \tag{1.53}$$

As with the basis vectors, the vector product is orthogonal to both of the individual vectors, such as

$$\mathbf{P} \cdot (\mathbf{P} \times \mathbf{Q}) = (P^d \, \hat{\boldsymbol{e}}_d) \cdot (P^a \, Q^b \, \epsilon_{abc} \, \hat{\boldsymbol{e}}^c) \tag{1.54a}$$

$$= P^a \, P^d \, Q^b \, \epsilon_{abc} \, \delta^c{}_d \tag{1.54b}$$

$$= P^a \, P^c \, Q^b \, \epsilon_{abc} \tag{1.54c}$$

$$= 0, \tag{1.54d}$$

where the final equality follows since the product, $P^c \, P^a$, is symmetric on the labels ac ($P^c \, P^a = P^a \, P^c$), whereas ϵ_{abc} is anti-symmetric on these same labels.

1.6.5 Geometric interpretation of the vector cross product

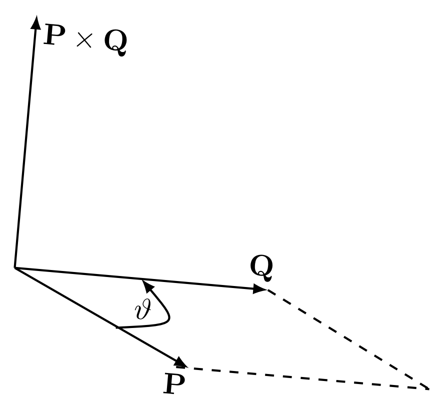


FIGURE 1.3: The magnitude of the vector product between two vectors is given by the product of their magnitudes and the sine of the angle between them, $|\mathbf{P} \times \mathbf{Q}| = |\mathbf{P}| \, |\mathbf{Q}| \sin\vartheta$. This magnitude equals to the area of the parallelogram formed by the two vectors. The vector cross product is a one-form that is directed perpendicular to the plane determined by the two vectors and oriented according to the right hand rule. The right hand rule is found by placing the fingers of the right hand along the first vector, $\mathbf{P}$. Closing the fingers in the direction of the second vector, $\mathbf{Q}$ (as here depicted by the arrow on the arc for the angle ϑ), then ensures that the thumb provides the orientation for the vector cross product, $\mathbf{P} \times \mathbf{Q}$.

The expression (1.52d) leads to the identity

$$|\mathbf{P} \times \mathbf{Q}|^2 = |\mathbf{P}|^2 \, |\mathbf{Q}|^2 - (\mathbf{P} \cdot \mathbf{Q})^2 \tag{1.55a}$$

$$= |\mathbf{P}|^2 \, |\mathbf{Q}|^2 \, (1 - \cos^2\vartheta) \tag{1.55b}$$

$$= |\mathbf{P}|^2 \, |\mathbf{Q}|^2 \sin^2\vartheta, \tag{1.55c}$$

where we used the scalar product expression (1.20) to introduce the angle subtended by the two vectors. Trigonometry indicates that the area of the parallelogram defined by the two vectors, $\mathbf{P}$ and $\mathbf{Q}$, is given by $|\mathbf{P}| \, |\mathbf{Q}| \sin\vartheta$. Hence, the vector cross product has a magnitude given by this area

$$\text{area}(\mathbf{P},\mathbf{Q}) = |\mathbf{P}|\,|\mathbf{Q}|\sin\vartheta = |\mathbf{P}\times\mathbf{Q}|. \tag{1.56}$$

Since $\mathbf{P}\times\mathbf{Q}$ is a one-form orthogonal to the plane defined by $\mathbf{P}$ and $\mathbf{Q}$, we can write the vector product in the purely geometric manner

$$\mathbf{P}\times\mathbf{Q} = \hat{\boldsymbol{n}}\,\text{area}(\mathbf{P},\mathbf{Q}) = \hat{\boldsymbol{n}}\,|\mathbf{P}|\,|\mathbf{Q}|\sin\vartheta, \tag{1.57}$$

where $\hat{\boldsymbol{n}}$ is a *unit normal one-form* (also known as the *normal direction*) that points normal to the area and in a direction given by the right hand rule. This formula is illustrated in Figure 1.3.

1.6.6 Decomposing the vector cross product

To further our geometric interpretation of the vector cross product using Cartesian coordinates, let $\boldsymbol{P}$ be the vertical unit vector, $\boldsymbol{P}=\hat{\boldsymbol{z}}$. The vector cross product then produces

$$\begin{aligned}
\boldsymbol{Q}^{\hat{\boldsymbol{z}}\perp} &= \hat{\boldsymbol{z}}\times\boldsymbol{Q} && (1.58\text{a})\\
&= \hat{\boldsymbol{z}}\times[\boldsymbol{Q}-(\hat{\boldsymbol{z}}\cdot\boldsymbol{Q})\,\hat{\boldsymbol{z}}] && (1.58\text{b})\\
&= (\hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}})\,(\hat{\boldsymbol{x}}\cdot\boldsymbol{Q}) + (\hat{\boldsymbol{z}}\times\hat{\boldsymbol{y}})\,(\hat{\boldsymbol{y}}\cdot\boldsymbol{Q}) && (1.58\text{c})\\
&= \hat{\boldsymbol{y}}\,(\hat{\boldsymbol{x}}\cdot\boldsymbol{Q}) - \hat{\boldsymbol{x}}\,(\hat{\boldsymbol{y}}\cdot\boldsymbol{Q}). && (1.58\text{d})
\end{aligned}$$

By construction, $\boldsymbol{Q}^{\hat{\boldsymbol{z}}\perp}$ is in the horizontal plane and it is perpendicular to the horizontal projection of $\boldsymbol{Q}$

$$\boldsymbol{Q}^{\hat{\boldsymbol{z}}\perp}\cdot\hat{\boldsymbol{z}} = 0 \quad\text{and}\quad \boldsymbol{Q}^{\hat{\boldsymbol{z}}\perp}\cdot\boldsymbol{Q} = 0 \Longrightarrow \boldsymbol{Q}^{\hat{\boldsymbol{z}}\perp}\cdot[\boldsymbol{Q}-(\hat{\boldsymbol{z}}\cdot\boldsymbol{Q})\,\hat{\boldsymbol{z}}] = 0. \tag{1.59}$$

Hence, $\boldsymbol{Q}^{\hat{\boldsymbol{z}}\perp}$ is geometrically computed by rotating the horizontal component of $\boldsymbol{Q}$ by $\pi/2$ radians counter-clockwise about the $\hat{\boldsymbol{z}}$ axis. That interpretation holds for all coordinate directions:

$$\begin{aligned}
\hat{\boldsymbol{x}}\times\boldsymbol{Q} &= \hat{\boldsymbol{z}}\,(\hat{\boldsymbol{y}}\cdot\boldsymbol{Q}) - \hat{\boldsymbol{y}}\,(\hat{\boldsymbol{z}}\cdot\boldsymbol{Q}) = \text{project } \boldsymbol{Q} \text{ to y-z plane} + \text{rotate } \pi/2 \text{ CCW around } \hat{\boldsymbol{x}} && (1.60\text{a})\\
\hat{\boldsymbol{y}}\times\boldsymbol{Q} &= \hat{\boldsymbol{x}}\,(\hat{\boldsymbol{z}}\cdot\boldsymbol{Q}) - \hat{\boldsymbol{z}}\,(\hat{\boldsymbol{x}}\cdot\boldsymbol{Q}) = \text{project } \boldsymbol{Q} \text{ to z-x plane} + \text{rotate } \pi/2 \text{ CCW around } \hat{\boldsymbol{y}} && (1.60\text{b})\\
\hat{\boldsymbol{z}}\times\boldsymbol{Q} &= \hat{\boldsymbol{y}}\,(\hat{\boldsymbol{x}}\cdot\boldsymbol{Q}) - \hat{\boldsymbol{x}}\,(\hat{\boldsymbol{y}}\cdot\boldsymbol{Q}) = \text{project } \boldsymbol{Q} \text{ to x-y plane} + \text{rotate } \pi/2 \text{ CCW around } \hat{\boldsymbol{z}}. && (1.60\text{c})
\end{aligned}$$

This geometric interpretation may appear a bit arcane. However, it serves to couple the analytical to the geometrical, and so supports our understanding. We also encounter it in various forms, particularly when studying vorticity in VOLUME 3.

1.7 Measuring volume

The vector cross product offers a means to measure area defined by two vectors. We now see how it can be of use to measure the volume determined by three non-parallel vectors. This result has particular relevance to the volume element used for integration over space. As for the vector cross product in Section 1.6, we require a bit more nuance for the case of general tensors and present details in Section 3.10. So the material in the present section assumes Cartesian tensors.

1.7.1 Volume defined by three vectors

Consider the scalar product of an arbitrary vector, $\mathbf{R}$, with the vector product of two vectors, $(\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R}$. This scalar product projects that portion of the vector, $\mathbf{R}$, onto the direction parallel to the normal to the plane defined by $\mathbf{P} \times \mathbf{Q}$. Given that $|\mathbf{P} \times \mathbf{Q}|$ is the area of the parallelogram defined by $\mathbf{P}$ and $\mathbf{Q}$, we see that $(\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R}$ is the volume of the parallelepiped defined by the three vectors.

The sign of the volume depends on the orientation of the three vectors, $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R}$. We assign a convention so that the volume is positive if the vectors are oriented in a right hand sense. Ignoring this convention, we can simply place an absolute value around the triple product to ensure a positive volume.

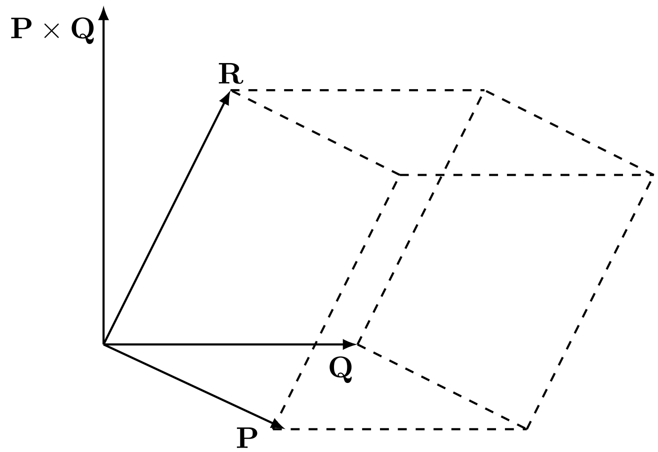


FIGURE 1.4: Three linearly independent vectors determine a volume given by $|(\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R}| = |(\mathbf{R} \times \mathbf{P}) \cdot \mathbf{Q}| = |(\mathbf{Q} \times \mathbf{R}) \cdot \mathbf{P}|$. We see this identity geometrically by noting that the area of the base is given by $|\mathbf{P} \times \mathbf{Q}|$, so that the volume of the parallelepiped is given by the base area times the height, $|(\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R}|$. Cyclic permutation then leads to the equality with the other two expressions.

We prove cyclic symmetry of $(\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R}$ through the following manipulations

$$
\begin{aligned}
(\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R} &= (P^a\, \hat{\boldsymbol{e}}_a \times Q^b\, \hat{\boldsymbol{e}}_b) \cdot R^d\, \hat{\boldsymbol{e}}_d && \text{expose coordinates and basis vectors} && (1.61a)\\
&= P^a\, Q^b\, (\hat{\boldsymbol{e}}_a \times \hat{\boldsymbol{e}}_b) \cdot R^d\, \hat{\boldsymbol{e}}_d && \text{rearrange} && (1.61b)\\
&= P^a\, Q^b\, (\epsilon_{abc}\, \hat{\boldsymbol{e}}^c) \cdot \hat{\boldsymbol{e}}_d\, R^d && \text{vector product as per equation (1.46)} && (1.61c)\\
&= P^a\, Q^b\, \epsilon_{abc}\, (\hat{\boldsymbol{e}}^c \cdot \hat{\boldsymbol{e}}_d)\, R^d && \text{rearrange} && (1.61d)\\
&= P^a\, Q^b\, \epsilon_{abc}\, \delta^c{}_d\, R^d && \text{duality: } \hat{\boldsymbol{e}}^c \cdot \hat{\boldsymbol{e}}_d = \delta^c{}_d = \delta_d{}^c && (1.61e)\\
&= P^a\, Q^b\, \epsilon_{abc}\, R^c && d \text{ index is contracted: } \delta^c{}_d\, R^d = R^c && (1.61f)\\
&= R^c\, P^a\, Q^b\, \epsilon_{abc} && \text{rearrange} && (1.61g)\\
&= R^a\, P^b\, Q^c\, \epsilon_{bca} && \text{relabel: } a \to b \text{ and } b \to c \text{ and } c \to a && (1.61h)\\
&= R^a\, P^b\, Q^c\, \epsilon_{abc} && \text{even permutation: } \epsilon_{bca} = \epsilon_{abc} && (1.61i)\\
&= (\mathbf{R} \times \mathbf{P}) \cdot \mathbf{Q} && \text{redintroduce boldface notation.} && (1.61j)
\end{aligned}
$$

This identity yields the geometric result illustrated in Figure 1.4

$$
\text{volume}(\mathbf{P}, \mathbf{Q}, \mathbf{R}) = (\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R} = (\mathbf{R} \times \mathbf{P}) \cdot \mathbf{Q} = (\mathbf{Q} \times \mathbf{R}) \cdot \mathbf{P}. \tag{1.62}
$$

1.7.2 Cartesian volume element for integration

We need the volume of an infinitesimal region when performing an integration over space. When making use of Cartesian coordinates we need the volume of a rectangular prism defined by infinitesimal distances along each of the Cartesian coordinate axes. We thus set

$$\mathbf{P} = \hat{\boldsymbol{x}}\,\mathrm{d}x \quad \text{and} \quad \mathbf{Q} = \hat{\boldsymbol{y}}\,\mathrm{d}y \quad \text{and} \quad \mathbf{R} = \hat{\boldsymbol{z}}\,\mathrm{d}z, \tag{1.63}$$

in which case the volume element is

$$\mathrm{d}V = (\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R} = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,(\hat{\boldsymbol{x}} \times \hat{\boldsymbol{y}}) \cdot \hat{\boldsymbol{z}} = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z. \tag{1.64}$$

This expression for the volume element could have been written down without the formalism of a vector triple product. However, in Chapter 3 we find the expression, $(\mathbf{P} \times \mathbf{Q}) \cdot \mathbf{R}$, provides a useful starting point to derive the volume element using arbitrary coordinates.

1.7.3 $n-$space volumes and the Levi-Civita tensor

Let us now combine the geometric specification (1.57) of the vector product as a means to measure area, with the algebraic specification (1.52d), and do so by writing

$$\text{2-volume} = \epsilon(\mathbf{P}, \mathbf{Q}) = \epsilon_{ab}\,P^a\,Q^b = \det \begin{bmatrix} P^1 & Q^1 \\ P^2 & Q^2 \end{bmatrix}. \tag{1.65}$$

In this equation, ϵ_{ab} is the totally anti-symmetric second order tensor, whose Cartesian component expression can be organized as a 2×2 matrix

$$\begin{bmatrix} \epsilon_{11} & \epsilon_{12} \\ \epsilon_{21} & \epsilon_{22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \tag{1.66}$$

In words, the first equality in equation (1.65) states that the ϵ-tensor in two dimensions takes two vectors as its argument and produces a 2-volume (i.e., an area). The three dimensional generalization yields

$$\text{3-volume} = \epsilon(\mathbf{P}, \mathbf{Q}, \mathbf{R}) = \epsilon_{abc}\,P^a\,Q^b\,R^c = \det \begin{bmatrix} P^1 & Q^1 & R^1 \\ P^2 & Q^2 & R^2 \\ P^3 & Q^3 & R^3 \end{bmatrix}. \tag{1.67}$$

Suppressing the first vector argument in the 3-volume produces the surface area one-form defined by the other two vectors

$$\text{surface area one-form} = \epsilon(\quad, \mathbf{Q}, \mathbf{R}). \tag{1.68}$$

By construction, the vectorial surface area is orthogonal to both $\mathbf{Q}$ and $\mathbf{R}$ since

$$\epsilon(\mathbf{Q}, \mathbf{Q}, \mathbf{R}) = \epsilon(\mathbf{R}, \mathbf{Q}, \mathbf{R}) = 0. \tag{1.69}$$

1.7.4 Vector identities derived with the Levi-Civita tensor

The Levi-Civita tensor is a versatile tool for deriving identities in tensor algebra, and we illustrated some of these features in the previous discussion. In Exercises 1.4, 1.5, and 1.6 we work through further examples that expose elements of the index gymnastics involved with tensor manipulations. The best way to learn index gymnastics is to do it, with these exercises a great way to start.

1.8 Coordinate transforming tensor components

The orthonormal Cartesian basis vectors are mutually orthogonal, and once their orientation is chosen they are fixed in space. However, the choice of orientation is arbitrary.[11] We can consider an alternative specification to the Cartesian basis vectors by performing a linear transformation according to

$$\hat{\boldsymbol{e}}_{\overline{a}} = \mathcal{R}^{a}{}_{\overline{a}}\,\hat{\boldsymbol{e}}_{a}. \tag{1.70}$$

This expression introduced components to the transformation matrix, $\mathcal{R}^{a}{}_{\overline{a}}$, that moves the basis vectors between the unbarred and the barred Cartesian coordinates. In Cartesian tensor analysis, the transformation is independent of space, whereas the transformation matrix is a function of space and time for general tensors. Much of the formalism developed here for a bulk rotation of Cartesian tensors has a straightforward generalization to the general tensors as considered in Chapter 3.

Although the transformation in equation (1.70) carries two indices, it is not a tensor. Instead, it is a matrix operator used to transform from one set of basis vectors to another.[12] Indeed, the slightly rightward placement of the lower index for the transformation in equation (1.70) is a convention used to express the transformation as a matrix

$$\mathcal{R}^{a}{}_{\overline{a}} \longleftrightarrow \mathcal{R} = \begin{bmatrix} \mathcal{R}^{1}{}_{\overline{1}} & \mathcal{R}^{1}{}_{\overline{2}} & \mathcal{R}^{1}{}_{\overline{3}} \\ \mathcal{R}^{2}{}_{\overline{1}} & \mathcal{R}^{2}{}_{\overline{2}} & \mathcal{R}^{2}{}_{\overline{3}} \\ \mathcal{R}^{3}{}_{\overline{1}} & \mathcal{R}^{3}{}_{\overline{2}} & \mathcal{R}^{3}{}_{\overline{3}} \end{bmatrix}. \tag{1.71}$$

Note that typically we dispense with the double-headed arrow correspondence symbol, and simply think of $\mathcal{R}^{a}{}_{\overline{a}}$ as $\mathcal{R}$. In that way we have the transpose of the transformation written as

$$(\mathcal{R}^{\mathrm{T}})^{a}{}_{\overline{a}} = \mathcal{R}_{\overline{a}}{}^{a} = \begin{bmatrix} \mathcal{R}^{1}{}_{\overline{1}} & \mathcal{R}^{2}{}_{\overline{1}} & \mathcal{R}^{3}{}_{\overline{1}} \\ \mathcal{R}^{1}{}_{\overline{2}} & \mathcal{R}^{2}{}_{\overline{2}} & \mathcal{R}^{3}{}_{\overline{2}} \\ \mathcal{R}^{1}{}_{\overline{3}} & \mathcal{R}^{2}{}_{\overline{3}} & \mathcal{R}^{3}{}_{\overline{3}} \end{bmatrix}. \tag{1.72}$$

We have occasion to work with the transpose transformation matrix since, as we show in Section 1.8.2, transformations between Cartesian coordinates with orthonormal basis vectors are orthogonal transformations, which means that the inverse of the transformation matrix equals to the transpose.

1.8.1 Inverse transformation

Assuming the transformation is invertible leads to the inverse transformation

$$\hat{\boldsymbol{e}}_{a} = (\mathcal{R}^{-1})^{\overline{a}}{}_{a}\,\hat{\boldsymbol{e}}_{\overline{a}}. \tag{1.73}$$

[11] The choice for origin is also arbitrary. Here we assume the origins to be the same, so that it is only the orientation that differs between the two Cartesian coordinate systems.

[12] We have more to say about the distinction between a matrix and tensor in Section 1.8.7.

As a self-consistency check we combine this relation with equation (1.70) to render

$$\hat{\boldsymbol{e}}_a = (\mathcal{R}^{-1})^{\overline{a}}{}_a \, \hat{\boldsymbol{e}}_{\overline{a}} = (\mathcal{R}^{-1})^{\overline{a}}{}_a \, \mathcal{R}^c{}_{\overline{a}} \, \hat{\boldsymbol{e}}_c. \tag{1.74}$$

This relation holds since

$$(\mathcal{R}^{-1})^{\overline{a}}{}_a \, \mathcal{R}^c{}_{\overline{a}} = \delta^c{}_a, \tag{1.75}$$

or as a matrix identity

$$\mathcal{R}^{-1} \, \mathcal{R} = I. \tag{1.76}$$

1.8.2 Orthogonal transformation

We now assume that the two sets of Cartesian basis vectors are orthonormal, which leads to the following constraint on the transformation matrix

$$\begin{aligned}
\delta_{\overline{a}\overline{b}} &= \hat{\boldsymbol{e}}_{\overline{a}} \cdot \hat{\boldsymbol{e}}_{\overline{b}} && && (1.77a)\\
&= \mathcal{R}^a{}_{\overline{a}} \, \hat{\boldsymbol{e}}_a \cdot \mathcal{R}^b{}_{\overline{b}} \, \hat{\boldsymbol{e}}_b && \text{relate barred to unbarred basis vectors} && (1.77b)\\
&= \mathcal{R}^a{}_{\overline{a}} \, \mathcal{R}^b{}_{\overline{b}} \, \hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b && \text{rearrange} && (1.77c)\\
&= \mathcal{R}^a{}_{\overline{a}} \, \mathcal{R}^b{}_{\overline{b}} \, \delta_{ab} && \text{orthonormality of basis vectors} && (1.77d)\\
&= (\mathcal{R}^{\mathrm{T}})_{\overline{a}}{}^a \, \delta_{ab} \, \mathcal{R}^b{}_{\overline{b}} && \text{rearrange,} && (1.77e)
\end{aligned}$$

where $\mathcal{R}^{\mathrm{T}}$ is the transpose of the transformation matrix whose components are (note the positioning of the indices as per equation (1.72))

$$(\mathcal{R}^{\mathrm{T}})_{\overline{a}}{}^a = \mathcal{R}^a{}_{\overline{a}}. \tag{1.78}$$

Written as a matrix equation, the identity (1.77e) means that

$$\mathcal{R}^{\mathrm{T}} \, \mathcal{R} = I. \tag{1.79}$$

This identity defines an orthogonal transformation, whereby the inverse matrix equals to the matrix transpose

$$\mathcal{R}^{-1} = \mathcal{R}^{\mathrm{T}}. \tag{1.80}$$

Consequently, the Cartesian basis vectors transform under rotations according to

$$\hat{\boldsymbol{e}}_{\overline{a}} = \mathcal{R}^a{}_{\overline{a}} \, \hat{\boldsymbol{e}}_a \quad \text{and} \quad \hat{\boldsymbol{e}}_a = (\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_a \, \hat{\boldsymbol{e}}_{\overline{a}}. \tag{1.81}$$

These transformation rules have implications for how components to vectors and tensors transform.

1.8.3 Geometric interpretation of orthogonal transformations

Orthogonal transformations convert one set of Cartesian coordinates to another. Geometrically, an orthogonal transformation corresponds to a rotation so long as the determinant of the transformation is $+1$, with Figure 1.5 illustrating this axis rotation in two dimensions.[13] For this two dimensional example, the rotation matrix can be written in terms of the cosine of the

[13] If the determinant of the transformation is -1, then the transformation involves a reflection in addition to a rotation, with the reflection changing the handedness of the basis vectors. We only consider right handed basis vectors in this book, so that we only consider orthogonal transformations of Cartesian tensors.

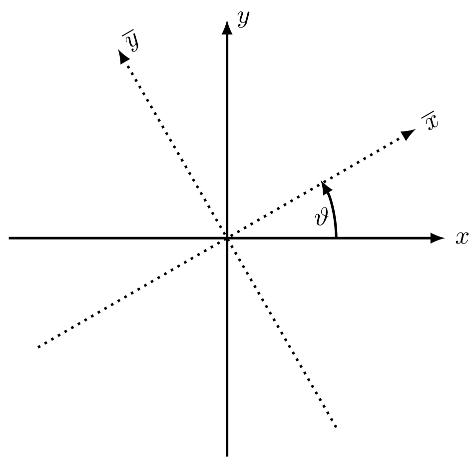


FIGURE 1.5: Counter-clockwise rotation through an angle ϑ of a right-handed horizontal Cartesian axes. The coordinate transformation is $x^{\overline{a}} = (\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_{a}\, x^{a}$, so that $\overline{x} = x\cos\vartheta - y\sin\vartheta$ and $\overline{y} = x\sin\vartheta + y\cos\vartheta$.

angles between the unit vectors; i.e., the *direction cosines*

$$\mathcal{R}^{a}{}_{\overline{a}} = \begin{bmatrix} \cos\vartheta & \sin\vartheta \\ -\sin\vartheta & \cos\vartheta \end{bmatrix} = \begin{bmatrix} \cos\vartheta & \cos(\pi/2-\vartheta) \\ \cos(\pi/2+\vartheta) & \cos\vartheta \end{bmatrix} = \begin{bmatrix} \hat{\boldsymbol{e}}_1\cdot\hat{\boldsymbol{e}}_{\overline{1}} & \hat{\boldsymbol{e}}_1\cdot\hat{\boldsymbol{e}}_{\overline{2}} \\ \hat{\boldsymbol{e}}_2\cdot\hat{\boldsymbol{e}}_{\overline{2}} & \hat{\boldsymbol{e}}_2\cdot\hat{\boldsymbol{e}}_{\overline{2}} \end{bmatrix}. \tag{1.82}$$

The final form of the rotation matrix reveals that it is built from the projection of the rotated basis vectors onto the original basis vectors. This result holds for rotations in three dimensions as well, thus leading to

$$\mathcal{R}^{a}{}_{\overline{a}} = \begin{bmatrix} \hat{\boldsymbol{e}}_1\cdot\hat{\boldsymbol{e}}_{\overline{1}} & \hat{\boldsymbol{e}}_1\cdot\hat{\boldsymbol{e}}_{\overline{2}} & \hat{\boldsymbol{e}}_1\cdot\hat{\boldsymbol{e}}_{\overline{3}} \\ \hat{\boldsymbol{e}}_2\cdot\hat{\boldsymbol{e}}_{\overline{1}} & \hat{\boldsymbol{e}}_2\cdot\hat{\boldsymbol{e}}_{\overline{2}} & \hat{\boldsymbol{e}}_2\cdot\hat{\boldsymbol{e}}_{\overline{3}} \\ \hat{\boldsymbol{e}}_3\cdot\hat{\boldsymbol{e}}_{\overline{1}} & \hat{\boldsymbol{e}}_3\cdot\hat{\boldsymbol{e}}_{\overline{2}} & \hat{\boldsymbol{e}}_3\cdot\hat{\boldsymbol{e}}_{\overline{3}} \end{bmatrix}. \tag{1.83}$$

With the basis vectors all normalized, elements of the rotation matrix are given by the cosine of the angle between the respective basis vectors. As such, one sometimes refers to this rotation matrix as the direction cosines matrix.

1.8.4 Transforming the coordinate representation of a vector

We introduced the transformation (1.73) according to how it acts on the basis vectors. Now consider how it acts on the coordinate representation of an arbitrary vector, as revealed by moving around brackets

$$\mathbf{P} = P^{a}\,\hat{\boldsymbol{e}}_{a} = P^{a}\,(\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_{a}\,\hat{\boldsymbol{e}}_{\overline{a}} \equiv P^{\overline{a}}\,\hat{\boldsymbol{e}}_{\overline{a}}. \tag{1.84}$$

The second equality made use of equation (1.81) for the transformation of the basis vector under an orthogonal transformation. The final equality reveals the transformation of the contravariant components according to

$$P^{\overline{a}} = P^{a}\,(\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_{a} = P^{a}\,\mathcal{R}_{a}{}^{\overline{a}}. \tag{1.85}$$

1.8.5 Form invariance of the scalar product

The above properties of an orthogonal transformation ensure that the scalar product takes on the same form regardless the choice of Cartesian coordinates

$$
\begin{aligned}
\mathbf{P}\cdot\mathbf{Q} &= P^{a}\,Q^{b}\,\delta_{ab} && \text{expose tensor indices \& Kronecker delta} && (1.86a)\\
&= P^{a}\,Q^{b}\,\hat{\boldsymbol{e}}_{a}\cdot\hat{\boldsymbol{e}}_{b} && \text{orthonormal basis vectors: } \hat{\boldsymbol{e}}_{a}\cdot\hat{\boldsymbol{e}}_{b}=\delta_{ab} && (1.86b)\\
&= P^{a}\,Q^{b}\,(\mathcal{R}^{\mathrm{T}})^{\bar{a}}{}_{a}\,\hat{\boldsymbol{e}}_{\bar{a}}\cdot(\mathcal{R}^{\mathrm{T}})^{\bar{b}}{}_{b}\,\hat{\boldsymbol{e}}_{\bar{b}} && \text{rotate basis vectors to bar frame} && (1.86c)\\
&= P^{a}\,(\mathcal{R}^{\mathrm{T}})^{\bar{a}}{}_{a}\,Q^{b}\,(\mathcal{R}^{\mathrm{T}})^{\bar{b}}{}_{b}\,\hat{\boldsymbol{e}}_{\bar{a}}\cdot\hat{\boldsymbol{e}}_{\bar{b}} && \text{rearrange} && (1.86d)\\
&= P^{\bar{a}}\,Q^{\bar{b}}\,\hat{\boldsymbol{e}}_{\bar{a}}\cdot\hat{\boldsymbol{e}}_{\bar{b}} && \text{transform vector components (1.84)} && (1.86e)\\
&= P^{\bar{a}}\,Q^{\bar{b}}\,\delta_{\bar{a}\bar{b}} && \text{orthonormal basis vectors: } \hat{\boldsymbol{e}}_{\bar{a}}\cdot\hat{\boldsymbol{e}}_{\bar{b}}=\delta_{\bar{a}\bar{b}}. && (1.86f)
\end{aligned}
$$

1.8.6 Transforming the coordinate representation of a second order tensor

We determine how the Cartesian coordinate components of a second order tensor, T^{ab}, transform by following the now familiar procedure for transforming the basis vectors. The key new facet is that we now have two basis vectors to carry around rather than just one

$$
\begin{aligned}
\mathbf{T} &= T^{ab}\,\hat{\boldsymbol{e}}_{a}\otimes\hat{\boldsymbol{e}}_{b} && \text{expose indices and basis vectors} && (1.87a)\\
&= T^{ab}\,(\mathcal{R}^{\mathrm{T}})^{\bar{a}}{}_{a}\,\hat{\boldsymbol{e}}_{\bar{a}}\otimes(\mathcal{R}^{\mathrm{T}})^{\bar{b}}{}_{b}\,\hat{\boldsymbol{e}}_{\bar{b}} && \text{rotate basis vectors to barred frame} && (1.87b)\\
&= T^{ab}\,(\mathcal{R}^{\mathrm{T}})^{\bar{a}}{}_{a}\,(\mathcal{R}^{\mathrm{T}})^{\bar{b}}{}_{b}\,\hat{\boldsymbol{e}}_{\bar{a}}\otimes\hat{\boldsymbol{e}}_{\bar{b}} && \text{rearrange} && (1.87c)\\
&\equiv T^{\bar{a}\bar{b}}\,\hat{\boldsymbol{e}}_{\bar{a}}\otimes\hat{\boldsymbol{e}}_{\bar{b}} && \text{define transformed tensor components.} && (1.87d)
\end{aligned}
$$

The final equality introduced the transformed components to the second order tensor

$$T^{\bar{a}\bar{b}} = T^{ab}\,(\mathcal{R}^{\mathrm{T}})^{\bar{a}}{}_{a}\,(\mathcal{R}^{\mathrm{T}})^{\bar{b}}{}_{b}. \tag{1.88}$$

Transformation of the $(1,1)$ representation of a second order tensor is given by

$$T^{\bar{a}}{}_{\bar{b}} = T^{a}{}_{b}\,(\mathcal{R}^{\mathrm{T}})^{\bar{a}}{}_{a}\,\mathcal{R}^{b}{}_{\bar{b}}, \tag{1.89}$$

which compares to equation (1.88) for the fully contravariant representation, $T^{\bar{a}\bar{b}}$. The transformation of the components to higher order tensors follows analogously by carrying around further basis vectors or basis one-forms.

1.8.7 Distinguishing between tensors and matrices

Matrices are useful for organizing the coordinate components to a tensor. For example, the $(1,0)$ coordinate representation of a first-order tensor

$$\mathbf{F} = F^{1}\,\hat{\boldsymbol{e}}_{1} + F^{2}\,\hat{\boldsymbol{e}}_{2} + F^{3}\,\hat{\boldsymbol{e}}_{3}, \tag{1.90}$$

has components that can be organized into a *row vector*

$$\boldsymbol{F} = (F^{1}, F^{2}, F^{3}). \tag{1.91}$$

We are justified in assigning the name "vector" in this context since we know that the array elements comprise the coordinate representation of the vector, $\mathbf{F}$. However, if we just see an array of numbers, say (Q^1, Q^2, Q^3), on its own, then we generally have no idea whether the elements of that array are related to each other, or if the list is just an ordering of numbers. If merely a list of numbers, then nothing necessarily changes when we alter coordinates. But if the array is the coordinate representation of a vector, then we know that the elements of the row vector are related, and we know how they change when the coordinates are rotated.

Consider two examples of objects that are not tensors. First, the rotation matrix, $\mathcal{R}^a{}_{\overline{a}}$, has elements built from the direction cosines according to equation (1.83). Although carrying two indices, this matrix is not the coordinate representation of a second order tensor. Rather, it is simply the matrix used to transform the components of tensors from one coordinate system to another. The rotation matrix is not a geometric object like a tensor. Rather, it is an ordered array that contains information for how a coordinate transformation alters the representation of geometric objects.

Although jumping ahead somewhat, we mention one further example encountered when studying general tensors in Chapter 3. Namely, the Christoffel symbols encountered in Section 3.15 are built from the covariant derivatives of the basis vectors. Although these derivatives are zero for Cartesian basis vectors, they are not necessarily zero when using general coordinates. The Christoffel symbols carry three indices, and yet they are not elements of a third order tensor since they do not transform as elements to a tensor. One way to understand this property is to note that a tensor that vanishes in one coordinate system is zero for all coordinate systems. It turns out that all of the Christoffel symbols vanish for Euclidean space using Cartesian coordinates, because the Cartesian basis vectors are spatially constant. However, there are some nonzero Christoffel symbols when representing Euclidean space with spherical coordinates or other general coordinates. By this example, we conclude that the Christoffel symbols cannot be components of a tensor.

The key point we re-emphasize is that a tensor is a geometric object that can be represented using any arbitrary set of coordinates. Since the tensor has an objective existence independent of coordinates, its coordinate components are constrained to transform in a precise manner when changing coordinates. These properties of tensors are generally not shared with arbitrary matrices, hence the importance of making the distinction between tensors and matrices.[14]

1.8.8 Coordinates are not tensors

Coordinates provide a numerical representation for points in space. Hence, they provide the means to assign numbers to tensors, which are otherwise purely geometrical objects. We here emphasize a conceptual point about coordinates that might seem pedantic, but it is fundamental to how we work with tensors, particularly with general tensors. So it is important to raise this point here to suppress a common confusion.

The transformation of Cartesian tensors between two Cartesian coordinate systems is realized as a rotation (when keeping the coordinate origins the same). This is a linear transformation. Furthermore, we can write the corresponding coordinate transformation in the same form as the transformation of tensor components

$$x^a = \mathcal{R}^a{}_{\overline{a}}\, x^{\overline{a}} \quad \text{and} \quad x^{\overline{a}} = (\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_a\, x^a. \tag{1.92}$$

[14] See this video or this video for pedagogical illustrations why it is important to distinguish a matrix from a tensor.

We might, therefore, be tempted to consider coordinates as the representation of a tensor, which is furthermore suggested by the use of tensor indices on coordinates. For Cartesian tensors this way of thinking does not lead to any algebraic errors, again since the transformation of coordinates is identical to that of tensors. However, this identification fails for the general tensors studied Chapter 3, where coordinate transformations are generally nonlinear.

To be more specific, note that the rotation matrix for Cartesian coordinates is a constant in space, so that the transformation between Cartesian coordinates is a linear operation and the rotation matrix equals to the matrix of partial derivatives

$$\partial x^a / \partial x^{\overline{a}} = \mathcal{R}^a{}_{\overline{a}} \quad \text{and} \quad \partial x^{\overline{a}} / \partial x^a = (\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_a. \tag{1.93}$$

This relation between partial derivatives and transformation matrix extends to the case of general tensors studied in Chapter 3, where the transformation is typically not a constant rotation matrix. Indeed, for general tensors the coordinate transformation is nonlinear, as exemplfied by the nonlinear between Cartesian coordinates and spherical coordinates. The resulting transformation matrix, built from partial derivatives, is a function of space, and the coordinate transformation is not in the form of a linear matrix operation as in equation (1.92). These distinctions highlight why coordinates are not components to tensors, even though they carry indices. Rather, coordinates are used to represent tensors, thus allowing us to assign numbers to tensors.

1.9 Homogeneity and isotropy of second order tensors

We have many occasions to consider symmetry properties of tensor fields, such as whether they are homogeneous (the same at each point in space) or display isotropy (have no dependence on orientation). When determining these properties for second order Cartesian tensors, we make use of the $(1,1)$ tensor representation

$$T^a{}_b = T^{ac}\, \mathbb{g}_{cb}, \tag{1.94}$$

which is the natural representation of a second order tensor (Section 1.5.1). We also follow this approach in Section 1.10 when decomposing a second order tensor into its irreducible parts.[15]

1.9.1 Transpose of a second order tensor

When represented as a matrix, the transpose of a second order tensor is obtained by swapping the rows and columns. In a similar manner we generate the transpose through the following

$$T_a{}^b = (T^b{}_a)^{\mathrm{T}} = \mathbb{g}_{ac}\, T^c{}_d\, \mathbb{g}^{db} = \mathbb{g}_{ac}\, T^{cb} \tag{1.95a}$$

$$T^b{}_a = (T_a{}^b)^{\mathrm{T}} = \mathbb{g}^{bc}\, T_c{}^d\, \mathbb{g}_{da} = T^{bd}\, \mathbb{g}_{da}. \tag{1.95b}$$

The transpose for the sharp and flat representations of the second order tensor follow a bit more simply

$$(T^{ab})^{\mathrm{T}} = T^{ba} \quad \text{and} \quad (T_{ab})^{\mathrm{T}} = T_{ba}. \tag{1.96}$$

[15]We certainly can perform such decompositions with other representations of a tensor. But we choose to focus on the $(1,1)$ representation here.

1.9.2 Trace of a tensor

The trace of a second order tensor is given by the contraction

$$\mathrm{trace}(\mathbf{T}) = T^{ac}\, \mathbb{g}_{ca} = T^{a}{}_{a} = T^{1}{}_{1} + T^{2}{}_{2} + T^{3}{}_{3}, \tag{1.97}$$

which is the sum of the diagonal components of the natural representation.

1.9.3 Homogeneous tensors

A tensor field is homogeneous if it possesses the same value at each point in space. For example, a uniform temperature field is homogeneous, as is a uniform velocity field. As defined, a homogeneous tensor field has no spatial dependence and thus it does not provide any means to distinguish points in space. Likewise, a time independent tensor is said to be homogeneous in time.

1.9.4 Isotropic tensors

A tensor field possesses isotropy if its representation remains independent of coordinate basis. A scalar tensor is, by definition, isotropic since it has no information about spatial directions. A nonzero vector field cannot be isotropic since it points in a particular direction and so its representation is dependent on the orientation of the basis vectors.

A second order Cartesian tensor, $\mathbf{J}$, is isotropic if its components are unchanged when undergoing rotation, so that

$$J^{\overline{a}}{}_{\overline{b}} = J^{a}{}_{b}\, \mathcal{R}^{b}{}_{\overline{b}}\, (\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_{a} \equiv J^{a}{}_{b}, \tag{1.98}$$

where we used equation (1.89) for the component transformation. For nonzero tensors, equation (1.98) is satisfied only if we can write

$$J^{a}{}_{b} = \lambda\, \delta^{a}{}_{b}, \tag{1.99}$$

with λ an arbitrary scalar. We verify this property through noting that

$$\delta^{a}{}_{b}\, \mathcal{R}^{b}{}_{\overline{b}}\, (\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_{a} = \mathcal{R}^{a}{}_{\overline{b}}\, (\mathcal{R}^{\mathrm{T}})^{\overline{a}}{}_{a} = \delta^{\overline{a}}{}_{\overline{b}}, \tag{1.100}$$

where the final equality made use of orthogonality of the rotation matrix. Hence, the most general second order isotropic tensor is proportional to the Kronecker (identity) tensor.

1.9.5 Decomposition into isotropic and anisotropic tensors

We often find it useful to decompose an arbitrary second order tensor into its anisotropic and isotropic components according to

$$T^{a}{}_{b} = \underbrace{T^{a}{}_{b} - \frac{T^{p}{}_{p}\, \delta^{a}{}_{b}}{N}}_{\text{anisotropic}} + \underbrace{\frac{T^{p}{}_{p}\, \delta^{a}{}_{b}}{N}}_{\text{isotropic}} \tag{1.101}$$

where

$$N = \delta^{p}{}_{p} \tag{1.102}$$

is the number of space dimensions, which equals to the trace of the Kronecker tensor. By extracting the isotropic portion, we know that the remaining portion is anisotropic by construction.

1.10 Irreducible parts of a second order tensor

Second order tensors have nine degrees of freedom when working in $N = 3$ dimensional space. It can be useful to decompose these tensors into their *irreducible tensorial parts* as presented here, with these parts generally representing distinct kinematic properties that help to understand the physical nature of the tensor. The decomposition shares much with the decomposition into anisotropic and isotropic tensors in Section 1.9.5, and with the following results holding for general tensors.

Any second order tensor can be decomposed into its symmetric and anti-symmetric parts according to

$$\mathbf{T} = (\mathbf{T} + \mathbf{T}^{\mathrm{T}})/2 + (\mathbf{T} - \mathbf{T}^{\mathrm{T}})/2 = \mathbf{S} + \mathbf{A}. \tag{1.103}$$

Working with the components in the $(1,1)$ representation of the tensor leads to

$$T^{a}{}_{b} = \underbrace{(T^{a}{}_{b} + T_{b}{}^{a})/2}_{\text{symmetric}} + \underbrace{(T^{a}{}_{b} - T_{b}{}^{a})/2}_{\text{anti-symmetric}} \equiv S^{a}{}_{b} + A^{a}{}_{b}. \tag{1.104}$$

The symmetric tensor, $\mathbf{S}$, is so-named because it satisfies

$$\mathbf{S} = \mathbf{S}^{\mathrm{T}} \longleftrightarrow S^{a}{}_{b} = S_{b}{}^{a}. \tag{1.105}$$

That is, swapping rows and columns leaves elements of a symmetric tensor unchanged. In three space dimensions, a symmetric second order tensor has six (rather than nine) degrees of freedom. The anti-symmetric tensor (also called the skew-symmetric tensor) satisfies

$$\mathbf{A} = -\mathbf{A}^{\mathrm{T}} \longleftrightarrow A^{a}{}_{b} = -A_{b}{}^{a}, \tag{1.106}$$

which means it has three degrees of freedom (for $N = 3$ space dimensions) and has zero elements along its diagonal

$$A^{1}{}_{1} = A^{2}{}_{2} = A^{3}{}_{3} = 0. \tag{1.107}$$

The final irreducible part of a tensor is the trace, which we introduced in Section 1.9.2 and it is given by the sum of the diagonal elements

$$T^{a}{}_{a} = T^{1}{}_{1} + T^{2}{}_{2} + T^{3}{}_{3} = S^{a}{}_{a}, \tag{1.108}$$

with the second equality following since the skew symmetric tensor, $\mathbf{A}$, has zero for its diagonal elements. We are thus led to the irreducible decomposition of an arbitrary second order tensor

$$T^{a}{}_{b} = \underbrace{S^{p}{}_{p}\,\delta^{a}{}_{b}/N}_{\text{trace}} + \underbrace{(S^{a}{}_{b} - S^{p}{}_{p}\,\delta^{a}{}_{b}/N)}_{\text{deviator}} + \underbrace{A^{a}{}_{b}}_{\text{skew}}. \tag{1.109}$$

The combination,

$$(S^{\mathrm{dev}})^{a}{}_{b} = S^{a}{}_{b} - S^{p}{}_{p}\,\delta^{a}{}_{b}/N, \tag{1.110}$$

is known as the deviatoric tensor, which, by construction, has zero trace

$$(S^{\text{dev}})^{a}{}_{a} = 0. \tag{1.111}$$

1.11 Dot and double-dot notation for contractions

As seen many times in this chapter, we often use a dot notation for the contraction of two first order tensors, such as for the scalar product, $\mathbf{F} \cdot \mathbf{G}$, encountered in equation (1.14). The are additional occasions to contract higher order tensors, yet in this case we must follow a convention to know what indices participate in the contraction. Following *Tromp* (2025a) and others, we define the dot as a contraction between the last index of the first tensor and the first index of the second tensor. For example, let $\mathbf{T}$ be a second order tensor and $\mathbf{A}$ a first order tensor. Their dot product is thus equal to a first order tensor

$$\mathbf{T} \cdot \mathbf{A} = \mathbf{F}, \tag{1.112}$$

which can be written in components as

$$T^{a}{}_{b}\, A^{b} = T^{ab}\, A_{b} = F^{a}. \tag{1.113}$$

Alternatively, consider the dot product, $\mathbf{A} \cdot \mathbf{T} = \mathbf{G}$, which has components

$$A^{b}\, T_{b}{}^{a} = A^{b}\, (T^{a}{}_{b})^{\mathrm{T}} = A_{b}\, (T^{ab})^{\mathrm{T}} = A_{b}\, T^{ba} = G^{a}. \tag{1.114}$$

Evidently, $\mathbf{F} = \mathbf{G}$, only if the second order tensor is symmetric, $\mathbf{T} = \mathbf{T}^{\mathrm{T}}$.

There are occasions to contract both indices between two second order tensors, in which we make use of the colon with the following index convention

$$\mathbf{D} \colon \mathbf{T} = D^{ab}\, T_{ab} = D_{ab}\, T^{ab}. \tag{1.115}$$

When performed between two symmetric tensors, the ordering of the indices does not matter, whereas index placement is important when considering anti-symmetric tensors.

1.12 Exercises

EXERCISE 1.1: PRODUCT OF SYMMETRIC MATRICES
Let $A = A^{\mathrm{T}}$ and $B = B^{\mathrm{T}}$ be two symmetric matrices. Under what condition is their product also symmetric: $A\, B = (A\, B)^{\mathrm{T}}$?

EXERCISE 1.2: PRODUCT OF SYMMETRIC AND ANTI-SYMMETRIC MATRICES AND TENSORS

(a) Let $A = -A^{\mathrm{T}}$ be an anti-symmetric matrix, and $S = S^{\mathrm{T}}$ be a symmetric matrix. Show that the trace of their product vanishes: $\mathrm{trace}(A\, S) = 0$.

(b) In terms of tensors, show that the double contraction of an anti-symmetric tensor with a symmetric tensor vanishes: $A^{m}{}_{n}\, S^{m}{}_{n} = 0$.

EXERCISE 1.3: PROJECTION OPERATOR
Consider an arbitrary direction in space specified by the unit direction, $\hat{\boldsymbol{n}}$, with components,

$\hat{n}_a$. Define the symmetric tensor

$$P_{ab} = \mathbb{g}_{ab} - \hat{n}_a\,\hat{n}_b, \tag{1.116}$$

so that

$$P_{ab}\,T^{bc} = T_a{}^c - T^{bc}\,\hat{n}_a\,\hat{n}_b. \tag{1.117}$$

Show that $P_{ab}\,T^{bc}$ is the projection of $\mathbf{T}$ onto the plane perpendicular to the direction $\hat{\boldsymbol{n}}$. Do so by showing that

$$P_{ab}\,T^{bc}\,(\mathbb{g}^{ad}\,\hat{n}_d) = 0 \quad \forall\, c. \tag{1.118}$$

This result motivates referring to P_{ab} as a *projection operator*.

EXERCISE 1.4: CONTRACTIONS OF THE LEVI-CIVITA TENSOR
In this exercise we verify some standard identities involving contractions with the Levi-Civita tensor.

(a) Show, through explicit display of a few specific cases, that the following identity holds

$$\delta^{df}\,\epsilon_{bcf}\,\epsilon_{aed} = \epsilon_{bc}{}^{d}\,\epsilon_{aed} = \delta_{ba}\,\delta_{ce} - \delta_{be}\,\delta_{ca}. \tag{1.119}$$

(b) Verify that if $c = e$ then equation (1.119) reduces to

$$\epsilon_{be}{}^{d}\,\epsilon_{aed} = 2\,\delta_{ba}. \tag{1.120}$$

EXERCISE 1.5: TRIPLE VECTOR PRODUCT IDENTITY
Use the Levi-Civita tensor to derive the triple cross product identity

$$\mathbf{P} \times (\mathbf{Q} \times \mathbf{R}) = (\mathbf{P}\cdot\mathbf{R})\,\mathbf{Q} - (\mathbf{P}\cdot\mathbf{Q})\,\mathbf{R}, \tag{1.121}$$

EXERCISE 1.6: SCALAR PRODUCT OF TWO VECTOR PRODUCTS
Make use of the Levi-Civita identity (1.119) to derive the following identity for the scalar product of two vector cross products

$$(\mathbf{P}\times\mathbf{Q})\cdot(\mathbf{R}\times\mathbf{S}) = (\mathbf{P}\cdot\mathbf{R})\,(\mathbf{Q}\cdot\mathbf{S}) - (\mathbf{P}\cdot\mathbf{S})\,(\mathbf{Q}\cdot\mathbf{R}). \tag{1.122}$$

EXERCISE 1.7: MAGNITUDE IDENTITIES
Prove the following two identities.

(a) $|\boldsymbol{A}\cdot\boldsymbol{B}| \le |\boldsymbol{A}|\,|\boldsymbol{B}|$
(b) $|\boldsymbol{A}+\boldsymbol{B}| \le |\boldsymbol{A}| + |\boldsymbol{B}|$.

Chapter 2

Vector calculus

In this chapter we build from the tensor algebra of Chapter 1 to here develop elements of Cartesian tensor calculus, also known as vector calculus. This material is central to nearly every chapter in this book.

CHAPTER GUIDE

The material in this chapter can be found in various forms in nearly all books on calculus with analytic geometry. Particular treatments, with applications to physics, are given in the following. Recall from Chapter 1 that we are mindful of the distinction between a tensor and its coordinate representation. We retain that distinction in this chapter, though mostly focus on particular Cartesian coordinate representations since we are concerned with analytical/operational aspects of tensor fields.

- Feynman Lectures: Chapters 2 and 3 in Volume II of the Feynman Lectures offers insightful discussions of vector differential calculus. Although written for students of electrodynamics, many of Feynman's examples are drawn from fluid mechanics.
- Div, Grad, Curl and all That (*Schey*, 2004): This text presents the methods and theorems of vector calculus in a manner that greatly assists the development of intuition.
- Chapter 2 in *Segel* (1987) provides a lucid review of vector calculus using Cartesian tensors.
- Theory and Problems of Vector Analysis (*Spiegel*, 1974a): This "Schaum's Outline Series" book has nearly 500 worked exercises and provides a useful resource to develop problem solving in vector calculus. Some of the exercises in Section 2.9 at the end of this chapter are drawn from *Spiegel* (1974a).
- This video from 3Blue1Brown provides some compelling graphics to support intuition for the divergence and curl operators.
- This Youtube channel from Steve Brunton offers some pedagogical lectures on vector calculus.

2.1 Gradient of a scalar field

Consider a real valued scalar field defined on Euclidean space that is a function of Cartesian coordinates, $\psi(\boldsymbol{x})$. We can estimate the value of the field at an adjacent point an infinitesimal distance away, $\boldsymbol{x} + \mathrm{d}\boldsymbol{x}$, through use of a truncated Taylor series

$$\psi(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) = \psi(\boldsymbol{x}) + \frac{\partial \psi}{\partial x^1}\,\mathrm{d}x^1 + \frac{\partial \psi}{\partial x^2}\,\mathrm{d}x^2 + \frac{\partial \psi}{\partial x^3}\,\mathrm{d}x^3 + \mathcal{O}\left(\mathrm{d}\boldsymbol{x} \cdot \mathrm{d}\boldsymbol{x}\right) \tag{2.1a}$$

$$\approx \left[1 + \mathrm{d}x^a\,\partial_a\right]\psi(\boldsymbol{x}), \tag{2.1b}$$

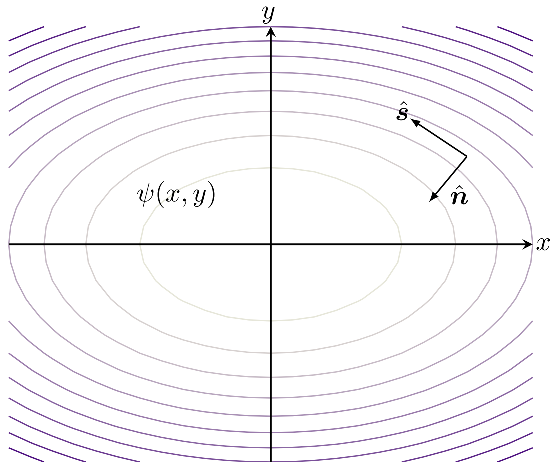


FIGURE 2.1: Contours of a scalar field $\psi(x, y) = -x^2/2 - y^2$. At any point in space, the gradient, $\nabla\psi = -(x\,\hat{\boldsymbol{x}} + 2\,y\,\hat{\boldsymbol{y}})$, points in the direction of steepest increase (ascent) and orients the unit normal, $\hat{\boldsymbol{n}} = |\nabla\psi|^{-1}\,\nabla\psi$. The unit tangent, $\hat{\boldsymbol{s}}$, points in a direction tangent to a ψ isosurface so that it is everywhere orthogonal to the direction of steepest ascent: $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{s}} = 0$. We follow the convention in which the unit normal is oriented to the left of the unit tangent when facing in the tangent direction.

where we dropped higher order terms to reach the final approximate expression. We also introduced the shorthand notation for the partial derivative operator

$$\partial_a = \frac{\partial}{\partial x^a}, \tag{2.2}$$

which is a notation used throughout this book. Notice how the index on the partial derivative naturally sits downstairs, in the covariant position. We can thus introduce the Cartesian gradient operator according to

$$\nabla = \hat{\boldsymbol{x}}\,\partial_x + \hat{\boldsymbol{y}}\,\partial_y + \hat{\boldsymbol{z}}\,\partial_z = \boldsymbol{e}^a\,\partial_a, \tag{2.3}$$

in which case

$$\psi(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) \approx (1 + \mathrm{d}\boldsymbol{x} \cdot \nabla)\,\psi(\boldsymbol{x}). \tag{2.4}$$

The second equality in equation (2.3) made use of the basis one-forms in Cartesian coordinates from Section 1.2.3. Evidently, the gradient operator naturally appears as a one-form.

2.1.1 Direction of steepest ascent

Using the approximate relation (2.4), and the geometric expression (1.20) for the scalar product, renders

$$\psi(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) - \psi(\boldsymbol{x}) \approx |\mathrm{d}\boldsymbol{x}|\,|\nabla\psi|\cos\vartheta, \tag{2.5}$$

where ϑ is the angle between the differential increment, $\mathrm{d}\boldsymbol{x}$, and the gradient, $\nabla\psi$. Orienting the increment $\mathrm{d}\boldsymbol{x}$ so that $\vartheta = 0$ ensures that $\psi(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) - \psi(\boldsymbol{x})$ is maximal. Consequently, $\nabla\psi$ points in the direction of *steepest ascent* across constant ψ isosurfaces (Figure 2.1). The opposite direction is that of *steepest descent*, where $\vartheta = \pi$.

2.1.2 Tangent to an isosurface

Consider a family of isosurfaces defined by points satisfying

$$\psi(\boldsymbol{x}) = \text{constant}. \tag{2.6}$$

Figure 2.1 shows a two dimensional example where the isosurfaces are lines where ψ is a constant. As another example, consider $\psi(\boldsymbol{x}) = \psi(r)$, where $r^2 = \boldsymbol{x} \cdot \boldsymbol{x}$ is the squared radius of a sphere. Isosurfaces for this spherically symmetric function are spherical surfaces of radius r.

In general, moving along an isosurface keeps the scalar field unchanged. Let $\hat{\boldsymbol{s}}$ have magnitude unity and point in the direction tangent to the isosurface at any point determined by the Cartesian coordinate, $\boldsymbol{x}$. By construction[1]

$$\psi(\boldsymbol{x} + \hat{\boldsymbol{s}}\, \mathrm{d}s) - \psi(\boldsymbol{x}) = 0, \tag{2.7}$$

where $\mathrm{d}s$ is an infinitesimal arc length along the tangent direction. In words, this identity says that if we move an infinitesimal distance in the direction tangent to the isosurface, then the function ψ does not change its value. Now expanding this identity in a Taylor series leads to the vanishing of the tangential partial derivative

$$\hat{\boldsymbol{s}} \cdot \nabla \psi = \partial_s \psi = 0. \tag{2.8}$$

That is, isosurfaces of a function ψ are defined by directions along which the partial derivative of the function vanishes. For the spherically symmetric function, $\psi(\boldsymbol{x}) = \psi(r)$, the tangent vector points in either of the two angular directions along the spherical surface.

2.1.3 Unit normal to an isosurface

We may normalize the direction of maximal ascent, in which case we define the unit normal (a one-form)

$$\hat{\boldsymbol{n}} = |\nabla \psi|^{-1} \nabla \psi. \tag{2.9}$$

By construction, the gradient computed in the $\hat{\boldsymbol{n}}$ direction yields the maximum change for the function so that the *normal derivative* is given by

$$\hat{\boldsymbol{n}} \cdot \nabla \psi = |\nabla \psi|. \tag{2.10}$$

For the spherically symmetric example,

$$\hat{\boldsymbol{n}} = \frac{\boldsymbol{x}}{|\boldsymbol{x}|} = \hat{\boldsymbol{r}}, \tag{2.11}$$

where $\hat{\boldsymbol{r}}$ is the unit vector pointing radially outward from the origin. In this case the normal derivative is equal to the radial derivative

$$\hat{\boldsymbol{n}} \cdot \nabla \psi = \partial_r \psi \qquad \text{spherically symmetric } \psi. \tag{2.12}$$

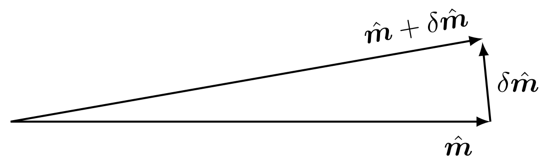


FIGURE 2.2: The infinitesimal change to a unit direction, $\delta\hat{\boldsymbol{m}}$, is orthogonal to itself: $\delta\hat{\boldsymbol{m}} \cdot \hat{\boldsymbol{m}} = 0$. The reason is that the unit direction is constrained to retain its unit length, so that the only way that it can change is to change its direction. In this image we have $|\hat{\boldsymbol{m}} + \delta\hat{\boldsymbol{m}}| = |\hat{\boldsymbol{m}}| = 1$, which requires $\delta\hat{\boldsymbol{m}} \cdot \hat{\boldsymbol{m}} = 0$. Evidently, $\delta\hat{\boldsymbol{m}}$ is orthogonal to $\hat{\boldsymbol{m}}$ in the limit that $\delta\hat{\boldsymbol{m}}$ gets tiny.

2.1.4 Unit directions change only via rotation

Consider an arbitrary unit direction, $\hat{\boldsymbol{m}}$, that is generally a function of space. The unit direction has unit magnitude,

$$\hat{\boldsymbol{m}} \cdot \hat{\boldsymbol{m}} = 1, \tag{2.13}$$

with this property holding at each point in space. Unit directions can only be modified through changes in their orientation since their magnitude is everywhere fixed at unity. Hence, they differ in space only through rotations. An important consequence of this property is that the infinitesimal spatial change to a unit direction is perpendicular to unit direction itself (see Figure 2.2). We see this property through considering an arbitrary infinitesimal change, symbolized by δ, in which

$$0 = \delta(1) = \delta\,(\hat{\boldsymbol{m}} \cdot \hat{\boldsymbol{m}}) = 2\,\hat{\boldsymbol{m}} \cdot \delta\hat{\boldsymbol{m}}. \tag{2.14}$$

In Section 11.3, we formally show that the constraint

$$\delta\hat{\boldsymbol{m}} \cdot \hat{\boldsymbol{m}} = 0 \tag{2.15}$$

means that unit direction changes can only arise from rotations. Even so, the above assertion should make intuitive sense, with Figure 2.2 illustrating this property.

2.1.5 Showing that $\delta\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}} = 0$

As an illustration of the constraint (2.15), we verify that it holds for the special case of a unit normal defined according to equation (2.9) for surfaces of constant scalar field

$$\hat{\boldsymbol{n}} = |\nabla\psi|^{-1}\,\nabla\psi. \tag{2.16}$$

The proof follows first by writing

$$\delta\hat{\boldsymbol{n}} = |\nabla\psi|^{-1}\,\left[\delta(\nabla\psi) - \hat{\boldsymbol{n}}\,\delta|\nabla\psi|\right], \tag{2.17}$$

so that

$$|\nabla\psi|\,\hat{\boldsymbol{n}} \cdot \delta\hat{\boldsymbol{n}} = \hat{\boldsymbol{n}} \cdot \delta(\nabla\psi) - \delta(|\nabla\psi|) = \frac{\nabla\psi \cdot \delta(\nabla\psi)}{|\nabla\psi|} - \delta|\nabla\psi|. \tag{2.18}$$

[1]Sometimes in this book we write $\hat{\boldsymbol{t}}$ for the unit tangent.

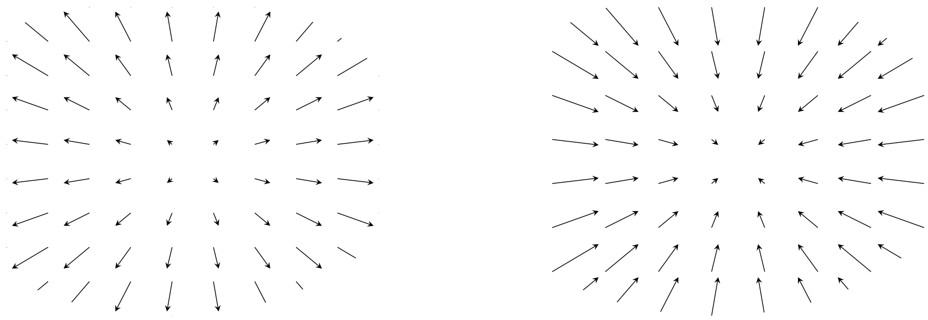


FIGURE 2.3: Two vector fields with a non-zero horizontal divergence. Left panel: The vector field $\mathbf{F} = x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}}$ has a spatially constant positive divergence at each point with $\nabla \cdot \mathbf{F} = 2$. We thus say that the vector field is "diverging from each point." Right panel: with the opposite sign, the vector field $\mathbf{G} = -\mathbf{F} = -x\,\hat{\boldsymbol{x}} - y\,\hat{\boldsymbol{y}}$ has a spatially constant negative divergence at each point with $\nabla \cdot \mathbf{G} = -2$. We thus say that the vector field is "converging to each point." Note that these two vector fields have zero curl, $\nabla \times \mathbf{F} = \nabla \times \mathbf{G} = 0$.

We now make use of the identity

$$\delta(|\nabla\psi|) = \delta(\sqrt{\nabla\psi \cdot \nabla\psi}) = \frac{1}{2\sqrt{\nabla\psi \cdot \nabla\psi}}\delta(\nabla\psi \cdot \nabla\psi) = \frac{\nabla\psi \cdot \delta(\nabla\psi)}{|\nabla\psi|}, \tag{2.19}$$

in which case we have shown that $\delta\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}} = 0$.

2.1.6 Notation for the derivative operators

The discussion in this section made use of the Cartesian partial derivative operator, ∂_a, as well as the gradient operator, ∇. The Cartesian gradient operator becomes the covariant gradient operator when moving to general tensors, with the covariant gradient also written as ∇. When acting on first or higher order tensors, the covariant gradient picks up some extra terms beyond the familiar partial derivatives. To facilitate a translation of the equations in this chapter to those in general tensors, we commonly make use of the ∇_a symbol rather than ∂_a. Again, they are identical for Cartesian tensors.

2.2 Divergence of a vector field

The divergence of a vector field, $\mathbf{F}$, is the scalar product of the gradient operator with the vector

$$\operatorname{div}(\mathbf{F}) = \nabla \cdot \mathbf{F} = \nabla_a F^a \begin{cases} > 0 \Rightarrow \text{diverging vector field} \\ < 0 \Rightarrow \text{converging vector field} \\ = 0 \Rightarrow \text{divergence-free (or non-divergent) vector field.} \end{cases} \tag{2.20}$$

If the vector field in the surrounding neighborhood of a point is directed away from that point, then the vector field is diverging as if there is a source at the point (Figure 2.3). In this case the divergence of the vector field is positive. The converse occurs for a vector field converging to a point as if there is a sink.

If the vector field under consideration is the velocity field of a moving fluid, then these considerations are directly related to mass conservation as studied in VOLUME 2. That discussion motivates us to consider a positive divergence for a vector field as representing the creation

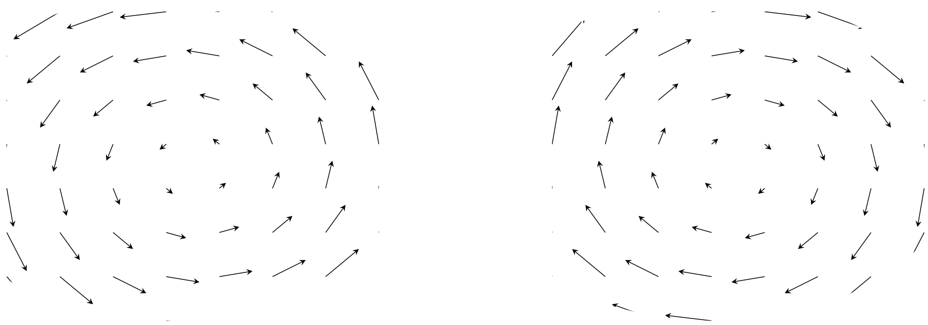


FIGURE 2.4: A horizontal vector field with a constant curl and zero divergence. Left panel: $\mathbf{F} = -y\,\hat{\boldsymbol{x}} + x\,\hat{\boldsymbol{y}}$, $\Longrightarrow \nabla \times \mathbf{F} = 2\,\hat{\boldsymbol{z}}$ and $\nabla \cdot \mathbf{F} = 0$. Right panel: $\mathbf{G} = -\mathbf{F}$ so that $\nabla \times \mathbf{G} = -2\,\hat{\boldsymbol{z}}$.

of "stuff" at a point where there is a positive divergence. Again for the case of a fluid velocity, there is a net divergence if more fluid leaves a point than enters, and the converse holds if the velocity field is converging. We further discuss these ideas as part of our study of the divergence theorem in Section 2.8.

2.2.1 Divergence of a scalar field times a vector field

We have many opportunities to make use of properties of the divergence operator following from application of the chain rule. For example, use of the chain rule indicates that the divergence of a scalar field times a vector field is given by

$$\nabla \cdot (\phi\,\mathbf{F}) = \nabla_a(\phi\,F^a) \tag{2.21a}$$

$$= F^a\,\nabla_a\phi + \phi\,\nabla_a F^a \tag{2.21b}$$

$$= \mathbf{F} \cdot \nabla\phi + \phi\,\nabla \cdot \mathbf{F}. \tag{2.21c}$$

2.2.2 Laplacian of a scalar field

The Laplacian of a scalar field is the divergence of the gradient

$$\nabla^2\psi = \nabla \cdot \nabla\psi. \tag{2.22}$$

Scalar fields that have a vanishing Laplacian are said to be *harmonic*

$$\nabla^2\psi = 0 \qquad \text{harmonic function.} \tag{2.23}$$

The name *harmonic* originates from the relation of harmonic functions to characteristic vibrational modes of a taut string such as those found on musical instruments (when played with skill). Furthermore, harmonic functions play a central role in complex analysis.

2.3 Curl of a vector field

The curl characterizes how a vector field spins around each point in space. For example, in VOLUME 3 we study the vorticity field, which is the curl of the velocity.

2.3.1 Computing the curl

We measure the curl of a vector by computing the cross product of the gradient operator with the vector field. Hence, just like the cross product from Section 1.6, the curl is specified by both a magnitude and a direction

$$
\begin{aligned}
\operatorname{curl}(\mathbf{F}) &= \nabla \times \mathbf{F} && (2.24\text{a})\\
&= \boldsymbol{e}^a \, \nabla_a \times \boldsymbol{e}_b \, F^b && \text{coordinate representation} \quad (2.24\text{b})\\
&= \boldsymbol{e}^a \, \times \nabla_a \, (\boldsymbol{e}_b \, F^b) && \text{move derivative operator} \quad (2.24\text{c})\\
&= (\boldsymbol{e}^a \times \boldsymbol{e}_b) \, \nabla_a F^b + F^b \, (\boldsymbol{e}^a \times \nabla_a \boldsymbol{e}_b) && \text{product rule} \quad (2.24\text{d})\\
&= (\boldsymbol{e}^a \times \boldsymbol{e}_b) \, \nabla_a F^b && \nabla_a \boldsymbol{e}_b = 0 \text{ for Cartesian coordinates} \quad (2.24\text{e})\\
&= \delta^{ad} \, (\boldsymbol{e}_d \times \boldsymbol{e}_b) \, \nabla_a F^b && \delta^{ad} \, \boldsymbol{e}_d = \boldsymbol{e}^a = 0 \quad (2.24\text{f})\\
&= (\epsilon_{dbg} \, \nabla^d \, F^b) \, \boldsymbol{e}^g && \delta^{ad} \, \nabla_a = \nabla^d . \quad (2.24\text{g})
\end{aligned}
$$

To reach this result we set $\nabla_a \boldsymbol{e}_b = 0$ since the Cartesian basis vectors are spatially constant. We also made use of the relation (1.46) for the cross product of basis vectors. Expanding the final expression, using the Cartesian identity $\nabla^d = \nabla_d = \partial_d$, leads to

$$
\operatorname{curl}(\boldsymbol{F}) = \nabla \times \boldsymbol{F} = \left[\frac{\partial F^3}{\partial x^2} - \frac{\partial F^2}{\partial x^3} \right] \boldsymbol{e}^1 + \left[\frac{\partial F^1}{\partial x^3} - \frac{\partial F^3}{\partial x^1} \right] \boldsymbol{e}^2 + \left[\frac{\partial F^2}{\partial x^1} - \frac{\partial F^1}{\partial x^2} \right] \boldsymbol{e}^3, \tag{2.25}
$$

which can also be written as a determinant

$$
\nabla \times \boldsymbol{F} = \det \begin{bmatrix} \boldsymbol{e}^1 & \boldsymbol{e}^2 & \boldsymbol{e}^3 \\ \partial_1 & \partial_2 & \partial_3 \\ F^1 & F^2 & F^3 \end{bmatrix} . \tag{2.26}
$$

The horizontal vector field $\mathbf{F} = x \, \hat{\boldsymbol{x}} + y \, \hat{\boldsymbol{y}}$ shown in Figure 2.3 has zero curl yet non-zero divergence. Figure 2.4 shows another vector field, $\mathbf{F} = -y \, \hat{\boldsymbol{x}} + x \, \hat{\boldsymbol{y}}$, with zero divergence yet nonzero curl $\nabla \times \mathbf{F} = 2 \, \hat{\boldsymbol{z}}$.

2.3.2 Curl-free vector fields

There are some cases of physically relevant vector fields that have a vanishing curl

$$
\nabla \times \boldsymbol{F} = 0. \tag{2.27}
$$

We sometimes refer to such curl-free vectors as *irrotational*. In fluid mechanics a curl-free velocity field has zero vorticity, which is a property maintained by linear gravity waves in the absence of rotation (see VOLUME 4). We illustrate a curl-free vector field in Figure 2.5.

The curl of a gradient vanishes, which follows from

$$
\nabla \times \nabla \psi = \boldsymbol{e}^a \, \nabla_a \times \boldsymbol{e}^b \, \nabla_b \psi = (\boldsymbol{e}^a \times \boldsymbol{e}^b) \, \nabla_a \nabla_b \psi = 0, \tag{2.28}
$$

where the final equality follows since $\boldsymbol{e}^a \times \boldsymbol{e}^b$ is anti-symmetric on the labels ab

$$
\boldsymbol{e}^a \times \boldsymbol{e}^b = -\boldsymbol{e}^b \times \boldsymbol{e}^a, \tag{2.29}
$$

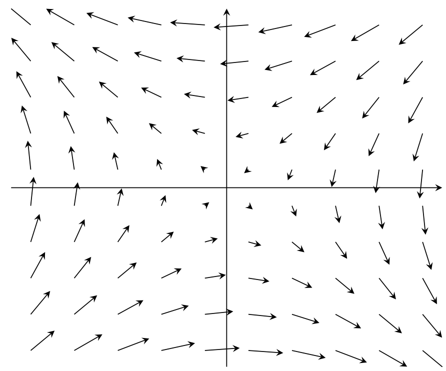


FIGURE 2.5: A horizontal vector field with a zero curl, where $\mathbf{F} = -\nabla\psi$ with the scalar potential given by $\psi = \sin(x/10)\,\sin(y/10)$.

whereas $\nabla_a \nabla_b$ is symmetric on these labels

$$\nabla_a \nabla_b = \nabla_b \nabla_a. \tag{2.30}$$

This property allows us to introduce a scalar field whose gradient equals to the curl-free vector field

$$\mathbf{F} = -\nabla\psi \qquad \text{scalar potential.} \tag{2.31}$$

The scalar ψ is known as the *scalar potential*. We may be familiar with the scalar potential for the gravitational force, in which ψ is called the gravitational potential (see Section 11.7).

2.3.3 Vector fields that are both curl-free and divergence-free

Consider a vector field that has zero curl *and* zero divergence. The curl-free property means that

$$\nabla \times \mathbf{F} = 0 \Longrightarrow \mathbf{F} = -\nabla\psi. \tag{2.32}$$

The divergence-free property means that ψ is a harmonic function (Section 2.2.2)

$$\nabla \cdot \nabla\psi = \nabla^2\psi = 0. \tag{2.33}$$

The velocity field arising from a linear non-rotating gravity wave (see VOLUME 4) in a Boussinesq ocean (see VOLUME 2) maintains zero vorticity and zero divergence. Furthermore, curl-free and divergence-free velocity fields are commonly encountered in engineering applications such as aerodynamics (e.g., see *Acheson* (1990) for many elementary examples).

2.3.4 Index gymnastics and practice with curl identities

We close this section by deriving a suite of identities involving the curl operator. These identities arise in various places within this book, particulely when developing dynamical equations for vorticity. Furthermore, by making use of the rules for general tensor analysis developed in Chapter 3, these identities take on the same form regardless the coordinate choice. The derivations are presented in detail to facilitate understanding of the various steps involving index gymnastics. After some practice, many of the steps can be readily skipped.

Vanishing divergence of the curl

Making use of equation (2.24g) for the curl leads to its zero divergence

$$
\begin{aligned}
\nabla \cdot (\nabla \times \mathbf{F}) &= \boldsymbol{e}^{a}\, \nabla_{a} \cdot [\epsilon_{dbg}\, \nabla^{d}\, F^{b})\, \boldsymbol{e}^{g}] && \text{equation (2.24g)} && (2.34a)\\
&= \boldsymbol{e}^{a} \cdot \boldsymbol{e}^{g}\, \epsilon_{dbg}\, \nabla_{a}\, \nabla^{d}\, F^{b} && \nabla_{a} \cdot (\epsilon_{dbg}\, \boldsymbol{e}^{g}) = 0 && (2.34b)\\
&= \delta^{ag}\, \epsilon_{dbg}\, \nabla_{a}\, \nabla^{d}\, F^{b} && \text{orthonormality, } \boldsymbol{e}^{a} \cdot \boldsymbol{e}^{g} = \delta^{ag} && (2.34c)\\
&= (\epsilon_{dbg}\, \nabla^{g}\, \nabla^{d})\, F^{b} && \delta^{ag}\, \nabla_{a} = \nabla^{g} && (2.34d)\\
&= 0 && \epsilon_{dbg}\, \nabla^{g}\, \nabla^{d} = 0. && (2.34e)
\end{aligned}
$$

The final equality holds since $\nabla^{g}\, \nabla^{d}$ is symmetric on the indices, gd, whereas ϵ_{dbg} is anti-symmetric on these same two indices.

Divergence of the cross product of two gradients vanishes

In a similar manner we find that the divergence acting on the cross product of two gradients vanishes:

$$
\begin{aligned}
\nabla \cdot (\nabla\phi \times \nabla\psi) &= \boldsymbol{e}^{a}\, \nabla_{a} \cdot (\boldsymbol{e}^{b}\, \nabla_{b}\phi \times \boldsymbol{e}^{c}\, \nabla_{c}\psi) && \text{basis one-forms and indices} && (2.35a)\\
&= \boldsymbol{e}^{a} \cdot (\boldsymbol{e}^{b} \times \boldsymbol{e}^{c})\, \nabla_{a}\, (\nabla_{b}\phi\, \nabla_{c}\psi) && \nabla_{a}\boldsymbol{e}^{b} = 0 && (2.35b)\\
&= \epsilon^{abc}\, \nabla_{a}\, (\nabla_{b}\phi\, \nabla_{c}\psi) && \boldsymbol{e}^{a} \cdot (\boldsymbol{e}^{b} \times \boldsymbol{e}^{c}) = \boldsymbol{e}^{a} \cdot \epsilon^{bcd}\, \boldsymbol{e}_{d} = \epsilon^{bca} && (2.35c)\\
&= 0, && && (2.35d)
\end{aligned}
$$

where the result vanishes since $\nabla_{a}\nabla_{b}\phi$ and $\nabla_{a}\nabla_{c}\psi$ are both symmetric on their indices, whereas ϵ^{abc} is anti-symmetric on these indices. This derivation can be streamlined by dropping the basis vectors and basis one-forms:

$$
\nabla \cdot (\nabla\phi \times \nabla\psi) = \nabla_{a}(\epsilon^{abc}\, \nabla_{b}\phi\, \nabla_{c}\psi) = \epsilon^{abc}\, (\nabla_{ab}\phi\, \nabla_{c}\psi + \nabla_{b}\phi\, \nabla_{ac}\psi) = 0. \qquad (2.36)
$$

Divergence of a cross product

The divergence of a cross product is determined through the following manipulations:

$$
\begin{aligned}
\nabla \cdot (\mathbf{F} \times \mathbf{E}) &= \boldsymbol{e}^{a}\, \nabla_{a} \cdot (F^{b}\, \boldsymbol{e}_{b} \times E^{c}\, \boldsymbol{e}_{c}) && \text{expose indices and basis vectors} && (2.37a)\\
&= \boldsymbol{e}^{a} \cdot (\boldsymbol{e}_{b} \times \boldsymbol{e}_{c})\, \nabla_{a}(F^{b}\, E^{c}) && \nabla_{a}\boldsymbol{e}_{b} = 0 && (2.37b)\\
&= \boldsymbol{e}^{a} \cdot (\epsilon_{bcd}\, \boldsymbol{e}^{d})\, \nabla_{a}(F^{b}\, E^{c}) && \text{cross product equation (1.46)} && (2.37c)\\
&= \delta^{ad}\, \epsilon_{bcd}\, \nabla_{a}(F^{b}\, E^{c}) && \text{orthonormality, } \delta^{ad} = \boldsymbol{e}^{a} \cdot \boldsymbol{e}^{d} && (2.37d)\\
&= \epsilon_{bcd}\, \nabla^{d}(F^{b}\, E^{c}) && \text{raise index: } \delta^{ad}\, \nabla_{a} = \nabla^{d} && (2.37e)\\
&= \epsilon_{bcd}\, (F^{b}\, \nabla^{d}E^{c} + E^{c}\, \nabla^{d}F^{b}) && \text{product rule} && (2.37f)\\
&= -\mathbf{F} \cdot (\nabla \times \mathbf{E}) + \mathbf{E} \cdot (\nabla \times \mathbf{F}) && \text{equation (2.24g) for curl.} && (2.37g)
\end{aligned}
$$

Curl of a scalar times a vector

The curl of a scalar times a vector, $\psi\, \mathbf{F}$, is determined by

$$
\begin{aligned}
\nabla \times (\psi\, \mathbf{F}) &= \epsilon_{dbg}\, \nabla^{d}(\psi\, F^{b})\, \boldsymbol{e}^{g} && \text{equation (2.24g) for curl} && (2.38a)\\
&= \boldsymbol{e}^{g}\, \epsilon_{dbg}\, [(\nabla^{d}\psi)\, F^{b} + \psi\, \nabla^{d}F^{b}] && \text{zero derivative for basis one-forms and } \epsilon && (2.38b)
\end{aligned}
$$

$$= \psi \, \nabla \times \mathbf{F} + \nabla\psi \times \mathbf{F} \qquad \text{reintroduce boldface notation.} \tag{2.38c}$$

Curl of a cross product

The curl of a cross product of two vectors is given by

$$\begin{aligned}
\nabla \times (\mathbf{F} \times \mathbf{E}) &= \boldsymbol{e}^a \, \nabla_a \times (\boldsymbol{e}_b \, F^b \times \boldsymbol{e}_c \, E^c) && (2.39a)\\
&= \boldsymbol{e}^a \times (\boldsymbol{e}_b \times \boldsymbol{e}_c) \, \nabla_a (F^b \, E^c) && (2.39b)\\
&= \boldsymbol{e}^a \times (\epsilon_{bcd} \, \boldsymbol{e}^d) \, \nabla_a (F^b \, E^c) && (2.39c)\\
&= \epsilon^{ade} \, \epsilon_{bcd} \, \boldsymbol{e}_e \, \nabla_a (F^b \, E^c) && (2.39d)\\
&= -\epsilon^{aed} \, \epsilon_{bcd} \, \boldsymbol{e}_e \, \nabla_a (F^b \, E^c) && (2.39e)\\
&= -(\delta^a{}_b \, \delta^e{}_c - \delta^a{}_c \, \delta^e{}_b) \, \boldsymbol{e}_e \, \nabla_a (F^b \, E^c) && (2.39f)\\
&= (-\delta^a{}_b \, \boldsymbol{e}_c + \delta^a{}_c \, \boldsymbol{e}_b) \, \nabla_a (F^b \, E^c) && (2.39g)\\
&= \mathbf{F} \, (\nabla \cdot \mathbf{E}) + (\mathbf{E} \cdot \nabla) \, \mathbf{F} - \mathbf{E} \, (\nabla \cdot \mathbf{F}) - (\mathbf{F} \cdot \nabla) \, \mathbf{E}, && (2.39h)
\end{aligned}$$

where we made use of the identity (1.119) for the contraction of two Levi-Civita tensors.

Curl of a curl

The curl of a curl is given by

$$\nabla \times (\nabla \times \mathbf{F}) = \boldsymbol{e}^a \times (\boldsymbol{e}^b \times \boldsymbol{e}_c) \, \nabla_a \nabla_b \, F^c. \tag{2.40}$$

The double cross product is computed by

$$\begin{aligned}
\boldsymbol{e}^a \times (\boldsymbol{e}^b \times \boldsymbol{e}_c) &= \delta_{cd} \, \boldsymbol{e}^a \times (\boldsymbol{e}^b \times \boldsymbol{e}^d) && (2.41a)\\
&= \delta_{cd} \, \epsilon^{bde} \, \boldsymbol{e}^a \times \boldsymbol{e}_e && (2.41b)\\
&= \delta_{cd} \, \delta^{af} \, \epsilon^{bde} \, \epsilon_{feg} \, \boldsymbol{e}^g && (2.41c)\\
&= \delta_{cd} \, \delta^{af} \, (-\delta^b{}_f \, \delta^d{}_g + \delta^b{}_g \, \delta^d{}_f) \, \boldsymbol{e}^g && (2.41d)\\
&= -\delta^{ab} \, \boldsymbol{e}_c + \delta^a{}_c \, \boldsymbol{e}^b, && (2.41e)
\end{aligned}$$

where we made use of the identity (1.119) for the contraction of two Levi-Civita tensors. We are then left with

$$\begin{aligned}
\nabla \times (\nabla \times \mathbf{F}) &= (-\delta^{ab} \, \boldsymbol{e}_c + \delta^a{}_c \, \boldsymbol{e}^b) \, \nabla_a \nabla_b \, F^c && (2.42a)\\
&= -\nabla^b \, \nabla_b (\boldsymbol{e}_c \, F^c) + \nabla_c \, (\boldsymbol{e}^b \, \nabla_b F^c) && (2.42b)\\
&= -\nabla^2 \mathbf{F} + \nabla(\nabla \cdot \mathbf{F}). && (2.42c)
\end{aligned}$$

This identity is particularly useful for non-divergent vector fields, in which

$$\nabla \times (\nabla \times \mathbf{F}) = -\nabla^2 \mathbf{F} \quad \text{if } \nabla \cdot \mathbf{F} = 0. \tag{2.43}$$

Relating advection, curl, and kinetic energy

In Exercise 2.6, we apply some of the previous results to derive a relation required for the vorticity equation studied in VOLUME 3. In particular, we show that

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \omega \times \mathbf{v} + \nabla(\mathbf{v} \cdot \mathbf{v})/2, \tag{2.44}$$

where

$$\omega = \nabla \times \mathbf{v} \tag{2.45}$$

is the vorticity, $\mathbf{v} \cdot \mathbf{v}/2$ is the kinetic energy per mass, and $\mathbf{v}$ is the fluid velocity field.

2.4 Analytic geometry of curves and surfaces

In this section we introduce some of the basics of curves and surfaces embedded in Euclidean space. Although the curves and surfaces encountered in geophysical fluid mechanics are commonly moving as part of the fluid flow, we are concerned in this section with describing their instantaneous spatial properties. Hence, time does not appear in this section. Furthermore, although curves and surfaces can overturn and intersect themselves, we restrict attention to orientable curves and surfaces whose normal direction has a nonzero projection onto the vertical; i.e., they have no overhangs and no wrapping (Figure 2.6). This constraint is satisfied by the surfaces of constant generalized vertical coordinates (e.g., isopycnal surfaces) as detailed in VOLUME 3.

2.4.1 Definitions and notation

We assume the notion of a curve and surface embedded in Euclidean space to be self-evident, offering analytical expressions for curves in Section 2.4.3 and surfaces in Section 2.4.4. Given such, we here define some related notions used in our study of fluid mechanics.

- ORIENTABLE: An *orientable curve* is a curve that allows for normal and tangent directions to specify directions and sides to the curve. Likewise, an *orientable surface* has two sides, allowing one to choose a positive side and a negative side. A Möbius strip is the canonical surface that is not orientable since it only has one side. Likewise, the boundary of a Möbius strip is a non-orientable curve. We only consider orientable objects in this book.
- PATH OR CONTOUR: A path or contour is a continuous piecewise smooth oriented curve. A *simple path* or *simple contour* does not cross itself. We already encountered contours when considering path integrals in Chapter 2.
- CIRCUIT: A circuit is a path that closes, and a *simple circuit* is a circuit that does not cross itself. Finally, a *reducible circuit* is a circuit that can be continuously deformed to a point within the domain without leaving the domain. For example, a circuit within the ocean that encloses an island or continent cannot be deformed to a point since doing so requires the circuit to cross onto land and thus to leave the ocean.

We write the Cartesian position of a point on a surface as

$$\boldsymbol{S} = x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}} + \eta(x,y)\,\hat{\boldsymbol{z}} \qquad \text{position on surface,} \tag{2.46}$$

with the vertical position written as

$$z = \eta(x,y) \qquad \text{vertical position on surface.} \tag{2.47}$$

If we are instead referring to a point on a planar curve in the x-z-plane, then we drop the y-dependence to have

$$\boldsymbol{S} = x\,\hat{\boldsymbol{x}} + \eta(x)\,\hat{\boldsymbol{z}} \qquad \text{position on planar curve.} \tag{2.48}$$

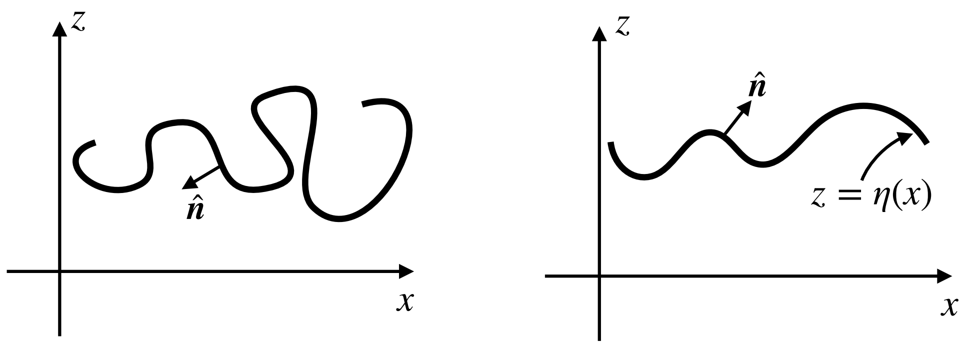


FIGURE 2.6: Two sample curves on the x-z plane. The left panel shows a curve whose outward unit normal, $\hat{\boldsymbol{n}}$, encounters points where $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{z}} = 0$ and where $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{z}}$ changes sign. This curve, and its generalization to a surface, are not treated in this chapter. The right panel shows a more gently undulating curve where $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{z}} \neq 0$ everywhere, and thus where $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{z}}$ is single signed. For curves such as the right panel, we can express the vertical position as a 1-to-1 function of the horizontal position, $z = \eta(x)$. Again, this curve has its natural generalization to a gently undulating surface whereby $z = \eta(x, y)$ provides a unique mapping between horizontal position and vertical. The assumption regarding no overhanging curves and surfaces is consistent with our study of surfaces defined by a constant generalized vertical coordinate (e.g., isopycnals or isentropes) in VOLUME 3.

We assume the outward normal direction on the curve or the surface has a nonzero projection into the vertical as shown in Figure 2.6. Indeed, we are only able to write the vertical position as $z = \eta(x, y)$ so long as there are no overturns in the surface, in which case the outward unit normal direction is

$$\hat{\boldsymbol{n}} = \frac{\nabla(z - \eta)}{|\nabla(z - \eta)|} = \frac{\hat{\boldsymbol{z}} - \nabla\eta}{\sqrt{1 + |\nabla\eta|^2}}. \tag{2.49}$$

Figure 2.7 provides an example surface along with the notation.

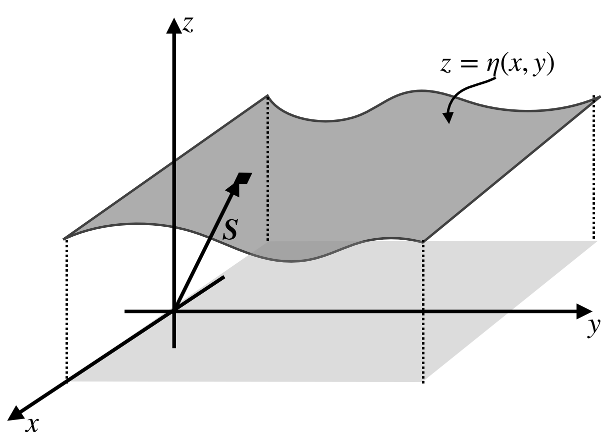


FIGURE 2.7: Example two-dimensional surface embedded in Euclidean space. The position of a point on the surface is given by the Cartesian position vector $\boldsymbol{S} = x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}} + \eta(x, y)\,\hat{\boldsymbol{z}}$. The relation $z = \eta(x, y)$ provides a 1-to-1 mapping between the horizontal position and the vertical position of a point on the surface. Correspondingly, the surface is uniquely specified by finding the envelope of points where $z - \eta(x, y) = 0$. The lightly shaded region represents the projection of the curved surface onto the flat horizontal x-y plane below.

2.4.2 Surfaces with $x = \gamma(y, z)$ or $y = \psi(x, z)$

We generally find it most useful to specify a point on a surface according to equation (2.46), whereby we write $z = \eta(x, y)$ for the vertical position as a function of the horizontal position. This approach is typical for our applications since the surfaces we encounter most commonly in stratified geophysical flows have a normal direction that has a non-zero projection into the $\hat{\boldsymbol{z}}$ direction. However, there are occasions where it is more convenient to define a point on a surface according to

$$\boldsymbol{S} = \gamma(y, z)\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}} + z\,\hat{\boldsymbol{z}} \qquad \text{alternative specification.} \tag{2.50}$$

Here, we specify the x position on the surface as a function of y and z via the function $\gamma(y, z)$. The unit normal is thus specified by

$$\hat{\boldsymbol{n}} = \frac{\nabla(x - \gamma)}{|\nabla(x - \gamma)|} = \frac{\hat{\boldsymbol{x}} - \hat{\boldsymbol{y}}\,\partial_y\gamma - \hat{\boldsymbol{z}}\,\partial_z\gamma}{1 + \sqrt{(\partial_y\gamma)^2 + (\partial_z\gamma)^2}}. \tag{2.51}$$

Hence, this specification is useful for those surfaces with normal direction having a non-zero $\hat{\boldsymbol{x}}$ component everywhere on the surface, in which case we are afforded the ability to define $\gamma(y, z)$. Alternatively, if the surface instead has a normal direction with a non-zero $\hat{\boldsymbol{y}}$ component everywhere, then we find it more suitable to define a point on the surface according to

$$\boldsymbol{S} = x\,\hat{\boldsymbol{x}} + \psi(x, z)\,\hat{\boldsymbol{y}} + z\,\hat{\boldsymbol{z}}, \tag{2.52}$$

where $y = \psi(x, z)$ provides the y position of a point on the surface as a function of x, z.

2.4.3 Planar curves in 2D Euclidan space

We here describe the geometry of a curve on the x-z-plane (a *planar curve*) as depicted in Figure 2.8. These curves are one-dimensional objects living in a two-dimensional Euclidean space. Extensions to curves on non-Euclidean surfaces, such as the sphere or an isopycnal, are straightforward when those surfaces are embedded in the background Euclidean space assumed in this book.

Differential increments along the curve

As a one-dimensional geometric object, an arbitrary curve can be parameterized by a single coordinate, referred to here as φ. Let $\boldsymbol{S}(\varphi)$ specify the position of a point along the curve. Correspondingly, the differential increment between two infinitesimally close points on the curve is given by

$$\boldsymbol{S}(\varphi + \mathrm{d}\varphi) - \boldsymbol{S}(\varphi) = \mathrm{d}\boldsymbol{S} = \frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}\varphi}\,\mathrm{d}\varphi \equiv \boldsymbol{t}\,\mathrm{d}\varphi, \tag{2.53}$$

where

$$\boldsymbol{t} = \frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}\varphi} \tag{2.54}$$

is tangent to the curve. If $\varphi = s$ is the arc length along the curve, then $\boldsymbol{t} = \hat{\boldsymbol{t}}$ is a unt vector

$$\hat{\boldsymbol{t}} \cdot \hat{\boldsymbol{t}} = \frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}s} \cdot \frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}s} = 1. \tag{2.55}$$

In some treatments in this book we also write

$$\hat{\boldsymbol{s}} = \hat{\boldsymbol{t}} \tag{2.56}$$

to correspond to s for arc length.

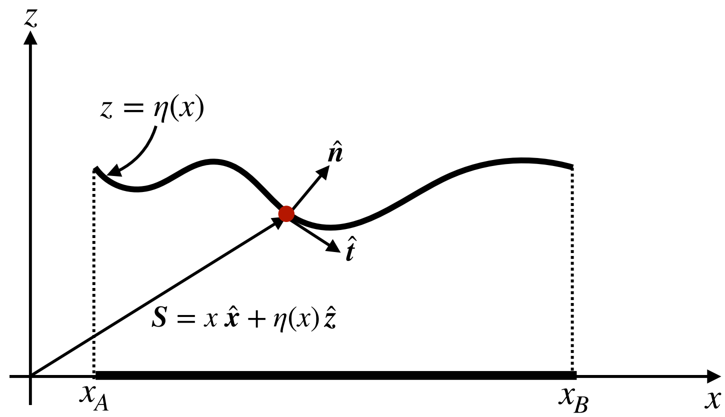


FIGURE 2.8: An orientable path in the x-z plane defined by a planar curve that does not intersect itself. The Cartesian position of a point on the curve is given by $\boldsymbol{S} = x\,\hat{\boldsymbol{x}} + \eta(x)\,\hat{\boldsymbol{z}}$, where $z = \eta(x)$ provides the vertical position of the point as a function of the horizontal position. The projection of the curve onto the horizontal x-axis occupies a range $x_A \le x \le x_B$. One way to define the curve is by finding the envelope of points where $z - \eta(x) = 0$, in which case we can readily find the unit normal direction pointing upward as $\hat{\boldsymbol{n}} = \nabla(z-\eta)/|\nabla(z-\eta)| = [\hat{\boldsymbol{z}} - (\mathrm{d}\eta/\mathrm{d}x)\,\hat{\boldsymbol{x}}]\,[1 + (\mathrm{d}\eta/\mathrm{d}x)^2\,]^{-1/2}$, and the unit tangent vector $\hat{\boldsymbol{t}} = [\hat{\boldsymbol{x}} + (\mathrm{d}\eta/\mathrm{d}x)\,\hat{\boldsymbol{z}}]\,[1 + (\mathrm{d}\eta/\mathrm{d}x)^2\,]^{-1/2}$.

Length along the curve

As in equation (2.48) we can represent the position of a point along the curve using Cartesian coordinates

$$\boldsymbol{S} = x\,\hat{\boldsymbol{x}} + \eta(x)\,\hat{\boldsymbol{z}}. \tag{2.57}$$

Hence, letting $\varphi = x$ parameterize the curve leads to the representation of the tangent direction

$$\boldsymbol{t} = \frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}x} = \hat{\boldsymbol{x}} + \frac{\mathrm{d}\eta}{\mathrm{d}x}\,\hat{\boldsymbol{z}}, \tag{2.58}$$

which has the magnitude

$$\boldsymbol{t} \cdot \boldsymbol{t} = 1 + (\mathrm{d}\eta/\mathrm{d}x)^2, \tag{2.59}$$

so that the unit tangent vector is

$$\hat{\boldsymbol{t}} = \frac{\hat{\boldsymbol{x}} + (\mathrm{d}\eta/\mathrm{d}x)\,\hat{\boldsymbol{z}}}{\sqrt{1 + (\mathrm{d}\eta/\mathrm{d}x)^2}}. \tag{2.60}$$

Likewise, the curve's unit normal vector is given by

$$\hat{\boldsymbol{n}} = \frac{\nabla(z-\eta)}{|\nabla(z-\eta)|} = \frac{\hat{\boldsymbol{z}} - (\mathrm{d}\eta/\mathrm{d}x)\,\hat{\boldsymbol{x}}}{\sqrt{1 + (\mathrm{d}\eta/\mathrm{d}x)^2}}, \tag{2.61}$$

with orthogonality manifest

$$\hat{\boldsymbol{t}} \cdot \hat{\boldsymbol{n}} = 0. \tag{2.62}$$

The squared length of an infinitesimal segment along the curve is given by

$$(\mathrm{d}s)^2 = \mathrm{d}\boldsymbol{S} \cdot \mathrm{d}\boldsymbol{S} = \left[\frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}x} \cdot \frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}x}\right] \mathrm{d}x\,\mathrm{d}x, \tag{2.63}$$

so that the finite length of the curve is determined by the integral

$$L = \int_0^L \mathrm{d}s = \int_{x_A}^{x_B} |\mathrm{d}\boldsymbol{S}/\mathrm{d}x|\,\mathrm{d}x = \int_{x_A}^{x_B} \sqrt{1 + (\mathrm{d}\eta/\mathrm{d}x)^2}\,\mathrm{d}x, \tag{2.64}$$

where $x_A \le x \le x_B$ is the range over which x runs for the projection of the curve onto the x-axis (see Figure 2.8).

Curvature of a curve

Curvature measures the amount that the unit normal changes along the curve. For a planar curve, the curvature at a point equals to the inverse radius of a circle that shares the same tangent plane to the curve at the point (see Figure 2.9). We refer to the radius as the *radius of curvature* and the corresponding circle as the *curvature circle*. To formulate an analytic expression for the radius of curvature at a point on a curve, orient the Cartesian coordinate axes so that the point is at the origin and the tangent plane sits along the x-axis as in Figure 2.9. Consequently, the outward unit normal, $\hat{\boldsymbol{n}}$, is parallel to the $\hat{\boldsymbol{z}}$ direction.

A Taylor series expansion about the origin tells us that the vertical position of a point along the curve and near to the origin can be written

$$\eta(x) = \eta(0) + x \left[\frac{\mathrm{d}\eta}{\mathrm{d}x}\right]_{x=0} + \frac{x^2}{2}\left[\frac{\mathrm{d}^2\eta}{\mathrm{d}x^2}\right]_{x=0} + \mathcal{O}(x^3) \tag{2.65a}$$

$$= \frac{x^2}{2}\left[\frac{\mathrm{d}^2\eta}{\mathrm{d}x^2}\right]_{x=0} + \mathcal{O}(x^3). \tag{2.65b}$$

This result follows since we placed the origin so that $\eta(0) = 0$, and aligned the x-axis so that it is tangent at the origin, in which case $\mathrm{d}\eta/\mathrm{d}x = 0$ at $x = 0$. Hence, η has a quadratic behavior near the origin.

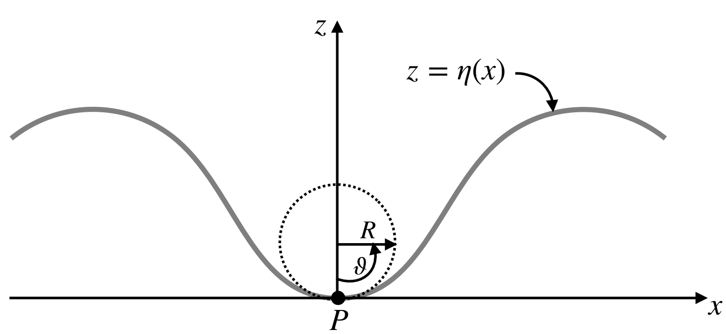


FIGURE 2.9: The radius of curvature at a point on a curve, P, equals to the radius of the curvature circle that shares the same tangent plane as the curve at the point P. When constructing the curvature circle we make use of the angle, ϑ, to measure the height of a point along the circle, $h(x) = R\,(1 - \cos\vartheta) \approx R\,\vartheta^2/2 \approx x^2/(2R)$. Setting $R^{-1} = \mathrm{d}^2\eta/\mathrm{d}x^2$ provides a second order accurate fit of the curvature circle to the curve at the point P.

Now place a circle with center along the z-axis so that it is tangent to the curve at the origin, as depicted in Figure 2.9. What is the radius, R, of the circle that best fits the curve at the origin? To answer this question note that the height of a point on the circle is given by $h(x) = R\,(1 - \cos\vartheta)$, where $\vartheta = 0$ for a point at the origin and $\vartheta = \pi$ at the diametrically opposite point. For small ϑ this height takes the form

$$h(x) \approx R\,[1 - 1 + \vartheta^2/2] = x^2/(2R), \tag{2.66}$$

where $\vartheta = x/R$ near the origin. For the height of a point on the curve (equation (2.65b)) to match the height along the circle, to second order accuracy, requires us to set the circle's radius to

$$\frac{1}{R} = \frac{\mathrm{d}^2\eta}{\mathrm{d}x^2}. \tag{2.67}$$

Equation (2.67) thus provides an expression for the radius of curvature, R, whose inverse is the curvature

$$\text{curvature} = \frac{1}{R}. \tag{2.68}$$

This result supports our expectation that the second derivative measures the curvature. As the radius, R, gets larger, the curvature decreases, which means that local regions along the circle appear more flat. In the opposite limit the curvature grows as R decreases. Note that we could have chosen to orient the circle on the opposite side of the tangent (on the convex side), in which case the radius of curvature is negative. That is, $R > 0$ when the unit normal points towards the concave side (side where the curve rises towards $\hat{\boldsymbol{n}}$), whereas $R < 0$ when the unit normal points towards the convex side (side where the curve falls away from $\hat{\boldsymbol{n}}$).

Observe that

$$-\nabla \cdot \hat{\boldsymbol{n}} = \frac{\mathrm{d}^2\eta/\mathrm{d}x^2}{[1 + (\mathrm{d}\eta/\mathrm{d}x)^2\,]^{3/2}}. \tag{2.69}$$

When evaluated at the point of interest along the curve, we set $\mathrm{d}\eta/\mathrm{d}x = 0$ so that

$$-\nabla \cdot \hat{\boldsymbol{n}} = \frac{\mathrm{d}^2\eta}{\mathrm{d}x^2} = \frac{1}{R}. \tag{2.70}$$

This result supports our earlier statement that curvature measures the change in the normal direction along the curve. In fact, the identity

$$-\nabla \cdot \hat{\boldsymbol{n}} = \frac{1}{R} \tag{2.71}$$

holds for an arbitrary point along the curve since it is a coordinate invariant statement.

Equation (2.70) provides a useful means to check the sign of the radius of curvature. Namely, the point of interest in Figure 2.9 is a local minimum, so that $\mathrm{d}^2\eta/\mathrm{d}x^2 > 0$ and the radius of curvature is thus positive, $R > 0$. In contrast, the radius of curvature is negative at points where $\mathrm{d}^2\eta/\mathrm{d}x^2 < 0$.

2.4.4 Surfaces embedded in 3D Euclidean space

We now extend the previous discussion to a two-dimensional surface embedded in three-dimensional Euclidean space such as in Figure 2.7. In general, a 2D surface in 3D space can be parameterized by two variables, φ_1 and φ_2, so that infinitesimal increments along the surface satisfy

$$\mathrm{d}\boldsymbol{S} = \frac{\partial \boldsymbol{S}}{\partial \varphi_1}\,\mathrm{d}\varphi_1 + \frac{\partial \boldsymbol{S}}{\partial \varphi_2}\,\mathrm{d}\varphi_2 = \boldsymbol{t}_1\,\mathrm{d}\varphi_1 + \boldsymbol{t}_2\,\mathrm{d}\varphi_2. \tag{2.72}$$

The vectors $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$ are tangent to the surface at the point (φ_1, φ_2), and yet they are not generally orthogonal to one another.

Making use of the Cartesian expression (2.46)

$$\boldsymbol{S} = x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}} + \eta(x,y)\,\hat{\boldsymbol{z}} \tag{2.73}$$

brings the two tangent directions and the unit tangent directions into the form

$$\boldsymbol{t}_1 = \frac{\partial \boldsymbol{S}}{\partial x} = \hat{\boldsymbol{x}} + \frac{\partial \eta}{\partial x}\,\hat{\boldsymbol{z}} \quad \text{and} \quad \hat{\boldsymbol{t}}_1 = \frac{\hat{\boldsymbol{x}} + (\partial\eta/\partial x)\,\hat{\boldsymbol{z}}}{\sqrt{1 + (\partial\eta/\partial x)^2}} \tag{2.74a}$$

$$\boldsymbol{t}_2 = \frac{\partial \boldsymbol{S}}{\partial y} = \hat{\boldsymbol{y}} + \frac{\partial \eta}{\partial y}\,\hat{\boldsymbol{z}} \quad \text{and} \quad \hat{\boldsymbol{t}}_2 = \frac{\hat{\boldsymbol{y}} + (\partial\eta/\partial y)\,\hat{\boldsymbol{z}}}{\sqrt{1 + (\partial\eta/\partial y)^2}}. \tag{2.74b}$$

Likewise, the surface unit normal vector is given by

$$\hat{\boldsymbol{n}} = \frac{\nabla(z-\eta)}{|\nabla(z-\eta)|} = \frac{\hat{\boldsymbol{z}} - \nabla\eta}{|\hat{\boldsymbol{z}} - \nabla\eta|} = \frac{\hat{\boldsymbol{z}} - \nabla\eta}{\sqrt{1 + |\nabla\eta|^2}}, \tag{2.75}$$

and it is straightforward to show orthogonality with the two tangent vectors

$$\hat{\boldsymbol{t}}_1 \cdot \hat{\boldsymbol{n}} = \hat{\boldsymbol{t}}_2 \cdot \hat{\boldsymbol{n}} = 0. \tag{2.76}$$

Area on the surface

Recall from Section 1.6.5 that the magnitude of a vector product of two vectors equals to the area of the parallelogram subtended by the vectors. Hence, the area of an infinitesimal surface element with sides $\mathrm{d}\varphi_1$ and $\mathrm{d}\varphi_2$ is given by

$$\mathrm{d}\mathcal{S} = \left| \frac{\partial \boldsymbol{S}}{\partial \varphi_1} \times \frac{\partial \boldsymbol{S}}{\partial \varphi_2} \right| \mathrm{d}\varphi_1\,\mathrm{d}\varphi_2. \tag{2.77}$$

Making use of Cartesian coordinates brings the area element to

$$\mathrm{d}\mathcal{S} = \sqrt{1 + |\nabla\eta|^2}\,\mathrm{d}x\,\mathrm{d}y = \sqrt{1 + |\nabla\eta|^2}\,\mathrm{d}A, \tag{2.78}$$

where

$$\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y \tag{2.79}$$

is the area of the surface projected onto the horizontal plane. Hence, the area of a finite region is given by the integral

$$\mathcal{S} = \int \mathrm{d}\mathcal{S} = \int |\nabla(z-\eta)|\,\mathrm{d}A = \int \sqrt{1 + |\nabla\eta|^2}\,\mathrm{d}x\,\mathrm{d}y, \tag{2.80}$$

where the second and third integrals extend over the region defined by the projection of the surface onto the horizontal (see Figure 2.7).

Curvature of a surface

We now seek an expression for the curvature of a point on the surface. Since the surface has two dimensions, we expect the curvature to be measured by two numbers rather than the single curvature for the curve discussed in Section 2.4.3. The method for developing the curvature is analogous to that used for a curve, yet with a bit more mathematics needed to allow for the extra dimension. Figure 2.10 depicts the situation.

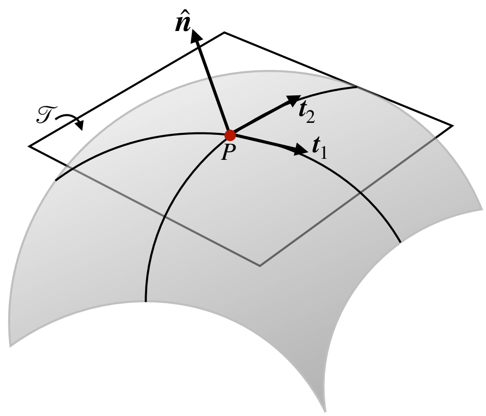


FIGURE 2.10: Depicting the elements needed to construct the curvature of a surface at an arbitrary point, P. The local unit normal direction is given by $\hat{\boldsymbol{n}}$, along with the two tangent vectors $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$. The tangent vectors span the space of the tangent plane, $\mathcal{T}$, shown as a flat surface that is tangent at the chosen point on the surface. In this case the surface falls away from the normal direction, as per a convex surface, so that the two radii of curvature are negative. Other surfaces can be concave, whereby both radii of curvature are positive, or hyperbolic (saddle), whereby one is positive and another negative.

Let $\boldsymbol{x} = (x_1, x_2) = (x, y)$ be Cartesian coordinates on a tangent plane local to an arbitrary point on the surface, with the origin of the coordinate system taken at the point. Near to the point, we can estimate the vertical distance of a point on the surface from the tangent plane according to the quadratic form

$$\eta \approx \frac{1}{2} x_m \, \mathbb{C}^{mn} \, x_n, \tag{2.81}$$

where $\mathbb{C}$ is the matrix of second partial derivatives evaluated at the point

$$\mathbb{C} = \begin{bmatrix} \dfrac{\partial^2 \eta}{\partial x_1^2} & \dfrac{\partial^2 \eta}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 \eta}{\partial x_1 \partial x_2} & \dfrac{\partial^2 \eta}{\partial x_2^2} \end{bmatrix}. \tag{2.82}$$

As a symmetric matrix, $\mathbb{C}$ is diagonalizable and it has two eigenvalues, R_1^{-1} and R_2^{-1}, along with its associated eigenvectors, $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$. The quadratic form (2.81) can thus be written as

$$\eta \approx \frac{1}{2} R_1^{-1} \, (\boldsymbol{x} \cdot \boldsymbol{c}_1)^2 + \frac{1}{2} R_2^{-1} \, (\boldsymbol{x} \cdot \boldsymbol{c}_2)^2. \tag{2.83}$$

R_1 and R_2 are the principle radii of curvature for the surface at the point P. They correspond, respectively, to the radii of the curvature circles in the $\hat{\boldsymbol{n}}$-$\boldsymbol{c}_1$ and $\hat{\boldsymbol{n}}$-$\boldsymbol{c}_2$ planes. If the radius of curvature, R_m, is positive, then the surface curves towards $\hat{\boldsymbol{n}}$ along the $\hat{\boldsymbol{n}}$-$\boldsymbol{c}_i$ plane, and conversely if R_m is negative. The surface takes the shape of a saddle when the two radii of curvature have opposite signs.

There are two scalar invariants of the matrix $\mathbb{C}$ that commonly arise in applications.

- $\mathrm{Tr}(\mathbb{C}) = R_1^{-1} + R_2^{-1}$, which is twice the mean curvature for the surface. With the unit normal vector given by equation (2.49), one can show that

$$-\nabla \cdot \hat{\boldsymbol{n}} = \frac{\nabla^2 \eta}{[1 + (\nabla \eta)^2 \,]^{3/2}}. \tag{2.84}$$

A bit of algebra leads us to conclude that

$$-\nabla \cdot \hat{\boldsymbol{n}} = \frac{\nabla^2 \eta}{[1 + (\nabla \eta)^2]^{3/2}} = \frac{1}{R_1} + \frac{1}{R_2}, \tag{2.85}$$

for any point along the surface, thus generalizing the result (2.70) found for a curve. Furthermore, if $|\nabla \eta| \ll 1$ then we have

$$\nabla^2 \eta \approx \frac{1}{R_1} + \frac{1}{R_2}, \tag{2.86}$$

with this result proving useful in the study of surface tension in VOLUME 2.

- $\det(\mathbb{C}) = 1/(R_1 R_2)$ is known as the *Gaussian curvature*, which is the product of the two curvatures.

2.4.5 Curves on the surface $z = \eta(x, y)$

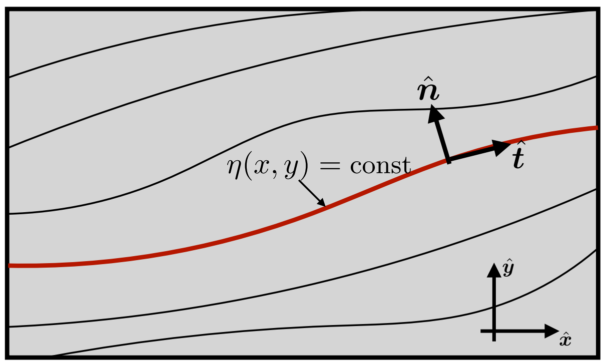


FIGURE 2.11: Geometry depicting a contour along a particular line of constant $z = \eta(x, y)$, such as for a constant elevation path along a mountain or valley. The along-contour unit tangent direction is $\hat{\boldsymbol{t}} = \mathrm{d}\boldsymbol{x}/\mathrm{d}s$, with s the arc length along the contour. The unit normal direction pointing to the left of the contour direction is $\hat{\boldsymbol{n}}$, with $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{t}} = 0$ and $\hat{\boldsymbol{t}} \times \hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}$. Both $\hat{\boldsymbol{n}}$ and $\hat{\boldsymbol{t}}$ are horizontal unit vectors.

We now consider a curve, as in Section 2.4.3, defined along a two dimensional surface, $z = \eta(x, y)$, with the curve defined by lines of constant $z = \eta(x, y)$. For example, if $\eta(x, y)$ is the solid earth topography, then lines of constant η are contours of constant topography. By definition, these contours have no projection into the vertical direction (i.e., they do not go uphill or downhill), and they are determined by

$$\mathrm{d}\eta = 0 = \nabla \eta \cdot \mathrm{d}\boldsymbol{x}, \tag{2.87}$$

where $\mathrm{d}\boldsymbol{x} = \hat{\boldsymbol{x}}\,\mathrm{d}x + \hat{\boldsymbol{y}}\,\mathrm{d}y$ is the horizontal space increment along the contour. Following the introduction of arc length in Section 2.4.3, we write

$$\mathrm{d}\eta = 0 = \nabla \eta \cdot \mathrm{d}\boldsymbol{x} = \nabla \eta \cdot \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s}\,\mathrm{d}s = \nabla \eta \cdot \hat{\boldsymbol{t}}\,\mathrm{d}s, \tag{2.88}$$

where $\hat{\boldsymbol{t}}$ is a unit vector pointing in the direction of the contour. To build an orthogonal triad of coordinates, we then define an orthogonal unit vector, $\hat{\boldsymbol{n}}$, that points to the left of the contour direction so that

$$\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{t}} = 0 \quad \text{and} \quad \hat{\boldsymbol{t}} \times \hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}, \tag{2.89}$$

as depicted in Figure 2.11. We provide an example use of this formalism in Exercise 2.9.

2.5 Path integral of a scalar function

Consider the integral of a scalar function, ψ, over an arbitrary one-dimensional path in space, $\mathcal{C}$

$$\mathcal{I} = \int_A^B \psi(\alpha)\,\mathrm{d}\alpha. \tag{2.90}$$

Since any path is a one-dimensional curve, a point along the path can be specified by a single parameter, denoted here by α with endpoints $\alpha = A$ and $\alpha = B$.[2] We now consider some explicit examples of how to parameterize a curve to thus enable an explicit evaluation of the integral.[3]

2.5.1 Cartesian coordinates

Lay down a Cartesian coordinate system with an arbitrary origin, in which case the Cartesian coordinate representation of a point along the path is written

$$\boldsymbol{x}(\alpha) = \hat{\boldsymbol{x}}\,x(\alpha) + \hat{\boldsymbol{y}}\,y(\alpha) + \hat{\boldsymbol{z}}\,z(\alpha), \tag{2.91}$$

along with the endpoints along the path

$$\boldsymbol{x}(A) = \boldsymbol{x}_A \quad \text{and} \quad \boldsymbol{x}(B) = \boldsymbol{x}_B. \tag{2.92}$$

In this way the path integral is written

$$\mathcal{I} = \int_A^B \psi(\alpha)\,\mathrm{d}\alpha = \int_A^B \psi[\boldsymbol{x}(\alpha)]\,\mathrm{d}\alpha. \tag{2.93}$$

To bring the integral (2.93) fully into a Cartesian parameterized form requires a coordinate transformation from α to $\boldsymbol{x}$ along the curve. For this purpose, consider two points on the path that are separated by an infinitesimal parameter difference, in which the difference in their Cartesian coordinates is given by

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{x}(\alpha + \mathrm{d}\alpha) - \boldsymbol{x}(\alpha) = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\alpha}\,\mathrm{d}\alpha. \tag{2.94}$$

We thus have

$$(\mathrm{d}\alpha)^2 = \frac{\mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x}}{\mathrm{d}\boldsymbol{x}/\mathrm{d}\alpha\cdot\mathrm{d}\boldsymbol{x}/\mathrm{d}\alpha}. \tag{2.95}$$

Assuming $\mathrm{d}\alpha > 0$ then leads to the integral (2.93) taking on the rather clumsy, but nonetheless general, form

$$\mathcal{I} = \int_A^B \psi(\alpha)\,\mathrm{d}\alpha = \int_A^B \psi[\boldsymbol{x}(\alpha)]\,\mathrm{d}\alpha = \int_A^B \psi[\boldsymbol{x}(\alpha)]\sqrt{\frac{\mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x}}{\mathrm{d}\boldsymbol{x}/\mathrm{d}\alpha\cdot\mathrm{d}\boldsymbol{x}/\mathrm{d}\alpha}}. \tag{2.96}$$

[2] We have more to say about the geometry of paths in Sections 2.4.1 and 2.4.3.

[3] In the more general language of differential forms, the evaluation of a path integral requires one to parameterize points along the path so to then write the path integral as a Riemann integral. This process is known as *pulling back* the path integral to a Riemann integral. *Shifrin* (2004) provies a treatment accessible to those having studied undergraduate calculus.

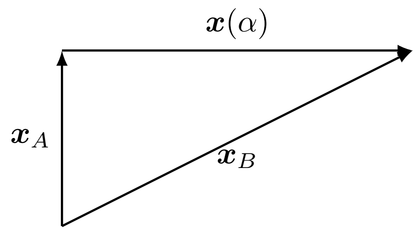


FIGURE 2.12: A linear path, $\boldsymbol{x}(\alpha)$ extending from $\boldsymbol{x}_A$ to $\boldsymbol{x}_B$ can be parameterized by a non-dimensional parameter $\alpha \in [0, 1]$ via $\boldsymbol{x}(\alpha) = \boldsymbol{x}_A + (\boldsymbol{x}_B - \boldsymbol{x}_A)\,\alpha$. Alternatively, it can be parameterized by the arc-length along the path via $\boldsymbol{x}(s) = \boldsymbol{x}_A + \hat{\boldsymbol{s}}\,s$ with $s \in [0, L]$, $L = |\boldsymbol{x}_B - \boldsymbol{x}_A|$, and $\hat{\boldsymbol{s}}$ the unit tangent vector pointing from $\boldsymbol{x}_A$ to $\boldsymbol{x}_B$.

2.5.2 Arc length parameterization

Now consider a common special case for path parameterization where $\alpha = s$ is the arc length along the path[4]

$$\mathcal{I} = \int_A^B \psi(\alpha)\,\mathrm{d}\alpha = \int_{s_A}^{s_B} \psi[\boldsymbol{x}(s)]\,\mathrm{d}s. \tag{2.97}$$

For Euclidean space using Cartesian coordinates, the differential increment of arc length is given by

$$\mathrm{d}s = \sqrt{\mathrm{d}\boldsymbol{x} \cdot \mathrm{d}\boldsymbol{x}}. \tag{2.98}$$

Inserting $\boldsymbol{x}(s)$ into this expression renders

$$\mathrm{d}s = \sqrt{\mathrm{d}\boldsymbol{x} \cdot \mathrm{d}\boldsymbol{x}} = \mathrm{d}s\,\sqrt{\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s} \cdot \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s}}. \tag{2.99}$$

This expression is self-consistent if

$$\sqrt{\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s} \cdot \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s}} = 1, \tag{2.100}$$

which is merely a rewrite of the defining expression (2.98). Note that the derivative of the curve with respect to the arc-length, $\mathrm{d}\boldsymbol{x}/\mathrm{d}s$, defines a unit tangent vector to the curve

$$\hat{\boldsymbol{s}} = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s} \Longrightarrow \hat{\boldsymbol{s}} \cdot \hat{\boldsymbol{s}} = 1. \tag{2.101}$$

2.5.3 Linear path example

As a specific example, consider a straight line between two points, $\boldsymbol{x}_A$ and $\boldsymbol{x}_B$, as in Figure 2.12. We can parameterize the line using a dimensionless parameter α according to

$$\boldsymbol{x}(\alpha) = \boldsymbol{x}_A + (\boldsymbol{x}_B - \boldsymbol{x}_A)\,\alpha \qquad \alpha \in [0, 1]. \tag{2.102}$$

Alternatively, we can parameterize using the arc length

$$\boldsymbol{x}(s) = \boldsymbol{x}_A + \hat{\boldsymbol{s}}\,s \qquad s \in [0, L], \tag{2.103}$$

[4]We offer a more focused discussion of curves and tangents in Section 2.4.3 (see in particular equation (2.54)).

where $L = \int_A^B \mathrm{d}s = |\boldsymbol{x}_B - \boldsymbol{x}_A|$ is the total arc length of the line, and where $\hat{\boldsymbol{s}}$ is the unit tangent vector pointing along the path from $\boldsymbol{x}_A$ to $\boldsymbol{x}_B$

$$\hat{\boldsymbol{s}} = \frac{\boldsymbol{x}'(s)}{|\boldsymbol{x}'(s)|} = \frac{\boldsymbol{x}_B - \boldsymbol{x}_A}{|\boldsymbol{x}_B - \boldsymbol{x}_A|}. \tag{2.104}$$

As defined we have $|\boldsymbol{x}'(s)| = |\hat{\boldsymbol{s}}| = 1$, so that the path integral is given by

$$\mathcal{I} = \int_{\mathcal{C}} \psi(\alpha)\,\mathrm{d}\alpha = \int_0^L \psi[\boldsymbol{x}(s)]\,\mathrm{d}s. \tag{2.105}$$

2.6 Path integral of a vector field

Generalizing to a vector field, $\mathbf{F}(\mathbf{x})$, we could conceivably integrate each component of the vector along the curve independently, making use of the approach for scalars functions in Section 2.5. In practice, however, that quantity rarely appears in physics.[5] Instead, we more commonly wish to integrate that component of $\mathbf{F}$ that projects onto a curve

$$\int_{\mathcal{C}} \mathbf{F}\cdot\mathrm{d}\boldsymbol{x} = \int_{\mathcal{C}} \mathbf{F}\cdot\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s}\,\mathrm{d}s = \int_{\mathcal{C}} \mathbf{F}\cdot\hat{\boldsymbol{s}}\,\mathrm{d}s, \tag{2.106}$$

where $\hat{\boldsymbol{s}} = \mathrm{d}\boldsymbol{x}/\mathrm{d}s$ is tangent to the curve given by equation (2.101). A common example for the path integral of a vector concerns the work performed by a force field applied to a physical system that is moving along a path (this example is studied in Section 11.2.4).

2.6.1 Circulation

For the case of a closed curve or a circuit (see Section 2.4.1), we refer to the path integral as the *circulation* and use the convention of putting an arrowed circle on the integral sign

$$\text{circulation of vector field} = \oint_{\mathcal{C}} \mathbf{F}\cdot\mathrm{d}\boldsymbol{x}. \tag{2.107}$$

The arrow indicates that we conventionally traverse the closed path in a counter-clockwise (right hand) manner when looking down on the path from above. Note that by choosing a viewpoint as "above", we necessarily allow for an unambiguous definition of "counter-clockwise", thus chosing an *orientation*.

2.6.2 Circulation example

Consider the vector field, $\mathbf{F}$, expressed using Cartesian coordinates by the following vector function

$$\boldsymbol{F}(\boldsymbol{x}) = -y\,\hat{\boldsymbol{x}} + x\,\hat{\boldsymbol{y}}, \tag{2.108}$$

as shown in Figure 2.4. What is the circulation computed around a circle of radius r whose center is the origin? To compute this circulation we make use of plane polar coordinates, in which $x = r\cos\alpha$ and $y = r\sin\alpha$, with $\alpha \in [0, 2\pi]$ the polar angle measured from the positive x-axis. The position of a point on the circle is thus written $\boldsymbol{x}(\alpha) = r\,(\hat{\boldsymbol{x}}\cos\alpha + \hat{\boldsymbol{y}}\sin\alpha)$, and

[5] The mathematical reason it does not appear is that for a general manifold, the addition of vectors is only defined locally within a tangent space. This limitation prevents us from integrating general tensors over a volume.

the tangent to the circle is $\mathrm{d}\boldsymbol{x}(\alpha)/\mathrm{d}\alpha = r\,(-\hat{\boldsymbol{x}}\,\sin\alpha + \hat{\boldsymbol{y}}\,\cos\alpha)$. The integrand to the circulation (2.106) is given by

$$\boldsymbol{F} \cdot \frac{\mathrm{d}\boldsymbol{x}(\alpha)}{\mathrm{d}\alpha} = r\,(y\,\sin\alpha + x\cos\alpha) = r^2. \tag{2.109}$$

Hence, the circulation around the constant radius circle is twice the area of the circle

$$\oint_{\mathcal{C}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = 2\,\pi\,r^2. \tag{2.110}$$

We apply this result in to geophysical fluids when computing the vorticity induced by the rotating planet in VOLUME 3.

2.6.3 Fundamental theorem of calculus

The special case of $\mathbf{F} = -\nabla\psi$ for a scalar field ψ recovers the fundamental theorem of calculus

$$\psi(\boldsymbol{x}_B) - \psi(\boldsymbol{x}_A) = \int_{\boldsymbol{x}_A}^{\boldsymbol{x}_B} \mathrm{d}\psi = \int_{\boldsymbol{x}_A}^{\boldsymbol{x}_B} \nabla\psi \cdot \mathrm{d}\boldsymbol{x}. \tag{2.111}$$

It follows that for any closed curve with $\boldsymbol{x}_A = \boldsymbol{x}_B$, the circulation of $\nabla\psi$ vanishes

$$\oint_{\mathcal{C}} \mathrm{d}\psi = \oint_{\mathcal{C}} \nabla\psi \cdot \mathrm{d}\boldsymbol{x} = 0. \tag{2.112}$$

2.7 Stokes' curl theorem

Stokes' curl theorem relates the integral of a vector field, projected onto the tangent of a boundary around an orientable surface, to the integral of the unit normal projected onto the curl of the vector over the area of the surface.[6] The geometry of Stokes' theorem is illustrated in Figure 2.13. This theorem is used extensively in our study of circulation and vorticity in VOLUME 3.

2.7.1 Statement of Stokes' theorem

For an oriented[7] two-dimensional surface, $\mathcal{S}$, with a closed boundary, $\partial\mathcal{S}$, Stokes' theorem says that the circulation around the boundary equals to the area integrated curl projected onto the surface outward unit normal

$$\oint_{\partial\mathcal{S}} \mathbf{F} \cdot \mathrm{d}\boldsymbol{x} = \int_{\mathcal{S}} (\nabla \times \mathbf{F}) \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}. \tag{2.113}$$

In this equation,

$$\mathrm{d}\boldsymbol{x} = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s}\,\mathrm{d}s = \hat{\boldsymbol{s}}\,\mathrm{d}s \tag{2.114}$$

is the vector line element along the closed path (circuit), $\hat{\boldsymbol{s}}$ is the unit tangent vector along the path, and s is the arc-distance along the path. For the surface integral we have the outward

[6] As noted in a footnote on page 13 of *Truesdell* (1954), Lord Kelvin and later Hankel independently discovered what we here refer to as Stokes' theorem. Stokes' name became attached to the theorem since he included its derivation on an examination for students.

[7] For a surface to be orientable means that we can unambiguously describe its two sides, thus allowing us to determine a positive (top) side and negative (bottom) side. We are here only concerned with surfaces that are orientable, thus precluding non-orientable surfaces such as the Möbius strip.

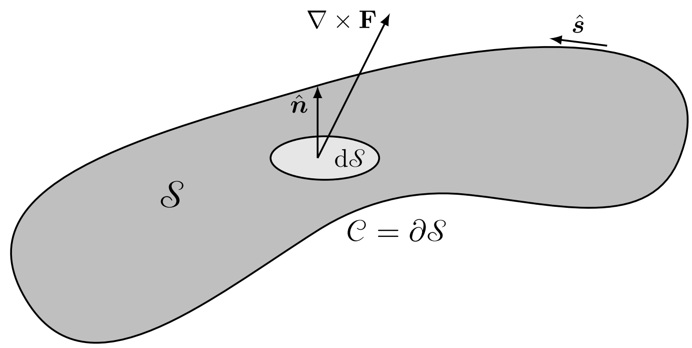


FIGURE 2.13: Illustrating the geometry of Stokes' theorem (2.113). The unit outward normal, $\hat{\boldsymbol{n}}$, points outward from the surface, $\mathcal{S}$, thus defining the positive or up direction. The integrand $(\nabla \times \boldsymbol{F}) \cdot \hat{\boldsymbol{n}}$ is the projection of the curl of a vector field onto the surface outward normal. The boundary of the area, $\mathcal{C} = \partial\mathcal{S}$, is traversed counterclockwise following the unit tangent vector, $\hat{\boldsymbol{s}}$, when computing the circulation. Counter-clockwise is oriented relative to the positive side of the surface as defined by the outward normal.

unit normal, $\hat{\boldsymbol{n}}$, and $\mathrm{d}\mathcal{S}$ is the infinitesimal surface area element. The orientation of the outward normal determines, through the right hand rule, the counter-clockwise direction for the path integral.

2.7.2 Stokes' theorem for a rectangular region

To build experience with Stokes' theorem, consider the case of a rectangle in the x-y plane with dimensions L_x and L_y, and make use of Cartesian coordinates. In this case the outward normal is $\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}$, so that

$$(\nabla \times \boldsymbol{F}) \cdot \hat{\boldsymbol{z}} = \frac{\partial F^2}{\partial x} - \frac{\partial F^1}{\partial y}, \tag{2.115}$$

in which case the right hand side of Stokes' theorem reduces to

$$\int_{\mathcal{S}} (\nabla \times \boldsymbol{F}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S = \int_0^{L_x} \int_0^{L_y} \left(\frac{\partial F^2}{\partial x} - \frac{\partial F^1}{\partial y} \right) \mathrm{d}x \, \mathrm{d}y. \tag{2.116}$$

Integration around the rectangle then leads to a direct verification of Stokes' theorem

$$\int_0^{L_x} \int_0^{L_y} \left(\frac{\partial F^2}{\partial x} - \frac{\partial F^1}{\partial y} \right) \mathrm{d}x \, \mathrm{d}y = \int_0^{L_y} F^2(x,y) \Big|_{x=0}^{x=L_x} \mathrm{d}y - \int_0^{L_x} F^1(x,y) \Big|_{y=0}^{y=L_y} \mathrm{d}x \tag{2.117a}$$

$$= \int_0^{L_x} F^1(x,0) \, \mathrm{d}x + \int_0^{L_y} F^2(L_x,y) \, \mathrm{d}y + \int_{L_x}^0 F^1(x,L_y) \, \mathrm{d}x + \int_{L_y}^0 F^2(0,y) \, \mathrm{d}y \tag{2.117b}$$

$$= \oint_{\partial\mathcal{S}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x}. \tag{2.117c}$$

We can generalize this result to verify Stokes' theorem for an arbitrary surface. We do so by breaking the surface into a lattice of tiny rectangles. Integrating around the tiny rectangles and summing their contributions leads to a cancellation of the line integrals over all interior boundaries. The cancellation occurs since an internal edge of a rectangle is integrated once in each direction thus cancelling its contribution. The only nonzero contribution comes from

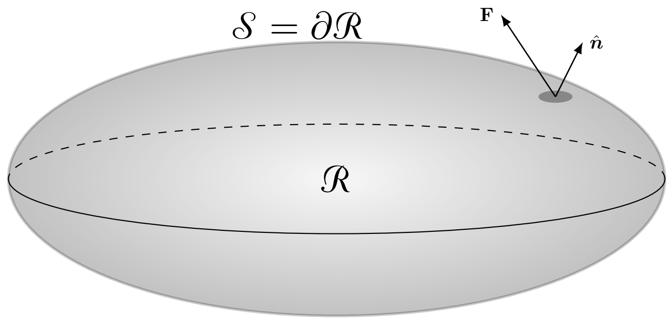


FIGURE 2.14: Illustrating the geometry of Gauss's divergence theorem for an ellipsoidal volume, $\mathcal{R}$, with closed boundary surface, $\mathcal{S} = \partial\mathcal{R}$. The outward unit normal along the boundary, $\hat{\boldsymbol{n}}$, is projected onto the vector field, $\mathbf{F}$, via the scalar product, $\mathbf{F} \cdot \hat{\boldsymbol{n}}$. Gauss's divergence theorem says that integrating $\mathbf{F} \cdot \hat{\boldsymbol{n}}$ over the closed boundary surface, $\mathcal{S}$, yields the same answer as computing the volume integral of the divergence, $\nabla \cdot \mathbf{F}$, over the closed region, $\mathcal{R}$, bounded by the closed surface $\mathcal{S}$.

integration over the boundary, $\partial\mathcal{S}$.

2.7.3 Stokes' theorem for a second order tensor

We now prove an expression of Stokes' theorem that holds for second order tensors. For this purpose, write the vector, $\mathbf{F}$, as

$$\mathbf{F} = \mathbf{c} \cdot \mathbf{T}, \tag{2.118}$$

where $\mathbf{c}$ is a spatially constant vector and $\mathbf{T}$ is a second order tensor. We thus have the curl given by

$$\nabla \times \mathbf{F} = \nabla \times (\mathbf{c} \cdot \mathbf{T}) = (\nabla \times \mathbf{T}) \cdot \mathbf{c}, \tag{2.119}$$

so that Stokes' theorem (2.113) takes the form

$$\left[\int_{\mathcal{S}} (\nabla \times \mathbf{T}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S}\right] \cdot \mathbf{c} = \left[\oint_{\partial\mathcal{S}} \mathbf{T} \cdot \mathrm{d}\boldsymbol{x}\right] \cdot \mathbf{c}. \tag{2.120}$$

Since $\mathbf{c}$ is an arbitrary constant vector, this equality holds in general

$$\int_{\mathcal{S}} (\nabla \times \mathbf{T}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \oint_{\partial\mathcal{S}} \mathbf{T} \cdot \mathrm{d}\boldsymbol{x}. \tag{2.121}$$

In Section 2.8 we make use of a similar trick with constant vectors to derive corollaries to the divergence theorem.

2.8 Gauss's divergence theorem

For a continuously differentiable vector field $\mathbf{F}$, Gauss's divergence theorem states that

$$\int_{\mathcal{R}} \nabla \cdot \mathbf{F} \, \mathrm{d}V = \oint_{\partial\mathcal{R}} \mathbf{F} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S}, \tag{2.122}$$

where $\hat{\boldsymbol{n}}$ is the outward unit normal to the boundary surface, $\partial\mathcal{R}$, and $\mathrm{d}\mathcal{S}$ is the surface area element on the boundary. In words, the left hand side of Gauss's theorem is the volume integral of the divergence of a continuously differentiable vector field over a connected volume, $\mathcal{R}$. The right hand side is the unit normal projection of the vector field that is area integrated over the closed surface, $\partial\mathcal{R}$, bounding the volume. We follow the convention that $\oint_{\partial\mathcal{R}}$ refers to a surface integral over a closed surface that bounds a volume. This notation contrasts with the surface integral, $\int_{\mathcal{S}}$, that generally does not enclose a volume. Figure 2.14 illustrates the geometry of Gauss's divergence theorem. In physics jargon, we say that the divergence of a vector field, $\nabla \cdot \mathbf{F}$, integrated over a volume equals to the flux of that vector field, $\mathbf{F} \cdot \hat{\boldsymbol{n}}$, integrated over the area bounding the volume.

2.8.1 An example rectangular volume

To build intuition for Gauss's divergence theorem, consider a rectangular volume with dimensions L_x, L_y, and L_z and make use of Cartesian coordinates. The volume integral on the left hand side of equation (2.122) gives

$$\int_{\mathcal{R}} \left[\frac{\partial F^1}{\partial x} + \frac{\partial F^2}{\partial y} + \frac{\partial F^3}{\partial z} \right] \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z. \tag{2.123}$$

Focusing on just the leftmost term, integration in x gives

$$\int_{\mathcal{R}} \frac{\partial F^1}{\partial x} \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z = \int_{y=0}^{y=L_y} \int_{z=0}^{z=L_z} [F^1(L_x, y, z) - F^1(0, y, z)] \, \mathrm{d}y \, \mathrm{d}z \tag{2.124a}$$

$$= \int_{\mathcal{S}_1 + \mathcal{S}_2} \boldsymbol{F} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S}, \tag{2.124b}$$

where $\mathcal{S}_1$ is the rectangle's face with outward unit normal $\hat{\boldsymbol{n}} = \hat{\boldsymbol{x}}$ and $\mathcal{S}_2$ is the rectangle's face with unit normal $\hat{\boldsymbol{n}} = -\hat{\boldsymbol{x}}$. Repeating this procedure on the other terms in equation (2.123) gives the area integrated flux through the full boundary. To verify the theorem for a general volume V, we take the approach used to prove Stokes' theorem. First, divide the volume into many rectangular sub-volumes. Then apply the above result to each sub-volume and sum up the result. The area integrated fluxes through internal rectangular faces cancel exactly. Therefore, the sum of all the area integrated fluxes equals to just the flux integrated over the external boundary, thus yielding the divergence theorem.

2.8.2 Gradient theorem for the volume integral of scalar fields

We consider various corollaries of the divergence theorem, the first of which arises from considering the special case of a vector field $\mathbf{F} = \phi \, \mathbf{c}$ with $\mathbf{c}$ an arbitrary *constant* vector. Substitution into the divergence theorem (2.122) yields

$$\oint_{\partial\mathcal{R}} \phi \, \mathbf{c} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \int_{\mathcal{R}} \nabla \cdot (\phi \, \mathbf{c}) \, \mathrm{d}V. \tag{2.125}$$

Pulling the constant vector out of the integrals and rearrangement leads to

$$\mathbf{c} \cdot \left[\oint_{\partial\mathcal{R}} \phi \, \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} - \int_{\mathcal{R}} \nabla\phi \, \mathrm{d}V \right] = 0. \tag{2.126}$$

Since $\mathbf{c}$ is an arbitrary constant vector, this equality is generally true if and only if

$$\oint_{\partial\mathcal{R}} \phi \, \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \int_{\mathcal{R}} \nabla\phi \, \mathrm{d}V. \tag{2.127}$$

In words, this result says that the integral of a scalar field over the boundary of a closed surface, when weighted by the outward unit normal to the surface, equals to the volume integral of the gradient of the scalar field integrated over the region bounded by the closed surface. We make use of this result in VOLUME 2 when studying how stresses contribute to the motion of a fluid element, with particular application to the case of pressure.

2.8.3 Surface integral of the outward unit normal

A corollary of equation (2.127) can be found by setting the scalar field, ϕ, to a constant so that $\nabla\phi = 0$. We thus find that the area integral of the outward unit normal vanishes when integrated over the surface of a closed volume

$$\oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = 0. \tag{2.128}$$

This identity also holds for two-dimensions, so that the outward normal has a zero line integral around a closed curve.

An example of the identity (2.128) can be seen by integrating over a closed rectangular volume, whereby the area integrals cancel component-wise. Another example is the sphere, where $\hat{\boldsymbol{n}} = \hat{\boldsymbol{r}}$ is the radial outward unit vector, so that integration of the radial unit vector over the spherical surface identically vanishes. For some purposes we can take equation (2.128) as the definition of a *simply closed volume* (or simply closed curve for the two dimensional case).

2.8.4 Integral of the curl of a vector field

Another identity follows from Gauss's theorem by setting

$$\mathbf{F} = \mathbf{c}\times\mathbf{v} \tag{2.129}$$

where $\mathbf{c}$ is a constant vector. As a result,

$$\nabla\cdot\mathbf{F} = -\mathbf{c}\cdot(\nabla\times\mathbf{v}), \tag{2.130}$$

which follows from the identity (2.37g). We are thus led to the following identities

$$\int_{\mathcal{R}} \nabla\cdot\mathbf{F}\,\mathrm{d}V = -\mathbf{c}\cdot\int_{\mathcal{R}} \nabla\times\mathbf{v}\,\mathrm{d}V \quad \text{equation (2.130)} \tag{2.131a}$$

$$= \oint_{\partial\mathcal{R}} (\mathbf{c}\times\mathbf{v})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} \quad \text{divergence theorem (2.122)} \tag{2.131b}$$

$$= \boldsymbol{c}\cdot\oint_{\partial\mathcal{R}} (\mathbf{v}\times\hat{\boldsymbol{n}})\,\mathrm{d}\mathcal{S} \quad \text{since } \boldsymbol{c} \text{ is a constant.} \tag{2.131c}$$

Since these identities hold for arbitrary $\mathbf{c}$, we are led to

$$\int_{\mathcal{R}} (\nabla\times\mathbf{v})\,\mathrm{d}V = \oint_{\partial\mathcal{R}} (\hat{\boldsymbol{n}}\times\mathbf{v})\,\mathrm{d}\mathcal{S}. \tag{2.132}$$

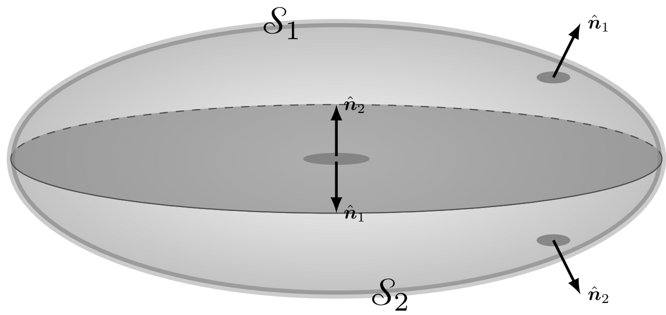


FIGURE 2.15: The integral of the unit normal component of the curl of a function vanishes when integrated over an oriented closed surface, $\partial\mathcal{R}$, which forms the boundary to a closed volume, $\mathcal{R}$. This result follows from both Gauss's divergence theorem as well as Stokes' curl theorem. For Gauss's theorem we find $\oint_{\partial\mathcal{R}}(\nabla\times\mathbf{F})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}}\nabla\cdot(\nabla\times\mathbf{F})\,\mathrm{d}V = 0$, which follows since the divergence of the curl vanishes. For Stokes' theorem we split the closed volume into two so that its boundary surface has also been split into two, $\partial\mathcal{R} = \mathcal{S}_1 + \mathcal{S}_2$. Separately applying Stokes' theorem to $\mathcal{S}_1$ and $\mathcal{S}_2$ leads to the calculation of the circulation around the common boundary along the equatorial plane. Since orientation of the outward unit normal along the equatorial plane is opposite for the two regions, the two circulations exactly cancel since they are computed in opposite directions.

2.8.5 First and second form of Green's identities

The further corollary to the divergence theorem arises from considering another special vector field

$$\mathbf{F} = \psi\,\nabla\phi, \tag{2.133}$$

with ψ and ϕ scalar fields. Substitution into the divergence theorem (2.122) leads to

$$\oint_{\partial\mathcal{R}}\psi\,\frac{\partial\phi}{\partial n}\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}}[\nabla\psi\cdot\nabla\phi + \psi\,\nabla^2\phi]\,\mathrm{d}V \qquad \text{Green's first integral identity.} \tag{2.134}$$

We can make this result more symmetric by swapping ψ and ϕ and then subtracting, to render

$$\oint_{\partial\mathcal{R}}\left[\psi\,\frac{\partial\phi}{\partial n} - \phi\,\frac{\partial\psi}{\partial n}\right]\mathrm{d}\mathcal{S} = \int_{\mathcal{R}}[\psi\,\nabla^2\phi - \phi\,\nabla^2\psi]\,\mathrm{d}V \qquad \text{Green's second integral identity.} \tag{2.135}$$

Setting $\phi = 1$ yields

$$\oint_{\partial\mathcal{R}}\frac{\partial\psi}{\partial n}\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}}\nabla^2\psi\,\mathrm{d}V \Longleftrightarrow \oint_{\partial\mathcal{R}}\nabla\psi\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}}\nabla\cdot\nabla\psi\,\mathrm{d}V. \tag{2.136}$$

We make use of these identities when studying the Green's function method for solving linear partial differential equations.

2.8.6 Integral of a curl over a closed surface

Application of Gauss's divergence theorem leads us to conclude that the following integral vanishes

$$\oint_{\partial\mathcal{R}}(\nabla\times\mathbf{F})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}}\nabla\cdot(\nabla\times\mathbf{F})\,\mathrm{d}V = 0, \tag{2.137}$$

where the final equality follows since the divergence of a curl vanishes. Hence, the integral of the unit normal projection of the curl of a vector field, as computed over an oriented closed surface, vanishes. We can understand this result geometrically by splitting the closed volume into two regions and then applying Stokes' theorem separately to the two regions (see Figure

2.15)

$$\oint_{\partial\mathcal{R}} (\nabla\times\mathbf{F})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = \int_{\mathcal{S}_1} (\nabla\times\mathbf{F})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}_1 + \int_{\mathcal{S}_2} (\nabla\times\mathbf{F})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}_2 \tag{2.138a}$$

$$= \oint_{\partial\mathcal{S}_1} \mathbf{F}\cdot\mathrm{d}\boldsymbol{x} - \oint_{\partial\mathcal{S}_2} \mathbf{F}\cdot\mathrm{d}\boldsymbol{x} \tag{2.138b}$$

$$= 0. \tag{2.138c}$$

The minus sign appearing in front of $\oint_{\partial\mathcal{S}_2}$ occurs since the orientation of the circulation integral is opposite that for $\oint_{\partial\mathcal{S}_1}$. We are thus left with a cancellation of the circulations. When applied to the vorticity of fluid flow (VOLUME 3), we see that

$$\oint_{\partial\mathcal{R}} \omega\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}} \nabla\cdot\omega\,\mathrm{d}V = 0, \tag{2.139}$$

where $\omega = \nabla\times\mathbf{v}$ is the vorticity and $\boldsymbol{v}$ is the fluid velocity.

2.8.7 The domain integral of a non-divergent Cartesian vector field

Consider a vector field that has zero divergence everywhere within a domain, $\mathcal{R}$. Consequently, $\int_{\mathcal{R}} \nabla\cdot\mathbf{F}\,\mathrm{d}V = \oint_{\partial\mathcal{R}} \mathbf{F}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = 0$. Now what can we say about $\int_{\mathcal{R}} \mathbf{F}\,\mathrm{d}V$? One might be tempted to say that it vanishes. But that is generally an incorrect statement as we now show.[8]

Making use of Cartesian coordinates, and the non-divergence property, know that

$$0 = \int_{\mathcal{R}} x^a\,\nabla\cdot\boldsymbol{F}\,\mathrm{d}V = \int_{\mathcal{R}} [\nabla\cdot(x^a\,\boldsymbol{F}) - F^a]\,\mathrm{d}V, \tag{2.140}$$

where we used $\partial_b x^a = \delta^a{}_b$. Use of the divergence theorem leads to

$$\int_{\mathcal{R}} F^a\,\mathrm{d}V = \int_{\mathcal{R}} \nabla\cdot(x^a\,\boldsymbol{F})\,\mathrm{d}V = \oint_{\partial\mathcal{R}} x^a\,(\hat{\boldsymbol{n}}\cdot\boldsymbol{F})\,\mathrm{d}\mathcal{S}. \tag{2.141}$$

The right hand side does not generally vanish since $\hat{\boldsymbol{n}}\cdot\boldsymbol{F}$ does not generally vanish at each point along $\partial\mathcal{R}$. Hence, we find that $\int_{\mathcal{R}} \boldsymbol{F}\,\mathrm{d}V = 0$ only for those domains where $\hat{\boldsymbol{n}}\cdot\boldsymbol{F} = 0$ at each point along $\partial\mathcal{R}$.

2.8.8 Gradient tensor theorem for $\nabla\otimes\boldsymbol{F}$

Certain elements from Stokes' theorem in Section 2.7 and the divergence theorem from Section 2.8 can be summarized in the following *gradient tensor theorem* that originates from *Gibbs* (1884) (equation (2), page 65, §161), and has been further discussed by *Lilly et al.* (2024) for applications in fluid mechanics. For this purpose, recall the definition of the *tensor product* from Section 1.5. We are led to the tensor product of the gradient operator and a vector field, here using Cartesian coordinates

$$\nabla\otimes\boldsymbol{F} = \boldsymbol{e}^a\partial_a\otimes\boldsymbol{e}_b\,F^b = (\boldsymbol{e}^a\otimes\boldsymbol{e}_b)\,\partial_a F^b. \tag{2.142}$$

[8] As a technical note, we observe that the integral of a vector field is only well defined in Euclidean space and using Cartesian coordinates. A general manifold requires extra formalism for the purpose of comparing vectors at two different points.

Integrating over a domain leads to

$$\begin{aligned}
\int_{\mathcal{R}} \nabla \otimes \boldsymbol{F} \, \mathrm{d}V &= \int_{\mathcal{R}} \boldsymbol{e}^a \partial_a \otimes \boldsymbol{e}_b \, F^b \, \mathrm{d}V && \text{expose tensor indices} && (2.143a) \\
&= \int_{\mathcal{R}} \partial_a (\boldsymbol{e}^a \otimes \boldsymbol{e}_b \, F^b) \, \mathrm{d}V && \text{Cartesian basis is constant} && (2.143b) \\
&= \left[\int_{\mathcal{R}} \partial_a (\boldsymbol{e}^a \, F^b) \, \mathrm{d}V \right] \otimes \boldsymbol{e}_b && \text{Cartesian basis is constant} && (2.143c) \\
&= \left[\oint_{\partial\mathcal{R}} \hat{n}_a \, \boldsymbol{e}^a \, F^b \, \mathrm{d}\mathcal{S} \right] \otimes \boldsymbol{e}_b && \text{divergence theorem} && (2.143d) \\
&= \oint_{\partial\mathcal{R}} \hat{n}_a \, \boldsymbol{e}^a \otimes \boldsymbol{e}_b \, F^b \, \mathrm{d}\mathcal{S} && \text{Cartesian basis is constant} && (2.143e) \\
&= \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \otimes \boldsymbol{F} \, \mathrm{d}\mathcal{S} && \text{boldface notation.} && (2.143f)
\end{aligned}$$

Use of the divergence theorem in equation (2.143d) is an application of the scalar gradient theorem (2.127) separately to each component of F^b, so that

$$\oint_{\partial\mathcal{R}} F^b \, \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \int_{\mathcal{R}} \nabla F^b \, \mathrm{d}V. \tag{2.144}$$

The assumption of Cartesian tensors is seemingly basic to the above derivation of the *gradient tensor theorem*

$$\int_{\mathcal{R}} \nabla \otimes \mathbf{F} \, \mathrm{d}V = \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \otimes \mathbf{F} \, \mathrm{d}\mathcal{S}. \tag{2.145}$$

In fact, when making use of the *covariant derivative* of a vector as defined in Section 3.15, as well as the *metricity condition* from Section 3.16, the derivation also follows for arbitrary coordinates. The only modification is to interpret the derivative operator, ∇, as a covariant derivative operator rather than a partial derivative operator. However, while offering a variety of uses of the gradient tensor theorem for fluids moving on flat surfaces and described by arbitrary coordinates, *Lilly et al.* (2024) also shows that the theorem cannot be generalized to arbitrary curved surfaces.

2.9 Exercises

Throughout these exercises we consider a point whose position, relative to an arbitrary origin, is represented using Cartesian coordinates according to

$$\boldsymbol{x} = x \, \hat{\boldsymbol{x}} + y \, \hat{\boldsymbol{y}} + z \, \hat{\boldsymbol{z}}, \tag{2.146}$$

and whose squared distance from the origin is

$$r^2 = \boldsymbol{x} \cdot \boldsymbol{x} = |\boldsymbol{x}|^2 = x^2 + y^2 + z^2. \tag{2.147}$$

Also note that the radial unit direction is

$$\hat{\boldsymbol{r}} = \nabla r = \boldsymbol{x}/|\boldsymbol{x}|. \tag{2.148}$$

Unless otherwise noted, all variables are non-dimensional.

EXERCISE 2.1: PRACTICE WITH THE GRADIENT OPERATOR
Prove the following identities:

(a) $\nabla(|\boldsymbol{x}|) = \nabla r = \boldsymbol{x}\,|\boldsymbol{x}|^{-1} \equiv \hat{\boldsymbol{r}}$
(b) $\nabla \ln|\boldsymbol{x}| = \boldsymbol{x}\,|\boldsymbol{x}|^{-2} = |\boldsymbol{x}|^{-1}\,\hat{\boldsymbol{r}}$
(c) $\nabla|\boldsymbol{x}|^{-1} = -\boldsymbol{x}\,|\boldsymbol{x}|^{-3} = -\hat{\boldsymbol{r}}\,|\boldsymbol{x}|^{-2}$
(d) $\partial_n\partial_m|\boldsymbol{x}|^{-1} = |\boldsymbol{x}|^{-3}\,(3\,\hat{\boldsymbol{r}}_m\,\hat{\boldsymbol{r}}_n - \delta_{mn})$

EXERCISE 2.2: PRACTICE WITH THE LAPLACIAN OPERATOR
Show that the Laplacian of the function

$$\psi = \frac{z\,x^2}{r^2} \tag{2.149}$$

is given by

$$\nabla^2\psi = \frac{2\,z\,(r^2 - 5\,x^2)}{r^4}. \tag{2.150}$$

Perform the derivation using both Cartesian coordinates as well as spherical coordinates (see Figure 4.2), making use of the following expressions for Laplacian operator acting on a scalar field

$$\nabla^2\psi(x,y,z) = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} \tag{2.151a}$$

$$\nabla^2\psi(\lambda,\phi,r) = \frac{1}{r^2\,\cos\phi}\left[\frac{1}{\cos\phi}\frac{\partial^2\psi}{\partial\lambda^2} + \frac{\partial}{\partial\phi}\left(\cos\phi\frac{\partial\psi}{\partial\phi}\right) + \cos\phi\,\frac{\partial}{\partial r}\left(r^2\,\frac{\partial\psi}{\partial r}\right)\right]. \tag{2.151b}$$

EXERCISE 2.3: MORE PRACTICE WITH OPERATORS
Prove the following identities with $r \neq 0$:

(a) $\nabla^2 r^{-1} = 0$
(b) $\nabla\cdot(\boldsymbol{x}/r^3) = 0$
(c) $\nabla\cdot(\boldsymbol{A}\times\boldsymbol{x}) = \boldsymbol{x}\cdot(\nabla\times\boldsymbol{A})$ for an arbitrary vector field $A(\boldsymbol{x})$.
(d) $\nabla\times[\boldsymbol{x}\,f(r)] = 0$ for an arbitrary function $f(r) = f(|\boldsymbol{x}|)$.

EXERCISE 2.4: RIGID-BODY ROTATION
Define a velocity field according to

$$\boldsymbol{v} = \boldsymbol{\Omega}\times\boldsymbol{x} \tag{2.152}$$

with $\boldsymbol{\Omega}$ a spatially constant angular rotation velocity (e.g., rotation of the earth). This velocity field describes rigid-body rotation as discussed in Section 11.3. Show that $2\,\boldsymbol{\Omega} = \nabla\times\boldsymbol{v}$.

EXERCISE 2.5: DIVERGENCE-FREE AND IRROTATIONAL VECTOR
Let Φ be a harmonic function so that $\nabla^2\Phi = 0$. Show that $\boldsymbol{v} = -\nabla\Phi$ satisfies

(a) $\nabla\cdot\boldsymbol{v} = 0$
(b) $\nabla\times\boldsymbol{v} = 0$.

In this way we prove that all harmonic scalar fields correspond to a divergence-free and curl-free vector field.

EXERCISE 2.6: RELATING ADVECTION, CURL, AND KINETIC ENERGY
Derive the following equation

$$(\mathbf{v}\cdot\nabla)\mathbf{v} = \omega\times\mathbf{v} + \nabla(\mathbf{v}\cdot\mathbf{v})/2, \tag{2.153}$$

where

$$\omega = \nabla \times \mathbf{v} \tag{2.154}$$

is the vorticity, $\mathbf{v} \cdot \mathbf{v}/2$ is the kinetic energy per mass, and $\mathbf{v}$ is the fluid velocity field. This equation is key for the study of vorticity in fluid mechanics.

EXERCISE 2.7: CONSERVATIVE VECTOR FIELD AND THE SCALAR POTENTIAL
Show that the curl, $\nabla \times \boldsymbol{F}$, of the following vector field vanishes

$$\boldsymbol{F} = 2\,x\,z\,\hat{\boldsymbol{x}} + 2\,y\,z^2\,\hat{\boldsymbol{y}} + (x^2 + 2\,y^2\,z - 1)\,\hat{\boldsymbol{z}}. \tag{2.155}$$

Hence, deduce that $\boldsymbol{F}$ is a conservative vector field, meaning that it can be written as the gradient of a scalar potential ψ according to $\boldsymbol{F} = -\nabla\psi$, where (to within an arbitrary constant)

$$\psi = -[x^2\,z + (y\,z)^2 - z]. \tag{2.156}$$

EXERCISE 2.8: PRODUCT RULE IDENTITIES
Prove the following identities with $\boldsymbol{F}$ a Cartesian vector in $\mathbb{R}^3$:

(a) $\boldsymbol{F} = \partial_n\,(F^n\,\boldsymbol{x}) - \boldsymbol{x}\,\nabla\cdot\boldsymbol{F}$
(b) $2\,F^m = [\boldsymbol{x}\times(\nabla\times\boldsymbol{F})]^m - \partial_m(\boldsymbol{x}\cdot\boldsymbol{F}) + \nabla\cdot(\boldsymbol{x}\,F^m)$.
(c) $\partial_m\,\partial_n(x^k\,x^l) = \delta^k{}_m\,\delta^l{}_n + \delta^l{}_m\,\delta^k{}_n$.

Make use of Cartesian tensors and show all relevant steps, including use of the Levi-Civita tensor from Section 1.6.1 for the vector cross product.

As discussed in Section 12.4.1 of *Bühler* (2014), the first two product rule identities have use for the study of impulses imparted by a body force per volume, $\boldsymbol{F}$, to a fluid on an unbounded domain where the force has compact support (i.e., the force vanishes outside a finite domain). In that case the above product rule identities allow us to make use of the corresponding integral identities

$$\int \boldsymbol{F}\,\mathrm{d}V = -\int \boldsymbol{x}\,\nabla\cdot\boldsymbol{F}\,\mathrm{d}V = \frac{1}{2}\int \boldsymbol{x}\times(\nabla\times\boldsymbol{F})\,\mathrm{d}V. \tag{2.157}$$

EXERCISE 2.9: JACOBIAN EVALUATED ALONG A CONTOUR
Consider the vector cross product of two functions, $\psi(x,y)$ and $Q(x,y)$

$$\hat{\boldsymbol{z}}\cdot(\nabla\psi\times\nabla Q) = \partial_x\psi\,\partial_y Q - \partial_y\psi\,\partial_x Q \equiv J(\psi,Q), \tag{2.158}$$

where the final equality defined the Jacobian operator. Show that when evaluated along a contour of constant Q, the Jacobian is given by

$$J(\psi,Q) = -(\hat{\boldsymbol{n}}\cdot\nabla Q)\,(\hat{\boldsymbol{t}}\cdot\nabla\psi) \tag{2.159}$$

where $\hat{\boldsymbol{t}}$ is the unit tangent along the contour and $\hat{\boldsymbol{n}}$ is a unit vector pointing to the left of the tangent (e.g., see Figure 2.11).

EXERCISE 2.10: CURVATURE OF A CIRCLE
Make use of equation (2.71), as well as the polar coordinates from Section 4.3, to show that for a circle of radius r we have

$$-\nabla\cdot\hat{\boldsymbol{n}} = \frac{1}{r}, \tag{2.160}$$

where $\hat{\boldsymbol{n}} = -\hat{\boldsymbol{r}}$ points to the left when moving around the circle in a counterclockwise direction. Hence, the radius of curvature for the circle equals to the circle's radius.

EXERCISE 2.11: BELTRAMI FLOW

Beltrami flow is defined by velocity and vorticity fields satisfying

$$\nabla \cdot \boldsymbol{v} = 0 \tag{2.161a}$$

$$\boldsymbol{\omega} = \nabla \times \boldsymbol{v} = \lambda\, \boldsymbol{v} \tag{2.161b}$$

where λ is a constant with dimensions of inverse length. Show that the following velocity field is a Beltrami flow

$$\boldsymbol{v} = [A \sin(\lambda\, z) + C \cos(\lambda\, y)]\, \hat{\boldsymbol{x}} + [B \sin(\lambda\, x) + A \cos(\lambda\, z)]\, \hat{\boldsymbol{y}} + [C \sin(\lambda\, y) + B \cos(\lambda\, x)]\, \hat{\boldsymbol{z}}, \tag{2.162}$$

where A, B, C are constants with dimensions of length per time. Hint: the solution follows directly from computing

$$\lambda\, u = \partial_y w - \partial_z v \quad \text{and} \quad \lambda\, v = \partial_z u - \partial_x w \quad \text{and} \quad \lambda\, w = \partial_x v - \partial_y u. \tag{2.163}$$

EXERCISE 2.12: PRACTICE WITH PATH INTEGRALS

Consider the vector field written using Cartesian coordinates,

$$\boldsymbol{F} = x\, y^2\, \hat{\boldsymbol{x}} + 2\, \hat{\boldsymbol{y}} + x\, \hat{\boldsymbol{z}}, \tag{2.164}$$

where all variables are non-dimensional. Let $\mathcal{L}$ be a path parameterized by

$$x = c\, t \qquad y = c/t \qquad z = d \qquad t \in [1, 2], \tag{2.165}$$

where c and d are non-dimensional constants and t is a non-dimensional parameter. Show that the following identities hold

(a) $\int_{\mathcal{L}} \boldsymbol{F}\, \mathrm{d}t = c^3 \ln 2\, \hat{\boldsymbol{x}} + 2\, \hat{\boldsymbol{y}} + \frac{3\, c}{2}\, \hat{\boldsymbol{z}}$

(b) $\int_{\mathcal{L}} \boldsymbol{F}\, \mathrm{d}y = -\frac{3\, c^4}{8}\, \hat{\boldsymbol{x}} - c\, \hat{\boldsymbol{y}} - c^2 \ln 2\, \hat{\boldsymbol{z}}$

(c) $\int_{\mathcal{L}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = c^4 \ln 2 - c$,

where $\mathrm{d}\boldsymbol{x} = \hat{\boldsymbol{x}}\, \mathrm{d}x + \hat{\boldsymbol{y}}\, \mathrm{d}y + \hat{\boldsymbol{z}}\, \mathrm{d}z$. Although all three integrals are computed along the same path, they are not necessarily of the same type. In particular, the first two integrals are vector fields, whereas the third integral is a scalar.

EXERCISE 2.13: AREA OF A SIMPLY CONNECTED REGION

Consider a simply connected region, $\mathcal{S}$, on the plane. Show that twice the area of this region is given by

$$2 \oiint_{\mathcal{S}} \mathrm{d}\mathcal{S} = \oint_{\partial \mathcal{S}} (x\, \mathrm{d}y - y\, \mathrm{d}x), \tag{2.166}$$

where $\partial \mathcal{S}$ is the simple closed curve that bounds the area $\mathcal{S}$. Hint: use Stokes' theorem on the plane, also known as Green's theorem. Check your answer for the special case with $\mathcal{S}$ a circle of radius R.

EXERCISE 2.14: STOKES' THEOREM ON A PLANE

Show that

$$\mathcal{I} = \oint_{\partial \mathcal{S}} \left[y\, (4\, x^2 + y^2)\, \mathrm{d}x + x\, (2\, x^2 + 3\, y^2)\, \mathrm{d}y \right] = \pi\, a^3\, b/2 \tag{2.167}$$

when integrating around the boundary of an ellipse $\mathcal{S}$ defined by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \tag{2.168}$$

where a, b are constants. Hint: make use of Stokes' theorem on a plane, otherwise known as Green's Theorem. Then map the ellipse to a circle and use polar coordinates.

exercise 2.15: Practice with Gauss's divergence theorem

We here demonstrate the validity of Gauss's divergence theorem for a particular vector field

$$\boldsymbol{F} = \frac{\alpha\,\boldsymbol{x}}{(r^2 + a^2)^{3/2}}, \tag{2.169}$$

where α and a are constants and $r^2 = \boldsymbol{x} \cdot \boldsymbol{x}$ is the squared radial distance to a point. Using fluid mechanics jargon, we think of $\boldsymbol{F}$ as a matter flux with physical dimensions of $\mathrm{M\ L^{-2}\ T^{-1}}$ (mass length^{-2} time^{-1}).

(a) Compute the transport of $\boldsymbol{F}$ through a spherical surface, $\mathcal{S}$, of radius $|\boldsymbol{x}| = a\sqrt{3}$

$$\Phi = \oint_{|\boldsymbol{x}|=a\sqrt{3}} \boldsymbol{F} \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = \frac{3\,\pi\,\alpha\sqrt{3}}{2}. \tag{2.170}$$

With $\boldsymbol{F}$ a matter flux then Φ has physical dimensions of $\mathrm{M\ T^{-1}}$, so that it is the *mass transport* through the spherical surface.

(b) Show that this transport is equal to the integral of the divergence over the volume of the sphere

$$\Phi = \int_{|\boldsymbol{x}|=a\sqrt{3}} \nabla \cdot \boldsymbol{F}\,\mathrm{d}V. \tag{2.171}$$

We thus verify, for this particular vector field, the divergence theorem

$$\int_{\mathcal{R}} \nabla \cdot \boldsymbol{F}\,\mathrm{d}V = \int_{\partial\mathcal{R}} \boldsymbol{F} \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}, \tag{2.172}$$

where $\hat{\boldsymbol{n}}$ is the outward unit normal on the bounding surface $\mathcal{S}$.

exercise 2.16: More practice with Gauss's divergence theorem

Prove the following identities, which are readily shown using Gauss's divergence theorem.

(a) $\oint_{\partial\mathcal{R}} \boldsymbol{x} \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = 3\int_{\mathcal{R}} \mathrm{d}V = 3\,V$, where $\mathcal{R}$ is a closed region bounded by $\partial\mathcal{R}$ and with volume $\int_{\mathcal{R}} \mathrm{d}V = V$.

(b) $\oint_{\partial\mathcal{R}} (\hat{\boldsymbol{n}} \times \boldsymbol{F})\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}} \nabla \times \boldsymbol{F}\,\mathrm{d}V$, for an arbitrary vector field $\boldsymbol{F}$ and with $\hat{\boldsymbol{n}}$ the outward unit normal on the bounding surface $\partial\mathcal{R}$. Hint: in a manner similar to the gradient theorem proven in Section 2.8.2, make use of Gauss's theorem with $\boldsymbol{A} = \boldsymbol{F} \times \boldsymbol{C}$ where $\boldsymbol{C}$ is a constant vector.

(c) Let $\partial\mathcal{R}$ be a closed surface bounding a simply connected region, $\mathcal{R}$, and let $\boldsymbol{x}$ denote the position vector of a point measured from an arbitrary origin. Prove the following

$$\oint_{\partial\mathcal{R}} \frac{\hat{\boldsymbol{n}} \cdot \boldsymbol{x}}{r^3}\,\mathrm{d}\mathcal{S} = \begin{cases} 0 & \text{if the origin lies outside of } \partial\mathcal{R} \\ 4\pi & \text{if the origin lies inside of } \partial\mathcal{R}. \end{cases} \tag{2.173}$$

exercise 2.17: A curious feature of spheres in high dimensions

The volume of a sphere of radius r in n dimensions is given by

$$V^{(n)}(r) = \frac{\pi^{n/2}\,r^n}{\Gamma(1 + n/2)}, \tag{2.174}$$

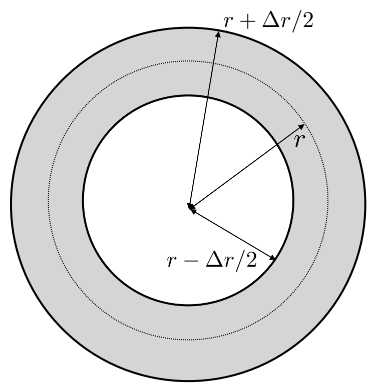


FIGURE 2.16: Schematic for Exercise 2.17 where we consider the volume within a spherical shell (gray region), $V(r + \Delta r/2) - V(r - \Delta r/2)$, as compared to the volume, $V(r)$, of the sphere straddled by the shell.

where the Gamma function satisfies

$$\Gamma(1/2) = \sqrt{\pi} \quad \text{and} \quad \Gamma(n+1) = n! \quad \text{for integer } n. \tag{2.175}$$

Show that the ratio,

$$\Delta V/V = \frac{V(r + \Delta r/2) - V(r - \Delta r/2)}{V(r)}, \tag{2.176}$$

is huge unless $\Delta r/r \approx n^{-1}$. That is, the volume of a spherical shell is far larger than the volume of the sphere straddled by the shell, except for tiny shells whose radial extent scales as n^{-1}. This result is particularly useful in the study of many body mechanics, as encountered in statistical mechanics, where n is on the order of Avogadro's number, $n \approx A^{\mathrm{v}} = 6.022 \times 10^{23}$.

Chapter 3

General tensors

Vector calculus, as formalized by the Cartesian tensor analysis of Chapters 1 and 2, is sufficient for many areas of geophysical fluid mechanics. However, there are a number of topics where general tensors helps the physics to shine through the maths. First, it supports our understanding of general curvilinear coordinates such as cylindrical-polar and spherical considered in Chapter 4. Next, and more importantly, general tensors are very useful in the study of fluid kinematics from the Lagrangian reference frame as considered in VOLUME 2. We also require general tensors when using Lagrangian kinematics to formulate Hamilton's principle for fluids in VOLUME 4. General tensors are quite useful in the study of generalized vertical coordinates from VOLUME 3. For the reader uninterested in any of these topics, then this chapter can be readily skipped.

READER'S GUIDE TO THIS CHAPTER

We restrict attention to tensor analysis on spatial manifolds endowed with a metric, thus touching on the rudiments of Riemannian differential geometry. When considering tensor analysis on curved manifolds, such as the sphere or an isopycnal, we assume these manifolds are embedded in Euclidean space, which provides a direct connection to the Cartesian tensor analysis from Chapters 1 and 2.[a] Indeed, some material from those two earlier chapters is revisited here as we generalize to arbitrary coordinates. In particular, Sections 3.1 through 3.5 focus on foundations and notation. Sections 3.6 through 3.13 then focus on algebra, and the remaining sections are concerned with calculus.

Our treatment is consistent with the fluid mechanics book by *Aris* (1962), the physics book by *Thorne and Blandford* (2017), and the geophysics and tensor analysis books of *Tromp* (2025a) and *Tromp* (2025b). There are a relatively large number of footnotes in this chapter. Many offer mathematical details meant for the aficionado, and they can be readily skipped by the more casual reader.

[a]Given that all manifolds of concern in this book are embedded in Euclidean space, it follows that all manifolds have zero torsion, which is also assumed in Einstein's treatment of gravity (e.g., *Misner et al.* (1973)). See *Tromp* (2025a) and *Tromp* (2025b) for applications of nonzero torsion in continuum mechanics.

3.1 Our use for general tensors

There is a nontrivial payoff for learning the formalism of general tensor analysis. Namely, it enables us to formulate the fluid mechanical equations for motion on a rotating spheroidal planet; to garner insights into the foundations of Lagrangian fluid mechanics and Hamilton's principle; and to understand the many facets of generalized vertical coordinates used for stratified fluid mechanics. The good news (for those typically eschewing mathematical formalisms) is that our needs for general tensors in geophysical fluid mechanics are rather modest, thus requiring only a minor degree of new tools beyond those for Cartesian tensors presented in Chapters 1 and 2. The following reasons allow for our relatively shallow dive into general tensor analysis.

- Geophysical fluids are embedded within the same Euclidean space used for classical particle mechanics. Notably, Euclidean space has zero intrinsic curvature so it is a flat space. Through this embedding, the local geometry inherits features from the Euclidean space such as the Kronecker metric, δ_{ab}. For these reasons we can make great use of Cartesian tensors for many purposes in this book.
- We make use of universal Newtonian time. Hence, time is measured the same by all observers and reference frames. Consequently, all observers agree on past, present, and future. So although the spatial coordinates used by geophysical fluid mechanics can be a function of time, the time coordinate is always independent of space.
- For many purposes, we make use of the position of matter relative to an origin, typically assumed to be the center of the planet. In so doing, we retain the notion of a position as a directed line segment starting from an origin, just as for Cartesian tensor analysis in Section 1.1. We do introduce a more general treatment that dispenses with the notion of an origin and corresponding position vector. Such treatments are essential for many purposes in continuum mechanics, particularly Lagrangian kinematics, where there is no special point that can serve as the origin. Yet for most purposes, we retain the notion of a coordinate origin since it is needed to discuss motion on a rotating planet. Namely, on a rotating planet the planet's center provides a natural coordinate origin and the rotational axis breaks the isotropy of space. It follows that to understand the planetary Coriolis acceleration and planetary centrifugal acceleration requires an acknowledgement of the

rotational axis and planetary center.

Each of these features of the space and time used for geophysical fluid mechanics means that our mathematical needs are less than the general relativist who studies fluids moving in strong gravity fields and/or over galactic distances, or for the solid mechanic who works extensively with Lagrangian coordinates.

3.2 Galilean covariance and coordinate invariance

Physics is independent of the subjective choices made by physicists who describe physical phenomena. This principle motivates us to seek mathematical expressions that include objects whose meaning transcends a particular coordinate representation. At its most basic level, there is no *a priori* notion of the underlying geometry, with an insistence on such generalities leading to general relativity and the notion of general covariance. General covariance means that the physical equations take on the same form regardless the coordinates, even though when unpacked into coordinate components the terms in these equations generally are effected by coordinate choice. We do not need general covariance in this book, given that we work on a background Euclidean space on which the fluid moves, and we make use of a universal Newtonian time measured by all observers. That is, we work with Galilean space-time. Nonetheless, within the restricted class of Galilean space-time, we do insist that the mathematical expression of a physical relation in an inertial reference frame be independent of the coordinate choice.

We refer to the above property of the physical equations as covariance, which means the equations are coordinate invariant within the bounds of Galilean relativity. This property is ensured when the equations of geophysical fluid mechanics are relations between geometric objects such as points, vectors, and tensors. In this chapter we develop operational tools for expressing the continuum fluid equations in a form that exposes their underlying geometric foundation.

Although physics does not care about coordinates, the practicing physicist does. Namely, it is convenient, and sometimes necessary, to work with specific coordinates suited to the symmetry of the physical system, particularly when comparing theory to experiment or when formulating a numerical model. After deriving a physical law in one set of coordinates, it is often of interest to establish the form of the law in another set of coordinates. How does the physical law, typically represented as a differential equation, transform into other coordinates? So long as the equations are written in a proper tensorial form then the equations are form invariant. What constitutes "proper tensorial form" is a central topic for this chapter. It follows that we can establish the validity of a physical relation in any convenient set of coordinates (including Cartesian), and then extend that relation to all coordinates so long as we respect the basic rules of tensor analysis.[1]

3.2.1 Covariance as distinct from covariant

The term covariance is here used as a noun, referring to the coordinate invariant form of equations under a particular set of coordinate transformations (here Galilean transformations). In tensor analysis we also use covariant as an adjective, which refers to how a particular

[1] Equations written in a non-inertial reference frame do not manifest Galilean covariance. Even so, terrestrial observers are in a non-inertial reference frame and so it is important to confront physics within this reference frame. Hence, throughout this book, we study the accelerations that result from viewing physical systems from an accelerating reference frame, with such accelerations (planetary centrifugal and planetary Coriolis) fundamental to geophysical fluid flows.

coordinate representation of a tensor transforms, as signaled by use of a downstairs covariant index placement. In particular, the downstairs index is used for the coordinate representations of a one-form, which is sometimes referred to as a "covariant vector." The covariant terminology was introduced into the early days of tensor analysis given that the transformation properties of a one-form align with the components to the metric tensor; i.e., they "covary".

3.2.2 Tensor operations

Ensuring a mathematical physics equation is form invariant requires the equations to respect certain tensor rules. In brief, all tensor indices are properly matched and each derivative is covariant as specified in Section 3.15. The elegance and power rendered by coordinate invariance is the key reason that tensor analysis is ubiquitous in theoretical physics.

To ensure an equation respects coordinate invariance requires us to understand certain properties of tensors and operations with tensors that produce components of new tensors. We here summarize the specific properties characterizing coordinate invariance (taken after page 153 of *Schutz*, 1985).

1. Manipulations of tensor components are called *permissible tensor operations* if they produce components of new tensors. The following are permissible operations:
   (a) Multiplication of a tensor by a scalar produces a new tensor of the same type.
   (b) Addition of components of two tensors of the same type gives components of a new tensor of the same type. In particular, only tensors of the same type can be equal.
   (c) Multiplication of components of two tensors of arbitrary type gives components of a new tensor whose type is given by the sum of the types for the individual tensors. This operation is called the tensor product and is denoted by the operator $\otimes$. For example, if $\mathbf{A}$ and $\mathbf{B}$ are two vectors, then $\mathbf{A} \otimes \mathbf{B}$ is a second order tensor built from the outer product of two vectors. The discussion in Section 1.5 of tensor products largely holds for general tensors as well.
   (d) The covariant derivative operator (Sections 3.15 and 3.15.2) increases by one the order of a tensor, with the covariant derivative operator denoted by $\boldsymbol{\nabla}$.
   (e) Contraction on a pair of indices of the components of a tensor reduces by one the order of the tensor.
   (f) A corollary of the multiplication rule is that if the contraction of two objects yields a tensor, and if one of these objects is a tensor, then so too is the other. This result is known as the quotient rule.
2. If two tensors of the same type have equal components in a given coordinate system, then they have equal components in all coordinate systems. Hence, the tensors are identical. As a corollary, if a tensor is zero in one coordinate system, then it is zero in all coordinate systems. Conversely, if an object vanishes in one set of coordinates but is nonzero in another, then that object is *not* a tensor.
3. If a mathematical equation consists of tensors combined only by the permissible tensor operations, and if the equation is true in one coordinate system, then it is true in any coordinate system. If the equations involve covariant derivatives, then the equations remain form invariant under coordinate changes. For the partial differential equations of geophysical fluid mechanics, covariant differentiation is the key to coordinate invariance.

3.2.3 Comments

In the remainder of this chapter we provide details needed to unpack the notions of coordinate invariance and to specify tensor operations. Even without penetrating these details, the reader should be able to appreciate why coordinate invariance is so central to physics. Namely, if we needed to rederive physical equations each time we converted to a new coordinate system, then the task of deriving and understanding physical relations would be far more tedious and onerous.

3.3 Notational conventions

We here summarize notational conventions associated with tensor manipulations, sometimes referred to as index gymnastics. Many of these conventions are familiar from our study of Cartesian tensors in Chapters 1 and 2, so that the presentation here is relatively terse.

3.3.1 Covariant, contravariant, and Einstein summation

For general tensors, the Einstein summation convention assumes that tensor labels (also called tensor indices) are summed over their range when a lower covariant index matches an upper contravariant index. In this way we have an arbitrary vector represented as

$$\mathbf{F} = \sum_{a=1}^{3} F^a \, \boldsymbol{e}_a = F^a \, \boldsymbol{e}_a. \tag{3.1}$$

By extension, the placement of tensor labels (covariant versus contravariant) has specific meaning in general tensor analysis. In particular, it is necessary to ensure conservation of indices across an equal sign, balancing across both upstairs (contravariant) indices and downstairs (covariant) indices.

The names *covariant* and *contravariant* originate from their relation to the labels placed on the coordinate basis vectors introduced in Section 3.6, and thus how they change under coordinate transformations relative to how basis vectors change (Section 3.7). The covariant tensor label accords with the downstairs label placement for a coordinate basis, whereas the upstairs contravariant label is contrary to the coordinate basis (e.g., see Section 2.26 of *Schutz* (1980). A useful mneumonic is "co-low" to signal that the covariant label is downstairs ("low").

Although we make use of the names covariant and contravariant when useful, these terms are used infrequently in modern tensor analysis. The reason for their infrequent use is that modern treatments consider tensors as geometric objects (e.g., Section 3.6), so that such treatments are not primarily concerned with the coordinate representations of tensors. Additionally, when allowing for a metric tensor (Section 3.7), then the metric can move tensor indices up and down, thus blurring the distinction between covariant and contravariant representations.

3.3.2 Upright and slanted notation

As detailed in Section 1.1.2 we use a bold upright symbol for a geometric object living in Euclidean space. The position, $\mathbf{x} \in \mathbb{E}^3$, is an example, as are tensors written as $\mathbf{F}$. In contrast, a slanted bold symbol is used for a coordinate representation of the position, $\boldsymbol{x}$, or of a tensor, $\boldsymbol{F}$, with coordinate representations living in the space of real numbers. Each coordinate representation is realized by specifying a subjectively chosen set of coordinates. The tensor does not change when changing coordinates, but its coordinate representation does change. Tensor analysis provides a systematic means to transform between coordinate representations.

The upright versus slanted notation is fundamental conceptually, since it is important to appreciate that tensors are geometric objects that are not subject to the whims of a particular coordinate choice. Correspondingly, physically robust differential and integral equations are coordinate invariant. Even so, the upright-slanted notation can be softly adhered to without much cause for concern, so long as we are careful to write the coordinate equations using rules of tensor analysis. In that case, the coordinate equations are unaltered in form when changing coordinates; i.e., they are tensor equations.

3.3.3 Physical dimensions

When representing a tensor, such as a vector, in terms of a particular coordinate basis,

$$\mathbf{F} = F^a \, \boldsymbol{e}_a, \tag{3.2}$$

the physical dimensions of the basis vectors, $\boldsymbol{e}_a$, determine those of the coordinate representation, F^a. Also, as noted in Section 3.6.4, the physical dimensions generally differ when evaluating a tensor at a point on a manifold versus when we are considering the tensor as a field. These notions about physical dimensions are often ignored in the mathematics literature, where physical dimensions are of no concern. Hence, care should be exercised when translating math equations to a physics application. In this book, we are concerned with physics, so that mathematical equations must have physically consistent dimensions, and the corresponding examination of dimensional consistency is a very useful means to find bugs in mathematical manipulations.

3.3.4 Space-time notation

As introduced in Section 3.4.3, we make use of a Greek label when incorporating time to the tensor indices, with $\alpha = 0$ denoting the time coordinate. Time is universal in the Newtonian world of geophysical fluid mechanics, so that the time coordinate is independent of space. However, many spatial coordinates are functions of both space and time. Therefore, the time derivative of a tensor field computed in one set of spatial coordinates generally differs from another set of spatial coordinates.

Following equation (3.6) used for coordinates, we make use of the following index notation and ordered list for contravariant components of space-time 4-vectors

$$F^\alpha = (F^0, F^1, F^2, F^3) = (F^0, F^a) = (F^0, \boldsymbol{F}). \tag{3.3}$$

The time component of the velocity 4-vector is unity

$$v^\alpha = (1, v^1, v^2, v^3) = (1, v^a) = (1, \boldsymbol{v}) \qquad \text{4-velocity in arbitrary coordinates.} \tag{3.4}$$

There are a variety of points in this book where the space-time formalism is convenient. For example, we use the space-time formalism in Section 3.14 when performing the transformation of partial derivatives, and we use this formalism for the study of Galilean transformations in VOLUME 2. This notation is especially useful when studying classical field theory using Hamilton's principle in VOLUME 4.

3.4 Trajectories and coordinates

In section 1.1 we introduced the notion of points and trajectories. We here revisit that discussion while extending it for the more general purposes of this chapter.

3.4.1 Trajectories

Consider a point in space, $\mathcal{P}$, at a particular time, τ. If this point represents the position of a point particle that moves, then as time increases the point traces out a curve in space-time. We call that curve the particle trajectory. A trajectory through space-time could be determined by a point particle satisfying Newton's laws, as discussed in Chapter 11. Or it could be that of a fluid particle within a continuum, whose motion defines the Lagrangian reference frame studied in VOLUME 2. As the trajectory is a one-dimensional curve, it is specified mathematically by a single parameter (see Section 2.4.3). We choose the time as measured by an observer on the trajectory for this parameter. This time is referred to as proper time in special relativity. Yet since we are working with Newtonian time, there is no ambiguity concerning past, present, and future. In this manner, time is monotonically increasing and it has the same value at all points in space.

3.4.2 Time as a parameter and time as a coordinate

In special and general relativity, there is a mixing of space and time that warrants the use of four-dimensional space-time tensor analysis and with the physical equations manifesting Lorentz covariance rather than Galilean covariance.[2] In contrast, for classical mechanics forming the foundation of geophysical fluid mechanics, time is a monotonically increasing parameter that is numerically the same value throughout all of space. We thus make use of the same universal (or Newtonian) time since the fluid velocity and wave speeds are far smaller than the speed of light. Hence, for our studies we generally restrict attention to the space + time formalism of classical mechanics rather than the space-time formalism of relativity. Even so, we have some occasions for using the space-time formalism, such as mentioned in Section 3.3.4.

The time parameter, τ, specifies a point along a trajectory in space. When making use of the Lagrangian reference frame for continuum mechanics, we refer to τ as T, the Lagrangian time coordinate. An alternative measure of time is given by t, which measures time at positions fixed throughout space, which is referred to as the Eulerian reference frame in continuum mechanics.

This distinction between the two times is pedantic given that $\tau = T = t$ (up to a constant offset) in Galilean relativity. Nonetheless, it is convenient to make the distinction when measuring how fluid properties change since these changes are subject to motion of the observer. For example, changes following a trajectory, found by computing the trajectory time derivative $\partial/\partial\tau = \partial/\partial T$, are generally distinct from changes found by computing the time derivative $\partial/\partial t$, in which the spatial coordinates are held fixed.

When the trajectory is defined by a fluid particle, we refer to $\partial/\partial T$ as the Lagrangian time derivative, or equivalently the material time derivative. This time derivative is the same as when working with classical particle mechanics as in Chapter 11. In contrast, if the spatial coordinates are fixed in space, then $\partial/\partial t$ is an Eulerian time derivative. When alternative

[2] Maxwell's equations for electrodynamics do not satisfy Galilean covariance. Instead, they satisfy Lorentz covariance. This observation formed a central motivation for Einstein to develop special relativity, which is the mechanics of particles that satisfy Lorentz covariance.

spatial coordinates are used, some of which can move (see Section 3.4.3), then $\partial/\partial t$ can be a mixture of Lagrangian and Eulerian or perhaps neither. We return to these time derivatives in VOLUME 2 when discussing fluid kinematics.

3.4.3 The importance of index placement

In Chapters 1 and 2 we introduced the covariant index placement (downstairs) and the contravariant index placement (upstairs) as part of a representation of tensors. That distinction is unimportant for Cartesian tensors since the Cartesian representation of the metric tensor is given by the Kronecker symbol, δ_{ab}. In contrast, for general tensors the position of a tensor label has significance.

We are inspired by the choice for index placement by the placement of indices on coordinates. For that purpose we choose to express each of the coordinates in an ordered list according to

$$\xi^\alpha = (\xi^0, \xi^1, \xi^2, \xi^3), \tag{3.5}$$

with ξ^α representing a general coordinate. The α index is a coordinate label; it is not an exponent or power. We use a convention whereby Greek labels run from $\alpha = 0, 1, 2, 3$ with $\alpha = 0$ the time coordinate and $\alpha = a = 1, 2, 3$ the three labels for locating a point in space. The following shorthand notations are commonly used in this book

$$\xi^\alpha = (\xi^0, \xi^1, \xi^2, \xi^3) = (\xi^0, \xi^a) = (\xi^0, \boldsymbol{\xi}). \tag{3.6}$$

3.5 Coordinate descriptions

Throughout this book, the time coordinate is independent of space (we use universal Newtonian time), whereas in many cases we make use of space coordinates that are a function of time. We here offer spatial coordinate examples used for describing geophysical fluid systems, with the coordinate label, $a = 1, 2, 3$.

3.5.1 Orthogonal Eulerian coordinates

In Chapter 4 we detail three sets lof commonly used locally orthogonal Eulerian coordinates: Cartesian, cylindrical-polar, and spherical. Each makes use of orthonormal basis vectors. These coordinates are all time independent, which means they are make use of the Eulerian reference frame for describing amotion of the continuum.

In Cartesian coordinates, (x, y, z), the position for a point in space is written

$$\boldsymbol{x} = x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}} + z\,\hat{\boldsymbol{z}} \qquad \text{Cartesian.} \tag{3.7}$$

For spherical coordinates, (λ, ϕ, r), the position is (see Figure 4.2)

$$\boldsymbol{x} = r\,\hat{\boldsymbol{r}} \qquad \text{spherical,} \tag{3.8}$$

and cylindrical-polar coordinates, (r, ϑ, z), (see Figure 4.1) we have

$$\boldsymbol{x} = r\,\hat{\boldsymbol{r}} + z\,\hat{\boldsymbol{z}} \qquad \text{cylindrical-polar.} \tag{3.9}$$

Note that in spherical coordinates, r is the distance from the origin to the point and $\hat{\boldsymbol{r}}$ points from the origin to the point. In contrast, for cylindrical-polar coordinates, r is the distance from the z-axis and $\hat{\boldsymbol{r}}$ is the horizontal vector pointing from the z-axis to the point. Each of these coordinate representations identify positions in space relative to a fixed coordinate origin.

Cartesian coordinates have orthonormal basis vectors that maintain a fixed direction throughout space. This feature lends simplicity to Cartesian coordinates and its corresponding Cartesian tensor analysis (Chapters 1 and 2). In contrast, the orthonormal spherical basis vectors are spatially dependent. Likewise, the radial and angular basis vectors for cylindrical coordinates are spatially dependent, whereas the vertical direction is the same as the Cartesian vertical direction. Additionally, the spherical and cylindrical coordinates do not all have the same physical dimensions. Each of these features of spherical and cylindrical coordinates places them outside the purview of Cartesian tensor analysis.

3.5.2 Lagrangian (material) coordinates

We can conceive of a fluid as a continuum of fluid particles that are distinguished by continuum labels. These labels can then serve as coordinates for space, so long as there is a one-to-one mapping between points in the matter continuum and points in Euclidean space. The initial position for a fluid particle offers a suitable (and common) choice for these material coordinates or Lagrangian coordinates.

The fluid dynamical equations of motion can be formulated using material coordinates, so long as the material coordinates maintain a one-to-one invertible relation to points in space. This kinematical framework is termed the Lagrangian reference frame. The resulting dynamical equations share much in common with Newtonian particle mechanics, though with the added feature of contact forces acting between the fluid elements. We introduce Lagrangian coordinates in VOLUME 2, with Lagrangian descriptions used throughout this book. Transformations between Lagrangian coordinates and Eulerian coordinates are enabled by the methods of general tensor analysis.

3.5.3 Isopycnal coordinates: quasi-Eulerian/Lagrangian

In geophysical fluids that are stably stratified in the vertical, it can be physically useful to measure the vertical position of a fluid element by specifying its specific entropy (or potential temperature), global Archimedean buoyancy, or potential density depending on the application. Borrowing from the ocean physics community, we generically refer to this description as an isopycnal coordinate description, and write the spatial coordinates as

$$\xi^a = (x, y, b) \qquad \text{with} \qquad b = b(x, y, z, t) \qquad \text{isopycnal coordinates,} \tag{3.10}$$

where $b = b(x, y, z, t)$ is a generic symbol for this generalized vertical coordinate. Specific entropy, buoyancy, and potential density are materially invariant for perfect fluid flow (flow absent irreversible processes). Hence, all fluid particle motion in a perfect stratified fluid occurs on surfaces of constant b. Under such perfect fluid conditions, isopycnal coordinates are of great use for describing fluid mechanics of stably stratified geophysical flows. In particular, we can make use of the isopycnal coordinate system to determine the vertical position so long as there is a one-to-one invertible relation between z and b at each horizontal point and at each time instance.

The isopycnal coordinates, (x, y, b), are generally not orthogonal since the direction normal to a buoyancy surface is not generally vertical. Hence, even if the horizontal coordinates are Cartesian, the use of b to measure the vertical precludes the use of Cartesian tensor analysis.

Furthermore, we note the distinct physical dimensions of the three spatial coordinates (x, y, b), again necessitating the use of general tensor analysis.

The horizontal coordinates, x and y, are fixed in space, and so are Eulerian, whereas b is moving. For perfect fluids, b marks a particular material surface and so it is Lagrangian, whereas for real fluids b is not attached to material surfaces. Evidently, the isopycnal coordinates are neither fully Eulerian nor fully Lagrangian. This situation contrasts to most treatments in continuum mechanics (e.g., *Tromp* (2025a)), which means we must exercise care when translating the many results from continuum mechanics when working with such coordinates. In VOLUME 3we develop the mathematical physics of geophysical fluid flows using a generalized vertical coordinate description.

3.5.4 Tracer coordinates

Consider a triplet of linearly independent tracer concentrations, $C^a = C^a(x, y, z, t)$, that spans $\mathbb{R}^3$. Hence, at any point in space there is a unique intersection of three tracer isosurfaces, so that we can uniquely determine a point in space by specifying the value for the three tracer concentrations. Correspondingly, we can use tracer concentrations as the spatial coordinates

$$\xi^a = (C^1, C^2, C^3). \tag{3.11}$$

In some cases there are only two linearly independent tracers, in which case the two may be used in combination with a third spatial coordinate such as depth or pressure. Furthermore, the case of one tracer coordinate formally reduces to the isopycnal coordinate system from Section 3.5.3.[3]

3.5.5 Distinguishing coordinates from tensors

When labels are placed on a coordinate, such as in equation (3.6), we do not refer to a as a tensor index. Rather, a is simply a label that delineates the distinct spatial coordinates used to locate a point on a manifold. We make this distinction since, as noted in Section 1.8.8, coordinates are not tensors. Rather, coordinates are used to label the location of points on a manifold, whereas tensors are geometric objects that live on the manifold.[4] When we choose a set of coordinates, we say that we are choosing a chart to cover an open region of a manifold with grid lines for the purpose of supporting computations and measurements. Most manifolds require more than a single chart to cover the manifold (e.g., the surface of a sphere), and the accumulation of charts is referred to as an atlas.

We can transform between coordinates (movement from one chart to another chart), with coordinate transformations inducing changes to the representation of tensors. Cartesian coordinate transformations are restricted to rigid rotations; i.e., all Cartesian coordinate transformations result in the same rotation of the Cartesian coordinates at each point in space. Hence, coordinate transformations between Cartesian coordinates are linear transformations. In contrast, coordinate transformations can generally be nonlinear, such as the transformation between Cartesian and spherical coordinates, and the transformation between Eulerian and Lagrangian coordinates. Nonlinear coordinate transformations require the tools of general tensor analysis to transform the coordinate representation of tensors.

[3] See *Nurser et al.* (2022) for a description of fluid flow in the space defined by arbitrary continuous properties such as tracers.

[4] For our purposes, a manifold is a set of points such that any open subset of the manifold can be mapped the real numbers. In this manner, a manifold is locally Euclidean.

3.6 Vectors

In this section we introduce vectors, which are first order tensors.[5] We motivate the discussion by considering the velocity vector, $\mathbf{v}$, as defined using a particle trajectory, with the position along a trajectory, $\mathbf{x}$, specified relative to an origin. This approach follows the Cartesian tensor approach from Chapter 1, in which the Cartesian basis vectors are fixed directions in space. We then pursue a more general approach that makes use of the geometry of curves on a manifold. The general approach dispenses with trajectories and coordinate origins, instead considering vectors as tangents to curves intrinsic to a manifold. The general approach reduces to the trajectory-based approach when evaluating vectors at the position of a point on the manifold.[6]

3.6.1 Velocity as the time derivative of a trajectory

Referring to Figure 3.1, consider two points on a spatial manifold that sit along a particular trajectory separated by a time increment, $\Delta\tau$. Write $\mathcal{P}(\tau - \Delta\tau/2)$ and $\mathcal{P}(\tau + \Delta\tau/2)$ for the two points, whose time argument parameterizes the trajectory, which is a spatial curve. Let $\mathbf{x}(\tau - \Delta\tau/2)$ and $\mathbf{x}(\tau + \Delta\tau/2)$ be the corresponding position for the point as defined relative to a chosen origin.[7] The velocity for this trajectory is defined by the finite difference[8]

$$\mathbf{v}(\tau) = \lim_{\Delta\tau \to 0} \frac{\mathbf{x}(\tau + \Delta\tau/2) - \mathbf{x}(\tau - \Delta\tau/2)}{\Delta\tau} = \frac{\mathrm{d}\mathbf{x}(\tau)}{\mathrm{d}\tau}. \tag{3.12}$$

The velocity points in the direction determined by the difference between points at two time instances, in the limit as $\Delta\tau$ vanishes. Consequently, the velocity at a point in space is tangent to the trajectory.

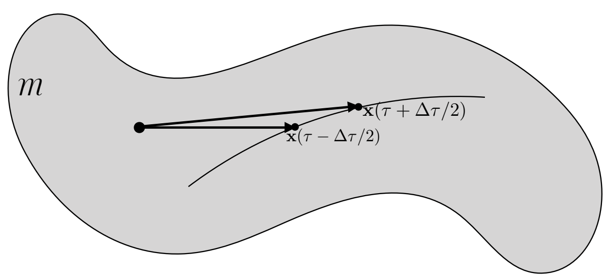


FIGURE 3.1: Illustrating the definition of the velocity vector, $\mathbf{v}$, as the time derivative of a trajectory represented by a position relative to an origin, $\mathbf{x}(\tau)$. The trajectory takes place on a manifold, $\mathcal{M}$.

We now introduce a set of coordinates, ξ^a, to locate points in on a manifold. These

[5] Scalars, such as temperature and kinetic energy, are zero order tensors.

[6] Many aspects of mechanics need no specification of a coordinate specification. By isolating those aspects, we streamline our view of the physics while offering a more general and parsimonious mathematical description. Readers raised on a coordinate-first approach to physics must be patient as they learn to remove the coordinate-based clothing that can obscure the essential physics.

[7] In this book, we sometimes choose to use the lowercase, $\mathbf{x}(\tau)$, for the particle position along a trajectory, and sometimes choose the uppercase, $\mathbf{X}(\tau)$.

[8] Note the use of centered time differences to define the derivative in equation (3.12). This specification is one order more accurate than the forward difference, $[\mathbf{x}(\tau + \Delta\tau) - \mathbf{x}(\tau)]/\Delta\tau$, typically found in other treatments. Since we are taking the limit, $\Delta\tau \to 0$, there is no fundamental reason to be concerned with the rate that the finite difference approaches the derivative. Even so, our numerical methods brain-muscle is more satisfied by the use of centered difference, both for its accuracy and symmetry.

coordinates are used to label the spatial position of the trajectory according to

$$\mathbf{x}(\tau) = \mathbf{x}[\xi^a(\tau)], \tag{3.13}$$

where $\xi^a(\tau)$ is the spatial coordinate of the trajectory at time, τ. The coordinate chart induces a coordinate representation for the velocity vector through use of the chain rule

$$\mathbf{v}(\tau) = \frac{\mathrm{d}\mathbf{x}(\tau)}{\mathrm{d}\tau} = \frac{\mathrm{d}\xi^a}{\mathrm{d}\tau}\frac{\partial \mathbf{x}}{\partial \xi^a} \equiv v^a\, \boldsymbol{e}_a(\mathbf{x}). \tag{3.14}$$

We here wrote the time derivative of the coordinates as

$$v^a = \frac{\mathrm{d}\xi^a}{\mathrm{d}\tau}, \tag{3.15}$$

which provides a coordinate representation for the velocity vector. The physical dimensions of v^a depend on the physical dimensions of the coordinates. For example, Cartesian coordinates have each velocity component, v^a, dimensioned length per time. In contrast, spherical coordinates have length per time for the radial coordinate, $\xi^a = r$, and per time for the longitude and latitude, $\xi^a = \lambda, \phi$.

For each velocity component in equation (3.14), we have a corresponding basis vector, $\boldsymbol{e}_a$, whose value at the position, $\mathbf{x}$, is given by

$$\boldsymbol{e}_a(\mathbf{x}) = \frac{\partial \mathbf{x}}{\partial \xi^a}. \tag{3.16}$$

In general, the basis vectors are functions of space and time. Cartesian coordinates are a notable exception, whose orthonormal basis vectors, $\hat{\boldsymbol{e}}_a$, are space-time constants as introduced in Section 1.1.3, so that the position of a point is represented by

$$\mathbf{x} = x^a\, \hat{\boldsymbol{e}}_a. \tag{3.17}$$

Consequently, the basis vectors for Cartesian coordinates are simply given by

$$\boldsymbol{e}_a(\mathbf{x}) = \frac{\partial(\hat{\boldsymbol{e}}_b\, x^b)}{\partial x^a} = \hat{\boldsymbol{e}}_b \frac{\partial x^b}{\partial x^a} = \hat{\boldsymbol{e}}_b\, \delta^b{}_a = \hat{\boldsymbol{e}}_a. \tag{3.18}$$

The spherical coordinates (λ, ϕ, r) from Section 4.4 are also constants in time. However, they are functions of space. The position for a point in space relative to an origin is given by $\mathbf{x} = r\, \hat{\boldsymbol{r}}$, with r the radial distance from the origin and $\hat{\boldsymbol{r}}$ the unit vector pointing from the origin to the point. We show in Section 4.4.2 that

$$\partial_\lambda(r\, \hat{\boldsymbol{r}}) = \boldsymbol{e}_\lambda \quad \text{and} \quad \partial_\phi(r\, \hat{\boldsymbol{r}}) = \boldsymbol{e}_\phi \quad \text{and} \quad \partial_r(r\, \hat{\boldsymbol{r}}) = \boldsymbol{e}_r. \tag{3.19}$$

3.6.2 Vectors as elements of a tangent space

In the previous approach to defining the velocity vector, we made use of the position relative to an origin, $\mathbf{x}$, and followed that position as time moved forward. But note that the coordinate origin dropped out when computing the velocity. As we show here, we can define a vector, including the velocity vector, without introducing a coordinate origin or coordinate position. That is, we present a definition of the velocity vector (indeed all vectors), along with a set of basis vectors, that dispenses with the origin. This approach is more general and more parsimonious than the coordinate-based trajectory approach. Our presentation follows Section 4 of *Tromp* (2025b).

Referring to Figure 3.2, consider a point on a manifold, $\mathcal{P} \in \mathcal{M}$, and introduce the tangent space, $T_{\mathcal{P}}$. The tangent space is a linear vector space[9] and it can be considered a local patch of (flat) Euclidean space fixed to a point, $\mathcal{P}$, on the generally (curved) non-Euclidean manifold. Elements of the tangent space are vectors, $\mathbf{v}(\mathcal{P}) \in T_{\mathcal{P}}$. Extending this specification to the full manifold leads to a distinct tangent space living at each point of the manifold.[10] In this manner we have introduced vectors at each point of a manifold, and yet there has been no need to specify a coordinate origin. Consequently, we no longer consider a vector to be an arrow pointing between two points. Instead, we consider a vector to be a member of the tangent space at each point of a manifold, and pictured as a line with an arrow at the point as in Figure 3.2.

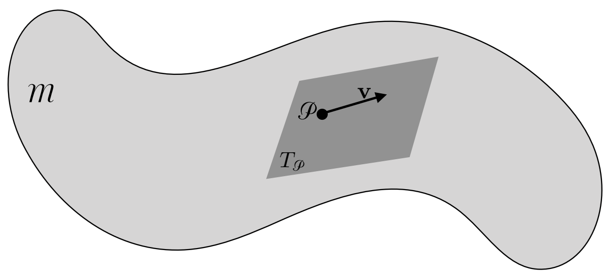


FIGURE 3.2: Illustrating the definition of a vector, $\mathbf{v}$, as a member of the tangent space, $T_{\mathcal{P}}$, to a point on a manifold, $\mathcal{P} \in \mathcal{M}$. Each point on the manifold has a distinct tangent space, with the accumulation of points and tangent spaces comprising the tangent bundle. A vector field living on the manifold serves to map points on the manifold to points in the tangent space at each point of the manifold. Comparison of two vectors (and by extension any higher order tensor) can only occur if they are within the same tangent space, thus requiring special treatment for differentiation and integration of tensors on general manifolds.

3.6.3 Vectors as derivative operators

We now use the geometric ideas from Figure 3.2 to formulate vectors in a manner suitable for analysis. For that purpose, introduce a chart with coordinates, ξ^a, defined within an open region of the manifold, and let $f(\xi^a)$ be a differentiable scalar function. Now restrict attention to a smooth curve whose coordinate values are $\xi^a(\lambda)$, with λ the curve's parameter.[11] Restricting f to those values defined along the curve allows us to introduce a function of the curve's parameter

$$g(\lambda) \equiv f[\xi^a(\lambda)]. \tag{3.20}$$

Observe that g is defined only along the curve, $\xi^a(\lambda)$, so that g is a function of the curve's parameter, λ. The derivative, $\mathrm{d}g/\mathrm{d}\lambda$, provides the directional derivative of f along the curve, with the chain rule yielding

$$\frac{\mathrm{d}g}{\mathrm{d}\lambda} = \frac{\mathrm{d}\xi^a}{\mathrm{d}\lambda}\frac{\partial f}{\partial \xi^a}. \tag{3.21}$$

[9]See section 4 of *Tromp* (2025b).

[10]The tangent bundle is the product space of a manifold with the tangent spaces defined at each point of the manifold. A vector field serves to map each point of a manifold to the tangent bundle. As we need to specify a point and corresponding tangent vector to specify an element of the tangent bundle, the dimension of the tangent bundle twice that of the manifold. The configuration space defined by particle positions and their velocities forms a tangent bundle used in the Lagrangian mechanics studied in VOLUME 4.

[11]We choose $\lambda = \tau$ for curves defined by particle trajectories moving through space, as in our study of mechanics later in this volume.

Now we acknowledge the arbitrary nature of f and its restriction to the curve, g. As such, we extract f and g from equation (3.21) to identify the operator equality

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} = \frac{\mathrm{d}\xi^a}{\mathrm{d}\lambda} \frac{\partial}{\partial \xi^a}. \tag{3.22}$$

The right hand side provides a coordinate representation of the directional derivative operator, decomposed in terms of the directional derivative of the coordinates along the curve and contracted with the partial derivative operator along the coordinate directions.

As illustrated in Figure 3.3, we identify the directional derivative operator with the vector whose integral curve is $\xi^a(\lambda)$,

$$\mathbf{v} \equiv \mathrm{d}/\mathrm{d}\lambda = v^a \, \partial_a. \tag{3.23}$$

The integral curve is everywhere tangent to the vector. A vector field has a corresponding manifold filling set of integral curves referred to as a congruence.[12] Furthermore, we introduced the coordinate components for the vector, which are given from equation (3.22) as

$$v^a \equiv \frac{\mathrm{d}\xi^a}{\mathrm{d}\lambda}. \tag{3.24}$$

Finally, the corresponding basis vectors are given by the coordinate partial derivative operators

$$\boldsymbol{e}_a \equiv \partial/\partial \xi^a = \partial_a. \tag{3.25}$$

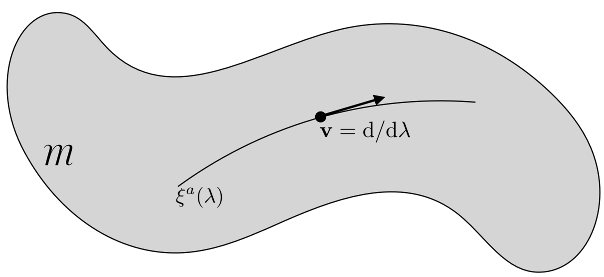


FIGURE 3.3: Illustrating the definition of a vector, $\mathbf{v}$, as a directional derivative operator along a curve on a manifold, $\mathcal{M}$. Positions on the curve are parameterized by λ, and with coordinates along the curve written as $\xi^a(\lambda)$. At each point along the curve, the directional derivative operator, $\mathrm{d}/\mathrm{d}\lambda$, points along the curve's tangent, and it defines the vector, $\mathbf{v} = \mathrm{d}/\mathrm{d}\lambda$, whose value at a particular point is given by $\mathbf{v}(\mathbf{x}) = \mathrm{d}\mathbf{x}/\mathrm{d}\lambda$. The curve thus serves as the integral curve for the vector. Given a vector field defined over a manifold, we can find an integral curve (section 4.1.3 of *Tromp* (2025b) and 2.12 of *Schutz* (1980)) at each point of the manifold, by integrating the first order differential equation for the vector components, $v^a = \mathrm{d}\xi^a/\mathrm{d}\lambda$. The manifold-filling set of integral curves is said to be a congruence.

3.6.4 Comments

The treatment in Section 3.6.1, by starting with the trajectory as a primary object, led us to define vectors and basis vectors according to their location on a manifold as specified by the position, $\mathbf{x}$. In the more general approach, we focus on vectors as tangents to integral curves defined on a manifold, with these curves defined using parameters intrinsic to the curve

[12] In Section 2.12 of *Schutz* (1980), he shows that for each vector field over a manifold, there is a unique integral curve passing through each point. For fluid mechanics, this result means there is a unique fluid particle trajectory passing through each point at any chosen time instance.

and having no connection to a chosen coordinate origin. Evidently, the curve-based geometric approach enables us to define a vector without any need to consider the position, which proves to be a more general and parsimonious approach.

Equation (3.23) is somewhat puzzling on first encounter, particularly for the physicist raised on Cartesian coordinates and Cartesian tensors. However, experience builds trust with the conceptual and practical use of this definition. One aspect of this definition that can introduce confusion concerns the distinct physical dimensionality of the vector written as an operator, versus the vector evaluated at a point, $\mathbf{x}$, on the manifold:

$$\mathbf{v} = v^a\,\partial_a \quad \text{and} \quad \mathbf{v}(\mathbf{x}) = v^a\,\partial_a\mathbf{x}. \tag{3.26}$$

For example, let $\mathbf{v}$ be the velocity vector and choose Cartesian coordinates so that $\xi^a = x^a$ and $\mathbf{x} = x^a\,\hat{\boldsymbol{e}}_a$. In this case, the velocity as an operator, $\mathbf{v} = v^a\,\partial_a$, has physical dimensions of inverse time, whereas velocity evaluated at a point, $\mathbf{v}(\mathbf{x}) = v^a\,\partial_a\mathbf{x} = v^a\,\hat{\boldsymbol{e}}_a$, has dimensions of length per time.

3.6.5 Further reading

The treatment here is consistent with Section 2.3 of *Misner et al.* (1973), chapter 2 of *Schutz* (1980), chapter 4 of *Tromp* (2025b), and appendix C.3 of *Tromp* (2025a). In the remainder of this book, we keep in mind the connection between derivative operators and vector fields. Even so, we maintain the generic notion, $\boldsymbol{e}_a$, for the basis vectors, rather than write ∂_a.

After working through vectors, *Tromp* (2025b) then moves onto one-forms, which are the duals to vectors. The same approach was considered when writing this chapter, as it provides a natural next step for building the infrasture of general tensors in continuum mechanics. Yet we chose not to move along that path. Instead, we introduce one-forms only after introducing the metric tensor. Our choice reflects an intention to not enter into a study of differential forms in this book. The interested reader is encouraged to consult *Tromp* (2025b) for that purpose.

3.7 The metric tensor and coordinate transformations

We find the need to measure the distance between two points in space at a particular time instance. Since we assume all points live on a smooth and orientable manifold (e.g., a sphere, an isopycnal in a stably stratified fluid, the Lagrangian manifold defined by fluid particle labels), it is sufficient to consider the distance between two infinitesimally close points and use integration to measure finite distances. The measurement of distance requires a metric tensor, which is the subject of this section.

3.7.1 Cartesian coordinates in Euclidean space

Consider a Cartesian coordinate representation for the spatial position of two points, with point $\mathcal{P}$ having space coordinates $\xi^a = x^a$ and the other point $\mathcal{Q}$ an infinitesimal distance away at $x^a + \mathrm{d}x^a$. Furthermore, let

$$\mathrm{d}\boldsymbol{x} = \mathrm{d}x^a\,\boldsymbol{e}_a \tag{3.27}$$

be the infinitesimal space vector pointing from $\mathcal{P}$ to $\mathcal{Q}$. Since the space is Euclidean, the squared distance between the two points is based on the Euclidean norm; i.e., the familiar scalar or dot product (Section 1.3)

$$\mathrm{d}s^2 = \mathrm{d}\boldsymbol{x} \cdot \mathrm{d}\boldsymbol{x} = \boldsymbol{e}_a \cdot \boldsymbol{e}_b \, \mathrm{d}x^a \mathrm{d}x^b = \delta_{ab} \, \mathrm{d}x^a \mathrm{d}x^b. \tag{3.28}$$

In this expression,

$$(\mathrm{d}s)^2 \equiv \mathrm{d}s^2 \tag{3.29}$$

is the squared infinitesimal arc-length separating the two points. The Kronecker symbol, δ_{ab}, is symmetric

$$\delta_{ab} = \delta_{ba}, \tag{3.30}$$

and vanishes when $a \neq b$ and is unity when $a = b$

$$\delta_{ab} = \begin{cases} 0 & \text{if } a \neq b \\ 1 & \text{if } a = b. \end{cases} \tag{3.31}$$

The Kronecker symbol is a representation of the unit tensor.

3.7.2 The metric as a symmetric second order tensor

As defined by equation (3.28), δ_{ab} forms the Cartesian representation of the metric tensor for Euclidean space. The metric is a second order tensor, meaning that its coordinate representation carries two tensor labels. Contracting the metric tensor with two vectors leads to a number, namely the squared distance between the two points. Hence, the metric establishes the means to measure the distance between two points that live on a manifold.

We write this distance-measuring property of the metric tensor in a geometric manner through

$$\text{distance}(\mathbf{P}, \mathbf{Q}) = \sqrt{\mathbb{g}(\mathbf{P}, \mathbf{Q})}. \tag{3.32}$$

Here, $\mathbb{g}$ is the metric tensor with coordinate representation $\mathbb{g}_{ab}$ and $\mathbf{P}$, $\mathbf{Q}$ are infinitesimally close vectors with coordinate representations

$$\mathbf{P} = \xi^a \, \boldsymbol{e}_a \quad \text{and} \quad \mathbf{Q} = \mathbf{P} + \mathrm{d}\xi^a \, \boldsymbol{e}_a. \tag{3.33}$$

Equation (3.32) indicates that the metric tensor takes two vectors as argument and produces a scalar. Furthermore, since

$$\text{distance}(\mathbf{P}, \mathbf{Q}) = \text{distance}(\mathbf{Q}, \mathbf{P}) \geq 0, \tag{3.34}$$

the metric tensor is a symmetric and positive tensor that produces zero only when $\mathbf{P} = \mathbf{Q}$.

3.7.3 Coordinate representation of the metric tensor

Given the geometric expression (3.32) for the metric, we determine its representation in an arbitrary coordinate system by considering the squared distance between the coordinate basis vectors

$$\text{distance}(\boldsymbol{e}_a, \boldsymbol{e}_b) = \sqrt{\mathbb{g}(\boldsymbol{e}_a, \boldsymbol{e}_b)}. \tag{3.35}$$

This relation determines the coordinate components of the metric tensor

$$\mathbb{g}(\boldsymbol{e}_a, \boldsymbol{e}_b) \equiv \mathbb{g}_{ab}. \tag{3.36}$$

Furthermore, for any coordinates in Euclidean space, this relation is written

$$\mathbb{g}_{ab} = \boldsymbol{e}_a \cdot \boldsymbol{e}_b = \sum_{i=1}^{3} (\boldsymbol{e}_a)^i \, (\boldsymbol{e}_b)^i, \tag{3.37}$$

with the Euclidean dot product defining the scalar product between the two basis vectors. In this manner, we see that the metric tensor components are determined by computing the Euclidean scalar product between the basis vectors. We also see that if the basis vectors are orthogonal, then the metric tensor coordinate representation has vanishing components for $a \neq b$.

3.7.4 Transforming the coordinate representation of the metric tensor

We find opportunities to represent the metric tensor in various coordinate systems. Here, we consider the transformation from Cartesian coordinates, $\xi^a = x^a$, to arbitrary coordinates, $\xi^{\overline{a}}$. Use of the chain rule along with index gymnastics leads to the equivalent expression for the squared infinitesimal distance between two points

$$\begin{aligned}
\mathrm{d}s^2 &= \delta_{ab} \, \mathrm{d}\xi^a \, \mathrm{d}\xi^b && \text{squared distance with Cartesian coordinates} && (3.38a)\\
&= \delta_{ab} \, \frac{\partial \xi^a}{\partial \xi^{\overline{a}}} \frac{\partial \xi^b}{\partial \xi^{\overline{b}}} \, \mathrm{d}\xi^{\overline{a}} \, \mathrm{d}\xi^{\overline{b}} && \text{chain rule to new coordinates} && (3.38b)\\
&\equiv \delta_{ab} \Lambda^a{}_{\overline{a}} \, \Lambda^b{}_{\overline{b}} \, \mathrm{d}\xi^{\overline{a}} \, \mathrm{d}\xi^{\overline{b}} && \text{define the transformation matrix, } \Lambda && (3.38c)\\
&\equiv \mathbb{g}_{\overline{a}\overline{b}} \, \mathrm{d}\xi^{\overline{a}} \, \mathrm{d}\xi^{\overline{b}}, && \text{define the new coordinate components, } \mathbb{g}_{\overline{a}\overline{b}}, && (3.38d)
\end{aligned}$$

where

$$\mathbb{g}_{\overline{a}\overline{b}} = \delta_{ab} \, \Lambda^a{}_{\overline{a}} \, \Lambda^b{}_{\overline{b}} \tag{3.39}$$

defines the components to the metric tensor as represented by the coordinates $\xi^{\overline{a}}$.

In equation (3.38c) we introduced elements to the transformation matrix

$$\Lambda^a{}_{\overline{a}} = \frac{\partial \xi^a}{\partial \xi^{\overline{a}}}. \tag{3.40}$$

As seen in Section 1.8 , the transformation matrix has constant elements when transforming between two sets of Cartesian coordinates, for which case case the transformation matrix is orthogonal and so represents a bulk rotation. More generally, coordinate transformations are nonlinear and so the transformation matrix is a function of space and time. This matrix of partial derivatives has a non-zero entry when a coordinate in one representation changes while moving along the direction of a coordinate in the other representation. As for any partial derivative, the complement coordinates are held fixed when performing the derivative. Although carrying indices, it is crucial to note that the matrix elements, $\Lambda^a{}_{\overline{a}}$, are *not* components to a second order tensor. Instead, they are components of a matrix used to transform tensor representations from one coordinate system to another.

Organized as a matrix, we follow a convention whereby the row is denoted by the label closest to Λ, here being a, whereas the column is denoted by the label furthest from Λ, here denoted by $\overline{a}$, so that

$$\Lambda^{a}{}_{\overline{a}} = \begin{bmatrix} (\partial\xi^1/\partial\xi^{\overline{1}})_{\overline{2},\overline{3}} & (\partial\xi^1/\partial\xi^{\overline{2}})_{\overline{1},\overline{3}} & (\partial\xi^1/\partial\xi^{\overline{3}})_{\overline{1},\overline{2}} \\ (\partial\xi^2/\partial\xi^{\overline{1}})_{\overline{2},\overline{3}} & (\partial\xi^2/\partial\xi^{\overline{2}})_{\overline{1},\overline{3}} & (\partial\xi^2/\partial\xi^{\overline{3}})_{\overline{1},\overline{2}} \\ (\partial\xi^3/\partial\xi^{\overline{1}})_{\overline{2},\overline{3}} & (\partial\xi^3/\partial\xi^{\overline{2}})_{\overline{1},\overline{3}} & (\partial\xi^3/\partial\xi^{\overline{3}})_{\overline{1},\overline{2}} \end{bmatrix}. \tag{3.41}$$

Again, the lower index is displaced to the right to delineate which index refers to the column. The extra labels denote those coordinates held fixed when performing the partial derivatives. The transformation matrix is nonsingular for one-to-one invertible coordinate transformations, in which case its determinant, called the Jacobian of transformation, is nonvanishing and single signed. Finally, we sometimes find it useful to write the un-barred coordinates as an ordered list in a column,

$$\boldsymbol{\xi} = (\xi^1, \xi^2, \xi^3)^{\mathsf{T}}, \tag{3.42}$$

in which case the transformation matrix takes on the abbreviated form

$$\Lambda^{a}{}_{\overline{a}} = \begin{bmatrix} (\partial\boldsymbol{\xi}/\partial\xi^{\overline{1}})_{\overline{2},\overline{3}} & (\partial\boldsymbol{\xi}/\partial\xi^{\overline{2}})_{\overline{1},\overline{3}} & (\partial\boldsymbol{\xi}/\partial\xi^{\overline{3}})_{\overline{1},\overline{2}} \end{bmatrix}. \tag{3.43}$$

Use of this expression for the transformation matrix leads to the arbitrary coordinate representation of the metric tensor

$$\mathbb{g}_{\overline{a}\overline{b}} = \boldsymbol{e}_{\overline{a}} \cdot \boldsymbol{e}_{\overline{b}} = \begin{bmatrix} \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}} & \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}} & \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}} \\ \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}} & \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}} & \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}} \\ \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}} & \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}} & \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}} \cdot \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}} \end{bmatrix}. \tag{3.44}$$

The dot appearing in this equation is the usual Cartesian scalar product, which we are afforded since the background space is Euclidean.

So in summary, the transformation matrix is the central tool needed to perform coordinate transformations, which is one of the key practical uses of tensor analysis. It is defined by the partial derivatives of one coordinate with respect to another. The transformation matrix is not a tensor and so it is not a geometric object. Rather, it is dependent on how the two coordinate systems are related, and it provides the means to transform coordinate representations of tensors between these coordinates. Given our assumption of motion in the background Euclidean space, we commonly assume one of the two coordinates to be Cartesian. Yet this assumption is not necessary.

3.7.5 Transformation of the basis vectors

We transform the basis vectors from Cartesian into arbitrary coordinates through the transformation

$$\boldsymbol{e}_{\overline{a}} = \Lambda^{a}{}_{\overline{a}}\, \boldsymbol{e}_{a}. \tag{3.45}$$

Use of the transformation matrix (3.43) renders the arbitrary coordinate basis vectors, as evaluated at a position in space,

$$\boldsymbol{e}_{\overline{1}} = \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}} \qquad \text{and} \qquad \boldsymbol{e}_{\overline{2}} = \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}} \qquad \text{and} \qquad \boldsymbol{e}_{\overline{3}} = \frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}}, \tag{3.46}$$

which corresponds to the metric tensor written as in equation (3.44).

3.7.6 Finite distance between points

Once the metric is determined, the distance along a curve between two finitely separated points is given by integration

$$L = \int \sqrt{\mathrm{d}s^2} = \int_{\varphi_1}^{\varphi_2} \left| \mathbb{g}_{ab} \frac{\mathrm{d}\xi^a}{\mathrm{d}\varphi} \frac{\mathrm{d}\xi^b}{\mathrm{d}\varphi} \right|^{1/2} \mathrm{d}\varphi, \tag{3.47}$$

where φ is a parameter specifying the curve (e.g., the arc length as in Section 2.5), with φ_1 and φ_2 specifying the endpoints of the curve.

3.8 One-forms

The metric tensor, $\mathbb{g}$, is a function of two vectors, so that when the metric eats the two vectors the result is the scalar distance between the vectors (equation (3.32))

$$\text{distance}(\mathbf{A}, \mathbf{B}) = \sqrt{\mathbb{g}(\mathbf{A}, \mathbf{B})}. \tag{3.48}$$

If the metric only eats one vector, then the resulting geometric object is known as a one-form, $\mathbb{g}(\mathbf{A},\ )$. We can determine the coordinate representation of a one-form by eating a basis vector

$$\mathbb{g}(\mathbf{A}, \boldsymbol{e}_b) = \mathbb{g}(A^a\, \boldsymbol{e}_a, \boldsymbol{e}_b) = \mathbb{g}(\boldsymbol{e}_a, \boldsymbol{e}_b)\, A^a = \mathbb{g}_{ab} A^a. \tag{3.49}$$

To reach this result we pulled the coordinate representation, A^a, outside of the metric tensor since the tensor eats vectors rather than numbers. This equation defines the coordinate representation of the one-form as

$$A_b = \mathbb{g}_{ab} A^a. \tag{3.50}$$

Evidently, contraction with the metric tensor provides the operational means to lower an index on the representation of a vector, thus producing the representation of a one-form.

3.8.1 Basis one-forms and the biorthogonality relation

Just as for vectors, we find use for a basis of one-forms to specify their coordinate representation. The basis of one-forms, $\boldsymbol{e}^a$, are defined through the biorthogonality relation (sometimes also called the duality condition)

$$\boldsymbol{e}^a \cdot \boldsymbol{e}_b \equiv \mathbb{g}(\boldsymbol{e}^a, \boldsymbol{e}_b) = \delta^a{}_b, \tag{3.51}$$

where

$$\delta^a{}_b = \mathbb{g}^{ac}\, \mathbb{g}_{cb} \tag{3.52}$$

are components to the Kronecker tensor, taking the value of unity when $a = b$ and zero otherwise

$$\delta^a{}_b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{3.53}$$

Note that it is only for Cartesian coordinates that we have

$$\delta^a{}_c = \mathbb{g}^{ab}\,\delta_{bc} \qquad \text{Cartesian coordinates,} \tag{3.54}$$

which follows since $\mathbb{g}^{ab} = \delta^{ab}$ in Cartesian coordinates (recall Section 1.2.4).

We can obtain an explicit expression for the basis one-forms in arbitrary coordinates by transforming from Cartesian coordinates through use of the inverse transformation

$$\boldsymbol{e}^{\overline{a}} = \Lambda^{\overline{a}}{}_a\,\boldsymbol{e}^a, \tag{3.55}$$

which renders

$$\boldsymbol{e}^{\overline{1}} = \hat{\boldsymbol{x}}\,\frac{\partial \xi^{\overline{1}}}{\partial x} + \hat{\boldsymbol{y}}\,\frac{\partial \xi^{\overline{1}}}{\partial y} + \hat{\boldsymbol{z}}\,\frac{\partial \xi^{\overline{1}}}{\partial z} = \nabla \xi^{\overline{1}} \tag{3.56a}$$

$$\boldsymbol{e}^{\overline{2}} = \hat{\boldsymbol{x}}\,\frac{\partial \xi^{\overline{2}}}{\partial x} + \hat{\boldsymbol{y}}\,\frac{\partial \xi^{\overline{2}}}{\partial y} + \hat{\boldsymbol{z}}\,\frac{\partial \xi^{\overline{2}}}{\partial z} = \nabla \xi^{\overline{2}} \tag{3.56b}$$

$$\boldsymbol{e}^{\overline{3}} = \hat{\boldsymbol{x}}\,\frac{\partial \xi^{\overline{3}}}{\partial x} + \hat{\boldsymbol{y}}\,\frac{\partial \xi^{\overline{3}}}{\partial y} + \hat{\boldsymbol{z}}\,\frac{\partial \xi^{\overline{3}}}{\partial z} = \nabla \xi^{\overline{3}}. \tag{3.56c}$$

In Section 3.14.3 we verify that the basis one-forms satisfy the duality condition (3.51) with the basis vectors

$$\boldsymbol{e}^{\overline{a}} \cdot \boldsymbol{e}_{\overline{b}} = \delta^{\overline{a}}{}_{\overline{b}}. \tag{3.57}$$

3.8.2 Metric as a mapping between vectors and one-forms

We can contract the expression (3.50) with components of the inverse metric tensor, $\mathbb{g}^{ab}$, to render

$$\mathbb{g}^{ab} A_b = \mathbb{g}^{ab}\,\mathbb{g}_{bc} A^c = \delta^a{}_c A^c = A^a. \tag{3.58}$$

This identity, as well as equation (3.50), show that the metric provides a map between coordinate representations of one-forms and vectors. That is, contraction with the metric tensor and its inverse provides the operational means to raise and lower indices on the coordinate representations of a tensor.

To every vector there is a corresponding one-form. We say that the one-forms and vectors are dual, with the mapping between one-forms and vectors rendered by the metric tensor. Abstractly, we say that a vector at a point on a manifold lives on the tangent space at that point, whereas its one-form dual lives on the cotangent space. The metric tensor provides a link between the tangent space and cotangent space. This link blurs the distinction between one-forms and vectors, and more generally between covariant and contravariant representations of tensors. Furthermore, as for Cartesian tensors, we construct an inner product by contracting one-forms and vectors to produce a scalar. Finally, the duality relation given by equation (3.50) offers us the means to raise and lower tensor indices in a manner akin to the transpose operation in linear algebra that produces a row vector from a column vector.

3.8.3 Transformation of the coordinate representation

The transformation matrix (3.40) provides the means to convert any arbitrary coordinate representation of a tensor from one coordinate system to another. For example, consider the coordinate representation of a vector, which is realized by letting the vector eat one of the basis one-forms

$$\mathbf{A}(\boldsymbol{e}^{a}) = A^{a}. \tag{3.59}$$

Now consider another coordinate system with basis one-forms $\boldsymbol{e}^{\overline{a}}$, so that the vector has a representation

$$\mathbf{A}(\boldsymbol{e}^{\overline{a}}) = A^{\overline{a}}. \tag{3.60}$$

Transforming the basis one-form using the transformation matrix leads to

$$A^{\overline{a}} = \mathbf{A}(\boldsymbol{e}^{\overline{a}}) = \mathbf{A}(\Lambda^{\overline{a}}{}_{a}\, \boldsymbol{e}^{a}) = \Lambda^{\overline{a}}{}_{a}\, \mathbf{A}(\boldsymbol{e}^{a}) = \Lambda^{\overline{a}}{}_{a}\, A^{a}. \tag{3.61}$$

The transformation of an arbitrary one-form representation takes place with the inverse transformation matrix

$$A_{\overline{a}} = \mathbf{A}(\boldsymbol{e}_{\overline{a}}) = \mathbf{A}(\Lambda^{a}{}_{\overline{a}}\, \boldsymbol{e}_{a}) = \Lambda^{a}{}_{\overline{a}}\, \mathbf{A}(\boldsymbol{e}_{a}) = \Lambda^{a}{}_{\overline{a}}\, A_{a}. \tag{3.62}$$

3.8.4 Arbitrary coordinate representation of inverse metric

The inverse metric tensor has an arbitrary coordinate representation given by

$$\mathbb{g}^{\overline{a}\overline{b}} = \boldsymbol{e}^{\overline{a}} \cdot \boldsymbol{e}^{\overline{b}} = \begin{bmatrix} \nabla\xi^{\overline{1}} \cdot \nabla\xi^{\overline{1}} & \nabla\xi^{\overline{1}} \cdot \nabla\xi^{\overline{2}} & \nabla\xi^{\overline{1}} \cdot \nabla\xi^{\overline{3}} \\ \nabla\xi^{\overline{2}} \cdot \nabla\xi^{\overline{1}} & \nabla\xi^{\overline{2}} \cdot \nabla\xi^{\overline{2}} & \nabla\xi^{\overline{2}} \cdot \nabla\xi^{\overline{3}} \\ \nabla\xi^{\overline{3}} \cdot \nabla\xi^{\overline{1}} & \nabla\xi^{\overline{3}} \cdot \nabla\xi^{\overline{2}} & \nabla\xi^{\overline{3}} \cdot \nabla\xi^{\overline{3}} \end{bmatrix}. \tag{3.63}$$

Proof that $\mathbb{g}^{\overline{a}\overline{b}}\, \mathbb{g}_{\overline{b}\overline{c}} = \delta^{\overline{a}}{}_{\overline{c}}$ requires use of the chain rule relations derived in Section 3.14.3.

3.8.5 Comments

The more general approach to studying one-forms (and higher n-forms) has no need for a metric, thus allowing one to develop a calculus with differential forms absent a metric tensor. Additionally, we chose not to define the one-form basis as the exterior derivative of the coordinate, $\boldsymbol{e}^{a} = \mathrm{d}\xi^{a}$, along with the corresponding geometric picture of the one-form basis elements as sheets perpendicular to the coordinate axis (e.g., see Figure 4.7 of *Tromp* (2025b)). In this more general approach, the biorthogonality relation takes the form

$$\mathbb{g}(\boldsymbol{e}^{a}, \boldsymbol{e}_{b}) = \mathbb{g}(\partial_{a}, \mathrm{d}\xi^{b}) = \delta_{a}{}^{b} \quad \text{and} \quad \mathbb{g}(\boldsymbol{e}^{b}, \boldsymbol{e}_{a}) = \mathbb{g}(\mathrm{d}\xi^{b}, \partial_{a}) = \delta^{b}{}_{a}. \tag{3.64}$$

Again, we chose not delve deeply into the mathematics of differential forms, thus motivating us to eschew a serious study of these more general ideas. The interested reader is encouraged to study *Tromp* (2025b) for a lucid presentation targeting physicists.

3.9 Scalar product between vectors

In Section 1.3.3 we defined the scalar product between two Cartesian vectors. The natural generalization is given by

$$\mathbf{P} \cdot \mathbf{Q} = P^{a}\, Q^{b}\, \boldsymbol{e}_{a} \cdot \boldsymbol{e}_{b} = P^{a}\, Q^{b}\, \mathbb{g}_{ab} = P^{a}\, Q_{a} = P_{b}\, Q^{b}, \tag{3.65}$$

where the second equality made use of the metric tensor representation given by equation (3.37). We can conceive of the scalar product in a somewhat more general manner by recalling that a one-form, $\mathbf{P}$, operates on a vector, $\mathbf{Q}$, and conversely, a vector operates on a one-form. Exposing components leads to

$$\mathbf{P}(\mathbf{Q}) = \mathbf{P}(Q^a\,\boldsymbol{e}_a) = Q^a\,\mathbf{P}(\boldsymbol{e}_a) = Q^a\,P_a, \tag{3.66}$$

which equals to

$$\mathbf{Q}(\mathbf{P}) = \mathbf{Q}(P_a\,\boldsymbol{e}^a) = P_a\,\mathbf{Q}(\boldsymbol{e}^a) = P_a\,Q^a. \tag{3.67}$$

The scalar product is invariant to coordinate changes, as seen through

$$\mathbf{Q}(\mathbf{P}) = \mathbf{Q}(P_a\,\boldsymbol{e}^a) = P_a\,Q^a = \mathbf{Q}(P_{\overline{a}}\,\boldsymbol{e}^{\overline{a}}) = P_{\overline{a}}\,Q^{\overline{a}}. \tag{3.68}$$

The invariance is also revealed by working just with the coordinate representations and introducing the transformation matrix elements

$$P_a\,Q^a = (\Lambda^{\overline{a}}{}_a\,P_{\overline{a}})\,(\Lambda^a{}_{\overline{b}}\,Q^{\overline{b}}) = \Lambda^{\overline{a}}{}_a\,\Lambda^a{}_{\overline{b}}\,P_{\overline{a}}\,Q^{\overline{b}} = \delta^{\overline{a}}{}_{\overline{b}}\,P_{\overline{a}}\,Q^{\overline{b}} = P_{\overline{a}}\,Q^{\overline{a}}. \tag{3.69}$$

3.10 The volume element and Jacobian of transformation

Recall from Section 1.7.2 that we derived an expression for the volume of an infinitesimal region of Euclidean space, $\mathbb{R}^3$, using Cartesian coordinates

$$\mathrm{d}V = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,(\hat{\boldsymbol{x}}\times\hat{\boldsymbol{y}})\cdot\hat{\boldsymbol{z}} = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z. \tag{3.70}$$

This volume element is used for integrating over a region of $\mathbb{R}^3$ using Cartesian coordinates. We now generalize this result to arbitrary coordinates.

3.10.1 Jacobian of transformation

From multi-variate calculus, the relation between $\mathrm{d}\xi^1\,\mathrm{d}\xi^2\,\mathrm{d}\xi^3$ and $\mathrm{d}\xi^{\overline{1}}\,\mathrm{d}\xi^{\overline{2}}\,\mathrm{d}\xi^{\overline{3}}$ for two sets of coordinates is given by

$$\begin{aligned}
\mathrm{d}\xi^1\,\mathrm{d}\xi^2\,\mathrm{d}\xi^3 &= \frac{\partial\boldsymbol{\xi}}{\partial\overline{\boldsymbol{\xi}}}\,\mathrm{d}\xi^{\overline{1}}\,\mathrm{d}\xi^{\overline{2}}\,\mathrm{d}\xi^{\overline{3}} && \text{(3.71a)}\\
&= \frac{\partial(\xi^1,\xi^2,\xi^3)}{\partial(\xi^{\overline{1}},\xi^{\overline{2}},\xi^{\overline{3}})}\,\mathrm{d}\xi^{\overline{1}}\,\mathrm{d}\xi^{\overline{2}}\,\mathrm{d}\xi^{\overline{3}} && \text{(3.71b)}\\
&= \left[\frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{1}}}\times\frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{2}}}\right]\cdot\frac{\partial\boldsymbol{\xi}}{\partial\xi^{\overline{3}}}\,\mathrm{d}\xi^{\overline{1}}\,\mathrm{d}\xi^{\overline{2}}\,\mathrm{d}\xi^{\overline{3}} && \text{(3.71c)}\\
&= \det(\Lambda^a{}_{\overline{a}})\,\mathrm{d}\xi^{\overline{1}}\,\mathrm{d}\xi^{\overline{2}}\,\mathrm{d}\xi^{\overline{3}}, && \text{(3.71d)}
\end{aligned}$$

where $\det(\Lambda^a{}_{\overline{a}})$ is the determinant of the transformation matrix, also known as the *Jacobian of transformation*, and each of the equalities (3.71a)-(3.71d) provides alternative notations for the Jacobian. The coordinate transformation is well defined, in the sense that it is 1-to-1 and invertible, so long as the Jacobian does not vanish (or equivalently the Jacobian is single-signed). We maintain labels on the transformation matrix inside the determinant symbol to help indicate the sense for the transformation, with this notation also supporting the conservation of tensor indices.

3.10.2 Jacobian related to the determinant of the metric

Recall the expression (3.38d) for the transformation of the metric

$$\mathbb{g}_{\bar{a}\bar{b}} = \Lambda^{a}{}_{\bar{a}}\Lambda^{b}{}_{\bar{b}}\,\mathbb{g}_{ab} = (\Lambda^{\top})_{\bar{a}}{}^{a}\,\mathbb{g}_{ab}\,\Lambda^{b}{}_{\bar{b}}. \tag{3.72}$$

Taking determinants of both sides yields[13]

$$\det(\mathbb{g}_{\bar{a}\bar{b}}) = \det((\Lambda^{\top})_{\bar{a}}{}^{a})\,\det(\mathbb{g}_{ab})\,\det(\Lambda^{b}{}_{\bar{b}}) = [\det(\Lambda)]^{2}\,\det(\mathbb{g}_{ab}). \tag{3.73}$$

To reach this result we used the property of determinants that $\det(A\,B) = \det(A)\det(B)$ for any two matrices, and determinant of a matrix equals to the determinant of its transpose so that, $\det(\Lambda^{\top}) = \det(\Lambda)$. Consequently,

$$\det(\Lambda^{a}{}_{\bar{a}}) = \frac{\sqrt{\det(\mathbb{g}_{\bar{a}\bar{b}})}}{\sqrt{\det(\mathbb{g}_{ab})}}. \tag{3.74}$$

3.10.3 Invariant/covariant volume element

Use of equation (3.74) in equation (3.71d) leads to the equivalent expressions for the volume element

$$\mathrm{d}V \equiv \sqrt{\det(\mathbb{g}_{ab})}\,\mathrm{d}\xi^{1}\,\mathrm{d}\xi^{2}\,\mathrm{d}\xi^{3} = \sqrt{\det(\mathbb{g}_{\bar{a}\bar{b}})}\,\mathrm{d}\xi^{\bar{1}}\,\mathrm{d}\xi^{\bar{2}}\,\mathrm{d}\xi^{\bar{3}}. \tag{3.75}$$

This relation provides a general coordinate expression for the volume element, which we refer to as the *invariant volume element*, or equivalently the *covariant volume element*. For the special case when the unbarred coordinates are Cartesian, $\mathbb{g}_{ab} = \delta_{ab}$, so that $\det(\mathbb{g}_{ab}) = 1$ and

$$\det(\Lambda^{a}{}_{\bar{a}}) = \sqrt{\det(\mathbb{g}_{\bar{a}\bar{b}})} \qquad \text{unbarred coordinates are Cartesian.} \tag{3.76}$$

This is a rather useful expression for our purposes, since we can always use Cartesian as the unbarred coordinates given that geophysical fluids move in a background Euclidean space.

We find that $\sqrt{\det(\mathbb{g}_{ab})}$ appears throughout general tensor analysis, so that it is useful to introduce the shorthand

$$\mathsf{g} = \sqrt{\det(\mathbb{g}_{ab})}. \tag{3.77}$$

The indices are mere placeholders and play no role in the summation convention.

3.11 Properties of the Jacobian determinant

As discussed in Section 1.6.1, the Cartesian components of the Levi-Civita tensor are given by the permutation symbol, ϵ_{abc}. To help determine the general coordinate representation of the Levi-Civita tensor in Section 3.12, we here develop some identities satisfied by the determinant of the transformation matrix.

[13] As in equation (3.71d) for the Jacobian, we here leave the indices exposed on the metric tensor when inside of the determinant operator. These indices are not subject to index conservation, with the determinant a number and so carrying no indices. Rather, the indices are kept to remind us what coordinates are used to represent the metric tensor.

3.11.1 Connecting the permutation symbol to the determinant

Consider a two-dimensional space with a transformation matrix, $\Lambda^a{}_{\overline{a}}$, between two sets of coordinates. The determinant of the transformation matrix is given by

$$\det(\Lambda^a{}_{\overline{a}}) = \Lambda^1{}_{\overline{1}}\,\Lambda^2{}_{\overline{2}} - \Lambda^1{}_{\overline{2}}\,\Lambda^2{}_{\overline{1}}. \tag{3.78}$$

Introducing the permutation symbol, ϵ_{ab}, allows us to write this expression in the tidy manner

$$\det(\Lambda^a{}_{\overline{a}}) = \epsilon_{ab}\,\Lambda^a{}_{\overline{1}}\,\Lambda^b{}_{\overline{2}} \tag{3.79}$$

with

$$\epsilon_{12} = 1 \qquad \text{and} \qquad \epsilon_{21} = -1. \tag{3.80}$$

The permutation symbol is defined to have numerically the same values whether the labels are raised or lowered, $\epsilon^{ab} = \epsilon_{ab}$, and for any coordinate set, $\epsilon^{\overline{a}\overline{b}} = \epsilon_{\overline{a}\overline{b}}$.

We can generalize equation (3.79) to any number of dimensions, each of which adds one more label to the permutation symbol and one more number added to the permutation string. We already encountered the three dimensional version in Section 1.6.1 when discussing the vector cross product, in which case the permutation symbol is

$$\epsilon_{123} = 1 \tag{3.81a}$$

$$\epsilon_{abc} = \begin{cases} 0 & \text{if any two labels are the same,} \\ 1 & \text{if } a,b,c \text{ is an even permutation of } 1,2,3, \\ -1 & \text{if } a,b,c \text{ is an odd permutation of } 1,2,3. \end{cases} \tag{3.81b}$$

Likewise, the determinant of the transformation matrix takes the form

$$\det(\Lambda^a{}_{\overline{a}}) = \frac{\partial \boldsymbol{\xi}}{\partial \overline{\boldsymbol{\xi}}} = \frac{\partial(\xi^1,\xi^2,\xi^3)}{\partial(\xi^{\overline{1}},\xi^{\overline{2}},\xi^{\overline{3}})} = \epsilon_{abc}\,\Lambda^a{}_{\overline{1}}\,\Lambda^b{}_{\overline{2}}\,\Lambda^c{}_{\overline{3}}. \tag{3.82}$$

3.11.2 Further identities satisfied by the determinant

The following identity in two dimensions can be readily verified through enumeration

$$\epsilon_{ab}\,\Lambda^a{}_{\overline{a}}\,\Lambda^b{}_{\overline{b}} = \epsilon_{\overline{a}\overline{b}}\,\det(\Lambda^a{}_{\overline{a}}), \tag{3.83}$$

which follows directly from the definition of the determinant and can be explicity verified since the permutation symbol, $\epsilon_{\overline{a}\overline{b}}$, is numerically identical to ϵ_{ab}. Now contract both sides of this relation with $\epsilon^{\overline{a}\overline{b}}$ to isolate the determinant

$$\frac{1}{2}\,\epsilon^{\overline{a}\overline{b}}\,\epsilon_{ab}\,\Lambda^a{}_{\overline{a}}\,\Lambda^b{}_{\overline{b}} = \det(\Lambda^a{}_{\overline{a}}), \tag{3.84}$$

where we used

$$\epsilon^{\overline{a}\overline{b}}\,\epsilon_{\overline{a}\overline{b}} = \epsilon^{\overline{1}\overline{2}}\,\epsilon_{\overline{1}\overline{2}} + \epsilon^{\overline{2}\overline{1}}\,\epsilon_{\overline{2}\overline{1}} = 2. \tag{3.85}$$

The three dimensional version takes the form

$$\epsilon_{abc}\,\Lambda^a{}_{\overline{a}}\,\Lambda^b{}_{\overline{b}}\,\Lambda^c{}_{\overline{c}} = \epsilon_{\overline{a}\overline{b}\overline{c}}\,\det(\Lambda^a{}_{\overline{a}}), \tag{3.86}$$

so that

$$\frac{1}{3!}\,\epsilon^{\overline{a}\overline{b}\overline{c}}\,\epsilon_{abc}\,\Lambda^{a}{}_{\overline{a}}\,\Lambda^{b}{}_{\overline{b}}\,\Lambda^{c}{}_{\overline{c}} = \det(\Lambda^{a}{}_{\overline{a}}). \tag{3.87}$$

The identity (3.87) is an elegant means to write the determinant and it serves many needs. Here is another relation that can be of use

$$\Lambda^{\overline{a}}{}_{a}\,\det(\Lambda^{a}{}_{\overline{a}}) = \frac{1}{2!}\,\epsilon^{\overline{a}\overline{b}\overline{c}}\,\epsilon_{abc}\,\Lambda^{b}{}_{\overline{b}}\,\Lambda^{c}{}_{\overline{c}}. \tag{3.88}$$

The right hand side is $(-1)^{\overline{a}+a}$ times the determinant of the 2×2 matrix built from excluding the $\overline{a}$ and a elements from $\Lambda^{a}{}_{\overline{a}}$, with this reduced determinant known as the *cofactor*. To prove equation (3.88) we contract both sides by $\Lambda^{a}{}_{\overline{a}}$ and use the identity $\Lambda^{a}{}_{\overline{a}}\,\Lambda^{\overline{a}}{}_{a} = 3$, in which case we recover the expression for the determinant in equation (3.87).

3.11.3 Derivative of the Jacobian with respect to a matrix element

There are occasions in which we need to compute the derivative of the Jacobian determinant with respect to an element of the transformation matrix, such as in the study of Hamilton's principle for fluid mechanics in VOLUME 4. For this purpose we make use of the identity

$$\frac{\partial \Lambda^{a}{}_{\overline{a}}}{\partial \Lambda^{d}{}_{\overline{d}}} = \delta^{a}{}_{d}\,\delta^{\overline{d}}{}_{\overline{a}}, \tag{3.89}$$

which then leads to

$$\frac{\partial \det(\Lambda^{a}{}_{\overline{a}})}{\partial \Lambda^{d}{}_{\overline{d}}} = \frac{1}{3!}\,\epsilon^{\overline{a}\overline{b}\overline{c}}\,\epsilon_{abc}\,\frac{\partial}{\partial \Lambda^{d}{}_{\overline{d}}}\left[\Lambda^{a}{}_{\overline{a}}\,\Lambda^{b}{}_{\overline{b}}\,\Lambda^{c}{}_{\overline{c}}\right] = \frac{1}{2}\,\epsilon^{\overline{d}\overline{b}\overline{c}}\,\epsilon_{dbc}\,\Lambda^{b}{}_{\overline{b}}\,\Lambda^{c}{}_{\overline{c}} = \Lambda^{\overline{d}}{}_{d}\,\det(\Lambda^{a}{}_{\overline{a}}), \tag{3.90}$$

where the final equality made use of equation (3.88). Evidently, the derivative of the determinant with respect to a matrix element equals to the determinant multiplied by the element of the inverse matrix.

3.12 The volume element and Levi-Civita tensor

The metric tensor introduced in Section 3.7 provides a means to measure distance between two points. The Levi-Civita tensor allows us to compute volumes (or areas for two dimensional manifolds). We make particular use of this tensor to compute the volume element used for integration. This section generalizes the Cartesian coordinate discussion provided in Section 1.7.3.

3.12.1 General coordinate representation of the Levi-Civita tensor

The relation (3.83) indicates that the permutation symbol, ϵ_{ab}, *does not* transform as the components to a second order covariant tensor, unless the determinant of the transformation is unity. The same can be said for the permutation symbol, ϵ_{abc}, with equation (3.86) indicating that it does not transform as the components to a third order covariant tensor, unless the determinant of the transformation is unity. Unit determinants occur for special transformations, such as rotations (i.e., Cartesian to Cartesian coordinate transformation as in Chapter 1) and the identity transformation. Indeed, we have already noted that the permutation symbol has the same representation regardless the coordinate choice. As we now show, the permutation symbol is the Cartesian coordinate representation of the Levi-Civita tensor.

The above relations for the determinant motivate us to introduce the general coordinate form of the *Levi-Civita tensor*

$$\varepsilon_{abc} \equiv \sqrt{\det(\mathbb{g}_{ab})}\, \epsilon_{abc}. \tag{3.91}$$

We highlight the distinct symbols in this definition, with ε the Levi-Civita tensor and ϵ the permutation symbol. By construction, the components to the Levi-Civita tensor transform as

$$\Lambda^{a}{}_{\overline{a}}\, \Lambda^{b}{}_{\overline{b}}\, \Lambda^{c}{}_{\overline{c}}\, \varepsilon_{abc} = \Lambda^{a}{}_{\overline{a}}\, \Lambda^{b}{}_{\overline{b}}\, \Lambda^{c}{}_{\overline{c}} \sqrt{\det(\mathbb{g}_{ab})}\, \epsilon_{abc} \tag{3.92a}$$

$$= \sqrt{\det(\mathbb{g}_{ab})}\, \epsilon_{\overline{a}\overline{b}\overline{c}}\, \det(\Lambda^{a}{}_{\overline{a}}) \tag{3.92b}$$

$$= \sqrt{\det(\mathbb{g}_{\overline{a}\overline{b}})}\, \epsilon_{\overline{a}\overline{b}\overline{c}} \tag{3.92c}$$

$$= \varepsilon_{\overline{a}\overline{b}\overline{c}}, \tag{3.92d}$$

where equations (3.74) and (3.91) were used. Therefore, ε_{abc} transforms as components to a third order tensor, here written in the $(0,3)$ representation. Likewise,

$$\varepsilon^{abc} = \frac{\epsilon^{abc}}{\sqrt{\det(\mathbb{g}_{ab})}} \tag{3.93}$$

transforms as the $(3,0)$ components to the tensor. These transformation rules allow us to identify ε as a third order tensor.

3.12.2 The volume element

As a third order tensor, the Levi-Civita tensor can eat three vectors or three one-forms. For example, for three infinitesimal vectors we have

$$\varepsilon(\boldsymbol{e}_1\, \mathrm{d}\xi^1, \boldsymbol{e}_2\, \mathrm{d}\xi^2, \boldsymbol{e}_3\, \mathrm{d}\xi^3) = \mathrm{d}\xi^1\, \mathrm{d}\xi^2\, \mathrm{d}\xi^3\, \varepsilon(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) \tag{3.94a}$$

$$= \mathrm{d}\xi^1\, \mathrm{d}\xi^2\, \mathrm{d}\xi^3\, \varepsilon_{123} \tag{3.94b}$$

$$= \mathrm{d}\xi^1\, \mathrm{d}\xi^2\, \mathrm{d}\xi^3 \sqrt{\det(\mathbb{g}_{ab})}\, \epsilon_{123} \tag{3.94c}$$

$$= \mathrm{d}V, \tag{3.94d}$$

where we introduced the invariant volume element from equation (3.75) for the final step. Geometrically, this result means that the Levi-Civita tensor measures the volume defined by three vectors

$$\varepsilon(\mathbf{A}, \mathbf{B}, \mathbf{C}) = \mathrm{volume}(\mathbf{A}, \mathbf{B}, \mathbf{C}). \tag{3.95}$$

This interpretation accords with the Cartesian coordinate discussion of the Levi-Civita tensor in Section 1.7.3.

3.13 Vector cross product

From Section 1.6 we introduced the vector cross product for two Cartesian basis vectors

$$\hat{\boldsymbol{x}} \times \hat{\boldsymbol{y}} = \hat{\boldsymbol{z}} \qquad \text{and cyclic permutations.} \tag{3.96}$$

The coordinate invariant generalization of this relation is given by

$$\boldsymbol{e}_a \times \boldsymbol{e}_b \equiv \varepsilon_{abc}\, \boldsymbol{e}^c, \tag{3.97}$$

so that the vector cross product of two vectors leads to a one-form. We are thus led to the general coordinate expression for the vector cross product of two arbitrary vectors

$$\mathbf{P} \times \mathbf{Q} = P^a \, Q^b \, \boldsymbol{e}_a \times \boldsymbol{e}_b = P^a \, Q^b \, \varepsilon_{abc} \, \boldsymbol{e}^c. \tag{3.98}$$

3.14 Coordinate transformation of partial derivatives

Throughout this book, the background space is Euclidean and time is universal (i.e., we maintain the notion of absolute simultaneity). We are thus concerned with space tensors rather than the space-time tensors of special and general relativity. Nonetheless, our description of space generally makes use of coordinates that are time dependent, such as for Lagrangian cordinates and generalized vertical coordinates. Time dependence of these coordinates motivates the space-time formulation from Section 3.3.4, in particular for the purpose of transforming the partial time derivative operator. In this section we establish some properties of the space-time transformation matrix and then make use of this matrix for transforming space and time partial time derivatives.

3.14.1 The space-time transformation matrix

As discussed in Section 3.7.4, transformations between coordinate representations are enabled by the transformation matrix built from partial derivatives of the coordinate transformations. The transformation matrix with a universal Newtonian time plus spatial coordinates (that are functions of space and time) takes on the form

$$\Lambda^{\alpha}{}_{\overline{\alpha}} = \frac{\partial \xi^{\alpha}}{\partial \xi^{\overline{\alpha}}} = \begin{bmatrix} \frac{\partial \xi^0}{\partial \xi^{\overline{0}}} & \frac{\partial \xi^0}{\partial \xi^{\overline{1}}} & \frac{\partial \xi^0}{\partial \xi^{\overline{2}}} & \frac{\partial \xi^0}{\partial \xi^{\overline{3}}} \\ \frac{\partial \xi^1}{\partial \xi^{\overline{0}}} & \frac{\partial \xi^1}{\partial \xi^{\overline{1}}} & \frac{\partial \xi^1}{\partial \xi^{\overline{2}}} & \frac{\partial \xi^1}{\partial \xi^{\overline{3}}} \\ \frac{\partial \xi^2}{\partial \xi^{\overline{0}}} & \frac{\partial \xi^2}{\partial \xi^{\overline{1}}} & \frac{\partial \xi^2}{\partial \xi^{\overline{2}}} & \frac{\partial \xi^2}{\partial \xi^{\overline{3}}} \\ \frac{\partial \xi^3}{\partial \xi^{\overline{0}}} & \frac{\partial \xi^3}{\partial \xi^{\overline{1}}} & \frac{\partial \xi^3}{\partial \xi^{\overline{2}}} & \frac{\partial \xi^3}{\partial \xi^{\overline{3}}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{\partial \xi^1}{\partial \xi^{\overline{0}}} & \frac{\partial \xi^1}{\partial \xi^{\overline{1}}} & \frac{\partial \xi^1}{\partial \xi^{\overline{2}}} & \frac{\partial \xi^1}{\partial \xi^{\overline{3}}} \\ \frac{\partial \xi^2}{\partial \xi^{\overline{0}}} & \frac{\partial \xi^2}{\partial \xi^{\overline{1}}} & \frac{\partial \xi^2}{\partial \xi^{\overline{2}}} & \frac{\partial \xi^2}{\partial \xi^{\overline{3}}} \\ \frac{\partial \xi^3}{\partial \xi^{\overline{0}}} & \frac{\partial \xi^3}{\partial \xi^{\overline{1}}} & \frac{\partial \xi^3}{\partial \xi^{\overline{2}}} & \frac{\partial \xi^3}{\partial \xi^{\overline{3}}} \end{bmatrix}. \tag{3.99}$$

The final equality made use of our assumption that $\xi^{\overline{0}} = \xi^0$ since the time coordinate remains universal. Hence, when computing $\partial \xi^0 / \partial \xi^{\overline{a}}$ we keep $\xi^{\overline{0}}$ fixed so that the derivative vanishes as in the specific case of

$$\left[\frac{\partial \xi^0}{\partial \xi^{\overline{1}}}\right]_{\xi^{\overline{0}}, \xi^{\overline{2}}, \xi^{\overline{3}}} = \left[\frac{\partial \xi^0}{\partial \xi^{\overline{1}}}\right]_{\xi^0, \xi^{\overline{2}}, \xi^{\overline{3}}} = 0. \tag{3.100}$$

Zero elements in the first row of the transformation matrix (3.99) reveals that time is not a function of space

$$\frac{\partial \xi^0}{\partial \xi^{\overline{a}}} = 0, \tag{3.101}$$

which is expected since we are assuming universal *Newtonian time* in which time is independent of space. In contrast, nonzero elements in the first column indicate that our description of space is generally a function of time

$$\frac{\partial \xi^a}{\partial \xi^{\overline{0}}} \neq 0. \tag{3.102}$$

We see the same overall structure in the inverse space-time transformation matrix

$$\Lambda^{\overline{\alpha}}{}_{\alpha} = \frac{\partial \xi^{\overline{\alpha}}}{\partial \xi^{\alpha}} = \begin{bmatrix} \frac{\partial \xi^{\overline{0}}}{\partial \xi^{0}} & \frac{\partial \xi^{\overline{0}}}{\partial \xi^{1}} & \frac{\partial \xi^{\overline{0}}}{\partial \xi^{2}} & \frac{\partial \xi^{\overline{0}}}{\partial \xi^{3}} \\ \frac{\partial \xi^{\overline{1}}}{\partial \xi^{0}} & \frac{\partial \xi^{\overline{1}}}{\partial \xi^{1}} & \frac{\partial \xi^{\overline{1}}}{\partial \xi^{2}} & \frac{\partial \xi^{\overline{1}}}{\partial \xi^{3}} \\ \frac{\partial \xi^{\overline{2}}}{\partial \xi^{0}} & \frac{\partial \xi^{\overline{2}}}{\partial \xi^{1}} & \frac{\partial \xi^{\overline{2}}}{\partial \xi^{2}} & \frac{\partial \xi^{\overline{2}}}{\partial \xi^{3}} \\ \frac{\partial \xi^{\overline{3}}}{\partial \xi^{0}} & \frac{\partial \xi^{\overline{3}}}{\partial \xi^{1}} & \frac{\partial \xi^{\overline{3}}}{\partial \xi^{2}} & \frac{\partial \xi^{\overline{3}}}{\partial \xi^{3}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{\partial \xi^{\overline{1}}}{\partial \xi^{0}} & \frac{\partial \xi^{\overline{1}}}{\partial \xi^{1}} & \frac{\partial \xi^{\overline{1}}}{\partial \xi^{2}} & \frac{\partial \xi^{\overline{1}}}{\partial \xi^{3}} \\ \frac{\partial \xi^{\overline{2}}}{\partial \xi^{0}} & \frac{\partial \xi^{\overline{2}}}{\partial \xi^{1}} & \frac{\partial \xi^{\overline{2}}}{\partial \xi^{2}} & \frac{\partial \xi^{\overline{2}}}{\partial \xi^{3}} \\ \frac{\partial \xi^{\overline{3}}}{\partial \xi^{0}} & \frac{\partial \xi^{\overline{3}}}{\partial \xi^{1}} & \frac{\partial \xi^{\overline{3}}}{\partial \xi^{2}} & \frac{\partial \xi^{\overline{3}}}{\partial \xi^{3}} \end{bmatrix}. \tag{3.103}$$

3.14.2 Determinant of the transformation matrix

The determinant of the space-time transformation and its inverse remain identical to the determinant of their purely space portions

$$\det(\Lambda^{\alpha}{}_{\overline{\alpha}}) = \det(\Lambda^{a}{}_{\overline{a}}) \qquad \text{and} \qquad \det(\Lambda^{\overline{\alpha}}{}_{\alpha}) = \det(\Lambda^{\overline{a}}{}_{a}), \tag{3.104}$$

which follows since the first row in both transformations has only a single non-zero value, $\Lambda^{0}{}_{\overline{0}} = 1$ and $\Lambda^{\overline{0}}{}_{0} = 1$. Hence, the relations developed in Sections 3.10 and 3.11 for the volume element and Jacobian of transformation remain unchanged when adding the universal time coordinate.

3.14.3 Multiplying the transformation matrix and its inverse

We here verify that the transformation matrix (3.99) indeed has its inverse given by (3.103). For this purpose we must prove the space-time duality relations

$$\delta^{\alpha}{}_{\beta} = \Lambda^{\alpha}{}_{\overline{\beta}}\,\Lambda^{\overline{\beta}}{}_{\beta} \qquad \text{and} \qquad \delta^{\overline{\alpha}}{}_{\overline{\beta}} = \Lambda^{\overline{\alpha}}{}_{\beta}\,\Lambda^{\beta}{}_{\overline{\beta}}, \tag{3.105}$$

where $\delta^{\alpha}{}_{\beta}$ and $\delta^{\overline{\alpha}}{}_{\overline{\beta}}$ are components to the identity tensor. The proof relies on writing the space-time coordinate transformation as a composite function

$$\xi^{\overline{\alpha}} = \xi^{\overline{\alpha}}(\xi^{\alpha}) = \xi^{\overline{\alpha}}[\xi^{\alpha}(\xi^{\overline{\beta}})]. \tag{3.106}$$

Taking partial derivatives and using the chain rule renders

$$\delta^{\overline{\alpha}}{}_{\overline{\beta}} = \frac{\partial \xi^{\overline{\alpha}}}{\partial \xi^{\overline{\beta}}} = \frac{\partial \xi^{\overline{\alpha}}}{\partial \xi^{\alpha}}\frac{\partial \xi^{\alpha}}{\partial \xi^{\overline{\beta}}} = \Lambda^{\overline{\alpha}}{}_{\alpha}\,\Lambda^{\alpha}{}_{\overline{\beta}} \qquad \text{and} \qquad \delta^{\alpha}{}_{\beta} = \frac{\partial \xi^{\alpha}}{\partial \xi^{\beta}} = \frac{\partial \xi^{\alpha}}{\partial \xi^{\overline{\alpha}}}\frac{\partial \xi^{\overline{\alpha}}}{\partial \xi^{\beta}} = \Lambda^{\alpha}{}_{\overline{\alpha}}\,\Lambda^{\overline{\alpha}}{}_{\beta}. \tag{3.107}$$

Furthermore, the space subcomponents decouple from time, which can be seen by considering a few representative cases

$$1 = \delta^{0}{}_{0} = \Lambda^{0}{}_{\overline{\alpha}}\,\Lambda^{\overline{\alpha}}{}_{0} = \Lambda^{0}{}_{\overline{0}}\,\Lambda^{\overline{0}}{}_{0} \tag{3.108a}$$
$$1 = \delta^{1}{}_{1} = \Lambda^{1}{}_{\overline{\alpha}}\,\Lambda^{\overline{\alpha}}{}_{1} = \Lambda^{1}{}_{\overline{a}}\,\Lambda^{\overline{a}}{}_{1} \tag{3.108b}$$
$$0 = \delta^{0}{}_{1} = \Lambda^{0}{}_{\overline{\alpha}}\,\Lambda^{\overline{\alpha}}{}_{1} = \Lambda^{0}{}_{\overline{0}}\,\Lambda^{\overline{0}}{}_{1} \tag{3.108c}$$
$$0 = \delta^{1}{}_{2} = \Lambda^{1}{}_{\overline{\alpha}}\,\Lambda^{\overline{\alpha}}{}_{2} = \Lambda^{1}{}_{\overline{a}}\,\Lambda^{\overline{a}}{}_{2}. \tag{3.108d}$$

Consequently, the spatial components satisfy

$$\delta^{a}{}_{b} = \Lambda^{a}{}_{\overline{b}} \Lambda^{\overline{b}}{}_{b} \qquad \text{and} \qquad \delta^{\overline{a}}{}_{\overline{b}} = \Lambda^{\overline{a}}{}_{b} \Lambda^{b}{}_{\overline{b}}, \tag{3.109}$$

which allows for a splitting of the spatial components from the time component.

3.14.4 Transformation of space and time partial derivatives

Application of the chain rule leads to the transformation of the partial derivative operator

$$\partial_{\overline{\alpha}} = \frac{\partial}{\partial \xi^{\overline{\alpha}}} = \frac{\partial \xi^{\alpha}}{\partial \xi^{\overline{\alpha}}} \frac{\partial}{\partial \xi^{\alpha}} = \Lambda^{\alpha}{}_{\overline{\alpha}} \, \partial_{\alpha}. \tag{3.110}$$

Extracting the time and space components from the transformation matrix (3.99) yields

$$\partial_{\overline{0}} = \Lambda^{\alpha}{}_{\overline{0}} \, \partial_{\alpha} = \partial_{0} + \Lambda^{a}{}_{\overline{0}} \, \partial_{a} \tag{3.111a}$$

$$\partial_{\overline{a}} = \Lambda^{\alpha}{}_{\overline{a}} \, \partial_{\alpha} = \Lambda^{a}{}_{\overline{a}} \, \partial_{a}. \tag{3.111b}$$

Notably, the time derivative operator in one coordinate system transforms into both space and time derivative operators in the new coordinate system. We expect this result since the time derivative in one coordinate system is computed with its spatial coordinates held fixed, but these spatial coordinates are generally moving with respect to the other coordinate system. In contrast, the spatial components to the partial derivative operator transform among just the other spatial components; there is no mixing with the time derivative operator. This property of the spatial derivative operator follows from the use of universal Newtonian time. It allows us to focus on space tensors in the following sections.

3.15 Covariant derivative operator

The partial differentiation discussed in Section 3.14 is sufficient for Euclidean space using Cartesian coordinates. Namely, the derivative of any Cartesian tensor is found merely by computing the component-wise partial derivative. Furthermore, this operation transforms under rotations according to the rules of Cartesian tensor analysis. What we seek in this section is an operator that satisfies the same properties, only now when using arbitrary coordinates and on arbitrary spaces. We call this operator the covariant derivative and write it as $\boldsymbol{\nabla}$ (`{\bm \nabla}` in LaTeX). When operating on an arbitrary (p, q) tensor, $\mathbf{T}$, we assume $\boldsymbol{\nabla}\mathbf{T}$ is a $(p, q+1)$ tensor and that it takes the component expression

$$\boldsymbol{\nabla}\mathbf{T} = (\boldsymbol{\nabla} T)^{i_1 \cdots i_p}{}_{j_1 \cdots j_q m} \, \boldsymbol{e}^{m} \otimes \boldsymbol{e}_{i_1} \otimes \cdots \boldsymbol{e}_{i_p} \otimes \boldsymbol{e}^{j_1} \otimes \cdots \boldsymbol{e}^{j_q}. \tag{3.112}$$

The term $(\boldsymbol{\nabla} T)^{i_1 \cdots i_p}{}_{j_1 \cdots j_q m}$ provides the coordinate representation of the covariant derivative, and the $\boldsymbol{e}^{m} \otimes \boldsymbol{e}_{i_1} \otimes \cdots \boldsymbol{e}_{i_p} \otimes \boldsymbol{e}^{j_1} \otimes \cdots \boldsymbol{e}^{j_q}$ tensor product manifests the $(p, q+1)$ tensorial order. Since the tensor product is not commutative (Section 1.5.2), the ordering of the basis vectors and one-forms is important. In particular, note the $\boldsymbol{e}^{m}$ basis one-form is placed at the front of the tensor product.

3.15.1 Covariant derivative acting on a scalar

Equation (3.112) is rather formidible so it helps to consider examples. To start, consider a scalar field (a $(0,0)$ tensor) whose covariant derivative is defined as the gradient

$$\boldsymbol{\nabla}\mathrm{F} = (\boldsymbol{\nabla} F)_m \, \boldsymbol{e}^m \equiv (\partial_m F) \, \boldsymbol{e}^m. \tag{3.113}$$

This definition indeed produces a $(0,1)$ tensor, whose components, $\partial_m F$, transform according to the partial derivative operator discussed in Section 3.14. We can refer to equation (3.113) as either the gradient of the scalar field or the covariant derivative of the scalar field.

3.15.2 Covariant derivative acting on a $(1,0)$ tensor

Here we determine the covariant derivative of a first order tensor, which requires more work than for the scalar. We separately consider the expression for the covariant derivative when acting on the $(1,0)$ representation of the first order tensor (i.e., a vector), and thereafter when acting on the $(0,1)$ representation (i.e., a one-form).

Following the notation from equation (3.112), we have the covariant derivative of the $(1,0)$ representation of a first order tensor, $\mathbf{F} = F^a \, \boldsymbol{e}_a$, given by the following $(1,1)$ tensor

$$\boldsymbol{\nabla}\mathbf{F} \equiv (\boldsymbol{\nabla}\boldsymbol{F})^a{}_m \, \boldsymbol{e}^m \otimes \boldsymbol{e}_a, \tag{3.114}$$

where $(\boldsymbol{\nabla}\boldsymbol{F})^a{}_m$ is the $(1,1)$ coordinate representation of $\boldsymbol{\nabla}\mathbf{F}$. To determine $(\boldsymbol{\nabla}\boldsymbol{F})^a{}_m$ we expand the left hand side of equation (3.114) according to

$$\boldsymbol{\nabla}\mathbf{F} = \boldsymbol{e}^m \nabla_m \otimes (F^a \, \boldsymbol{e}_a) = \boldsymbol{e}^m \otimes [(\nabla_m F^a) \, \boldsymbol{e}_a + F^a \, \nabla_m \boldsymbol{e}_a], \tag{3.115}$$

where the first equality used

$$\boldsymbol{\nabla} = \boldsymbol{e}^m \nabla_m \quad \text{and} \quad \mathbf{F} = F^a \, \boldsymbol{e}_a, \tag{3.116}$$

and the second equality used the product rule assumed to hold for the covariant derivative. We next assume that the covariant derivative of the basis vectors, $\nabla_m \boldsymbol{e}_a$, produces an object that is itself represented by a linear sum of basis vectors. That is, we assume $\nabla_m \boldsymbol{e}_a$ lives in the same space as $\boldsymbol{e}_a$, thus allowing us to define the Levi-Civita connection as

$$\nabla_m \boldsymbol{e}_a = \Gamma^d_{ma} \, \boldsymbol{e}_d, \tag{3.117}$$

where the Γ^d_{ma} expansion coefficients are known as Christoffel symbols. We determine an explicit expression for the Christoffel symbols, in terms of the metric tensor, in Section 3.17.[14]

Use of the Christoffel symbols (3.117) brings the covariant derivative from equation (3.115) to

$$\boldsymbol{\nabla}\mathbf{F} = \boldsymbol{e}^m \otimes [(\nabla_m F^a) \, \boldsymbol{e}_a + F^a \, \Gamma^d_{ma} \boldsymbol{e}_d] = \boldsymbol{e}^m \otimes \boldsymbol{e}_a \, (\nabla_m F^a + \Gamma^a_{md} \, F^d), \tag{3.118}$$

where the second equality follows from rearranging indices. For the final step we note that as a stand-alone operation,

$$\nabla_m F^a = \partial_m F^a, \tag{3.119}$$

[14]Equation (21.7) of *Griffies* (2004) defined the Christoffel symbols in terms of the partial derivatives (rather than covariant derivatives) of the basis vectors. This definition is not general and so is best avoided without the necessary caveats. See equation (5.44) of *Schutz* (2009).

since in this equation, F^a appears alone as an array of scalar functions, so that the covariant derivative appears as the partial derivative as it is acting on scalars. We are thus led to the $(1,1)$ representation of the covariant derivative of a first order tensor

$$\begin{aligned}\boldsymbol{\nabla}\mathbf{F} &= (\boldsymbol{\nabla}\mathbf{F})^a{}_m\, \boldsymbol{e}^m \otimes \boldsymbol{e}_a && (3.120a)\\ &= (\partial_m F^a + \Gamma^a_{md}\, F^d)\, \boldsymbol{e}^m \otimes \boldsymbol{e}_a && (3.120b)\\ &= F^a{}_{;m}\, \boldsymbol{e}^m \otimes \boldsymbol{e}_a. && (3.120c)\end{aligned}$$

The final equality introduced the semi-colon shorthand

$$F^a{}_{;m} = \partial_m F^a + \Gamma^a_{md}\, F^d, \tag{3.121}$$

which is used in many physics texts. Furthermore, it is common to introduce yet another symbol for the covariant derivative,

$$\boldsymbol{\nabla}\mathbf{F} = (\boldsymbol{\nabla}\mathbf{F})^a{}_m\, \boldsymbol{e}^m \otimes \boldsymbol{e}_a = F^a{}_{;m}\, \boldsymbol{e}^m \otimes \boldsymbol{e}_a = \nabla_m F^a\, \boldsymbol{e}^m \otimes \boldsymbol{e}_a. \tag{3.122}$$

The final expression is an abuse of notation since it is easily confused with equation (3.119). Confusion is reduced by agreeing to always write $\nabla_m F^a$ along with the basis one-forms and vectors, $\boldsymbol{e}^m \otimes \boldsymbol{e}_a$, thus emphasizing that $\nabla_m F^a$ is meant to represent the $(1,1)$ components to the covariant derivative. Unfortunately, that convention is often ignored, in which case we must accept the overloaded usage of $\nabla_m F^a$.

3.15.3 Covariant derivative acting on the $(0,1)$ representation

We here consider the $(0,1)$ representation of a first order tensor (i.e., a one-form), $\mathbf{F} = F_a\, \boldsymbol{e}^a$, and compute its covariant derivative

$$\boldsymbol{\nabla}\mathbf{F} \equiv (\boldsymbol{\nabla}\mathbf{F})_{am}\, \boldsymbol{e}^m \otimes \boldsymbol{e}^a = F_{a;m}\, \boldsymbol{e}^m \otimes \boldsymbol{e}^a = (\nabla_m F_a)\, \boldsymbol{e}^a \otimes \boldsymbol{e}^m. \tag{3.123}$$

Evidently, the covariant derivative of the $(0,1)$ representation yields a $(0,2)$ tensor. In Exercise 3.4 we show that the $(0,2)$ representation of the covariant derivative is given by

$$F_{a;m} = \nabla_m F_a = \partial_m F_a - \Gamma^c_{ma} F_c. \tag{3.124}$$

Results from equations (3.122) and (3.123) lead to the covariant derivative acting on a first order tensor

$$\boldsymbol{\nabla}\mathbf{F} = (\partial_m F^a + \Gamma^a_{md}\, F^d)\, \boldsymbol{e}^m \otimes \boldsymbol{e}_a = (\partial_m F_a - \Gamma^c_{ma} F_c)\, \boldsymbol{e}^a \otimes \boldsymbol{e}^m. \tag{3.125}$$

3.15.4 Christoffel symbols and the Levi-Civita connection

Recall from elementary calculus that the derivative of a function is computed by comparing the function at two points in space, dividing by the distance between those points, and taking the limit as the points get infinitesimally close. This operation is well defined for scalar fields on arbitrary manifolds. However, it is problematic for vectors since the vectors live on distinct tangent spaces and so cannot be directly compared (see Section 3.2). For example, how do we compare two vectors at distinct points on a sphere? To do so we must provide a method to move one vector to the position of the other before comparing. As seen through the above discussion of covariant derivative of a vector, the Levi-Civita connection, along with the corresponding

Christoffel symbols, provide the means to move vectors. Namely, they connect the two vectors by carrying information about how the basis vectors change in space. It is for this reason that some refer to the Levi-Civita connection as the metric connection, with the Christoffel symbols the associated connection coefficients.

The Christoffel symbols are coordinate dependent. For example, the Christoffel symbols all vanish in Euclidean space when using Cartesian coordinates, whereas they are nonzero with other coordinates. As discussed in Section 3.2, a tensor that vanishes in one coordinate system remains zero for all coordinate systems. We thus conclude that the Christoffel symbols are not components to a tensor. Rather, they carry information regarding the partial derivatives of the coordinate basis vectors and as such they are fundamentally tied to a chosen coordinate system.

3.16 Covariant derivative of the metric tensor

The metric is a second order tensor that can be represented either as $(0,2)$ or $(2,0)$ tensors

$$\mathbb{g} = \mathbb{g}_{ab}\, \boldsymbol{e}^a \otimes \boldsymbol{e}^b = \mathbb{g}^{ab}\, \boldsymbol{e}_a \otimes \boldsymbol{e}_b, \tag{3.126}$$

with $\mathbb{g}_{ab}$ typically referred to as "the metric" and $\mathbb{g}^{ab}$ the inverse metric. When written in Cartesian coordinates, the covariant derivative of components to the metric tensor for Euclidean space vanishes trivially,

$$\nabla\mathbb{g} = (\nabla_m \mathbb{g}_{ab})\, \boldsymbol{e}^m \otimes \boldsymbol{e}^a \otimes \boldsymbol{e}^b = (\partial_m \delta_{ab})\, \boldsymbol{e}^m \otimes \boldsymbol{e}^a \otimes \boldsymbol{e}^b = 0. \tag{3.127}$$

That is, the Cartesian representation of the Euclidean metric is the Kronecker symbol, δ_{ab}, and this is a constant tensor. Previous results establish the tensorial nature of the covariant derivative. Hence,

$$\nabla_m \mathbb{g}_{ab} = 0 \tag{3.128}$$

holds when representing the metric in any coordinates.

Equation (3.128) is called the metricity condition. It represents a self-consistency condition required for the manifolds considered in this book. In this manner, we say that the Levi-Civita connection is compatible with the metric, so that the metric has a zero covariant derivative. Importantly, this property holds only so long as the covariant derivative and the metric tensor are represented by the same coordinates. Namely, if we consider an alternative coordinate system to represent the covariant derivative, say $\nabla_{\overline{m}}$, then we generally have $\nabla_{\overline{m}} \mathbb{g}_{ab} \neq 0$.

3.17 Christoffel symbols in terms of the metric tensor

In Exercise 3.5 we derive an expression for the covariant derivative when acting on the $(0,2)$ representation of a second order tensor. When applied to the metric tensor, its vanishing covariant derivative (equation (3.127)) then leads to the identity

$$0 = \nabla_m \mathbb{g}_{ab} = \partial_m \mathbb{g}_{ab} - \Gamma^d_{ma}\, \mathbb{g}_{db} - \Gamma^d_{mb}\, \mathbb{g}_{ad}. \tag{3.129}$$

In Exercise 3.6 we solve this equation for the Christoffel symbols

$$\Gamma^c_{ab} = \frac{1}{2}\, \mathbb{g}^{cd}\, (\partial_b\, \mathbb{g}_{da} + \partial_a\, \mathbb{g}_{db} - \partial_d\, \mathbb{g}_{ab}). \tag{3.130}$$

This expression exhibits the symmetry property of the lower two indices on the Christoffel symbols

$$\Gamma^{c}_{ab} = \Gamma^{c}_{ba}, \tag{3.131}$$

which was a property assumed when deriving equation (3.130). This symmetry holds for spaces with zero torsion, such as the Euclidean space.[15]

3.17.1 Christoffel symbols and the transformation matrix

The identity (3.130) provides a very useful expression for the Christoffel symbols in terms of the metric tensor. In Exercise 3.8 we provide yet another expression in terms of the transformation matrix between Cartesian coordinates and arbitrary coordinates. Namely, let the unbarred coordinates be Cartesian and the barred coordinates be arbitrary. Since the Christoffel symbols vanish for Euclidean space as represented with Cartesian coordinates, we find the expression (again, this equation is derived in Exercise 3.8)

$$\Gamma^{\overline{a}}_{\overline{b}\overline{d}} = \Lambda^{\overline{a}}{}_{a}\, \partial_{\overline{b}}\, \Lambda^{a}{}_{\overline{d}}. \tag{3.132}$$

This expression proves of use in Section 3.20.1 for expressing the covariant divergence of a symmetric second order tensor.

3.17.2 A remark on torsion

As an aside, we remark on the notation used for the Christoffel symbols. For that purpose, introduce the torsion tensor,[16]

$$\mathcal{T}_{ab}{}^{c} = \Gamma^{c}_{ab} - \Gamma^{c}_{ba}. \tag{3.133}$$

For Euclidean space, or any smooth manifold embedded in Eucldean space, the torsion tensor vanishes identically. That is, regardless the coordinates, each element of the torsion tensor is zero, $\mathcal{T}_{ab}{}^{c} = 0$. Having $\mathcal{T}_{ab}{}^{c} = 0$ for all coordinates assures us that the torsion is indeed a tensor. As a tensor we make use of the tensor notation with the upstairs c on the torsion tensor displaced to the right, $\mathcal{T}_{ab}{}^{c}$. In contrast, the non-tensorial Christoffel symbols have the c vertically aligned and so not displaced.

3.18 Covariant divergence of a $(1,0)$ tensor

The covariant divergence to the $(1,0)$ components of a first order tensor (i.e., a vector) results in a scalar

$$\boldsymbol{\nabla} \cdot \mathbf{F} = \nabla_{a} F^{a} = \partial_{a} F^{a} + \Gamma^{a}_{ab} F^{b}. \tag{3.134}$$

We now bring this expression into a form more convenient for practical calculations.

3.18.1 Contraction of the Christoffel symbols

Expression (3.130) for the Christoffel symbols yields the contraction

$$\Gamma^{a}_{ab} = \frac{1}{2}\, \mathbb{g}^{ad} (\partial_{b}\, \mathbb{g}_{da} + \partial_{a}\, \mathbb{g}_{db} - \partial_{d}\, \mathbb{g}_{ab}) = \frac{1}{2}\, \mathbb{g}^{ad}\, \partial_{b}\, \mathbb{g}_{ad} \tag{3.135}$$

[15] See *Tromp* (2025b) for more on torsion and its uses in gravity and continuum mechanics.

[16] The expression (3.133) for the torsion assumes the coordinate basis is holonomic or coordinate based, If we instead use an anholonomic basis, then an extra term appears as discussed in Section 7.1.5 of *Tromp* (2025b).

where symmetry of both the metric tensor and its inverse was used.

3.18.2 Exponential of the determinant

For the matrix representation of a symmetric positive definite tensor, such as the metric tensor, we can write

$$\begin{aligned}
\det(A) &= e^{\ln\det(A)} && \text{identity} && (3.136a)\\
&= e^{\ln(\Pi_i \Lambda_i)} && \text{determinant related to product of eigenvalues} && (3.136b)\\
&= e^{\Sigma_i \ln \Lambda_i} && \text{identity} && (3.136c)\\
&= e^{\mathrm{Tr}(\ln A)} && \text{sum of eigenvalues related to trace of matrix.} && (3.136d)
\end{aligned}$$

Each of these identities is trivial to verify using a set of coordinates in which the matrix is diagonal. For any symmetric and positive definite matrix, such a set of coordinates always exists, in which case

$$\partial_c \ln\det(A) = \partial_c[\mathrm{Tr}(\ln A)] = \mathrm{Tr}(\partial_c \ln A) = \mathrm{Tr}(A^{-1}\partial_c A). \tag{3.137}$$

With A now set equal to the metric tensor, $\mathbb{g}_{ab}$, this result yields

$$\partial_c \ln\det(\mathbb{g}_{ab}) = \mathbb{g}^{ab}\,\partial_c\,\mathbb{g}_{ab}, \tag{3.138}$$

which in turn yields for the contracted Christoffel symbol

$$\Gamma^a_{ac} = \partial_c \ln(\sqrt{\det(\mathbb{g}_{ab})}) = \partial_c \ln \mathsf{g}. \tag{3.139}$$

This result brings the covariant divergence to the form

$$\boldsymbol{\nabla}\cdot\mathbf{F} = \nabla_a\,F^a = \partial_a F^a + F^a\,\partial_a \ln \mathsf{g} = \mathsf{g}^{-1}\,\partial_a(\mathsf{g}\,F^a). \tag{3.140}$$

This is a very convenient result. In particular, it only requires partial derivatives in the chosen coordinate system, with all the coordinate dependent properties summarized by $\mathsf{g} = \sqrt{\det(\mathbb{g}_{ab})}$.

3.19 Covariant Laplacian of a scalar

Making use of equation (3.140) with

$$F^a = \mathbb{g}^{ab}\,\partial_b\psi \tag{3.141}$$

leads to the covariant Laplacian of a scalar field

$$\nabla_a\,(\mathbb{g}^{ab}\,\partial_b\,\psi) = \frac{1}{\sqrt{\det(\mathbb{g}_{ab})}}\,\partial_a\,[\sqrt{\det(\mathbb{g}_{ab})}\,\mathbb{g}^{ab}\,\partial_b\psi] = \mathsf{g}^{-1}\,\partial_a\,(\mathsf{g}\;\mathbb{g}^{ab}\,\partial_b\psi). \tag{3.142}$$

This expression is sometimes referred to as the Laplace-Beltrami operator.

3.20 Covariant divergence of a second order tensor

Following the methods used for deriving the covariant derivative of a vector in Section 3.18, and for the covariant derivative of a $(0,2)$ tensor in Exercise 3.5, we find the covariant divergence of the $(2,0)$ (sharp) representation of a second order tensor

$$\nabla_a T^{ab} = \partial_a T^{ab} + \Gamma^b_{ad} T^{da} + \Gamma^a_{ad} T^{bd}. \tag{3.143}$$

Following our discussion of the irreducible parts of a second order tensor in Section 1.10, we are motivated to split the $(2,0)$ tensor components into their symmetric and antisymmetric parts

$$S^{ab} = (T^{ab} + T^{ba})/2 \quad \text{and} \quad R^{ab} = (T^{ab} - T^{ba})/2. \tag{3.144}$$

3.20.1 Covariant divergence of a symmetric tensor

The covariant divergence of the symmetric tensor is

$$\nabla_a S^{ab} = \mathsf{g}^{-1} \partial_a (\mathsf{g}\, S^{ab}) + \Gamma^b_{ad} S^{ad}, \tag{3.145}$$

where we used equation (3.139) for the contraction of the Christoffel symbol, Γ^a_{ad}. If the space is Euclidean and the coordinates are Cartesian, then $\Gamma^a_{ad} = 0$ and $\mathsf{g} = 1$. Now let use introduce general coordinates denoted by the overbar, and use equation (3.132) for the Christoffel symbols so that

$$\nabla_{\bar{a}} S^{\bar{a}\bar{b}} = \mathsf{g}^{-1} \partial_{\bar{a}} (\mathsf{g}\, S^{\bar{a}\bar{b}}) + \Gamma^{\bar{b}}_{\bar{a}\bar{d}} S^{\bar{a}\bar{d}} = \mathsf{g}^{-1} \partial_{\bar{a}} (\mathsf{g}\, S^{\bar{a}\bar{b}}) + \Lambda^{\bar{b}}{}_{a} \, (\partial_{\bar{a}} \Lambda^{a}{}_{\bar{d}}) \, S^{\bar{a}\bar{d}}. \tag{3.146}$$

Contracting both sides with $\Lambda^{d}{}_{\bar{b}}$ leads to

$$\Lambda^{d}{}_{\bar{b}} \nabla_{\bar{a}} S^{\bar{a}\bar{b}} = \Lambda^{d}{}_{\bar{b}}\, \mathsf{g}^{-1} \partial_{\bar{a}} (\mathsf{g}\, S^{\bar{a}\bar{b}}) + \delta^{d}{}_{a} \, (\partial_{\bar{a}} \Lambda^{a}{}_{\bar{d}}) \, S^{\bar{a}\bar{d}}, \tag{3.147}$$

with an index relabel giving the rather compact expression

$$\Lambda^{d}{}_{\bar{b}} \nabla_{\bar{a}} S^{\bar{a}\bar{b}} = \mathsf{g}^{-1} \partial_{\bar{a}} (\mathsf{g}\, \Lambda^{d}{}_{\bar{b}} S^{\bar{a}\bar{b}}). \tag{3.148}$$

Again, this formula holds for the case with the unbarred labels representing Cartesian coordinates, whereas the barred coordinates are arbitrary.

3.20.2 Covariant divergence of an anti-symmetric tensor

For the anti-symmetric tensor, the $\Gamma^b_{ad} R^{ad}$ term in equation (3.143) drops out since Γ^b_{ad} is symmetric on the indices a, d, whereas R^{ad} is anti-symmetric. We are thus led to the covariant divergence of an anti-symmetric tensor

$$\nabla_a R^{ab} = \mathsf{g}^{-1} \, \partial_a (\mathsf{g}\, R^{ab}). \tag{3.149}$$

This relation is analogous to the covariant divergence of a vector given by equation (3.140). In particular, the vector components

$$F^b \equiv \nabla_a R^{ab} \tag{3.150}$$

are divergence-free since

$$\nabla_b F^b = \mathsf{g}^{-1} \, \partial_b (\mathsf{g}\, F^b) = \mathsf{g}^{-1} \, \partial_b (\mathsf{g} \, \nabla_a R^{ab}) = \mathsf{g}^{-1} \, \partial_b \partial_a (\mathsf{g}\, R^{ab}) = 0, \tag{3.151}$$

which follows from anti-symmetry of the components R^{ab} under interchange of a, b and symmetry of $\partial_b \partial_a$.

3.21 Covariant curl of a vector

The Levi-Civita tensor from Section 3.12

$$\varepsilon_{abc} = \sqrt{\det(\mathbb{g}_{ab})}\, \epsilon_{abc} = \mathsf{g}\, \epsilon_{abc}, \tag{3.152}$$

as well as its inverse

$$\varepsilon^{abc} = (1/\sqrt{\det(\mathbb{g}_{ab})})\, \epsilon^{abc} = (1/\mathsf{g})\, \epsilon^{abc}, \tag{3.153}$$

provide the pieces needed to generalize the curl operation. For this purpose we define the covariant curl according to the coordinate invariant expression

$$\text{curl}(\mathbf{F}) = \boldsymbol{e}_a\, \varepsilon^{abc}\, (\nabla_b F_c) = \boldsymbol{e}^a\, \varepsilon_{abc}\, (\nabla^b F^c). \tag{3.154}$$

This expression simplifies by making use of equation (3.124) for the covariant derivative, $\nabla_b F_c = \partial_b F_c - \Gamma^a_{cb}\, F_a$. Conveniently, the contraction $\varepsilon^{abc}\, \Gamma^a_{cb}$ vanishes identically since $\varepsilon^{abc} = -\varepsilon^{acb}$ whereas $\Gamma^a_{cb} = \Gamma^a_{bc}$. Hence, we are left with a general expression for the covariant curl that involves just the partial derivatives

$$\text{curl}(\mathbf{F}) = \boldsymbol{e}_a\, \varepsilon^{abc}\, \partial_b F_c = \boldsymbol{e}_a\, \varepsilon^{abc}\, \partial_b(\mathbb{g}_{cd} F^d) = \boldsymbol{e}_a\, \mathsf{g}^{-1}\, \epsilon^{abc}\, \partial_b(\mathbb{g}_{cd} F^d), \tag{3.155}$$

where the second equality made use of the identity $F_c = \mathbb{g}_{cd} F^d$.

3.22 Gauss's divergence theorem

The integral theorems from Cartesian vector analysis transform in a straightforward manner to arbitrary coordinates in arbitrary smooth and oriented spaces. An easy way to prove the theorems is to invoke the ideas of general coordinate invariance from Section 3.2, in which the integral theorems are written in a tensorially proper manner with partial derivatives changed to covariant derivatives. The divergence theorem offers a particularly simple example, and one that we make use of throughout this book. For this purpose, make use of the volume element (3.75)

$$\mathrm{d}V = \sqrt{\det(\mathbb{g}_{ab})}\, \mathrm{d}\xi^1\, \mathrm{d}\xi^2\, \mathrm{d}\xi^3, \tag{3.156}$$

multiplied by the covariant divergence (3.140). Hence, the volume integral of the divergence is given by

$$\int_{\mathcal{R}} \boldsymbol{\nabla} \cdot \mathbf{F}\, \mathrm{d}V = \int_{\mathcal{R}} (\nabla_a\, F^a)\, \mathrm{d}V = \int_{\mathcal{R}} \partial_a [\sqrt{\det(\mathbb{g}_{ab})}\, F^a]\, \mathrm{d}\xi^1\, \mathrm{d}\xi^2\, \mathrm{d}\xi^3 = \oint_{\partial\mathcal{R}} F^a\, \hat{n}_a\, \mathrm{d}\mathcal{S}. \tag{3.157}$$

In this equation, $\hat{\boldsymbol{n}}$ is the outward *normal one-form* for the boundary, $\partial\mathcal{R}$, with $\mathrm{d}\mathcal{S}$ the invariant area element on the boundary, and $\hat{n}_a$ the covariant components of the outward normal.

3.23 Stokes' curl theorem

The Cartesian form of Stokes' Theorem from Section 2.7 is generalized in a manner similar to the divergence theorem

$$\oint_{\partial\mathcal{S}} \mathbf{F}\cdot \mathrm{d}\mathbf{x} = \int_{\mathcal{S}} \mathrm{curl}(\mathbf{F})\cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}, \tag{3.158}$$

where $\mathrm{d}\boldsymbol{x}$ is the vector line element along the path and $\partial\mathcal{S}$ is the closed path defining the boundary to a simply connected two-dimensional surface, $\mathcal{S}$. For the circulation on the left hand side we have a particular coordinate realization

$$\mathbf{F}\cdot \mathrm{d}\mathbf{x} = F^{a}\,\boldsymbol{e}_{a}\cdot\boldsymbol{e}_{b}\,\mathrm{d}x^{b} = F_{b}\,\mathrm{d}x^{b} = F_{\overline{b}}\,\mathrm{d}\xi^{\overline{b}}. \tag{3.159}$$

For the curl on the right hand side we have

$$\mathrm{curl}(\mathbf{F})\cdot\hat{\boldsymbol{n}} = \varepsilon^{abc}\,(\partial_{b}F_{c})\,\boldsymbol{e}_{a}\cdot\hat{\boldsymbol{n}} = \varepsilon^{abc}\,(\partial_{b}F_{c})\,\hat{n}_{a} = \varepsilon^{\overline{a}\overline{b}\overline{c}}\,(\partial_{\overline{b}}F_{\overline{c}})\,\hat{n}_{\overline{a}}, \tag{3.160}$$

thus leading to the expression of Stokes' theorem in arbitrary coordinates

$$\oint_{\partial\mathcal{S}} F_{\overline{b}}\,\mathrm{d}\xi^{\overline{b}} = \int_{\mathcal{S}} \varepsilon^{\overline{a}\overline{b}\overline{c}}\,(\partial_{\overline{b}}F_{\overline{c}})\,\hat{n}_{\overline{a}}\,\mathrm{d}\mathcal{S}. \tag{3.161}$$

3.24 Exercises

EXERCISE 3.1: OBLIQUE COORDINATES

In this exercise we derive some properties of oceanic oblique coordinates for the x-z plane as illustrated in Figure 3.1 and specified by the basis vectors

$$\boldsymbol{e}_{\overline{1}} = \boldsymbol{e}_{1}\cos\varphi + \boldsymbol{e}_{3}\sin\varphi = \hat{\boldsymbol{x}}\cos\varphi + \hat{\boldsymbol{z}}\sin\varphi \qquad \text{and} \qquad \boldsymbol{e}_{\overline{3}} = \boldsymbol{e}_{3} = \hat{\boldsymbol{z}}. \tag{3.162}$$

The oblique coordinate basis vectors, $\boldsymbol{e}_{\overline{a}}$, are orthogonal when the angle $\varphi = 0,\pi$; otherwise they are non-orthogonal. Also note that if $\varphi = \pm\pi/2$ then $\boldsymbol{e}_{\overline{1}} = \pm\boldsymbol{e}_{\overline{3}}$, in which case the vectors no longer form a basis for the x-z plane. So in this exercise we assume $-\pi/2 < \varphi < \pi/2$. We also ignore the 2 direction and just work within the x-z plane. Hence, tensor indices carry values 1 and 3, with 2 ignored. Note that oblique coordinate offer a pedagogical step towards the slightly more complex case of generalized vertical coordinate studied in VOLUME 3.

(a) Determine the transformation matrix, $\Lambda^{a}{}_{\overline{a}}$, and its inverse, $\Lambda^{\overline{a}}{}_{a}$.
(b) Determine the basis one-forms, $\boldsymbol{e}^{\overline{a}}$, for the oblique coordinates. Hint: Make use of the transformation equation $\boldsymbol{e}^{\overline{a}} = \Lambda^{\overline{a}}{}_{a}\,\boldsymbol{e}^{a}$, with $\boldsymbol{e}^{a}$ the Cartesian basis one-forms. Also check that the bi-orthogonality relation (3.51) is satisfied: $\boldsymbol{e}_{\overline{a}}\cdot\boldsymbol{e}^{\overline{b}} = \delta^{\overline{b}}{}_{\overline{a}}$.
(c) Determine the expression for the contravariant components, $P^{\overline{a}}$, of a vector. Hint: use the transformation equation $P^{\overline{a}} = \Lambda^{\overline{a}}{}_{a}\,P^{a}$.
(d) Determine the covariant components via $P_{\overline{a}} = \Lambda^{a}{}_{\overline{a}}\,P_{a}$.
(e) Determine $(0,2)$ representation of the metric tensor using oblique coordinate. Hint: make use of the transformation equation

$$g_{\overline{a}\,\overline{b}} = \delta_{ab}\,\Lambda^{a}{}_{\overline{a}}\,\Lambda^{b}{}_{\overline{b}}, \tag{3.163}$$

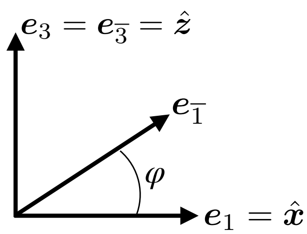


FIGURE 3.4: Oblique basis vectors for the x-z plane where $\boldsymbol{e}_{\overline{1}} = \boldsymbol{e}_1 \cos\varphi + \boldsymbol{e}_3 \sin\varphi$ and $\boldsymbol{e}_{\overline{3}} = \boldsymbol{e}_3$, with $\boldsymbol{e}_1 = \hat{\boldsymbol{x}}$ and $\boldsymbol{e}_3 = \hat{\boldsymbol{z}}$. In Exercise 3.1 we examine properties of these coordinates. Note that the oblique basis vectors $\boldsymbol{e}_{\overline{1}}$ and $\boldsymbol{e}_{\overline{3}}$ are related (but not identical) to those used for the mathematics of generalized vertical coordinates in VOLUME 3.

where $\mathbb{g}_{ab} = \delta_{ab}$ is the representation of the metric using Cartesian coordinates.

(f) Determine the $(2,0)$ representation of the metric tensor (i.e., the inverse metric tensor) using oblique coordinates.

(g) Expand the expression for the squared magnitude of a vector using oblique coordinates, and explicitly verify the form invariance for the scalar product,

$$P_{\overline{b}} \, P^{\overline{b}} = P_a \, P^a \tag{3.164}$$

EXERCISE 3.2: COORDINATE TRANSFORMATION FOR OBLIQUE COORDINATES

In Exercise 3.1 we only made use of the basis vectors, $\boldsymbol{e}_{\overline{a}}$, and never exposed the coordinate transformation. That approach is sufficient for some purposes. But we generally want the coordinate transformation, if indeed there is one. What is it for the oblique coordinates?

EXERCISE 3.3: PRODUCT OF TWO JACOBIANS

Consider two coordinate transformations that are each 1-to-1 and invertible. Summarize these transformations as

$$\xi^a \xrightarrow{\Lambda^a{}_{a'}} \xi^{a'} \xrightarrow{\Lambda^{a'}{}_{\overline{a}}} \xi^{\overline{a}}, \tag{3.165}$$

with the corresponding transformation matrix elements

$$\Lambda^a{}_{a'} = \frac{\partial \xi^a}{\partial \xi^{a'}} \quad \text{and} \quad \Lambda^{a'}{}_{\overline{a}} = \frac{\partial \xi^{a'}}{\xi^{\overline{a}}}. \tag{3.166}$$

Prove the determinant identity

$$\det(\Lambda^a{}_{\overline{a}}) = \det(\Lambda^a{}_{a'}) \det(\Lambda^{a'}{}_{\overline{a}}), \tag{3.167}$$

where

$$\xi^a \xrightarrow{\Lambda^a{}_{\overline{a}}} \xi^{\overline{a}}, \tag{3.168}$$

is the coordinate transformation going directly from ξ^a to $\xi^{\overline{a}}$.

EXERCISE 3.4: COVARIANT DERIVATIVE OF A $(0,1)$ TENSOR

Prove equation (3.124) for the covariant derivative of the $(0,1)$ representation of a tensor

$$F_{a;m} = \nabla_m F_a = \partial_m F_a - \Gamma^c_{ma} F_c. \tag{3.169}$$

Hint: note that the product of a one-form and a vector is a scalar so that

$$\nabla_m(\mathbf{E}\cdot\mathbf{F}) = \partial_m\,(E_a\,F^a). \tag{3.170}$$

Expand the right hand side and rearrange, making use of equation (3.121) for the covariant derivative acting on the $(1,0)$ representation.

EXERCISE 3.5: COVARIANT DERIVATIVE OF A $(0,2)$ TENSOR
Consider a second order tensor, $\mathbf{T}$. Show that its $(0,2)$ representation has the following expression for the covariant derivative

$$\nabla_m T_{ab} = T_{ab;m} = \partial_m T_{ab} - \Gamma^d_{ma}\,T_{db} - \Gamma^d_{mb}\,T_{ad}. \tag{3.171}$$

Hint: Introduce a $(1,0)$ tensor, $\mathbf{F}$, and define the scalar $T_{ab}\,F^a\,F^b$. The covariant derivative of this scalar is given by the partial derivative. Perform this derivative, use the product rule, and use the known covariant derivative of F^a to deduce $\nabla_m T_{ab}$.

EXERCISE 3.6: CHRISTOFFEL SYMBOLS IN TERMS OF THE METRIC TENSOR
Derive equation (3.130) that expresses the Christoffel symbols in terms of derivatives of the metric tensor

$$\Gamma^c_{ab} = \frac{1}{2}\,\mathbb{g}^{cd}\,(\partial_b\,\mathbb{g}_{da} + \partial_a\,\mathbb{g}_{db} - \partial_d\,\mathbb{g}_{ab}). \tag{3.172}$$

Hint: write the metricity equation (3.129) along with equation (3.129) with a and m interchanged, and then again equation (3.129) with b and m interchanged. Then rearrange and contract with the inverse metric tensor. Assume zero torsion, as we do throughout this book.

EXERCISE 3.7: METRICITY CONDITION FOR THE CHRISTOFFEL SYMBOLS
Equation (3.37) provides a coordinate expression for the metric tensor in terms of the scalar product of basis vectors

$$\mathbb{g}_{ab} = \boldsymbol{e}_a\cdot\boldsymbol{e}_b. \tag{3.173}$$

This expression holds for the Euclidean space used in this book. We also know from Section 3.16 that the covariant derivative of the metric tensor vanishes

$$\nabla_m\mathbb{g}_{ab} = 0, \tag{3.174}$$

which is known as the metricity condition. Derive the following implication of metricity in terms of the Christoffel symbols

$$\mathbb{g}_{bd}\,\Gamma^d_{ma} + \mathbb{g}_{ad}\,\Gamma^d_{mb} = 0. \tag{3.175}$$

EXERCISE 3.8: TRANSFORMATION OF THE CHRISTOFFEL SYMBOLS
In this exercise we establish the transformation rules for the Christoffel symbol, and then identify a useful implication of that transformation.

(a) We noted in Section 3.15 that the Christoffel symbols do not transform as components to a tensor. Derive the following expression for the transformation of the Christoffel symbols

$$\Gamma^{\overline{a}}_{\overline{b}\overline{d}} = \Lambda^a{}_{\overline{d}}\,\Lambda^b{}_{\overline{b}}\,\Lambda^{\overline{a}}{}_d\,\Gamma^d_{ba} - \Lambda^a{}_{\overline{d}}\,\partial_{\overline{b}}\,\Lambda^{\overline{a}}{}_a. \tag{3.176}$$

It is because of the second term that the Christoffel symbols do not transform as components to a tensor. Hint: start by noting that the covariant derivative of a $(1,0)$ representation of a tensor defines a $(1,1)$ representation of a tensor, so that its components transform according to

$$\nabla_{\overline{b}} F^{\overline{a}} = \Lambda^{b}{}_{\overline{b}} \Lambda^{\overline{a}}{}_{a} \, \nabla_{b} F^{a}. \tag{3.177}$$

Expand the left hand and right hand sides and rearrange. Note that this exercise requires some adept use of index gymnastics.

(b) Now choose the unbarred coordinates to be Cartesian, and recall we are concerned with Euclidean space. Hence, the Christoffel symbols, Γ^{d}_{ba}, all vanish. Show then that the transformation rule (3.176) results in

$$\Gamma^{\overline{a}}_{\overline{b}\overline{d}} = -\Lambda^{a}{}_{\overline{d}} \, \partial_{\overline{b}} \, \Lambda^{\overline{a}}{}_{a} = \Lambda^{\overline{a}}{}_{a} \, \partial_{\overline{b}} \, \Lambda^{a}{}_{\overline{d}}. \tag{3.178}$$

Chapter 4

Orthogonal coordinates

In this chapter we provide a handbook of commonly used equations using Cartesian coordinates, cylindrical-polar coordinates, and spherical coordinates. In so doing, we provide specific examples of the formulations provided in Chapter 3, thus exposing common formula arising in continuum mechanics. Each of these coordinates specifies a fixed point in space. We refer to these as Eulerian coordinates, as they connect to the fixed or Eulerian reference frame used in fluid kinematics from volume 1. As a result, the components to the metric tensor are independent of time. These Eulerian coordinates are distinguished from time dependent coordinates, such as the generalized vertical coordinate studied in volume 4, as well as the Lagrangian coordinates studied in volume 1.

reader's guide to this chapter

Equations from this chapter are referred to throughout this book, and they are commonly encountered in numerical model applications.

4.1 Coordinate and non-coordinate bases

When using orthogonal curvilinear coordinates to describe Euclidean space, we commonly make use of non-coordinate basis vectors (also called anholonomic basis). These basis vectors are orthonormal, which is a convenient property that allows for such coordinates to be treated very much like Cartesian tensors. In this section we summarize non-coordinate bases and point to their pros and cons. In section 4.5 we further a discussion of non-coordinate bases by considering basic properties of generalized orthogonal coordinates.

4.1.1 The case of polar coordinates on the plane

To illustrate the notion of non-coordinate basis vectors, consider polar coordinates for the plane. Following the material presented in Section 4.3, we write the polar coordinates $(\xi^{\overline{1}}, \xi^{\overline{2}}) = (r, \vartheta)$, which are related to Cartesian coordinates, $(\xi^1, \xi^2) = (x, y)$, through the coordinate transformation

$$\xi^1 = \xi^{\overline{1}} \cos \xi^{\overline{2}} \quad \text{and} \quad \xi^2 = \xi^{\overline{1}} \sin \xi^{\overline{2}}. \tag{4.1}$$

The corresponding basis vectors, $\boldsymbol{e}_{\overline{a}}$, and basis one-forms, $\boldsymbol{e}^{\overline{a}}$, are found through the chain rule

$$\boldsymbol{e}_{\overline{a}} = (\partial \xi^a / \partial \xi^{\overline{a}}) \, \hat{\boldsymbol{e}}_a = \Lambda^a{}_{\overline{a}} \, \hat{\boldsymbol{e}}_a \quad \text{and} \quad \boldsymbol{e}^{\overline{a}} = (\partial \xi^{\overline{a}} / \partial \xi^a) \, \hat{\boldsymbol{e}}^a = \Lambda^{\overline{a}}{}_a \, \hat{\boldsymbol{e}}^a, \tag{4.2}$$

with the Cartesian basis given by the orthonormal directions

$$\hat{\boldsymbol{e}}_a = \hat{\boldsymbol{e}}^a = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}), \tag{4.3}$$

as discussed in Chapter 1. For consistency with the notation to follow, we move the hat sitting above the Cartesian basis vector onto the tensor index, so that we write the Cartesian basis

$$\boldsymbol{e}_{\hat{a}} = \boldsymbol{e}^{\hat{a}} = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}), \tag{4.4}$$

and the corresponding transformation to the polar coordinate basis

$$\boldsymbol{e}_{\overline{a}} = \Lambda^{\hat{a}}{}_{\overline{a}}\, \boldsymbol{e}_{\hat{a}} \quad \text{and} \quad \boldsymbol{e}^{\overline{a}} = \Lambda^{\overline{a}}{}_{\hat{a}}\, \hat{\boldsymbol{e}}^{\hat{a}}. \tag{4.5}$$

The reason we use this notation is to be consistent with the following set of basis vectors and basis one-forms that we refer to as a non-coordinate orthonormal polar basis

$$\boldsymbol{e}_{\hat{\overline{1}}} = \boldsymbol{e}^{\hat{\overline{1}}} = \hat{\boldsymbol{r}} \quad \text{and} \quad \boldsymbol{e}_{\hat{\overline{2}}} = \boldsymbol{e}^{\hat{\overline{2}}} = \hat{\boldsymbol{\phi}}. \tag{4.6}$$

This basis is appealing due to its orthonormal property and given the identity between the vector and one-form bases. Correspondingly, the metric tensor and its inverse are represented by the Kronecker delta

$$\mathbb{g}_{\hat{\overline{a}}\hat{\overline{b}}} = \boldsymbol{e}_{\hat{\overline{a}}} \cdot \boldsymbol{e}_{\hat{\overline{b}}} = \delta_{\hat{\overline{a}}\hat{\overline{b}}} \quad \text{and} \quad \mathbb{g}^{\hat{\overline{a}}\hat{\overline{b}}} = \boldsymbol{e}^{\hat{\overline{a}}} \cdot \boldsymbol{e}^{\hat{\overline{b}}} = \delta^{\hat{\overline{a}}\hat{\overline{b}}}, \tag{4.7}$$

which accords with their representation using the orthonormal Cartesian basis, $\boldsymbol{e}_{\hat{a}}$, in which

$$\mathbb{g}_{\hat{a}\hat{b}} = \boldsymbol{e}_{\hat{a}} \cdot \boldsymbol{e}_{\hat{b}} = \delta_{\hat{a}\hat{b}} \quad \text{and} \quad \mathbb{g}^{\hat{a}\hat{b}} = \boldsymbol{e}^{\hat{a}} \cdot \boldsymbol{e}^{\hat{b}} = \delta^{\hat{a}\hat{b}}. \tag{4.8}$$

In this way, tensor algebra using the orthonormal polar bases is very much like that for Cartesian tensors.

However, there is a price to pay. Namely, there are no coordinates, $\xi^{\hat{\overline{a}}}$, that correspond to the orthonormal bases. We thus say that the orthonormal basis, $(\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}})$, is a non-coordinate basis for the plane. Without coordinates, there is no transformation matrix, $\Lambda^{\hat{\overline{a}}}{}_{a}$. Hence, there is no systematic process for transforming coordinate components of tensors. In fact, for generalized orthogonal coordinates discussed in Section 4.5, we provide a systematic transformation in Section 4.5.4, which is indeed non-tensorial.

Non-coordinate bases are commonly used in this book for spherical and cylindrical-polar coordinates, as well as generalized orthogonal coordinates. This usage motivates us to expose their elements in this section, with further examples given in Section 4.5 for generalized orthogonal coordinates.

4.1.2 Proof there are no coordinates for $\boldsymbol{e}_{\hat{\overline{a}}}$

To prove there are no coordinates, $\xi^{\hat{\overline{a}}}$, we follow a proof by contradiction. Namely, if there are coordinates then there must be a tranformation matrix, $\Lambda^{\hat{\overline{a}}}{}_{\hat{a}}$, that transforms from the Cartesian basis to the orthonormal polar basis

$$\boldsymbol{e}_{\hat{\overline{a}}} \stackrel{?}{=} \Lambda^{\hat{a}}{}_{\hat{\overline{a}}}\, \boldsymbol{e}_{\hat{a}} \quad \text{and} \quad \boldsymbol{e}^{\hat{\overline{a}}} \stackrel{?}{=} \Lambda^{\hat{\overline{a}}}{}_{\hat{a}}\, \boldsymbol{e}^{\hat{a}}. \tag{4.9}$$

So our job is to show that there is, in fact, no transformation matrix.

For orientation, recall we are considering three sets of basis vectors: the orthonormal Cartesian basis

$$\boldsymbol{e}_{\hat{a}} = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}), \tag{4.10}$$

the unnormalized polar coordinate basis

$$\boldsymbol{e}_{\overline{a}} = (\boldsymbol{e}_{\overline{1}}, \boldsymbol{e}_{\overline{2}}) = (\boldsymbol{e}_{r}, \boldsymbol{e}_{\vartheta}) = (\hat{\boldsymbol{r}}, r\, \hat{\boldsymbol{\vartheta}}), \tag{4.11}$$

and the orthonormal polar basis

$$\boldsymbol{e}_{\hat{\hat{a}}} = (\boldsymbol{e}_{\hat{\hat{1}}}, \boldsymbol{e}_{\hat{\hat{2}}}) = (\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}}), \tag{4.12}$$

with the relation between the orthonormal polar basis and orthonormal Cartesian basis given by

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}} \cos\vartheta + \hat{\boldsymbol{y}} \sin\vartheta \quad \text{and} \quad \hat{\boldsymbol{\vartheta}} = -\hat{\boldsymbol{x}} \sin\vartheta + \hat{\boldsymbol{y}} \cos\vartheta. \tag{4.13}$$

In Section 4.3 we work through the relationship between Cartesian coordinates and polar coordinates. In particular, we know that the polar basis vectors are related to the orthonormal Cartesian basis vectors via the transformation matrix

$$\boldsymbol{e}_{\overline{a}} = \Lambda^{\hat{a}}{}_{\overline{a}}\, \boldsymbol{e}_{\hat{a}} = (\partial \xi^{\hat{a}} / \partial \xi^{\overline{a}})\, \boldsymbol{e}_{\hat{a}}. \tag{4.14}$$

The question is whether there is a set of coordinates, $\xi^{\hat{\hat{a}}}$, that allows us to compute the orthonormal polar basis vectors according to

$$\boldsymbol{e}_{\hat{\hat{a}}} = \Lambda^{\hat{a}}{}_{\hat{\hat{a}}}\, \boldsymbol{e}_{\hat{a}} = (\partial \xi^{\hat{a}} / \partial \xi^{\hat{\hat{a}}})\, \boldsymbol{e}_{\hat{a}}. \tag{4.15}$$

It turns out to be simpler to work with the one-form basis to examine the equivalent question for one-forms. Namely, consider

$$\boldsymbol{e}^{\hat{a}} = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}) \quad \text{and} \quad \boldsymbol{e}^{\overline{a}} = (\hat{\boldsymbol{r}}, r^{-1}\, \hat{\boldsymbol{\vartheta}}) \quad \text{and} \quad \boldsymbol{e}^{\hat{\hat{a}}} = (\boldsymbol{e}^{\hat{\hat{1}}}, \boldsymbol{e}^{\hat{\hat{2}}}) = (\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}}), \tag{4.16}$$

in which we ask whether there is a transformation matrix rendering

$$\boldsymbol{e}^{\hat{\hat{a}}} \stackrel{?}{=} \Lambda^{\hat{\hat{a}}}{}_{\hat{a}}\, \boldsymbol{e}^{\hat{a}} = \Lambda^{\hat{\hat{a}}}{}_{1}\, \hat{\boldsymbol{x}} + \Lambda^{\hat{\hat{a}}}{}_{2}\, \hat{\boldsymbol{y}}. \tag{4.17}$$

With $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\vartheta}}$ given by equation (4.13), a set of coordinates exists if they satisfy

$$\partial \xi^{\hat{\hat{1}}} / \partial x = \cos\vartheta \qquad \text{and} \quad \partial \xi^{\hat{\hat{1}}} / \partial y = \sin\vartheta \tag{4.18a}$$

$$\partial \xi^{\hat{\hat{2}}} / \partial x = -\sin\vartheta \quad \text{and} \quad \partial \xi^{\hat{\hat{2}}} / \partial y = \cos\vartheta. \tag{4.18b}$$

For this set of equations to hold, we must have equal mixed second partial derivatives. Focusing on $\xi^{\hat{\hat{2}}}$ we must have the mixed partial derivatives equal

$$\frac{\partial}{\partial x} \frac{\partial \xi^{\hat{\hat{2}}}}{\partial y} = \frac{\partial}{\partial y} \frac{\partial \xi^{\hat{\hat{2}}}}{\partial x}. \tag{4.19}$$

However, this identity does not hold for the orthonormal basis since

$$\frac{\partial(-\sin\vartheta)}{\partial y} \neq \frac{\partial \cos\vartheta}{\partial x} \quad \text{or equivalently} \quad -\frac{\partial}{\partial x} \frac{x}{\sqrt{x^2 + y^2}} \neq \frac{\partial}{\partial y} \frac{y}{\sqrt{x^2 + y^2}}. \tag{4.20}$$

We conclude that no coordinates, $\xi^{\hat{\hat{a}}}$, exist that correspond to the orthonormal basis $(\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}})$, in which we say that $(\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}})$ is a non-coordinate basis.

4.1.3 How we use non-coordinate bases

We commonly make use of non-coordinate bases for spherical coordinates and cyclindrical-polar coordinates. For spherical coordinates (Section 4.4) we have the spherical coordinate basis vectors, $\boldsymbol{e}_{\overline{a}}$, which are not normalized, and we have the corresponding orthonormal spherical non-coordinate basis,

$$\boldsymbol{e}_{\hat{\overline{a}}} = (\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}}). \tag{4.21}$$

The spherical coordinate and spherical non-coordinate basis vectors are related by

$$\boldsymbol{e}_{\overline{1}} = r\cos\phi\, \boldsymbol{e}_{\hat{\overline{1}}} = r\cos\phi\, \hat{\boldsymbol{\lambda}} \quad \text{and} \quad \boldsymbol{e}_{\overline{2}} = r\, \boldsymbol{e}_{\hat{\overline{2}}} = r\, \hat{\boldsymbol{\phi}} \quad \text{and} \quad \boldsymbol{e}_{\overline{3}} = \boldsymbol{e}_{\hat{\overline{3}}} = \hat{\boldsymbol{r}}. \tag{4.22}$$

In this manner we can represent an arbitrary vector, $\mathbf{F}$, in any of the following equivalent manners

$$\begin{aligned}
\mathbf{F} &= F^{1}\, \hat{\boldsymbol{x}} + F^{2}\, \hat{\boldsymbol{y}} + F^{3}\, \hat{\boldsymbol{z}} && \text{orthonormal Cartesian representation} && (4.23a)\\
&= F^{\overline{1}}\, \boldsymbol{e}_{\overline{1}} + F^{\overline{2}}\, \boldsymbol{e}_{\overline{2}} + F^{\overline{3}}\, \boldsymbol{e}_{\overline{3}} && \text{spherical coordinate representation} && (4.23b)\\
&= F^{\overline{1}}\, (r\cos\phi\, \hat{\boldsymbol{\lambda}}) + F^{\overline{2}}\, (r\, \hat{\boldsymbol{\phi}}) + F^{\overline{3}}\, \hat{\boldsymbol{r}} && \text{spherical coordinate representation} && (4.23c)\\
&= (F^{\overline{1}}\, r\cos\phi)\, \hat{\boldsymbol{\lambda}} + (F^{\overline{2}}\, r)\, \hat{\boldsymbol{\phi}} + F^{\overline{3}}\, \hat{\boldsymbol{r}} && \text{rearrange brackets} && (4.23d)\\
&= F^{\hat{\overline{1}}}\, \hat{\boldsymbol{\lambda}} + F^{\hat{\overline{2}}}\, \hat{\boldsymbol{\phi}} + F^{\hat{\overline{3}}}\, \hat{\boldsymbol{r}} && \text{spherical non-coordinate representation} && (4.23e)\\
&= (\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{F})\, \hat{\boldsymbol{\lambda}} + (\hat{\boldsymbol{\phi}} \cdot \boldsymbol{F})\, \hat{\boldsymbol{\phi}} + (\hat{\boldsymbol{r}} \cdot \boldsymbol{F})\, \hat{\boldsymbol{r}} && \text{spherical non-coordinate representation,} && (4.23f)
\end{aligned}$$

where the final expression defined the coordinate representation in terms of the non-coordinate orthonormal spherical basis as well as the Cartesian representation, $\boldsymbol{F}$,

$$F^{\hat{\overline{1}}} = F^{\overline{1}}\, r\cos\phi = \hat{\boldsymbol{\lambda}} \cdot \boldsymbol{F} \quad \text{and} \quad F^{\hat{\overline{2}}} = F^{\overline{2}}\, r = \hat{\boldsymbol{\phi}} \cdot \boldsymbol{F} \quad \text{and} \quad F^{\hat{\overline{3}}} = F^{\overline{3}} = \hat{\boldsymbol{r}} \cdot \boldsymbol{F}. \tag{4.24}$$

4.1.4 Physical components of tensors

Following Section 7.41 of *Aris* (1962), we refer to $F^{\hat{\overline{a}}}$ as the physical tensor components. That name is motivated since each component has the same physical dimension. Furthermore, we often refer to the non-coordinate representation in equation (4.24), $F^{\hat{\overline{a}}}$ as the spherical coordinate representation, rather than the coordinate representation, $F^{\overline{a}}$. The meaning is clear by the choice of basis vectors; i.e., whether we use coordinate basis vectors, $\boldsymbol{e}_{\overline{a}}$, or non-coordinate basis unit vectors, $\boldsymbol{e}_{\hat{\overline{a}}}$. We build on the notion of a physical tensor component in Section 4.5.4.

4.1.5 Further study

This section is based on Section 5.5 of *Schutz* (2009), with a similar discussion also provided in example 4.5 of *Tromp* (2025b).

4.2 Cartesian coordinates

Whenever developing a general tensor relation it is useful to check its validity by considering Cartesian coordinates, in which case we can make use of familiar rules from vector calculus. We here summarize some results from our discussion of Cartesian tensors in Chapters 1 and 2.

4.2.1 The basics

We start by expressing the position of a particle in space at time, τ, which defines its trajectory

$$\mathbf{x}(\tau) = \hat{\boldsymbol{e}}_1\, x(\tau) + \hat{\boldsymbol{e}}_2\, y(\tau) + \hat{\boldsymbol{e}}_3\, z(\tau) = \hat{\boldsymbol{x}}\, x(\tau) + \hat{\boldsymbol{y}}\, y(\tau) + \hat{\boldsymbol{z}}\, z(\tau), \tag{4.25}$$

with the orthonormal basis vectors written

$$\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{x}} \quad \text{and} \quad \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{y}} \quad \text{and} \quad \hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{z}}. \tag{4.26}$$

These expressions for the basis vectors correspond to the definition (3.16) for the basis vectors evaluated at a point

$$\boldsymbol{e}_a(\mathbf{x}) = \frac{\partial \mathbf{x}}{\partial \xi^a}, \tag{4.27}$$

whereas the basis vectors as partial derivative operators, as per equation (3.25), are given by

$$\hat{\boldsymbol{e}}_x = \partial_x \quad \text{and} \quad \hat{\boldsymbol{e}}_y = \partial_y \quad \text{and} \quad \hat{\boldsymbol{e}}_z = \partial_z. \tag{4.28}$$

The basis vectors for Cartesian coordinates satisfy the orthonormality condition,

$$\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b = \delta_{ab}. \tag{4.29}$$

Furthermore, the basis vectors are identical to the basis one-forms

$$\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{e}}^1 = \hat{\boldsymbol{x}} \quad \text{and} \quad \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}^2 = \hat{\boldsymbol{y}} \quad \text{and} \quad \hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{e}}^3 = \hat{\boldsymbol{z}}, \tag{4.30}$$

so that there is no importance placed on whether a tensor index is up or down. Since the orthonormal Cartesian basis vectors are independent of both space and time, we compute the coordinate representation of the velocity vector through taking the time derivative as

$$\boldsymbol{v}(\tau) = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\tau}, \tag{4.31}$$

which takes on the expanded expressions

$$\begin{aligned} \boldsymbol{v}(\tau) &= \hat{\boldsymbol{e}}_1 \frac{\mathrm{d}x(\tau)}{\mathrm{d}\tau} + \hat{\boldsymbol{e}}_2 \frac{\mathrm{d}y(\tau)}{\mathrm{d}\tau} + \hat{\boldsymbol{e}}_3 \frac{\mathrm{d}z(\tau)}{\mathrm{d}\tau} && (4.32a) \\ &= \hat{\boldsymbol{x}}\, v^1(\tau) + \hat{\boldsymbol{y}}\, v^2(\tau) + \hat{\boldsymbol{z}}\, v^3(\tau). && (4.32b) \\ &= \hat{\boldsymbol{x}}\, u(\tau) + \hat{\boldsymbol{y}}\, v(\tau) + \hat{\boldsymbol{z}}\, w(\tau), && (4.32c) \end{aligned}$$

where (u, v, w) is the notation commonly used in this book for the three Cartesian velocity components.

4.2.2 Concerning the horizontal gradient operator

Geophysical fluids are affected by gravity, and gravity breaks the spatial symmetry between the locally vertical (i.e., radial) and horizontal (i.e., local tangent plane) for fluid motion on a planet. We thus have many occasions where the horizontal is treated distinctly from the vertical. For that purpose we often find it useful to decompose the gradient operator into its horizontal and vertical components, writing the operator in one of the following manners

$$\nabla = \nabla_{\mathrm{h}} + \nabla_z = \nabla_{\mathrm{h}} + \hat{\boldsymbol{z}}\, \partial_z, \tag{4.33}$$

where the horizontal gradient operator, using Cartesian coordinates, is

$$\nabla_{\mathsf{h}} = \hat{\boldsymbol{x}}\,\nabla_x + \hat{\boldsymbol{y}}\,\nabla_y. \tag{4.34}$$

In many publications, we find the horizontal gradient operator written as ∇_z rather than ∇_{h}. In this alternative notation, the subscript z indicates that the derivative acts along surfaces of constant z. However, this notation is easily confused in this book. In particular, the ∇ operator has tensor labels that distinguish its components, in which case the vertical component to the Cartesian gradient operator is $\nabla_z = \hat{\boldsymbol{z}}\,\partial_z$. Herein lies the source of much confusion. To avoid that confusion we have chosen to use the ∇_{h} in this book for the horizontal gradient operator. To further reduce potential for confusion, we make use of the upright and sans serif h label. This label is not a tensor index but instead indicates that the operator is computed along a locally constant horizontal direction, and so is part of the operator's name. The h label is also placed quite close to the ∇ symbol, nearly attached, in order to further distinguish it from a tensor index.

4.2.3 Summary

In Cartesian coordinates, mathematical operators and integral theorems take their familiar form from vector calculus. We here list those encountered throughout this book.

$$\boldsymbol{x} = (x^1, x^2, x^3) = (x, y, z) \qquad \text{Cartesian coordinates} \tag{4.35}$$

$$\boldsymbol{F} = \hat{\boldsymbol{x}}F^1 + \hat{\boldsymbol{y}}F^2 + \hat{\boldsymbol{z}}F^3 = \hat{\boldsymbol{x}}F_1 + \hat{\boldsymbol{y}}F_2 + \hat{\boldsymbol{z}}F_3 \qquad \text{covariant} = \text{contravariant} \tag{4.36}$$

$$\frac{\partial}{\partial x^a} = \partial_a \;\text{ or }\; (\partial_x, \partial_y, \partial_z) \qquad \text{partial derivative operator} \tag{4.37}$$

$$\nabla = \hat{\boldsymbol{x}}\,\partial_x + \hat{\boldsymbol{y}}\,\partial_y + \hat{\boldsymbol{z}}\,\partial_z \qquad \text{gradient} = \text{covariant derivative} \tag{4.38}$$

$$\nabla_{\mathsf{h}} = \hat{\boldsymbol{x}}\,\partial_x + \hat{\boldsymbol{y}}\,\partial_y \qquad \text{horizontal gradient operator} \tag{4.39}$$

$$\nabla \cdot \boldsymbol{F} = \partial_x F^x + \partial_y F^y + \partial_z F^z \qquad \text{divergence of a vector} \tag{4.40}$$

$$\nabla_{\mathsf{h}} \cdot \boldsymbol{F} = \partial_x F^x + \partial_y F^y \qquad \text{horizontal divergence of a vector} \tag{4.41}$$

$$(\nabla \times \boldsymbol{F})_a = \epsilon_{abc}\,\partial^b F^c \qquad \text{components to the curl} \tag{4.42}$$

$$\nabla \cdot \nabla\psi = \nabla^2\psi = (\partial_{xx} + \partial_{yy} + \partial_{zz})\,\psi \qquad \text{Laplacian of a scalar} \tag{4.43}$$

$$\int_{\mathcal{R}} \nabla \cdot \boldsymbol{F}\,\mathrm{d}V = \oint_{\partial\mathcal{R}} \boldsymbol{F} \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} \qquad \text{Gauss's divergence theorem} \tag{4.44}$$

$$\oint_{\partial\mathcal{S}} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = \int_{\mathcal{S}} (\nabla \times \boldsymbol{F}) \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}. \qquad \text{Stokes' curl theorem.} \tag{4.45}$$

4.3 Cylindrical-polar coordinates

Many physical systems exhibit circular symmetry in two-dimensions or cylindrical symmetry in three-dimensions. Motion of liquid in a rotating circular tank provides the primary physical example encountered in this book. In the following, we emulate the discussion presented for the spherical coordinates in Section 4.4, here focusing on the cylindrical-polar coordinates shown in Figure 4.1. Our task is somewhat simpler than for the spherical coordinates since the vertical/axial position, z, remains unchanged from its Cartesian value. In a slight corruption of notation, we use the symbol r for the radial distance from the vertical axis in cylindrical-polar coordinates (Figure 4.1), which is distinct from the radial distance, r, used to measure the distance from the origin in spherical coordinates (Figure 4.2).[1]

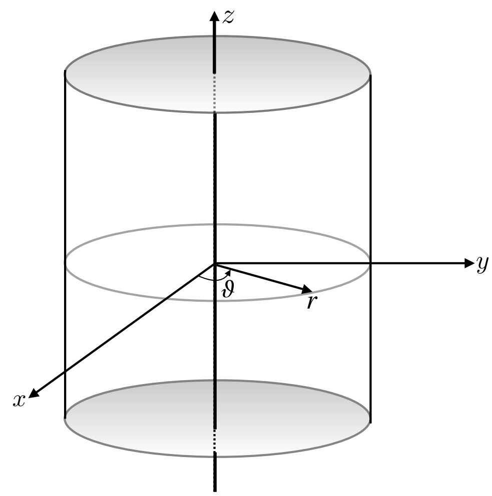


FIGURE 4.1: This schematic illustrates the geometry and notation for cylindrical-polar coordinates. The Cartesian triad of orthonormal basis vectors, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$ points along the orthogonal axes. The cylindrical-polar triad of orthonormal basis vectors, $(\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}}, \hat{\boldsymbol{z}})$, makes use of the radial unit vector $\hat{\boldsymbol{r}}$, which points outward from the vertical axis, the angular unit vector, $\hat{\boldsymbol{\vartheta}}$, which points in the counter-clockwise direction around the circle (when looking down in the $-\hat{\boldsymbol{z}}$ direction), and the vertical unit vector $\hat{\boldsymbol{z}}$. Note that the radial unit vector used for cylindrical-polar coordinates is distinct from that radial vector used in spherical coordinates shown in Figure 4.2.

The coordinate transformation between Cartesian coordinates and cylindrical-polar coordinates is given by

$$x = r\cos\vartheta \equiv \xi^{\bar{1}}\cos\xi^{\bar{2}} \tag{4.46a}$$

$$y = r\sin\vartheta \equiv \xi^{\bar{1}}\sin\xi^{\bar{2}} \tag{4.46b}$$

$$z = \xi^{\bar{3}}. \tag{4.46c}$$

The radial coordinate for cylindrical-polar coordinates

$$r = \sqrt{x^2 + y^2} \tag{4.47}$$

measures the distance from the vertical z-axis, and the angular coordinate $0 \leq \vartheta \leq 2\pi$ measures the angle counter-clockwise from the positive x-axis. We introduce the unbarred and barred labels for the Cartesian and cylindrical polar coordinates

$$(x, y, z) = (\xi^1, \xi^2, \xi^3) \equiv \xi^a \quad \text{and} \quad (r, \vartheta, z) = (\xi^{\bar{1}}, \xi^{\bar{2}}, \xi^{\bar{3}}) \equiv \xi^{\bar{a}}. \tag{4.48}$$

Although the vertical coordinate, z, remains the same in both Cartesian and cylindrical-polar coordinates, and it is orthogonal to the other coordinates, we find it useful to introduce a distinct symbols, ξ^3 and $\xi^{\bar{3}}$, to specify what other coordinates are held fixed when performing partial derivative operations.

[1] Many mathematics texts use ρ for the cylindrical radial distance. We choose not to follow that convention, since ρ is reserved in this book for mass density.

4.3.1 Cartesian and cylindrical-polar transformation

The coordinate transformation (4.46a)-(4.46c) leads to the transformation matrix

$$\Lambda^{a}{}_{\overline{a}} = \begin{bmatrix} \partial\xi^1/\partial\xi^{\overline{1}} & \partial\xi^1/\partial\xi^{\overline{2}} & \partial\xi^1/\partial\xi^{\overline{3}} \\ \partial\xi^2/\partial\xi^{\overline{1}} & \partial\xi^2/\partial\xi^{\overline{2}} & \partial\xi^2/\partial\xi^{\overline{3}} \\ \partial\xi^3/\partial\xi^{\overline{1}} & \partial\xi^3/\partial\xi^{\overline{2}} & \partial\xi^3/\partial\xi^{\overline{3}} \end{bmatrix} = \begin{bmatrix} \cos\vartheta & -r\sin\vartheta & 0 \\ \sin\vartheta & r\cos\vartheta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{4.49}$$

and the inverse transformation is given by

$$\Lambda^{\overline{a}}{}_{a} = \frac{1}{r}\begin{bmatrix} r\cos\vartheta & r\sin\vartheta & 0 \\ -\sin\vartheta & \cos\vartheta & 0 \\ 0 & 0 & r \end{bmatrix}. \tag{4.50}$$

The determinant of the transformation (Jacobian) is given by

$$\det(\Lambda^{a}{}_{\overline{a}}) = r, \tag{4.51}$$

which vanishes along the vertical axis, which is where the transformation is singular. Otherwise, the coordinate transformation is one-to-one and invertible.

4.3.2 Basis vectors

The cylindrical-polar coordinate basis vectors, $\boldsymbol{e}_{\overline{a}}$, are related to the orthonormal Cartesian coordinate basis vectors, $\hat{\boldsymbol{e}}_a$, through the transformation $\boldsymbol{e}_{\overline{a}} = \Lambda^{a}{}_{\overline{a}}\,\hat{\boldsymbol{e}}_a$. The transformation matrix (4.49) leads to

$$\boldsymbol{e}_r = \hat{\boldsymbol{x}}\cos\vartheta + \hat{\boldsymbol{y}}\sin\vartheta \tag{4.52a}$$
$$\boldsymbol{e}_\vartheta = r\,(-\hat{\boldsymbol{x}}\sin\vartheta + \hat{\boldsymbol{y}}\cos\vartheta) \tag{4.52b}$$
$$\boldsymbol{e}_{\overline{z}} = \hat{\boldsymbol{z}}. \tag{4.52c}$$

It is convenient to introduce the following orthonormal cylindrical-polar basis vectors (Section 4.1) $(\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}}, \hat{\boldsymbol{z}})$, which point in directions of increasing r, ϑ, and z, and which are related to the cylindrical-polar coordinate basis vectors via

$$\boldsymbol{e}_r = \hat{\boldsymbol{r}} \quad \text{and} \quad \boldsymbol{e}_\vartheta = r\,\hat{\boldsymbol{\vartheta}} \quad \text{and} \quad \boldsymbol{e}_{\overline{z}} = \hat{\boldsymbol{z}}, \tag{4.53}$$

along with the inverse relations

$$\hat{\boldsymbol{x}} = \hat{\boldsymbol{r}}\cos\vartheta - \hat{\boldsymbol{\vartheta}}\sin\vartheta \tag{4.54a}$$
$$\hat{\boldsymbol{y}} = \hat{\boldsymbol{r}}\sin\vartheta + \hat{\boldsymbol{\vartheta}}\cos\vartheta \tag{4.54b}$$
$$\hat{\boldsymbol{z}} = \hat{\boldsymbol{z}}. \tag{4.54c}$$

These expressions for the basis vectors correspond to the definition (3.16), in which the basis vectors at a point in space are the spatial derivatives of the position

$$\boldsymbol{e}_{\overline{a}}(\mathbf{x}) = \frac{\partial\mathbf{x}}{\partial\xi^{\overline{a}}}. \tag{4.55}$$

Namely, with the position written in cylindrical-polar coordinates, $\mathbf{x} = r\,\hat{\boldsymbol{r}} + z\,\hat{\boldsymbol{z}}$, we have

$$\boldsymbol{e}_r(\mathbf{x}) = \frac{\partial(r\,\hat{\boldsymbol{r}} + z\,\hat{\boldsymbol{z}})}{\partial r} = \hat{\boldsymbol{r}} \tag{4.56a}$$

$$\boldsymbol{e}_\vartheta(\mathbf{x}) = \frac{\partial(r\,\hat{\boldsymbol{r}} + z\,\hat{\boldsymbol{z}})}{\partial \vartheta} = r\,\hat{\boldsymbol{\vartheta}} \tag{4.56b}$$

$$\boldsymbol{e}_{\overline{z}}(\mathbf{x}) = \frac{\partial(r\,\hat{\boldsymbol{r}} + z\,\hat{\boldsymbol{z}})}{\partial z} = \hat{\boldsymbol{z}}, \tag{4.56c}$$

whereas the coordinate basis vectors written as partial derivative operators, as per equation (3.25), are given by

$$\boldsymbol{e}_r = \partial_r \quad \text{and} \quad \boldsymbol{e}_\vartheta = \partial_\vartheta \quad \text{and} \quad \boldsymbol{e}_{\overline{z}} = \partial_{\overline{z}}. \tag{4.57}$$

4.3.3 Basis one-forms

Since cylindrical-polar coordinates are orthogonal, we can readily derive the one-form basis through inverting the vector basis. At a point in space we have

$$\boldsymbol{e}^r(\mathbf{x}) = \hat{\boldsymbol{r}} \quad \text{and} \quad \boldsymbol{e}^\vartheta(\mathbf{x}) = r^{-1}\hat{\boldsymbol{\vartheta}} \quad \text{and} \quad \boldsymbol{e}^{\overline{z}}(\mathbf{x}) = \hat{\boldsymbol{z}}, \tag{4.58}$$

and as exact differentials

$$\boldsymbol{e}^r = \mathrm{d}r \quad \text{and} \quad \boldsymbol{e}^\vartheta = \mathrm{d}\vartheta \quad \text{and} \quad \boldsymbol{e}^{\overline{z}} = \mathrm{d}\overline{z}. \tag{4.59}$$

In either form, it is straightforward to verify the biorthogonality relation (Section 3.8.1) between the basis one-forms and basis vectors

$$\boldsymbol{e}^{\overline{b}} \cdot \boldsymbol{e}_{\overline{a}} = \delta^{\overline{b}}{}_{\overline{a}}. \tag{4.60}$$

4.3.4 Position and velocity

In cylindrical-polar coordinates, the position of a point in space, parameterized by its time, is specified by its radial plus vertical coordinate

$$\boldsymbol{x}(\tau) = r\,\boldsymbol{e}_r + z\,\boldsymbol{e}_{\overline{z}}. \tag{4.61}$$

The velocity requires all three coordinates since the radial basis vector is a function of the angular positions, which are in turn functions of time. Use of the chain rule renders

$$\boldsymbol{v}(\tau) = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\tau} \tag{4.62a}$$

$$= \boldsymbol{e}_r\,\frac{\mathrm{d}r}{\mathrm{d}\tau} + r\,\frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}\tau} + \boldsymbol{e}_{\overline{z}}\,\frac{\mathrm{d}z}{\mathrm{d}\tau} \tag{4.62b}$$

$$= \boldsymbol{e}_r\,\frac{\mathrm{d}r}{\mathrm{d}\tau} + r\,\frac{\partial\boldsymbol{e}_r}{\partial\vartheta}\,\frac{\mathrm{d}\vartheta}{\mathrm{d}\tau} + \boldsymbol{e}_{\overline{z}}\,\frac{\mathrm{d}z}{\mathrm{d}\tau} \tag{4.62c}$$

$$= \boldsymbol{e}_r\,\frac{\mathrm{d}r}{\mathrm{d}\tau} + \boldsymbol{e}_\vartheta\,\frac{\mathrm{d}\vartheta}{\mathrm{d}\tau} + \boldsymbol{e}_{\overline{z}}\,\frac{\mathrm{d}z}{\mathrm{d}\tau} \tag{4.62d}$$

$$= \boldsymbol{e}_r\,v^r + \boldsymbol{e}_\vartheta\,v^\vartheta + \boldsymbol{e}_{\overline{z}}\,v^{\overline{z}}. \tag{4.62e}$$

To reach this result we made use of the identity

$$\boldsymbol{e}_\vartheta = r\,\frac{\partial \boldsymbol{e}_r}{\partial \vartheta} = r\,\hat{\boldsymbol{\vartheta}}. \tag{4.63}$$

4.3.5 Metric tensor

Cylindrical-polar coordinates are orthogonal, so that the metric tensor and its inverse are represented by diagonal matrices

$$\mathbb{g}_{\overline{a}\overline{b}} = \boldsymbol{e}_{\overline{a}} \cdot \boldsymbol{e}_{\overline{b}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbb{g}^{\overline{a}\overline{b}} = \boldsymbol{e}^{\overline{a}} \cdot \boldsymbol{e}^{\overline{b}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4.64}$$

We can readily verify that the metric tensor raises and lowers indices, thus connecting the basis vectors to the basis one-forms

$$\boldsymbol{e}_{\overline{a}} = \mathbb{g}_{\overline{a}\overline{b}}\,\boldsymbol{e}^{\overline{b}} \quad \text{and} \quad \boldsymbol{e}^{\overline{a}} = \mathbb{g}^{\overline{a}\overline{b}}\,\boldsymbol{e}_{\overline{b}}. \tag{4.65}$$

For example, we have

$$\boldsymbol{e}_{\overline{2}} = \mathbb{g}_{\overline{2}\overline{2}}\,\boldsymbol{e}^{\overline{2}} = r^2\,(r^{-1}\,\hat{\boldsymbol{\vartheta}}) = r\,\hat{\boldsymbol{\vartheta}}, \tag{4.66}$$

and for the inverse

$$\boldsymbol{e}^{\overline{2}} = \mathbb{g}^{\overline{2}\overline{2}}\,\boldsymbol{e}_{\overline{2}} = r^{-2}\,(r\,\hat{\boldsymbol{\vartheta}}) = r^{-1}\,\hat{\boldsymbol{\vartheta}}. \tag{4.67}$$

4.3.6 Volume element and Levi-Civita tensor

The square root of the determinant of the metric tensor written in cylindrical-polar coordinates (from equation (4.64)) is given by

$$\sqrt{\det(\mathbb{g}_{\overline{a}\overline{b}})} = r \tag{4.68}$$

so that the volume element is

$$\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}z. \tag{4.69}$$

The covariant Levi-Civita tensor has the cylindrical-polar representation

$$\varepsilon_{\overline{a}\overline{b}\overline{c}} = r\,\epsilon_{\overline{a}\overline{b}\overline{c}}. \tag{4.70}$$

4.3.7 Components of a vector field

A vector field, $\mathbf{F}$, has Cartesian components, F^a, related to its cylindrical-polar components, $F^{\overline{a}}$, via the transformation matrix, $F^{\overline{a}} = \Lambda^{\overline{a}}{}_a\,F^a$. This transformation leads to

$$F^{\overline{1}} = F^x \cos\vartheta + F^y \sin\vartheta \tag{4.71a}$$

$$F^{\overline{2}} = r^{-1}\,[-F^x \sin\vartheta + F^y \cos\vartheta] \tag{4.71b}$$

$$F^{\overline{3}} = F^z. \tag{4.71c}$$

Introducing the cylindrical-polar unit vectors (4.53) leads to the relations between the coordinate and non-coordinate basis representations

$$F^{\overline{1}} = \hat{\boldsymbol{r}} \cdot \boldsymbol{F} = F^{\hat{\overline{1}}} \tag{4.72a}$$

$$r\,F^{\overline{2}} = \hat{\boldsymbol{\vartheta}} \cdot \boldsymbol{F} = F^{\hat{\overline{2}}} \tag{4.72b}$$

$$F^{\overline{3}} = \hat{\boldsymbol{z}} \cdot \boldsymbol{F} = F^{\hat{\overline{3}}}, \tag{4.72c}$$

where $\boldsymbol{F}$ is the Cartesian representation of the vector, and where $F^{\hat{\overline{a}}}$ are the physical components to the vector as introduced in Section 4.1.4.

4.3.8 Differential operators

In cylindrical-polar coordinates, the gradient operator $\nabla = \boldsymbol{e}^a \partial_a$ is given by

$$\nabla = \hat{\boldsymbol{r}}\,\frac{\partial}{\partial r} + \frac{\hat{\boldsymbol{\vartheta}}}{r}\,\frac{\partial}{\partial \vartheta} + \hat{\boldsymbol{z}}\,\frac{\partial}{\partial z} \tag{4.73}$$

and the covariant divergence of a vector field is

$$\begin{aligned}
\nabla_{\overline{a}} F^{\overline{a}} &= r^{-1}\,\partial_{\overline{a}}\,(r\,F^{\overline{a}}) && (4.74a)\\
&= r^{-1}\left(\partial_r[r\,F^{\overline{1}}] + \partial_\vartheta[r\,F^{\overline{2}}] + \partial_{\overline{z}}[r\,F^{\overline{3}}]\right) && (4.74b)\\
&= \frac{1}{r}\,\frac{\partial(r\,\hat{\boldsymbol{r}}\cdot\boldsymbol{F})}{\partial r} + \frac{1}{r}\frac{\partial\,(\hat{\boldsymbol{\vartheta}}\cdot\boldsymbol{F})}{\partial\vartheta} + \frac{\partial(\hat{\boldsymbol{z}}\cdot\boldsymbol{F})}{\partial z}, && (4.74c)
\end{aligned}$$

where $\boldsymbol{F}$ is the Cartesian coordinate representation. The covariant Laplacian of a scalar, $\nabla^2\psi = \nabla\cdot\nabla\psi$, is

$$\begin{aligned}
\nabla^2\psi &= \nabla\cdot\left[\hat{\boldsymbol{r}}\,\frac{\partial\psi}{\partial r} + \frac{\hat{\boldsymbol{\vartheta}}}{r}\,\frac{\partial\psi}{\partial\vartheta} + \hat{\boldsymbol{z}}\,\frac{\partial\psi}{\partial z}\right] && (4.75a)\\
&= \frac{1}{r}\,\frac{\partial}{\partial r}\left[r\,\frac{\partial\psi}{\partial r}\right] + \frac{1}{r^2}\,\frac{\partial^2\psi}{\partial\vartheta^2} + \frac{\partial^2\psi}{\partial z^2}. && (4.75b)
\end{aligned}$$

The covariant curl (Section 3.21) is

$$\begin{aligned}
(\mathrm{curl}\,\mathbf{F})^{\overline{1}} &= r^{-1}\,[\partial_\vartheta\,F^{\overline{3}} - \partial_{\overline{z}}\,(r^2\,F^{\overline{2}})] && (4.76a)\\
(\mathrm{curl}\,\mathbf{F})^{\overline{2}} &= r^{-1}\,[\partial_{\overline{z}}\,F^{\overline{1}} - \partial_r\,F^{\overline{3}}] && (4.76b)\\
(\mathrm{curl}\,\mathbf{F})^{\overline{3}} &= r^{-1}\,[\partial_r\,(r^2\,F^{\overline{2}}) - \partial_\vartheta\,F^{\overline{1}})], && (4.76c)
\end{aligned}$$

which can be written more conventionally as

$$\begin{aligned}
(\mathrm{curl}\,\mathbf{F})^{\overline{1}} &= \frac{1}{r}\,\frac{\partial(\hat{\boldsymbol{z}}\cdot\boldsymbol{F})}{\partial\vartheta} - \frac{\partial(\hat{\boldsymbol{\vartheta}}\cdot\boldsymbol{F})}{\partial z} && (4.77a)\\
r\,(\mathrm{curl}\,\mathbf{F})^{\overline{2}} &= \frac{\partial(\hat{\boldsymbol{r}}\cdot\boldsymbol{F})}{\partial z} - \frac{\partial(\hat{\boldsymbol{z}}\cdot\boldsymbol{F})}{\partial r} && (4.77b)\\
(\mathrm{curl}\,\mathbf{F})^{\overline{3}} &= \frac{1}{r}\,\frac{\partial(r\,\hat{\boldsymbol{\vartheta}}\cdot\boldsymbol{F})}{\partial r} - \frac{1}{r}\,\frac{\partial(\hat{\boldsymbol{r}}\cdot\boldsymbol{F})}{\partial\vartheta}. && (4.77c)
\end{aligned}$$

4.3.9 Summary

We here summarize the cylindrical coordinate version of some common mathematical operators.

$$(r, \vartheta, z) = (x^{\overline{1}}, x^{\overline{2}}, x^{\overline{3}}) \tag{4.78}$$

$$F^{\overline{1}} = \hat{\boldsymbol{r}} \cdot \boldsymbol{F} \quad r\, F^{\overline{2}} = \hat{\boldsymbol{\vartheta}} \cdot \boldsymbol{F} \quad F^{\overline{3}} = \hat{\boldsymbol{z}} \cdot \boldsymbol{F} \tag{4.79}$$

$$\nabla = \hat{\boldsymbol{r}}\, \frac{\partial}{\partial r} + \frac{\hat{\boldsymbol{\vartheta}}}{r}\, \frac{\partial}{\partial \vartheta} + \hat{\boldsymbol{z}}\, \frac{\partial}{\partial z} \tag{4.80}$$

$$\nabla_{\overline{a}} F^{\overline{a}} = \frac{1}{r}\, \frac{\partial (r\, \hat{\boldsymbol{r}} \cdot \boldsymbol{F})}{\partial r} + \frac{1}{r} \frac{\partial\, (\hat{\boldsymbol{\vartheta}} \cdot \boldsymbol{F})}{\partial \vartheta} + \frac{\partial (\hat{\boldsymbol{z}} \cdot \boldsymbol{F})}{\partial z}. \tag{4.81}$$

$$\nabla^2 \psi = \frac{1}{r}\, \frac{\partial}{\partial r} \left[r\, \frac{\partial \psi}{\partial r} \right] + \frac{1}{r^2}\, \frac{\partial^2 \psi}{\partial \vartheta^2} + \frac{\partial^2 \psi}{\partial z^2} \tag{4.82}$$

$$(\nabla \times \boldsymbol{F})_{\overline{a}} = \varepsilon_{\overline{a}\overline{b}\overline{c}} \partial_{\overline{b}} F^{\overline{c}} \quad \text{see equations (4.76a)} - \text{(4.76c)} \tag{4.83}$$

Note that we wrote the vector components in terms of the non-coordinate orthonormal basis vectors, $(\hat{\boldsymbol{r}}, \hat{\boldsymbol{\vartheta}}, \hat{\boldsymbol{z}})$, which accords with usage in most of the geophysical fluids literature.

4.4 Spherical coordinates

We now consider spherical coordinates defined by Figure 4.2, which are related to Cartesian coordinates through the coordinate transformation

$$x = r \cos\phi\, \cos\lambda = r \sin\varphi\, \cos\lambda \tag{4.84a}$$

$$y = r \cos\phi\, \sin\lambda = r \sin\varphi\, \sin\lambda \tag{4.84b}$$

$$z = r \sin\phi = r \cos\varphi. \tag{4.84c}$$

The radial coordinate

$$r = |\boldsymbol{x}| = \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{x^2 + y^2 + z^2} \tag{4.85}$$

measures the distance from the center of the sphere to the position of the point. The spherical angle coordinates,

$$0 \le \lambda \le 2\pi \quad \text{longitude} \tag{4.86}$$

$$-\pi/2 \le \phi \le \pi/2 \quad \text{latitude} \tag{4.87}$$

$$0 \le \varphi \le \pi \quad \text{co-latitude,} \tag{4.88}$$

specify the longitude, measuring the radians of the position east of the prime meridian, the latitude, measuring the radians north ($\phi > 0$) and south ($\phi < 0$) of the equator, and the co-latitude, φ, measuring the radians relative to the north pole. In most of our applications we make use of the latitude, thus corresponding to common nomenclature in geophysical studies. However, much of the maths and physics communities make use of the co-latitude. Conversion between the two is straightforward via

$$\varphi = \pi/2 - \phi \quad \text{and} \quad \cos\phi = \sin\varphi \quad \text{and} \quad \sin\phi = \cos\varphi. \tag{4.89}$$

To streamline notation in the following, we introduce the unbarred and barred labels for the Cartesian and spherical coordinates, respectively

$$(x, y, z) = (\xi^1, \xi^2, \xi^3) \equiv \xi^a \quad \text{and} \quad (\lambda, \phi, r) = (\xi^{\overline{1}}, \xi^{\overline{2}}, \xi^{\overline{3}}) \equiv \xi^{\overline{a}}. \tag{4.90}$$

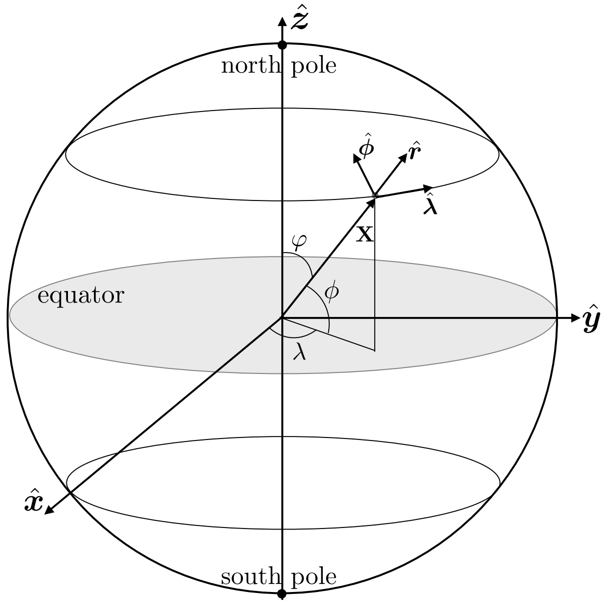


FIGURE 4.2: Geometry of the planetary spherical coordinates and planetary Cartesian coordinates. The planetary Cartesian triad of orthonormal basis vectors, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$, points along the orthogonal axes. The planetary spherical triad of non-coordinate orthonormal basis vectors, $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, makes use of the longitudinal unit vector, $\hat{\boldsymbol{\lambda}}$, which points in the longitudinal direction (positive eastward), the latitudinal unit vector, $\hat{\boldsymbol{\phi}}$, which points in the latitudinal direction (positive northward) and the radial unit vector, $\hat{\boldsymbol{r}}$, which point in the radial direction (positive away from the center). We also show the co-latitude, $\varphi = \pi/2 - \phi$, which measures the meridional angle relative to the north pole. In this book we mostly use the latitude, whereas the co-latitude is commonly used in the maths and physics communities.

4.4.1 Cartesian and spherical transformation

Following the general discussion in Section 3.7.4, we consider the infinitesimal distance along one of the Cartesian coordinate axes, $\mathrm{d}\xi^a$. The chain rule allows us to relate this distance to those along the axes of the spherical coordinate system

$$\mathrm{d}\xi^a = \frac{\partial \xi^a}{\partial \xi^{\overline{a}}}\,\mathrm{d}\xi^{\overline{a}} = \Lambda^a{}_{\overline{a}}\,\mathrm{d}\xi^{\overline{a}}. \tag{4.91}$$

The partial derivatives $\partial\xi^a/\partial\xi^{\overline{a}}$ form components to the transformation matrix that transforms between coordinate representations. For the coordinate transformation (4.84a)-(4.84c), the transformation matrix is given by

$$\Lambda^a{}_{\overline{a}} = \begin{bmatrix} \partial\xi^1/\partial\xi^{\overline{1}} & \partial\xi^1/\partial\xi^{\overline{2}} & \partial\xi^1/\partial\xi^{\overline{3}} \\ \partial\xi^2/\partial\xi^{\overline{1}} & \partial\xi^2/\partial\xi^{\overline{2}} & \partial\xi^2/\partial\xi^{\overline{3}} \\ \partial\xi^3/\partial\xi^{\overline{1}} & \partial\xi^3/\partial\xi^{\overline{2}} & \partial\xi^3/\partial\xi^{\overline{3}} \end{bmatrix} \tag{4.92a}$$

$$= \begin{bmatrix} -r\cos\phi\sin\lambda & -r\sin\phi\,\cos\lambda & \cos\phi\cos\lambda \\ r\cos\phi\cos\lambda & -r\sin\phi\,\sin\lambda & \cos\phi\sin\lambda \\ 0 & r\cos\phi & \sin\phi \end{bmatrix}. \tag{4.92b}$$

The determinant of the transformation (Jacobian) is given by

$$\det(\Lambda^a{}_{\overline{a}}) = r^2\cos\phi. \tag{4.93}$$

The Jacobian vanishes at the north and south poles ($\phi = \pm\pi/2$), where the transformation is singular. Otherwise, the transformation is one-to-one and invertible. Methods familiar from linear algebra render the inverse transformation matrix

$$\Lambda^{\overline{a}}{}_{a} = \frac{1}{r^2\cos\phi}\begin{bmatrix} -r\sin\lambda & r\cos\lambda & 0 \\ -r\cos\phi\,\sin\phi\,\cos\lambda & -r\cos\phi\sin\phi\sin\lambda & r\cos^2\phi \\ r^2\cos^2\phi\cos\lambda & r^2\,\cos^2\phi\sin\lambda & r^2\cos\phi\sin\phi \end{bmatrix}, \tag{4.94}$$

so that

$$\Lambda^{\overline{a}}{}_{a}\,\Lambda^{a}{}_{\overline{b}} = \delta^{\overline{a}}{}_{\overline{b}} \quad \text{and} \quad \Lambda^{a}{}_{\overline{b}}\Lambda^{\overline{b}}{}_{b} = \delta^{a}{}_{b}. \tag{4.95}$$

4.4.2 Basis vectors

The spherical coordinate basis vectors, $\boldsymbol{e}_{\overline{a}}$, are related to the orthonormal Cartesian coordinate basis vectors, $\boldsymbol{e}_a$, through the transformation

$$\boldsymbol{e}_{\overline{a}} = \Lambda^{a}{}_{\overline{a}}\,\hat{\boldsymbol{e}}_a. \tag{4.96}$$

The transformation matrix (4.92b) leads to

$$\boldsymbol{e}_\lambda = r\cos\phi\,(-\hat{\boldsymbol{x}}\sin\lambda + \hat{\boldsymbol{y}}\cos\lambda) \tag{4.97a}$$

$$\boldsymbol{e}_\phi = r(-\hat{\boldsymbol{x}}\sin\phi\,\cos\lambda - \hat{\boldsymbol{y}}\sin\phi\,\sin\lambda + \hat{\boldsymbol{z}}\cos\phi) \tag{4.97b}$$

$$\boldsymbol{e}_r = \hat{\boldsymbol{x}}\cos\phi\,\cos\lambda + \hat{\boldsymbol{y}}\cos\phi\,\sin\lambda + \hat{\boldsymbol{z}}\sin\phi. \tag{4.97c}$$

It is convenient to introduce the orthonormal non-coordinate unit vectors (Section 4.1) $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, which point in directions of increasing λ, ϕ, and r, so that the basis vectors are written

$$\boldsymbol{e}_\lambda = r\cos\phi\,\hat{\boldsymbol{\lambda}} \quad \text{and} \quad \boldsymbol{e}_\phi = r\,\hat{\boldsymbol{\phi}} \quad \text{and} \quad \boldsymbol{e}_r = \hat{\boldsymbol{r}}, \tag{4.98}$$

in which case we have the relations

$$\hat{\boldsymbol{\lambda}} = -\hat{\boldsymbol{x}}\sin\lambda + \hat{\boldsymbol{y}}\cos\lambda \tag{4.99a}$$

$$\hat{\boldsymbol{\phi}} = -\hat{\boldsymbol{x}}\cos\lambda\,\sin\phi - \hat{\boldsymbol{y}}\sin\lambda\,\sin\phi + \hat{\boldsymbol{z}}\cos\phi \tag{4.99b}$$

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}}\,\cos\lambda\,\cos\phi + \hat{\boldsymbol{y}}\sin\lambda\,\cos\phi + \hat{\boldsymbol{z}}\sin\phi \tag{4.99c}$$

along with their inverse

$$\hat{\boldsymbol{x}} = -\hat{\boldsymbol{\lambda}}\sin\lambda - \hat{\boldsymbol{\phi}}\cos\lambda\sin\phi + \hat{\boldsymbol{r}}\,\cos\lambda\,\cos\phi \tag{4.100a}$$

$$\hat{\boldsymbol{y}} = \hat{\boldsymbol{\lambda}}\cos\lambda - \hat{\boldsymbol{\phi}}\,\sin\lambda\,\sin\phi + \hat{\boldsymbol{r}}\sin\lambda\,\cos\phi \tag{4.100b}$$

$$\hat{\boldsymbol{z}} = \hat{\boldsymbol{\phi}}\,\cos\phi + \hat{\boldsymbol{r}}\,\sin\phi. \tag{4.100c}$$

Furthermore, note that these expressions for the basis vectors correspond to the definition (3.16), in which the basis vectors evaluated at a point are the spatial derivatives of the position

$$\boldsymbol{e}_{\overline{a}}(\mathbf{x}) = \frac{\partial\mathbf{x}}{\partial\xi^{\overline{a}}}. \tag{4.101}$$

Namely, with the position written in spherical coordinates, $\mathbf{x} = r\,\hat{\boldsymbol{r}}$, we have

$$\boldsymbol{e}_\lambda(\mathbf{x}) = \frac{\partial(r\,\hat{\boldsymbol{r}})}{\partial\lambda} = r\cos\phi\,\hat{\boldsymbol{\lambda}} \quad\text{and}\quad \boldsymbol{e}_\phi(\mathbf{x}) = \frac{\partial(r\,\hat{\boldsymbol{r}})}{\partial\phi} = r\,\hat{\boldsymbol{\phi}} \quad\text{and}\quad \boldsymbol{e}_r(\mathbf{x}) = \frac{\partial(r\,\hat{\boldsymbol{r}})}{\partial r} = \hat{\boldsymbol{r}}, \tag{4.102}$$

whereas the coordinate basis vectors written as partial derivative operators, as per equation (3.25), are given by

$$\boldsymbol{e}_\lambda = \partial_\lambda \quad\text{and}\quad \boldsymbol{e}_\phi = \partial_\phi \quad\text{and}\quad \boldsymbol{e}_r = \partial_r. \tag{4.103}$$

4.4.3 Basis one-forms

Since spherical coordinates are locally orthogonal, we can readily derive the one-form basis through inverting the vector basis. When evaluated at a point in space we have

$$\boldsymbol{e}^\lambda(\mathbf{x}) = (r\cos\phi)^{-1}\,\hat{\boldsymbol{\lambda}} \quad\text{and}\quad \boldsymbol{e}^\phi(\mathbf{x}) = r^{-1}\hat{\boldsymbol{\phi}} \quad\text{and}\quad \boldsymbol{e}^r(\mathbf{x}) = \hat{\boldsymbol{r}}, \tag{4.104}$$

and as exact differentials

$$\boldsymbol{e}^\lambda = \mathrm{d}\lambda \quad\text{and}\quad \boldsymbol{e}^\phi = \mathrm{d}\phi \quad\text{and}\quad \boldsymbol{e}^r = \mathrm{d}r. \tag{4.105}$$

In either form it is straightforward to verify the biorthogonality relation between the basis one-forms and basis vectors (Section 3.8.1)

$$\boldsymbol{e}^{\bar{b}} \cdot \boldsymbol{e}_{\bar{a}} = \delta^{\bar{b}}{}_{\bar{a}}. \tag{4.106}$$

4.4.4 Position and velocity

In spherical coordinates, the position of a point is fully specified by its radial position

$$\boldsymbol{x}(\tau) = r\,\boldsymbol{e}_r = r\,\hat{\boldsymbol{r}}. \tag{4.107}$$

However, the velocity requires all three spherical coordinates since the radial basis vector is a function of the angular positions, which are functions of time. Use of the chain rule renders

$$\boldsymbol{v}(\tau) = \frac{\mathrm{d}\boldsymbol{x}(\tau)}{\mathrm{d}\tau} \tag{4.108a}$$
$$= \boldsymbol{e}_r\,\frac{\mathrm{d}r}{\mathrm{d}\tau} + r\,\frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}\tau} \tag{4.108b}$$
$$= \boldsymbol{e}_r\,\frac{\mathrm{d}r}{\mathrm{d}\tau} + r\,\frac{\partial\boldsymbol{e}_r}{\partial\lambda}\,\frac{\mathrm{d}\lambda}{\mathrm{d}\tau} + r\,\frac{\partial\boldsymbol{e}_r}{\partial\phi}\,\frac{\mathrm{d}\phi}{\mathrm{d}\tau} \tag{4.108c}$$
$$\equiv \boldsymbol{e}_r\,\frac{\mathrm{d}r}{\mathrm{d}\tau} + \boldsymbol{e}_\lambda\,\frac{\mathrm{d}\lambda}{\mathrm{d}\tau} + \boldsymbol{e}_\phi\,\frac{\mathrm{d}\phi}{\mathrm{d}\tau} \tag{4.108d}$$
$$= \boldsymbol{e}_r\,v^r + \boldsymbol{e}_\lambda\,v^\lambda + \boldsymbol{e}_\phi\,v^\phi. \tag{4.108e}$$

To reach this result we made use of the identities satisfied by the spherical basis vectors

$$\boldsymbol{e}_\lambda = r\,\frac{\partial\boldsymbol{e}_r}{\partial\lambda} \quad\text{and}\quad \boldsymbol{e}_\phi = r\,\frac{\partial\boldsymbol{e}_r}{\partial\phi}, \tag{4.109}$$

which can be readily verified by equations (4.97a)-(4.97c).

4.4.5 Metric tensor

Since spherical coordinates are orthogonal, the metric tensor is diagonal and it is given by

$$\mathbb{g}_{\bar{a}\bar{b}} = \boldsymbol{e}_{\bar{a}} \cdot \boldsymbol{e}_{\bar{b}} = \begin{bmatrix} (r\cos\phi)^2 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{4.110}$$

along with the inverse metric tensor

$$\mathbb{g}^{\bar{a}\bar{b}} = \boldsymbol{e}^{\bar{a}} \cdot \boldsymbol{e}^{\bar{b}} = \begin{bmatrix} (r\cos\phi)^{-2} & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4.111}$$

We can readily verify that the metric tensor raises and lowers indices, thus connecting the basis vectors to the basis one-forms

$$\boldsymbol{e}_{\bar{a}} = \mathbb{g}_{\bar{a}\bar{b}}\, \boldsymbol{e}^{\bar{b}} \quad \text{and} \quad \boldsymbol{e}^{\bar{a}} = \mathbb{g}^{\bar{a}\bar{b}}\, \boldsymbol{e}_{\bar{b}}. \tag{4.112}$$

For example, we have

$$\boldsymbol{e}_{\bar{1}} = \mathbb{g}_{\bar{1}\bar{1}}\, \boldsymbol{e}^{\bar{1}} = (r\cos\phi)^2\,(r\cos\phi)^{-1}\,\hat{\boldsymbol{\lambda}} = r\cos\phi\,\hat{\boldsymbol{\lambda}}, \tag{4.113}$$

and for the inverse

$$\boldsymbol{e}^{\bar{1}} = \mathbb{g}^{\bar{1}\bar{1}}\, \boldsymbol{e}_{\bar{1}} = (r\cos\phi)^{-2}\,(r\cos\phi\,\hat{\boldsymbol{\lambda}}) = (r\cos\phi)^{-1}\,\hat{\boldsymbol{\lambda}}. \tag{4.114}$$

Volume element and Levi-Civita tensor

The square root of the determinant of the metric tensor written in spherical coordinates (from equation (4.110)) is given by

$$\sqrt{\det(\mathbb{g}_{\bar{a}\bar{b}})} = r^2\cos\phi \tag{4.115}$$

so that the volume element is

$$\mathrm{d}V = r^2\cos\phi\,\mathrm{d}r\,\mathrm{d}\lambda\,\mathrm{d}\phi. \tag{4.116}$$

The covariant Levi-Civita tensor has the spherical representation

$$\varepsilon_{\bar{a}\bar{b}\bar{c}} = (r^2\cos\phi)\,\epsilon_{\bar{a}\bar{b}\bar{c}}, \tag{4.117}$$

where $\epsilon_{\bar{a}\bar{b}\bar{c}}$ are components to the permutation symbol (i.e., the Cartesian components to the Levi-Civita tensor) from Section 1.6.1.

4.4.6 Components of a vector field

A vector field, $\mathbf{F}$, has Cartesian components, F^a, related to its spherical components, $F^{\bar{a}}$, via the transformation matrix, $F^{\bar{a}} = \Lambda^{\bar{a}}{}_a\, F^a$. This transformation leads to

$$F^{\bar{1}} = (r\cos\phi)^{-1}\,[-F^x\sin\lambda + F^y\cos\lambda] \tag{4.118a}$$

$$F^{\bar{2}} = r^{-1}\,[-F^x\sin\phi\cos\lambda - F^y\sin\phi\sin\lambda + F^z\cos\phi] \tag{4.118b}$$

$$F^{\bar{3}} = F^x\cos\phi\cos\lambda + F^y\cos\phi\sin\lambda + F^z\sin\phi. \tag{4.118c}$$

Making use of the spherical unit vector (4.99a)-(4.99c) leads to the more tidy relations

$$(r\cos\phi)\,F^{\overline{1}} = \hat{\boldsymbol{\lambda}}\cdot\boldsymbol{F} \tag{4.119a}$$

$$r\,F^{\overline{2}} = \hat{\boldsymbol{\phi}}\cdot\boldsymbol{F} \tag{4.119b}$$

$$F^{\overline{3}} = \hat{\boldsymbol{r}}\cdot\boldsymbol{F}, \tag{4.119c}$$

where $\boldsymbol{F} = F^{a}\,\hat{\boldsymbol{e}}_{a}$ is the Cartesian representation.

4.4.7 Differential operators

In spherical coordinates the gradient operator $\nabla = \boldsymbol{e}^{a}\partial_{a}$ takes on the form

$$\nabla = \hat{\boldsymbol{\lambda}}\,(r\;\cos\phi)^{-1}\,\partial_{\lambda} + \hat{\boldsymbol{\phi}}\,r^{-1}\,\partial_{\phi} + \hat{\boldsymbol{r}}\,\partial_{r}, \tag{4.120}$$

and the covariant divergence of a vector field is given by

$$\nabla_{\overline{a}}F^{\overline{a}} = (r^{2}\cos\phi)^{-1}\partial_{\overline{a}}[r^{2}\cos\phi\,F^{\overline{a}}] \tag{4.121a}$$

$$= (r^{2}\cos\phi)^{-1}\left(\partial_{\lambda}[r^{2}\cos\phi\,F^{\overline{1}}] + \partial_{\phi}[r^{2}\cos\phi\,F^{\overline{2}}] + \partial_{r}[r^{2}\cos\phi\,F^{\overline{3}}]\right) \tag{4.121b}$$

$$= \frac{1}{r\cos\phi}\,\frac{\partial(\hat{\boldsymbol{\lambda}}\cdot\boldsymbol{F})}{\partial\lambda} + \frac{1}{r\cos\phi}\,\frac{\partial\,(\hat{\boldsymbol{\phi}}\cdot\boldsymbol{F}\;\cos\phi)}{\partial\phi} + \frac{1}{r^{2}}\,\frac{\partial\,(\hat{\boldsymbol{r}}\cdot\boldsymbol{F}\,r^{2})}{\partial r}. \tag{4.121c}$$

The covariant Laplacian of a scalar, $\nabla^{2}\psi = \nabla\cdot\nabla\psi$, is given by

$$\nabla^{2}\psi = \nabla\cdot\left[\hat{\boldsymbol{\lambda}}\,(r\cos\phi)^{-1}\,\partial_{\lambda}\psi + \hat{\boldsymbol{\phi}}\,r^{-1}\,\partial_{\phi}\psi + \hat{\boldsymbol{r}}\,\partial_{r}\psi\right] \tag{4.122a}$$

$$= \frac{1}{(r\cos\phi)^{2}}\,\frac{\partial^{2}\psi}{\partial\lambda^{2}} + \frac{1}{r^{2}\cos\phi}\,\frac{\partial}{\partial\phi}\left[\cos\phi\,\frac{\partial\psi}{\partial\phi}\right] + \frac{1}{r^{2}}\,\frac{\partial}{\partial r}\left[r^{2}\,\frac{\partial\psi}{\partial r}\right]. \tag{4.122b}$$

The covariant curl (Section 3.21) takes the form

$$(\mathrm{curl}\,\boldsymbol{F})^{\overline{1}} = (r^{2}\cos\phi)^{-1}\,[\partial_{\phi}\,F^{\overline{3}} - \partial_{r}\,(r^{2}\,F^{\overline{2}})] \tag{4.123a}$$

$$(\mathrm{curl}\,\boldsymbol{F})^{\overline{2}} = (r^{2}\cos\phi)^{-1}\,[\partial_{r}\,(r^{2}\cos^{2}\phi\,F^{\overline{1}}) - \partial_{\lambda}\,F^{\overline{3}}] \tag{4.123b}$$

$$(\mathrm{curl}\,\boldsymbol{F})^{\overline{3}} = (r^{2}\cos\phi)^{-1}\,[\partial_{\lambda}\,(r^{2}\,F^{\overline{2}}) - \partial_{\phi}\,(r^{2}\cos^{2}\phi\,F^{\overline{1}})], \tag{4.123c}$$

which can be written in the more conventional form[2]

$$r\cos\phi\,(\mathrm{curl}\,\boldsymbol{F})^{\overline{1}} = \frac{1}{r}\left[\frac{\partial(\hat{\boldsymbol{r}}\cdot\boldsymbol{F})}{\partial\phi} - \frac{\partial(r\,\hat{\boldsymbol{\phi}}\cdot\boldsymbol{F})}{\partial r}\right] \tag{4.124a}$$

$$r\,(\mathrm{curl}\,\boldsymbol{F})^{\overline{2}} = \frac{1}{r}\left[\frac{\partial(r\,\hat{\boldsymbol{\lambda}}\cdot\boldsymbol{F})}{\partial r} - \frac{1}{\cos\phi}\,\frac{\partial(\hat{\boldsymbol{r}}\cdot\boldsymbol{F})}{\partial\lambda}\right] \tag{4.124b}$$

$$(\mathrm{curl}\,\boldsymbol{F})^{\overline{3}} = \frac{1}{r\cos\phi}\left[\frac{\partial(\hat{\boldsymbol{\phi}}\cdot\boldsymbol{F})}{\partial\lambda} - \frac{\partial(\cos\phi\,\hat{\boldsymbol{\lambda}}\cdot\boldsymbol{F})}{\partial\phi}\right]. \tag{4.124c}$$

[2]Equations (4.124a)-(4.124c) correspond to equation (2.33) of *Vallis* (2017).

4.4.8 Summary

We here summarize the spherical coordinate version of some common mathematical operators:

$$(\lambda, \phi, r) = (x^{\bar{1}}, x^{\bar{2}}, x^{\bar{3}}) \tag{4.125}$$

$$(r\cos\phi)\, F^{\bar{1}} = \hat{\boldsymbol{\lambda}} \cdot \boldsymbol{F} \quad r\, F^{\bar{2}} = \hat{\boldsymbol{\phi}} \cdot \boldsymbol{F} \quad F^{\bar{3}} = \hat{\boldsymbol{r}} \cdot \boldsymbol{F} \tag{4.126}$$

$$\nabla = \hat{\boldsymbol{\lambda}}\, (r\, \cos\phi)^{-1}\, \partial_\lambda + \hat{\boldsymbol{\phi}}\, r^{-1}\, \partial_\phi + \hat{\boldsymbol{r}}\, \partial_r \tag{4.127}$$

$$\nabla_{\mathrm{h}} = \hat{\boldsymbol{\lambda}}\, (r\, \cos\phi)^{-1}\, \partial_\lambda + \hat{\boldsymbol{\phi}}\, r^{-1}\, \partial_\phi \tag{4.128}$$

$$\nabla_{\bar{a}} F^{\bar{a}} = \frac{1}{r\cos\phi} \frac{\partial(\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{F})}{\partial\lambda} + \frac{1}{r\cos\phi} \frac{\partial\,(\hat{\boldsymbol{\phi}} \cdot \boldsymbol{F}\, \cos\phi)}{\partial\phi} + \frac{1}{r^2} \frac{\partial\,(\hat{\boldsymbol{r}} \cdot \boldsymbol{F}\, r^2)}{\partial r} \tag{4.129}$$

$$\nabla^2\psi = \frac{1}{(r\cos\phi)^2} \frac{\partial^2\psi}{\partial\lambda^2} + \frac{1}{r^2\cos\phi} \frac{\partial}{\partial\phi} \left[\cos\phi\, \frac{\partial\psi}{\partial\phi}\right] + \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2\, \frac{\partial\psi}{\partial r}\right] \tag{4.130}$$

$$(\nabla \times \boldsymbol{F})_{\bar{a}} = \varepsilon_{\bar{a}\bar{b}\bar{c}} \partial_{\bar{b}} F^{\bar{c}} \quad \text{see equations (4.124a)} - \text{(4.124c)}. \tag{4.131}$$

Note that we wrote the vector components in terms of the non-coordinate orthonormal basis vectors, $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, which accords with usage in most of the geophysical fluids literature.

4.5 Generalized orthogonal coordinates

In this section we consider generalized orthogonal coordinates, which occur frequently in numerical models of the ocean and atmosphere, particularly in the horizontal directions.

4.5.1 Lamé coefficents for stretching/scaling

Generalized orthogonal coordinates provide a generalization of the Cartesian coordinates, spherical coordinates, and cylindrical-polar coordinates already considered in this chapter. They do so by making use of a nonsingular and orthogonal set of coordinates defined such that the metric tensor and its inverse are diagonal

$$\mathrm{g}_{\bar{a}\bar{b}} = \boldsymbol{e}_{\bar{a}} \cdot \boldsymbol{e}_{\bar{b}} = \begin{bmatrix} (h_{\bar{1}})^2 & 0 & 0 \\ 0 & (h_{\bar{2}})^2 & 0 \\ 0 & 0 & (h_{\bar{3}})^2 \end{bmatrix} \tag{4.132a}$$

$$\mathrm{g}^{\bar{a}\bar{b}} = \boldsymbol{e}^{\bar{a}} \cdot \boldsymbol{e}^{\bar{b}} = \begin{bmatrix} (h^{\bar{1}})^2 & 0 & 0 \\ 0 & (h^{\bar{2}})^2 & 0 \\ 0 & 0 & (h^{\bar{3}})^2 \end{bmatrix}. \tag{4.132b}$$

In these expressions, $h_{\bar{a}} > 0$ are positive and time-independent stretching or scaling functions that are dependent on the spatial position

$$h_{\bar{a}} = h_{\bar{a}}(\xi^{\bar{a}}) = h_{\bar{a}}(\xi^{\bar{1}}, \xi^{\bar{2}}, \xi^{\bar{3}}), \tag{4.133}$$

and their inverses are given by

$$h^{\bar{a}} = 1/h_{\bar{a}}. \tag{4.134}$$

The corresponding squared line element is given by

$$(\mathrm{d}s)^2 = \mathbb{g}_{ab}\,\mathrm{d}\xi^a\,\mathrm{d}\xi^b = \mathbb{g}_{\overline{a}\overline{b}}\,\mathrm{d}\xi^{\overline{a}}\,\mathrm{d}\xi^{\overline{b}} = \sum_{\overline{a}=\overline{1}}^{\overline{3}}(h_{\overline{a}}\,\mathrm{d}\xi^{\overline{a}})^2, \tag{4.135}$$

where we introduced the coordinates ξ^a and corresponding metric tensor representation, $\mathbb{g}_{ab}$.

The strecthing functions, $h_{\overline{a}}$, are referred to as Lamé coefficients in the differential geometry literature. In this section, we assume that the Lamé coefficients are time-independent, so that they are direct generalizations of spherical and cylindrical-polar coordinates.

Volume element

The volume element is expressed as

$$\mathrm{d}V = \mathsf{g}\,\mathrm{d}\xi^{\overline{1}}\,\mathrm{d}\xi^{\overline{2}}\,\mathrm{d}\xi^{\overline{3}} = h_{\overline{1}}\,h_{\overline{2}}\,h_{\overline{3}}\,\mathrm{d}\xi^{\overline{1}}\,\mathrm{d}\xi^{\overline{2}}\,\mathrm{d}\xi^{\overline{3}}, \tag{4.136}$$

where the square root of the metric tensor determinant is

$$\mathsf{g} = \sqrt{\det \mathbb{g}_{\overline{a}\overline{b}}} = h_{\overline{1}}\,h_{\overline{2}}\,h_{\overline{3}}, \tag{4.137}$$

thus determining how the volume of a region of space is measured.

Basis vectors

Generalized orthogonal coordinates have an orthogonal set of basis vectors

$$\boldsymbol{e}_{\overline{a}} = h_{\overline{a}}\,\boldsymbol{e}_{\hat{\overline{a}}} \quad \text{no implied sum}, \tag{4.138}$$

where we introduced the dimensionless non-coordinate unit directions, $\boldsymbol{e}_{\hat{\overline{a}}}$, from Section 4.1. The corresponding one-form basis is given by

$$\boldsymbol{e}^{\overline{a}} = (h_{\overline{a}})^{-1}\,\boldsymbol{e}^{\hat{\overline{a}}}, \tag{4.139}$$

where, again as in Section 4.1, there is no distinction between raised and lowered indices on the unit directions

$$\boldsymbol{e}_{\hat{\overline{a}}} = \boldsymbol{e}^{\hat{\overline{a}}}. \tag{4.140}$$

4.5.2 Spherical coordinates as an example

The spherical coordinates, $\xi^{\overline{a}} = (\lambda, \phi, r)$, from Section 4.4 provide an example set of orthogonal curvilinear coordinates, in which case we have the Lamé coefficients

$$h_{\overline{1}} = r\,\cos\phi \quad \text{and} \quad h_{\overline{2}} = r \quad \text{and} \quad h_{\overline{3}} = 1, \tag{4.141}$$

the volume element

$$\mathrm{d}V = h_{\overline{1}}\,h_{\overline{2}}\,h_{\overline{3}}\,\mathrm{d}\lambda\,\mathrm{d}\phi\,\mathrm{d}r = r^2\cos\phi\,\mathrm{d}\lambda\,\mathrm{d}\phi\,\mathrm{d}r, \tag{4.142}$$

the basis vectors

$$\boldsymbol{e}_{\overline{1}} = r\,\cos\phi\,\hat{\boldsymbol{\lambda}} \quad \text{and} \quad \boldsymbol{e}_{\overline{2}} = r\,\hat{\boldsymbol{\phi}} \quad \text{and} \quad \boldsymbol{e}_{\overline{3}} = \hat{\boldsymbol{r}}, \tag{4.143}$$

and basis one-forms

$$\boldsymbol{e}^{\overline{1}} = (r\,\cos\phi)^{-1}\,\hat{\boldsymbol{\lambda}} \quad \text{and} \quad \boldsymbol{e}^{\overline{2}} = r^{-1}\,\hat{\boldsymbol{\phi}} \quad \text{and} \quad \boldsymbol{e}^{\overline{3}} = \hat{\boldsymbol{r}}. \tag{4.144}$$

For spherical coordinates, there is no stretching in the vertical, so that $h_{\overline{3}} = 1$, in which case we commonly refer to spherical coordinates as generalized orthogonal horizontal (lateral) coordinates. A nontrivial vertical stretching is rarely considered for orthogonal coordinates in geophysical fluid mechanics. Instead, we more commonly use a time-dependent generalized vertical coordinate, which forms a non-orthogonal set of coordinates with the horizontal directions, and with the associated mathematical physics treated in VOLUME 4. For global models with a generalized vertical coordinate, they are typically combined with generalized orthogonal horizontal coordinates, thus making the mathematics of the present section relevant for those cases.

4.5.3 Lamé coefficients relative to spherical coordinates

Let us label the spherical coordinates with ξ^a, so that $\xi^a = (\lambda, \phi, r)$, and now let $\xi^{\overline{a}}$ be another set of orthogonal coordinates, such as those used in the ocean modeling community to remove the spherical coordinate singularies from the ocean domain (e.g., *Murray* (1996) and *Madec and Imbard* (1996)). Introduce the transformation matrix,

$$\Lambda^{a}{}_{\overline{a}} = \frac{\partial \xi^a}{\partial \xi^{\overline{a}}}, \tag{4.145}$$

so that the generalized orthogonal representation of the metric tensor is related to the spherical representation by

$$\mathbb{g}_{\overline{a}\overline{b}} = \Lambda^{a}{}_{\overline{a}}\,\Lambda^{b}{}_{\overline{b}}\,\mathbb{g}_{ab}. \tag{4.146}$$

Since both $\mathbb{g}_{\overline{a}\overline{b}}$ and $\mathbb{g}_{ab}$ are diagonal, we can write the Lamé coefficients as

$$(h_{\overline{1}})^2 = g_{ab}\,\Lambda^{a}{}_{\overline{1}}\,\Lambda^{b}{}_{\overline{1}} \tag{4.147a}$$
$$= g_{11}\,\Lambda^{1}{}_{\overline{1}}\,\Lambda^{1}{}_{\overline{1}} + g_{22}\,\Lambda^{2}{}_{\overline{1}}\,\Lambda^{2}{}_{\overline{1}} + g_{33}\,\Lambda^{3}{}_{\overline{1}}\,\Lambda^{3}{}_{\overline{1}} \tag{4.147b}$$
$$= (r\cos\phi\,\Lambda^{1}{}_{\overline{1}})^2 + (r\,\Lambda^{2}{}_{\overline{1}})^2 + (\Lambda^{3}{}_{\overline{1}})^2. \tag{4.147c}$$

Similarly, the other Lamé coefficients are

$$(h_{\overline{2}})^2 = (r\cos\phi\,\Lambda^{1}{}_{\overline{2}})^2 + (r\,\Lambda^{2}{}_{\overline{2}})^2 + (\Lambda^{3}{}_{\overline{2}})^2 \tag{4.148a}$$
$$(h_{\overline{3}})^2 = (r\cos\phi\,\Lambda^{1}{}_{\overline{3}})^2 + (r\,\Lambda^{2}{}_{\overline{3}})^2 + (\Lambda^{3}{}_{\overline{3}})^2. \tag{4.148b}$$

In this manner, a specification of the coordinate transformation

$$\xi^{\overline{a}} = \xi^{\overline{a}}(\lambda, \phi, r) \tag{4.149}$$

provides the means to compute the Lamé coefficients.

4.5.4 Physical components of tensors

We here further the notion of physical tensor components from Section 4.1.4. We emphasize that the physical components all have the same physical dimensions, thus motivating their names. Consequently, physical components to tensors they are often the representations that appear in applications, such as in numerical models.

Contravariant and covariant physical components

In terms of a locally orthogonal basis of unit directions, $\boldsymbol{e}_{\hat{\overline{a}}}$, an arbitrary vector, $\mathbf{F}$, takes the form

$$\mathbf{F} = F^{\overline{a}}\,\boldsymbol{e}_{\overline{a}} = (F^{\overline{a}}\,h_{\overline{a}})\,(h^{\overline{a}}\,\boldsymbol{e}_{\overline{a}}) \equiv F^{\hat{\overline{a}}}\,\boldsymbol{e}_{\hat{\overline{a}}}. \tag{4.150}$$

Importantly, there is no summation performed on the $F^{\overline{a}}\,h_{\overline{a}}$ product nor the $h^{\overline{a}}\,\boldsymbol{e}_{\overline{a}}$ product when defining the physical components and the unit vectors

$$F^{\hat{\overline{a}}} = F^{\overline{a}}\,h_{\overline{a}} \quad \text{and} \quad \boldsymbol{e}_{\hat{\overline{a}}} = h^{\overline{a}}\,\boldsymbol{e}_{\overline{a}} \quad \text{no summation.} \tag{4.151}$$

More explicitly, introducing the summation symbol we find that equation (4.150) is given by

$$\mathbf{F} = \sum_{\overline{a}=\overline{1}}^{\overline{3}} F^{\hat{\overline{a}}}\,\boldsymbol{e}_{\hat{\overline{a}}} = (F^{\overline{1}}\,h_{\overline{1}})\,(h^{\overline{1}}\,\boldsymbol{e}_{\overline{1}}) + (F^{\overline{2}}\,h_{\overline{2}})\,(h^{\overline{2}}\,\boldsymbol{e}_{\overline{2}}) + (F^{\overline{3}}\,h_{\overline{3}})\,(h^{\overline{3}}\,\boldsymbol{e}_{\overline{3}}). \tag{4.152}$$

In the same manner, the covariant physical representation of a tensor is introduced according to

$$\mathbf{F} = \sum_{\overline{a}=\overline{1}}^{\overline{3}} F_{\overline{a}}\,\boldsymbol{e}^{\overline{a}} = (F_{\overline{1}}\,h^{\overline{1}})\,\boldsymbol{e}^{\hat{\overline{1}}} + (F_{\overline{2}}\,h^{\overline{2}})\,\boldsymbol{e}^{\hat{\overline{2}}} + (F_{\overline{3}}\,h^{\overline{3}})\,\boldsymbol{e}^{\hat{\overline{3}}} = \sum_{\overline{a}=\overline{1}}^{\overline{3}}(F_{\overline{a}}\,h^{\overline{a}})\,\boldsymbol{e}^{\hat{\overline{a}}} \equiv \sum_{\overline{a}=\overline{1}}^{\overline{3}} F_{\hat{\overline{a}}}\,\boldsymbol{e}^{\hat{\overline{a}}}. \tag{4.153}$$

As noted in Section 4.1.4, the unit vectors, $\boldsymbol{e}_{\hat{\overline{a}}}$, are all dimensionless, which means that the non-coordinate components, $F^{\hat{\overline{a}}}$, each have the same physical dimensions. This property motivates us to refer to $F^{\hat{\overline{a}}}$ as the contravariant physical components to $\mathbf{F}$. In the same manner, we refer to $F_{\hat{\overline{a}}}$ as the covariant physical components.

Transformation of a tensor's physical components

The physical components to a tensor transform in a manner distinct from true tensor components, so that for the contravariant components we have

$$F^{\hat{\overline{a}}} = F^{\overline{a}}\,h_{\overline{a}} = (\Lambda^{\overline{a}}{}_{a}\,F^{a})\,h_{\overline{a}} = \Lambda^{\overline{a}}{}_{a}\,(F^{\hat{a}}/h_{a})\,h_{\overline{a}} = \Lambda^{\overline{a}}{}_{a}\,F^{\hat{a}}\,(h_{\overline{a}}/h_{a}). \tag{4.154}$$

The indices are a bit confusing, so let us reintroduce the summation symbol, which ranges over $a = 1, 2, 3$ (and with $\overline{a}$ fixed) according to

$$F^{\hat{\overline{a}}} = \sum_{a=1}^{3} \Lambda^{\overline{a}}{}_{a}\,F^{\hat{a}}\,(h_{\overline{a}}/h_{a}) = \Lambda^{\overline{a}}{}_{1}\,F^{\hat{1}}\,(h_{\overline{a}}/h_{1}) + \Lambda^{\overline{a}}{}_{2}\,F^{\hat{2}}\,(h_{\overline{a}}/h_{2}) + \Lambda^{\overline{a}}{}_{3}\,F^{\hat{3}}\,(h_{\overline{a}}/h_{3}). \tag{4.155}$$

The covariant physical representation of a tensor has the transformation properties

$$F_{\hat{\overline{a}}} = F_{\overline{a}}\,h^{\overline{a}} = (\Lambda^{a}{}_{\overline{a}}\,F_{a})\,h^{\overline{a}} = \Lambda^{a}{}_{\overline{a}}\,(F_{a}/h^{a})\,h^{\overline{a}} = \Lambda^{a}{}_{\overline{a}}\,F_{\hat{a}}\,(h^{\overline{a}}/h^{a}), \tag{4.156}$$

with the summation convention holding just like for the contravariant components in equation (4.155).

In the contravariant case (4.155) and the covariant case (4.156), the ratio of stretching functions distinguishes the transformation properties of physical components from true tensor components. Notably, if the coordinates, ξ^{a}, are Cartesian, then $h_{a} = 1$, in which case the transformations reduce to

$$F^{\hat{\overline{a}}} = h_{\overline{a}}\, \Lambda^{\overline{a}}{}_{a}\, F^{\hat{a}} \quad \text{and} \quad F_{\hat{\overline{a}}} = h^{\overline{a}}\, \Lambda^{a}{}_{\overline{a}}\, F_{\hat{a}} \quad \Longleftarrow \xi^{a} \text{ are Cartesian.} \tag{4.157}$$

Physical components to a second order tensor

The previous results for the contravariant and covariant representation of a first order tensor, $\mathbf{F}$, generalize to an arbitrary tensor, $\mathbf{T}$. For example, if one expresses a second order tensor in terms of its $(2,0)$ contravariant components, these components take the following form when using orthogonal coordinates.

$$\mathbf{T} = T^{\overline{a}\overline{b}}\, \boldsymbol{e}_{\overline{a}} \otimes \boldsymbol{e}_{\overline{b}} = (h_{\overline{a}}\, h_{\overline{b}}\, T^{\overline{a}\overline{b}})\, (h^{\overline{a}}\, h^{\overline{b}}\, \boldsymbol{e}_{\overline{a}} \otimes \boldsymbol{e}_{\overline{b}}) \equiv T^{\hat{\overline{a}}\hat{\overline{b}}}\, \boldsymbol{e}_{\hat{\overline{a}}} \otimes \boldsymbol{e}_{\hat{\overline{b}}}. \tag{4.158}$$

Each of the physical components, $T^{\hat{\overline{a}}\hat{\overline{b}}}$, has the same physical dimensions.

4.5.5 Pseudo-Cartesian coordinates

Orthogonal coordinates afford an added feature that is quite useful to exploit when discretizing the equations for a numerical model. Since the generalized coordinates are orthogonal, a coordinate increment in one direction is independent of the other direction. For example,

$$\partial_{\overline{a}}(\mathrm{d}\xi^{\overline{b}}) = \delta^{\overline{b}}{}_{\overline{a}}. \tag{4.159}$$

We can thus introduce pseudo-Cartesian increments (with dimensions of length)

$$\mathrm{d}x = h_{\overline{1}}\, \mathrm{d}\xi^{\overline{1}} \quad \text{and} \quad \mathrm{d}y = h_{\overline{2}}\, \mathrm{d}\xi^{\overline{2}} \quad \text{and} \quad \mathrm{d}z = h_{\overline{3}}\, \mathrm{d}\xi^{\overline{3}}. \tag{4.160}$$

These increments are convenient since they determine the distance factors required to discretize the equations of a numerical model.

4.5.6 Gradient operator

The gradient operator, ∇, is naturally represented in terms of its covariant components. Using orthogonal coordinates it can be represented as

$$\nabla = \boldsymbol{e}^{\overline{a}}\, \partial_{\overline{a}} = (h_{\overline{a}}\, \boldsymbol{e}^{\overline{a}})\, (h^{\overline{a}}\, \partial_{\overline{a}}) = \boldsymbol{e}^{\hat{\overline{a}}}\, \partial_{\hat{\overline{a}}}, \tag{4.161}$$

where we introduced the physical components

$$\partial_{\hat{\overline{a}}} = h^{\overline{a}}\, \partial_{\overline{a}} \quad \text{no summation.} \tag{4.162}$$

For example, with the spherical coordinates of Section 4.4, the gradient operator is

$$\nabla = \hat{\boldsymbol{\lambda}}\, (r\, \cos\phi)^{-1}\, \partial_{\lambda} + \hat{\boldsymbol{\phi}}\, r^{-1}\, \partial_{\phi} + \hat{\boldsymbol{r}}\, \partial_{r}. \tag{4.163}$$

Following from Section 4.5.5, we write the physical components of the partial derivative in their pseudo-Cartesian form

$$\partial_{\hat{\overline{a}}} = (\partial_x, \partial_y, \partial_z). \tag{4.164}$$

The nonflat nature of the sphere manifests itself in the nonvanishing commutator

$$[\partial_x, \partial_y] = \partial_x \, \partial_y - \partial_y \, \partial_x = (\partial_x \ln \mathrm{d}y) \, \partial_y - (\partial_y \ln \mathrm{d}x) \, \partial_x, \tag{4.165}$$

4.5.7 Comments on the Lamé coefficients

A feature common to non-Cartesian coordinates concerns the presence of spatially dependent factors in front of certain components of the gradient operators. Indeed, recall the gradient for the three coordinates discussed in this chapter:

$$\nabla = \hat{\boldsymbol{x}} \, \frac{\partial}{\partial x} + \hat{\boldsymbol{y}} \, \frac{\partial}{\partial y} + \hat{\boldsymbol{z}} \, \frac{\partial}{\partial z} \qquad \text{Cartesian} \tag{4.166a}$$

$$\nabla = \hat{\boldsymbol{r}} \, \frac{\partial}{\partial r} + \frac{\hat{\boldsymbol{\vartheta}}}{r} \, \frac{\partial}{\partial \vartheta} + \hat{\boldsymbol{z}} \, \frac{\partial}{\partial z} \qquad \text{cylindrical-polar} \tag{4.166b}$$

$$\nabla = \frac{\hat{\boldsymbol{\lambda}}}{r \, \cos\phi} \, \frac{\partial}{\partial \lambda} + \frac{\hat{\boldsymbol{\phi}}}{r} \, \frac{\partial}{\partial \phi} + \hat{\boldsymbol{r}} \, \frac{\partial}{\partial r} \qquad \text{spherical,} \tag{4.166c}$$

where we recall that r for the spherical coordinates measures the distance from the origin, whereas r for cylindrical-polar coordinates measures the distance from the vertical axis. Observe the r^{-1} dependence for the angular components of the gradient for both spherical and cylindrical-polar coordinates. This dependence arises since as r increases, the surfaces of constant r become flatter, a result of their larger radius of curvature.

To further help understand the r^{-1} factor, consider two points with the same r but distinct angular coordinates. As r increases, so too does the distance between the two points, even though their angular positions remain unchanged. Speaking prosaically, this property is why the outer portion of a pizza pie is bigger than the inner portion. Correspondingly, when measuring the derivative of a function in the angular direction, that derivative must be weighted by r^{-1} to account for the difference in the arc-distance (more precisely the geodesic distance) between the two points. The same idea explains the $\cos\phi$ factor on the longitudinal derivative in spherical coordinates. Each of these notions are encapsulated in the spatial dependence of the basis vectors for non-Cartesian coordinates.

A practical implication of the r and $r \cos\phi$ factors appearing in the spherical coordinate gradient operator arises when performing a vertical integral of the fluid mechanical equations. Such integrals are often used to study reduced order versions of the full three-dimensional equations. But to be used in practice, such integrals must be done either with Cartesian coordinates or for those cases where it is acceptable to set the r^{-1} factor to a constant, such as occurs for the shallow fluid approximation examined in VOLUME 2. Indeed, the vertically integrated equations are of most practical use only when making a shallow fluid approximation. If this point seems obscure now, it will be clarified when studying the depth integrated mechanical energy, the depth integrated momentum equation, the depth integrated angular momentum, all in VOLUME 2, as well as the depth integrated vorticity equations in VOLUME 3.

4.5.8 Further study

This section is based on Section 21.11 of *Griffies* (2004). Similar treatments can also be found at the end of chapter 1 in *Morse and Feshbach* (1953), various sections in *Aris* (1962), and Appendix 2 of *Batchelor* (1967). Exercise 4.6 specializes to orthogonal coordinates the expressions for the covariant divergence, Laplacian, and curl.

4.6 The velocity gradient tensor

In this section we offer a case study that exemplifies much of the coordinate work presented in this chapter, with this example also connecting to fluid kinematics studied in VOLUME 2. Namely, we derive coordinate representations for the velocity gradient tensor

$$\mathbf{G} = \boldsymbol{\nabla}\mathbf{v}, \tag{4.167}$$

where $\mathbf{v}$ is the velocity vector field for the fluid, and $\boldsymbol{\nabla}$ is the covariant derivative. Exposing tensor labels for a $(1,1)$ coordinate representation yields

$$\mathbf{G} = \boldsymbol{e}^{m} \otimes \nabla_{m}(\boldsymbol{e}_{n}\, v^{n}) = \boldsymbol{e}^{\overline{m}} \otimes \nabla_{\overline{m}}(\boldsymbol{e}_{\overline{n}}\, v^{\overline{n}}), \tag{4.168}$$

where m and $\overline{m}$ are tensor labels for two different coordinate systems. For example, using a Cartesian coordinate system yields the representation

$$\mathbf{G} = \boldsymbol{e}^{m} \otimes \boldsymbol{e}_{n}\, \partial_{m} v^{n} \equiv \boldsymbol{e}^{m} \otimes \boldsymbol{e}_{n}\, G^{n}{}_{m}. \tag{4.169}$$

Here, we noted that the Cartesian basis vectors, $\boldsymbol{e}_{n}$, are independent of space, so that they can be pulled outside of the derivative. More care is needed with general coordinates, as pursued in the following. Also, we chose a convention for $G^{n}{}_{m}$ whereby the nearest label corresponds to the velocity label, and the furthest label to the derivative.

Since the covariant derivative, $\boldsymbol{\nabla}$, has physical dimensions of inverse length, and the velocity field, $\mathbf{v}$, has dimensions of length per time, we see that $\mathbf{G}$ has physical dimensions of inverse time; i.e., a rate. Indeed, as studied in VOLUME 2, the strain rate tensor, $\mathbf{S}$, is the symmetric part of $\mathbf{G}$, and the rotation rate tensor, $\mathbf{R}$, is the anti-symmetric portion:

$$\mathbf{S} \equiv \tfrac{1}{2}(\mathbf{G} + \mathbf{G}^{\top}) \quad \text{and} \quad \mathbf{R} \equiv \tfrac{1}{2}(\mathbf{G} - \mathbf{G}^{\top}). \tag{4.170}$$

These two tensors appear throughout fluid mechanics.

4.6.1 Physical component representation

The velocity components are given by the coordinate components

$$v^{\overline{m}} = \dot{\xi}^{\overline{m}}, \tag{4.171}$$

so that the physical dimension for each velocity component is determined by the physical dimension of the coordinates. In turn, the elements of a coordinate representation of $\mathbf{G}$ generally have distinct physical dimensions. It is only when building the full tensor that we get an inverse time dimension for any particular coordinate representation, which we can write as

$$\mathbf{G} = \boldsymbol{e}^{\overline{m}} \otimes \boldsymbol{e}_{\overline{n}}\, G^{\overline{n}}{}_{\overline{m}} = \boldsymbol{e}^{\hat{\overline{m}}} \otimes \boldsymbol{e}_{\hat{\overline{n}}}\, G^{\hat{\overline{n}}}{}_{\hat{\overline{m}}}. \tag{4.172}$$

In the second equality we followed the approach from Section 4.1 by introducing non-coordinate basis vectors, $\boldsymbol{e}_{\hat{\overline{n}}}$, and one-forms, $\boldsymbol{e}^{\hat{\overline{m}}}$, each of which are dimensionless. Correspondingly, we introduced the physical representation of the velocity gradient tensor, $G^{\hat{\overline{n}}}{}_{\hat{\overline{m}}}$, with each component having physical dimensions of inverse time.

4.6.2 The covariant derivative of a $(1,0)$ vector

We make use of methods from Section 3.15.2 to compute the covariant derivative of the $(1,0)$ representation of the velocity vector

$$
\begin{aligned}
\mathbf{G} &= \boldsymbol{\nabla}(\boldsymbol{e}_{\overline{n}}\, v^{\overline{n}}) && \text{equation (4.167)} && (4.173a)\\
&= \boldsymbol{e}^{\overline{m}} \otimes \nabla_{\overline{m}}(\boldsymbol{e}_{\overline{n}}\, v^{\overline{n}}) && \boldsymbol{\nabla} = \boldsymbol{e}^{\overline{m}}\, \nabla_{\overline{m}} && (4.173b)\\
&= \boldsymbol{e}^{\overline{m}} \otimes (\nabla_{\overline{m}}\boldsymbol{e}_{\overline{n}})\, v^{\overline{n}} + \boldsymbol{e}^{\overline{m}} \otimes \boldsymbol{e}_{\overline{n}}\, \nabla_{\overline{m}} v^{\overline{n}} && \text{product rule} && (4.173c)\\
&= \boldsymbol{e}^{\overline{m}} \otimes \boldsymbol{e}_{\overline{p}}\, \Gamma^{\overline{p}}_{\overline{m}\overline{n}}\, v^{\overline{n}} + \boldsymbol{e}^{\overline{m}} \otimes \boldsymbol{e}_{\overline{n}}\, \partial_{\overline{m}} v^{\overline{n}} && \nabla_{\overline{m}}\boldsymbol{e}_{\overline{n}} = \Gamma^{\overline{p}}_{\overline{m}\overline{n}}\, \boldsymbol{e}_{\overline{p}} \text{ and Section 3.15.2} && (4.173d)\\
&= \boldsymbol{e}^{\overline{m}} \otimes \boldsymbol{e}_{\overline{n}} \left[\partial_{\overline{m}} v^{\overline{n}} + \Gamma^{\overline{n}}_{\overline{m}\overline{p}}\, v^{\overline{p}}\right] && \text{relabel and rearrange} && (4.173e)\\
&\equiv \boldsymbol{e}^{\overline{m}} \otimes \boldsymbol{e}_{\overline{n}}\, (\boldsymbol{\nabla}\mathbf{v})^{\overline{n}}{}_{\overline{m}} && \text{equation (3.120b)} && (4.173f)\\
&\equiv \boldsymbol{e}^{\overline{m}} \otimes \boldsymbol{e}_{\overline{n}}\, G^{\overline{n}}{}_{\overline{m}} && \text{components to velocity gradient tensor.} && (4.173g)
\end{aligned}
$$

Operationally, for any chosen coordinate system we compute the Christoffel symbols by making use of equation (3.130)

$$
\Gamma^{\overline{p}}_{\overline{m}\overline{n}} = \tfrac{1}{2}\, \mathbb{g}^{\overline{p}\overline{d}}\, (\partial_{\overline{n}}\, \mathbb{g}_{\overline{d}\overline{m}} + \partial_{\overline{m}}\, \mathbb{g}_{\overline{d}\overline{n}} - \partial_{\overline{d}}\, \mathbb{g}_{\overline{m}\overline{n}}), \tag{4.174a}
$$

and then compute the velocity gradient tensor components

$$
G^{\overline{n}}{}_{\overline{m}} = \partial_{\overline{m}} v^{\overline{n}} + \Gamma^{\overline{n}}_{\overline{m}\overline{p}}\, v^{\overline{p}}. \tag{4.175}
$$

In the following, we pursue this calculation for cylindrical-polar coordinates and spherical coordinates. The derivations are straightforward, yet tedious. At each stage it is very useful to be mindful of physical dimensions, as doing so greatly assists in debugging the maths.

4.6.3 Cylindrical-polar coordinate representation

We here consider the cylindrical-polar coordinate system discussed in Section 4.3. In particular, we need the cylindrical-polar representation of the metric tensor, and its inverse, from Section 4.3.5

$$
\mathbb{g}_{\overline{m}\overline{q}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbb{g}^{\overline{a}\overline{b}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4.176}
$$

The corresponding basis vectors, $\boldsymbol{e}_{\overline{m}}$, are given by equation (4.53) in terms of the non-coordinate basis

$$
\boldsymbol{e}_r = \hat{\boldsymbol{r}} \quad \text{and} \quad \boldsymbol{e}_\vartheta = r\, \hat{\boldsymbol{\vartheta}} \quad \text{and} \quad \boldsymbol{e}_z = \hat{\boldsymbol{z}}, \tag{4.177}
$$

and the basis one-forms, $\boldsymbol{e}^{\overline{m}}$, are given by equation (4.58)

$$
\boldsymbol{e}^r = \hat{\boldsymbol{r}} \quad \text{and} \quad \boldsymbol{e}^\vartheta = r^{-1}\hat{\boldsymbol{\vartheta}} \quad \text{and} \quad \boldsymbol{e}^z = \hat{\boldsymbol{z}}, \tag{4.178}
$$

Making use of equation (4.174a) (see also Exercise 4.5), we find three non-zero Christoffel symbols

$$
\Gamma^{\overline{2}}_{\overline{1}\overline{2}} = \Gamma^{\vartheta}_{r\vartheta} = 1/r \quad \text{and} \quad \Gamma^{\overline{2}}_{\overline{2}\overline{1}} = \Gamma^{\vartheta}_{\vartheta r} = 1/r \quad \text{and} \quad \Gamma^{\overline{1}}_{\overline{2}\overline{2}} = \Gamma^{r}_{\vartheta\vartheta} = -r. \tag{4.179}
$$

The velocity gradient tensor in cylindrical-polar coordinates

We are led to the following cylindrical-polar coordinate components to the velocity gradient tensor

$$G^{\overline{1}}{}_{\overline{1}} = G^{r}{}_{r} = \partial_r v^r \tag{4.180a}$$
$$G^{\overline{1}}{}_{\overline{2}} = G^{r}{}_{\vartheta} = \partial_\vartheta v^r - r\, v^\vartheta \tag{4.180b}$$
$$G^{\overline{1}}{}_{\overline{3}} = G^{r}{}_{z} = \partial_z v^r \tag{4.180c}$$
$$G^{\overline{2}}{}_{\overline{1}} = G^{\vartheta}{}_{r} = \partial_r v^\vartheta + r^{-1}\, v^\vartheta \tag{4.180d}$$
$$G^{\overline{2}}{}_{\overline{2}} = G^{\vartheta}{}_{\vartheta} = \partial_\vartheta v^\vartheta + r^{-1}\, v^r \tag{4.180e}$$
$$G^{\overline{2}}{}_{\overline{3}} = G^{\vartheta}{}_{z} = \partial_z v^\vartheta \tag{4.180f}$$
$$G^{\overline{3}}{}_{\overline{1}} = G^{z}{}_{r} = \partial_r v^z \tag{4.180g}$$
$$G^{\overline{3}}{}_{\overline{2}} = G^{z}{}_{\vartheta} = \partial_\vartheta v^z \tag{4.180h}$$
$$G^{\overline{3}}{}_{\overline{3}} = G^{z}{}_{z} = \partial_z v^z. \tag{4.180i}$$

As a check, note that the trace of this coordinate representation yields the velocity divergence

$$G^{\overline{m}}{}_{\overline{m}} = \frac{1}{r}\,\frac{\partial(r\, v^r)}{\partial r} + \frac{\partial v^\vartheta}{\partial \vartheta} + \frac{\partial v^z}{\partial z}, \tag{4.181}$$

which agrees with equation (4.74b).

Physical component representation of the velocity gradient tensor

The velocity components for cylindrical-polar coordinates are given by

$$v^{\overline{m}} = (\dot{r}, \dot{\vartheta}, \dot{z}), \tag{4.182}$$

which means that some elements of $G_{\overline{m}}{}^{\overline{n}}$ have physical dimensions of inverse time, inverse (length-time), and length per time. We follow the discussion in Section 4.6.1 to identify the physical components

$$\mathbf{G} = \boldsymbol{e}^{\hat{\overline{m}}} \otimes \boldsymbol{e}_{\hat{\overline{n}}}\, G^{\hat{\overline{n}}}{}_{\hat{\overline{m}}}. \tag{4.183}$$

Making use of the basis vectors and one-forms from equations (4.177) and (4.178) yields

$$G^{\hat{\overline{1}}}{}_{\hat{\overline{1}}} = G^{\hat{r}}{}_{\hat{r}} = \partial_r \dot{r} \tag{4.184a}$$
$$G^{\hat{\overline{1}}}{}_{\hat{\overline{2}}} = G^{\hat{r}}{}_{\hat{\vartheta}} = r^{-1}\,\partial_\vartheta \dot{r} - \dot{\vartheta} \tag{4.184b}$$
$$G^{\hat{\overline{1}}}{}_{\hat{\overline{3}}} = G^{\hat{r}}{}_{\hat{z}} = \partial_z \dot{r} \tag{4.184c}$$
$$G^{\hat{\overline{2}}}{}_{\hat{\overline{1}}} = G^{\hat{\vartheta}}{}_{\hat{r}} = r\,\partial_r \dot{\vartheta} + \dot{\vartheta} \tag{4.184d}$$
$$G^{\hat{\overline{2}}}{}_{\hat{\overline{2}}} = G^{\hat{\vartheta}}{}_{\hat{\vartheta}} = \partial_\vartheta \dot{\vartheta} + r^{-1}\,\dot{r} \tag{4.184e}$$
$$G^{\hat{\overline{2}}}{}_{\hat{\overline{3}}} = G^{\hat{\vartheta}}{}_{\hat{z}} = r\,\partial_z \dot{\vartheta} \tag{4.184f}$$
$$G^{\hat{\overline{3}}}{}_{\hat{\overline{1}}} = G^{\hat{z}}{}_{\hat{r}} = \partial_r \dot{z} \tag{4.184g}$$
$$G^{\hat{\overline{3}}}{}_{\hat{\overline{2}}} = G^{\hat{z}}{}_{\hat{\vartheta}} = r^{-1}\,\partial_\vartheta \dot{z} \tag{4.184h}$$

$$G^{\hat{\bar{3}}}{}_{\hat{\bar{3}}} = G^{\hat{z}}{}_{\hat{z}} = \partial_z \dot{z}. \tag{4.184i}$$

As a final step, we introduce the physical components to the cylindrical-polar coordinate representation of the velocity vector

$$v^{\hat{r}} = \dot{r} \quad \text{and} \quad v^{\hat{\vartheta}} = r\,\dot{\vartheta} \quad \text{and} \quad v^{\hat{z}} = \dot{z}, \tag{4.185}$$

so that the physical components to the velocity gradient tensor take the form

$$G^{\hat{\bar{1}}}{}_{\hat{\bar{1}}} = G^{\hat{r}}{}_{\hat{r}} = \partial_r v^{\hat{r}} \tag{4.186a}$$
$$G^{\hat{\bar{1}}}{}_{\hat{\bar{2}}} = G^{\hat{r}}{}_{\hat{\vartheta}} = r^{-1}\,\partial_\vartheta v^{\hat{r}} - r^{-1}\,v^{\hat{\vartheta}} \tag{4.186b}$$
$$G^{\hat{\bar{1}}}{}_{\hat{\bar{3}}} = G^{\hat{r}}{}_{\hat{z}} = \partial_z v^{\hat{r}} \tag{4.186c}$$
$$G^{\hat{\bar{2}}}{}_{\hat{\bar{1}}} = G^{\hat{\vartheta}}{}_{\hat{r}} = \partial_r v^{\hat{\vartheta}} \tag{4.186d}$$
$$G^{\hat{\bar{2}}}{}_{\hat{\bar{2}}} = G^{\hat{\vartheta}}{}_{\hat{\vartheta}} = r^{-1}\,\partial_\vartheta v^{\hat{\vartheta}} + r^{-1}\,v^{\hat{r}} \tag{4.186e}$$
$$G^{\hat{\bar{2}}}{}_{\hat{\bar{3}}} = G^{\hat{\vartheta}}{}_{\hat{z}} = \partial_z v^{\hat{\vartheta}} \tag{4.186f}$$
$$G^{\hat{\bar{3}}}{}_{\hat{\bar{1}}} = G^{\hat{z}}{}_{\hat{r}} = \partial_r v^{\hat{z}} \tag{4.186g}$$
$$G^{\hat{\bar{3}}}{}_{\hat{\bar{2}}} = G^{\hat{z}}{}_{\hat{\vartheta}} = r^{-1}\,\partial_\vartheta v^{\hat{z}} \tag{4.186h}$$
$$G^{\hat{\bar{3}}}{}_{\hat{\bar{3}}} = G^{\hat{z}}{}_{\hat{z}} = \partial_z v^{\hat{z}}. \tag{4.186i}$$

As a check, note that the trace of this tensor is given by the velocity divergence

$$G^{\hat{\overline{m}}}{}_{\hat{\overline{m}}} = \frac{1}{r}\frac{\partial(r\,v^{\hat{r}})}{\partial r} + \frac{1}{r}\frac{\partial\,v^{\hat{\vartheta}}}{\partial\vartheta} + \frac{\partial v^{\hat{z}}}{\partial z}, \tag{4.187}$$

which agrees with equation (4.81).

4.6.4 Spherical coordinate representation

We now derive the spherical coordinate version of the velocity gradient tensor, making use of the spherical coordinates, (λ, ϕ, r), from Section 4.4 and following the same procedure as in Section 4.6.3 for cylindrical-polar coordinates.

The spherical coordinate representation of the metric tensor, and its inverse, are taken from Section 4.4.5

$$\mathbb{g}_{\overline{mq}} = \begin{bmatrix} (r\cos\phi)^2 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbb{g}^{\overline{mn}} = \begin{bmatrix} (r\cos\phi)^{-2} & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4.188}$$

The corresponding basis vectors, $\boldsymbol{e}_{\overline{m}}$, are given by equation (4.98) in terms of the non-coordinate basis

$$\boldsymbol{e}_\lambda = r\cos\phi\,\hat{\boldsymbol{\lambda}} \quad \text{and} \quad \boldsymbol{e}_\phi = r\,\hat{\boldsymbol{\phi}} \quad \text{and} \quad \boldsymbol{e}_r = \hat{\boldsymbol{r}}, \tag{4.189}$$

and the basis one-forms, $\boldsymbol{e}^{\overline{m}}$, are given by equation (4.104)

$$\boldsymbol{e}^\lambda = (r\cos\phi)^{-1}\,\hat{\boldsymbol{\lambda}} \quad \text{and} \quad \boldsymbol{e}^\phi = r^{-1}\hat{\boldsymbol{\phi}} \quad \text{and} \quad \boldsymbol{e}^r = \hat{\boldsymbol{r}}. \tag{4.190}$$

Making use of equation (4.174a) we find the following non-zero Christoffel symbols

$$\Gamma^{\overline{1}}_{\overline{21}} = \Gamma^{\lambda}_{\phi\lambda} = -\tan\phi \tag{4.191a}$$
$$\Gamma^{\overline{1}}_{\overline{12}} = \Gamma^{\lambda}_{\lambda\phi} = -\tan\phi \tag{4.191b}$$
$$\Gamma^{\overline{1}}_{\overline{13}} = \Gamma^{\lambda}_{\lambda r} = r^{-1} \tag{4.191c}$$
$$\Gamma^{\overline{1}}_{\overline{31}} = \Gamma^{\lambda}_{r\lambda} = r^{-1} \tag{4.191d}$$
$$\Gamma^{\overline{2}}_{\overline{11}} = \Gamma^{\phi}_{\lambda\lambda} = \cos\phi\sin\phi \tag{4.191e}$$
$$\Gamma^{\overline{2}}_{\overline{23}} = \Gamma^{\phi}_{\phi r} = r^{-1} \tag{4.191f}$$
$$\Gamma^{\overline{2}}_{\overline{32}} = \Gamma^{\phi}_{r\phi} = r^{-1} \tag{4.191g}$$
$$\Gamma^{\overline{3}}_{\overline{11}} = \Gamma^{r}_{\lambda\lambda} = -r\cos^2\phi \tag{4.191h}$$
$$\Gamma^{\overline{3}}_{\overline{22}} = \Gamma^{r}_{\phi\phi} = -r. \tag{4.191i}$$

The velocity gradient tensor in spherical coordinates

We are led to the following spherical coordinate components to the velocity gradient tensor

$$G^{\overline{1}}{}_{\overline{1}} = G^{\lambda}{}_{\lambda} = \partial_\lambda\dot{\lambda} - \dot{\phi}\tan\phi + r^{-1}\,\dot{r} \tag{4.192a}$$
$$G^{\overline{1}}{}_{\overline{2}} = G^{\lambda}{}_{\phi} = \partial_\phi\dot{\lambda} - \dot{\lambda}\tan\phi \tag{4.192b}$$
$$G^{\overline{1}}{}_{\overline{3}} = G^{\lambda}{}_{r} = \partial_r\dot{\lambda} + r^{-1}\,\dot{\lambda} \tag{4.192c}$$
$$G^{\overline{2}}{}_{\overline{1}} = G^{\phi}{}_{\lambda} = \partial_\lambda\dot{\phi} + \dot{\lambda}\cos\phi\sin\phi \tag{4.192d}$$
$$G^{\overline{2}}{}_{\overline{2}} = G^{\phi}{}_{\phi} = \partial_\phi\dot{\phi} + r^{-1}\,\dot{r} \tag{4.192e}$$
$$G^{\overline{2}}{}_{\overline{3}} = G^{\phi}{}_{r} = \partial_r\dot{\phi} + r^{-1}\,\dot{\phi} \tag{4.192f}$$
$$G^{\overline{3}}{}_{\overline{1}} = G^{r}{}_{\lambda} = \partial_\lambda\dot{r} - r\,\dot{\lambda}\cos^2\phi \tag{4.192g}$$
$$G^{\overline{3}}{}_{\overline{2}} = G^{r}{}_{\phi} = \partial_\phi\dot{r} - r\,\dot{\phi} \tag{4.192h}$$
$$G^{\overline{3}}{}_{\overline{3}} = G^{r}{}_{r} = \partial_r\dot{r}. \tag{4.192i}$$

As a check, note that the trace of this spherical coordinate representation yields the velocity divergence

$$G^{\overline{m}}{}_{\overline{m}} = \frac{\partial\dot{\lambda}}{\partial\lambda} + \frac{1}{\cos\phi}\,\frac{\partial(\dot{\phi}\cos\phi)}{\partial\phi} + \frac{1}{r^2}\,\frac{\partial(\dot{r}\,r^2)}{\partial r}, \tag{4.193}$$

which agrees with equation (4.121b).

Physical component representation of the velocity gradient tensor

The velocity components for spherical coordinates are given by

$$v^{\overline{m}} = (\dot{\lambda}, \dot{\phi}, \dot{r}), \tag{4.194}$$

which means that some elements of $G^{\overline{m}}{}_{\overline{n}}$ have physical dimensions of inverse time or length per time. We can express the physical components, each with dimensions of inverse time, using spherical coordinates by including the basis vectors (4.189) and one-forms (4.190), just like we did for the cylindrical-polar coordinates in Section 4.6.3. in which case we identify the physical components, written as a matrix

$$G^{\hat{\bar{m}}}{}_{\hat{\bar{n}}} = \begin{bmatrix} \partial_\lambda \dot{\lambda} - \dot{\phi} \tan\phi + r^{-1}\,\dot{r} & \cos\phi\,(\partial_\phi \dot{\lambda} - \dot{\lambda}\tan\phi) & (r\cos\phi)\,(\partial_r \dot{\lambda} + r^{-1}\,\dot{\lambda}) \\ (1/\cos\phi)\,(\partial_\lambda \dot{\phi} + \dot{\lambda}\cos\phi\sin\phi) & \partial_\phi \dot{\phi} + r^{-1}\,\dot{r} & r\,\partial_r \dot{\phi} + \dot{\phi} \\ (1/r\cos\phi)\,(\partial_\lambda \dot{r} - r\,\dot{\lambda}\cos^2\phi) & r^{-1}\,\partial_\phi \dot{r} - \dot{\phi} & \partial_r \dot{r} \end{bmatrix}. \tag{4.195}$$

As a final step, we introduce the physical components to the spherical velocity vector

$$v^{\hat{\lambda}} = (r\cos\phi)\,\dot{\lambda} \quad \text{and} \quad v^{\hat{\phi}} = r\,\dot{\phi} \quad \text{and} \quad v^{\hat{r}} = \dot{r}, \tag{4.196}$$

so that the physical components to the velocity gradient tensor take the form

$$G^{\hat{\bar{1}}}{}_{\hat{\bar{1}}} = G^{\hat{\lambda}}{}_{\hat{\lambda}} = \frac{1}{r\cos\phi}\frac{\partial v^{\hat{\lambda}}}{\partial\lambda} - \frac{\tan\phi}{r}\,v^{\hat{\phi}} + \frac{1}{r}\,v^{\hat{r}} \tag{4.197a}$$

$$G^{\hat{\bar{1}}}{}_{\hat{\bar{2}}} = G^{\hat{\lambda}}{}_{\hat{\phi}} = \frac{\cos\phi}{r}\frac{\partial(v^{\hat{\lambda}}/\cos\phi)}{\partial\phi} - \frac{\tan\phi}{r}\,v^{\hat{\lambda}} \tag{4.197b}$$

$$G^{\hat{\bar{1}}}{}_{\hat{\bar{3}}} = G^{\hat{\lambda}}{}_{\hat{r}} = r\,\frac{\partial(v^{\hat{\lambda}}/r)}{\partial r} + \frac{1}{r}\,v^{\hat{\lambda}} \tag{4.197c}$$

$$G^{\hat{\bar{2}}}{}_{\hat{\bar{1}}} = G^{\hat{\phi}}{}_{\hat{\lambda}} = \frac{1}{r\cos\phi}\frac{\partial v^{\hat{\phi}}}{\partial\lambda} + \frac{\tan\phi}{r}\,v^{\hat{\lambda}} \tag{4.197d}$$

$$G^{\hat{\bar{2}}}{}_{\hat{\bar{2}}} = G^{\hat{\phi}}{}_{\hat{\phi}} = \frac{1}{r}\frac{\partial v^{\hat{\phi}}}{\partial\phi} + \frac{1}{r}v^{\hat{r}} \tag{4.197e}$$

$$G^{\hat{\bar{2}}}{}_{\hat{\bar{3}}} = G^{\hat{\phi}}{}_{\hat{r}} = r\,\frac{\partial(v^{\hat{\phi}}/r)}{\partial r} + \frac{1}{r}\,v^{\hat{\phi}} \tag{4.197f}$$

$$G^{\hat{\bar{3}}}{}_{\hat{\bar{1}}} = G^{\hat{r}}{}_{\hat{\lambda}} = \frac{1}{r\cos\phi}\frac{\partial v^{\hat{r}}}{\partial\lambda} - \frac{1}{r}\,v^{\hat{\lambda}} \tag{4.197g}$$

$$G^{\hat{\bar{3}}}{}_{\hat{\bar{2}}} = G^{\hat{r}}{}_{\hat{\phi}} = \frac{1}{r}\frac{\partial v^{\hat{r}}}{\partial\phi} - \frac{1}{r}\,v^{\hat{\phi}} \tag{4.197h}$$

$$G^{\hat{\bar{3}}}{}_{\hat{\bar{3}}} = G^{\hat{r}}{}_{\hat{r}} = \frac{\partial v^{\hat{r}}}{\partial r}. \tag{4.197i}$$

As a check, note that the trace of this tensor is given by the divergence of the velocity

$$G^{\hat{\bar{m}}}{}_{\hat{\bar{m}}} = \frac{1}{r\cos\phi}\frac{\partial v^{\hat{\lambda}}}{\partial\lambda} + \frac{1}{r\cos\phi}\frac{\partial(v^{\hat{\phi}}\cos\phi)}{\partial\phi} + \frac{1}{r^2}\frac{\partial(r^2\,v^{\hat{r}})}{\partial r}, \tag{4.198}$$

which agrees with equation (4.129).

4.7 Exercises

EXERCISE 4.1: VECTOR CROSS PRODUCT OF CYLINDRICAL BASIS VECTORS
As a check on the formalism for vector cross products, verify the relation (3.97),

$$\boldsymbol{e}_{\bar{a}} \times \boldsymbol{e}_{\bar{b}} = \varepsilon_{\bar{a}\bar{b}\bar{c}}\,\boldsymbol{e}^{\bar{c}}, \tag{4.199}$$

for the cross product of two basis vectors using cylindrical-polar coordinates.

EXERCISE 4.2: VECTOR CROSS PRODUCT OF SPHERICAL BASIS VECTORS
As a check on the formalism for vector cross products, verify the relation (3.97),

$$\boldsymbol{e}_{\overline{a}} \times \boldsymbol{e}_{\overline{b}} = \varepsilon_{\overline{a}\overline{b}\overline{c}}\, \boldsymbol{e}^{\overline{c}}, \tag{4.200}$$

for the cross product of two basis vectors using spherical coordinates.

EXERCISE 4.3: LIE BRACKET
The Lie bracket is given by the commutator $[\mathbf{u}, \mathbf{v}]$. Making use of the differential operator specification of a vector according to Section 3.6, show that

$$[\mathbf{u}, \mathbf{v}] = (u^m \, \partial_m v^n - v^m \, \partial_m u^n) \, \partial_n. \tag{4.201}$$

Hint: introduce a test function, f, and compute $[\mathbf{u}, \mathbf{v}]\, f$, being careful with the product rule for partial derivatives. Also, see Section 4.4 of *Tromp* (2025b).

EXERCISE 4.4: LIE BRACKET FOR NON-COORDINATE SPHERICAL BASIS VECTORS
Make use of the Lie bracket in equation (4.201) and the non-coordinate spherical basis from Section 4.4 to compute

$$[\hat{\boldsymbol{r}}, \hat{\boldsymbol{\lambda}}] \quad \text{and} \quad [\hat{\boldsymbol{r}}, \hat{\boldsymbol{\phi}}] \quad \text{and} \quad [\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}]. \tag{4.202}$$

EXERCISE 4.5: CHRISTOFFEL SYMBOLS FOR POLAR COORDINATES
Compute the Christoffel symbols for polar coordinates, making use of equation (3.130) for the Christoffel symbols in terms of the metric tensor. Hint: this exercise only considers two-dimensional polar coordinates, (r, ϑ). Also note that the solution is given in Section 4.6.3.

EXERCISE 4.6: OPERATORS WITH GENERALIZED ORTHOGONAL COORDINATES
In this exercise we specialize formulas for the covariant divergence, Laplacian, and curl by making use of generalized orthogonal coordinates, $\xi^{\overline{a}}$, studied in Section 4.5.

(a) Write the expression for the covariant divergence of a $(1, 0)$ vector, $\boldsymbol{\nabla} \cdot \mathbf{F} = \nabla_{\overline{a}} F^{\overline{a}}$. Start from the general expression given by equation (3.140) and written here as

$$\boldsymbol{\nabla} \cdot \mathbf{F} = \nabla_{\overline{a}}\, F^{\overline{a}} = \mathsf{g}^{-1}\, \partial_{\overline{a}}(\mathsf{g}\, F^{\overline{a}}), \tag{4.203}$$

where equation (4.137) provides the square root of the metric determinant,

$$\mathsf{g} = \sqrt{\det \mathbb{g}_{\overline{a}\overline{b}}} = h_{\overline{1}}\, h_{\overline{2}}\, h_{\overline{3}}. \tag{4.204}$$

(b) Write the covariant Laplacian of a scalar, ψ, starting from the general expression for the Laplace-Beltrami operator given by equation (3.142) and here written as

$$\nabla_a(\mathbb{g}^{ab}\, \partial_b\, \psi) = \mathsf{g}^{-1}\, \partial_a(\mathsf{g}\, \mathbb{g}^{ab}\, \partial_b \psi). \tag{4.205}$$

(c) Write the covariant curl of a vector, starting from the general expression given by equation (3.155), here written as

$$\operatorname{curl}(\mathbf{F}) = \boldsymbol{e}_{\overline{a}}\, \mathsf{g}^{-1}\, \epsilon^{\overline{a}\overline{b}\overline{c}}\, \partial_{\overline{b}}(\mathbb{g}_{\overline{c}\overline{d}} F^{\overline{d}}), \tag{4.206}$$

where $\epsilon^{\overline{a}\overline{b}\overline{c}}$ is the permutation symbol.

Chapter 5

The Dirac delta

We formulate the equations of Newtonian mechanics by focusing on the mechanics of a point mass, where a point mass is the idealization of a small unit of matter in the limit that the volume goes to zero while the mass remains finite. The Dirac delta allows one to rationally conceive of the corresponding density for point objects. The Dirac delta also plays a central role in many areas of theoretical physics, as well as for the solution to linear equations of mathematical physics via the Green's function. In this chapter we present salient properties of the Dirac delta. Although somewhat magical on first encounter, use soon allows them to become part of our trusted mathematical toolbox.

We use the nomenclature "Dirac delta", rather than the commonly used "Dirac delta function", as a hint that this mathematical object is not a typical function. Rather, the Dirac delta has the very peculiar property of vanishing everywhere except at a single point, where it is infinite. As a result, it has a nonzero spatial integral that is normalized to unity. Furthermore, when multiplied by an arbitrary smooth test function and then integrated over the point where the Dirac delta "fires", the result is the test function evaluated at the point. This behavior is referred to as the sifting property. When the test function is unity, then the sifting property reduces to the normalization, so that the sifting property serves as the defining property of the Dirac delta.

READER'S GUIDE TO THIS CHAPTER

We present properties of the Dirac delta that find use in our studies. This material is placed within the Newtonian mechanics part of the book due to its relation to point matter. It also provides support to our study of Newtonian gravity in Chapter 11. There are many treatments of the Dirac delta within the mathematical physics literature. Chapter 5 of *Stakgold* (2000b) provides a mathematical presentation accessible to physicists, with Stakgold referring to the Dirac delta as a *generalized function* or a *distribution*. Our presentation is consistent with the physics treatments given in Appendix A of *Gasiorowicz* (1974) and Appendix C of *Pope* (2000).

Our treatment of the Dirac delta is physically formal. Namely, we offer a deductive formulation that is motivated from a physical perspective rather than serving the needs for a mathematically rigorous presentation.[a] The physically formal treatment is supported by heuristic arguments taken from Newtonian gravity (Section 11.7), electrostatics (e.g., *Jackson* (1975)), and the diffusion of temperature and matter within a continuous media (VOLUME 4).

[a]The term physically formal is used here as a complement to mathematically rigorous. A mathematically rigorous treatment for the topics of this chapter require an array of mathematical apparatus, particularly in

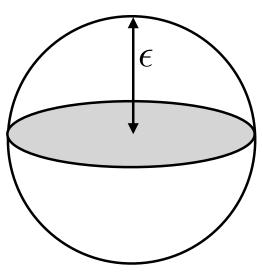
topics of functional analysis, that are outside our scope.

5.1 Motivation from Newtonian gravity

In our study of Newtonian gravity in Section 11.7, we encounter the Poisson equation for the Newtonian gravitational potential, Φ_{N}, arising from an arbitrary mass density, ρ

$$\nabla^2 \Phi_{\mathsf{N}} = 4\,\pi\,G\,\rho, \tag{5.1}$$

with G Newton's gravitational constant. The gravitational potential for an arbitrary spherically symmetric mass, when sampled at a point outside the mass, equals to the potential of a point mass located at the origin. Making precise the notion of a "point mass" provides a physical venue for introducing the Dirac delta, which we interpret as the mass density for a point mass.



$$V_\epsilon = \frac{4}{3}\,\pi\,\epsilon^3$$

$$\delta^{(\epsilon)}(\boldsymbol{x}) \equiv \begin{cases} V_\epsilon^{-1} & \text{if } |\boldsymbol{x}| \le \epsilon \\ 0 & \text{if } |\boldsymbol{x}| > \epsilon. \end{cases}$$

$$\rho(\boldsymbol{x}) = M\,\delta^{(\epsilon)}(\boldsymbol{x})$$

FIGURE 5.1: Depicting a spherical region of radius, $\epsilon > 0$, and with fixed mass, M, whose volume is given by $V_\epsilon = \frac{4}{3}\,\pi\,\epsilon^3$. The Dirac delta is given by the limit as the radius tends to zero, $\delta(\boldsymbol{x}) \equiv \lim_{\epsilon \to 0} \delta^{(\epsilon)}(\boldsymbol{x})$. That is, the Dirac delta is the mass density for a point mass.

For that purpose, consider a mass, M, distributed uniformly within a sphere of radius, $\epsilon > 0$, and volume,

$$V_\epsilon = \frac{4}{3}\,\pi\,\epsilon^3, \tag{5.2}$$

and let the sphere be centered at the origin of a coordinate system (see Figure 5.1). The mass distribution has a mass density,

$$\rho(\boldsymbol{x}) = M\,\delta^{(\epsilon)}(\boldsymbol{x}), \tag{5.3}$$

where we introduced the $\epsilon-$distribution

$$\delta^{(\epsilon)}(\boldsymbol{x}) \equiv \begin{cases} V_{\epsilon}^{-1} & \text{if } |\boldsymbol{x}| \leq \epsilon \\ 0 & \text{if } |\boldsymbol{x}| > \epsilon. \end{cases} \tag{5.4}$$

By construction, an integral over a domain, $\mathcal{R}$, that fully encompasses the sphere yields the mass

$$\int_{\mathcal{R}} \rho\,\mathrm{d}V = M \int_{\mathcal{R}} \delta^{(\epsilon)}(\boldsymbol{x})\,\mathrm{d}V = M, \tag{5.5}$$

with this result holding even as the radius of the sphere becomes arbitrarily small, $\epsilon \to 0$. We define the Dirac delta as the limiting ϵ-distribution

$$\delta(\boldsymbol{x}) \equiv \lim_{\epsilon \to 0} \delta^{(\epsilon)}(\boldsymbol{x}). \tag{5.6}$$

Evidently, the Dirac delta is the mass density for a mass source that is zero everywhere in space except at a single point. Hence, the Dirac delta distribution is infinite at the location of the point source. This dual property, namely an object defined only at a single point and yet having an infinite value at that point, allows the Dirac delta to have a nonzero integral, which is normalized according to

$$\int_{\mathcal{R}} \delta(\boldsymbol{x})\,\mathrm{d}V = 1, \tag{5.7}$$

where the region, $\mathcal{R}$, includes the point $\boldsymbol{x} = 0$ where the Dirac delta "fires". Concerns with how to interpret the infinite value of $\delta(\boldsymbol{x})$ at $\boldsymbol{x} = 0$ are ameliorated by recognizing that $\delta(\boldsymbol{x})$ is evaluated only as part of an integral. Connecting to other physical analogs beyond the point mass source, we can consider the Dirac delta as the charge density (charge per volume) for a point charge in electrostatics, or the mass density for a point source of trace matter within a fluid.

5.2 Sifting property of the Dirac delta

Multiply an ϵ-distribution by an arbitrary smooth function, $\psi(\boldsymbol{x})$. Since the ϵ-distribution has support only within the ϵ-sphere surrounding the origin, an integral of $\delta^{(\epsilon)}(\boldsymbol{x})\,\psi(\boldsymbol{x})$ over the sphere, in the limit that $\epsilon \to 0$, leads to the sifting property

$$\lim_{\epsilon \to 0} \int_{\mathcal{R}} \delta^{(\epsilon)}(\boldsymbol{x})\,\psi(\boldsymbol{x})\,\mathrm{d}V = \int_{\mathcal{R}} \delta(\boldsymbol{x})\,\psi(\boldsymbol{x})\,\mathrm{d}V = \psi(\boldsymbol{x} = 0). \tag{5.8}$$

Evidently, the Dirac delta sifts out the smooth function as evaluated at the location of the Dirac delta source. The normalization property (5.7) and the sifting property (5.8) are the two defining features of the Dirac delta. To derive further properties implied by normalization and sifting, we find it useful to write

$$\psi(\boldsymbol{x}) = \int_{\mathcal{R}} \delta(\boldsymbol{x} - \boldsymbol{y})\,\psi(\boldsymbol{y})\,\mathrm{d}V, \tag{5.9}$$

which reduces to the normalization property (5.7) when $\psi = 1$.

The sifting property (5.9) holds whether the Dirac delta has argument $\boldsymbol{x} - \boldsymbol{y}$ or $\boldsymbol{y} - \boldsymbol{x}$

$$\psi(\boldsymbol{x}) = \int_{\mathcal{R}} \delta(\boldsymbol{x} - \boldsymbol{y})\, \psi(\boldsymbol{y})\, \mathrm{d}V = \int_{\mathcal{R}} \delta(\boldsymbol{y} - \boldsymbol{x})\, \psi(\boldsymbol{y})\, \mathrm{d}V, \tag{5.10}$$

in which case we conclude that the Dirac delta is a symmetric or even distribution

$$\delta(\boldsymbol{x}) = \delta(-\boldsymbol{x}). \tag{5.11}$$

5.3 Dirac delta carries physical dimensions

The Dirac delta carries physical dimensions given by the inverse dimension of its argument. For example, $\delta(x)$ has dimensions inverse length (here, x is a Cartesian space coordinate), $\delta(t)$ has dimensions inverse time (t is time), and $\delta(\phi)$ is non-dimensional (ϕ is latitude in radians). The dimensional properties of a Dirac delta are manifest from its corresponding sifting property. It is notable that many treatments, particularly in the maths literature, ignore the physical dimensions of the Dirac delta. Hence, it is important to exercise care when transferring Dirac delta identities from a maths source to a physics context.

5.4 Example $\delta^{(\epsilon)}(x)$ distributions

In one-dimension, the construction of the Dirac delta following equation (5.6) is given by

$$\delta^{(\epsilon)}(x) = \epsilon^{-1} \quad \text{for } |x| < \epsilon/2 \text{ and } 0 \text{ for } |x| > \epsilon/2, \tag{5.12}$$

with this function depicted in Figure 5.2.

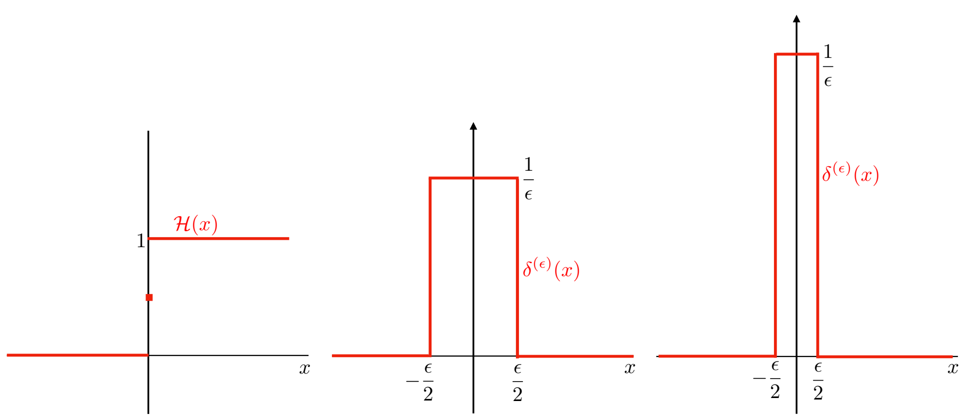


FIGURE 5.2: Left panel: Heaviside step function, $\mathcal{H}(x)$, as given by equation (5.20). Middle and right panels: the square pulse function, $\delta^{(\epsilon)}(x)$, given by equation (5.12), with the far right panel having a smaller value for $\epsilon > 0$ than the middle panel. We see that $\lim_{\epsilon\to 0} \delta^{(\epsilon)}(x) = \delta(x) = \mathrm{d}\mathcal{H}(x)/\mathrm{d}x$, with $\int_{\mathcal{R}} \delta^{(\epsilon)}(x)\, \mathrm{d}x = 1$ for each ϵ.

There are many other ϵ-distributions whose limiting behavior also result in a Dirac delta, as defined by the unit normalization and sifting properties

$$\lim_{\epsilon\to 0} \delta^{(\epsilon)}(x) = \delta(x). \tag{5.13}$$

Indeed, any even function that is normalized to unity and whose central peak's width is infinitesimal can be used to define a suitable $\delta^{(\epsilon)}(x)$, with the following examples (see Figure

5.3) appearing in applications ($\epsilon > 0$ for each example)

$$\delta^{(\epsilon)}(x) = \frac{\mathrm{e}^{-|x|/\epsilon}}{2\,\epsilon} \tag{5.14a}$$

$$\delta^{(\epsilon)}(x) = \frac{\mathrm{e}^{-x^2/\epsilon^2}}{\epsilon\,\sqrt{\pi}} \tag{5.14b}$$

$$\delta^{(\epsilon)}(x) = \frac{\epsilon}{\pi\,(x^2+\epsilon^2)} \tag{5.14c}$$

$$\delta^{(\epsilon)}(x) = \frac{\sin(x/\epsilon)}{\pi\,x} \tag{5.14d}$$

$$\delta^{(\epsilon)}(x) = \frac{\epsilon\sin^2(x/\epsilon)}{\pi\,x^2}. \tag{5.14e}$$

With x corresponding to a spatial position, note how each of these one-dimensional functions has physical dimensions of inverse length. The sifting property holds since each function is even and becomes infinitely peaked yet infinitesimally narrow as $\epsilon \to 0$. We verify the normalization condition for equation (5.14a) through

$$\frac{1}{2\,\epsilon}\int_{-\infty}^{\infty}\mathrm{e}^{-|x|/\epsilon}\,\mathrm{d}x = \frac{1}{2\,\epsilon}\int_{-\infty}^{0}\mathrm{e}^{x/\epsilon}\,\mathrm{d}x + \frac{1}{2\,\epsilon}\int_{0}^{\infty}\mathrm{e}^{-x/\epsilon}\,\mathrm{d}x = \frac{1}{2}+\frac{1}{2} = 1. \tag{5.15}$$

The other expressions for $\delta^{(\epsilon)}(x)$ likewise have unit integrals over the real line and so satisfy the normalization property. The expression (5.14b) is related to the causal free space Green's function for the diffusion equation studied in VOLUME 4. The expression (5.14d) is notable for its connection to Fourier analysis studied in VOLUME 4, in which we have the integral expression

$$\delta(x) = \frac{1}{2\,\pi}\lim_{\epsilon\to 0}\int_{-1/\epsilon}^{1/\epsilon}\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k = \frac{1}{2\,\pi\,\mathrm{i}\,x}\lim_{\epsilon\to 0}(\mathrm{e}^{\mathrm{i}\,x/\epsilon}-\mathrm{e}^{-\mathrm{i}\,x/\epsilon}) = \lim_{\epsilon\to 0}\frac{\sin(x/\epsilon)}{\pi\,x}. \tag{5.16}$$

Equivalently, we have the identity

$$\delta(x) = \frac{1}{2\,\pi}\int_{-\infty}^{\infty}\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k = \frac{1}{\pi}\int_{-\infty}^{\infty}\cos(k\,x)\,\mathrm{d}k, \tag{5.17}$$

which follows from the Euler identity,

$$\mathrm{e}^{\mathrm{i}\alpha} = \cos\alpha + \mathrm{i}\sin\alpha, \tag{5.18}$$

and anti-symmetry of the sin function

$$\int_{-\infty}^{\infty}\sin(k\,x)\,\mathrm{d}k = 0. \tag{5.19}$$

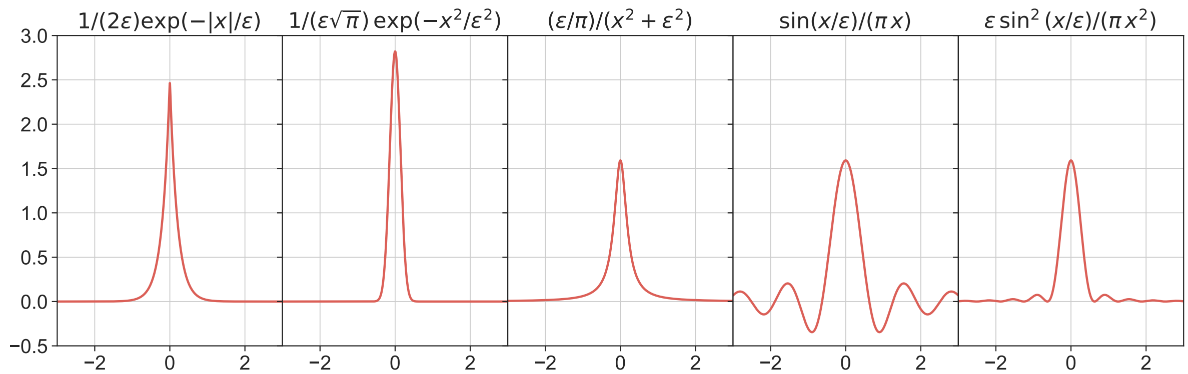


FIGURE 5.3: Plots of the functions $\delta^{(\epsilon)}(x)$ from equations (5.14a)-(5.14e) for $\epsilon = 0.2$, each of which converges to the Dirac delta as $\epsilon \to 0$.

5.5 Connection to the Heaviside step function

The Heaviside step function (Figure 5.2) is given by[1]

$$\mathcal{H}(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0, \end{cases} \tag{5.20}$$

and it is related to the sgn function

$$\operatorname{sgn}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0, \end{cases} \tag{5.21}$$

according to

$$\operatorname{sgn}(x) = 2\,\mathcal{H}(x) - 1. \tag{5.22}$$

Both the Heaviside and sgn functions are piecewise continuous and have infinite derivatives at $x = 0$. In particular, the derivative of the Heaviside step function equals to the Dirac delta

$$\frac{\mathrm{d}\mathcal{H}(x)}{\mathrm{d}x} = \delta(x). \tag{5.23}$$

This identity is most apparent by considering the square pulse approximation to the Dirac delta given by equation (5.12), in which we find

$$\delta(x) = \lim_{\epsilon \to 0} \delta^{(\epsilon)}(x) = \lim_{\epsilon \to 0} \frac{\mathcal{H}(x + \epsilon/2) - \mathcal{H}(x - \epsilon/2)}{\epsilon} = \frac{\mathrm{d}\mathcal{H}(x)}{\mathrm{d}x}. \tag{5.24}$$

To be convinced of the second equality, separately consider the cases with $x < -\epsilon/2$, $x > \epsilon/2$, and $|x| < \epsilon/2$. Equivalently, we see that the Heaviside step function is the cumulative distribution of the Dirac delta

$$\int_{-\infty}^{x} \delta(y)\,\mathrm{d}y = \int_{-\infty}^{x} \frac{\mathrm{d}\mathcal{H}(y)}{\mathrm{d}y}\,\mathrm{d}y = \mathcal{H}(x) - \mathcal{H}(-\infty) = \mathcal{H}(x). \tag{5.25}$$

[1] We do not define the Heaviside at $x = 0$, though some authors give it a value of $\mathcal{H}(0) = 1/2$. For our purposes, the properties of the Heaviside step function remain unchanged whether it is defined at $x = 0$ or not. See footnote on page 20 of *Stakgold* (2000a) for more details.

Connecting to the language of probability theory, the Dirac delta corresponds to a probability density function peaked over an infinitesimal region, whereas the Heaviside step function is the corresponding probability distribution function.

There are many other functions whose derivatives yield the Dirac delta. In particular, the Green's function is a continuous function whose first derivative has a jump and second derivative equals to a Dirac delta. Just as there is no unique function, $\delta^{(\epsilon)}(\boldsymbol{x})$, whose limit yields a Dirac delta, there is no unique function whose derivative (or 2nd derivative) equals to a Dirac delta. The key point is that any "function" that respects the sifting property (5.9) is a legitimate Dirac delta.

5.6 Scaling property of the Dirac delta

Write the sifting property (5.9) in the form

$$\psi(x) = \int_{-L}^{L} \delta(x-y)\,\psi(y)\,\mathrm{d}y \qquad \text{with } |x| < L, \tag{5.26}$$

where we introduced the constant, L, to keep track of the integration limits. Now scale the coordinates by a non-dimensional constant, $\alpha \neq 0$, so that $x = \alpha\,\xi$, $y = \alpha\,\eta$, and $\mathrm{d}y = \alpha\,\mathrm{d}\eta$, in which case the sifting property (5.26) becomes

$$\psi(\alpha\,\xi) = \int_{-L/\alpha}^{L/\alpha} \alpha\,\delta(\alpha\,\xi - \alpha\,\eta)\,\psi(\alpha\,\eta)\,\mathrm{d}\eta. \tag{5.27}$$

The sifting property (5.26) also implies

$$\psi(\alpha\,\xi) = \int_{-L/\alpha}^{L/\alpha} \delta(\xi - \eta)\,\psi(\alpha\,\eta)\,\mathrm{d}\eta, \tag{5.28}$$

so that equating the two expressions (5.27) and (5.28) suggests that $\delta(x) = \alpha\,\delta(\alpha\,x)$. However, this expression fails for $\alpha < 0$. Namely, from the symmetry property (5.11) we know that $\delta(x) = \delta(-x)$ so that $\delta(\alpha\,x) = \delta(-\alpha\,x)$. Hence, the scaling property holds for an arbitrary nonzero constant in the following form with the absolute value of the constant,

$$\delta(x) = |\alpha|\,\delta(\alpha\,x). \tag{5.29}$$

Stated more simply, the symmetry property (5.11), $\delta(x) = \delta(-x)$, is a special case of the scaling property with $\alpha = -1$. For this connection to hold requires application of the absolute value operation, $|-1| = 1$, thus leading to the expression (5.29).

5.7 Dirac delta with a function argument

Consider the integral

$$\int_{\mathcal{R}} \psi(x)\,\delta[f(x)]\,\mathrm{d}x, \tag{5.30}$$

for some smooth function, $f(x)$. The argument to the Dirac delta vanishes where the function vanishes, so this integral is nonzero only if $f(x)$ has at least one root somewhere. Assume there are N roots of $f(x)$, where $f(x_n) = 0$. Let us furthermore assume the roots are simple, so that

$f(x)$ has a non-zero first derivative at each root, $f'(x_n) \neq 0$. Consequently, near any of the simple roots we can write the Taylor expansion

$$f(x) \approx f'(x_n)\,(x - x_n). \tag{5.31}$$

We are thus led to

$$\int_{\mathcal{R}} \psi(x)\,\delta[f(x)]\,\mathrm{d}x = \sum_{n=1}^{N} \int_{\mathcal{R}} \psi(x)\,\delta[f'(x_n)\,(x - x_n)]\,\mathrm{d}x \tag{5.32a}$$

$$= \sum_{n=1}^{N} \frac{1}{|f'(x_n)|} \int_{\mathcal{R}} \psi(x)\,\delta(x - x_n)\,\mathrm{d}x \tag{5.32b}$$

$$= \sum_{n=1}^{N} \frac{\psi(x_n)}{|f'(x_n)|}, \tag{5.32c}$$

where the second equality made use of the scaling property (5.29). Evidently, we we find the generalized scaling property for the Dirac delta

$$\delta[f(x)] = \sum_{n=1}^{N} \frac{\delta(x - x_n)}{|f'(x_n)|} \quad \text{for } f(x) \text{ having } N \text{ simple roots, } f(x_n) = 0, \text{ and with } f'(x_n) \neq 0. \tag{5.33}$$

As an example, consider the quadratic function, $f(x) = x^2 - \alpha^2$, which has two simple roots, $x_1 = \alpha$ and $x_2 = -\alpha$, and corresponding derivatives, $f'(x_1) = 2\alpha$ and $f'(x_2) = -2\alpha$. The generalized scaling property (5.33) leads to

$$\delta(x^2 - \alpha^2) = \frac{\delta(x - \alpha) + \delta(x + \alpha)}{2\,|\alpha|}, \tag{5.34}$$

and the corresponding sifting result

$$\int_{\mathcal{R}} \psi(x)\,\delta(x^2 - \alpha^2)\,\mathrm{d}x = \frac{\psi(\alpha) + \psi(-\alpha)}{2\,|\alpha|}. \tag{5.35}$$

5.8 Equivalence classes of Dirac deltas

We commonly write a variety of equations satisfied by the Dirac that are outside of an integral sign, such as equations (5.33) and (5.34). Those equations are placeholders or shorthands for relations that hold inside of integrals and when multiplied by a smooth test function. For example, consider

$$\psi(a)\,A(a) = \int_{\mathcal{R}} \psi(x)\,A(x)\,\delta(x - a)\,\mathrm{d}x, \tag{5.36}$$

which is equivalent to

$$\psi(a)\,A(a) = \int_{\mathcal{R}} \psi(x)\,A(a)\,\delta(x - a)\,\mathrm{d}x. \tag{5.37}$$

We conclude that

$$A(x)\,\delta(x - a) = A(a)\,\delta(x - a). \tag{5.38}$$

As a corollary, let $A(a) = 1$ so that

$$A(x)\,\delta(x-a) = \delta(x-a). \tag{5.39}$$

We can thus consider the Dirac delta as an equivalance class

$$[A(x)\,\delta(x)] \sim [\delta(x)]. \tag{5.40}$$

That is, if a Dirac delta within an integral is multiplied by a non-dimensional function, $A(x)$, satisfying $A(0) = 1$, then we can disregard $A(x)$ since it does not modify how the Dirac delta acts within the integral. That is, $A(x)$ does not modify the sifting property of $\delta(x)$, and so it has no impact on how $\delta(x)$ affects test functions inside of integrals.

5.9 Taylor series decomposition of the Dirac delta

The first derivative of a Dirac delta represents an idealization of a dipole (e.g., see exercise 1.14 of *Stakgold* (2000a)). How does a Dirac dipole act on a test function? To answer this question, make use of the identity

$$\int_{-\epsilon}^{\epsilon} \frac{\mathrm{d}[\psi(x)\,\delta(x)]}{\mathrm{d}x}\,\mathrm{d}x = \psi(\epsilon)\,\delta(\epsilon) - \psi(-\epsilon)\,\delta(-\epsilon). \tag{5.41}$$

Each term on the right hand side vanishes when $\epsilon > 0$ since the Dirac delta never fires. Applying the product rule inside of the integral renders the identity

$$\int_{-\epsilon}^{\epsilon} \frac{\mathrm{d}[\psi(x)\,\delta(x)]}{\mathrm{d}x}\,\mathrm{d}x = \int_{-\epsilon}^{\epsilon} \left[\psi(x)\,\frac{\mathrm{d}\delta(x)}{\mathrm{d}x} + \delta(x)\,\frac{\mathrm{d}\psi(x)}{\mathrm{d}x}\right]\mathrm{d}x = 0, \tag{5.42}$$

so that

$$\int_{-\epsilon}^{\epsilon} \psi(x)\,\frac{\mathrm{d}\delta(x)}{\mathrm{d}x}\,\mathrm{d}x = -\left[\frac{\mathrm{d}\psi(x)}{\mathrm{d}x}\right]_{x=0}. \tag{5.43}$$

Consequently, the Dirac dipole acts to sift minus the first derivative of the test function. For example, let $\psi(x) = x$, in which case

$$\int_{-\epsilon}^{\epsilon} x\,\frac{\mathrm{d}\delta(x)}{\mathrm{d}x}\,\mathrm{d}x = -1 \Longrightarrow \int_{-\epsilon}^{\epsilon} \left[x\,\frac{\mathrm{d}\delta(x)}{\mathrm{d}x} + \delta(x)\right]\mathrm{d}x = 0, \tag{5.44}$$

which can be formally written

$$x\,(\mathrm{d}\delta(x)/\mathrm{d}x) = -\delta(x). \tag{5.45}$$

We follow the same procedure to derive the sifting property of the second derivative of the Dirac delta

$$\int_{\mathcal{R}} \psi\,\frac{\mathrm{d}^2\delta}{\mathrm{d}x^2}\,\mathrm{d}x = \int_{\mathcal{R}} \left[\frac{\mathrm{d}}{\mathrm{d}x}\left(\psi\,\frac{\mathrm{d}\delta}{\mathrm{d}x}\right) - \frac{\mathrm{d}\psi}{\mathrm{d}x}\,\frac{\mathrm{d}\delta}{\mathrm{d}x}\right]\mathrm{d}x. \tag{5.46}$$

The right hand side first term integrates to boundary contributions, each of which vanish if we assume the Dirac delta is in the interior of the domain, and we assume the test function is well behaved on the boundaries. Integration by parts one more time renders

$$\int_{\mathcal{R}} \psi\,\frac{\mathrm{d}^2\delta}{\mathrm{d}x^2}\,\mathrm{d}x = -\int_{\mathcal{R}} \frac{\mathrm{d}\psi}{\mathrm{d}x}\,\frac{\mathrm{d}\delta}{\mathrm{d}x}\,\mathrm{d}x = -\int_{\mathcal{R}} \frac{\mathrm{d}}{\mathrm{d}x}\left(\delta\,\frac{\mathrm{d}\psi}{\mathrm{d}x}\right)\mathrm{d}x + \int_{\mathcal{R}} \delta\,\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\,\mathrm{d}x = \int_{\mathcal{R}} \delta\,\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\,\mathrm{d}x, \tag{5.47}$$

where the boundary term vanishes from the penultimate equation. This procedure readily generalizes to the identity holding for any integer $n > 0$

$$\int_{\mathcal{R}} \psi(x) \, \frac{\mathrm{d}^n \delta(x)}{\mathrm{d}x^n} \, \mathrm{d}x = (-1)^n \int_{\mathcal{R}} \delta(x) \, \frac{\mathrm{d}^n \psi(x)}{\mathrm{d}x^n} \, \mathrm{d}x = (-1)^n \left[\frac{\mathrm{d}^n \psi(x)}{\mathrm{d}x^n} \right]_{x=0}. \tag{5.48}$$

Hence, the $n'th$ derivative of the Dirac delta acts to sift the $n'th$ derivative of a test function as multiplied by $(-1)^n$.

The sifting property in the form of equation (5.48) allows us to define the Taylor series of a Dirac delta as follows. Consider the expression for the test function at an arbitrary point $x = a$, written using both a Taylor series and the sifting property

$$\psi(a) = \psi(0) + \sum_{n=1}^{\infty} \frac{a^n}{n!} \, \psi^{(n)}(0) = \int_{\mathcal{R}} \psi(x) \, \delta(x - a) \, \mathrm{d}x, \tag{5.49}$$

where we introduced the shorthand

$$\psi^{(n)}(0) = \left[\frac{\mathrm{d}^n \psi(x)}{\mathrm{d}x^n} \right]_{x=0}. \tag{5.50}$$

Making use of the sifting property (5.48) also yields

$$\psi(0) + \sum_{n=1}^{\infty} \frac{a^n}{n!} \, \psi^{(n)}(0) = \int_{\mathcal{R}} \psi(x) \left[\delta(x) + \sum_{n=1}^{\infty} \frac{a^n}{n!} \, (-1)^n \, \delta^{(n)}(x) \right] \mathrm{d}x, \tag{5.51}$$

where we introduced the shorthand

$$\delta^{(n)}(x) = \frac{\mathrm{d}^n \delta(x)}{\mathrm{d}x^n}. \tag{5.52}$$

We are thus led to the identity

$$\psi(a) = \int_{\mathcal{R}} \psi(x) \, \delta(x - a) \, \mathrm{d}x = \int_{\mathcal{R}} \psi(x) \left[\delta(x) + \sum_{n=1}^{\infty} \frac{a^n}{n!} \, (-1)^n \, \delta^{(n)}(x) \right] \mathrm{d}x, \tag{5.53}$$

which renders the Taylor series

$$\delta(x - a) = \delta(x) + \sum_{n=1}^{\infty} \frac{a^n}{n!} \, (-1)^n \, \delta^{(n)}(x). \tag{5.54}$$

5.10 Spectral decomposition of the Dirac delta

Consider a discrete set of orthonormal polynomials, $P_n(x)$, that satisfy

$$\int_{\mathcal{R}} w(x) \, P_n(x) \, P_m(x) \, \mathrm{d}x = \delta_{mn}, \tag{5.55}$$

where m, n are integers, $w(x)$ is a weight function, and δ_{mn} is the Kronecker delta. We assume these polynomials form a complete basis for the domain, $\mathcal{R}$, with the Legendre polynomials, Hermite polynomials, and Laguerre polynomials examples that arise in physics.

Now consider an arbitrary smooth test function, $\psi(x)$, defined over the domain $\mathcal{R}$. Completeness of the polynomials means that we can represent $\psi(x)$ as the infinite series, which is sometimes referred to as a spectral decomposition

$$\psi(x) = \sum_n \Phi_n \, P_n(x) = \sum_n \left[\int_{\mathcal{R}} w(y) \, P_n(y) \, \psi(y) \, \mathrm{d}y \right] P_n(x), \tag{5.56}$$

where the expansion coefficients, Φ_n, are determined by projecting ψ onto the polynomials along with the weighting function. Rearranging the spectral decomposition (5.56) allows us to write

$$\psi(x) = \int_{\mathcal{R}} \left[\sum_m w(y) \, P_m(y) \, P_m(x) \right] \psi(y) \, \mathrm{d}y = \int_{\mathcal{R}} \delta(x-y) \, \psi(y) \, \mathrm{d}y, \tag{5.57}$$

where the second equality made use of the sifting property (5.9) of the Dirac delta. We thus identify the spectral decomposition of the Dirac delta

$$\delta(x-y) = w(y) \sum_m P_m(y) \, P_m(x). \tag{5.58}$$

This equation is referred to as completeness in the quantum mechanics literature (e.g., *Cohen-Tannoudji et al.* (1977)).

5.11 The product of Dirac deltas

The product of two Dirac deltas is not defined if the deltas act on the same space dimension. For example, one possible definition of the product of two x-space Dirac deltas could be

$$\delta(x) \, \delta(x) \stackrel{?}{=} \lim_{\epsilon_1 \to 0} \delta^{(\epsilon_1)}(x) \, \lim_{\epsilon_2 \to 0} \delta^{(\epsilon_2)}(x). \tag{5.59}$$

If we take ϵ_1 to zero before ϵ_2 then

$$\delta(x) \, \delta(x) \stackrel{?}{=} \delta(x) \, \lim_{\epsilon_2 \to 0} \delta^{(\epsilon_2)}(x). \tag{5.60}$$

The problem is seen by taking the integral along the real line,

$$\int \delta(x) \, \delta(x) \, \mathrm{d}x \stackrel{?}{=} \int \delta(x) \, \lim_{\epsilon_2 \to 0} \delta^{(\epsilon_2)}(x) \, \mathrm{d}x = \lim_{\epsilon_2 \to 0} \delta^{(\epsilon_2)}(0) = \delta(0) = \infty. \tag{5.61}$$

Evidently, the product of two Dirac deltas, when acting on the same dimension, is not defined since the integral is not finite. However, when the two Dirac deltas act over distinct coordinate dimensions, such as the x and y Cartesian coordinates, then there is no problem since each dimension has a corresponding integration

$$\int_{\mathcal{R}} \delta(x) \, \delta(y) \, \mathrm{d}x \, \mathrm{d}y = \left[\int_{\mathcal{R}_x} \delta(x) \, \mathrm{d}x \right] \left[\int_{\mathcal{R}_y} \delta(y) \, \mathrm{d}y \right] = 1, \tag{5.62}$$

where the domains $\mathcal{R}_x$ and $\mathcal{R}_y$ contain the origin. We pursue this point in Section 5.12 when decomposing the Dirac delta for three space dimensions into its coordinate components.

5.12 Cartesian, spherical, and cylindrical-polar coordinates

We here display the form of the Dirac delta for three space dimensions using Cartesian, spherical, and cylindrical coordinates.

5.12.1 Cartesian coordinates

We can decompose the three-dimensional Dirac delta according to the Cartesian coordinates

$$\delta(\boldsymbol{x}) = \delta(x)\,\delta(y)\,\delta(z), \tag{5.63}$$

so that, with the domain $\mathcal{R}$ containing the origin, we have

$$\int_{\mathcal{R}} \delta(\boldsymbol{x})\mathrm{d}V = \int_{\mathcal{R}} \delta(x)\,\delta(y)\,\delta(z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = 1. \tag{5.64}$$

Each of the Dirac deltas, $\delta(x)$, $\delta(y)$, and $\delta(z)$ has physical dimensions of inverse length, so that their product, $\delta(x)\,\delta(y)\,\delta(z)$, has the dimension inverse volume.

5.12.2 Spherical coordinates

We sometimes find it useful to make use of the spherical coordinates from Section 4.4, in which the Dirac delta takes the form

$$\delta(\boldsymbol{x}) = \frac{\delta(r)\,\delta(\phi)\,\delta(\lambda)}{r^2\cos\phi}, \tag{5.65}$$

so that

$$\int_{\mathcal{R}} \delta(\boldsymbol{x})\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \int_{\mathcal{R}} \delta(\boldsymbol{x})\,r^2\cos\phi\,\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}\lambda = \int_{\mathcal{R}} \delta(r)\,\delta(\phi)\,\delta(\lambda)\,\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}\lambda = 1. \tag{5.66}$$

Notice how the dimensions of a particular Dirac delta equals to the inverse dimensions of its argument, so that both $\delta(\phi)$ and $\delta(\lambda)$ are dimensionless whereas $\delta(r)$ has the dimension of inverse length.

5.12.3 Cylindrical-polar coordinates

When written using the cylindrical-polar coordinates from Section 4.3, we have

$$\delta(\boldsymbol{x}) = \frac{\delta(r)\,\delta(\vartheta)\,\delta(z)}{r}, \tag{5.67}$$

so that

$$\int_{\mathcal{R}} \delta(\boldsymbol{x})\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \int_{\mathcal{R}} \delta(\boldsymbol{x})\,r\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}z = \int_{\mathcal{R}} \delta(r)\,\delta(\vartheta)\,\delta(z)\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}z = 1. \tag{5.68}$$

Again, the dimensions of a particular Dirac delta equals to the inverse dimensions of its argument, so that $\delta(\vartheta)$ is dimensionless whereas $\delta(r)$ and $\delta(z)$ have dimensions of inverse length.

5.13 Temporal Dirac delta and impulses

Poisson's equation (5.1) for the gravitational potential is an elliptic partial differential equation, in which there is no time derivative. We now introduce the temporal Dirac delta to support the study of evolution equations. The temporal Dirac delta is a point source that is turned on just at one time instance and it is normalized according to

$$\int_{\mathcal{T}} \delta(t)\,\mathrm{d}t = 1, \tag{5.69}$$

where $\mathcal{T}$ is a time interval containing the source time, $t = 0$. This normalization means that $\delta(t)$ has dimensions of inverse time. The temporal Dirac delta also possesses the sifting property from Section 5.2, in which

$$\int_{\mathcal{T}} \delta(t)\,\psi(t)\,\mathrm{d}t = \psi(t = 0). \tag{5.70}$$

In the study of transient behavior of dynamical systems, it is often of interest to examine the response of the system to an idealized force, $\mathcal{F}(t)$, where the force occurs over a small time increment. The time integral of this force is referred to as the impulse

$$I(\tau) = \int_{-\tau}^{\tau} \mathcal{F}(t)\,\mathrm{d}t. \tag{5.71}$$

If the force is further idealized to occur just at a single moment in time, and it is normalized to unity, then we have the unit impulse, which is the integral of the Dirac delta

$$I(\tau) = 1 = \int_{-\tau}^{\tau} \delta(t)\,\mathrm{d}t. \tag{5.72}$$

The corresponding response of the dynamical system is referred to as the impulse response function. If the dynamical system is linear, then the impulse response function equals to the Green's function for the initial value problem.

5.14 Shifting the space-time position of the source

We can arbitrarily place the Dirac source at $(\boldsymbol{x}_0, t_0)$, in which case the Dirac delta is written

$$\delta(\boldsymbol{x} - \boldsymbol{x}_0)\,\delta(t - t_0) = \delta(x - x_0)\,\delta(y - y_0)\,\delta(z - z_0)\,\delta(t - t_0). \tag{5.73}$$

Defining the region, $\mathcal{R}$, to now encompass the source point in space, $\boldsymbol{x} = \boldsymbol{x}_0$, and the time increment $\mathcal{T}$ to encompasses the source time, $t = t_0$, we have the normalization condition

$$\int_{\mathcal{R}} \int_{\mathcal{T}} \delta(\boldsymbol{x} - \boldsymbol{x}_0)\,\delta(t - t_0)\,\mathrm{d}V\,\mathrm{d}t = 1, \tag{5.74}$$

as well as the sifting property

$$\int_{\mathcal{R}} \int_{\mathcal{T}} \psi(\boldsymbol{x}, t)\,\delta(\boldsymbol{x} - \boldsymbol{x}_0)\,\delta(t - t_0)\,\mathrm{d}V\,\mathrm{d}t = \psi(\boldsymbol{x}_0, t_0). \tag{5.75}$$

5.15 Exercises

EXERCISE 5.1: PRACTICE WITH THE SIFTING PROPERTY

Evaluate the following integrals, making use of the sifting property of the Dirac delta.

(a) $\int_{-\infty}^{\infty} \delta(2x-1) \cos x \, \mathrm{d}x$
(b) $\int_{0}^{\infty} \delta(x^2-4) \, \mathrm{e}^{-x} \, \mathrm{d}x$

EXERCISE 5.2: PRACTICE WITH THE DIRAC DERIVATIVE

Evaluate the following integral, making use of properties of the Dirac derivative from Section 5.9

$$\int_{-\infty}^{\infty} \delta'(x-1) \, \mathrm{e}^{-x^2} \, \mathrm{d}x. \tag{5.76}$$

EXERCISE 5.3: IMPULSE FORCING

Consider a point particle with mass, m, free to move on the real line without friction. Assume it is at rest for times, $t \leq t_0$. At time $t = 0$ apply an impulse, $J \, \delta(t)$, where $J > 0$ is the strength (dimensions M L of the impulse. What is the velocity of the particle for times $t > 0$? Hint: write Newton's law of motion for this particle, and then time integrate.

EXERCISE 5.4: DIRAC AND THE SECOND DERIVATIVE OF $|x|$

Show that

$$\frac{\mathrm{d}^2|x|}{\mathrm{d}x^2} = 2 \, \delta(x). \tag{5.77}$$

Hint: focus on the first derivative and its connection to the Heaviside step function, and then make use of the relation between the Heaviside and Dirac.

EXERCISE 5.5: CIRCULAR MASS DISTRIBUTION

Consider a mass that is uniformly distributed in a ring with radius $r = R$ at vertical position $z = 0$ in three-dimensional space. Assume the total mass of the ring is M. Write the mass density, ρ, in cylindrical coordinates. Verify that the space integral over the ring indeed yields the total mass, so that $\int \rho \, \mathrm{d}V = M$.

Chapter 6

FOURIER ANALYSIS

In this chapter we survey salient features of Fourier analysis, which represents the canonical spectral decomposition of a signal into orthogonal components, here given by trigonometric functions. Conversely, Fourier analysis allows one to synthesize a signal by specifying the amplitudes of the component modes. Fourier analysis is particularly relevant to the study of wave mechanics, thus motivating the use of wave mechanics terminology in this chapter. We assume Cartesian coordinates throughout this chapter, given that our treatment of Fourier analysis assumes flat Euclidean space.[1] Furthermore, we are only concerned in this book with physical systems, in which all physical quantities are real numbers. Hence, our use of complex numbers, common in Fourier analysis, is solely for convenience.

The following lists our conventions for Fourier analysis, with further context for these conventions given later in the chapter.

- P is the period for functions represented using a Fourier series. Note that some treatments instead write the period as $P = 2\,L$.
- A factor of $1/2\pi$ is placed on the inverse Fourier transform as per equation (6.68c), whereas there is unity for the Fourier transform. An alternative convention is commonly used in the quantum mechanics literature whereby $1/\sqrt{2\pi}$ appears on both the Fourier transform and its inverse. We have more to say about this convention in Section 6.3.7. Note that the $1/2\pi$ factor for the inverse Fourier transform in one-dimension becomes $(1/2\pi)^n$ for n-dimensions. In this chapter we focus on $n = 1$ since generalizations to higher dimensions are straightforward.
- Fourier space is referred to as $\boldsymbol{k}$-space or wavevector space, which is a dual vector space (through the Fourier integral transform) to $\boldsymbol{x}$-space or position space. Some treatments refer to $\boldsymbol{x}$-space as "physical space". We avoid that language since for describing a physical system, $\boldsymbol{x}$-space is no more or less physical than $\boldsymbol{k}$-space. Instead, they emphasize complementary features and both offer physical insights.

[1]Spherical harmonics offers a generalization of Cartesian Fourier analysis to the sphere.

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The following references offer compatible treatments to that here, with many also providing far more details: Chapter 12 of *Elmore and Heald* (1969), Chapters 2 and 5 of *Spiegel* (1974b), Appendix A of *Gasiorowicz* (1974), Appendix I of *Cohen-Tannoudji et al.* (1977), Sections 5.10-5.15 of *Hildebrand* (1976), Section 5.6 of *Stakgold* (2000b), and Section 6.4.2 of *Thorne and Blandford* (2017). However, there are slight inconsistencies in conventions that warrant care if taking results from the literature, such as integral tables for Fourier transforms. This video from 3Blue1Brown provides an insightful introduction to Fourier series, and this video, also from 3Blue1Brown provides a corresponding introduction to Fourier transforms.

6.1 Complex numbers

All physical fields are real in the physics considered in this book. Even so, we sometimes make use of complex numbers for convenience, particularly when studying linear waves since manipulating exponentials is simpler than the alternative sines and cosines. We here summarize a few salient points for working with complex numbers in physics.

6.1.1 Modulus and phase

A complex number, $\mathcal{A}$, is the sum of its real and imaginary parts, which we write as[2]

$$\mathcal{A} = \mathrm{Re}[\mathcal{A}] + \mathrm{i}\ \mathrm{Im}[\mathcal{A}], \tag{6.1}$$

and its complex conjugate is written

$$\mathcal{A}^* = \mathrm{Re}[\mathcal{A}] - \mathrm{i}\ \mathrm{Im}[\mathcal{A}]. \tag{6.2}$$

Making use of the Euler identity,[3]

$$\mathrm{e}^{\mathrm{i}\alpha} = \cos\alpha + \mathrm{i}\sin\alpha, \tag{6.3}$$

allows us to write a complex number as

$$\mathcal{A} = \sqrt{\mathcal{A}\,\mathcal{A}^*}\,\mathrm{e}^{\mathrm{i}\,\alpha} = |\mathcal{A}|\,(\cos\alpha + \mathrm{i}\sin\alpha) \quad \text{with } \tan\alpha = \mathrm{Im}[\mathcal{A}]/\,\mathrm{Re}[\mathcal{A}]. \tag{6.4}$$

The term

$$|\mathcal{A}| = \sqrt{\mathcal{A}\,\mathcal{A}^*} = \sqrt{\mathrm{Re}[\mathcal{A}]^2 + \mathrm{Im}[\mathcal{A}]^2} \tag{6.5}$$

is the modulus of the complex number $\mathcal{A}$. The notation $|\mathcal{A}|$ is motivated since the modulus is a generalization of the absolute value used for real numbers. The angle, α, is called the *argument* in the maths literature, whereas we refer to it as the phase to correspond to its name in wave mechanics. We illustrate these formulae in Figure 6.1 within the complex plane.

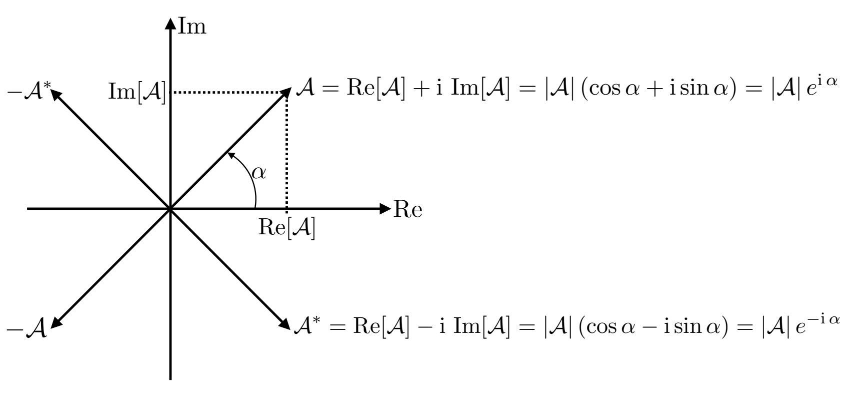

FIGURE 6.1: Illustrating the complex plane and its representation of a complex number, $\mathcal{A} = \mathrm{Re}[\mathcal{A}] + \mathrm{i}\ \mathrm{Im}[\mathcal{A}] = |\mathcal{A}|\,(\cos\alpha + \mathrm{i}\sin\alpha) = |\mathcal{A}|\,\mathrm{e}^{\mathrm{i}\,\alpha}$, along with its complex congugate, $\mathcal{A}^* = |\mathcal{A}|\,\mathrm{e}^{-\mathrm{i}\,\alpha}$, as well as $-\mathcal{A}^*$ and $-\mathcal{A}$.

The complex conjugate of the product of two complex numbers is the product of the

[2]In this book, we generally use the LaTeX *mathcal* notation for complex numbers.
[3]There are few equations in mathematics that are more elegant and powerful than Euler's identity.

conjugate, which can be shown by

$$\mathcal{A}\,\mathcal{B} = |\mathcal{A}|\,|\mathcal{B}|\,\mathrm{e}^{\mathrm{i}\,(\alpha+\beta)} \tag{6.6a}$$

$$\mathcal{A}^*\,\mathcal{B}^* = |\mathcal{A}|\,|\mathcal{B}|\,\mathrm{e}^{-\mathrm{i}\,(\alpha+\beta)} = (\mathcal{A}\,\mathcal{B})^*. \tag{6.6b}$$

When we write a complex number in this book, it is the real part that is of physical interest

$$\mathrm{Re}[\mathcal{A}] = \sqrt{\mathcal{A}\,\mathcal{A}^*}\cos\alpha = |\mathcal{A}|\cos\alpha. \tag{6.7}$$

Hence, for most purposes we evaluate the product, $\mathcal{A}\,\mathcal{A}$, as the product of the real parts

$$\mathrm{Re}[\mathcal{A}]\,\mathrm{Re}[\mathcal{A}] = |\mathcal{A}|^2\cos^2\alpha. \tag{6.8}$$

Likewise, with $\mathcal{B} = |\mathcal{B}|\,\mathrm{e}^{\mathrm{i}\,\beta}$, we interpret the product $\mathcal{A}\,\mathcal{B}$ as

$$\mathrm{Re}[\mathcal{A}]\,\mathrm{Re}[\mathcal{B}] = |\mathcal{A}|\,|\mathcal{B}|\cos\alpha\cos\beta. \tag{6.9}$$

These considerations are particularly important when computing energetics of wave fields. Namely, energetics involves the product of real fields rather than the product of complex fields. So if we work with the complex representation of a wave, then we must compute its real part prior to computing the product; i.e., the real part of the product of two complex fields is not equal to the product of the real parts

$$\mathrm{Re}[\mathcal{A}\,\mathcal{B}] \neq \mathrm{Re}[\mathcal{A}]\,\mathrm{Re}[\mathcal{B}]. \tag{6.10}$$

The following identities are particularly useful when computing energetics of wave fields (see VOLUME 4)

$$\mathrm{Re}[\mathrm{i}\,\mathcal{A}] = \mathrm{Re}[\mathrm{i}\,\mathrm{Re}[\mathcal{A}] - \mathrm{Im}[\mathcal{A}]] = -\,\mathrm{Im}[\mathcal{A}] \tag{6.11a}$$

$$\mathrm{Re}[\mathcal{A}^*\,\mathcal{B}] = |\mathcal{A}|\,|\mathcal{B}|\cos(-\alpha+\beta) = \mathrm{Re}[\mathcal{A}\,\mathcal{B}^*] \tag{6.11b}$$

$$\mathrm{Im}[\mathcal{A}^*\,\mathcal{B}] = |\mathcal{A}|\,|\mathcal{B}|\sin(-\alpha+\beta) = -|\mathcal{A}|\,|\mathcal{B}|\sin(\alpha-\beta) = -\,\mathrm{Im}[\mathcal{A}\,\mathcal{B}^*]. \tag{6.11c}$$

6.1.2 Phase averaging

There are many occasions in the study of waves where it is useful to average over the extent of a wave, either in space or time. Doing so provides a measure for the "wave averaged" properties. To add precision to this notion, consider two real wave fields written in the form of a traveling plane wave

$$A = \mathrm{Re}[\mathcal{A}\,\mathrm{e}^{\mathrm{i}\,(\boldsymbol{k}\cdot\boldsymbol{x}-\omega\,t)}] = |\mathcal{A}|\cos(\boldsymbol{k}\cdot\boldsymbol{x}-\omega\,t+\alpha) \tag{6.12a}$$

$$B = \mathrm{Re}[\mathcal{B}\,\mathrm{e}^{\mathrm{i}\,(\boldsymbol{k}\cdot\boldsymbol{x}-\omega\,t)}] = |\mathcal{B}|\cos(\boldsymbol{k}\cdot\boldsymbol{x}-\omega\,t+\beta), \tag{6.12b}$$

where $\boldsymbol{k}\cdot\boldsymbol{x}-\omega\,t$ is the space-time dependence found for a traveling plane wave, with $\boldsymbol{k}$ the wavevector and ω the angular frequency. We wrote the complex wave amplitudes as

$$\mathcal{A} = |\mathcal{A}|\,\mathrm{e}^{\mathrm{i}\,\alpha} \quad \text{and} \quad \mathcal{B} = |\mathcal{B}|\,\mathrm{e}^{\mathrm{i}\,\beta}, \tag{6.13}$$

where α and β are phase shifts relative to $\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t$. To introduce the notion of a phase average, introduce a constant phase shift, φ, according to

$$\alpha = \alpha' + \varphi \quad \text{and} \quad \beta = \beta' + \varphi, \tag{6.14}$$

in which case we define a phase average as

$$\langle \cdots \rangle \equiv \frac{1}{2\pi} \int_0^{2\pi} (\cdots)\, \mathrm{d}\varphi, \tag{6.15}$$

so that a phase average is computed by sampling over the extent of a single wavelength or single wave period. The phase average of a traveling plane wave is zero

$$\langle A \rangle = \frac{1}{2\pi} \int_0^{2\pi} |\mathcal{A}| \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \alpha' + \varphi)\, \mathrm{d}\varphi = 0, \tag{6.16}$$

whereas the phase average for the product of two real wave fields does not generally vanish

$$\langle A\, B \rangle = \frac{1}{2\pi} \int_0^{2\pi} |\mathcal{A}|\, |\mathcal{B}| \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \alpha' + \varphi) \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \beta' + \varphi)\, \mathrm{d}\varphi \tag{6.17a}$$

$$= \frac{1}{4\pi} \int_0^{2\pi} |\mathcal{A}|\, |\mathcal{B}|\, [\cos(\alpha' - \beta') + \cos(2\, \boldsymbol{k} \cdot \boldsymbol{x} - 2\, \omega\, t + 2\, \varphi + \alpha' - \beta')]\, \mathrm{d}\varphi \tag{6.17b}$$

$$= \frac{1}{2}\, |\mathcal{A}|\, |\mathcal{B}| \cos(\alpha' - \beta') \tag{6.17c}$$

$$= \frac{1}{2}\, \mathrm{Re}[\mathcal{A}^*\, \mathcal{B}] \tag{6.17d}$$

$$= \frac{1}{2}\, \mathrm{Re}[\mathcal{A}\, \mathcal{B}^*]. \tag{6.17e}$$

We see that the phase average of a product acts to remove information about the common phase, leaving only information about the modulus of the wave amplitudes and their phase difference. For example,

$$\langle A^2 \rangle = |\mathcal{A}|^2/2, \tag{6.18}$$

so that the phase average for the square of a real wave field is one-half the squared modulus of the wave amplitude. Similarly, if $\mathcal{B} = \mathrm{i}\, \mathcal{A}$ then

$$\langle A\, B \rangle = \frac{1}{2\pi} \int_0^{2\pi} |\mathcal{A}|\, |\mathcal{B}|\, \cos(\pi/2)\, \mathrm{d}\varphi = 0, \tag{6.19}$$

in which we say that the real wave fields A and B are out of phase.

6.1.3 Euler's identity and complex numbers

As noted earlier, there is no physical reason to introduce complex numbers in classical physics. However, there are many mathematical reasons. In particular, it is generally more convenient to use complex Fourier analysis than real Fourier analysis when establishing various theoretical results in wave mechanics.

The central reason to employ complex numbers is the shear elegance and power of Euler's identity (6.3). We encountered examples in Section 6.1.2 when discussing phase averaging. One more example concerns the sum of two waves, such as the sum of the two real wave fields

(6.12a) and (6.12b)

$$C = A + B = |\mathcal{A}| \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \alpha) + |\mathcal{B}| \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \beta). \tag{6.20}$$

It is certainly possible to determine an expression for the combined wave, C, through the use of trigonometric identities. However, it is far less tedious to compute the sum via Euler's identity, in which case

$$\begin{aligned}
C &= |\mathcal{A}| \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \alpha) + |\mathcal{B}| \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \beta) && \text{(6.21a)}\\
&= \mathrm{Re}[|\mathcal{A}|\, \mathrm{e}^{\mathrm{i}\,(\boldsymbol{k}\cdot\boldsymbol{x} - \omega\, t + \alpha)} + |\mathcal{B}|\, \mathrm{e}^{\mathrm{i}\,(\boldsymbol{k}\cdot\boldsymbol{x} - \omega\, t + \beta)}] && \text{(6.21b)}\\
&= \mathrm{Re}[(|\mathcal{A}|\, \mathrm{e}^{\mathrm{i}\,\alpha} + |\mathcal{B}|\, \mathrm{e}^{\mathrm{i}\,\beta})\, \mathrm{e}^{\mathrm{i}\,(\boldsymbol{k}\cdot\boldsymbol{x} - \omega\, t)}] && \text{(6.21c)}\\
&\equiv \mathrm{Re}[|\mathcal{C}|\, \mathrm{e}^{\mathrm{i}\,(\boldsymbol{k}\cdot\boldsymbol{x} - \omega\, t + \gamma)}] && \text{(6.21d)}\\
&= |\mathcal{C}| \cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega\, t + \gamma), && \text{(6.21e)}
\end{aligned}$$

where we introduced the complex amplitude

$$|\mathcal{C}|\, \mathrm{e}^{\mathrm{i}\,\gamma} \equiv |\mathcal{A}|\, \mathrm{e}^{\mathrm{i}\,\alpha} + |\mathcal{B}|\, \mathrm{e}^{\mathrm{i}\,\beta}. \tag{6.22}$$

We can further use Euler's identity and some trigonometric identities to determine the real amplitude and phase shift for the combined wave

$$\begin{aligned}
|\mathcal{C}|^2 &= |\mathcal{A}|^2 + |\mathcal{B}|^2 + 2\, |\mathcal{A}|\, |\mathcal{B}| \cos(\alpha - \beta) && \text{(6.23a)}\\
\tan\gamma &= \frac{|\mathcal{A}| \sin\alpha + |\mathcal{B}| \sin\beta}{|\mathcal{A}| \cos\alpha + |\mathcal{B}| \cos\beta}. && \text{(6.23b)}
\end{aligned}$$

The curious reader is encouraged to use just real analysis to compute the expression (6.21e), along with the real amplitude (6.23a) and phase shift (6.23b).

6.2 Fourier series

Consider an arbitrary real function, $F(x)$, defined over the real line and that is piecewise continuous on every finite interval, as is its first derivative.[4] We also assume the function satisfies the periodicity condition[5]

$$F(x + P) = F(x), \tag{6.24}$$

where $P > 0$ is the period.[6] The periodicity condition (6.24) holds for every point on the real line so that

$$F(x + n\, P) = F(x) \quad \text{for arbitrary positive or negative integer } n. \tag{6.25}$$

How general are periodic functions defined over the full real line? To answer this question, consider a function defined only over a finite domain, $x \in [\alpha, \beta]$, and that is periodic over that

[4] These properties for $F(x)$ are commonly satisfied for functions considered in this book.

[5] We use the notation x as for a spatial coordinate. Yet everything that follows holds if x is interpreted as time.

[6] Some treatments, such as Chapter 2 of *Spiegel* (1974b), assume the function has period $F(x) = F(x + 2L)$, in which case $P = 2L$. In general, care is needed when making use of expressions from the literature to account for slight differences in conventions.

domain

$$G(\alpha) = G(\beta). \tag{6.26}$$

Quite trivially, we can construct the extended function, $F(x)$, that equals to $G(x)$ for $x \in [\alpha, \beta]$ and that is periodic over the real line, $F(x) = F(x + P)$, with period $P = \beta - \alpha$. For this reason, it is sufficient to focus on periodic functions defined over the full real line, $x \in (-\infty, \infty)$.

6.2.1 Fourier series of sines and cosines

A Fourier series decomposes an arbitrary periodic function in terms of the function's mean value over a period, plus an infinite series of sines and cosines[7]

$$F(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(k_n\, x) + b_n \sin(k_n\, x)], \tag{6.27}$$

where we introduced the discrete wavenumber

$$k_n = 2\,\pi\, n/P. \tag{6.28}$$

Clearly the Fourier series (6.27) satisfies the periodicity condition, $F(x) = F(x + n\, P)$. Mathematically, the Fourier series is enabled by completeness of the trigonometric functions as a basis for periodic functions over a finite interval. Furthermore, the real coefficients, a_n and b_n, are determined through use of orthonormality conditions satisfied by the sine and cosine functions

$$\frac{2}{P} \int_{x_\circ}^{x_\circ + P} \cos(k_n\, x) \sin(k_m\, x)\, \mathrm{d}x = 0 \tag{6.29a}$$

$$\frac{2}{P} \int_{x_\circ}^{x_\circ + P} \cos(k_n\, x) \cos(k_m\, x)\, \mathrm{d}x = \delta_{mn} \tag{6.29b}$$

$$\frac{2}{P} \int_{x_\circ}^{x_\circ + P} \sin(k_n\, x) \sin(k_m\, x)\, \mathrm{d}x = \delta_{mn}, \tag{6.29c}$$

where δ_{mn} is the Kronecker delta, which is unity if $m = n$ and vanishes otherwise.[8] Furthermore, note that the constant, $x_\circ$, is arbitrary and with the equalities holding for any value. We can understand this arbitrariness by noting that the integral over a single period of a periodic function is unchanged when shifting the start point of the integration. All that matters is that we extend over a single period.[9]

The orthonormality conditions (6.29a)-(6.29c) directly lead to the following expressions for the Fourier series expansion coefficients

$$a_0 = \frac{1}{P} \int_{x_\circ}^{x_\circ + P} F(x)\, \mathrm{d}x \tag{6.30a}$$

[7]There are certain mathematical conditions required of $F(x)$ that allow for the Fourier series to be an identity. These conditions are typically satisfied by functions encountered in physics. See Section 5.10 of *Hildebrand* (1976) for more details.

[8]As seen in Section 1.3, the Kronecker delta is the representation of the Euclidean metric when represented by Cartesian coordinates. For equations (6.29b) and (6.29c), we use the Kronecker delta merely as a signal of orthonormality of the trigonometric functions.

[9]We make use of this arbitrariness in Section 6.3.1 where we choose a particularly useful value for developing the notion of a Fourier integral.

$$a_{n>0} = \frac{2}{P} \int_{x_\circ}^{x_\circ+P} \cos(k_n\, x)\, F(x)\, \mathrm{d}x \tag{6.30b}$$

$$b_0 = 0 \tag{6.30c}$$

$$b_{n>0} = \frac{2}{P} \int_{x_\circ}^{x_\circ+P} \sin(k_n\, x)\, F(x)\, \mathrm{d}x. \tag{6.30d}$$

Evidently, a_0 is the mean value of $F(x)$ over a single period. As an exercise, we verify equation (6.30b) by multiplying the Fourier series (6.27) by $\cos(k_m\, x)$ for $m > 0$ and integrating over a period

$$\int_{x_\circ}^{x_\circ+P} \cos(k_m\, x)\, F(x)\, \mathrm{d}x = \int_{x_\circ}^{x_\circ+P} \cos(k_m\, x) \left[a_0 + \sum_{n=1}^{\infty} [a_n \cos(k_n\, x) + b_n \sin(k_n\, x)] \right] \mathrm{d}x \tag{6.31a}$$

$$= (P/2)\, a_n\, \delta_{mn}, \tag{6.31b}$$

where the second equality used the orthogonality conditions (6.29a)-(6.29c). We use analogous steps to verify equation (6.30d),

$$\int_{x_\circ}^{x_\circ+P} \sin(k_m\, x)\, F(x)\, \mathrm{d}x = \int_{x_\circ}^{x_\circ+P} \sin(k_m\, x) \left[a_0 + \sum_{n=1}^{\infty} [a_n \cos(k_n\, x) + b_n \sin(k_n\, x)] \right] \mathrm{d}x \tag{6.32a}$$

$$= (P/2)\, b_n\, \delta_{mn}. \tag{6.32b}$$

6.2.2 Functions with a particular parity

Any function, $F(x)$, can be written as the sum

$$F(x) = \frac{F(x) + F(-x)}{2} + \frac{F(x) - F(-x)}{2} = F^{\text{even}}(x) + F^{\text{odd}}(x). \tag{6.33}$$

The function $F^{\text{odd}}(x)$ is said to have odd parity since it swaps sign upon the transformation $x \to -x$

$$F^{\text{odd}}(-x) = -F^{\text{odd}}(x), \tag{6.34}$$

whereas F^{even} has even parity since it maintains the same value upon the transformation $x \to -x$

$$F^{\text{even}}(-x) = F^{\text{even}}(x). \tag{6.35}$$

The cosine and sine functions used in the Fourier series have even and odd parities, respectively

$$\cos(k_n\, x) = \cos(-k_n\, x) \quad \Longrightarrow \text{cosine has even parity} \tag{6.36a}$$

$$\sin(k_n\, x) = -\sin(-k_n\, x) \Longrightarrow \text{sine has odd parity.} \tag{6.36b}$$

As we show next, the Fourier series expansion of F^{even} and F^{odd} involve only a portion of the full series (6.27).

6.2.3 Fourier series of periodic functions with parity

Consider an even parity periodic function

$$F^{\text{even}}(x) = F^{\text{even}}(x+P), \tag{6.37}$$

so that this function has a Fourier series representation (equation (6.27))

$$F^{\text{even}}(x) = a_0 + \sum_{n=1}^{\infty}[a_n \cos(k_n\, x) + b_n \sin(k_n\, x)]. \tag{6.38}$$

Making use of the even parity property, $F^{\text{even}}(-x) = F^{\text{even}}(x)$, on the left hand side, as well as the parity properties (6.36a) and (6.36b) of the cosine and sine functions on the right hand side, leads to

$$F^{\text{even}}(x) = a_0 + \sum_{n=1}^{\infty}[a_n \cos(k_n\, x) - b_n \sin(k_n\, x)]. \tag{6.39}$$

For the series expansions (6.38) and (6.39) to be consistent for arbitrary even functions requires $b_n = 0$ for all n. Evidently, an even periodic function only has a Fourier cosine series expansion

$$F^{\text{even}}(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n\, x), \tag{6.40}$$

An analogous result holds for an odd parity periodic function, in which $a_n = 0$ so that F^{odd} only has sine functions for its Fourier expansion

$$F^{\text{odd}}(x) = \sum_{n=1}^{\infty} b_n \sin(k_n\, x). \tag{6.41}$$

As a self-consistency check on the cosine expansion (6.40), compute the b_n coefficients according to equation (6.30d) whereby (setting $x_\circ = -P/2$ for convenience)

$$\begin{aligned}
b_{n>0} &= \frac{2}{P}\int_{-P/2}^{P/2} \sin(k_n\, x)\, F^{\text{even}}(x)\, \mathrm{d}x && \text{equation (6.30d)} && (6.42\text{a})\\
&= \frac{2}{P}\int_{P/2}^{-P/2} \sin(-k_n\, y)\, F^{\text{even}}(-y)\, (-\mathrm{d}y) && \text{let } x = -y && (6.42\text{b})\\
&= \frac{2}{P}\int_{P/2}^{-P/2} \sin(k_n\, y)\, F^{\text{even}}(-y)\, \mathrm{d}y && \sin(-k_n y) = -\sin(k_n y) && (6.42\text{c})\\
&= -\frac{2}{P}\int_{-P/2}^{P/2} \sin(k_n\, y)\, F^{\text{even}}(-y)\, \mathrm{d}y. && \text{swap integration limits} && (6.42\text{d})\\
&= -\frac{2}{P}\int_{-P/2}^{P/2} \sin(k_n\, y)\, F^{\text{even}}(y)\, \mathrm{d}y. && F^{\text{even}}(y) = F^{\text{even}}(-y). && (6.42\text{e})
\end{aligned}$$

The only way to satisfy this equation is for $b_n = 0$ for each n, thus verifying the expansion (6.40). The same sort of argument is used to prove that $a_n = 0$ when computing the Fourier series expansion of an odd function, thus verifying the Fourier sine expansion (6.41) (see Exercise).

6.2.4 Complex Fourier series of exponentials

Rather than a sum of sines and cosines, we can use the Euler identity (6.3) to decompose a periodic function into a series of exponentials by introducing the complex expansion coefficients

$$
\begin{aligned}
c_0 &= a_0 && (6.43a)\\
c_{-n} &= (a_n + \mathrm{i}\, b_n)/2 \quad \text{for } n > 0 && (6.43b)\\
c_n &= (a_n - \mathrm{i}\, b_n)/2 \quad \text{for } n > 0. && (6.43c)
\end{aligned}
$$

Making use of the expressions (6.30b) and (6.30d) yields

$$
c_n = \frac{1}{P} \int_{x_\circ}^{x_\circ + P} e^{-\mathrm{i}\, k_n\, x}\, F(x)\, \mathrm{d}x. \tag{6.44}
$$

The reality condition, commonly referred to as *conjugate symmetry*,

$$
c_{-n} = c_n^*, \tag{6.45}
$$

holds since we are only considering real functions, $F(x)$. Inverting the relations (6.43a)-(6.43c)

$$
\begin{aligned}
a_0 &= c_0 && (6.46a)\\
a_n &= c_n + c_{-n} \qquad \text{for } n > 0 && (6.46b)\\
b_n &= \mathrm{i}\,(c_n - c_{-n}) \quad \text{for } n > 0 && (6.46c)
\end{aligned}
$$

then allows us to write the Fourier series (6.27) as

$$
\begin{aligned}
F(x) &= a_0 + \sum_{n=1}^{\infty} [a_n \cos(k_n\, x) + b_n \sin(k_n\, x)] && (6.47a)\\
&= c_0 + \sum_{n=1}^{\infty} [(c_n + c_{-n}) \cos(k_n\, x) + \mathrm{i}\,(c_n - c_{-n}) \sin(k_n\, x)] && (6.47b)\\
&= c_0 + \sum_{n=1}^{\infty} c_n\, [\cos(k_n\, x) + \mathrm{i} \sin(k_n\, x)] + \sum_{n=1}^{\infty} c_{-n}\, [\cos(k_n\, x) - \mathrm{i} \sin(k_n\, x)] && (6.47c)\\
&= c_0 + \sum_{n=1}^{\infty} c_n\, e^{\mathrm{i}\, k_n\, x} + \sum_{n=1}^{\infty} c_{-n}\, e^{-\mathrm{i}\, k_n\, x} && (6.47d)\\
&= c_0 + \sum_{n=1}^{\infty} c_n\, e^{\mathrm{i}\, k_n\, x} + \sum_{n=1}^{\infty} c_{-n}\, e^{\mathrm{i}\, k_{-n}\, x} && (6.47e)\\
&= \sum_{n=-\infty}^{\infty} c_n\, e^{\mathrm{i}\, k_n\, x}, && (6.47f)
\end{aligned}
$$

where the penultimate step made use of equation (6.28) for the wavenumber, whereby

$$
k_n = 2\, \pi\, n/P = -k_{-n}. \tag{6.48}
$$

The exponential expression in equation (6.47f) is commonly more convenient for manipulations, with the conjugate symmetry condition (6.45) of fundamental importance when working with real functions. Importantly, note that the real coefficients, a_n and b_n, are defined for $n \geq 0$ (with $b_0 = 0$) , whereas the complex coefficients, c_n, are defined for all integers.

6.2.5 Bessel-Parseval relations

Making use of the conjugate symmetry condition (6.45) allows us to write the square of a real periodic function

$$
\begin{aligned}
F(x)\,F(x) &= \left[\sum_{n=-\infty}^{\infty} c_n\,\mathrm{e}^{\mathrm{i}\,k_n\,x}\right]\left[\sum_{m=-\infty}^{\infty} c_m\,\mathrm{e}^{\mathrm{i}\,k_m\,x}\right] && \text{equation (6.47f)} && (6.49\text{a})\\
&= \left[\sum_{n=-\infty}^{\infty} c_n\,\mathrm{e}^{\mathrm{i}\,k_n\,x}\right]\left[\sum_{m=-\infty}^{\infty} c_{-m}\,\mathrm{e}^{-\mathrm{i}\,k_m\,x}\right] && \text{swap } m \text{ summation order} && (6.49\text{b})\\
&= \left[\sum_{n=-\infty}^{\infty} c_n\,\mathrm{e}^{\mathrm{i}\,k_n\,x}\right]\left[\sum_{m=-\infty}^{\infty} c_m^*\,\mathrm{e}^{-\mathrm{i}\,k_m\,x}\right] && \text{conjugate symmetry (6.45)} && (6.49\text{c})\\
&= \sum_{n,m=-\infty}^{\infty} c_n\,c_m^*\,\mathrm{e}^{\mathrm{i}\,(k_n-k_m)\,x} && \text{combine sums.} && (6.49\text{d})
\end{aligned}
$$

The *Bessel-Parseval relation*, sometimes just referred to as Parseval's identity, results from performing an average of equation (6.49d) over a single period

$$
\frac{1}{P}\int_{x_\circ}^{x_\circ+P} F(x)\,F(x)\,\mathrm{d}x = \sum_{n=-\infty}^{\infty} c_n\,c_n^* = \sum_{n=-\infty}^{\infty} |c_n|^2 = \sum_{n=-\infty}^{\infty} c_n\,c_{-n}. \tag{6.50}
$$

To reach this identity we used the orthonormality condition holding for the exponentials

$$
\frac{1}{P}\int_{x_\circ}^{x_\circ+P} \mathrm{e}^{\mathrm{i}\,(k_n-k_m)x}\,\mathrm{d}x = \delta_{mn}, \tag{6.51}
$$

and the final equality in equation (6.50) follows from conjugate symmetry (6.45) that holds since F is real. The analogous form of Parseval's identity holds when using the sine/cosine version of the Fourier series (6.27), whereby

$$
F(x)\,F(x) = \left[a_0 + \sum_{n=1}^{\infty}[a_n\cos(k_n\,x) + b_n\sin(k_n\,x)]\right]\left[a_0 + \sum_{m=1}^{\infty}[a_m\cos(k_m\,x) + b_m\sin(k_m\,x)]\right], \tag{6.52}
$$

with integration and the orthonormality relations (6.29a)-(6.29c) resulting in

$$
\frac{1}{P}\int_{x_\circ}^{x_\circ+P} F(x)\,F(x)\,\mathrm{d}x = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2). \tag{6.53}
$$

We find it useful to double-check this form of Parseval's identity while confirming we are consistently using both the real and complex forms of the Fourier series. For this purpose, substitute the expressions (6.43a)-(6.43c) for c_n into the final form of Parsevel's identity (6.50) to yield

$$\sum_{n=-\infty}^{\infty} c_n\, c_{-n} = c_0^2 + \sum_{n=-\infty}^{-1} c_n\, c_{-n} + \sum_{n=1}^{\infty} c_n\, c_{-n} = c_0^2 + 2\sum_{n=1}^{\infty} c_n\, c_{-n} = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2), \tag{6.54a}$$

which agrees with equation (6.53).

We can go one further step by introducing a second real and periodic function, $H(x) = H(x + m\, P)$, in which case

$$H(x) = \sum_{n=-\infty}^{\infty} d_n\, \mathrm{e}^{\mathrm{i}\, k_n\, x}. \tag{6.55}$$

The Bessel-Parseval relation (6.50) can thus be generalized to

$$\frac{1}{P}\int_{x_\circ}^{x_\circ+P} F(x)\, H(x)\, \mathrm{d}x = \sum_{n=-\infty}^{\infty} c_n\, d_n^* = \sum_{n=-\infty}^{\infty} c_n\, d_{-n}. \tag{6.56}$$

6.3 Fourier integrals

We develop *Fourier integrals* as the limit of a Fourier series and then derive some properties of these integrals. Our treatment is heuristic.[10]

6.3.1 Fourier's integral theorem

Consider a real periodic function, $F(x) = F(x + n\, P)$ with n an integer, and decompose it according to the Fourier series (6.27). Also, insert the expressions (6.30a)-(6.30d) for the expansion coefficients and specify the arbitrary constant $x_\circ = -P/2$ in order to symmetrize the integral limits. The resulting Fourier series is given by

$$F(x) = \frac{2}{P}\int_{-P/2}^{P/2} F(u)\left[\frac{1}{2} + \sum_{n=1}^{\infty}[\cos(k_n\, u)\cos(k_n\, x) + \sin(k_n\, u)\sin(k_n\, x)]\right]\mathrm{d}u. \tag{6.57}$$

We now consider the limit as $P \to \infty$, in which case the period becomes infinite so that the function, $F(x)$, is no longer periodic over a finite interval. Precisely, we make the following assumptions and conversions.

- FINITE INTEGRAL: We assume the absolute value of the function has a finite integral for arbitrary P, which in turn means that its average tends to zero as P becomes infinite:

$$\int_{-P/2}^{P/2} |F(u)|\, \mathrm{d}u < \infty \Longrightarrow \lim_{P\to\infty}\frac{1}{P}\int_{-P/2}^{P/2} F(u)\, \mathrm{d}u = 0. \tag{6.58}$$

- INFINITESIMAL WAVENUMBER INCREMENT: We introduce the wavenumber increment

$$k_{n+1} - k_n = \Delta k = 2\pi/P, \tag{6.59}$$

[10] The derivation of the Fourier transform equations in this section follows that given by Chapter 5 of *Spiegel* (1974b), particularly worked exercise 5.11, yet note that L in *Spiegel* (1974b) is given by $L = P/2$.

which becomes infinitesimal as P becomes large. As a result, the infinite sum over discrete wavenumbers transitions to an integral over continuous wavenumbers

$$\lim_{P\to\infty}\sum_{n=1}^{\infty}\frac{2}{P}=\frac{1}{\pi}\int_0^{\infty}\mathrm{d}k. \tag{6.60}$$

Making use of these results in equation (6.57) leads to

$$F(x)=\frac{1}{\pi}\int_0^{\infty}\left[\cos(k\,x)\int_{-\infty}^{\infty}F(u)\cos(k\,u)\,\mathrm{d}u+\sin(k\,x)\int_{-\infty}^{\infty}F(u)\sin(k\,u)\,\mathrm{d}u\right]\mathrm{d}k, \tag{6.61}$$

which is known as *Fourier's integral theorem.* We can bring Fourier's integral theorem a bit closer to the discrete Fourier sum (6.27) by introducing the real amplitude functions

$$A(k)\equiv\frac{1}{\pi}\int_{-\infty}^{\infty}F(u)\cos(k\,u)\,\mathrm{d}u \quad\text{and}\quad B(k)\equiv\frac{1}{\pi}\int_{-\infty}^{\infty}F(u)\sin(k\,u)\,\mathrm{d}u, \tag{6.62}$$

thus resulting in the tidy expression of the Fourier integral theorem (6.61)

$$F(x)=\int_0^{\infty}[A(k)\cos(k\,x)+B(k)\sin(k\,x)]\,\mathrm{d}k. \tag{6.63}$$

This version of Fourier's integral theorem forms the foundation for the remainder of this section, which largely consists of rewriting this identity in a variety of forms. Before doing so, we emphasize the following points.

All physical fields produce real numbers

As noted at the start of this chapter, all physical fields in this book are real. Hence, every term in equation (6.63) is real, including the amplitude functions, $A(k)$ and $B(k)$. So although many versions of Fourier analysis involve complex numbers, as we describe in Section 6.3.2, the introduction of complex numbers is based on mathematical convenience and so it is not physically motivated.

Concerning the functional degrees of freedom

For each function, $F(x)$, there are two amplitude functions, $A(k)$ and $B(k)$. However, there is no explosion of functional degrees of freedom since $F(x)$ is defined over the full real line, $-\infty<x<\infty$ whereas $A(k)$ and $B(k)$ are defined with $k\geq 0$. The equal functional degrees of freedom manifest when using the complex version of the Fourier integral theorem in Section 6.3.2.

Parity properties of the amplitude functions

The parity properties of the amplitude functions, $A(k)$ and $B(k)$, are directly inherited from the cosine and sine functions appearing in their definition (6.62), so that we find

$$A(k)=A(-k) \quad\text{and}\quad B(k)=-B(-k). \tag{6.64}$$

As shown in Section 6.3.5, these properties are reflected in the nature of the integral theorem when representing functions, $F(x)$, with a particular parity.

Allowing the wavenumber to range over the full real line

The integral over wavenumber, k, in equation (6.62) extends from $k = 0$ to $k = \infty$. When moving to a complex representation that involves the Fourier transform in Section 6.3.2, the integral is extended to $-\infty < k < \infty$. The extension to $k < 0$ can be interpreted as a plane wave moving in the direction opposite to the wave with $k > 0$.

6.3.2 Fourier transform

Introducing the *Fourier transform*, $\mathcal{F}(k)$, allows us to connect to the complex Fourier series from Section (6.2.4)

$$\mathcal{F}(k) \equiv \pi\,[A(k) - \mathrm{i}\,B(k)] = \int_{-\infty}^{\infty} F(u)\,\mathrm{e}^{-\mathrm{i}\,k\,u}\,\mathrm{d}u \quad \Longleftarrow \text{Fourier transform.} \tag{6.65}$$

We can write the Fourier integral theorem (6.63) in terms of $\mathcal{F}(k)$ according to

$$F(x) = \int_{0}^{\infty} [A(k)\cos(k\,x) + B(k)\sin(k\,x)]\,\mathrm{d}k \tag{6.66a}$$

$$= \frac{1}{2\pi}\int_{0}^{\infty} \Big[\cos(k\,x)\,[\mathcal{F}(k) + \mathcal{F}^{*}(k)] + \mathrm{i}\,\sin(k\,x)\,[\mathcal{F}(k) - \mathcal{F}^{*}(k)]\,\Big]\,\mathrm{d}k \tag{6.66b}$$

$$= \frac{1}{2\pi}\int_{0}^{\infty} \left[\mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x} + \left[\mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\right]^{*}\right]\mathrm{d}k. \tag{6.66c}$$

From its definition (6.65), we know that the complex conjugate of the Fourier transform satisfies the *conjugate symmetry* identity

$$\mathcal{F}^{*}(k) = \int_{-\infty}^{\infty} F(u)\,\mathrm{e}^{\mathrm{i}\,k\,u}\,\mathrm{d}u = \mathcal{F}(-k). \tag{6.67}$$

This identity is directly analogous to the discrete Fourier transform identity (6.45), both of which hold due to the reality of the function, $F(x)$. We are thus led to the complex version of the Fourier integral theorem (6.63)

$$F(x) = \frac{1}{2\pi}\int_{0}^{\infty} \mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k + \frac{1}{2\pi}\int_{0}^{\infty} \mathcal{F}(-k)\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}k \tag{6.68a}$$

$$= \frac{1}{2\pi}\int_{0}^{\infty} \mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k - \frac{1}{2\pi}\int_{0}^{-\infty} \mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k \tag{6.68b}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k. \tag{6.68c}$$

With $\mathcal{F}(k)$ referred to as the Fourier transform we sometimes refer to $F(x)$ as the *inverse Fourier transform*. Correspondingly, $\mathcal{F}(k)$ and $F(x)$ are *Fourier transform pairs*.

6.3.3 A comment on conjugate symmetry

As noted above, reality of the function, $F(x)$, is ensured by a Fourier transform that satisfies the conjugate symmetry property (6.67). We make use of this property in much of our analysis in this book, such as when studying wave packets in VOLUME 4. However, there are reasons to sometimes dispense with conjugate symmetry.[11] Evidently, the identity (6.66c) ensures that $F(x)$ is real, even if the Fourier amplitudes, $\mathcal{F}(k)$, do not satisfy conjugate symmetry. That is, conjugate symmetry is a sufficient condition to ensure $F(x)$ is real, but it is not a necessary condition.

6.3.4 Integrals over a symmetric interval

Consider the integral of a function, $F(x)$, defined over an arbitrary, but symmetric, interval $x \in [-L, L]$. Decomposing this function into its even an odd components leads to

$$\int_{-L}^{L} F(x)\,\mathrm{d}x = \frac{1}{2}\int_{-L}^{L}[F(x)+F(-x)]\,\mathrm{d}x + \frac{1}{2}\int_{-L}^{L}[F(x)-F(-x)]\,\mathrm{d}x \tag{6.69a}$$

$$\equiv \int_{-L}^{L} F^{\mathsf{even}}(x)\mathrm{d}x + \int_{-L}^{L} F^{\mathsf{odd}}(x)\,\mathrm{d}x. \tag{6.69b}$$

The integral of the even parity component can be written

$$\int_{-L}^{L} F^{\mathsf{even}}(x)\,\mathrm{d}x = \int_{-L}^{0} F^{\mathsf{even}}(x)\,\mathrm{d}x + \int_{0}^{L} F^{\mathsf{even}}(x)\,\mathrm{d}x \tag{6.70a}$$

$$= -\int_{L}^{0} F^{\mathsf{even}}(-x)\,\mathrm{d}x + \int_{0}^{L} F^{\mathsf{even}}(x)\,\mathrm{d}x \tag{6.70b}$$

$$= 2\int_{0}^{L} F^{\mathsf{even}}(x)\,\mathrm{d}x. \tag{6.70c}$$

Evidently, the integral of an even parity function over a symmetric interval equals to twice the integral over half the interval

$$\int_{-L}^{L} F^{\mathsf{even}}(x)\,\mathrm{d}x = 2\int_{0}^{L} F^{\mathsf{even}}(x)\,\mathrm{d}x. \tag{6.71}$$

In a similar manner we find that the integral of an odd parity function over a symmetric interval vanishes

$$\int_{-L}^{L} F^{\mathsf{odd}}(x)\,\mathrm{d}x = \int_{-L}^{0} F^{\mathsf{odd}}(x)\,\mathrm{d}x + \int_{0}^{L} F^{\mathsf{odd}}(x)\,\mathrm{d}x \tag{6.72a}$$

$$= -\int_{L}^{0} F^{\mathsf{odd}}(-x)\,\mathrm{d}x + \int_{0}^{L} F^{\mathsf{odd}}(x)\,\mathrm{d}x \tag{6.72b}$$

$$= \int_{0}^{L} [F^{\mathsf{odd}}(-x) + F^{\mathsf{odd}}(x)]\,\mathrm{d}x \tag{6.72c}$$

$$= 0. \tag{6.72d}$$

We thus find that the integral of an arbitrary function over a symmetric interval is given by twice the integral of the even component over half the interval

$$\int_{-L}^{L} F(x)\,\mathrm{d}x = 2\int_{0}^{L} F_{\mathrm{even}}(x)\,\mathrm{d}x = \int_{0}^{L}[F(x)+F(-x)]\,\mathrm{d}x. \tag{6.73}$$

[11] One example is provided in VOLUME 4 when providing information about a wave packet's initial tendency rather than its initial position.

We make use of these relations in developing parity conditions for Fourier transforms, in which case the integral limits are infinity, $L = \infty$.

6.3.5 Fourier cosine and sine transforms

The Fourier integral theorem simplifies when acting on functions of a particular parity. As any function can be decomposed into its even and odd parity components (see Section 6.3.4), it is common to make use of the simplification by introducing the cosine and sine transforms. These transforms allow one to work with all real valued functions when convenient, rather than the complex valued functions used with the standard Fourier transform of Section 6.3.2.

Fourier cosine transform for even functions: $F(x) = F(-x)$

Consider an even function, $F(x) = F(-x)$, and expand it in terms of the Fourier integral theorem (6.63)

$$F(x) = \int_0^\infty [A(k)\cos(k\,x) + B(k)\sin(k\,x)]\,\mathrm{d}k. \tag{6.74}$$

Since $F(x) = F(-x)$ it follows that

$$F(x) = F(-x) = \int_0^\infty [A(k)\cos(k\,x) - B(k)\sin(k\,x)]\,\mathrm{d}k, \tag{6.75}$$

which follows from the odd parity of the sine function. For this relation to hold requires $B(k) = 0$, which indeed follows from its definition

$$\pi\,B(k) = \int_{-\infty}^{\infty} F(u)\sin(k\,u)\,\mathrm{d}u \tag{6.76a}$$

$$= \int_{-\infty}^{0} F(u)\sin(k\,u)\,\mathrm{d}u + \int_0^\infty F(u)\sin(k\,u)\,\mathrm{d}u \tag{6.76b}$$

$$= \int_0^\infty [-F(u) + F(u)]\sin(k\,u)\,\mathrm{d}u \tag{6.76c}$$

$$= 0. \tag{6.76d}$$

These simplification of Fourier's integral theorem motivate us to define the *Fourier cosine transform* for functions with even parity

$$F_{\mathrm{C}}(k) \equiv (\pi/2)\,A(k) = \frac{1}{2}\int_{-\infty}^{\infty} F(u)\cos(k\,u)\,\mathrm{d}u = \int_0^\infty F(u)\cos(k\,u)\,\mathrm{d}u, \tag{6.77}$$

so that Fourier's integral theorem (6.75) for an even parity function is

$$F(x) = \int_0^\infty A(k)\cos(k\,x)\,\mathrm{d}k = \frac{2}{\pi}\int_0^\infty F_{\mathrm{C}}(k)\cos(k\,x)\,\mathrm{d}k. \tag{6.78}$$

Correspondingly, the Fourier cosine transform of an odd parity function vanishes.

Fourier sine transform for odd functions: $F(x) = -F(-x)$

We now consider the case of an odd function, $F(x) = -F(-x)$, and expand it in terms of the Fourier integral theorem (6.63)

$$F(x) = \int_0^\infty [A(k)\cos(k\,x) + B(k)\sin(k\,x)]\,\mathrm{d}k = \int_0^\infty [-A(k)\cos(k\,x) + B(k)\sin(k\,x)]\,\mathrm{d}k, \tag{6.79}$$

where the second equality follows since $F(x) = -F(-x)$ and the parity of the cosine and sine functions. This relation is consistent only if $A(k) = 0$ for odd functions, which indeed follows from its definition

$$\pi\,A(k) = \int_{-\infty}^{\infty} F(u)\cos(k\,u)\,\mathrm{d}u \tag{6.80a}$$

$$= \int_{-\infty}^{0} F(u)\cos(k\,u)\,\mathrm{d}u + \int_0^\infty F(u)\cos(k\,u)\,\mathrm{d}u \tag{6.80b}$$

$$= \int_0^\infty [F(-u) + F(u)]\cos(k\,u)\,\mathrm{d}u \tag{6.80c}$$

$$= 0. \tag{6.80d}$$

We are thus motivated to define the *Fourier sine transform* for functions with odd parity

$$F_{\mathrm{S}}(k) \equiv (\pi/2)\,B(k) = \frac{1}{2}\int_{-\infty}^{\infty} F(u)\sin(k\,u)\,\mathrm{d}u = \int_0^\infty F(u)\sin(k\,u)\,\mathrm{d}u, \tag{6.81}$$

so that Fourier's integral theorem (6.79) for an odd parity function takes the form

$$F(x) = \int_0^\infty B(k)\sin(k\,x)\,\mathrm{d}k = \frac{2}{\pi}\int_0^\infty F_{\mathrm{S}}(k)\sin(k\,x)\,\mathrm{d}k. \tag{6.82}$$

Correspondingly, the Fourier sine transform of an even parity function vanishes.

6.3.6 Parseval-Plancherel formulas

In Section 6.2.5 we showed that a Fourier series expansion of a periodic function satisfies the Parseval identities (6.50) and (6.56). Here we derive the analog for Fourier integral transforms. For this purpose, consider the integral of the product of two real functions

$$\int_{-\infty}^{\infty} G(x)\,F(x)\,\mathrm{d}x = \frac{1}{2\pi}\int_{-\infty}^{\infty} G(x)\left[\int_{-\infty}^{\infty} \mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k\right]\mathrm{d}x, \tag{6.83}$$

where we expressed $F(x)$ in terms of its inverse Fourier transform (6.68c). Now swap the integration order and rearrange to render

$$\int_{-\infty}^{\infty} G(x)\,F(x)\,\mathrm{d}x = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} G(x)\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}x\right]^*\mathcal{F}(k)\,\mathrm{d}k, \tag{6.84}$$

where we noted that

$$\left[\mathrm{e}^{\mathrm{i}\,k\,x}\right]^* = \mathrm{e}^{-\mathrm{i}\,k\,x} \quad \text{and} \quad G^* = G. \tag{6.85}$$

We recognize the bracketed term in equation (6.84) as the complex conjugate Fourier transform of G, thus bringing us to the identity

$$\int_{-\infty}^{\infty} G(x)\,F(x)\,\mathrm{d}x = \frac{1}{2\pi}\int_{-\infty}^{\infty} [\mathcal{G}(k)]^{*}\,\mathcal{F}(k)\,\mathrm{d}k. \tag{6.86}$$

For the special case with $G = F$ we recover the *Parseval-Plancherel formula*, commonly referred to as Parseval's identity

$$\int_{-\infty}^{\infty} [F(x)]^{2}\,\mathrm{d}x = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\mathcal{F}(k)|^{2}\,\mathrm{d}k. \tag{6.87}$$

If $[F(x)]^2$ is proportional to the energy, then Parseval's identity expresses the identity of energy when expressed in either $\boldsymbol{x}$-space or $\boldsymbol{k}$-space.[12]

6.3.7 Concerning the placement of $1/2\pi$

There is no universal convention for the pre-factors appearing in the Fourier transform pairs (6.65) and (6.68c). That is, the unity for the Fourier transform pre-factor in equation (6.65), versus the $1/2\pi$ pre-factor for the inverse Fourier transform (6.68c), could just as well be swapped, or alternatively they could both be equal to $1/\sqrt{2\pi}$. However these factors are chosen, their product must equal to $1/2\pi$.

Furthermore, the $1/2\pi$ factor in the inverse Fourier transform (6.68c) can be eliminated through use of the *reduced wavenumber*, $\bar{k}$, defined as

$$\bar{k} = k/2\pi, \tag{6.88}$$

in which case the inverse Fourier transform (6.68c) is given by

$$F(x) = \int_{-\infty}^{\infty} \mathcal{F}(\bar{k})\,\mathrm{e}^{\mathrm{i}\,2\,\pi\,\bar{k}x}\,\mathrm{d}\bar{k}. \tag{6.89}$$

An analogous approach is used when working in the time-frequency domain and using the frequency, f, rather than the angular frequency, ω, where $f = \omega/2\pi$ (see Section 6.4). Even though $\bar{k}$ is somewhat more elegant, we generally work with the wavenumber, k, thus necessitating care to properly place the $1/2\pi$ factor.

6.3.8 Fourier transforms and derivatives

Here we examine some properties of the Fourier transform of derivatives. Note that for all of the results, we must assume that the function and its Fourier transform decay to zero at infinity at a rate sufficient to ensure that all integrals are all bounded.

Each derivative operation multiplies by $(\mathrm{i}\,k)^p$

Consider a function expressed as the inverse Fourier transform (6.68c)

$$F(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathcal{F}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k, \tag{6.90}$$

[12] The $1/2\pi$ factor in the Parsevel identity (6.87) can be eliminated by distributing a $1/\sqrt{2\pi}$ equally between the Fourier transform and the inverse Fourier transform. That convention is generally followed in the quantum mechanics literature but not in the fluid mechanics literature. See Section 6.3.7 for more on the 2π factor.

where $\mathcal{F}(k)$ is the Fourier transform of $F(x)$ as per equation (6.65). Now take the p'th derivative of $F(x)$ and note that the derivative operator commutes with the $\boldsymbol{k}$-space integral

$$\frac{\mathrm{d}^p F(x)}{\mathrm{d}x^p} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(k)\,(\mathrm{i}\,k)^p\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k, \tag{6.91}$$

where we set

$$\frac{\mathrm{d}^p\,(\mathrm{e}^{\mathrm{i}\,k\,x})}{\mathrm{d}x^p} = (\mathrm{i}\,k)^p\,\mathrm{e}^{\mathrm{i}\,k\,x}. \tag{6.92}$$

Equation (6.91) indicates that each derivative increases the power of $(\mathrm{i}\,k)$ inside the $\boldsymbol{k}$-space integral, in which case the relative contributions from higher wavenumbers are enhanced relative to lower wavenumbers.[13]

Furthering the above result

To further the above remarks, consider a function, $G(x)$, that is expressed as an inverse Fourier transform

$$G(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{G}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k, \tag{6.93}$$

and define $G(x)$ as the p'th derivative of $F(x)$

$$G^{(p)}(x) = \frac{\mathrm{d}^p F(x)}{\mathrm{d}x^p}. \tag{6.94}$$

From equation (6.91) we make the identification

$$\mathcal{G}^{(p)}(k) = (\mathrm{i}\,k)^p\,\mathcal{F}(k). \tag{6.95}$$

Hence, the Fourier transform of the derivative of a function results in a power of $(\mathrm{i}\,k)$ for each derivative, so that, for example,

$$\mathcal{G}^{(1)}(k) = \mathrm{i}\,k\,\mathcal{F}(k) \tag{6.96a}$$

$$\mathcal{G}^{(2)}(k) = -k^2\,\mathcal{F}(k). \tag{6.96b}$$

For $p = 1$ we see that the Fourier transform of the derivative of a function has a $\pi/2$ phase shift relative to the function, whereas for $p = 2$ there is a π phase shift.

Directly computing the Fourier transform of the derivative of a function

There is yet another way to compute the Fourier transform of the derivative of a function, through use of integration by parts

$$\mathcal{G}^{(1)}(k) = \int_{-\infty}^{\infty} \frac{\mathrm{d}F}{\mathrm{d}x}\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}x \tag{6.97a}$$

$$= \int_{-\infty}^{\infty} \left[\frac{\mathrm{d}(F\,\mathrm{e}^{-\mathrm{i}\,k\,x})}{\mathrm{d}x} - F\,\frac{\mathrm{d}(\mathrm{e}^{-\mathrm{i}\,k\,x})}{\mathrm{d}x}\right]\mathrm{d}x \tag{6.97b}$$

[13] As noted at the start of this subsection, we must assume a finite wavenumber cutoff to ensure the integral (6.91) is bounded. That is, we must assume a length scale below which there is no structure in $F(x)$, which then means that $\mathcal{F}(k)$ vanishes (or is exponentially small) for wavenumbers above some finite wavenumber. This assumption is typically satisfied by physical fields.

$$= \int_{-\infty}^{\infty} \left[\frac{\mathrm{d}(F\,\mathrm{e}^{-\mathrm{i}\,k\,x})}{\mathrm{d}x} + (\mathrm{i}\,k)\,F(x)\,\mathrm{e}^{-\mathrm{i}\,k\,x} \right] \mathrm{d}x. \tag{6.97c}$$

If we assume $F(x)$ decays to zero as $x \to \pm\infty$, then the total derivative vanishes, in which case we are left with

$$\mathcal{G}^{(1)}(k) = (\mathrm{i}\,k) \int_{-\infty}^{\infty} F(x)\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}x = (\mathrm{i}\,k)\,\mathcal{F}(k). \tag{6.98}$$

This result agrees with equation (6.96a) derived earlier. Higher derivatives can be computed likewise.

Parseval's identity for derivatives of a function

Write Parseval's identity (6.87) as

$$\int_{-\infty}^{\infty} |G(x)|^2\,\mathrm{d}x = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\mathcal{G}(k)|^2\,\mathrm{d}k, \tag{6.99}$$

and let $G(x)$ be the p'th derivative of $F(x)$ as per equation (6.94), whose Fourier transform, $\mathcal{G}^{(p)}(k)$, is related to $\mathcal{F}(k)$ by equation (6.95). Hence, Parseval's identity becomes an identity for derivatives of a function

$$\int_{-\infty}^{\infty} \left| \frac{\mathrm{d}^p F(x)}{\mathrm{d}x^p} \right|^2 \mathrm{d}x = \frac{1}{2\pi} \int_{-\infty}^{\infty} k^{2p}\,|\mathcal{F}(k)|^2\,\mathrm{d}k, \tag{6.100}$$

with the first derivative identity used in deriving the uncertainty inequality in Section 6.6

$$\int_{-\infty}^{\infty} \left| \frac{\mathrm{d}F(x)}{\mathrm{d}x} \right|^2 \mathrm{d}x = \frac{1}{2\pi} \int_{-\infty}^{\infty} k^2\,|\mathcal{F}(k)|^2\,\mathrm{d}k. \tag{6.101}$$

A Fourier series example with two Fourier components

To further illustrate the role of derivatives and how they enhance the relative contribution of small scales versus large scales, consider a finite sized one-dimensional periodic domain. For a function, $F(x)$, that is periodic on this domain we can represent it according to its Fourier series (6.27), and assume there are just two non-zero wavenumbers contributing to the function

$$F(x) = a_1 \cos(k_1\,x) + a_{10} \cos(k_{10}\,x), \tag{6.102}$$

with $a_1/a_{10} = 0.95/0.05 = 19$. As illustrated in Figure 6.2, the function $F(x)$ is the sum of a low wavenumber Fourier component, $a_1 \cos(k_1\,x)$, plus a high wavenumber component, $a_{10} \cos(k_{10}\,x)$. The second derivative of this function is given by

$$\mathrm{d}^2 F(x)/\mathrm{d}x^2 = -a_1\,k_1^2 \cos(k_1\,x) - a_{10}\,k_{10}^2 \cos(k_{10}\,x), \tag{6.103}$$

so that each wavenumber contribution is multiplied by its respective wavenumber squared. Evidently, the contribution from $n = 10$ is enhanced relative to the $n = 1$ contribution by the ratio $(k_{10}/k_1)^2 = (10/1)^2 = 100$. In general, any nonzero a_n coefficient in the expansion of a function is amplified by its respective wavenumber when taking a derivative, so that higher wavenumber features are enhanced relative to lower wavenumber features.

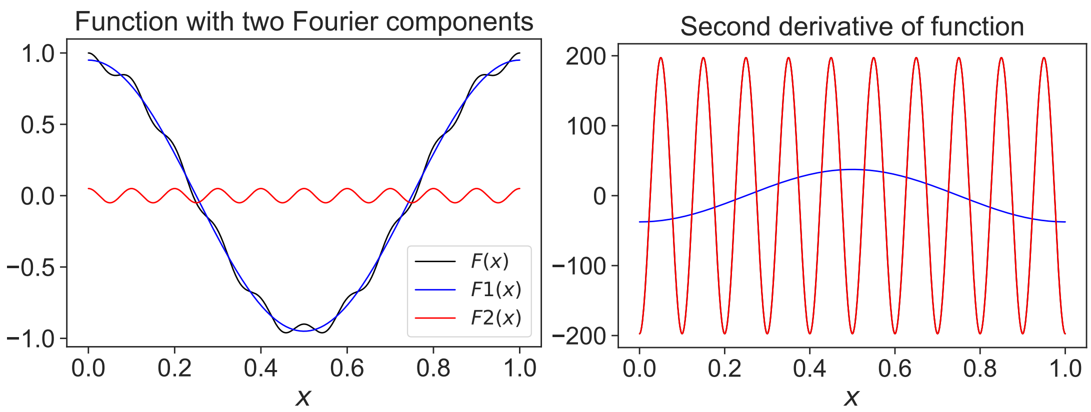

FIGURE 6.2: Left panel: a function comprised of two Fourier components, $F(x) = F_1(x) + F_2(x) = a_1 \cos(k_1\, x) + a_{10} \cos(k_{10}\, x)$, with a low wavenumber $k_1 = 2\,\pi/L$, and a high wavenumber, $k_{10} = 2\,\pi/(L/10)$, along with the amplitudes $a_1 = 0.95$ and $a_{10} = 0.05$. We choose $L = 1$ in arbitrary units. Given that $a_1/a_{10} = 19$, the low wavenumber component dominates $F(x)$. Right panel: the second derivative of the function, $\mathrm{d}^2 F(x)/\mathrm{d}x^2 = -a_1\, k_1^2 \cos(k_1\, x) - a_{10}\, k_{10}^2 \cos(k_{10}\, x)$, which is dominated by the high wavenumber component, with the high wavenumber component having an amplitude that is roughly five times larger than the low wavenumber component, $a_{10}\, k_{10}^2/(a_1\, k_1^2) \approx 5$.

Streamfunction and vorticity example

An example from fluid mechanics occurs with a horizontally non-divergent barotropic fluid (VOLUME 3). In this case the vorticity, ζ, is the Laplacian of the streamfunction, ψ, so that

$$\zeta = \nabla^2 \psi. \tag{6.104}$$

Taking the Fourier transform of this equation reveals that the Fourier transform of the vorticity equals to $-k^2$ times the Fourier transform of the streamfunction. Hence, if there is any structure in the streamfunction at scale k, then the vorticity at that same scale is amplified by k^2. Consequently, the vorticity field has higher wavenumber features (i.e., smaller spatial scales) than the streamfunction field.

6.4 Time-frequency Fourier transforms

Our notation has thus far been based on space, x, and wavenumber, k. However, all formula hold whatever interpretation one gives to these coordinates, with time and frequency commonly used as well as space and wavenumber. Even so, for the time/frequency Fourier analysis we follow the typical physics convention of swapping sign in the exponents. Doing so accords with conventions in wave mechanics as pursued in VOLUME 4. With that motivation we define the time/frequency Fourier transform and its inverse according to

$$\mathcal{F}(\omega) = \int_{-\infty}^{\infty} F(t)\, \mathrm{e}^{\mathrm{i}\,\omega\, t}\, \mathrm{d}t \tag{6.105a}$$

$$F(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\omega)\, \mathrm{e}^{-\mathrm{i}\,\omega\, t}\, \mathrm{d}\omega, \tag{6.105b}$$

where ω is the angular frequency. The time integral extends from $-\infty$ to ∞, meaning that it extends arbitrarily far in the past and arbitrarily far into the future. The negative angular frequency arises from the same symmetrization of the integral limits made when moving to the complex Fourier transform in Section 6.3.2. As in our discussion of wave mechanics in VOLUME

4, we here find it important to interpret the negative angular frequency, $\omega < 0$, which we do next.

6.4.1 How to interpret negative frequency for waves

As noted at the start of this chapter, we generally make use of Fourier analysis for the study of waves in VOLUME 4. When working in $\boldsymbol{x}$-space and $\boldsymbol{k}$-space, we can consider the components of a wavevector to be positive or negative given that the wavevector determines the direction of a wave. In our study of waves, we let information about the direction of the wave be fully carried by the wavevector. Consequently, we consider the angular frequency of a wave to be a non-negative number, $\omega \geq 0$, which corresponds to a non-negative wave period, $2\pi/\omega$. Hence, when working with waves in VOLUME 4, we interpret the expression (6.105b) for the inverse Fourier transform, with its negative frequency integral limits, as a convenient mathematical means to bring the Fourier sine and cosine functions together into a single integral. Yet we do not give any physical meaning to $\omega < 0$.

To provide details to support the above comments, return to the Fourier integral theorem (6.63), only now interpreted for time and angular frequency

$$F(t) = \int_0^\infty [A(\omega)\cos(\omega\, t) + B(\omega)\sin(\omega\, t)]\mathrm{d}\omega. \tag{6.106}$$

Note that the frequency integral extends only over non-negative frequencies, consistent with ω as a frequency. In contrast, the amplitude functions are built from integrals over all time according to equation (6.62)

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} F(t)\cos(\omega\, t)\,\mathrm{d}t \quad \text{and} \quad B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} F(t)\sin(\omega\, t)\,\mathrm{d}t, \tag{6.107}$$

so that[14]

$$\mathcal{F}(\omega) = \pi\,[A(\omega) + \mathrm{i}\,B(\omega)]. \tag{6.108}$$

We have no problem with negative time simply because the origin of time is arbitrary. In contrast, we ascribe no physical meaning to a negative frequency, but instead interpret a negative frequency as a mathematical expedient enabling us to write equation (6.105b) in a compact form. We emphasize this point by writing

$$\begin{aligned}
F(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}\,\omega\, t}\,\mathcal{F}(\omega)\,\mathrm{d}\omega && \text{Fourier eq. (6.105b)} && (6.109\text{a})\\
&= \frac{1}{2}\int_{-\infty}^{\infty} [\cos(\omega\, t) - \mathrm{i}\sin(\omega\, t)]\,[A(\omega) + \mathrm{i}\,B(\omega)]\mathrm{d}\omega && \text{Euler + eq. (6.107)} && (6.109\text{b})\\
&= \frac{1}{2}\int_{-\infty}^{\infty} [A(\omega)\cos(\omega\, t) + B(\omega)\sin(\omega\, t)]\mathrm{d}\omega && \text{parity properties} && (6.109\text{c})\\
&= \int_0^{\infty} [A(\omega)\cos(\omega\, t) + B(\omega)\sin(\omega\, t)]\mathrm{d}\omega && \text{parity properties.} && (6.109\text{d})
\end{aligned}$$

So again, we compute the inverse Fourier transform as in equation (6.105b) with a frequency interval including negative frequencies. Yet there is no physical meaning given to the negative

[14]Compare equation (6.108) to equation (6.65). The sign difference arises from the sign convention: we use $\mathrm{e}^{-\mathrm{i}\,\omega\, t}$ for the time-domain inverse Fourier transform (6.105b), whereas $\mathrm{e}^{\mathrm{i}\,k\,x}$ for the space-domain inverse Fourier transform (6.68c).

frequencies. Instead, they simply offer the means to unify the Fourier cosine and sine amplitude functions according to the above identities.

6.4.2 Frequency versus angular frequency

As noted in Section 6.3.7, we can eliminate the $1/2\pi$ factor appearing in the inverse Fourier transform (6.105b) through using the frequency, f (cycles per time), rather than the angular frequency, ω (radians per time),

$$f = \omega/2\pi, \tag{6.110}$$

in which case the Fourier transform partners (6.105a) and (6.105b) take the form

$$\mathcal{F}(f) = \int_{-\infty}^{\infty} F(t)\, \mathrm{e}^{\mathrm{i}\, 2\pi\, f\, t}\, \mathrm{d}t \tag{6.111a}$$

$$F(t) = \int_{-\infty}^{\infty} \mathcal{F}(f)\, \mathrm{e}^{-\mathrm{i}\, 2\pi\, f\, t}\, \mathrm{d}f. \tag{6.111b}$$

Even though f is rather elegant from this perspective, we generally use the angular frequency, ω (along with the angular wavenumber, k), thus necessitating care with the placement of the $1/2\pi$ factor for the Fourier transform pair given by equations (6.105a) and (6.105b).

6.5 Example Fourier transform pairs

In this section we offer examples of Fourier transform pairs along with their corresponding *uncertainty relation* that manifests the *complementarity* between $\boldsymbol{x}$-space and $\boldsymbol{k}$-space.

6.5.1 Dirac delta in $\boldsymbol{x}$-space

Consider the Dirac delta from Chapter 5,

$$F(x) = \delta(x). \tag{6.112}$$

Following from its sifting property (5.8), we find that the Fourier transform (equation (6.65)) of the Dirac delta is given by

$$\mathcal{F}(k) = \int_{-\infty}^{\infty} \delta(x)\, \mathrm{e}^{-\mathrm{i}\, k\, x}\, \mathrm{d}x = 1. \tag{6.113}$$

This result provides an extreme example of the complementarity relation maintained between the Fourier transform and its inverse. Namely, if a function is concentrated in $\boldsymbol{x}$-space then its Fourier transform is broadly distributed in $\boldsymbol{k}$-space. The Dirac delta is infinitely concentrated in $\boldsymbol{x}$-space at the origin, whereas it has a uniform and constant distribution in $\boldsymbol{k}$-space. That is, the $\boldsymbol{x}$-space distribution is infinitesimally narrow whereas the $\boldsymbol{k}$-space distribution is infinitely broad. As a consistency check, we note the physical dimensionality of the terms in equation (6.113). The Dirac delta, $\delta(x)$, has dimensions of inverse length (Section 5.12), so that $\delta(x)\,\mathrm{d}x$ is non-dimensional. We thus expect its Fourier transform, $\mathcal{F}(k)$, to be non-dimensional.

It follows that the inverse Fourier transform (equation (6.68c)) of the Dirac delta is given by

$$\delta(x - x_{\circ}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\, k\, (x - x_{\circ})}\, \mathrm{d}k \tag{6.114}$$

where we introduced an arbitrary shift from the origin, $x_\circ$. Note that

$$\int_{-\infty}^{\infty} \sin[k\,(x - x_\circ)]\,\mathrm{d}k = 0, \tag{6.115}$$

which follows since the $\boldsymbol{k}$-space integral is over a symmetric domain yet the integrand has odd $\boldsymbol{k}$-space parity. Consequently, we have

$$\delta(x - x_\circ) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \cos[k\,(x - x_\circ)]\,\mathrm{d}k = \frac{1}{\pi}\int_{0}^{\infty} \cos[k\,(x - x_\circ)]\,\mathrm{d}k. \tag{6.116}$$

This equation is a rather remarkable expression that can be heuristically explained by noting that for all values of $x - x_\circ \neq 0$, the cosine function oscillates so that any positive values are exactly matched by negative values, thus yielding a zero integral. However, when $x - x_\circ = 0$, the integral diverges to infinity.

6.5.2 Dirac delta in $\boldsymbol{k}$-space

We now assume exact information about the wave number by setting the Fourier transform to

$$\mathcal{F}(k) = \delta(k - k_\circ), \tag{6.117}$$

which has physical dimensions of length. The resulting inverse Fourier transform is given by the dimensionless function

$$F(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(k - k_\circ)\,\mathrm{e}^{\mathrm{i}\,k\,x}\,\mathrm{d}k = \frac{\mathrm{e}^{\mathrm{i}\,k_\circ\,x}}{2\pi}. \tag{6.118}$$

So as a complement to the case in Section 6.5.1, we here find that exact information about the $\boldsymbol{k}$-space location (i.e., the wavenumber is exactly $k_\circ$) results in zero information about the $\boldsymbol{x}$-space position.

6.5.3 Gaussian distribution

Consider a normalized Gaussian distribution,

$$F(x) = (4\,\pi\,\sigma)^{-1/2}\,\mathrm{e}^{-x^2/4\sigma} \quad \text{with} \quad \int_{-\infty}^{\infty} F(x)\,\mathrm{d}x = 1, \tag{6.119}$$

and where the mean and variance are

$$\langle x \rangle = \int_{-\infty}^{\infty} x\,F(x)\,\mathrm{d}x = 0 \quad \text{and} \quad \langle x^2 \rangle = \int_{-\infty}^{\infty} x^2\,F(x)\,\mathrm{d}x = 2\,\sigma. \tag{6.120}$$

Its Fourier transform is given by

$$\mathcal{F}(k) = (4\,\pi\,\sigma)^{-1/2}\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/4\sigma}\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}x = \mathrm{e}^{-\sigma\,k^2}, \tag{6.121}$$

with evaluation of this integral following from the calculus of residues.[15] We expect the Fourier transform to be real since $F(x)$ has even parity so that its Fourier sine transform vanishes (see

[15]In the wave packet chapter in VOLUME 4 we discuss a few of the necessary steps for evaluating the integral (6.121), as part of our study of Gaussian wave packets.

Section 6.3.5). What is remarkable, however, is that the Fourier transform is also a Gaussian. This property is unique to the Gaussian and it is illustrated in Figure 6.3.

To compute the variance of the Fourier transform (in $\boldsymbol{k}$-space) requires us to first normalize it according to

$$(\sigma/\pi)^{1/2} \int_{-\infty}^{\infty} \mathrm{e}^{-\sigma k^2} \, \mathrm{d}k = 1, \tag{6.122}$$

so that its variance is

$$\langle k^2 \rangle = (\sigma/\pi)^{1/2} \int_{-\infty}^{\infty} k^2 \, \mathrm{e}^{-\sigma k^2} \, \mathrm{d}k = (2\,\sigma)^{-1}. \tag{6.123}$$

We thus find that the product of the variances is unity

$$\langle x^2 \rangle \langle k^2 \rangle = 1. \tag{6.124}$$

Hence, if the Gaussian is narrow in $\boldsymbol{x}$-space, as per a small value of σ, then it is broad in $\boldsymbol{k}$-space, and vice versa. This behavior offers another example of the uncertainty relation that manifests the complementarity property of Fourier transform pairs.

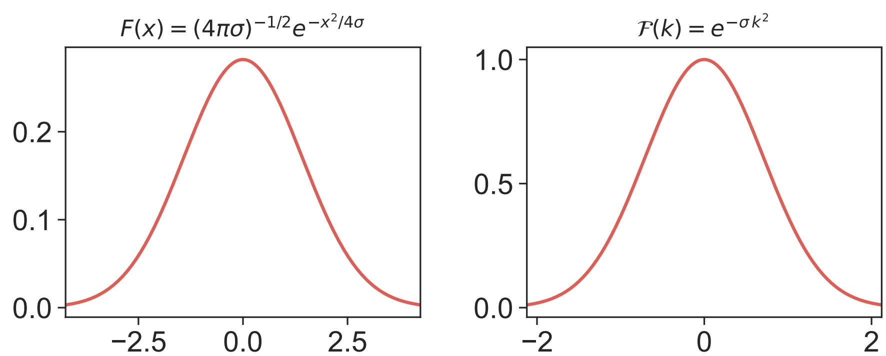

FIGURE 6.3: Left panel: the Gaussian function, $F(x) = (4\,\pi\,\sigma)^{-1/2}\,\mathrm{e}^{-x^2/4\sigma}$, which has a variance, $\langle x^2 \rangle = 2\,\sigma$, with the horizontal axes having a range $\pm 3\,\sqrt{\langle x^2 \rangle} = \pm 3\,(2\,\sigma)^{1/2}$. Right panel: the Fourier transform, $\mathcal{F}(k) = \mathrm{e}^{-\sigma\,k^2}$, which has a variance $\langle k^2 \rangle = (2\,\sigma)^{-1}$ and the horizontal axes range $\pm 3\,\sqrt{\langle k^2 \rangle} = \pm 3\,(2\,\sigma)^{-1/2}$. We set $\sigma = 1$ in arbitrary units.

6.5.4 Lorentzian distribution

The Lorentzian distribution

$$F(x) = \frac{1}{\pi} \frac{\gamma}{(x - x_\circ)^2 + \gamma^2}, \tag{6.125}$$

appears in spectroscopy and resonance phenomena, and we also encounter the Lorentzian in Section 5.4 when studying the Dirac delta. Like the Gaussian distribution from Section 6.5.3, the Lorentzian is symmetric around its peak (here given by $x = x_\circ$), and yet the Lorentzian has a broader tail than the Gaussian. As a result of the broader tail, the Lorentzian's mean and variance are undefined. Even so, 2γ measures the full-width-at-half-maximum, which is an analog to the standard deviation (squared variance) of the Gaussian. That is, the Lorentzian is maximized at $x = x_\circ$ whereas it reaches its half-maximum at $x = x_\circ \pm \gamma$:

$$F(x = x_\circ) = 1/(\pi\,\gamma) \quad \text{and} \quad F(x = x_\circ \pm \gamma) = 1/(2\,\pi\,\gamma). \tag{6.126}$$

The Fourier transform of the Lorentzian distribution is given by

$$\mathcal{F}(k) = \frac{\gamma}{\pi} \int_{-\infty}^{\infty} \frac{\mathrm{e}^{-\mathrm{i}\,k\,x}}{(x - x_\circ)^2 + \gamma^2}\,\mathrm{d}x = \frac{\gamma\,\mathrm{e}^{-\mathrm{i}\,k\,x_\circ}}{\pi} \int_{-\infty}^{\infty} \frac{\mathrm{e}^{-\mathrm{i}\,k\,y}}{y^2 + \gamma^2}\,\mathrm{d}y = \mathrm{e}^{-\mathrm{i}\,k\,x_\circ - \gamma\,|k|}, \tag{6.127}$$

so that the $x_\circ$ shift in x-space leads to a phase shift in k-space.[16] Furthermore, broadening the distribution in x-space by increasing γ renders a sharpening of the distribution in k-space, thus manifesting $\boldsymbol{x}$-space and $\boldsymbol{k}$-space complementarity. Exercise 6.5 computes the uncertainty relation for the Lorentzian distribution by following the approach used in Section 6.5.3 for the Gaussian. In Figure 6.4 we illustrate the Lorentzian and its Fourier transform.

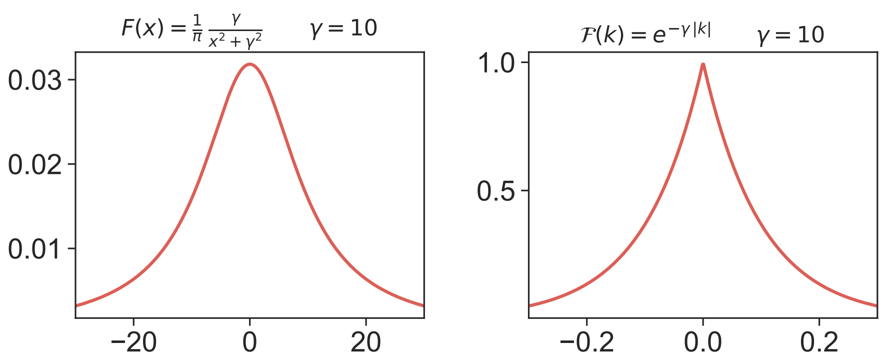

FIGURE 6.4: Left panel: the Lorentzian distribution, $F(x) = \gamma/[\pi\,(x^2 + \gamma^2)]$. Right panel: the Fourier transform, $\mathcal{F}(k) = \mathrm{e}^{-\gamma\,|k|}$. We set $\gamma = 10$ in arbitrary units. Notice how the Lorentzian in the left panel has a relatively low amplitude but broad span in x-space, whereas its k-space Fourier transform in the right panel is sharper in k-space but with a relatively large amplitude, with these properties manifesting $\boldsymbol{x}$-space and $\boldsymbol{k}$-space complementarity.

6.6 A general uncertainty inequality

The uncertainty relations derived in Section 6.5 originate from a fundamental property of Fourier transform pairs. Our goal in this section to expose this property by deriving a general uncertainty inequality that reveals that the uncertainty has a lower bound independent of the function, $F(x)$.

6.6.1 Width of a function determined by squared modulus

There is no single method to define the width of a function in $\boldsymbol{x}$-space or $\boldsymbol{k}$-space. In Section 6.5 we defined the width as the root-mean-square based on the normalized distributions in $\boldsymbol{x}$-space and $\boldsymbol{k}$-space. Here we define the width as the root-mean-square with weighting by the squared modulus of the distribution. When studying waves in VOLUME 4 of this book, the energy of a wave is proportional to the squared wave amplitude, which provides some motivation for using this approach to defining width of a distribution.

Hence, we define the squared spread of the x-space and k-space distributions as

$$(\Delta x)^2 = \frac{\int_{-\infty}^{\infty} (x - \overline{x})^2\,|F(x)|^2\,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2\,\mathrm{d}x} = \frac{\int_{-\infty}^{\infty} (x^2 - \overline{x}^2)\,|F(x)|^2\,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2\,\mathrm{d}x} \tag{6.128a}$$

[16]This property holds in general, as shown in Exercise 6.1.

$$(\Delta k)^2 = \frac{\int_{-\infty}^{\infty} (k - \overline{k})^2 \, |\mathcal{F}(k)|^2 \, \mathrm{d}k}{\int_{-\infty}^{\infty} |\mathcal{F}(k)|^2 \, \mathrm{d}k} = \frac{\int_{-\infty}^{\infty} (k^2 - \overline{k}^2) \, |\mathcal{F}(k)|^2 \, \mathrm{d}k}{\int_{-\infty}^{\infty} |\mathcal{F}(k)|^2 \, \mathrm{d}k}, \tag{6.128b}$$

where the centroids for $F(x)$ and $\mathcal{F}(k)$ are given by

$$\overline{x} = \frac{\int_{-\infty}^{\infty} x \, |F(x)|^2 \, \mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2 \, \mathrm{d}x} \quad \text{and} \quad \overline{k} = \frac{\int_{-\infty}^{\infty} k \, |\mathcal{F}(k)|^2 \, \mathrm{d}k}{\int_{-\infty}^{\infty} |\mathcal{F}(k)|^2 \, \mathrm{d}k}. \tag{6.129}$$

In this section we derive the general uncertainty inequality,

$$\Delta x \, \Delta k \geq 1/2, \tag{6.130}$$

which holds for an arbitrary function, $F(x)$, so long as it has a well defined Fourier transformation, $\mathcal{F}(k)$. The uncertainty inequality (6.130) manifests complementarity between $\boldsymbol{x}$-space and $\boldsymbol{k}$-space, and it is fundamental to wave mechanics. Furthermore, it appears in quantum mechanics where it is referred to as the Heisenberg uncertainty principle.

6.6.2 Cauchy-Schwarz inequality

To prove the uncertainty inequality (6.130), we take a diversion to prove the *Cauchy-Schwarz inequality*. For that purpose, start with the self-evident inequality

$$\begin{aligned} 0 &\leq \int_{-\infty}^{\infty} |G(x) + \epsilon \, H(x)|^2 \, \mathrm{d}x & (6.131\text{a}) \\ &= \int_{-\infty}^{\infty} [G(x) + \epsilon \, H(x)] \, [G^*(x) + \epsilon \, H^*(x)] \, \mathrm{d}x & (6.131\text{b}) \\ &= \epsilon^2 \int_{-\infty}^{\infty} |H|^2 \, \mathrm{d}x + \epsilon \int_{-\infty}^{\infty} (G \, H^* + G^* \, H) \, \mathrm{d}x + \int_{-\infty}^{\infty} |G|^2 \, \mathrm{d}x, & (6.131\text{c}) \end{aligned}$$

where ϵ is an arbitrary real number and $G(x)$ and $H(x)$ are square integrable functions

$$\int_{-\infty}^{\infty} |G|^2 \, \mathrm{d}x < \infty \quad \text{and} \quad \int_{-\infty}^{\infty} |H|^2 \, \mathrm{d}x < \infty. \tag{6.132}$$

Equation (6.131c) can be written as a non-negative quadratic function

$$0 \leq \Phi(\epsilon) = a \, \epsilon^2 + b \, \epsilon + c, \tag{6.133}$$

with real coefficients

$$a = \int_{-\infty}^{\infty} |H|^2 \, \mathrm{d}x \quad \text{and} \quad b = \int_{-\infty}^{\infty} (G \, H^* + G^* \, H) \, \mathrm{d}x \quad \text{and} \quad c = \int_{-\infty}^{\infty} |G|^2 \, \mathrm{d}x. \tag{6.134}$$

Since $a \geq 0$, we know that $\Phi(\epsilon) \geq 0$ is a parabola opening upwards. Furthermore, $\Phi \geq 0$ for all ϵ means that is has at most a single zero, $\Phi = 0$, which occurs when the discriminant, $b^2 - 4 \, a \, c$, vanishes. Alternatively, we can have $\Phi > 0$, which means that the discriminant is negative so that there are no real roots. We thus have the inequality for the discriminant

$$b^2 - 4 \, a \, c \leq 0, \tag{6.135}$$

which, when substituting for a, b, c from equation (6.134), leads to the *Cauchy-Schwarz inequality*

$$\left[\int_{-\infty}^{\infty} |H|^2 \,\mathrm{d}x\right] \left[\int_{-\infty}^{\infty} |G|^2 \,\mathrm{d}x\right] \geq \frac{1}{4} \left[\int_{-\infty}^{\infty} (G\,H^* + G^*\,H) \,\mathrm{d}x\right]^2. \tag{6.136}$$

When G and H are real functions, the Cauchy-Schwarz inequality becomes

$$\left[\int_{-\infty}^{\infty} |H|^2 \,\mathrm{d}x\right] \left[\int_{-\infty}^{\infty} |G|^2 \,\mathrm{d}x\right] \geq \left[\int_{-\infty}^{\infty} G\,H \,\mathrm{d}x\right]^2. \tag{6.137}$$

6.6.3 Uncertainty inequality

From equations (6.128a) and (6.128b) we have the product

$$(\Delta x)^2\,(\Delta k)^2 = \left[\frac{\int_{-\infty}^{\infty} (x^2 - \overline{x}^2)\,|F(x)|^2 \,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2 \,\mathrm{d}x}\right] \left[\frac{\int_{-\infty}^{\infty} (k^2 - \overline{k}^2)\,|\mathcal{F}(k)|^2 \,\mathrm{d}k}{\int_{-\infty}^{\infty} |\mathcal{F}(k)|^2 \,\mathrm{d}k}\right]. \tag{6.138}$$

Without loss of generality we can define the x coordinate so that the x-centroid vanishes, $\overline{x} = 0$. Furthermore, since $F(x)$ is real, we can invoke Exercise 6.3 to find that the k-centroid is zero, $\overline{k} = 0$. Finally, we make use of Parseval's identity (6.87) and its extension to first derivatives in equation (6.101) to transform the k-space integrals to their x-space version, thus rendering

$$(\Delta x)^2\,(\Delta k)^2 = \left[\frac{\int_{-\infty}^{\infty} x^2\,|F(x)|^2 \,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2 \,\mathrm{d}x}\right] \left[\frac{\int_{-\infty}^{\infty} |\mathrm{d}F(x)/\mathrm{d}x|^2 \,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2 \,\mathrm{d}x}\right]. \tag{6.139}$$

Now introduce the real functions

$$G(x) = x\,F(x) \quad \text{and} \quad H(x) = \mathrm{d}F(x)/\mathrm{d}x, \tag{6.140}$$

in which case

$$(\Delta x)^2\,(\Delta k)^2 = \left[\frac{\int_{-\infty}^{\infty} |G(x)|^2 \,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2 \,\mathrm{d}x}\right] \left[\frac{\int_{-\infty}^{\infty} |H(x)|^2 \,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2 \,\mathrm{d}x}\right] \geq \left[\frac{\int_{-\infty}^{\infty} G(x)\,H(x) \,\mathrm{d}x}{\int_{-\infty}^{\infty} |F(x)|^2 \,\mathrm{d}x}\right]^2, \tag{6.141}$$

where we made use of the Cauchy-Schwarz inequality (6.137). For the next step we write

$$\int_{-\infty}^{\infty} G\,H \,\mathrm{d}x = \int_{-\infty}^{\infty} x\,F\,(\mathrm{d}F/\mathrm{d}x) \,\mathrm{d}x \tag{6.142a}$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} x\,(\mathrm{d}F^2/\mathrm{d}x) \,\mathrm{d}x \tag{6.142b}$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} [\mathrm{d}(x\,F^2)/\mathrm{d}x] \,\mathrm{d}x - \frac{1}{2} \int_{-\infty}^{\infty} F^2 \,\mathrm{d}x. \tag{6.142c}$$

For the physical fields considered in this book, such as spatially localized wave packets, we can safely assume F^2 goes to zero at infinity faster than x, so that

$$\lim_{x \to \pm\infty} x\,F^2 = 0. \tag{6.143}$$

We are thus left with
$$\int_{-\infty}^{\infty} G\,H\,\mathrm{d}x = -\frac{1}{2}\int_{-\infty}^{\infty} F^2\,\mathrm{d}x, \tag{6.144}$$
which, when brought into equation (6.141), renders the uncertainty inequality
$$(\Delta x)^2\,(\Delta k)^2 \geq 1/4 \Longrightarrow \Delta x\,\Delta k \geq 1/2. \tag{6.145}$$
This inequality sets a lower bound for the uncertainty, with most distributions having larger uncertainties. In Exercise 6.6 we show that the Gaussian distribution saturates this inequality ($\Delta x\,\Delta k = 1/2$), so that it provides the least amount of uncertainty available to a distribution.

6.6.4 Further study

The derivation in this section follows that given in Appendix C of *Elmore and Heald* (1969). For readers versed in quantum mechanics, see also Section 1.4.5 of *Sakurai and Napolitano* (2020).

6.7 Exercises

EXERCISE 6.1: SHIFT THEOREM FOR FOURIER ANALYSIS
Let $\mathcal{F}(k)$ be the Fourier transform of $F(x)$. Show that $\mathrm{e}^{-\mathrm{i}\,k\,x_\circ}\,\mathcal{F}(k)$ is the Fourier transform of $F(x - x_\circ)$, with $x_\circ$ a constant. Hint: see the discussion of the Lorentzian distribution in Section 6.5.4.

EXERCISE 6.2: SIMILARITY RELATION FOR FOURIER TRANSFORM PAIRS
Let $\mathcal{F}(k)$ be the Fourier transform of $F(x)$. Show that $a^{-1}\,\mathcal{F}(k/a)$ is the Fourier transform of $F(a\,x)$, with a an arbitrary constant.

EXERCISE 6.3: EVEN NATURE OF $|\mathcal{F}(k)|^2$
Let $\mathcal{F}(k)$ be the Fourier transform of a real function, $F(x)$.

(a) Show that
$$|\mathcal{F}(k)|^2 = |\mathcal{F}(-k)|^2. \tag{6.146}$$

(b) Since $|\mathcal{F}(k)|^2 = |\mathcal{F}(-k)|^2$, show that this implies the $\overline{k} = 0$, where $\overline{k}$ is the wavenumber centroid from equation (6.129).

EXERCISE 6.4: BESSEL-PARSEVAL IDENTITY FOR TWO FUNCTIONS
Show all steps necessary to derive the generalized Bessel-Parseval identity (6.56)
$$\frac{1}{P}\int_{x_\circ}^{x_\circ+P} F(x)\,H(x)\,\mathrm{d}x = \sum_{n=-\infty}^{\infty} c_n\,d_n^* = \sum_{n=-\infty}^{\infty} c_n\,d_{-n}. \tag{6.147}$$

EXERCISE 6.5: UNCERTAINTY RELATION FOR THE LORENTZIAN DISTRIBUTION
Following the approach used for the Gaussian distribution in Section 6.5.3, compute the uncertainty relation, $\sqrt{\langle x^2\rangle\,\langle k^2\rangle}$, for the Lorentzian distribution from Section 6.5.4. Use the full-width-at-half-maximum for the root-mean-square x-space spread, $\sqrt{\langle x^2\rangle}$. Hint: remember to normalize the k-space distribution before computing $\langle k^2\rangle$. Hint: let $x_\circ = 0$ for simplicity and without loss of generality.

EXERCISE 6.6: UNCERTAINTY RELATION FOR THE GAUSSIAN DISTRIBUTION
We derived the uncertainty relation, $\sqrt{\langle x^2 \rangle \, \langle k^2 \rangle}$, for the Gaussian distribution in Section 6.5.3. In this exercise we derive the modified uncertainty relation, $\Delta x \, \Delta k$, following the definition of Δx in equation (6.128a) and Δk in equation (6.128b). In particular, for the Gaussian distribution from equation (6.119)

$$F(x) = (4\,\pi\,\sigma)^{-1/2}\, \mathrm{e}^{-x^2/4\sigma} \quad \text{with} \quad \int_{-\infty}^{\infty} F(x)\, \mathrm{d}x = 1, \tag{6.148}$$

Show that

$$\Delta x \, \Delta k = 1/2, \tag{6.149}$$

thus realizing the lowest uncertainty available according to the uncertainty inequality (6.145).

EXERCISE 6.7: SOLVING AN ORDINARY DIFFERENTIAL EQUATION
In this exercise we find a general expression for a forced ordinary differential equation using Fourier transform methods to determine the Green's function.

(a) Use a Fourier transform to derive an integral expression for the solution to the one-dimensional ordinary differential equation

$$\frac{\mathrm{d}^2 H(x)}{\mathrm{d}x^2} - \lambda^2\, H(x) = -F(x), \tag{6.150}$$

for $\lambda > 0$ and assuming that $H(x)$ vanishes as $|x| \to \infty$. Express your answer in the form of a Green's function, $G(x, x')$, so that the solution can be written

$$H(x) = \int_{-\infty}^{\infty} G(x, x')\, F(x')\, \mathrm{d}x'. \tag{6.151}$$

(b) Find an explicit expression for the Green's function. Hint: feel free to use an integral table.

(c) What are the physical dimensions of the Green's function, G? Relative to F, what are the dimensions of H?

(d) Discuss the limit $\lambda \to 0$.

Chapter 7

CALCULUS OF VARIATIONS

The calculus of variations provides a mathematical framework to determine a function that extremizes an integral. The integral is a function of functions, and it returns a real number, in which case we refer to the integral as a functional. As discussed by *Yourgrau and Mandelstam* (1968), the intellectual origins of variational methods date from extremum questions posed by the ancient Greeks, which then found their realization in the work of Fermat in optics (e.g., see Chapter 1 in *Tracy et al.* (2014)), and in the mechanics of Newton, the Bernoulli brothers, Euler, Legendre, Lagrange, and Hamilton. Variational methods are central to much of 20th and 21st century physics, including classical and quantum field theory, as well as optimization methods, information theory, and data assimilation.

We build our study of the calculus of variations as per treatments given in classical mechanics texts, such as Chapter 2 of *Goldstein* (1980), Chapter 5 of *Marion and Thornton* (1988), Chapter 2 of *Basdevant* (2007), Appendix B of *Tracy et al.* (2014), and Chapter 7 of *Tong* (2025). We first work through the derivation of the Euler equation of variational calculus and then provide a few analytical examples. Through the introduction of Lagrange multipliers, variational methods can handle constraints that affect the solution to the extremum problem. However, we do not explicitly consider constraints in this chapter, instead leaving constraints for our study of analytical mechanics in Chapter 14, and then for fluid mechanics in VOLUME 4.

READER'S GUIDE TO THIS CHAPTER

The calculus of variations is fundamental to those parts of the book that make use of Hamilton's principle, such as through analytical mechanics in Chapter 14, classical field theory and fluid mechanics in VOLUME 4, and wave mechanics in VOLUME 4. In these mechanical examples, the integrand is known as the Lagrangian and the integral is called the action. We sometimes make use of that language, even when not concerned with mechanics. After providing a mathematical formulation in Section 7.2, we then study a number of examples to further our conceptual and practical understanding.

7.1 Loose threads

- Rayleigh-Ritz method for estimating eigenvalues.
- Consider the quadratic form (7.32b) in the context of thermodynamics and entropy and stability.

7.2 Mathematical formulation

To introduce variational methods, let (x, y) be the Cartesian coordinates for a point in $\mathbb{E}^2$ (i.e., the two-dimensional flat Euclidean plane), and assume that $y = y(x)$ specifies y as a differentiable function of x. In this chapter we focus on functions of one variable, with extensions to higher dimensions straightforward and presented where needed later in this book in our study of mechanics. Now consider an integral of the form

$$\mathcal{J} = \int_{x_1}^{x_2} L[y(x), y'(x), x] \, \mathrm{d}x, \tag{7.1}$$

where L is a function of the functions, $y(x)$ and its derivative,

$$y'(x) = \frac{\mathrm{d}y}{\mathrm{d}x} \tag{7.2}$$

as well as the independent variable, x. We seek a function, y, that extremizes the integral, $\mathcal{J}$, for a given L, and with specified end points, x_1 and x_2. Given a function, y, the integral returns a number, with such functions of functions referred to as functionals. The calculus of variations provides a systematic method to determine this extremal function. We offer the following remarks.

- In physics applications via Hamilton's principle, we do not encounter higher than the first derivative as part of the integrand, L. Consequently, a functional of the form (7.1) is sufficient for our needs. We refer to L as the Lagrangian, which is a name borrowed from mechanics.

- We write the independent variable as x, which typically means space point in this book. However, in this chapter it can also mean time, t, which is the typical case in mechanics.
- The Lagrangian, L, depends implicitly on x as realized through its dependence on $y(x)$ and $y'(x)$. It can also depend explicitly on x, as indicated by the final argument in equation (7.1). This tandem implicit and explicit dependence can lead to confusion, so it is essential to clearly state how derivatives of L are taken. This point is further explained in Section 7.2.4.

7.2.1 Variation of a function

Assume that $y(x)$ is the function that extremizes $\mathcal{J}$, and assume this function is unique. As a result, any function distinct from $y(x)$ is not an extremum of $\mathcal{J}$. One means to consider such non-extremal functions is by introducing a non-dimensional parameter, α, that defines

$$Y(x, \alpha) = y(x) + \alpha\, \eta(x). \tag{7.3}$$

The function $\eta(x)$ is an arbitrary smooth function (with same physical dimensions as $y(x)$) that vanishes at the endpoints

$$\eta(x_1) = \eta(x_2) = 0. \tag{7.4}$$

The specific form of η is not important, so long as it is smooth and satisfies the endpoint conditions (7.4). Specifying these endpoint conditions means that $Y(x, \alpha)$ equals to the extremum function at the endpoints

$$Y(x_1, \alpha) = y(x_1) \quad \text{and} \quad Y(x_2, \alpha) = y(x_2). \tag{7.5}$$

Furthermore, by definition,

$$Y(x, 0) = y(x) \tag{7.6}$$

is the extremum function for all x. We say that $Y(x, \alpha)$ is a function of the spatial point, x, and the parameter, α. It follows that varying α allows $Y(x, \alpha)$ to sample the space of functions that are close, but not identical, to the extreme function, $y(x)$. We emphasize that the variation is quite arbitrary, so long as it satisfies the endpoint conditions and is smooth. Otherwise, it can be anything.

7.2.2 The δ variation operator acting on a function

As a notational shorthand, it is very useful to make use of the non-dimensional variation operator, δ, defined so that

$$Y(x, \alpha) = y(x) + \alpha\, \eta(x) \equiv y(x) + \delta y(x) = (1 + \delta)\, y(x), \tag{7.7}$$

where $\delta y(x)$ defines the variation of the function, which is smooth and vanishes at the endpoints. Importantly, the variation operator, δ, acts only on the function and not the space point, so that x remains unchanged upon acting with δ on y. Correspondingly, δ commutes with the derivative operator

$$\delta[y'(x)] = \delta[\mathrm{d}y(x)/\mathrm{d}x] = \mathrm{d}[\delta y(x)]/\mathrm{d}x. \tag{7.8}$$

As a final shorthand, we often write

$$y(x) \to y(x) + \delta y(x), \tag{7.9}$$

to indicate that a function, $y(x)$, is to be replaced by $(1+\delta)\,y(x)$ each place it is found. In Figure 7.1 we illustrate $y(x) + \delta y(x)$, where $y(x)$ is a straight line whereas $\delta y(x)$ are arbitrary curved paths for which $\delta y(x_1) = \delta y(x_2) = 0$.

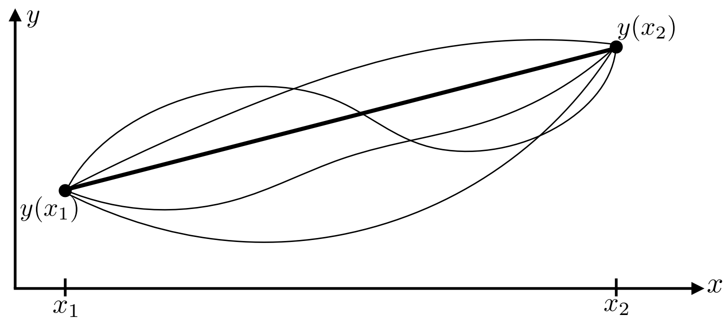

FIGURE 7.1: Illustrating some variations to the function, $y(x)$, here depicted as a straight line whereas $y(x)+\delta y(x)$ are curved paths. Note that each variation satisfies $\delta y(x_1) = \delta y(x_2) = 0$.

7.2.3 Derivative of the integral

Introducing the function, $Y(x, \alpha)$, into the integral (7.1) yields

$$\mathcal{J}(\alpha) = \int_{x_1}^{x_2} L[Y(x,\alpha), Y'(x,\alpha), x] \, \mathrm{d}x, \tag{7.10}$$

where we exposed the α dependence of the integral by writing $\mathcal{J}(\alpha)$, and where

$$Y'(x,\alpha) = \frac{\partial Y(x,\alpha)}{\partial x} = y'(x) + \alpha\, \eta'(x). \tag{7.11}$$

By construction, the integral, $\mathcal{J}(\alpha)$, is an extremum at $\alpha = 0$, which means that its derivative vanishes there

$$\left. \frac{\mathrm{d}\mathcal{J}}{\mathrm{d}\alpha} \right|_{\alpha=0} = 0. \tag{7.12}$$

The integration limits in equation (7.10) are independent of α, so that the chain rule renders

$$\frac{\mathrm{d}\mathcal{J}(\alpha)}{\mathrm{d}\alpha} = \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial Y} \frac{\partial Y}{\partial \alpha} + \frac{\partial L}{\partial Y'} \frac{\partial Y'}{\partial \alpha} \right] \mathrm{d}x. \tag{7.13}$$

Making use of the identities

$$\frac{\partial Y}{\partial \alpha} = \eta \quad \text{and} \quad \frac{\partial Y'}{\partial \alpha} = \eta' \tag{7.14}$$

leads to

$$\frac{\mathrm{d}\mathcal{J}(\alpha)}{\mathrm{d}\alpha} = \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial Y} \eta + \frac{\partial L}{\partial Y'} \eta' \right] \mathrm{d}x \tag{7.15a}$$

$$= \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial Y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial Y'} \right) \right] \eta \, \mathrm{d}x + \int_{x_1}^{x_2} \frac{\mathrm{d}}{\mathrm{d}x} \left[\frac{\partial L}{\partial Y'} \eta \right] \mathrm{d}x, \tag{7.15b}$$

where the second equality follows from integration by parts. Since $\eta(x_1) = \eta(x_2) = 0$, the total derivative term vanishes from equation (7.15b), thus leaving

$$\frac{\mathrm{d}\mathcal{J}(\alpha)}{\mathrm{d}\alpha} = \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial Y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial Y'} \right) \right] \eta \, \mathrm{d}x. \tag{7.16}$$

We emphasize the importance of distinguishing the derivative operators as they appear in equation (7.16), whereby

$$\frac{\partial L}{\partial Y'} = \left(\frac{\partial L}{\partial Y'} \right)_{Y,x} \tag{7.17a}$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \left[\frac{\partial L}{\partial Y'} \right] = \left[\frac{\partial Y}{\partial x} \frac{\partial}{\partial Y} \Big)_{Y',x} + \frac{\partial Y'}{\partial x} \frac{\partial}{\partial Y'} \Big)_{Y,x} + \frac{\partial}{\partial x} \Big)_{Y,Y'} \right] \frac{\partial L}{\partial Y'}. \tag{7.17b}$$

The ∂_Y and $\partial_{Y'}$ terms in equation (7.17b) arise from the implicit x dependence through Y and Y', with the explicit x dependence leading to the ∂_x term. Being mindful of these operations greatly reduces confusion when manipulating the equations of variational calculus. In the following, we typically drop the subscripts to reduce notational clutter. Yet where confusion arises it is useful to return to the above two equations for clarification.

7.2.4 Variation of the integral and the Fréchet derivative

Rather than computing the derivative of the integral as in Section 7.2.3, it is common in the physics literature to compute the variation of the integral through use of the δ operator, which in turn motivates defining the functional derivative, or sometimes called the Fréchet derivative. In this approach we write

$$\delta \mathcal{J} = \delta \int_{x_1}^{x_2} L[y(x), y'(x), x] \, \mathrm{d}x = \int_{x_1}^{x_2} \delta L[y(x), y'(x), x] \, \mathrm{d}x, \tag{7.18}$$

where the variation operator commutes with the spatial integral since, as noted in Section 7.2.2, δ does not touch the space coordinate. To compute the variation of L, we make use of the chain rule and by noting that $\delta x = 0$, so that

$$\delta \mathcal{J} = \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial y} \, \delta y + \frac{\partial L}{\partial y'} \delta y' \right] \mathrm{d}x \tag{7.19a}$$

$$= \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \right) \right] \delta y \, \mathrm{d}x + \int_{x_1}^{x_2} \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \, \delta y \right) \mathrm{d}x \tag{7.19b}$$

$$= \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \right) \right] \delta y \, \mathrm{d}x \tag{7.19c}$$

$$= \alpha \left[\frac{\mathrm{d}\mathcal{J}(\alpha)}{\mathrm{d}\alpha} \right]_{\alpha=0}, \tag{7.19d}$$

where the penultimate equality holds since $\delta y(x_1) = \delta y(x_2) = 0$, and the final equality set $\delta y = \alpha \, \eta$ and used equation (7.16) for $\mathrm{d}\mathcal{J}(\alpha)/\mathrm{d}\alpha$. Evidently, equation (7.19d) provides a direct connection between the variation of the integral to the derivative of the integral via

$$\delta \mathcal{J} = \alpha \left[\frac{\mathrm{d}\mathcal{J}(\alpha)}{\mathrm{d}\alpha} \right]_{\alpha=0}. \tag{7.20}$$

The variation operator formalism in equation (7.19c) suggests we define the functional derivative, (also known as the Fréchet derivative), via

$$\frac{\delta \mathcal{J}}{\delta y(x^*)} = \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \right) \right] \frac{\delta y(x)}{\delta y(x^*)} \, \mathrm{d}x = \left[\frac{\partial L}{\partial y(x^*)} - \frac{\mathrm{d}}{\mathrm{d}x^*} \left(\frac{\partial L}{\partial y'(x^*)} \right) \right] \mathrm{d}x^*, \tag{7.21}$$

where

$$\frac{\delta y(x)}{\delta y(x^*)} = \delta(x - x^*) \, \mathrm{d}x^*, \tag{7.22}$$

with $\delta(x - x^*)$ the Dirac delta studied in Chapter 5, and with x^* an arbitrary space point.[1] It is notable that the $\mathrm{d}x^*$ factor appearing in equation (7.22) is typically ignored in most treatments, since it plays no role in the Euler equation of variational calculus that follows from setting $\delta \mathcal{J}/\delta y(x^*) = 0$ (Section 7.2.5). However, this factor is necessary for dimensional consistency since the Dirac delta has dimensions of inverse length. So in summary, the Fréchet derivative of the integral is given by

$$\frac{\delta \mathcal{J}}{\delta y} = \left[\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \right) \right] \mathrm{d}x. \tag{7.23}$$

The Fréchet derivative provides the functional analog to the gradient operator, with this derivative vanishing at an extremum of $\mathcal{J}$.

7.2.5 The Euler equation of variational calculus

The derivative (7.16) vanishes when $\alpha = 0$, by construction, in which case

$$0 = \left. \frac{\mathrm{d}\mathcal{J}(\alpha)}{\mathrm{d}\alpha} \right|_{\alpha=0} = \int_{x_1}^{x_2} \left[\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \right) \right] \eta \, \mathrm{d}x, \tag{7.24}$$

where the derivatives inside the integral are now taken with respect to y since $Y(0, x) = y(x)$. Since $\eta(x)$ is an arbitrary function, this equation is satisfied only if the integrand vanishes, which leads to the Euler equation of variational calculus

$$\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \right) = 0 \Longleftrightarrow y(x) \text{ extremizes } \mathcal{J}. \tag{7.25}$$

Euler's equation is equivalently found by setting the Fréchet derivative (7.23) to zero

$$\frac{\delta \mathcal{J}}{\delta y} = 0 \Longrightarrow \left[\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\partial L}{\partial y'} \right) \right] = 0. \tag{7.26}$$

In our study of mechanics in Chapter 14, we refer to the Euler equation as the Euler-Lagrange equation, given work done by Lagrange to extend Euler's work on variational methods to mechanics.

7.2.6 The second form of Euler's equation

The chain rule leads to the total derivative of the Lagrangian

$$\frac{\mathrm{d}L}{\mathrm{d}x} = \left[\frac{\partial L}{\partial x} \right]_{y,y'} + \frac{\mathrm{d}y}{\mathrm{d}x} \left[\frac{\partial L}{\partial y} \right]_{y',x} + \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} \left[\frac{\partial L}{\partial y'} \right]_{y,x} \tag{7.27a}$$

[1] Use of the δ symbol for both the variational operator as well as the Dirac delta is unfortunate but universal.

$$= \frac{\partial L}{\partial x} + y' \frac{\partial L}{\partial y} + y'' \frac{\partial L}{\partial y'}, \tag{7.27b}$$

where the second equality dropped the subscript notation to reduce clutter. In turn, we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[y' \frac{\partial L}{\partial y'}\right] = y'' \frac{\partial L}{\partial y'} + y' \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\partial L}{\partial y'}\right], \tag{7.28}$$

with substitution from equation (7.27b) rendering

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[L - y' \frac{\partial L}{\partial y'}\right] = \frac{\partial L}{\partial x} + y'\left[\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial L}{\partial y'}\right)\right]. \tag{7.29}$$

The final term on the right hand side vanishes due to Euler's equation (7.25), thus leading to the second form of Euler equation of variational calculus

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[L - y' \frac{\partial L}{\partial y'}\right] = \frac{\partial L}{\partial x}. \tag{7.30}$$

This equation is particularly useful when the Lagrangian, L, has no explicit dependence on x, in which case the right hand side vanishes. Furthermore, when applied to mechanics, the term $y'\,\partial L/\partial y' - L$ is the Hamiltonian (e.g., see Section 14.8.4).

7.2.7 Nature of the extremum

We have established that the Euler equation of variational calculus, equation (7.25), has a solution, $y(x)$, that is an extremum to the integral $\mathcal{J}[y(x), y'(x), x]$. In this manner, we connect a question about the extremum of an integral to the solution of a differential equation. However, is the extremum a minimum, maximum, or saddle? To answer that question requires taking the second derivative of the integral (equivalently, the second variation),

$$\delta^2 \mathcal{J} = \alpha^2 \left[\frac{\mathrm{d}^2 \mathcal{J}(y + \alpha\,\eta)}{\mathrm{d}\alpha^2}\right]_{\alpha=0}. \tag{7.31}$$

Here we consider α as a bookkeeping parameter, with the second variation the coefficient associated with α^2 in a Taylor series expansion. In many, though not all, cases from physics, $y(x)$ minimizes $\mathcal{J}[y(x), y'(x), x]$, and the following arguments provide reasons.

To derive the second derivative we return to the manipulations of Section 7.2.3, only now keep terms out to order α^2. For this purpose we consider the second variation of the Lagrangian

$$\delta^2 L = \frac{1}{2} \frac{\partial^2 L}{\partial y^2} (\delta y)^2 + \frac{\partial^2 L}{\partial y\,\partial y'} \delta y\, \delta y' + \frac{1}{2} \frac{\partial^2 L}{\partial y'^2} (\delta y')^2 \tag{7.32a}$$

$$= \frac{1}{2}\left[\frac{\partial^2 L}{\partial y^2} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial^2 L}{\partial y\,\partial y'}\right)\right](\delta y)^2 + \frac{1}{2} \frac{\partial^2 L}{\partial y'^2} (\delta y')^2 + \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\partial^2 L}{\partial y\,\partial y'} (\delta y)^2\right]. \tag{7.32b}$$

The total derivative term integrates to a boundary contribution, and that vanishes if we assume δy vanishes at the boundaries, just like we did for the first variation. For the remaining terms, we can only make a modestly general statement, but one that is of use in many cases. Namely, a necessary condition for the action to be a local minimum is for the following weak Legendre condition to hold

$$\frac{\partial^2 L}{\partial y'^2} > 0 \quad \Longleftarrow \text{necessary for local minumum.} \tag{7.33}$$

This term appears in the second variation equation (7.32b), and it is multiplied by the non-negative $(\delta y')^2 \geq 0$. So if $\partial^2 L/\partial y'^2 < 0$, then we can find a test function in which $(\delta y')^2$ pulls $\delta^2 L$ down to a negative value, no matter what the value of the first term in equation (7.32b), in which case the extremum cannot be a minimum. So although $\partial^2 L/\partial y'^2 > 0$ is not sufficient to ensure $\delta^2 L > 0$ (in which case the integral is minimized), it is certainly necessary.

Consider an example with a Lagrangian that only depends on $y'(x)$

$$L = \tfrac{1}{2}(y')^2, \tag{7.34}$$

in which case

$$\delta^2 L = \tfrac{1}{2}(\delta y')^2 > 0. \tag{7.35}$$

Hence, the solution, $y(x)$, provides a local minimum to the functional, $\mathcal{J}[y(x), y'(x), x]$. Indeed, for this Lagrangian, the Euler equation (7.25) means that y' is a constant along the path. This case is relevant to our study of geodesics in Sections 7.3 and 7.4, with free particles in mechanics following geodesics paths.

Even for those cases where we cannot rely on the Legendre condition (7.33), the nature of the extremum is often apparent by evaluating the functional with the solution to the Euler equation. In these cases, we do not need to evaluate the second variation.

7.3 Geodesics in Euclidean space

In this section we make use of the calculus of variations to determine a path in space known as a geodesic. The simplest geodesic is a straight line, which provides the minimum distance between two points in Euclidean space. Geodesics provide a generalization of the notion of straight line for arbitrary smooth Riemannian manifolds. For example, in Section 7.4 we show that geodesics on the surface of a sphere are the shorter of the two arcs living on the great circle connecting the two points, When applied in the context of mechanics, geodesics are trajectories taken by a free particle; i.e., a particle that experiences no force field so that its energy is purely kinetic.[2] Given the connection to mechanics, we introduce some mechanics language in this section, even though the ideas hold for general curves in space that are unrelated to particle trajectories.

7.3.1 Shortest distance between two points on a plane

Let $(x, y) \in \mathbb{R}^2$ be the Cartesian coordinates for a point in $\mathbb{E}^2$ (i.e., a flat plane), where $y = y(x)$ specifies y as a differentiable function of x. We seek an expresion for the function, $y(x)$, that represents the shortest path between two arbitrary points, A and B, in $\mathbb{E}^2$. From elementary Euclidean geometry we know that the answer is a straight line

$$y = y_{\mathsf{A}} + \frac{x - x_{\mathsf{A}}}{x_{\mathsf{B}} - x_{\mathsf{A}}}\,(y_{\mathsf{B}} - y_{\mathsf{A}}) = \frac{y_{\mathsf{A}}\,(x_{\mathsf{B}} - x) + y_{\mathsf{B}}\,(x - x_{\mathsf{A}})}{x_{\mathsf{B}} - x_{\mathsf{A}}}, \tag{7.36}$$

[2] It can experience forces of constraint, such as those keeping the particle moving on a particular surface. But such forces act perpendicular to the motion, so they do no work on the particle, and as such they do not alter the particle's kinetic energy. We consider forces of constraint as part of our study of analytical mechanics in Chapter 14.

with $y_A = y(x_A)$ and $y_B = y(x_B)$. We now derive the straight line geodesics using the calculus of variations, thus garnering experience with the formalism for a problem where we know the answer.

In $\mathbb{E}^2$, the distance between two infinitesimally close points is given by Pythagoras' theorem

$$\mathrm{d}s = [(\mathrm{d}x)^2 + (\mathrm{d}y)^2]^{1/2} = \sqrt{1+(y')^2}\,\mathrm{d}x, \tag{7.37}$$

so that the finite distance between two points is given by the integral

$$\mathcal{J} = \int_A^B \mathrm{d}s = \int_{x_A}^{x_B} \sqrt{1+(y')^2}\,\mathrm{d}x = \int_{x_A}^{x_B} L\,\mathrm{d}x, \tag{7.38}$$

where the Lagrangian is

$$L = \sqrt{1+(y')^2}. \tag{7.39}$$

Terms in the Euler equation (7.25) are thus given by

$$\frac{\partial L}{\partial y} = 0 \quad \text{and} \quad \frac{\partial L}{\partial y'} = \frac{y'}{\sqrt{1+(y')^2}}, \tag{7.40}$$

so that the Euler equation is

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{y'}{\sqrt{1+(y')^2}}\right] = 0 \Longrightarrow y'' = \frac{y''\,(y')^2}{1+(y')^2}. \tag{7.41}$$

If $y'' \neq 0$ then we have the inconsistent result that $0 = 1$, so that $y'' = 0$ (a line) is the only self-consistent solution.

7.3.2 Parameterized curves in $\mathbb{E}^n$

We saw in Section 2.4 that all simple curves in Euclidean space can be parameterized by a one-dimensional parameter, φ, that monotonically increases (or decreases) along the path. Let $\boldsymbol{x}(\varphi)$ be the Cartesian position for a point in n-dimensional Euclidean space, $\mathbb{E}^n$. In this case the squared infinitesimal arc-length is given by

$$(\mathrm{d}s)^2 = \delta_{ab}\,\mathrm{d}x^a\,\mathrm{d}x^b = \delta_{ab}\,\frac{\mathrm{d}x^a}{\mathrm{d}\varphi}\,\frac{\mathrm{d}x^b}{\mathrm{d}\varphi}\,(\mathrm{d}\varphi)^2 = \delta_{ab}\,\dot{x}^a\,\dot{x}^b\,(\mathrm{d}\varphi)^2, \tag{7.42}$$

where we introduced the dot notation as a shorthand for the velocity[3]

$$\dot{x}^a = \frac{\mathrm{d}x^a}{\mathrm{d}\varphi}. \tag{7.43}$$

As a result, the distance between two points, $\boldsymbol{x}_A$ and $\boldsymbol{x}_B$, takes the form

$$\mathcal{J} = \int_{\boldsymbol{x}_A}^{\boldsymbol{x}_B} \mathrm{d}s = \int_{\varphi_A}^{\varphi_B} \sqrt{\delta_{ab}\,\dot{x}^a\,\dot{x}^b}\,\mathrm{d}\varphi = \int_{\varphi_A}^{\varphi_B} |\dot{\boldsymbol{x}}|\,\mathrm{d}\varphi = \int_{\varphi_A}^{\varphi_B} L[\dot{x}^a(\varphi)]\,\mathrm{d}\varphi. \tag{7.44}$$

[3] The velocity here refers to the rate of change of the position with respect to the parameter, φ. When φ is the time, then $\dot{\boldsymbol{x}}(t)$ is the velocity of a particle along its trajectory.

In the penultimate equality we introduced the speed

$$|\dot{\boldsymbol{x}}| = \sqrt{\delta_{ab}\, \dot{x}^a\, \dot{x}^b}, \tag{7.45}$$

so that the distance is the integral of the speed along the curve. Notice how the distance is invariant to the choice of parameter, φ, so long as the new parameter maintains a monotonic relation with φ.

7.3.3 Euler equation for the straight line geodesic

As before, the Euler equation is relatively simple because the Lagrangian has no explicit dependence on either the parameter, φ, nor the trajectory, x^a

$$\frac{\partial L}{\partial \varphi} = 0 \quad \text{and} \quad \frac{\partial L}{\partial x^a} = 0. \tag{7.46}$$

The Lagrangian only depends on the velocity, $\dot{x}^a$, in which case the Euler equation is

$$\frac{\mathrm{d}}{\mathrm{d}\varphi}\left[\frac{\dot{x}^a\, \delta_{ac}}{|\dot{\boldsymbol{x}}|}\right] = 0 \tag{7.47}$$

This Euler equation for the geodesic suggests that we introduce the unit direction,

$$T_c \equiv \frac{\dot{x}^a\, \delta_{ac}}{|\dot{\boldsymbol{x}}|} \tag{7.48}$$

which points in the direction of the velocity along the path. Evidently, the Euler equation (7.47) says that all components to the velocity direction remain fixed along the geodesic

$$\frac{\mathrm{d}T_c}{\mathrm{d}\varphi} = 0. \tag{7.49}$$

Consequently, the geodesic is a straight line, along which

$$\dot{x}^a\, \delta_{ac} = T_c\, |\dot{\boldsymbol{x}}|, \tag{7.50}$$

with T_c a constant.

Performing the derivative on the Euler equation (7.47), and some rearrangement, yields

$$\delta_{ac}\, \ddot{x}^a = \frac{1}{|\dot{\boldsymbol{x}}|}\frac{\mathrm{d}|\dot{\boldsymbol{x}}|}{\mathrm{d}\varphi}\, \delta_{ac}\dot{x}^a \equiv F(\varphi)\, \delta_{ac}\dot{x}^a, \tag{7.51}$$

where we identified the scaling function

$$F(\varphi) = \frac{1}{|\dot{\boldsymbol{x}}|}\frac{\mathrm{d}|\dot{\boldsymbol{x}}|}{\mathrm{d}\varphi} = \frac{\mathrm{d}\ln|\dot{\boldsymbol{x}}|}{\mathrm{d}\varphi}. \tag{7.52}$$

Equation (7.51) says that the velocity and the acceleration are along the same direction, as expected for a straight line motion.

7.3.4 Using the arc length as the geodesic parameter

We again emphasize that the parameter, φ, for the geodesic is arbitrary so long as it is monotonic along the geodesic. Even so, it is useful to simplify the previous formulas by choosing the arc-length from Section 2.5.2,

$$\varphi = s, \tag{7.53}$$

in which case the squared speed is unity

$$|\dot{\boldsymbol{x}}|^2 = \delta_{ab} \frac{\mathrm{d}x^a}{\mathrm{d}s} \frac{\mathrm{d}x^b}{\mathrm{d}s} = 1, \tag{7.54}$$

so that the geodesic equation (7.47) simplifies to

$$\delta_{ac} \, \ddot{x}^a = 0. \tag{7.55}$$

Equivalently, the scaling function, (7.52), vanishes since the speed is a constant. Again, we recover the equation for a straight line, only now in the more familiar form

$$x^a(s) = s \, M^a + B^a, \tag{7.56}$$

where M^a is a constant measuring the slope and B^a is a constant measuring the intercept.

7.3.5 Affine and non-affine geodesic parameters

For the special case of $\varphi = \alpha \, s + \beta$, with constants α and β, we refer to φ as an affine parameter, in which case the curve parameter is a linear function of the arc-length, and the speed is a constant along a geodesic. A non-affine parameter, $\varphi = \varphi(s)$, is a nonlinear and monotonic function of the arc-length. For non-affine parameters, the speed is not constant along the geodesic since

$$|\dot{\boldsymbol{x}}|^2 = \delta_{ab} \frac{\mathrm{d}x^a}{\mathrm{d}\varphi} \frac{\mathrm{d}x^b}{\mathrm{d}\varphi} \tag{7.57a}$$

$$= \delta_{ab} \frac{\mathrm{d}x^a}{\mathrm{d}s} \frac{\mathrm{d}x^b}{\mathrm{d}s} \frac{\mathrm{d}s}{\mathrm{d}\varphi} \frac{\mathrm{d}s}{\mathrm{d}\varphi} \tag{7.57b}$$

$$= \left[\frac{\mathrm{d}s}{\mathrm{d}\varphi} \right]^2, \tag{7.57c}$$

where we made use of the unit speed condition (7.54) for the arc-length parameter. Geometrically, the geodesic is the shortest path between two points, and it is independent of how the geodesic curve is parameterized. The unit-speed condition (7.54) refers to a specific choice of the arc-length parameter. Changing to a more general non-affine parameter keeps the geodesic curve unchanged, but it produces a speed (defined relative to the non-affine parameter) that is generally not a constant.

7.3.6 Extremizing the kinetic energy

Consider the integral

$$\mathcal{J} = \frac{1}{2} \int_{\varphi_{\mathsf{A}}}^{\varphi_{\mathsf{B}}} \delta_{ab} \, \dot{x}^a \, \dot{x}^b \, \mathrm{d}\varphi = \frac{1}{2} \int_{\varphi_{\mathsf{A}}}^{\varphi_{\mathsf{B}}} |\dot{\boldsymbol{x}}|^2 \, \mathrm{d}\varphi, \tag{7.58}$$

where the Lagrangian,

$$L = \tfrac{1}{2}\delta_{ab}\,\dot{x}^a\,\dot{x}^b = \tfrac{1}{2}|\dot{\boldsymbol{x}}|^2, \tag{7.59}$$

measures the kinetic energy (per mass) along the path.[4] The Euler equation for the kinetic energy (7.59) is given by

$$\delta_{ac}\,\ddot{x}^a = 0. \tag{7.60}$$

Evidently, the kinetic energy produces the straight line geodesic found with the distance functional (7.44), only now it is restricted to the affine parameter. One way to further support this conclusion is to take the scalar product of $\delta_{ac}\,\ddot{x}^a$ with $\dot{x}^c$, so have

$$\tfrac{1}{2}\,\dot{x}^c\,\delta_{ac}\,\ddot{x}^a = \frac{\mathrm{d}|\dot{\boldsymbol{x}}|}{\mathrm{d}\varphi} = 0, \tag{7.61}$$

so that the speed is a constant, as expected with φ an affine parameter.

7.4 Geodesics on curved manifolds

We now generalize the discussion from Section 7.3 to the case of arbitrary coordinates. We furthermore allow the manifold to be curved. In this case the geodesic equation requires a bit more work to derive. For example, we may be interested in geodesics on the surface of a sphere embedded in Euclidean space, in which case spherical coordinates are suited and a geodesic is the shortest arc along a great circle connecting two points. We may also be interested in geodesics on stratification surfaces (i.e., an isopycnal surface) encountered in stratified fluid mechanics studied in later volumes of this book. Relative to the Euclidean example, the only substantial change needed is to introduce the metric tensor, $\mathbb{g}_{ab}$, from Section 3.7 that measures distances between points on the manifold. We also extremize the kinetic energy from Section 7.3.6 in order to simplify the algebra by focusing on the affine parameter as given by the arc length parameter,

$$\varphi = s. \tag{7.62}$$

We thus consider the functional

$$\mathcal{J} = \frac{1}{2}\int_{\varphi_{\mathsf{A}}}^{\varphi_{\mathsf{B}}} \mathbb{g}_{ab}\,\dot{\xi}^a\,\dot{\xi}^b\,\mathrm{d}\varphi = \frac{1}{2}\int_{\varphi_{\mathsf{A}}}^{\varphi_{\mathsf{B}}} L[\xi^a,\dot{\xi}^a]\,\mathrm{d}\varphi, \tag{7.63}$$

where ξ^a are arbitrary spatial coordinates. We furthermore assume that the metric tensor is an explicit function only of the spatial coordinates[5]

$$\mathbb{g}_{ab} = \mathbb{g}_{ab}(\boldsymbol{\xi}). \tag{7.64}$$

7.4.1 Euler equation for the geodesic

Terms appearing in the Euler equation (7.25) are given by

$$\frac{\partial L}{\partial \xi^c} = \tfrac{1}{2}\,\partial_c \mathbb{g}_{ab}\,\dot{\xi}^a\,\dot{\xi}^b \tag{7.65a}$$

---

[4] Again, we make use of mechanics language even though this is the kinetic energy only when the parameter is time, $\varphi = t$.

[5] When using generalized vertical coordinates from VOLUME 4, we must extend the formalism to allow for a time dependent metric tensor.

$$\frac{\partial L}{\partial \dot{\xi}^c} = \tfrac{1}{2} \mathbb{g}_{cb} \, \dot{\xi}^b + \tfrac{1}{2} \mathbb{g}_{ca} \, \dot{\xi}^a, \tag{7.65b}$$

so that the Euler equation is

$$\partial_c \mathbb{g}_{ab} \, \dot{\xi}^a \, \dot{\xi}^b = \frac{\mathrm{d}(\mathbb{g}_{cb} \, \dot{\xi}^b + \mathbb{g}_{ca} \, \dot{\xi}^a)}{\mathrm{d}\varphi}. \tag{7.66}$$

Application of the chain rule leads to

$$\frac{\mathrm{d}(\mathbb{g}_{cb} \, \dot{\xi}^b)}{\mathrm{d}\varphi} = \partial_e \mathbb{g}_{cb} \, \dot{\xi}^e \, \dot{\xi}^b + \mathbb{g}_{cb} \, \ddot{\xi}^b \quad \text{and} \quad \frac{\mathrm{d}(\mathbb{g}_{ca} \, \dot{\xi}^a)}{\mathrm{d}\varphi} = \partial_e \mathbb{g}_{ca} \, \dot{\xi}^e \, \dot{\xi}^a + \mathbb{g}_{ca} \, \ddot{\xi}^a, \tag{7.67}$$

so that

$$\frac{\mathrm{d}}{\mathrm{d}\varphi} \left[\frac{\partial L}{\partial \dot{\xi}^c} \right] = 2 \, \mathbb{g}_{ca} \, \ddot{\xi}^a + \partial_e \mathbb{g}_{cb} \, \dot{\xi}^e \, \dot{\xi}^b + \partial_e \mathbb{g}_{ca} \, \dot{\xi}^e \, \dot{\xi}^a \tag{7.68a}$$

$$= 2 \, \mathbb{g}_{ca} \, \ddot{\xi}^a + \partial_a \mathbb{g}_{cb} \, \dot{\xi}^a \, \dot{\xi}^b + \partial_b \mathbb{g}_{ca} \, \dot{\xi}^b \, \dot{\xi}^a \tag{7.68b}$$

$$= 2 \, \mathbb{g}_{ca} \, \ddot{\xi}^a + (\partial_a \mathbb{g}_{cb} + \partial_b \mathbb{g}_{ca}) \, \dot{\xi}^a \, \dot{\xi}^b. \tag{7.68c}$$

The above index gymnastics are motivated by the desire to bring the nonlinear terms $\dot{\xi}^a \, \dot{\xi}^b$ together, in which case the Euler equation (7.66) is

$$\mathbb{g}_{ca} \, \ddot{\xi}^a + \tfrac{1}{2} (\partial_a \mathbb{g}_{cb} + \partial_b \mathbb{g}_{ca} - \partial_c \mathbb{g}_{ab}) \, \dot{\xi}^a \, \dot{\xi}^b = 0. \tag{7.69}$$

Contracting with the inverse metric, $\mathbb{g}^{ec}$, and using

$$\mathbb{g}^{ec} \, \mathbb{g}_{ca} = \delta^e{}_a, \tag{7.70}$$

leads to the geodesic equation

$$\ddot{\xi}^e + \Gamma^e_{ab} \, \dot{\xi}^a \, \dot{\xi}^b = 0, \tag{7.71}$$

where we introduced the Christoffel symbols from equation (3.130)

$$\Gamma^e_{ab} = \tfrac{1}{2} \, \mathbb{g}^{ed} \, (\partial_b \, \mathbb{g}_{da} + \partial_a \, \mathbb{g}_{db} - \partial_d \, \mathbb{g}_{ab}). \tag{7.72}$$

We again encounter the geodesic equation (7.71) in our study of analytical mechanics in Chapter 14, as well as in a Lagrangian treatment of fluid mechanics in VOLUME 4, yet with added terms arising from potential energy in addition to the kinetic energy

7.4.2 Geodesics on the sphere

For the case of Cartesian coordinates in Euclidean space, the Christoffel symbols all vanish, so that we recover the results from Section 7.3, in which the geodesic equation (7.71) leads to straight lines satisfying the affine parameter geodesic equation $\ddot{x}^a = 0$. A less trivial example concerns geodesics on the surface of a sphere with fixed radius, R. Given the spherical symmetry, we determine geodesics by using the spherical longitude and latitude coordinates, (λ, ϕ), from Section 4.4 and illustrated in Figure 4.2.

Deriving the geodesic equations

For points restricted to the surface of the sphere, the spherical metric tensor and its inverse are given by

$$\mathbb{g}_{ab} = R^2 \begin{bmatrix} \cos^2\phi & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbb{g}^{ab} = R^{-2} \begin{bmatrix} \cos^{-2}\phi & 0 \\ 0 & 1 \end{bmatrix} \quad \text{with} \quad a, b = \lambda, \phi. \tag{7.73}$$

We find the following nonzero Christoffel symbols

$$\Gamma^{\lambda}_{\lambda\phi} = \Gamma^{\lambda}_{\phi\lambda} = -\tan\phi \quad \text{and} \quad \Gamma^{\phi}_{\lambda\lambda} = \cos\phi\sin\phi, \tag{7.74}$$

which then leads to the two geodesic equations

$$\ddot{\lambda} - 2\,\dot{\lambda}\,\dot{\phi}\tan\phi = 0 \quad \text{and} \quad \ddot{\phi} + (\dot{\lambda})^2\cos\phi\sin\phi = 0, \tag{7.75}$$

where the overdot is shorthand for

$$\dot{\lambda} = \frac{\mathrm{d}\lambda}{\mathrm{d}\varphi} \quad \text{and} \quad \dot{\phi} = \frac{\mathrm{d}\phi}{\mathrm{d}\varphi}, \tag{7.76}$$

with φ an affine parameter along the geodesic. There are two invariants (relations between zeroth and first order derivatives of λ and ϕ) that are implied by the geodesic equations (7.75). These invariants are (using the language of mechanics) the kinetic energy per mass (which is non-negative) and axial angular momentum per mass

$$K = \tfrac{1}{2}[(R\,\dot{\lambda}\cos\phi)^2 + (R\,\dot{\phi})^2] \quad \text{and} \quad l^z = (R\cos\phi)^2\,\dot{\lambda} \quad \text{with} \quad \frac{\mathrm{d}K}{\mathrm{d}\varphi} = 0 \quad \text{and} \quad \frac{\mathrm{d}l^z}{\mathrm{d}\varphi} = 0. \tag{7.77}$$

Note that the kinetic energy per mass is an obvious invariant since we are making use of an affine parameter so that the speed remains constant along a geodesic. The angular momentum invariant holds due to rotational symmetry around the axis defined by z. It is very useful to express the geodesics in terms of these constants. For example, a rearrangement allows us to eliminate $\dot{\lambda}$ to render

$$(R\,\dot{\phi})^2 = 2\,K - [l^z/(R\cos\phi)]^2. \tag{7.78}$$

We thus find that the path of a geodesic extends only so far as to ensure

$$\cos^2\phi \geq \left[\frac{l^z}{R\sqrt{2\,K}}\right]^2. \tag{7.79}$$

That is, for a given kinetic energy and axial angular momentum, this inequality defines the maximum latitude reached by the geodesic. Larger kinetic energy allows the geodesic to reach closer to the poles, whereas larger axial angular momentum keeps the geodesics closer to the equator.

Integrating the geodesic equation

The kinetic energy and axial angular momentum invariants allow us to eliminate the affine parameter, thus obtaining an expression for the change in latitude as longitude changes along a geodesic

$$\left[\frac{\mathrm{d}\phi}{\mathrm{d}\lambda}\right]^2 = \frac{[2\,K\cos^2\phi - (l^z/R)^2]\,R^2\cos^2\phi}{(l^z)^2}. \tag{7.80}$$

This equation can be integrated to determine the relation between the latitude and longitude along a geodesic. For that purpose, it is useful to introduce the non-dimensional constant

$$B = \frac{l^z}{R\sqrt{2\,K}}, \tag{7.81}$$

in which case the inequality (7.79) becomes

$$B^2 \leq \cos^2\phi \Longleftrightarrow B \in [0,1], \tag{7.82}$$

and the geodesic equation (7.80) is

$$\left[\frac{\mathrm{d}\phi}{\mathrm{d}\lambda}\right]^2 = \frac{\cos^2\phi\,(\cos^2\phi - B^2)}{B^2}. \tag{7.83}$$

Now transform to a new coordinate

$$w = \tan\phi \Longrightarrow \cos^2\phi = 1/(1+w^2) \quad \text{and} \quad \mathrm{d}\phi = (\cos^2\phi)\,\mathrm{d}w, \tag{7.84}$$

in which case equation (7.83) becomes

$$\left[\frac{\mathrm{d}w}{\mathrm{d}\lambda}\right]^2 = A^2 - w^2, \tag{7.85}$$

where

$$A^2 = \frac{1-B^2}{B^2} = \frac{2\,K\,R^2}{(l^z)^2} - 1 \quad \text{and} \quad w^2 < A^2. \tag{7.86}$$

The solution to equation (7.85) is given by

$$w = \tan\phi = A\cos(\lambda - \lambda_\circ), \tag{7.87}$$

where $\lambda_\circ$ is a constant. This equation relates the latitude and longitude along a geodesic. Furthermore, as per equation (7.79) we find that the geodesic reaches its maximum poleward latitude, ϕ_{max}, at the longitude, $\lambda = \lambda_\circ$,

$$\tan\phi_{\mathsf{max}} = A = \pm\sqrt{2\,K\,R^2/(l^z)^2 - 1}. \tag{7.88}$$

So in summary, the geodesic equation (7.87) has two constants, $\lambda_\circ$ and A. The constant, A, is set by the sphere's radius, R, the kinetic energy per mass, K, and the axial angular momentum per mass, l^z, whereas $\lambda_\circ$ is the longitude at the maximum latitude.

Great circle geometry of geodesics

We now connect the geodesic equation (7.87) to that of a great circle, with reference to Figure 7.2. For that purpose, consider a vertical plane with horizontal outward normal, $\hat{\boldsymbol{x}}\cos\lambda_\circ + \hat{\boldsymbol{y}}\sin\lambda_\circ$, and assume this plane passes through the center of the sphere with radius R. The intersection of this plane with the surface of the sphere defines a great circle that connects points aligned along a constant longitude. To connect points that have different longitudes requires us to tilt the plane so that its outward normal now has a projection in the $\hat{\boldsymbol{z}}$ direction. Let us choose the following normal

$$\boldsymbol{n} = \hat{\boldsymbol{x}}\cos\lambda_\circ + \hat{\boldsymbol{y}}\sin\lambda_\circ - A^{-1}\,\hat{\boldsymbol{z}}, \tag{7.89}$$

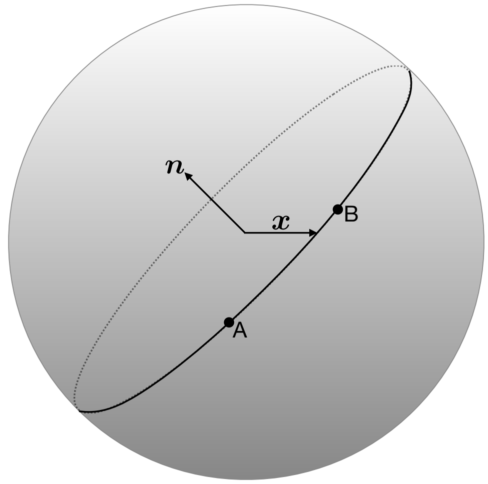

FIGURE 7.2: Illustrating the great circle that passes through two points, A and B, on the surface of the sphere. The circle is created from the intersection of the sphere and a plane that passes through the sphere's center. The plane's normal direction, $\boldsymbol{n} = \hat{\boldsymbol{x}} \cos\lambda_\circ + \hat{\boldsymbol{y}} \sin\lambda_\circ - A^{-1}\,\hat{\boldsymbol{z}}$, is set by the two constants, $\lambda_\circ$ and A, both of which are set by the kinetic energy per mass, K, axial angular momentum per mass, l^z, and radius of the sphere, R. By construction, any point on the great circle has a position, $\boldsymbol{x}$, that is perpendicular to the normal, $\boldsymbol{n} \cdot \boldsymbol{x} = 0$. The geodesic is the shortest arc along the great circle that connects points A and B.

so that the intersection of the plane with points on the surface of the sphere consist of all points that satisfy

$$\boldsymbol{x} \cdot \boldsymbol{n} = R\,(\hat{\boldsymbol{x}} \cos\phi\,\cos\lambda + \hat{\boldsymbol{y}} \cos\phi\,\sin\lambda + \hat{\boldsymbol{z}} \sin\phi) \cdot (\hat{\boldsymbol{x}} \cos\lambda_\circ + \hat{\boldsymbol{y}} \sin\lambda_\circ - A^{-1}\,\hat{\boldsymbol{z}}) = 0. \tag{7.90}$$

Trigonometry thus renders

$$\tan\phi = A\cos(\lambda - \lambda_\circ), \tag{7.91}$$

which is the same as the geodesic equation (7.87). We have thus shown that the equation for a geodesic is identical to that of a great circle, where a great circle consists of the intersection of the sphere with a plane whose normal direction is $\boldsymbol{n}$. So for any two points along the sphere, there is a great circle that connects these points. The geodesic is the shorter of the two arcs along the great circle that connects the two points (see Figure 7.2).

Connection to angular momentum

In Exercise 7.3 we show that the angular momentum per mass,

$$\boldsymbol{l} = \boldsymbol{x} \times \dot{\boldsymbol{x}} \tag{7.92}$$

is parallel to the normal direction, $\boldsymbol{n}$, so that

$$\boldsymbol{l} \times \boldsymbol{n} = 0. \tag{7.93}$$

This orientation further connects the geodesics to the mechanics of a point particle freely moving on the sphere, whereby the particle motion is in a plane perpendicular to the angular momentum vector.

7.4.3 Comments on constraints

The example of a geodesic on a sphere (embedded in Euclidean space) reveals something rather subtle but very important when extending these ideas to analytical mechanics in Chapter 14. Namely, the geodesic is constrained to be on the surface of the sphere, with the constraint handled by our choice to use spherical coordinates with the assumption that we are only interested in geodesics that maintain a constant radius. Alternatively, we could have used Cartesian coordinates with the constraint that $x^2+y^2+z^2 = R^2$, with the contraint maintained through the introduction of a Lagrange multiplier. Constraints in mechanics are imposed by a force. In some cases, we are uninterested in such forces of constraints, in which case we can often choose suitable coordinates that build in the constraint *a priori*, just like we did here by choosing spherical coordinates. In other cases, we are interested in the forces of constraint, so that we then introduce the Lagrange multiplier and find that it is proportional to the constraint force. We detail how to compute forces of constraint in Section 14.7, which reveals a rather feature of variational methods in mechanics.

7.4.4 Further study

Section 8.3 of *Rajeev* (2013) provides a reasonably thorough discussion of geodesics on the sphere, consistent with that presented here.

7.5 Extremum shapes

In this section we provide examples where the calculus of variations is used to determine shapes that extremize a particular functional.

7.5.1 The brachistochrone problem

The brachistochrone problem asks what is the curve upon which a frictionless bead moves under the effects from a uniform gravity field and travels in the shortest time between two points, A and B, where these two points are not underneath one another.[6] We illustrate the system in Figure 7.3. The integral expression for time of the excursion is given by

$$T = \int_{\mathsf{A}}^{\mathsf{B}} \mathrm{d}t = \int_{s_{\mathsf{A}}}^{s_{\mathsf{B}}} \frac{\mathrm{d}s}{v(s)}, \tag{7.94}$$

where s is the arc length and $v(s)$ is the speed of the bead as a function of the arc length. The minimal time is achieved by the particle moving in a single plane, which we set as the x-z plane without loss of generality. In this case the differential arc length is given by

$$\mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}z)^2} = \sqrt{1+(z')^2}\,\mathrm{d}x, \tag{7.95}$$

where we assumed that the vertical position, z, is a monotonic function of the horizontal position, x, thus enabling a functional expression $z = z(x)$ for the curve.[7]

[6] If the two points A and B are vertically oriented, then the brachistochrone is the straight vertical line connecting these two points.

[7] Some treatments, such as Example 5.2 of *Marion and Thornton* (1988), solve for $x = x(z)$, which is possible since the is a one-to-one relation between x and z.

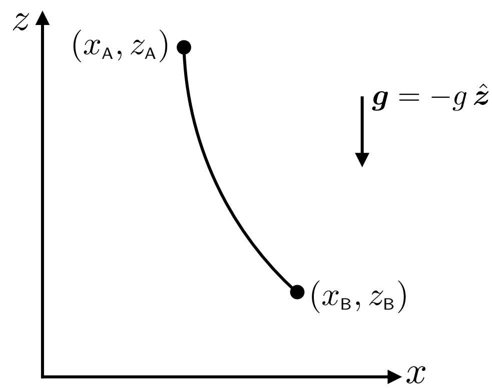

FIGURE 7.3: A bead falls along a frictionless wire from point A to point B. The brachistochrone problem seeks the shape of the curve that allows the bead to traverse this path in the least amount of time. Using variational methods we determine the shape to be a cycloid.

Mechanical energy of the bead is constant since the bead is frictionless.[8] If the bead starts from rest at point A, then the mechanical energy at this point is just given by the gravitational potential energy, $g\,m\,z_{\mathsf{A}}$, where m is the mass of the bead, z_{A} is its vertical position, and we choose the zero for potential energy at $z = 0$. We thus find that the speed at any point along the path is given by

$$v^2/2 + g\,z = g\,z_{\mathsf{A}} \Longrightarrow v = \sqrt{2\,g\,(z_{\mathsf{A}} - z)}, \tag{7.96}$$

in which the time integral (7.94) becomes

$$T = \int_{s_{\mathsf{A}}}^{s_{\mathsf{B}}} \frac{\mathrm{d}s}{v(s)} = \int_{x_{\mathsf{A}}}^{x_{\mathsf{B}}} \frac{\sqrt{1 + (z')^2}}{\sqrt{2\,g\,(z_{\mathsf{A}} - z)}}\,\mathrm{d}x. \tag{7.97}$$

The Lagrangian appearing in the time integral (7.97) is given by

$$L = L[z(x), z'(x)] = \frac{\sqrt{1 + (z')^2}}{\sqrt{2\,g\,(z_{\mathsf{A}} - z)}}, \tag{7.98}$$

and it has dimensions of inverse speed. This Lagrangian differs from the arc-length from our study of geodesics by a position-dependent weight, reflecting the conversion of potential energy into kinetic energy. With no explicit dependence on x (the system has translational invariance in x), the second form of Euler's equation of variational calculus, given by equation (7.30), leads to the identity

$$L - z'\,\frac{\partial L}{\partial z'} = \frac{1}{\sqrt{2\,g\,(z_{\mathsf{A}} - z)\,(1 + (z')^2)}} = c^{-1}, \tag{7.99}$$

where c is a constant with dimensions of speed ($\mathrm{L\ T^{-1}}$). We can write the solution to this equation by introducing the angle parameter θ, so that

$$x = [c^2/(4g)]\,(\sin\theta - \theta) \quad \text{and} \quad z_{\mathsf{A}} - z = [c^2/(4g)]\,(1 - \cos\theta) \tag{7.100a}$$

$$\mathrm{d}x/\mathrm{d}\theta = [c^2/(4g)]\,(\cos\theta - 1) \quad \text{and} \quad \mathrm{d}(z_{\mathsf{A}} - z)/\mathrm{d}\theta = [c^2/(4g)]\sin\theta. \tag{7.100b}$$

These parameteric equations describe a cycloid, which is the curve shown in Figure 7.4 that is

[8]In Section 11.2.5 we prove this property of conservative classical discrete systems.

traced by a point on a circle that is rolling (without slipping) along a straight line on a plane.[9]

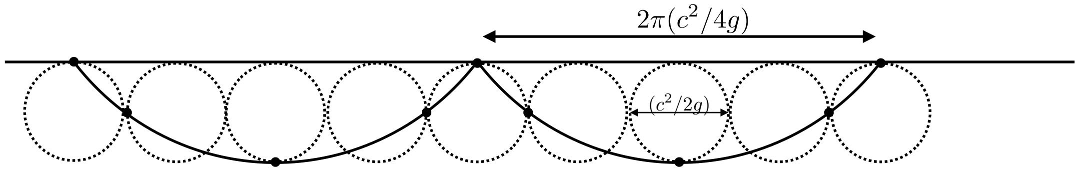

FIGURE 7.4: A cycloid is the curve traced by a point on a circle that is rolling (without slipping) along a straight line on a plane. As per equation (7.100a), the radius of the circle is $c^2/(4g)$ so that its circumference is $2\,\pi\,c^2/(4g)$. We depict the circle each quarter rotation as it moves along the line, with the black dot fixed relative to the circle and tracing out the cycloid.

7.5.2 Shape of a hanging massive string

Consider a string or cable with uniform mass per length, σ, which is suspended from each end as depicted in Figure 7.5. What is the shape of the string when in mechanical equilibrium? To answer this question one can make use of mechanics through studying the tensile forces acting within the string and the gravitational forces acting on the string, with equilibrium realized when the net forces and torques sum to zero. Alternatively, we can assume the shape minimizes the gravitational potential energy of the string, in which case there is no need to determine any of the forces. Instead we make use of variational calculus to determine the minimum potential energy state.

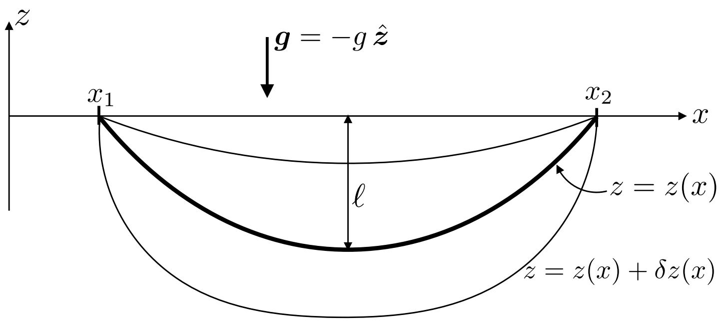

FIGURE 7.5: A massive string or cable is supported at its two ends at a vertical position $z_1 = z(x_1) = z_2 = z(x_2)$. The string is placed in a vertically directed gravitational field with acceleration, $\boldsymbol{g} = -g\,\hat{\boldsymbol{z}}$, where g is the constant gravitational acceleration. Variational methods determine the shape of the string as given by the hyperbolic cosine in equation (7.106), which is known as a catenary. This is the ideal shape taken by stationary power lines, spider webs, clothes lines, and chains, for example.

The variational problem arises from writing the gravitational potential energy as

$$P = g\,\sigma \int_{s(x_1)}^{s(x_2)} z(x)\,\mathrm{d}s, \tag{7.101}$$

where s is the arc distance along the curve and g is the constant gravitational acceleration. Along the curve, the arc-length differential is

$$\mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}z)^2} = \sqrt{1 + (z')^2}\,\mathrm{d}x, \tag{7.102}$$

[9]See Example 5-2 of *Marion and Thornton* (1988) for more details of the cycloid.

where $z'(x) = \mathrm{d}z/\mathrm{d}x$ is the derivative. Hence, the potential energy is the integral

$$P = g\,\sigma \int_{x_1}^{x_2} z(x)\,\sqrt{1+(z')^2}\,\mathrm{d}x = g\,\sigma \int_{x_1}^{x_2} L[z(x), z'(x)]\,\mathrm{d}x, \tag{7.103}$$

where we introduced the Lagrangian

$$L[z(x), z'(x)] = z(x)\,\sqrt{1+(z')^2}. \tag{7.104}$$

This Lagrangian is reminiscent of that found for the shortest path problem as given by equation (7.39), yet here there is an extra $z(x)$ factor multiplying the square root term.

Since the Lagrangian (7.104) has no explicit x dependence, we make use of the second form of the Euler equation, in which

$$L - y'\,\frac{\partial L}{\partial y'} = z\,\sqrt{1+(z')^2}\left[1 - \frac{(z')^2}{1+(z')^2}\right] = \ell \Longrightarrow z/\ell = 1+(z')^2, \tag{7.105}$$

where ℓ is a constant of integration with dimensions of length. The solution to this differential equation that satisfies the two endpoint conditions $z(x_1) = z(x_2)$ is given by

$$z = \ell\cosh[(2\,x - x_1 - x_2)/(2\ell)], \tag{7.106}$$

which is referred to as a catenary and is depicted in Figure 7.5. The constant, ℓ, is the vertical position of the midpoint of the curve, which is also the lowest point

$$z = \ell \quad \text{at } x = (x_1 + x_2)/2. \tag{7.107}$$

7.6 Fermat's principle and Snell's law

Consider a light ray (or any wave approximated as a ray) traveling in a material with a spatially dependent index of refraction, n, so that that speed of the wave is given by

$$v = c/\mathsf{n}, \tag{7.108}$$

where c is the speed with $\mathsf{n} = 1$. Following Fermat, we formulate the variational problem for the least time that a ray travels between two points

$$T = \int_{\mathsf{A}}^{\mathsf{B}} \mathrm{d}t = \int_{s_{\mathsf{A}}}^{s_{\mathsf{B}}} \frac{\mathrm{d}s}{v(s)} = \frac{1}{c}\int_{s_{\mathsf{A}}}^{s_{\mathsf{B}}} \mathsf{n}(s)\,\mathrm{d}s, \tag{7.109}$$

and examine the rays with given spatial dependence for the index of refraction.

7.6.1 Vertically dependent index of refraction

Consider the case of an index of refraction that is a function only of the vertical position, and assume the ray path is in the x-z plane. Additionally, assume that the vertical position along the ray can be uniquely specified by knowing x, so that we can write $z = z(x)$. The ray time is given by

$$T = \frac{1}{c}\int_{x_{\mathsf{A}}}^{x_{\mathsf{B}}} \mathsf{n}(z)\,\sqrt{(\mathrm{d}x)^2 + (\mathrm{d}z)^2} = \frac{1}{c}\int_{x_{\mathsf{A}}}^{x_{\mathsf{B}}} \mathsf{n}(z)\,\sqrt{1+[z'(x)]^2}\,\mathrm{d}x. \tag{7.110}$$

The Lagrangian

$$L = c^{-1}\,\mathsf{n}(z)\,\sqrt{1 + [z'(x)]^2}, \tag{7.111}$$

is of the same form as the geodesics for Euclidean space as studied in Section 7.3, but with the added feature of an index of refraction. Consequently, as we expect, the ray paths are straight lines when the index of refraction is constant, but they are curved when the index is a function of space.

Since $\partial L/\partial x = 0$, we make use of the second form of Euler equation (7.30) to identify an invariant along the ray

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[L - z'\,\frac{\partial L}{\partial z'}\right] = 0, \tag{7.112}$$

which renders

$$\frac{\mathsf{n}(z)}{c\,\sqrt{1 + [z'(x)]^2}} = k, \tag{7.113}$$

where k is a constant. For a given expression of the index of refraction, this equation provides an equation for the ray path, $z(x)$, given a value of k at any point along the path. For a geometrical perspective, we set

$$z'(x) = \frac{\mathrm{d}z}{\mathrm{d}x} = \tan\alpha, \tag{7.114}$$

where α is the angle the ray makes relative to the horizontal. With this angle we bring equation (7.113) to

$$\mathsf{n}\cos\alpha = k \Longleftrightarrow \mathsf{n}\sin\theta = k, \tag{7.115}$$

where θ is the angle the ray makes relative to the vertical. In regions of larger index of refraction, the ray turns more vertical so that θ becomes smaller.

7.6.2 Snell's law for refraction at the boundary of two uniform media

Now specialize to the case depicted in Figure 7.6, in which there are two uniform mediums (such as air and water) with $\mathsf{n} = \mathsf{n}_1$ for $z < 0$ and $\mathsf{n} = \mathsf{n}_2$ for $z > 0$. In this case, equation (7.115) becomes Snell's law from optics

$$\mathsf{n}_1\cos\alpha_1 = \mathsf{n}_2\cos\alpha_2 \Longleftrightarrow \mathsf{n}_1\sin\theta_1 = \mathsf{n}_2\sin\theta_2. \tag{7.116}$$

Again we see that a media with a larger index of refraction causes the ray to bend closer to the vertical direction (see Figure 7.6).

There are many examples of refraction of light rays in everyday life. For example, consider a desert road on a hot day, in which the road in the distance appears to have water. This image arises from refraction of light rays in air with a vertical temperature gradient. Hot air near the surface has a lower refractive index, bending light upward from the sky.

7.6.3 Rays in a rotationally invariant media

Consider rays on the horizontal x-y plane and assume the index of refraction is a function of the radial distance from the origin,

$$\mathsf{n} = \mathsf{n}(r). \tag{7.117}$$

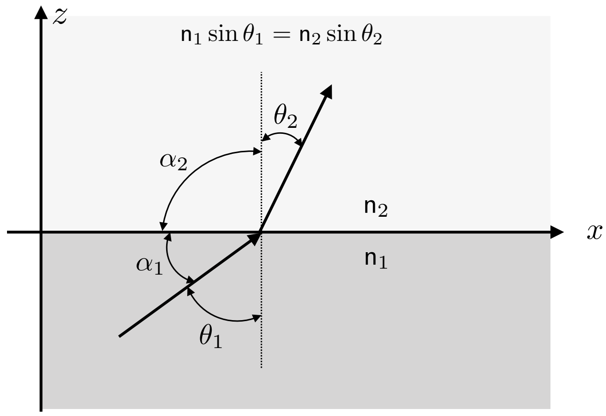

FIGURE 7.6: Illustrating Snell's law, $\mathsf{n}_1 \sin\theta_1 = \mathsf{n}_2 \sin\theta_2$, where $\mathsf{n} > 0$ is the index of refraction for the media, assumed here to be uniform in the lower and upper half-planes and with $\mathsf{n}_1 < \mathsf{n}_2$. A media with larger the index of refraction causes the ray to bend closer to the normal direction, here depicted by the vertical.

Invariants along the ray

Making use of plane polar coordinates (Section 4.3),

$$x = r\cos\vartheta \quad \text{and} \quad y = r\sin\vartheta, \tag{7.118}$$

leads to the time integral (7.109) taking the form

$$T = \frac{1}{c}\int_{s_\mathsf{A}}^{s_\mathsf{B}} \mathsf{n}(r)\,\mathrm{d}s, \tag{7.119}$$

where

$$\mathrm{d}s = \sqrt{(\mathrm{d}r)^2 + r^2\,(\mathrm{d}\vartheta)^2}. \tag{7.120}$$

Now assume that the polar angle is a function of the radial coordinate along the ray, $\vartheta = \vartheta(r)$, in which

$$T = \frac{1}{c}\int_{s_\mathsf{A}}^{s_\mathsf{B}} \mathsf{n}\,[1 + (r\,\vartheta')^2]^{1/2}\,\mathrm{d}r, \tag{7.121}$$

where we introduced the shorthand

$$\vartheta'(r) = \frac{\mathrm{d}\vartheta}{\mathrm{d}r}. \tag{7.122}$$

With the Lagrangian,

$$L = \mathsf{n}\,[1 + (r\,\vartheta')^2]^{1/2}, \tag{7.123}$$

we find the Euler equation (7.25) leads to the invariant

$$\frac{\mathsf{n}\,r^2\,\vartheta'}{\sqrt{1 + (r\,\vartheta')^2}} = \mathsf{n}\,r^2\,\frac{\mathrm{d}\vartheta}{\mathrm{d}s} = \gamma, \tag{7.124}$$

where γ is a constant with dimensions of length, and we used equation (7.120) for the first equality. We can interpret the identity (7.124) as an angular momentum invariant, which is admitted due to rotational symmetry of the index of refraction.

A Snell's law expression

It follows from the arc length increment in equation (7.120) that

$$1 = r^2\,\dot{\vartheta}^2 + \dot{r}^2 \quad \text{with} \quad \dot{\vartheta} = \frac{\mathrm{d}\vartheta}{\mathrm{d}s} \quad \text{and} \quad \dot{r} = \frac{\mathrm{d}r}{\mathrm{d}s}, \tag{7.125}$$

in which case equation (7.124) allows us to identify

$$\dot{r}^2 = 1 - r^2\,\dot{\vartheta}^2 = 1 - \gamma^2/(\mathsf{n}\,r)^2. \tag{7.126}$$

Since $\dot{r}^2 \geq 0$ we know that $r^2\,\dot{\vartheta}^2 \leq 1$, so that we can write

$$\sin^2\psi = r^2\,\dot{\vartheta}^2 = \gamma^2/(\mathsf{n}\,r)^2, \tag{7.127}$$

which allows us to bring the angular momentum invariant (7.124) into the form of Snell's law

$$\mathsf{n}(r)\,r\sin\psi = \gamma. \tag{7.128}$$

We can interpret ψ in equation (7.128) by introducing the position along a ray

$$\boldsymbol{x}(s) = r(s)\,[\hat{\boldsymbol{x}}\cos\vartheta(s) + \hat{\boldsymbol{y}}\sin\vartheta(s)], \tag{7.129}$$

and the corresponding velocity (equivalently the tangent vector)

$$\dot{\boldsymbol{x}} = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s} = \dot{r}\,\hat{\boldsymbol{r}} + r\,\dot{\vartheta}\,\hat{\boldsymbol{\vartheta}}, \tag{7.130}$$

where we introduced the polar coordinate unit directions from Section 4.3.2

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}}\cos\vartheta + \hat{\boldsymbol{y}}\sin\vartheta \quad \text{and} \quad \hat{\boldsymbol{\vartheta}} = -\hat{\boldsymbol{x}}\sin\vartheta + \hat{\boldsymbol{y}}\cos\vartheta. \tag{7.131}$$

The velocity has a unit magnitude by definition of the arc length parameter,

$$\dot{\boldsymbol{x}}\cdot\dot{\boldsymbol{x}} = (\dot{r})^2 + (r\,\dot{\vartheta})^2 = 1, \tag{7.132}$$

so that $\dot{r}$ and $r\,\dot{\vartheta}$ each have magnitudes less than unity. Consequently, we can write

$$\dot{\boldsymbol{x}} = \dot{r}\,\hat{\boldsymbol{r}} + r\,\dot{\vartheta}\,\hat{\boldsymbol{\vartheta}} = \cos\psi\,\hat{\boldsymbol{r}} + \sin\psi\,\hat{\boldsymbol{\vartheta}}, \tag{7.133}$$

where

$$\cos\psi = \dot{r} \quad \text{and} \quad \sin\psi = r\,\dot{\vartheta}. \tag{7.134}$$

We thus see that ψ is the angle between the radial direction and the ray's tangent direction

$$\dot{\boldsymbol{x}}\cdot\hat{\boldsymbol{r}} = \cos\psi, \tag{7.135}$$

as depicted in Figure 7.7.

Refraction around a circle

Consider the special case of an index of refraction

$$\mathsf{n}(r) = \alpha/r, \tag{7.136}$$

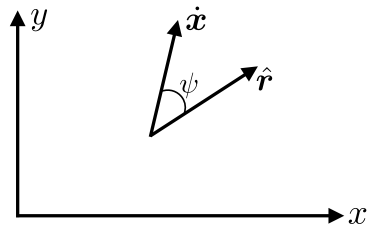

FIGURE 7.7: The direction of a ray, as defined by the unit tangent, $\dot{\boldsymbol{x}}$, which makes an angle, ψ, relative to the radial direction, so the $\dot{\boldsymbol{x}} \cdot \hat{\boldsymbol{r}} = \cos\psi$.

where α is a constant. For this index of refraction, equation (7.126) reveals that

$$\dot{r} = 1 - (\gamma/\alpha)^2, \tag{7.137}$$

which vanishes if the index of refraction matches the invariant

$$\dot{r} = 0 \quad \text{if} \quad \alpha = \gamma. \tag{7.138}$$

In this special case, the ray path is circular with constant radius, $r = R$, and we have

$$\dot{r} = 0 \Longrightarrow r = R \tag{7.139a}$$

$$R\,\dot{\vartheta} = 1 \Longrightarrow s = R\,\vartheta. \tag{7.139b}$$

This motion is analogous to circular motion of a particle in a central force field.

7.7 Exercises

EXERCISE 7.1: STRAIGHT LINE GEODESICS USING POLAR COORDINATES
In Section 7.3 we made use of Cartesian coordinates to derive the straight line geodesics in Euclidean space. Cartesian coordinates are the best choice for unconstrained motion in Euclidean space. But geodesics are geometric objects that have no dependence on coordinates. Here, we consider the two-dimensional case and make use of polar coordinates (Section 4.3)

$$x = r\cos\vartheta \quad \text{and} \quad y = r\sin\vartheta. \tag{7.140}$$

(A) Derive the geodesic equation (7.71) for two-dimensional Euclidean space using (r, ϑ) polar coordinates. Make use of the affine parameter, $\varphi = s$, where s is the arc distance.
(B) Show that straight lines are solutions to the polar coordinate geodesic equations.

EXERCISE 7.2: STABILITY OF THE LINE GEODESIC IN EUCLIDEAN SPACE
In Section 7.2.7 we derived an expression (7.32b) for the second variation. Use that expression to determine whether the straight line geodesic from Section 7.3 is a maximum, minimum, or saddle.

EXERCISE 7.3: GREAT CIRCLE AND ANGULAR MOMENTUM
For the great circle geodesics in Section 7.4.2, show that the normal direction, $\boldsymbol{n} = \hat{\boldsymbol{x}}\cos\lambda_\circ + \hat{\boldsymbol{y}}\sin\lambda_\circ - A^{-1}\,\hat{\boldsymbol{z}}$, is parallel to the angular momentum per mass, $\boldsymbol{l} = \boldsymbol{x} \times \dot{\boldsymbol{x}}$, so that $\boldsymbol{l} \times \boldsymbol{n} = 0$. There is a relatively quick way to prove this result through use of some vector identities for

points on the great circle. Alternatively, we can make use of some trignometric identities along with the equations satisfied by the geodesic, with the alternative approach requiring far more algebraic manipulations.

EXERCISE 7.4: MINIMAL SURFACE OF REVOLUTION

Figure 7.8 shows an axially symmetric surface that is bounded by two circles with distinct radii and that are aligned along the $\hat{\boldsymbol{z}}$ axis. This surface is formed by connecting two points in the x-z plane, (x_1, z_1) and (x_2, z_2), and then revolving the curve around the z-axis. Show that a catenoid is the shape of the surface that results in the minimum surface area, with the cross-section of a cetenoid consisting of the catenaries found in Section 7.5.2.

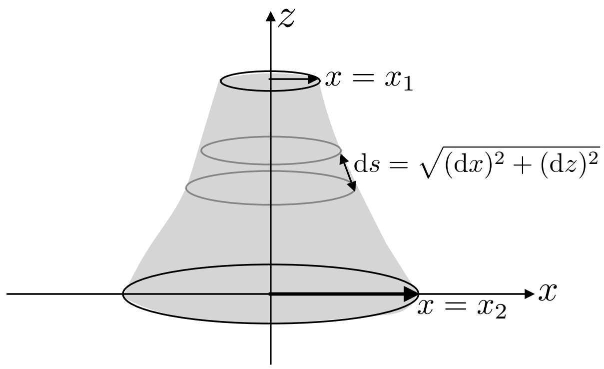

FIGURE 7.8: This figure shows an axially symmetric surface that connects two circles aligned along the $\hat{\boldsymbol{z}}$ axis. The thickness of a strip along the surface and within the x-z plane is given by $\mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}z)^2} = \sqrt{1 + (z')^2}\,\mathrm{d}x$. The shape of the minimal surface that connects the two circles is known as a catenoid. The catenoid is built from connecting the two circles with a catenary and then rotating the catenary around the vertical axis. The potential energy of soap bubbles is proportional to the surface tension times the surface area of the bubble. So the catenoid minimal surface area is a minimal potential energy surface for soap bubbles.

EXERCISE 7.5: BRACHISTOCHRONE TIME

How much time does it take for the bead to fall along the wire for the Brachistochrone?

EXERCISE 7.6: INVERSE SINH FUNCTION

Consider the functional

$$\mathcal{J} = \int_{x_{\mathsf{A}}}^{x_{\mathsf{B}}} x\,[1 - (y')^2]^{1/2}\,\mathrm{d}x, \tag{7.141}$$

with the first derivative written

$$y'(x) = \frac{\mathrm{d}y}{\mathrm{d}x}, \tag{7.142}$$

and

$$(y')^2 \le 1. \tag{7.143}$$

Show that $\mathcal{J}$ is an extrema with a $\sinh^{-1}$ function.

EXERCISE 7.7: ISOPERIMETER PROBLEM

Consider a string of length, ℓ, attached at one end to the origin of the x-y horizonal plane (see Figure 7.9). Attach the other end somewhere on the x axis so that the strong then encloses an area with the x-axis along one side. Show that a semi-circle forms the maximum area enclosed by the string with the x-axis. Hint: assume that we can write $y = y(x)$ along the shape, so that the area is $A = \int y\,\mathrm{d}x$. To build in the constraint that the string has fixed length, ℓ, express $\mathrm{d}x$ and y in terms of the arc-length along the string, and then to make use of the Euler equation, whose solution extremizes the area.

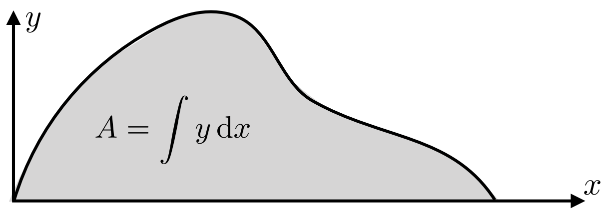

FIGURE 7.9: The isoperimeter problem seeks the shape of a string of fixed length that encloses the maximum area between the string and the x-axis. The semi-circle is the solution to the problem, which is found through variational methods.

EXERCISE 7.8: REFRACTION IN A ROTATIONALLY INVARIANT MEDIA

In Section 7.6.3 we developed properties of refraction within a media that is rotationally invariant relative to a fixed point. That is, we considered rays on the horizontal x-y plane with an ndex of refraction is a function of the radial distance from the origin,

$$\mathsf{n} = \mathsf{n}(r). \tag{7.144}$$

In that discussion we wrote the equations for the angle, $\vartheta = \vartheta(r)$. In this exercise we work through the problem to develop the radial position as a function of the polar angle,

$$r = r(\vartheta). \tag{7.145}$$

(A) Determine a general expression for the invariant property maintained along a ray.
(B) Assume

$$\mathrm{d}r/\mathrm{d}\vartheta = 0 \quad \text{at} \quad r = R, \tag{7.146}$$

in which case we can make use of the invariant to show that the radial position along a ray satisfies

$$\frac{1}{r^2}\left[\frac{\mathrm{d}r}{\mathrm{d}\vartheta}\right]^2 = -1 + \frac{\mathsf{n}^2(r)\, r^2}{\mathsf{n}^2(R)\, R^2}. \tag{7.147}$$

Hint: make use of the second form of the Euler equation of variational calculus (7.30).

EXERCISE 7.9: MINIMUM VALUE FOR AN INTEGRAL

Consider the integral

$$\mathcal{J} = \int_0^1 [(x^2\, f'')^2 + (2\, x\, f')^2]\, \mathrm{d}x, \tag{7.148}$$

where $f = f(x)$, $f' = \mathrm{d}f/\mathrm{d}x$, and $f'' = \mathrm{d}^2 f/\mathrm{d}x^2$.

(A) Determine the function, $f(x)$, that extremizes the integral, assuming that $f(1) = f'(1) = 1$ and $f(0)$ is non-singular.
(B) Making use of results from Section 7.2.7, show that the $f(x)$ found in the previous part minimizes the integral.

Chapter 8

ELLIPTIC EQUATIONS

In this chapter we explore the rudiments of partial differential equations (PDEs), with linear PDEs categorized into elliptic, parabolic, and hyperbolic through the use of method of characteristics. We then focus on elliptic partial differential equations, their boundary value problems, their Green's functions, and their eigenvalue problems. Elliptic operators are encountered in a variety of places in our study, such as Newton's gravity in Chapter 11, pressure in a non-divergent flow in VOLUME 2, and vorticity and potential vorticity in VOLUME 3.

CHAPTER GUIDE

There are many resources devoted to the theory and application of partial differential equations throughout physics, engineering, and applied mathematics. Chapter IV of *Dennery and Krzywicki* (1967) and chapters 8 and 9 of *Hildebrand* (1976) offer a pedagogical starting point, whereas *Morse and Feshbach* (1953), *Greenberg* (1971), and *Stakgold* (2000a,b) thoroughly develop the theory and methods available for elliptic boundary value problems encountered in physics and engineering, including Green's function solutions. *Duchateau and Zachmann* (1986) concisely summarize partial differential equations and offer many worked examples.
This is the first of three chapters focusing the mathematics of linear partial differential equations, with Chapter 9 studying parabolic equations as arise for diffusion, and Chapter 10 studying hyperbolic equations as arise in non-dispersive waves. The present chapter is the longest of the three as it establishes some foundational topics that support our study in the later chapters.

8.1 Introducing characteristic curves

In this section we develop the method of characteristics formalism that determines the functional dependence of solutions to partial differential equations. It does so by determining characteristic curves, which are curves in the space of independent variables, along which a partial differential equation reduces to an ordinary differential equation. In this manner, a solution to the PDE can, in principle, be constructed by following these curves.

We introduce characteristic curves in the context of the following first order PDE,

$$P(x,t,\psi)\,\partial_x\psi + Q(x,t,\psi)\,\partial_t\psi = R(x,t,\psi), \tag{8.1}$$

where P, Q, and R are arbitrary smooth functions, x, t are the independent variables, and ψ is the unknown dependent function. This partial differential equation is linear if P, Q, and R are independent of ψ, and quasi-linear if P and Q are independent of ψ and R is at most a linear function of ψ.

8.1.1 General formulation

In the first order partial differential equation given by equation (8.1), assume there is a functional relation,

$$\Upsilon(x,t,\psi) = \text{constant}, \tag{8.2}$$

that determines the dependent function, ψ, consistent with equation (8.1). We refer to Υ as an integral surface that specifies a solution to the partial differential equation. If Υ specifies an integral surface according to equation (8.2), then it must have a zero differential

$$\mathrm{d}\Upsilon = 0 = \frac{\partial\Upsilon}{\partial x}\,\mathrm{d}x + \frac{\partial\Upsilon}{\partial t}\,\mathrm{d}t + \frac{\partial\Upsilon}{\partial\psi}\,\mathrm{d}\psi, \tag{8.3}$$

which holds since the differential of a constant vanishes. For the two dependent variables, x and t, the condition (8.3) leads to the differential constraints[1]

$$\frac{\mathrm{d}\Upsilon}{\mathrm{d}t} = 0 = \frac{\partial\Upsilon}{\partial\psi}\frac{\partial\psi}{\partial t} + \frac{\partial\Upsilon}{\partial t} \tag{8.4a}$$

$$\frac{\mathrm{d}\Upsilon}{\mathrm{d}x} = 0 = \frac{\partial\Upsilon}{\partial\psi}\frac{\partial\psi}{\partial x} + \frac{\partial\Upsilon}{\partial x}. \tag{8.4b}$$

[1]We are not here considering space-time trajectories in which $x = x(t)$. Rather, we assume that x and t are independent variables. For that reason we set $\mathrm{d}x/\mathrm{d}t = 0$ and $\mathrm{d}t/\mathrm{d}x = 0$ to reach equations (8.4a) and (8.4b).

Assuming $\partial\Upsilon/\partial\psi \neq 0$ allows us to write

$$\frac{\partial \psi}{\partial t} = -\frac{\partial \Upsilon/\partial t}{\partial \Upsilon/\partial \psi} \quad \text{and} \quad \frac{\partial \psi}{\partial x} = -\frac{\partial \Upsilon/\partial x}{\partial \Upsilon/\partial \psi} \tag{8.5}$$

so that the first order partial differential equation (8.1) takes on the equivalent form

$$P\,\frac{\partial \Upsilon}{\partial x} + Q\,\frac{\partial \Upsilon}{\partial t} + R\,\frac{\partial \Upsilon}{\partial \psi} = 0. \tag{8.6}$$

So by assuming there exists an integral surface, Υ, we are able to symmetrize the appearance of the functions P, Q, R appearing in the partial differential equation (8.1).

Consistency of equation (8.6) with

$$\mathrm{d}\Upsilon = \frac{\partial \Upsilon}{\partial x}\,\mathrm{d}x + \frac{\partial \Upsilon}{\partial t}\,\mathrm{d}t + \frac{\partial \Upsilon}{\partial \psi}\,\mathrm{d}\psi = 0, \tag{8.7}$$

requires

$$P = \rho\,\mathrm{d}x \quad \text{and} \quad Q = \rho\,\mathrm{d}t \quad \text{and} \quad R = \rho\,\mathrm{d}\psi, \tag{8.8}$$

for some non-dimensional function, $\rho(x, t, \psi)$. Eliminating the unknown function, ρ, renders the ordinary differential equations

$$\frac{\mathrm{d}x}{P} = \frac{\mathrm{d}t}{Q} = \frac{\mathrm{d}\psi}{R}. \tag{8.9}$$

Paths in (x, t, ψ) space that satisfy these differential equations are known as characteristic curves. Note that if any one of the functions P, Q, or R vanish, then we merely remove that term from these relations. It is notable that the first order differential equations (8.9) have the same appearance as those equations used in VOLUME 2 to find streamlines in a non-divergent flow.

8.1.2 Example characteristic curves

We now determine characteristic curves for some specific examples.

Advection with a constant advection velocity

First consider the linear advection equation with constant advection velocity

$$(\partial_t + U\,\partial_x)\psi = 0, \tag{8.10}$$

in which we identify $P = U$, $Q = 1$, and $R = 0$. The single ordinary differential equation defining the characteristic curve is given by

$$\frac{\mathrm{d}x}{U} = \frac{\mathrm{d}t}{1}, \tag{8.11}$$

so that characteristics are given by the family of space-time lines

$$\mathrm{d}(x - U\,t) = 0 \Longrightarrow x - U\,t = \alpha, \tag{8.12}$$

with α an arbitrary constant. These lines determine the paths in space-time along which advective signals are transmitted.

Advection with a constant advection velocity and a constant source

Now add a constant source to the linear advection equation

$$(\partial_t + U\,\partial_x)\psi = R. \tag{8.13}$$

The two ordinary differential equations defining the characteristic curve are

$$\frac{\mathrm{d}x}{U} = \frac{\mathrm{d}t}{1} = \frac{\mathrm{d}\psi}{R}. \tag{8.14}$$

In addition to the relation $x - U\,t = \alpha_1$ determined from the homogeneous case, we also have $\psi - R\,t = \alpha_2$ for α_2 an arbitrary constant. Hence, the characteristic equations render the general solution of the form

$$\Gamma[x - U\,t, \psi - R\,t] = \text{constant}, \tag{8.15}$$

for Γ an arbitrary function. One example solution is given by

$$\psi = f(x - U\,t) + R\,t, \tag{8.16}$$

for an arbitrary smooth function, f. This solution has the form of a traveling signal, $f(x - U\,t)$, plus a signal growing linearly with time, $R\,t$.

Linear advection with non-constant coefficients and source

For the final example, consider the linear advection equation with non-constant coefficients and non-constant source

$$x\,\frac{\partial\psi}{\partial x} + t\,\frac{\partial\psi}{\partial t} = \psi, \tag{8.17}$$

in which the ordinary differential equations determining the characteristics are given by

$$\frac{\mathrm{d}x}{x} = \frac{\mathrm{d}t}{t} = \frac{\mathrm{d}\psi}{\psi}. \tag{8.18}$$

The first equality leads to the identity

$$\mathrm{d}(\ln x) = \mathrm{d}(\ln t) \Longrightarrow t/x = \alpha_1. \tag{8.19}$$

Similarly, by equating the first and third terms in equation (8.18) we have

$$\psi/x = \alpha_2. \tag{8.20}$$

Hence, the general solution of the partial differential equation (8.17) is given by

$$\Gamma(t/x, \psi/x) = 0 \Rightarrow \psi = x\,F(t/x) \tag{8.21}$$

for an arbitrary smooth function F.

8.1.3 Arbitrary functions resulting from PDEs

The solution to a partial differential equation is typically given in terms of an arbitrary function with a specified dependence on the independent variables. The function itself is unspecified without additional information from initial and/or boundary conditions. The arbitrary functional degree of freedom is a generalization of the case with ordinary differential equations, whose solutions are determined up to constants that are specified by initial and/or boundary conditions.

8.2 Classifying second order differential equations

There are many second order partial differential equations appearing in fluid mechanics, a general form of which is given by

$$A\,\frac{\partial^2\psi}{\partial x^2} + B\,\frac{\partial^2\psi}{\partial x\,\partial t} + C\,\frac{\partial^2\psi}{\partial t^2} = \Lambda. \tag{8.22}$$

For linear partial differential equations, A, B, C are arbitrary functions of space and time, but they are each independent of ψ. Furthermore, Λ is a function of space and time and at most a linear function of ψ and its derivatives. The most general solution to a linear partial differential equation consists of the sum of any *particular solution* plus a solution to the homogeneous problem (where $\Lambda = 0$).

The terms involving second derivatives in equation (8.22) determine the character of the solutions, with importance placed on the sign of the discriminant $B^2 - 4\,A\,C$. By analogy with conic sections we classify second order partial differential equations as follows:

$$\text{PDE form} = \begin{cases} \text{hyperbolic} & B^2 - 4\,A\,C > 0 \\ \text{elliptic} & B^2 - 4\,A\,C < 0 \\ \text{parabolic} & B^2 - 4\,A\,C = 0. \end{cases} \tag{8.23}$$

We can further motivate this terminology by considering the case of a homogeneous constant coefficient partial differential equation and an assumed solution of the form

$$\psi(x, t) = f(m\,x + t). \tag{8.24}$$

Plugging into the second order partial differential equation (8.22) with $\Lambda = 0$ leads to

$$A\,m^2 + B\,m + C = 0. \tag{8.25}$$

The two solutions, m_1 and m_2, are both real for the hyperbolic case, complex conjugate for the elliptic case, and a perfect square for the parabolic case. In the remainder of this chapter we focus on the elliptic case, leaving study of parabolic partial differential equations for Chapter 9 and hyperbolic partial differential equations for Chapter 10.

8.3 Properties of elliptic equations

The elliptic operator from Section 8.2 has discriminant $B^2 - 4\,A\,C < 0$, in which case there are two imaginary characteristics. The canonical example elliptic equation is Laplace's equation,

$$(\partial_{xx} + \partial_{yy})\,\psi = 0, \tag{8.26}$$

where we converted from the time coordinate, t, to the space coordinate y, and assumed Cartesian coordinates. Formally, this transition is realized by setting $t = \mathrm{i}\,y$, where $\mathrm{i} = \sqrt{-1}$.

Poisson's equation is another common elliptic equation, which results from adding a source to Laplace's equation

$$(\partial_{xx} + \partial_{yy})\,\psi = \Lambda. \tag{8.27}$$

As a frequent shorthand, we define the Laplacian operator (for three space-dimensions)

$$\nabla^2 = \partial_{xx} + \partial_{yy} + \partial_{zz}. \tag{8.28}$$

This expression generalizes to arbitrary coordinates, with the following spherical and cylindrical examples taken from Chapter 4:

$$\nabla^2\psi = \frac{1}{r}\frac{\partial}{\partial r}\left[r\,\frac{\partial\psi}{\partial r}\right] + \frac{1}{r^2}\frac{\partial^2\psi}{\partial\vartheta^2} + \frac{\partial^2\psi}{\partial z^2} \qquad \text{cylindrical} \tag{8.29}$$

$$\nabla^2\psi = \frac{1}{(r\cos\phi)^2}\frac{\partial^2\psi}{\partial\lambda^2} + \frac{1}{r^2\cos\phi}\frac{\partial}{\partial\phi}\left[\cos\phi\,\frac{\partial\psi}{\partial\phi}\right] + \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\,\frac{\partial\psi}{\partial r}\right] \qquad \text{spherical.} \tag{8.30}$$

8.3.1 Absence of time in elliptic equations

There is no time variable in elliptic partial differential equations. Hence, information is transmitted instantaneously so that the structure of the solution is determined by boundary conditions. Instantaneous information transfer is not physical since all physical signals have a finite propagation speed no greater than the speed of light. Even so, an infinite speed can be a useful mathematical construct motivated by the physics. For example, acoustic wave signals (VOLUME 4) propagate in liquids much faster than other waves and fluid particle speeds. So for many purposes of large-scale planetary fluid mechanics, it is suitable to assume acoustic speeds are infinite, and in so doing we filter out acoustic waves from of the dynamical equations.[2] In the process, the hyperbolic equation describing non-dispersive acoustic waves is converted into an elliptic equation.

8.3.2 Harmonic functions

Solutions to Laplace's equation, $\nabla^2\psi = 0$, are known as harmonic functions. The name "harmonic" originates from the study of fundamental modes of oscillation, such as arise from musical instruments or more generally the motion of linear oscillators. Such fundamental modes are solutions to the Laplace equation with boundary conditions depending on the instrument. On domains with certain symmetries, the solution to Laplace's equation can be written in terms of the sine and cosine trignometric functions. Notably, these functions are solutions to the one-dimensional Helmholtz equation encountered in our study of waves in VOLUME 4. As a result, the sine and cosine functions, though not satisfying Laplace's equation, provide building blocks to the harmonic functions that do satisfy Laplace's equation.

Here are some example harmonic functions for two space dimensions:

$$\psi = x^3 - 3\,x\,y^2 \qquad \psi = \ln(x^2 + y^2) \qquad \psi = e^{\gamma\,x}\cos(\gamma\,y) \qquad \psi = a\,x + b\,y, \tag{8.31}$$

[2] A scuba diver feeling the beat of a ship underwater, or an audience member at a rock concert may question why we wish to filter out acoustic waves from the dynamics. Even so, in VOLUME 4 in our study of acoustic waves, we see that acoustic energy is in fact tiny relative to planetary waves and gravity waves, and they are negligible for studies of large scale geophysical fluid motions.

with arbitrary constants a, b, γ. Furthermore, with

$$\nabla^2(\psi\,\phi) = \psi\,\nabla^2\phi + 2\,\nabla\phi\cdot\nabla\psi + \phi\,\nabla^2\psi, \tag{8.32}$$

we see that the product of two harmonic functions ($\nabla^2\psi = \nabla^2\phi = 0$) is itself harmonic if and only if their gradients are orthogonal, $\nabla\psi\cdot\nabla\phi = 0$. In the remainder of this section we present some properties of harmonic functions and develop self-consistency for the boundary conditions.

8.3.3 Mean value property of harmonic functions

The mean value property says that the value of a harmonic function, ψ, at a point $\boldsymbol{x}_\circ$ within an open region of $\mathcal{R}$ equals to the average of ψ taken over the surface of a sphere within $\mathcal{R}$ that is centered at $\boldsymbol{x}_\circ$. In equations this property states that[3]

$$\psi(\boldsymbol{x}_\circ) = \frac{\oint_{\mathcal{S}_R}\psi(\boldsymbol{x})\,\mathrm{d}\mathcal{S}}{\oint_{\mathcal{S}_R}\mathrm{d}\mathcal{S}}, \tag{8.33}$$

where $\mathcal{S}_R$ is a sphere with radius R centered at $\boldsymbol{x}_\circ$ with "area" given by

$$\oint_{\mathcal{S}_R}\mathrm{d}\mathcal{S} = \begin{cases} 2\pi R & n = 2 \text{ space dimensions} \\ 4\pi R^2 & n = 3 \text{ space dimensions.} \end{cases} \tag{8.34}$$

We illustrate this property in Figure 8.1.

The mean-value property of harmonic functions holds anywhere within the domain where $\nabla^2\psi = 0$, so long as the sphere is fully contained within that domain. Furthermore, it holds for a sphere of any radius with the point at the center. This property implies that there can be no extrema of ψ within the domain, for if there was an extrema then we could no longer guarantee that the average value around all surrounding spheres equals to the value at the extrema. Hence, extrema of harmonic functions must live on the domain boundary. This property lends mathematical support for considering harmonic functions to be solutions to continuous physical systems that are in equilibrium or a steady state. As a physical example, consider a temperature field, $T(\boldsymbol{x})$, in a region with zero heat sources and zero fluid flow. The steady state temperature satisfies $\nabla^2 T = 0$, and as such it is harmonic and hence has no extrema within the domain.

8.3.4 Laplace's boundary value problem

Laplace's equation requires boundary conditions to fully specify the harmonic function. We here consider Laplace's boundary value problem in the form

$$\nabla^2\psi = 0 \qquad \boldsymbol{x}\in\mathcal{R} \tag{8.35a}$$

$$\alpha\,\psi + \beta\,\hat{\boldsymbol{n}}\cdot\nabla\psi = \sigma \qquad \boldsymbol{x}\in\partial\mathcal{R}, \tag{8.35b}$$

where $\mathcal{R}$ is a smooth and simply connected open region[4] (also called the domain), $\partial\mathcal{R}$ is the boundary of $\mathcal{R}$ with outward unit normal, $\hat{\boldsymbol{n}}$, and with α, β, and σ prescribed boundary conditions. This setup of Laplace's boundary value problem holds for other operators. Namely, the differential equation is satisfied on the open domain, $\mathcal{R}$, whereas the specified boundary

[3] We prove the mean value property in Section 8.9.10 using methods from Green's function theory.

[4] An open region is a set or domain that includes none of its boundary points.

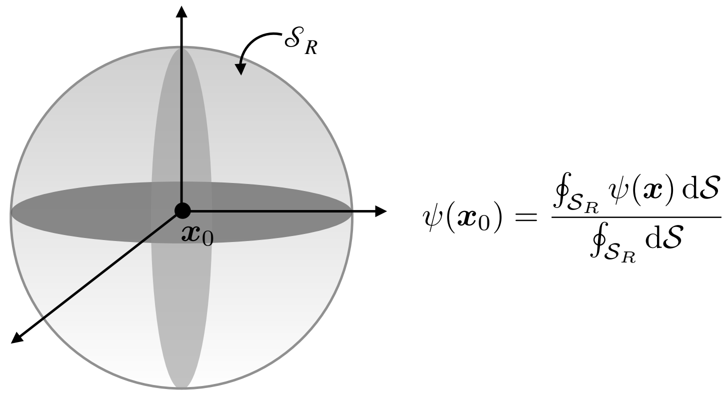

FIGURE 8.1: The value of a harmonic function at a point $\boldsymbol{x}_\circ$ equals to the area average of the function over a sphere of arbitrary radius (so long as $\nabla^2\psi = 0$ inside the sphere) centered at $\boldsymbol{x}_\circ$. We here illustrate this property for 3-dimensions, with this property holding for arbitrary space dimensions, $n \geq 1$. The mean value property for harmonic functions implies that there can be no extrema of ψ within the domain, for if there was an extrema then we could no longer guarantee that the average value around all surrounding spheres equals to the value at the extrema. Hence, extrema of harmonic functions live on the domain boundary.

conditions are set on the domain boundary, $\partial\mathcal{R}$. In this manner, it is the job of the differential operator to establish the structure of the function within $\mathcal{R}$, given boundary data on $\partial\mathcal{R}$.

We can establish constraints on the boundary conditions that lead to a self-consistent boundary value problem (8.35a)-(8.35b). We do so through the use of Gauss's divergence theorem (Section 2.8) in which integration over the full domain leads to

$$0 = \int_{\mathcal{R}} \nabla^2\psi \, \mathrm{d}V = \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla\psi \, \mathrm{d}\mathcal{S}. \tag{8.36}$$

In physical applications the boundary condition (8.35b) usually appears with either $\alpha = 0$ or $\beta = 0$, and these two cases are associated with distinct self-consistency constraints.

Dirichlet boundary condition

The case with $\beta = 0$ is referred to as a Dirichlet boundary condition, so that

$$\psi = \sigma \qquad \boldsymbol{x} \in \partial\mathcal{R}, \tag{8.37}$$

where we set $\alpha = 1$ without loss of generality. In this case all boundary data result in a self-consistent Laplacian boundary value problem so there is no constraint on σ. Thinking again about temperature, this boundary condition specifies the temperature on the domain boundary to equal $T = \sigma$, with all boundary functions σ consistent with a harmonic temperature distribution within the domain interior.

Neumann boundary condition

The case with $\alpha = 0$ results in a Neumann boundary condition. Without loss of generality we set $\beta = 1$ and reach the following self-consistency condition

$$\oint_{\partial\mathcal{R}} \sigma \, \mathrm{d}\mathcal{S} = 0. \tag{8.38}$$

A self-consistent boundary condition for Laplace's equation with the Neumann boundary condition must satisfy this surface integral constraint. For the temperature example, this boundary condition says that to realize a steady state harmonic temperature distribution within a region, we can at most apply a zero area averaged boundary heating. If the boundary constraint (8.38) is not satisfied, then the interior temperature field cannot be harmonic so that it will not be in a steady state.

8.3.5 Poisson's equation

The generic boundary value problem for Poisson's equation takes the form

$$\nabla^2\psi = \Lambda \qquad \boldsymbol{x} \in \mathcal{R} \tag{8.39a}$$

$$\alpha\,\psi + \beta\,\hat{\boldsymbol{n}}\cdot\nabla\psi = \sigma \qquad \boldsymbol{x} \in \partial\mathcal{R}, \tag{8.39b}$$

where $\Lambda(\boldsymbol{x})$ is a specified source function. We here present, without proof, some general properties of solutions to Poisson's equation and develop self-consistency conditions for the boundary conditions appearing in equation (8.39b).

8.3.6 Extended max-min principle for Poisson's equation

A subharmonic function, ψ, satisfies the Poisson equation with a non-negative source

$$\nabla^2\psi = \Lambda \geq 0 \qquad \boldsymbol{x} \in \mathcal{R}. \tag{8.40}$$

Here, the source function makes the curvature of a subharmonic function positive. Correspondingly, every point within $\mathcal{R}$ satisfies the minimum principle

$$\psi(\boldsymbol{x}_\circ) \leq \frac{\oint_{\mathcal{S}_R}\psi(\boldsymbol{x})\,\mathrm{d}\mathcal{S}}{\oint_{\mathcal{S}_R}\mathrm{d}\mathcal{S}}, \tag{8.41}$$

for spheres, $\mathcal{S}_R$, that are fully within $\mathcal{R}$ and where $\nabla^2\psi = \Lambda$. The signs switch for a super harmonic function whereby $\nabla^2\psi \leq 0$ for $\boldsymbol{x} \in \mathcal{R}$

Returning to the temperature example, consider a temperature field in a region with a positive heat source, $\Lambda > 0$. The steady state temperature with constant diffusivity in the presence of zero fluid flow satisfies Poisson's equation, $\nabla^2 T = \Lambda \geq 0$, for regions with the heat source. The minimum principle (8.41) means that the temperature at any point within the heating region is less than the temperature averaged over a sphere centered on the point, so long as the sphere remains within the region of heating. It is only in the absence of a heat source or sink, where $\nabla^2 T = 0$, that we recover the mean-value property of harmonic functions given by equation (8.33).

8.3.7 Poisson's boundary value problem

We follow the method in Section 8.3.4 to develop constraints on the boundary conditions applied as part of the Poisson boundary value problem (8.39a)-(8.39b). Use of Gauss's divergence theorem leads to the constraint on the area integrated boundary flux and the volume integrated source

$$\oint_{\partial\mathcal{R}}\hat{\boldsymbol{n}}\cdot\nabla\psi\,\mathrm{d}\mathcal{S} = \int_{\mathcal{R}}\Lambda\,\mathrm{d}V. \tag{8.42}$$

We return in Section 8.9.1 for a discussion of this constraint. Next, we separately consider Dirichlet and Neumann boundary conditions.

Dirichlet boundary condition

The Dirichlet condition with $\beta = 0$ leads to

$$\psi = \sigma \quad \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.43}$$

Just as for Laplace's boundary value problem, all boundary values result in a self-consistent Poisson boundary value problem, so there is no constraint on σ. Thinking again about temperature, this boundary condition specifies the temperature on the domain boundary to equal $T = \sigma$, with all boundary functions, σ, consistent with the interior heating, Λ, and a temperature field satisfying $\nabla^2 T = \Lambda$ within the interior.

Self-consistent Neumann boundary condition

The Neumann boundary condition leads to the following self-consistency condition

$$\oint_{\partial\mathcal{R}} \sigma \, \mathrm{d}\mathcal{S} = \int_{\mathcal{R}} \Lambda \, \mathrm{d}V. \tag{8.44}$$

A self-consistent boundary condition for Poisson's equation with a Neumann boundary condition must satisfy this surface integral constraint. For the temperature example, this boundary condition says that the area integrated boundary data must be consistent with the volume integrated source function in order for the temperature to satisfy Poisson's equation. Otherwise, the temperature field will evolve in time and thus not be in a steady state. Compatibility of the boundary conditions ensures the existence of a solution to the Poisson equation that is unique up to an arbitrary constant.

8.4 Laplace's eigenvalue problem

In this section we consider Laplace's eigenvalue problem

$$\nabla^2 \Psi = \lambda \, \Psi \quad \text{for} \quad \boldsymbol{x} \in \mathcal{R} \tag{8.45a}$$

$$\Psi = 0 \quad \text{for} \quad \boldsymbol{x} \in \partial\mathcal{R} \quad \text{or} \quad \hat{\boldsymbol{n}} \cdot \nabla\Psi = 0 \quad \text{for} \quad \boldsymbol{x} \in \partial\mathcal{R}, \tag{8.45b}$$

where λ is a number, the eigenvalue, and Ψ is the corresponding eigenfunction. Details of the eigenvalues and eigenfunctions depend on the boundary conditions and geometry of the region. We focus on boundary conditions that ensure Laplace's eigenvalue problem is self-adjoint, which results in eigenfunctions that form a complete orthonormal basis. In working through the example in this section, we make use of the separation of variables method that reduces a partial differential equation to a series of ordinary differential equations.

8.4.1 Dirichlet eigenfunctions on a rectangle

Consider Laplace's eigenvalue problem on a domain with homogeneous Dirichlet boundary conditions

$$\nabla^2 \Psi = \lambda \, \Psi \quad \boldsymbol{x} \in \mathcal{R} \tag{8.46a}$$

$$\Psi = 0 \quad \boldsymbol{x} \in \partial \mathcal{R}, \tag{8.46b}$$

where $\mathcal{R}$ is an open and finite domain. Assuming $\mathcal{R}$ is a rectangular domain suggests we make use of Cartesian coordinates

$$(\partial_{xx} + \partial_{yy}) \Psi = \lambda \, \Psi \qquad 0 < x < L_x \qquad 0 < y < L_y \tag{8.47a}$$

$$\Psi = 0 \qquad x = 0, x = L_x \qquad y = 0, y = L_y. \tag{8.47b}$$

The separation of variables method proceeds by seeking solutions of the form

$$\Psi(x, y) = A(x) \, B(y), \tag{8.48}$$

which, when substituted into equation (8.47a), lead to

$$\frac{1}{A} \frac{\mathrm{d}^2 A}{\mathrm{d}x^2} + \frac{1}{B} \frac{\mathrm{d}^2 B}{\mathrm{d}y^2} = \lambda. \tag{8.49}$$

Since $A(x)$ and $B(y)$ have distinct independent variables, this equation has solutions only if

$$(\mathrm{d}^2/\mathrm{d}x^2 + \lambda_{\mathsf{A}}) \, A = 0, \quad 0 < x < L_x \quad \text{and} \quad A = 0 \quad \text{for} \quad x = 0, x = L_x \tag{8.50a}$$

$$(\mathrm{d}^2/\mathrm{d}y^2 + \lambda_{\mathsf{B}}) \, B = 0, \quad 0 < y < L_y \quad \text{and} \quad B = 0 \quad \text{for} \quad y = 0, y = L_y, \tag{8.50b}$$

where λA and λ_{B} are two separation constants and

$$\lambda = \lambda_{\mathsf{A}} + \lambda_{\mathsf{B}}. \tag{8.51}$$

The solutions are readily seen to be sine functions

$$A_m = \sin(m \, \pi \, x / L_x) \qquad \lambda_{\mathsf{A}} = -(m \, \pi / L_x)^2 \tag{8.52a}$$

$$B_n = \sin(n \, \pi \, y / L_y) \qquad \lambda_{\mathsf{B}} = -(n \, \pi / L_y)^2, \tag{8.52b}$$

where m, n are positive integers. We are thus led to the normalized eigenfunctions and corresponding eigenvalues

$$\Psi_{mn} = \frac{2}{\sqrt{L_x \, L_y}} \sin(m \, \pi \, x / L_x) \sin(n \, \pi \, y / L_y) \tag{8.53a}$$

$$\lambda_{mn} = -(m \, \pi / L_x)^2 - (n \, \pi / L_y)^2, \tag{8.53b}$$

with the eigenfunctions forming an orthonormal set of functions over the rectangular domain

$$\int_0^{L_x} \int_0^{L_y} \Psi_{mn}(x, y) \, \Psi_{m'n'}(x, y) \, \mathrm{d}x \, \mathrm{d}y = \delta_{mm'} \, \delta_{nn'}. \tag{8.54}$$

We make use of these eigenfunctions in Exercise 8.9 as a means to express the Green's function for the Poisson boundary value problem with Dirichlet boundary conditions.

8.4.2 Self-adjoint linear operators

We consider a linear operator to be a linear differential operator plus linear boundary conditions. Determining whether a linear operator is self-adjoint requires us to define an inner product, the differential operator, and the boundary conditions. Since the Laplacian differential operator is separable on the rectangular domain from Section 8.4.1, we focus on a single space dimension, in which the relevant inner product is given by integration over the domain

$$\langle f, g \rangle \equiv \int_0^L f(x)\, g(x)\, \mathrm{d}x. \tag{8.55}$$

The one-dimensional Laplacian differential operator,

$$\mathcal{L} = \mathrm{d}^2/\mathrm{d}x^2, \tag{8.56}$$

along with either homogeneous Dirichlet or homogeneous Neumann boundaries, forms a linear operator. We establish self-adjointness of the linear operator by considering two arbitrary differentiable functions, f and g, in which case

$$\int_0^L f\, \frac{\mathrm{d}^2 g}{\mathrm{d}x^2}\, \mathrm{d}x = \int_0^L \frac{\mathrm{d}}{\mathrm{d}x}\left[f\, \frac{\mathrm{d}g}{\mathrm{d}x} - g\, \frac{\mathrm{d}f}{\mathrm{d}x}\right] \mathrm{d}x + \int_0^L \frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\, g\, \mathrm{d}x. \tag{8.57}$$

The boundary terms drop out if f and g satisfy either homogeneous Dirichlet or homogeneous Neumann boundary conditions. The ability to move the differential operator from g to f, without picking up any extra boundary terms, defines the self-adjoint property of the linear operator.[5] Finally, we can make use of the more concise notation introduced in equation (8.55) to write

$$\langle f, \mathcal{L} g \rangle = \int_0^L f(x)\, [\mathcal{L} g(x)]\, \mathrm{d}x = \int_0^L [\mathcal{L} f(x)]\, g(x)\, \mathrm{d}x = \langle \mathcal{L} f, g \rangle. \tag{8.58}$$

We state, without proof, a key theorem for self-adjoint operators.[6] Namely, their eigenfunctions form a complete basis for functions that satisfy the same boundary conditions as the eigenfunctions. In the present case, the Dirichlet boundaries lead to the Fourier sine series for functions that vanish at the domain boundaries

$$f(x) = \sum_{n=1}^{\infty} f_n \sin(n\, \pi\, x/L) \quad \text{with} \quad f_n = \frac{2}{L} \int_0^L f(x) \sin(n\, \pi\, x/L)\, \mathrm{d}x. \tag{8.59}$$

Similarly, we are led to a Fourier cosine series for functions satisfying the homogeneous Neumann boundaries

$$g(x) = \overline{g} + \sum_{n=1}^{\infty} g_n \cos(n\, \pi\, x/L) \tag{8.60a}$$

$$\overline{g} = \frac{1}{L} \int_0^L g(x)\, \mathrm{d}x \tag{8.60b}$$

$$g_n = \frac{2}{L} \int_0^L g(x) \cos(n\, \pi\, x/L)\, \mathrm{d}x. \tag{8.60c}$$

[5] We use the identical manipulations in Section 8.9.6 when establishing the reciprocity condition for Green's functions. These manipulations are particular realizations of Green's first and second identities derived in Section 2.8.5.

[6] See, for example, *Dennery and Krzywicki* (1967), *Hildebrand* (1976), *Boyce and DiPrima* (2009).

Note the presence of the average value, $\overline{g}$, which results from the nontrivial null space for the Laplacian operator with Neumann boundary conditions. We encounter consequences of this null space when studying solutions to the Neumann boundary value problem in Sections 8.10, 8.12, and 8.13. Finally, for functions that are periodic over the domain, $h(x) = h(x+L)$, we make use of both the cosine and sine eigenfunctions, along with the average, so that we have the full Fourier series expansion

$$h(x) = \overline{h} + \sum_{n=1}^{\infty} [a_n \sin(n\,\pi\,x/L) + b_n \sin(n\,\pi\,x/L)] \tag{8.61a}$$

$$a_n = \frac{2}{L} \int_0^L h(x) \sin(n\,\pi\,x/L)\,\mathrm{d}x \tag{8.61b}$$

$$b_n = \frac{2}{L} \int_0^L h(x) \cos(n\,\pi\,x/L)\,\mathrm{d}x. \tag{8.61c}$$

8.4.3 Comments

As noted in Section IV.9 of *Dennery and Krzywicki* (1967), if there is a nonzero eigenfunction corresponding to the zero eigenvalue, then there is a nontrivial null space and the absence of a Green's function. For Laplace's eigenvalue problem with Neumann boundaries, zero is an eigenvalue corresponding to a constant eigenfunction, and this explains the mean value present in the cosine Fourier series (8.60a). We thus find no Green's function for the Poisson equation with Neumann boundaries, motivating us to develop the modified Green's function in Section 8.10. Conversely, if there is only the zero eigenfunction corresponding to the zero eigenvalue (i.e., trivial null space), then we can construct a Green's function, as exemplified by Poisson's equation with Dirichlet boundaries studied in Section 8.9. These ideas are further studied in Sections 8.12 and 8.13, where we derive eigenfunction expansions for Green's functions.

8.5 Harmonic functions on the sphere

We now consider Laplace's equation on sphere, with the spherical coordinate Laplacian operator given by equation (4.130) so that we seek solutions to

$$\frac{1}{(r\cos\phi)^2}\frac{\partial^2\Psi}{\partial\lambda^2} + \frac{1}{r^2\cos\phi}\frac{\partial}{\partial\phi}\left[\cos\phi\,\frac{\partial\Psi}{\partial\phi}\right] + \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\,\frac{\partial\Psi}{\partial r}\right] = 0. \tag{8.62}$$

Solutions to this equation are harmonic functions. Furthermore, the spherical portion leads to an eigenvalue problem whose solutions are known as spherical harmonics.

There are many conventions used to express spherical harmonics. We follow those from the physics community as detailed in Chapter 3 of *Jackson* (1975). For that purpose we find it useful to rewrite the spherical Laplace equation (8.62) in an equivalent form to accord with *Jackson* (1975). First, we rewrite the radial derivative term in the equivalent form

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\,\frac{\partial\Psi}{\partial r}\right] = \frac{1}{r}\,\frac{\partial^2(r\,\Psi)}{\partial r^2}. \tag{8.63}$$

Second, we make use of co-latitude, φ, rather than latitude, ϕ

$$\varphi = \pi/2 - \phi \Longrightarrow \cos\phi = \sin\varphi. \tag{8.64}$$

With these changes, our goal in this section is to outline the derivation of solutions to Laplace's equation written in the form

$$\frac{1}{(r\sin\varphi)^2}\frac{\partial^2\Psi}{\partial\lambda^2}+\frac{1}{r^2\sin\varphi}\frac{\partial}{\partial\varphi}\left[\sin\varphi\,\frac{\partial\Psi}{\partial\varphi}\right]+\frac{1}{r}\frac{\partial^2(r\,\Psi)}{\partial r^2}=0. \tag{8.65}$$

8.5.1 Separation of variables solution

Consider a separation of variables ansatz of the form

$$\Psi(\lambda,\varphi,r)=r^{-1}\,U(r)\,P(\varphi)\,Q(\lambda), \tag{8.66}$$

which brings Laplace's equation (8.65) to

$$\frac{U\,P}{(r\sin\varphi)^2}\frac{\mathrm{d}^2Q}{\mathrm{d}\lambda^2}+\frac{U\,Q}{r^2\sin\varphi}\frac{\mathrm{d}}{\mathrm{d}\varphi}\left[\sin\varphi\,\frac{\mathrm{d}P}{\mathrm{d}\varphi}\right]+P\,Q\,\frac{\mathrm{d}^2U}{\mathrm{d}r^2}=0, \tag{8.67}$$

and with further multiplication by $(r\sin\varphi)^2/(U\,P\,Q)$ yielding

$$\underbrace{\frac{1}{Q}\frac{\mathrm{d}^2Q}{\mathrm{d}\lambda^2}}_{\text{function of }\lambda}+\underbrace{\frac{\sin\varphi}{P}\frac{\mathrm{d}}{\mathrm{d}\varphi}\left[\sin\varphi\,\frac{\mathrm{d}P}{\mathrm{d}\varphi}\right]+\frac{(r\sin\varphi)^2}{U}\frac{\mathrm{d}^2U}{\mathrm{d}r^2}}_{\text{function of }(\varphi,r)}=0. \tag{8.68}$$

Longitudinal periodicity

The longitudinal dependence is isolated to the first term of equation (8.68), whereas the remaining terms are functions of radial position and co-latitude. To be self-consistent, we require the longitudinal term to be equal to a constant,

$$\frac{1}{Q}\frac{\mathrm{d}^2Q}{\mathrm{d}\lambda^2}=-m^2\Longrightarrow Q_m(\lambda)=\mathrm{e}^{\pm\mathrm{i}\,m\,\lambda}, \tag{8.69}$$

with m an integer in order to respect periodicity when moving around the sphere at a fixed latitude. The $Q_m(\lambda)$ functions form a complete set of periodic functions on the domain $0\le\lambda\le2\pi$.

Power law radial behavior

Substituting equation (8.69) into Laplace's equation (8.68), and dividing by $\sin^2\varphi$, leads to

$$\underbrace{-\frac{m^2}{\sin^2\varphi}+\frac{1}{P\sin\varphi}\frac{\mathrm{d}}{\mathrm{d}\varphi}\left[\sin\varphi\,\frac{\mathrm{d}P}{\mathrm{d}\varphi}\right]}_{\text{function }\varphi}+\underbrace{\frac{r^2}{U}\frac{\mathrm{d}^2U}{\mathrm{d}r^2}}_{\text{function }r}=0. \tag{8.70}$$

We introduce another real separation constant, written in the form $l\,(l+1)$, in which case

$$-\frac{m^2}{\sin^2\varphi}+\frac{1}{P\sin\varphi}\frac{\mathrm{d}}{\mathrm{d}\varphi}\left[\sin\varphi\,\frac{\mathrm{d}P}{\mathrm{d}\varphi}\right]=-l\,(l+1) \tag{8.71a}$$

$$\frac{r^2}{U}\frac{\mathrm{d}^2U}{\mathrm{d}r^2}=l\,(l+1). \tag{8.71b}$$

The radial equation has the power law solution

$$U_l(r) = A\, r^{l+1} + B\, r^{-l}, \tag{8.72}$$

where A and B are constants.

Associated Legendre functions over the co-latitude domain

A slight reorganization of equation (8.71a) leads to

$$\frac{1}{\sin\varphi}\frac{\mathrm{d}}{\mathrm{d}\varphi}\left[\sin\varphi\,\frac{\mathrm{d}P}{\mathrm{d}\varphi}\right] + \left[l\,(l+1) - \frac{m^2}{\sin^2\varphi}\right] P = 0. \tag{8.73}$$

Introducing the coordinate change

$$w = \cos\varphi \in [-1,1] \Longrightarrow \mathrm{d}w = -\sin\varphi\,\mathrm{d}\varphi, \tag{8.74}$$

brings equation (8.73) to the generalized Legendre equation

$$\frac{\mathrm{d}}{\mathrm{d}w}\left[(1-w^2)\,\frac{\mathrm{d}P}{\mathrm{d}w}\right] + \left[l\,(l+1) - \frac{m^2}{1-w^2}\right] P = 0. \tag{8.75}$$

The longitudinally symmetric case occurs with $m = 0$, which yields the ordinary Legendre equation whose solutions are Legendre polynomials, P_l. With longitudinal variations, in order for the solutions to be finite for $w = \pm 1$ (the north and south pole), requires l to be a non-negative integer, and m is an integer bounded by $\pm l$, so that

$$l = 0, 1, 2, \ldots \quad \text{and} \quad m = -l, -(l-1), \ldots, 0, \ldots, (l-1), l. \tag{8.76}$$

The corresponding solutions to equation (8.75) are known as associated Legendre functions, $P_l^m(w)$. These functions are related to the ordinary Legendre polynomials via

$$P_l^m(w) = (-1)^m\,(1-w^2)^{m/2}\,\frac{\mathrm{d}^m P_l(w)}{\mathrm{d}^m w}, \tag{8.77}$$

and they satisfy the following orthogonality relation[7]

$$\int_{-1}^{1} P_l^m(w)\,P_{l'}^m(w)\,\mathrm{d}w = \frac{2}{2\,l+1}\,\frac{(l+m)!}{(l-m)!}\,\delta_{l,l'} \tag{8.78}$$

8.5.2 Spherical harmonics

It is quite useful to combine the latitudinal and longitudinal functions to build a complete set of orthonormal basis functions over the unit sphere. These spherical harmonics are written

$$Y_{l,m}(\varphi,\lambda) = \sqrt{\frac{(2l+1)\,(l-m)!}{4\,\pi\,(l+m)!}}\,P_l^m(\cos\varphi)\,\mathrm{e}^{\mathrm{i}\,m\,\lambda}, \tag{8.79}$$

[7] Recall that $0! = 1$.

and they are eigenfunctions for the spherical portion of the Laplacian operator that satisfy the eigenvalue problem on a sphere with constant radius[8]

$$r^2\,\nabla^2 Y_{l.m} = \frac{1}{\sin^2\varphi}\,\frac{\partial^2 Y_{l,m}}{\partial\lambda^2} + \frac{1}{\sin\varphi}\,\frac{\partial}{\partial\varphi}\left[\sin\varphi\,\frac{\partial Y_{l,m}}{\partial\varphi}\right] = -l\,(l+1)\,Y_{l,m}. \tag{8.80}$$

Spherical harmonics satisfy the identity

$$Y_{l,-m}(\varphi,\lambda) = (-1)^m\,Y^*_{l,m}(\varphi,\lambda), \tag{8.81}$$

which stems from the longitudinal exponential eigenfunction as well as the identities of the associated Legengre polynomials

$$(\mathrm{e}^{\mathrm{i}\,m\,\lambda})^* = \mathrm{e}^{-\mathrm{i}\,m\,\lambda} \quad \text{and} \quad P_l^{-m}(w) = (-1)^m\,\frac{(l-m)!}{(l+m)!}\,P_l^m(w). \tag{8.82}$$

Spherical harmonics are normalized on the unit sphere according to

$$\int_0^{2\pi} \mathrm{d}\lambda \int_0^{\pi} Y_{l,m}(\varphi,\lambda)\,Y^*_{l',m'}(\varphi,\lambda)\sin\varphi\,\mathrm{d}\varphi = \delta_{l,l'}\,\delta_{m,m'}, \tag{8.83}$$

and they form a complete basis over the unit sphere, thus ensuring they satisfy the completeness relation

$$\sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y^*_{l,m}(\varphi',\lambda')\,Y_{l,m}(\varphi,\lambda) = \frac{\delta(\lambda-\lambda')\,\delta(\varphi-\varphi')}{\sin\varphi}. \tag{8.84}$$

The lowest spherical harmonic is a constant

$$Y_{0,0} = 1/\sqrt{4\pi}, \tag{8.85}$$

whereas spherical harmonics with $l = 1$ are given by[9]

$$Y_{1,0} = \frac{1}{2}\sqrt{\frac{3}{\pi}}\cos\varphi \tag{8.86a}$$

$$Y_{1,1} = -\frac{1}{2}\sqrt{\frac{3}{2\pi}}\,\mathrm{e}^{\mathrm{i}\lambda}\sin\varphi, \tag{8.86b}$$

and those with $l = 2$ are

$$Y_{2,0} = \frac{1}{4}\sqrt{\frac{5}{\pi}}\,(3\cos^2\varphi - 1) \tag{8.87a}$$

$$Y_{2,1} = -\frac{1}{2}\sqrt{\frac{15}{2\pi}}\,\mathrm{e}^{\mathrm{i}\lambda}\sin\varphi\cos\varphi \tag{8.87b}$$

$$Y_{2,2} = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\,\mathrm{e}^{2\mathrm{i}\lambda}\sin^2\varphi, \tag{8.87c}$$

[8]Despite the name, spherical harmonics are not harmonic functions. Instead, they are eigenfunctions of the spherical Laplacian on the unit sphere. It is only when they are combined with the radial dependence, $U(r) = A\,r^{l+1} + B\,r^{-l}$, then the product, $U_l(r)\,Y_{l,m}(\varphi,\lambda)$, form harmonic functions: $\nabla^2(U_l\,Y_l^m) = 0$.

[9]We only list harmonics for $m \geq 0$ since $m < 0$ can be found through the identity (8.81).

and those with $l = 3$ are

$$Y_{3,0} = \frac{1}{4}\sqrt{\frac{7}{\pi}}\,(5\cos^3\varphi - 3\cos\varphi) \tag{8.88a}$$

$$Y_{3,1} = -\frac{1}{8}\sqrt{\frac{21}{\pi}}\,\mathrm{e}^{\mathrm{i}\lambda}\sin\varphi\,(5\cos^2\varphi - 1) \tag{8.88b}$$

$$Y_{3,2} = \frac{1}{4}\sqrt{\frac{105}{2\pi}}\,\mathrm{e}^{2\mathrm{i}\lambda}\sin^2\varphi\cos\varphi, \tag{8.88c}$$

$$Y_{3,3} = -\frac{1}{8}\sqrt{\frac{35}{\pi}}\,\mathrm{e}^{3\mathrm{i}\lambda}\sin^3\varphi. \tag{8.88d}$$

In Figure 8.2 we depict these spherical harmonic patterns as polar plots on the sphere.

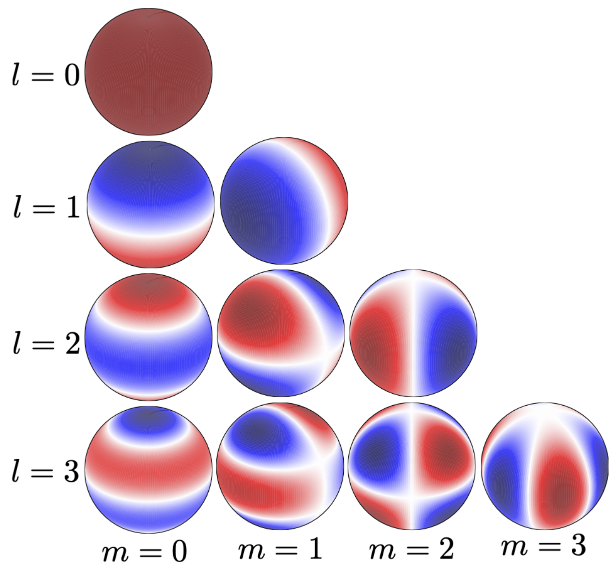

FIGURE 8.2: Depicting the real part of the first few spherical harmonics, shown as polar plots on a spherical projection viewed from 30 degrees north of the equator ($\phi = \pi/2 - \varphi = \pi/6$) and 45 degrees east longitude ($\lambda = \pi/4$). The first row shows $Y_{0,0}$; the second row shows $Y_{1,0}, Y_{1,1}$; the third row shows $Y_{2,0}, Y_{2,1}, Y_{2,2}$, and the fourth row shows $Y_{3,0}, Y_{3,1}, Y_{3,2}, Y_{3,3}$. White areas straddle zero, with red and blue having opposite signs.

8.5.3 Further study

Our treatment follows Chapter 3 of *Jackson* (1975), with a compatible treatment also given in Section 31 of *Dennery and Krzywicki* (1967). There are a variety of conventions for spherical harmonics, with appendix A.8 of *Wunsch* (2015) providing a summary. There are many galleries that can be found on the internet visualize spherical harmonics.

8.6 Introducing Green's functions

The mathematical problem of classical continuum mechanics concerns a field (scalar, vector, or tensor field), that represents a physical property such as material tracer concentration, velocity streamfunction, temperature, Newtonian gravitational potential, velocity. The space-time structure of the field is the result of a space-time distributed source present within the domain (e.g., tracer source, potential vorticity distribution, mass distribution) along with differential operators (e.g., time derivative, advection, Laplacian diffusion, wave operator) that serve to connect the field across points in space-time, as well as to the initial and boundary conditions along selected space-time regions.[10] If the equations are linear, then we can generally find an integral expression for the solution in terms of the Green's function. In this section we develop the theory of Green's functions for elliptic boundary value problems, restricting our focus to scalar fields such as tracers.

8.6.1 The central role of the superposition principle

The conceptual foundation of the Green's function method is based on observing that if the field equations are linear, then the field, ψ, can be constructed by the convolution of boundary/source data with the Green's function. The Green's function is the solution to the field equations yet with point-sized portions of the distributed source (including sources on the space-time boundaries). More precisely, we write $G(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t)$ as the field at an observation space-time point, $(\boldsymbol{x}, t)$ (the field point), that is caused by a point source at $(\boldsymbol{x}_{\circ}, t_0)$ (the source point). We then observe that $\psi(\boldsymbol{x}, t)$ caused by a source that is distributed through space and time is the product of G multiplied by the source and integrated over the space-time domain. Similar superpositions are made for the initial and boundary conditions. The function, G, is referred to as the Green's function in honor of the 19th century mathematical physicist who introduced the method.

The above arguments rely on the superposition principle that holds for linear systems. It does so by finding a particular solution (the Green's function) to a linear initial-boundary value problem with a singular point source (the Dirac delta from Chapter 5) and homogeneous boundary conditions. The solution to the original problem is found by integrating (convolving) the Green's function with the boundary conditions, initial conditions, and distributed sources. The Green's function is generally simpler to determine than the solution to the original initial-boundary value problem, thus providing the key practical reason to pursue the method. Furthermore, the Green's function provides a formal inverse to the linear partial differential operator in a manner reminiscent of matrix inversion used to solve a matrix-vector problem. Just as for the matrix-vector problem, once we have the Green's function we can write the solution to any of the associated initial-boundary value problems regardless the details of the distributed source, initial data, or boundary data. Herein lies the power of the Green's function method and why it has found much use across mathematical physics.

8.6.2 Some notation conventions

In the remainder of this chapter, we develop the Green's function method for a variety of linear elliptic boundary value problems. Use of the common symbol, G, for the Green's function, and $\mathcal{G}$ for the free-space Green's function, minimizes the adornments otherwise needed to distinguish Green's functions derived for each of the distinct equations. Confusion is avoided by noting that properties of any particular Green's function are specific to the section where the function is discussed. The modified Green's function, $\widetilde{G}$, is the one place we do provide extra adornment, with this function encountered in our study of the Poisson equation with Neumann boundary conditions. Furthermore, it is useful on certain occasions to write the gradient operator as either $\nabla_{\boldsymbol{x}}$ or $\nabla_{\boldsymbol{x}_{\circ}}$, depending on what argument of a Green's function is being differentiated (i.e., the field point or source point). This notation is particularly helpful in organizing manipulations in

[10]In some cases, sources are present only along spatial boundaries.

this chapter.[11]

8.7 Free space Green's functions for Laplace's operator

We study the Dirac delta in Chapter 5 by considering the Newtonian gravitational potential in the presence of a point mass source. This physical example also serves to introduce the notion of a Green's function, in this case a particularly simple Green's function known as a free-space Green's function for Laplace's equation. The free space Green's function serves an important role in the analytical theory of Green's functions, and it offers a pedagogical introduction to Green's functions. We here derive explicit expressions for the Laplace free space Green's function in one, two, and three space dimensions.

Additionally, for the study of Green's functions on bounded domains, the Green's function can be decomposed (following the superposition principle) into a singular part and non-singular part, with the singular part simply the free space Green's function. Consequently, the free space Green's function is useful even for realistic problems encountered in geophysical applications.

8.7.1 Gravitational potential from a point mass source

Consider the gravitational potential, Φ, in three space dimensions and in the presence of a point mass source at $\boldsymbol{x} = \boldsymbol{x}_\circ$ in the absence of boundaries (i.e., in free space). In the study of Green's functions this potential is referred to as the fundamental solution to Laplace's equation.[12] In other contexts it is referred to as the free-space Green's function, which is the terminology we choose since it emphasizes the absence of any spatial boundaries. From our discussion of Newtonian gravity in Section 5.1, the free space Green's function for Newtonian gravity satisfies the Poisson equation with a singular mass source

$$\nabla^2_{\boldsymbol{x}} \Phi(\boldsymbol{x}|\boldsymbol{x}_\circ) = 4\,\pi\,G^{\text{grv}}\,M\,\delta(\boldsymbol{x}-\boldsymbol{x}_\circ), \tag{8.89}$$

where we wrote the singular mass density in terms of the Dirac delta

$$\rho(\boldsymbol{x}|\boldsymbol{x}_\circ) = M\,\delta(\boldsymbol{x}-\boldsymbol{x}_\circ). \tag{8.90}$$

We added the "grv" label to Newton's gravitational constant, G^{grv}, rather than use the more common nomenclature, G, in order to distinguish the gravitational constant from a Green's function.

The gravitational potential in equation (8.89) has two arguments, $\Phi(\boldsymbol{x}|\boldsymbol{x}_\circ)$. The first argument, $\boldsymbol{x}$, is the point where the field is sampled and is referred to as the field point or sometimes the observation point. The second argument, $\boldsymbol{x}_\circ$, is where the Dirace source is located and is thus referred to as the source point. Figure 8.3 summarizes the notation. The Laplacian operator acts on the field point, $\boldsymbol{x}$, so that it is useful to write it with an $\boldsymbol{x}$ subscript, $\nabla^2_{\boldsymbol{x}}$, for clarity. The dimensonal multiplier, $4\,\pi\,G^{\text{grv}}\,M$, acting on the Dirac delta in equation (8.89) is specific to the physical problem, here being for Newtonian gravity. In this case the Green's function has dimensions of $\text{L}^2\ \text{T}^{-2}$ since it is a gravitational potential. In other cases,

[11] Much in this material is based on the treatments found in Chapter 7 of *Morse and Feshbach* (1953), *Greenberg* (1971), and *Stakgold* (2000a,b). These treatments are highly recommended for those wishing to pursue Green's function methods in more depth and to see many further applications.

[12] For example, see Section 5.8 of *Stakgold* (2000b) where he provides a discussion of a variety of such free space or fundamental solutions.

the Green's function has different dimensions, whereas the Dirac delta retains with the same dimensions.[13]

The gravitational potential for a point mass source, in the absence of any boundaries, is the free space Green's function for Newtonian gravity. Anticipating our study in Section 11.7.4, we write this free space Green's function in the form

$$\Phi(\boldsymbol{x}|\boldsymbol{x}_\circ) = -\frac{M\, G^{\text{grv}}}{|\boldsymbol{x}-\boldsymbol{x}_\circ|}. \tag{8.91}$$

This function is singular when sampling the field at the source location, $\boldsymbol{x} = \boldsymbol{x}_\circ$, and it decreases according to the inverse distance when moving away from the source.

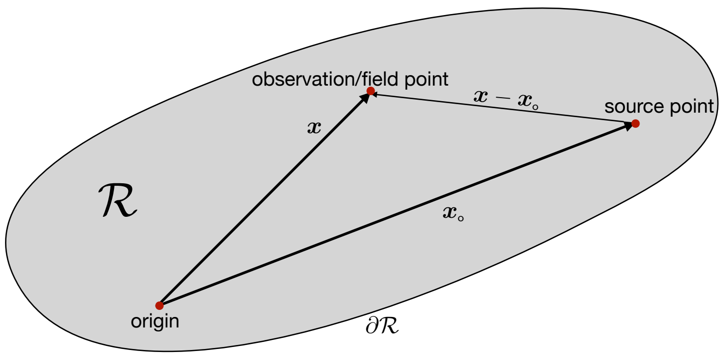

FIGURE 8.3: Depicting the geometry of a typical function problem in space, shown here for a finite domain, $\mathcal{R}$, with boundary $\partial\mathcal{R}$. Note that the free-space Green's function has no prescribed boundary conditions, whereas other Green's functions typically do. The field point (also observation point), $\boldsymbol{x}$, is where the Green's function is sampled, whereas the source point, $\boldsymbol{x}_\circ$, is where the Dirac delta source is located. The origin is at an arbitrary position within the domain. The Green's function, $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$, is the response of the field as sampled at $\boldsymbol{x}$ arising from the Dirac delta placed at $\boldsymbol{x}_\circ$.

8.7.2 Free space Green's functions

Abstracting the previous discussion motivates us to define the free space Green's function, $\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)$, for the Laplace operator as the solution to the singular Poisson equation

$$-\nabla^2_{\boldsymbol{x}}\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta(\boldsymbol{x}-\boldsymbol{x}_\circ). \tag{8.92}$$

It is conventional in many treatments to place a minus sign on the Laplacian operator to correspond to how it appears in the diffusion equation studied in VOLUME 4. In one, two and three space dimensions the free space Green's function is given by

$$\mathcal{G}(x|x_\circ) = -|x-x_\circ|/2 \qquad \text{for } x, x_\circ \in \mathbb{R}^1 \tag{8.93a}$$
$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = -(2\,\pi)^{-1}\ln|\boldsymbol{x}-\boldsymbol{x}_\circ|, \qquad \text{for } \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathbb{R}^2 \tag{8.93b}$$
$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = (4\,\pi\,|\boldsymbol{x}-\boldsymbol{x}_\circ|)^{-1} \qquad \text{for } \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathbb{R}^3. \tag{8.93c}$$

When considering the Dirac delta as the mass density for a point mass, the corresponding gravitational acceleration is proportional to the gradient of the free space Green's function.

[13] As emphasized throughout this book, checking for physical dimensional consistency offers a powerful means to ensure that a mathematical expression makes physical sense. If the equation is not dimensionally consistent, then something is wrong either with the maths or the physics. This rule is particularly useful when making use of Green's functions.

Evidently, for one space dimension, a point mass yields a gravity field that is a constant on either side of the mass point, with the acceleration pointing towards the mass. For two space dimensions, a point mass source produces a gravitational acceleration that points to the mass, and with a magnitude that falls off as the inverse distance from the source. Finally, for three space dimensions, the gravitational acceleration points towards the mass and falls off as the inverse squared distance.

We examine the three-dimensional Green's function in Section 8.7.3 and provide a derivation in Section 8.7.4. We then derive the two-dimensional Green's function (8.93b) in Section 8.7.5, and make use of it for studying flow around a point vortex in VOLUME 3. Finally, we encounter the one-dimensional Green's function in Exercise 8.12 when studying the one-dimensional Poisson equation. Indeed, we already encountered the one-dimensional Green's function in Exercise 5.4, where we showed that $\mathrm{d}^2|x|/\mathrm{d}x^2 = 2\,\delta(x)$.

8.7.3 Properties of the three-dimensional Green's function

We here verify that the expression (8.93c) is indeed the free space Green's function for $\mathbb{R}^3$, with this discussion providing exposure to some of the manipulations often encountered with Dirac deltas and Green's functions.

To simplify notation, place the Dirac delta at $\boldsymbol{x}_\circ = 0$ so that

$$\mathcal{G}(\boldsymbol{x}|0) = \frac{1}{4\,\pi\,r}, \tag{8.94}$$

where $|\boldsymbol{x}| = r$ is the radial distance from the origin. Introduce the continuous and non-singular function

$$G^{(\epsilon)}(\boldsymbol{x}) = \frac{1}{4\pi}\begin{cases} r^{-1} & \text{for } r > \epsilon \\ \epsilon^{-1} & \text{for } r \le \epsilon, \end{cases} \tag{8.95}$$

with $\epsilon > 0$. By constructing $G^{(\epsilon)}(\boldsymbol{x})$ we have removed the singularity at $r = 0$. Now consider the integral

$$\mathcal{I}(\epsilon) = -\int_{r\le\epsilon} \psi(\boldsymbol{x})\,\nabla^2 G^{(\epsilon)}(\boldsymbol{x})\,\mathrm{d}V, \tag{8.96}$$

for an arbitrary smooth function, ψ, and for a spherical region of radius $r = \epsilon$ centered on the origin. Since ψ is a smooth function, taking the limit as $\epsilon \to 0$ allows us to remove ψ from the integral and evaluate it at the origin

$$\lim_{\epsilon\to 0}\mathcal{I}(\epsilon) = -\psi(r=0)\,\lim_{\epsilon\to 0}\left[\int_{r\le\epsilon} \nabla^2 G^{(\epsilon)}(\boldsymbol{x})\,\mathrm{d}V\right]. \tag{8.97}$$

Making use of the divergence theorem brings the volume integral to a surface integral over the ϵ-sphere

$$\lim_{\epsilon\to 0}\mathcal{I}(\epsilon) = -\psi(r=0)\,\lim_{\epsilon\to 0}\left[\int_{r=\epsilon} \hat{\boldsymbol{r}}\cdot\nabla(1/r)\,r^2\,\mathrm{d}r\right] = \psi(r=0), \tag{8.98}$$

where we introduced spherical coordinates from Section 4.4. This result establishes both the sifting property and normalization property for $-\nabla^2 G^{(\epsilon)}(\boldsymbol{x})$. These two properties are the defining features of a Dirac delta, in which case we have established the identity

$$\lim_{\epsilon\to 0}\left[-\nabla^2 G^{(\epsilon)}(\boldsymbol{x})\right] = -\nabla^2(4\,\pi\,|\boldsymbol{x}|)^{-1} = \delta(\boldsymbol{x}), \tag{8.99}$$

so that

$$\lim_{\epsilon \to 0} G^{(\epsilon)}(\boldsymbol{x}) = \mathcal{G}(\boldsymbol{x}|0). \tag{8.100}$$

8.7.4 Deriving the three-dimensional Green's function

As a means to further understand the free space Green's function (8.93c), we here offer a derivation using Fourier transform methods.[14] This approach presumes an understanding of Fourier analysis from VOLUME 4.

We introduce the Fourier representation of the Dirac delta so that

$$-\nabla^2_{\boldsymbol{x}} \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) = \frac{1}{(2\pi)^3} \int e^{\mathrm{i}\,\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}_\circ)} \, \mathrm{d}\boldsymbol{k}, \tag{8.101}$$

where $\mathrm{d}\boldsymbol{k}$ is the $\boldsymbol{k}$-space volume element. It is straightforward to verify that

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{1}{(2\pi)^3} \int \frac{e^{\mathrm{i}\,\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}_\circ)}}{|\boldsymbol{k}|^2} \, \mathrm{d}\boldsymbol{k} \tag{8.102}$$

solves the Green's function equation (8.101), thus providing a Fourier integral expression for the free space Green's function.

Now evaluate the integral (8.102) to derive a closed form expression by introducing spherical coordinates in $\boldsymbol{k}$-space, whereby[15]

$$k_x = |\boldsymbol{k}| \cos\phi \cos\lambda \quad \text{and} \quad k_y = |\boldsymbol{k}| \cos\phi \sin\lambda \quad \text{and} \quad k_z = |\boldsymbol{k}| \sin\phi. \tag{8.103}$$

We are free to orient the axes as suits the integration, with the following choice very convenient

$$\boldsymbol{k} \cdot (\boldsymbol{x} - \boldsymbol{x}_\circ) = |\boldsymbol{k}| \, |\boldsymbol{x} - \boldsymbol{x}_\circ| \sin\phi, \tag{8.104}$$

thus bringing the integral (8.102) to

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{1}{(2\pi)^3} \int_0^\infty \mathrm{d}|\boldsymbol{k}| \int_{-\pi/2}^{\pi/2} \cos\phi \, \mathrm{d}\phi \int_0^{2\pi} e^{\mathrm{i}\,|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|\sin\phi} \, \mathrm{d}\lambda. \tag{8.105}$$

The λ integral trivially picks up a factor of 2π since the integrand is independent of λ. Performing the ϕ integral takes just a bit more work

$$\int_{-\pi/2}^{\pi/2} e^{\mathrm{i}\,|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|\sin\phi} \cos\phi \, \mathrm{d}\phi = \int_{-\pi/2}^{\pi/2} e^{\mathrm{i}\,|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|\sin\phi} \, \mathrm{d}(-\sin\phi) \tag{8.106a}$$

$$= -\int_{-1}^{1} e^{\mathrm{i}\,|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|\alpha} \, \mathrm{d}\alpha \tag{8.106b}$$

$$= \frac{2\sin[|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|]}{|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|}, \tag{8.106c}$$

so that the Green's function is

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{4\pi}{(2\pi)^3} \int_0^\infty \frac{\sin[|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|]}{|\boldsymbol{k}|\,|\boldsymbol{x}-\boldsymbol{x}_\circ|} \, \mathrm{d}|\boldsymbol{k}| = \frac{4\pi}{(2\pi)^3\,|\boldsymbol{x}-\boldsymbol{x}_\circ|} \int_0^\infty \frac{\sin\alpha}{\alpha} \, \mathrm{d}\alpha. \tag{8.107}$$

[14] This approach follows Section 28 of *Dennery and Krzywicki* (1967).

[15] See Section 4.4 for a discussion of spherical coordinates.

The integral in equation (8.107) can be readily found in an integral table. Nonetheless, we use this integral as an opportunity to introduce a version of Feynman's trick, which for this integral proceeds by considering the Laplace transform

$$F(s) = \int_0^\infty \mathrm{e}^{-s\,\alpha} \frac{\sin\alpha}{\alpha} \,\mathrm{d}\alpha, \tag{8.108}$$

where $s \geq 0$ is a parameter. The desired integral is $F(0)$, but we first determine $F(s)$ by taking its derivative

$$\frac{\mathrm{d}F(s)}{\mathrm{d}s} = -s \int_0^\infty \mathrm{e}^{-s\,\alpha} \sin\alpha \,\mathrm{d}\alpha. \tag{8.109}$$

Integrating by parts two times leads to

$$-\frac{\mathrm{d}F(s)}{\mathrm{d}s} = \frac{1}{1+s^2}, \tag{8.110}$$

which can be directly integrated to find

$$-F(s) = \tan^{-1}(s) + C. \tag{8.111}$$

To determine the constant of integration, C, take the limit $s \to \infty$, in which case $F(s)$ as given by equation (8.108) vanishes so that

$$C = -\lim_{s\to\infty} \tan^{-1}(s) = -\pi/2, \tag{8.112}$$

in which case

$$F(s) = -\tan^{-1}(s) + \pi/2 = \int_0^\infty \mathrm{e}^{-s\,\alpha} \frac{\sin\alpha}{\alpha} \,\mathrm{d}\alpha. \tag{8.113}$$

Setting $s = 0$ and noting that $\tan^{-1}(s=0) = 0$ leads to the desired integral

$$\int_0^\infty \mathrm{e}^{-s\,\alpha} \frac{\sin\alpha}{\alpha} \,\mathrm{d}\alpha = \pi/2, \tag{8.114}$$

and then to the free space Green's function (8.93c)

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = (4\,\pi\,|\boldsymbol{x} - \boldsymbol{x}_\circ|)^{-1}. \tag{8.115}$$

8.7.5 The two-dimensional free space Green's function

To simplify the derivation of the two-dimensional free space Green's function (8.93b), place the Dirac delta at the origin so that $\boldsymbol{x}_\circ = 0$. Furthermore, the Dirac delta induces no angular asymmetry, so that it is convenient to use polar coordinates since the Green's function is independent of the polar angle. As detailed in Section 4.3, the two-dimensional Laplace operator acting on a circularly symmetric function leads to the free-space Green's function equation

$$-\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left[r\,\frac{\mathrm{d}\mathcal{G}}{\mathrm{d}r}\right] = \frac{\delta(r)\,\delta(\vartheta)}{r}, \tag{8.116}$$

where we made use of equation (5.67) for the polar coordinate form of the Dirac delta. Integrating over the angle to remove $\delta(\vartheta)$ renders

$$-\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left[r\,\frac{\mathrm{d}\mathcal{G}}{\mathrm{d}r}\right] = -\frac{\delta(r)}{2\,\pi\,r}. \tag{8.117}$$

One radial integral that includes the origin yields

$$\frac{\mathrm{d}\mathcal{G}}{\mathrm{d}r} = -\frac{1}{2\,\pi\,r}, \tag{8.118}$$

with a second radial integral leading to

$$\mathcal{G}(r|0) - \mathcal{G}(r_0|0) = -(2\pi)^{-1}\ln(r/r_0), \tag{8.119}$$

where $r_0 \neq 0$ is an arbitrary reference radius. Without loss of generality we set $\mathcal{G}(r_0|0) = -(2\pi)^{-1}\ln r_0$ so that

$$\mathcal{G}(r|0) = -(2\pi)^{-1}\ln r, \tag{8.120}$$

whose more general expression is given by equation (8.93b).

8.8 The Helmholtz equation

We study the wave equation in Chapter 10, which is given by equation (10.35a) space

$$(\partial_{tt} - c^2\,\nabla^2)\,\psi = \Lambda. \tag{8.121}$$

When working on a unbounded spatial domain it is useful to consider a time-frequency Fourier transforms, $\tilde{\psi}(\boldsymbol{x},\omega)$ and $\tilde{\Lambda}(\boldsymbol{x},\omega)$ (see equations (6.105a) and (6.105b)) whereby

$$\psi(\boldsymbol{x},t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\tilde{\psi}(\boldsymbol{x},\omega)\,\mathrm{e}^{-\mathrm{i}\,\omega\,t}\,\mathrm{d}\omega \quad \text{and} \quad h(\boldsymbol{x},t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\tilde{\Lambda}(\boldsymbol{x},\omega)\,\mathrm{e}^{-\mathrm{i}\,\omega\,t}\,\mathrm{d}\omega, \tag{8.122}$$

thus yielding the frequency domain version of the wave equation

$$[\nabla^2 + (\omega/c)^2]\,\Psi = -\tilde{\Lambda}/c^2. \tag{8.123}$$

This is the Helmholtz equation, and it is an elliptic partial differential equation that it arises when interested in the steady state wave field that is fully established. That is, we are not interested in the development of waves via an initial value problem.

The Helmholtz equation has a free space Green's function the satisfies the singular elliptic problem

$$[\nabla^2 + (\omega/c)^2]\,\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ;\omega) = -\delta(\boldsymbol{x}-\boldsymbol{x}_\circ). \tag{8.124}$$

Observe that the Helmholtz operator, $\nabla^2 + (\omega/c)^2$, differs from the screened Poisson operator, $\nabla^2 - \mu^2$, considered in Exercise 8.1, due to the differing signs on the constant. We can relate the two by setting

$$\mu = -\mathrm{i}\,\omega/c. \tag{8.125}$$

This connection proves rather important for both the physics and the maths, with details involving aspects of causality and complex analysis that are beyond our scope.[16] It is satisfying

[16] *Fetter and Walecka* (2003) present a lucid discussion of the role of causality in their Section 50. See also

that the details do confirm that the Helmholtz free space Green's function is related to the screened Poisson Green's function (8.338) simply by setting $\mu = -\mathrm{i}\,\omega/c$, so that

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ;\omega) = \frac{\mathrm{e}^{\mathrm{i}\,|\boldsymbol{x}-\boldsymbol{x}_\circ|\,\omega/c}}{4\,\pi\,|\boldsymbol{x}-\boldsymbol{x}_\circ|}. \tag{8.126}$$

8.9 Poisson equation with Dirichlet boundaries

In this section we develop the Green's function method for Poisson's equation with a Dirichlet boundary condition

$$-\nabla^2\psi = \Lambda, \quad \text{for } \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \qquad \psi = \sigma, \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.127}$$

As noted at the start of Section 8.3.4, the region, $\mathcal{R}$, is an open domain, and it is here that the field, ψ, satisfies Poisson's equation with a distributed source, Λ. The region's boundary, $\partial\mathcal{R}$, is where boundary data is prescribed, here in the form of the Dirichlet boundary data with $\psi = \sigma$. We focus on the three-dimensional case, though the results hold for one and two dimensions as well. Our mathematical goal is to determine an integral expression for ψ in terms of the source, $\Lambda(\boldsymbol{x})$, and boundary data, $\sigma(\boldsymbol{x})$, with each convolved with the Green's function.

8.9.1 Constraints on the source and boundary normal derivative

Assuming there exists a solution to the boundary value problem (8.127), we can derive a constraint on the normal derivative of ψ along the domain boundary.[17] This constraint is revealed by integrating the Poisson equation (8.127) over the spatial domain, with the left hand side yielding

$$-\int_{\mathcal{R}} \nabla^2\psi\,\mathrm{d}V = -\oint_{\partial\mathcal{R}} \nabla\psi\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}, \tag{8.128}$$

where we made use of the divergence theorem and with $\partial\mathcal{R}$ the closed boundary of the domain, $\mathcal{R}$. Equating this result to the integral of the source, Λ, leads to the constraint

$$\oint_{\partial\mathcal{R}} \nabla\psi\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} = -\int_{\mathcal{R}} \Lambda\,\mathrm{d}V. \tag{8.129}$$

We can physically interpret the constraint (8.129) by invoking a steady state tracer diffusion interpretation of the Poisson equation, with $\nabla\psi\cdot\hat{\boldsymbol{n}}$ interpreted as a flux of ψ-stuff through the boundary. For a steady state solution to exist in the presence of specified boundary tracer concentration as well as with an interior source function, there must be a balance between the boundary integrated flux and the volume integrated source. In particular, this balance is required to maintain the specified Dirichlet boundary values in the presence of an interior source. Absent this balance, there will be depletion or accumulation of ψ within the domain that leads to a transient adjustment, thus breaking the steady state assumption.[18] Notably, the constraint (8.129) can be realized for arbitrary Dirichlet boundary data, σ.

Section 5.8 and 7.12 of *Stakgold* (2000b) for a detailed presentation of the mathematical derivations.

[17] To be self-contained, we here repeat discussion of the constraint (8.129) that is also provided in Section 8.3.7.

[18] An analogous interpretation holds when ψ is the gravitational potential resulting from the mass source, Λ. For ψ to be a specified value along the domain boundary requires the volume integrated mass source to balance a gravitational acceleration integrated along the boundary.

8.9.2 Uniqueness of the Dirichlet solution

To establish uniqueness of the solution to the boundary value problem (8.127), consider two functions, ψ_{a} and ψ_{b}, each satisfying the same boundary value problem (8.127). In turn, their difference,

$$\Psi = \psi_{\mathsf{a}} - \psi_{\mathsf{b}}, \tag{8.130}$$

is a harmonic function that satisfies a homogeneous Dirichlet boundary condition

$$-\nabla^2\Psi = 0, \quad \text{for } \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \qquad \Psi = 0, \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.131}$$

We offer a proof by contradiction to establish that $\Psi = 0$ is the only solution to this boundary value problem. Namely, we assume $\Psi \neq 0$ and then show that to be an inconsistent assumption. First consider the case where Ψ is a constant. A constant is certainly a harmonic function, $\nabla^2\Psi = 0$. But only the zero constant, $\Psi = 0$, satisfies the homogeneous Dirichlet boundary condition. Next, assume Ψ has spatial dependence so that $\nabla\Psi \neq 0$ and examine the non-negative integral over the full domain

$$\mathcal{I} = \int_{\mathcal{R}} \nabla\Psi \cdot \nabla\Psi \, \mathrm{d}V. \tag{8.132}$$

Since $\nabla^2\Psi = 0$ we have $\nabla\Psi \cdot \nabla\Psi = \nabla \cdot (\Psi \, \nabla\Psi)$ so that

$$\mathcal{I} = \int_{\mathcal{R}} \nabla \cdot (\Psi \, \nabla\Psi) \, \mathrm{d}V = \oint_{\partial\mathcal{R}} \Psi \, \nabla\Psi \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S}, \tag{8.133}$$

where we used the divergence theorem for the second equality. But since $\Psi = 0$ on $\partial\mathcal{R}$ we find that $\mathcal{I} = 0$. Since the integrand of $\mathcal{I}$ is non-negative, $\mathcal{I} = 0$ can only be realized by $\nabla\Psi = 0$ everywhere, which in turn means that Ψ is a constant. But we just saw that the only constant that satisfies the homogeneous Dirichlet boundary condition is $\Psi = 0$ everywhere. We thus conclude that $\Psi = 0$ is the only harmonic function with zero Dirichlet boundary conditions, in which case the solution, ψ, to the boundary value problem (8.127) is indeed unique.

The above proof assumes that $\nabla\Psi \cdot \hat{\boldsymbol{n}}$ is non-singular on the boundary of the domain, which is not the case for the Green's functions, as shown by equation (8.161) derived below. An more general proof of uniqueness makes use of the mean value property from Section 8.3.3. Namely, the harmonic function, Ψ, has no extrema within the domain interior, and since Ψ vanishes on the domain boundary, is also vanishes within the domain interior.

8.9.3 The Green's function problem

The Green's function, $G(\boldsymbol{x}|\boldsymbol{x}_{\circ})$, corresponding to the boundary value problem (8.127) is the solution to the Poisson equation with a Dirac delta source (rather than Λ) and a homogeneous Dirichlet boundary condition (rather than σ)

$$-\nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = \delta(\boldsymbol{x} - \boldsymbol{x}_{\circ}) \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_{\circ} \in \mathcal{R} \quad \text{and} \quad G(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = 0 \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.134}$$

In three space dimensions, the Dirac delta has dimensions of inverse volume (Section 5.3), so that the Green's function has dimensions of inverse length. By construction, the Green's function is harmonic everywhere except at the location of the Dirac delta source, $\boldsymbol{x} = \boldsymbol{x}_{\circ}$. Furthermore, the Green's function satisfies a homogenous Dirichlet boundary condition whenever the field point, $\boldsymbol{x}$, is on the boundary and for any source position, $\boldsymbol{x}_{\circ}$.

Although the Green's function cares about the existence of spatial boundaries, it is independent of the boundary data, σ, and the distributed source, Λ, that appear in the boundary value problem (8.127) for ψ. In that manner, the Green's function is connected to the original boundary value problem only through the differential operator (here the Laplacian) and the type of boundary condition (here a Dirichlet condition).

8.9.4 Decomposing the Green's function

Linearity enables us to decompose the Green's function into the sum of the free space Green's function, $\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)$, plus a harmonic function

$$G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) + H(\boldsymbol{x}|\boldsymbol{x}_\circ), \tag{8.135}$$

where the free space Green's function and harmonic function satisfy

$$-\nabla^2_{\boldsymbol{x}}\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta(\boldsymbol{x}-\boldsymbol{x}_\circ) \quad \text{for} \quad \boldsymbol{x},\boldsymbol{x}_\circ \in \mathbb{R}^n \tag{8.136a}$$

$$-\nabla^2_{\boldsymbol{x}}H(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0 \quad \text{for} \quad \boldsymbol{x},\boldsymbol{x}_\circ \in \mathcal{R} \tag{8.136b}$$

$$-H(\boldsymbol{x}|\boldsymbol{x}_\circ) = \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) \quad \text{for} \quad \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.136c}$$

Given the free space Green's function for the corresponding space dimension from equations (8.93a), (8.93b), or (8.93c), then the mathematical problem of finding $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$ reduces to finding the harmonic function, $H(\boldsymbol{x}|\boldsymbol{x}_\circ)$, satisfying the inhomogeneous Dirichlet boundary condition (8.136c). Hence, the decomposition (8.135) is quite useful in practice. Namely, it serves to encapsulate in the free-space Green's function the non-smooth behavior induced by Dirac delta, whereas the harmonic function is smooth.

8.9.5 Jump condition induced by the Dirac source

Following the consistency condition established in Section 8.9.1, we integrate the Green's function partial differential equation (8.134) over a volume, $\mathcal{R}_\circ$, that encloses the Dirac delta source point at $\boldsymbol{x}_\circ \in \mathcal{R}_\circ$. Doing so leads to

$$\int_{\mathcal{R}} \nabla_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ = \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ = -1, \tag{8.137}$$

where we used the divergence theorem for the first equality. This result means that for any domain enclosing the Dirac source point, the normal derivative of the Green's function must satisfy this integral jump condition across the region boundary. Decomposing the Green's function per equation (8.135) reveals that it is the free space Green's function that contributes to the jump

$$\int_{\mathcal{R}} \nabla_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}} \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ = \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ = -1, \tag{8.138}$$

whereas the harmonic contribution has no jump

$$\int_{\mathcal{R}} \nabla_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}} H(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ = \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} H(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ = 0. \tag{8.139}$$

8.9.6 Reciprocity of the Green's function

The Green's function satisfies the following reciprocity condition

$$G(\boldsymbol{x}|\boldsymbol{x}_\circ) = G(\boldsymbol{x}_\circ|\boldsymbol{x}). \tag{8.140}$$

Reciprocity says that the Green's function at the field point, $\boldsymbol{x}$, arising from a Dirac delta source at the source point, $\boldsymbol{x}_\circ$, is identical to the Green's function at the field point, $\boldsymbol{x}_\circ$, arising from a Dirac delta source at point, $\boldsymbol{x}$. By inspection, the free space Green's functions (8.93a)-(8.93c) satisfy reciprocity. Hence, by implication the harmonic function, $H(\boldsymbol{x}|\boldsymbol{x}_\circ)$, also satisfies reciprocity. We here offer a direct derivation of reciprocity by using steps similar to those used for establishing the second form of Green's integral identity in equation (2.135).

Proof of reciprocity

Consider two Green's functions, $G(\boldsymbol{x}|\boldsymbol{a})$ and $G(\boldsymbol{x}|\boldsymbol{b})$, arising from Dirac delta sources at points $\boldsymbol{a} \in \mathcal{R}$ and $\boldsymbol{b} \in \mathcal{R}$

$$-\nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{a}) = \delta(\boldsymbol{x} - \boldsymbol{a}) \quad \text{and} \quad -\nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{b}) = \delta(\boldsymbol{x} - \boldsymbol{b}). \tag{8.141}$$

Multiply the first equation by $G(\boldsymbol{x}, \boldsymbol{b})$ and the second by $G(\boldsymbol{x}, \boldsymbol{a})$, then subtract to find

$$-G(\boldsymbol{x}|\boldsymbol{b})\, \nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{a}) + G(\boldsymbol{x}|\boldsymbol{a})\, \nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{b}) = G(\boldsymbol{x}|\boldsymbol{b})\, \delta(\boldsymbol{x} - \boldsymbol{a}) - G(\boldsymbol{x}|\boldsymbol{a})\, \delta(\boldsymbol{x} - \boldsymbol{b}). \tag{8.142}$$

Now integrate this equation over the region, $\mathcal{R}$, with the right hand side rendering

$$\int_{\mathcal{R}} [G(\boldsymbol{x}|\boldsymbol{b})\, \delta(\boldsymbol{x} - \boldsymbol{a}) - G(\boldsymbol{x}|\boldsymbol{a})\, \delta(\boldsymbol{x} - \boldsymbol{b})]\, \mathrm{d}V = G(\boldsymbol{a}, \boldsymbol{b}) - G(\boldsymbol{b}, \boldsymbol{a}), \tag{8.143}$$

where we made use of the sifting property (5.8). Integrating the left hand side of equation (8.142) leads to

$$\int_{\mathcal{R}} [-G(\boldsymbol{x}|\boldsymbol{b})\, \nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{a}) + G(\boldsymbol{x}|\boldsymbol{a})\, \nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{b})]\, \mathrm{d}V \tag{8.144a}$$

$$= \int_{\mathcal{R}} [-\nabla_{\boldsymbol{x}} \cdot [G(\boldsymbol{x}|\boldsymbol{b})\, \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{a})] + \nabla_{\boldsymbol{x}} \cdot [G(\boldsymbol{x}|\boldsymbol{a})\, \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{b})]]\, \mathrm{d}V \tag{8.144b}$$

$$= \oint_{\partial\mathcal{R}} [-G(\boldsymbol{x}|\boldsymbol{b})\, \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{a}) + G(\boldsymbol{x}|\boldsymbol{a})\, \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{b})] \cdot \hat{\boldsymbol{n}}\, \mathrm{d}\mathcal{S}, \tag{8.144c}$$

where we made use of the divergence theorem for the second equality. The final result vanishes when making use of the homogeneous Dirichlet boundary condition (8.134). We are thus led to

$$G(\boldsymbol{a}, \boldsymbol{b}) = G(\boldsymbol{b}, \boldsymbol{a}), \tag{8.145}$$

which is the reciprocity condition (8.140).

Comments on self-adjoint operators

The reciprocity relation (8.140) for Green's functions holds for self-adjoint differential boundary value problems.[19] Self-adjointness reflects properties of both the differential operator and the boundary conditions, with the discussion in this section revealing that the Laplacian operator

[19]See Theorem 8.10 of *Duchateau and Zachmann* (1986).

is self-adjoint with Dirichlet boundary conditions. Operators that are not self-adjoint, such as the diffusion operator (VOLUME 4), satisfy a slightly more general reciprocity relation that involve the adjoint Green's function.

The absence of self-adjointness reflects some form of symmetry breaking either through the operator itself or through the boundary and/or initial conditions. The diffusion operator has a single time derivative, ∂_t, which breaks symmetry between past and future thus leading to the absence of self-adjointness. Furthermore, self-adjointness is a property that depends on the nature of the chosen inner product, with the inner product in the present discussion defined by integration over the domain, $\mathcal{R}$. See *Stakgold* (2000a,b) for a thorough discussion accessible to physicists.

8.9.7 The integral solution

We have the elements in place to determine ψ as an integral expression involving $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$ along with the prescribed source, Λ, and boundary data, σ. To do so we follow steps similar to those used to establish reciprocity. Recall the Poisson boundary value problem (8.127) for ψ and the associated Green's function problem (8.134),

$$-\nabla^2_{\boldsymbol{x}}\psi(\boldsymbol{x}) = \Lambda(\boldsymbol{x}) \tag{8.146a}$$

$$-\nabla^2_{\boldsymbol{x}}G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta(\boldsymbol{x}-\boldsymbol{x}_\circ). \tag{8.146b}$$

Multiply the $\psi(\boldsymbol{x})$ equation by $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$ and the $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$ equation by $\psi(\boldsymbol{x})$ to find

$$-G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\nabla^2_{\boldsymbol{x}}\psi(\boldsymbol{x}) = G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\Lambda(\boldsymbol{x}) \tag{8.147a}$$

$$-\psi(\boldsymbol{x})\,\nabla^2_{\boldsymbol{x}}G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \psi(\boldsymbol{x})\,\delta(\boldsymbol{x}-\boldsymbol{x}_\circ), \tag{8.147b}$$

and then subtract these two equations

$$-G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\nabla^2_{\boldsymbol{x}}\psi(\boldsymbol{x}) + \psi(\boldsymbol{x})\,\nabla^2_{\boldsymbol{x}}G(\boldsymbol{x}|\boldsymbol{x}_\circ) = G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\Lambda(\boldsymbol{x}) - \psi(\boldsymbol{x})\,\delta(\boldsymbol{x}-\boldsymbol{x}_\circ). \tag{8.148}$$

Now integrate over observational points, $\boldsymbol{x}$, sampled over the region $\mathcal{R}$, in which case the right hand side becomes

$$\int_{\mathcal{R}} [G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\Lambda(\boldsymbol{x}) - \psi(\boldsymbol{x})\,\delta(\boldsymbol{x}-\boldsymbol{x}_\circ)]\,\mathrm{d}V = -\psi(\boldsymbol{x}_\circ) + \int_{\mathcal{R}} G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\Lambda(\boldsymbol{x})\,\mathrm{d}V, \tag{8.149}$$

where we made use of the sifting property (5.8) to expose the function, ψ, at the location of the Dirac source, $\boldsymbol{x}_\circ$. Integrating the left hand side of equation (8.148) yields

$$\int_{\mathcal{R}} [-G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\nabla^2_{\boldsymbol{x}}\psi(\boldsymbol{x}) + \psi(\boldsymbol{x})\,\nabla^2_{\boldsymbol{x}}G(\boldsymbol{x}|\boldsymbol{x}_\circ)]\,\mathrm{d}V \tag{8.150a}$$

$$= \int_{\mathcal{R}} \nabla_{\boldsymbol{x}}\cdot[-G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}) + \psi(\boldsymbol{x})\,\nabla_{\boldsymbol{x}}G(\boldsymbol{x}|\boldsymbol{x}_\circ)]\,\mathrm{d}V \tag{8.150b}$$

$$= \oint_{\partial\mathcal{R}} [-G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}) + \psi(\boldsymbol{x})\,\nabla_{\boldsymbol{x}}G(\boldsymbol{x}|\boldsymbol{x}_\circ)]\cdot\hat{\boldsymbol{n}}_{\boldsymbol{x}}\,\mathrm{d}\mathcal{S}. \tag{8.150c}$$

Bringing together the two sides of equation (8.148) leads to

$$\psi(\boldsymbol{x}_\circ) = \int_{\mathcal{R}} \Lambda(\boldsymbol{x})\,G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V + \oint_{\partial\mathcal{R}} [G(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}) - \psi(\boldsymbol{x})\,\nabla_{\boldsymbol{x}}G(\boldsymbol{x}|\boldsymbol{x}_\circ)]\cdot\hat{\boldsymbol{n}}_{\boldsymbol{x}}\,\mathrm{d}\mathcal{S}. \tag{8.151}$$

As a final step, we make use of the Dirichlet boundary conditions (homogeneous for the Green's function and prescribed function for ψ), relabel $\boldsymbol{x}_\circ \leftrightarrow \boldsymbol{x}$, and make use of reciprocity, $G(\boldsymbol{x}|\boldsymbol{x}_\circ) = G(\boldsymbol{x}_\circ|\boldsymbol{x})$, so that

$$\psi(\boldsymbol{x}) = \underbrace{\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, G(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ}_{\text{volume integral over } \mathcal{R}} - \underbrace{\oint_{\partial\mathcal{R}} [\sigma(\boldsymbol{x}_\circ)\, \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ)] \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}\, \mathrm{d}\mathcal{S}_\circ}_{\text{boundary area integral over } \partial\mathcal{R}}, \tag{8.152}$$

where we inserted the boundary data, $\psi(\boldsymbol{x}) = \sigma(\boldsymbol{x})$, for $\boldsymbol{x} \in \partial\mathcal{R}$. We have thus established ψ as a volume integral of the Green's function with the source, Λ, plus a boundary integral of the Green's function with the boundary data. Note that the Green's function is independent of the source, Λ, and the boundary data, so that $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$ can be used to express ψ for arbitrary source functions and boundary data.

8.9.8 Properties of the Dirichlet solution

We refer to the inward normal gradient of the Green's function as the boundary Green's function

$$G^{\mathsf{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) \equiv -\hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \cdot \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ) \quad \text{for} \quad \boldsymbol{x}_\circ \in \partial\mathcal{R}, \tag{8.153}$$

in which case the Dirichlet solution (8.152) takes the form

$$\psi(\boldsymbol{x}) = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, G(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ)\, G^{\mathsf{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ. \tag{8.154}$$

This form for the solution, which is depicted in Figure 8.4, emphasizes the role of the Green's function in the region interior as the mediator between the distributed source, Λ, and all surrounding points $\boldsymbol{x} \in \mathcal{R}$, along with the boundary Green's function acting as mediator between information prescribed along the boundary, $\boldsymbol{x} \in \partial\mathcal{R}$, and interior points. In this subsection we summarize certain properties of the solution (8.152) and (8.154), and infer (through insisting on self-consistency) corresponding properties of the Green's function and boundary Green's function.

Verifying the partial differential equation for $\boldsymbol{x} \in \mathcal{R}$

We derived the Dirichlet solution (8.152) with manipulations that are reversible; i.e., equal signs at every step. Hence, we know that the expression (8.152) indeed satisfies the Dirichlet boundary value problem (8.127). Even so, the exercise of directly verifying the solution reveals insights into properties of the Green's function.

As a starting point, decompose the Green's function into the free-space Green's function and a harmonic function, as per equation (8.135), to bring the solution (8.135) to the form

$$\psi(\boldsymbol{x}) = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, [\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) + H(\boldsymbol{x}|\boldsymbol{x}_\circ)]\, \mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ)\, G^{\mathsf{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ. \tag{8.155}$$

Now assume the field point, $\boldsymbol{x}$, is located at a point within the the open region, $\boldsymbol{x} \in \mathcal{R}$, and operate with $-\nabla^2_{\boldsymbol{x}}$ on equation (8.155) to find

$$-\nabla^2_{\boldsymbol{x}}\psi(\boldsymbol{x}) = -\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \nabla^2_{\boldsymbol{x}}[\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) + H(\boldsymbol{x}|\boldsymbol{x}_\circ]\, \mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ)\, [-\nabla^2_{\boldsymbol{x}} G^{\mathsf{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)]\, \mathrm{d}\mathcal{S}_\circ. \tag{8.156}$$

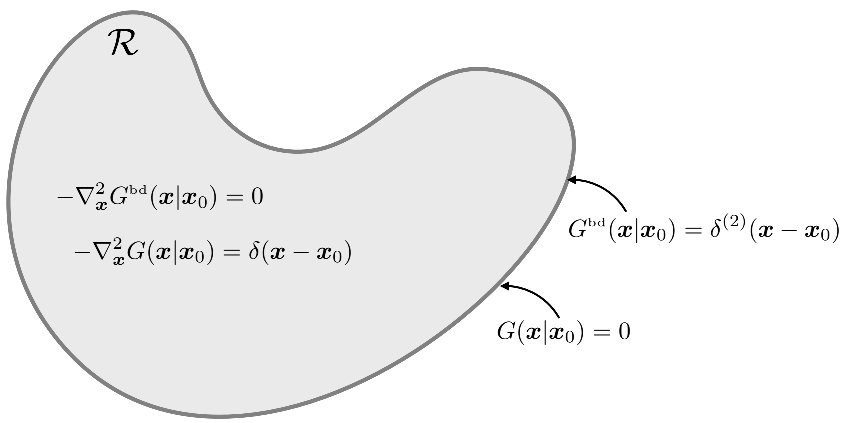

FIGURE 8.4: Decomposing the Green's function solution (8.152) for a Poisson boundary value problem with Dirichlet boundary conditions. The decomposition is written in terms of an interior Green's function, $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$, that satisfies equation (8.134) with both the field point, $\boldsymbol{x}$, and Dirac source point, $\boldsymbol{x}_\circ$, within the domain, $\boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R}$. The second piece of the solution arises from the boundary Green's function, $G^{\mathrm{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)$, with the source point on the boundary, $\boldsymbol{x}_\circ \in \partial\mathcal{R}$. The boundary Green's function satisfies the boundary value problem (8.163a)-(8.163b), in which G^{bd} is harmonic for $\boldsymbol{x} \in \mathcal{R}$, and it is a Dirac delta when sampled on the boundary, $\boldsymbol{x} \in \partial\mathcal{R}$.

We brought the Laplacian operator, $\nabla_{\boldsymbol{x}}^2$, inside the integrals since both integrations are over the source points, $\boldsymbol{x}_\circ$. For the first term on the right hand side of equation (8.156), set $\nabla_{\boldsymbol{x}}^2 H(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0$ as per equation (8.136b), and use the free space Green's function identity $-\nabla_{\boldsymbol{x}}^2 \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta(\boldsymbol{x}-\boldsymbol{x}_\circ)$. Then use the sifting property of the Dirac delta, $\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\,\delta(\boldsymbol{x}-\boldsymbol{x}_0)\,\mathrm{d}V_\circ = \Lambda(\boldsymbol{x})$. To show that the boundary contribution vanishes, make use of the following

$$
\begin{aligned}
\nabla_{\boldsymbol{x}}^2 G^{\mathrm{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) &= \nabla_{\boldsymbol{x}}^2[-\nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}] && (8.157a)\\
&= \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \cdot \nabla_{\boldsymbol{x}_\circ}[-\nabla_{\boldsymbol{x}}^2 G(\boldsymbol{x}|\boldsymbol{x}_\circ)] && (8.157b)\\
&= \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \cdot \nabla_{\boldsymbol{x}_\circ} \delta(\boldsymbol{x}-\boldsymbol{x}_\circ). && (8.157c)
\end{aligned}
$$

With the Dirac source on the boundary, $\boldsymbol{x}_\circ \in \partial\mathcal{R}$, and the field point within the domain interior, $\boldsymbol{x} \in \mathcal{R}$ (recall that $\mathcal{R}$ is an open region), then we see that the Dirac delta never fires, thus eliminating the boundary contribution,

$$
-\nabla_{\boldsymbol{x}}^2 G^{\mathrm{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0 \qquad \boldsymbol{x} \in \mathcal{R} \quad \text{and} \quad \boldsymbol{x}_\circ \in \partial\mathcal{R}. \tag{8.158}
$$

We have thus verified that $-\nabla^2\psi = \Lambda$ for points $\boldsymbol{x} \in \mathcal{R}$.

Verifying the Dirichlet boundary condition for $\boldsymbol{x} \in \partial\mathcal{R}$

To verify that the boundary conditions are satisfied by the Dirichlet solution (8.152), bring the field point, $\boldsymbol{x}$, onto the boundary

$$
\boldsymbol{x} \to \boldsymbol{x}_{\partial\mathcal{R}} \in \partial\mathcal{R}. \tag{8.159}
$$

For such boundary points, the volume integral in the solution (8.152) vanishes since the Dirichlet Green's function vanishes on the boundary, $G(\boldsymbol{x}_{\partial\mathcal{R}}|\boldsymbol{x}_\circ) = 0$. Self-consistency with the Dirichlet boundary data, $\psi(\boldsymbol{x}_{\partial\mathcal{R}}) = \sigma(\boldsymbol{x}_{\partial\mathcal{R}})$, leads to the identity

$$
\sigma(\boldsymbol{x}) = \oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ)\, G^{\mathrm{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ = -\oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ)\,\nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}_{\partial\mathcal{R}}|\boldsymbol{x}_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}\,\mathrm{d}\mathcal{S}_\circ \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.160}
$$

This integral equation is consistent so long as $G^{\text{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)$ satisfies the boundary condition

$$G^{\text{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) = -\hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \cdot \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}, \tag{8.161}$$

where $\delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ)$ is the surface Dirac delta with physical dimensions of inverse area. This property of the boundary Green's function is consistent with the normalization condition (8.164) found below. Furthermore, it leads to the corresponding integral identity

$$\oint_{\partial\mathcal{R}} G^{\text{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ = \oint_{\partial\mathcal{R}} \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ = 1 \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}. \tag{8.162}$$

We emphasize[20] that the expressions (8.161) and (8.162) are found by first placing the source point, $\boldsymbol{x}_\circ$, which indeed is where the source point is located for the boundary Green's function. Thereafter, we move the field point to the boundary, $\boldsymbol{x} \to \boldsymbol{x}_{\partial\mathcal{R}} \in \partial\mathcal{R}$. We return to the normalization (8.162) in Section 8.9.9 when discussing the partial differential equation satisfied by the boundary Green's function.

Linear superposition

The Dirichlet solution (8.152) manifests the linear superposition principle by writing $\psi = \psi_1 + \psi_2$ as given by Table 8.1. By construction, ψ_1 satisfies Poisson's equation with homogeneous Dirichlet boundary conditions, whereas ψ_2 satisfies Laplace's equation with inhomogeneous Dirichlet boundary conditions.

PDE: $\boldsymbol{x} \in \mathcal{R}$	BC: $\boldsymbol{x} \in \partial\mathcal{R}$	SOLUTION $\psi = \psi_1 + \psi_2$
$-\nabla^2\psi_1 = \Lambda$	$\psi_1 = 0$	$\psi_1(\boldsymbol{x}) = \int_{\mathcal{R}} G(\boldsymbol{x}\|\boldsymbol{x}_\circ)\, \Lambda(\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ$
$-\nabla^2\psi_2 = 0$	$\psi_2 = \sigma$	$\psi_2(\boldsymbol{x}) = \oint_{\partial\mathcal{R}} G^{\text{bd}}(\boldsymbol{x}\|\boldsymbol{x}_\circ)\, \sigma(\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ$

TABLE 8.1: Decomposing the Dirichlet solution (8.152) into $\psi = \psi_1 + \psi_2$, with ψ_1 and ψ_2 satisfying the properties shown in this table. For the boundary contribution we made use of the boundary Green's function, $G^{\text{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) = -\nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}$, as given by equation (8.153) and whose properties are further developed in Section 8.9.9.

8.9.9 Boundary Green's function

Boundary value problem for the boundary Green's function

As part of the development in Section 8.9.8, we revealed that the boundary Green's function satisfies the following boundary value problem

$$-\nabla^2_{\boldsymbol{x}} G^{\text{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0 \qquad \text{for} \quad \boldsymbol{x} \in \mathcal{R} \quad \text{and} \quad \boldsymbol{x}_\circ \in \partial\mathcal{R} \tag{8.163a}$$

$$G^{\text{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \text{for} \quad \boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}. \tag{8.163b}$$

So in summary, the equation (8.134) reveals that the Dirichlet Green's function, $G(\boldsymbol{x}|\boldsymbol{x}_\circ)$, arises from a Dirac source in the interior of the domain, $\boldsymbol{x}_\circ \in \mathcal{R}$, and it satisfies homogeneous Dirichlet boundary conditions for $\boldsymbol{x} \in \partial\mathcal{R}$. As a complement, the boundary Green's function, $G^{\text{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ)$, arises from a Dirac source placed on the domain boundary, $\boldsymbol{x}_\circ \in \partial\mathcal{R}$. The boundary Green's function is harmonic for all field points within the domain, $\boldsymbol{x} \in \mathcal{R}$, and it equals to

[20] As per page 801 of *Morse and Feshbach* (1953).

the Dirac source when sampled along the boundary, $\boldsymbol{x} \in \partial\mathcal{R}$. By construction, the boundary Green's function incorporates boundary information into the region as part of the Dirichlet solution (8.154). We make further use of the boundary Green's function in VOLUME 4 when studying the diffusion equation and the advection-diffusion equation, in which case the boundary Green's function is referred to as the boundary propagator.

Normalization of the boundary Green's function

Consider the special case of constant boundary data, $\sigma = \sigma_{\mathsf{const}}$, in which case the harmonic function in Table 8.1 is itself a constant, $\psi_B(\boldsymbol{x}) = \sigma_{\mathsf{const}}$. Consequently, the boundary Green's function satisfies

$$-\oint_{\partial\mathcal{R}} \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \,\mathrm{d}\mathcal{S}_\circ = \oint_{\partial\mathcal{R}} G^{\mathsf{bd}}(\boldsymbol{x}|\boldsymbol{x}_\circ) \,\mathrm{d}\mathcal{S}_\circ = 1 \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.164}$$

This is the same normalization derived above in equation (8.162). It is built by placing Dirac delta sources along the boundary, $\boldsymbol{x}_\circ \in \partial\mathcal{R}$, and then area integrating over the boundary area. It holds for any field point placed on the boundary, $\boldsymbol{x} \in \partial\mathcal{R}$. Although derived by considering the special case of constant boundary data, equation (8.164) holds in general since the Green's function is independent of the boundary data.

To help understand the identity (8.164), consider the Green's function to be the steady state temperature or tracer concentration resulting from a Dirac delta source located within the domain interior. The area integrated condition (8.164) acts to maintain the homogeneous Dirichlet boundary condition, $G(\boldsymbol{x}_{\partial\mathcal{R}}|\boldsymbol{x}_\circ) = 0$, for every point $\boldsymbol{x}_{\partial\mathcal{R}} \in \partial\mathcal{R}$. It does so by providing a boundary flux, via the normal gradient, whose area integral precisely cancels the unit positive source from the Dirac delta.

8.9.10 Proving the mean value property for harmonic functions

Recall the discussion in Section 8.3.3 of the mean value property satisfied by harmonic functions. This property says that the value of a harmonic function, ψ, at a point $\boldsymbol{x}$ within an open region of $\mathcal{R}$ equals to the average of ψ taken over the surface of a sphere within $\mathcal{R}$ that is centered at $\boldsymbol{x}$. We here prove the mean value property by using Green's function methods. For this purpose, specialize the Dirichlet solution (8.152) to the case of zero source, $\Lambda = 0$, so that the harmonic field is determined solely through the boundary Green's function and boundary data

$$\psi(\boldsymbol{x}) = -\oint_{\partial\mathcal{R}} \psi(\boldsymbol{x}_\circ) \,\nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \,\mathrm{d}\mathcal{S}_\circ. \tag{8.165}$$

Recall the discussion in Section (8.9.4) whereby the Green's function is decomposed into the sum of the free space Green's function plus a harmonic function that equals to minus the free space Green's function on the boundary. Furthermore, to simplify the algebra, without loss in generality, set the coordinate system so that the field point is at the origin, $\boldsymbol{x} = 0$, and let $\boldsymbol{y}$ be the source point, in which case

$$\psi(0) = -\oint_{\partial\mathcal{R}} \psi(\boldsymbol{y}) \,\nabla_{\boldsymbol{y}} G(0|\boldsymbol{y}) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{y}} \,\mathrm{d}\mathcal{S}_{\boldsymbol{y}}. \tag{8.166}$$

Two dimensions

Making use of the free space Green's function (8.93b) leads to the solution for two space dimensions

$$\psi(0) = \frac{1}{2\pi} \oint_{\partial\mathcal{R}} \psi(\boldsymbol{y}) \, \nabla_{\boldsymbol{y}} \left[\ln |\boldsymbol{y}| - 2\,\pi\, H(0|\boldsymbol{y}) \right] \cdot \hat{\boldsymbol{n}}_{\boldsymbol{y}} \, \mathrm{d}\ell \tag{8.167}$$

Now specialize the region, $\mathcal{R}$, to be a circle with center at $\boldsymbol{x} = 0$, radius R, and make use of polar coordinates so that $\mathrm{d}\ell = R\,\mathrm{d}\vartheta$ (Section 4.3). For this domain the free space Green's function is a constant on the circle's perimeter so that

$$\oint_{\partial\mathcal{R}} \psi(\boldsymbol{y}) \, (\nabla_{\boldsymbol{y}} \ln |\boldsymbol{y}|) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{y}} \, \mathrm{d}\ell = \int_0^{2\pi} \psi(R, \vartheta) \, (\partial_r \ln r) \, R \, \mathrm{d}\vartheta = \int_0^{2\pi} \psi(R, \vartheta) \, \mathrm{d}\vartheta, \tag{8.168}$$

in which case

$$\psi(0) = \frac{1}{2\,\pi} \int_0^{2\pi} \psi(R, \vartheta) \, \mathrm{d}\vartheta - \int_0^{2\pi} \partial_r H(0|R, \vartheta) \, \mathrm{d}\vartheta. \tag{8.169}$$

The harmonic function, H, satisfies

$$-\nabla^2_{\boldsymbol{y}} H(0|\boldsymbol{y}) = 0 \quad \text{for } |\boldsymbol{y}| < R \quad \text{and} \quad H(0|\boldsymbol{y}) = (1/2\pi) \ln R \quad \text{for } |\boldsymbol{y}| = R. \tag{8.170}$$

The unique solution is given by the constant

$$H(0|\boldsymbol{y}) = (1/2\pi) \ln R, \tag{8.171}$$

which has a zero derivative. We thus find that equation (8.169) reduces to the mean value expression

$$\psi(0) = \frac{1}{2\,\pi} \int_0^{2\pi} \psi(R, \vartheta) \, \mathrm{d}\vartheta = \frac{\int_0^{2\pi} \psi(R, \vartheta) \, \mathrm{d}\vartheta}{\int_0^{2\pi} \mathrm{d}\vartheta}. \tag{8.172}$$

That is, the value of the harmonic function, ψ, at the center of a circle of arbitrary radius, R, equals to its average integrated over the circle's perimeter.

Three dimensions

The expression for three dimensions is given in Exercise 8.8, which follows from the two dimensional arguments yet making use of the three-dimensional free space Green's function (8.93c) and spherical coordinates.

8.10 Poisson equation with Neumann boundaries

Consider the Poisson equation with Neumann boundary conditions

$$-\nabla^2 \psi = \Lambda, \quad \text{for } \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \qquad \hat{\boldsymbol{n}} \cdot \nabla \psi = \Sigma, \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.173}$$

Rather than specifying the value of ψ along the boundary, the Neumann condition specifies the normal derivative. As we see, this slight change in the boundary conditions leads to some rather basic distinctions from the Dirichlet problem.

8.10.1 Constraints on the source and boundary data

As in our discussion of Dirichlet boundary conditions in Section 8.9.1, we can realize a solution to the boundary value problem (8.173) only so long as the constraint (8.129) is satisfied. For Neumann boundary conditions, we specify the normal derivative along the boundary so that the constraint (8.129) is now imposed on the volume source and boundary data[21]

$$\oint_{\partial\mathcal{R}} \Sigma \, \mathrm{d}\mathcal{S} = -\int_{\mathcal{R}} \Lambda \, \mathrm{d}V. \tag{8.174}$$

If the source and boundary data do not satisfy this constraint, then there is no solution to the Poisson problem (8.173). If the Poisson problem arises physically from steady state tracer diffusion, then the constraint (8.174) imposes a balance between the diffusive flux integrated around the boundary (left hand side) with the volume integrated tracer source (right hand side). In the absence of this balance, there is no solution to the Poisson problem thus indicating the presence of transients (e.g., time dependent diffusion). If the Poisson problem arises from Newtonian gravity, then the condition (8.174) means that the area integrated gravitational acceleration specified on the boundary must be consistent with the volume integrated mass source distributed within the domain.

8.10.2 Uniqueness of the solution up to a constant

As for the Dirichlet problem in Section 8.9.2, consider the uniqueness of the solution to the boundary value problem (8.173). Do so by considering two functions, ψ_{a} and ψ_{b}, each satisfying the boundary value problem (8.173) and noting that their difference, $\Psi = \psi_{\mathsf{a}} - \psi_{\mathsf{b}}$, satisfies the homogeneous problem

$$-\nabla^2 \Psi = 0, \quad \text{for } \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \qquad \hat{\boldsymbol{n}} \cdot \nabla \Psi = 0, \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.175}$$

The same arguments we used in Section 8.9.2 lead us to conclude that $\nabla\Psi = 0$. But for Neumann boundaries, $\nabla\Psi = 0$ is consistent with Ψ being an arbitrary spatial constant. We can understand this arbitrariness since the Neumann boundary condition involves a derivative, with the derivative of a constant vanishing. We thus conclude that the solution to the boundary value problem (8.173) is unique up to an arbitrary constant.

8.10.3 Modified Green's function problem

The standard Green's function corresponding to the Poisson boundary value problem (8.173) is the solution to the Poisson equation with a Dirac delta source and a homogeneous Neumann boundary condition

$$-\nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R} \tag{8.176a}$$

$$\hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0, \qquad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.176b}$$

However, there is no solution to the Green's function problem (8.176a)-(8.176b) since the self-consistency condition (8.174) is not satisfied. Namely, an integral over the Dirac delta source leads to unity, which is not consistent with the homogeneous Neumann boundary condition (8.176b).

[21] To be self-contained, we here repeat the discussion of the constraint (8.174) that is also provided in Section 8.3.7.

We are led to consider the modified Green's function that satisfies[22]

$$-\nabla^2_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta(\boldsymbol{x}-\boldsymbol{x}_\circ) - 1/V \quad \text{for } \boldsymbol{x},\boldsymbol{x}_\circ \in \mathcal{R} \tag{8.177a}$$

$$\hat{\boldsymbol{n}}\cdot\nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0, \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.177b}$$

Introducing the region volume to the source in equation (8.177a)

$$V = \int_{\mathcal{R}} \mathrm{d}V, \tag{8.178}$$

ensures that the self-consistency condition (8.174) is satisfied by $\widetilde{G}$

$$-\int_{\mathcal{R}} \nabla^2_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_{\boldsymbol{x}} = -\oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}}_{\boldsymbol{x}}\cdot\nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_{\boldsymbol{x}} = 0 \tag{8.179a}$$

$$\int_{\mathcal{R}} \delta(\boldsymbol{x}-\boldsymbol{x}_\circ)\,\mathrm{d}V - \frac{1}{V}\int_{\mathcal{R}}\mathrm{d}V = 0. \tag{8.179b}$$

Making use of the method from Section 8.9.7, we readily derive an integral expression for the solution, ψ, in terms of the modified Green's function,

$$\psi(\boldsymbol{x}) - \langle\psi\rangle = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\,\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\,\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ, \tag{8.180}$$

where $\Sigma = \hat{\boldsymbol{n}}\cdot\nabla\psi$ is the Neumann boundary data given in the problem statement (8.173), and we introduced the domain average

$$\langle\psi\rangle = \frac{1}{V}\int_{\mathcal{R}} \psi\,\mathrm{d}V. \tag{8.181}$$

As expected, the solution is unspecified up to a constant, here given by the domain average.

8.10.4 Reciprocity of the modified Green's function

The reciprocity condition (8.140) satisfied by the Dirichlet Green's function is a very useful property. What is required to respect this property for the modified Green's function? To answer that question, consider the Green's function problem (8.177a)-(8.177b) for two source points, $\boldsymbol{a}\in\mathcal{R}$ and $\boldsymbol{b}\in\mathcal{R}$,

$$-\nabla^2_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{a}) = \delta(\boldsymbol{x}-\boldsymbol{a}) - 1/V \quad \text{and} \quad \hat{\boldsymbol{n}}\cdot\nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{a}) = 0 \tag{8.182a}$$

$$-\nabla^2_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{b}) = \delta(\boldsymbol{x}-\boldsymbol{b}) - 1/V \quad \text{and} \quad \hat{\boldsymbol{n}}\cdot\nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{b}) = 0. \tag{8.182b}$$

Multiplying equation (8.182a) by $\widetilde{G}(\boldsymbol{x}|\boldsymbol{b})$ and equation (8.182b) by $\widetilde{G}(\boldsymbol{x}|\boldsymbol{a})$, subtracting, then integrating over the domain, and using the homogeneous Neumann boundary conditions leads to

$$\widetilde{G}(\boldsymbol{x}|\boldsymbol{b}) - \widetilde{G}(\boldsymbol{x}|\boldsymbol{a}) = \frac{1}{V}\int_{\mathcal{R}} [\widetilde{G}(\boldsymbol{x}|\boldsymbol{b}) - \widetilde{G}(\boldsymbol{x}|\boldsymbol{a})]\,\mathrm{d}V. \tag{8.183}$$

[22] See exercise 6.9 of *Greenberg* (1971), exercise 6.46 of *Stakgold* (2000a), or exercise 8.10 of *Duchateau and Zachmann* (1986). *Hildebrand* (1976) in his Section 11.11 refers to the solution of equation (8.177a) (8.177b) as *Hilbert's function*. *Kellogg* (1953) on his page 246 refers to $G^{(1)}$ as the Green's function of the second kind. For simplicity in nomenclature we refer to $\widetilde{G}$ as the modified Green's function. Both *Hildebrand* (1976) and Section 1.10 of *Jackson* (1975) consider the complement modification of the Green's function, whereby the Neumann boundary condition is not homogeneous whereas the source remains a Dirac delta.

Reciprocity holds if the modified Green's function has a constant integral over the domain,

$$\int_{\mathcal{R}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V = \text{constant } \forall\, \boldsymbol{x}_\circ \in \mathcal{R}, \tag{8.184}$$

with zero a convenient constant that we use in the following. In words, we see that if the domain integral of the modified Green's function is the independent of where the Dirac delta is placed, then the modified Green's function satisfies the reciprocity condition

$$\int_{\mathcal{R}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V = 0 \Longrightarrow \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \widetilde{G}(\boldsymbol{x}_\circ|\boldsymbol{x}). \tag{8.185}$$

8.10.5 Decomposing the modified Green's function

Motivated by the decomposition (8.135) for the Dirichlet Green's function, we linearly decompose the modified Green's function as the sum of the free space Green's function, $\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)$, plus another function

$$\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) + \widetilde{H}(\boldsymbol{x}|\boldsymbol{x}_\circ), \tag{8.186}$$

where

$$-\nabla^2_{\boldsymbol{x}} \widetilde{H}(\boldsymbol{x}|\boldsymbol{x}_\circ) = -1/V \qquad \text{for} \quad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R} \tag{8.187a}$$

$$-\hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} \widetilde{H}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) \quad \text{for} \quad \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.187b}$$

This decomposition is self-consistent since the free space Green's function satisfies the global boundary constraint

$$\oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_{\boldsymbol{x}} = -1, \tag{8.188}$$

and likewise, equation (8.187a) means that

$$\oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla_{\boldsymbol{x}} \widetilde{H}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_{\boldsymbol{x}} = 1. \tag{8.189}$$

Hence, the local boundary condition (8.187b) is consistent with these two global boundary constraints.

8.10.6 Verifying the solution

It is a useful exercise to verify that the solution (8.180) indeed solves the Poisson boundary value problem with Neumann boundary conditions (equation (8.173)). First consider a point in the interior of the region, $\boldsymbol{x} \in \mathcal{R}$, and apply $-\nabla^2_{\boldsymbol{x}}$ to equation (8.180) to find

$$-\nabla^2_{\boldsymbol{x}} \psi(\boldsymbol{x}) = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, [-\nabla^2_{\boldsymbol{x}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)]\, \mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\, [-\nabla^2_{\boldsymbol{x}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)]\, \mathrm{d}\mathcal{S}_\circ \tag{8.190a}$$

$$= \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, [\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) - 1/V]\, \mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\, [\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) - 1/V]\, \mathrm{d}\mathcal{S}_\circ \tag{8.190b}$$

$$= \Lambda(\boldsymbol{x}) - \frac{1}{V}\left[\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ\right] + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ. \tag{8.190c}$$

For field points, $\boldsymbol{x} \in \mathcal{R}$, within the domain interior, the boundary integral with the Dirac delta never fires since $\boldsymbol{x} \notin \partial\mathcal{R}$, so that $\oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ$ is zero. Furthermore, the $1/V$ term drops out due to the consistency condition (8.174). We have thus verified that $-\nabla^2\psi = \Lambda$ holds for field points in the domain interior.

To verify that the boundary condition is met, act with the gradient, $\nabla_{\boldsymbol{x}}$, on the solution (8.180)

$$\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}) = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\,[\nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)]\,\mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\,[\nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)]\,\mathrm{d}\mathcal{S}_\circ. \tag{8.191}$$

Now move the field point onto the boundary, $\boldsymbol{x} \to \boldsymbol{x}_{\boldsymbol{x}_{\partial\mathcal{R}}} \in \partial\mathcal{R}$, and project the gradient onto the unit normal direction, $\hat{\boldsymbol{n}}_{\boldsymbol{x}}$, at the point $\boldsymbol{x}_{\partial\mathcal{R}}$. This projection leads to $\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}_{\boldsymbol{x}_{\partial\mathcal{R}}}) = \Sigma(\boldsymbol{x}_{\boldsymbol{x}_{\partial\mathcal{R}}})$ on the left hand side, and it annihilates the volume term on the right hand side since $\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}_{\boldsymbol{x}_{\partial\mathcal{R}}}|\boldsymbol{x}_\circ) = 0$. We are thus left with

$$\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}) = \Sigma(\boldsymbol{x}) = \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\,[\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)]\,\mathrm{d}\mathcal{S}_\circ \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.192}$$

We emphasize that both arguments of the Green's function are on the boundary, $\boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}$, with the Dirac source point, $\boldsymbol{x}_\circ$, integrated around the boundary whereas the field point, $\boldsymbol{x}$, is an arbitrary point on the boundary. Equation (8.160) is the analogous integral equation for the solution with Dirichlet boundary conditions. We are ensured a solution to the integral equation (8.192) if the modified Green's function satisfies

$$\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}, \tag{8.193}$$

which takes on the integral expression

$$\oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ = \oint_{\partial\mathcal{R}} \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.194}$$

Table 8.2 compares the boundary properties of the Green's functions for the Dirichlet and Neumann problems.

BOUNDARY CONDITION	GREEN'S FUNCTION PROPERTY	EQUATION
Dirichlet	$\hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \cdot \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ) = -\delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ)$	(8.161)
Neumann	$\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ)$	(8.193)

TABLE 8.2: Comparing the boundary normal derivatives for the Poisson equation Green's function with Dirichlet and Neumann boundary conditions. Each point is on the boundary, $\boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}$, and the Dirac delta is two-dimensional so it has dimensions of inverse area. The properties satisfied by these Green's functions are realized by first placing the source point, $\boldsymbol{x}_\circ$, on the boundary and then moving the field point to the boundary, $\boldsymbol{x} \to \boldsymbol{x}_{\partial\mathcal{R}} \in \partial\mathcal{R}$. For the Dirichlet condition, the normal derivative is computed at the source point, whereas the Neumann condition computes the normal derivative at the field point.

8.10.7 Extracting a harmonic portion to the solution

The analysis in Section 8.10.6 reveals that we can write the Neumann solution (8.180) in the form

$$\psi(\boldsymbol{x}) - \langle\psi\rangle = \underbrace{\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\,\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ}_{\psi^{\Lambda}} + \underbrace{\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\,\widetilde{H}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\,\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ}_{\psi^{\mathsf{harmonic}}}. \tag{8.195}$$

The function, ψ^{Λ}, is directly computable since both the source function, Λ, and free space Green's function, $\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)$, are known. The Neumann boundary value problem (8.173) has thus been reduced to finding the harmonic function, ψ^{harmonic}, which satisfies the Neumann boundary value problem

$$-\nabla^2\psi^{\mathsf{harmonic}}(\boldsymbol{x}) = 0 \qquad \text{for } \boldsymbol{x} \in \mathcal{R} \tag{8.196a}$$

$$\hat{\boldsymbol{n}} \cdot \nabla\psi^{\mathsf{harmonic}}(\boldsymbol{x}) = \Sigma(\boldsymbol{x}) - \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ \qquad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.196b}$$

We verify that the boundary value problem (8.196a)-(8.196b) is self-consistent by integrating the boundary condition (8.196b) around the domain boundary

$$\int_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla\psi^{\mathsf{harmonic}}(\boldsymbol{x})\, \mathrm{d}\mathcal{S}_{\boldsymbol{x}} = \int_{\partial\mathcal{R}} \Sigma(\boldsymbol{x})\, \mathrm{d}\mathcal{S}_{\boldsymbol{x}} - \int_{\partial\mathcal{R}} \left[\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ\right] \mathrm{d}\mathcal{S}_{\boldsymbol{x}}. \tag{8.197}$$

In the final term we can interchange the surface integral over $\boldsymbol{x}$ with the volume integral over $\boldsymbol{x}_\circ$ to have

$$\int_{\partial\mathcal{R}} \left[\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ\right] \mathrm{d}\mathcal{S}_{\boldsymbol{x}} = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ) \left[\int_{\partial\mathcal{R}} \hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{x}}\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_{\boldsymbol{x}}\right] \mathrm{d}V_\circ \tag{8.198a}$$

$$= -\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ, \tag{8.198b}$$

where the second equality made use of the identity (8.188) satisfied by the free space Green's function. Making use of the constraint (8.174) satisfied by the source and boundary data leads to

$$\int_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \nabla\psi^{\mathsf{harmonic}}(\boldsymbol{x})\, \mathrm{d}\mathcal{S}_{\boldsymbol{x}} = 0, \tag{8.199}$$

as required for a harmonic function.

8.10.8 The Dirac delta sheet and boundary data

The Dirichlet solution (8.152) has the Green's function in the volume integral but its normal gradient in the surface integral. In contrast, the Neumann solution (8.180) involves the modified Green's function, $\widetilde{G}$, within both the volume integral and the surface integral. For the Neumann problem we can thus consider a slightly reformulation for how the boundary data is incorporated. This reformulation yields equivalent results, but it can be more useful for some purposes and it is commonly pursued in geophysical fluid mechanical applications.

To motivate this reformulation, assume the only nontrivial Neumann boundary data appears along the flat plane surface, $z = z_{\mathsf{b}}$, with all other boundaries maintaining the homogeneous boundary condition, $\hat{\boldsymbol{n}} \cdot \nabla\psi = 0$. In this special case the Neumann solution (8.180) takes the form

$$\psi(\boldsymbol{x}) - \langle\psi\rangle = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ + \int_{z=z_{\mathsf{b}}} \Sigma(x_\circ, y_\circ)\, \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}x_\circ\, \mathrm{d}y_\circ \tag{8.200a}$$

$$= \int_{\mathcal{R}} [\Lambda(\boldsymbol{x}_\circ) + \Sigma(x_\circ, y_\circ)\, \delta(z_\circ - z_{\mathsf{b}})]\, \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ \tag{8.200b}$$

$$\equiv \int_{\mathcal{R}} \Lambda^*(\boldsymbol{x}_\circ)\, \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ. \tag{8.200c}$$

These manipulations have absorbed the non-homogeneous Neumann boundary condition into a modified source, $\Lambda^*(\boldsymbol{x})$. We are thus led to two equivalent formulations for the Poisson boundary value problem with Neumann conditions. The first is given by equation (8.173), whereby ψ is the solution to the Poisson equation with source, Λ, and with inhomogeneous Neumann boundary data, Σ. The second formulation considers ψ to be the solution to Poisson's equation with homogeneous Neumann boundary conditions yet with a modified source function

$$\Lambda^*(\boldsymbol{x}) = \Lambda(\boldsymbol{x}) + \Sigma(x,y)\,\delta(z - z_{\mathsf{b}}) = \Lambda(\boldsymbol{x}) + \partial_z \psi(\boldsymbol{x})\,\delta(z - z_{\mathsf{b}}). \tag{8.201}$$

As a check, note that the physical dimensionality of $\Lambda^*(\boldsymbol{x})$ is indeed correct since the Dirac delta, $\delta(z - z_{\mathsf{b}})$, has dimensions of inverse length. We interpret the term, $\Sigma(x,y)\,\delta(z - z_{\mathsf{b}})$, as a flux or Dirac delta sheet that sits just inside the boundary (at $z = z_{\mathsf{b}} - \epsilon$ with $\epsilon \to 0$). In this manner, the boundary data is incorporated to the volume source rather than in the boundary condition. The specific form (8.201) can be generalized to the expression

$$\Lambda^*(\boldsymbol{x}) = \Lambda(\boldsymbol{x}) + \Sigma(\boldsymbol{x})\,\delta[\hat{\boldsymbol{n}} \cdot (\boldsymbol{x} - \boldsymbol{x}_{\partial\mathcal{R}})] = \Lambda(\boldsymbol{x}) + \hat{\boldsymbol{n}} \cdot \nabla\psi(\boldsymbol{x})\,\delta[\hat{\boldsymbol{n}} \cdot (\boldsymbol{x} - \boldsymbol{x}_{\partial\mathcal{R}})], \tag{8.202}$$

where the argument to the Dirac delta picks out the coordinates in the direction of the outward unit normal. The transformed Neumann problem thus takes the generic form

$$-\nabla^2\psi = \Lambda^*, \quad \text{for } \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \qquad \hat{\boldsymbol{n}} \cdot \nabla\psi = 0, \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.203}$$

Again, the solution to the boundary value problem (8.203) is identical to the solution of the original problem (8.173). We have simply rearranged how the boundary data is handled.

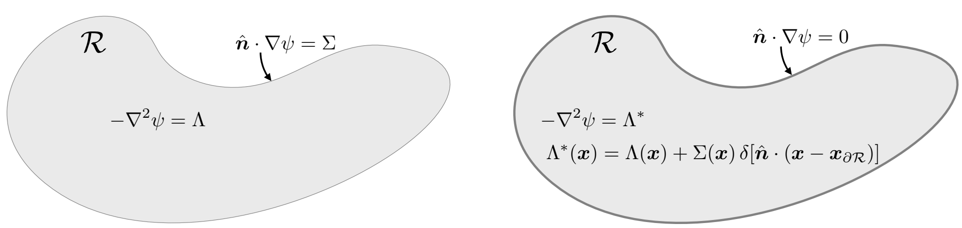

FIGURE 8.5: Use of Neumann boundary conditions for the Poisson boundary value problem allows for two equivalent formulations of how the boundary data is handled. The first, shown in the left, considers the Poisson equation $-\nabla^2\psi = \Lambda$ satisfied within the domain, $\boldsymbol{x} \in \mathcal{R}$, along with boundary data along $\boldsymbol{x} \in \partial\mathcal{R}$, where $\hat{\boldsymbol{n}} \cdot \nabla\psi = \Sigma$. The second formulation, shown on the right, absorbs the boundary data into the domain source via a Dirac delta sheet that is placed infinitesimally close, on the inside, of the boundary. In this approach the Poisson equation has a modified source, $\Lambda^*(\boldsymbol{x}) = \Lambda(\boldsymbol{x}) + \Sigma(\boldsymbol{x})\,\delta[\hat{\boldsymbol{n}} \cdot (\boldsymbol{x} - \boldsymbol{x}_{\partial\mathcal{R}})]$, and the Neumann boundary condition is now homogeneous, $\hat{\boldsymbol{n}} \cdot \nabla\psi = 0$. The delta sheet is here depicted by the slightly thicker line placed around the boundary.

The same boundary data, Σ, is needed for both equations (8.203) and (8.173), so there is nothing fundamentally special or efficient about one formulation or the other. Rather, it is a matter of convenience. For example, the formulation using Λ^* has found some favor in the study of quasi-geostrophic potential vorticity (*Bretherton*, 1966), and we pursue this approach in VOLUME 3. We also make use of a similar construct in VOLUME 4 for studies of boundary buoyancy fluxes crossing the ocean surface.

8.11 Method of images

In this section we illustrate the method of images, which allows us to derive Green's functions for highly symmetric geometries by making use of selected placements of free-space Green's functions. The power of the method relies on the uniqueness theorems derived in Section 8.9.2 for the Dirichlet boundary value problem and Section 8.10.2 for the Neumann problem.

8.11.1 Dirichlet Green's function for the sphere

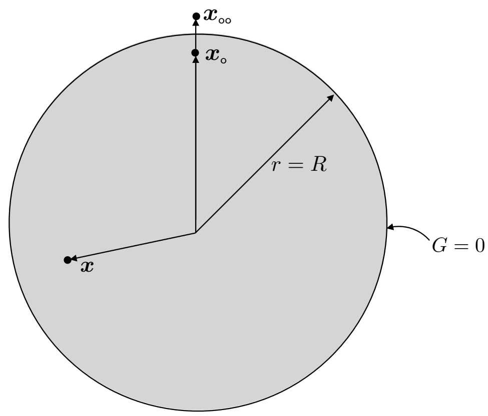

FIGURE 8.6: Illustrating the method of images for a sphere, in which the Green's function is singular at the source point, $\boldsymbol{x}_{\circ}$, located within the sphere. The image source point at $\boldsymbol{x}_{\circ\circ}$ is located outside of the sphere. It combines with the interior source to yield a Green's function that vanishes when the field point, $\boldsymbol{x}$, is evaluated on the boundary, $r = R$.

The method is readily described by working through an example, such as illustrated in Figure 8.6. Here, we consider the Dirichlet Green's function for the Poisson equation

$$-\nabla^2_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = \delta(\boldsymbol{x} - \boldsymbol{x}_{\circ}) \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_{\circ} \in \mathcal{R} \quad \text{and} \quad G(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = 0 \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}, \tag{8.204}$$

where the region, $\mathcal{R}$, is the sphere of radius R, so that

$$\mathcal{R} = \{r = |\boldsymbol{x}| < R\}. \tag{8.205}$$

In the absence of the sphere, the Green's function equals to the free space Green's function (8.93c)

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = (4\,\pi\,|\boldsymbol{x} - \boldsymbol{x}_{\circ}|)^{-1} \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_{\circ} \in \mathbb{R}^3, \tag{8.206}$$

which is harmonic everywhere except at the source point, $\boldsymbol{x} = \boldsymbol{x}_{\circ}$. In the presence of the sphere, we assume the source is placed within the sphere, $|\boldsymbol{x}_{\circ}| < R$, and we make use of the decomposition from Section 8.9.4 to write the Green's function as

$$G(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_{\circ}) + H(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = (4\,\pi\,|\boldsymbol{x} - \boldsymbol{x}_{\circ}|)^{-1} + H(\boldsymbol{x}|\boldsymbol{x}_{\circ}). \tag{8.207}$$

Here, H is harmonic within the sphere, and it is constructed to satisfy the homogeneous Dirichlet boundary condition

$$G(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_{\circ}) + H(\boldsymbol{x}|\boldsymbol{x}_{\circ}) = 0 \quad \text{for } |\boldsymbol{x}| = R. \tag{8.208}$$

We furthermore write H in the form of a free space Green's function, but with its source point located outside of the sphere (so that $\nabla^2 H = 0$ inside the sphere), and with its strength set to ensure $G(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0$ on the sphere boundary. For this purpose we write

$$H(\boldsymbol{x}|\boldsymbol{x}_\circ) = \beta\,(4\,\pi\,|\boldsymbol{x} - \boldsymbol{x}_{\circ\circ}|)^{-1}, \tag{8.209}$$

with β a constant and $|\boldsymbol{x}_{\circ\circ}| > R$. Furthermore, symmetry suggests that we place $\boldsymbol{x}_{\circ\circ}$ along the same radial vector as $\boldsymbol{x}_\circ$,

$$\boldsymbol{x}_{\circ\circ} = \boldsymbol{x}_\circ\,|\boldsymbol{x}_{\circ\circ}|/|\boldsymbol{x}_\circ|, \tag{8.210}$$

so that the Green's function is

$$G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{1}{4\,\pi\,|\boldsymbol{x} - \boldsymbol{x}_\circ|} + \frac{\beta}{4\,\pi\,|\boldsymbol{x} - \boldsymbol{x}_\circ\,(|\boldsymbol{x}_{\circ\circ}|/|\boldsymbol{x}_\circ|)|}. \tag{8.211}$$

Making use of spherical coordinates with the origin at the center of the sphere, we have each of the position vectors represented radial vectors, so that

$$G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{1}{4\,\pi\,|r - |\boldsymbol{x}_\circ||} + \frac{\beta}{4\,\pi\,|r - |\boldsymbol{x}_{\circ\circ}||}. \tag{8.212}$$

In this form we see that by setting

$$|\boldsymbol{x}_{\circ\circ}| = R^2/|\boldsymbol{x}_\circ| \quad \text{and} \quad \beta = -R/|\boldsymbol{x}_\circ| = -|\boldsymbol{x}_{\circ\circ}|/R \Longrightarrow \boldsymbol{x}_{\circ\circ} = \boldsymbol{x}_\circ\,(R/|\boldsymbol{x}_\circ|)^2 \tag{8.213}$$

we are led to the Green's function

$$4\,\pi\,G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{1}{|\boldsymbol{x} - \boldsymbol{x}_\circ|} - \frac{R/|\boldsymbol{x}_\circ|}{|\boldsymbol{x} - \boldsymbol{x}_\circ\,(R/|\boldsymbol{x}_\circ|)^2|}, \tag{8.214}$$

which properly vanishes for fields points on the sphere's boundary, $\boldsymbol{x} = R\,\hat{\boldsymbol{r}}$. Evidently, the Green's function is built from two point charges with opposite sign and distinct magnitudes.

8.11.2 Green's functions for the half-plane

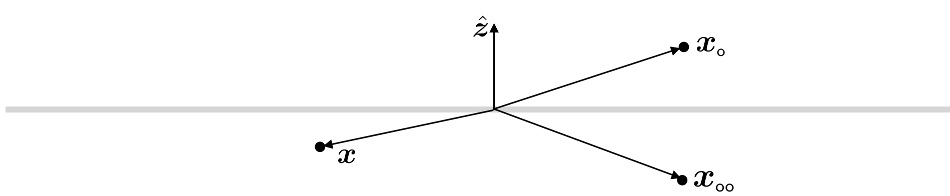

FIGURE 8.7: Illustrating the method of images for the half-plane, in which the Green's function is singular at the source point, $\boldsymbol{x}_\circ = \hat{\boldsymbol{x}}\,x_\circ + \hat{\boldsymbol{y}}\,y_\circ + \hat{\boldsymbol{z}}\,z_\circ$, located in the upper half-plane ($z_\circ > 0$). The image source point at $\boldsymbol{x}_{\circ\circ} = \hat{\boldsymbol{x}}\,x_\circ + \hat{\boldsymbol{y}}\,y_\circ - \hat{\boldsymbol{z}}\,z_\circ$ is located in the lower half-plane. For the Dirichlet problem the sources have the opposite sign, thus ensuring that the Green's function vanishes for field points, $\boldsymbol{x}$, evaluated on the boundary at $z = 0$. For the Neumann problem, both sources have the same sign, which renders $\partial_z G = 0$ for field points on the boundary at $z = 0$.

The method of images involves an element of intuition about the physics of point charges and their ability to satisfy boundary conditions in selected geometries. Here, we seek the

Dirichlet Green's function on the upper half-plane, $z > 0$, which is immediately found to be

$$4\,\pi\,G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{1}{|\boldsymbol{x} - \boldsymbol{x}_\circ|} - \frac{1}{|\boldsymbol{x} + \boldsymbol{x}_{\circ\circ}|}, \tag{8.215}$$

with the upper half-plane source at $\boldsymbol{x}_\circ$ so that

$$\boldsymbol{x}_{\circ\circ} = \hat{\boldsymbol{x}}\,x_\circ + \hat{\boldsymbol{y}}\,y_\circ - \hat{\boldsymbol{z}}\,z_\circ \tag{8.216}$$

is in the lower half-plane. It is straightforward to verify that for field points at $z = 0$ then the Green's function properly vanishes. Likewise, the Neumann Green's function is given by

$$4\,\pi\,G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{1}{|\boldsymbol{x} - \boldsymbol{x}_\circ|} + \frac{1}{|\boldsymbol{x} - \boldsymbol{x}_{\circ\circ}|}, \tag{8.217}$$

which satisfies

$$\partial_z G = 0 \quad \text{at } z = 0. \tag{8.218}$$

8.11.3 Further study

The method of images is commonly used in classical electromagnetic field theory, with *Jackson* (1975) providing many applications. Section 30 of *Dennery and Krzywicki* (1967) presents a pedagogical introduction to the method, and *Greenberg* (1971) makes extensive use of the method throughout his book.

8.12 Eigenfunctions with Dirichlet boundaries

Self-adjoint linear operators have orthogonal eigenfunctions with real eigenvalues that form a complete basis for functions satisfying the same boundary conditions as the eigenfunctions. This is a very powerful property of such functions, making them of great use for physical problems. In this section we explore eigenfunctions of the generalized Laplacian differential operator with Dirichlet boundary conditions. We use the complete set of eigenfunctions to form a spectral decomposition of the Green's function satisfying homogeneous Dirichlet boundary conditions. Doing so offers a practical method to compute the Green's function, particularly in domains where the method of images is not suited. It also offers an intuitive depiction of the Green's function and its connection to the linear operator. We consider the Neumann problem in Section 8.13, which requires some special considerations given its nontrivial null space. An eigenfunction decomposition of the Laplace operator is of great use for deriving the Green's function for the diffusion equation (Chapter 9) and wave equation (Chapter 10), both of which contain the Laplace operator.

8.12.1 Setup of the eigenvalue problem

Building from the material in Section 8.4, we consider the following eigenvalue problem with homogeneous Dirichlet boundary conditions

$$-\nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi) = \lambda\,\Phi, \quad \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \quad \Phi = 0, \quad \boldsymbol{x} \in \partial\mathcal{R}, \tag{8.219}$$

with $\boldsymbol{K}(\boldsymbol{x})$ a symmetric tensor so that $K_{ij} = K_{ji}$, and with $\mathcal{R}$ a simply connected and bounded domain. In a geophysical context, the generalized Laplacian differential operator,

$$\mathcal{L}_{\mathsf{glap}} = -\nabla \cdot (\boldsymbol{K} \cdot \nabla), \tag{8.220}$$

arises when considering steady solutions to the diffusion equation (Chapter 9), in which $\boldsymbol{K}$ is the kinematic diffusion tensor with dimensions $\mathrm{L}^2\ \mathrm{T}^{-1}$ so that $\mathcal{L}_{\mathsf{glap}}$ has dimensions T^{-1}. The constant, λ, in equation (8.219) is the eigenvalue corresponding to the eigenfunction, Φ. The linear operator is self-adjoint with the homogeneous Dirichlet boundary conditions. Furthermore, for bounded domains the eigenvalues are discrete ("quantized") and form an infinite set, as we saw for the rectangle in Section 8.4.1 and surface of the sphere in Section 8.5.

8.12.2 Orthogonality of the eigenfunctions

Let λ_a and λ_b be eigenvalues, with a and b integers that label the discrete eigenvalues, and let Φ_a and Φ_b be the corresponding eigenfunctions. The discrete labels, a, b, are shorthand for double indices for two-dimensional problems, as in Section 8.4.1 for the rectangle and Section 8.5 for the surface of the sphere, or triple subscripts for three-dimensional domains.

Multiply Φ_b times the Φ_a eigenvalue equation, and vice versa, to render

$$\Phi_b \nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_a) + \lambda_a\, \Phi_b\, \Phi_a = 0 \tag{8.221a}$$
$$\Phi_a \nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_b) + \lambda_b\, \Phi_a\, \Phi_b = 0, \tag{8.221b}$$

and then take their difference

$$\Phi_b \nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_a) - \Phi_a \nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_b) + (\lambda_a - \lambda_b)\, \Phi_b\, \Phi_a = 0. \tag{8.222}$$

Now make use of the identities

$$\Phi_b \nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_a) = \nabla \cdot (\Phi_b\, \boldsymbol{K} \cdot \nabla \Phi_a) - \nabla \Phi_b \cdot \boldsymbol{K} \cdot \nabla \Phi_a \tag{8.223a}$$
$$\Phi_a \nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_b) = \nabla \cdot (\Phi_a\, \boldsymbol{K} \cdot \nabla \Phi_b) - \nabla \Phi_a \cdot \boldsymbol{K} \cdot \nabla \Phi_b, \tag{8.223b}$$

and note that since $\boldsymbol{K}$ is symmetric then

$$\nabla \Phi_b \cdot \boldsymbol{K} \cdot \nabla \Phi_a = \nabla \Phi_a \cdot \boldsymbol{K} \cdot \nabla \Phi_b, \tag{8.224}$$

so that equation (8.222) can be written

$$\nabla \cdot (\Phi_b\, \boldsymbol{K} \cdot \nabla \Phi_a - \Phi_a\, \boldsymbol{K} \cdot \nabla \Phi_b) + (\lambda_a - \lambda_b)\, \Phi_b\, \Phi_a = 0. \tag{8.225}$$

Now integrate over the domain and make use of the divergence theorem to convert the total divergence term to a boundary integral. The homogeneous Dirichlet boundary conditions ensure the boundary contributions vanish, in which case we are led to

$$(\lambda_a - \lambda_b) \int_{\mathcal{R}} \Phi_b\, \Phi_a\, \mathrm{d}V = 0. \tag{8.226}$$

This result establishes the orthogonality property for distinct eigenfunctions

$$\int_{\mathcal{R}} \Phi_b\, \Phi_a\, \mathrm{d}V = 0 \quad \text{if } \lambda_a \neq \lambda_b. \tag{8.227}$$

For convenience in the following, we assume the eigenfunctions are normalized so that

$$\int_{\mathcal{R}} \Phi_b \, \Phi_a \, \mathrm{d}V = \delta_{ab}. \tag{8.228}$$

8.12.3 Positivity of the eigenvalues

Multiply the eigenvalue problem (8.219) by the eigenfunction, Φ, and then rearrange to find

$$\lambda \, \Phi^2 = \nabla\Phi \cdot \boldsymbol{K} \cdot \nabla\Phi - \nabla \cdot (\Phi \, \boldsymbol{K} \cdot \nabla\Phi). \tag{8.229}$$

Now integrate over the domain and use the homogeneous Dirichlet boundary condition, in which

$$\lambda \int_{\mathcal{R}} \Phi^2 \, \mathrm{d}V = \int_{\mathcal{R}} \nabla\Phi \cdot \boldsymbol{K} \cdot \nabla\Phi \, \mathrm{d}V. \tag{8.230}$$

Since $\boldsymbol{K}$ is a symmetric and positive tensor, the right hand side is positive, which then means that the eigenvalues are positive

$$\lambda = \frac{\int_{\mathcal{R}} \nabla\Phi \cdot \boldsymbol{K} \cdot \nabla\Phi \, \mathrm{d}V}{\int_{\mathcal{R}} \Phi^2 \, \mathrm{d}V} > 0. \tag{8.231}$$

8.12.4 Eigenfunction expansion of the Green's function

Now consider the Green's function for the generalized Laplacian operator with homogeneous Dirichlet boundary conditions

$$-\nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}|\boldsymbol{x}_\circ)] = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \text{for} \quad \boldsymbol{x} \in \mathcal{R} \tag{8.232a}$$

$$G(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0 \quad \text{for} \quad \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.232b}$$

Because the linear operator is self-adjoint on the bounded domain, $\mathcal{R}$, with Dirichlet boundary conditions, the eigenfunctions, Φ_a, form a complete orthonormal basis for square-integrable functions that vanish on the domain boundary. Hence, any such function, including the Green's function, admits an eigenfunction decomposition

$$G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \sum_a C_a(\boldsymbol{x}_\circ) \, \Phi_a(\boldsymbol{x}), \tag{8.233}$$

Determining the expansion coefficients

To determine the expansion coefficients, $C_a(\boldsymbol{x}_\circ)$, apply the linear differential operator, $-\nabla_{\boldsymbol{x}} \cdot (\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}})$, to equation (8.233)

$$\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) = \sum_a C_a(\boldsymbol{x}_\circ) \, \lambda_a \, \Phi_a(\boldsymbol{x}), \tag{8.234}$$

where we used the Green's function equation (8.232a) for the left hand side, and the eigenvalue equation (8.219) for the right hand side. Now multiply by the eigenfunction, $\Phi_b(\boldsymbol{x})$, and integrate over the domain

$$\int_{\mathcal{R}} \Phi_b(\boldsymbol{x}) \, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \, \mathrm{d}V_{\boldsymbol{x}} = \int_{\mathcal{R}} \sum_a C_a(\boldsymbol{x}_\circ) \, \lambda_a \, \Phi_a(\boldsymbol{x}) \, \Phi_b(\boldsymbol{x}) \mathrm{d}V_{\boldsymbol{x}}. \tag{8.235}$$

The sifting property of the Dirac delta, and orthonormality (8.228) of the eigenfunctions, lead to

$$\Phi_b(\boldsymbol{x}_\circ) = \lambda_b \, C_b(\boldsymbol{x}_\circ) \Longrightarrow C_b(\boldsymbol{x}_\circ) = \Phi_b(\boldsymbol{x}_\circ)/\lambda_b, \tag{8.236}$$

where the second equality holds since there are no zero eigenmodes for the Dirichlet problem.

Physical dimensions

In working through the derivations in this section, it is useful to be explicit with the physical dimensions of various quantities as a means to double-check the correctness of the formulations. Here are the dimensions in the equations presented thus far:

$$[\nabla] = \mathrm{L}^{-1} \quad [\boldsymbol{K}] = \mathrm{L}^2 \, \mathrm{T}^{-1} \quad [\lambda] = \mathrm{T}^{-1} \quad [\Phi] = \mathrm{L}^{-3/2} \quad [G] = \mathrm{T} \, \mathrm{L}^{-3}. \tag{8.237}$$

The normalization condition (8.228) implies the dimensions for the eigenfunction, which in turn leads to the dimension for the eigenvalue. The Green's function equation (8.232a) determines the dimensions for the Green's function.

Green's function and completeness

Making use of the expansion coefficients (8.236) in the Green's function spectral decomposition (8.233) leads to

$$G(\boldsymbol{x}|\boldsymbol{x}_\circ) = \sum_a \frac{\Phi_a(\boldsymbol{x}_\circ) \, \Phi_a(\boldsymbol{x})}{\lambda_a}, \tag{8.238}$$

which is an elegant manifestation of the reciprocity condition from Section 8.9.6. Similarly, making use of the expansion coefficients in equation (8.234) renders the completeness relation for the Dirichlet eigenfunctions

$$\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) = \sum_a \Phi_a(\boldsymbol{x}_\circ) \, \Phi_a(\boldsymbol{x}). \tag{8.239}$$

Note that in equation (8.237) we noted that the eigenfunctions have dimensions of $[\Phi] = \mathrm{L}^{-3/2}$, with the completeness relation (8.239) rendering dimensions of inverse volume for the Dirac delta, $[\delta] = \mathrm{L}^{-3}$, which are indeed the proper dimensions.

8.12.5 Dirichlet boundary value problem

Now consider the Dirichlet boundary value problem

$$-\nabla \cdot (\boldsymbol{K} \cdot \nabla \psi) = \Lambda, \quad \text{for } \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \qquad \psi = \sigma, \quad \text{for } \boldsymbol{x} \in \partial \mathcal{R}, \tag{8.240}$$

with the source function and boundary data having physical dimensions

$$[\Lambda] = [\psi] \, \mathrm{T}^{-1} \quad [\sigma] = [\psi]. \tag{8.241}$$

It is straightforward to generalize the Green's function solution (8.152) to now include the diffusion tensor

$$\psi(\boldsymbol{x}) = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ) \, G(\boldsymbol{x}|\boldsymbol{x}_\circ) \, \mathrm{d}V_\circ - \oint_{\partial \mathcal{R}} [\sigma(\boldsymbol{x}_\circ) \, \boldsymbol{K}(\boldsymbol{x}_\circ) \cdot \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}|\boldsymbol{x}_\circ)] \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \, \mathrm{d}\mathcal{S}_\circ. \tag{8.242}$$

Making use of the eigenfunction expansion (8.238) within this Green's function solution renders

$$\psi(\boldsymbol{x}) = \sum_a \frac{\Phi_a(\boldsymbol{x})}{\lambda_a} \left[\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \Phi_a(\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ - \oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ)\, \boldsymbol{K}(\boldsymbol{x}_\circ) \cdot \nabla_{\boldsymbol{x}_\circ} \Phi_a(\boldsymbol{x}_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}\, \mathrm{d}\mathcal{S}_\circ \right]. \quad (8.243)$$

This equation is the spectral decomposition of ψ in terms of the projection of the volume source, Λ, onto the eigenfunctions (first right hand side term), plus the projection of the boundary data onto the boundary normal gradient of the eigenfunctions as contracted with the diffusion tensor. As a check on the formalism, consider the special case of $\Lambda = \lambda_b\, \Phi_b$, in which case the orthonormality condition (8.228) leads to

$$\psi(\boldsymbol{x}) = \Phi_b(\boldsymbol{x}) - \sum_a \frac{\Phi_a(\boldsymbol{x})}{\lambda_a} \left[\oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ)\, \boldsymbol{K}(\boldsymbol{x}_\circ) \cdot \nabla_{\boldsymbol{x}_\circ} \Phi_a(\boldsymbol{x}_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}\, \mathrm{d}\mathcal{S}_\circ \right]. \quad (8.244)$$

With homogeneous boundary data ($\sigma = 0$), then $\psi = \Phi_b$, as expected.

8.13 Eigenfunctions with Neumann boundaries

We follow the procedure from Section 8.12, applied now to the Neumann problem[23]

$$-\nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi) = \lambda\, \Phi, \quad \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \quad \hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla \Phi = 0, \quad \boldsymbol{x} \in \partial\mathcal{R}. \quad (8.245)$$

The Neumann eigenvalue problem admits a constant (nonzero) eigenfunction,

$$\Phi_\circ = k \quad \text{and} \quad \lambda_\circ = 0, \quad (8.246)$$

with k an arbitrary constant. As a result, the linear operator cannot be inverted since it has a nontrivial null space. Correspondingly, there is no Green's function for the Neumann problem. Even so, as seen in Section 8.10, we can derive a pseudo-inverse to produce a modified Green's function, and we develop the details in this section from the perspective of an eigenfunction decomposition.

Observe that for the case of non-zero modes, the volume integral of the Neumann eigenfunction vanishes as a result of the homogeneous Neumann boundary condition

$$\int_{\mathcal{R}} \lambda\, \Phi\, \mathrm{d}V = -\int_{\mathcal{R}} \nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi)\, \mathrm{d}V = -\oint_{\mathcal{R}} \hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla \Phi\, \mathrm{d}\mathcal{S} = 0. \quad (8.247)$$

This property is not satisfied by the zero mode, $\Phi_\circ$, further emphasizing the need for special considerations to handle this mode. Finally, note that equation (8.231) used to deduce that $\lambda > 0$ for the Dirichlet eigenvalues here implies that the Neumann eigenvalues are non-negative

$$\lambda = \frac{\int_{\mathcal{R}} \nabla \Phi \cdot \boldsymbol{K} \cdot \nabla \Phi\, \mathrm{d}V}{\int_{\mathcal{R}} \Phi^2\, \mathrm{d}V} \geq 0, \quad (8.248)$$

with $\lambda_\circ = 0$ for the constant eigenfunction, $\nabla \Phi_\circ = 0$.

[23] To reduce the number of new symbols, we continue to write λ and Φ for the eigenvalue and corresponding eigenfunction, yet in this section they refer to those arising with Neumann boundary conditions rather than Dirichlet boundaries.

8.13.1 Orthogonality of the eigenfunctions

For the Neumann eigenfunctions, we start from the step (8.225) used for the Dirichlet eigenfunctions to write

$$\nabla \cdot (\Phi_b \, \boldsymbol{K} \cdot \nabla \Phi_a - \Phi_a \, \boldsymbol{K} \cdot \nabla \Phi_b) + (\lambda_a - \lambda_b) \, \Phi_b \, \Phi_a = 0. \tag{8.249}$$

Integration over the domain, and use of the homogeneous Neumann boundary conditions, renders

$$(\lambda_a - \lambda_b) \int_{\mathcal{R}} \Phi_b \, \Phi_a \, \mathrm{d}V = 0, \tag{8.250}$$

which means that the eigenfunctions are orthogonal. This property holds even for the zero eigenfunction, $\Phi_\circ = k$, since each of the nonzero eigenfunctions have a zero volume integral as we established in equation (8.247).

8.13.2 Spectral decomposition and the Fredholm alternative

Consider the Neumann boundary value problem

$$-\nabla \cdot (\boldsymbol{K} \cdot \nabla \psi) = \Lambda, \quad \text{for } \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \qquad \hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla \psi = \Sigma, \quad \text{for } \boldsymbol{x} \in \partial \mathcal{R}, \tag{8.251}$$

along with the integral consistency condition

$$-\oint_{\partial \mathcal{R}} \Sigma \, \mathrm{d}\mathcal{S} = \int_{\mathcal{R}} \Lambda \, \mathrm{d}V. \tag{8.252}$$

Solvability and the Fredholm alternative

The Fredholm alternative states that a linear differential equation of the form, $\mathcal{L}\psi = \Lambda$, has a solution only so long as the volume forcing and/or boundary forcing are orthogonal to all functions within the null space of the linear operator's adjoint. Alternatively, we say that if forcing excites a zero eigenmode (here the constant, $\Phi_\circ$), then there is no solution. To understand how the Fredholm alternative works for the self-adjoint Neumann problem, multiply the differential equation (8.251) by an eigenfunction and integrate to find

$$-\oint_{\partial \mathcal{R}} \Phi \, \Sigma \, \mathrm{d}\mathcal{S} + \int_{\mathcal{R}} \nabla \Phi \cdot \boldsymbol{K} \cdot \nabla \psi \, \mathrm{d}V = \int_{\mathcal{R}} \Phi \, \Lambda \, \mathrm{d}V, \tag{8.253}$$

where we made use of the Neumann boundary conditions. Now the zero mode, with $\nabla \Phi_\circ = 0$, spans the operator's null space, and it furthermore has $\int_{\mathcal{R}} \nabla \Phi_\circ \cdot \boldsymbol{K} \cdot \nabla \psi \, \mathrm{d}V = 0$. So insisting that the volume and boundary forcings are orthogonal to the null space (Fredholm's alternative) is equivalent to the self-consistency condition (8.252). That is, when the self-consistency condition is satisfied, the volume and boundary forcing live within the range of the operator. In turn, the pseudo-inverse (i.e., the modified Green's function) can be applied to construct a solution to the boundary value problem (8.251), which is unique up to an additive constant.

Modified Green's function solution

The modified Green's function problem (Section 8.10.3) corresponding to equation (8.251) is given by

$$-\nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)] = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) - 1/V \quad \text{for } \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R} \tag{8.254a}$$

$$\hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = 0 \qquad \text{for } \boldsymbol{x} \in \partial\mathcal{R} \tag{8.254b}$$

$$\int_{\mathcal{R}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V = 0 \qquad \text{for reciprocity: } \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \widetilde{G}(\boldsymbol{x}_\circ|\boldsymbol{x}), \tag{8.254c}$$

with an integral solution to the boundary value problem (8.251) given by

$$\psi(\boldsymbol{x}) - \langle\psi\rangle = \int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\,\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\,\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}\mathcal{S}_\circ. \tag{8.255}$$

As a check, we verify that the following physical dimensions are consistent with the above formulas

$$[\Sigma] = [\psi]\ \mathrm{L}\ \mathrm{T}^{-1} \quad [\widetilde{G}] = \mathrm{T}\ \mathrm{L}^{-3}. \tag{8.256}$$

Spectral decomposition of the Dirac delta and the modified Green's function

Completeness of the Neumann eigenfunctions, Φ_a (including the zero mode) ensures that we have the following spectral decomposition of the Dirac delta[24]

$$\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) = \sum_{a=0}^{\infty} \Phi_a(\boldsymbol{x})\,\Phi_a(\boldsymbol{x}_\circ) = \Phi_\circ\,\Phi_\circ + \sum_{a=1}^{\infty} \Phi_a(\boldsymbol{x})\,\Phi_a(\boldsymbol{x}_\circ). \tag{8.257}$$

Since the $a > 0$ Neumann eigenfunctions have zero domain integral according to equation (8.247), we must choose the following normalization of the zero mode

$$\int_{\mathcal{R}} \Phi_\circ\,\Phi_\circ\,\mathrm{d}V = 1 \Longrightarrow \Phi_\circ = V^{-1/2}, \tag{8.258}$$

where $V = \int_{\mathcal{R}} \mathrm{d}V$ is the domain volume. With this normalization, the completeness relation (8.257) becomes

$$\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) = 1/V + \sum_{a=1}^{\infty} \Phi_a(\boldsymbol{x})\,\Phi_a(\boldsymbol{x}_\circ). \tag{8.259}$$

The spectral decomposition of the modified Green's function is given by

$$\widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \sum_{a=1}^{\infty} \frac{\Phi_a(\boldsymbol{x}_\circ)\,\Phi_a(\boldsymbol{x})}{\lambda_a}, \tag{8.260}$$

where it is notable that the summation does not include the zero eigenmode. To verify the consistency of this expression, apply the linear operator, $-\nabla \cdot (\boldsymbol{K} \cdot \nabla)$, and make use of the modified Green's function equation (8.254a) for the left hand side, and the eigenfunction equation (8.245) to the right hand side, in which case we recover the statement of completeness given by equation (8.259). Furthermore, note that equation (8.254c) is satisfied by the spectral decomposition (8.260) since the non-zero eigenfunctions have a zero domain integral according to equation (8.247)

$$\int_{\mathcal{R}} \widetilde{G}(\boldsymbol{x}|\boldsymbol{x}_\circ)\,\mathrm{d}V = \sum_{a=1}^{\infty} \frac{\Phi_a(\boldsymbol{x}_\circ)}{\lambda_a} \int_{\mathcal{R}} \Phi_a(\boldsymbol{x})\,\mathrm{d}V = 0. \tag{8.261}$$

[24] We write the summation as $\sum_{a=0}^{\infty}$, noting that a is a shorthand for two indices (two-dimensional $\mathcal{R}$) or three indices (three-dimensional $\mathcal{R}$). The mode with $a = 0$ specifies the zero mode, with all non-zero eigenmodes signified by $a > 0$.

Spectral decomposition of ψ

Making use of the modified Green's function (8.260) in the integral solution (8.255) leads to the spectral decomposition of ψ given by

$$\psi(\boldsymbol{x}) = \langle\psi\rangle + \sum_{a=1}^{\infty} \frac{\Phi_a(\boldsymbol{x})}{\lambda_a} \left[\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_\circ)\, \Phi_a(\boldsymbol{x}_\circ)\, \mathrm{d}V_\circ + \oint_{\partial\mathcal{R}} \Sigma(\boldsymbol{x}_\circ)\, \Phi_a(\boldsymbol{x}_\circ)\, \mathrm{d}\mathcal{S}_\circ \right], \tag{8.262}$$

which compares to the Dirichlet solution (8.243).

8.14 Steady temperature within an annulus

To exemplify certain of the methods and ideas introduced thus far in this chapter, we here derive the steady state temperature distribution within a circular annulus region as depicted in Figure 8.8. Mathematically, we seek a solution to the Dirichlet problem

$$\nabla^2 T = 0 \qquad r_1 < r < r_2 \tag{8.263a}$$
$$T(r,\vartheta) = f(\vartheta) \quad r = r_1 \tag{8.263b}$$
$$T(r,\vartheta) = g(\vartheta) \quad r = r_2, \tag{8.263c}$$

where $f(\vartheta)$ and $g(\vartheta)$ are prescribed temperatures that provide the boundary data as a function of the polar angle, ϑ. The domain is two-dimensional and circular, so that we make use of polar coordinates from Section 4.3, in which Laplace's equation takes the form

$$\nabla^2 T = \frac{1}{r}\frac{\partial}{\partial r}\left[r\,\frac{\partial T}{\partial r}\right] + \frac{1}{r^2}\frac{\partial^2 T}{\partial \vartheta^2} = 0. \tag{8.264}$$

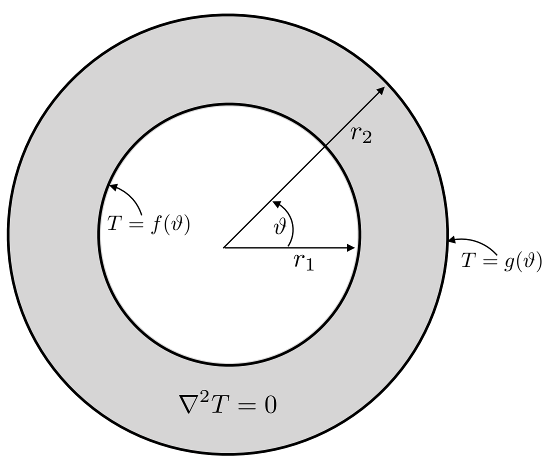

FIGURE 8.8: Depicting the setup for deriving the steady temperature distribution within a circular annulus. Within the annulus, $r_1 < r < r_2$, the temperature is harmonic, $\nabla^2 T = 0$, whereas it is prescribed at the boundaries by two Dirichlet conditions $T(r_1,\vartheta) = f(\vartheta)$ and $T(r_2,\vartheta) = g(\vartheta)$.

8.14.1 Separation of variables

We seek solutions of the separable form

$$T(r,\vartheta) = R(r)\,\Theta(\vartheta), \tag{8.265}$$

with substitution into equation (8.264) leading to

$$\frac{r\,\partial_r(r\,R')}{R} + \frac{\Theta''}{\Theta} = 0, \tag{8.266}$$

where the prime is a shorthand for the derivative, $R' = \mathrm{d}R/\mathrm{d}r$ and $\Theta' = \mathrm{d}\Theta/\mathrm{d}\vartheta$. This equation is self-consistent if

$$r\,\frac{\mathrm{d}}{\mathrm{d}r}\left[r\,\frac{\mathrm{d}R}{\mathrm{d}r}\right] = k^2\,R \quad \text{and} \quad \frac{\mathrm{d}^2\Theta}{\mathrm{d}\vartheta^2} = -k^2\,\Theta, \tag{8.267}$$

where $-k^2 \leq 0$ is a non-dimensional separation constant. We chose the separation constant to ensure that the angular function, Θ, is periodic, thus maintaining symmetry around the circle. Notably, it is necessary to separately consider the case of $k = 0$ versus $k \neq 0$.

8.14.2 Solutions to the separated equations

Solutions to the radial and angular equations (8.267) are given by

$$R(r) = A_k\,r^k + B_k\,r^{-k} \qquad k \neq 0 \tag{8.268a}$$

$$R(r) = A_0 + B_0\ln(r/\ell) \qquad k = 0, \tag{8.268b}$$

and

$$\Theta(\vartheta) = C_k\cos(k\vartheta) + D_k\sin(k\vartheta) \qquad k \neq 0 \tag{8.269a}$$

$$\Theta(\vartheta) = C_0 + D_0\,\vartheta \qquad k = 0. \tag{8.269b}$$

The constant, ℓ, is an arbitrary length that we insert to ensure the argument of the natural log is non-dimensional. We thus have the temperature distribution

$$\begin{aligned} T(r,\vartheta) = a_0 + b_0\ln(r/\ell) + (c_0 + d_0\ln r/\ell)\,\vartheta \\ + \sum_k [(a_k\,r^k + b_k\,r^{-k})\cos(k\,\vartheta) + (c_k\,r^k + d_k\,r^{-k})\sin(k\,\vartheta)]. \end{aligned} \tag{8.270}$$

To ensure the temperature is a periodic function of the angular coordinate requires $c_0 = d_0 = 0$ and for the separation constant, k, to be a positive integer

$$k = n = 1, 2, 3, \ldots. \tag{8.271}$$

Hence, the temperature is given by

$$T(r,\vartheta) = a_0 + b_0\ln(r/\ell) + \sum_{n=1}^{\infty}[(a_n\,r^n + b_n\,r^{-n})\cos(n\,\vartheta) + (c_n\,r^n + d_n\,r^{-n})\sin(n\,\vartheta)]. \tag{8.272}$$

Note that the following combinations of coefficients each have dimensions of temperature

$$a_0, b_0, a_n\,r^n, b_n\,r^{-n}, c_n\,r^n, d_n\,r^{-n}. \tag{8.273}$$

8.14.3 Satisfying the boundary conditions

Evaluating the temperature (8.272) at the inner radius, $r = r_1$, leads to

$$f(\vartheta) = a_0 + b_0 \ln(r_1/\ell) + \sum_{n=1}^{\infty} [(a_n \, r_1^n + b_n \, r_1^{-n}) \cos(n \, \vartheta) + (c_n \, r_1^n + d_n \, r_1^{-n}) \sin(n \, \vartheta)]. \quad (8.274)$$

Making use of the orthogonality properties of sines and cosines (e.g., see Section 8.4.1) leads to

$$a_0 + b_0 \ln(r_1/\ell) = \frac{1}{2\pi} \int_0^{2\pi} f(\vartheta) \, \mathrm{d}\vartheta \quad (8.275a)$$

$$a_n \, r_1^n + b_n \, r_1^{-n} = \frac{1}{\pi} \int_0^{2\pi} f(\vartheta) \cos(m \, \vartheta) \, \mathrm{d}\vartheta \quad (8.275b)$$

$$c_n \, r_1^n + d_n \, r_1^{-n} = \frac{1}{\pi} \int_0^{2\pi} f(\vartheta) \sin(m \, \vartheta) \, \mathrm{d}\vartheta, \quad (8.275c)$$

and likewise for the outer radius

$$a_0 + b_0 \ln(r_2/\ell) = \frac{1}{2\pi} \int_0^{2\pi} g(\vartheta) \, \mathrm{d}\vartheta \quad (8.276a)$$

$$a_n \, r_2^n + b_n \, r_2^{-n} = \frac{1}{\pi} \int_0^{2\pi} g(\vartheta) \cos(m \, \vartheta) \, \mathrm{d}\vartheta \quad (8.276b)$$

$$c_n \, r_2^n + d_n \, r_2^{-n} = \frac{1}{\pi} \int_0^{2\pi} g(\vartheta) \sin(m \, \vartheta) \, \mathrm{d}\vartheta. \quad (8.276c)$$

8.14.4 Solution for the interior of a circle

Focus on the case with $r_1 = 0$ and evaluate the temperature in the region $0 \leq r \leq r_2$ (we consider the temperature for $r \geq r_2$ in Exercise 8.11). To maintain regularity at the origin we must set

$$b_0 = b_n = d_n = 0, \quad (8.277)$$

so that the temperature distribution is

$$T(r, \vartheta) = a_0 + \sum_{n=1}^{\infty} a_n \, r^n \cos(n \, \vartheta) + \sum_{n=1}^{\infty} c_n \, r^n \sin(n \, \vartheta). \quad (8.278)$$

Notice that evaluating the temperature at the origin leads to

$$T(r = 0, \vartheta) = a_0 = \frac{1}{2\pi} \int_0^{2\pi} g(\vartheta) \, \mathrm{d}\vartheta, \quad (8.279)$$

which manifests the mean-value property of harmonic functions studied in Section 8.3.3.

To expose the physical dimensions of terms in the series expansion, it is convenient to introduce the expansion coefficients, each with dimensions of temperature,

$$E_0 = a_0 = \frac{1}{2\pi} \int_0^{2\pi} g(\vartheta) \, \mathrm{d}\vartheta \quad (8.280a)$$

$$E_n = a_n \, r_2^n = \frac{1}{\pi} \int_0^{2\pi} g(\vartheta) \cos(n \, \vartheta) \, \mathrm{d}\vartheta \quad (8.280b)$$

$$F_n = c_n \, r_2^n = \frac{1}{\pi} \int_0^{2\pi} g(\vartheta) \sin(n\,\vartheta) \, \mathrm{d}\vartheta, \tag{8.280c}$$

so that the temperature is

$$T(r, \vartheta) = E_0 + \sum_{n=1}^{\infty} E_n \, (r/r_2)^n \cos(n\,\vartheta) + \sum_{n=1}^{\infty} F_n \, (r/r_2)^n \sin(n\,\vartheta). \tag{8.281}$$

Inserting the expansion coefficients (8.280a)-(8.280c) to this sum leads to

$$T(r, \vartheta) = \frac{1}{\pi} \int_0^{2\pi} g(\vartheta) \left[\frac{1}{2} + \sum_{n=1}^{\infty} (r/r_2)^n [\cos(n\,\vartheta)\cos(n\,\theta) + \sin(n\,\vartheta)\sin(n\,\theta)] \right] \mathrm{d}\theta \tag{8.282a}$$

$$= \frac{1}{\pi} \int_0^{2\pi} g(\vartheta) \left[\frac{1}{2} + \sum_{n=1}^{\infty} (r/r_2)^n \cos[n\,(\theta - \vartheta)] \, \mathrm{d}\theta \right] \tag{8.282b}$$

$$= \frac{1}{\pi} \int_0^{2\pi} g(\vartheta) \, \mathrm{Re} \left[\frac{1}{2} + \sum_{n=1}^{\infty} [(r/r_2) \, \mathrm{e}^{\mathrm{i}\,(\theta - \vartheta)}]^n \, \mathrm{d}\theta \right], \tag{8.282c}$$

where the final equality introduced the real-part operator, Re, while converting the cosine to the exponential. The sum is written in the form of a convergent geometric series with

$$\gamma = (r/r_2) \, \mathrm{e}^{\mathrm{i}\,(\theta - \vartheta)} \quad \text{with} \quad |\gamma| = |r/r_2| < 1, \tag{8.283}$$

so that

$$\sum_{n=1}^{\infty} [(r/r_2) \, \mathrm{e}^{\mathrm{i}\,(\theta-\vartheta)}]^n = \sum_{n=1}^{\infty} \gamma^n = -\gamma^0 + \sum_{n=0}^{\infty} \gamma^n = \frac{\gamma}{1-\gamma}, \tag{8.284}$$

and

$$\frac{1}{2} + \frac{\gamma}{1-\gamma} = \frac{1+\gamma}{2\,(1-\gamma)} = \frac{1 - |\gamma|^2 + 2\,\mathrm{i}\ \mathrm{Im}(\gamma)}{2\,(1 - 2\ \mathrm{Re}(\gamma) + |\gamma|^2)}, \tag{8.285}$$

whose real part is

$$\mathrm{Re}\left[\frac{1}{2} + \frac{\gamma}{1-\gamma}\right] = \frac{1 - |\gamma|^2}{2\,(1 - 2\ \mathrm{Re}(\gamma) + |\gamma|^2)} = \frac{1 - (r/r_2)^2}{2\,[1 - 2\,(r/r_2)\cos(\theta - \vartheta) + (r/r_2)^2]}. \tag{8.286}$$

We are thus led to the rather remarkable result, originally due to Poisson, in which the temperature distribution is

$$T(r, \vartheta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{T(r_2, \vartheta)\,(r_2^2 - r^2)}{r_2^2 - 2\,r\,r_2 \cos(\theta - \vartheta) + r^2} \mathrm{d}\theta \quad \text{for } r < r_2. \tag{8.287}$$

This formula expresses the harmonic temperature distribution, everywhere within the circle, $r < r_2$, as an integral of the temperature over the circle's perimeter, $r = r_2$. Finally, note that when the temperature is a uniform value over the boundary, then

$$1 = \frac{1}{2\pi} \int_0^{2\pi} \frac{r_2^2 - r^2}{r_2^2 - 2\,r\,r_2 \cos(\theta - \vartheta) + r^2} \mathrm{d}\theta, \tag{8.288}$$

which follows from the integral

$$\int_0^{2\pi} \frac{\mathrm{d}\theta}{1 - A\cos\theta} = \frac{2\pi}{\sqrt{1-A^2}} \quad \text{for } |A| < 1. \tag{8.289}$$

8.15 Helmholtz decomposition

In characterizing the kinematic properties of vector fields, such as the velocity vector for a moving fluid, *Helmholtz* (1867) introduced a method to decompose any vector into two distinct component vectors whose properties are readily analyzed: one component has a zero divergence and the second has a zero curl. This Helmholtz decomposition has extensive applications throughout fluid mechanics. See also *Bhatia et al.* (2013) for a review, as well as an earlier treatment in Section 26 of *Serrin* (1959). We make particular use of the Helmholtz decomposition when introducing a scalar or vector potential for studying the kinematics of a non-divergent flow in VOLUME 2. In this section we provide further insights into features of the Helmholtz decomposition, with much of the discussion following *Denaro* (2003) and *Bhatia et al.* (2013). Our treatment makes use of Cartesian vector analysis, with more general treatments, such as Chapter 14 of *Frankel* (2012), outside our scope.

This section is placed within this chapter since we make use of some ideas from elliptic partial differential equations, including free space Green's functions and self-consistency conditions.

8.15.1 Introducing the Helmholtz decomposition

To introduce the Helmholtz decomposition, consider a vector field, $\boldsymbol{F}$, written using Cartesian coordinates and in free space (i.e., no boundaries), and assume the field decays to zero outside of a finite domain. In this case we can decompose $\boldsymbol{F}$ into

$$\boldsymbol{F} = -\nabla D + \nabla \times \boldsymbol{R}, \tag{8.290}$$

where

$$D(\boldsymbol{x}) = \frac{1}{4\pi} \int_{\mathcal{R}} \frac{\nabla \cdot \boldsymbol{F}(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} \, \mathrm{d}V' \tag{8.291a}$$

$$\boldsymbol{R}(\boldsymbol{x}) = \frac{1}{4\pi} \int_{\mathcal{R}} \frac{\nabla \times \boldsymbol{F}(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} \, \mathrm{d}V'. \tag{8.291b}$$

The scalar potential, D, and vector potential, $\boldsymbol{R}$, are here given by the Green's function solutions derived in Section 8.9.7.

The Helmholtz decomposition (8.291a) and (8.291b) is both elegant and straightforward. It offers motivation to seek a similar decomposition for domains of relevance to geophysical fluids, where boundaries play a fundamental role. However, there are many mathematical nuances associated with Helmholtz decompositions in more general situations, each depending on the domain topology and nature of the prescribed boundary conditions. Our goal in this section is to explore the mathematics for a few common boundary conditions for a fluid on simply connected manifolds.[25] We offer rather brief comments for the more general non-simply

[25] A domain is simply connected if a closed curve can be continuously shrunk to a point while remaining in the domain. For example, the domain of the global ocean on a water covered planet is a simply connected manifold. Adding land masses in the form of islands or continents makes the ocean domain non-simply connected.

connected case as the mathematics is beyond our scope.[26]

8.15.2 Summarizing the Helmholtz decomposition

The Helmholtz decomposition on a simply connected manifold, $\mathcal{R}$, states that a vector field, $\boldsymbol{F}$, is fully determined by specifying its divergence, $\nabla \cdot \boldsymbol{F}$, and curl, $\nabla \times \boldsymbol{F}$, for points $\boldsymbol{x} \in \mathcal{R}$, along with the normal component, $\hat{\boldsymbol{n}} \cdot \boldsymbol{F}$, or the tangential component,[27] $\hat{\boldsymbol{n}} \times \boldsymbol{F}$, along the domain boundary, $\partial\mathcal{R}$. Here, $\hat{\boldsymbol{n}}$ is the outward normal along the boundary. Expressed with equations, the Helmholtz decomposition is given by

$$\boldsymbol{F} = -\nabla D + \nabla \times \boldsymbol{R} = \text{gradient term (curl-free)} + \text{rotation term (divergence-free)}. \tag{8.292}$$

The scalar potential, D, is arbitrary up to a constant, and the vector potential, $\boldsymbol{R}$, is arbitrary up to the gradient of a scalar field. This ambiguity in specifying D and $\boldsymbol{R}$ is referred to as gauge symmetry, and we encounter it in other contexts within this book. For present purposes, gauge freedom allows us to choose the Coulomb gauge that is commonly used in electromagnetics (*Jackson*, 1975), whereby $\boldsymbol{R}$ is prescribed to be divergence-free

$$\nabla \cdot \boldsymbol{R} = 0. \tag{8.293}$$

As seen next, this constraint simplifies the boundary value problem satisfied by $\boldsymbol{R}$.

Taking the divergence of (8.292) reveals that the scalar potential satisfies Poisson's equation

$$-\nabla^2 D = \nabla \cdot \boldsymbol{F} \quad \text{for } \boldsymbol{x} \in \mathcal{R}. \tag{8.294}$$

Taking the curl of the decomposition (8.292), and using the curl identity (2.42c) along with the Coulomb gauge (8.293)

$$\nabla \times (\nabla \times \boldsymbol{R}) = \nabla(\nabla \cdot \boldsymbol{R}) - \nabla^2 \boldsymbol{R} = -\nabla^2 \boldsymbol{R}, \tag{8.295}$$

leads to the vector Poisson equation

$$-\nabla^2 \boldsymbol{R} = \nabla \times \boldsymbol{F} \quad \text{for } \boldsymbol{x} \in \mathcal{R}. \tag{8.296}$$

The two elliptic equations, (8.294) and (8.296), are supplemented by boundary conditions for points $\boldsymbol{x} \in \partial\mathcal{R}$. The choice of boundary conditions depend on information available about the vector $\boldsymbol{F}$ along the boundary. We here consider the following three sets of boundary conditions:

$$\begin{aligned}
&\text{normal component} && -\hat{\boldsymbol{n}} \cdot \nabla D = \hat{\boldsymbol{n}} \cdot \boldsymbol{F} && \text{and} && \hat{\boldsymbol{n}} \cdot (\nabla \times \boldsymbol{R}) = 0 && (8.297\text{a})\\
&\text{tangential component} && \hat{\boldsymbol{n}} \times \nabla D = 0 && \text{and} && \hat{\boldsymbol{n}} \times (\nabla \times \boldsymbol{R}) = \hat{\boldsymbol{n}} \times \boldsymbol{F} && (8.297\text{b})\\
&\text{vanishing boundary} && \hat{\boldsymbol{n}} \cdot \nabla D = 0 && \text{and} && \hat{\boldsymbol{n}} \times (\nabla \times \boldsymbol{R}) = 0 \Longrightarrow \boldsymbol{F} = 0. && (8.297\text{c})
\end{aligned}$$

The homogeneous Neumann boundary condition $\hat{\boldsymbol{n}} \cdot (\nabla \times \boldsymbol{R}) = 0$ in equation (8.297a) means

[26] Section 5.2 of *Bhatia et al.* (2013) provides a few comments on the non-simply connected case, whereas chapter 14 of *Frankel* (2012) provides details for the mathematically experienced reader.

[27] *Helmholtz* (1867) only considered vector fields with a specified normal component along boundaries. This situation is most common in fluid mechanical applications. *Denaro* (2003) showed how specification of the tangential component along the boundary also allows for a Helmholtz decomposition. We explore both boundary conditions in this section.

that $(\nabla \times \boldsymbol{R})$ is parallel to the boundary. Likewise, $\hat{\boldsymbol{n}} \times \nabla D = 0$ in equation (8.297b) means that ∇D is parallel to the boundary. If $\boldsymbol{F} = \boldsymbol{v}$ is the fluid velocity field, then $\boldsymbol{v} \cdot \hat{\boldsymbol{n}} = 0$ is the no-normal flow kinematic boundary condition that holds for static solid boundaries. Furthermore, the vanishing boundary condition (8.297c) holds for the velocity with the no-slip boundary condition.

8.15.3 Concerning a harmonic contribution

Consider a vector $\boldsymbol{H}$ that is both divergent-free and curl-free

$$\nabla \cdot \boldsymbol{H} = 0 \quad \text{and} \quad \nabla \times \boldsymbol{H} = 0. \tag{8.298}$$

$\boldsymbol{H}$ is a harmonic vector function (Section 8.3.2), which is seen by noting that with $\nabla \cdot \boldsymbol{H} = 0$ then (see equation (2.42c)) $\boldsymbol{H}$ satisfies the vector Laplace equation

$$\nabla \times (\nabla \times \boldsymbol{H}) = -\nabla^2 \boldsymbol{H} = 0, \tag{8.299}$$

so that each component of $\boldsymbol{H}$ is a harmonic function.

An arbitrary vector, $\boldsymbol{F}$, generally contains a portion that is harmonic, in which case we consider the three-component decomposition

$$\boldsymbol{F} = -\nabla E + \nabla \times \boldsymbol{S} + \boldsymbol{H} = \text{gradient term} + \text{rotation term} + \text{harmonic term}. \tag{8.300}$$

However, as we show in this section, there are many physically interesting cases in which the original two-component Helmholtz decomposition (8.292) is sufficient. That is, for many cases the two-component Helmholtz decomposition is able to include contributions from the harmonic portion of $\boldsymbol{F}$ as part of either ∇D or $\nabla \times \boldsymbol{R}$.

Normal component boundary condition

The constraint $\nabla \times \boldsymbol{H} = 0$ can be satisfied by writing $\boldsymbol{H} = -\nabla \phi$, which then brings the decomposition (8.300) into the form

$$\boldsymbol{F} = -(\nabla E + \nabla \phi) + \nabla \times \boldsymbol{S}. \tag{8.301}$$

We connect this decomposition to the orginal form of the Helmholtz decomposition in equation (8.292) by identifying

$$\nabla E + \nabla \phi \equiv \nabla D \quad \text{and} \quad \nabla \times \boldsymbol{S} \equiv \nabla \times \boldsymbol{R}. \tag{8.302}$$

In so doing, the harmonic term, here captured by $\nabla \phi$, is absorbed into the scalar potential, D. This method for absorbing the harmonic term is suited to the normal component boundary condition (8.297a), in which

$$-\hat{\boldsymbol{n}} \cdot \boldsymbol{F} = \hat{\boldsymbol{n}} \cdot (\nabla E + \nabla \phi) = \hat{\boldsymbol{n}} \cdot \nabla D \Longrightarrow \hat{\boldsymbol{n}} \cdot \nabla E = -\hat{\boldsymbol{n}} \cdot (\boldsymbol{F} + \nabla \phi). \tag{8.303}$$

In practice, we determine D and $\boldsymbol{R}$ as per the two-component decomposition (8.292), with the harmonic contribution to $\boldsymbol{F}$ contained as part of the scalar potential, D.

Tangential component boundary condition

The constraint $\nabla \cdot \boldsymbol{H} = 0$ is satisfied by writing $\boldsymbol{H} = \nabla \times \boldsymbol{A}$, so that the decomposition (8.300) takes on the form

$$\boldsymbol{F} = -\nabla E + \nabla \times (\boldsymbol{S} + \boldsymbol{A}). \tag{8.304}$$

We connect this decomposition to the orginal form of the Helmholtz decomposition of equation (8.292) by identifying

$$\nabla E \equiv \nabla D \quad \text{and} \quad \nabla \times (\boldsymbol{S} + \boldsymbol{A}) \equiv \nabla \times \boldsymbol{R}. \tag{8.305}$$

In so doing, the harmonic term, here captured by $\nabla \times \boldsymbol{A}$, is included in the vector potential, $\boldsymbol{R}$. This method for absorbing the harmonic term is suited for the tangential component boundary condition (8.297b), in which case

$$\hat{\boldsymbol{n}} \times \boldsymbol{F} = \hat{\boldsymbol{n}} \times (\boldsymbol{S} + \boldsymbol{A}) = \hat{\boldsymbol{n}} \times \boldsymbol{R} \Longrightarrow \hat{\boldsymbol{n}} \times \boldsymbol{S} = \hat{\boldsymbol{n}} \times (\boldsymbol{F} - \boldsymbol{A}). \tag{8.306}$$

Non-divergent velocity vector with no-normal flow boundary condition

Many geophysical fluid flows maintain a non-divergent flow field. Such velocity fields also maintain the no-normal flow kinematic boundary condition at static material boundaries. For this vector field the scalar potential is harmonic and satisfies the vanishing Neumann boundary condition

$$-\nabla^2 D = 0 \quad \text{for} \quad \boldsymbol{x} \in \mathcal{R} \tag{8.307a}$$

$$\hat{\boldsymbol{n}} \cdot \nabla D = 0 \quad \text{for} \quad \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.307b}$$

The solution to this boundary value problem is a spatially constant D, so that the velocity field is fully specified by a vector potential

$$\boldsymbol{F} = \nabla \times \boldsymbol{R}. \tag{8.308}$$

In the presence of a time dependent boundary condition, such as a fluctuating free surface, the scalar potential is not generally a spatial constant. We encounter an example when studying surface gravity waves in VOLUME 4.

Vanishing boundary condition

The case with $\boldsymbol{F} = 0$ on the boundary means that

$$\boldsymbol{F} \cdot \hat{\boldsymbol{n}} = 0 \quad \text{and} \quad \hat{\boldsymbol{n}} \times \boldsymbol{F} = 0 \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.309}$$

We proceed with both of the previous boundary condition constraints whereby

$$\nabla E + \nabla \phi = \nabla D \quad \text{and} \quad \nabla \times \boldsymbol{S} = \nabla \times \boldsymbol{R} \tag{8.310a}$$

$$\nabla E = \nabla D \quad \text{and} \quad \nabla \times (\boldsymbol{A} + \boldsymbol{S}) = \nabla \times \boldsymbol{R}. \tag{8.310b}$$

These two sets of constraints are mutually satisfied only when

$$\nabla \phi = 0 \quad \text{and} \quad \nabla \times \boldsymbol{A} = 0, \tag{8.311}$$

so that the harmonic term vanishes identically for the case of $\boldsymbol{F} = 0$ along the boundary.

Summary comments

We have shown that for a simply connected domain, it is possible to absorb an arbitrary harmonic portion of $\boldsymbol{F}$ into either the scalar potential D (for the normal component boundary conditions (8.297a)) or vector potential, $\boldsymbol{R}$ (for tangential component boundary conditions (8.297b)). Furthermore, the harmonic term vanishes altogether for the vanishing boundary condition (e.g., no-slip boundary condition).

We infer from this discussion that if both the normal and tangential components of $\boldsymbol{F}$ are specified nonzero values on the boundary, then the harmonic term cannot be fully absorbed into one of the potentials D or $\boldsymbol{R}$. Rather, the harmonic term must be explictly computed, in which case the three-component decomposition (8.300) is referred to as the *Helmholtz-Hodge decomposition*.

These considerations are analogous to the Cauchy-Stokes decomposition theorem (VOLUME 2). That is, Cauchy-Stokes decomposes the motion of a fluid element into three processes: a deformation plus a rigid rotation plus a uniform translation. The deformation corresponds to the curl-free vector in the Helmholtz decomposition ($-\nabla D$), the rotation corresponds to the divergent free vector ($\nabla \times \boldsymbol{R}$), and the uniform translation corresponds to the harmonic vector.

8.15.4 Self-consistency conditions

We here establish existence conditions for the scalar potential, D, and vector potential, $\boldsymbol{R}$, that satisfy the decomposition (8.292). We separately do so for the two sets of boundary conditions (8.297a) and (8.297b). The proof consists of showing that the source term on the right hand side of the Poisson equation is self-consistent with the boundary condition.

Normal component boundary condition

Consider the scalar Poisson equation with Neumann boundary conditions

$$-\nabla^2 \chi = \Lambda \quad \text{for } \boldsymbol{x} \in \mathcal{R} \tag{8.312a}$$

$$-\hat{\boldsymbol{n}} \cdot \nabla \chi = \sigma \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}. \tag{8.312b}$$

As discussed in Section 8.3.7, this elliptic boundary value problem has a solution so long as the source, Λ, and boundary condition, σ, satisfy a self-consistency condition given by equation (8.44)

$$\int_{\mathcal{R}} \Lambda \, \mathrm{d}V = \oint_{\partial\mathcal{R}} \sigma \, \mathrm{d}\mathcal{S}. \tag{8.313}$$

Now specialize to the Helmholtz decomposition (8.292) whereby $\chi = D$ is the scalar potential with source $\Lambda = \nabla \cdot \boldsymbol{F}$ and boundary condition $\sigma = \hat{\boldsymbol{n}} \cdot \boldsymbol{F}$. Gauss's divergence theorem (2.122) readily verifies self-consistency

$$\int_{\mathcal{R}} \Lambda \, \mathrm{d}V = \int_{\mathcal{R}} \nabla \cdot \boldsymbol{F} \, \mathrm{d}V = \oint_{\partial\mathcal{R}} \boldsymbol{F} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \oint_{\partial\mathcal{R}} \sigma \, \mathrm{d}\mathcal{S}. \tag{8.314}$$

As noted in Section 8.3.7, self-consistency ensures the existence of a solution to the Neumann problem that is unique up to an arbitrary constant. Hence, ∇D is unique so that

$$\nabla \times \boldsymbol{R} = \boldsymbol{F} + \nabla D \tag{8.315}$$

is also unique (we further discuss uniqueness in Section 8.15.6). Finally, the inhomogeneous Neumann boundary condition satisfied by the scalar potential, $-\hat{\boldsymbol{n}} \cdot \nabla D = \hat{\boldsymbol{n}} \cdot \boldsymbol{F}$, means that the curl satisfies the homogeneous Neumann boundary condition

$$\hat{\boldsymbol{n}} \cdot (\nabla \times \boldsymbol{R}) = \hat{\boldsymbol{n}} \cdot (\boldsymbol{F} + \nabla D) = 0, \tag{8.316}$$

so that $(\nabla \times \boldsymbol{R})$ is parallel to the boundary.

Tangential component boundary condition

Consider the vector Poisson equation

$$-\nabla^2 \boldsymbol{\Psi} = \boldsymbol{\Lambda} \qquad \text{for } \boldsymbol{x} \in \mathcal{R} \tag{8.317a}$$

$$\hat{\boldsymbol{n}} \times (\nabla \times \boldsymbol{\Psi}) = \hat{\boldsymbol{n}} \times \boldsymbol{\Sigma} \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}, \tag{8.317b}$$

with an assumed non-divergence condition (Coulomb gauge) placed on $\boldsymbol{\Psi}$

$$\nabla \cdot \boldsymbol{\Psi} = 0. \tag{8.318}$$

The region $\mathcal{R}$ is assumed to be a bounded three-dimensional volume with a closed boundary surface, $\partial\mathcal{R}$. Now let $\mathcal{S}$ be an arbitrary simply connected portion of $\partial\mathcal{R}$, and write $\partial\mathcal{S}$ for the closed contour that bounds $\mathcal{S}$. Integrate the normal projection of the vector Poisson equation over $\mathcal{S}$ and make use of Stokes' theorem (Section 2.7) to render

$$\int \boldsymbol{\Lambda} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = -\int \nabla^2 \boldsymbol{\Psi} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \int \nabla \times (\nabla \times \boldsymbol{\Psi}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \oint_{\partial\mathcal{S}} (\nabla \times \boldsymbol{\Psi}) \cdot \hat{\boldsymbol{t}} \, \mathrm{d}s, \tag{8.319}$$

with $\hat{\boldsymbol{t}}$ the unit tangent vector pointing counter-clockwise along the closed contour, and $\mathrm{d}s$ the increment of arc-length along the contour (see Section 2.5). The assumed boundary condition (8.317b) means that for $\boldsymbol{x} \in \partial\mathcal{R}$ we can write

$$\nabla \times \boldsymbol{\Psi} = \boldsymbol{\Sigma} + \boldsymbol{m}, \tag{8.320}$$

where $\boldsymbol{m}$ is parallel to $\hat{\boldsymbol{n}}$ so that $\hat{\boldsymbol{n}} \times \boldsymbol{m} = 0$. In turn, with $\boldsymbol{m}$ parallel to $\hat{\boldsymbol{n}}$ then it is also perpendicular to the unit tangent vector, $\hat{\boldsymbol{t}} \cdot \boldsymbol{m} = 0$, so that

$$\oint_{\partial\mathcal{S}} (\nabla \times \boldsymbol{\Psi}) \cdot \hat{\boldsymbol{t}} \, \mathrm{d}s = \oint_{\partial\mathcal{S}} \boldsymbol{\Sigma} \cdot \hat{\boldsymbol{t}} \, \mathrm{d}s. \tag{8.321}$$

We are thus led to the self-consistency condition between the source, $\boldsymbol{\Lambda}$, and boundary condition, $\boldsymbol{\Sigma}$

$$\int \boldsymbol{\Lambda} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \oint_{\partial\mathcal{S}} \boldsymbol{\Sigma} \cdot \hat{\boldsymbol{t}} \, \mathrm{d}s, \tag{8.322}$$

with this condition holding for any arbitrary simply connected region, $\mathcal{S}$, that lives on the boundary, $\partial\mathcal{R}$.

Now apply the self-consistency condition (8.322) to the Helmholtz decomposition (8.292), in which case $\boldsymbol{\Psi} = \boldsymbol{R}$, $\boldsymbol{\Lambda} = \nabla \times \boldsymbol{F}$, and $\boldsymbol{\Sigma} = \boldsymbol{F}$. Making use of Stokes' theorem readily verifies self-consistency

$$\int \boldsymbol{\Lambda} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \int (\nabla \times \boldsymbol{F}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \oint_{\partial\mathcal{S}} \boldsymbol{F} \cdot \hat{\boldsymbol{t}} \, \mathrm{d}s = \oint_{\partial\mathcal{S}} \boldsymbol{\Sigma} \cdot \hat{\boldsymbol{t}} \, \mathrm{d}s. \tag{8.323}$$

We appeal to the scalar Poisson equation to conclude that self-consistency between the source and boundary conditions ensures the existence of a vector potential, $\boldsymbol{R}$, that is unique up to the gradient of a scalar. Hence, the gradient of the scalar potential is itself unique. Finally, we note that the assumed boundary condition (8.297b) for the vector potential, $\hat{\boldsymbol{n}} \times (\nabla \times \boldsymbol{R}) = \hat{\boldsymbol{n}} \times \boldsymbol{F}$, means that the scalar potential satisfies the homogeneous boundary condition

$$\hat{\boldsymbol{n}} \times \boldsymbol{F} = \hat{\boldsymbol{n}} \times (\nabla \times \boldsymbol{R}) = \hat{\boldsymbol{n}} \times (\boldsymbol{F} - \nabla D) \Longrightarrow \hat{\boldsymbol{n}} \times \nabla D = 0. \tag{8.324}$$

We thus see that ∇D is aligned parallel to $\hat{\boldsymbol{n}}$ along the boundary.

8.15.5 Orthogonality with the L^2 inner product

Consider an L^2 inner product for vector functions defined according to the volume integral of their scalar product

$$\langle \boldsymbol{A}, \boldsymbol{B} \rangle \equiv \int_{\mathcal{R}} \boldsymbol{A} \cdot \boldsymbol{B} \, \mathrm{d}V. \tag{8.325}$$

We here show that $\langle \nabla D, (\nabla \times \boldsymbol{R}) \rangle = 0$. That is, the Helmholtz decomposition (8.292) serves to decompose a vector into two orthogonal component vectors, where orthogonality is defined over the L^2 inner product (8.325). This orthogonality property is of great importance for the practical use of the Helmholtz decomposition.

Orthogonality with the normal component boundary condition

Assuming the normal component boundary conditions (8.297a), we readily find that the two vectors ∇D and $\nabla \times \boldsymbol{R}$ have a vanishing inner product

$$\langle \nabla D, \nabla \times \boldsymbol{R} \rangle = \int_{\mathcal{R}} \nabla D \cdot (\nabla \times \boldsymbol{R}) \, \mathrm{d}V = \int_{\mathcal{R}} \nabla \cdot (D \, \nabla \times \boldsymbol{R}) \, \mathrm{d}V = \int_{\partial \mathcal{R}} D \, (\nabla \times \boldsymbol{R}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = 0, \tag{8.326}$$

where the final equality made use of the homogeneous boundary condition $(\nabla \times \boldsymbol{R}) \cdot \hat{\boldsymbol{n}} = 0$.

Orthogonality with the tangential boundary condition

Now assume the tangential component boundary conditions (8.297b). For this case we make use of the following identities holding for Cartesian tensors

$$\begin{aligned}
\nabla D \cdot (\nabla \times \boldsymbol{R}) &= \partial_i D \, (\nabla \times \boldsymbol{R})_i && \text{expose Cartesian tensor indices} && (8.327\text{a}) \\
&= \partial_i D \, \epsilon^{ijk} \, \partial_j R_k && \text{vector product as per equation (1.52d)} && (8.327\text{b}) \\
&= \partial_j (\epsilon^{ijk} \, \partial_i D \, R_k) && \text{since } \epsilon^{ijk} \partial_i \partial_j D = 0 \text{ and } \partial_j \epsilon^{ijk} = 0 && (8.327\text{c}) \\
&= -\partial_j (\epsilon^{jik} \, \partial_i D \, R_k) && \epsilon^{ijk} = -\epsilon^{jik} && (8.327\text{d}) \\
&= -\nabla \cdot (\nabla D \times \boldsymbol{R}) && \text{reintroduce Cartesian vector notation.} && (8.327\text{e})
\end{aligned}$$

As a result we have orthogonality

$$\langle \nabla D, \nabla \times \boldsymbol{R} \rangle = \int_{\mathcal{R}} \nabla \cdot (\boldsymbol{R} \times \nabla D) \, \mathrm{d}V = \int_{\mathcal{R}} (\boldsymbol{R} \times \nabla D) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S} = \int_{\partial \mathcal{R}} (\nabla D \times \hat{\boldsymbol{n}}) \cdot \boldsymbol{R} \, \mathrm{d}\mathcal{S} = 0, \tag{8.328}$$

where the final equality made use of the homogeneous boundary condition $(\nabla D \times \hat{\boldsymbol{n}}) = 0$.

Comments

We have shown that the set of boundary conditions (8.297a) and (8.297b) are sufficient to produce an L^2-orthogonal Helmholtz decomposition on a simply connected domain. However, we have *not* shown that these boundary conditions are necessary for orthogonality, with other boundary conditions generally available. Since orthogonality is central to the practical use of the Helmholtz decomposition, it is important to verify whether orthogonality property holds when using alternative boundary conditions.

8.15.6 Uniqueness of the decomposition

We already commented on the uniqueness of the scalar and vector potentials in Section 8.15.4. We here further that discussion by offering a uniqueness proof following a "proof by contradiction" method. For this approach we write the Helmholtz decomposition as

$$\boldsymbol{F} = -\nabla D_1 + \nabla \times \boldsymbol{R}_1 = -\nabla D_2 + \nabla \times \boldsymbol{R}_2, \tag{8.329}$$

and show that the only consistent solution has $D_1 = D_2$ and $\boldsymbol{R}_1 = \boldsymbol{R}_2$. Note that uniqueness is a function of the boundary conditions. That is, the normal component boundary conditions (8.297a) generally lead to a decomposition that is distinct from the tangential component boundary conditions (8.297b).

Uniqueness with the normal component boundary condition

From the assumed relation (8.329) we have

$$0 = -\nabla(D_1 - D_2) + \nabla \times (\boldsymbol{R}_1 - \boldsymbol{R}_2). \tag{8.330}$$

Taking the scalar product of this equality with $\nabla \times (\boldsymbol{R}_1 - \boldsymbol{R}_2)$ and then integrating over the domain leads to

$$0 = -\int_{\mathcal{R}} [\nabla(D_1 - D_2) \cdot \nabla \times (\boldsymbol{R}_1 - \boldsymbol{R}_2)]\,\mathrm{d}V + \int_{\mathcal{R}} [\nabla \times (\boldsymbol{R}_1 - \boldsymbol{R}_2)]^2\,\mathrm{d}V. \tag{8.331}$$

The first integral vanishes, as seen by

$$-\int_{\mathcal{R}} [\nabla(D_1 - D_2) \cdot \nabla \times (\boldsymbol{R}_1 - \boldsymbol{R}_2)]\,\mathrm{d}V = \int_{\mathcal{R}} [\nabla D_1 \cdot (\nabla \times \boldsymbol{R}_2) + \nabla D_2 \cdot (\nabla \times \boldsymbol{R}_1)]\,\mathrm{d}V \tag{8.332a}$$

$$= \int_{\mathcal{R}} \nabla \cdot [D_1\,(\nabla \times \boldsymbol{R}_2) + D_2\,(\nabla \times \boldsymbol{R}_1)]\,\mathrm{d}V \tag{8.332b}$$

$$= \int_{\partial\mathcal{R}} [D_1\,(\nabla \times \boldsymbol{R}_2) + D_2\,(\nabla \times \boldsymbol{R}_1)] \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S} \tag{8.332c}$$

$$= 0, \tag{8.332d}$$

where the first equality follows from orthogonality as per equation (8.326); the second equality holds since the divergence of the curl vanishes (Section 2.3.4); the third equality follows from Gauss's divergence theorem (Section 2.8); and the fourth equality follows from the homogeneous boundary conditions, $\hat{\boldsymbol{n}} \cdot (\nabla \times \boldsymbol{R}_1) = \hat{\boldsymbol{n}} \cdot (\nabla \times \boldsymbol{R}_2) = 0$, satisfied with the normal component boundary conditions (8.297a). We are thus left with the equality

$$\int_{\mathcal{R}} [\nabla \times (\boldsymbol{R}_1 - \boldsymbol{R}_2)]^2 \, \mathrm{d}V = 0, \tag{8.333}$$

which is generally satisfied only when $\nabla \times \boldsymbol{R}_1 = \nabla \times \boldsymbol{R}_2$ so that $\nabla D_1 = \nabla D_2$. We have thus shown that the Helmholtz decomposition $\boldsymbol{F} = -\nabla D + \nabla \times \boldsymbol{R}$ is unique with the normal component boundary conditions (8.297a).

Uniqueness with the tangential component boundary condition

Now consider the tangential component boundary conditions (8.297b). The proof proceeds much like above, only now we take the scalar product of equation (8.330) with $\nabla(D_1 - D_2)$ and integrate over the domain. In doing so we encounter the term

$$-\int_{\mathcal{R}} [\nabla(D_1 - D_2) \cdot \nabla \times (\boldsymbol{R}_1 - \boldsymbol{R}_2)] \, \mathrm{d}V = \int_{\mathcal{R}} [\nabla D_1 \cdot (\nabla \times \boldsymbol{R}_2) + \nabla D_2 \cdot (\nabla \times \boldsymbol{R}_1)] \, \mathrm{d}V \tag{8.334a}$$

$$= \int_{\mathcal{R}} \nabla \cdot [\nabla D_1 \times \boldsymbol{R}_2 + \nabla D_2 \times \boldsymbol{R}_1] \, \mathrm{d}V \tag{8.334b}$$

$$= \int_{\partial \mathcal{R}} [\boldsymbol{R}_2 \cdot (\hat{\boldsymbol{n}} \times \nabla D_1) + \boldsymbol{R}_1 \cdot (\hat{\boldsymbol{n}} \times \nabla D_2)] \tag{8.334c}$$

$$= 0, \tag{8.334d}$$

where the second equality made use of the identity (8.327e) derived when examining orthogonality, the third equality made of the divergence theorem, and the fourth equality holds according to the homogenous boundary conditions $\hat{\boldsymbol{n}} \times \nabla D = 0$ following from equation (8.297b). We are thus led to

$$\int_{\mathcal{R}} [\nabla(D_1 - D_2)]^2 \, \mathrm{d}V = 0, \tag{8.335}$$

which generally holds only if $\nabla D_1 = \nabla D_2$ and hence $\nabla \times \boldsymbol{R}_1 = \nabla \times \boldsymbol{R}_2$. We have thus shown that the Helmholtz decomposition is unique with the tangential component boundary conditions (8.297b).

8.16 Exercises

EXERCISE 8.1: GREEN'S FUNCTION FOR THE SCREENED POISSON EQUATION

Introduce an extension of the Coulomb potential of electrostatics by considering the following elliptic equation

$$(-\nabla^2 + \mu^2) \, \psi = \Lambda, \tag{8.336}$$

where $\mu^2 > 0$ is a constant with dimesion L^{-2}. As noted in Section 50 of *Fetter and Walecka* (2003), this equation occurs in the Yukawa theory of mesons and the Debye-Hückel screening of ionic solutions, hence the name screened Poisson equation. We see in Section 8.8 that there is a direct connection to the Helmholtz equation, which is important for the geophysical wave theory studied in VOLUME 4.

Consider the screened Poisson equation (8.336) in the absence of boundaries, so that its corresponding free-space Green's function satisfies

$$(\nabla^2 - \mu^2)\, \mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = -\delta(\boldsymbol{x} - \boldsymbol{x}_\circ). \tag{8.337}$$

Show that

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ) = \frac{\mathrm{e}^{-\mu\,|\boldsymbol{x}-\boldsymbol{x}_\circ|}}{4\pi\,|\boldsymbol{x}-\boldsymbol{x}_\circ|}, \tag{8.338}$$

Hint: make use of the Fourier method from Section 8.7.4.

EXERCISE 8.2: TRANSFORMATION TO THE SCREENED POISSON EQUATION
Consider the elliptic equation

$$\nabla \cdot (\nabla \psi + 2\,\boldsymbol{c})\psi = 0, \tag{8.339}$$

where $\boldsymbol{c}$ is a constant with dimensions of inverse length. Show that the transformation

$$\psi = \mathrm{e}^{-\boldsymbol{c}\cdot\boldsymbol{x}}\, U, \tag{8.340}$$

leads to the screened Poisson equation considered in Exercise (8.1)

$$(\nabla^2 - |\boldsymbol{c}|^2)U = 0. \tag{8.341}$$

EXERCISE 8.3: LAPLACIAN AND BIHARMONIC EQUATIONS
If $\nabla^2\psi = 0$, then show that $\nabla^4 U = 0$ where

$$U = (\boldsymbol{c} \cdot \boldsymbol{x})\,\psi, \tag{8.342}$$

with $\boldsymbol{c}$ a constant. The equation, $\nabla^4 U = 0$, is known as the biharmonic equation.

EXERCISE 8.4: BIHARMONIC FREE SPACE GREEN'S FUNCTION IN TWO-DIMENSIONS
In Section 8.7 we studied free space Green's functions for the Laplace operator. Show that in two-dimensions that the free space Green's function for the biharmonic equation is given by

$$\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ = 0) = \frac{r^2\,(1 - \ln r)}{8\pi}, \tag{8.343}$$

where we take the source point at the origin, $\boldsymbol{x}_\circ = 0$, for simplicity and to expose circular symmetry in the plane. That is, show that

$$-\nabla^4_{\boldsymbol{x}}\mathcal{G}(\boldsymbol{x}|\boldsymbol{x}_\circ = 0) = \delta(\boldsymbol{x}), \tag{8.344}$$

EXERCISE 8.5: BIHARMONIC BOUNDARY VALUE PROBLEM WITH A SOURCE
Consider the biharmonic equation boundary value problem

$$-\nabla^4\psi = \Lambda \quad \text{for } \boldsymbol{x} \in \mathcal{R} \tag{8.345a}$$

$$\nabla^2\psi = \sigma \quad \text{and} \quad \hat{\boldsymbol{n}} \cdot \nabla(\nabla^2\psi) = \gamma \quad \text{for } \boldsymbol{x} \in \partial\mathcal{R}, \tag{8.345b}$$

with Λ a prescribed source function and σ, γ known boundary data.

(a) Derive the integral self-consistency condition required for the boundary data and the source function, Λ.
(b) Is there a Green's function for this boundary value problem? Discuss.

EXERCISE 8.6: HARMONIC FUNCTION AND ITS GRADIENT

If $\nabla^2\psi = 0$, then show that each component to the gradient in Cartesian coordinates is also a harmonic function

$$\nabla^2(\nabla\psi) = 0. \tag{8.346}$$

EXERCISE 8.7: HARMONIC FUNCTION IN CYLINDRICAL COORDINATES

If $\nabla^2\psi = 0$, then show that using cylindrical-polar coordinates, (r, ϑ, z), from Section 4.3 that

$$\nabla^2(\partial_\vartheta\psi) = 0 \quad \text{and} \quad \nabla^2(\partial_z\psi) = 0 \quad \text{and} \quad \nabla^2(\partial_r\psi) \neq 0. \tag{8.347}$$

EXERCISE 8.8: MEAN VALUE PROPERTY OF HARMONIC FUNCTIONS IN 3D

Following the proof in Section 8.9.10 for two dimensions, express the mean value property of harmonic functions in three dimensions.

EXERCISE 8.9: POISSON'S BOUNDARY VALUE PROBLEM ON A RECTANGLE

Consider the Poisson boundary value problem with Dirichlet conditions

$$-\nabla^2\psi = \Lambda \quad \text{for } \boldsymbol{x} \in \mathcal{R} \tag{8.348a}$$

$$\psi = \sigma \quad \text{for } x \in \partial\mathcal{R}, \tag{8.348b}$$

where the domain is given by the rectangle

$$0 \leq x \leq L_x \quad \text{and} \quad 0 \leq y \leq L_y. \tag{8.349}$$

(a) Write ψ in terms of the corresponding Green's function.

(b) Write the Green's function in terms of the complete set of eigenfunctions of the Laplace operator from Section 8.4.1. Check that the solution agrees with the general results from Section 8.12.

EXERCISE 8.10: TIME-INDEPENDENT SCHRÖDINGER EQUATION

Consider the eigenvalue problem for the time-independent Schrödinger equation

$$-\nabla \cdot (\boldsymbol{K} \cdot \nabla\Phi) + \mathcal{V}\,\Phi = \lambda\,\Phi \quad \text{for } \boldsymbol{x} \in \mathcal{R}, \tag{8.350}$$

with Φ satisfying either Dirichlet or Neumann boundary conditions, and $\mathcal{V}(\boldsymbol{x}) > 0$ is a non-negative function. The case of $\mathcal{V} = 0$ reduces to the Poisson equation considered in Sections 8.12 and 8.13. Prove that all eigenvalues are non-negative. Discuss any cases with zero eigenmodes. That is, are there any nontrivial eigenfunctions that have zero eigenvalues?

EXERCISE 8.11: HARMONIC TEMPERATURE EXTERNAL TO A CIRCLE

In Section 8.14.4 we derived the steady temperature distribution inside of a circle of radius $r = r_2$. In this exercise, derive the temperature outside of the circle for $r \geq r_2$.

EXERCISE 8.12: 1D POISSON EQUATION WITH DIRICHLET BOUNDARIES

Consider the one-dimensional Poisson equation on a finite domain with Dirichlet boundary conditions

$$-\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = \Lambda \qquad \text{for } -L < x < L \tag{8.351a}$$

$$\psi(x) = \sigma^{\pm} \qquad \text{for } x = \pm L, \tag{8.351b}$$

where $\Lambda(x)$ is an interior source and $\sigma^{\pm}$ is prescribed boundary data. The corresponding

Green's function satisfies

$$-\frac{\mathrm{d}^2 G(x|x_\circ)}{\mathrm{d}x^2} = \delta(x - x_\circ) \qquad \text{for } -L < x < L \tag{8.352a}$$

$$G(x|x_\circ) = 0 \qquad \text{for } x = \pm L. \tag{8.352b}$$

Following the general methods from Section 8.9, derive the Green's function, $G(x|x_\circ)$, and the integral expression for the field, ψ, being mindful to satisfy the conditions at $x = x_\circ$. Sketch the Green's function for a chosen source point, $x_\circ$, and sketch the Green's function with the source at $-x_\circ$. Evaluate $G(x_\mathsf{A}|x_\circ)$ and show it equals, as per reciprocity, to $G(x_\circ|x_\mathsf{A})$, where x_A and $x_\circ$ are some chosen values. Working through the details to the somewhat long exercise offers great insight into the Green's function method.

EXERCISE 8.13: STEPS IN DERIVING RECIPROCITY FOR THE MODIFIED GREEN'S FUNCTION
We skipped a few steps in deriving reciprocity of the modified Green's function in Section 8.10.4. Work through those steps for this exercise.

EXERCISE 8.14: 1D POISSON EQUATION WITH NEUMANN BOUNDARIES
To exemplify the results from Section 8.10, consider the one-dimensional Poisson equation with Neumann boundary conditions

$$-\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = \Lambda \qquad \text{for } -L < x < L \tag{8.353a}$$

$$\frac{\mathrm{d}\psi}{\mathrm{d}x} = \Sigma^\pm \qquad \text{for } x = \pm L, \tag{8.353b}$$

with $\Sigma^\pm$ prescribed boundary data. The corresponding modified Green's function, $\widetilde{G}(x|x_\circ)$, satisfies the boundary value problem

$$-\frac{\mathrm{d}^2\widetilde{G}(x|x_\circ)}{\mathrm{d}x^2} = \delta(x - x_\circ) - \frac{1}{2\,L} \qquad \text{for } -L < x, x_\circ < L, \tag{8.354a}$$

$$\frac{\mathrm{d}\widetilde{G}(x|x_\circ)}{\mathrm{d}x} = 0 \qquad \text{for } x = \pm L. \tag{8.354b}$$

Determine the modified Green's function and write the solution to the boundary value problem. Sketch the Green's function for a chosen source point, $x_\circ$, and sketch the Green's function with the source at $-x_\circ$. Evaluate $G(x_\mathsf{A}|x_\circ)$ and show it equals, as per reciprocity, to $G(x_\circ|x_\mathsf{A})$, where x_A and $x_\circ$ are specific values chosen for illustration purposes.

EXERCISE 8.15: METHOD OF IMAGES IN TWO DIMENSIONS
Use the method of images to construct the two-dimensional Green's functions for the half-plane associated with Dirichlet and Neumann boundary conditions.

EXERCISE 8.16: HELMHOLTZ DECOMPOSITION FOR CORIOLIS ACCELERATION
Consider a vector field

$$\boldsymbol{F} = 2\,\boldsymbol{\Omega} \times \boldsymbol{v}, \tag{8.355}$$

where $\boldsymbol{\Omega}$ is a spatial constant and $\nabla \cdot \boldsymbol{v} = 0$. In the context of geophysical fluid mechanics, $\boldsymbol{F}$ is minus the Coriolis acceleration. Since the velocity is non-divergent, there exists a vector potential so that

$$\boldsymbol{v} = \nabla \times \boldsymbol{B}. \tag{8.356}$$

Show that we can perform a Helmholtz decomposition of the Coriolis acceleration so that

$$\boldsymbol{F} = 2\,\boldsymbol{\Omega} \times \boldsymbol{v} = -\nabla\Psi + \nabla \times \boldsymbol{A}, \tag{8.357}$$

where

$$-\nabla^2\Psi = -2\,\boldsymbol{\Omega} \cdot (\nabla \times \boldsymbol{v}) \tag{8.358a}$$

$$\nabla \times \boldsymbol{A} = (\boldsymbol{\Omega} \cdot \nabla)\,\boldsymbol{B} + \nabla\lambda, \tag{8.358b}$$

with λ an arbitrary gauge function.

Chapter 9

Diffusion equation

The diffusion equation is the canonical example of a parabolic partial differential equation. It appears throughout geophysical fluid applications, such as in the transport of enthalpy (heat) and tracers. In this chapter, we examine salient mathematical properties of the diffusion equation, including a study of Green's functions for both the Dirichlet boundary condition and Neumann boundary condition. We also isolate the temporal aspect of the diffusion equation by focusing on response functions that describe the linear response of a dynamical system to either an impulse forcing or a step forcing.

Solutions to the diffusion equation are smoothed in time, thus reflecting the irreversible dissipative physics modeled by this equation. Furthermore, the solution to a parabolic initial value problem depends on all the initial data, no matter where that data is placed in space. This unbounded domain of influence indicates there is an infinite propagation speed for information transmitted by the diffusion equation. Even so, the amplitude of the information decays exponentially in space. So although parabolic equations possess an unphysically unbounded domain of influence, the ability to make use of all that information is exponentially small. We have more to say on this point in Section 9.1.4.

READER'S GUIDE TO THIS CHAPTER

We make use of material from *Duchateau and Zachmann* (1986) and *Hildebrand* (1976) for general properties of parabolic equations. The treatment of Green's functions in this chapter is based on the treatments found in Chapter 7 of *Morse and Feshbach* (1953) as well as *Greenberg* (1971) and *Stakgold* (2000a,b). Use of the common symbol, G, for the Green's function, minimizes the adornments otherwise needed to distinguish Green's functions derived for each of the distinct boundary conditions. Furthermore, it is useful on certain occasions to write the gradient operator as either $\nabla_{\boldsymbol{x}}$ or $\nabla_{\boldsymbol{x}_\circ}$, depending on what argument of a Green's function is being differentiated (i.e., the field point or source point). This notation is particularly helpful for organizing manipulations in this chapter.

9.1 Parabolic operators

In our classification of partial differential equations in Section 8.2, we note that parabolic equations contain a single real characteristic. The diffusion equation (also the heat equation) is the canonical example of a parabolic equation and it takes the form in one-space dimension

$$\partial_t \psi = \kappa \, \partial_{xx} \psi, \tag{9.1}$$

where $\kappa > 0$ is a constant kinematic diffusivity with dimensions of squared length per time. We later generalize to a space-time dependent kinematic diffusivity, but keep it constant for now to facilitate analytical solutions.

9.1.1 Initial and initial-boundary value problems

The Cauchy problem is the name given to the initial value problem for the diffusion equation over all space and for times $t \geq t_{\mathsf{init}}$

$$\partial_t \psi = \kappa \, \nabla^2 \psi \qquad \boldsymbol{x} \in \mathbb{R}^n, t > t_{\mathsf{init}} \tag{9.2a}$$
$$\psi(\boldsymbol{x}, t_{\mathsf{init}}) = \sigma(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathbb{R}^n \tag{9.2b}$$
$$|\psi(\boldsymbol{x}, t)| < \infty \qquad \boldsymbol{x} \in \mathbb{R}^n, t > t_{\mathsf{init}}. \tag{9.2c}$$

The corresponding initial-boundary value problem posed on a finite and open spatial domain, $\mathcal{R}$, takes the form

$$\partial_t \psi = \kappa \nabla^2 \psi \qquad \boldsymbol{x} \in \mathcal{R}, t > t_{\mathsf{init}} \tag{9.3a}$$

$$\psi(\boldsymbol{x}, t_{\mathsf{init}}) = \sigma(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathcal{R} \tag{9.3b}$$

$$\alpha(\boldsymbol{x})\, \psi(\boldsymbol{x}, t) + \beta(\boldsymbol{x})\, \hat{\boldsymbol{n}} \cdot \nabla \psi(\boldsymbol{x}, t) = g(\boldsymbol{x}, t) \qquad \boldsymbol{x} \in \partial\mathcal{R}, t > t_{\mathsf{init}}, \alpha\, \beta \geq 0, \tag{9.3c}$$

with $\partial\mathcal{R}$ the domain boundary. Following from the study of Poisson's equation in Chapter 8, choices for the boundary functions, α and β, impact on the character of the boundary conditions and the associated solution. The Neumann boundary condition is most commonly applied to set the flux of a tracer or temperature at the boundaries. The alternative Dirichlet boundary condition is commonly applied for idealized tracers in geophysical fluid applications (see *Haine et al.* (2025) for a review of such idealized ocean tracers).

9.1.2 Smoothing property

As we prove in Section 9.1.3, a version of the max-min principle from Chapter 8 holds also for the diffusion equation, which is consistent with solutions to the diffusion equation generally decaying their initial condition towards a constant by reducing the amplitude of all extrema. Hence, no extrema are introduced in the interior of the domain by the diffusion operator. Rather, extrema only arise via boundary and/or initial conditions. Furthermore, the steady state limit of the diffusion equation (with a constant diffusivity) is a harmonic function, and so solves Laplace's equation and possesses the mean value property from Section 8.3.3.

Illustrating the smoothing property in a finite domain

We illustrate the smoothing property for the specific case of the one-dimensional initial-boundary value problem with homogeneous Dirichlet boundary conditions

$$\partial_t \psi = \kappa\, \partial_{xx} \psi \qquad 0 < x < L,\ t > t_{\mathsf{init}} \tag{9.4a}$$

$$\psi(x, t_{\mathsf{init}}) = I(x) \qquad 0 < x < L \tag{9.4b}$$

$$\psi(x = 0, t) = \psi(x = L, t) = 0 \qquad t > 0. \tag{9.4c}$$

In Exercise 9.9 we make use of separation of variables to derive the following Fourier series solution

$$\psi(x, t \geq t_{\mathsf{init}}) = \sum_{n=1}^{\infty} I_n\, e^{-\kappa\, (t - t_{\mathsf{init}})\, (n\pi/L)^2} \sin(n\, \pi\, x/L) \tag{9.5a}$$

$$I_n = \frac{2}{L} \int_0^L I(x) \sin(n\, \pi\, x/L)\, \mathrm{d}x. \tag{9.5b}$$

Observe how the amplitude of each Fourier mode decays exponentially as time increases $t > t_{\mathsf{init}}$, thus damping all initial structure towards zero. Note that we are given no information about times $t < t_{\mathsf{init}}$, so that this solution is only known for $t \geq t_{\mathsf{init}}$. This point is formalized later in this section via the causality condition.

Smoothing property for an initial value problem on the real line

Now consider the one-dimensional diffusion equation on the real line, with the only boundary conditions being regularity at infinity

$$\partial_t \psi = \kappa \, \partial_{xx} \psi \qquad -\infty < x < \infty, \; t > t_{\text{init}} \tag{9.6a}$$

$$\psi(x, t) = I(x) \qquad -\infty < x < \infty, \; t = t_{\text{init}}, \tag{9.6b}$$

One can show by direct differentiation that the following Gaussian is a solution[1]

$$\psi(x, t) = \frac{1}{\sqrt{4 \, \pi \, \kappa \, (t - t_{\text{init}})}} \int_{-\infty}^{\infty} I(\xi) \, \mathrm{e}^{-(x-\xi)^2/[4 \, \kappa \, (t - t_{\text{init}})]} \, \mathrm{d}\xi. \tag{9.7}$$

Again, this function smooths/damps the initial condition function, $I(x)$, as time increases.

9.1.3 Minimum-maximum principle for diffusion

Following Chapter 4 of *Duchateau and Zachmann* (1986), we here prove the minimum-maximum principle for diffusion, which says that the solution to the diffusion equation, with constant diffusivity, has values between the minimum and maximum found either on the region boundary or prescribed at the initial condition. As a result, diffusion cannot create extrema. Evidently, this principle formalizes the smoothing property discussed in Section 9.1.2.

Statement of the minimum-maximum principle

Let an open spatial region, $\mathcal{R}$, have a smooth boundary, $\partial\mathcal{R}$, and write $\overline{\mathcal{R}}$ for the closed region, $\overline{\mathcal{R}} = \mathcal{R} + \partial\mathcal{R}$. Let $[0, T]$ be a closed time interval and $\psi(\boldsymbol{x}, t)$ be a continuous function on $\overline{\mathcal{R}} \times [0, T]$. Let $M_{\partial\mathcal{R}}$ be the maximum of ψ along $\partial\mathcal{R}$; M_{init} the maximum of ψ within $\mathcal{R}$ at the initial time; and let M be their maximum

$$M_{\partial\mathcal{R}} = \max\{\psi(\boldsymbol{x}, t) : \boldsymbol{x} \in \partial\mathcal{R}, t \in [0, T]\} \tag{9.8a}$$

$$M_{\text{init}} = \max\{\psi(\boldsymbol{x}, t) : \boldsymbol{x} \in \mathcal{R}, t = 0\} \tag{9.8b}$$

$$M = \max(M_{\partial\mathcal{R}}, M_{\text{init}}). \tag{9.8c}$$

Furthermore, let $m_{\partial\mathcal{R}}, m_{\text{init}}, m$ be the corresponding minimum values. We prove the following three statements:

$$\text{If } \partial_t \psi \leq \kappa \nabla^2 \psi \text{ in } \mathcal{R} \times (0, T), \text{ then } \psi \leq M \text{ in } \overline{\mathcal{R}} \times [0, T]. \tag{9.9a}$$

$$\text{If } \partial_t \psi \geq \kappa \nabla^2 \psi \text{ in } \mathcal{R} \times (0, T), \text{ then } \psi \geq m \text{ in } \overline{\mathcal{R}} \times [0, T]. \tag{9.9b}$$

$$\text{If } \partial_t \psi = \kappa \nabla^2 \psi \text{ in } \mathcal{R} \times (0, T), \text{ then } m \leq \psi \leq M \text{ in } \overline{\mathcal{R}} \times [0, T]. \tag{9.9c}$$

Statement (9.9c) is the minimum-maximum principle for solutions to the diffusion equation, and this result follows from statements (9.9a) and (9.9b).

Proof of the minimum-maximum principle

The proof is by contradiction, and we start by proving statement (9.9a). For that purpose, introduce

$$\Psi = \psi + \epsilon \, |\boldsymbol{x}|^2 \quad \text{with} \quad \epsilon > 0, \tag{9.10}$$

[1] In Section 9.2 we develop Green's function methods to derive the integral solution (9.7).

so that

$$\psi \leq \Psi \quad \text{and} \quad \partial_t \Psi - \kappa \nabla^2 \Psi = -6\,\epsilon < 0. \tag{9.11}$$

Now assume that Ψ has its maximum at some point, $\boldsymbol{x}_\circ \in \mathcal{R}$ during a time $t_\circ \in (0, T]$. When Ψ is evaluated at $(\boldsymbol{x}, t) = (\boldsymbol{x}_\circ, t_\circ)$, it satisfies $\nabla^2 \Psi < 0$ (since it is a maximum at $\boldsymbol{x} = \boldsymbol{x}_\circ$) and $\partial_t \Psi = 0$ (since it is an extrema at $t = t_\circ$). But these properties then mean that $\partial_t \Psi - \kappa \nabla^2 \Psi > 0$, which contradicts the inequality (9.11). To resolve the contradiction requires us to concede that the maximum of Ψ cannot be realized within $\mathcal{R}$ or during a time after the initial condition. Rather, it can only be realized either at the initial condition, $t = 0$, or on the boundary, $\partial\mathcal{R}$. Since $\psi \leq \Psi \leq M$, it follows that $\psi \leq M$ for $(\boldsymbol{x}, t) \in \overline{\mathcal{R}} \times [0, T]$.

Proof of statement (9.9b) follows using the same logic but just with minima rather than maxima. Finally, for $\partial_t \psi = \kappa \nabla^2 \psi$, statements (9.9a) and (9.9b) lead to the minimum-maximum principle (9.9c) for solutions to the diffusion equation.

Comments about non-constant diffusivities

The minimum-maximum principle (9.9c) holds only for constant diffusivities. No such principle holds for the geophysically relevant case of non-constant diffusivities, such as when using a space and time dependent diffusion tensor as studied in Section 9.2. Indeed, in some cases the space-time dependent diffusivities can lead to the local growth of extrema in scalar fields, even while it dissipates global variance.[2] Even so, the minimim-maximum principle for the case of constant diffusivity establishes an important baseline from which to judge the evolution of scalar fields with a general diffusion tensor.

9.1.4 Infinite domain of influence

As noted in the abstract to this chapter, the solution to a parabolic initial value problem at any spatial point, $\boldsymbol{x}$, depends on all the initial data no matter how far away. We can see this property from the solution (9.7) for the diffusion equation on the real line

$$\psi(x, t) = \frac{1}{\sqrt{4\,\pi\,\kappa\,(t - t_{\mathsf{init}})}} \int_{-\infty}^{\infty} I(\xi)\, \mathrm{e}^{-(x-\xi)^2/[4\,\kappa\,(t - t_{\mathsf{init}})]}\, \mathrm{d}\xi. \tag{9.12}$$

For example, consider an initial condition that is a Dirac delta at the origin,

$$I(x) = X\,\delta(x), \tag{9.13}$$

where X is a constant with dimensions of $[\psi]$ L. In this case the scalar field is

$$\psi(x, t) = \frac{X\,\mathrm{e}^{-x^2/[4\,\kappa\,(t - t_{\mathsf{init}})]}}{\sqrt{4\,\pi\,\kappa\,(t - t_{\mathsf{init}})}}, \tag{9.14}$$

which reveals that $\psi(x, t > t_{\mathsf{init}}) \neq 0$, no matter how far from the origin. Evidently, the initial condition is propagated infinitely fast. However, the amplitude of the information decays exponentially away from the origin. So although parabolic equations possess an unphysically unbounded domain of influence, the ability to detect the initial condition is exponentially small outside of the region $x = \pm 2\,\sqrt{\kappa\,(t - t_{\mathsf{init}})}$.

[2] One example, the Phillip's layering instability, is pedagogically treated in Section 12.2 of *Smyth and Carpenter* (2019).

9.1.5 Duhamel's superposition integral

Consider a scalar field that starts from zero initial conditions and evolves according to the diffusion equation in the presence of a source

$$\partial_t \Psi = \kappa \nabla^2 \Psi + f(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t > 0 \tag{9.15a}$$

$$\Psi(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t = 0. \tag{9.15b}$$

Now consider the converse, in which another scalar field evolves without a source and yet is initialized according to the source

$$\partial_t \psi = \kappa \nabla^2 \psi \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t > 0 \tag{9.16a}$$

$$\psi(\boldsymbol{x}, t) = f(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t = 0. \tag{9.16b}$$

Remarkably, these two scalar fields are related by Duhamel's superposition integral

$$\Psi(\boldsymbol{x}, t) = \int_0^t \psi(\boldsymbol{x}, t - \tau)\, \mathrm{d}\tau. \tag{9.17}$$

We verify the connection by direct differentiation

$$\partial_t \Psi(\boldsymbol{x}, t) = \psi(\boldsymbol{x}, 0) + \int_0^t \frac{\partial \psi(\boldsymbol{x}, t - \tau)}{\partial t}\, \mathrm{d}\tau = f(\boldsymbol{x}) + \kappa \nabla^2\, \Psi(\boldsymbol{x}, t). \tag{9.18}$$

Duhamel's superposition integral allows us to move the source between the partial differential operator and the initial condition. It reveals that the forced solution, $\Psi(\boldsymbol{x}, t)$, is built by time integrating the "retarded" values of the unforced solution, ψ, from the initial time, $t = 0$, to the current time, t. A more general presentation allows for the source function to be a function of time, $f(\boldsymbol{x}, t)$, in which case we develop a family of solutions, $\psi_f(\boldsymbol{x}, t; \tau)$, generated by reinitializing $\psi_f(\boldsymbol{x}, t = \tau; \tau) = f(\boldsymbol{x}, \tau)$ and then superposing the members of this family to generate $\Psi(\boldsymbol{x}, t)$. This more general solution method is encompassed by the Green's functions of Section 9.2.

9.1.6 Uniqueness of the solution

In section 9.2 we study the following parabolic equations that satisfy either Neumann boundary conditions

$$\partial_t \psi - \nabla \cdot (\boldsymbol{K} \cdot \nabla \psi) = \Lambda \quad \boldsymbol{x} \in \mathcal{R} \tag{9.19a}$$

$$\hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla \psi = \Sigma \quad \boldsymbol{x} \in \partial \mathcal{R} \tag{9.19b}$$

$$\psi(\boldsymbol{x}, t = t_{\mathsf{init}}) = I \quad \boldsymbol{x} \in \mathcal{R}, \tag{9.19c}$$

or Dirichlet boundary conditions

$$\partial_t \psi - \nabla \cdot (\boldsymbol{K} \cdot \nabla \psi) = \Lambda \quad \boldsymbol{x} \in \mathcal{R} \tag{9.20a}$$

$$\psi = \sigma \quad \boldsymbol{x} \in \partial \mathcal{R} \tag{9.20b}$$

$$\psi(\boldsymbol{x}, t = t_{\mathsf{init}}) = I \quad \boldsymbol{x} \in \mathcal{R}. \tag{9.20c}$$

In these equations, $\boldsymbol{K}$ is a symmetric and non-negative second order diffusion tensor[3] that arises in many applications considered in VOLUME 4, whereas σ, Σ, I are prescribed boundary and initial data. After deriving a solution, such as through the Green's function methods in Section 9.2, how do we know the solution is unique?

We answer the uniqueness question in a manner similar to that used in Chapter 8 for elliptic equations. Namely, assume there are two solutions, ψ_1 and ψ_2, and define their difference as

$$\Psi = \psi_1 - \psi_2. \tag{9.21}$$

The initial-boundary value problem satisfied by Ψ is given by the homogeneous system

$$\partial_t \Psi - \nabla \cdot (\boldsymbol{K} \cdot \nabla \Psi) = 0 \quad \boldsymbol{x} \in \mathcal{R} \tag{9.22a}$$

$$\hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla \Psi = 0 \quad \text{or} \quad \Psi = 0 \quad \boldsymbol{x} \in \partial \mathcal{R} \tag{9.22b}$$

$$\Psi(\boldsymbol{x}, t = t_{\mathsf{init}}) = 0 \quad \boldsymbol{x} \in \mathcal{R}. \tag{9.22c}$$

Now consider the non-negative integral

$$J(t) = \int_{\mathcal{R}} \Psi^2 \, \mathrm{d}V \geq 0, \tag{9.23}$$

which starts from a zero initial condition, $J(t_{\mathsf{init}}) = 0$, and has a time derivative

$$\frac{1}{2} \frac{\mathrm{d}J}{\mathrm{d}t} = \int_{\mathcal{R}} \Psi \, \partial_t \Psi \, \mathrm{d}V = \int_{\mathcal{R}} \Psi \, \nabla \cdot (\boldsymbol{K} \, \nabla \Psi) \, \mathrm{d}V \tag{9.24a}$$

$$= \int_{\mathcal{R}} \nabla \cdot (\Psi \, \boldsymbol{K} \, \nabla \Psi) \, \mathrm{d}V - \int_{\mathcal{R}} \nabla \Psi \cdot \boldsymbol{K} \cdot \nabla \Psi \, \mathrm{d}V \tag{9.24b}$$

$$= - \int_{\mathcal{R}} \nabla \Psi \cdot \boldsymbol{K} \cdot \nabla \Psi \, \mathrm{d}V, \tag{9.24c}$$

where we used the divergence theorem and the homogeneous boundary conditions (9.22b) to drop the total divergence term. The diffusion tensor is a symmetric and non-negative second order tensor, so that the quadratic form is non-negative, $\nabla \Psi \cdot \boldsymbol{K} \cdot \nabla \Psi \geq 0$. Hence, the integral, $J(t) \geq 0$, has a negative time tendency and it starts from a zero initial condition. We conclude that it is identically zero for all time, $J(t) = 0$, which in turn requires that $\nabla \Psi = 0$ in $\mathcal{R}$. For the homogeneous Dirichlet problem, with $\Psi = 0$ on $\partial \mathcal{R}$, then $\Psi = 0$ everywhere, thus establishing uniqueness of the Dirichlet solution. In contrast, the solution to the Neumann boundary condition problem is unique up to an arbitrary constant, since the constant plays no role in the Neumann boundary condition. We understand this arbitrary constant by noting that prescribing the flux of a scalar field at the boundary only establishes the value of the scalar in the interior relative to an arbitrary constant offset.

9.2 Green's functions for the diffusion equation

In this section we study a variety of Green's functions for the diffusion equation. In various places in the following development, we compare the diffusion equation results to those from Poisson's equation studied in Chapter 8. There are many similarities since both equations involve a Laplacian operator, and since the steady diffusion equation satisfies a Poisson equation.

[3]A symmetric and non-negative second order tensor has non-negative eigenvalues.

Yet time evolution presents a fundamentally new element to the diffusion equation relative to Poisson's equation. In particular, it introduces the need for setting an initial condition, enforcing the causality condition, and deriving a modified reciprocity condition.

9.2.1 Time independence of the spatial domain

Certain geophysical applications of interest in this book make use of spatial regions, $\mathcal{R}$, that have time dependent boundaries. The ocean free surface is the key example. Indeed, we consider time dependent boundaries when studying the tracer advection-diffusion equation in VOLUME 4. However, a time dependent boundary adds conceptual and technical details that are best confronted only after development of our Green's function brain muscle. For that reason, we assume $\mathcal{R}$ is static in this section.

9.2.2 Causal free space Green's function

Consider a Dirac delta source of tracer at the space-time point, $(\boldsymbol{x}_\circ, t_\circ)$, and assume a continuous media where there is no advection (e.g., a stagnant fluid or elastic solid). The simplest solution to the diffusion equation is known as the causal free space Green's function, $\mathcal{G}$, which is defined for $t \in (-\infty, \infty)$ and satisfies

$$(\partial_t - \kappa \nabla^2_{\boldsymbol{x}})\, \mathcal{G}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(t - t_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \qquad \boldsymbol{x} \in \mathbb{R}^n \tag{9.25a}$$

$$\mathcal{G}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) \to 0 \qquad |\boldsymbol{x} - \boldsymbol{x}_\circ| \to \infty \tag{9.25b}$$

$$\mathcal{G}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad t < t_\circ, \tag{9.25c}$$

where $\mathbb{R}^n$ is the real numbers with $n = 1, 2, 3$. Additionally, $\kappa > 0$ is a constant kinematic diffusivity (dimensions $\mathrm{L}^2\ \mathrm{T}^{-1}$). Equation (9.25b) ensures that the free space Green's function decays as the field point gets further away from the source point. The causality condition (9.25c) means that the Green's function vanishes throughout all of Euclidean space for times prior to the time, $t_\circ$, at which the Dirac source occurs. Given the dimensions of the Dirac source, we see that the Green's function has dimensions of L^{-n}.

The causal free-space Green's function is given by[4]

$$\mathcal{G}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \frac{\mathcal{H}(t - t_\circ)}{[4\,\pi\,\kappa\,(t - t_\circ)]^{n/2}} \mathrm{e}^{-(\boldsymbol{x}-\boldsymbol{x}_\circ)^2/[4\,\kappa\,(t-t_\circ)]}, \tag{9.26}$$

where the Heaviside step function, $\mathcal{H}$, enforces causality. The amplitude of the Green's function exponentially decays when moving away from the source location, thus satisfying the condition (9.25b). Additionally, as time progresses beyond the source time, the Green's function decays according to the pre-factor, $(t - t_\circ)^{-n/2}$. As seen in our discussion of the Dirac delta in Chapter 5, for one dimension we have $\delta^{(\epsilon)}(x)$ becoming a Dirac delta as $\epsilon \to 0$

$$\delta^{(\epsilon)}(x) = \frac{\mathrm{e}^{-x^2/\epsilon^2}}{\epsilon\,\sqrt{\pi}} \xrightarrow{\epsilon \to 0} \delta(x). \tag{9.27}$$

Consequently, as time approaches the Dirac delta time, $t \downarrow t_\circ$, the Green's function (9.26)

[4] See Section 5.8 of *Stakgold* (2000b) for a derivation of the free space solution (9.26). Additionally, see Exercise 9.9 for a related derivation.

becomes a Dirac delta in space[5]

$$\lim_{t \downarrow t_\circ} \mathcal{G}(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ). \tag{9.28}$$

This property is shared with the causal Green's functions with boundary conditions, as seen by equation (9.36).

9.2.3 Causal Green's function

We here introduce the causal Green's function for the diffusion equation.

Neumann boundary conditions

The causal Green's function is defined for $t \in (-\infty, \infty)$ and it satisfies the following equations when assuming Neumann boundary conditions

$$\partial_t [G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)] - \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)] = \delta(t - t_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \qquad \boldsymbol{x} \in \mathcal{R} \tag{9.29a}$$

$$\hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R}, \boldsymbol{x}_\circ \notin \partial\mathcal{R} \tag{9.29b}$$

$$G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \qquad t < t_\circ. \tag{9.29c}$$

The Neumann boundary condition is relevant when one knows the boundary flux of heat or tracer concentration. For the free space solution from Section 9.2.2 we assumed a constant diffusivity. In contrast, we here allow for a space-time dependent diffusion tensor, $\boldsymbol{K}(\boldsymbol{x}, t)$, which is a symmetric second order tensor with dimensions $\mathrm{L}^2\ \mathrm{T}^{-1}$.

There is a nicety regarding the placement of the field and source points for the boundary condition (9.29b). Namely, the homogeneous Neumann boundary condition (9.29b) holds when the field point, $\boldsymbol{x}$, is on the boundary and yet the source point, $\boldsymbol{x}_\circ$, is not on the boundary. We later extend the boundary condition (see equation (9.79)) to allow both $\boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}$.

Dirichlet boundary conditions

The Dirichlet boundary condition is used when knowing the boundary values for the field, in which case the corresponding causal Green's function satisfies

$$\partial_t [G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)] - \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)] = \delta(t - t_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \qquad \boldsymbol{x} \in \mathcal{R} \tag{9.30a}$$

$$G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R} \tag{9.30b}$$

$$G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \qquad t < t_\circ. \tag{9.30c}$$

The Dirichlet condition is particularly relevant for passive tracers in the atmosphere and ocean, as studied in VOLUME 4.

Some properties of the causal Green's function

Both the Neumann and Dirichlet Green's functions have dimensions of L^{-n}, as implied since the Dirac delta source has dimensions of $\mathrm{L}^{-n}\ \mathrm{T}^{-1}$, where n is the space dimension. Furthermore, in the presence of spatial boundaries, the spatial position of the Dirac delta source impacts the value of the causal Green's function. In contrast, the causality condition (9.29c) means that the Green's function is dependent only on the time since the introduction of the source, $t - t_\circ$.

[5] See also exercise 5.3 in *Stakgold* (2000b) for the one-dimensional case.

Hence, there is no added generality afforded by setting the source time, $t_\circ$, to be distinct from $t_\circ = 0$. Even so, we retain $t_\circ$ to maintain symmetry with the spatial location of the source, $\boldsymbol{x}_\circ$. Doing so also helps to distinguish the Dirac source time, $t_\circ$, from the initial time, t_{init}, with the initial time introduced in Section 9.4.1.

Finally, notice that the causality conditions (9.29c) and (9.30c) strictly hold for $t < t_\circ$. As seen in equation (9.76) disussed later, when the source time is evaluated at $t_\circ = t_{\mathsf{init}}$, then the limit as the field time goes to the initial time, $t \downarrow t_{\mathsf{init}}$, results in a spatial Dirac delta, $\delta(\boldsymbol{x} - \boldsymbol{x}_\circ)$. That is, the Green's function is initialized by a Dirac delta source at $\boldsymbol{x} = \boldsymbol{x}_\circ$.

9.2.4 An integral consistency condition

It is notable that there is a Green's function for the diffusion equation in the presence of homogeneous Neumann conditions. In contrast, as shown in Section 8.10, there is no Green's function for the Poisson equation with Neumann boundaries. Time dependence in the diffusion equation is the key distinction that allows for a Neumann Green's function.

To determine a self-consistency condition for the Green's function partial differential equation (9.29a), integrate it over the static spatial domain, $\mathcal{R}$, and make use of the divergence theorem to find

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathcal{R}} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) \, \mathrm{d}V_{\boldsymbol{x}} = \delta(t - t_\circ) + \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \boldsymbol{K}(\boldsymbol{x}) \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) \, \mathrm{d}\mathcal{S}_{\boldsymbol{x}}. \tag{9.31}$$

The boundary integral term on the right hand side vanishes for the homogeneous Neumann boundary conditions (9.29b), in which the domain integrated Green's function evolves only as a result of the Dirac delta source firing at $t = t_\circ$. For the case of homogeneous Dirichlet boundary conditions (9.30b), the domain integrated Green's function evolves both from the Dirac source and from the generally nonzero fluxes crossing the domain boundaries.

Now integrate equation (9.31) over a time intervel, $t_\circ - \epsilon \leq t \leq t_\circ + \epsilon$, thus resulting in

$$\int_{\mathcal{R}} G(\boldsymbol{x}, t = t_\circ + \epsilon|\boldsymbol{x}_\circ, t_\circ) \, \mathrm{d}V_{\boldsymbol{x}} = 1 + \int_{t_\circ - \epsilon}^{t_\circ + \epsilon} \oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}} \cdot \boldsymbol{K}(\boldsymbol{x}) \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) \, \mathrm{d}\mathcal{S}_{\boldsymbol{x}} \, \mathrm{d}t, \tag{9.32}$$

where we used causality (9.29c) to set $G(\boldsymbol{x}, t = t_\circ - \epsilon|\boldsymbol{x}_\circ, t_\circ) = 0$. The unit impulse on the right hand side results from the integrated Dirac delta source that fires at $t = t_\circ$. For homogeneous Neumann boundaries, the domain integrated Green's function is unity at all times $t > t_\circ$

$$\int_{\mathcal{R}} G(\boldsymbol{x}, t = t_\circ + \epsilon|\boldsymbol{x}_\circ, t_\circ) \, \mathrm{d}V_{\boldsymbol{x}} = 1 \qquad \text{for Neumann boundaries.} \tag{9.33}$$

That is, the Green's function is normalized to unity when integrated over the full domain. This result follows from the conservation of domain integrated tracer, which holds in the absence of boundary fluxes (homogeneous Neumann conditions) and for all times.

9.2.5 A temporal consistency condition

We can develop another consistency condition by integrating the partial differential equation (9.29a) over a time interval that straddles the Dirac source time, $t \in [t_\circ - \epsilon, t_\circ + \epsilon]$, in which case

$$G(\boldsymbol{x}, t_\circ + \epsilon|\boldsymbol{x}_\circ, t_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) + \int_{t_\circ - \epsilon}^{t_\circ + \epsilon} \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)] \, \mathrm{d}t, \tag{9.34}$$

where we used causality (9.29c) to set $G(\boldsymbol{x}, t_\circ - \epsilon | \boldsymbol{x}_\circ, t_\circ) = 0$. The divergence operator commutes with the time integral so that the right hand side term can be written

$$\int_{t_\circ-\epsilon}^{t_\circ+\epsilon} \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)] \, \mathrm{d}t = \nabla_{\boldsymbol{x}} \cdot \int_{t_\circ-\epsilon}^{t_\circ+\epsilon} \boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) \, \mathrm{d}t. \tag{9.35}$$

Now take the limit as $\epsilon \to 0$, and assume the integrand is smooth in time so that the integral vanishes,[6] thus leaving the relation

$$\lim_{\epsilon \to 0} G(\boldsymbol{x}, t_\circ + \epsilon | \boldsymbol{x}_\circ, t_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ). \tag{9.36}$$

Evidently, the Green's function, when evaluated at the source time $t = t_\circ$, equals to the spatial Dirac delta source. This property is shared with the causal free space Green's function in Section 9.2.2 (see equation (9.28)).

9.2.6 Adjoint causal Green's function

We make use of the adjoint causal Green's function for solving initial-boundary value problems for the diffusion equation. The adjoint causal Green's function is defined for $t \in (-\infty, \infty)$ and satisfies the following boundary value problem for the Neumann conditions, with these equations representing the adjoint to the Green's function equations (9.29a)-(9.29c)

$$-\frac{\partial [G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)]}{\partial t} - \nabla_{\boldsymbol{x}} \cdot \Big[\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)\Big] = \delta(t - t_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \boldsymbol{x} \in \mathcal{R} \tag{9.37a}$$

$$\hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \quad \boldsymbol{x} \in \partial\mathcal{R}, \boldsymbol{x}_\circ \notin \partial\mathcal{R} \tag{9.37b}$$

$$G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \quad t > t_\circ. \tag{9.37c}$$

Similarly, the adjoint Green's function satisfying Dirichlet boundary conditions is determined by

$$-\frac{\partial [G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)]}{\partial t} - \nabla_{\boldsymbol{x}} \cdot \Big[\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ)\Big] = \delta(t - t_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \boldsymbol{x} \in \mathcal{R} \tag{9.38a}$$

$$G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \quad \boldsymbol{x} \in \partial\mathcal{R} \tag{9.38b}$$

$$G^\ddagger(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) = 0 \quad t > t_\circ. \tag{9.38c}$$

Note the sign change on the time derivative in equations (9.37a) and (9.38a) relative to equations (9.29a) and (9.30a). This change results since the single partial time derivative is not a self-adjoint operator, reflecting the lack of time symmetry of the diffusion equation (i.e., the diffusion equation distinguishes between past and future). Also note the backward causal condition, equations (9.37c) and (9.38c). Physically we can interpret the adjoint Green's function as a solution to the diffusion equation run backwards in time.

9.2.7 Reciprocity of the Green's function and its adjoint

When studying Poisson's equation we made use of the reciprocity condition satisfied by its Green's function. We desire a corresponding reciprocity condition for the causal Green's

[6]This assumption is not fully satisfactory. Even so, we infer the property (9.36) when studying properties of the Green's function solution in Section 9.4 to the diffusion equation initial-boundary value problem. It is also directly confirmed by the eigenfunction expansion in Section 9.3.1.

function for the diffusion equation. Deriving reciprocity requires a bit more work for the diffusion equation due to the added time derivative term, which renders the adjoint diffusion operator distinct from the diffusion operator. That is, the differential diffusion operator is not self-adjoint due to sign change on the time derivative, whereas the Laplacian operator is self-adjoint.

Before starting this derivation, note that we did not derive the adjoint Green's function equation in Section 9.2.6, instead we merely wrote it down. However, introduction of the adjoint Green's function is largely motivated by the following derivation of the reciprocity relation. In this derivation we find that the Green's function for the diffusion equation satisfies a reciprocity relation with the adjoint Green's function (equation (9.47) below). Hence, as part of the following derivation we indirectly see how to construct the adjoint problem for the diffusion equation.

Setting up the derivation

To derive reciprocity, consider the partial differential equation (9.29a) with a Dirac delta source, $\delta(t-t_1)\,\delta(\boldsymbol{x}-\boldsymbol{x}_1)$, and the adjoint partial differential equation (9.37a) with a distinct Dirac delta source, $\delta(t-t_2)\,\delta(\boldsymbol{x}-\boldsymbol{x}_2)$. Multiply each of these equations by the complement Green's function and subtract

$$\begin{aligned} & G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)(\partial_t G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1) - \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)]) \\ & -G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)(-\partial_t G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2) - \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)]) \\ & \quad = G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,\delta(t-t_1)\,\delta(\boldsymbol{x}-\boldsymbol{x}_1) - G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\,\delta(t-t_2)\,\delta(\boldsymbol{x}-\boldsymbol{x}_2). \end{aligned} \tag{9.39}$$

The Dirac delta source locations, $\boldsymbol{x}_1 \in \mathcal{R}$ and $\boldsymbol{x}_2 \in \mathcal{R}$, are arbitrary points within the spatial domain $\mathcal{R}$. Likewise, the source times, t_1 and t_2, are arbitrary. In the following, we find it useful for organizational purposes to introduce an arbitrarily large time, T, so that

$$-T < t_1, t_2 < T, \tag{9.40}$$

with T later dropping out from the results through use of the causality conditions.

Integration and use of the sifting property

An integral of the right hand side of equation (9.39) over the spatial domain, $\boldsymbol{x} \in \mathcal{R}$, and over the time domain, $t \in [-T, T]$, leads to

$$\begin{aligned} \int_{-T}^{T} \int_{\mathcal{R}} \Big[G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,\delta(t-t_1)\,\delta(\boldsymbol{x}-\boldsymbol{x}_1) - G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\,\delta(t-t_2)\,\delta(\boldsymbol{x}-\boldsymbol{x}_2) \Big] \,\mathrm{d}V\,\mathrm{d}t \\ = G^{\ddagger}(\boldsymbol{x}_1,t_1|\boldsymbol{x}_2,t_2) - G(\boldsymbol{x}_2,t_2|\boldsymbol{x}_1,t_1). \end{aligned} \tag{9.41}$$

As shown in the following, the same integral of the left hand side of equation (9.39) vanishes, which then establishes the reciprocity property between the Green's function and the adjoint Green's function.

Moving the time derivative from G to $G^{\ddagger}$ and picking up a minus sign

The left hand side of equation (9.39) requires us to massage just the first term since, as we will show, this term equals to the second so that the left hand side of equation (9.39) vanishes. To prove this assertion, start by examining the time derivative. Since the spatial domain, $\mathcal{R}$, is assumed to be static,[7] we can swap the time and space derivatives to find

$$\int_{-T}^{T} G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,\partial_t\,G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\;\mathrm{d}t = \int_{-T}^{T} \left[\partial_t \left(G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\right) - \partial_t G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\;G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\right]\mathrm{d}t. \quad (9.42)$$

We now set

$$G(\boldsymbol{x},t=-T|\boldsymbol{x}_1,t_1) = 0 \quad \text{and} \quad G^{\ddagger}(\boldsymbol{x},t=+T|\boldsymbol{x}_2,t_2) = 0, \quad (9.43)$$

which result from the forward causality condition (9.29c) and the backward causality condition (9.37c). Hence, in moving the time derivative from G to $G^{\ddagger}$ we pick up a minus sign, which, again, means that the time derivative is not a self-adjoint differential operator

$$\int_{-T}^{T} G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,\partial_t\,G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\;\mathrm{d}t = -\int_{-T}^{T} \partial_t G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\;G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\,\mathrm{d}t. \quad (9.44)$$

The Laplacian is self-adjoint with a symmetric diffusion tensor

Consider next the spatial derivative term on the left hand side of equation (9.39)

$$-\int_{\mathcal{R}} G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,\nabla_{\boldsymbol{x}}\cdot[\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)]\,\mathrm{d}V = \int_{\mathcal{R}} \left[-\nabla_{\boldsymbol{x}}\cdot\left(G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\right) + \nabla_{\boldsymbol{x}}G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\cdot\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\right]\mathrm{d}V. \quad (9.45)$$

Use of the divergence theorem and either the homogeneous Neumann boundary condition (9.29b) or homogeneous Dirichlet condition (9.30b) allow us to drop the total derivative term on the right hand side. The same manipulation, with either the Neumann condition (9.37b) or Dirichlet condition (9.38b) satisfied by the adjoint Green's function $G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)$, allows us to seamlessly move the Laplacian operator from $G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)$ onto $G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)$, thus manifesting the self-adjoint nature of the Laplacian operator (with appropriate boundary conditions) even in the presence of a symmetric diffusion tensor

$$\int_{\mathcal{R}} G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\,\nabla_{\boldsymbol{x}}\cdot[\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)]\,\mathrm{d}V = \int_{\mathcal{R}} \nabla_{\boldsymbol{x}}\cdot\left[\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_2,t_2)\right]G(\boldsymbol{x},t|\boldsymbol{x}_1,t_1)\,\mathrm{d}V. \quad (9.46)$$

Reciprocity of the Green's function and the adjoint Green's function

The above manipulations show that the space-time integral for the left hand side of equation (9.39) vanishes. Consequently, we are left with the reciprocity relation satisfied by the Green's

[7] In our study of the advection-diffusion equation for passive tracers in VOLUME 4, we dispense with the assumption of a time independent spatial domain. Doing so requires extra care, both physically and mathematically, thus motivating us to postpone that discussion until we have more experience.

function and adjoint Green's function for the diffusion equation

$$G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ}) = G(\boldsymbol{x}_{\circ}, t_{\circ}|\boldsymbol{x}, t). \tag{9.47}$$

The Green's function, $G(\boldsymbol{x}_{\circ}, t_{\circ}|\boldsymbol{x}, t)$, results from placing a Dirac delta source at $(\boldsymbol{x}, t)$ and using the forward diffusion equation to determine the response at $(\boldsymbol{x}_{\circ}, t_{\circ})$ with $t < t_{\circ}$. The adjoint Green's function, $G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ})$, results from placing a Dirac delta source at $(\boldsymbol{x}_{\circ}, t_{\circ})$ and using the adjoint diffusion equation to determine the response at $(\boldsymbol{x}, t)$, again with $t < t_{\circ}$. The reciprocity condition (9.47) shows that the two responses are identical. Furthermore, the backward time nature of the adjoint problem allows us to write the equivalent expression of reciprocity

$$G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ}) = G(\boldsymbol{x}_{\circ}, t_{\circ}|\boldsymbol{x}, t) = G(\boldsymbol{x}, -t|\boldsymbol{x}_{\circ}, -t_{\circ}). \tag{9.48}$$

In general, we emphasize that reciprocity emerges from properties of the differential operator, the boundary conditions, and the causality condition.

9.2.8 The importance of reciprocity

Reciprocity in the form of equations (9.47) or (9.48) means we have no practical need to solve the adjoint Green's function equation for $G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ})$ since the adjoint Green's function equals to the Green's function after swapping the space-time points for the field and source. Hence, it is sufficient to determine the Green's function, G and then use reciprocity to determine $G^{\ddagger}$. Reciprocity then means, as found in Section 9.4.1, that the integral solutions for ψ, given by equations (9.73) and (9.74), only need the Green's function.

9.2.9 Composition property of the Green's function

Following from the reciprocity relation derived in Section 9.2.7, we derive in Exercise 9.6 the composition property satisfied by the Green's functions for the diffusion equation

$$G(\boldsymbol{x}_2, t_2|\boldsymbol{x}_1, t_1) = \int_{\mathcal{R}} G(\boldsymbol{x}_2, t_2|\boldsymbol{x}, \tau)\, G(\boldsymbol{x}, \tau|\boldsymbol{x}_1, t_1)\, \mathrm{d}V \quad \text{if } t_1 < \tau < t_2. \tag{9.49}$$

This property connects the Green's function to Markov processes, in which the composition property is known as the Chapman-Kolmogorov relation (*Gardiner*, 1985). The derivation closely follows that given in Section 9.2.7 for reciprocity, though with some extra nuance related to the time integration. *Larson* (1999) and *Holzer* (2009) discuss the composition property in the context of the advection-diffusion equation, and we return to it in VOLUME 4. *Holzer* (2009) also provides further connections to probability theory, thus promoting the interpretation of the Green's function as a transition probability.

The Green's function, $G(\boldsymbol{x}_2, t_2|\boldsymbol{x}_1, t_1)$, on the left hand side of the composition property (9.49) is the response from a Dirac delta source diffused from $(\boldsymbol{x}_1, t_1)$ and measured at the field space-time point $(\boldsymbol{x}_2, t_2)$. The right hand side says that $G(\boldsymbol{x}_2, t_2|\boldsymbol{x}_1, t_1)$ is identical to the composition of a Green's function feeling the source at $(\boldsymbol{x}_1, t_1)$ but now sampled at an intermediate space-time position, $(\boldsymbol{x}, \tau)$, and then further diffused to $(\boldsymbol{x}_2, t_2)$, with integration over all possible intermediate positions $\boldsymbol{x}$. Furthermore, note that the intermediate sampling can occur at an arbitrary intermediate time, τ, so long as $t_1 < \tau < t_2$. The composition property allows us to conceive of a long-time interval Green's function as the composition of an arbitrary number of shorter time interval Green's functions.

9.3 Spectral decomposition of the Green's function

In Sections 8.12 and 8.13 we provided a spectral decomposition of the generalized Laplace operator in the presence of either Dirichlet or Neumann boundary conditions. We here make use of that decomposition to further our understanding of the diffusion equation Green's functions.

9.3.1 Dirichlet boundary conditions

We start with the Dirichlet boundary problem, in which the Green's function satisfies equations (9.30a)-(9.30c), repeated here for convenience

$$\partial_t[G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ)] - \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ)] = \delta(t-t_\circ)\,\delta(\boldsymbol{x}-\boldsymbol{x}_\circ) \qquad \boldsymbol{x} \in \mathcal{R} \tag{9.50a}$$

$$G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R} \tag{9.50b}$$

$$G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ) = 0 \qquad t < t_\circ. \tag{9.50c}$$

The corresponding eigenvalue problem for the generalized Laplace operator is given by equation (8.219), here written as

$$-\nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_a) = \lambda_a\,\Phi_a, \quad \boldsymbol{x} \in \mathcal{R} \quad \text{and} \quad \Phi_a = 0, \quad \boldsymbol{x} \in \partial\mathcal{R}. \tag{9.51}$$

The discrete set of spatial eigenfunctions, $\Phi_a(\boldsymbol{x})$, form a complete basis over the space of functions on $\mathcal{R}$ that satisfy homogeneous Dirichlet boundary conditions. Hence, we can expand the Green's function according to the eigenfunction decomposition

$$G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ) = \sum_a C_a(t|\boldsymbol{x}_\circ,t_\circ)\,\Phi_a(\boldsymbol{x}), \tag{9.52}$$

thus separating out the $\boldsymbol{x}$ dependence. Making use of this expansion in the Green's function equation (9.50a) leads to

$$\sum_a \Phi_a(\boldsymbol{x})\,[\partial_t C_a + \lambda_a\,C_a] = \delta(t-t_\circ)\,\delta(\boldsymbol{x}-\boldsymbol{x}_\circ). \tag{9.53}$$

Completeness of the eigenfunctions ensures that the spatial Dirac delta can be spectrally decomposed according to equation (8.239)

$$\delta(\boldsymbol{x}-\boldsymbol{x}_\circ) = \sum_a \Phi_a(\boldsymbol{x}_\circ)\,\Phi_a(\boldsymbol{x}), \tag{9.54}$$

which then brings equation (9.53) to

$$\sum_a \Phi_a(\boldsymbol{x})[\partial_t C_a + \lambda_a\,C_a - \delta(t-t_\circ)\,\Phi_a(\boldsymbol{x}_\circ)] = 0. \tag{9.55}$$

Satisfaction of this equation requires the bracket term to vanish, which then renders

$$C_a(t|\boldsymbol{x}_\circ,t_\circ) = \mathcal{H}(t-t_\circ)\,\mathrm{e}^{-\lambda_a\,(t-t_\circ)}\,\Phi_a(\boldsymbol{x}_\circ). \tag{9.56}$$

In verifying this solution we must use the identity

$$\partial_t C_a + \lambda_a\,C_a = \delta(t-t_\circ)\,\mathrm{e}^{-\lambda_a\,(t-t_\circ)}\,\Phi_a(\boldsymbol{x}_\circ) = \delta(t-t_\circ)\,\Phi_a(\boldsymbol{x}_\circ), \tag{9.57}$$

which follows from our discussion of equivalence classes of Dirac deltas in Section 5.8. Namely, the exponential, $e^{-\lambda_a (t-t_\circ)}$, is a non-dimensional function that equals to unity at the source time, $t = t_\circ$, and as such it has no affect on the Dirac delta and so it is irrelevant.

Making use of the C_a coefficients (9.56) in equation (9.52) leads to the spectral decomposition of the Green's function for the diffusion equation with Dirichlet boundaries

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \mathcal{H}(t - t_\circ) \sum_a e^{-\lambda_a (t-t_\circ)} \Phi_a(\boldsymbol{x}_\circ) \Phi_a(\boldsymbol{x}). \tag{9.58}$$

This is a remarkably elegant expression that manifests properties of the Green's function derived in Section 9.2. For example, in addition to manifesting causaility via the Heaviside, it manifests property (9.36), whereby the Green's function evaluated at the source time equals to the spatial Dirac delta

$$\lim_{\epsilon \to 0} G(\boldsymbol{x}, t_\circ + \epsilon|\boldsymbol{x}_\circ, t_\circ) = \sum_a \Phi_a(\boldsymbol{x}_\circ) \Phi_a(\boldsymbol{x}) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ). \tag{9.59}$$

We are thus left with the picture of the Green's function as a Dirac pulse located at $\boldsymbol{x} = \boldsymbol{x}_\circ$ that fires at time $t = t_\circ$, with the Green's function at this time spectrally decomposed into an equally weighted set of all eigenfunctions. Afterward, for times $t > t_\circ$, each term in the eigenfunction expansion (9.58) is exponentially damped. Furthermore, the higher eigenmodes decay faster given that the eigenvalues, λ_a, are a non-decreasing set. As a result, the lowest mode is the dominant pattern for times $t > t_\circ$ as a result of the diffusion process acting to more quickly deplete the amplitude of higher modes.

9.3.2 Neumann boundary conditions

In Section 8.13, we studied the Neumann eigenfunctions, Φ_a, for the Laplace operator, which satisfy

$$-\nabla \cdot (\boldsymbol{K} \cdot \nabla \Phi_a) = \lambda_a \Phi_a, \quad \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \quad \hat{\boldsymbol{n}} \cdot \boldsymbol{K} \cdot \nabla \Phi_a = 0, \quad \boldsymbol{x} \in \partial \mathcal{R}. \tag{9.60}$$

Note that we continue to use Φ_a to denote the eigenfunctions, where in this subsection they are eigenfunctions for the Neumann problem rather than the Dirichlet problem. The lowest Neumann eigenmode has a constant eigenfunction and zero eigenvalue, thus requiring extra effort to properly account for this nontrivial null space. We present the necessary analysis in Section 8.13 to account for this zero mode, and make use of that analysis here.

We follow through the recipe used for Dirichlet boundary in Section 9.3.1, only now considering the Neumann boundaries. That is, we expand the Neumann Green's function satisfying equations (9.29a)-(9.29c) in terms of the Neumann eigenfunctions. Indeed, all steps follow directly from the Dirichlet case, so that we can immediately write down the Neumann Green's function

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \mathcal{H}(t - t_\circ) \sum_a e^{-\lambda_a (t-t_\circ)} \Phi_a(\boldsymbol{x}_\circ) \Phi_a(\boldsymbol{x}) \tag{9.61a}$$

$$= \mathcal{H}(t - t_\circ) \left[V^{-1} + \sum_{a=1}^{\infty} e^{-\lambda_a (t-t_\circ)} \Phi_a(\boldsymbol{x}_\circ) \Phi_a(\boldsymbol{x}) \right], \tag{9.61b}$$

where we separated out the zero mode, $\Phi_0 = V^{-1/2}$, with V the domain volume. Evidently, the zero mode remains in the long time limit, even after all higher modes are damped,

$$\lim_{t\to\infty} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ) = V^{-1}. \tag{9.62}$$

To help interpret this result, let us we look ahead to the solution (9.73) for the Neumann initial-boundary value problem, and interpret these results in terms of tracer concentration. In the case with zero boundary fluxes and zero interior source, the infinite time limit for the tracer concentration equals to the domain average of the initial condition. That is, a tracer concentration with zero boundary flux and zero interior source has an equilibrium (infinite time limit) given by a the domain average of the initial tracer concentration. We expect this result based on physical arguments alone, so that it is satisfying to see how it also appears from the mathematical analysis.

9.4 Green's functions and solutions to the diffusion equation

In this section we build from the variety of Green's functions developed in Section 9.2 to now derive integral expressions for the solutions to diffusion initial-boundary value problems.

9.4.1 An integral solution to the diffusion equation

Having established reciprocity for the Green's function in equation (9.47), we are ready to derive an integral solution for the field, $\psi(\boldsymbol{x},t)$, satisfying the diffusion equation initial-boundary value problem with either the Neumann boundary conditions

$$\partial_t\psi(\boldsymbol{x},t) - \nabla_{\boldsymbol{x}}\cdot[\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x},t)] = \Lambda(\boldsymbol{x},t) \quad \boldsymbol{x}\in\mathcal{R} \tag{9.63a}$$

$$\hat{\boldsymbol{n}}\cdot\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x},t) = \Sigma(\boldsymbol{x},t) \quad \boldsymbol{x}\in\partial\mathcal{R} \tag{9.63b}$$

$$\psi(\boldsymbol{x},t=t_{\mathsf{init}}) = I(\boldsymbol{x}) \qquad \boldsymbol{x}\in\mathcal{R}, \tag{9.63c}$$

or Dirichlet boundary conditions

$$\partial_t\psi(\boldsymbol{x},t) - \nabla_{\boldsymbol{x}}\cdot[\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x},t)] = \Lambda(\boldsymbol{x},t) \quad \boldsymbol{x}\in\mathcal{R} \tag{9.64a}$$

$$\psi(\boldsymbol{x},t) = \sigma(\boldsymbol{x},t) \quad \boldsymbol{x}\in\partial\mathcal{R} \tag{9.64b}$$

$$\psi(\boldsymbol{x},t=t_{\mathsf{init}}) = I(\boldsymbol{x}) \qquad \boldsymbol{x}\in\mathcal{R}. \tag{9.64c}$$

In these equations we introduced the initial time, $t=t_{\mathsf{init}}$, which is distinct from the Dirac delta source time, $t=t_\circ$. We are interested in the evolution of ψ after specification of the initial data, $\psi(\boldsymbol{x},t=t_{\mathsf{init}}) = I(\boldsymbol{x})$. Correspondingly, t_{init} defines the lower limit on time integrals in the following. Correspondingly, the Dirac delta source is fired after the initial time,

$$t_{\mathsf{init}} < t_\circ. \tag{9.65}$$

Use of the $\nabla_{\boldsymbol{x}}$ notation is not needed for these equations, since there is no source point, $\boldsymbol{x}_\circ$, in any of the expressions. However, writing $\nabla_{\boldsymbol{x}}$ helps us remain organized during the following manipulations. Finally, note that the diffusion tensor appearing in these equations means that the source function, $\Lambda(\boldsymbol{x},t)$, in equations (9.63a) and (9.64a), as well as the boundary data, $\Sigma(\boldsymbol{x},t)$, in equation (9.63b), have different physical dimensions from their counterparts found in the Poisson boundary value problems studied in Chapter 8.

The following derivation emulates that for the Poisson equation in Chapter 8, yet with distinct features arising from time evolution and the corresponding need to use the adjoint causal Green's function, $G^{\ddagger}$. We expose many details to reveal important notions arising with Green's function methods for initial-boundary value problems.

Setting up the derivation

To start the derivation, multiply the diffusion equation (9.63a) by the adjoint Green's function, $G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ})$, and the adjoint Green's function equation (9.37a) by $\psi(\boldsymbol{x}, t)$. Subtracting and rearranging leads to

$$\partial_t(G^{\ddagger}\,\psi)+\nabla\cdot[\psi\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G^{\ddagger}-G^{\ddagger}\,\boldsymbol{K}\cdot\nabla\psi] = G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ})\,\Lambda(\boldsymbol{x}, t)-\psi(\boldsymbol{x}, t)\,\delta(\boldsymbol{x}-\boldsymbol{x}_{\circ})\,\delta(t-t_{\circ}), \tag{9.66}$$

where we temporarily suppressed arguments on the left hand side for brevity. Since the spatial domain is assumed to be static, we can integrate this equation over space and time without concern for the order of integration.

Time integration

A time integral of the first left hand side term in equation (9.66) leads to[8]

$$\int_{t_{\mathsf{init}}}^{T} \partial_t[G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ})\,\psi(\boldsymbol{x}, t)]\,\mathrm{d}t = G^{\ddagger}(\boldsymbol{x}, T|\boldsymbol{x}_{\circ}, t_{\circ})\,\psi(\boldsymbol{x}, T) - G^{\ddagger}(\boldsymbol{x}, t_{\mathsf{init}}|\boldsymbol{x}_{\circ}, t_{\circ})\,\psi(\boldsymbol{x}, t_{\mathsf{init}}) \tag{9.67a}$$

$$= -G^{\ddagger}(\boldsymbol{x}, t_{\mathsf{init}}|\boldsymbol{x}_{\circ}, t_{\circ})\,I(\boldsymbol{x}), \tag{9.67b}$$

where we made use of the backward causal condition (9.37c) satisfied by the adjoint Green's function to set $G^{\ddagger}(\boldsymbol{x}, t = T|\boldsymbol{x}_{\circ}, t_{\circ}) = 0$, and used the initial condition (9.63c) to introduce the initial data, $\psi(\boldsymbol{x}, t_{\mathsf{init}}) = I(\boldsymbol{x})$.

Space integration

A space integral over all observation points, and use of the divergence theorem, brings the divergence term on the left side of equation (9.66) into

$$\int_{\mathcal{R}} \nabla\cdot[\psi(\boldsymbol{x}, t)\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ}) - G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ})\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}, t)]\,\mathrm{d}V$$

$$= \oint_{\partial\mathcal{R}} \Big[\psi(\boldsymbol{x}, t)\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ}) - G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_{\circ}, t_{\circ})\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x}, t)\Big]\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}. \tag{9.68}$$

We keep both boundary terms pending specification of whether the fields satisfy Dirichlet or Neumann boundary conditions. This boundary integral has the same appearance as found for Poisson's equation in Chapter 8, with the added feature here of the diffusion tensor. Also, recall that the diffusion tensor is generally a function of space and time, $\boldsymbol{K}(\boldsymbol{x}, t)$, with the dependence supressed for brevity.

[8] Recall that T is an arbitrarily large but finite time. It drops out from the final answer, but proves useful for organizational purposes in intermediate steps.

Integrating the right hand side of equation (9.66)

A space and time integral for the right hand side of equation (9.66), along with the sifting properties of the Dirac delta, render

$$\int_{t_{\text{init}}}^{T}\left[\int_{\mathcal{R}}[G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\Lambda(\boldsymbol{x},t)-\psi(\boldsymbol{x},t)\,\delta(\boldsymbol{x}-\boldsymbol{x}_{\circ})\,\delta(t-t_{\circ})]\,\mathrm{d}V\right]\mathrm{d}t$$
$$=-\psi(\boldsymbol{x}_{\circ},t_{\circ})+\int_{t_{\text{init}}}^{T}\left[\int_{\mathcal{R}}G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\Lambda(\boldsymbol{x},t)\mathrm{d}V\right]\mathrm{d}t. \quad (9.69)$$

Notice how integration over all field space-time points, $(\boldsymbol{x},t)$, serves to pick out the field, ψ, evaluated at the space-time point, $(\boldsymbol{x}_{\circ},t_{\circ})$, which is the point where the Dirac delta source is located.

Rearrangement and use of reciprocity

Bringing the above results together leads to the expression

$$\psi(\boldsymbol{x}_{\circ},t_{\circ})=\int_{\mathcal{R}}G^{\ddagger}(\boldsymbol{x},t_{\text{init}}|\boldsymbol{x}_{\circ},t_{\circ})\,I(\boldsymbol{x})\,\mathrm{d}V+\int_{t_{\text{init}}}^{t_{\circ}}\left[\int_{\mathcal{R}}G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\Lambda(\boldsymbol{x},t)\,\mathrm{d}V\right]\mathrm{d}t$$
$$+\int_{t_{\text{init}}}^{t_{\circ}}\left[\oint_{\partial\mathcal{R}}\left[G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x},t)-\psi(\boldsymbol{x},t)\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\right]\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}\right]\mathrm{d}t. \quad (9.70)$$

The time integrals are restricted to the range $t\in[t_{\text{init}},t_{\circ}]$ through use of the causality condition (9.37c) for the adjoint Green's function. Hence, the arbitrary time, T, drops out from the solution and there is no dependence on fields at times later than $t_{\circ}$ nor before t_{init}.

Use of reciprocity (9.47) allows us to replace the adjoint Green's function with the Green's function to thus bring equation (9.70) to

$$\psi(\boldsymbol{x}_{\circ},t_{\circ})=\int_{\mathcal{R}}G(\boldsymbol{x}_{\circ},t_{\circ}|\boldsymbol{x},t_{\text{init}})\,I(\boldsymbol{x})\,\mathrm{d}V+\int_{t_{\text{init}}}^{t_{\circ}}\left[\int_{\mathcal{R}}G(\boldsymbol{x}_{\circ},t_{\circ}|\boldsymbol{x},t)\,\Lambda(\boldsymbol{x},t)\,\mathrm{d}V\right]\mathrm{d}t$$
$$+\int_{t_{\text{init}}}^{t_{\circ}}\left[\oint_{\partial\mathcal{R}}[G(\boldsymbol{x}_{\circ},t_{\circ}|\boldsymbol{x},t)\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}\psi(\boldsymbol{x},t)-\psi(\boldsymbol{x},t)\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}}G(\boldsymbol{x}_{\circ},t_{\circ}|\boldsymbol{x},t)]\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}\right]\mathrm{d}t. \quad (9.71)$$

Finally, it is convenient to relabel $(\boldsymbol{x}_{\circ},t_{\circ})\leftrightarrow(\boldsymbol{x},t)$ to write

$$\psi(\boldsymbol{x},t)=\underbrace{\int_{\mathcal{R}}G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\text{init}})\,I(\boldsymbol{x}_{\circ})\,\mathrm{d}V_{\circ}}_{\text{space integral of }G\text{ times }I\text{ over }\mathcal{R}}+\underbrace{\int_{t_{\text{init}}}^{t}\left[\int_{\mathcal{R}}G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\Lambda(\boldsymbol{x}_{\circ},t_{\circ})\,\mathrm{d}V_{\circ}\right]\mathrm{d}t_{\circ}}_{\text{space-time integral of }G\text{ times }\Lambda\text{ over }\mathcal{R}}$$
$$+\underbrace{\int_{t_{\text{init}}}^{t}\left[\oint_{\partial\mathcal{R}}[G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}_{\circ}}\psi(\boldsymbol{x}_{\circ},t_{\circ})-\psi(\boldsymbol{x}_{\circ},t_{\circ})\,\boldsymbol{K}\cdot\nabla_{\boldsymbol{x}_{\circ}}G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})]\cdot\hat{\boldsymbol{n}}_{\boldsymbol{x}_{\circ}}\,\mathrm{d}\mathcal{S}_{\circ}\right]\mathrm{d}t_{\circ}.}_{\text{space-time integral of boundary terms over }\partial\mathcal{R}}$$
$$(9.72)$$

Specializing to Neumann boundary conditions leads to

$$\psi^{\text{Neumann}}(\boldsymbol{x},t)=\int_{\mathcal{R}}G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\text{init}})\,I(\boldsymbol{x}_{\circ})\,\mathrm{d}V_{\circ}+\int_{t_{\text{init}}}^{t}\left[\int_{\mathcal{R}}G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\Lambda(\boldsymbol{x}_{\circ},t_{\circ})\,\mathrm{d}V_{\circ}\right]\mathrm{d}t_{\circ}$$

$$+\int_{t_{\mathsf{init}}}^{t}\left[\oint_{\partial\mathcal{R}} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ)\,\Sigma(\boldsymbol{x}_\circ,t_\circ)\,\mathrm{d}\mathcal{S}_\circ\right]\mathrm{d}t_\circ, \quad (9.73)$$

whereas the solution with Dirichlet boundary conditions is

$$\psi^{\mathsf{Dirichlet}}(\boldsymbol{x},t) = \int_{\mathcal{R}} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_{\mathsf{init}})\,I(\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ + \int_{t_{\mathsf{init}}}^{t}\left[\int_{\mathcal{R}} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ)\,\Lambda(\boldsymbol{x}_\circ,t_\circ)\,\mathrm{d}V_\circ\right]\mathrm{d}t_\circ$$
$$-\int_{t_{\mathsf{init}}}^{t}\left[\oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ,t_\circ)\,\boldsymbol{K}(\boldsymbol{x}_\circ,t_\circ)\cdot\nabla_{\boldsymbol{x}_\circ}G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}\,\mathrm{d}\mathcal{S}_\circ\right]\mathrm{d}t_\circ. \quad (9.74)$$

9.4.2 Properties of the solution: initial conditions

Many properties of the solution (9.72) are also reflected in the Poisson equation solutions from Chapter 8. In particular, it manifests the superposition principle shared by linear systms, with the solution given by the sum of three terms arising from the initial conditions, distributed volume source, and spatial boundary conditions. We expect to have this connection given that the steady state diffusion equation satisfies a generalized Poisson equation (generalized by the presence of a diffusion tensor). Uniqueness of the solution (9.72) also follows as in the discussion of the Poisson equation. Namely, consider two solutions to the diffusion equation and take their difference, $\Psi = \psi_{\mathsf{a}} - \psi_{\mathsf{b}}$. We readily see that Ψ satisfies the homogeneous diffusion equation with homogeneous boundary conditions along with a zero initial condition. Ψ thus remains zero for both the Dirichlet and Neumann cases, thus proving that the solution to both problems is unique.

By sampling the solution (9.72) as time decreases towards the initial time, $t \downarrow t_{\mathsf{init}}$, and noting the initial condition $\psi(\boldsymbol{x},t_{\mathsf{init}}) = I(\boldsymbol{x})$, we are led to[9]

$$\lim_{t\downarrow t_{\mathsf{init}}} \psi(\boldsymbol{x},t) = I(\boldsymbol{x}) = \lim_{t\downarrow t_{\mathsf{init}}} \int_{\mathcal{R}} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_{\mathsf{init}})\,I(\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ. \quad (9.75)$$

This temporal sampling of the field time is distinguished from the source time, which here is fixed at the initial time, t_{init}. Self-consistency in equation (9.75) implies that the Green's function for both Neumann and Dirichlet boundary conditions satisfies the initial condition

$$\lim_{t\downarrow t_{\mathsf{init}}} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_{\mathsf{init}}) = \delta(\boldsymbol{x}-\boldsymbol{x}_\circ) \quad \text{with } \boldsymbol{x},\boldsymbol{x}_\circ \in \mathcal{R}. \quad (9.76)$$

That is, the Green's function is initialized as a Dirac delta pulse at the source point, $\boldsymbol{x}_\circ$, which is a result already derived in Section 9.2.5 (equation (9.36)) through use of causality. This result then leads to the identity

$$\lim_{t\downarrow t_{\mathsf{init}}} \int_{\mathcal{R}} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_{\mathsf{init}})\,I(\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ = \int_{\mathcal{R}} \delta(\boldsymbol{x}-\boldsymbol{x}_\circ)\,I(\boldsymbol{x}_\circ)\,\mathrm{d}V_\circ = I(\boldsymbol{x}). \quad (9.77)$$

9.4.3 Properties of the solution with Neumann boundary conditions

Acting with $\boldsymbol{K}(\boldsymbol{x},t)\cdot\nabla_{\boldsymbol{x}}$ on the Neumann solution (9.73); evaluating the expression on the boundary $\boldsymbol{x} = \boldsymbol{x}_{\partial\mathcal{R}} \in \partial\mathcal{R}$; and then projecting onto the outward unit normal, $\hat{\boldsymbol{n}}_{\boldsymbol{x}}$, serves to annihilate the volume integrals as per the homogeneous Neumann condition satisfied by the

[9] Since t_{init} is the initial time, the limit $t \downarrow t_{\mathsf{init}}$ means that $t = t_{\mathsf{init}} + \epsilon$ with $\epsilon \to 0$.

Green's function (9.29b). We are thus left with

$$\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \boldsymbol{K}(\boldsymbol{x},t) \cdot \nabla_{\boldsymbol{x}}\psi(\boldsymbol{x},t) = \Sigma(\boldsymbol{x},t) \tag{9.78a}$$

$$= \int_{t_{\mathsf{init}}}^{t} \left[\oint_{\partial\mathcal{R}} \hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \boldsymbol{K}(\boldsymbol{x},t) \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\, \Sigma(\boldsymbol{x}_{\circ},t_{\circ})\, \mathrm{d}\mathcal{S}_{\circ} \right] \mathrm{d}t_{\circ}. \tag{9.78b}$$

Self-consistency implies that the Green's function for the Neumann problem, when evaluated on the spatial boundary, satisfies

$$\hat{\boldsymbol{n}}_{\boldsymbol{x}} \cdot \boldsymbol{K}(\boldsymbol{x},t) \cdot \nabla_{\boldsymbol{x}} G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ}) = \delta(t-t_{\circ})\, \delta^{(2)}(\boldsymbol{x}-\boldsymbol{x}_{\circ}) \quad \text{with } \boldsymbol{x},\boldsymbol{x}_{\circ} \in \partial\mathcal{R}, \tag{9.79}$$

where $\delta^{(2)}(\boldsymbol{x}-\boldsymbol{x}_{\circ})$ is a surface Dirac delta with physical dimensions inverse area. The boundary condition (9.79) generalizes the analogous property holding for the Poisson equation Green's function. Furthermore, it extends the homogeneous Neumann condition (9.29b) holding when $\boldsymbol{x} \in \partial\mathcal{R}$ and $\boldsymbol{x}_{\circ} \notin \partial\mathcal{R}$ to now allow $\boldsymbol{x} \in \partial\mathcal{R}$ and $\boldsymbol{x}_{\circ} \in \partial\mathcal{R}$.

When studying the Poisson boundary value problem, we saw how to reformulate the Neumann boundary condition into tgihe interior by modifying the source function, thus introducing the construct of a Dirac delta sheet. The diffusion equation Neumann solution (9.73) allows for the same reformulation by writing

$$\psi^{\mathsf{Neumann}}(\boldsymbol{x},t) = \int_{\mathcal{R}} G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\mathsf{init}})\, I(\boldsymbol{x}_{\circ})\, \mathrm{d}V_{\circ} + \int_{t_{\mathsf{init}}}^{t} \left[\int_{\mathcal{R}} G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\, \Lambda^{*}(\boldsymbol{x}_{\circ},t_{\circ})\, \mathrm{d}V_{\circ} \right] \mathrm{d}t_{\circ}, \tag{9.80}$$

where the modified source function follows from that used for the Poisson equation

$$\Lambda^{*}(\boldsymbol{x}_{\circ},t_{\circ}) = \Lambda(\boldsymbol{x}_{\circ},t_{\circ}) + \Sigma(\boldsymbol{x}_{\circ},t_{\circ})\, \delta[\hat{\boldsymbol{n}} \cdot (\boldsymbol{x}_{\circ} - \boldsymbol{x}_{\partial\mathcal{R}})]. \tag{9.81}$$

9.4.4 Properties of the solution with Dirichlet boundary conditions

Evaluating the Dirichlet solution (9.74) on a spatial boundary, $\boldsymbol{x} = \boldsymbol{x}_{\partial\mathcal{R}} \in \partial\mathcal{R}$, eliminates both of the volume integrals so that we are left with

$$\psi^{\mathsf{Dirichlet}}(\boldsymbol{x}_{\partial\mathcal{R}},t) = \sigma(\boldsymbol{x}_{\partial\mathcal{R}},t) = -\int_{t_{\mathsf{init}}}^{t} \left[\oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_{\circ},t_{\circ})\, \boldsymbol{K} \cdot \nabla_{\boldsymbol{x}_{\circ}} G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ}) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_{\circ}}\, \mathrm{d}\mathcal{S}_{\circ} \right] \mathrm{d}t_{\circ}. \tag{9.82}$$

Self-consistency implies that the Green's function for the Dirichlet problem, when evaluated on the spatial boundary, satisfies

$$\hat{\boldsymbol{n}}_{\boldsymbol{x}_{\circ}} \cdot \boldsymbol{K}(\boldsymbol{x}_{\circ},t_{\circ}) \cdot \nabla_{\boldsymbol{x}_{\circ}} G(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ}) = -\delta(t-t_{\circ})\, \delta^{(2)}(\boldsymbol{x}-\boldsymbol{x}_{\circ}) \quad \text{with } \boldsymbol{x},\boldsymbol{x}_{\circ} \in \partial\mathcal{R}, \tag{9.83}$$

which is a generalization of the analogous property holding for the Poisson equation Green's function. It is notable that this boundary condition appears with the opposite sign to the analog (9.79) holding for the Neumann conditions.

9.4.5 The boundary propagator for the Dirichlet problem

In Chapter 8, we studied the boundary Green's function for the Poisson equation. Here we extend those ideas to the boundary propagator for the diffusion equation. The boundary propagator mediates the transfer of Dirichlet boundary information into the region interior. Boundary propagators for diffusion and advection-diffusion (VOLUME 4) have extensive use in geophysical fluids given that many tracers have no interior sources.

Defining the boundary propagator

To focus on the role of the boundary, consider a tracer whose initial conditions and interior source both vanish: $I(\boldsymbol{x}) = 0$ and $\Lambda(\boldsymbol{x}, t) = 0$. Assuming Dirichlet boundary conditions, the initial-boundary value problem (9.63a)-(9.63c) simplifies to

$$\partial_t \psi(\boldsymbol{x}, t) - \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} \psi(\boldsymbol{x}, t)] = 0 \qquad \boldsymbol{x} \in \mathcal{R} \tag{9.84a}$$

$$\psi(\boldsymbol{x}, t) = \sigma(\boldsymbol{x}, t) \qquad \boldsymbol{x} \in \partial\mathcal{R} \tag{9.84b}$$

$$\psi(\boldsymbol{x}, t = t_{\mathsf{init}}) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \tag{9.84c}$$

with the corresponding Dirichlet Green's function solution (9.74) taking the form

$$\psi(\boldsymbol{x}, t) = -\int_{t_{\mathsf{init}}}^{t} \left[\oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ, t_\circ)\, \boldsymbol{K}(\boldsymbol{x}_\circ, t_\circ) \cdot \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ}\, \mathrm{d}\mathcal{S}_\circ \right] \mathrm{d}t_\circ. \tag{9.85}$$

We define the boundary propagator as the kernel in this equation

$$G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) \equiv -\boldsymbol{K}(\boldsymbol{x}_\circ, t_\circ) \cdot \nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \quad \text{with } \boldsymbol{x}_\circ \in \partial\mathcal{R}, \tag{9.86}$$

with this definition giving G^{bp} the dimensions of $\mathrm{L}^{-2}\ \mathrm{T}^{-1}$. Use of the boundary propagator brings the Dirichlet solution (9.85) into the succinct form

$$\psi(\boldsymbol{x}, t) = \int_{t_{\mathsf{init}}}^{t} \left[\oint_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ, t_\circ)\, G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)\, \mathrm{d}\mathcal{S}_\circ \right] \mathrm{d}t_\circ. \tag{9.87}$$

Boundary value problem for the boundary propagator

If we know the Green's function, $G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)$, then we can compute the boundary propagator through the definition (9.86). Alternatively, we can directly determine the boundary propagator by solving its boundary value problem, which we now discuss. Following from the definition (9.86) and the boundary condition (9.83), we know that

$$G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(t - t_\circ)\, \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ) \quad \text{with } \boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}. \tag{9.88}$$

Hence, the boundary propagator, when evaluated along the boundary, is a Dirac delta source that fires at time $t = t_\circ$ at the location $\boldsymbol{x} = \boldsymbol{x}_\circ \in \partial\mathcal{R}$.

In Exercise 9.7 we derive the following causal boundary value problem for the boundary propagator

$$\partial_t G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) - \nabla_{\boldsymbol{x}} \cdot [\boldsymbol{K} \cdot \nabla_{\boldsymbol{x}} G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)] = 0 \qquad \boldsymbol{x} \in \mathcal{R} \tag{9.89a}$$

$$G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad t < t_\circ \tag{9.89b}$$

$$G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(t - t_\circ)\, \delta^{(2)}(\boldsymbol{x} - \boldsymbol{x}_\circ) \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \partial\mathcal{R}. \tag{9.89c}$$

In words, we see that upon firing the Dirac delta source on the boundary at time $t = t_\circ$ and point $\boldsymbol{x} = \boldsymbol{x}_\circ \in \partial\mathcal{R}$, the boundary propagator diffuses the Dirac source into the region interior. Whereas the Dirichlet Green's function, $G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)$, is zero along the boundary and yet feels the Dirac delta source within the interior, the boundary propagator, $G^{\mathsf{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)$, places the Dirac delta source on the boundary and feels no source within the interior. Just as the causality condition means that the Green's function is a function of $t - t_\circ$, so too is the boundary propagator. Furthermore, a focus on the boundary propagator rather than the Green's function allows us to dispense with the need to compute the normal gradient of the Green's function at the boundary.

Normalization of the boundary propagator

Consider the special case of a uniform constant Dirichlet boundary data, $\sigma = \sigma_{\mathsf{const}}$, in the solution (9.87). Diffusion acts on this constant boundary data to spread it throughout the region. After sufficient time, the solution reaches a steady state whereby $\psi = \sigma_{\mathsf{const}}$ at every point within the domain. This result means that the boundary propagator satisfies the normalization condition

$$\lim_{t_{\mathsf{init}} \to -\infty} \int_{t_{\mathsf{init}}}^{t} \left[\oint_{\partial \mathcal{R}} G^{\mathsf{bp}}(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ) \, \mathrm{d}\mathcal{S}_\circ \right] \mathrm{d}t_\circ = 1, \tag{9.90}$$

where the lower time limit is meant to indicate some arbitrary time sufficiently far in the past so that a steady state has been reached. This normalization condition holds for every point within the domain and for any time, t. It corresponds to the analogous normalization condition satisfied by the boundary Green's function for the Poisson equation.

9.5 Initial value problems and response functions

In this section we focus on initial value problems and study the response functions that help to characterize a dynamical system. For this purpose, consider the first order temporal ordinary differential equation

$$[\mathrm{d}/\mathrm{d}t + \lambda(t)] \, \psi(t) = F(t) \quad \text{with } \psi(t \le t_{\mathsf{init}}) = 0, \tag{9.91}$$

where ψ is some geophysical field, such as the anomalous sea surface temperature, λ is a feedback parameter that is positive for a damped system, and F is a forcing function such as that introduced by atmospheric variability on the surface ocean. This equation has found widespread use in the climate dynamics community, largely following the work of *Hasselmann* (1976).

In earlier sections of this chapter, we repeatedly pointed out the connection between the diffusion equation and Poisson's equation, with the connection holding through their common elliptic spatial operator. In this section we take the complement approach by focusing on the time aspect of the diffusion equation. Namely, we are here only concerned with temporal behavior, so that all spatial information is ignored. Hence, connection to the earlier material on the diffusion equation is solely through time evolution via the single time derivative.

9.5.1 Impulse response function

Consider equation (9.91) with $\lambda > 0$ a time-independent feedback parameter damping the system back to zero, and with the forcing given by a weighted Dirac delta

$$[\mathrm{d}/\mathrm{d}t + \lambda] \, G(t|t_\circ) = \alpha \, \delta(t - t_\circ) \quad \text{with } G(t|t_\circ) = 0 \text{ for } t < t_\circ, \tag{9.92}$$

where $\alpha > 0$ is a constant dimensionless scaling coefficient. We refer to the resulting causal Green's function, $G(t|t_\circ)$, as the impulse response function since it represents the response of the dynamical system to an impulse provided by the temporal Dirac delta.

Initial condition for the impulse response function

To determine the initial condition for the Green's function, integrate equation (9.92) over an interval containing the source time, $t_\circ$, to render ($\epsilon > 0$)

$$\lim_{\epsilon \to 0} \left[G(t_\circ + \epsilon | t_\circ) - G(t_\circ - \epsilon | t_\circ) + \int_{t_\circ - \epsilon}^{t_\circ + \epsilon} \lambda \, G(t|t_\circ) \, \mathrm{d}t \right] = \alpha. \tag{9.93}$$

We enforce causality by setting

$$G(t_\circ - \epsilon | t_\circ) = 0 \quad \text{for } \epsilon > 0, \tag{9.94}$$

in which case

$$\lim_{\epsilon \to 0} G(t_\circ + \epsilon | t_\circ) + \lim_{\epsilon \to 0} \int_{t_\circ}^{t_\circ + \epsilon} \lambda \, G(t|t_\circ) \, \mathrm{d}t = \alpha. \tag{9.95}$$

We assume that the integral vanishes in the limit of $\epsilon \to 0$, which is a sensible assumption since the only means to have a nonzero integral is if the Green's function had a singularity similar to a Dirac delta. We are thus led to the initial condition for the Green's function

$$G(t = t_\circ | t_\circ) = \alpha. \tag{9.96}$$

Solution for the impulse response function

The causality condition $G(t < t_\circ | t_\circ) = 0$ can be satisfied by introducing the Heaviside step function

$$G(t|t_\circ) = \mathcal{H}(t - t_\circ) \, g(t) \quad \text{with} \quad [\mathrm{d}/\mathrm{d}t + \lambda] \, g = 0 \quad \text{and} \quad g(t = t_\circ) = \alpha, \tag{9.97}$$

with the solution readily determined to be the damped exponential

$$G(t|t_\circ) = \mathcal{H}(t - t_\circ) \, \alpha \, \mathrm{e}^{-\lambda \, (t - t_\circ)}. \tag{9.98}$$

We verify this function satisfies the initial value problem (9.92) by noting that

$$\mathrm{d}G(t|t_\circ)/\mathrm{d}t = \alpha \, \delta(t - t_\circ) \, \mathrm{e}^{-\lambda \, (t - t_\circ)} - \lambda \, G(t|t_\circ) = \alpha \, \delta(t - t_\circ) - \lambda \, G(t|t_\circ). \tag{9.99}$$

To reach the second equality, we noted that the $\mathrm{e}^{-\lambda \, (t - t_\circ)}$ term multiplying the Dirac delta is unity at $t = t_\circ$, and so it does not alter the sifting property of the Dirac delta. Hence, following the discussion of equivalence classes of Dirac deltas in Chapter 5, we can drop $\mathrm{e}^{-\lambda \, (t - t_\circ)}$ from the Dirac. As illustrated in Figure 9.1, the impulse response function (9.98) has a particularly simple interpretation as the damped exponential response of the dynamical system to a Dirac impulse fired at $t = t_\circ$.

9.5.2 Step response function

Rather than hit the system at a particular moment in time with a Dirac delta, we now impose a force that turns on and remains on after some initial time, as per a Heaviside step function. The step response function, $S(t|t_\circ)$, measures the response of the dynamical system to this step forcing and it satisfies the differential equation

$$[\mathrm{d}/\mathrm{d}t + \lambda] \, S(t|t_\circ) = \alpha \, \mathcal{H}(t - t_\circ) \quad \text{with } S(t|t_\circ) = 0 \text{ for } t < t_\circ. \tag{9.100}$$

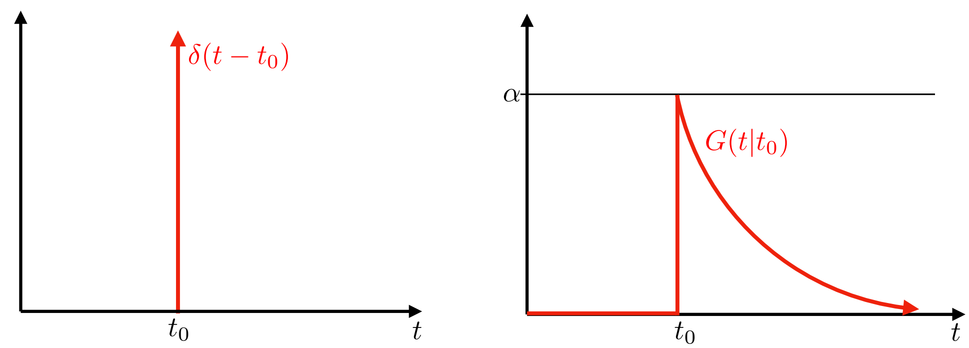

FIGURE 9.1: Left panel: Dirac delta that is fired at time $t = t_\circ$. Right panel: The impulse response function (9.98) resulting from the Dirac delta impulse as realized for the damped linear system (9.92).

Note that in the steady state at $t \to \infty$, the step response function asymptotes to the constant

$$\lim_{t\to\infty} S(t|t_\circ) = \alpha/\lambda. \tag{9.101}$$

Connection to the impulse response function

Taking the time derivative, $\mathrm{d}/\mathrm{d}t_\circ$, on the step response function equation (9.100) leads to

$$[\mathrm{d}/\mathrm{d}t + \lambda]\, \mathrm{d}S(t|t_\circ)/\mathrm{d}t_\circ = \alpha\, \mathrm{d}\mathcal{H}(t-t_\circ)/\mathrm{d}t_\circ. \tag{9.102}$$

The derivative of the Heaviside step function equals to the Dirac delta, in which

$$\mathrm{d}\mathcal{H}(t-t_\circ)/\mathrm{d}t_\circ = -\mathrm{d}\mathcal{H}(t-t_\circ)/\mathrm{d}t = -\delta(t-t_\circ). \tag{9.103}$$

Use of this result in equation (9.102), and comparison to the impulse response function equation (9.92), yields the identity

$$\frac{\mathrm{d}S(t|t_\circ)}{\mathrm{d}t_\circ} = -G(t|t_\circ). \tag{9.104}$$

So the time derivative of the step response function equals to the impulse response function. This identity holds even when the feedback parameter is a function of time, $\lambda = \lambda(t)$, since the time derivative operator, $\mathrm{d}/\mathrm{d}t_\circ$, has no affect on $\lambda(t)$.

Initial condition for the step response function

To determine the initial condition for the step response function, integrate equation (9.100) over an interval bounding $t_\circ$ and take the limit as that interval vanishes

$$\lim_{\epsilon\to 0}\left[S(t_\circ+\epsilon|t_\circ) - S(t_\circ-\epsilon|t_\circ) + \int_{t_\circ-\epsilon}^{t_\circ+\epsilon} \lambda\, S(t|t_\circ)\, \mathrm{d}t = \int_{t_\circ-\epsilon}^{t_\circ+\epsilon} \mathcal{H}(t-t_\circ)\, \mathrm{d}t\right]. \tag{9.105}$$

Causality means that $S(t_\circ - \epsilon|t_\circ) = 0$. Furthermore, the integral of the Heaviside is given by

$$\lim_{\epsilon\to 0}\int_{t_\circ-\epsilon}^{t_\circ+\epsilon} \mathcal{H}(t-t_\circ)\, \mathrm{d}t = \lim_{\epsilon\to 0}\int_{t_\circ}^{t_\circ+\epsilon} \mathcal{H}(t-t_\circ)\, \mathrm{d}t = \lim_{\epsilon\to 0} \epsilon = 0, \tag{9.106}$$

so that

$$\lim_{\epsilon\to 0} S(t_\circ+\epsilon|t_\circ) = \epsilon\,\lambda \Longrightarrow S(t=t_\circ|t_\circ) = 0. \tag{9.107}$$

That is, the step response function starts at zero and then grows in time in response to the Heaviside step function forcing.

Solution for the step response function

It is straightforward to show that the causal step response function is given by the saturating exponential

$$S(t|t_\circ) = \frac{\alpha}{\lambda}\left[1 - \mathrm{e}^{-\lambda\,(t-t_\circ)}\right]\mathcal{H}(t-t_\circ). \tag{9.108}$$

Figure 9.2 depicts this function along with the Heaviside step forcing. Furthermore, we verify the connection between $S(t|t_\circ)$ and $G(t|t_\circ)$ by computing

$$\mathrm{d}S(t|t_\circ)/\mathrm{d}t_\circ = -\alpha\,\mathcal{H}(t-t_\circ)\,\mathrm{e}^{-\lambda\,(t-t_\circ)} - \frac{\alpha}{\lambda}\left[1 - \mathrm{e}^{-\lambda\,(t-t_\circ)}\right]\delta(t-t_\circ) \tag{9.109a}$$

$$= -G(t|t_\circ) - \frac{\alpha}{\lambda}\left[1 - \mathrm{e}^{-\lambda\,(t-t_\circ)}\right]\delta(t-t_\circ). \tag{9.109b}$$

The second term on the right hand side vanishes since

$$\int_{t_\circ-\epsilon}^{t_\circ+\epsilon}\left[1 - \mathrm{e}^{-\lambda\,(t-t_\circ)}\right]\delta(t-t_\circ)\,\mathrm{d}t = 0, \tag{9.110}$$

in which case we have

$$\mathrm{d}S(t|t_\circ)/\mathrm{d}t_\circ = -G(t|t_\circ). \tag{9.111}$$

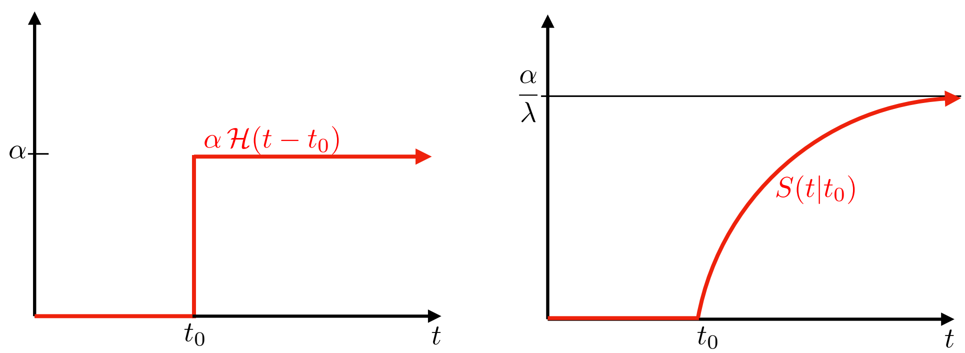

FIGURE 9.2: Left panel: Heaviside step function is fired at time $t = t_\circ$ and stays on afterward. Right panel: The step response function (9.108) resulting from the Heaviside step forcing impulse as realized for the damped linear system (9.92).

9.5.3 Reciprocity relation

We here develop a reciprocity condition for the impulse response function and its adjoint, following the procedure used for the diffusion equation in Section 9.2.7. Again, the impulse response function satisfies

$$[\mathrm{d}/\mathrm{d}t + \lambda]\,G(t|t_1) = \alpha\,\delta(t-t_1) \quad \text{with } G(t|t_1) = 0 \text{ for } t < t_1 \text{ and } G(t_1|t_1) = \alpha, \tag{9.112}$$

and the adjoint impulse response function satisfies

$$[-\mathrm{d}/\mathrm{d}t + \lambda]\,G^\ddagger(t|t_2) = \alpha\,\delta(t-t_2) \quad \text{with } G^\ddagger(t|t_2) = 0 \text{ for } t > t_2 \text{ and } G^\ddagger(t_2|t_2) = \alpha. \tag{9.113}$$

We here introduced two Dirac delta source times, t_1, t_2, which both occur after the initial time and before the end time

$$t_{\mathsf{init}} < t_1, t_2 < T. \tag{9.114}$$

As we will see, causality eliminates the final time, T, from the following. We retain it merely for bookeeping.

Determining the reciprocity relation between $G^{\ddagger}$ and G follows by multiplying equation (9.112) by $G^{\ddagger}(t|t_2)$ and multiplying equation (9.113) by $G(t|t_1)$ and then subtracting

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[G(t|t_1)\, G^{\ddagger}(t|t_2)\right] = \alpha\, [G^{\ddagger}(t|t_2)\, \delta(t - t_1) - G(t|t_1)\, \delta(t - t_2)]. \tag{9.115}$$

Now integrate this equation over the time range $t_{\mathsf{init}} \le t \le T$. For the right hand side we assume α to be a constant, which then leads to the difference $G^{\ddagger}(t_1|t_2) - G(t_2|t_1)$. For the left hand side, use of the causality conditions in equations (9.112) and (9.113) render

$$\int_{t_{\mathsf{init}}}^{T} \frac{\mathrm{d}}{\mathrm{d}t}\left[G(t|t_1)\, G^{\ddagger}(t|t_2)\right] \mathrm{d}t = 0, \tag{9.116}$$

thus yielding the reciprocity relation

$$G^{\ddagger}(t|t_{\circ}) = G(t_{\circ}|t). \tag{9.117}$$

The feedback parameter, λ, dropped out from the derivation, with the reciprocity relation holding even if λ is time dependent. However, we again needed to assume the coefficient α to be constant.

9.5.4 Response function for general forcing

We now return to the initial value problem (9.91) and determine the response, ψ, to a general forcing function, $F(t)$, that turns on at some initial time $t = t_{\circ} > t_{\mathsf{init}}$. As for our earlier discussions of Green's functions, we express the general response function as an integral over impulse responses, with Exercise 9.13 deriving the expression

$$\alpha\, \psi(t) = G(t|t_{\mathsf{init}})\, \psi(t_{\mathsf{init}}) + \int_{t_{\mathsf{init}}}^{t} G(t|t_{\circ})\, F(t_{\circ})\, \mathrm{d}t_{\circ}. \tag{9.118}$$

Evidently, the general response is written as an initial response plus the integral of the general forcing with the impulse response function. Causality ensures that ψ is dependent only on forcing that is active between the initial time, t_{init}, and current time, t. To garner further insights into the general expression (9.118), consider the special case of constant feedback parameter, λ, in which the impulse response function is (9.98) so that

$$\psi(t) = \mathrm{e}^{-\lambda\,(t - t_{\mathsf{init}})}\, \psi(t_{\mathsf{init}}) + \int_{t_{\mathsf{init}}}^{t} \mathrm{e}^{-\lambda\,(t - t_{\circ})}\, F(t_{\circ})\, \mathrm{d}t_{\circ}. \tag{9.119}$$

Hence, the feedback parameter acts to exponentially damp the initial condition, and it is convolved with the step forcing.

9.5.5 Connection to the boundary propagator

Recall our discussion in Section 9.4.5 of the boundary propagator, $G^{\text{bp}}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)$, for the diffusion equation, which solves the causal boundary value problem (9.89a)-(9.89c). Again, the boundary propogator measures the response of the system at $(\boldsymbol{x}, t)$ to a Dirac delta space-time source imposed along the surface boundary. The details of the diffusion process are encoded into the boundary propagator so that the propagator is able to build up the response, ψ, to a general boundary forcing function, σ, as per equation (9.87). The discussion in the present section thus prompts us to consider the boundary propagator as an impulse response function for spatially distributed sources whose influence through space and time is mediated by diffusion.

9.5.6 Comments and further study

Many applications of Green's function methods in geophysical fluid mechanics and climate dynamics do not make use of analytical methods to solve for the Green's function. Instead, they make use of numerical estimates based on time stepping passive tracers in ocean and atmosphere models, with source functions approximating Dirac delta sources. Many applications have focused on boundary propagators, which as shown in this section are equivalant to impulse response functions for boundary sources. We return to this point when discussing the passive tracer equation in VOLUME 4.

Hasselmann et al. (1993) introduced the impulse response function and step response function to the study of climate model drift. *Marshall et al.* (2014) and *Zanna et al.* (2019) presented further studies using this framework. Some of the mathematical formulation of impulse response and step response functions as presented here follow that offered in Exercise 1.52 of *Stakgold* (2000a).

9.6 Evolution of time averages

In geophysical fluid mechanics, we generically refer to an equation with a time derivative, such as a parabolic or hyperbolic equation, as a prognostic equation or an evolution equation. In the analysis of such equations, for example when analyzing simulation output or time series data, it is common to take the time average in order to focus on lower frequency behavior. This section provides a discussion concerning this time averaging operation, with the material here following *Bladwell et al.* (2026).

For this purpose, ignore all space coordinates and write a generic prognostic equation in the form

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \mathcal{B}. \tag{9.120}$$

For example, the quantity A might be the velocity or temperature at a point in space, and $\mathcal{B}$ might be the acceleration due to pressure or the heating due to temperature diffusion. We term $\mathrm{d}A/\mathrm{d}t$ the time tendency of the quantity A whereas $\mathcal{B}$ is the "forcing" that gives rise to the time tendency. In the analysis of fluid flows, we commonly wish to diagnose terms appearing in the evolution equations for the purpose of ascribing physical understanding to the flow regime; e.g., what forces are more active in certain regions. Although sitting a bit outside the scope of a chapter on partial differential equations, the material in this section exposes some common questions that arise when time averaging terms appearing in the prognostic equations of geophysical fluid mechanics.

Time integration of equation (9.120) leads to

$$A(t) = A(t_0) + \int_{t_0}^{t} \mathcal{B}(s)\,\mathrm{d}s, \tag{9.121}$$

thus providing an expression for the instantaneous value of A at an arbitrary time t, assuming knowledge of the initial value, $A(t_0)$, as well as the time integral of $\mathcal{B}$. In practice, particularly when working with numerical models, we typically have access to time averages over some time interval (e.g., days, months, years, decades) rather than instantaneous (snapshot) values of A. Furthermore, instantaneous snapshots can be prone to relatively large fluctuations that expose the diagnostic calculations to numerical precision errors (e.g., small differences between relatively large fluctuating values). We are thus interested in relating time averages of A to time averages of $\mathcal{B}$.

9.6.1 Time averages

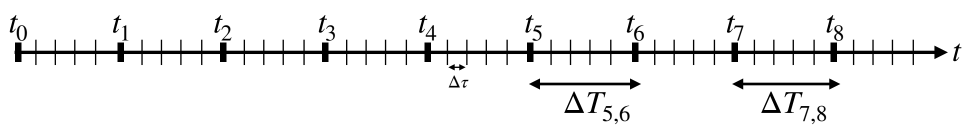

FIGURE 9.3: Example time axis for the discussion of time averaging. The labeled times, t_n, can represent, for example, days, months or years with the time interval, $\Delta T_{n,n+1} = t_{n+1} - t_n$, not necessarily the same (e.g., different number of days in a month or a leap year versus non-leap year). The smaller unlabeled time steps represent the time steps for the model's prognostic equations (e.g., days, hours, seconds, etc.), with fixed time step $\Delta\tau$.

Introduce a discrete partitioning of the time axis as in Figure 9.3 and define an unweighted time average over a chosen time interval $\Delta T_{n,n+1} = t_{n+1} - t_n > 0$

$$\overline{A}_{n,n+1} = \frac{1}{\Delta T_{n,n+1}} \int_{t_n}^{t_{n+1}} A(t)\,\mathrm{d}t \tag{9.122a}$$

$$\overline{\mathcal{B}}_{n,n+1} = \frac{1}{\Delta T_{n,n+1}} \int_{t_n}^{t_{n+1}} \mathcal{B}(t)\,\mathrm{d}t. \tag{9.122b}$$

These integrals are realized in practice as a discrete sum over the model time steps, with only the lower limit inclusive so as to not double-count endpoints; i.e., $[t_n, t_{n+1})$. We allow for non-constant time intervals, $\Delta T_{n,n+1}$, as arises in monthly and yearly (with leap-years) time averages.

Substituting expression (9.121) into the time mean (9.122a) renders

$$\overline{A}_{n,n+1} - A(t_0) = \frac{1}{\Delta T_{n,n+1}} \int_{t_n}^{t_{n+1}} \left[\int_{t_0}^{t} \mathcal{B}(s)\,\mathrm{d}s\right] \mathrm{d}t, \tag{9.123}$$

and a similar expression over a later time interval $[t_p, t_{p+1})$ with $p \geq n+1$ leads to the difference between time averages

$$\overline{A}_{p,p+1} - \overline{A}_{n,n+1} = \frac{1}{\Delta T_{p,p+1}} \int_{t_p}^{t_{p+1}} \left[\int_{t_0}^{t} \mathcal{B}(s)\,\mathrm{d}s\right] \mathrm{d}t - \frac{1}{\Delta T_{n,n+1}} \int_{t_n}^{t_{n+1}} \left[\int_{t_0}^{t} \mathcal{B}(s)\,\mathrm{d}s\right] \mathrm{d}t. \tag{9.124}$$

The formalism allows us to take differences between time averages over intervals that are separated, such as might be of interest in taking decadal means between the beginning and end of a century, for example. Importantly, the initial value, $A(t_0)$, is absent from the difference in time means so that there are only time integrated quantities appearing in equation (9.124).

9.6.2 Massaging the double time integrals

The double time integrals in equation (9.124) can be massaged into a simpler form. We start by making the following decomposition and noting that $t_n \leq t \leq t_{n+1}$

$$\int_{t_n}^{t_{n+1}} \left[\int_{t_0}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t = \int_{t_n}^{t_{n+1}} \left[\int_{t_0}^{t_n} \mathcal{B}(s)\,\mathrm{d}s + \int_{t_n}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t \tag{9.125a}$$

$$= \Delta T_{n,n+1} \int_{t_0}^{t_n} \mathcal{B}(s)\,\mathrm{d}s + \int_{t_n}^{t_{n+1}} \left[\int_{t_n}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t. \tag{9.125b}$$

We are thus led to the difference

$$\begin{aligned}
\overline{A}_{p,p+1} - \overline{A}_{n,n+1} &= \left[\int_{t_0}^{t_p} \mathcal{B}(s)\,\mathrm{d}s - \int_{t_0}^{t_n} \mathcal{B}(s)\,\mathrm{d}s \right] \\
&+ \frac{1}{\Delta T_{p,p+1}} \int_{t_p}^{t_{p+1}} \left[\int_{t_p}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t - \frac{1}{\Delta T_{n,n+1}} \int_{t_n}^{t_{n+1}} \left[\int_{t_n}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t \\
&= \int_{t_n}^{t_p} \mathcal{B}(t)\,\mathrm{d}t + \frac{1}{\Delta T_{p,p+1}} \int_{t_p}^{t_{p+1}} \left[\int_{t_p}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t - \frac{1}{\Delta T_{n,n+1}} \int_{t_n}^{t_{n+1}} \left[\int_{t_n}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t.
\end{aligned} \tag{9.126}$$

The double integrals in equation (9.126) take place over triangular time domains, such as shown in Figure 9.4.

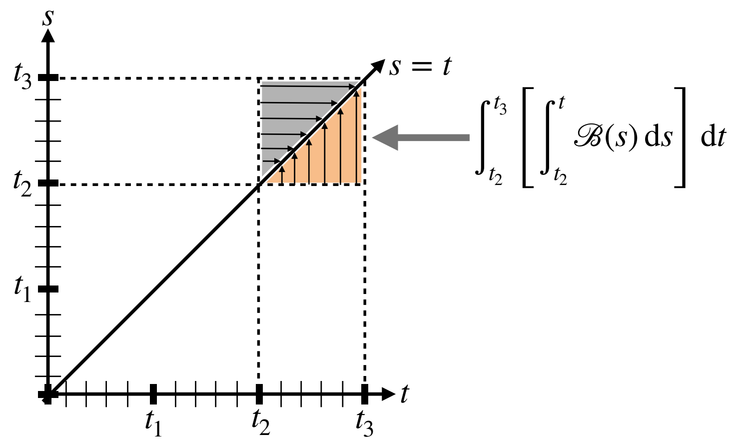

FIGURE 9.4: Gold region depicts the time integration domain used in one of the double integrals from equation (9.126) for the special case of $n = 2$. Note that the gray triangular region generally leads to a distinct integral.

9.6.3 Making use of Cauchy's double integral identity

We here derive an identity, originally due to *Cauchy* (1823), that reduces the double integral in equation (9.126) to a single integral to expose the underlying geometry of the time windowing. For this purpose we make use of the following identity

$$\int_{t_n}^{t_{n+1}} \left[\int_{t_n}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t = \int_{t_n}^{t_{n+1}} (t_{n+1} - t)\,\mathcal{B}(t)\,\mathrm{d}t. \tag{9.127}$$

To prove this identity we make the substitution $\mathcal{B}(s) = \mathrm{d}A/\mathrm{d}s$ from equation (9.120) and then show that both sides to equation (9.127) yield the same result. For the left hand side we have

$$\int_{t_n}^{t_{n+1}} \left[\int_{t_n}^{t} \mathcal{B}(s)\,\mathrm{d}s \right] \mathrm{d}t = \int_{t_n}^{t_{n+1}} \left[\int_{t_n}^{t} \frac{\mathrm{d}A(s)}{\mathrm{d}s}\,\mathrm{d}s \right] \mathrm{d}t \tag{9.128a}$$

$$= \int_{t_n}^{t_{n+1}} \left[\int_{t_n}^{t} \mathrm{d}A(s) \right] \mathrm{d}t \tag{9.128b}$$

$$= \int_{t_n}^{t_{n+1}} [A(t) - A(t_n)]\mathrm{d}t \tag{9.128c}$$

$$= \int_{t_n}^{t_{n+1}} A(t)\,\mathrm{d}t - (t_{n+1} - t_n)\,A(t_n), \tag{9.128d}$$

whereas the right hand side is

$$\int_{t_n}^{t_{n+1}} (t_{n+1} - t)\,\mathcal{B}(t)\,\mathrm{d}t = \int_{t_n}^{t_{n+1}} (t_{n+1} - t)\,\frac{\mathrm{d}A(t)}{\mathrm{d}t})\,\mathrm{d}t \tag{9.129a}$$

$$= \int_{t_n}^{t_{n+1}} (t_{n+1} - t)\,\mathrm{d}A(t) \tag{9.129b}$$

$$= \int_{t_n}^{t_{n+1}} \mathrm{d}[A(t)\,(t_{n+1} - t)] + A(t)\,\mathrm{d}t \tag{9.129c}$$

$$= -A(t_n)\,(t_{n+1} - t_n) + \int_{t_n}^{t_{n+1}} A(t)\,\mathrm{d}t, \tag{9.129d}$$

which is identical to the left hand side given by equation (9.128d). We have thus proven the double integral formula (9.127).

The right hand side of the double integral formula (9.127) can be written

$$\int_{t_n}^{t_{n+1}} (t_{n+1} - t)\,\mathcal{B}(t)\,\mathrm{d}t = t_{n+1}\,\Delta T_{n,n+1}\,\overline{\mathcal{B}}_{n,n+1} - \int_{t_n}^{t_{n+1}} t\,\mathcal{B}(t)\,\mathrm{d}t, \tag{9.130}$$

which might be useful in some contexts. However, it is awkward for our purposes since it exposes the absolute time, t_{n+1}, in the first term on the right hand side and the time, t, within the integral. Since we generally do not hold any initial time as special (i.e., the initial time, t_0, is arbitrary), it is preferable to retain the time differences throughout the formulation. Hence, when making use of the double integral identity (9.127) we bring equation (9.126) into the form

$$\overline{A}_{p,p+1} - \overline{A}_{n,n+1} = \int_{t_n}^{t_p} \mathcal{B}(t)\,\mathrm{d}t + \int_{t_p}^{t_{p+1}} \frac{(t_{p+1} - t)\,\mathcal{B}(t)}{t_{p+1} - t_p}\,\mathrm{d}t - \int_{t_n}^{t_{n+1}} \frac{(t_{n+1} - t)\,\mathcal{B}(t)}{t_{n+1} - t_n}\,\mathrm{d}t \tag{9.131a}$$

$$= \int_{t_n}^{t_{n+1}} \frac{(t - t_n)\,\mathcal{B}(t)}{t_{n+1} - t_n}\,\mathrm{d}t + \int_{t_{n+1}}^{t_p} \mathcal{B}(t)\,\mathrm{d}t + \int_{t_p}^{t_{p+1}} \frac{(t_{p+1} - t)\,\mathcal{B}(t)}{t_{p+1} - t_p}\,\mathrm{d}t. \tag{9.131b}$$

The first right hand side term is a weighted integral with a linearly increasing weight from zero to unity, whereas the final right hand side term has a linearly descreasing weight from unity to zero. The middle term has a unity weight throughout and it vanishes if $p = n + 1$, as when the averaging regions are adjacent. Figure 9.5. illustrates the time windowing used for equation (9.131b).

As a final means to write equation (9.131b), extend the middle term to the end of the time

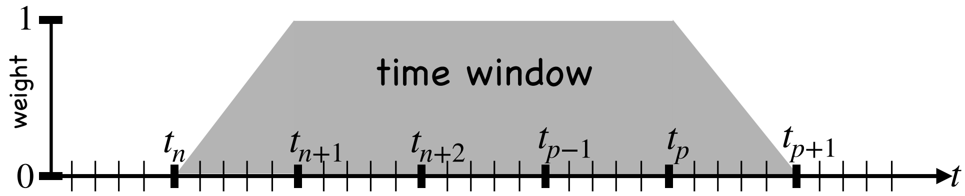

FIGURE 9.5: Illustrating the time window weighting used in computing the right hand side of equation (9.131b). Note that if $p = n+1$, then the middle term in equation (9.131b) and there is no plateau region of unit weight, in which case the window region becomes two adjacent triangles.

period and then subtract the extra piece and recombine to render

$$\overline{A}_{p,p+1} - \overline{A}_{n,n+1} = \int_{t_{n+1}}^{t_{p+1}} \mathcal{B}(t)\,\mathrm{d}t + \int_{t_n}^{t_{n+1}} \frac{(t-t_n)\,\mathcal{B}(t)}{t_{n+1}-t_n}\,\mathrm{d}t - \int_{t_p}^{t_{p+1}} \frac{(t-t_p)\,\mathcal{B}(t)}{t_{p+1}-t_p}\,\mathrm{d}t. \tag{9.132}$$

The first term on the right hand side is an unweighted integral from the end of the first interval to the end of the final interval, whereas the other two terms both have increasing weights over their respective integration intervals. This form allows for some advantages diagnostically since we only need to save unweighted integrals plus linearly increasing weighted integrals; there is no need to save decreasing weighted integrals.

9.7 Exercises

EXERCISE 9.1: TRANSFORMING AWAY A SOURCE FROM THE DIFFUSION EQUATION
Consider the following diffusion equation with a source

$$(\partial_t + b\,\kappa)\psi = \kappa\,\nabla^2\psi, \tag{9.133}$$

with κ a constant diffusivity and b a constant with dimensions of L^{-2}. Show that the transformation

$$\psi(\boldsymbol{x},t) = \mathrm{e}^{b\,\kappa\,t}\,\Psi(\boldsymbol{x},t), \tag{9.134}$$

leads to Ψ satisfying the diffusion equation

$$(\partial_t - \kappa\,\nabla^2)\Psi = 0. \tag{9.135}$$

EXERCISE 9.2: CONSTANT ADVECTION PLUS DIFFUSION TRANSFORMED TO DIFFUSION
Consider the advection-diffusion equation

$$(\partial_t + \boldsymbol{v}\cdot\nabla)\psi = \kappa\,\nabla^2\psi, \tag{9.136}$$

for the special case of a constant advection velocity, $\boldsymbol{v}$, and a constant diffusivity, κ. Show that the transformation

$$\psi(\boldsymbol{x},t) = \mathrm{e}^{-\boldsymbol{v}\cdot(\boldsymbol{v}\,t-2\,\boldsymbol{x})/(4\kappa)}\,\Psi(\boldsymbol{x},t), \tag{9.137}$$

means that Ψ satisfies the diffusion equation

$$\partial_t\Psi = \kappa\,\nabla^2\Psi. \tag{9.138}$$

Evidently, we can transform away the constant advection velocity, thus solving for Ψ using familiar diffusion equation methods, and then using the transformation (9.137) to recover ψ.

EXERCISE 9.3: PRODUCT LAW FOR THE DIFFUSION EQUATION
Let $A(x,t), B(y,t), C(z,t)$ each satisfy the one-dimensional diffusion equation with the same space-time dependent diffusivity, $\kappa(x,y,z,t) > 0$

$$\partial_t A = \partial_x(\kappa\,\partial_x A) \quad \text{and} \quad \partial_t B = \partial_y(\kappa\,\partial_y B) \quad \text{and} \quad \partial_t C = \partial_z(\kappa\,\partial_z C). \tag{9.139}$$

Show that $\psi(x,y,z,t) = A(x,t)\,B(y,t)\,C(z,t)$ satisfies the three-dimensional diffusion equation

$$\partial_t \psi = \nabla \cdot (\kappa\,\nabla\psi). \tag{9.140}$$

EXERCISE 9.4: A BIHARMONIC OPERATOR FOR DISSIPATION
A biharmonic operator is commonly employed for dissipation in numerical models due to its enhanced scale selectivity. However, biharmonic operators have some unphysical properties, such as those encountered in this exercise focused on the biharmonic mixing of scalars arising from a flux

$$\boldsymbol{F}^{\mathsf{bih}} = \sqrt{B}\,\nabla_{\mathsf{h}} L, \tag{9.141}$$

where

$$L = \nabla \cdot (\sqrt{B}\,\nabla\psi) \tag{9.142}$$

is the horizontal Laplacian operator acting on the scalar, and $B > 0$ is a biharmonic mixing coefficient with units $\mathrm{L}^4\,\mathrm{T}^{-1}$. The convergence of the biharmonic flux $\boldsymbol{F}^{\mathsf{bih}}$ yields the biharmonic mixing operator

$$R^{\mathsf{bih}} = -\nabla \cdot \boldsymbol{F}^{\mathsf{bih}}. \tag{9.143}$$

Examine the biharmonic operator's effects on scalar variance by computing the time tendency of the squared scalar over the full domain, ignoring all boundary contributions. Is the biharmonic scalar flux downgradient?

EXERCISE 9.5: DUHAMEL'S SUPERPOSITION INTEGRAL FOR THE DIFFUSION EQUATION
Consider the initial value problem for the diffusion equation on a line as given by equations (9.6a)-(9.6b)

$$\partial_t \psi = \kappa\,\partial_{xx}\psi \qquad -\infty < x < \infty,\ t > 0 \tag{9.144a}$$

$$\psi(x,t) = f(x) \qquad -\infty < x < \infty,\ t = 0. \tag{9.144b}$$

The solution to this problem is given by the Gaussian function in equation (9.7). Use Duhamel's superposition integral to determine the solution to the corresponding forced problem with zero initial condition. Hint: see Section 9.11 of *Hildebrand* (1976).

EXERCISE 9.6: COMPOSITION PROPERTY OF THE DIFFUSION GREEN'S FUNCTION
Derive the composition property (9.49) satisfied by the diffusion Green's function.

EXERCISE 9.7: BOUNDARY VALUE PROBLEM SATISFIED BY THE BOUNDARY PROPAGATOR
In Section 9.4.5. we defined the boundary propagator for the Dirichlet problem as

$$G^{\mathsf{bp}}(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ) \equiv -\boldsymbol{K}(\boldsymbol{x}_\circ,t_\circ)\cdot\nabla_{\boldsymbol{x}_\circ} G(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{x}_\circ} \quad \text{with } \boldsymbol{x}_\circ \in \partial\mathcal{R}, \tag{9.145}$$

Show that it satisfies the initial-boundary value problem (9.89a)–(9.89c).

EXERCISE 9.8: PROOF THAT THE GENERALIZED LAPLACIAN OPERATOR IS SELF-ADJOINT

In Section 9.2.7 we derived the reciprocity relation satisfied by the diffusion equation Green's function. As part of that derivation we showed by equation (9.46) that the generalized Laplacian operator is self-adjoint, where generalization arises from presence of a symmetric diffusion tensor. We asserted that the self-adjoint property holds since the diffusion tensor is symmetric. Expose tensor indices as part of the derivation of equation (9.46) to provide details for why diffusion tensor symmetry is needed.

EXERCISE 9.9: SOLUTION TO ONE-DIMENSIONAL DIFFUSION EQUATION

Consider the causal initial-boundary value problem for the diffusion equation with homogeneous Dirichlet boundary conditions and constant diffusivity

$$\partial_t \psi = \kappa\, \partial_{xx}\psi \qquad 0 < x < L,\ t > t_{\mathsf{init}} \tag{9.146a}$$
$$\psi(x, t_{\mathsf{init}}) = I(x) \qquad 0 < x < L \tag{9.146b}$$
$$\psi(x = 0, t) = \psi(x = L, t) = 0 \qquad t > 0 \tag{9.146c}$$
$$\psi(x, t < t_{\mathsf{init}}) = 0 \qquad \text{causality.} \tag{9.146d}$$

(a) Make use of separation of variables to find the causal version of the solution that we wrote in equation (9.5a).
(b) Write the solution in terms of the Dirichlet Green's function equation (9.74). Identify the Green's function.
(c) Explicity verify that the Green's function identified in the previous part satisfies the diffusion boundary value problem with a Dirac space-time source.
(d) Determine the adjoint causal Green's function, $G^{\ddagger}$, and verify that it satisfies the adjoint causal Green's function equation.

EXERCISE 9.10: INITIAL VALUE PROBLEM ON THE REAL LINE

Consider the causal initial value problem for the diffusion equation on the real line

$$\partial_t \psi = \kappa\, \partial_{xx}\psi \qquad -\infty < x < \infty,\ t > t_{\mathsf{init}} \tag{9.147a}$$
$$\psi(x, t_{\mathsf{init}}) = I(x) \qquad -\infty < x < \infty \tag{9.147b}$$
$$|\psi(x, t)| < \infty \qquad -\infty < x < \infty, t \geq t_{\mathsf{init}} \tag{9.147c}$$
$$\psi(x, t < t_{\mathsf{init}}) = 0 \qquad \text{causality.} \tag{9.147d}$$

(a) What is the Green's function for this initial value problem?
(b) Write the solution for ψ in terms of the Green's function.
(c) Consider the special case of an initial condition

$$I(x) = \psi_{\circ}\,(x/|x|) \quad \text{with } x \neq 0, \tag{9.148}$$

where $\psi_{\circ}$ is a constant. Write $\psi(x, t)$ in terms of the error function

$$\operatorname{erf}(\xi) = \frac{2}{\sqrt{\pi}} \int_0^{\xi} \mathrm{e}^{-\lambda^2}\, \mathrm{d}\lambda \qquad \operatorname{erf}(0) = 0 \quad \text{and} \quad \operatorname{erf}(\infty) = 1. \tag{9.149}$$

EXERCISE 9.11: INITIAL VALUE PROBLEMS ON THE REAL LINE

In Exercise 9.10 we solved the diffusion equation on the real line with a particular initial condition. Use that solution, and nothing else, to find the solutions to the following initial-boundary value problems, each for times $t > t_{\mathsf{init}} = 0$.

(a) Find a solution to the following initial-boundary value problem

$$\partial_t \psi = \kappa\, \partial_{xx}\psi \qquad 0 < x < \infty,\ t > 0 \tag{9.150a}$$
$$\psi(x=0,t) = 0 \qquad t > 0 \tag{9.150b}$$
$$\psi(x,t=0) = U \qquad 0 < x < \infty. \tag{9.150c}$$

(b) Find a solution to the following initial-boundary value problem

$$\partial_t \psi = \kappa\, \partial_{xx}\psi \qquad 0 < x < \infty,\ t > 0 \tag{9.151a}$$
$$\psi(x=0,t) = U \qquad t > 0 \tag{9.151b}$$
$$\psi(x,t=0) = 0 \qquad 0 < x < \infty. \tag{9.151c}$$

(c) Find a solution to the following initial value problem

$$\partial_t \psi = \kappa\, \partial_{xx}\psi \qquad -\infty < x < \infty,\ t > 0 \tag{9.152a}$$
$$\psi(x,t=0) = U\,\mathcal{H}(x) \qquad -\infty < x < \infty. \tag{9.152b}$$

(d) Find a solution to the following initial value problem

$$\partial_t \psi = \kappa\, \partial_{xx}\psi \qquad -\infty < x < \infty,\ t > 0 \tag{9.153a}$$
$$\psi(x,t=0) = U \qquad |x| < \ell \tag{9.153b}$$
$$\psi(x,t=0) = 0 \qquad |x| > \ell. \tag{9.153c}$$

EXERCISE 9.12: METHOD OF IMAGES AND DIFFUSIVE SIGNAL PROPAGATION

Consider the causal initial-boundary value problem for the diffusion equation on the half-line with a Dirichlet boundary condition at $x = 0$ and bounded values for $x \to \infty$

$$\partial_t \psi = \kappa\, \partial_{xx}\psi \qquad 0 < x < \infty,\ t > t_{\mathsf{init}} \tag{9.154a}$$
$$\psi(x, t_{\mathsf{init}}) = I(x) \qquad 0 < x < \infty \tag{9.154b}$$
$$\psi(x=0,t) = J(t) \qquad t > t_{\mathsf{init}} \tag{9.154c}$$
$$|\psi(x,t)| < \infty \qquad \forall x \geq 0, t \geq t_{\mathsf{init}} \tag{9.154d}$$
$$\psi(x, t < t_{\mathsf{init}}) = 0 \qquad \text{causality.} \tag{9.154e}$$

(a) Write the corresponding Green's function problem and Green's function solution for ψ.
(b) Use the method of images from Section 8.11 to find the Green's function, using the appropriate combination of free space Green's functions from Section 9.2.2.
(c) Consider the special case of zero initial condition ($I = 0$), and constant boundary condition, $J = J_\circ$, and write the solution, ψ, in terms of the complementary error function

$$\operatorname{erfc}(\xi) = \frac{2}{\sqrt{\pi}} \int_{\xi}^{\infty} \mathrm{e}^{-\lambda^2}\, \mathrm{d}\lambda. \tag{9.155}$$

(d) The analytical solution from the previous part is a diffusive signal moving outward (towards increasing x) from the Dirichlet boundary condition at $x = 0$. Compute the speed, $\mathrm{d}x/\mathrm{d}t > 0$, for a point following a fixed value for the scalar field, $\psi(x,t)$. That speed can be interpreted as measuring the speed of the diffusive signal.

EXERCISE 9.13: RESPONSE FUNCTION FOR GENERAL FORCING

Derive the general response function (9.118). Hint: make use of a method analogous to that used for the diffusion equation.

Chapter 10

Wave equation

In this chapter we study elements of hyperbolic partial differential equations by focusing on the wave equation $(\partial_{tt} - c^2 \nabla^2)\psi = 0$. Solutions to this equation are non-dispersive waves whose phase travels with speed, c. We study acoustic waves and shallow water gravity waves in VOLUME 4, each of which are nondispersive waves that satisfy this wave equation. However, many geophysical fluids support dispersive waves that satisfy other wave equations, and we study the physics of such waves in VOLUME 4. Our study in this chapter of the nondispersive wave equation offers a useful introduction to geophysical wave mechanics and the associated mathematical methods.

Solutions to hyperbolic partial differential equations are characterized by signals propagating with a finite propagation speed, $|c| < \infty$. Correspondingly, the wave signal at a point in space-time is dependent on information given at prior times over a cone whose extent is determined by the propagation speed. This cone is known as the domain of influence. Likewise, a signal at a point has a future domain of influence also given by a cone, now pointed to the future. Non-dispersive wave signals experience no dissipation if the media supporting the waves has no friction, thus reflecting the underlying reversible physics.

CHAPTER GUIDE

We make use of material from our study of elliptic partial differential equations in Chapter 8 as well as parabolic partial differential equations from Chapter 9. Further study of the wave equation and solution methods can be found in many textbooks, such as *Dennery and Krzywicki* (1967), *Greenberg* (1971), *Hildebrand* (1976), *Duchateau and Zachmann* (1986), and *Stakgold* (2000b).

10.1 Loose threads

- Section 7.11 and 7.12 of *Stakgold* (2000b) for dissipative waves and causality.

10.2 Wave equation basics

Referring to our classification of linear equations in Section 8.2, we found that hyperbolic partial differential equations have two real characteristics. The linear homogeneous wave equation is the canonical example of a hyperbolic partial differential equation

$$(\partial_{tt} - c^2 \, \partial_{xx})\psi = 0. \tag{10.1}$$

Solutions have the form of a signal moving at a finite speed in both directions and that move along the two characteristic curves

$$\psi(x, t) = A(x - ct) + B(x + c\,t), \tag{10.2}$$

where A and B are differentiable functions whose form is determined by the initial conditions. Note that we can factor the differential operator into the form

$$(\partial_t - c\,\partial_x)\,(\partial_t + c\,\partial_x)\psi = 0. \tag{10.3}$$

Consequently, if either one of the linear first-order partial differential equations are satisfied,

$$(\partial_t - c\,\partial_x)\psi = 0 \quad \text{or} \quad (\partial_t + c\,\partial_x)\psi = 0, \tag{10.4}$$

then ψ will satisfy the full wave equation. These first-order partial differential equations are the one-dimensional advection equations considered in Section 8.1 with opposite advection direction, and each of which has a single characteristic curve. In this manner, we can think of advection by constant velocity as the square root of the wave equation. Similarly, some disciplines refer to the linear advection equation (10.30), with constant advection speed, as the one way wave equation. It represents the simplest of the hyperbolic partial differential equations.

10.2.1 Initial value problem for the infinite-domain wave equation

Since there are two time derivatives, specification of a solution requires initial conditions for the field and its first time derivative. To illustrate the structure of a solution to the wave equation, we develop a solution to the Cauchy problem for the wave equation on the line. That is, we consider the initial value problem for the one-dimensional wave equation on the real line (infinite spatial domain so no boundary conditions)

$$(\partial_{tt} - c^2\,\partial_{xx})\psi = 0 \qquad -\infty < x < \infty,\ t > 0 \qquad \text{wave equation on a line} \tag{10.5a}$$

$$\psi = F(x) \qquad -\infty < x < \infty,\ t = 0 \qquad \text{initial condition} \tag{10.5b}$$

$$\partial_t\psi = E(x) \qquad -\infty < x < \infty,\ t = 0 \qquad \text{initial tendency,} \tag{10.5c}$$

where the initial condition data, E, F, are arbitrary functions of space and c is a constant wave speed. Following from the discussion of characteristic curves in Section 8.1, we are motivated to perform a linear transformation of the wave equation into wave characteristic coordinates

$$\xi = x + c\,t \quad \text{and} \quad \eta = x - c\,t \Longrightarrow (\xi+\eta)/2 = x \quad \text{and} \quad (\xi-\eta)/(2c) = t. \tag{10.6}$$

Wave signals propagate in directions defined by constant ξ and η, so that these coordinates isolate the signal transmission. Furthermore, as we will see, this coordinate transformation facilitates a direct integration of the wave equation.

Transformation to characteristic coordinates

To help organize the linear transformation to characteristic coordinates, write equation (10.6) as a matrix-vector equation

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} c & 1 \\ -c & 1 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix} \Longleftrightarrow \begin{bmatrix} t \\ x \end{bmatrix} = \frac{1}{2}\begin{bmatrix} c^{-1} & -c^{-1} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \xi \\ \eta \end{bmatrix}. \tag{10.7}$$

Furthermore, introduce tensor indices (see Chapter 1)

$$\mu = (0,1) \quad \text{and} \quad \overline{\mu} = (\overline{0}, \overline{1}), \tag{10.8}$$

where indices 0 and $\overline{0}$ label time and indices 1 and $\overline{1}$ label space, in which case the time and space coordinates are written

$$x^{\mu} = (x^0, x^1) = (t, x) \quad \text{and} \quad x^{\overline{\mu}} = (x^{\overline{0}}, x^{\overline{1}}) = (\xi, \eta). \tag{10.9}$$

Use of this index notation brings the linear transformation (10.7) into the tidy form

$$x^{\overline{\mu}} = \Lambda^{\overline{\mu}}{}_{\nu}\, x^{\nu} \Longleftrightarrow x^{\mu} = \Lambda^{\mu}{}_{\overline{\nu}}\, x^{\overline{\nu}}, \tag{10.10}$$

where the transformation matrices are given by the constant matrices[1]

$$\Lambda^{\overline{\mu}}{}_{\nu} = \begin{bmatrix} \partial x^{\overline{1}}/\partial x^1 & \partial x^{\overline{1}}/\partial x^2 \\ \partial x^{\overline{1}}/\partial x^2 & \partial x^{\overline{2}}/\partial x^2 \end{bmatrix} = \begin{bmatrix} c & 1 \\ -c & 1 \end{bmatrix} \tag{10.11}$$

[1]For linear coordinate transformations such as considered here, the space-time coordinates as well as tensors transform with the same transformation matrices. Such is not the case for nonlinear transformations, such as those appearing in general tensor analysis as studied in Chapter 3.

and

$$\Lambda^{\mu}{}_{\overline{\nu}} = \begin{bmatrix} \partial x^1/\partial x^{\overline{1}} & \partial x^1/\partial x^{\overline{2}} \\ \partial x^2/\partial x^{\overline{1}} & \partial x^2/\partial x^{\overline{2}} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} c^{-1} & -c^{-1} \\ 1 & 1 \end{bmatrix}. \tag{10.12}$$

Note the use of an upstairs position for the row index on the transformation matrix, which conforms to the use with general tensors from Chapter 3. The coordinate transformation (10.10) and the transformation matrices (10.11)-(10.12) then lead to the partial derivative relationship

$$\partial_{\overline{\nu}} = \Lambda^{\mu}{}_{\overline{\nu}}\, \partial_{\mu} \Longleftrightarrow \partial_{\nu} = \Lambda^{\overline{\mu}}{}_{\nu}\, x_{\overline{\mu}}, \tag{10.13}$$

so that

$$\frac{\partial^2}{\partial x^2} = \left[\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta}\right]^2 = \frac{\partial^2}{\partial \xi^2} + 2\,\frac{\partial}{\partial \xi}\,\frac{\partial}{\partial \eta} + \frac{\partial^2}{\partial \eta^2} \tag{10.14a}$$

$$c^{-2}\,\frac{\partial^2}{\partial t^2} = \left[\frac{\partial}{\partial \xi} - \frac{\partial}{\partial \eta}\right]^2 = \frac{\partial^2}{\partial \xi^2} - 2\,\frac{\partial}{\partial \xi}\,\frac{\partial}{\partial \eta} + \frac{\partial^2}{\partial \eta^2}. \tag{10.14b}$$

General solution for the initial value problem

The operator transformations (10.14a) and (10.14b) bring the initial value problem (10.5a)-(10.5c) into the form

$$\frac{\partial^2 \psi}{\partial \xi \partial \eta} = 0 \qquad -\infty < \eta < \xi < \infty \tag{10.15a}$$

$$\psi(\xi, \eta) = F(\xi) \qquad -\infty < \xi < \infty,\ \xi = \eta \tag{10.15b}$$

$$\frac{\partial \psi}{\partial \xi} - \frac{\partial \psi}{\partial \eta} = c^{-1}\, E(\xi) \qquad -\infty < \xi < \infty,\ \xi = \eta. \tag{10.15c}$$

Integrating equation (10.15a) leads to $\partial_{\xi}\psi = \theta(\xi)$ so that

$$\psi(\xi, \eta) = \Phi(\eta) + \int^{\xi} \theta(s)\,\mathrm{d}s \equiv \Phi(\eta) + \Theta(\xi), \tag{10.16}$$

for two functions $\Phi(\eta)$ and $\Theta(\xi)$. The initial conditions (10.15b) and (10.15c) determine relations between $\Phi(\eta)$ and $\Theta(\xi)$ and the initial data

$$\Theta(\xi) = \frac{1}{2}\left[F(\xi) + \frac{1}{c}\int^{\xi} E(s)\,\mathrm{d}s\right] \tag{10.17a}$$

$$\Phi(\eta) = \frac{1}{2}\left[F(\eta) - \frac{1}{c}\int^{\eta} E(s)\,\mathrm{d}s\right], \tag{10.17b}$$

in which case the general solution to the initial value problem (10.5a)-(10.5c) takes the form

$$\psi(x,t) = \frac{1}{2}\left[F(x + c\,t) + F(x - c\,t)\right] + \frac{1}{2c}\int_{x-c\,t}^{x+c\,t} E(s)\,\mathrm{d}s, \tag{10.18}$$

where we reintroduced the variables (x,t). This solution is known as the D'Alembert's formula for the wave equation. Note how the initial profile, $F(x)$, is propagated along the two characteristics, $\xi = x + c\,t$ and $\eta = x - c\,t$, without any change. In contrast, the initial tendency, $\partial_t \psi(x, t = 0) = E(x)$, is smoothed through the time integration. This behavior contrasts to the heat equation in Section 9.1, with its single time derivative resulting in a smoothing of the full solution.

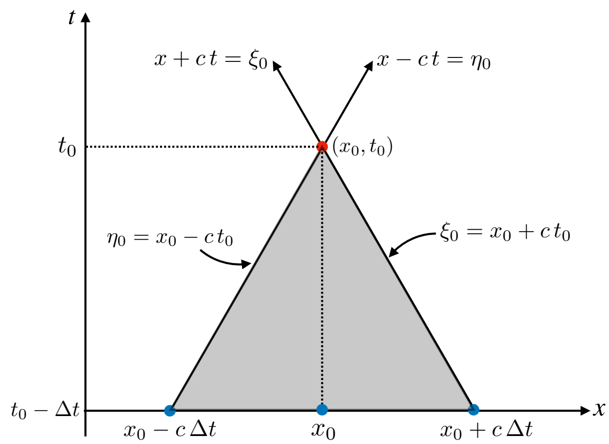

FIGURE 10.1: Equation (10.18) is the general solution to the wave equation initial value problem on a line, in which case an arbitrary space-time point, $(x_\circ, t_\circ)$, is causally connected via wave signals to all space-time points within the shaded region in its past. This domain of influence is bounded by the two wave characteristics, $\xi_\circ = x_\circ + c\,t_\circ$ and $\eta_\circ = x_\circ - c\,t_\circ$, with these characteristics the pathway for propagating information about the initial wave profile, $\psi(x, t = 0) = F(x)$. Points in between the characteristics are causally connected via the initial time tendency, $\partial_t \psi(x, t = 0) = E(x)$, which is integrated over the region $x_\circ - c\,t_\circ \leq x \leq x_\circ + c\,t_\circ$. Points outside the domain of influence are causally disconnected from the point $(x_\circ, t_\circ)$.

10.2.2 Domain of influence for wave signals

The wave solution (10.18) at a point in space time, $(x_\circ, t_\circ)$, depends on data to its past within a causality triangle, or domain of influence, as shown in Figure 10.1. The domain of influence is bounded by the two wave characteristics, $\xi_\circ = x_\circ + c\,t_\circ$ and $\eta_\circ = x_\circ - c\,t_\circ$. These characteristics are the pathways for propagating information about the initial wave profile, $\psi(x, t = 0) = F(x)$. Points in between the characteristics are causally connected via the initial time tendency, $\partial_t \psi(x, t = 0) = E(x)$, which is integrated over the region $x_\circ - c\,t_\circ \leq x \leq x_\circ + c\,t_\circ$. Points outside the domain of influence are causally disconnected from the point $(x_\circ, t_\circ)$.

10.2.3 Duhamel's superposition integral for the wave equation

We here present Duhamel's superposition integral for the wave equation, following from the similar discussion for the diffusion equation in Section 9.1.5. For this purpose, consider the forced wave equation with time independent forcing

$$(\partial_{tt} - c^2\,\nabla^2)\Psi = K(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t > 0 \tag{10.19a}$$
$$\Psi(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t = 0, \tag{10.19b}$$

and the corresponding unforced wave equation with inhomogenous initial time tendency

$$(\partial_{tt} - c^2\,\nabla^2)\psi = 0 \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t > 0 \tag{10.20a}$$
$$\psi(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t = 0. \tag{10.20b}$$
$$\partial_t \psi(\boldsymbol{x}, t) = K(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathbb{R}^n,\ t = 0. \tag{10.20c}$$

The two scalar fields are related by Duhamel's superposition integral

$$\Psi(\boldsymbol{x},t) = \int_0^t \psi(\boldsymbol{x},t-\tau)\,\mathrm{d}\tau. \tag{10.21}$$

We can verify this formula through direct differentiation

$$\partial_t \Psi(\boldsymbol{x},t) = \int_0^t \frac{\partial \psi(\boldsymbol{x},t-\tau)}{\partial t}\,\mathrm{d}\tau \tag{10.22a}$$

$$\partial_{tt} \Psi(\boldsymbol{x},t) = \partial_t \psi(\boldsymbol{x},0) + \int_0^t \partial_{tt}\psi(\boldsymbol{x},t-\tau)\,\mathrm{d}\tau = K(\boldsymbol{x}) + c^2\,\nabla^2\Psi(\boldsymbol{x},t). \tag{10.22b}$$

As an example, consider the initial value problem for wave equation on a line with time independent forcing

$$(\partial_{tt} - c^2\,\partial_{xx})\Psi = K(x) \qquad -\infty < x < \infty,\ t > 0 \tag{10.23a}$$

$$\Psi(x,t) = 0 \qquad -\infty < x < \infty,\ t = 0 \tag{10.23b}$$

$$\partial_t \Psi(x,t) = 0 \qquad -\infty < x < \infty,\ t = 0. \tag{10.23c}$$

Duhamel's superposition integral says that Ψ is related to the solution of the unforced wave equation with initial time tendency given by the forcing

$$(\partial_{tt} - c^2\,\partial_{xx})\psi = 0 \qquad -\infty < x < \infty,\ t > 0 \tag{10.24a}$$

$$\psi(x,t) = 0 \qquad -\infty < x < \infty, t = \tau \tag{10.24b}$$

$$\partial_t \psi(x,t) = K(x) \qquad -\infty < x < \infty,\ t = \tau > 0. \tag{10.24c}$$

We know from Section 10.2.1 that the solution, ψ, is given by the D'Alembert formula in equation (10.18), only here with the initial condition function set to zero

$$\psi(x,t) = \frac{1}{2c}\int_{x-c\,t}^{x+c\,t} K(s)\,\mathrm{d}s. \tag{10.25}$$

Hence, D'Alembert's formula says that the solution to the forced wave equation (10.23a)-(10.23c) is given by the superposition integral

$$\Psi(x,t) = \frac{1}{2c}\int_0^t \int_{x-c\,(t-\tau)}^{x+c\,(t-\tau)} K(s)\,\mathrm{d}s. \tag{10.26}$$

Introducing the anti-derivative function via

$$\mathcal{K}(s) = \int^s K(s')\,\mathrm{d}s' \Longleftrightarrow \frac{\partial \mathcal{K}(s)}{\partial s} = K(s) \tag{10.27}$$

allows us to interpret the solution (10.26) as the superposition of two oppositely traveling waves

$$\Psi(x,t) = \frac{1}{2c}\int_0^t \Big[\mathcal{K}[x + c\,(t-\tau)] - \mathcal{K}[x - c\,(t-\tau)]\Big]\,\mathrm{d}\tau. \tag{10.28}$$

10.2.4 The advection equation

Before closing this section, we mention the advection equation for the special case of a constant flow field. As noted at the start of this section, the advection equation, with a constant advection velocity, is the simplest hyperbolic equation, and it forms the basis for much intuition about advective transport in fluids.

Consider a scalar field, C, that is a function of space and time, and assume it satisfies the advection equation

$$(\partial_t + \boldsymbol{v} \cdot \nabla)C = 0. \tag{10.29}$$

The space and time derivatives acting on C are both first order, indicating that the advection equation is a first order partial differential equation. It is a nonlinear partial differential equation for those active tracers such as temperature, where active tracers affect the velocity, $\boldsymbol{v}$, through changes to density and hence to pressure. In contrast, the advection equation is linear for passive tracers (e.g., colored dye, dust), which are tracers whose effects on velocity are negligible. We limit the present discussion to passive tracers so that the advection equation is linear.

Consider one space dimension and let the advection velocity be constant in space and time,

$$(\partial_t + U\,\partial_x)C = 0, \tag{10.30}$$

where U is a constant velocity in the $\hat{\boldsymbol{x}}$ direction. This equation has the exact characteristic solution

$$C(x,t) = \Gamma(\alpha) = \Gamma(x - Ut), \tag{10.31}$$

with Γ is an arbitrary differentiable function that is determined by the initial conditions. Furthermore, there is only one argument to Γ, here written as $\alpha = x - Ut$. As seen in Section 8.1, $\alpha = x - Ut$ is the characteristic curve for the constant speed one-dimensional advection equation.

Verification of the solution (10.31) is readily found by noting

$$\frac{\partial C}{\partial x} = \frac{\mathrm{d}\Gamma(\alpha)}{\mathrm{d}\alpha}\,\frac{\partial \alpha}{\partial x} = \Gamma'(\alpha)\,\frac{\partial(x-Ut)}{\partial x} = \Gamma' \tag{10.32a}$$

$$\frac{\partial C}{\partial t} = \frac{\mathrm{d}\Gamma(\alpha)}{\mathrm{d}\alpha}\,\frac{\partial \alpha}{\partial t} = \Gamma'(\alpha)\,\frac{\partial(x-Ut)}{\partial t} = -\Gamma'\,U. \tag{10.32b}$$

We have thus verified that

$$(\partial_t + \boldsymbol{v} \cdot \nabla)C = -\Gamma'\,U + U\,\Gamma' = 0. \tag{10.33}$$

The functional dependence,

$$C(x,t) = \Gamma(\alpha) = \Gamma(x - Ut), \tag{10.34}$$

reveals that as time progresses with $U > 0$, an observer that moves in the positive $\hat{\boldsymbol{x}}$ direction with a speed U maintains a constant value for the tracer concentration. This behavior means that the tracer concentration is transported by advection with a speed U without changing its structure. We illustrate this behavior in Figure 10.2.

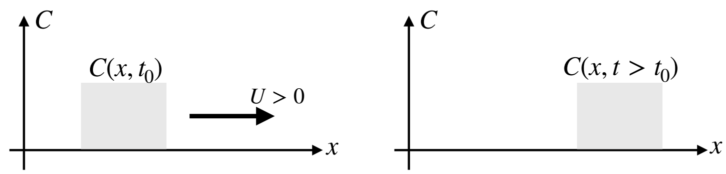

FIGURE 10.2: Illustrating the advection of a scalar field resulting from a constant advection velocity $\boldsymbol{v} = U\hat{\boldsymbol{x}}$ with $U > 0$. The initially square pulse of tracer concentration is translated, unchanged, by the constant advection velocity.

10.3 Initial-boundary value problem

In this section we introduce the initial-boundary value problem for the wave equation and prove the uniqueness of its solution. We return in Section 10.6 to express the solution in terms of a Green's function.

10.3.1 Wave equation initial-boundary value problem

Consider the following initial-boundary value problem for a wave function satisfying the linear non-dispersive wave equation

$$(\partial_{tt} - c^2\nabla^2)\psi(\boldsymbol{x}, t) = \Lambda(\boldsymbol{x}, t) \qquad \boldsymbol{x} \in \mathcal{R}, \quad t > t_{\mathsf{init}} \tag{10.35a}$$
$$\psi(\boldsymbol{x}, t) = I(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathcal{R}, \quad t = t_{\mathsf{init}} \tag{10.35b}$$
$$\partial_t\psi(\boldsymbol{x}, t) = J(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathcal{R}, \quad t = t_{\mathsf{init}} \tag{10.35c}$$
$$\psi(\boldsymbol{x}, t) = \sigma(\boldsymbol{x}, t) \qquad \boldsymbol{x} \in \partial\mathcal{R}, \tag{10.35d}$$

where $\mathcal{R}$ is an open spatial region, and with Λ, I, J, and σ known functions that provide the initial and boundary data. We choose to focus on the Dirichlet boundary condition here, though note that the Neumann boundary condition is handled similarly.[2] Compatibility between the boundary conditions and initial conditions is ensured if

$$\sigma(\boldsymbol{x}, t) = I(\boldsymbol{x}) \qquad \boldsymbol{x} \in \partial\mathcal{R}, \quad t = t_{\mathsf{init}}. \tag{10.36}$$

As for the diffusion equation in Section 9.4.1, we introduced the initial time, $t = t_{\mathsf{init}}$, which is distinct from the Dirac delta source time, $t = t_{\circ}$, appearing in the Green's function equations. Correspondingly, the Dirac delta source is fired after the initial time,

$$t_{\mathsf{init}} < t_{\circ}, \tag{10.37}$$

which follows since we are interested in evolution of ψ after specification of the initial data. Correspondingly, t_{init} defines the lower limit on time integrals in the following. For the upper limit we find it convenient (though not necessary) to introduce an arbitrary time $T > t_{\circ}$. This time drops out from the final answer due to causality, just like it did for the diffusion equation in Section 9.4.1.

[2]See Section 8.5 of *Duchateau and Zachmann* (1986) for the general Robin boundary condition.

10.3.2 Uniqueness of the solution

Before developing a solution to the wave equation, it is important to determine whether that solution is unique. If there are two distinct solutions, ψ_1 and ψ_2, that each solve the initial-boundary value problem (10.35a)–(10.35d), then their difference, $\Psi = \psi_1 - \psi_2$, must solve the following homogeneous system

$$(\partial_{tt} - c^2\nabla^2)\,\Psi(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \quad t > t_{\mathsf{init}} \tag{10.38a}$$

$$\Psi(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \quad t = t_{\mathsf{init}} \tag{10.38b}$$

$$\partial_t\Psi(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \quad t = t_{\mathsf{init}} \tag{10.38c}$$

$$\Psi(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R}. \tag{10.38d}$$

Now multiply the homogeneous wave equation (10.38a) by $\partial_t\Psi$ and integrate over the domain, with integration by parts leading to an energy equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathcal{R}}[(\partial_t\Psi)^2 + c^2\,\nabla\Psi\cdot\nabla\Psi]\,\mathrm{d}V = 2\,c^2\int_{\partial\mathcal{R}}(\partial_t\Psi)\,\nabla\Psi\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}. \tag{10.39}$$

Since $\Psi = 0$ on the boundary (equation (10.38d)), we know that $\partial_t\Psi = 0$ also holds on the boundary, so that the right hand side vanishes, meaning that the energy integral equals to a constant,

$$\int_{\mathcal{R}}[(\partial_t\Psi)^2 + c^2\,\nabla\Psi\cdot\nabla\Psi]\,\mathrm{d}V = \alpha. \tag{10.40}$$

At the initial time, both $\Psi = 0$ and $\partial_t\Psi = 0$, in which case $\nabla\Psi = 0$ as well, which means that the constant vanishes, $\alpha = 0$. Since the integrand is non-negative, we can only satisfy this constraint if Ψ is a space-time constant. With $\Psi = 0$ as the initial condition, then $\Psi = 0$ holds for all time, which then proves our assertion that the wave equation solution is unique.[3]

10.4 Green's functions for the wave equation

In this section we introduce the Green's function method for non-dispersive waves. Much of the development in this section emulates that for the diffusion equation in Section 9.2, in particular the details of causality that are shared between the diffusion equation and wave equation since they are both prognostic (evolution) equations.

10.4.1 Causal free space Green's function

The causal free space Green's function for the wave equation satisfies

$$(\partial_{tt} - c^2\nabla^2_{\boldsymbol{x}})\mathcal{G}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(t - t_\circ)\,\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathbb{R}^n \tag{10.41a}$$

$$\mathcal{G}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathbb{R}^n, \quad t \le t_\circ \tag{10.41b}$$

$$\partial_t\mathcal{G}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathbb{R}^n, \quad t = t_\circ. \tag{10.41c}$$

The initial condition (10.41c) on the time derivative is needed since the wave equation has two time derivatives, with this condition chosen as a Dirac delta source. Note that we do not impose a vanishing solution condition at spatial infinity, in contrast to equation (9.25b) for the

[3]The Green's function solution (10.97) offers an alternative proof of uniqueness. Namely, equation (10.97) identically vanishes for the case of an unforced non-dispersive wave with zero initial and boundary conditions.

diffusion equation free space Green's function. The reason is that we can conceive, at least in principle, of waves reaching out to infinity at time infinity. Finally, given the dimensions of the Dirac source, we see that the Green's function has dimensions of T L^{-n}, where n is the space dimension.

Consider the slightly simpler Green's function problem by focusing on times $t > t_{\circ}$ and setting $t_{\circ} = 0$ and $\boldsymbol{x}_{\circ} = 0$, in which case we have

$$
\begin{aligned}
(\partial_{tt} - c^2 \nabla^2)\, \}(\boldsymbol{x}, t) &= 0 & \boldsymbol{x} \in \mathbb{R}^n, \quad t > 0 && \text{(10.42a)}\\
\}(\boldsymbol{x}, t) &= 0 & \boldsymbol{x} \in \mathbb{R}^n, \quad t = 0+ && \text{(10.42b)}\\
\partial_t \}(\boldsymbol{x}, t) &= \delta(\boldsymbol{x}) & \boldsymbol{x} \in \mathbb{R}^n, \quad t = 0+, && \text{(10.42c)}
\end{aligned}
$$

where $t = 0+$ refers to a time that is arbitrarily close to, but greater than, $t = 0$. We now show that

$$
\mathcal{G}(\boldsymbol{x}, t | \boldsymbol{x}_{\circ} = 0, t_{\circ} = 0) = \mathcal{H}(t)\, \}(\boldsymbol{x}, t), \tag{10.43}
$$

or more generally

$$
\mathcal{G}(\boldsymbol{x}, t | \boldsymbol{x}_{\circ}, t_{\circ}) = \mathcal{H}(t - t_{\circ})\, \}(\boldsymbol{x}, t | \boldsymbol{x}_{\circ}, t_{\circ}). \tag{10.44}
$$

This result is not too surprising since, by causality, $\mathcal{G}(\boldsymbol{x}, t | \boldsymbol{x}_{\circ}, t_{\circ})$ vanishes for times prior to when the Dirac delta is fired, so we suspect it should be proportional to a Heaviside step function just like for the diffusion equation in Section 9.2.2.

A proof of equation (10.44) requires the time derivatives (dropping the $\boldsymbol{x}$ label for brevity)

$$
\begin{aligned}
\partial_t \mathcal{G}(t) &= \delta(t)\, \}(t) + \mathcal{H}(t)\, \partial_t \}(t) && \text{(10.45a)}\\
\partial_{tt} \mathcal{G}(t) &= (\mathrm{d}\delta(t)/\mathrm{d}t)\, \}(t) + 2\, \delta(t)\, \partial_t \}(t) + \mathcal{H}(t)\, \partial_{tt} \}(t). && \text{(10.45b)}
\end{aligned}
$$

To massage the $\partial_{tt}\mathcal{G}$ term, introduce a prime symbol for time derivative so that[4]

$$
\begin{aligned}
\delta'(t)\, \}(t) + \delta(t)\, \partial_t \}(t) &= \delta'(t)\, \}(t) - t\, \delta'(t)\, \partial_t \}(t) && \text{dipole identity (5.45)} && \text{(10.46a)}\\
&= \delta'(t)\, [\}(t) - t\, \partial_t \}(t)] && \text{reorganize} && \text{(10.46b)}\\
&= \delta'(t)\, g(0) + t\, \delta'(t)\, [\partial_t \}(0) - \partial_t \}(t)] && \text{Taylor series from } t = 0 && \text{(10.46c)}\\
&= \delta'(t)\, \}(0) - \delta(t)\, [\partial_t \}(0) - \partial_t \}(t)] && \text{dipole identity (5.45).} && \text{(10.46d)}
\end{aligned}
$$

Causality, along with the initial condition (10.42b), means that $\}(t = 0) = 0$, so that the first term vanishes. When multiplying a function and the Dirac delta, it is sufficient to evaluate that function at the place where the Dirac fires. For the second term, the Dirac delta fires at $t = 0$, which is where $\partial_t \}(0) - \partial_t \}(t) = 0$, so that second term vanishes. We are thus led to

$$
\partial_{tt} \mathcal{G}(t) = \delta(t)\, \partial_t \}(t) + \mathcal{H}(t)\, \partial_{tt} \}(t). \tag{10.47}
$$

Likewise, we find

$$
\delta(t)\, \partial_t \}(\boldsymbol{x}, t) = \delta(t)\, \partial_t \}(\boldsymbol{x}, 0) = \delta(t)\, \delta(\boldsymbol{x}), \tag{10.48}
$$

where the final equality made use of the initial condition (10.42c). Bringing all the pieces together leads to

$$
(\partial_{tt} - c^2\, \nabla^2_{\boldsymbol{x}}) \mathcal{G} = \mathcal{H}(t)\, (\partial_{tt} - c^2\, \nabla^2) \} + \delta(t)\, \delta(\boldsymbol{x}) = \delta(t)\, \delta(\boldsymbol{x}), \tag{10.49}
$$

[4] These manipulations are motivated from page 61 of *Stakgold* (2000b), including his footnote.

with the final equality holding since $(\partial_{tt} - c^2 \nabla^2)\} = 0$ as per equation (10.42a). We have thus shown that the causal free-space Green's function for the wave equation can be decomposed as in equation (10.44), in which case it is sufficient to solve equations (10.42a)-(10.42c).

10.4.2 Expressions for the free space Green's functions

Following the methods from Section IV-28.4 of *Dennery and Krzywicki* (1967) (summarized here in Section 10.4.3), Section 8.5 of *Duchateau and Zachmann* (1986), and Section 5.8 of *Stakgold* (2000b), the causal free-space Green's function for the wave equation is given by[5]

$$\mathcal{G}(x,t|x_\circ,t_\circ) = \frac{\mathcal{H}[c\,(t-t_\circ) - |x-x_\circ|]}{2c} \qquad \text{for } \mathbb{R}^1 \tag{10.50a}$$

$$\mathcal{G}(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ) = \frac{\mathcal{H}[c\,(t-t_\circ) - |\boldsymbol{x}-\boldsymbol{x}_\circ|]}{2\,\pi\,c\,\sqrt{[c\,(t-t_\circ)]^2 - |\boldsymbol{x}-\boldsymbol{x}_\circ|^2}} \qquad \text{for } \mathbb{R}^2 \tag{10.50b}$$

$$\mathcal{G}(\boldsymbol{x},t|\boldsymbol{x}_\circ,t_\circ) = \frac{\delta[c\,(t-t_\circ) - |\boldsymbol{x}-\boldsymbol{x}_\circ|]}{4\,\pi\,c\,|\boldsymbol{x}-\boldsymbol{x}_\circ|} \qquad \text{for } \mathbb{R}^3. \tag{10.50c}$$

These wave solutions manifest causality since they vanish for regions in space-time that are unconnected to the source, as accords with the domain of influence shown by Figure 10.1. Additionally, note that the three-dimensional Green's function needs no Heaviside to enforce causality since the argument to the Dirac delta never vanishes for times prior to the Dirac source, $t < t_\circ$.

There is an essential distinction between the three-dimensional Green's function (10.50c) relative to the one and two dimensional Green's functions (10.50a) and (10.50b). Namely, in three dimensions the Green's function is a spherical wave front that is infinitely sharp (i.e., a Dirac front) that propagates outward at speed c from the source point at $\boldsymbol{x} = \boldsymbol{x}_\circ$. The one and two dimensional Green's function fronts also move with speed c, yet these fronts are trailed by a wake as realized by the Heaviside step function. Furthermore, observe that the two dimensional wake decays according to the inverse distance from the front.

10.4.3 Deriving the 3d free space Green's function

We here present some details for deriving the three-dimensional free space Green's function (10.50c), with many of the manipulations appearing in a variety of contexts with wave equations. To reduce clutter it is useful to introduce the shorthand[6]

$$\boldsymbol{r} = \boldsymbol{x} - \boldsymbol{x}_\circ \quad \text{and} \quad \tau = t - t_\circ. \tag{10.51}$$

Also note that the Green's function (10.50c) has physical dimensions of time per volume, T L^{-3}. Debugging the maths is greatly served by tracking the physical dimensions, particularly when transforming between $(\boldsymbol{x}, t)$ and $(\boldsymbol{k}, \omega)$ as part of the Fourier transform solution method presented here.

[5]Recall from Section 5.5 that the Heaviside step function is a non-dimensional function, whereas the Dirac delta has dimensions given by the inverse dimensions of its argument. These properties are key to checking that the given Green's functions in equations (10.50a)-(10.50c) have dimensions T L^{-n}, where n is the space dimenson.

[6]Since space and time are homogeneous and isotropic for the free space problem, we could, without loss of generality, take the source space-time point to be the origin, $\boldsymbol{x}_\circ = 0$ and $t_\circ = 0$. Even so, we retain $\boldsymbol{x}_\circ$ and $t_\circ$ through use of $\boldsymbol{r}$ and τ.

Fourier solution

Introduce the inverse Fourier transform, $\widetilde{\mathcal{G}}(\boldsymbol{k},\omega)$ (see Section 6.3.2)

$$\mathcal{G}(\boldsymbol{x},\boldsymbol{x}_\circ|t,t_\circ) = \frac{1}{(2\pi)^4}\int \widetilde{\mathcal{G}}(\boldsymbol{k},\omega)\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{r}-\omega\tau)}\mathrm{d}^3k\,\mathrm{d}\omega, \tag{10.52}$$

where the integrals are over all $(\boldsymbol{k},\omega)$ values, and $\widetilde{\mathcal{G}}(\boldsymbol{k},\omega)$ has physical dimensions of T^2. The space-time wave operator acting on the Green's function takes the form

$$(\partial_{tt} - c^2\nabla^2_{\boldsymbol{x}})\mathcal{G}(\boldsymbol{x},\boldsymbol{x}_\circ|t,t_\circ) = \frac{1}{(2\pi)^4}\int (-\omega^2 + c^2\,\boldsymbol{k}^2)\,\widetilde{\mathcal{G}}(\boldsymbol{k},\omega)\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{r}-\omega\tau)}\mathrm{d}^3k\,\mathrm{d}\omega. \tag{10.53}$$

With the Dirac delta expressed according to the identity (6.114)

$$\delta(\boldsymbol{r}) = \frac{1}{(2\pi)^3}\int \mathrm{e}^{\mathrm{i}\,\boldsymbol{k}\cdot\boldsymbol{r}}\,\mathrm{d}^3k \quad \text{and} \quad \delta(\tau) = \frac{1}{2\pi}\int \mathrm{e}^{-\mathrm{i}\,\omega\tau}\,\mathrm{d}\omega, \tag{10.54}$$

we reach the Fourier space representation of the free space Green's function

$$\widetilde{\mathcal{G}}(\boldsymbol{k},\omega) = \frac{1}{-\omega^2 + c^2\,\boldsymbol{k}^2}, \tag{10.55}$$

so that the $(\boldsymbol{x},t)$ representation is given by the inverse Fourier transform (10.52)

$$\mathcal{G}(\boldsymbol{x},\boldsymbol{x}_\circ|t,t_\circ) = \frac{1}{(2\pi)^4}\int \frac{\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{r}-\omega\tau)}}{-\omega^2 + c^2\,\boldsymbol{k}^2}\,\mathrm{d}^3k\,\mathrm{d}\omega. \tag{10.56}$$

Inverting the frequency portion of the Fourier transform

Consider just the angular frequency integral

$$\Delta \equiv \int_{-\infty}^{\infty} \frac{\mathrm{e}^{-\mathrm{i}\,\omega\,\tau}}{-\omega^2 + c^2\,\boldsymbol{k}^2}\,\mathrm{d}\omega, \tag{10.57}$$

which has two singularities where the angular frequency equals to that of the wave, $\omega^2 = c^2\,\boldsymbol{k}^2$. To make physical sense of this integral requires standard results from contour integration in the complex plane. Rather than take a rather sizable tangent to describe the basics of complex analysis[7], we quote from Section IV-28 of *Dennery and Krzywicki* (1967) to write the *retarded* and *advanced* solutions

$$\Delta^{\mathsf{ret}} = \frac{2\pi\sin(c\,|\boldsymbol{k}|\,\tau)}{c\,|\boldsymbol{k}|}\begin{bmatrix} 1 & \text{for } \tau = t - t_\circ > 0 \\ 0 & \text{for } \tau = t - t_\circ < 0 \end{bmatrix} \tag{10.58a}$$

$$\Delta^{\mathsf{adv}} = \frac{2\pi\sin(c\,|\boldsymbol{k}|\,\tau)}{c\,|\boldsymbol{k}|}\begin{bmatrix} 0 & \text{for } \tau = t - t_\circ > 0 \\ -1 & \text{for } \tau = t - t_\circ < 0. \end{bmatrix} \tag{10.58b}$$

The retarded solution, Δ^{ret}, corresponds to a signal that is non-zero for times after firing the Dirac delta source, $t > t_\circ$. In contrast, the advanced solution, Δ^{adv}, is nonzero prior to firing the Dirac delta, $t < t_\circ$. We choose the causal solution, Δ^{ret}, in which case the Greens function

[7]See Chapter I of *Dennery and Krzywicki* (1967) for a lucid discussion.

expressed as equation (10.56) becomes

$$\mathcal{G}(\boldsymbol{x}, \boldsymbol{x}_\circ|t, t_\circ) = \frac{\mathcal{H}(\tau)}{(2\pi)^3\, c} \int \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\, \frac{\sin(c\,|\boldsymbol{k}|\,\tau)}{|\boldsymbol{k}|}\, \mathrm{d}^3 k. \tag{10.59}$$

Inverting the wavevector portion of the Fourier transform

Following the approach used in Section 8.7.4 for the Poisson equation free space Green's function, we here introduce spherical coordinates in wavevector space according to

$$k_x = |\boldsymbol{k}| \cos\phi \cos\lambda \quad \text{and} \quad k_y = |\boldsymbol{k}| \cos\phi \sin\lambda \quad \text{and} \quad k_z = |\boldsymbol{k}| \sin\phi. \tag{10.60}$$

We are free to orient the axes as suits the integration, with the following choice very convenient[8]

$$\boldsymbol{k} \cdot \boldsymbol{r} = |\boldsymbol{k}|\, |\boldsymbol{r}| \sin\phi, \tag{10.61}$$

thus bringing the integral (10.59) to

$$\mathcal{G}(\boldsymbol{x}, \boldsymbol{x}_\circ|t, t_\circ) = \frac{\mathcal{H}(\tau)}{(2\pi)^3\, c} \int_0^\infty \mathrm{d}|\boldsymbol{k}| \int_{-\pi/2}^{\pi/2} \mathrm{d}(\sin\phi) \int_0^{2\pi} \mathrm{e}^{\mathrm{i}|\boldsymbol{k}|\,|\boldsymbol{r}|\sin\phi}\, |\boldsymbol{k}| \sin(c\,|\boldsymbol{k}|\,\tau)\, \mathrm{d}\lambda. \tag{10.62}$$

The λ integral is trivial, and picks up a factor of 2π, whereas for the ϕ integral we set $w = \sin\phi$, in which case

$$\begin{aligned}
\mathcal{G}(\boldsymbol{x}, \boldsymbol{x}_\circ|t, t_\circ) &= \frac{\mathcal{H}(\tau)}{(2\pi)^2\, c} \int_0^\infty \int_{-1}^{1} \mathrm{e}^{\mathrm{i}|\boldsymbol{k}|\,|\boldsymbol{r}|\,w}\, |\boldsymbol{k}| \sin(c\,|\boldsymbol{k}|\,\tau)\, \mathrm{d}w\, \mathrm{d}|\boldsymbol{k}| && (10.63\text{a}) \\
&= \frac{\mathcal{H}(\tau)}{2\,\pi^2\, c\, |\boldsymbol{r}|} \int_0^\infty \sin(|\boldsymbol{k}|\,|\boldsymbol{r}|) \sin(c\,|\boldsymbol{k}|\,\tau)\, \mathrm{d}|\boldsymbol{k}|. && (10.63\text{b})
\end{aligned}$$

The integrand is an even function of $|\boldsymbol{k}|$, which then allows us to extend the integral to the full real line

$$\begin{aligned}
\mathcal{G}(\boldsymbol{x}, \boldsymbol{x}_\circ|t, t_\circ) &= \frac{\mathcal{H}(\tau)}{4\,\pi^2\, c\, |\boldsymbol{r}|} \int_{-\infty}^\infty \sin(\kappa\,|\boldsymbol{r}|) \sin(c\,\kappa\,\tau)\, \mathrm{d}\kappa && (10.64\text{a}) \\
&= -\frac{\mathcal{H}(\tau)}{8\,\pi^2\, c\, |\boldsymbol{r}|} \int_{-\infty}^\infty (\mathrm{e}^{\mathrm{i}\,(c\,\kappa\,\tau + \kappa\,|\boldsymbol{r}|)} - \mathrm{e}^{\mathrm{i}\,(c\,\kappa\,\tau - \kappa\,|\boldsymbol{r}|)})\, \mathrm{d}\kappa && (10.64\text{b}) \\
&= -\frac{\mathcal{H}(\tau)}{4\,\pi\, c\, |\boldsymbol{r}|} [\delta(c\,\tau + |\boldsymbol{r}|) - \delta(c\,\tau - |\boldsymbol{r}|)]. && (10.64\text{c})
\end{aligned}$$

For the causal Green's function we are only concerned with times $\tau = t - t_\circ > 0$, so that the Dirac delta, $\delta(|\boldsymbol{r}| + c\,\tau)$, never fires. We are thus led to the causal free space Green's function for the wave equation

$$\mathcal{G}(\boldsymbol{x}, \boldsymbol{x}_\circ|t, t_\circ) = \frac{\delta(c\,\tau - |\boldsymbol{r}|)}{4\,\pi\, c\, |\boldsymbol{r}|} = \frac{\delta(c\,(t - t_\circ) - |\boldsymbol{x} - \boldsymbol{x}_\circ|)}{4\,\pi\, c\, |\boldsymbol{x} - \boldsymbol{x}_\circ|}, \tag{10.65}$$

which agrees with equation (10.50c). Note that we do not need a Heaviside, $\mathcal{H}(t - t_\circ)$, to enforce causality since the argument to the Dirac delta never vanishes for $t < t_\circ$.

[8] Recall space is isotropic since this is a free-space problem, thus allowing for arbitrary orientation of the axes.

10.4.4 Causal Green's function for the wave equation

The causal Green's function corresponding to the initial-boundary value problem (10.35a)-(10.35d) is satisfies

$$(\partial_{tt} - c^2 \nabla^2_{\boldsymbol{x}}) G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ)\, \delta(t - t_\circ) \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R} \tag{10.66a}$$

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \quad t < t_\circ \tag{10.66b}$$

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R}. \tag{10.66c}$$

The following properties are of use for subsequent development.

Time dependence

Time dependence only appears as the difference, $t - t_\circ$, so that

$$\partial_t G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = -\partial_{t_\circ} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ). \tag{10.67}$$

Auxiliary Green's function problem

Through the same arguments as used in Section 10.4.1, we can determine the causal Green's function by solving the following initial-boundary value problem

$$(\partial_{tt} - c^2 \nabla^2_{\boldsymbol{x}}) \}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R}, \quad t > t_\circ \tag{10.68a}$$

$$\}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \quad t = t_\circ \tag{10.68b}$$

$$\partial_t \}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ) \qquad \boldsymbol{x} \in \mathcal{R}, \quad t = t_\circ \tag{10.68c}$$

$$\}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R}. \tag{10.68d}$$

As in equation (10.44) for the free space Green's function, we have

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \mathcal{H}(t - t_\circ)\, \}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ). \tag{10.69}$$

10.4.5 Adjoint or anti-causal Green's function

Swapping t and $t_\circ$ results in a swapped causality relation so that if $G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)$ is the causal Green's function that satisfies equations (10.66a)-(10.66c), then $G(\boldsymbol{x}_\circ, t_\circ|\boldsymbol{x}, t)$ represents an anti-causal Green's function that satisfies

$$(\partial_{tt} - c^2 \nabla^2_{\boldsymbol{x}})\, G(\boldsymbol{x}_\circ, t_\circ|\boldsymbol{x}, t) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ)\, \delta(t - t_\circ) \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R} \tag{10.70a}$$

$$G(\boldsymbol{x}_\circ, t_\circ|\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \quad t > t_\circ \tag{10.70b}$$

$$G(\boldsymbol{x}_\circ, t_\circ|\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R}. \tag{10.70c}$$

Analogous to the diffusion equation in Section 9.2.6, we refer to $G(\boldsymbol{x}_\circ, t_\circ|\boldsymbol{x}, t)$ as the adjoint causal Green's function

$$G^\ddagger(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = G(\boldsymbol{x}_\circ, t_\circ|\boldsymbol{x}, t), \tag{10.71}$$

that satisfies

$$(\partial_{tt} - c^2 \nabla^2_{\boldsymbol{x}})\, G^\ddagger(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \delta(\boldsymbol{x} - \boldsymbol{x}_\circ)\, \delta(t - t_\circ) \qquad \boldsymbol{x}, \boldsymbol{x}_\circ \in \mathcal{R} \tag{10.72a}$$

$$G^\ddagger(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \quad t > t_\circ \tag{10.72b}$$

$$G^\ddagger(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R}. \tag{10.72c}$$

10.5 Spectral decomposition of the Green's function

We here develop an eigenfunction decomposition of the Green's function, otherwise known as a spectral decomposition, following an analogous treatment in Sections 8.12 and 8.13 for the generalized Laplace operator, and Section 9.3 for the diffusion operator. We make frequent use of dimensional analysis to check the maths, starting from the dimensions, $\mathrm{T}\,\mathrm{L}^{-3}$, for the Green's function.

10.5.1 Dirichlet boundary conditions

The causal Green's function for the wave equation satisfies equations (10.66a)-(10.66c). Our goal is to decompose this Green's function in terms of the Dirichlet eigenfunctions, Φ_a, of the generalized Laplace operator from Section 8.12, in which we write

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \sum_a C_a(t|\boldsymbol{x}_\circ, t_\circ)\, \Phi_a(\boldsymbol{x}), \tag{10.73}$$

with the coefficients, C_a, determined below.

The eigenfunctions

The spatial eigenfunctions we need satisfy the eigenvalue problem (8.219) with the diffusion tensor set to unity, so that

$$-\nabla^2 \Phi_a = \lambda_a\, \Phi_a, \quad \boldsymbol{x} \in \mathcal{R} \quad \text{and} \quad \Phi_a = 0, \quad \boldsymbol{x} \in \partial\mathcal{R}, \tag{10.74}$$

and with $0 < \lambda_a$ the non-decreasing set of discrete eigenvalues with physical dimension of inverse area, L^{-2}. We are afforded the expansion (10.73) since the eigenfunctions for the Laplace operator with homogeneous Dirichlet boundaries form a complete basis for functions on the domain, $\mathcal{R}$, that also satisfy homogeneous Dirichlet boundary conditions.

Deriving the equation for the expansion coefficient

Use of the spectral decomposition (10.73) in the Green's function differential equation (10.66a) leads to

$$\sum_a (\partial_{tt} C_a + c^2\, \lambda_a\, C_a)\, \Phi_a(\boldsymbol{x}) = \delta(t - t_\circ)\, \delta(\boldsymbol{x} - \boldsymbol{x}_\circ). \tag{10.75}$$

The completeness property of the eigenfunctions allows us to decompose the spatial Dirac delta according to

$$\delta(\boldsymbol{x} - \boldsymbol{x}_\circ) = \sum_a \Phi_a(\boldsymbol{x})\, \Phi_a(\boldsymbol{x}_\circ), \tag{10.76}$$

which then brings equation (10.75) to

$$\sum_a [\partial_{tt} C_a + c^2\, \lambda_a\, C_a - \delta(t - t_\circ)\, \Phi_a(\boldsymbol{x}_\circ)]\, \Phi_a(\boldsymbol{x}) = 0. \tag{10.77}$$

A dimensional check

Before going further, we insert a check on the physical dimensions, starting from the dimensions of the Dirac deltas

$$[\delta(t - t_\circ)] = \mathrm{T}^{-1} \quad \text{and} \quad [\delta(\boldsymbol{x} - \boldsymbol{x}_\circ)] = \mathrm{L}^{-3}. \tag{10.78}$$

From the equations written thus far, we have

$$[G] = \mathrm{T\,L^{-3}} \quad \text{and} \quad [\Phi_a] = \mathrm{L^{-3/2}} \quad \text{and} \quad [\lambda_a] = \mathrm{L^{-2}} \quad \text{and} \quad [C_a] = \mathrm{T\,L^{-3/2}}. \tag{10.79}$$

This dimensional check can be particularly useful to ensure that all terms are properly sorted in equation (10.77) as well as its solution in equation (10.80).

The expansion coefficients

The expansion coefficients, C_a, satisfying equation (10.77) are given by

$$C_a(t|\boldsymbol{x}_\circ, t_\circ) = \frac{\mathcal{H}(t-t_\circ)\,\Phi_a(\boldsymbol{x}_\circ)\sin[(t-t_\circ)\,c\sqrt{\lambda_a}]}{c\sqrt{\lambda_a}}. \tag{10.80}$$

It is useful to expose some details in verifying that this expression for C_a actually satisfies equation (10.77), which requires the second time derivative

$$\Phi_a^{-1}\,\partial_{tt}C_a = \frac{\delta'(t-t_\circ)\sin[(t-t_\circ)\,c\sqrt{\lambda_a}]}{c\sqrt{\lambda_a}} + 2\,\delta(t-t_\circ)\cos[(t-t_\circ)\,c\sqrt{\lambda_a}] - c\sqrt{\lambda_a}\,\mathcal{H}(t-t_\circ)\sin[(t-t_\circ)\,c\sqrt{\lambda_a}]. \tag{10.81}$$

The $\delta'(t-t_\circ)$ term can be written as

$$\delta'(t-t_\circ)\sin[(t-t_\circ)\,c\sqrt{\lambda_a}] = \delta'(t-t_\circ)\,(t-t_\circ)\,c\sqrt{\lambda_a} = -\delta(t-t_\circ)\,c\sqrt{\lambda_a}, \tag{10.82}$$

where the first equality holds since the Dirac only fires when $t = t_\circ$ so that we can approximate the sine function by the first term in a Taylor series. The second equality follows from the identity (5.45) satisfied by the Dirac delta, which says that $x\,\delta'(x) = -\delta(x)$. Next, we note that

$$\delta(t-t_\circ)\cos[(t-t_\circ)\,c\sqrt{\lambda_a}] = \delta(t-t_\circ), \tag{10.83}$$

which follows from our discussion of equivalence classes of Dirac deltas in Section 5.8. Namely, the cosine factor is non-dimensional function that equals to unity at the source time, $t = t_\circ$, and as such it has no affect on the Dirac delta and so it is irrelevant. These result thus leads to

$$\partial_{tt}C_a + c^2\,\lambda_a\,C_a = \delta(t-t_\circ)\,\Phi_a(\boldsymbol{x}_\circ). \tag{10.84}$$

Spectral decomposition of the Green's function

Making use of the expansion coefficients (10.80) leads to the spectral decomposition of the Green's function

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \mathcal{H}(t-t_\circ)\sum_a \frac{\Phi_a(\boldsymbol{x})\,\Phi_a(\boldsymbol{x}_\circ)\sin[(t-t_\circ)\,c\sqrt{\lambda_a}]}{c\sqrt{\lambda_a}}. \tag{10.85}$$

Observe that the Green's function and its time derivative satisfy the $t = t_\circ$ conditions

$$\lim_{t\downarrow t_\circ} G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = 0 \tag{10.86a}$$

$$\lim_{t\downarrow t_\circ} \partial_t G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \sum_a \Phi_a(\boldsymbol{x})\,\Phi_a(\boldsymbol{x}_\circ) = \delta(\boldsymbol{x}-\boldsymbol{x}_\circ). \tag{10.86b}$$

We can make the causal Green's function (10.85) a bit more tidy by introducing the eigenfrequencies

$$\omega_a = c\sqrt{\lambda_a}, \tag{10.87}$$

so that the Green's function is written

$$G(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ) = \mathcal{H}(t - t_\circ)\sum_a \frac{\Phi_a(\boldsymbol{x})\,\Phi_a(\boldsymbol{x}_\circ)\sin[(t - t_\circ)\,\omega_a]}{\omega_a}. \tag{10.88}$$

10.5.2 Neumann boundary conditions

In Section 8.13, we studied the Neumann eigenfunctions, Φ_a, for the Laplace operator, which satisfy

$$-\nabla^2\Phi_a = \lambda_a\,\Phi_a, \quad \boldsymbol{x} \in \mathcal{R} \qquad \text{and} \quad \hat{\boldsymbol{n}}\cdot\nabla\Phi_a = 0, \quad \boldsymbol{x} \in \partial\mathcal{R}. \tag{10.89}$$

Note that we continue to use Φ_a to denote the eigenfunctions, where in this subsection they are eigenfunctions for the Neumann problem rather than the Dirichlet problem. The lowest Neumann eigenmode has a constant eigenfunction and zero eigenvalue, thus requiring extra effort to properly account for this nontrivial null space. We present the necessary analysis in Section 8.13 to account for this zero mode, and make use of that analysis here.

The derivation in Section 10.5.1 for the Dirichlet problem carries over to the Neumann problem, only noting that the eigenfunctions and eigenvalues differ. Additionally, since the time dependence of the spectral decomposition carries a sine function, the zero eigenmode does not contribute to the Green's function. As a result, the wave equation Green's function for the Neumann problem has the same expansion as that for the Dirichlet problem (10.85), only now using the Neumann eigenfunctions and eigenvalues.

10.5.3 Further study

The material in this section closely followed Section 7.7 of *Stakgold* (2000b), who provides a number of examples.

10.6 Green's function solution to the wave equation

We now derive the Green's function solution to the wave function that satisfies the initial-boundary value problem (10.35a)-(10.35d). For that purpose, multiply the adjoint causal Green's function equation (10.72a) by the wave function and integrate over the space-time domain[9]

$$\psi(\boldsymbol{x}_\circ, t_\circ) = \int_{t_{\text{init}}}^{T}\int_{\mathcal{R}} \psi(\boldsymbol{x}, t)\,(\partial_{tt} - c^2\nabla^2)\,G^{\ddagger}(\boldsymbol{x}, t|\boldsymbol{x}_\circ, t_\circ)\,\mathrm{d}V_{\boldsymbol{x}}\,\mathrm{d}t. \tag{10.90}$$

Just like we did for the diffusion equation in Section 9.4.1, we here introduced an arbitrary book-keeping time, T, that is larger than any other times in the problem. As we see in the following, T is eliminated from the final solution through use of the anti-causality property of $G^{\ddagger}$.

[9]We assume the spatial domain is time independent, so that the space and time integrations can be interchanged.

10.6.1 Integration by parts manipulations

Dropping arguments for brevity, we move the wave operator in equation (10.90) from the Green's function to the wave function, thus leading to

$$\psi\,(\partial_{tt} - c^2\nabla^2)G^{\ddagger} = G^{\ddagger}\,(\partial_{tt} - c^2\,\nabla^2)\psi + \partial_t(\psi\,\partial_t G^{\ddagger} - G^{\ddagger}\,\partial_t\psi) - c^2\,\nabla\cdot(\psi\,\nabla G^{\ddagger} - G^{\ddagger}\,\nabla\psi). \tag{10.91}$$

Making use of the partial differential equation (10.35a) satisfied by the wave function, ψ, brings the first right hand side term to

$$G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,(\partial_{tt} - c^2\,\nabla^2)\psi(\boldsymbol{x},t) = G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\Lambda(\boldsymbol{x},t). \tag{10.92}$$

The space and time integral of the spatial derivative term in equation (10.91) is given by

$$\int_{t_{\mathsf{init}}}^{T}\int_{\mathcal{R}} \nabla_{\boldsymbol{x}}\cdot(\psi\,\nabla_{\boldsymbol{x}}G^{\ddagger} - G^{\ddagger}\,\nabla_{\boldsymbol{x}}\psi)\,\mathrm{d}V_{\boldsymbol{x}}\,\mathrm{d}t = \int_{t_{\mathsf{init}}}^{t_{\circ}}\int_{\partial\mathcal{R}} \sigma(\boldsymbol{x},t)\,\nabla_{\boldsymbol{x}}G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}_{\boldsymbol{x}}\,\mathrm{d}t. \tag{10.93}$$

For this equality we used the Dirichlet boundary condition (10.35d) satisfied by ψ and the homogeneous Dirichlet condition (10.72c) satisfied by the adjoint Green's function. We also used the anti-causality condition (10.72b) to set the upper time limit to $t_{\circ}$.

For the time integral of the time derivative term in equation (10.91), we have

$$\int_{t_{\mathsf{init}}}^{T} \partial_t(\psi\,\partial_t G^{\ddagger} - G^{\ddagger}\,\partial_t\psi)\,\mathrm{d}t = \Big[\psi(\boldsymbol{x},t)\,\partial_t G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ}) - G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\partial_t\psi(\boldsymbol{x},t)\Big]\Big|_{t_{\mathsf{init}}}^{T}. \tag{10.94}$$

The initial conditions (10.35b) and (10.35c), along with the anti-causality condition (10.72b), render

$$\int_{t_{\mathsf{init}}}^{T} \partial_t(\psi\,\partial_t G^{\ddagger} - G^{\ddagger}\,\partial_t\psi)\,\mathrm{d}t = -I(\boldsymbol{x})\,\partial_t G^{\ddagger}(\boldsymbol{x},t_{\mathsf{init}}|\boldsymbol{x}_{\circ},t_{\circ}) + J(\boldsymbol{x})\,G^{\ddagger}(\boldsymbol{x},t_{\mathsf{init}}|\boldsymbol{x}_{\circ},t_{\circ}) \tag{10.95a}$$

$$= I(\boldsymbol{x})\,\partial_{t_{\circ}} G^{\ddagger}(\boldsymbol{x},t_{\mathsf{init}}|\boldsymbol{x}_{\circ},t_{\circ}) + J(\boldsymbol{x})\,G^{\ddagger}(\boldsymbol{x},t_{\mathsf{init}}|\boldsymbol{x}_{\circ},t_{\circ}), \tag{10.95b}$$

where the second equality used the property (10.67) of the time derivatives acting on the Green's function.

10.6.2 Bringing terms together

Bringing terms together leads to the Green's function solution to the wave equation

$$\begin{aligned}\psi(\boldsymbol{x}_{\circ},t_{\circ}) = &\int_{t_{\mathsf{init}}}^{t_{\circ}}\int_{\mathcal{R}} \Lambda(\boldsymbol{x},t)\,G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\,\mathrm{d}V_{\boldsymbol{x}}\,\mathrm{d}t\\ &- c^2\int_{t_{\mathsf{init}}}^{t_{\circ}}\int_{\partial\mathcal{R}} \sigma(\boldsymbol{x},t)\nabla_{\boldsymbol{x}}G^{\ddagger}(\boldsymbol{x},t|\boldsymbol{x}_{\circ},t_{\circ})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}_{\boldsymbol{x}}\,\mathrm{d}t\\ &+\int_{\mathcal{R}}\Big[I(\boldsymbol{x})\,\partial_{t_{\circ}}G^{\ddagger}(\boldsymbol{x},t_{\mathsf{init}}|\boldsymbol{x}_{\circ},t_{\circ}) + J(\boldsymbol{x})\,G^{\ddagger}(\boldsymbol{x},t_{\mathsf{init}}|\boldsymbol{x}_{\circ},t_{\circ})\Big]\,\mathrm{d}V_{\boldsymbol{x}},\end{aligned} \tag{10.96}$$

with a notation swap yielding

$$\psi(\boldsymbol{x},t) = \int_{t_{\mathsf{init}}}^{t}\int_{\mathcal{R}} \Lambda(\boldsymbol{x}_{\circ},t_{\circ})\,G^{\ddagger}(\boldsymbol{x}_{\circ},t_{\circ}|\boldsymbol{x},t)\,\mathrm{d}V_{\boldsymbol{x}_{\circ}}\,\mathrm{d}t_{\circ}$$

$$
\begin{aligned}
&- c^2 \int_{t_{\mathrm{init}}}^{t} \int_{\partial\mathcal{R}} \sigma(\boldsymbol{x}_\circ, t_\circ) \nabla_{\boldsymbol{x}_\circ} G^\ddagger(\boldsymbol{x}_\circ, t_\circ | \boldsymbol{x}, t) \cdot \hat{\boldsymbol{n}}_\circ \, \mathrm{d}\mathcal{S}_{\boldsymbol{x}_\circ} \, \mathrm{d}t_\circ \\
&+ \int_{\mathcal{R}} \Big[I(\boldsymbol{x}_\circ) \, \partial_t G^\ddagger(\boldsymbol{x}_\circ, t_{\mathrm{init}} | \boldsymbol{x}, t) + J(\boldsymbol{x}_\circ) \, G^\ddagger(\boldsymbol{x}_\circ, t_{\mathrm{init}} | \boldsymbol{x}, t) \Big] \, \mathrm{d}V_{\boldsymbol{x}_\circ}.
\end{aligned} \tag{10.97}
$$

The Green's function solution (10.97) only depends on processes happening between the initial time, t_{init}, and current time, t, thus manifesting causality. Furthermore, as advertised, the arbitrary time, T, drops out due to the anti-causality condition satisfied by $G^\ddagger$, where again

$$
G^\ddagger(\boldsymbol{x}_\circ, t_\circ | \boldsymbol{x}, t) = G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_\circ). \tag{10.98}
$$

The solution (10.97) shares much with the diffusion equation in Section 9.2, but here with the additional factor of a second initial condition due to the second order time derivative appearing in the wave equation. Additionally, note that $\psi = 0$ for the case of zero initial and boundary data, which accords with the uniqueness arguments from Section 10.3.2.

10.6.3 Further reading

Our formulation of the Green's function expression for the wave function closely follows Section 7.7 of *Stakgold* (2000b).

10.7 Initial value problem in free space

As an exercise to further our exposure to the formalism, we follow Section IV-29 of *Dennery and Krzywicki* (1967) by considering the source-free initial value problem for the wave equation in free space ($\mathcal{R} = \mathbb{R}^3$)

$$
\begin{aligned}
(\partial_{tt} - c^2 \nabla^2) \psi(\boldsymbol{x}, t) &= 0 && \boldsymbol{x} \in \mathbb{R}^3, \quad t > t_{\mathrm{init}} && (10.99\mathrm{a}) \\
\psi(\boldsymbol{x}, t) &= I(\boldsymbol{x}) && \boldsymbol{x} \in \mathbb{R}^3, \quad t = t_{\mathrm{init}} && (10.99\mathrm{b}) \\
\partial_t \psi(\boldsymbol{x}, t) &= J(\boldsymbol{x}) && \boldsymbol{x} \in \mathbb{R}^3, \quad t = t_{\mathrm{init}}, && (10.99\mathrm{c})
\end{aligned}
$$

so that the wave function solution (10.97) is

$$
\psi(\boldsymbol{x}, t) = \int \Big[I(\boldsymbol{x}_\circ) \, \partial_t G^\ddagger(\boldsymbol{x}_\circ, t_{\mathrm{init}} | \boldsymbol{x}, t) + J(\boldsymbol{x}_\circ) \, G^\ddagger(\boldsymbol{x}_\circ, t_{\mathrm{init}} | \boldsymbol{x}, t) \Big] \, \mathrm{d}V_{\boldsymbol{x}_\circ}. \tag{10.100}
$$

Since we are working in free space, we make use of the causal free-space Green's function (10.50c) so that

$$
G^\ddagger(\boldsymbol{x}_\circ, t_{\mathrm{init}} | \boldsymbol{x}, t) = G(\boldsymbol{x}, t | \boldsymbol{x}_\circ, t_{\mathrm{init}}) = \frac{\delta[c\,(t - t_{\mathrm{init}}) - |\boldsymbol{x} - \boldsymbol{x}_\circ|]}{4\,\pi\,c\,|\boldsymbol{x} - \boldsymbol{x}_\circ|}, \tag{10.101}
$$

with the corresponding integral expression for the wave function

$$
\begin{aligned}
&\psi(\boldsymbol{x}, t) = \\
&\quad \frac{1}{4\,\pi\,c} \int \left[\frac{I(\boldsymbol{x}_\circ) \, \partial_t \delta(c\,(t - t_{\mathrm{init}}) - |\boldsymbol{x} - \boldsymbol{x}_\circ|) + J(\boldsymbol{x}_\circ) \, \delta(c\,(t - t_{\mathrm{init}}) - |\boldsymbol{x} - \boldsymbol{x}_\circ|)}{|\boldsymbol{x} - \boldsymbol{x}_\circ|} \right] \mathrm{d}V_{\boldsymbol{x}_\circ}.
\end{aligned} \tag{10.102}
$$

There are no boundaries so that space is homogeneous and isotropic, in which case we can take the origin of the coordinates based on convenience. We choose to define spherical coordinates according to equations (4.84a)-(4.84c)[10]

$$x_\circ - x = r\cos\phi\,\cos\lambda \tag{10.103a}$$
$$y_\circ - y = r\cos\phi\,\sin\lambda \tag{10.103b}$$
$$z_\circ - z = r\sin\phi \tag{10.103c}$$

in which case equation (10.102) becomes

$$\psi(\boldsymbol{x},t) = \frac{1}{4\,\pi\,c}\int_0^\infty r\,\mathrm{d}r\int_{-1}^{1}\mathrm{d}(\sin\phi)\int_0^{2\pi}\mathrm{d}\phi\,[I(\boldsymbol{x}_\circ)\,\partial_t\delta(c\,(t-t_{\mathsf{init}})-r)+J(\boldsymbol{x}_\circ)\,\delta(c\,(t-t_{\mathsf{init}})-r)]\,, \tag{10.104}$$

where

$$-\pi/2 \le \phi \le \pi/2 \Longrightarrow -1 \le \sin\phi \le 1. \tag{10.105}$$

Rather than consider the derivative of the Dirac delta, it is more convenient to use the identity

$$\partial_t\delta(c\,(t-t_{\mathsf{init}})-r) = -c\,\partial_r\delta(c\,(t-t_{\mathsf{init}})-r), \tag{10.106}$$

which then allows us to integrate by parts according to

$$\int_0^\infty r\,I\,\partial_t\delta(c\,(t-t_{\mathsf{init}})-r)\,\mathrm{d}r = -\int_0^\infty c\,r\,I\,\partial_r\delta(c\,(t-t_{\mathsf{init}})-r)\,\mathrm{d}r \tag{10.107a}$$
$$= \int_0^\infty c\,\partial_r(r\,I)\,\delta(c\,(t-t_{\mathsf{init}})-r)\,\mathrm{d}r, \tag{10.107b}$$

where the boundary terms vanish. Setting $t_{\mathsf{init}} = 0$ to reduce clutter, we are led to the wave function (10.104) taking the form

$$\psi(\boldsymbol{x},t) = \frac{1}{4\,\pi\,c}\int_0^\infty \delta(c\,t-r)\,\mathrm{d}r\int_{-1}^{1}\mathrm{d}w\int_0^{2\pi}\mathrm{d}\phi\,[\partial_r[c\,r\,I(\boldsymbol{x}_\circ)]+r\,J(\boldsymbol{x}_\circ)]\,. \tag{10.108}$$

The radial integral fires at $r = c\,t$, so that

$$\psi(\boldsymbol{x},t) = \frac{1}{4\,\pi}\int_{-1}^{1}\mathrm{d}w\int_0^{2\pi}\mathrm{d}\phi\,[\partial_t[t\,I(\boldsymbol{x}_\circ)]+t\,J(\boldsymbol{x}_\circ)]\,, \tag{10.109}$$

where we evaluate the initial conditions along the moving wave front at

$$x_\circ = x + c\,t\cos\phi\,\cos\lambda \tag{10.110a}$$
$$y_\circ = y + c\,t\cos\phi\,\sin\lambda \tag{10.110b}$$
$$z_\circ = z + c\,t\sin\phi. \tag{10.110c}$$

[10] Note that λ here is the longitude coordinate, not an eigenvalue.

10.8 Exercises

EXERCISE 10.1: RELATING MODIFIED INITIAL CONDITION SOLUTIONS
Consider the following initial-boundary value problem

$$(\partial_{tt} - c^2\nabla^2)\, u(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \;\; t > t_{\mathsf{init}} \tag{10.111a}$$
$$u(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \;\; t = t_{\mathsf{init}} \tag{10.111b}$$
$$\partial_t u(\boldsymbol{x}, t) = F(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathcal{R}, \;\; t = t_{\mathsf{init}} \tag{10.111c}$$
$$u(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R}, \;\; t \geq 0, \tag{10.111d}$$

as well as the slightly modified problem

$$(\partial_{tt} - c^2\nabla^2)\, v(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \;\; t > t_{\mathsf{init}} \tag{10.112a}$$
$$v(\boldsymbol{x}, t) = F(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathcal{R}, \;\; t = t_{\mathsf{init}} \tag{10.112b}$$
$$\partial_t v(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \mathcal{R}, \;\; t = t_{\mathsf{init}} \tag{10.112c}$$
$$v(\boldsymbol{x}, t) = 0 \qquad \boldsymbol{x} \in \partial\mathcal{R},, \;\; t \geq 0 \tag{10.112d}$$

where the only difference appears in the initial conditions. Compatibility between the initial condition and boundary condition requires that $F(\boldsymbol{x}) = 0$ on $\partial\mathcal{R}$. Show that

$$v = \partial_t u, \tag{10.113}$$

so that finding u is sufficient for finding v.

EXERCISE 10.2: ONE-DIMENSIONAL STANDING WAVES
Consider the unforced scalar wave equation on a one-dimensional finite domain with homogeneous Dirichlet boundary conditions

$$(\partial_{tt} - c^2\, \partial_{xx})\psi = 0 \qquad 0 < x < L \tag{10.114a}$$
$$\psi(x, t) = \psi(x, t) = 0 \qquad x = 0 \quad \text{and} \quad x = L \tag{10.114b}$$
$$\psi(x, t) = I(x) \quad t = 0 \tag{10.114c}$$
$$\partial_t \psi(x, t) = 0 \qquad t = 0. \tag{10.114d}$$

This initial-boundary value problem can represent a string with ψ the lateral displacement. The homogeneous Dirichlet conditions arise when the string has fixed ends. The initial condition corresponds to an initial deflection, $f(x)$, and with zero initial time derivative (i.e., string is released from rest).

(a) Use separation of variables to determine a solution, $\psi(x, t)$, as a Fourier series expansion.
(b) Show that the solution can be written as the sum of two waves moving in opposite direction so that their sum is a standing wave.
(c) Identify the Green's function, $G(x, t|x_\circ, t_\circ)$, and check that it corresponds to the eigenfunction expansion (10.85).

EXERCISE 10.3: DAMPED STANDING WAVES
Consider the unforced scalar wave equation on a one-dimensional finite domain with homogeneous Dirichlet boundary conditions and with damping

$$(\partial_{tt} + 2\,\gamma\,\partial_t - c^2\,\partial_{xx})\psi = 0 \qquad 0 < x < L \tag{10.115a}$$

$$\psi(x,t) = \psi(x,t) = 0 \qquad x = 0 \quad \text{and} \quad x = L \tag{10.115b}$$
$$\psi(x,t) = I(x) \quad t = 0 \tag{10.115c}$$
$$\partial_t \psi(x,t) = 0 \qquad t = 0. \tag{10.115d}$$

This initial-boundary value problem can represent a string with ψ the lateral displacement, and with $\gamma^{-1} > 0$ a time scale for damping the wave kinetic energy. When $\gamma = 0$ this problem reduces to the undamped case considered in Exercise 10.2.

(a) UNDER-DAMPED WAVES: Use separation of variables to determine a solution, $\psi(x,t)$, as a Fourier series expansion. Assume that the dissipation is smaller than any of the undamped eigenfrequencies from Exercise 10.2 so that

$$0 < \gamma < \omega_n. \tag{10.116}$$

(b) OVER-DAMPED $n = 1$ WAVE Now increase the dissipation so that

$$0 < \omega_1 < \gamma < \omega_{n>1}. \tag{10.117}$$

The $n = 1$ mode is now over-damped and becomes exponential in time. Derive its solution and discuss.

EXERCISE 10.4: STANDING WAVES PRESCRIBED AT TWO TIMES
Consider the unforced scalar wave equation,

$$(\partial_{tt} - c^2\, \partial_{xx})\psi = 0 \quad 0 < x < L, \quad 0 < t < T \tag{10.118a}$$
$$\psi(x,0) = \psi(x,T) = 0 \quad 0 < x < L \tag{10.118b}$$
$$\psi(0,t) = \psi(L,t) = 0 \quad 0 < t < T. \tag{10.118c}$$

The novel element here is that the wave function has a value prescribed at the two times, $t = 0$ and $t = T$, as well as at the spatial boundaries. Find the solution to this equation that is normalized over the spatial range.

EXERCISE 10.5: GREEN'S FUNCTION ON A LINE SEGMENT
Consider the Green's function problem for the wave equation on a line segment

$$(\partial_{tt} - c^2\, \partial_{xx})G(x,t|x_\circ,t_\circ) = \delta(x - x_\circ)\, \delta(t - t_\circ) \quad -L/2 < x < L/2 \tag{10.119a}$$
$$G(x,t|x_\circ,t_\circ) = 0 \qquad x = \pm L/2 \tag{10.119b}$$
$$G(x,t|x_\circ,t_\circ) = 0 \qquad t < t_\circ. \tag{10.119c}$$

(a) Construct the Green's function using an eigenfunction expansion.

EXERCISE 10.6: SPHERICALLY SYMMETRIC WAVES
Consider the unforced scalar wave equation,

$$(\partial_{tt} - c^2\, \nabla^2)\psi = 0, \tag{10.120}$$

in three-dimensional space without boundaries.

(a) Write the Laplacian operator in spherical coordinates.
(b) Assume the initial conditions are spherically symmetric. Write the corresponding spherically symmetric wave equation.
(c) Show that the most general expression for a spherically symmetric wave is given by an

extension of D'Alembert's formula (10.18).

Part II

Classical mechanics

Classical mechanics lives in Euclidean space and Newtonian time, with their marriage referred to as Galilean space-time. Euclidean space has no curvature (flat space), so that Cartesian coordinates are commonly sufficient (though not always the most convenient) for developing the equations of motion. With universal or Newtonian time, an event occuring at one point in space at time, t, also occurs at the same time everywhere else in space.[11] That is, classical mechanics is based on absolute simultaneity holding throughout all of Euclidean space. The marriage of Euclidean space and Newtonian time is referred to as Galilean space-time.

In this part of the book we treat the fundamentals of classical particle mechanics with a perspective and emphasis on what proves central to the study of geophysical fluid mechanics. In particular, we encounter Newtonian mechanics, trajectories, linear momentum, angular momentum, body forces, accelerations due to the rotating reference frame (planetary Coriolis and planetary centrifugal), space-time symmetries and conservation laws, center of mass coordinates, planetary Cartesian coordinates, and planetary spherical coordinates. We start by surveying foundational elements of Newtonian particle mechanics in Chapter 11. We here also introduce some salient kinematic properties of non-inertial reference frames.

In Chapters 12 and 13 we focus on the geophysically relevant case of a classical point particle moving around a rotating and gravitating planet as described from the rotating terrestrial reference frame. Planetary rotation and gravitation are two defining features of geophysical fluid mechanics. Furthermore, geophysical flows have a speed that is small relative to that of the rotating planet. We thus say that geophysical fluids are in near rigid-body motion, making it convenient to use a rotating (non-inertial) terrestrial reference frame to describe the fluid motion. Rotating and gravitating motion around a planet introduce new physical ideas beyond more familiar non-rotating physical systems. By introducing these ideas in the context of particle mechanics, we can introduce the novel, and sometimes confusing, aspects of rotating physics before also needing to explore facets of continuum mechanics needed for fluid motion.

In Chapters 14 and 15 we explore the fundamentals and practices of analytical mechanics, where analytical mechanics is comprised of Lagrangian mechanics and Hamiltonian mechanics. Both formulations of mechanics are logically equal to Newtonian mechanics, and yet they offer further conceptual insights and practical advantages beyond the Newtonian approach. For example, rather than the Newtonian focus on forces and accelerations (vectors), energy (a scalar) plays the central role in Lagrangian and Hamiltonian mechanics. Additionally, the Newtonian approach starts with Newton's second law, which is a differential equation, whereas Lagrangian and Hamiltonian mechanics are based on Hamilton's principle of stationary action, with the action the time integral of a scalar function (the Lagrangian). An allure of variational principles, such as Hamilton's principle, is that they can be extended to many areas of mechanics beyond just the reformulation of Newtonian particle mechanics. In particular, we make use of Hamilton's principle in our study of fluid mechanics in VOLUME 4 and wave mechanics in VOLUME 4. Indeed, variational principles have been extended well outside the bounds of classical mechanics, with variational principles at the foundation for nearly all areas of theoretical physics, such as electromagnetism, general relativity, and quantum field theory.

MATHEMATICS IN THIS PART

Mathematical methods in this part of the book relies on the Cartesian tensor analysis and vector calculus, along with a bit of general tensors for treating cylindrical and spherical

[11] For two observers in Galilean spacetime, their time label can differ by an arbitrary constant. We set that constant to zero for simplicity, and without losing any generality.

coordinates. We try to remain self-contained, though some exposure to the tensor analysis in Part I can be useful.

Chapter 11

NEWTONIAN MECHANICS AND GRAVITY

Classical physics is concerned with describing the motion of matter through Galilean space-time, with Newtonian classical mechanics based on using Newton's laws to describe how forces cause accelerations. A mechanical description of motion is decomposed into kinematics and dynamics. Kinematics is concerned with the nature of motion (how motion occurs) and dynamics is concerned with the causes of motion (why motion occurs). More specifically, kinematics is concerned with the position, velocity, and acceleration, whereas dynamics, as formulated via Newton's laws, provides the means to determine how motion is altered in the presence of forces.

In this chapter we study Newtonian particle mechanics for a single massive point particle as well as for a system of such particles. We also introduce salient ideas from Newtonian gravitation, including basics of tidal producing forces. We encounter concepts such as space, time, mass, force, and rotation. Within the bounds of classical physics, we must remain satisfied with common experience definitions for many concepts that are not deduced from more fundamental principles.

CHAPTER GUIDE

Our study of point particle motion serves to set the conceptual and technical stage for our later studies of fluid motion. Every classical mechanics textbook has some form of the material presented in this chapter. The books from *Goldstein* (1980) and *Fetter and Walecka* (2003) are targeted at the entering physics graduate student, whereas *French* (1971), *Symon* (1971), *Marion and Thornton* (1988), and *Taylor* (2005) are targeted at second or third year undergraduates.

11.1 A synopsis of Newton's laws of mechanics

We here offer a brief overview of Newton's laws and the nature of classical mechanics.

11.1.1 Classical matter

Classical matter is organized into three categories: (i) massive point particles; (ii) extended and continuous rigid media (rigid bodies); (iii) extended and continuous non-rigid media (e.g., elastic solids, fluids). A point particle has zero dimensions and so it occupies a single point in space. A rigid-body occupies a finite region of space and is comprised of a continuum of matter whose elements maintain a fixed relative position, thus constraining the media to maintain constant mass and constant volume. Matter comprising an extended non-rigid continuous media can exhibit relative motion, so that the media maintains constant mass though with a generally changing volume.[1] In this chapter we focus on Newtonian mechanics of point particles as a venue to introduce rudimentary features of classical mechanics, all of which are relevant to our study of geophysical fluid mechanics.

The decomposition of matter into point particles, rigid bodies, and continuous media, directly corresponds to the kinematic decomposition of motion into translations, rotations, and dilations.[2] Namely, point particles move by translations through space, with this space-time motion referred to as a trajectory. The motion of a rigid body consists of its center of mass translations plus rotations of the body relative to the center of mass. Finally, the motion of a continuous media consists of the translations and rotations of a rigid body, plus changes in the relative positions of the particles that can lead to changes in the volume of the media, with such motions referred to as dilatations.

11.1.2 Newton's three laws of mechanics

The mechanics of Newton are summarized by three laws of motion:

- Newton's first law: In an inertial reference frame, every massive body remains at rest or in uniform motion unless acted on by a net force. This law is sometimes referred to as the *law of inertia*.
- Newton's second law: In an inertial reference frame, application of a net force to a body causes the body's linear momentum to change.
- Newton's third law: To each action there is an equal and oppositely directed reaction.

The first and second laws offer definitions for an inertial reference frame and for a force. The third law provides a statement about how forces act, thus providing physical substance to the first and second laws.[3] The third law holds for central forces, such as arise in Newtonian gravity and electrostatics. However, it does not hold for all forces, such as the Lorentz force acting on a moving charged particle, nor to the surface tension acting on the interface between two fluids.

11.1.3 Dynamics affects changes to kinematics

Newton's second law for particle motion reveals how dynamic properties (forces, torques, work) cause time changes in kinematic properties (linear momentum, angular momentum, kinetic energy).

1. The rate of change of linear momentum equals to the net force, which is familiar form of the second law.
2. The rate of change of angular momentum (moment of the momentum) equals to the net torque (moment of the forces). This relation arises from taking the moment of the linear momentum balance, and so it is a corollary to the second law.
3. The rate of change of kinetic energy equals to the power (the rate of work by forces). This relation is known as the work-energy theorem, and it too is a corollary to the second law.

These three relations are detailed in this chapter for classical particles, and they are developed in VOLUME 2 for continuous matter.

[1] We typically abbreviate "extended non-rigid continuous media" as "continuous media".

[2] This decomposition is referred to as the Cauchy-Stokes decomposition theorem and we encounter it in our study of fluid kinematics VOLUME 2.

[3] For more on this perspective of Newton's laws, see Chapter 1 of *Symon* (1971) or Chapter 2 of *Marion and Thornton* (1988).

11.1.4 Concerning the practical need for the continuum approximation

Classical matter satisfies Newton's equations of motion. So from a fundamental perspective, Newton found the key to a deductive and mechanistic description of classical motion. Formulating these laws for mechanics was a huge intellectual achievement that has, arguably, not been surpassed by any subsequent development in physics. Even so, Newton's method becomes difficult in practice when there are more than a few dynamical degrees of freedom. In these cases, the equations of motion become a set of coupled second order differential equations, one for each matter component, with the numbers of dynamical degrees of freedom reaching beyond Avogadro's number for many purposes, such as the study of fluid motion. Furthermore, it is not always possible to know the forces acting between the matter constituents. Additionally, determinstic chaos makes a solution to the equations of motion less useful than one might hope, since the tiniest of perturbation to the initial conditions leads to a distinct evolution. We are led for these, and other, reasons to make use of the continuum approximation for our studies of fluid motions, thus reducing the effective number of degrees of freedom and allowing us to focus on bulk properties of the motion.[4]

11.2 Mechanics of a point particle

In this section we provide a synopsis of the Newtonian mechanics of a single point particle with constant mass, m.

11.2.1 Position, velocity, and acceleration

Let $\mathbf{X}(t)$ be the position of a point particle at a Newtonian time instance. The spatial position is measured relative to an arbitrary origin, so that $\mathbf{X}(t)$ points from the origin to the position of the particle. Universal Newtonian time, t, is a monotonically increasing parameter that allows for the unambiguous distinction between past, present, and future. Mathematically, we see that time acts to parameterize the particle's position in Euclidean space, so that $\mathbf{X}(t)$ is the mathematical representation for the trajectory in space. Galilean space-time is the marriage of Euclidean space and Newtonian universal time, with Galilean space-time used in our study of the non-relativistic motion of matter embedded in a background Euclidean space

The time derivative of the position defines the velocity, and the time derivative of the velocity is the acceleration

$$\mathbf{V}(t) = \frac{\mathrm{d}\mathbf{X}(t)}{\mathrm{d}t} = \dot{\mathbf{X}}(t) \quad \text{and} \quad \mathbf{A}(t) = \frac{\mathrm{d}\mathbf{V}(t)}{\mathrm{d}t} = \dot{\mathbf{V}}(t) = \ddot{\mathbf{X}}(t), \tag{11.1}$$

where we introduced the dot notation for the particle's time derivative. By definition, the velocity is instantaneously tangent to the trajectory, and the acceleration is instantaneously tangent to the velocity. As discussed in Section 1.2, the velocity and acceleration are tensors since they are objective physical properties of the particle trajectory (i.e., they have no dependence on coordinates or the origin of any coordinate system). Newton's law connects the acceleration to the net force acting on the particle, so that mechanics has no concern for third or higher time derivatives.

[4]Determinstic chaos survives the continuum approximation. Indeed, the pioneering study of determinstic chaos from *Lorenz* (1963) is based on an idealized model of convection in a fluid.

11.2.2 Linear momentum and Newton's second law

The linear momentum of the point particle equals to the mass of the particle, m, times its velocity

$$\mathbf{P} = m\,\dot{\mathbf{X}} = m\,\mathbf{V}. \tag{11.2}$$

The linear momentum of a particle changes when the particle experiences a net force. Writing $\mathbf{F}$ as the vector sum of all forces, then Newton's second law of motion states that there exists an inertial reference frame where motion of the particle is described by the differential equation[5]

$$\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} = \mathbf{F}. \tag{11.3}$$

If the particle mass is fixed, as assumed in this chapter, then the equation of motion becomes a second order differential equation for the particle position as a function of time

$$\mathbf{A} = \frac{\mathrm{d}\mathbf{V}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{X}}{\mathrm{d}t^2} = \mathbf{F}/m. \tag{11.4}$$

Additionally, a time integral of the equation of motion (11.3) renders

$$\mathbf{P}(t_2) - \mathbf{P}(t_1) = \int_{t_1}^{t_2} \mathbf{F}(t)\,\mathrm{d}t, \tag{11.5}$$

so that the time integral of the force gives the change in momentum between the time end points. The integral of a force over a time interval is referred to as the impulse delivered by the force.

Many conclusions in mechanics are expressed in terms of conservation laws (Chapter 13), which provide relations or conditions whereby mechanical properties of a physical system remain time invariant (i.e., time independent). Newton's second law provides our first conservation law since, in the absence of a net force, the linear momentum remains unchanged: $\mathrm{d}\mathbf{P}/\mathrm{d}t = 0$. Depending on the nature of the forces, this conservation law might hold for one, two, or all three of the vector components to the linear momentum. Note that for a constant mass particle, a time independent linear momentum means that the velocity remains constant, which is a statement of Newton's first law (the law of inertia).

11.2.3 Galilean covariance and inertial reference frames

If the forces acting on the particle are not directly dependent on the particle velocity, then the inertial frame equation of motion (11.4) is unchanged by shifting the velocity by a constant. Such velocity-independent forces (such as Newton's gravitational force) are commonly found in the conservative (non-dissipative) motion of charge-free particles. This arbitrariness in the velocity reflects a symmetry respected by Newton's equation of motion, where symmetry refers to an operation that can be performed on the physical system without changing any physics.[6] We refer to this particular symmetry as Galilean covariance.

[5] In Section 11.2.3 we show that there are an infinite number of inertial reference frames that differ by a constant velocity.

[6] We consider two versions of a symmetry transformation. An active transformation alters the physical system (e.g., move an experiment from one point in space to another), whereas a passive transformation alters the mathematical description of the physics (e.g., change coordinates). We have far more to say on these concepts in VOLUME 4when studying Hamilton's principle and Noether's theorem.

Galilean covariance holds for all inertial reference frames in non-relativistic mechanics. It is thus notable that there is no *absolute reference frame*, since we cannot mechanically distinguish any of the inertial reference frames that differ by an arbitrary constant velocity. That is, there is no classical experiment that can distinguish between two arbitrary inertial reference frames, so long as the experiments are described by Newton's equation of motion (11.4) and the force is independent of the velocity. The operational reason we cannot make a distinction is that the equation of motion is indistinguishable in the two inertial frames. As a corollary, two inertial reference frames can at most be moving relative to one another by a constant velocity. Otherwise, at least one of the reference frames is accelerating, which in turn means that this accelerating reference frame is not an inertial frame.

We mathematically represent the Galilean transformation as the following coordinate transformation[7]

$$\bar{t} = t \qquad \text{and} \qquad \overline{\boldsymbol{X}} = \boldsymbol{X} + \boldsymbol{U} t, \tag{11.6}$$

where the barred position is measured in the moving reference frame. We choose to use the same Newtonian universal time in both reference frames, so that it remains unchanged. In contrast, the coordinate position for the particle in the new frame equals to that in the original reference frame plus a contribution from the constant velocity, $\boldsymbol{U}$. We may sometimes refer to the barred reference frame as the result of a Galilean boost. The particle velocity in the moving (boosted) reference frame is given by

$$\overline{\boldsymbol{V}} = \frac{\mathrm{d}\overline{\boldsymbol{X}}}{\mathrm{d}\bar{t}} = \frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} + \frac{\mathrm{d}\,(\boldsymbol{U} t)}{\mathrm{d}t} = \boldsymbol{V} + \boldsymbol{U}, \tag{11.7}$$

where we set $\mathrm{d}\boldsymbol{U}/\mathrm{d}t = 0$ since $\boldsymbol{U}$ has a fixed magnitude and direction (as per our assumption that it is a constant vector). As expected, the velocity is shifted by the constant reference frame velocity, $\boldsymbol{U}$, whereas the acceleration in the two reference frames is identical

$$\overline{\boldsymbol{A}} = \frac{\mathrm{d}^2\overline{\boldsymbol{X}}}{\mathrm{d}\bar{t}^2} = \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}t} = \boldsymbol{A}. \tag{11.8}$$

11.2.4 Kinetic energy and mechanical work

Return to the equation of motion in the form (11.4) for a constant mass particle, and take the scalar product with the velocity so that

$$\frac{m}{2}\,\frac{\mathrm{d}(\mathbf{V}\cdot\mathbf{V})}{\mathrm{d}t} = \mathbf{F}\cdot\mathbf{V}. \tag{11.9}$$

Defining the kinetic energy as

$$K = m\,\mathbf{V}\cdot\mathbf{V}/2, \tag{11.10}$$

and integrating equation (11.9) over a time interval renders

$$K(t_2) - K(t_1) = \int_{t_1}^{t_2} \mathbf{F}\cdot\mathbf{V}\,\mathrm{d}t = \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{F}\cdot\mathrm{d}\mathbf{x}. \tag{11.11}$$

[7] Recall from Section 1.1.2 that we use a slanted boldface for coordinate representations of the position and tensors, hence the slanted font in the equations of this subsection.

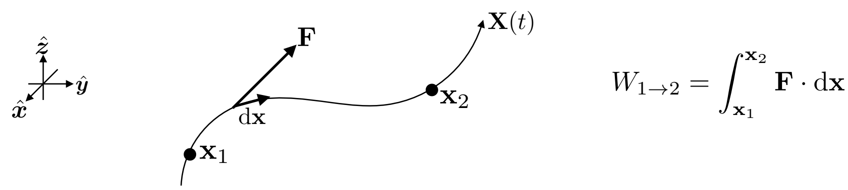

FIGURE 11.1: Illustrating equation (11.14) for the mechanical work done on a particle as it moves through space and is acted on by a force.

In the second equality we introduced the spatial positions for the trajectory endpoints at times t_1 and t_2,

$$\mathbf{x}_1 = \mathbf{X}(t = t_1) \quad \text{and} \quad \mathbf{x}_2 = \mathbf{X}(t = t_2), \tag{11.12}$$

and noted that $\mathrm{d}\mathbf{x}$ is the differential vector increment along the particle's trajectory, so that

$$\mathrm{d}\mathbf{x} = \mathbf{V}\,\mathrm{d}t. \tag{11.13}$$

In equation (11.11) we identify the integrals as the mechanical work done by the force on the particle during the time interval (see Figure 11.1)

$$W_{1\to 2} = \int_{t_1}^{t_2} \mathbf{F} \cdot \mathbf{V}\,\mathrm{d}t = \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{F} \cdot \mathrm{d}\mathbf{x}. \tag{11.14}$$

We also define the integrand of the first expression as the power supplied to the particle by the forcer

$$\text{power} = \mathbf{F} \cdot \mathbf{V}, \tag{11.15}$$

so that the mechanical work is the time integral of the power. Evidently, equation (11.11) says that the change in kinetic energy between two times is given by the work done to the particle over the time interval

$$W_{1\to 2} = K(t_2) - K(t_1). \tag{11.16}$$

This result is called the work-energy theorem.

11.2.5 Conservative forces and mechanical energy conservation

There are many forces encountered in physics. Some forces are an explicit function of the spatial position of the particle, the velocity of the particle, and time. However, there is an important type of force, known as a conservative force, that is an explicit function solely of the spatial position

$$\mathbf{F} = \mathbf{F}(\mathbf{x}). \tag{11.17}$$

The gravitational force acting on a massive particle in a static gravitational field provides a canonical example of a conservative force. For such forces we are afforded the ability to introduce a scalar function, P, that we refer to as the potential energy (for reasons to be clarified soon)

$$P(\mathbf{x}) - P(\mathbf{x}_1) = -\int_{\mathbf{x}_1}^{\mathbf{x}} \mathbf{F} \cdot \mathrm{d}\mathbf{x}. \tag{11.18}$$

Consequently, the conservative force vector is given by minus the gradient of the potential energy

$$\mathbf{F} = -\nabla P. \tag{11.19}$$

There are important consequences of being able to write the force as the gradient of the potential energy. First, we see that the work done by a conservative force acting on a particle moving from $\mathbf{x}_1$ to $\mathbf{x}_2$ is independent of the path taken between these two points, which follows since the work is simply given by the difference in the potential energy

$$W_{1\to 2} = \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{F} \cdot \mathrm{d}\mathbf{x} = -\int_{\mathbf{x}_1}^{\mathbf{x}_2} \nabla P \cdot \mathrm{d}\mathbf{x} = P(\mathbf{x}_1) - P(\mathbf{x}_2). \tag{11.20}$$

Evidently, $W_{1\to 2} > 0$ if the potential energy decreases when moving from $\mathbf{x}_1$ to $\mathbf{x}_2$. Furthermore, it follows that $W_{1\to 2} = -W_{2\to 1}$, and that the work done by a conservative force vanishes for a particle traversing a closed spatial trajectory

$$W = \oint_{\mathcal{C}} \mathbf{F} \cdot \mathrm{d}\mathbf{x} = 0, \tag{11.21}$$

where $\mathcal{C}$ is an arbitrary closed trajectory. Finally, inserting the potential energy, P, into the work equation (11.16) leads to an expression of mechanical energy conservation

$$W_{1\to 2} = \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{F} \cdot \mathrm{d}\mathbf{x} = \Delta K = -\Delta P \Longrightarrow K(t_2) + P(t_2) = K(t_1) + P(t_1). \tag{11.22}$$

That is, the sum of the kinetic energy plus potential energy remains constant for a particle moving in a conservative force field, with this invariance referred to as the conservation of mechanical energy.

11.2.6 Effective gravity as a conservative force

The Earth's gravitational field as well as the planetary centrifugal acceleration (due to planetary rotation) both give rise to conservative forces. We discuss these ideas in Section 12.8.3, where we see that the combined gravitational and planetary centrifugal accelerations are encapsulated by the gradient of the geopotential, Φ. In this case, the effective gravity force acting on a point particle of mass, m, is given by

$$\mathbf{F}_{\text{geo}} = -m \, \nabla \Phi. \tag{11.23}$$

Consequently, the work done on the particle by the effective gravitational field is

$$W_{1\to 2} = \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{F}_{\text{geo}} \cdot \mathrm{d}\mathbf{x} = -m \int_{\mathbf{x}_1}^{\mathbf{x}_2} \nabla \Phi \cdot \mathrm{d}\mathbf{x} = -m \, [\Phi(\mathbf{x}_2) - \Phi(\mathbf{x}_1)]. \tag{11.24}$$

The effective gravitational force field does positive work on the particle if $\Phi(\mathbf{x}_1) > \Phi(\mathbf{x}_2)$. That is, work is applied to the particle as it moves (i.e., falls) from a high geopotential at $\mathbf{x}_1$ to a lower geopotential at $\mathbf{x}_2$, in which case the work-energy theorem means that the kinetic energy increases. Conversely, if $\Phi(\mathbf{x}_1) < \Phi(\mathbf{x}_2)$, then gravity does negative work on the particle. In this case the potential energy of the particle increases as it moves (i.e., rises) to a higher geopotential, while, through the work-energy theorem, the kinetic energy of the particle decreases.

11.2.7 Rayleigh drag as an example non-conservative force

Friction is the canonical non-conservative force that depends on the velocity. For example, Rayleigh drag takes on the following form

$$\mathbf{F}^{\text{Rayleigh}} = -\gamma\, m\, \mathbf{V}, \tag{11.25}$$

where $\gamma > 0$ is a parameter with dimensions of inverse time. Newton's equation of motion with Rayleigh drag (and no other forces) is given by

$$\frac{\mathrm{d}\mathbf{V}}{\mathrm{d}t} = -\gamma\, \mathbf{V}. \tag{11.26}$$

Notably, Rayleigh drag is not Galilean invariant since it is dependent on the velocity. We can understand this lack of Galilean invariance by noting that the friction force identifies the state of rest ($\mathbf{V} = 0$) as a special reference frame.

If the Rayleigh drag parameter, γ, is a constant, then the solution to the first order ordinary differential equation (11.26) is the exponential decay

$$\mathbf{V}(t) = \mathbf{V}(0)\, e^{-\gamma\, t}, \tag{11.27}$$

with $\mathbf{V}(0)$ the velocity at time $t = 0$. We thus see that Rayleigh drag exponentially drives the velocity towards zero. Correspondingly, Rayleigh drag dissipates the kinetic energy according to twice the exponential decay

$$\frac{\mathrm{d}K}{\mathrm{d}t} = m\, \mathbf{V} \cdot \frac{\mathrm{d}\mathbf{V}}{\mathrm{d}t} = -2\, \gamma\, K \Longrightarrow K(t) = K(0)\, e^{-2\, \gamma\, t}, \tag{11.28}$$

where $K(0)$ is the kinetic energy at the initial time. The power from friction is the non-positive number,

$$\text{power} = \mathbf{F} \cdot \mathbf{V} = -m\, \gamma\, \mathbf{V} \cdot \mathbf{V} = -2\, \gamma\, K, \tag{11.29}$$

so that the work done on the particle is non-positive

$$W_{1 \to 2} = -2 \int_{t_1}^{t_2} \gamma\, K\, \mathrm{d}t \le 0. \tag{11.30}$$

11.3 Kinematics of rigid-body rotational motion

The motion of a point particle is described by its trajectory, $\mathbf{X}(t)$, which provides the particle location in Euclidean space as time progresses. At each time instance the trajectory's velocity, $\mathrm{d}\mathbf{X}/\mathrm{d}t$, can be decomposed into radial motion towards or away from an arbitrary fixed origin, plus rotation about an instantaneously defined axis through the origin. In this section we focus on the rotational motion, so that we assume the particle moves on a trajectory that maintains a fixed distance from the origin, $\mathrm{d}|\mathbf{X}|/\mathrm{d}t = 0$. Since $|\mathbf{X}|$ is fixed, our study here makes use of some rudiments of rigid-body mechanics, which is concerned with the motion of finite sized bodies where each material component remains in a position fixed relative to one another (and thus relative to the center of mass for the rigid body). We thus refer to this particle motion as rigid-body rotational motion.[8] Besides offering insights into the nature of motion, rigid-body

[8] Some prefer to the term solid-body motion rather than rigid-body motion. We use rigid-body since that accurately exposes the assumption that all particle positions remain rigidly fixed relative to one another. In

rotations have direct application to the rigid-body rotating reference frame used by observers on a rotating planet. We pick up on that geophysical application in Section 11.5.

11.3.1 Why we prefer radians versus degrees

In many applications of rotational kinematics, one finds the use of either degrees and radians, with no obvious reasons for prefering one or the other. However, radians has an objective definition whereas degrees is subject to the arbitrary choice for partitioning the angles around a circle. Namely, the difference in angle, $\Delta\vartheta$, for displacements along the perimeter of a circle (less than one full rotation), is given by the non-dimensional ratio

$$\Delta\vartheta = s/R, \tag{11.31}$$

where s is the arc-distance along the circle's perimeter, and R is the circle's radius. This ratio defines the angle, $\Delta\vartheta$, in radians, with one revolution around the circle equaling 2π radians. Now 2π radians also equals to $360°$, but that equivalence arises from the subjective choice to partition the circle into 360 units, with other choices available. In contrast, the ratio, s/R, is objective since it is independent of the units chosen for measuring the arc-distance and radius.

11.3.2 Instantaneous angular velocity vector

A particle that moves on a trajectory, $\mathbf{X}(t)$, that is a fixed distance from an origin can only exhibit rotational motion around an axis that extends through the origin. In general, the rotational axis has an evolving orientation in space. Yet at any time instance, the particle motion is specified by the axis around which rotation occurs, as well as the angular rate of rotation. Bringing the rotation rate and the axis orientation together leads to the notion of the angular velocity vector (dimensions of radians per time)

$$\boldsymbol{\Omega} = |\boldsymbol{\Omega}|\,\hat{\boldsymbol{\Omega}}, \tag{11.32}$$

where $\hat{\boldsymbol{\Omega}}$ is the unit vector specifying the direction of the rotational axis, and $|\boldsymbol{\Omega}|$ is the angular speed of the rotation. We choose the direction of the rotational axis according to the right hand rule. Although an arbitrary choice (i.e., we could just as well work with the left hand rule), the right hand rule establishes a universal convention for the kinematics of rotational motion. Figure 11.2 provides an example of rotational motion of a particle around an axis passing through a fixed origin.

11.3.3 Velocity of rigid-body rotation

Since the particle moves along a trajectory that has a fixed distance from the origin, its velocity is given by

$$\frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} = |\mathbf{X}|\,\frac{\mathrm{d}\hat{\mathbf{X}}}{\mathrm{d}t}, \tag{11.33}$$

where $\mathrm{d}\hat{\mathbf{X}}/\mathrm{d}t$ is the time derivative of the unit vector that points from the origin to the particle. Following from Figure 11.2, we see that the unit vector evolves under such rigid-body rotation

contrast, the particles comprising a solid remain relatively rigid only over certain time scales, with the Deborah number providing a non-dimensional measure. For example, rocks are solids on human time scales but they deform and flow on the far longer geological time scales.

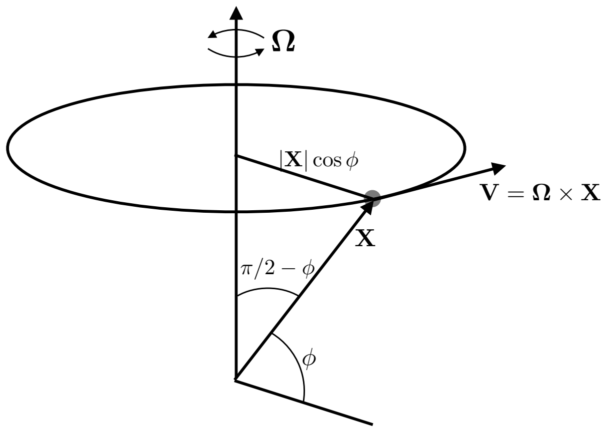

FIGURE 11.2: Kinematics of the rotational motion of a particle moving a fixed distance from an origin, whereby the motion at each instance can be represented as a rotation around an instantaneously determined axis that passes through the origin. Rotational motion is characterized by the angular velocity, $\boldsymbol{\Omega}$, that is oriented according to the right hand rule, so that the rotation depicted here is counter-clockwise relative to $\hat{\boldsymbol{\Omega}}$. The velocity corresponding to this rigid-body rotation is $\mathbf{V} = \boldsymbol{\Omega} \times \mathbf{X}$, which is perpendicular to both the particle position, $\mathbf{X}$, and the angular velocity, $\boldsymbol{\Omega}$. The angle, ϕ, is the angle relative to plane perpendicular to the rotation axis that passes through the origin. This choice for angle is motivated from the latitude used in geophysics. In contrast, it is conventional in the mechanics literature to use the co-latitude, $\pi/2 - \phi$, which is the angle relative to the rotational axis. This figure anticipates our study of motion viewed from the rotating terrestrial reference frame in Chapter 12.

according to

$$\frac{\mathrm{d}\hat{\mathbf{X}}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \hat{\mathbf{X}}, \tag{11.34}$$

with this time derivative perpendicular to both $\boldsymbol{\Omega}$ and $\hat{\mathbf{X}}$. Multiplying by the constant, $|\mathbf{X}|$, renders the rigid-body velocity vector

$$\frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} = \mathbf{V} = \boldsymbol{\Omega} \times \mathbf{X}, \tag{11.35}$$

with this velocity perpendicular to both the position and the rotational axis (see Figure 11.2)

$$\mathbf{V} \cdot \boldsymbol{\Omega} = 0 \quad \text{and} \quad \mathbf{V} \cdot \mathbf{X} = 0. \tag{11.36}$$

We are also led to the corresponding linear momentum

$$\mathbf{P} = m\,\mathbf{V} = m\,\boldsymbol{\Omega} \times \mathbf{X}. \tag{11.37}$$

Evidently, motion that is further away from the rotational axis has a larger velocity magnitude (speed) for a given angular velocity, and thus a larger linear momentum. This property accords with our experience on a merry-go-round, in which there is a higher speed at the outer rim of the merry-go-round relative to points near the center.

11.3.4 Rigid-body rotation of an arbitrary vector

The results from Section 11.3.3, in particular the key result in equation (11.35), hold for any vector undergoing rigid-body rotation with a fixed origin. This result appears many times in the study of rotational physics, thus motivating us to here discuss it a bit further from slightly complementary perspectives. Indeed, we offer yet another derivation in Section 11.5.2 when studying rotating reference frames. Although we focus here on the position, $\mathbf{X}(t)$, the derivations in this subsection hold for an arbitrary vector of fixed magnitude that rotates about a fixed origin.

Proof that $\mathbf{X} \cdot \mathrm{d}\mathbf{X}/\mathrm{d}t = 0$ for rigid-body rotations

The rigid-body rotation of the position does not change the distance from the origin, so that

$$|\mathbf{X}(t)| = |\mathbf{X}(t + \delta t)|, \tag{11.38}$$

where δt is a small time increment during which the position experiences a rigid-body rotation. The condition (11.38) can be written in the equivalent form

$$\frac{\mathrm{d}(\mathbf{X} \cdot \mathbf{X})}{\mathrm{d}t} = 0, \tag{11.39}$$

which then leads to the constraint

$$\mathbf{X} \cdot \frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} = 0. \tag{11.40}$$

Evidently, the time derivative of the position undergoing a rigid-body rotation about a fixed origin is itself perpendicular to the position.

Geometric derivation of $\mathrm{d}\mathbf{X}/\mathrm{d}t = \boldsymbol{\Omega} \times \mathbf{X}$ for rigid-body rotations

In Figure 11.3 we consider the position at two time instances, $\mathbf{X}(t + \delta t)$ and $\mathbf{X}(t)$, with the change in the position generated by an infinitesimal rotation around the axis defined by the angular velocity vector, $\boldsymbol{\Omega}$. Figure 11.3 reveals that the infinitesimal difference, $\mathbf{X}(t+\delta t) - \mathbf{X}(t)$, equals to the cross product of the angular velocity with the position

$$\mathbf{X}(t + \delta t) - \mathbf{X}(t) = \delta t \, \boldsymbol{\Omega} \times \mathbf{X}(t). \tag{11.41}$$

Dividing by δt leads to

$$\frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \mathbf{X}. \tag{11.42}$$

This time derivative satisfies the constraint (11.40) since $\mathbf{X} \cdot (\boldsymbol{\Omega} \times \mathbf{X}) = 0$, meaning that the magnitude of the position indeed remains fixed. It is via equation (11.42) that we say $\boldsymbol{\Omega}$ generates rotations of $\mathbf{X}$.

Analytical derivation of $\mathrm{d}\mathbf{X}/\mathrm{d}t = \boldsymbol{\Omega} \times \mathbf{X}$ for rigid-body rotations

Let the direction of the angular velocity, $\boldsymbol{\Omega}$, be vertical, and let $\mathbf{X}$ be confined to the horizontal plane. In a time increment, δt, we find that $\mathbf{X}$ is rotated by an angle

$$\delta\vartheta = |\boldsymbol{\Omega}| \, \delta t. \tag{11.43}$$

In the limit of small $\delta\vartheta$, the difference vector, $\mathbf{X}(t + \delta t) - \mathbf{X}(t)$, is perpendicular to $\mathbf{X}(t)$ and is of magnitude equal to the arc length

$$\delta s = |\mathbf{X}(t)| \, \delta\vartheta = |\mathbf{X}(t)| \, |\boldsymbol{\Omega}| \, \delta t. \tag{11.44}$$

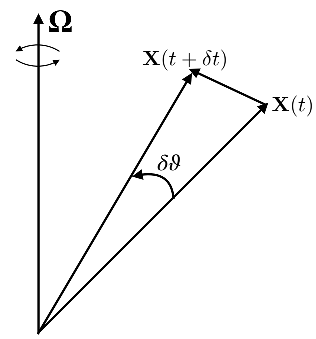

FIGURE 11.3: The change in the position under a pure rotation leaves its origin fixed and the magnitude unchanged, $|\mathbf{X}(t)| = |\mathbf{X}(t+\delta t)|$. Only the direction changes by the angle $\delta\vartheta = \Omega\,\delta t$. Infinitesimal changes generated by the angular velocity, $\mathbf{\Omega}$, lead to the difference, $\mathbf{X}(t+\delta t) - \mathbf{X}(t) = \delta t\,\mathbf{\Omega} \times \mathbf{X}(t)$. We say that $\mathbf{\Omega}$ is the generator of rotation around its axis.

Observe that $\mathbf{\Omega} \times \mathbf{X}(t)$ points in the same direction as the vector, $\mathbf{X}(t+\delta t) - \mathbf{X}(t)$, and is of length $|\mathbf{X}(t)|\,|\mathbf{\Omega}|$. We conclude that

$$\mathbf{X}(t+\delta t) - \mathbf{X}(t) = \mathbf{\Omega} \times \mathbf{X}(t)\,\delta t. \tag{11.45}$$

Dividing through by δt and taking the limit $\delta t \to 0$ gives

$$\frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} = \mathbf{\Omega} \times \mathbf{X}. \tag{11.46}$$

The proof for the case in which $\mathbf{\Omega}$ has a component along $\mathbf{X}$ is a straightforward generalization. The rotation of the trajectory is still confined to a plane, but only the component of $\mathbf{\Omega}$ normal to $\mathbf{X}$ generates rotation.

11.3.5 Angular momentum of rigid-body rotations

The angular momentum is the moment of the linear momentum relative to a chosen origin

$$\mathbf{L} = \mathbf{X} \times \mathbf{P}. \tag{11.47}$$

Newton's second law (11.3) leads to the time derivative

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \mathbf{X} \times \mathbf{F}, \tag{11.48}$$

where $\dot{\mathbf{X}} \times \mathbf{P} = \dot{\mathbf{X}} \times m\,\dot{\mathbf{X}} = 0$. The cross product, $\mathbf{X} \times \mathbf{F}$, is the torque acting on the system relative to the chosen origin, with torques having dimensions of a force times a length. Equation (11.48) leads to our second conservation law: a particle has a constant angular momentum when experiencing zero torques, with this statement dependent on the choice of origin for the angular momentum and the corresponding torques.

Inserting the rigid-body velocity (11.35) into the angular momentum (11.47) renders

$$\mathbf{L} = \mathbf{X} \times \mathbf{P} = m\,\mathbf{X} \times (\mathbf{\Omega} \times \mathbf{X}) = m\,[\mathbf{\Omega}\,|\mathbf{X}|^2 - \mathbf{X}\,(\mathbf{\Omega} \cdot \mathbf{X})] \equiv \mathbf{M} \cdot \mathbf{\Omega}. \tag{11.49a}$$

In the final equality we introduced the moment of inertia tensor (a symmetric second order tensor) for a point particle, represented using Cartesian coordinates, is given by

$$M^{pq} = m\,(\boldsymbol{X}\cdot\boldsymbol{X}\,\delta^{pq} - X^p\,X^q). \tag{11.50}$$

The moment of inertia tensor measures the inertia appropriate for determining angular momentum relative to a rotational axis.

The utility and relevance of angular momentum stems from its conservation for systems exhibiting rotational symmetry about special points or special directions. For example, motion on a smooth sphere exhibits rotational symmetry with respect to the center of the sphere. Consequently, all components of angular momentum for a particle are constant in the absence of externally applied torques. For motion around a smooth rotating sphere, the rotational axis breaks the three dimensional isotropy so that we only have a single angular momentum conservation; namely, the axial angular momentum. We study the connection between symmetry and conservation laws in Chapter 13, with particular focus on axial angular momentum in Section 13.5.

11.3.6 Rotations, Lie groups, and Lie algebras

We close this section by mentioning the mathematics associated with rotations. Namely, the finite rotation of a rigid-body around the $\hat{\boldsymbol{x}}$-axis and then the $\hat{\boldsymbol{y}}$-axis is not the same as the rotations taken in the opposite sense.[9] Non-commutivity of finite rotations is a property of three-dimensional space, and has far reaching implications for kinematics as formalized through the special orthogonal Lie group $SO(3)$. To illustrate this non-commutivity consider the rotational matrices for rotations by an angle, ϑ, around the $\hat{\boldsymbol{x}}$-axis and the $\hat{\boldsymbol{y}}$-axis

$$R^{\hat{\boldsymbol{x}}}(\vartheta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\vartheta & -\sin\vartheta \\ 0 & \sin\vartheta & \cos\vartheta \end{bmatrix} \quad \text{and} \quad R^{\hat{\boldsymbol{y}}}(\vartheta) = \begin{bmatrix} \cos\vartheta & 0 & \sin\vartheta \\ 0 & 1 & 0 \\ -\sin\vartheta & 0 & \cos\vartheta \end{bmatrix}, \tag{11.51}$$

in which case we have the commutation relation (with $s = \sin\vartheta$ and $c = \cos\vartheta$)

$$R^{\hat{\boldsymbol{x}}}(\vartheta)\,R^{\hat{\boldsymbol{y}}}(\vartheta) - R^{\hat{\boldsymbol{y}}}(\vartheta)\,R^{\hat{\boldsymbol{x}}}(\vartheta) = \begin{bmatrix} c & 0 & s \\ s^2 & c & -s\,c \\ -s\,c & s & c^2 \end{bmatrix} - \begin{bmatrix} c & s^2 & s\,c \\ 0 & c & -s \\ -s & s\,c & c^2 \end{bmatrix} \tag{11.52a}$$

$$= \begin{bmatrix} 0 & -\sin^2\vartheta & \sin\vartheta(1-\cos\vartheta) \\ \sin^2\vartheta & 0 & \sin\vartheta(1-\cos\vartheta) \\ \sin\vartheta(1-\cos\vartheta) & \sin\vartheta(1-\cos\vartheta) & 0 \end{bmatrix}. \tag{11.52b}$$

The nonzero commutation means that rotations by a finite angle do not commute; order matters. However, when assuming an infinitesimal rotation we have a zero commutation up to second order in the rotation angle

$$R^{\hat{\boldsymbol{x}}}(\vartheta)\,R^{\hat{\boldsymbol{y}}}(\vartheta) - R^{\hat{\boldsymbol{y}}}(\vartheta)\,R^{\hat{\boldsymbol{x}}}(\vartheta) = \vartheta^2 \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \mathcal{O}(\vartheta^3). \tag{11.53}$$

[9] This notion is nicely described in Section 1.2.2 of *Seitaridou and Farris* (2023), in particular see their Figure 1.7.

Evidently, rotations commute in the infinitesimal limit, in which case order does not matter so we are afforded the ability to define the instantaneous rotational velocity as a vector.[10] Equivalently, we note that the angular velocity arises from the time derivative of the particle trajectory, and as such it lives in the tangent space of the trajectory at a particular point, with the tangent space forming a linear vector space.

11.3.7 Further study

A thorough discussion of rotational kinematics can be found in Section 4-6 of *Goldstein* (1980). *Jeevanjee* (2016) provides an accessible presentation of Lie groups and Lie algebras for physicists.

11.4 An accelerated non-rotating reference frame

An inertial reference frame has a special status in Newtonian mechanics since it is here that Newton's second law holds as per equations (11.3) or (11.4). Inertial reference frames are not accelerating, and as such they are related to one another by a Galilean transformation as considered in Section 11.2.3. Inertial reference frames are an idealization that are only approximately realized in practice. For motion taking place within terrestrial laboratories, the Earth or *laboratory reference frame* provides a good approximation to an inertial reference frame for the study of many forms of motion. However, when the size of the physical system increases, and/or the length scale of the motion increases, then we must account for the Earth's rotation. This is the situation holding for much of the geophysical fluid motions of concern in this book.

In this section we initiate our study of non-inertial reference frames by working in a frame that is accelerating along a straight line without turning (i.e., no rotation), such as when observing motion from a train accelerating on a straight track. We follow this study in Section 11.5 by examining motion from a rotating reference frame that has a fixed origin, thus allowing us to make use of the rigid-body rotating mechanics from Section 11.3. Although the rigid rotating frame in Section 11.5 is more directly connected to the description of geophysical motion from a rotating terrestrial reference frame, it is useful to warm up to the study of non-inertial reference frames by first studying the non-rotating case.

11.4.1 Reference frame induced accelerations and forces

Let $\mathcal{R}$ be an inertial reference frame so that Newton's law of motion takes on the familiar form

$$m\,\dot{\mathbf{V}} = \mathbf{F}, \tag{11.54}$$

where $\mathbf{F}$ is the net force vector, with such forces arising from force fields (e.g., gravity or electromagnetic) or from contact forces that arise from interactions between bodies (e.g., friction). Now consider another reference frame, $\mathcal{R}^*$, that moves with velocity, $\mathbf{V}^{\mathsf{frame}}$, relative to $\mathcal{R}$, and let $\mathbf{V}^*$ be the velocity of a particle measured relative to the $\mathcal{R}^*$. Accordingly, the velocity of a particle is decomposed into two terms

$$\mathbf{V} = \mathbf{V}^{\mathsf{frame}} + \mathbf{V}^*. \tag{11.55}$$

[10] Mathematically, we say that the Lie group has infinitesimal generators that form a Lie algebra, with the Lie algebra living at the tangent space to the identity transformation.

Making use of this relation in Newton's law (11.54), and rearranging, leads to

$$m\,\dot{\mathbf{V}}^{*} = m\,(\dot{\mathbf{V}} - \dot{\mathbf{V}}^{\mathsf{frame}}) = \mathbf{F} - m\,\dot{\mathbf{V}}^{\mathsf{frame}}. \tag{11.56}$$

If $\dot{\mathbf{V}}^{\mathsf{frame}}$ vanishes, then Newton's law (11.56) remains form invariant and we refer to $\mathcal{R}^{*}$ as another inertial reference frame that is realized via a Galilean transformation from the original. For the more general case with a non-vanishing $\dot{\mathbf{V}}^{\mathsf{frame}}$, we encounter an extra force in equation (11.56) when working in a non-inertial reference frame

$$\mathbf{F}^{\mathsf{frame}} = -m\,\mathbf{A}^{\mathsf{frame}} \equiv -m\,\dot{\mathbf{V}}^{\mathsf{frame}} \tag{11.57}$$

That is, by choosing to write equation (11.56) in the form of Newton's law, but from the perspective of the non-inertial reference frame, we must include a force that arises solely due to acceleration of the reference frame

$$m\,\dot{\mathbf{V}}^{*} = \mathbf{F} + \mathbf{F}^{\mathsf{frame}}. \tag{11.58}$$

To appreciate the minus sign in equation (11.57), consider the backward force felt on the driver when accelerating forward in a car.

Equation (11.57) is somewhat overloaded. Namely, when the non-inertial reference frame is rotating, then the coordinate basis vectors are time dependent, and that time dependence must be considered when transforming vectors. As a result, we only make direct use of equation (11.57) when studying linear accelerating frames in Section 11.4.3, since in this case the basis vectors in the two reference frames are the same. The case of rotating reference frames in Section 11.5 requires a bit more work.

11.4.2 A comment on terminology

The reference frame induced force, $\mathbf{F}^{\mathsf{frame}}$, from equation (11.57) is sometimes called a "fictitious force" or a "pseudo-force". We eschew that terminology since reference frame induced forces can play a dominant role in describing motion viewed from the non-inertial reference frame. So from the non-inertial reference frame perspective, there is nothing fictional about $\mathbf{F}^{\mathsf{frame}}$.

Chapter 12 of *French* (1971) refers to $\mathbf{F}^{\mathsf{frame}}$ as an *inertial force* along with the corresponding *inertial acceleration*. As noted on pages 493-494 of *French* (1971), such forces that are experienced by an object due to the acceleration of a reference frame are "forces of inertia" that allow one to return Newton's law back to its familiar form found in inertial reference frames. We find this terminology prone to confusion since these so-called inertial forces arise due to non-inertial motion of the reference frame, and as such are more clearly be referred to as "non-inertial terms" or "non-inertial forces" (see Section 9.3 of *Marion and Thornton* (1988)). We aim to avoid confusion by being somewhat pedantic when discussing forces induced by motion of a non-inertial reference frame.

11.4.3 Simple pendulum in an accelerating train

Consider a pendulum with constant mass, m, on a massless string of length L. Hang the pendulum from the top of a train car in a constant gravity field and assume the train has an acceleration, $\dot{\mathbf{V}}^{\mathsf{frame}}$, along a straight track. When viewed from the inertial reference frame, the

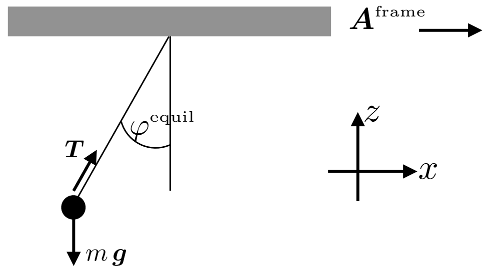

FIGURE 11.4: A pendulum oscillating from the ceiling of a train car, with the train moving with velocity $\mathbf{V}^{\text{frame}} = \hat{\boldsymbol{x}}\,|\mathbf{V}^{\text{frame}}|$. Forces acting on the pendulum arise from the string tension, $\mathbf{T}$, and the weight, $\mathbf{g}\,m = -\hat{\boldsymbol{z}}\,g\,m$. In the accelerating (non-inertial) reference frame, the additional force appears, $-\hat{\boldsymbol{x}}\,m\,|\dot{\mathbf{V}}^{\text{frame}}|$. When viewed in the accelerating reference frame, the pendulum reaches its equilibrium ($\dot{\mathbf{V}}^{*} = 0$) at an angle satisfying $\tan\varphi^{\text{equil}} = |\dot{\mathbf{V}}^{\text{frame}}|/g$.

pendulum satisfies Newton's law in the familiar form

$$m\,\dot{\mathbf{V}} = \mathbf{F}. \tag{11.59}$$

The force, $\mathbf{F}$, arises from the pendulum's weight, $m\,\mathbf{g}$, as well as the string tension, $\mathbf{T}$, so that

$$m\,\dot{\mathbf{V}} = \mathbf{T} + m\,\mathbf{g}. \tag{11.60}$$

When viewed in the accelerating reference frame the equation of motion takes the form

$$m\,\dot{\mathbf{V}}^{*} = \mathbf{T} + m\,\mathbf{g} - m\,\dot{\mathbf{V}}^{\text{frame}} = \mathbf{T} + m\,(\mathbf{g} - \dot{\mathbf{V}}^{\text{frame}}). \tag{11.61}$$

The principle of equivalence says that the same mass, m, is used to compute Newton's gravitational force as well as the inertial force due to the accelerating reference frame. As such, we can define an effective gravity by writing

$$m\,\dot{\mathbf{V}}^{*} = \mathbf{T} + m\,\mathbf{g}^{\text{eff}} \quad \text{with} \quad \mathbf{g}^{\text{eff}} = \mathbf{g} - \dot{\mathbf{V}}^{\text{frame}}. \tag{11.62}$$

The effective gravity is not vertical when the reference frame accelerates. For example, if the reference frame accelerates in the $+\hat{\boldsymbol{x}}$ direction, then

$$\mathbf{g}^{\text{eff}} = -g\,\hat{\boldsymbol{z}} - \dot{V}^{\text{frame}}\,\hat{\boldsymbol{x}} = -g\,(\hat{\boldsymbol{z}} + \hat{\boldsymbol{x}}\,\dot{V}^{\text{frame}}/g). \tag{11.63}$$

We depict this situation in Figure 11.4.

What happens when the acceleration of the pendulum vanishes in the inertial reference frame? In this case the string tension exactly balances the weight so that from equation (11.60) we find

$$\mathbf{T} = -m\,\mathbf{g} \quad \Longleftarrow \quad \text{no acceleration in inertial frame, } \mathcal{R}. \tag{11.64}$$

Evidently, the pendulum comes to rest in the inertial frame with a string tension that exactly balances the vertical gravitational force acting on the pendulum mass. Hence, the pendulum hangs vertically.

The same question posed from the non-inertial reference frame leads to the distinct string tension

$$\mathbf{T} = -m\,\mathbf{g}^{\text{eff}} \quad \Longleftarrow \quad \text{no acceleration in non-inertial frame, } \mathcal{R}^{*}. \tag{11.65}$$

That is, the string tension now balances the effective gravity, so that the resting pendulum is not vertical but instead it is aligned with the effective gravity. We can readily find the angle of the resting pendulum, relative to the vertical, through the expression (11.63) for the effective gravity, so that

$$\tan\varphi^{\mathsf{equil}} = |\dot{\mathbf{V}}^{\mathsf{frame}}|/g. \tag{11.66}$$

11.5 Rigid-body rotating reference frames

We now consider a rotating reference frame. For simplicity, as well as to connect to the terrestrial case, assume rotation is around a fixed axis that passes through a fixed origin, such as that depicted in Figure 11.5. In this manner, the reference frame rotates as a rigid-body, thus enabling use of the kinematics from Section 11.3.[11]

In this chapter we make use of Cartesian coordinates for both the inertial reference frame, $\mathcal{R}$, and the rigid-body rotating reference frame, $\mathcal{R}^*$. We orient the motion so that the basis vectors of the rotating frame rotate counter-clockwise around the vertical axis, $\hat{\mathbf{z}}$ when looking down from above. As we see in this section, a proper accounting for the basis vector rotation is critical for computing non-inertial accelerations arising from rotation.

11.5.1 Incorrect calculation of Coriolis acceleration

Directly plugging into the expression (11.57) leads to the force in the non-inertial reference frame

$$\begin{aligned}
\mathbf{F}^{\mathsf{frame}} &= -m\,\dot{\mathbf{V}}_{\mathsf{rigid}} && (11.67\text{a})\\
&= -m\,\frac{\mathrm{d}(\boldsymbol{\Omega}\times\mathbf{X})}{\mathrm{d}t} && (11.67\text{b})\\
&= -m\,\boldsymbol{\Omega}\times\mathbf{V} && (11.67\text{c})\\
&= -m\,\boldsymbol{\Omega}\times(\mathbf{V}_{\mathsf{rigid}}+\mathbf{V}^*). && (11.67\text{d})
\end{aligned}$$

The first term is the correct expression for the centrifugal acceleration (see Figure 11.6), yet the second term is missing a factor of two for the Coriolis acceleration. As hinted at following equation (11.57), the error arises since the coordinate basis vectors change as the reference frame rotates, and this time dependence must be considered when computing the acceleration seen in a non-inertial reference frame.

11.5.2 Rigid-body rotation of an arbitrary vector

We here offer yet another means to derive the key result from Section 11.3.4 concerning the time derivative of a vector undergoing rigid-body rotation around a fixed origin. For this purpose, consider a vector that is measured in the rotating reference frame and represented using Cartesian coordinates

$$\mathbf{Q} = Q^{a*}\,\boldsymbol{e}_{a*}. \tag{11.68}$$

[11] A rigid-body rotating reference frame is what one generally means when studying rotating physics. Even so, there are more general forms of rotation, whereby angular rotation is a function of space and time. We do not consider time dependent rotating frames in this book (though see Exercise 11.1). However, the rigid-body frame of the Earth's rotation is experienced as a latitudinally dependent rotation rate that arises from sphericity of the planet. Such dependence of the rotational rate is referred to as differential rotation. Spatial dependence of the rotating terrestrial frame is a fundamental feature of planetary fluid motions, such as the planetary Rossby waves studied in VOLUME 4.

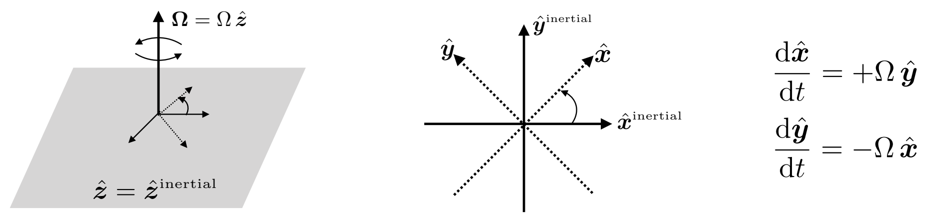

FIGURE 11.5: Depicting the coordinate axes in an inertial reference frame, $(\hat{\boldsymbol{x}}^{\text{inertial}}, \hat{\boldsymbol{y}}^{\text{inertial}}, \hat{\boldsymbol{z}}^{\text{inertial}})$, and a corresponding rigid-body rotating non-inertial reference frame, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$. The origin for the two reference frames is the same, and the rotation is around the vertical axis so that $\hat{\boldsymbol{z}} = \hat{\boldsymbol{z}}^{\text{inertial}}$ and $\mathbf{\Omega} = \Omega\,\hat{\boldsymbol{z}}$. Left panel: perspective view of the inertial and rotating reference frames. Middle panel: horizontal (plan view) looking down from above, and rotation of the horizontal basis vectors in the rotating reference frame. Right panel: expressing the time derivatives of the horizontal basis vectors according to the rigid-body rotation equation (11.69).

We here introduced the rotating reference frame Cartesian basis vectors, $\boldsymbol{e}_{1*} = \hat{\boldsymbol{x}}$, $\boldsymbol{e}_{2*} = \hat{\boldsymbol{y}}$, and $\boldsymbol{e}_{3*} = \hat{\boldsymbol{z}}$, which are rigidly rotating with the reference frame. Following the discussion in Section 11.3.3, rigid-body rotation of the basis vectors leads to a time derivative according to equation (11.34), so that

$$\frac{\mathrm{d}\boldsymbol{e}_{a*}}{\mathrm{d}t} = \mathbf{\Omega} \times \boldsymbol{e}_{a*}. \tag{11.69}$$

Hence, the time derivative of $\mathbf{Q}$, when represented in the rotating reference frame, is given by two terms,

$$\begin{aligned}
\frac{\mathrm{d}\mathbf{Q}}{\mathrm{d}t} &= \frac{\mathrm{d}Q^{a*}}{\mathrm{d}t}\,\boldsymbol{e}_{a*} + Q^{a*}\,\frac{\mathrm{d}\boldsymbol{e}_{a*}}{\mathrm{d}t} && (11.70\text{a})\\
&= \frac{\mathrm{d}Q^{a*}}{\mathrm{d}t}\,\boldsymbol{e}_{a*} + Q^{a*}\,(\mathbf{\Omega} \times \boldsymbol{e}_{a*}) && (11.70\text{b})\\
&= \frac{\mathrm{d}Q^{a*}}{\mathrm{d}t}\,\boldsymbol{e}_{a*} + \mathbf{\Omega} \times \mathbf{Q}. && (11.70\text{c})
\end{aligned}$$

The first expression on the right hand side is the time derivative of the vector's components as measured in the rotating reference frame, which we write as

$$\dot{\mathbf{Q}}^{*} \equiv \frac{\mathrm{d}Q^{a*}}{\mathrm{d}t}\,\boldsymbol{e}_{a*}. \tag{11.71}$$

The second term arises from rigid-body rotation of the rotating reference frame's coordinate basis.

11.5.3 Coriolis and centrifugal accelerations

We can make use of equation (11.70c) for the particle's position, in which the velocity of the particle is represented using Cartesian coordinates as

$$\frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} = \frac{\mathrm{d}X^{a*}}{\mathrm{d}t}\,\boldsymbol{e}_{a*} + \mathbf{\Omega} \times \mathbf{X} = \mathbf{V}^{*} + U_{\text{rigid}}. \tag{11.72}$$

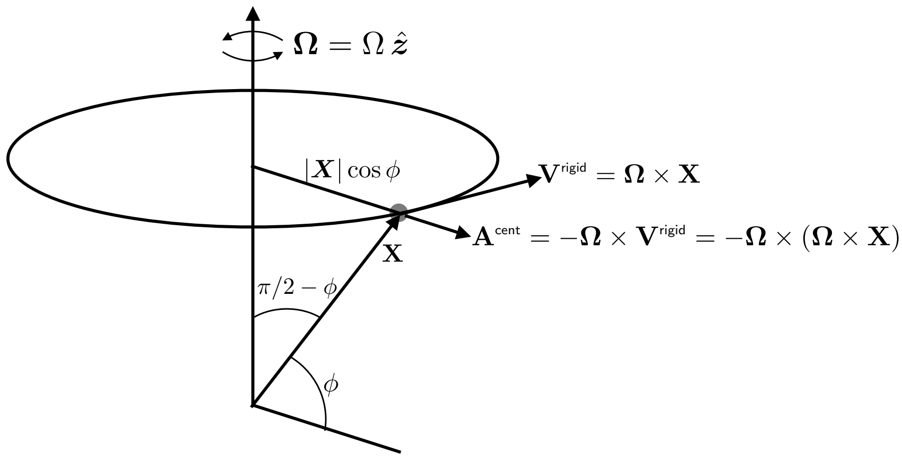

FIGURE 11.6: Depicting the kinematics of a rigid-body rotating reference frame, with rotation around a fixed vertical axis that passes through a fixed origin. The orientation is geophysically motivated as per Figure 4.2. In addition to the angular velocity, $\boldsymbol{\Omega}$, and rigid-body velocity, $\mathbf{V}_{\text{rigid}} = \boldsymbol{\Omega} \times \mathbf{X}$, both depicted in Figure 11.2, we also show the centrifugal acceleration, $\mathbf{A}_{\text{centrifugal}} = -\boldsymbol{\Omega} \times \mathbf{V}_{\text{rigid}} = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X})$, that points outward away from the rotational axis.

Evidently, the velocity as measured in the inertial frame equals to the velocity measured in the non-inertial reference frame, $\mathbf{V}^*$, plus the velocity of the rigid-body rotation,

$$U_{\text{rigid}} = \boldsymbol{\Omega} \times \mathbf{X}. \tag{11.73}$$

Taking the second derivative of the position leads to the decomposition of the acceleration measured in the inertial reference frame[12]

$$\begin{aligned}
\mathbf{A} &= \frac{\mathrm{d}^2\mathbf{X}}{\mathrm{d}t^2} && (11.74a)\\
&= \frac{\mathrm{d}^2 X^{a*}}{\mathrm{d}t^2}\,\boldsymbol{e}_{a*} + \frac{\mathrm{d}X^{a*}}{\mathrm{d}t}\,(\boldsymbol{\Omega} \times \boldsymbol{e}_{a*}) + \boldsymbol{\Omega} \times \frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} && (11.74b)\\
&= \mathbf{A}^* + \boldsymbol{\Omega} \times \mathbf{V}^* + \boldsymbol{\Omega} \times (\mathbf{V}^* + \mathbf{V}_{\text{rigid}}) && (11.74c)\\
&= \mathbf{A}^* + 2\,\boldsymbol{\Omega} \times \mathbf{V}^* + \boldsymbol{\Omega} \times \mathbf{V}_{\text{rigid}}. && (11.74d)
\end{aligned}$$

Hence, the acceleration as measured in the rigid-body rotating reference frame is given by

$$\mathbf{A}^* = \mathbf{A} \ \underbrace{-2\,\boldsymbol{\Omega} \times \mathbf{V}^*}_{\text{Coriolis}} \ \underbrace{-\boldsymbol{\Omega} \times \mathbf{V}_{\text{rigid}}}_{\text{centrifugal}}. \tag{11.75}$$

We here identified the Coriolis and centrifugal accelerations

$$\mathbf{A}_{\text{coriolis}} = -2\,\boldsymbol{\Omega} \times \mathbf{V}^* \quad \text{and} \quad \mathbf{A}_{\text{centrifugal}} = -\boldsymbol{\Omega} \times \mathbf{V}_{\text{rigid}} = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{X}), \tag{11.76}$$

with both of these accelerations arising from the rigid-body rotation of the non-inertial reference frame. Figure 11.6 provides a schematic of these accelerations, which prove central to describing terrestrial motion within the Earth's rotating reference frame.

[12]Equation (11.74d) is Coriolis' theorem for the special case of a constant rotating reference frame. In Exercise 11.1 we extend this result to allow for $\mathrm{d}\boldsymbol{\Omega}/\mathrm{d}t \neq 0$.

11.6 Newtonian mechanics for a system of particles

We here consider the mechanics for a system of N point particles with mass, $m_{(i)}$, position, $\mathbf{X}_{(i)}(t)$, and velocity, $\mathbf{V}_{(i)}(t) = \mathrm{d}\mathbf{X}_{(i)}/\mathrm{d}t$, where $i = 1, N$ is an integer that labels the particles,[13] and we assume the particle mass remains fixed in time. In the following, we find it useful to introduce the total mass of the system as well as the position for the center of mass according to

$$M = \sum_{i=1}^{N} m_{(i)} \quad \text{and} \quad \mathbf{X} = \frac{1}{M} \sum_{i=1}^{N} m_{(i)} \, \mathbf{X}_{(i)}. \tag{11.77}$$

11.6.1 Forces on the particles

We conceive of two contributions to the forces acting on each particle. The first arises from external forces, $\mathbf{F}^{(\mathrm{ext})}_{(i)}$, that are independent of the particles. For example, we may consider a system of particles moving in the gravitational field of the planet, with the planet's gravitational field assumed to be independent of the particles. The second force arises from the interactions between the particles, which we refer to as an internal force. We write the internal force vector as $\mathbf{F}_{(ji)}$, which is taken to be the force on particle i due to interactions with particle j. We furthermore assume that there are no self-forces so that

$$\mathbf{F}_{(ii)} = 0. \tag{11.78}$$

The net force acting on particle i is thus written

$$\mathbf{F}_{(i)} = \mathbf{F}^{(\mathrm{ext})}_{(i)} + \sum_{j=1}^{N} \mathbf{F}_{(ji)}, \tag{11.79}$$

with $\sum_{j=1}^{N} \mathbf{F}_{(ji)}$ being the summation of all forces of interaction acting on particle i. It follows that the net force acting on the full N particle system is

$$\sum_{i=1}^{N} \mathbf{F}_{(i)} = \sum_{i=1}^{N} \mathbf{F}^{(\mathrm{ext})}_{(i)} + \sum_{i=1}^{N} \sum_{j=1}^{N} \mathbf{F}_{(ji)}. \tag{11.80}$$

The final sum can be written

$$\sum_{i=1}^{N} \sum_{j=1}^{N} \mathbf{F}_{(ji)} = \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} (\mathbf{F}_{(ji)} + \mathbf{F}_{(ij)}), \tag{11.81}$$

where, again, $\mathbf{F}_{(ji)}$ is the force on particle i due to interactions with particle j, and $\mathbf{F}_{(ij)}$ is the force on particle j due to interactions with particle i. This identity follows since the sums are finite so that limits on double sums can be swapped. In the next subsection we see why it is useful to write the internal forces in this manner.

[13]We place the particle index within parentheses to emphasize that it is not a tensor index. Correspondingly, particle indices do not follow the summation convention of tensor indices.

11.6.2 Weak and strong form of Newton's third law

Newton's third law states that the force acting on particle i due to particle j is equal to, yet oppositely directed, the force acting on particle j due to particle i

$$\mathbf{F}_{(ij)} = -\mathbf{F}_{(ji)}. \tag{11.82}$$

For those forces satisfying Newton's third law, the total force acting on the N particle system arises from just the external force, since the internal forces cancel pairwise

$$\sum_{i=1}^{N}\sum_{j=1}^{N} \mathbf{F}_{(ij)} = \sum_{i=1}^{N} \mathbf{F}^{(\text{ext})}_{(i)} = \mathbf{F}^{(\text{ext})}. \tag{11.83}$$

As discussed in Section 11.7.4, the gravitational force acting between two point masses offers an example force that satisfies Newton's third law, in which case the force is given by the inverse square expression

$$\mathbf{F}_{(ji)} = -\frac{G\, m_{(i)}\, m_{(j)}\, (\mathbf{X}_{(i)} - \mathbf{X}_{(j)})}{|\mathbf{X}_{(i)} - \mathbf{X}_{(j)}|^3} = -\mathbf{F}_{(ji)}, \tag{11.84}$$

where G is Newton's gravitational constant discussed in Section 11.7.2. Notice how the gravitational force acts along the line connecting the two particles, so that

$$(\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)} = 0. \tag{11.85}$$

The electrostatic force acting on charged particles in an electric field is given by Coulombs law, which is also an inverse squared force and that acts along the line between the two particles.

Newton's gravity force and the Coulomb electrostatic force are known as central forces, with central forces acting on the line separating two particles. Central forces are said to satisfy the *strong form* of Newton's third law, in that both $\mathbf{F}_{(ij)} = -\mathbf{F}_{(ji)}$ and $(\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)} = 0$. Slightly more general forces arise that satisfy $\mathbf{F}_{(ji)} = -\mathbf{F}_{(ij)}$ and yet $(\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)} \neq 0$, with such forces said to satisfy the *weak form* of Newton's third law:

$$\mathbf{F}_{(ij)} = -\mathbf{F}_{(ji)} \quad \text{and} \quad (\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)} = 0 \Longrightarrow \text{central force; strong third} \tag{11.86a}$$

$$\mathbf{F}_{(ij)} = -\mathbf{F}_{(ji)} \quad \text{and} \quad (\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)} \neq 0 \Longrightarrow \text{non-central force; weak third.} \tag{11.86b}$$

Any force that depends on the velocity of the particle is not a central force.[14] The distinction between the weak and strong form of Newton's third law becomes important when studying particle angular momentum in Section 11.6.4.

11.6.3 Newton's second law of motion

The linear momentum of a particle is its mass times the velocity

$$\mathbf{P}_{(i)} = m_{(i)}\, \mathbf{V}_{(i)} \quad \text{no implied summation.} \tag{11.87}$$

[14] The force acting on a classical charged particle moving in an electromagnetic field is known as the Lorentz force. The magnetic portion of the Lorentz force depends on the velocity of the charged particle and so it is not a central force, with the Lorentz force only satisfying the weak form of Newton's third law. See page 45 of *Marion and Thornton* (1988) for more discussion.

Assuming the mass of the particle is constant, the corresponding equation of motion is given by Newton's second law

$$\dot{\mathbf{P}}_{(i)} = m_{(i)}\,\dot{\mathbf{V}}_{(i)} = \mathbf{F}^{(\text{ext})}_{(i)} + \sum_{j=1}^{N} \mathbf{F}_{(ji)}. \tag{11.88}$$

Assuming the internal forces satisfy either the weak or strong form of Newton's third law as in equations (11.86a)-(11.86b), we find that the total momentum for the N particle system evolves according to just the total external force

$$\dot{\mathbf{P}} = \sum_{i=1}^{N} \dot{\mathbf{P}}_{(i)} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} \sum_{i=1}^{N} m_{(i)}\,\mathbf{X}_{(i)} = M\,\ddot{\mathbf{X}} = \mathbf{F}^{(\text{ext})}. \tag{11.89}$$

That is, the time change in the total linear momentum is given by the total mass times the acceleration of the center of mass, which equals, through Newton's second and third laws, to the total external force acting on the system. We thus find that the center of mass maintains a fixed velocity if there is no net external force acting on the many particle system. This result accords with common experience whereby we cannot lift ourselves up by our own bootstraps.

11.6.4 Angular momentum

The total angular momentum for the many particle system is given by

$$\mathbf{L} = \sum_{i=1}^{N} \mathbf{L}_{(i)} = \sum_{i=1}^{N} \mathbf{X}_{(i)} \times \mathbf{P}_{(i)} = \sum_{i=1}^{N} m_{(i)}\,\mathbf{X}_{(i)} \times \mathbf{V}_{(i)}, \tag{11.90}$$

and its time derivative is given by

$$\dot{\mathbf{L}} = \sum_{i=1}^{N} m_{(i)}\,\mathbf{X}_{(i)} \times \dot{\mathbf{V}}_{(i)} = \sum_{i=1}^{N} \mathbf{X}_{(i)} \times \mathbf{F}^{(\text{ext})}_{(i)} + \sum_{i=1}^{N}\sum_{j=1}^{N} \mathbf{X}_{(i)} \times \mathbf{F}_{(ji)}, \tag{11.91}$$

where we set $\dot{\mathbf{X}}_{(i)} \times \mathbf{V}_{(i)} = \dot{\mathbf{X}}_{(i)} \times \dot{\mathbf{X}}_{(i)} = 0$, and made use of the particle equation of motion (11.88). We identify the term

$$\boldsymbol{\Gamma}^{(\text{ext})} \equiv \sum_{i=1}^{N} \mathbf{X}_{(i)} \times \mathbf{F}^{(\text{ext})}_{(i)} \tag{11.92}$$

as the total torque acting on the N particle system arising from the external force acting on each particle. The internal torque contribution can be written

$$\sum_{i=1}^{N}\sum_{j=1}^{N} \mathbf{X}_{(i)} \times \mathbf{F}_{(ji)} = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} (\mathbf{X}_{(i)} \times \mathbf{F}_{(ji)} + \mathbf{X}_{(j)} \times \mathbf{F}_{(ij)}) \tag{11.93a}$$

$$= \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} (\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)}, \tag{11.93b}$$

where the first equality follows from interchanging particle indices, and the second equality follows from the weak form of Newton's third law given by equation (11.86b). If we furthermore assume the force satisfies the strong form of Newton's third law (11.86a), as for a central force, then the total angular momentum evolves according to just the total external torque

$$\dot{\mathbf{L}} = \mathbf{\Gamma}^{(\text{ext})}. \tag{11.94}$$

That is, we require the strong form of Newton's third law to ensure that the total angular momentum is only affected by the torques created by external forces. If the forces only satisfy the weak form of Newton's third law, then evolution of the total angular momentum is generally affected by internal torques. Such torques arise in the presence of nonmechanical forces, such as a magnetic force between charged particles when the electromagnetic field contains intrinsic angular momentum. We have no occasion to study internal torques in this book, so that all physical systems are assumed to satisfy the strong form of Newton's third law.

11.6.5 Center of mass coordinates

It is sometimes preferable to move to an internal set of coordinates that dispenses with the arbitrary fixed origin. In this case we place the coordinate origin at the moving center of mass,

$$\mathbf{X}_{(i)} = \mathbf{X} + \mathbf{X}'_{(i)}, \tag{11.95}$$

with a corresponding velocity expression

$$\mathbf{V}_{(i)} = \mathbf{V} + \mathbf{V}'_{(i)}. \tag{11.96}$$

Making use of the definition of the center of mass coordinate (11.77), we readily find that the relative position and relative velocity each have a vanishing mass weighted sum over all the particles

$$\sum_{i=1}^{N} m_{(i)}\, \mathbf{X}'_{(i)} = 0 \quad \text{and} \quad \sum_{i=1}^{N} m_{(i)}\, \mathbf{V}'_{(i)} = 0. \tag{11.97}$$

11.6.6 Angular momentum in center of mass coordinates

The total angular momentum of the N particle system, expressed in the center of mass coordinates, is given by

$$\mathbf{L} = \sum_{i=1}^{N} m_{(i)}\, \mathbf{X}_{(i)} \times \mathbf{V}_{(i)} \tag{11.98a}$$

$$= \sum_{i=1}^{N} m_{(i)}\, (\mathbf{X} + \mathbf{X}'_{(i)}) \times (\mathbf{V} + \mathbf{V}'_{(i)}) \tag{11.98b}$$

$$= M\, \mathbf{X} \times \mathbf{V} + \sum_{i=1}^{N} m_{(i)}\, \mathbf{X}'_{(i)} \times \mathbf{V}'_{(i)} \tag{11.98c}$$

$$= \mathbf{L}^{\text{cm}} + \mathbf{L}', \tag{11.98d}$$

where we used equation (11.97) to reach the third equality. The first term is the angular momentum of the center of mass relative to the fixed origin

$$\mathbf{L}^{\text{cm}} = M\, \mathbf{X} \times \mathbf{V}, \tag{11.99}$$

whereas the second term is the internal angular momentum measured relative to the center of mass

$$\mathbf{L}' = \sum_{i=1}^{N} m_{(i)}\, \mathbf{X}'_{(i)} \times \mathbf{V}'_{(i)}. \tag{11.100}$$

Evolution of the center of mass angular momentum takes on the form

$$\dot{\mathbf{L}}^{\text{cm}} = M\, \mathbf{X} \times \dot{\mathbf{V}} = \mathbf{X} \times \mathbf{F}^{(\text{ext})}, \tag{11.101}$$

with the right hand side the torque from the net external force computed relative to the center of mass position. Correspondingly, evolution of the total angular momentum is

$$\dot{\mathbf{L}} = \sum_{i=1}^{N} m_{(i)}\, \mathbf{X}_{(i)} \times \dot{\mathbf{V}}_{(i)} \tag{11.102a}$$

$$= \sum_{i=1}^{N} (\mathbf{X} + \mathbf{X}'_{(i)}) \times \mathbf{F}^{\text{ext}}_{(i)} \tag{11.102b}$$

$$= \mathbf{X} \times \mathbf{F}^{\text{ext}} + \sum_{i=1}^{N} \mathbf{X}'_{(i)} \times \mathbf{F}^{\text{ext}}_{(i)} \tag{11.102c}$$

$$= \dot{\mathbf{L}}^{\text{cm}} + \dot{\mathbf{L}}'. \tag{11.102d}$$

We thus see that the rate of change for the angular momentum computed relative to the center of mass is given by the torques from the external forces applied to each particle, computed relative to the center of mass

$$\dot{\mathbf{L}}' = \sum_{i=1}^{N} \mathbf{X}'_{(i)} \times \mathbf{F}^{\text{ext}}_{(i)}. \tag{11.103}$$

11.6.7 Energy in center of mass coordinates

Making use of the identity (11.97), we find that the total kinetic energy is given by

$$K = \frac{1}{2} \sum_{i=1}^{N} m_{(i)}\, \mathbf{V}_{(i)} \cdot \mathbf{V}_{(i)} = \frac{M}{2} \mathbf{V} \cdot \mathbf{V} + \frac{1}{2} \sum_{i=1}^{N} m_{(i)}\, \mathbf{V}'_{(i)} \cdot \mathbf{V}'_{(i)} = K^{(\text{cm})} + K', \tag{11.104}$$

which is the sum of the center of mass kinetic energy, $K^{(\text{cm})}$, plus the kinetic energy of the internal motions relative to the center of mass, K'.

Consider two configurations of the N particle system, configuration A at time t_A and configuration B at time t_B. Let $\mathbf{X}_{(iA)}$ be the position of particle i in configuration A, and $\mathbf{X}_{(iB)}$ the position of this particle in configuration B, and assume the two configurations are connected by trajectories determined by the equation of motion for each particle. Extending our discussion of work in Section 11.2.4, we see that the work done by forces acting on the N particle system as it moves between these two configurations is given by

$$W_{AB} = \sum_{i=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \mathbf{F}_{(i)} \cdot \mathrm{d}\mathbf{x}_{(i)} \tag{11.105a}$$

$$= \sum_{i=1}^{N} \int_{t_A}^{t_B} \mathbf{F}_{(i)} \cdot \mathbf{V}_{(i)}\, \mathrm{d}t \tag{11.105b}$$

$$= \sum_{i=1}^{N} \int_{t_A}^{t_B} m_{(i)} \, \dot{\mathbf{V}}_{(i)} \cdot \mathbf{V}_{(i)} \, \mathrm{d}t \tag{11.105c}$$

$$= \frac{1}{2} \sum_{i=1}^{N} \int_{t_A}^{t_B} m_{(i)} \, \frac{\mathrm{d}}{\mathrm{d}t} (\mathbf{V}_{(i)} \cdot \mathbf{V}_{(i)}) \, \mathrm{d}t \tag{11.105d}$$

$$= K_A - K_B, \tag{11.105e}$$

where K_A and K_B are the kinetic energies in the two configurations.

Rather than making use of Newton's law of motion to introduce the kinetic energy, we can decompose the force vector in the expression for work, thus giving

$$W_{AB} = \sum_{i=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \mathbf{F}_{(i)} \cdot \mathrm{d}\mathbf{x}_{(i)} \tag{11.106a}$$

$$= \sum_{i=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \mathbf{F}^{(\text{ext})}_{(i)} \cdot \mathrm{d}\mathbf{x}_{(i)} + \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \mathbf{F}_{(ji)} \cdot (\mathrm{d}\mathbf{x}_{(i)} - \mathrm{d}\mathbf{x}_{(j)}), \tag{11.106b}$$

where the minus sign on the final term arises from Newton's third law satisfied by the internal forces. Now assume that both the internal and external forces are conservative, so that they can separately be written as the gradient of respective potential energies. For the external potential energy acting on particle i, we assume it is a function just of the position of that particle

$$\mathbf{F}^{\text{ext}}_{(i)} = -\nabla_i P^{\text{ext}}(\mathbf{X}_{(i)}), \tag{11.107}$$

where ∇_i is the gradient operator acting on the position in space, $\mathbf{x}_{(i)}$, of the trajectory at a particular time, $\mathbf{x}_{(i)} = \mathbf{X}_{(i)}(t)$. Furthermore, assume the interparticle potential energy between particles i and j is a function just of the distance between the two particles (such as for a central force). In this case we have

$$\mathbf{F}_{(ji)} = -\nabla_{ij} P(|\mathbf{X}_{(i)} - \mathbf{X}_{(j)}|), \tag{11.108}$$

where ∇_{ij} is the gradient operator acting on the relative position, $\mathbf{x}_{(i)} - \mathbf{x}_{(j)}$, of the two particles at a particular time instance. These assumptions then bring the work into the form

$$W_{AB} = -\sum_{i=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \nabla_i P^{(\text{ext})} \cdot \mathrm{d}\mathbf{x}_{(i)} - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \nabla_{ij} P \cdot (\mathrm{d}\mathbf{x}_{(i)} - \mathrm{d}\mathbf{x}_{(j)}) \tag{11.109a}$$

$$= -\sum_{i=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \mathrm{d}P^{(\text{ext})}_{(i)} - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \int_{\mathbf{X}_{(iA)}}^{\mathbf{X}_{(iB)}} \mathrm{d}P(|\mathbf{X}_{(i)} - \mathbf{X}_{(j)}|) \tag{11.109b}$$

$$\equiv -[P^{(\text{tot})}(A) - P^{(\text{tot})}(B)], \tag{11.109c}$$

where the total potential energy is

$$P^{(\text{tot})} = \sum_{i=1}^{N} P^{(\text{ext})}_{(i)} + \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} P(|\mathbf{X}_{(i)} - \mathbf{X}_{(j)}|). \tag{11.110}$$

Combining this result and the expression of work in terms of kinetic energy leads to the conservation of mechanical energy for the conservative N particle system

$$K + P^{(\mathrm{tot})} = \text{constant}. \tag{11.111}$$

11.6.8 Comments and further reading

This section largely follows Section 2 of *Fetter and Walecka* (2003). When considering a continuous fluid rather than N point particles, the interparticle forces manifest as pressure and friction, whereas the external force is given by gravitation, planetary centrifugal, and planetary Coriolis. Furthermore, we always assume the fluids in this book satisfy the strong form of Newton's third law.

11.7 Newtonian gravity

In this section we provide a synopsis of Newton's theory for gravity, which plays a leading role in the mechanics of particles and fluids moving around a massive planet. Indeed, as seen in Chapter 12, the gravitational force acting on a classical point particle is the only force felt by the particle when viewed in an inertial reference frame.

11.7.1 The gravitational body force

The gravitational acceleration vector, $\mathbf{g}$, is continuous in both space and time and is represented by a continuous vector function, $\boldsymbol{g}(\boldsymbol{x}, t)$. When placed within a gravitational field, the gravitational force (or more consisely, the gravity) acting on an infinitesimal matter distribution with mass $\rho\,\mathrm{d}V$, is

$$\mathrm{d}\mathbf{F} = \mathbf{g}\,\rho\,\mathrm{d}V. \tag{11.112}$$

Gravitational forces are body forces and thus act on all points within the matter distribution. Furthermore, gravitational forces are conservative, with the gradient of the Newtonian gravitational potential, Φ_{N}, rendering the gravitational acceleration

$$\boldsymbol{g} = -\boldsymbol{\nabla}\Phi_{\mathsf{N}}, \tag{11.113}$$

with the minus sign conventional. The gravitational potential has dimensions of $\mathrm{L}^2\ \mathrm{T}^{-2}$ so that its gradient has the dimensions of acceleration, $\mathrm{L}\ \mathrm{T}^{-2}$.

11.7.2 The gravity field from mass sources

In addition to feeling the effects from gravity, matter with a mass density, ρ, provides a source for the gravity field. Namely, the gravitational field has a convergence given by

$$-\boldsymbol{\nabla}\cdot\boldsymbol{g} = \boldsymbol{\nabla}^2\Phi_{\mathsf{N}} = 4\,\pi\,G\,\rho, \tag{11.114}$$

where

$$G = 6.674\times10^{-11}\ \mathrm{N\ m^2\ kg^{-2}} = 6.674\times10^{-11}\ \mathrm{m^3\ kg^{-1}\ s^{-2}} \tag{11.115}$$

is Newton's gravitational constant. Equation (11.114) is known as Poisson's equation for the gravitational potential. The gravitational acceleration has a nonzero convergence only for those regions with nonzero matter density. Since matter only has a non-negative mass density, $\rho \geq 0$, there are no regions where the gravitational field diverges, in which case there are no repulsive gravity fields.[15] Integrating over a region, $\mathcal{R}$, and use of the divergence theorem leads to

$$\int_{\mathcal{R}} \nabla \cdot \mathbf{g} \, \mathrm{d}V = \oint_{\partial \mathcal{R}} \mathbf{g} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}V = -4\,\pi\,G \int_{\mathcal{R}} \rho \, \mathrm{d}V = 4\,\pi\,G\,M, \tag{11.116}$$

where M is the total mass enclosed by the region. Evidently, the surface integral of the gravitational field, computed over a closed region, is nonzero only if the region contains matter. The Poisson equation (11.114) is an elliptic partial differential equation whose resulting gravitational potential responds instantaneously everywhere to any changes in the mass distribution.[16] It took a couple of centuries, and the genius of Einstein, to provide a local connection between mass and gravity as rendered by space-time curvature and gravitational waves.[17]

11.7.3 Newton's gravitational law

Newton's law of gravity follows from the assertion that the gravitational acceleration from a point mass is given by

$$\boldsymbol{g}(\boldsymbol{x}) = -G\,M\,\boldsymbol{x}/|\boldsymbol{x}|^3 = -\nabla(G\,M/|\boldsymbol{x}|) = -\nabla\Phi_{\mathsf{N}}, \tag{11.117}$$

thus pointing radially from the coordinate origin, which is where the point mass is located. This gravitational acceleration manifests the inverse squared gravitational force arising from Newtonian gravity. Furthermore, the gravitational acceleration has a convergence given by

$$-\nabla \cdot \boldsymbol{g} = \nabla^2 \Phi_{\mathsf{N}} = 4\,\pi\,G\,M\,\delta(\boldsymbol{x}), \tag{11.118}$$

where we wrote the point mass density in terms of the Dirac delta

$$\rho^{\mathsf{point}}(\boldsymbol{x}) = M\,\delta(\boldsymbol{x}), \tag{11.119}$$

as discussed in Section 5.1.

With a continuous mass density, ρ, the Newtonian gravitational potential can be written as the continuous accumulation of the potential from a point charge source that is weighted by the matter density[18]

$$\Phi_{\mathsf{N}}(\boldsymbol{x}) = -G \int_{\mathcal{R}} \frac{\rho(\boldsymbol{x}_{\circ})}{|\boldsymbol{x} - \boldsymbol{x}_{\circ}|} \, \mathrm{d}V_{\circ} \tag{11.120}$$

so that the gravitational acceleration is

$$\boldsymbol{g}(\boldsymbol{x}) = -\nabla\Phi_{\mathsf{N}} = -G \int_{\mathcal{R}} \frac{\rho(\boldsymbol{x}_{\circ})\,(\boldsymbol{x} - \boldsymbol{x}_{\circ})}{|\boldsymbol{x} - \boldsymbol{x}_{\circ}|^3} \, \mathrm{d}V_{\circ}. \tag{11.121}$$

[15] This property of gravity is distinct from the electrostatic forces, which arise from both positive and negative electrical charges, thus giving rise to both attractive and repulsive electric fields.

[16] This instantaneous transfer of information is a property of ellitical boundary value problems such as those arising from Poisson's equation.

[17] See *Thorne and Blandford* (2017) for a study of general relativity and gravitational waves.

[18] In our study of Green's function for the Laplacian operator in Chapter 8, we refer to the potential from a point charge as the free-space Green's function.

11.7.4 Gravitational field outside a spherical Earth

In this book we assume the mass density of the planet to be spherically symmetric, in which case the Earth's gravitational potential, $\Phi_{\rm e}$, is a function only of the radial distance from the center of the mass distribution, $\Phi_{\rm e} = \Phi_{\rm e}(r)$. Letting the integration region, $\mathcal{R}$, be a sphere of radius $r > R_{\rm e}$, where $R_{\rm e}$ is the radius of the planet, and making use of spherical coordinates from Section 4.4.8, brings the Poisson equation (11.114) to

$$4\,\pi\,r^2\,\frac{\partial \Phi_{\rm e}}{\partial r} = 4\,\pi\,G\,M_{\rm e}, \tag{11.122}$$

where $M_{\rm e}$ is the planet mass. Integration leads to the gravitational potential for an arbitrary point outside the spherically symmetric mass distribution[19]

$$\Phi_{\rm e} = -\frac{G\,M_{\rm e}}{r}, \tag{11.123}$$

where we set the integration constant to zero. Evidently, when sampling the gravity field at a radius equal to or larger than the spherical planet radius, the gravitational potential is identical to that of a point mass at the origin. The gradient of the gravitational potential (11.123) yields the inverse-squared dependence of the gravitational acceleration

$$\boldsymbol{g}_{\rm e} = -\nabla\Phi_{\rm e} = -\frac{G\,M\,\hat{\boldsymbol{r}}}{r^2} = -\frac{G\,M\,\boldsymbol{r}}{r^3}, \tag{11.124}$$

along with the gravitational force acting on a point particle of mass m

$$\mathbf{F}^{\rm gravity} = m\,\mathbf{g}_{\rm e} = -m\,\boldsymbol{\nabla}\Phi_{\rm e}. \tag{11.125}$$

Furthermore, the gravitational potential energy of the particle (dimensions M L^2 T^{-2}) is

$$P = m\,\Phi_{\rm e}. \tag{11.126}$$

11.7.5 Gravitational acceleration near the Earth's surface

For most applications of atmospheric and oceanic fluid mechanics, it is sufficient to assume the gravitational acceleration is constant and equal to its value at the Earth's surface. This assumption holds so long as the radial position of the particle is a distance from the Earth surface that is small relative to the Earth radius. We generally make this assumption throughout this book, though with the exception of astronomically driven tides as introduced in Section 11.8. This assumption considers the earth gravitational acceleration, $g_{\rm e}$, is a constant so that

$$\boldsymbol{g}_{\rm e} = -g_{\rm e}\,\hat{\boldsymbol{r}}, \tag{11.127}$$

where

$$g_{\rm e} = \frac{G\,M_{\rm e}}{R_{\rm e}^2} \approx 9.8 \text{ m s}^{-2}. \tag{11.128}$$

To reach this value, we assumed a sphere of mass equal to the Earth mass

$$M_{\rm e} = 5.977 \times 10^{24} \text{ kg}, \tag{11.129}$$

[19] In Exercise 11.8 we derive the gravitational potential inside of a spherical body.

and radius

$$R_e = 6.371 \times 10^6 \text{ m} \tag{11.130}$$

determined so that the sphere has the same volume as the Earth.

The corresponding gravitational potential for the particle is given by

$$\Phi_e = g_e \, r, \tag{11.131}$$

with the gravitational acceleration

$$\boldsymbol{g}_e = -\nabla \Phi_e = -g_e \, \hat{\boldsymbol{r}}, \tag{11.132}$$

and the gravitational potential energy

$$m \, \Phi_e = m \, g_e \, r. \tag{11.133}$$

We emphasize that the expression for the gravitational potential, (11.131), and potential energy, (11.133), are accurate only so long as the radial position of the particle is a distance from the Earth surface that is small relative to the Earth radius. Furthermore, note that the approximate gravitational potential (11.131) is positive whereas the unapproximated potential (11.123) is negative. However, the absolute zero of the potential has no physical significance. Instead, what is important is the change between two points in space, with both expressions for the gravitational potential increasing when moving away from the Earth center.

11.7.6 Further study

Newton's gravitational law is standard material from freshman physics. The presentation given here is consistent with *Symon* (1971), *Marion and Thornton* (1988), and *Taylor* (2005), and chapters 3 and 6 of *Lautrup* (2005). Some commonly used physical properties of the Earth are summarized in Appendix Two of *Gill* (1982).

11.8 Gravitational tidal accelerations

Newtonian gravity provides the theoretical framework for describing ocean and solid-Earth tides, and we here describe the rudiments. For simplicity we focus just on the Earth-Moon system, though recognize that the sun also plays an analogous role for observed tidal motion. Note that the Earth's atmosphere exhibits tidal motions. However, as developed by *Chapman and Lindzen* (1970), the dominant cause of atmospheric tidal motion is the diurnal solar heating of the atmosphere rather than the gravitational field. In contrast, the ocean tides and solid-Earth tides are caused by gravitational effects.

11.8.1 Tidal acceleration as differential gravity

Before considering the Earth-Moon system, we introduce the notion of tidal acceleration, which is felt by a finite sized body placed within a non-uniform gravitational field. Figure 11.7 depicts this situation where the finite sized body is a narrow rod whose axis points toward the center of a spherically symmetric massive body. One end of the rod experiences a different gravitational acceleration than the other since the gravitational field falls off as the inverse squared distance from the center of the sphere. It is this differential gravitational acceleration that we refer to as the tidal acceleration. As shown in this section, the tidal acceleration falls off as the inverse cube of the distance rather than the more familiar inverse square.

To develop a mathematical expression for the tidal acceleration, focus on the spherically symmetric gravitational field in which the gravitational acceleration at a point is given by

$$\boldsymbol{g} = -\frac{G\, M\, \hat{\boldsymbol{r}}}{r^2}. \tag{11.134}$$

In this equation, M is the mass of the sphere, $r = |\boldsymbol{x}|$ is the distance from the sphere's center, $\hat{\boldsymbol{r}}$ is the radial unit vector, with the minus sign indicating that the gravitational acceleration points toward the center of the massive sphere. For the rod in Figure 11.7, the difference between the gravitational acceleration acting at a point nearest to the sphere (point B) and a point furthest from the sphere (point A) is given by

$$\boldsymbol{g}(r_{\mathsf{B}}) - \boldsymbol{g}(r_{\mathsf{A}}) = \boldsymbol{g}(r_{\circ} - L/2) - \boldsymbol{g}(r_{\circ} + L/2), \tag{11.135}$$

where $r_{\circ}$ is the distance from the sphere's center to the center of the rod. Assuming the rod is not long, we can expand this difference in a Taylor series about the rod center at $r_{\circ}$, thus leading to an expression for the tidal acceleration

$$\boldsymbol{g}(r_{\mathsf{B}}) - \boldsymbol{g}(r_{\mathsf{A}}) \approx -L\, \left.\frac{\partial \boldsymbol{g}}{\partial r}\right|_{r=r_{\circ}} = -2L\, \frac{G\, M\, \hat{\boldsymbol{r}}}{r_{\circ}^3} = (2\, L/r_{\circ})\, \boldsymbol{g}(r_{\circ}). \tag{11.136}$$

Evidently, the tidal acceleration is proportional to the inverse cube of the distance to the center of the sphere. We see this property again when considering in Section 11.8.3 the gravitational acceleration generated from a remote body (e.g., the Moon) acting on the surface of a sphere (e.g., the Earth).

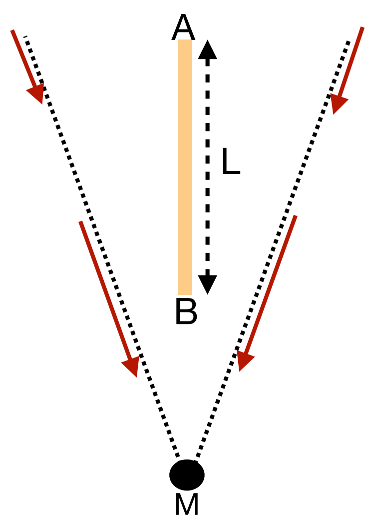

FIGURE 11.7: Tidal acceleration is the acceleration that acts on a finite sized object placed in a non-uniform or differential gravitational field. The finite object is here depicted as a narrow rod of length L placed in the gravity field of a spherically symmetric body of mass M. That portion of the rod closer to the gravitating sphere (end B) experiences a stronger gravitational acceleration than the end that is further away (end A). The gradient in the gravitational acceleration constitutes the tidal acceleration acting on the rod.

11.8.2 Heuristics of tidal acceleration on the surface of a sphere

We now consider the tidal acceleration acting on the surface of a smooth massive sphere due to a spherically symmetric gravitational field generated by a neighboring massive body. Figure 11.8 depicts this physical system, which we consider an idealized Earth-Moon system where each body is assumed homogeneous and spherical. Given that they gravitationally attract one another, it is not astronomically possible for the two bodies to remain spatially fixed. Instead, they orbit around their common center of mass while conserving their angular momentum.

A central question of tidal studies is why there are generally two tides per day (semi-diurnal tides) rather than just one (diurnal tides). We here offer two complementary arguments. The first is based on extending the tidal acceleration discussion of Section 11.8.1, whereas the second follows the more traditional account by considering a balance between gravitational and centrifugal accelerations.

General ideas

Every point on the surface of the Earth is attracted to the Earth's center by the Earth's gravitational field. For a spherical Earth, this attractive body force is purely radial, so that it cannot lead to lateral motion on the surface of the perfect sphere. We conclude that the radial gravitational field is not the cause of tidal motion. Instead, tidal motion arises from a non-radial gravitational field.

The Earth-Moon gravitational field accelerates the Earth and Moon toward one another along the axis connecting their centers. Additionally, the spatial dependence of the Moon's gravitational field over the Earth leads to lateral forces along the Earth's surface, thus providing the necessary ingredient for gravitationally induced tidal motion. To capture the essence of this force, we examine how the Moon's gravitational field acts at a point on the Earth relative to its action at the center of the Earth.

Sample tidal accelerations on the sphere

Again, we are tasked with computing the tidal acceleration from the Moon's gravitational field for selected points on the Earth, and we are computing these accelerations relative to the Earth center. As for the rod in Figure 11.7, the tidal acceleration at point B in Figure 11.8, relative to the center of the Earth, is given by

$$\boldsymbol{g}(r_{\mathsf{B}}) - \boldsymbol{g}(R_{\mathsf{em}}) = (2R_{\mathsf{e}}/R_{\mathsf{em}})\, \boldsymbol{g}(R_{\mathsf{em}}). \tag{11.137}$$

This acceleration points towards the Moon. In contrast, the tidal acceleration at point A relative to the center of the Earth is given by

$$\boldsymbol{g}(r_{\mathsf{A}}) - \boldsymbol{g}(R_{\mathsf{em}}) = -(2R_{\mathsf{e}}/R_{\mathsf{em}})\, \boldsymbol{g}(R_{\mathsf{em}}), \tag{11.138}$$

which is of equal magnitude but points away from the Moon.

The tidal accelerations at points A and B act radially away from the Earth's center. Hence, as noted above, these radial forces do not directly lead to tidal motion at those points. However, through symmetry of the configuration, points on the surface of the sphere between A and B have a tidal acceleration from the Moon's gravitational field with a nonzero lateral component. These lateral forces lead to the accumulation of water at points A and B. We can compute the gravitational acceleration at intermediate points. However, the trigonometry is somewhat complex and we prefer to compute the forces in Section 11.8.3 through use of the gravitational

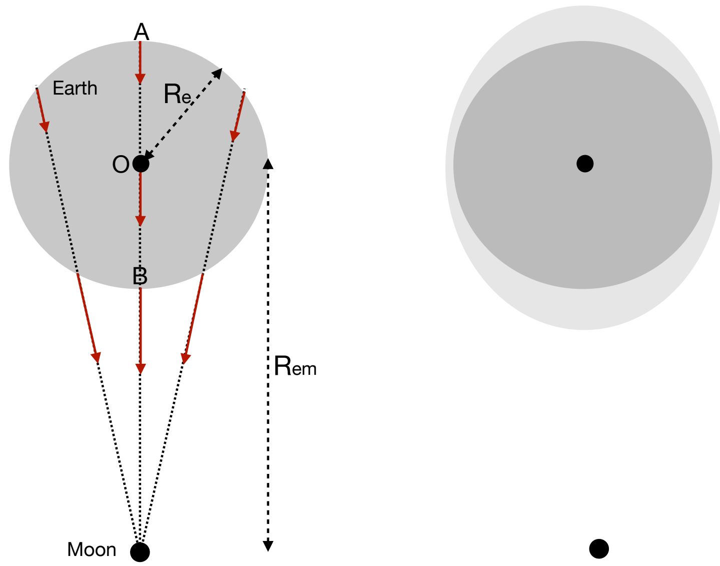

FIGURE 11.8: Illustrating the tidal force on the surface of a sphere. The sphere is an idealized depiction of the Earth and the smaller massive object is the Moon. The distance between the center of the Earth and Moon is R_{em}, and the radius of the Earth is R_e. The left panel shows representative Moon-generated gravitational field lines. Two points along these field lines on the surface of the Earth represent the two ends of an imaginary rod as depicted in Figure 11.7. The tidal acceleration acting at point B, relative to the Earth's center, points toward the Moon (equation (11.137)). In contrast, the tidal acceleration at point A, relative to the Earth's center, points in the opposite direction (equation (11.138)). Points on the Earth surface between A and B have tidal accelerations with a non-zero component directed along the surface of the Earth. Symmetry of the configuration allows us to conclude that a layer of water on the surface of the sphere will accumulate to produce two bulges as shown in the right panel. It is the lateral component of the gravitational acceleration that causes the water to accumulate to produce tidal bulges at points A and B. In contrast, the radial component to the Moon's gravitational field has no contribution to the tides. Note that as shown in Section 11.8.3, the bulge shown in the right panel is greatly exaggerated.

potential. For the current discussion we appeal to symmetry to conclude that the lateral tidal accelerations act to pile up water at both points A and B as depicted in the second panel of Figure 11.8. This argument, though heuristic, provides the means to understand how a water covered spherical planet has two bulges, rather than one, due to spatial gradients in the Moon's gravitational field. We confirm this argument in Section 11.8.3 by explicitly computing the gravitational potential for this idealized Earth-Moon system and then taking the gradient to compute the gravitational acceleration (see Figure 11.10).

Including orbital motion

Thus far we have ignored the orbital motion of the Earth-Moon system around their common center of mass. As we will see, there are no fundamental changes to the above arguments when allowing for orbital motion.

In the absence of dissipation, as assumed here, the Earth-Moon distance remains constant due to their angular momentum conserving orbital motion. From a force-balance perspective, the two spherical bodies remain in a fixed orbit since the gravitational acceleration acting at their centers is balanced by their respective centrifugal accelerations, where the centrifugal acceleration is computed relative to the center of mass of the two-body system. The gravitational acceleration from the Moon, acting at the center of the Earth, is given by the free fall value $\boldsymbol{g}(R_{em})$, which has magnitude $G\, M_m/R_{em}^2$ and is directed along the axis connecting the Earth and Moon centers.

Furthermore, when a body exhibits orbital motion, each point on the body exhibits the same orbital motion and has the same linear velocity. Consequently, each point on the Earth possess the same centrifugal acceleration

$$\boldsymbol{a}_{\text{orbital centrifugal}} = -\boldsymbol{g}(R_{\mathsf{em}}). \tag{11.139}$$

This property of orbital motion is distinct from the spinning motion of a planet rotating about its axis, whereby points further from the rotational axis have larger centrifugal acceleration (see Section 11.7). To help understand orbital motion, move your hand in a circle while maintaining the arm in a single direction so that the hand exhibits an orbital motion rather than a spinning motion. Notice that all parts of the hand move with the same linear velocity and exhibit the same orbital motion. Hence, each point on the hand has the same centrifugal acceleration.

We can now ask about the acceleration felt by a point on the surface of the Earth. The acceleration giving rise to tidal motions is the sum of the gravitational acceleration from the Moon plus the centrifugal acceleration due to orbital motion. However, this calculation is identical to that considered previously, which led, for example, to the tidal accelerations for points B and A as given by equations (11.137) and (11.138). We are thus led to the same result as before.

11.8.3 Gravitational potential for an idealized Earth-Moon system

We now perform a more thorough calculation of the gravitational acceleration by computing the gradient of the gravitational potential. First recall the discussion of Newton's gravitational law in Section 11.7.4, whereby the gravitational potential for a point at distance r from the center of a spherical Earth is given by

$$\Phi_{\mathsf{e}}(r) = -\frac{G M_{\mathsf{e}}}{r}, \tag{11.140}$$

where M_{e} is the mass of the Earth. The corresponding radial gravitational acceleration is given by

$$\boldsymbol{g}_{\mathsf{e}} = -\nabla\Phi_{\mathsf{e}} = -\frac{G M_{\mathsf{e}}\,\hat{\boldsymbol{r}}}{r^2}. \tag{11.141}$$

The same considerations hold for the Moon's gravitational potential. Hence, referring to Figure 11.9, the Moon's gravitational potential evaluated at a distance L from the Moon's center is given by

$$\Phi_{\mathsf{m}}(L) = -\frac{G M_{\mathsf{m}}}{L}. \tag{11.142}$$

Trigonometry leads to the law of cosines relation

$$L^2 = (R_{\mathsf{em}} - r\cos\psi)^2 + (r\sin\psi)^2 = R_{\mathsf{em}}^2 + r^2 - 2\,r\,R_{\mathsf{em}}\cos\psi, \tag{11.143}$$

where r is the distance to the Earth's center and ψ is the polar angle relative to the $\hat{\boldsymbol{x}}$ axis pointing between the Earth and Moon centers (see Figure 11.9).

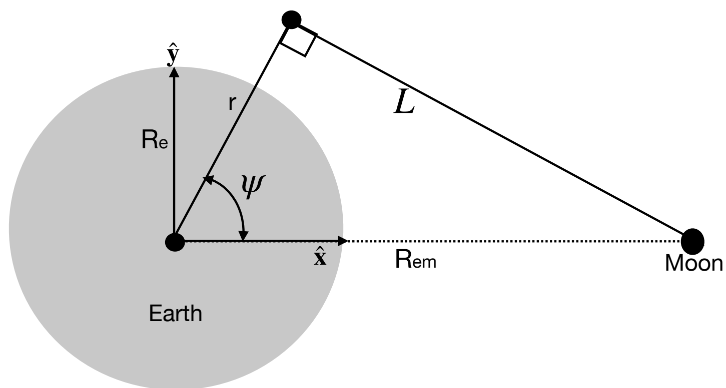

FIGURE 11.9: Geometry of an idealized Earth-Moon system. The center of the Earth is a distance R_{em} from the center of the Moon; the Moon has a mass M_{m}; and the Earth has a radius R_{e}. An arbitrary test point is shown a distance L from the center of the Moon, r from the center of the Earth, and with a polar angle ψ relative to the $\hat{\boldsymbol{x}}$ axis, where the $\hat{\boldsymbol{x}}$ axis points from the Earth center to the Moon center. Relative to the Earth's center, the test point has Cartesian coordinates $(x, y) = r(\cos\psi, \sin\psi)$, using equations from Section 4.3 for relating polar and Cartesian coordinates.

Identifying the leading order contributions

Assuming the test point in Figure 11.9 is closer to the Earth than to the Moon, we can perform a Taylor series expansion in the small parameter r/R_{em} to render

$$\Phi_{\text{m}}(L) = -\frac{G M_{\text{m}}}{L} = -\frac{G M_{\text{m}}}{R_{\text{em}}} \left[1 + \frac{r\cos\psi}{R_{\text{em}}} + \frac{r^2}{2R_{\text{em}}^2}(3\cos^2\psi - 1) + \mathcal{O}(r/R_{\text{em}})^3 \right]. \tag{11.144}$$

We thus identify the leading three terms to the geopotential

$$\Phi_{\text{m}}^{(0)} = -\frac{G M_{\text{m}}}{R_{\text{em}}} \tag{11.145}$$

$$\Phi_{\text{m}}^{(1)} = -\frac{G M_{\text{m}}}{R_{\text{em}}^2}\, r\cos\psi \tag{11.146}$$

$$\Phi_{\text{m}}^{(2)} = -\frac{G M_{\text{m}}}{2R_{\text{em}}^3}\, r^2 (3\cos^2\psi - 1). \tag{11.147}$$

Assuming the distance between the Earth and Moon remains fixed, the zeroth order term, $\Phi_{\text{m}}^{(0)}$, is a spatial constant and thus leads to no gravitational acceleration. We now examine the gravitational accelerations from the other two terms.

Acceleration maintaining the orbiting Earth-Moon system

For the first order term, $\Phi_{\text{m}}^{(1)}$, we introduce the Cartesian coordinate as in Figure 11.9 to write

$$\Phi_{\text{m}}^{(1)} = -\frac{G M_{\text{m}}\, x}{R_{\text{em}}^2}, \tag{11.148}$$

where $x = r\cos\psi$ is the distance along $\hat{\boldsymbol{x}}$. Hence, the gradient of $\Phi_{\text{m}}^{(1)}$ leads to the gravitational acceleration

$$\boldsymbol{g}_{\text{m}}^{(1)} = -\nabla\Phi_{\text{m}}^{(1)} = \hat{\boldsymbol{x}}\, \frac{G M_{\text{m}}}{R_{\text{em}}^2}. \tag{11.149}$$

This gravitational acceleration has a constant magnitude at every point in space and it everywhere points in a direction parallel to the Earth-Moon axis. Furthermore, the magnitude of $\boldsymbol{g}_{\mathsf{m}}^{(1)}$ equals to that of the Moon's gravitational acceleration, $\boldsymbol{g}_{\mathsf{m}}$, when evaluated at the Earth's center. As seen in Section 11.8.2, the acceleration, $\boldsymbol{g}_{\mathsf{m}}^{(1)}$, maintains the Earth in orbit about the center of mass for the Earth-Moon system; i.e., this is the free fall acceleration towards the Moon. Notably, at the Earth's surface, the magnitude of $\boldsymbol{g}_{\mathsf{m}}^{(1)}$ is tiny relative to the gravitational acceleration from the Earth itself, with their ratios given by

$$\frac{M_{\mathsf{m}}/R_{\mathsf{em}}^2}{M_{\mathsf{e}}/R_{\mathsf{e}}^2} \approx 3.4 \times 10^{-6}, \tag{11.150}$$

where we set

$$M_{\mathsf{e}} = 5.97 \times 10^{24}\ \text{kg} \qquad M_{\mathsf{m}} = 7.35 \times 10^{22}\ \text{kg} = (1/81.2)\, M_{\mathsf{e}} \tag{11.151a}$$

$$R_{\mathsf{e}} = 6.367 \times 10^{6}\ \text{m} \qquad R_{\mathsf{em}} = 3.84 \times 10^{8}\ \text{m} = 60.3\, R_{\mathsf{e}}. \tag{11.151b}$$

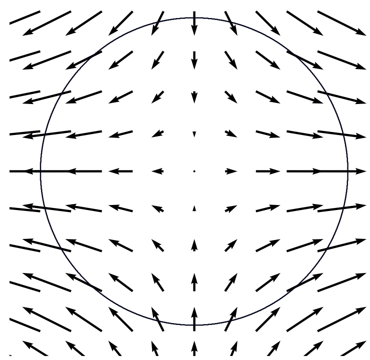

FIGURE 11.10: The tide producing gravitational acceleration $\boldsymbol{g}_{\mathsf{m}}^{(2)}$ given by equation (11.155). The Moon is assumed to be positioned in the equatorial plane of the earth.

Tide producing geopotential

The main tide producing acceleration results from $\Phi_{\mathsf{m}}^{(2)}$. Introducing the second Cartesian coordinate, $y = r \sin\psi$, leads to

$$\Phi_{\mathsf{m}}^{(2)} = -\frac{G M_{\mathsf{m}}}{2 R_{\mathsf{em}}^3}\, r^2\, (3\cos^2\psi - 1) = -\frac{G M_{\mathsf{m}}}{2 R_{\mathsf{em}}^3}\, (2\, x^2 - y^2). \tag{11.152}$$

The corresponding perturbed geopotential height field is given by

$$h = -\frac{\Phi_{\mathsf{m}}^{(2)}}{g} = \frac{R_{\mathsf{e}}^2}{2\, R_{\mathsf{em}}^3}\, \frac{M_{\mathsf{m}}}{M_{\mathsf{e}}}\, r^2\, (3\cos^2\psi - 1). \tag{11.153}$$

Placing the test point on the Earth surface, $r = R_{\mathsf{e}}$, renders

$$h = \frac{R_{\mathsf{e}}^4}{2\, R_{\mathsf{em}}^3}\, \frac{M_{\mathsf{m}}}{M_{\mathsf{e}}}\, (3\cos^2\psi - 1) \approx 2.8 \times 10^{-8}\, R_{\mathsf{e}}\, (3\cos^2\psi - 1). \tag{11.154}$$

Plugging in numbers for the Earth-Moon system suggests that the maximum perturbation to the geopotential height arising from the Moon's gravity field is roughly 36 cm. Correspondingly, the bulge shown in Figure 11.8 is greatly exaggerated. Note that ocean tidal amplitudes can get much larger (order meters) than this equilibrium tide amplitude due to resonances from ocean geometry, with the Bay of Fundy in Nova Scotia a particularly striking example.

Tide producing acceleration

The gravitational acceleration arising from the tidal potential is determined by the gradient of the tidal geopotential

$$\boldsymbol{g}_{\mathsf{m}}^{(2)} = -\nabla \Phi_{\mathsf{m}}^{(2)} = \frac{G M_{\mathsf{m}}}{R_{\mathsf{em}}^{3}} \, (2\, x\, \hat{\boldsymbol{x}} - y\, \hat{\boldsymbol{y}}). \tag{11.155}$$

We illustrate the vector field $\boldsymbol{g}_{\mathsf{m}}^{(2)}$ in Figure 11.10. Note how the accelerations lead to two bulges on opposite sides of the planet. We can write this acceleration using polar coordinates by introducing the polar unit vectors $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\psi}}$ according to Section 4.3.2

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}} \cos\psi + \hat{\boldsymbol{y}} \sin\psi \tag{11.156a}$$

$$\hat{\boldsymbol{\psi}} = -\hat{\boldsymbol{x}} \sin\psi + \hat{\boldsymbol{y}} \cos\psi \tag{11.156b}$$

thus rendering

$$\boldsymbol{g}_{\mathsf{m}}^{(2)} = \frac{G M_{\mathsf{m}} R_{\mathsf{e}}}{R_{\mathsf{em}}^{3}} \left[\hat{\boldsymbol{r}}\, (3\cos^{2}\psi - 1) - (3/2)\, \hat{\boldsymbol{\psi}}\, \sin 2\psi \right], \tag{11.157}$$

where we evaluated the acceleration at the Earth surface so that $r = R_{\mathsf{e}}$. Evaluating the acceleration at $\psi = 0, \pi$ verifies the heuristic calculation performed in Section 11.8.2 for points on the Earth surface nearest and furthest from the Moon. We can further gauge the magnitude of the tidal acceleration by introducing the acceleration due to the Earth's gravity field

$$\boldsymbol{g}_{\mathsf{m}}^{(2)} = g_{\mathsf{e}} \frac{M_{\mathsf{m}}}{M_{\mathsf{e}}} \frac{R_{\mathsf{e}}^{3}}{R_{\mathsf{em}}^{3}} \left[\hat{\boldsymbol{r}}\, (3\cos^{2}\psi - 1) - (3/2)\, \hat{\boldsymbol{\psi}}\, \sin 2\psi \right], \tag{11.158}$$

where $g_{\mathsf{e}} = G\, M_{\mathsf{e}}/R_{\mathsf{e}}^{2}$ is the acceleration at the Earth's surface from the Earth's gravity field. The dimensional prefactor has magnitude $\approx 5.6 \times 10^{-8}\, g_{\mathsf{e}}$, so that the tidal acceleration is tiny relative to that from the Earth's gravity field. It is for this reason that the radial component of the tidal acceleration is largely irrelevant since it is dominated by the far larger radial component of the Earth's gravity field. However, the angular component of the tidal acceleration, although small relative to the Earth's radial gravitational acceleration, is able to move water along the surface of the planet as indicated by Figure 11.10, thus leading to tidal motion.

11.8.4 Concerning realistic ocean tides

Our discussion of tides has been rather terse, aiming to identify key aspects of the tidal accelerations but giving little attention to details that impact real ocean or solid-earth tides. Here are a few points that must be considered for these purposes.

- As the Earth spins under the tidal bulges, there are two high and two low tides per day. Additional orbital motion of the Moon adds roughly 50 minutes per day to the diurnal (daily) tide and 25 minutes to the semi-diurnal (twice daily).
- The Moon orbits the Earth at a latitude of roughly $28.5°N$ rather than within the equatorial plane, so that the tidal bulges are offset from the equator. As the Earth spins under the bulges, one of the high tides is generally larger than the other due to the offset. This offset in turn introduces a diurnal component to the tides in addition to the semi-diurnal.

- The Sun contributes to tides in a manner similar to the Moon. The sun is more massive than the Moon, yet it is further away, so that the ratio of the magnitudes for the tidal producing accelerations is given by

$$\frac{\text{Moon tidal acceleration}}{\text{Sun tidal acceleration}} = \frac{M_{\mathsf{m}}/R_{\mathsf{em}}^3}{M_{\mathsf{s}}/R_{\mathsf{es}}} \approx 2.2 \tag{11.159}$$

where we set

$$M_{\mathsf{s}} = 1.99 \times 10^{30}\ \text{kg} \qquad R_{\mathsf{es}} = 23460\, R_{\mathsf{e}}. \tag{11.160}$$

Hence, the Moon has an impact on Earth tides that is a bit more than twice that of the Sun.
- The gravitational acceleration that leads to the tidal bulge moves around the mid-latitudes at roughly 330 m s^{-1}, which is faster than the $\approx$ 200 m s^{-1} wave speed for shallow water gravity waves. Hence, the ocean tidal motion is never equilibrated to the equilibrium tide defined by the tidal acceleration. In contrast, solid-Earth waves are much faster and so the solid-Earth tidal motions are mostly equilibrated with the equilibrium tidal acceleration. Solid-Earth tides have an amplitude on the order of 10 cm with wavelengths spanning the planet. Hence, an accurate treatment of ocean tides must take into account the solid-Earth tides.
- The movement of ocean mass modifies the Earth's gravity field and the domain within which the ocean moves, and these effects are referred to as self-attraction and loading (SAL). The loading term arises from alterations in the mass felt by the solid Earth that leads the crust to expand or compress. Self-attraction arises from modifications to the gravity field due to self-gravity of both the load-deformed solid Earth and the ocean tide itself. Locally, SAL can contribute to roughly 20% of the ocean tide amplitude, so that accurate tide modeling must include SAL. *Barton et al.* (2022) provide an example of global ocean tide modeling that includes a discussion of how to compute the SAL terms online during a computer simulation.
- Geometry of the ocean plays a leading role in determining tides at a particular location. We have incomplete information about the geometry of ocean basins, such as the presence of deep ridges, troughs, and seamounts. In lieu of such information we can garner useful information based on the analysis of past tides, with that information used to fit sinusoidal waves to the measured time series for use in projecting forward in time.

11.8.5 Comments

We emphasize that the tidal producing acceleration is the lateral (along-Earth) component of the Moon's tidal gravitational acceleration that produces the Earth's tides. These lateral forces cause water to accumulate at the point nearest to and furthest from the Moon (points A and B in Figure 11.8), thus producing the characteristic double-bulge pattern. Notably, many common literature presentations make it appear that it is the radial (i.e., pointing to the Earth's center) component of the Moon's gravitational force, and its gradient across the Earth, that leads to the Earth's tidal bulges. But as discussed in Section 11.8.2, radial gravitational forces cannot lead to tidal motions; what is needed is a force that leads to lateral motion. These key notions are nicely emphasized in this Space Time video.

Further presentations of tidal accelerations, at the level presented here, can be found in Chapter 3 of *Pugh* (1987), Section 5.15 of *Apel* (1987), Chapter 2 of *Brown* (1999), Section 17.4 of *Stewart* (2008) and Chapter 7 of *Lautrup* (2005).

11.9 Exercises

EXERCISE 11.1: ACCELERATION WITH $\mathrm{d}\boldsymbol{\Omega}/\mathrm{d}t \neq 0$

Geophysical applications warrant taking the planetary rotation to be a constant vector, $\mathrm{d}\boldsymbol{\Omega}/\mathrm{d}t = 0$, and that assumption is made throughout this book. However, for some applications, such as for rotating machines, we cannot make that assumption. What extra term appears in the acceleration vector (11.57) when $\mathrm{d}\boldsymbol{\Omega}/\mathrm{d}t \neq 0$? Hint: rework the derivation of equation (11.74d) retaining $\mathrm{d}\boldsymbol{\Omega}/\mathrm{d}t \neq 0$.

EXERCISE 11.2: KINETIC ENERGY OF RIGID-BODY ROTATIONS

Consider a particle undergoing rigid-body rotational motion so that $\mathbf{V} = \boldsymbol{\Omega} \times \mathbf{X}$ (equation (11.35)). Show that the kinetic energy for the particle is given by

$$K = \tfrac{1}{2}\, \boldsymbol{\Omega} \cdot \mathbf{M} \cdot \boldsymbol{\Omega} = \tfrac{1}{2}\, \Omega_m\, M^{mn}\, \Omega_n, \tag{11.161}$$

with $\mathbf{M}$ the moment of inertia tensor with components (11.50).

EXERCISE 11.3: FREE LINEAR OSCILLATOR

Consider one-dimensional motion with a linear restoring force

$$F = -k\, X, \tag{11.162}$$

where $k > 0$ is a constant. For small amplitude oscillations of a spring, k is referred to as the spring constant.

(a) Write Newton's equation of motion for this force when acting on a particle of constant mass, m.
(b) What is the position of the particle that started at the origin at time $t = 0$ with velocity $v_\circ$?
(c) What is the angular frequency of the motion?
(d) What is the potential energy for the particle?
(e) What is the kinetic energy for the particle?
(f) What is the mechanical energy for the particle?

EXERCISE 11.4: TIME DEPENDENT FORCING

Consider one-dimensional motion with a forcing that is an explicit function of time

$$F = F(t). \tag{11.163}$$

Derive an expression for the particle position, $X(t)$ relative to its initial position, $X(t_0) = x_\circ$ and initial velocity, $\dot{X}(t_0) = u_\circ$. Hint: the solution should be written in the form of a double time integral.

EXERCISE 11.5: FALLING BODY WITH RAYLEIGH DRAG

Consider a particle falling vertically ($-\hat{\boldsymbol{z}}$ direction) under the effects from a constant gravitational field as well as Rayleigh drag friction, so that the force acting on the particle is

$$F = -m\, g - m\, \gamma\, \dot{Z}, \tag{11.164}$$

where $g > 0$ is a constant gravitational acceleration, and $\gamma > 0$ is the constant Rayleigh drag coefficient.

(a) Write Newton's equation of motion for this force when acting on a particle of constant mass, m.
(b) Determine the velocity, $\dot{Z}$, for a particle starting at rest at time $t = 0$.
(c) Write the the velocity for short times, $t \ll \gamma^{-1}$, to second order accuracy (i.e., high enough order to see the role of γ).
(d) What is the terminal velocity, defined as the velocity at long times, $t \gg \gamma^{-1}$? Only go to leading order.
(e) What is the particle trajectory, $Z(t)$?
(f) Write the particle trajectory for short times, $t \ll \gamma^{-1}$? Expand the exponential sufficiently to include contributions from γ.

EXERCISE 11.6: PARTICLE FALLING INTO A MINE SHAFT
Consider a particle falling radially into a mine shaft on a non-rotating and spherical Earth. Assume the particle starts from rest at the Earth surface, and assume the Earth has a homogeneous composition. Hint: make use of the gravitational potential from Exercise 11.8.

(A) What is the time it takes for the particle to reach the center of the Earth.
(B) What is the speed of the particle when it reaches the center?
(C) Describe the motion of the particle after it passes the Earth center.

EXERCISE 11.7: GRAVITATIONAL FIELD IN A UNIFORM DENSITY UNIVERSE
Consider the case of a Newtonian universe everywhere filled with a constant density matter. What is the gravitational potential and gravitational acceleration for this universe? Note that this universe is clearly unrealistic. Even so, it provides a useful starting point in the study of Newtonian cosmology.

EXERCISE 11.8: GRAVITATIONAL POTENTIAL INSIDE A CONSTANT DENSITY SPHERE
In Section 11.7.4 we found the gravitational, Φ_{e}, for a point outside of the spherical Earth. Here we find the gravitational potential inside of the Earth, assumed to have a constant mass density ρ. That is, solve the Poisson equation for the gravitational potential,

$$\nabla^2 \Phi_{\text{e}} = 4\,\pi\,G\,\rho, \tag{11.165}$$

for a radial position $r < R_{\text{e}}$. Hint: follow the approach in Section 11.7.2 and then match the potential at $r = R_{\text{e}}$ to that for the region with $r > R_{\text{e}}$. Hint: the solution can be found in many undergraduate physics texts.

EXERCISE 11.9: GRAVITATIONAL FORCE ACTING ON A MASS DISTRIBUTION
Equation (11.112) provides an expression for the gravitational force acting on an infinitesimal mass distribution. Here we consider two mass distributions, $\rho_1(\boldsymbol{x}_1)$ and $\rho_2(\boldsymbol{x}_2)$. Let $\boldsymbol{g}$ be the gravitational acceleration generated by the mass distribution $\rho_1(\boldsymbol{x}_1)$. Write the gravitational force acting on a finite region of mass distribution ρ_2. Verify that this force satisfies Newton's third law. Assume Cartesian coordinates.

EXERCISE 11.10: GRAVITATIONAL FIELD WITH A POWER LAW MASS DISTRIBUTION
Consider a spherically symmetric mass distribution given by the power law expression

$$\rho(r) = \begin{bmatrix} A\,r^{\alpha} & r \le R \\ 0 & r > R, \end{bmatrix} \tag{11.166}$$

where $A > 0$ is a constant. The case with $\alpha = 0$ corresponds to the constant mass density considered in Exercise 11.8.

(A) What is the mass of a sphere, $M(r)$, as a function of r? What is the value for α that ensures a finite mass?

(B) Determine the gravitational potential for the region $r > R$, assuming it vanishes as $r \to \infty$. Determine the corresponding gravitational acceleration.

(C) Determine the gravitational potential and gravitational acceleration for the region $0 \leq r \leq R$. Be sure that the potential matches at $r = R$ with the potential computed for $r \geq R$. Assume $\alpha \neq -2$.

(D) Determine the gravitational potential and gravitational acceleration for the region $0 \leq r \leq R$, and set $\alpha = -2$. Be sure that the potential matches at $r = R$ with the potential computed for $r \geq R$.

(E) Are there values for α that lead to a gravitational acceleration that increases moving toward the center?

Chapter 12

Motion on a rotating planet

In this chapter we study the Newtonian mechanics of a point particle of fixed mass that moves around a rotating and massive planet. We assume motion of the planet is prescribed with a prescribed angular momentum around a fixed axis of rotation. Hence, we are only concerned with mechanics of the moving particle. We generally ignore friction and other non-gravitational forces, so that the only inertial force acting on the particle (force as viewed in an inertial reference frame) is that from the gravitational field produced by the massive planet. Our description of the motion is examined from the rigid-body rotating (non-inertial) reference frame (Section 11.5). This non-inertial reference frame is natural for an observer fixed on the planet surface, and it is central to our description of geophysical fluid motion. We must include the planetary centrifugal acceleration and planetary Coriolis acceleration when working within the rotating terrestrial reference frame.

CHAPTER GUIDE

We build from the Newtonian mechanics of Chapter 11, specializing to the geophysically relevant case of a particle moving around a rotating planet and as viewed from the rotating planet reference frame. We use planetary Cartesian coordinates and planetary spherical coordinates, both with their coordinate origin at the planet center and both that rotate with the planet. Furthermore, we use elements of tensor analysis from Part I, though here provide a synopsis of the salient features to keep the mathematical presentation reasonably self-contained. The mechanics in this chapter are fundamental to the mechanics of the earth's atmosphere and ocean as described by the frame of a planetary observer.

12.1 The rotating Earth

The Earth's angular velocity is comprised of two main contributions: the spinning motion of the Earth about its axis and the orbital motion of the Earth about the sun (see Figure 12.1). Other astronomical motions can be neglected for our purposes in studying geophysical fluid mechanics. Therefore, in the course of a single period of 24 hours, or $24 \times 3600 = 86400$ seconds, the Earth experiences an angular rotation of $(2\pi + 2\pi/365.24)$ radians. As such, the angular rotation rate is given by

$$\Omega = \frac{2\pi + 2\pi/365.24}{86400 \text{ s}} = \left[\frac{\pi}{43082}\right] \text{s}^{-1} = 7.2921 \times 10^{-5} \text{ s}^{-1}. \tag{12.1}$$

The Earth's angular velocity, both its direction through the polar axis and its magnitude, is assumed constant in time for our studies

$$\frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} = 0. \tag{12.2}$$

To get a sense for the magnitude of the angular velocity (12.1), consider a terrestrial

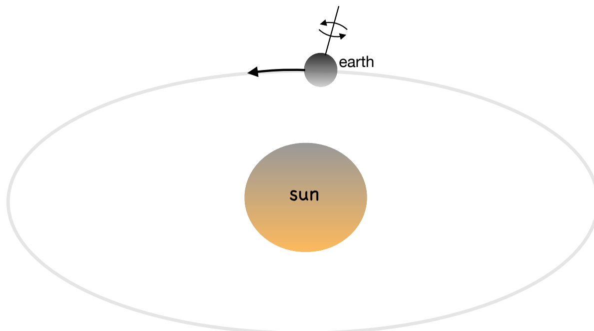

FIGURE 12.1: The angular velocity of the Earth arises from the spin about the polar axis plus the orbit of the planet around the sun. This angular velocity determines the strength of the Coriolis acceleration and the planetary centrifugal acceleration. The polar axis has an angle of roughly 23.4° relative to the orbital/ecliptic plane with respect to the sun. This angle is referred to as the *obliquity*, and the obliquity is the reason for the seasonal cycle (e.g., the summer season occurs in the hemisphere receiving more solar radiation). For an interesting perspective on planetary rotation, see this video from Veritasium.

reference frame at the surface of the Earth surface that undergoes rigid-body rotation with speed

$$U_{\text{rigid}}(\phi) = \Omega\, R_{\text{e}} \cos\phi \approx 465 \text{ m s}^{-1} \cos\phi = 1672 \text{ km hr}^{-1} \cos\phi, \tag{12.3}$$

where we set the Earth's radius to

$$R_{\text{e}} = 6.371 \times 10^{6}\,\text{m}. \tag{12.4}$$

This speed is quite large relative to a fixed frame outside the planet (see Figure 12.2)

$$U_{\text{rigid}}(0^{\circ}) = 1672 \text{ km hr}^{-1} \tag{12.5a}$$
$$U_{\text{rigid}}(30^{\circ}) = 1448 \text{ km hr}^{-1} \tag{12.5b}$$
$$U_{\text{rigid}}(60^{\circ}) = 826 \text{ km hr}^{-1}. \tag{12.5c}$$

Motion of the atmosphere and ocean are generally close to a rigid-body rotation, so that their speeds relative to the planet are much smaller than the speed of the rotating planet itself. So in addition to being directly relevant to terrestrial observers, we find it most interesting to study geophysical motion in the rotating planetary frame (a non-inertial frame) rather than a frame fixed relative to the stars (an approximate inertial frame).

12.2 Changes to basis vectors under rigid-body rotation

In Section 11.3, we studied the kinematics of a vector with fixed length that is rotating about a fixed origin. The key kinematic result (11.42) exposes how rotation generates a time change to the rigid-body rotation of a vector

$$\frac{\mathrm{d}\mathbf{Q}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \mathbf{Q}, \tag{12.6}$$

where $\mathbf{Q}$ is an arbitrary vector. We here make use of this relation to determine how the basis vectors for Cartesian coordinates, cylindrical-polar coordinates, and spherical coordinates change under rotation. For each of these coordinate choices, we assume the vertical axis is aligned with the rotation axis and with the $\hat{\boldsymbol{z}}$ direction pointing out from the north pole as in Figure 12.2.

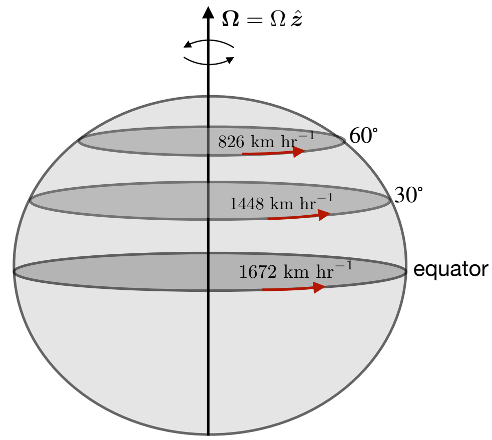

FIGURE 12.2: For an observer at rest on the Earth's surface, the rigid-body speed is $\Omega\,R_{e}\cos\phi = 7.2921 \times 10^{-5}\ \mathrm{s}^{-1} \times 6.371 \times 10^{6}\ \mathrm{m}\cos\phi \approx 465\ \mathrm{m\ s}^{-1}\cos\phi = 1672\ \mathrm{km\ hr}^{-1}\cos\phi$. We here display rigid-body speeds for motion at the equator, 30° latitude, and 60° latitude. These speeds are much larger than motion of the ocean and atmosphere relative to the rotating Earth, thus motivating a description of geophysical fluid motion from the rotating terrestrial frame moving in rigid-body rotation.

12.2.1 Changes to Cartesian basis vectors under rotation

With $\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}$ as for a rotating spherical Earth, equation (12.6) says that the change in planetary Cartesian unit vectors under rigid-body rotation is given by

$$\frac{\mathrm{d}\hat{\boldsymbol{x}}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\hat{\boldsymbol{x}} = \Omega\,\hat{\boldsymbol{y}} \quad \text{and} \quad \frac{\mathrm{d}\hat{\boldsymbol{y}}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\hat{\boldsymbol{y}} = -\Omega\,\hat{\boldsymbol{x}} \quad \text{and} \quad \frac{\mathrm{d}\hat{\boldsymbol{z}}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\hat{\boldsymbol{z}} = 0. \tag{12.7}$$

12.2.2 Changes to cylindrical-polar coordinate unit vectors under rotation

Now consider the polar coordinates from Section 4.3 with rotation continuing to be oriented about the vertical axis, $\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}$. In the horizontal plane we have the radial and angular orthonormal anholonomic basis vectors

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}}\cos\vartheta + \hat{\boldsymbol{y}}\sin\vartheta \tag{12.8a}$$
$$\hat{\boldsymbol{\vartheta}} = -\hat{\boldsymbol{x}}\sin\vartheta + \hat{\boldsymbol{y}}\cos\vartheta. \tag{12.8b}$$

We can directly verify that the time derivative of these unit vectors is given by

$$\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t} = (\Omega + \dot{\vartheta})\,\hat{\boldsymbol{\vartheta}} = (\Omega + \dot{\vartheta})\,(\hat{\boldsymbol{z}}\times\hat{\boldsymbol{r}}) \tag{12.9a}$$
$$\frac{\mathrm{d}\hat{\boldsymbol{\vartheta}}}{\mathrm{d}t} = -(\Omega + \dot{\vartheta})\,\hat{\boldsymbol{r}} = (\Omega + \dot{\vartheta})\,(\hat{\boldsymbol{z}}\times\hat{\boldsymbol{\vartheta}}). \tag{12.9b}$$

Notice how these unit vectors change both due to rotation of the reference frame, Ω, plus changes in the angular position relative to the reference frame, $\dot{\vartheta}$. We thus see that cylindrical-polar unit vectors change due to the rigid-body rotation, just like the Cartesian unit vectors. Additionally, non-Cartesian unit vectors change if their orientation is modified relative to the Cartesian coordinate axes at a rate distinct from the rigid-body (i.e., from $\dot{\vartheta} \neq 0$). This situation occurs when a particle trajectory moves relative to the rigid-body, with further discussion for spherical coordinates given in Sections 12.7.2 and 12.9.

12.2.3 Changes to spherical coordinate unit vectors under rotation

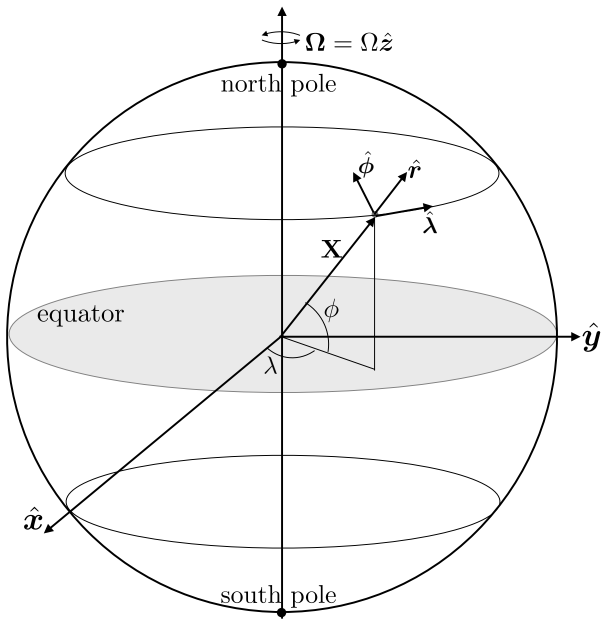

FIGURE 12.3: Geometry and notation for motion around a rotating sphere using planetary spherical and planetary Cartesian coordinates. For geophysical applications, the sphere rotates counter-clockwise when looking down from the north polar axis and it has an angular speed $\Omega > 0$. We assume that Ω accouts for both the spinning motion of the planet around its axis, plus the smaller contribution from the orbital motion of the planet around the sun (see Figure 12.1). The planetary Cartesian triad of orthonormal basis vectors, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$, points along the orthogonal axes and rotates with the sphere. The planetary spherical triad (also rotating with the sphere) of orthonormal non-coordinate basis vectors, $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, makes use of the longitudinal unit vector $\hat{\boldsymbol{\lambda}}$, which points in the longitudinal direction (positive eastward), the latitudinal unit vector $\hat{\boldsymbol{\phi}}$, which points in the latitudinal direction (positive northward) and the radial unit vector $\hat{\boldsymbol{r}}$, which point in the radial direction (positive away from the center).

As a third example, consider the rotation of spherical coordinate orthonormal anholonomic basis vectors depicted in Figure 12.3, whose form is derived in Section 4.4.2

$$\hat{\boldsymbol{\lambda}} = -\hat{\boldsymbol{x}} \sin\lambda + \hat{\boldsymbol{y}} \cos\lambda \tag{12.10a}$$

$$\hat{\boldsymbol{\phi}} = -\hat{\boldsymbol{x}} \cos\lambda \, \sin\phi - \hat{\boldsymbol{y}} \sin\lambda \, \sin\phi + \hat{\boldsymbol{z}} \cos\phi \tag{12.10b}$$

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}} \, \cos\lambda \, \cos\phi + \hat{\boldsymbol{y}} \sin\lambda \, \cos\phi + \hat{\boldsymbol{z}} \sin\phi. \tag{12.10c}$$

Again, each of the spherical basis vectors, $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, are normalized, so that their evolution arises just from rotations.

Start with deriving the time derivative of the radial unit vector, $\hat{\boldsymbol{r}}$. We know that it can rotate either through rigid-body motion of the rotating reference frame, or through changes in the spherical angles, (λ, ϕ), relative to the rotating reference frame. We see these two forms of time changes by taking the time derivative

$$\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\hat{\boldsymbol{x}} \, \cos\lambda \, \cos\phi + \hat{\boldsymbol{y}} \sin\lambda \, \cos\phi + \hat{\boldsymbol{z}} \sin\phi). \tag{12.11}$$

Expanding the right hand side leads to

$$\frac{\mathrm{d}}{\mathrm{d}t}(\hat{\boldsymbol{x}} \, \cos\lambda \, \cos\phi) = \frac{\mathrm{d}\hat{\boldsymbol{x}}}{\mathrm{d}t} \cos\lambda \cos\phi - \hat{\boldsymbol{x}} \, \dot{\lambda} \sin\lambda \cos\phi - \hat{\boldsymbol{x}} \, \dot{\phi} \cos\lambda \sin\phi \tag{12.12a}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\hat{\boldsymbol{y}}\sin\lambda\,\cos\phi) = \frac{\mathrm{d}\hat{\boldsymbol{y}}}{\mathrm{d}t}\sin\lambda\cos\phi + \hat{\boldsymbol{y}}\,\dot{\lambda}\cos\lambda\cos\phi - \hat{\boldsymbol{y}}\,\dot{\phi}\sin\lambda\sin\phi \tag{12.12b}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\hat{\boldsymbol{z}}\sin\phi) = \hat{\boldsymbol{z}}\,\dot{\phi}\cos\phi. \tag{12.12c}$$

Making use of equation (12.7) for the change in planetary Cartesian unit vectors due to rotation, and substituting the expressions (4.99a)-(4.99c) for the spherical unit vectors, leads to

$$\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t} = \hat{\boldsymbol{\lambda}}\,(\dot{\lambda}+\Omega)\cos\phi + \hat{\boldsymbol{\phi}}\,\dot{\phi}. \tag{12.13}$$

We thus confirm that changes to $\hat{\boldsymbol{r}}$ arise just from changes to its angular orientation. To help understand this expression, take the case of rigid-body motion, in which $\dot{\lambda} = \dot{\phi} = 0$, so that $\hat{\boldsymbol{r}}$ changes only through motion of the rotating reference frame. Any additional longitudinal motion arising from $\dot{\lambda} \neq 0$ renders further changes in the $\hat{\boldsymbol{\lambda}}$ direction. Finally, meridional motion through $\dot{\phi} \neq 0$ modifies the unit vector in the $\hat{\boldsymbol{\phi}}$ direction. Similar manipulations and considerations lead to the time derivatives for the angular unit vectors

$$\frac{\mathrm{d}\hat{\boldsymbol{\phi}}}{\mathrm{d}t} = -\hat{\boldsymbol{\lambda}}\,(\Omega+\dot{\lambda})\sin\phi - \hat{\boldsymbol{r}}\,\dot{\phi} \tag{12.14a}$$

$$\frac{\mathrm{d}\hat{\boldsymbol{\lambda}}}{\mathrm{d}t} = (\Omega+\dot{\lambda})\,(\hat{\boldsymbol{\phi}}\sin\phi - \hat{\boldsymbol{r}}\cos\phi). \tag{12.14b}$$

Further details for deriving equations (12.13), (12.14a), and (12.14b) are given in Exercise 12.3.

12.3 A synopsis of tensor algebra

We here summarize salient points from tensor algebra that are relevant to the present chapter. Further details can be found in the tensor analysis from Chapters 1 through 4.

12.3.1 The coordinate representation of a tensor

An arbitrary tensor, $\mathbf{Q}$, can be represented in terms of an arbitrary set of coordinates through decomposing the tensor into components aligned according to a set of basis vectors or basis one-forms. For a $(1,0)$ tensor, otherwise known as a vector, we write

$$\mathbf{Q} = Q^{a}\,\boldsymbol{e}_{a}, \tag{12.15}$$

where the Einstein summation convention is followed, whereby repeated indices are summed over their range. In this equation,

$$\boldsymbol{e}_{a} = (\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) \tag{12.16}$$

is a set of linearly independent basis vectors, and Q^{a} is the corresponding coordinate representation of the tensor, $\mathbf{Q}$. We can write the tensor using an alternative set of coordinates, in which case we write

$$\mathbf{Q} = Q^{a}\,\boldsymbol{e}_{a} = Q^{\overline{a}}\,\boldsymbol{e}_{\overline{a}}, \tag{12.17}$$

where

$$\boldsymbol{e}_{\overline{a}} = (\boldsymbol{e}_{\overline{1}}, \boldsymbol{e}_{\overline{2}}, \boldsymbol{e}_{\overline{3}}) \tag{12.18}$$

are the basis vectors in the alternative coordinates, and $Q^{\overline{a}}$ is the corresponding coordinate representation. Note that $\mathbf{Q}$, as a tensor, is a geometric object that remains unchanged upon changing coordinates, whereas the coordinate representation and coordinate basis vectors change.

The basis vectors may be normalized to unit magnitude, as in the case of Cartesian coordinates

$$\boldsymbol{e}_a = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}), \tag{12.19}$$

whereas the spherical coordinate basis

$$\boldsymbol{e}_\lambda = r \cos\phi \, \hat{\boldsymbol{\lambda}} \qquad \text{and} \qquad \boldsymbol{e}_\phi = r \, \hat{\boldsymbol{\phi}} \qquad \text{and} \qquad \boldsymbol{e}_r = \hat{\boldsymbol{r}}, \tag{12.20}$$

is unnormalized (see Section 4.4.2). However, we commonly also use the orthonormal anholonomic basis vectors $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, with details provided in Section 12.3.6.

12.3.2 Notation

We use an upright notation for a tensor, such as F for a scalar (zeroth order or $(0,0)$ tensor), and $\mathbf{Q}$ for a higher order tensor. In contrast, we use the slanted notation for the coordinate representation of a tensor, such as a scalar function, F, or a vector function, $\boldsymbol{Q}$. Basis vectors are always associated with a coordinate system, and so they are slanted as in $\boldsymbol{e}_a$. We have more to say in Section 1.1.2 concerning this notation convention.

The basis vectors in equation (12.15) have a lower index while the coordinate representation of a vector has an upper index. Why? For non-Cartesian coordinates (e.g., spherical), we make a distinction between a coordinate representation with the contravariant index upstairs versus the downstairs covariant index. The metric tensor, $\mathbb{g}$, provides the means to move between the covariant and contravariant representations, so that, for example, $Q^a = \mathbb{g}^{ab} \, Q_b$.

When working with general coordinates, it is necessary to distinguish between a basis vector, $\boldsymbol{e}_a$, and its dual partner known as a one-form, $\boldsymbol{e}^a$. Duality is here defined by the familiar (Euclidean) inner product that leads to the biorthogonality relation

$$\boldsymbol{e}_a \cdot \boldsymbol{e}^b = \delta^b{}_a = \delta_a{}^b, \tag{12.21}$$

with $\delta^b{}_a$ the Kronecker delta tensor (unit tensor) with components that are independent of coordinate system

$$\delta^b{}_a = \begin{cases} 1 & \text{if } b = a \\ 0 & \text{if } b \neq a. \end{cases} \tag{12.22}$$

In linear algebra, a row vector is dual to its column vector, with that analog appropriate for the present context. Cartesian basis vectors equal to the basis one-forms, in which case there is no distinction between contravariant and covariant. However, the distinction is important for general coordinates.

12.3.3 Cartesian and spherical coordinates

Let us write ξ^a for an arbitrary coordinate, and let the unbarred coordinates be planetary Cartesian coordinates,

$$(\xi^1, \xi^2, \xi^3) = (x, y, z), \tag{12.23}$$

and let the barred coordinates be the planetary spherical coordinates

$$(\xi^{\bar{1}}, \xi^{\bar{2}}, \xi^{\bar{3}}) = (\lambda, \phi, r). \tag{12.24}$$

These two sets of coordinates are related by the coordinate transformation (Section 4.4 and Figure 12.3)

$$x = r\cos\phi\,\cos\lambda \qquad \text{and} \qquad y = r\cos\phi\,\sin\lambda \qquad \text{and} \qquad z = r\sin\phi. \tag{12.25}$$

12.3.4 Transformation between coordinate representations

The coordinate representation of a tensor is a subjective analytical description of an objective geometric object. That is, the subjectively chosen coordinate representation of a vector, such as that given by equation (12.15), does not affect the vector as a geometric object. Hence, as noted in Section 12.3.1, we can represent the vector using any arbitrary set of coordinates

$$\mathbf{Q} = Q^{a}\,\boldsymbol{e}_{a} = Q^{\bar{a}}\,\boldsymbol{e}_{\bar{a}}. \tag{12.26}$$

In this equation, $Q^{\bar{a}}$ is the representation of the vector, $\mathbf{Q}$, using a chosen coordinate system specified by the basis vector $\boldsymbol{e}_{\bar{a}}$, whereas Q^{a} is the representation of the vector using the unbarred coordinate system with basis vectors $\boldsymbol{e}_{a}$.

The coordinate representation of a vector transforms when changing coordinates. The transformation is operationally realized by the transformation matrix, which is the matrix composed of the partial derivatives of one coordinate system with respect to the other. This procedure provides an organization of the chain rule for coordinate changes. For example, the relations between the coordinate representation of the velocity vector and acceleration vector, as well as the coordinate basis vectors, are given by

$$V^{\bar{a}} = \Lambda^{\bar{a}}{}_{a}\,V^{a} \qquad \text{and} \qquad A^{\bar{a}} = \Lambda^{\bar{a}}{}_{a}\,A^{a} \qquad \text{and} \qquad \boldsymbol{e}_{\bar{a}} = \Lambda^{a}{}_{\bar{a}}\,\boldsymbol{e}_{a}. \tag{12.27}$$

The general expression for the transformation matrix is given by the matrix of partial derivatives

$$\Lambda^{a}{}_{\bar{a}} = \begin{bmatrix} \partial\xi^{1}/\partial\xi^{\bar{1}} & \partial\xi^{1}/\partial\xi^{\bar{2}} & \partial\xi^{1}/\partial\xi^{\bar{3}} \\ \partial\xi^{2}/\partial\xi^{\bar{1}} & \partial\xi^{2}/\partial\xi^{\bar{2}} & \partial\xi^{2}/\partial\xi^{\bar{3}} \\ \partial\xi^{3}/\partial\xi^{\bar{1}} & \partial\xi^{3}/\partial\xi^{\bar{2}} & \partial\xi^{3}/\partial\xi^{\bar{3}} \end{bmatrix}. \tag{12.28}$$

12.3.5 Planetary spherical and planetary Cartesian coordinates

In this chapter we choose the unbarred coordinates as planetary Cartesian, and the barred coordinates are planetary spherical, in which case we have from Section 4.4.1 the transformation matrix and its inverse

$$\Lambda^{a}{}_{\bar{a}} = \begin{bmatrix} -r\cos\phi\sin\lambda & -r\sin\phi\,\cos\lambda & \cos\phi\cos\lambda \\ r\cos\phi\cos\lambda & -r\sin\phi\,\sin\lambda & \cos\phi\sin\lambda \\ 0 & r\cos\phi & \sin\phi \end{bmatrix} \tag{12.29}$$

$$\Lambda^{\bar{a}}{}_{a} = \frac{1}{r^{2}\cos\phi} \begin{bmatrix} -r\sin\lambda & r\cos\lambda & 0 \\ -r\cos\phi\,\sin\phi\,\cos\lambda & -r\cos\phi\sin\phi\sin\lambda & r\cos^{2}\phi \\ r^{2}\cos^{2}\phi\cos\lambda & r^{2}\,\cos^{2}\phi\sin\lambda & r^{2}\cos\phi\sin\phi \end{bmatrix}. \tag{12.30}$$

It is straightforward to verify the orthonormality relations satisfied by these matrices

$$\Lambda^{\bar{a}}{}_{a}\,\Lambda^{a}{}_{\bar{b}} = \delta^{\bar{a}}{}_{\bar{b}} \qquad \text{and} \qquad \Lambda^{a}{}_{\bar{a}}\,\Lambda^{\bar{a}}{}_{b} = \delta^{a}{}_{b}. \tag{12.31}$$

Hence, if we have the Cartesian representation of a vector, such as the velocity or acceleration, then we can use these transformation rules to derive the spherical representation through matrix multiplication, as per equation (12.27).

12.3.6 Coordinate and non-coordinate basis vectors

As discussed in Section 4.1, it is common when working with curvilinear coordinates on Euclidean space to make use of non-coordinate basis vectors that are orthonormal, and we make use of this basis in this chapter. These basis vectors are often referred to as an anholonomic basis in the literature. For spherical coordinates we have the spherical coordinate basis vectors, $\boldsymbol{e}_{\bar{a}}$, which are not normalized, as well as the orthonormal spherical non-coordinate (anholonomic) basis,

$$\boldsymbol{e}_{\hat{\bar{a}}} = (\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}}). \tag{12.32}$$

The spherical coordinate and spherical non-coordinate basis vectors are related by

$$\boldsymbol{e}_{\bar{1}} = r \cos\phi \, \boldsymbol{e}_{\hat{\bar{1}}} \quad \text{and} \quad \boldsymbol{e}_{\bar{2}} = r \, \boldsymbol{e}_{\hat{\bar{2}}} \quad \text{and} \quad \boldsymbol{e}_{\bar{3}} = \boldsymbol{e}_{\hat{\bar{3}}}, \tag{12.33}$$

and spherical non-coordinate basis is related to the planetary Cartesian basis through

$$\hat{\boldsymbol{\lambda}} = -\hat{\boldsymbol{x}} \sin\lambda + \hat{\boldsymbol{y}} \cos\lambda \tag{12.34a}$$

$$\hat{\boldsymbol{\phi}} = -\hat{\boldsymbol{x}} \cos\lambda \, \sin\phi - \hat{\boldsymbol{y}} \sin\lambda \, \sin\phi + \hat{\boldsymbol{z}} \cos\phi \tag{12.34b}$$

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}} \, \cos\lambda \, \cos\phi + \hat{\boldsymbol{y}} \sin\lambda \, \cos\phi + \hat{\boldsymbol{z}} \sin\phi. \tag{12.34c}$$

In this manner we can represent an arbitrary vector, $\mathbf{F}$, in any of the following equivalent manners

$$\begin{aligned}
\mathbf{F} &= F^{1} \, \boldsymbol{e}_{1} + F^{2} \, \boldsymbol{e}_{2} + F^{3} \, \boldsymbol{e}_{3} && \text{Cartesian representation} && (12.35a)\\
&= F^{\bar{1}} \, \boldsymbol{e}_{\bar{1}} + F^{\bar{2}} \, \boldsymbol{e}_{\bar{2}} + F^{\bar{3}} \, \boldsymbol{e}_{\bar{3}} && \text{spherical representation} && (12.35b)\\
&= F^{\bar{1}} \, (r \cos\phi \, \hat{\boldsymbol{\lambda}}) + F^{\bar{2}} \, (r \, \hat{\boldsymbol{\phi}}) + F^{\bar{3}} \, \hat{\boldsymbol{r}} && \text{spherical expanded} && (12.35c)\\
&= (F^{\bar{1}} \, r \cos\phi) \, \hat{\boldsymbol{\lambda}} + (F^{\bar{2}} \, r) \, \hat{\boldsymbol{\phi}} + F^{\bar{3}} \, \hat{\boldsymbol{r}} && \text{move brackets} && (12.35d)\\
&= F^{\hat{\bar{1}}} \, \hat{\boldsymbol{\lambda}} + F^{\hat{\bar{2}}} \, \hat{\boldsymbol{\phi}} + F^{\hat{\bar{3}}} \, \hat{\boldsymbol{r}} && \text{spherical non-coordinate representation,} && (12.35e)
\end{aligned}$$

where the final expression defined the coordiante representation in terms of the non-coordinate spherical basis

$$F^{\hat{\bar{1}}} = F^{\bar{1}} \, r \cos\phi \quad \text{and} \quad F^{\hat{\bar{2}}} = F^{\bar{2}} \, r \quad \text{and} \quad F^{\hat{\bar{3}}} = F^{\bar{3}}. \tag{12.36}$$

Note that for brevity, we often refer to the spherical non-coordinate representation, $F^{\hat{\bar{a}}}$, as simply the spherical coordinate representation. The meaning is clarified by the choice of basis vectors.

12.3.7 Two meanings for the same symbol

We highlight the need to hold two meanings in mind for (x, y, z) and (λ, ϕ, r), or for any set of coordinates. One meaning specifies a point in Euclidean space. The other meaning refers to the position in space for a moving material particle, in which case the coordinate position is a function of time. We commonly write the second meaning using the capital, $\mathbf{X}(t)$, to denote the particle trajectory, and $\boldsymbol{X}(t)$ for a particular coordinate realization. To accord with common usage for the Cartesian and spherical coordinates, we write the coordinate components to $\mathbf{X}(t)$ using the smaller case, so that the Cartesian position of the particle is

$$\boldsymbol{X}(t) = \hat{\boldsymbol{x}}\, x(t) + \hat{\boldsymbol{y}}\, y(t) + \hat{\boldsymbol{z}}\, z(t), \tag{12.37}$$

whereas the spherical position is

$$\boldsymbol{X}(t) = \hat{\boldsymbol{r}}(t)\, r(t). \tag{12.38}$$

Time dependence of the unit vectors depends on the reference frame. For an inertial reference frame, the Cartesian unit vectors are fixed in space, whereas for the rigid-body rotating frame (e.g., rotating terrestrial reference frame), the Cartesian basis vectors rotate around the vertical axis. Likewise, the spherical unit vector rotates with the rotating reference frame. Yet even in an inertial reference frame, $\hat{\boldsymbol{r}}$ is a function of time since it points from the origin towards the moving particle.

12.4 The velocity vector

The velocity is the time derivative of the position

$$\mathbf{V} = \frac{\mathrm{d}\mathbf{X}}{\mathrm{d}t}. \tag{12.39}$$

When expressing this relation using a particular coordinate representation, observe that time dependence lives with both the coordinate representation of the position as well as the basis vectors

$$\boldsymbol{V} = \frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \frac{\mathrm{d}\,(\xi^a\, \boldsymbol{e}_a)}{\mathrm{d}t} = \frac{\mathrm{d}\xi^a}{\mathrm{d}t}\, \boldsymbol{e}_a + \xi^a\, \frac{\mathrm{d}\boldsymbol{e}_a}{\mathrm{d}t}. \tag{12.40}$$

12.4.1 Coordinate velocity

The first term on the right hand side of equation (12.40) is the velocity as measured within the chosen reference frame using the chosen coordinates

$$\boldsymbol{V}_{\mathsf{coord}} \equiv \frac{\mathrm{d}\xi^a}{\mathrm{d}t}\, \boldsymbol{e}_a. \tag{12.41}$$

This is the contribution to velocity as measured in the reference frame that moves with the basis vectors. In the context of geophysical motions, this is the velocity measured in the rotating terrestrial reference frame using Cartesian, spherical, or another chosen coordinate system.

12.4.2 Temporal changes to the basis vectors

The second term on the right hand side of equation (12.40) arises from changes to the basis vectors. There are three means for a basis vector to have a time derivative, and we encounter them when considering coordinate representations later in this chapter. Two changes arise from rotations, each of which change the basis vector's direction without changing its magnitude. The third change leads to a modification of the basis vector's magnitude.

- RIGID-BODY ROTATION: For a rigid-body rotation of the reference frame, the rigid-body velocity is given by (see Section 11.3.3)

$$\mathbf{U}_{\mathsf{rigid}} = \boldsymbol{\Omega} \times \boldsymbol{X}. \tag{12.42}$$

- ROTATION RELATIVE TO RIGID-BODY: A vector can also rotate at a rate that differs from the rigid-body.
- CHANGE IN MAGNITUDE: Finally, if the basis vectors are not normalized to have unit magnitude, then they can change their magnitude during motion.

12.5 Decomposition of the inertial frame acceleration

The acceleration measured in an inertial reference frame is given by the time derivative of the velocity measured in this frame, so that the acceleration is the second derivative of the position vector

$$\mathbf{A} = \frac{\mathrm{d}\mathbf{V}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{X}}{\mathrm{d}t^2}. \tag{12.43}$$

This is a tensor equation, so that it is independent of any coordinate representation, manifesting the physical and geometrical content. When introducing a coordinate representation, the resulting expression becomes subject to details of the chosen coordinates and those details possibly obscure the underlying geometric meaning (particularly with spherical coordinates). Hence, it is important to keep the geometric expression in mind when offering an interpretation for coordinate dependent terms.

Introducing a coordinate representation for the position, $\boldsymbol{X} = \xi^a\, \boldsymbol{e}_a$, into the acceleration (12.43), and making use of the chain rule, leads to

$$\begin{aligned}
\boldsymbol{A} &= \frac{\mathrm{d}}{\mathrm{d}t}\,\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} && (12.44\mathrm{a})\\
&= \frac{\mathrm{d}}{\mathrm{d}t}\,\frac{\mathrm{d}\,(\xi^a\,\boldsymbol{e}_a)}{\mathrm{d}t} && (12.44\mathrm{b})\\
&= \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\mathrm{d}\,\xi^a}{\mathrm{d}t}\,\boldsymbol{e}_a + \xi^a\,\frac{\mathrm{d}\,\boldsymbol{e}_a}{\mathrm{d}t}\right] && (12.44\mathrm{c})\\
&= \frac{\mathrm{d}^2\xi^a}{\mathrm{d}t^2}\,\boldsymbol{e}_a + 2\,\frac{\mathrm{d}\xi^a}{\mathrm{d}t}\,\frac{\mathrm{d}\boldsymbol{e}_a}{\mathrm{d}t} + \xi^a\,\frac{\mathrm{d}^2\boldsymbol{e}_a}{\mathrm{d}t^2}. && (12.44\mathrm{d})
\end{aligned}$$

The first term on the right hand side is the acceleration of the coordinate representation as measured in the rotating reference frame

$$\boldsymbol{A}_{\mathsf{coord}} \equiv \frac{\mathrm{d}^2\xi^a}{\mathrm{d}t^2}\,\boldsymbol{e}_a, \tag{12.45}$$

which is the acceleration measured by an observer in the rotating frame defined by the basis, $\boldsymbol{e}_a$, and using coordinates, ξ^a. The remaining two terms in equation (12.44d) arise from changes to the basis vectors associated with the rotating reference frame, resulting in the Coriolis acceleration and planetary centrifugal acceleration. In non-Cartesian coordinates, they also give rise to a metric acceleration arising from changes in direction of unit vectors associated with motion of the particle relative to the rotating reference frame (Section 12.4.2). Note that the factor of two on the middle term (the Coriolis term) in equation (12.44d) results from the two time derivatives that act on the basis vectors that arise when computing the acceleration. We already encountered this factor of two when deriving the Cartesian expression of the Coriolis acceleration in Section 11.5.3.

12.6 Representing the position vector

We make use of some results from Section 4.4 relating Cartesian and spherical coordinates and as defined by Figure 12.3. In particular, we use the *planetary* Cartesian basis vectors, $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$, and corresponding spherical basis vectors, $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, thus leading to the coordinate representations of the position relative to the center of a rotating sphere

$$\begin{aligned}
\mathbf{X} &= x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}} + z\,\hat{\boldsymbol{z}} && (12.46a)\\
&= (r\cos\phi\,\cos\lambda)\,\hat{\boldsymbol{x}} + (r\cos\phi\,\sin\lambda)\,\hat{\boldsymbol{y}} + (r\sin\phi)\,\hat{\boldsymbol{z}} && (12.46b)\\
&= r\,\hat{\boldsymbol{r}} && (12.46c)\\
&= |\mathbf{X}|\,\hat{\boldsymbol{r}}. && (12.46d)
\end{aligned}$$

The expression for the position is quite simple when written in spherical coordinates, as it is merely the distance from the origin with a direction that points radially from the origin to the particle.

12.7 Representing the velocity vector

As seen in Section 12.4, the inertial frame velocity vector has a coordinate representation written as

$$\boldsymbol{V} = \frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \frac{\mathrm{d}(\xi^a\,\boldsymbol{e}_a)}{\mathrm{d}t} = \frac{\mathrm{d}\xi^a}{\mathrm{d}t}\,\boldsymbol{e}_a + \xi^a\,\frac{\mathrm{d}\boldsymbol{e}_a}{\mathrm{d}t}. \tag{12.47}$$

Contributions arise from both the time changes in the coordinates, ξ^a, and time changes to the basis vectors, $\boldsymbol{e}_a$. We now consider the Cartesian and spherical forms for these changes.

12.7.1 Planetary Cartesian coordinate representation

The basis vectors for the planetary Cartesian coordinates, $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$, are normalized, so they do not change their magnitude. Furthermore, they move only through rigid-body motion of the rotating reference frame. We refer to these coordinates as planetary Cartesian coordinates since they are oriented according to the rotating planet with origin at the planet's center.[1]

The angular velocity is oriented around the polar axis

$$\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}, \tag{12.48}$$

so that the rigid-body velocity of the rotating reference frame only has components in the $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$ directions

$$\boldsymbol{U}_{\mathsf{rigid}} = \boldsymbol{\Omega} \times \boldsymbol{X} = \Omega\,(-\hat{\boldsymbol{x}}\,y + \hat{\boldsymbol{y}}\,x). \tag{12.49}$$

The inertial frame velocity thus has the following representation in terms of planetary Cartesian

[1] In VOLUME 2 of this book we introduce the distinct tangent plane Cartesian coordinates. Tangent plane Cartesian coordinates are also moving with the rotating planet. Yet they are defined according to a tangent plane at a point on the surface of the planet. It is important to distinguish these two uses for Cartesian coordinates in geophysical fluid mechanics.

coordinates

$$
\begin{aligned}
\boldsymbol{V} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[\hat{\boldsymbol{x}}\, x + \hat{\boldsymbol{y}}\, y + \hat{\boldsymbol{z}}\, z\right] && \text{(12.50a)}\\
&= \hat{\boldsymbol{x}}\, \frac{\mathrm{d}x}{\mathrm{d}t} + \hat{\boldsymbol{y}}\, \frac{\mathrm{d}y}{\mathrm{d}t} + \hat{\boldsymbol{z}}\, \frac{\mathrm{d}z}{\mathrm{d}t} + x\, \frac{\mathrm{d}\hat{\boldsymbol{x}}}{\mathrm{d}t} + y\, \frac{\mathrm{d}\hat{\boldsymbol{y}}}{\mathrm{d}t} + z\, \frac{\mathrm{d}\hat{\boldsymbol{z}}}{\mathrm{d}t} && \text{(12.50b)}\\
&= \left[-y\,\Omega + \frac{\mathrm{d}x}{\mathrm{d}t}\right]\hat{\boldsymbol{x}} + \left[x\,\Omega + \frac{\mathrm{d}y}{\mathrm{d}t}\right]\hat{\boldsymbol{y}} + \frac{\mathrm{d}z}{\mathrm{d}t}\,\hat{\boldsymbol{z}} && \text{(12.50c)}\\
&= \boldsymbol{V}_{\mathsf{cartesian}} + \boldsymbol{\Omega} \times \boldsymbol{X} && \text{(12.50d)}\\
&= \boldsymbol{V}_{\mathsf{cartesian}} + \boldsymbol{U}_{\mathsf{rigid}}, && \text{(12.50e)}
\end{aligned}
$$

where we defined the Cartesian velocity vector

$$
\boldsymbol{V}_{\mathsf{cartesian}} \equiv \frac{\mathrm{d}x}{\mathrm{d}t}\,\hat{\boldsymbol{x}} + \frac{\mathrm{d}y}{\mathrm{d}t}\,\hat{\boldsymbol{y}} + \frac{\mathrm{d}z}{\mathrm{d}t}\,\hat{\boldsymbol{z}}, \tag{12.51}
$$

which is the velocity as measured in the rotating reference frame when using planetary Cartesian coordinates. We also made use of equation (12.7) to express the time rate of change for the planetary Cartesian unit vectors, with this change arising solely from the planetary rotation. Note that the results here are valid for $\boldsymbol{\Omega}$ varying in time.

12.7.2 Planetary spherical coordinate representation

The position in the planetary spherical coordinates representation is given by

$$
\boldsymbol{X} = r\,\hat{\boldsymbol{r}}. \tag{12.52}
$$

The basis vector $\hat{\boldsymbol{r}}$ is normalized (it is one of the spherical anholonomic basis vectors), so that its evolution arises just from rotations. We determined its time evolution in equation (12.13)

$$
\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t} = \hat{\boldsymbol{\lambda}}\,(\dot{\lambda} + \Omega)\cos\phi + \hat{\boldsymbol{\phi}}\,\dot{\phi}. \tag{12.53}
$$

Consequently, the velocity as viewed in the inertial frame has the following spherical coordinate representation

$$
\begin{aligned}
\boldsymbol{V} &= \frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} && \text{(12.54a)}\\
&= \frac{\mathrm{d}(r\,\hat{\boldsymbol{r}})}{\mathrm{d}t} && \text{(12.54b)}\\
&= r\,\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t} + \frac{\mathrm{d}r}{\mathrm{d}t}\,\hat{\boldsymbol{r}} && \text{(12.54c)}\\
&= r_{\perp}\,(\dot{\lambda} + \Omega)\,\hat{\boldsymbol{\lambda}} + r\,\dot{\phi}\,\hat{\boldsymbol{\phi}} + \dot{r}\,\hat{\boldsymbol{r}} && \text{(12.54d)}\\
&= (u + r_{\perp}\,\Omega)\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}} && \text{(12.54e)}\\
&= \boldsymbol{V}_{\mathsf{spherical}} + \boldsymbol{U}_{\mathsf{rigid}}. && \text{(12.54f)}
\end{aligned}
$$

In this equation we introduced the velocity components written in both the spherical coordinate basis, $\boldsymbol{e}_{\overline{a}}$, and spherical non-coordinate basis, $\boldsymbol{e}_{\hat{\overline{a}}}$, following Section 12.3.6

$$
\boldsymbol{V}_{\mathsf{spherical}} = v^{\overline{1}}\,\boldsymbol{e}_{\overline{1}} + v^{\overline{2}}\,\boldsymbol{e}_{\overline{2}} + v^{\overline{3}}\,\boldsymbol{e}_{\overline{3}} \quad \text{spherical coordinate basis} \tag{12.55a}
$$

$$= v^{\hat{\bar{1}}}\, \boldsymbol{e}_{\hat{\bar{1}}} + v^{\hat{\bar{2}}}\, \boldsymbol{e}_{\hat{\bar{2}}} + v^{\hat{\bar{3}}}\, \boldsymbol{e}_{\hat{\bar{3}}} \quad \text{spherical non-coordinate basis,} \tag{12.55b}$$

where

$$(v^{\bar{1}}, v^{\bar{2}}, v^{\bar{3}}) = (\dot{\lambda}, \dot{\phi}, \dot{r}) \qquad \text{spherical coordinate velocity components} \tag{12.56a}$$

$$(v^{\hat{\bar{1}}}, v^{\hat{\bar{2}}}, v^{\hat{\bar{3}}}) = (r \cos\phi\, \dot{\lambda}, r\, \dot{\phi}, \dot{r}) \quad \text{spherical non-coordinate velocity components.} \tag{12.56b}$$

We commonly write the spherical non-coordinate velocity components as

$$r \cos\phi\, \hat{\boldsymbol{\lambda}} + r\, \dot{\phi}\, \hat{\boldsymbol{\phi}} + \dot{r}\, \hat{\boldsymbol{r}} = u\, \hat{\boldsymbol{\lambda}} + v\, \hat{\boldsymbol{\phi}} + w\, \hat{\boldsymbol{r}}, \tag{12.57}$$

so that

$$u \equiv r_{\perp} \frac{\mathrm{d}\lambda}{\mathrm{d}t} \quad \text{and} \quad v \equiv r \frac{\mathrm{d}\phi}{\mathrm{d}t} \quad \text{and} \quad w \equiv \frac{\mathrm{d}r}{\mathrm{d}t}, \tag{12.58}$$

and with

$$r_{\perp} = |\boldsymbol{X}| \cos\phi = r \cos\phi \tag{12.59}$$

the distance to the polar axis. We also noted that the rigid-body velocity has the spherical coordinate representation

$$\boldsymbol{U}_{\mathsf{rigid}} = \boldsymbol{\Omega} \times \boldsymbol{X} = \Omega\, \boldsymbol{e}_{\bar{1}} = r_{\perp}\, \Omega\, \hat{\boldsymbol{\lambda}}. \tag{12.60}$$

That is, the rigid-body velocity is purely zonal.

12.7.3 Transforming from Cartesian to spherical

We can make use of the tensor analysis from Section 12.3 to transform from the Cartesian representation of the velocity vector to the spherical coordinate representation. This approach leads to an equivalent result as that pursued thus far in this section, but it is somewhat more systematic and it offers useful experience with the formalities of coordinate transformations. In particular, we make use of the transformation rule (12.27) along with the transformation matrix (12.29) and its inverse (12.30) to have

$$V^{\bar{1}} = \Lambda^{\bar{1}}{}_{1}\, V^{1} + \Lambda^{\bar{1}}{}_{2}\, V^{2} + \Lambda^{\bar{1}}{}_{3}\, V^{3} \tag{12.61a}$$

$$V^{\bar{2}} = \Lambda^{\bar{2}}{}_{1}\, V^{1} + \Lambda^{\bar{2}}{}_{2}\, V^{2} + \Lambda^{\bar{2}}{}_{3}\, V^{3} \tag{12.61b}$$

$$V^{\bar{3}} = \Lambda^{\bar{3}}{}_{1}\, V^{1} + \Lambda^{\bar{3}}{}_{2}\, V^{2} + \Lambda^{\bar{3}}{}_{3}\, V^{3}, \tag{12.61c}$$

where the Cartesian components are

$$V^{1} = \dot{x} - \Omega\, y = \dot{r} \cos\phi \cos\lambda - r\, \dot{\phi} \sin\phi \cos\lambda - r\, (\dot{\lambda} + \Omega) \cos\phi \sin\lambda \tag{12.62a}$$

$$V^{2} = \dot{y} + \Omega\, x = \dot{r} \cos\phi \sin\lambda - r\, \dot{\phi} \sin\phi \sin\lambda + r\, (\dot{\lambda} + \Omega) \cos\phi \cos\lambda \tag{12.62b}$$

$$V^{3} = \dot{z} = \dot{r} \sin\phi + r\, \dot{\phi} \cos\phi. \tag{12.62c}$$

Making use of the inverse transformation matrix components, $\Lambda^{\bar{a}}{}_{b}$, given by equation (12.30), as well as the relation (12.27) between the coordinate basis vectors, allows us to confirm that

$$\mathbf{V} = V^{a}\, \boldsymbol{e}_{a} = V^{\bar{a}}\, \boldsymbol{e}_{\bar{a}} = V^{\hat{\bar{a}}}\, \boldsymbol{e}_{\hat{\bar{a}}}. \tag{12.63}$$

12.7.4 Axial angular momentum

As seen in Section 13.5, the zonal component of the inertial frame velocity, times the moment arm (distance from the rotation axis), equals to the axial angular momentum per unit mass

$$L^z = m\, r_\perp\, \hat{\boldsymbol{\lambda}} \cdot \boldsymbol{V} = m\, r_\perp\, (u + r_\perp\, \Omega). \tag{12.64}$$

The distance to the rotational axis is given by $r_\perp$, and this is the moment arm for the axial angular momentum. For cases with rotational symmetry around polar axis, as for motion of a particle around a smooth sphere, the axial angular momentum is a constant of the motion. As discussed in Section 13.5, this conservation law provides a constraint on the particle trajectory and it plays a role in the motion of geophysical fluids.

12.8 Planetary Cartesian representation of acceleration

The acceleration measured in an inertial reference frame is given by the second time derivative of the position vector

$$\mathbf{A} = \frac{\mathrm{d}\mathbf{V}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{X}}{\mathrm{d}t^2}. \tag{12.65}$$

We here consider its representation using planetary Cartesian coordinates, (x, y, z), and the Cartesian basis, $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$. The results here were anticipated by the general discussion in Section 11.5.3, and we here go through the derivation with a focus on geophysical applications.

12.8.1 Planetary Cartesian representation

For the study of geophysical fluid motion, we assume the planetary angular velocity, $\boldsymbol{\Omega}$, is a constant in time

$$\frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} = 0. \tag{12.66}$$

Making use of the results from Section 12.2.1 leads to the time derivative of the Cartesian basis vector,

$$\frac{\mathrm{d}\boldsymbol{e}_a}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{e}_a, \tag{12.67}$$

and the second time derivative

$$\frac{\mathrm{d}^2\boldsymbol{e}_a}{\mathrm{d}t^2} = \frac{\mathrm{d}}{\mathrm{d}t}\,(\boldsymbol{\Omega} \times \boldsymbol{e}_a) = \boldsymbol{\Omega} \times \frac{\mathrm{d}\boldsymbol{e}_a}{\mathrm{d}t} = \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{e}_a), \tag{12.68}$$

which then yields the planetary Cartesian expression for acceleration

$$\begin{aligned}
\boldsymbol{A} &= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} && (12.69\text{a})\\
&= \frac{\mathrm{d}^2\xi^a}{\mathrm{d}t^2}\,\boldsymbol{e}_a + 2\,\frac{\mathrm{d}\xi^a}{\mathrm{d}t}\,(\boldsymbol{\Omega} \times \boldsymbol{e}_a) + \xi^a\,\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{e}_a) && (12.69\text{b})\\
&= \hat{\boldsymbol{x}}\left[\ddot{x} - 2\,\Omega\,\dot{y} - \Omega^2 x\right] + \hat{\boldsymbol{y}}\left[\ddot{y} + 2\,\Omega\,\dot{x} - \Omega^2 y\right] + \hat{\boldsymbol{z}}\,\ddot{z} && (12.69\text{c})\\
&= \ddot{x}\,\hat{\boldsymbol{x}} + \ddot{y}\,\hat{\boldsymbol{y}} + \ddot{z}\,\hat{\boldsymbol{z}} + 2\,\Omega\,(-\dot{y}\,\hat{\boldsymbol{x}} + \dot{x}\,\hat{\boldsymbol{y}}) - \Omega^2\,(x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}}) && (12.69\text{d})\\
&= \boldsymbol{A}_{\text{cartesian}} + 2\,\boldsymbol{\Omega} \times \boldsymbol{V}_{\text{cartesian}} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{X}) && (12.69\text{e})\\
&= \boldsymbol{A}_{\text{cartesian}} - \boldsymbol{A}_{\text{coriolis}} - \boldsymbol{A}_{\text{centrifugal}}. && (12.69\text{f})
\end{aligned}$$

The inertial frame acceleration is thus decomposed into three terms that we now discuss.

Coordinate acceleration in the rotating reference frame

The first acceleration on the right hand side of equation (12.69f) is

$$\boldsymbol{A}_{\text{cartesian}} = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}\,\hat{\boldsymbol{x}} + \frac{\mathrm{d}^2 y}{\mathrm{d}t^2}\,\hat{\boldsymbol{y}} + \frac{\mathrm{d}^2 z}{\mathrm{d}t^2}\,\hat{\boldsymbol{z}} = \ddot{x}\,\hat{\boldsymbol{x}} + \ddot{y}\,\hat{\boldsymbol{y}} + \ddot{z}\,\hat{\boldsymbol{z}}, \tag{12.70}$$

which is the coordinate acceleration measured in the rotating frame using planetary Cartesian coordinates.

Planetary Coriolis acceleration

The second term on the right hand side of equation (12.69f) is the planetary Coriolis acceleration

$$\boldsymbol{A}_{\text{coriolis}} = -2\,\boldsymbol{\Omega} \times \boldsymbol{V}_{\text{cartesian}}. \tag{12.71}$$

One key feature of the Coriolis acceleration is that it vanishes when there is no motion relative to the rotating reference frame. The Coriolis acceleration plays a fundamental role in geophysical fluid mechanics and is central to our studies in this book.

Planetary centrifugal/centripetal acceleration

The third contribution to acceleration in equation (12.69f) is the planetary centrifugal acceleration, which is also minus the planetary centripetal acceleration

$$\boldsymbol{A}_{\text{centrifugal}} = -\boldsymbol{A}_{\text{centripetal}} = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{X}). \tag{12.72}$$

The planetary centrifugal acceleration points outward from (perpendicular to) the polar axis of rotation whereas the planetary centripetal acceleration points inward; they are action/reaction pairs. They can be written as the gradient of a scalar potential

$$\boldsymbol{A}_{\text{centrifugal}} = -\nabla\Phi_{\text{centrifugal}} \tag{12.73}$$

with

$$-\Phi_{\text{centrifugal}} \equiv \frac{(\boldsymbol{\Omega} \times \boldsymbol{x}) \cdot (\boldsymbol{\Omega} \times \boldsymbol{x})}{2} = \frac{\Omega^2\, r_{\perp}^2}{2} = \frac{(\Omega\, r \cos\phi)^2}{2} = \frac{\Omega^2\,(x^2 + y^2)}{2}, \tag{12.74}$$

so that

$$\boldsymbol{A}_{\text{centrifugal}} = -\nabla\Phi_{\text{centrifugal}} = \Omega^2\,\nabla(x^2 + y^2)/2. \tag{12.75}$$

The planetary centripetal acceleration (pointing towards the rotational axis) keeps the rotating particle from flying outward away from the rotational axis. On a massive rotating sphere, the planetary centripetal acceleration acting on the moving particle is provided by that component of the gravitational acceleration (Section 11.7) that is directed towards the rotation axis. The planetary centripetal acceleration's opposing partner, the planetary centrifugal acceleration, arises from the rotating planet and it accounts for the slight equatorial bulge of the Earth. For a human scale example, note that the centrifugal acceleration pulls a person outward from the center of a rotating merry-go-round, whereas the person's arms provide the opposing centripetal acceleration to keep from from flying outward. In summary, the centrifugal acceleration arises from rotation of the non-inertial reference frame, and as such is a non-inertial acceleration, whereas the centripetal acceleration arises from the force that balances the centrifugal acceleration.

12.8.2 Summary

For the purpose of formulating the equation of motion in the rotating terrestrial frame, we write the rigid-body rotating frame acceleration as

$$\boldsymbol{A}_{\text{cartesian}} = \boldsymbol{A} + \boldsymbol{A}_{\text{coriolis}} + \boldsymbol{A}_{\text{centrifugal}} \tag{12.76a}$$

$$= \boldsymbol{A} - 2\,\boldsymbol{\Omega} \times \boldsymbol{V}_{\text{cartesian}} - \nabla\Phi_{\text{centrifugal}}, \tag{12.76b}$$

and summarize the following accelerations (force per unit mass).

- INERTIAL: Newton's law of motion is formulated within an inertial reference frame making use of the acceleration, $\boldsymbol{A}$, that is directly affected by forces such as gravitation.
- PLANETARY CORIOLIS: The planetary Coriolis acceleration,

$$\boldsymbol{A}_{\text{coriolis}} = -2\,\boldsymbol{\Omega} \times \boldsymbol{V}_{\text{cartesian}} = -2\,\Omega\,\hat{\boldsymbol{z}} \times \boldsymbol{V}_{\text{cartesian}} = -2\,\Omega \left[-\frac{\mathrm{d}y}{\mathrm{d}t}\,\hat{\boldsymbol{x}} + \frac{\mathrm{d}x}{\mathrm{d}t}\,\hat{\boldsymbol{y}} \right], \tag{12.77}$$

  arises from the choice to describe motion within the rotating terrestrial reference frame. The Coriolis acceleration is nonzero only when there is a nonzero velocity, $\boldsymbol{V}_{\text{cartesian}}$, measured within the non-inertial reference frame. It has components only in the horizontal planetary Cartesian plane spanned by the unit vectors $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$. That is, the Coriolis acceleration occurs in a plane parallel to the equatorial plane. We expect this orientation since the Coriolis acceleration arises from rotation about the $\hat{\boldsymbol{z}}$ axis, so that there can be no component of the Coriolis acceleration aligned with $\hat{\boldsymbol{z}}$.
- PLANETARY CENTRIFUGAL: The planetary centrifugal acceleration,

$$\boldsymbol{A}_{\text{centrifugal}} = -\nabla\Phi_{\text{centrifugal}} = \Omega^2\,\boldsymbol{r}_\perp = \Omega^2\,(x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}}), \tag{12.78}$$

  is the second term arising from the rotating reference frame. As for the Coriolis acceleration, the planetary centrifugal acceleration has components only in the horizontal planetary Cartesian plane spanned by the unit vectors $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$; i.e., the planetary centrifugal acceleration occurs in a plane parallel to the equatorial plane, which is again expected since the planetary centrifugal acceleration arises from rotation of the planet about the polar axis. $\boldsymbol{A}_{\text{centrifugal}}$ is directed outward from (perpendicular to) to the polar axis of rotation. We see this orientation in Figure 12.5 to be discussed later. Furthermore, $\boldsymbol{A}_{\text{centrifugal}}$ is nonzero even when the particle is fixed relative to the rotating planet.

  The planetary centrifugal acceleration can be written as the gradient of a scalar potential, $\boldsymbol{A}_{\text{centrifugal}} = -\nabla\Phi_{\text{centrifugal}}$ where $\Phi_{\text{centrifugal}} = -\Omega^2\,(x^2 + y^2)/2$ (equation (12.74)). Hence, the planetary centrifugal acceleration can be combined with the gravitational acceleration in the equation of motion (see Section 11.7). The resulting effective gravity leads to a conservative force field that is modified relative to the central gravitational field of the non-rotating spherical planet. We detail these points in Section 12.8.3.

12.8.3 Effective gravitational acceleration from the geopotential

Combining the potential for the planetary centrifugal acceleration as given by equation (12.74) with the gravitational potential (11.123), leads to the geopotential

$$\Phi = r\,[g_{\mathrm{e}} - |\boldsymbol{U}_{\mathrm{rigid}}|^2/(2r)] \quad \text{with} \quad \boldsymbol{U}_{\mathrm{rigid}} = r\cos\phi\,\Omega\,\hat{\boldsymbol{\lambda}}, \tag{12.79}$$

where $\boldsymbol{U}_{\mathrm{rigid}}$ is the rigid body velocity of the planet, as per equation (12.60). The expression (12.79) for the geopotential is relevant for motion that is close enough to the Earth surface that we can assume the Earth's gravitational acceleration, g_{e}, is constant (see discussion in Section 11.7.5). In Exercise 12.8 we derive the geopotential for the more general case when this assumption is not made. The acceleration from the effective gravity is given by

$$\boldsymbol{g} = -\nabla\Phi = -(g_{\mathrm{e}} - r\,\Omega^2\cos^2\phi)\,\hat{\boldsymbol{r}} - \Omega^2\cos\phi\sin\phi\,\hat{\boldsymbol{\phi}}. \tag{12.80}$$

The $\hat{\boldsymbol{\phi}}$ contribution adds a non-radial component to the acceleration, acting to accelerate particles on a sphere meridionally toward the equator. We have more to say about this effect when discussing the spherical coordinate representation of Newton's law in Section 12.9, as well as the geopotential coordinate representation in Section 12.10.3.

We estimate the radial contribution from the planetary centrifugal term in the effective gravitational acceleration (12.80) by making use of terrestrial values, in which $R = R_{\mathrm{e}} = 6.371 \times 10^6$ m (equation (11.130)), and $\Omega_{\mathrm{e}} = 7.292 \times 10^{-5}\ \mathrm{s}^{-1}$ (Section 12.1). The upper bound on the planetary centrifugal term occurs at the equator, $\phi = 0$, where

$$(R_{\mathrm{e}}\,\Omega)^2 \approx (465.1\ \mathrm{m\ s^{-1}})^2 = 216318\ \mathrm{m^2\ s^{-2}} \tag{12.81a}$$

$$g_{\mathrm{e}}\,R_{\mathrm{e}} \approx 6.24 \times 10^7\ \mathrm{m^2\ s^{-2}}, \tag{12.81b}$$

so that the ratio of the Earth's gravitational acceleration at the equator to the planetary centrifugal accelerations is

$$\frac{g_{\mathrm{e}}}{R_{\mathrm{e}}\,\Omega^2} \approx 1/289. \tag{12.82}$$

The effective gravitational acceleration is thus dominated by the Earth's gravitational field. Even so, the planetary centrifugal acceleration leads to a slight equatorial bulge on the Earth. To account for this slight non-sphericity, geophysical fluid models generally interpret the radial direction, $\hat{\boldsymbol{r}}$, as pointing parallel to $\nabla\Phi$ rather than parallel to $\nabla\Phi_{\mathrm{e}}$. We have more to say on this topic of geopotential coordinates in Section 12.10.3.

12.8.4 Further study

We build up our understanding of the Coriolis acceleration as the book develops, such as in Section 13.6 where we offer a thorough examination of its facets. Further related study can be found in Section 3.5 of *Apel* (1987). Visualizations from rotating tank experiments are useful to garner an experiential understanding of the Coriolis acceleration, with the first few minutes of this video from Prof. Fultz of the University of Chicago particularly useful for this purpose. Section 7.4 of *Lautrup* (2005) provides a pedagogical discussion of the shape of the planet, with the oblate spheroidal shape arising from the planetary centrifugal acceleration.

12.9 Spherical representation of acceleration

The spherical non-coordinate (anholonomic) representation of the velocity viewed in an inertial reference frame is given by equation (12.54f)

$$\boldsymbol{V} = \frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = (u + r_{\perp}\,\Omega)\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}} = \boldsymbol{V}_{\text{spherical}} + r_{\perp}\,\Omega\,\hat{\boldsymbol{\lambda}}, \tag{12.83}$$

where we introduced the spherical velocity from equation (12.58)

$$\boldsymbol{V}_{\text{spherical}} \equiv u\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}}. \tag{12.84}$$

We also make use of the notation for the zonal component of the velocity as measured in the inertial reference frame,

$$u_{\text{I}} = u + r_{\perp}\,\Omega. \tag{12.85}$$

Just as for computing the velocity vector, the corresponding acceleration measured in the inertial reference frame must take into account changes in both the spherical coordinates and spherical non-coordinate basis vectors. We again choose to work with the spherical non-coordinate basis vectors, $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$, in which case the inertial-frame acceleration takes the form

$$\boldsymbol{A} = \frac{\mathrm{d}}{\mathrm{d}t}\left(u_{\text{I}}\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}}\right) \tag{12.86a}$$

$$= \frac{\mathrm{d}u_{\text{I}}}{\mathrm{d}t}\,\hat{\boldsymbol{\lambda}} + \frac{\mathrm{d}v}{\mathrm{d}t}\,\hat{\boldsymbol{\phi}} + \frac{\mathrm{d}w}{\mathrm{d}t}\,\hat{\boldsymbol{r}} + u_{\text{I}}\,\frac{\mathrm{d}\hat{\boldsymbol{\lambda}}}{\mathrm{d}t} + v\,\frac{\mathrm{d}\hat{\boldsymbol{\phi}}}{\mathrm{d}t} + w\,\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t}. \tag{12.86b}$$

The spherical unit vectors change due to both the rigid-body rotation of the rotating reference frame, plus motion of the particle relative to the rotating frame. Making use of equations (12.13), (12.14a), and (12.14b) for the changes to the spherical coordinate unit vectors leads to the spherical non-coordinate decomposition of the acceleration

$$\boldsymbol{A}\cdot\hat{\boldsymbol{\lambda}} = \frac{\mathrm{d}u_{\text{I}}}{\mathrm{d}t} + \left[\frac{\mathrm{d}\lambda}{\mathrm{d}t} + \Omega\right](w\cos\phi - v\sin\phi) \tag{12.87a}$$

$$\boldsymbol{A}\cdot\hat{\boldsymbol{\phi}} = \frac{\mathrm{d}v}{\mathrm{d}t} + \left[\frac{\mathrm{d}\lambda}{\mathrm{d}t} + \Omega\right]u_{\text{I}}\,\sin\phi + w\,\frac{\mathrm{d}\phi}{\mathrm{d}t} \tag{12.87b}$$

$$\boldsymbol{A}\cdot\hat{\boldsymbol{r}} = \frac{\mathrm{d}w}{\mathrm{d}t} - \left[\frac{\mathrm{d}\lambda}{\mathrm{d}t} + \Omega\right]u_{\text{I}}\,\cos\phi - v\,\frac{\mathrm{d}\phi}{\mathrm{d}t}. \tag{12.87c}$$

Use of the identities

$$u = r_{\perp}\,\frac{\mathrm{d}\lambda}{\mathrm{d}t} \qquad \text{and} \qquad u_{\text{I}} = u + r_{\perp}\,\Omega \qquad \text{and} \qquad \frac{\mathrm{d}u_{\text{I}}}{\mathrm{d}t} = \frac{\mathrm{d}u}{\mathrm{d}t} + \Omega\,(w\cos\phi - v\sin\phi), \tag{12.88}$$

along with some reorganization, then renders the spherical non-coordinate representation of the inertial frame acceleration

$$\begin{aligned}\boldsymbol{A} = \hat{\boldsymbol{\lambda}}&\left[\frac{\mathrm{d}u}{\mathrm{d}t} + \frac{u\,(w - v\tan\phi)}{r} + 2\,\Omega\,(w\cos\phi - v\sin\phi)\right]\\
+\,\hat{\boldsymbol{\phi}}&\left[\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v\,w + u^2\tan\phi}{r} + 2\,\Omega\,u\sin\phi + r_{\perp}\,\Omega^2\sin\phi\right]\\
+\,\hat{\boldsymbol{r}}&\left[\frac{\mathrm{d}w}{\mathrm{d}t} - \frac{u^2 + v^2}{r} - 2\,\Omega\,u\cos\phi - r_{\perp}\,\Omega^2\cos\phi\right].\end{aligned} \tag{12.89}$$

12.9.1 Transforming from Cartesian to spherical

As in Section 12.7.3, we can make use of the tensor algebra from Section 12.3 to transform from the Cartesian representation of the acceleration vector to the spherical representation. Following the same steps as for the velocity leads to

$$A^{\overline{a}} = \Lambda^{\overline{a}}{}_{a}\, A^{a} \tag{12.90}$$

where the Cartesian components to the acceleration are

$$A^1 = \ddot{x} - 2\,\Omega\,\dot{y} - \Omega^2\,x \tag{12.91a}$$
$$A^2 = \ddot{y} + 2\,\Omega\,\dot{x} - \Omega^2\,y \tag{12.91b}$$
$$A^3 = \ddot{z}. \tag{12.91c}$$

Making use of the coordinate transformation (12.25) allows us to express these Cartesian components of the acceleration in terms of spherical coordinates, and then to identify the spherical non-coordinate representation. Then we make use of the inverse transformation matrix components, $\Lambda^{\overline{a}}{}_{b}$, given by equation (12.30), as well as the relation (12.27) between the coordinate basis vectors, which leads to

$$\mathbf{A} = A^{a}\,\boldsymbol{e}_{a} = A^{\overline{a}}\,\boldsymbol{e}_{\overline{a}} = A^{\hat{\overline{a}}}\,\boldsymbol{e}_{\hat{\overline{a}}}, \tag{12.92}$$

with the spherical non-coordinate representation given by equation (12.89) as derived without the formalism of tensor algebra. Both approaches require algebraic manipulations, so that it is useful to have two approaches to double-check results.

12.9.2 Decomposing the acceleration

We decompose the inertial frame acceleration (12.89) into the following terms

$$\boldsymbol{A} = \boldsymbol{A}_{\text{spherical}} + \boldsymbol{A}_{\text{metric}} - \boldsymbol{A}_{\text{coriolis}} - \boldsymbol{A}_{\text{centrifugal}}, \tag{12.93}$$

with signs chosen so that in the non-inertial (rigid-body rotating) frame the acceleration is written

$$\underbrace{\boldsymbol{A}_{\text{spherical}} + \boldsymbol{A}_{\text{metric}}}_{\text{net spherical acceleration}} = \boldsymbol{A} + \boldsymbol{A}_{\text{coriolis}} + \boldsymbol{A}_{\text{centrifugal}}. \tag{12.94}$$

We identify the net spherical acceleration as the sum of the coordinate acceleration and metric acceleration. In the absence of reference frame rotation, this sum provides an expression for the acceleration as represented by spherical coordinates. For example, if we are describing the motion of a satellite from an inertial reference frame using spherical coordinates, then we would use $\boldsymbol{A}_{\text{spherical}} + \boldsymbol{A}_{\text{metric}}$.

12.9.3 Spherical coordinate acceleration

The spherical coordinate acceleration is given by the time change in the spherical non-coordinate velocity components

$$\boldsymbol{A}_{\text{spherical}} = \frac{\mathrm{d}u}{\mathrm{d}t}\,\hat{\boldsymbol{\lambda}} + \frac{\mathrm{d}v}{\mathrm{d}t}\,\hat{\boldsymbol{\phi}} + \frac{\mathrm{d}w}{\mathrm{d}t}\,\hat{\boldsymbol{r}}. \tag{12.95}$$

This term has no contribution from changes to the spherical unit vectors.

12.9.4 Metric acceleration

We define the metric acceleration as that contribution to the acceleration arising from time dependence of the unit vectors that appears when taking the time derivative of the velocity vector. For spherical coordinates we have

$$\boldsymbol{A}_{\text{metric}} = \hat{\boldsymbol{\lambda}} \left[\frac{u\,(w - v\tan\phi)}{r} \right] + \hat{\boldsymbol{\phi}} \left[\frac{v\,w + u^2 \tan\phi}{r} \right] - \hat{\boldsymbol{r}} \left[\frac{u^2 + v^2}{r} \right] \tag{12.96a}$$

$$= \hat{\boldsymbol{\lambda}} \left[\frac{u\,(w\cos\phi - v\sin\phi)}{r\cos\phi} \right] + \hat{\boldsymbol{\phi}} \left[\frac{v\,w\cos\phi + u^2 \sin\phi}{r\cos\phi} \right] - \hat{\boldsymbol{r}} \left[\frac{u^2 + v^2}{r} \right] \tag{12.96b}$$

$$= \frac{1}{r} \left[u\tan\phi\,(\hat{\boldsymbol{r}} \times \boldsymbol{V}_{\text{spherical}}) + w\,\boldsymbol{U}_{\text{spherical}} - \hat{\boldsymbol{r}}\,\boldsymbol{U}_{\text{spherical}} \cdot \boldsymbol{U}_{\text{spherical}} \right], \tag{12.96c}$$

where we wrote the horizontal (angular) and vertical (radial) components of the spherical velocity according to

$$\boldsymbol{V}_{\text{spherical}} = \boldsymbol{U}_{\text{spherical}} + \hat{\boldsymbol{r}}\,w = \hat{\boldsymbol{\lambda}}\,u + \hat{\boldsymbol{\phi}}\,v + \hat{\boldsymbol{r}}\,w. \tag{12.97}$$

Note that

$$\boldsymbol{V}_{\text{spherical}} \cdot \boldsymbol{A}_{\text{metric}} = 0, \tag{12.98}$$

so that the metric acceleration is orthogonal to the spherical velocity, in which case with (see equation (12.54f)) $\boldsymbol{V} = \boldsymbol{V}_{\text{spherical}} + r_{\perp}\,\Omega\,\hat{\boldsymbol{\lambda}}$ we have

$$\boldsymbol{V} \cdot \boldsymbol{A}_{\text{metric}} = \Omega\,u\,(-v\sin\phi + w\cos\phi). \tag{12.99}$$

Furthermore, the metric acceleration vanishes when the curvature of the sphere vanishes (i.e., $r \to \infty$), as per a flat plane.

12.9.5 Planetary centrifugal acceleration

The spherical coordinate representation of the planetary centrifugal acceleration is given by

$$\boldsymbol{A}_{\text{centrifugal}} = -\nabla\Phi_{\text{centrifugal}} = \Omega^2\,(x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}}) = r_{\perp}\,\Omega^2\,(-\hat{\boldsymbol{\phi}}\sin\phi + \hat{\boldsymbol{r}}\cos\phi). \tag{12.100}$$

The planetary centrifugal acceleration points outward from the axis of rotation (see Figure 12.5 to be discussed later), so that it has no component in the longitudinal direction. Furthermore, note that

$$\boldsymbol{V} \cdot \boldsymbol{A}_{\text{centrifugal}} = \boldsymbol{V}_{\text{spherical}} \cdot \boldsymbol{A}_{\text{centrifugal}} = r_{\perp}\,\Omega^2\,(-v\sin\phi + w\cos\phi). \tag{12.101}$$

12.9.6 Planetary Coriolis acceleration

The spherical coordinate representation of the planetary Coriolis acceleration makes use of the spherical representation of the Earth's angular velocity

$$\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}} = \Omega\,(\hat{\boldsymbol{\phi}}\,\cos\phi + \hat{\boldsymbol{r}}\,\sin\phi). \tag{12.102}$$

Although $\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}$ is fixed in absolute space, its representation using spherical coordinates is a function of latitudinal position on the sphere, with components in the local radial and local meridional directions. This spatial dependence is known as differential rotation.

We use the representation (12.102) to write the spherical coordinate representation of the

Coriolis acceleration

$$\begin{aligned}\boldsymbol{A}_{\text{coriolis}} &= -2\,\boldsymbol{\Omega} \times \boldsymbol{V}_{\text{spherical}} && (12.103\text{a})\\ &= -2\,\Omega\,(\hat{\boldsymbol{\phi}}\,\cos\phi + \hat{\boldsymbol{r}}\,\sin\phi) \times (u\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}}) && (12.103\text{b})\\ &= -2\,\Omega\left[\hat{\boldsymbol{\lambda}}\,(w\cos\phi - v\sin\phi) + \hat{\boldsymbol{\phi}}\,u\sin\phi - \hat{\boldsymbol{r}}\,u\cos\phi\right]. && (12.103\text{c})\end{aligned}$$

As a check on this result, note that

$$\boldsymbol{A}_{\text{coriolis}} \cdot \hat{\boldsymbol{z}} = 0, \tag{12.104}$$

as it must since there is no component of the Coriolis acceleration that is parallel to the rotation axis. Furthermore, as for the metric acceleration, we see that the Coriolis acceleration is orthogonal to the spherical velocity

$$\boldsymbol{V}_{\text{spherical}} \cdot \boldsymbol{A}_{\text{coriolis}} = 0, \tag{12.105}$$

so that

$$\boldsymbol{V} \cdot \boldsymbol{A}_{\text{coriolis}} = 2\,r_{\perp}\,\Omega^2\,(v\sin\phi - w\cos\phi). \tag{12.106}$$

We commonly find it convenient to introduce a shorthand notation

$$\boldsymbol{f} = (2\,\Omega\,\sin\phi)\,\hat{\boldsymbol{r}} \qquad \text{and} \qquad \boldsymbol{f}^{*} = (2\,\Omega\,\cos\phi)\,\hat{\boldsymbol{\phi}}, \tag{12.107}$$

so that the Coriolis acceleration is decomposed into two terms

$$\boldsymbol{A}_{\text{coriolis}} = -(\boldsymbol{f} + \boldsymbol{f}^{*}) \times \boldsymbol{V}_{\text{spherical}}. \tag{12.108}$$

These two terms take on the form

$$\boldsymbol{A}^{\boldsymbol{f}}_{\text{coriolis}} = -(2\,\Omega\,\sin\phi)\,\hat{\boldsymbol{r}} \times (u\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}}) = 2\,\Omega\sin\phi\,(v\,\hat{\boldsymbol{\lambda}} - u\,\hat{\boldsymbol{\phi}}), \tag{12.109}$$

which is purely in the spherical angular directions, and

$$\boldsymbol{A}^{\boldsymbol{f}^{*}}_{\text{coriolis}} = -(2\,\Omega\,\cos\phi)\,\hat{\boldsymbol{\phi}} \times (u\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}}) = 2\,\Omega\cos\phi\,(u\,\hat{\boldsymbol{r}} - w\,\hat{\boldsymbol{\lambda}}), \tag{12.110}$$

which has a zonal and radial component.

12.9.7 Centrifugal acceleration from particle plus planetary motion

The radial component to the inertial frame acceleration (12.89) can be written in the form

$$\boldsymbol{A} \cdot \hat{\boldsymbol{r}} = \frac{\mathrm{d}w}{\mathrm{d}t} - (u_{\mathsf{I}}^2 + v^2)/r. \tag{12.111}$$

Evidently, the contribution from $(u_{\mathsf{I}}^2 + v^2)/r$ leads to a vertical (radially outward) acceleration in the radial equation of motion (12.126) as discussed in Section 12.10.2. We identify it as the radial acceleration arising from the particle's centrifugal acceleration due to angular motion of the particle around the sphere, combined with the local vertical contribution from the planetary centrifugal acceleration of Section 12.9.5.

12.9.8 Coriolis acceleration for large-scale motions

Let us again write the Coriolis acceleration in equation (12.103c), now underlining two terms

$$\boldsymbol{A}_{\text{coriolis}} = -2\,\Omega \left[\hat{\boldsymbol{\lambda}}\,(\underline{w\cos\phi} - v\sin\phi) + \hat{\boldsymbol{\phi}}\,u\sin\phi - \hat{\boldsymbol{r}}\,\underline{u\cos\phi}\right]. \tag{12.112}$$

For many applications in geophysical fluid mechanics, the term $\hat{\boldsymbol{r}}\,(2\,\Omega\,u\cos\phi)$ is much smaller than the competing gravitational acceleration that also contributes to the radial acceleration, thus prompting $\hat{\boldsymbol{r}}\,(2\,\Omega\,u\cos\phi)$ to be dropped from the $\hat{\boldsymbol{r}}$ equation of motion.[2] Furthermore, the vertical velocity term is often much smaller than the horizontal velocity term appearing in the $\hat{\boldsymbol{\lambda}}$ component. Dropping these two terms results in the Coriolis acceleration used for large-scale dynamics,

$$\boldsymbol{A}^{\text{largescale}}_{\text{coriolis}} \equiv -2\,\Omega\sin\phi\,(-\hat{\boldsymbol{\lambda}}\,v + \hat{\boldsymbol{\phi}}\,u) \equiv -f\,\hat{\boldsymbol{r}} \times \boldsymbol{V}_{\text{spherical}}, \tag{12.113}$$

with this term arising when considering the hydrostatic primitive equations for geophysical fluids in VOLUME 2. For the last equality we introduced the Coriolis parameter, which is the twice the radial component of the Earth's planetary rotation vector

$$f \equiv 2\,\boldsymbol{\Omega}\cdot\hat{\boldsymbol{r}} = 2\,\Omega\sin\phi. \tag{12.114}$$

As illustrated in Figure 12.4, we see that it is the radial (i.e., the local vertical) component of the Earth's angular rotation that plays the most important role in large-scale geophysical fluid mechanics

$$\boldsymbol{\Omega}\cdot\hat{\boldsymbol{r}} = \Omega\,\hat{\boldsymbol{z}}\cdot\hat{\boldsymbol{r}} = \Omega\,(\hat{\boldsymbol{\phi}}\,\cos\phi + \hat{\boldsymbol{r}}\,\sin\phi)\cdot\hat{\boldsymbol{r}} = \Omega\sin\phi = f/2. \tag{12.115}$$

The Coriolis parameter is a monotonically increasing function of latitude, moving from its minimum, $f = -2\Omega$, at the south pole to its maximum, $f = 2\Omega$, at the north pole. This monotonically increasing property is of fundamental importance to ocean and atmosphere flows.

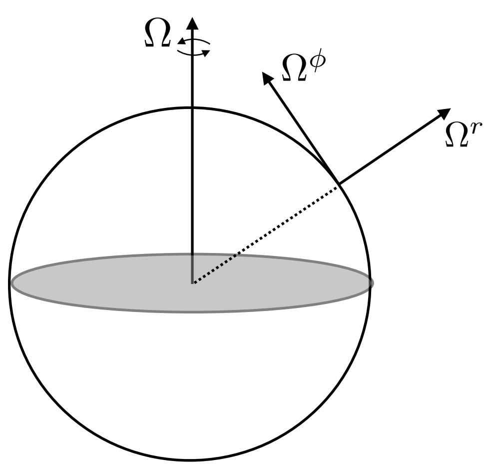

FIGURE 12.4: This figure illustrates the two components of the Earth's rigid-body rotational velocity, $\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}} = \Omega\,(\hat{\boldsymbol{\phi}}\cos\phi + \hat{\boldsymbol{r}}\sin\phi)$. The radial component (also known as the local vertical component), $\Omega^{r} = \Omega\sin\phi$, is generally more important for large scale geophysical fluid motions than the meridional component, $\Omega^{\phi} = \Omega\cos\phi$. For this reason, the Coriolis parameter, $f \equiv 2\,\boldsymbol{\Omega}\cdot\hat{\boldsymbol{r}} = 2\,\Omega\sin\phi$, plays an essential role in the dynamics of geophysical fluid motion.

[2] The term $\hat{\boldsymbol{r}}\,(2\,\Omega\,u\cos\phi)$ is called the Eötvös correction in the study of marine gravity.

12.10 Newton's law of motion

As seen in Section 11.2, Newton's law of motion says that in an inertial reference frame, the time derivative of the linear momentum arises only from externally applied forces. In our study, we are only concerned with the gravitational force that the constant mass particle feels in an inertial frame. In this case, Newton's equation of motion says that

$$m\,\mathbf{A} = -m\,\nabla\Phi_{\mathsf{e}}. \tag{12.116}$$

This is a relatively simple equation of motion written here as a tensor equation and so holding for all coordinates. To make use of this equation for practical purposes, we provide a coordinate representation and move to a non-inertial terrestrial reference frame, both of which can add algebraic complexity but do not remove any of the physics.[3]

12.10.1 Cartesian coordinate representation

The inertial frame acceleration using planetary Cartesian coordinates is decomposed according to equation (12.76b)

$$\boldsymbol{A} = \boldsymbol{A}_{\mathsf{cartesian}} - \boldsymbol{A}_{\mathsf{coriolis}} - \boldsymbol{A}_{\mathsf{centrifugal}} \tag{12.117a}$$

$$= \boldsymbol{A}_{\mathsf{cartesian}} + 2\,\boldsymbol{\Omega}\times\boldsymbol{V}_{\mathsf{cartesian}} + \nabla\Phi_{\mathsf{centrifugal}}. \tag{12.117b}$$

We thus have the equation of motion as viewed within the rigid-body rotating frame and using Cartesian coordinates

$$\boldsymbol{A}_{\mathsf{cartesian}} = \boldsymbol{A} - 2\,\boldsymbol{\Omega}\times\boldsymbol{V}_{\mathsf{cartesian}} - \nabla\Phi_{\mathsf{centrifugal}} \tag{12.118a}$$

$$= -\nabla\Phi_{\mathsf{e}} - 2\,\boldsymbol{\Omega}\times\boldsymbol{V}_{\mathsf{cartesian}} - \nabla\Phi_{\mathsf{centrifugal}} \tag{12.118b}$$

$$= -2\,\boldsymbol{\Omega}\times\boldsymbol{V}_{\mathsf{cartesian}} - \nabla\Phi, \tag{12.118c}$$

where the geopotential is the sum of the gravitational and planetary centrifugal potentials (equation (12.79))

$$\Phi = \Phi_{\mathsf{e}} + \Phi_{\mathsf{centrifugal}}. \tag{12.119}$$

We can write the equation of motion in the vector form within the rotating reference frame

$$\frac{\mathrm{d}^2\boldsymbol{X}}{\mathrm{d}t^2} + 2\,\boldsymbol{\Omega}\times\dot{\boldsymbol{X}} = -\nabla\Phi. \tag{12.120}$$

Note for this equation, the basis vectors are not time differentiated again since their rigid-body rotation has already been taken care of when exposing the Coriolis and planetary centrifugal accelerations. That is, the time derivatives are all computed within the rigid-body rotating reference frame.

Since the rotation of the reference frame is assumed to be constant in time, the equation of

[3] As emphasized by *Early* (2012), the acceleration from gravity, from the perspective of general relativity, cannot be distinguished from accelerations encountered when describing motion in a non-inertial reference frame. *Early* (2012) provides an accounting of such motion while clarifying what is meant by "inertial oscillations" for particles constrained to move around the Earth along a constant geopotential (we also consider inertial oscillations in VOLUME 2 and VOLUME 4). Even so, we here consider gravity as distinct from Coriolis and centrifugal since, in Newtonian mechanics, gravititational acceleration is present when the particle is viewed from an inertial reference frame, whereas the Coriolis and centrifugal accelerations are not.

motion (12.120) can be written

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{V}_{\text{cartesian}} + 2\,\boldsymbol{\Omega}\times\boldsymbol{X}) = -\nabla\Phi, \tag{12.121}$$

where we introduced the potential momentum per mass

$$\boldsymbol{M} = \boldsymbol{V}_{\text{cartesian}} + 2\,\boldsymbol{\Omega}\times\boldsymbol{X}. \tag{12.122}$$

Evidently, the potential momentum is a constant of the motion for particles moving along directions parallel to the geopotential (so long as the geopotential is a constant). We emphasize that the potential momentum per mass is distinct from the inertial frame velocity (12.50e). Namely, the factor of 2 in the potential momentum arises from the Coriolis acceleration, whereas the inertial frame velocity has $\boldsymbol{\Omega}\times\boldsymbol{X}$ arising from the rigid-body rotation of the planet. We further discuss potential momentum in Section 13.3.

12.10.2 Spherical coordinate representation

We now follow the spherical coordinate discussion in Section 12.9 by writing the inertial frame acceleration using planetary spherical coordinates and decomposing this acceleration into terms that arise in a rigid-body rotating non-inertial frame

$$\boldsymbol{A}_{\text{spherical}} + \boldsymbol{A}_{\text{metric}} = \boldsymbol{A}_{\text{coriolis}} + \boldsymbol{A} + \boldsymbol{A}_{\text{centrifugal}} = -2\,\boldsymbol{\Omega}\times\boldsymbol{V}_{\text{spherical}} - \nabla\Phi. \tag{12.123}$$

The effective gravitational force is not a central force due to the contribution from the planetary centrifugal acceleration. We see this fact more explicitly by using the equations in Section 12.9 to write the spherical equations of motion

$$\dot{u} + \frac{u\,(w - v\tan\phi)}{r} + 2\,\Omega\,(w\cos\phi - v\sin\phi) = 0 \tag{12.124}$$

$$\dot{v} + \frac{v\,w + u^2\tan\phi}{r} + 2\,\Omega\,u\sin\phi = -r_\perp\,\Omega^2\sin\phi \tag{12.125}$$

$$\dot{w} - \frac{u^2 + v^2}{r} - 2\,\Omega\,u\cos\phi = r_\perp\,\Omega^2\cos\phi - g_{\text{e}}, \tag{12.126}$$

where, again, $r_\perp = r\cos\phi$, which is the distance to the polar axis of rotation.

The Ω^2 term in both the meridional equation (12.125), and the radial equation (12.126), are the two components of the planetary centrifugal acceleration, $\boldsymbol{A}_{\text{centrifugal}} = r_\perp\,\Omega^2\,(-\hat{\boldsymbol{\phi}}\sin\phi + \hat{\boldsymbol{r}}\cos\phi)$. The planetary centrifugal acceleration is directed outward from the planetary axis of rotation, and it is balanced by an inward directed planetary centripetal acceleration provided by that portion of the gravitational acceleration directed oppositely to the planetary centrifugal. Notably, a particle initially at rest on a smooth spherical planet accelerates meridionally toward the equator due to the meridional component of the planetary centrifugal acceleration. The initial meridional acceleration for this particle is derived from equation (12.125), whereby $\dot{v} = -r_\perp\,\Omega^2\sin\phi$.

Imagine setting the planetary centripetal acceleration to zero, in which case the particle would still feel the central force from gravity but its trajectory would differ. As seen in Section 12.8.3, the Earth's gravitational acceleration is much larger than the planetary centrifugal acceleration, so that in the absence of the planetary centripetal acceleration the particle would still be bound to the planet. But in more extreme conditions where the rotational rate is much higher (e.g., a rotating neutron star), removing the centripetal acceleration causes a huge modification to the particle trajectory.

12.10.3 Geopotential coordinate representation

As we saw in Section 11.7.4, the radius of a sphere that best fits the volume of the Earth is given by $R_{\mathrm{e}} = 6.371 \times 10^6\,\mathrm{m}$. The non-central nature of the effective gravitational force (arising from central gravity plus planetary centrifugal) leads to an oblate spheroidal shape for planets such as the Earth. The result is a distinction between the Earth's equatorial and polar radii (Appendix Two of *Gill* (1982))

$$R_{\mathrm{equator}} = 6.378 \times 10^6\,\mathrm{m} \qquad \text{and} \qquad R_{\mathrm{pole}} = 6.357 \times 10^6\,\mathrm{m}, \tag{12.127}$$

with a corresponding ratio

$$1 - \frac{R_{\mathrm{pole}}}{R_{\mathrm{equator}}} \approx 3 \times 10^{-3}. \tag{12.128}$$

An oblate spheroid shape does a better job fitting the actual Earth shape than a sphere, thus motivating the use of oblate spheroid coordinates for describing planetary scale mechanics. In this case, the radial coordinate is constant on the oblate spheroid shaped geopotential, and the effective gravitational acceleration is precisely aligned with the geopotential direction. If we consider an ocean covered rotating spherical planet, then at static equilibrium the sea surface corresponds to an oblate spheroidal geopotential.

Even though oblate spheroidal coordinates are better than spherical for describing geopotentials, it is possible, to a high degree of accuracy, to describe the Earth's geometry as spherical (*Veronis*, 1973). Doing so simplifies the mathematics since oblate spheroidal coordinates are less convenient and less familiar than standard spherical coordinates. We are thus led to assume that the radial coordinate measures distances perpendicular to the geopotential, yet to use geometric/metric functions based on spherical coordinates. The error in this approach is small for the Earth, and well worth the price since there is no component to the effective gravitational force that is within the geopotential surface.

We illustrate the change in coordinates in Figure 12.5, with the figure caption also explaining how the force balances are reorganized. Absorbing the planetary centrifugal term into an effective gravitational potential then leads to the effective gravitational acceleration vector

$$-\nabla\Phi = -g\,\hat{\boldsymbol{r}}, \tag{12.129}$$

with g the effective gravitational acceleration. Using this convention, the particle equations of motion take the following form

$$\dot{u} + \frac{u\,(w - v\tan\phi)}{r} + 2\,\Omega\,(w\cos\phi - v\sin\phi) = 0 \tag{12.130a}$$

$$\dot{v} + \frac{v\,w + u^2\tan\phi}{r} + 2\,\Omega\,u\sin\phi = 0 \tag{12.130b}$$

$$\dot{w} - \frac{u^2 + v^2}{r} - 2\,\Omega\,u\cos\phi = -g. \tag{12.130c}$$

Notably, in geopotential coordinates the effective gravitational acceleration only impacts the radial equation of motion. There is no longer a component of the effective gravity pointing meridionally.

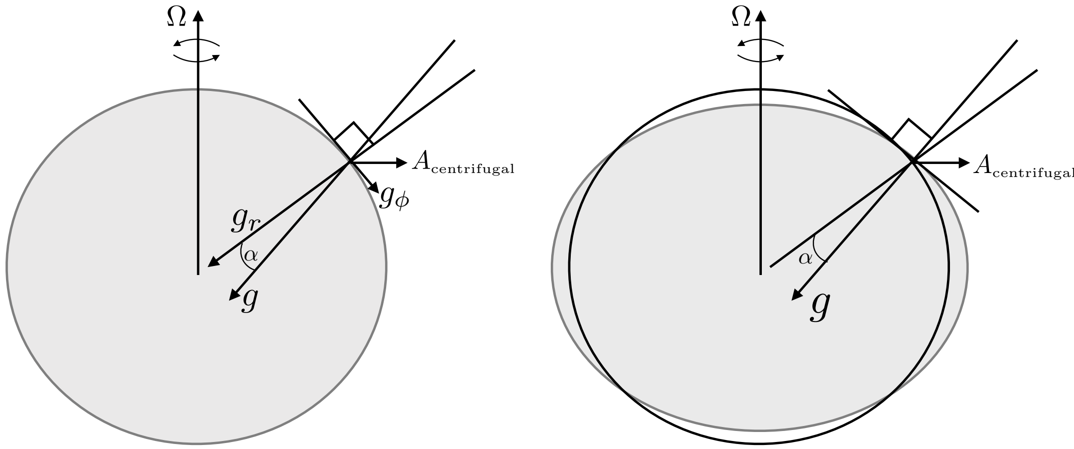

FIGURE 12.5: This figure illustrates the spherical (left panel) versus geopotential (right panel) coordinate systems used to study geophysical motions. The left panel shows the non-central nature of the effective gravitational acceleration, $\boldsymbol{g}$, on a rotating spherical planet, with the effective gravity given by the sum of the gravitational acceleration, $-g_{\mathsf{e}}\,\hat{\boldsymbol{r}}$, plus planetary centrifugal acceleration (equation (12.100)), $\boldsymbol{A}_{\mathsf{centrifugal}} = r_\perp\,\Omega^2\,(-\hat{\boldsymbol{\phi}}\sin\phi + \hat{\boldsymbol{r}}\cos\phi)$. The gravitational acceleration points radially to the center of the Earth whereas the planetary centrifugal acceleration points outward away from the polar axis of rotation, thus leading to an effective gravitational acceleration $\boldsymbol{g} = \hat{\boldsymbol{r}}\,(-g_{\mathsf{e}} + r_\perp\,\Omega^2\cos\phi) - \hat{\boldsymbol{\phi}}\,r_\perp\,\Omega^2\sin\phi = \hat{\boldsymbol{r}}\,g^r + \hat{\boldsymbol{\phi}}\,g^\phi$. The angle between the radial gravity and the effective gravity, α, is determined by a plumb line and is a function of the rotation rate, Earth radius, gravitational acceleration, and latitude. The expression for α is determined in Exercise 12.6. A particle initially at rest on a smooth spherical planet accelerates meridionally toward the equator due to the meridional component of the planetary centrifugal acceleration (from equation (12.125) we have $\dot{v} = -r_\perp\,\Omega^2\sin\phi$). The right panel shows a geopotential vertical coordinate, $r = R + z$, that measures the distance perpendicular to the oblate spheroid shaped geopotential surface. The geopotential vertical coordinate precisely aligns the effective gravitational force with the vertical coordinate, so that there is no component of the effective gravity along the surface directions ($g_\phi = 0$). Equivalently, the central gravitational acceleration now has a meridional component on the oblate spheroid that exactly balances the meridional component to the planetary centrifugal acceleration, leaving an effective gravity that is only vertical. Hence, a frictionless oblate spheroidal planet allows for a particle at rest on the planet's surface to remain at rest (see equation (12.130b)). Note that this figure is not drawn to scale, with the oblate nature highly exaggerated compared to the real Earth system (see equation (12.128)), and the planetary centrifugal acceleration much smaller than the gravitational (see equation (12.82)). This figure is adapted from Figure 2.2 of *Vallis* (2017) and Figure 2.8 of *Olbers et al.* (2012).

12.10.4 Comments

Figure 12.5, including its rather long caption, offers a view on the transition from spherical coordinates to geopotential coordinates. The use of geopotential coordinates is rather accurate and extremely convenient for most purposes of geophysical fluid mechanics, with a notable exception being the study of tides and sea level, in which detailed models of the Earth's mass distribution and gravity field are used (see *Gregory et al.* (2019) for a review).

Although precise, the transition to geopotential coordinates is somewhat subtle in principle since we are reorganizing how the planetary centrifugal acceleration appears. It is through this reorganization that we can largely ignore the planetary centrifugal acceleration in our studies of geophysical motions since it is absorbed by the geopotential. The single exception for our studies concerns rotating laboratory tank experiments, where we find it useful to expose the centrifugal acceleration.

12.10.5 Further study

Section 4.12 of *Gill* (1982) and section 2.2.1 of *Vallis* (2017) present the terrestrial scaling needed to justify spherical coordinates with a radial effective gravitational potential. *Morse and*

Feshbach (1953) and *Veronis* (1973) present details of oblate spheroidal coordinates. Chapter 14 in *Wells* (1967) summarizes many formula needed for a detailed treatment of gravity for the Earth. See also discussions by *Staniforth* (2022) and *Staniforth and White* (2025) for the treatment of Earth's gravity in numerical models of the atmosphere and ocean.

The transition from spherical to geopotential coordinates for geophysical fluid mechanics has been the topic of some confusion, as evidenced by the papers from *Stewart and McWilliams* (2022), *Chang and Wolfe* (2022), *Chang et al.* (2023), and *McWilliams* (2024). For the purposes of this book, the key point is that gravity is directed vertically and thus provides no acceleration in the local horizontal directions.

12.11 Exercises

EXERCISE 12.1: DISTANCE BETWEEN TWO POINTS

Throughout this book we assume motion within a background Euclidean space, so that the distance between two points is given by the usual Pythagorean expression. There are many occasions in which it is of interest to know the distance between two points on a spherical planet. Compute the distance between two such points in terms of their longtitude, λ, latitude, ϕ, and vertical distance relative to the Earth's radius, $r = z + R_{\mathrm{e}}$. Assume spherical coordinates are sufficiently accurate for this calculation.

EXERCISE 12.2: WORKING THROUGH THE SPHERICAL ACCELERATION

Convince yourself that the spherical form of the acceleration given by equation (12.89) is indeed correct.

EXERCISE 12.3: DERIVING THE SPHERICAL UNIT VECTOR TIME DERIVATIVES

In this exercise we work through details for deriving the time derivative of the spherical coordinate unit vector from Section 12.2.3.

(A) Fill in the details for deriving equation (12.13)
(B) Fill in the details for deriving equation (12.14a)
(C) Fill in the details for deriving equation (12.14b)

EXERCISE 12.4: VELOCITY AND ACCELERATION IN CYLINDRICAL-POLAR COORDINATES

In Section 4.3 we worked through the transformation from Cartesian coordinates to cylindrical-polar coordinates for describing motion in a rotating reference frame. We also made use of polar coordinates in Section 12.2.2 to illustrate how polar unit vectors change under rotations. The cylindrical-polar coordinates are also useful when describing physical systems with vertical axial symmetry, such as rotating fluid columns; e.g., fluids in a rotating circular tank or for cyclostrophically balanced flow as in ideal tornadoes. Here we work through the details using a cylindrical-polar coordinate system in the rigid-body rotating reference frame.

In particular, we here derive the cylindrical-polar coordinate representation of the velocity and acceleration vectors for a particle moving in a reference frame rigid-body rotating with a constant rate about the vertical axis ($\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}$). The derivation is directly analogous to the spherical coordinate representation presented in the chapter. Specifically, in Section 12.7.2 we determined a spherical coordinate representation of the velocity vector, and then in Section 12.9 we found the spherical coordinate representation of the acceleration vector.

(A) Determine the representation of the inertial frame velocity vector, $\boldsymbol{V} = \mathrm{d}\boldsymbol{X}/\mathrm{d}t$, in terms of cylindrical-polar coordinates. Hint: remember to include the solid body motion of the rotating reference frame, which was discussed in Section 12.2.2.

(B) Determine the representation of the inertial frame acceleration vector, $\boldsymbol{A} = \mathrm{d}\boldsymbol{V}/\mathrm{d}t$, in terms of cylindrical-polar coordinates.

(C) Writing the inertial frame acceleration in the form

$$\boldsymbol{A} = \boldsymbol{A}_{\mathsf{cylindrical}} + \boldsymbol{A}_{\mathsf{metric}} - \boldsymbol{A}_{\mathsf{centrifugal}} - \boldsymbol{A}_{\mathsf{coriolis}}, \tag{12.131}$$

give the mathematical expressions for these terms:

- $\boldsymbol{A}_{\mathsf{cylindrical}}$ = acceleration in the rotating reference frame using cylindrical-polar coordinates;
- $\boldsymbol{A}_{\mathsf{metric}}$ = acceleration due to motion of the cylindrical-polar unit vectors relative to the rotating reference frame;
- $\boldsymbol{A}_{\mathsf{centrifugal}}$ = centrifugal acceleration;
- $\boldsymbol{A}_{\mathsf{coriolis}}$ = Coriolis acceleration.

EXERCISE 12.5: VELOCITY PROJECTED ONTO ACCELERATION

The kinetic energy per mass of a particle is given by

$$\mathcal{K} = \boldsymbol{V} \cdot \boldsymbol{V}/2, \tag{12.132}$$

where $\boldsymbol{V}$ is the velocity of the particle viewed in the inertial reference frame. Hence, in an inertial reference frame it is trivial to show that

$$\frac{\mathrm{d}\mathcal{K}}{\mathrm{d}t} = \boldsymbol{V} \cdot \boldsymbol{A} \tag{12.133}$$

through use of the chain rule, where $\boldsymbol{A} = \mathrm{d}\boldsymbol{V}/\mathrm{d}t$ is the inertial frame acceleration. Verify that this identity also holds when viewing the motion in the rigid-body rotating reference frame. For simplicity make use of planetary Cartesian coordinates.

EXERCISE 12.6: ANGLE OF A PLUMB LINE

A plumb line defines the local vertical direction, which is parallel to the effective gravity. As illustrated in Figure 12.5, the planetary centrifugal acceleration causes the plumb line to not extend through the Earth center.

(A) What is the angle that the plumb line makes with a line that extends through the Earth center? Hint: read the caption to Figure 12.5, and orient your thinking according to the left panel of this figure. Make use of the spherical coordinate version of the equations of motion (12.124)–(12.126).

(B) Explain why the plumb line angle, α, vanishes at both the equator and the poles.

EXERCISE 12.7: GEOMETRY OF CONSTANT GEOPOTENTIAL SURFACES

Here we examine some properties of the geopotential given by equation (12.79), where the squared rigid-body speed is $U^2_{\mathsf{rigid}} = (\Omega\, r \cos\phi)^2$. We only consider geopotentials that are close to the radius of the planet, so that we can assume the gravitational acceleration, g_{e}, is constant and takes on its value at R_{e} as in Sections 11.7.4 and 12.8.3.

(A) Sketch surfaces of constant geopotential according to equation (12.79).

(B) By equating the geopotential going around the pole to that going around the equator, show that the polar radius is less than the equatorial radius when $\Omega > 0$.

(C) Taking the terrestrial values of g_{e}, R_{equator}, and Ω, what is the polar radius R_{pole}? Compare to the measured value of the polar radius given by equation (12.127).

EXERCISE 12.8: GENERAL FORM OF THE GEOPOTENTIAL
In Exercise 12.7, as in Sections 11.7.5 and 12.8.3, we only considered geopotentials that are close to the radius of the planet. Show that geopotentials have larger radius at the equator than at the poles even when not making this assumption. Hint: maintain the general form of the gravitational potential as given by equation (11.123), then add the potential for the planetary centrifugal acceleration (12.74). Evaluate the geopotential at the pole and then show that this same geopotential surface has a larger radial position anywhere equatorward of the pole.

EXERCISE 12.9: SCALING TO JUSTIFY USE OF GEOPOTENTIAL COORDINATES
Summarize the argument that justifies the use of geopotential coordinates while retaining the spherical geometry. Make use of your favorite textbook discussion such that given in Chapter 2 of *Vallis* (2017).

EXERCISE 12.10: ACCELERATIONS ACTING ON A RESTING PARTICLE
In this exercise we consider the accelerations acting on a particle at rest (in the rotating frame) on a smooth/frictionless rotating spherical planet and a rotating oblate spheroidal planet. We also consider similar questions in Section 13.8.

(A) Motion of a particle on a rotating spherical planet is described using the spherical coordinates from Section 12.10.2 with the corresponding equations of motion (12.124)-(12.126). Suppose that we drop a particle on a rotating sphere and that we describe its motion using the spherical coordinates. Discuss the initial acceleration of the particle that is released from rest.

(B) Motion of a particle on a rotating oblate spheroid planet is described using the geopotential coordinates from Section 12.10.3 with the corresponding equations of motion (12.130a)-(12.130c). Suppose that we drop a particle on a rotating oblate spheroid and that we describe its motion using the geopotential coordinates. Discuss the initial acceleration of the particle that is released from rest. Note that we return to this freely falling particle in Section 13.8.3.

EXERCISE 12.11: COMPARING $w = 0$ AND $\dot{w} = 0$ MOTION BETWEEN SPHERICAL AND GEOPOTENTIAL
In Section 12.10.2 we wrote the spherical coordinate version of the equations of motion, whereas in Section 12.10.3 we wrote the geopotential coordinate version. Compare the vertical velocity equation for these two coordinate for the special case of zero vertical motion ($w = 0$ and $\dot{w} = 0$) in their respective coordinates. Discuss.

Chapter 13

SYMMETRY AND CONSERVATION

A symmetry is a discrete or continuous transformation that leaves a physical system unchanged. For example, let $\mathbf{X}$ be a trajectory satisfying Newton's equation of motion and $\mathcal{A}$ be an operation. If $\mathcal{A}[\mathbf{X}]$ also satisfies Newton's equation of motion then $\mathcal{A}$ is a symmetry of the physical system. We consider two versions of a symmetry transformation. Active transformations alter the physical system (e.g., move an experiment from one point in space to another), whereas passive transformations alter the mathematical description of the physics (e.g., change coordinates). As we study in VOLUME 4, Noether's theorem connects a symmetry to a conservation law, which in turn provides a dynamical constraint on the motion. In this chapter we study a variety of dynamical constraints respected by particle motion. We then use these dynamical constraints as a means to study the nature of the motion and, in turn, to further our understanding. We are particularly interested in studying motion as observed from a rigid-body rotating reference frame as per a terrestrial observer on a rotating planet. As part of this study, we dive deeply into the nature of the Coriolis acceleration.

One of the key reasons to make use of conserved quantities concerns their additive nature. Namely, a conserved quantity for a system composed of several weakly interacting parts is given by the sum of the conserved quantity for the individual parts. As a result, symmetries and conservation laws are fundamental to how we garner a qualitative and quantitative understanding of motion in which, for many purposes, it is more useful to know the dynamically conserved properties of the motion than details of the trajectories.

CHAPTER GUIDE

This chapter relies on our study of particle mechanics in Chapter 12. Connecting symmetries to conservation laws is fundamental to theoretical physics, with this chapter providing a heuristic understanding of this connection. A more deductive approach is given in Chapter 14 when studying Noether's theorem.

13.1 Trajectories and dynamical constraints

Time integration (twice) of the equation of motion provides an expression for the trajectory of a particle. To assist in this integration we find it useful to make use of dynamical constraints respected by the motion. Dynamical constraints manifest as conservation laws and are generally essential for any practical determination of the trajectory. Dynamical constraints also provide predictive statements of value when studying the stability of motion and for developing numerical methods for simulations.

Knowledge of the trajectory is important if we are interested in details of a particular realization of the motion. For example, the particle might be an idealization of a satellite orbiting a planet, with an accurate trajectory needed to predict its location at a future time. However, for other purposes we might wish to know that all particles, no matter what trajectory, conserve axial angular momentum and mechanical energy. This information offers the ability to understand basic properties of the motion and to predict its response to perturbations, even without determining trajectories. Dynamical constraints imposed by conservation laws are especially relevant for fluids since it is rare to determine the analytical expression for the motion of fluid particles.

13.1.1 Connecting symmetries to conservation laws

The discovery of conservation laws often comes from inspired manipulations of the kinematical and dynamical equations of motion. However, there is a more robust and fundamental means to deduce conservation laws through their connection to symmetries, with a symmetry manifesting as an operation or transformation that leaves the physical system unchanged. For example, does the physical system remain unchanged when shifting the origin of time? If so, then mechanical energy is a constant of the motion. Likewise, if there is rotational symmetry around an axis, then the associated angular momentum is a constant of the motion.

The connection between symmetries (kinematics) and conservation laws (dynamics) was made by *Noether* (1918) (see *Noether and Tavel* (2018) for an English translation). We provide mathematical expressions of Noether's theorem in Chapter 14 as part of our study of analytical mechanics and Hamilton's principle. It is sufficient for this chapter to make use of this theorem as a conceptual framework for understanding conservation laws and their connections to symmetries. Quite simply, if there is a symmetry then there is a corresponding conservation law, and vice versa.

It is very useful to identify conserved quantities as a means to understand and to dynamically constrain the motion. This perspective holds even when the symmetries giving rise to conserved quantities are broken through the introduction of non-conservative forces such as friction. For example, as seen in Section 11.2.7, friction breaks time translation symmetry and so leads to the dissipation of mechanical energy. Nonetheless, understanding the conservative motion, and the associated energy conservation law, offers insights for the case with friction. In this chapter, we offer two examples to support this point: mechanical energy conservation and axial angular momentum conservation. These conservation laws formulated here for particles also hold, in a modified form, for the motion of geophysical fluids. Additional conservation properties also arise that are unique to the continuum, with conservation of potential vorticity the most notable one we encounter in geophysical fluids.

13.1.2 Further study

Conservation laws and symmetries in classical mechanics are lucidly discussed in Chapters 1 and 2 of *Landau and Lifshitz* (1976). Pedagogical presentations on these topics can be found in this online lecture from the Space Time series and this online lecture from Physics with Elliot. This essay about Emmy Noether provides insights into this mathematician whose work, conducted under some very unfortunate circumstances, forever connected symmetries to conservation laws, with this connection providing a foundation for much of 20th century theoretical physics.

13.2 Time reversal symmetry

As a warm-up to the ideas of symmetry, consider the question about time reversibility. A deterministic process is time-reversible if the time-reversed process satisfies the same dynamical equations as the forward-time process. That is, the dynamical equations are symmetric under a change in the sign of time so that the time-reversed evolution of one state is equivalent to the forward-time evolution of a corresponding state. We here discuss time reversal symmetry in the context of the point particle with trajectory, $\mathbf{X}(t)$, and velocity, $\mathbf{V}(t) = \mathrm{d}\mathbf{X}(t)/\mathrm{d}t$.

If $\mathbf{X}(t)$ is a solution to the equations of motion, then we seek conditions needed for

$$\mathbf{X}^*(t^*) = \mathbf{X}(-t) \quad \text{and} \quad \mathbf{V}^*(t^*) = \mathrm{d}\mathbf{X}^*(t^*)/\mathrm{d}t^* = -\mathrm{d}\mathbf{X}(-t)/\mathrm{d}t \quad \text{with} \quad t^* = -t, \tag{13.1}$$

to define the same trajectory traversed backwards in time. We answer this question by recalling from Section 12.10.1 that with planetary rotation a constant in time, then the equation of motion is given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\mathrm{d}\mathbf{X}(t)}{\mathrm{d}t} + 2\,\boldsymbol{\Omega}\times\mathbf{X}(t)\right] = -\nabla\Phi(t). \tag{13.2}$$

We are interested in the constraints that ensure the following equation is satisfied

$$\frac{\mathrm{d}}{\mathrm{d}t^*}\left[\frac{\mathrm{d}\mathbf{X}^*(t^*)}{\mathrm{d}t^*} + 2\,\boldsymbol{\Omega}^*\times\mathbf{X}^*(t^*)\right] = -\nabla\Phi^*(t^*). \tag{13.3}$$

The effective gravitational acceleration remains time reversible if

$$\Phi^*(t^*) = \Phi(t), \tag{13.4}$$

which is trivially satisfied for $\Phi = g\,z$. The Coriolis acceleration is velocity dependent so that it generally breaks time reversal symmetry. However, we can recover time symmetry by assuming that the rotation direction switches when time reverses so that

$$\boldsymbol{\Omega}^* = -\boldsymbol{\Omega}. \tag{13.5}$$

With this transformation, a trajectory, $\mathbf{X}(t)$, that solves the forward equation (13.2) yields a trajectory $\mathbf{X}(-t)$ that solves the same equation but with time (and $\boldsymbol{\Omega}$) reversed.

13.3 Potential momentum

In Section 12.10.1 we introduced the potential momentum, and here we study its conservation properties. Recall that potential momentum is a constant of the motion when a particle moves on a time independent geopotential in a direction where the geopotential does not change. That is, the conservation of potential momentum arises from a spatial symmetry of the geopotential.

13.3.1 Basics

Start with the planetary Cartesian coordinate equation of motion

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\dot{\boldsymbol{X}} + 2\,\boldsymbol{\Omega}\times\boldsymbol{X}\right] = -\nabla\Phi. \tag{13.6}$$

Now introduce the potential momentum per mass

$$\boldsymbol{M} \equiv \dot{\boldsymbol{X}} + 2\,\boldsymbol{\Omega}\times\boldsymbol{X} = \hat{\boldsymbol{x}}\,(u - 2\,\Omega\,y) + \hat{\boldsymbol{y}}\,(v + 2\,\Omega\,x) + \hat{\boldsymbol{z}}\,w, \tag{13.7}$$

in which case the equation of motion takes the form

$$\dot{\boldsymbol{M}} = -\nabla\Phi. \tag{13.8}$$

Now let $\hat{\boldsymbol{s}}$ be a unit vector tangent to the geopotential surface so that $\hat{\boldsymbol{s}}\cdot\nabla\Phi = 0$. Assuming the geopotential surface is time independent so that $\hat{\boldsymbol{s}}$ is also time independent, then the equation of motion (13.8) leads to

$$\frac{\mathrm{d}(\hat{\boldsymbol{s}}\cdot\boldsymbol{M})}{\mathrm{d}t} = 0. \tag{13.9}$$

That is, the projection of the potential momentum onto a static geopotential surface is a constant of motion. This dynamical constraint arises since we cannot distinguish one point on the geopotential from another; i.e., there is a symmetry associated with motion along the static geopotential. Noether's theorem then says that this geometric symmetry leads to a constant of the motion, here given by that component of potential momentum within the geopotential surface. We illustrate this situation in Figure 13.1 with a horizontal geopotential surface.

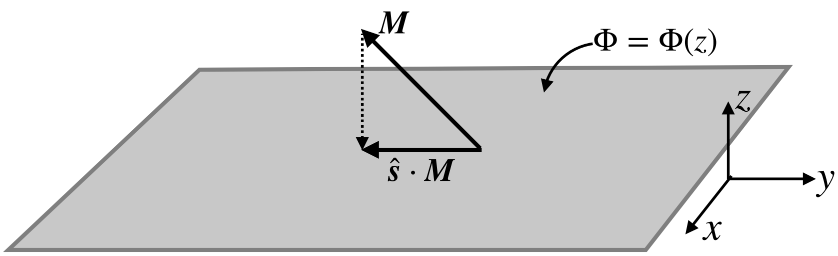

FIGURE 13.1: The projection of the potential momentum onto the geopotential surface is a constant of the motion, $\mathrm{d}(\hat{\boldsymbol{s}} \cdot \boldsymbol{M})/\mathrm{d}t = 0$. Here the geopotential surface is the x-y-plane so that $\hat{\boldsymbol{x}} \cdot \boldsymbol{M} = u - 2\,\Omega\,y$ and $\hat{\boldsymbol{y}} \cdot \boldsymbol{M} = v + 2\,\Omega\,x$ are the two conserved components of potential momentum.

Consider a particle with potential momentum, $\boldsymbol{M}$, and move it from an arbitrary point to a reference position with $\boldsymbol{X} = 0$. Upon reaching the reference position, the horizontal velocity of the particle must equal to $\boldsymbol{M}$ in order to maintain the same potential momentum. This example motivates the name "potential momentum", since $\boldsymbol{M}$ measures the potential for relative motion contained in the particle as it moves along a geopotential.

13.3.2 Comment about terminology

As noted on page 51 of *Markowski and Richardson* (2010), one might see potential momentum referred to as *pseudo angular momentum*, with some dropping the "pseudo" portion to the name. In either case, it is important to note that potential momentum is distinct from angular momentum. In particular, there is no moment-arm as part of the potential momentum, nor is there any axial symmetry corresponding to the conservation law.

Many authors use the term *absolute momentum* rather than potential momentum, perhaps in reference to the momentum measured in the absolute or inertial reference frame. However, that connection is incorrect since the inertial frame velocity is (Section 12.7.1)

$$\boldsymbol{V} = \boldsymbol{V}_{\text{cartesian}} + \boldsymbol{U}_{\text{rigid}} = \boldsymbol{V}_{\text{cartesian}} + \boldsymbol{\Omega} \times \boldsymbol{X}. \tag{13.10}$$

The factor of two multiplying the rotation rate in the potential momentum arises from the Coriolis acceleration. In contrast, the rigid-body rotation velocity contributes to the inertial frame velocity, and there is no factor of two multiplying the rotation rate.

13.4 Particle motion on the f-plane

For many purposes in geophysical fluid mechanics, it is useful to introduce the tangent plane approximation for motion on a rotating sphere. In this approximation, motion occurs on a geopotential surface with the surface approximated as a locally flat horizontal plane. Furthermore, we use local tangent plane Cartesian coordinates, which are distinct from the planetary Cartesian coordinates (origin at the planet's center) illustrated in Figure 12.3. The f-plane approximation sets the Coriolis parameter to a constant,

$$f = 2\,\Omega \sin\phi_{\circ}, \tag{13.11}$$

where $\phi_{\circ}$ is a chosen latitude. Consequently, a particle constrained to move on a constant geopotential under the f-plane approximation, yet without feeling any other forces, maintains a constant horizontal potential momentum

$$\frac{\mathrm{d}M_x}{\mathrm{d}t} = \frac{\mathrm{d}(u - f\,y)}{\mathrm{d}t} = 0 \tag{13.12a}$$

$$\frac{\mathrm{d}M_y}{\mathrm{d}t} = \frac{\mathrm{d}(v + f\,x)}{\mathrm{d}t} = 0, \tag{13.12b}$$

where we introduced the horizontal velocity components $(u, v) = (\dot{x}, \dot{y})$. The two conservation laws (13.12a) and (13.12b) provide dynamical constraints on the particle motion on a constant geopotential surface. It is important to recall from Section 12.8.3 that motion on a geopotential incorporates the acceleration from both the central gravitational field and the planetary centrifugal acceleration. Hence, there is no explicit appearance of a centrifugal acceleration for particle motion on the f-plane.

13.4.1 Oscillator equation for inertial oscillations

Taking the time derivative of the zonal equation (13.12a) and using the meridional equation (13.12b) leads to

$$\ddot{u} - f\,\dot{v} = \ddot{u} + f^2 u = 0. \tag{13.13}$$

Similar manipulations for the meridional velocity equation render the free oscillator equation for each component of the horizontal velocity

$$\ddot{u} + f^2 u = 0 \qquad \text{and} \qquad \ddot{v} + f^2 v = 0. \tag{13.14}$$

Motion that satisfies this equation is termed an inertial oscillation.

13.4.2 Particle trajectory and velocity

Time integrating the equation of motion (13.14) renders the Cartesian expression for the particle trajectory and its velocity

$$\boldsymbol{X}(t) = (U/f)\,[\hat{\boldsymbol{x}} \sin(f t) + \hat{\boldsymbol{y}} \cos(f t)] \tag{13.15a}$$

$$\boldsymbol{U}(t) = U\,[\hat{\boldsymbol{x}} \cos(f t) - \hat{\boldsymbol{y}} \sin(f t)], \tag{13.15b}$$

where $U > 0$ is the particle speed, which is a constant, and we assumed the initial conditions

$$\boldsymbol{X}(0) = (U/f)\,\hat{\boldsymbol{y}} \qquad \text{and} \qquad \boldsymbol{U}(0) = U\,\hat{\boldsymbol{x}}. \tag{13.16}$$

From the particle trajectory equation (13.15a), we see that the particle motion is circular with a radius

$$R = U/|f|. \tag{13.17}$$

As depicted in Figure 13.2, northern hemisphere ($f > 0$) inertial particle motion occurs in the clockwise direction whereas southern hemisphere motion is counter-clockwise. This motion

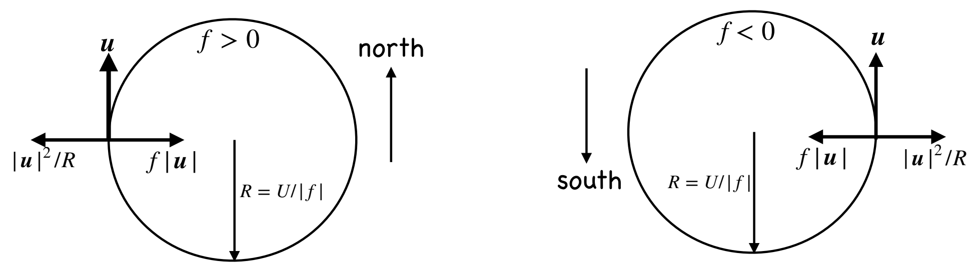

FIGURE 13.2: Inertial oscillation on a horizontal f-plane occurs when the particle's centrifugal acceleration (arising from the particle's curved motion) balances its Coriolis acceleration (arising from the rotating reference frame). Left panel: $f > 0$ for the northern hemisphere, revealing that the motion is an anti-cyclonic (clockwise) circular motion with radius $|R| = U/|f|$. The Coriolis acceleration is to the right, pointing into the center of the inertial circle, whereas the particle's centrifugal acceleration points away from the center. Right panel: Counterclockwise motion in the southern hemisphere with the same balance between Coriolis and centrifugal accelerations. Note that since we have taken and f-plane approximation, the planetary centrifugal acceleration is included in the effective gravitational acceleration.

arises from the rightward deflection by the Coriolis acceleration in the northern hemisphere and leftward deflection in the southern. Consequently, particle motion occurs in an anti-cyclonic sense (opposite to the sense of the rotating reference frame). This motion arises from a balance between the Coriolis acceleration of the rotating frame and the centrifugal acceleration arising from the particle's circular motion. The only way to realize this balance is for the particle to move anti-cyclonically, with the Coriolis acceleration pointing towards the inside of the circle and the centrifugal acceleration pointing outside. Finally, note that the potential momentum vanishes since

$$\boldsymbol{M}(t) = \boldsymbol{U}(t) + f\,\hat{\boldsymbol{z}} \times \boldsymbol{X}(t) = 0. \tag{13.18}$$

Adding an arbitrary constant to the initial position makes the potential momentum equal to a nonzero constant.

13.4.3 Period of the circular motion

The circular motion of the particle has a constant speed and moves around a circle with a period

$$T_{\text{inertial}} = \frac{2\,\pi}{f} = \frac{11.97}{|\sin\phi_{\circ}|}\,\text{hours}, \tag{13.19}$$

where we set $\Omega = 7.292 \times 10^{-5}\text{s}^{-1}$ (equation (12.1)). This period is the time it takes to go around the circle. It is smallest at the poles, where the latitude $\phi_{\circ} = \pm\pi/2$ and $T_{\text{smallest}} \approx 12\,\text{hour}$. At the equator, $\phi_{\circ} = 0$, so that the radius of the inertial circle is infinite and inertial oscillations are unavailable. Furthermore, T_{inertial} is the time for a Foucault pendulum to turn through π radians, so that T_{inertial} is sometimes referred to as one-half a pendulum day.

13.4.4 Inertial oscillations

The circular particle motion studied in this section is sometimes referred to as an inertial oscillation. Again, this circular motion occurs on a constant geopotential and on the f-plane when there is a balance between the Coriolis acceleration from planetary rotation and centrifugal acceleration arising from the curved motion of a particle on a constant geopotential. Importantly, the centrifugal acceleration here is not the planetary centrifugal acceleration (Section 12.8.3), with the planetary centrifugal acceleration absorbed into the effective gravity that is a fixed constant along a geopotential.

Furthermore, the name "inertial" does not refer to motion as observed in an inertial reference frame. Instead, it is used as a synonym for motion that is free while maintaining a constant geopotential.[1] More generally, there has been some confusion in the literature concerning the forces acting in inertial oscillations, particularly when not making a tangent plane approximation. *Early* (2012) clarifies the fundamental physics. We return to these matters in Chapter 14, making use of Lagrangian mechanics to examine the energetics and forces arising with motion of the particle.

13.4.5 Comments

Inertial oscillations of fluid elements are described by the above constant potential momentum equation of motion. Such oscillations are commonly measured by ocean current meters, especially in higher latitude regions where diurnal (day-night) variations in wind forcing have a strong projection onto the inertial period. This resonant forcing puts energy into inertial or near-inertial motions. It is quite amazing that such oscillations are indeed found in the ocean, given that we have ignored pressure and friction, which are two forces that impact on fluid motion. A key reason we can observe such motion is that upper ocean currents are often generated by winds even in the absence of horizontal pressure gradients. Hence, there are occasions when the motion is not strongly affected by pressure gradients or friction, thus allowing for the inertial oscillations to manifest.

13.5 Angular momentum

As introduced angular momentum in Section 11.3.5, angular momentum the moment of linear momentum and it is given by the following expression for the point particle

$$\mathbf{L} = m\,\mathbf{X} \times \mathbf{V}. \tag{13.20}$$

In this section we study the angular momentum computed relative to the center of a spherical planet; i.e., where $\mathbf{X}$ is the position of the particle with respect to the sphere's center. For a non-rotating sphere, there is spherical symmetry, meaning that rotations around any axis extending out from the sphere's center represent symmetry operations. Noether's theorem then says that the angular momentum projected onto any axis is a constant of the motion. For a rotating spherical planet, the polar rotational axis is distinguished from all other axes, so that the spherical symmetry of the non-rotating planet is broken down to symmetry around just the rotational axis. Indeed, for oblate spheroidal planets, the rotation axis is the only symmetry axis. Noether's theorem then says that it is only the axial angular momentum that is conserved for frictionless motion around a rotating planet.

To be more specific, we ask whether a point particle knows anything about the longitudinal angle, λ, on the rotating planet (Figure 13.3)? Assuming the planet is frictionless and smooth (i.e., no mountains), and the planet rotates around the polar axis (along the planetary $\hat{\boldsymbol{z}}$ direction), then there is indeed an arbitrariness in the value of the longitude. That is, the physical system remains unchanged if we shift the longitudinal angle by a constant. Noether's theorem then says that this symmetry leads to a corresponding conservation of angular momentum around the polar axis.

[1] See Section 11.4.2 for comments on the confusing terminology that is sometimes used in the literature that refers to the Coriolis acceleration and centrifugal acceleration as "inertial" accelerations.

We here display the manipulations that show axial angular momentum, $L^z = \hat{\boldsymbol{z}} \cdot \boldsymbol{L}$, is a constant of the motion for motion around the rotating sphere, and furthermore show that the other components to angular momentum, $L^x = \hat{\boldsymbol{x}} \cdot \boldsymbol{L}$ and $L^y = \hat{\boldsymbol{y}} \cdot \boldsymbol{L}$, are not constant.

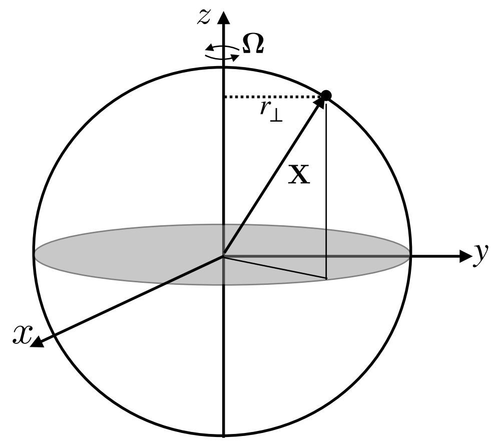

FIGURE 13.3: Axial angular momentum is the component of the angular momentum through the polar axis, $L^z = \hat{\boldsymbol{z}} \cdot \boldsymbol{L}$. It is the moment of the zonal momentum around the polar axis, with the moment-arm, $r_\perp = r \cos\phi$, the distance to the axis of rotation, whereas r is the radial distance to the center of the planet. The axial angular momentum can be written in the equivalent manners $L^z = m\, r_\perp\, (\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{V}) = m\, r_\perp^2\, (\dot{\lambda} + \Omega) = m\, r_\perp\, (u + r_\perp\, \Omega)$. It is a constant of the motion for the particle moving on a smooth rotating planet in the absence of friction.

13.5.1 Axial angular momentum

The axial angular momentum is the component of the angular momentum through the polar axis, and it can be written in the following equivalent manners

$$L^z = \boldsymbol{L} \cdot \hat{\boldsymbol{z}} = m\, (\boldsymbol{X} \times \boldsymbol{V}) \cdot \hat{\boldsymbol{z}} = m\, (\hat{\boldsymbol{z}} \times \boldsymbol{X}) \cdot \boldsymbol{V} = m\, r \cos\phi\, (\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{V}) = m\, r_\perp\, (\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{V}). \tag{13.21}$$

Evidently, the component of angular momentum along the polar axis equals to the component of the linear momentum in the longitudinal direction, multiplied by the distance to the polar rotational axis (the moment-arm)

$$r_\perp = r \cos\phi. \tag{13.22}$$

In deriving equation (13.21), we made use of the identity

$$\hat{\boldsymbol{z}} \times \boldsymbol{X} = r\, \hat{\boldsymbol{z}} \times \hat{\boldsymbol{r}} = r_\perp\, \hat{\boldsymbol{\lambda}}, \tag{13.23}$$

which we use below to prove that axial angular momentum is a constant of the motion.

We can write the axial angular momentum from equation (13.21) in terms of the rotating frame quantities. To do so, decompose the inertial frame velocity written using spherical coordinates according to equation (12.54f), which yields

$$L^z = m\, r_\perp\, (\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{V}) = m\, r_\perp^2\, (\dot{\lambda} + \Omega) = m\, r_\perp\, (u + r_\perp\, \Omega). \tag{13.24}$$

When measured from the rotating terrestrial frame, the axial angular momentum consists of two terms: one from the zonal velocity of the particle relative to the rotating planet and another from the rigid-body rotation of the planet.

13.5.2 Axial angular momentum is a constant of the motion

Making use of planetary Cartesian and planetary spherical coordinates, we find the time derivative of the axial angular momentum is given by

$$
\begin{aligned}
m^{-1}\,\mathrm{d}L^{z}/\mathrm{d}t &= \mathrm{d}/\mathrm{d}t\,[(\boldsymbol{X}\times\boldsymbol{V})\cdot\hat{\boldsymbol{z}}] && \text{definition of axial angular momentum} && (13.25\text{a})\\
&= \mathrm{d}/\mathrm{d}t\,[(\hat{\boldsymbol{z}}\times\boldsymbol{X})\cdot\boldsymbol{V}] && \text{cyclic permutation equation (1.61j)} && (13.25\text{b})\\
&= (\hat{\boldsymbol{z}}\times\boldsymbol{V})\cdot\boldsymbol{V} + (\hat{\boldsymbol{z}}\times\boldsymbol{X})\cdot\boldsymbol{A} && \dot{\boldsymbol{X}}=\boldsymbol{V},\ \dot{\boldsymbol{V}}=\boldsymbol{A},\ \text{and } \mathrm{d}\hat{\boldsymbol{z}}/\mathrm{d}t=0 && (13.25\text{c})\\
&= (\hat{\boldsymbol{z}}\times\boldsymbol{X})\cdot\boldsymbol{A} && (\hat{\boldsymbol{z}}\times\boldsymbol{V})\cdot\boldsymbol{V}=0 && (13.25\text{d})\\
&= r_{\perp}\,\hat{\boldsymbol{\lambda}}\cdot\boldsymbol{A}. && \hat{\boldsymbol{z}}\times\boldsymbol{X}=r_{\perp}\,\hat{\boldsymbol{\lambda}} \text{ from equation (13.23)}. && (13.25\text{e})
\end{aligned}
$$

The inertial frame acceleration arises just from the central-force gravitational field (equation (12.116))

$$
\boldsymbol{A} = -\nabla\Phi_{\mathrm{e}} = -g_{\mathrm{e}}\,\hat{\boldsymbol{r}}. \tag{13.26}
$$

Evidently, the central gravity force has no affect on the axial angular momentum since $\hat{\boldsymbol{\lambda}}\cdot\hat{\boldsymbol{r}}=0$. We are thus led to axial angular momentum conservation

$$
\frac{\mathrm{d}L^{z}}{\mathrm{d}t} = 0. \tag{13.27}
$$

Conservation of axial angular momentum provides an important dynamical constraint on the particle motion, and we explore implications in Section 13.6. In contrast, in Exercise 13.1 we show that neither L^{x} and L^{y} are constants.

13.6 Facets of the Coriolis acceleration

The Coriolis acceleration is a primary feature of large-scale motion in the ocean and atmosphere. In this section we offer a selection of thought experiments to help build understanding of the Coriolis acceleration. As part of those thought experiments, we show that acceleration induced by axial angular momentum conserving motion connects to the Coriolis acceleration appearing in the zonal momentum equation.

13.6.1 Two pictures to help frame the discussion

In Figure 13.4 we depict the motion of a particle on a rotating and frictionless flat table. We can observe this motion from the laboratory frame, which we assume approximates an inertial frame, or we can view the motion from the rotating table (preferably viewed by a camera on the table so as to avoid becoming dizzy). The figure caption details the motion from the two reference frames. Evidently, the rotating reference frame observer sees the particle motion deflected to the right, regardless the direction that the particle moves on the table. In contrast, the inertial reference frame observer views the table moving to the left according to the counter-clockwise motion of the table, whereas the particle moves along a straight line. We emphasize that the motion is objectively the same; it is the same particle moving through space along the same single trajectory. However, the perspectives of observers in the two reference frames differ.

The rotating table example in Figure 13.4 captures the essence of how the Coriolis acceleration, encountered in the non-inertial frame, acts to deflect particles moving around the rotating planet. In particular, it focuses on the Coriolis acceleration arising from the locally vertical component of the planetary rotation, in which each point on the planet is akin to a rotating

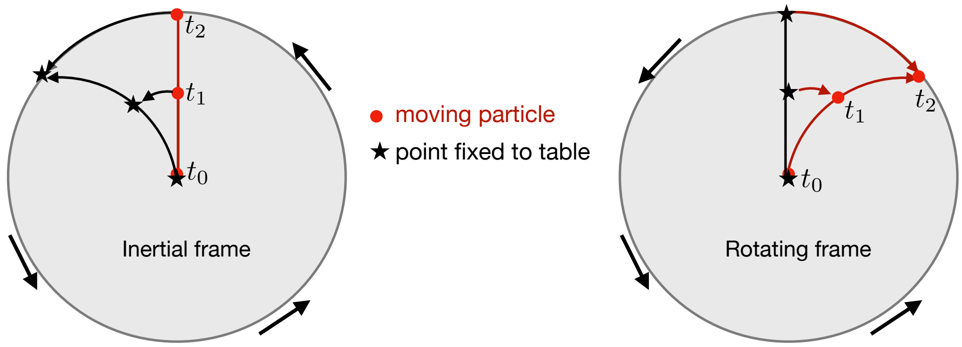

FIGURE 13.4: Illustrating the motion of a particle on a frictionless table (e.g., a perfectly slick ice rink), with the table rotating in the counter-clockwise sense around its center (corresponding to the planetary rotation as viewed from the northern hemisphere). The particle (red dot) moves outward from the center of the table at time t_0 to the edge at time t_2, with an intermediate position at time t_1. The left panel shows the motion viewed from an inertial (non-rotating) reference frame, with the particle observed to move along a straight line from the table center to the table edge. During this transit, the table moves underneath the particle. Stars designate points painted on the table, with the stars moving to the left as the table rotates. The right panel depicts the motion viewed within the frame at rest with the rotating table (rigid-body rotating non-inertial reference frame). Here, the positions on the table remain fixed (the stars do not move), whereas the particle moves along a rightward curved trajectory as it reaches the edge. The rotating frame observer interprets the rightward deflection as due to the Coriolis acceleration.

table. Importantly, the rotation rate (and sign) is a function of latitude since the imprint of the planetary rotation depends on the latitude. So although the planet rotates at a constant angular velocity around the polar axis, the imprint of that rotation has a latitudinal dependence due to the spherical geometry of the planet. This added feature of planetary Coriolis acceleration is referred to as differential rotation, and it provides a fundamental distinction from the Coriolis acceleration of the rotating flat table of Figure 13.4, where each point on the table feels the same angular rotation vector.

Figure 13.5 summarizes features of the Coriolis acceleration seen for motion on a rotating planet. Here, the particle trajectories are deflected to the right in the northern hemisphere, where the planetary rotation is counter-clockwise just as in the rotating table example. However, for the southern hemisphere the planetary rotation is clockwise, so that the Coriolis deflection is to the left. To help understand the sign switch, imagine viewing the planet from above the north pole, whereby the rotation of the planet is counter-clockwise. Now view the planet from below the south pole, in which case the earth is seen to rotate clockwise.[2] This sign swap reflects the axial vector property of the angular velocity vector, whereby the angular velocity changes sign upon changing the handedless of the axes (see Section 1.6.2).

13.6.2 Constraints from axial angular momentum conservation

There are various ways to express the constraints resulting from conservation of axial angular momentum, most notably whether we describe the motion from the inertial frame or rotating frame. Here we explore the constraints viewed by the rotating terrestrial observer. Since the mass, m, of the particle is constant, we find it a bit more convenient to work with the axial angular momentum per mass, which from Section 13.5.1 takes on the form

$$l^z = L^z/m = r_\perp \, \hat{\boldsymbol{\lambda}} \cdot \boldsymbol{V} = r_\perp^2 \, (\dot{\lambda} + \Omega) = r_\perp \, (u + r_\perp \, \Omega), \tag{13.28}$$

[2]Those who do not trust their mind's eye are encouraged to perform the thought experiment with a globe.

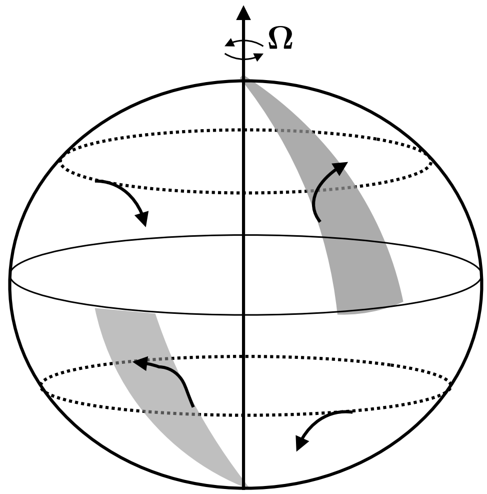

FIGURE 13.5: For a particle moving at constant radial distance from the planet center, the Coriolis acceleration (arising from the locally vertical component to the planetary rotation) provides an acceleration directed to the right in the northern hemisphere and to the left in the southern hemisphere. This figure summarizes key features of the Coriolis acceleration for large-scale geophysical motions as seen by terrestrial observers on the rotating planet. Notably, however, it is missing the vertical (radial) acceleration induced by the Coriolis acceleration. This vertical acceleration, as discussed in Section 13.6.8, is tiny compared to that from gravity and so it is commonly ignored.

where the zonal velocity component is given by equation (12.58)

$$u = r_\perp \, \dot{\lambda} = (r \cos\phi) \, \dot{\lambda}. \tag{13.29}$$

Note that for most geophysical applications, $l^z > 0$ since rigid-body motion dominates over the relative zonal velocity:

$$r_\perp \, \Omega > |u| \qquad \text{for most terrestrial motions,} \tag{13.30}$$

with $r_\perp \, \Omega = 465 \text{ m s}^{-1}$ at the equator (see Figure 12.2). In Exercise 13.2 we consider the interesting, but geophysically uncommon, case where $l^z \leq 0$.

Constraints arising from changes to the moment arm

To determine the axial angular momentum constraints, we set $\delta l^z = 0$, where δ refers to any small change brought about by some perturbation whose origins are not of concern. When writing $l^z = r_\perp^2 \, (\dot{\lambda} + \Omega)$ we have

$$\delta l^z = 2 \, r_\perp \, \delta r_\perp \, (\dot{\lambda} + \Omega) + r_\perp^2 \, \delta \dot{\lambda} = 2 \, l^z \, \frac{\delta r_\perp}{r_\perp} + r_\perp^2 \, \delta \dot{\lambda}, \tag{13.31}$$

where we set $\delta\Omega = 0$ since the earth's rotation rate is assumed to be fixed. Furthermore, we find it useful to express the perturbations in terms of l^z since it is a constant of the motion. Likewise, when writing $l^z = r_\perp \, (u + r_\perp \, \Omega)$ we have

$$\delta l^z = \delta r_\perp \, (u + r_\perp \, \Omega) + r_\perp \, (\delta u + \Omega \, \delta r_\perp) = \Omega \, r_\perp \, \delta r_\perp \left[1 + \frac{l^z}{\Omega \, r_\perp^2} \right] + r_\perp \, \delta u. \tag{13.32}$$

Setting $\delta l^z = 0$ leads to the constraints

$$\delta\dot{\lambda} = -\frac{2\,l^z}{r_\perp^2}\,\frac{\delta r_\perp}{r_\perp} \Longleftrightarrow \delta u = -\Omega\,\delta r_\perp \left[1 + \frac{l^z}{\Omega\,r_\perp^2}\right]. \tag{13.33}$$

As noted above, $l^z > 0$ is generally the case for geophysical fluid motion. Consequently, axial angular momentum conserving motion that brings the particle closer to the rotation axis ($\delta r_\perp < 0$) leads to an eastward velocity change (angular velocity $\delta\dot{\lambda} > 0$ and zonal velocity $\delta u > 0$). The opposite occurs for motion with $\delta r_\perp > 0$. These results hold in both the northern and southern hemispheres.

Unpacking the moment arm changes in terms of radius and latitude

Since $r_\perp = r\cos\phi$, the distance to the rotational axis can be perturbed through changing the radial distance from the planet center, or by changing the latitude

$$\delta r_\perp = (\cos\phi)\,\delta r - (r\sin\phi)\,\delta\phi. \tag{13.34}$$

Assuming these perturbations occur over a small time increment, δt, allows us to write

$$\delta r_\perp/\delta t = w\cos\phi - v\sin\phi, \tag{13.35}$$

where we made use of equation (12.58) to introduce the meridional and vertical velocity components

$$v = r\,\delta\phi/\delta t = r\,\dot{\phi} \quad \text{and} \quad w = \delta r/\delta t = \dot{r}. \tag{13.36}$$

For motion in the earth's atmosphere and ocean, perturbations to vertical distance, δr, are far smaller than the distance to the earth's center, $\delta r \ll r$, since the thickness of the ocean and atmosphere are very small compared to the earth's radius. In this case, when $\phi \neq 0$, then $\delta r_\perp$ is affected much more by meridional motion at constant radial position (second term on right hand side of equation (13.35)) than by vertical motion at constant latitude (first term on right hand side of equation (13.35)). We return to this observation in Section 13.6.9 when discussing the shallow fluid approximation used to develop the hydrostatic primitive equations for large-scale motion of the atmosphere and ocean.

13.6.3 Axial angular momentum conservation and Coriolis acceleration

Consider a particle at rest in the rotating frame so that its axial angular momentum equals to that arising from the rigid-body motion of the planet, $l^z = r_\perp^2\,\Omega$. If we perturb this initial state while conserving axial angular momentum, equation (13.33) says there must be an associated change in the zonal velocity given by

$$\delta\dot{\lambda} = -2\,\Omega\,\frac{\delta r_\perp}{r_\perp} \quad \text{and} \quad \delta u = -2\,\Omega\,\delta r_\perp. \tag{13.37}$$

If the perturbation occurs over a time increment, δt, then we have

$$\delta\dot{\lambda}/\delta t = \ddot{\lambda} = -2\,\Omega\,\frac{\delta r_\perp/\delta t}{r_\perp} = -2\,\Omega\,(w\cos\phi - v\sin\phi)/r_\perp \tag{13.38a}$$

$$\delta u/\delta t = \dot{u} = -2\,\Omega\,(\delta r_\perp/\delta t) = -2\,\Omega\,(w\cos\phi - v\sin\phi), \tag{13.38b}$$

where we used equation (13.35) for $\delta r_\perp/\delta t$. The expression for $\dot{u}$ is precisely the same as the Coriolis acceleration appearing in the zonal momentum equation (12.130a)

$$\dot{u} + \frac{u\,(w - v\tan\phi)}{r} = \underbrace{-2\,\Omega\,(w\cos\phi - v\sin\phi)}_{\text{Coriolis acceleration}}. \tag{13.39}$$

We thus find that the Coriolis acceleration appearing in the zonal momentum equation is identical to the zonal acceleration induced by constraining the motion to conserve axial angular momentum. That is, by unpacking the constraint of axial angular momentum conservation to reveal the zonal momentum equation, the Coriolis acceleration is revealed to be part of the package. Furthermore, we see that both vertical and meridional motion lead to zonal accelerations through the Coriolis acceleration or, equivalently, through axial angular momentum conservation.

13.6.4 Axial angular momentum conservation and zonal acceleration

The discussion in Section 13.6.3 can be formalized by analyzing how the conservation of axial angular momentum leads to an expression for the zonal acceleration, $\dot{u}$. For this purpose, compute the time derivative of the first form of the axial angular momentum in equation (13.28), in which case

$$\frac{\mathrm{d}l^z}{\mathrm{d}t} = \frac{\mathrm{d}[(r\cos\phi)^2\,(\dot{\lambda} + \Omega)]}{\mathrm{d}t} \tag{13.40a}$$
$$= 2\,(\dot{r}\cos\phi - r\,\dot{\phi}\sin\phi)\,(\dot{\lambda}\,r\cos\phi + r\,\Omega\cos\phi) + (r\cos\phi)^2\,\ddot{\lambda} \tag{13.40b}$$
$$= 2\,(w\cos\phi - v\sin\phi)\,(u + r\,\Omega\cos\phi) + (r\cos\phi)^2\,\ddot{\lambda}, \tag{13.40c}$$

where we introduced the (u, v, w) velocity components according to equation (12.58). With the zonal velocity, $u = \dot{\lambda}\,r_\perp = \dot{\lambda}\,r\cos\phi$, we have

$$r\cos\phi\,\ddot{\lambda} = \dot{u} + \frac{u}{r\cos\phi}\,(v\sin\phi - w\cos\phi)\,, \tag{13.41}$$

so that equation (13.40c) takes the form

$$\frac{\mathrm{d}l^z}{\mathrm{d}t} = 2\,(w\cos\phi - v\sin\phi)\,(u + r\,\Omega\cos\phi) + (r\cos\phi)^2\,\ddot{\lambda} \tag{13.42a}$$
$$= (w\cos\phi - v\sin\phi)\,(u + 2\,r\,\Omega\cos\phi) + \dot{u}\,r\cos\phi. \tag{13.42b}$$

Setting $\mathrm{d}l^z/\mathrm{d}t = 0$ and rearranging then leads to a prognostic equation for the zonal velocity

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \left[\frac{u}{r\cos\phi} + 2\,\Omega\right](v\sin\phi - w\cos\phi). \tag{13.43}$$

The first term in the bracket arises from curvature of the sphere (the metric acceleration) whereas the second term is the Coriolis acceleration.

The same result can be obtained by performing the time derivative on the second form of the axial angular momentum in equation (13.28), in which case

$$\frac{\mathrm{d}l^z}{\mathrm{d}t} = \frac{\mathrm{d}\,[u\,r\,\cos\phi + \Omega\,(r\,\cos\phi)^2]}{\mathrm{d}t} \tag{13.44a}$$
$$= \dot{u}\,r\,\cos\phi + u\,\dot{r}\,\cos\phi - u\,r\,\dot{\phi}\,\sin\phi + 2\,\Omega\,r\,\cos\phi\,(\dot{r}\,\cos\phi - r\,\dot{\phi}\,\sin\phi). \tag{13.44b}$$

Again, setting $\mathrm{d}l^z/\mathrm{d}t = 0$ and rearranging leads to the zonal acceleration (13.43).

13.6.5 Zonal acceleration induced by meridional motion

We now consider a few thought experiments to help understand how axial angular momentum conservation gives rise to the Coriolis acceleration appearing in the zonal momentum equation. Start by considering a particle moving meridionally ($\delta\phi \neq 0$) while maintaining a constant radial position ($\delta r = 0$). The axial angular momentum constraint (13.38b) induces a zonal acceleration

$$\dot{u} = 2\,\Omega\, v \sin\phi, \tag{13.45}$$

which, as seen by equation (13.39), is the Coriolis acceleration appearing in the zonal momentum equation arising from the meridional motion. For poleward motion in either hemisphere, the product $v \sin\phi$ is always positive.[3] Hence, axial angular momentum conserving motion towards either pole induces an eastward acceleration, whereas a westward acceleration is induced for equatorward motion. For the northern hemisphere, the induced acceleration deflects the particle to the right when looking downstream whereas in the southern hemisphere the induced acceleration deflects the particle to the left. These deflections are illustrated in Figure 13.5.

A rudimentary means to understand the deflections arising from meridional motion is to view a particle trajectory from an inertial reference frame off the planet. For a meridional trajectory, the spinning earth causes the trajectory to pick up zonal motion relative to the earth surface, zonally to the right in the northern hemisphere and zonally to the left in the southern hemisphere. Equivalently, consider a projectile starting from rest on the planet and shot poleward. Since the projectile started closer to the equator, it has a zonal velocity component that is larger than the more poleward ground underneath it as it flies away from the equator. Hence, as it moves poleward it also picks up an eastward velocity component, which is to the right of the poleward motion in the northern hemisphere and to the left in the southern. Figure 13.5 illustrates this thought experiment.

13.6.6 Zonal acceleration induced by radial (vertical) motion

Now consider a particle moving radially while holding the latitude fixed ($\delta r \neq 0$ and $\delta\phi = 0$). The axial angular momentum constraint (13.38b) induces a zonal acceleration

$$\dot{u} = -2\,\Omega\, w \cos\phi, \tag{13.46}$$

which, as seen by equation (13.39), is the Coriolis acceleration appearing in the zonal momentum equation arising from the vertical motion. Hence, for vertically downward motion ($w < 0$), axial angular momentum conservation induces a positive zonal acceleration, $\dot{u} > 0$, which we expect from axial angular momentum conservation since the particle is moving closer to the rotation axis.

13.6.7 Meridional acceleration from Coriolis

We showed in Section 13.6.4 that the zonal equation of motion determining $\dot{u}$ is another way of writing the conservation of axial angular momentum. The same, however, cannot be said for

[3]Recall from Figure 12.3 that $\sin\phi \leq 0$ in the Southern Hemisphere, where $-\pi/2 \leq \phi \leq 0$. In contrast, $\sin\phi \geq 0$ in the Northern Hemisphere, where $0 \leq \phi \leq \pi/2$.

the meridional equation of motion, which is given by equation (12.130b)

$$\dot{v} + \frac{v\,w + u^2 \tan\phi}{r} = -2\,\Omega\,u \sin\phi. \tag{13.47}$$

The Coriolis acceleration is on the right hand side. In the northern hemisphere ($\sin\phi > 0$), the Coriolis acceleration gives rise to a rightward (equatorward) acceleration, $-2\,\Omega\,u\sin\phi < 0$, when the particle is moving eastward, $u > 0$, thus inducing a negative meridional acceleration, $\dot{v} < 0$. Conversely, if the particle is moving to the west so that $u < 0$, then the Coriolis acceleration is again to the right, only this time it provides a poleward acceleration, $\dot{v} > 0$. The analogous considerations hold in the southern hemisphere where the particle is deflected to the left by the Coriolis acceleration. These motions are reflected in Figure 13.5. Once the particle picks up a meridional component to the motion, then we return to the axial angular momentum constraint (13.38b) to see how zonal flow is affected.

13.6.8 Vertical acceleration from Coriolis

As for the meridional Coriolis acceleration in Section 13.6.7 there is no angular momentum constraint that connects to the Coriolis acceleration appearing in the vertical velocity equation (12.130c)

$$\dot{w} - \frac{u^2 + v^2}{r} = 2\,\Omega\,u\cos\phi - g. \tag{13.48}$$

In both hemispheres the cosine factor is positive, $\cos\phi > 0$. Hence, the Coriolis acceleration is positive (upward) for eastward motion and negative (downward) for westward motion. Note that for typical geophysical motion, the Coriolis acceleration is tiny compared to that arising from gravity.

13.6.9 When lateral motions dominate vertical motions

We here consider two approximations relevant to large scale geophysical fluid motions.

1. The particle kinetic energy is dominated by lateral motions on the sphere (i.e., motion at constant radial position).
2. Vertical (radial) excursions are much smaller than the earth radius.

When applied to a fluid, the first assumption leads to the hydrostatic approximation, and the second assumption leads to the shallow fluid approximation. Self-consistency of the equations of motion means that these two assumptions must be applied together.

Dropping the vertical velocity component to the kinetic energy leads to

$$K \approx \frac{m}{2}\,[(u + r_{\perp}\,\Omega)^2 + v^2]. \tag{13.49}$$

The second assumption means that the axial angular momentum takes the approximate form

$$L^z \approx m\,R_{\perp}\,(u + \Omega\,R_{\perp}) = m\,R_{\perp}^2\,(\dot{\lambda} + \Omega), \tag{13.50}$$

where

$$r = R + z \approx R \qquad \text{and} \qquad R_{\perp} = R\cos\phi. \tag{13.51}$$

The approximate angular momentum (13.50) ignores contributions from vertical motion in changing the moment-arm. Indeed, as noted in Section 13.6.2, vertical movements within the atmosphere and ocean (relatively thin fluid layers over the earth's surface) lead to a relatively small modification to the moment-arm, so the assumption that $r_{\perp} \approx R\cos\phi$ is reasonable. With $r \approx R$, the zonal acceleration (13.43) is modified to the form

$$\frac{\mathrm{d}u}{\mathrm{d}t} = v\left[\frac{u\tan\phi}{R} + f\right] \qquad \text{where} \qquad f = 2\,\Omega\,\sin\phi. \tag{13.52}$$

That is, we dropped the vertical velocity component, w, from the general form of the acceleration (13.43). Correspondingly, the meridional momentum equation takes the form

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -u\left[\frac{u\tan\phi}{R} + f\right]. \tag{13.53}$$

13.6.10 Comments

A concise summary of many features of rotating physics is provided by this video from SciencePrimer.

13.7 Mechanical energy conservation

Since the angular velocity of the planet and the gravitational acceleration are both assumed constant in time, then shifting the time by an arbitrary constant leaves the physical system unaltered. That is, the physical system remains unchanged if we shift all clocks by a constant amount. Through Noether's theorem, time symmetry leads to mechanical energy conservation, which means that the particle's mechanical energy is a fixed constant. We here prove that mechanical energy is constant in time by manipulating the equations of motion.

13.7.1 Properties of kinetic energy

We here establish some basic properties of kinetic energy for a particle. As we saw in Section 11.2.4, changes in the kinetic energy of a particle equal to the mechanical work done on the particle as it moves along its trajectory

$$K(t_2) - K(t_1) = \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{F}\cdot\mathrm{d}\mathbf{x} = \int_{t_1}^{t_2} \mathbf{F}\cdot\mathbf{V}\,\mathrm{d}t, \tag{13.54}$$

where $\mathbf{V}\,\mathrm{d}t = \mathrm{d}\mathbf{x}$ defines the vector increment along the trajectory, and $\mathbf{x}_{1,2}$ are the endpoints of the trajectory at times $t_{1,2}$. The integrand, $\mathbf{F}\cdot\mathbf{V}$, is known as the power. Hence, equation (13.54) says that the time integral of the power equals to the difference in kinetic energy between the final and initial times.

The kinetic energy is *not* Galilean invariant since movement to another inertial reference frame leads to the kinetic energy change

$$\overline{\mathbf{V}} = \mathbf{V} + \mathbf{U} \Longrightarrow \overline{K} = K + \frac{m}{2}\,(2\,\mathbf{V}\cdot\mathbf{U} + \mathbf{U}\cdot\mathbf{U}), \tag{13.55}$$

where $\mathbf{U}$ is a constant boost velocity so that $\mathrm{d}\mathbf{U}/\mathrm{d}t = 0$. We do not expect kinetic energy to be Galilean invariant since kinetic energy measures energy of motion relative to a chosen reference frame. Even so, the time change of the kinetic energy in a different inertial frame is still given by the power as measured in the new frame

$$\frac{\mathrm{d}\overline{K}}{\mathrm{d}t} = \frac{\mathrm{d}K}{\mathrm{d}t} + m\,\mathbf{A}\cdot\mathbf{U} = \mathbf{F}\cdot\mathbf{V} + \mathbf{F}\cdot\mathbf{U} = \mathbf{F}\cdot\overline{\mathbf{V}}. \tag{13.56}$$

Hence, we can directly connect kinetic energy changes to forces within an arbitrary inertial reference frame.

Cartesian expression for kinetic energy

Consider the expression for kinetic energy when introducing the velocity of the rotating reference frame. Writing the inertial frame velocity in the planetary Cartesian form

$$\boldsymbol{V} = \boldsymbol{V}_{\text{cartesian}} + \boldsymbol{U}_{\text{rigid}}, \tag{13.57}$$

leads to

$$K = \frac{m}{2}\left[\boldsymbol{V}_{\text{cartesian}}\cdot\boldsymbol{V}_{\text{cartesian}} + 2\,\boldsymbol{V}_{\text{cartesian}}\cdot\boldsymbol{U}_{\text{rigid}} + \boldsymbol{U}_{\text{rigid}}\cdot\boldsymbol{U}_{\text{rigid}}\right] \tag{13.58}$$

The first term arises from motion of the particle relative to the rotating sphere; the second arises from coupling between relative velocity and rigid-body velocity; and the third arises from rigid-body rotation of the sphere.

Spherical expression for kinetic energy: Part I

To expose spherical symmetry of the physical system, we express the kinetic energy in terms of the planetary spherical coordinates defined in Figure 12.3. Doing so for the rigid-body velocity leads to equation (12.60)

$$\boldsymbol{U}_{\text{rigid}} = \Omega\, r\cos\phi\,\left(-\sin\lambda\,\hat{\boldsymbol{x}} + \cos\lambda\,\hat{\boldsymbol{y}}\right). \tag{13.59}$$

Likewise, the velocity components measured in the rotating frame are given by

$$\dot{X} = \frac{\mathrm{d}\,(r\cos\phi\cos\lambda)}{\mathrm{d}t} = \dot{r}\,\cos\phi\,\cos\lambda - r\dot{\phi}\,\sin\phi\,\cos\lambda - r\dot{\lambda}\,\cos\phi\,\sin\lambda \tag{13.60a}$$

$$\dot{Y} = \frac{\mathrm{d}\,(r\cos\phi\sin\lambda)}{\mathrm{d}t} = \dot{r}\,\cos\phi\,\sin\lambda - r\dot{\phi}\,\sin\phi\,\sin\lambda + r\dot{\lambda}\,\cos\phi\,\cos\lambda \tag{13.60b}$$

$$\dot{Z} = \frac{\mathrm{d}\,(r\sin\phi)}{\mathrm{d}t} = \dot{r}\,\sin\phi + r\dot{\phi}\,\cos\phi. \tag{13.60c}$$

Bringing terms together then leads to the kinetic energy in terms of spherical coordinates

$$K = \frac{m}{2}\left[(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{\lambda}^2\,r^2\cos^2\phi) + (2\,\Omega\,r^2\dot{\lambda}\,\cos^2\phi) + (\Omega\,r\,\cos\phi)^2\right]. \tag{13.61}$$

Spherical expression for kinetic energy: Part II

An alternative means for deriving the kinetic energy in equation (13.61) makes use of the spherical coordinate form of the inertial frame velocity given by equation (12.54f), in which case

$$\boldsymbol{V} = (u + r_{\perp}\,\Omega)\,\hat{\boldsymbol{\lambda}} + v\,\hat{\boldsymbol{\phi}} + w\,\hat{\boldsymbol{r}}, \tag{13.62}$$

so that

$$K = \frac{m}{2}\left[(u + r_{\perp}\,\Omega)^2 + v^2 + w^2\right], \tag{13.63}$$

where $r_\perp = r\cos\phi$. Additionally, as discussed in Section 13.5, the axial angular momentum is given by

$$L^z = m\, r_\perp\, (u + r_\perp\, \Omega) \equiv m\, l^z, \tag{13.64}$$

and this property is a constant of the motion when there is azimuthal (zonal) symmetry. It is thus convenient to write the kinetic energy as

$$K = \frac{m}{2}\, \left[(l^z/r_\perp)^2 + v^2 + w^2\right]. \tag{13.65}$$

Geopotential expression for mechanical energy

As seen in Section 13.7.4 below, the mechanical energy using geopotential coordinates takes on the form

$$M_{\text{geop}} = \frac{m}{2}\, \left[u^2 + v^2 + w^2 + 2\, g\, r\right], \tag{13.66}$$

where each of these symbols takes on their geopotential interpretation.

13.7.2 Planetary Cartesian mechanical energy

The time derivative of the kinetic energy is given by

$$\frac{\mathrm{d}K}{\mathrm{d}t} = m\, \boldsymbol{V} \cdot \frac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}t} = m\, \boldsymbol{V} \cdot \boldsymbol{A} = -m\, \boldsymbol{V} \cdot \nabla \Phi_{\mathrm{e}}. \tag{13.67}$$

For the final equality we introduced the gravitational potential given that the particle only feels an external force from gravity as per equation (12.116). The gravitational potential is given by (see equation (11.131))

$$\Phi_{\mathrm{e}} = g_{\mathrm{e}}\, r, \tag{13.68}$$

so that

$$\frac{\mathrm{d}K}{\mathrm{d}t} = -m\, \boldsymbol{V} \cdot \nabla \Phi_{\mathrm{e}} = -m\, g_{\mathrm{e}}\, \dot{r}. \tag{13.69}$$

This result means that kinetic energy is reduced when moving the particle away from the earth center ($\dot{r} > 0$). Moving away from the earth requires work to overcome the gravitational acceleration pointing towards the earth center. This work to overcome the gravitational attraction is taken away from the kinetic energy of the particle. Furthermore, the work is added to the gravitational potential energy, whose evolution is given by (see equation (11.133))

$$\frac{\mathrm{d}P_{\mathrm{e}}}{\mathrm{d}t} = m\, g_{\mathrm{e}}\, \dot{r}, \tag{13.70}$$

where we assumed a constant gravitational acceleration, g_{e}. Consequently, as the particle moves away from the earth center, its reduction in kinetic energy is exactly compensated by an increase in potential energy. Hence, the mechanical energy for the particle remains constant throughout the motion

$$\frac{\mathrm{d}(K + P_{\mathrm{e}})}{\mathrm{d}t} = 0, \tag{13.71}$$

where the mechanical energy is the sum of the inertial frame kinetic energy plus the gravitational potential energy

$$M = K + P_{\mathrm{e}} \tag{13.72a}$$

$$= \frac{m}{2}\,\boldsymbol{V}\cdot\boldsymbol{V} + m\,\Phi_{\rm e} \tag{13.72b}$$

$$= \frac{m}{2}\left[(u + r_{\perp}\,\Omega)^2 + v^2 + w^2\right] + m\,g_{\rm e}\,r \tag{13.72c}$$

$$= \frac{m}{2}\,\left[(l^z/r_{\perp})^2 + v^2 + w^2\right] + m\,g_{\rm e}\,r. \tag{13.72d}$$

13.7.3 Planetary spherical mechanical energy

It is physically revealing to expose the exchange of mechanical energy between kinetic and gravitational potential energies. Furthermore, knowledge of the total mechanical energy at any time affords knowledge for all time since the mechanical energy (in the absence of friction) remains constant. Following from the discussion in Section 13.6, where we examined the constraints on particle motion due to conservation of axial angular momentum, we here ask similar questions about mechanical energy conservation. We make use of the spherical form of the equations of motion, equations (12.124)-(12.126), which expose the planetary centrifugal and Coriolis accelerations, and pursue the calculation for geopotential coordinates in Section 13.7.4.

We find it convenient to write the momentum equations (12.124)-(12.126) in terms of the distance to the polar axis, $r_{\perp} = r\cos\phi$, and its time derivative, $\dot{r}_{\perp} = w\cos\phi - v\sin\phi$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[r_{\perp}\,u + \Omega\,r_{\perp}^2\right] = 0 \tag{13.73a}$$

$$\dot{v} = -\frac{v\,w}{r} - \frac{u\tan\phi}{r}\,(u + 2\,\Omega\,r_{\perp}) - r_{\perp}\,\Omega^2\sin\phi \tag{13.73b}$$

$$\dot{w} = \frac{u^2+v^2}{r} + 2\,\Omega\,u\,\cos\phi + r_{\perp}\,\Omega^2\cos\phi - g_{\rm e}. \tag{13.73c}$$

Equation (13.73a) expresses the conservation of axial angular momentum, $\dot{l}^z = 0$, where the axial angular momentum per mass is $l^z = r_{\perp}\,(u + r_{\perp}\,\Omega)$ (Section 13.5). For the mechanical energy, we have

$$\dot{M} = 0 \qquad \text{with} \qquad M = \frac{m}{2}\left[(l^z/r_{\perp})^2 + v^2 + w^2\right] + m\,g_{\rm e}\,r, \tag{13.74}$$

where $l^z/r_{\perp} = u + r_{\perp}\,\Omega$ is the zonal component to the inertial frame velocity (equation (12.54f)). We now show that $\dot{M} = 0$ arises from the momentum equations. Performing the time derivative, and setting $\mathrm{d}l^z/\mathrm{d}t = 0$, leads to

$$\frac{1}{m}\,\dot{M} = \frac{1}{2}\,\frac{\mathrm{d}}{\mathrm{d}t}\left[(l^z/r_{\perp})^2 + v^2 + w^2 + 2\,g_{\rm e}\,r\right] = -\frac{(l^z)^2\,\dot{r}_{\perp}}{(r_{\perp})^3} + v\,\dot{v} + w\,(\dot{w} + g_{\rm e}). \tag{13.75}$$

Use of the meridional momentum equation (13.73b) renders

$$v\,\dot{v} = -v\left[\frac{v\,w}{r} + \frac{u\tan\phi}{r}\,(u + 2\,\Omega\,r_{\perp}) + r_{\perp}\,\Omega^2\sin\phi\right] \tag{13.76a}$$

$$= -\frac{v^2\,w}{r} - \frac{v\tan\phi}{r}\,(u + r_{\perp}\,\Omega)^2. \tag{13.76b}$$

Likewise, the vertical momentum equation (13.73c) renders

$$w\,(\dot{w} + g_{\rm e}) = \frac{v^2\,w}{r} + \frac{w}{r}\left[u^2 + 2\,\Omega\,u\,r_{\perp} + (r_{\perp}\,\Omega)^2\right] \tag{13.77a}$$

$$= \frac{v^2\,w}{r} + \frac{w}{r}\,(u + r_\perp\,\Omega)^2, \tag{13.77b}$$

so that

$$v\,\dot{v} + w\,(\dot{w} + g_e) = r^{-1}\,(-v\tan\phi + w)\,(u + r_\perp\,\Omega)^2 = \frac{(l^z)^2\,\dot{r}_\perp}{(r_\perp)^3}. \tag{13.78}$$

Combining this result with equation (13.75) leads to the expected $\dot{M} = 0$. Hence, angular momentum conservation combined with the meridional and vertical momentum equations is equivalent to mechanical energy conservation.

13.7.4 Geopotential mechanical energy

In Exercise 13.4 we examine the mechanical energy conservation when using geopotential coordinates with the equations of motion (12.130a)-(12.130c). Since the coordinates are different (spherical versus geopotential), the velocity components (u, v, w) are slightly different in the two coordinate systems, as is the radial position. So care is needed when comparing the two expressions.

13.8 Sample particle trajectories

Here we examine some trajectories to illustrate how the dynamical properties discussed in this chapter, particularly angular momentum conservation, help to determine trajectories. The discussion complements that considered in Exercise 12.10.

13.8.1 Free fall starting from rest in the inertial reference frame

Consider a particle freely falling to the center of the sphere starting with zero axial angular momentum. Although the trajectory from an inertial reference frame is rather trivial, we need to do a bit more work in the non-inertial rotating frame, where we make use of the spherical coordinate equations to write

$$\dot{\lambda} = -\Omega \quad \text{and} \quad u = -r_\perp\,\Omega \quad \text{and} \quad v = 0 \quad \text{and} \quad \dot{w} = -g_e. \tag{13.79}$$

Evidently, the longitude, latitude, and vertical position are given by

$$(\lambda - \lambda_\circ) = -\Omega\,t \quad \text{and} \quad \phi = \phi_\circ \quad \text{and} \quad (w - w_\circ) = -g_e\,t, \tag{13.80}$$

where $\lambda_\circ$ is the initial longitude, $\phi_\circ$ is the initial latitude, and $w_\circ$ the initial vertical velocity. So the position with respect to the earth's longitude adjusts westward according to $(\lambda - \lambda_\circ) = -\Omega\,t$, which is needed to maintain a fixed absolute longitude. Likewise, the vertical free fall with $(w - w_\circ) = -g_e\,t$ accords with the free fall seen in the inertial reference frame. Integrating the vertical equation leads to the radial position

$$r - r_\circ = w_\circ\,t - g_e\,t^2/2. \tag{13.81}$$

If we drop the particle from rest, so that $w_\circ = 0$, then the particle falls a distance H within a time

$$T = (2\,H/g_e)^{1/2}. \tag{13.82}$$

During this time the particle deflects its longitude to the west by an amount

$$\lambda - \lambda_{\circ} = -\Omega\, T = -\Omega\, (2\, H/g_{e})^{1/2}. \tag{13.83}$$

It is useful to check self-consistency by verifying that the rotating reference frame description measures a zero axial angular momentum throughout the free fall. We first do so by noting that $l^{z} = r_{\perp}^{2}\, (\dot{\lambda} + \Omega) = 0$ since $\dot{\lambda} = -\Omega$. Taking the time derivative leads to

$$\dot{l}^{z} = r\, r_{\perp}\, \dot{r}_{\perp}\, (\Omega + \dot{\lambda}) + r_{\perp}^{2}\, \ddot{\lambda}. \tag{13.84}$$

Now with $\dot{\lambda} = -\Omega$ we find $\ddot{\lambda} = 0$, which then yields $\dot{l}^{z} = 0$ throughout the free fall. Furthermore, we can write $l^{z} = r_{\perp}\, (u + r_{\perp}\, \Omega) = 0$, which also means that

$$\dot{l}^{z} = \dot{r}_{\perp}\, (u + r_{\perp}\, \Omega) + r_{\perp}\, (\dot{u} + \dot{r}_{\perp}\, \Omega) = 0, \tag{13.85}$$

where we noted that $u = -r_{\perp}\, \Omega$ means that $\dot{u} = -\dot{r}_{\perp}\, \Omega$.

13.8.2 Falling with constant zonal velocity

Now consider a trajectory that maintains a constant zonal velocity, $\dot{u} = 0$, which, with $u = r_{\perp}\, \dot{\lambda}$, means that

$$\dot{u} = \dot{r}_{\perp}\, \dot{\lambda} + r_{\perp}\, \ddot{\lambda} = 0. \tag{13.86}$$

To maintain a constant axial angular momentum means

$$\dot{l}^{z} = 2\, r_{\perp}\, \dot{r}_{\perp}\, (\dot{\lambda} + \Omega) + r_{\perp}^{2}\, \ddot{\lambda} = r_{\perp}\, \dot{r}_{\perp}\, (\dot{\lambda} + 2\, \Omega), \tag{13.87}$$

where the second step made use of the assumed $\dot{u} = 0$ condition from equation (13.86). Equation (13.87) can be satisfied by either $\dot{r}_{\perp} = 0$ or $\dot{\lambda} = -2\, \Omega$. Yet setting one of these conditions means that the other must also hold in order to maintain $\dot{u} = 0$ from equation (13.86). We are thus led to

$$\dot{u} = 0 \quad \text{and} \quad \dot{l}^{z} = 0 \Longleftrightarrow \dot{\lambda} = -2\, \Omega \quad \text{and} \quad \dot{r}_{\perp} = 0 \quad \text{and} \quad l^{z} = -\Omega\, r_{\perp}^{2}. \tag{13.88}$$

Evidently, a particle moving westward with $\dot{\lambda} = -2\, \Omega$, and maintaining a fixed distance from the polar rotation axis, maintains both constant axial angular momentum and a constant zonal velocity

$$u = -2\, \Omega\, r_{\perp}. \tag{13.89}$$

Furthermore, observe that a constant distance from the rotation axis, $\dot{r}_{\perp} = 0$, means that

$$\dot{r}_{\perp} = 0 \Longrightarrow w \cos\phi = v \sin\phi. \tag{13.90}$$

This relation couples vertical and meridional motion in the rotating reference frame in a manner that keeps the distance to the rotational axis fixed. For example, the vertical velocity vanishes at the equator. Additionally, if the particle is falling so that $w \cos\phi < 0$, then there is an equatorward meridional velocity to keep the distance from the rotational axis fixed. In this manner, the particle is falling not towards the earth center but instead towards the equatorial plane, all while maintaining a fixed zonal velocity with $\dot{u} = 0$.

To garner further insight into the trajectory, we examine the meridional and vertical accelerations. For this purpose, recall the meridional equation of motion (12.125), written as

$$r\,\dot{v} + v\,w + u^2 \tan\phi + \Omega\,r \sin\phi\,(2\,u + r_\perp\,\Omega) = 0. \tag{13.91}$$

Now multiply by $\cos\phi$ and set $w \cos\phi = v \sin\phi$ to reach

$$r_\perp\,\dot{v} + (u^2 + v^2) \sin\phi + \Omega\,r_\perp \sin\phi\,(2\,u + r_\perp\,\Omega) = 0. \tag{13.92}$$

Next set $u = -2\,r_\perp\,\Omega$ to have

$$r_\perp\,\dot{v} + [(\Omega\,r_\perp)^2 + v^2] \sin\phi = 0, \tag{13.93}$$

which, when setting $l^z/r_\perp = -\Omega\,r_\perp$, leads to the meridional acceleration

$$r_\perp\,\dot{v} = -[(l^z/r_\perp)^2 + v^2] \sin\phi. \tag{13.94}$$

Evidently, the meridional acceleration is negative in the northern hemisphere (where $\sin\phi > 0$) and positive in the southern, so that the particle always accelerates toward the equator. Similarly, use of the vertical acceleration from equation (12.126) leads to

$$r\,(\dot{w} + g_{\text{e}}) = u^2 + v^2 + r_\perp\,\Omega\,u + (r_\perp\,\Omega)^2 = v^2 + (r_\perp\,\Omega)^2 = v^2 + (l^z/r_\perp)^2 > 0. \tag{13.95}$$

Hence, the vertical acceleration ($\dot{w} < 0$) is always smaller in magnitude than the earth's gravitational acceleration. Finally, combining the conditions (13.94) and (13.95) means that the meridional and vertical accelerations are related by

$$\dot{v} \cos\phi + (\dot{w} + g_{\text{e}}) \sin\phi = 0. \tag{13.96}$$

So when the particle reaches the equator at $\phi = 0$, we know from equation (13.90) that the vertical velocity vanishes, and from equation (13.96) we see that the merdional acceleration also vanishes. In contrast, at the poles where $\phi = \pm\pi/2$, the vertical acceleration is $\dot{w} = -g_{\text{e}}$ whereas equation (13.90) says that the meridional velocity vanishes.

13.8.3 Free fall from rest in the rotating frame

In Exercise 13.5 we examine the dynamics of a particle falling freely from rest in the rotating non-inertial reference frame.

13.9 Dynamical constraints from spatial symmetries

We close this chapter by summarizing the conservation laws that arise from spatial symmetries. These conservation laws complement the discussion in Section 13.7 for mechanical energy conservation arising from time symmetry.

13.9.1 Linear momentum conservation

As seen in Section 11.2.2, linear momentum remains constant for a particle moving without any forces acting on it; i.e., a free particle. This type of motion is not common in geophysics since matter feels gravity and so is not free. Even so, this limiting case is a useful point of comparison for the other conservation laws.

The conservation of linear momentum is best viewed within the inertial reference frame, where a vanishing inertial frame acceleration leads to a constant inertial frame velocity

$$\mathbf{A} = \mathrm{d}\mathbf{V}/\mathrm{d}t = 0. \tag{13.97}$$

In a Euclidean space, a vanishing acceleration means the particle is moving in a straight line with constant velocity.[4] When viewed from a rotating frame using Cartesian coordinates, a vanishing inertial frame acceleration means that the Cartesian acceleration balances planetary Coriolis and planetary centrifugal accelerations

$$\ddot{\mathbf{X}} = -2\,\boldsymbol{\Omega} \times \dot{\mathbf{X}} - \nabla\Phi_{\mathsf{centrifugal}}. \tag{13.98}$$

This equation is not a very useful means to describe unaccelerated free particle motion. We can provide a bit more compactness to this equation by introducing the potential momentum (13.7) so that

$$\mathrm{d}\mathbf{M}/\mathrm{d}t = -\nabla\Phi_{\mathsf{centrifugal}}. \tag{13.99}$$

While the inertial frame acceleration vanishes ($\mathbf{A} = \mathrm{d}\mathbf{V}/\mathrm{d}t = 0$), time changes to the potential momentum are balanced by the gradient of the centrifugal potential.

13.9.2 Potential momentum conservation

Conservation of potential momentum arises from symmetry of frictionless particle motion constrained to move on a constant geopotential surface. The conservation law is most readily viewed within the rotating rigid-body reference frame, whereby equation (13.9) is given by

$$\frac{\mathrm{d}(\hat{\boldsymbol{s}} \cdot \mathbf{M})}{\mathrm{d}t} = 0. \tag{13.100}$$

A geopotential is a two-dimensional surface so that this conservation law corresponds to two dynamical constraints such as shown in Figure 13.1.

13.9.3 Angular momentum conservation

As detailed in Section 13.5, the angular momentum computed with respect to the axis of rotation is a constant of the motion (Figure 13.3). This conservation law arises from symmetry of the system about the rotational axis. Axial (z-axis) angular momentum conservation takes the form

$$\mathrm{d}L^z/\mathrm{d}t = 0, \tag{13.101}$$

where the axial angular momentum is

$$L^z = m\,r_\perp^2\,(\dot{\lambda} + \Omega) \qquad \text{with} \qquad r_\perp = r\cos\phi. \tag{13.102}$$

The distance from the rotation axis, $r_\perp$, is the moment arm for the axial angular momentum. The longitude, λ, measures the angle in the counter-clockwise direction from the positive x-axis, and $\dot{\lambda}$ is the time change of the longitude.

[4] Recall that a zero acceleration as in equation (13.97) is an expression of Newton's first law (law of inertia), which was stated in the start of Chapter 11.

13.10 Exercises

EXERCISE 13.1: L^x AND L^y ARE NOT CONSTANTS OF THE MOTION
In Section 13.5.2 we showed that L^z is a constant of the motion for particles moving around a spherically symmetric planet rotating around the polar, $\hat{\boldsymbol{z}}$, axis. Make use of similar arguments to show that $L^x = \hat{\boldsymbol{x}} \cdot \boldsymbol{L}$ and $L^y = \hat{\boldsymbol{y}} \cdot \boldsymbol{L}$ are not constants of the motion.

EXERCISE 13.2: NEGATIVE AXIAL ANGULAR MOMENTUM
In Section 13.6 we assume the axial angular momentum is positive, which is geophysically the common situation since axial angular momentum from the rigid-body motion is so large relative to motion of geophysical fluids. But let us consider the uncommon case where the particle moves zonally westward at a speed greater than the planetary rotation speed so that

$$\dot{\lambda} + \Omega < 0 \Longleftrightarrow u + \Omega\, r_{\perp} < 0, \tag{13.103}$$

which means the axial angular momentum per mass of the particle is negative

$$l^z = r_{\perp}\,(u + r_{\perp}\,\Omega) = -|l^z| < 0. \tag{13.104}$$

Throughout this exercise we seek answers based on conservation of axial angular momentum. When checking to see whether an answer agrees with common sense, be careful since motion of this sort is not commonly experienced by terrestrial observers. Correspondingly, it is useful to check answers by viewing the motion from the perspective of an inertial reference frame rather than the rotating terrestrial reference frame. For example, the particular case where $\dot{\lambda} = -\Omega$, in which case the particle has zero angular momentum, corresponds to a particle that is stationary in the inertial reference frame while the planet rotates underneath.

(A) Discuss what happens to $\delta\dot{\lambda}$ for the particle that is deflected poleward with constant radius while conserving axial angular momentum. Hint: consider the first form of equation (13.33).
(B) Discuss what happens to δu for the particle as it is deflected poleward with constant radius while conserving axial angular momentum. Separately discuss the three cases where
  (i) $|l^z| = 0$
  (ii) $|l^z| < \Omega\, r_{\perp}^2$.
  (iii) $|l^z| > \Omega\, r_{\perp}^2$
  Hint: consider the second form of equation (13.33).
(C) Is fluid particle motion with $u + \Omega\, r_{\perp} < 0$ relevant for the terrestrial atmosphere and ocean? Why? To help answer this question, what is $\Omega\, r_{\perp}$ for $\phi = \pi/4$ and $r = R_{\mathrm{e}}$? Note, we already provided the result for the equator just after equation (13.30). Compare these speeds to that of a category 5 tropical cyclone.

EXERCISE 13.3: SPHERICAL COMPONENTS TO ANGULAR MOMENTUM
In Section 13.5 we studied angular momentum relative to the center of the sphere, $\boldsymbol{L} = m\,\boldsymbol{X} \times \boldsymbol{V}$. In particular, we considered the evolution equations for the planetary Cartesian compoents $\hat{\boldsymbol{x}} \cdot \boldsymbol{L}$, $\hat{\boldsymbol{y}} \cdot \boldsymbol{L}$, and $\hat{\boldsymbol{z}} \cdot \boldsymbol{L}$. Here we consider the projection of the angular momentum onto the planetary spherical directions $\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}$ and $\hat{\boldsymbol{r}}$. For a space that is spherically symmetric around the center of the sphere, we do not expect $\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{L}$ or $\hat{\boldsymbol{\phi}} \cdot \boldsymbol{L}$ to be constants of the motion. The reason is that neither $\hat{\boldsymbol{\lambda}}$ nor $\hat{\boldsymbol{\phi}}$ extend outward from the center of the sphere, so that rotations around either of these axes do not manifest a symmetry operation. Nevertheless, it is interesting to

explore properties of $\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{L}$ and $\hat{\boldsymbol{\phi}} \cdot \boldsymbol{L}$ as an exercise in the angular momentum formalism, and to display how their evolution relates to that of the linear momentum on the sphere.

(A) Write an expression for the zonal component to the angular momentum, $\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{L}$, in terms of $\dot{\phi}$ and r. Hint: check that the physical dimensions are correct, and that the sign accords with the right hand rule.
(B) Derive an expression for the time derivative, $\mathrm{d}(\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{L})/\mathrm{d}t$, in terms of $\dot{\phi}$ and $\dot{r}$.
(C) Write an alternative expression for the time derivative, $\mathrm{d}(\hat{\boldsymbol{\lambda}} \cdot \boldsymbol{L})/\mathrm{d}t$, in terms $u = r_{\perp}\,\dot{\lambda}$ and Ω. Hint: recall the spherical coordinate version of the meridional momentum equation (12.125).
(D) Derive an expression for the meridional component to the angular momentum, $\hat{\boldsymbol{\phi}} \cdot \boldsymbol{L}$, and write it in terms of r, Ω, $\dot{\lambda}$, and ϕ.
(E) Derive an expression for the time derivative, $\mathrm{d}(\hat{\boldsymbol{\phi}} \cdot \boldsymbol{L})/\mathrm{d}t$.
(F) Derive an expression for $\hat{\boldsymbol{r}} \cdot \boldsymbol{L}$.

EXERCISE 13.4: MECHANICAL ENERGY CONSERVATION USING GEOPOTENTIAL COORDINATES
In Section 13.7.3 we showed that the mechanical energy is conserved in the absence of friction, and we did so through use of the spherical coordinate equations of motion. Do the same here by making use of the geopotential vertical coordinate equations of motion (12.130a)-(12.130c).

EXERCISE 13.5: FREE FALL FROM REST IN THE ROTATING FRAME
We here consider the case of a particle initially at rest in the rotating reference frame that is allowed to freely fall. As in Exercise 12.10b, we make use of geopotential coordinates since the particle will initially fall parallel to the effective gravity direction; i.e., the plumb line. Consider the trajectory just in the zonal and vertical plane, assuming $v = 0$ throughout the free fall.

(A) Show that the zonal and vertical equations of motion are given by

$$\dot{u} + \frac{u\,w}{r} + 2\,\Omega\,w\cos\phi = 0 \tag{13.105a}$$

$$\dot{w} - \frac{u^2}{r} - 2\,\Omega\,u\cos\phi = -g. \tag{13.105b}$$

(B) Show that the linearized equations of motion are

$$\dot{u} = -2\,\Omega\,w\cos\phi \quad \text{and} \quad \dot{w} = 2\,\Omega\,u\cos\phi - g. \tag{13.106}$$

(C) Now write $w = w_0 + w_1$, where

$$\dot{w}_0 = -g \quad \text{and} \quad \dot{w}_1 = 2\,\Omega\,u\cos\phi, \tag{13.107}$$

so that $w_0 = -g\,t$. What is the velocity to first order in Ω?
(D) Determine the linearized expression for the particle trajectory.
(E) To leading order, what is the time it takes for the particle to fall a vertical distance H?
(F) During the time it takes to fall a distance, H, how far east has the particle deflected?

Chapter 14

Formulation of analytical mechanics

In this chapter we formulate the equations describing classical particle motion according to both Lagrangian mechanics and Hamiltonian mechanics. In both theories, we make use of Hamilton's principle of stationary action, which offers a basis for classical mechanics that is distinct from, though consistent with, Newton's laws of motion. That is, within classical mechanics, neither Newton's laws of motion nor Hamilton's principle are derived from more fundamental physical principles. The lens afforded by Lagrangian mechanics and Hamiltonian mechanics render insights into the nature of motion and dynamics, while also providing powerful tools for its quantitative and qualitative description.

Analytical mechanics is the name given to the formulation of mechanics based on Hamilton's variational principle. In analytical mechanics, a great deal of thought is placed up front as part of formulating the equations of motion, even before addressing specific applications. In so doing, we deepen our understanding of the conceptual foundations of mechanics, with this understanding extending to our studies of fluid mechanics in later volumes of this book. It is notable that analytical mechanics focuses on energy (a scalar) rather than forces (vectors). As a result, the Euler-Lagrange equation of motion, which results from extremizing the action via Hamilton's principle, remains the same regardless the coordinate choice. Furthermore, the switch from forces to energy has payoff for practical applications. Namely, as seen through the case studies in Chapter 15, the methods of analytical mechanics allow us to sidestep difficulties inherent in Newtonian mechanics that arise from the need to determine forces, including forces of constraint, which can be difficult if not impossible to determine *a priori*.

Lagrangian mechanics forms the foundations of both classical field theory and quantum field theory, and so it has extensions far beyond the classical point particle mechanics considered in the present chapter. In particular, we formulate the equations of perfect fluid mechanics in VOLUME 4 using Lagrangian methods with Hamilton's principle. We also use these methods in VOLUME 4 to add insights into the formulation of wave mechanics, including geometrical optics for geophysical waves. Hamiltonian mechanics further deepens our appreciation of the foundational elements of classical mechanics. It does so largely through examining motion within phase space, thus leading to general theorems satisfied by mechanical systems. Hamiltonian mechanics forms the natural starting point for quantum mechanics and statistical mechanics. We encounter Hamilton's equations of motion in VOLUME 4 when studying geometrical optics, following in the footsteps of Hamilton himself.

CHAPTER GUIDE

We assume an understanding of Newton's law of motion from Chapter 11. We also use a modest level of tensor notation for some of the manipulations. Book supplements to this chapter include intermediate and advanced treatments of classical mechanics, such as *Landau and Lifshitz* (1976), *Goldstein* (1980), *Marion and Thornton* (1988), *Fetter and Walecka* (2003), *José and Saletan* (1998), *Taylor* (2005), and *Tong* (2025).

14.1 Motivation for studying analytical mechanics

The formulation of analytical mechanics, as provided by both Lagrangian mechanics and Hamiltonian mechanics, requires a nontrivial effort in general formalism. Most of this effort is required to clearly account for the forces of constraint, and to then remove the need to know anything about forces, including forces of constraint. There are both intellectual and practical payoffs for the extra formalism needed beyond that of Newtonian mechanics. In particular, analytical mechanics provides an extremely straightforward means to derive the Euler-Lagrange equation of motion for physical systems. The key reason for the simplification is that with analytical mechanics we only need energies to derive the equation of motion. Energy is a scalar and thus is invariant under coordinate transformations, so that the Euler-Lagrange equations of motion take on the same form regardless the coordinates. In contrast, Newton's laws are simple to state in principle. However, they are vector equations since they require information about forces, and as such they can be very difficult to formulate for mechanical systems beyond the simplest.

Philosophically, we observe that the Newtonian approach is concerned with motion of a system arising from an outside agency, namely a force that brings about a change in the linear momentum; a torque that changes the angular momentum; or work that changes the kinetic energy. In contrast, the Lagrangian approach is concerned with properties of the physical system, namely its potential and kinetic energies. We propose that a complete understanding of a classical physical system comes from pursuing both the energy approaches of analytical mechanics (of Lagrange and Hamilton) and the force approach of Newtonian mechanics. Before diving into the details of analytical mechanics, we offer the following list of characteristics as further motivation for its study.

Generalized coordinates

As seen in Chapter 11, Newtonian mechanics is naturally expressed using Cartesian coordinates, whereas it takes some work (using tensor methods) to develop the equations using non-Cartesian coordinates (such as the spherical coordinates used in Chapter 12). In contrast, the equation of motion arising in Lagrangian mechanics, referred to as the Euler-Lagrange equation, has the same form using any coordinate system. Lagrangian mechanics is thus suited to the use of generalized coordinates, with such coordinates tailored to the particulars of each physical system. Hamilton's equations of motion are also form invariant, and they go one further step to render insights into the geometrical nature of classical motion in phase space.

Dynamical constraints and Noether's theorem

Noether's theorem states that for any symmetry of a physical system, there is a corresponding dynamical conservation law. For example, a classical particle system that exhibits Galilean symmetry maintains a constant linear momentum, constant mechanical energy, and constant angular momentum. A conservation law provides a dynamical constraint on the motion, with such constraints of great value for characterizing the motion and for deriving analytical solutions (when they exist). The deduction of dynamical constraints is naturally arrived at using the methods of analytical mechanics, particularly through Hamilton's principle of stationary action. Noether's theorem is one of the most important foundational elements in 20th century mathematical physics.

Forces of constraint

In addition to dynamical constraints, there are any number of further constraints that arise from details of the particular motion, such as the imposition of spatial boundaries, rods or springs connecting particles, or other geometric conditions. From the perspective of classical mechanics, such constraints might be considered purely kinematical. For example, the motion of a frictionless bead threaded by a wire is constrained to move along a trajectory defined by the shape of the wire. Such constraints reduce the spatial degrees of freedom of the particle motion. However, it is rarely simple, and often impossible in practice, to provide an *a priori* determination of the forces of constraint (also called reactive forces) that maintain the constraints. Since Newtonian mechanics requires all forces for determining the motion, it offers an impractical basis for studying systems where the forces of constraint are unknown *a priori*. In contrast, the equations of motion from analytical mechanics do not require forces, neither those from external fields nor the forces of constraint. As such, analytical mechanics provides the basis for the formulation of a huge range of phenomena. Indeed, it is for this reason that it forms the core of engineering mechanics.

Although Lagrangian mechanices does not need forces, it does provide the means to determine forces if so desired, including forces of constraint. It does so through the method of Lagrange multipliers.

Continuum mechanics

Each of the above motivations, developed in this chapter for particle mechanics, has its generalizations to continuum mechanics. That is, Lagrangian mechanics and Hamilton's principle are suited to continuum field theories such as geophysical fluid mechanics, as seen in VOLUME 4. Furthermore, Hamiltonian mechanics is directly connected to the ray theory perspective of linear wave propagation, which we study in VOLUME 4.

14.2 Constraints

A particle that moves in three dimensional Euclidean space has at most three independent degrees of freedom, meaning that three numbers are needed to specify the particle's spatial position at a particular time. If the particle moves without any forces then it is said to be a free particle. Even without forces on the particle, the motion of a free particle is constrained by one or more dynamical constraint that arises from each symmetry of the space and time through which the particle moves. For example, a free particle in Galilean space-time conserves its linear momentum, angular momentum, and mechanical energy (Section 14.8). In addition, there are further constraints that arise from forces of constraint, as we now discuss.

14.2.1 Holonomic constraints

When constraints are placed on the spatial position of the particles, then they are referred to as holonomic constraints. For example, consider the motion of a particle restricted to a frictionless two dimensional surface. As a result, its three Cartesian coordinates are related by a function of the form

$$\Psi[x(t), y(t), z(t)] = \mathcal{C}, \tag{14.1}$$

where $\mathcal{C}$ is a constant. If, for example, the particle is confined to move on the surface of a sphere of radius, R, then the constraint takes the form

$$\Psi[x(t), y(t), z(t)] = x(t)^2 + y(t)^2 + z(t)^2 = R^2. \tag{14.2}$$

By this restriction, the particle has two independent spatial degrees of freedom rather than the three available without the constraint.

Holonomic constraints are enforced by forces of constraint, which are also known as reactive forces. Forces of constraint reduce the degrees of freedom possessed by the motion. According to D'Alembert's principle, forces of constraint instantaneously perform no work on the physical system, which is the property that we use to define forces of constraint.[1] For the particle moving on a surface such as a sphere, we readily see that the forces of constraint that keep the particle on the surface perform no work since the forces act, at each time instance, in a direction that is perpendicular to the surface and hence perpendicular to the trajectory. Furthermore, note that the forces of constraint can be conservative (written as the gradient of a scalar potential energy function) or non-conservative (such as friction). Indeed, within the confines of classical mechanics, we generally have little fundamental knowledge of the forces of constraints (which can arise from electromagnetic or other fundamental forces). Rather, we only see their impact on the motion.

An equivalent expression for the algebraic constraint (14.1) is given by differential expressions

$$\mathrm{d}\Psi = 0 \quad \text{or} \quad \frac{\mathrm{d}\Psi}{\mathrm{d}t} = \frac{\partial \Psi}{\partial x^a}\,\dot{x}^a + \frac{\partial \Psi}{\partial t} = 0, \tag{14.3}$$

where we wrote $(x, y, z) = (x^1, x^2, x^3)$ and made use of the summation convention from tensor alegebra. We refer to holonomic constraints as integrable since the differential constraint (14.3) can be integrated to yield the algebraic constraint (14.1).

[1]Recall from Section 11.2.4 that mechanical work arises when a force acts in a direction aligned with the motion of a physical system. If the force is perpendicular to the motion, then that force performs no work and so does not alter the kinetic energy of the system. Hence, d'Alembert's principle says that forces of constraint act perpendicular to the motion. See Section 14 of *Fetter and Walecka* (2003) for more on d'Alembert's principle.

14.2.2 Non-holonomic constraints

There are physical systems that are affected by non-holonomic constraints. The canonical non-holonomic constraint involves both the position and velocity and so it is written

$$\Psi(\boldsymbol{X}, \dot{\boldsymbol{X}}, t) = \mathcal{C}. \tag{14.4}$$

The exact differential of this constraint includes the position, velocity, and time, so that

$$\mathrm{d}\Psi = \frac{\partial \Psi}{\partial x^a}\,\mathrm{d}x^a + \frac{\partial \Psi}{\partial \dot{x}^a}\,\mathrm{d}\dot{x}^a + \frac{\partial \Psi}{\partial t}\,\mathrm{d}t = 0, \tag{14.5}$$

which can be equivalently written as a statement about the total time derivative

$$\frac{\mathrm{d}\Psi}{\mathrm{d}t} = \frac{\partial \Psi}{\partial x^a}\,\dot{x}^a + \frac{\partial \Psi}{\partial \dot{x}^a}\,\ddot{x}^a + \frac{\partial \Psi}{\partial t} = 0. \tag{14.6}$$

This equation is not generally integrable in the sense of equation (14.3).

However, some special cases of non-holonomic constraints are integrable. For example, consider the constraint

$$H_a\,\dot{x}^a + I = 0, \tag{14.7}$$

where H_a and I are functions of time that can be written as

$$H_a = \frac{\partial \Psi}{\partial x^a} \quad \text{and} \quad I = \frac{\partial \Psi}{\partial t}. \tag{14.8}$$

For this special case, the constraint (14.7) takes the form of the differential holonomic constraint (14.3), in which we say that the constraint (14.7) is semi-holonomic. But more generally non-holonomic constraints are non-integrable and require methods that are not fully covered by the generalized coordinates or Lagrange multipliers considered in this chapter.

14.2.3 Dynamical constraints

At the start of this section we noted the possibility of a dynamical constraint. Such constraints typically involve both the position and velocity of the particle, and so might at first be considered non-holonomic. However, they do not arise from an external force of constraint. Rather, they arise from space and time symmetries. For example, in Section 14.8.4 we detail conditions under which the mechanical energy is a constant of the motion, whereby the sum of the kinetic and potential energies are constant. For a particle, this dynamical constraint takes the form

$$\tfrac{1}{2}\,m\,\dot{\boldsymbol{X}} \cdot \dot{\boldsymbol{X}} + P(\boldsymbol{X}) = \text{constant}. \tag{14.9}$$

Evidently, this constraint involves the position of the particle, through the potential energym $P(\boldsymbol{X})$, and the velocity, through the kinetic energy, $m\,\dot{\boldsymbol{X}} \cdot \dot{\boldsymbol{X}}/2$. It holds everywhere along the particle's trajectory as determined by the equation of motion

$$m\,\ddot{\boldsymbol{X}} = -\nabla P. \tag{14.10}$$

14.2.4 Comments

In this chapter we only consider dynamical constraints and holonomic constraints. The study of non-integrable (non-holonomic) constraints is outside our scope.

14.3 Coordinate description of motion

We here review the Cartesian coordinate formalism needed to describe the motion of a system of N particles, and then transform to generalized coordinates that are optimized for describing constrained motion.

14.3.1 Cartesian coordinates for N particles

Following the notation from Section 11.6, we specify the position of each discrete particle by its Cartesian position vector,

$$\boldsymbol{X}_{(i)} = \hat{\boldsymbol{x}}\, x^1_{(i)} + \hat{\boldsymbol{y}}\, x^2_{(i)} + \hat{\boldsymbol{z}}\, x^3_{(i)} = \boldsymbol{e}_a\, x^a_{(i)}, \tag{14.11}$$

where $i = 1, ..N$ is the particle index, $x^a_{(i)}$ is the a'th Cartesian coordinate for particle i, and $\boldsymbol{e}_a$ is the a'th Cartesian basis vector where $a = 1, 2, 3$. We place the particle index inside a parentheses to distinguish it from the coordinate indices, and we make use of the Einstein summation convention for the Cartesian index in the final equality. Each of the Cartesian coordinates for the particle position are functions of time, though we sometimes suppress time dependence for brevity.

The notation in equation (14.11) is somewhat cluttered on many occasions. To help declutter the notation we sometimes find it convenient to meld each of the Cartesian position vectors together into an ordered list of length $3N$

$$\boldsymbol{X} = (x^1, x^2, x^3, x^4, x^5, ..., x^{3N-2}, x^{3N-1}, x^{3N}). \tag{14.12}$$

This list is organized so that the first triplet, $n = 1, 2, 3$, contains the coordinates for the first particle, the second triplet, $n = 4, 5, 6$, is for the second particle, and so on up to the final triplet where $n = 3N-2, 3N-1, 3N$ is for particle N. In so ordering the indices we are able to drop the particle label since there is a clear mapping between particle number and position within the list.

14.3.2 An introduction to generalized coordinates

Cartesian coordinates are sufficient for describing motion through Euclidean space, and they offer a suitable description particularly when there are no constraints on the motion. However, when constraints reduce the degrees of freedom, we are motivated to choose each coordinate to directly correspond to an independent degree of freedom. We refer to such coordinates as generalized coordinates. Generalized coordinates can be any coordinate that specifies the spatial position of the matter comprising the physical system. For example, latitude and longitude provide generalized coordinates for a particle moving at a fixes radial position around a sphere, whereas the polar angle serves as a generalized coordinate for motion around the circumference of a planar circle. From these two examples, we see that generalized coordinates are typically not Cartesian coordinates.

A further example of the value of generalized coordinates arises in the description of rigid-body motion. A rigid body is comprised of a huge number of particles that are each constrained by inter-particle forces that keep the particles rigidly fixed relative to one another.[2] Internal forces of constraint that maintain the rigid spacing provide no net force nor net torqe on the

[2]In any real solid, the atoms are not rigidly fixed. But when concerned only with the macroscopic (classical) motion of a solid body, we can readily approximate its matter constituents as rigidly fixed relative to each other.

body as a whole, so that only externally applied forces and torques lead to time changes in the body's linear momentum and/or angular momentum. Correspondingly, the internal forces of constraint perform no net work on the body, which reflects d'Alembert's principle. Given that the particles are rigidly fixed relative to one another, the spatial configuration of the rigid body is specified by the position of the center of mass (three coordinates) and the orientation of the rigid body in space (three angles). The spatial configuration of a rigid body is thus fully described by just six degrees of freedom.[3]

14.3.3 Cartesian and generalized coordinates for N particles

Assume there are C holonomic constraints placed on the spatial coordinates of the system that are written in the form

$$\Psi_c(x^1, ..., x^{3N}, t) = \mathcal{C}_c, \quad \text{for} \quad c = 1, ..., C, \tag{14.13}$$

where $\mathcal{C}_c$ are constants. As a result, there are

$$D = 3N - C \tag{14.14}$$

spatial degrees of freedom. We can thus choose D generalized coordinates, $\xi^\sigma, \sigma = 1, 2, ..., D$, that are linearly independent and so fully describe the spatial position of the constrained system.[4] For example, if a single particle in three-dimensional space ($N = 3$) is constrained to move on a spherical shell of radius R, then the two ($D = 3 - C = 2$) spherical angular coordinates are suitable generalized coordinates corresponding to the single ($C = 1$) constraint equation, $r = R$. By choosing generalized coordinates, we have, in effect, integrated the constraints and so have no need to determine the corresponding forces of constraint.

With the spatial configuration of the physical system fully described by the generalized coordinates, we can determine the Cartesian position, $\boldsymbol{X}$, for the particles according to the functional relation

$$\boldsymbol{X} = \boldsymbol{X}(\xi^1, \xi^2, ..., \xi^D, t), \tag{14.15}$$

where the explicit time dependence in this relation offers generality that can be useful (e.g., motion on a spherical shell whose radius is time dependent). In turn, the exact differential of the Cartesian position is related to that of the generalized coordinates according to the chain rule

$$\mathrm{d}\boldsymbol{X} = \sum_{\sigma=1}^{D} \frac{\partial \boldsymbol{X}}{\partial \xi^\sigma} \,\mathrm{d}\xi^\sigma + \frac{\partial \boldsymbol{X}}{\partial t} \,\mathrm{d}t. \tag{14.16}$$

Note that we could choose to employ the summation convention in equation (14.16) to write

$$\mathrm{d}\boldsymbol{X} = \frac{\partial \boldsymbol{X}}{\partial \xi^\sigma} \,\mathrm{d}\xi^\sigma + \frac{\partial \boldsymbol{X}}{\partial t} \,\mathrm{d}t. \tag{14.17}$$

However, in this chapter we find it useful to expose the summation symbol to remind ourself that there are $D = 3N - C$ generalized coordinates, whereas there are $3N$ Cartesian coordinates. To further emphasize the distinction, we use a Greek letter for generalized coordinate indices and Latin letters for Cartesian coordinate indices.

[3] See any of the references at the start of this chapter for detailed analysis of rigid body motion.

[4] In mechanics texts, one typically finds generalized coordinates written as q^σ. We instead choose, ξ^σ.

14.3.4 Virtual displacements

When discussing forces of constraint in Section 14.2.1, we included the qualifier "instantaneous" or "time instance" when referring to the inability of a force of constraint to do work on the physical system. This qualifier is needed when the constraints are explicit functions of time, such as for motion on a spherical shell whose radius changes in time. We formalize this qualifier by introducing the concept of a virtual displacement. A virtual displacement is a tiny spatial displacement of a physical system that occurs at a fixed time instance. Furthermore, virtual displacements respect the forces of constraint, which means they are kinematically admissable displacements. However, they are not bound by dynamical constraints.

Conceptually, we imagine freezing the physical system at a specific time instance, and then probing alternative realizations of the physical system by considering kinematically admissable displacements, meaning that virtual displacements are consistent with the forces of constraints. By choosing a particular time instance, virtual displacements are not concerned with the dynamics since dynamics requires information about time changes. For a particle moving on a spherical shell, spherical angle displacements at a fixed time respect the constraint that the particle remains on the shell, so that tiny displacements of the spherical angles provide suitable virtual displacements.

Any perturbation of generalized coordinates represents a possible virtual displacement. In turn, a virtual displacement written in terms of Cartesian coordinates follows from the exact differential (14.16) with $\mathrm{d}t = 0$

$$\delta \boldsymbol{X} = \sum_{\sigma=1}^{D} \frac{\partial \boldsymbol{X}}{\partial \xi^{\sigma}} \, \delta\xi^{\sigma}. \tag{14.18}$$

In this expression we introduced the δ symbol for a virtual displacement as distinct from an exact differential. Virtual displacements hold a prominent role in analytical mechanics. In the context of Hamilton's principle, virtual displacements are generalized to the variations considered when probing alternative trajectories, with variations not assumed to be tiny.

Note that the δ symbol is commonly used in this book for tiny displacements or perturbations of a physical system that allow one to probe dynamical stability and examine alternative realizations of the motion. Not all cases where we use the δ symbol are virtual displacements. For example, in Section 13.6.2 we examined the constraints imposed by axial angular momentum conservation, with the use of δ in that context distinctly not virtual since the perturbations generally occur over a time interval.[5]

14.3.5 Particle moving on a spherical shell

As an example of the ideas presented in this section, consider a particle constrained to move a constant radial distance from a fixed center. This motion on a spherical shell is enabled through the imposition of forces of constaint that keep the particle from either moving radially outward or radially inward. The forces of constraint can perform no work on the particle since the forces act in the radial direction, whereas the motion is in the spherical angular directions (so the constraint forces are perpendicular to the velocity).

For a particular realization, consider particle motion on the surface of a solid massive and frictionless spherical shell. The gravitational force from the shell provides a radially inward force that keeps the particle from moving outward, whereas the solid surface of the shell provides

[5]The δ symbol is one of the more heavily overloaded symbols encountered in physics and mathematics.

the radially outward force that keeps the particle from moving inward.[6] It is as if the particle was confined to a narrow gap between two spherical surfaces that only allow motion in the angular directions. Neither of the constraining forces performs work on the particle since they act perpendicular to the particle's motion.

The coordinates used to specify the particle position are constrained to the spherical shell with fixed radius, R. Hence, the particle spatial position is fully specified by the latitude, ϕ, and longitude, λ, which are related to Cartesian coordinates via equations (4.84a)–(4.84c) (see also Figure 12.3)

$$x = R\cos\phi\,\cos\lambda \quad \text{and} \quad y = R\cos\phi\,\sin\lambda \quad \text{and} \quad z = R\sin\phi, \tag{14.19}$$

which manifests the constraint placed on the three Cartesian coordinates

$$\boldsymbol{X}(t)\cdot\boldsymbol{X}(t) = x^{a}(t)\,\delta_{ab}\,x^{b}(t) = x(t)^2 + y(t)^2 + z(t)^2 = R^2. \tag{14.20}$$

Note that the corresponding exact differentials

$$\mathrm{d}x = R\,(-\mathrm{d}\phi\sin\phi\cos\lambda - \mathrm{d}\lambda\cos\phi\sin\lambda) \tag{14.21a}$$
$$\mathrm{d}y = R\,(-\mathrm{d}\phi\sin\phi\sin\lambda + \mathrm{d}\lambda\cos\phi\cos\lambda) \tag{14.21b}$$
$$\mathrm{d}z = R\,\mathrm{d}\phi\cos\phi, \tag{14.21c}$$

are within the spherical shell since they are orthogonal to the radial unit vector

$$\hat{\boldsymbol{r}}\cdot(\hat{\boldsymbol{x}}\,\mathrm{d}x + \hat{\boldsymbol{y}}\,\mathrm{d}y + \hat{\boldsymbol{z}}\,\mathrm{d}z) = 0, \tag{14.22}$$

where the radial unit vector is given by equation (4.99c)

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}}\,\cos\lambda\,\cos\phi + \hat{\boldsymbol{y}}\sin\lambda\,\cos\phi + \hat{\boldsymbol{z}}\sin\phi. \tag{14.23}$$

14.4 Lagrange's equations of motion

In this section we step through the derivation of the Lagrange's equation of motion. This equation is equivalent to Newton's equation of motion, and yet it eliminates the forces of constraint and is written fully in terms of generalized coordinates. A key facet of the derivation concerns the need to articulate the functional dependency of the Cartesian position and velocity, $\boldsymbol{X}$ and $\dot{\boldsymbol{X}}$, on the generalized coordinates, generalized velocities, and time, $(\xi^{\sigma}, \dot{\xi}^{\sigma}, t)$.

14.4.1 Eliminating the forces of constraint

The inability of forces of constraint to perform work under virtual displacements, as per D'Alembert's principle, provides the key insight needed to eliminate the forces of constraint from the equations of motion. For this purpose, start by writing Newton's equation of motion using Cartesian coordinates

$$\dot{P}^{n} = F^{n} + F^{n}_{\text{constraint}} \quad \text{for } n = 1, ..., 3N, \tag{14.24}$$

[6]The constraint against moving inward arises from inter-atomic forces that keep the particle from penetrating into the solid shell. From the perspective of the classical particle motion, we are unconcerned with such inter-atomic forces, but instead only concerned with their effects on the classical motion of the particle.

where P^n is the n'th component of the linear momentum[7], F^n is the n'th component of the net applied force (e.g., gravity, electromagnetism) and $F^n_{\text{constraint}}$ is the n'th component of the force of constraint. It is convenient to make use of the $3N$ ordered list notation of equation (14.12), in which we organize the linear momentum time derivatives, and the applied forces, according to

$$\dot{\boldsymbol{P}} = (\dot{P}^1, \dot{P}^2, \dot{P}^3, \dot{P}^4, \dot{P}^5, ..., \dot{P}^{3N-2}, \dot{P}^{3N-1}, \dot{P}^{3N}) \tag{14.25a}$$

$$\boldsymbol{F} = (F^1, F^2, F^3, F^4, F^5, ..., F^{3N-2}, F^{3N-1}, F^{3N}), \tag{14.25b}$$

so that Newton's equation (14.24) for the N particle system becomes

$$\dot{\boldsymbol{P}} = \boldsymbol{F} + \boldsymbol{F}_{\text{constraint}}. \tag{14.26}$$

Through d'Alembert's principle we eliminate the forces of constraint by multiplying Newton's equation (14.26) by the vector of Cartesian virtual displacements, $\delta \boldsymbol{X}$, and summing over all $3N$ Cartesian coordinates

$$(\dot{\boldsymbol{P}} - \boldsymbol{F}) \cdot \delta \boldsymbol{X} = 0. \tag{14.27}$$

To reach this equation we set

$$\boldsymbol{F}_{\text{constraint}} \cdot \delta \boldsymbol{X} = 0, \tag{14.28}$$

due to d'Alembert's principle. To help appreciate d'Alembert's principle written as equation (14.28), again refer to the case of a particle moving on a spherical shell from Section 14.3.5, whereby the force of constraint is directed radially whereas a virtual displacement is directed along an angle within the shell.

Equation (14.27) is a statement of d'Alembert's principle using Cartesian coordinates. It succeeds in eliminating the forces of constraint, which is one of our goals. However, we cannot set each of the $3N$ terms individually to zero since each Cartesian component of the virtual displacement, $\delta \boldsymbol{X}$, is not generally independent of the other. To provide a more useful expression requires us to transform the Cartesian virtual displacements into generalized coordinate virtual displacements. This part of the derivation requires a few steps, and we do so by separately considering $\boldsymbol{F} \cdot \delta \boldsymbol{X}$ and $\dot{\boldsymbol{P}} \cdot \delta \boldsymbol{X}$.

14.4.2 Work by the applied forces acting on virtual displacements

Although the forces of constraint perform no work on the virtual displacements, the applied forces generally do perform work, as given by

$$\delta W = \boldsymbol{F} \cdot \delta \boldsymbol{X} = \boldsymbol{F} \cdot \sum_{\sigma=1}^{D} \frac{\partial \boldsymbol{X}}{\partial \xi^\sigma} \, \delta \xi^\sigma = \sum_{\sigma=1}^{D} Q_\sigma \, \delta \xi^\sigma. \tag{14.29}$$

The second equality made use of equation (14.18) to write the Cartesian virtual displacement in terms of the generalized coordinate displacements. The third equality defined the generalized force, which is the representation of the applied force using generalized coordinates

$$Q_\sigma = \boldsymbol{F} \cdot \frac{\partial \boldsymbol{X}}{\partial \xi^\sigma} \quad \text{for} \quad \sigma = 1, \ldots, D. \tag{14.30}$$

[7] In the context of analytical mechanics, we sometimes refer to the linear momentum as the mechanical momentum, with the generalized (or canonical) momentum introduced in Section 14.9.1.

We thus identify the product, $Q_\sigma \, \delta\xi^\sigma$ (no implied sum), as the work done by the generalized force, Q_σ, on the generalized virtual displacement, $\delta\xi^\sigma$. Correspondingly, the total work done by all generalized forces is given by the sum in equation (14.29). Observe that the generalized force can be computed from the defining expression (14.30), given knowledge of the force vector, $\boldsymbol{F}$, and coordinate transformation matrix, $\partial \boldsymbol{X}/\partial\xi^\sigma$. Alternatively, and often more conveniently, recall that the virtual displacements of the generalized coordinates are linearly independent so that we can compute the generalized force by computing the virtual work arising from a single virtual displacement, with all other displacements set to zero.

14.4.3 Massaging $\dot{\boldsymbol{P}} \cdot \delta\boldsymbol{X}$

As for equation (14.29), we transform the Cartesian virtual displacements into generalized coordinate virtual displacements according to

$$\dot{\boldsymbol{P}} \cdot \delta\boldsymbol{X} = \dot{\boldsymbol{P}} \cdot \sum_{\sigma=1}^{D} \frac{\partial \boldsymbol{X}}{\partial \xi^\sigma} \, \delta\xi^\sigma = \sum_{n=1}^{3N} \sum_{\sigma=1}^{D} \dot{p}^n \, \frac{\partial x^n}{\partial \xi^\sigma} \, \delta\xi^\sigma, \tag{14.31}$$

which takes on the following form if each particle has the same constant mass, m,

$$\dot{\boldsymbol{P}} \cdot \delta\boldsymbol{X} = m \sum_{n=1}^{3N} \sum_{\sigma=1}^{D} \ddot{x}^n \, \frac{\partial x^n}{\partial \xi^\sigma} \, \delta\xi^\sigma \tag{14.32a}$$

$$= m \sum_{n=1}^{3N} \sum_{\sigma=1}^{D} \left[\frac{\mathrm{d}}{\mathrm{d}t} \left(\dot{x}^n \, \frac{\partial x^n}{\partial \xi^\sigma} \right) - \dot{x}^n \, \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial x^n}{\partial \xi^\sigma} \right) \right] \delta\xi^\sigma. \tag{14.32b}$$

To massage this expression further requires us to be careful about the functional dependencies for the terms inside the square bracket.

Alternative expression for the transformation matrix elements

For the first right hand side term in equation (14.32b), recall that the functional dependence for the Cartesian expression of the particle velocity is found by making use of the exact differential (14.16), where division by $\mathrm{d}t$ leads to

$$\dot{\boldsymbol{X}} = \sum_{\sigma=1}^{D} \frac{\partial \boldsymbol{X}}{\partial \xi^\sigma} \, \dot{\xi}^\sigma + \frac{\partial \boldsymbol{X}}{\partial t}. \tag{14.33}$$

The Cartesian position, $\boldsymbol{X}$, is a function of the generalized coordinates and time as per equation (14.15), which then means that the partial derivatives, $\partial \boldsymbol{X}/\partial\xi^\sigma$ and $\partial \boldsymbol{X}/\partial t$, each have the same functional dependence. We thus conclude that the Cartesian velocity is a function of the generalized coordinates, the generalized velocities, and time

$$\dot{\boldsymbol{X}} = \dot{\boldsymbol{X}}(\xi^1, ..., \xi^D, \dot{\xi}^1, ..., \dot{\xi}^D, t). \tag{14.34}$$

Furthermore, equation (14.33) says that each element of $\dot{\boldsymbol{X}}$ is a linear function of the generalized velocities, $\dot{\xi}^\sigma$, from which it follows that

$$\frac{\partial \dot{x}^n}{\partial \dot{\xi}^\sigma} = \frac{\partial x^n}{\partial \xi^\sigma}. \tag{14.35}$$

This rather remarkable identity says that component-wise, the partial derivative of the Cartesian velocity, with respect to the generalized coordinate velocity, equals to the partial derivative of the Cartesian coordinate with respect to the generalized coordinate.

Making use of the identity (14.35) for the first right hand side term in equation (14.32b) leads to

$$m \sum_{n=1}^{3N} \sum_{\sigma=1}^{D} \frac{\mathrm{d}}{\mathrm{d}t} \left[\dot{x}^n \frac{\partial x^n}{\partial \xi^\sigma} \right] = m \sum_{n=1}^{3N} \sum_{\sigma=1}^{D} \frac{\mathrm{d}}{\mathrm{d}t} \left[\dot{x}^n \frac{\partial \dot{x}^n}{\partial \dot{\xi}^\sigma} \right] \tag{14.36a}$$

$$= \frac{m}{2} \sum_{n=1}^{3N} \sum_{\sigma=1}^{D} \frac{\mathrm{d}}{\mathrm{d}t} \left[\frac{\partial (\dot{x}^n \, \dot{x}^n)}{\partial \dot{\xi}^\sigma} \right] \tag{14.36b}$$

$$= \sum_{\sigma=1}^{D} \frac{\mathrm{d}}{\mathrm{d}t} \left[\frac{\partial K}{\partial \dot{\xi}^\sigma} \right], \tag{14.36c}$$

where we introduced the kinetic energy of the N particle system

$$K = \frac{m}{2} \sum_{n=1}^{3N} \dot{x}^n \, \dot{x}^n = \frac{m}{2} \sum_{i=1}^{N} \dot{\boldsymbol{X}}_{(i)} \cdot \dot{\boldsymbol{X}}_{(i)} = \frac{m}{2} \sum_{i=1}^{N} \dot{x}^a_{(i)} \, \delta_{ab} \, \dot{x}^b_{(i)}, \tag{14.37}$$

with the final equality introducing the Kronecker delta, δ_{ab}, which is the representation of the metric tensor for Euclidean space using Cartesian coordinates. Note that the functional dependence (14.34) for the Cartesian velocities is transferred to the kinetic energy so that

$$K = K(\xi^1, ..., \xi^D, \dot{\xi}^1, ..., \dot{\xi}^D, t). \tag{14.38}$$

Proving that $\mathrm{d}/\mathrm{d}t(\partial x^n / \partial \xi^\sigma) = \partial \dot{x}^n / \partial \xi^\sigma$

For the next step we prove the identity,

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial x^n}{\partial \xi^\sigma} = \frac{\partial}{\partial \xi^\sigma} \frac{\mathrm{d}x^n}{\mathrm{d}t}, \tag{14.39}$$

by evaluating both sides and showing they are equal. For the left hand side, recall from the derivation of equation (14.35) that the transformation matrix element, $\partial x^n / \partial \xi^\sigma$, is a function of the generalized coordinates and time, so that its total time derivative is

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial x^n}{\partial \xi^\sigma} = \left[\frac{\partial}{\partial t} + \sum_{\mu=1}^{D} \dot{\xi}^\mu \frac{\partial}{\partial \xi^\mu} \right] \frac{\partial x^n}{\partial \xi^\sigma}. \tag{14.40}$$

Next, evaluate the right hand side of equation (14.39) directly by expanding the total time derivative. Importantly, recall the functional dependence of the Cartesian velocity in equation (14.34), and note that the partial derivative, $\partial / \partial \xi^\sigma$, is computed while holding fixed the explicit time dependence and the dependence on the generalized velocity, $\dot{\xi}^\sigma$. We thus find

$$\frac{\partial}{\partial \xi^\sigma} \frac{\mathrm{d}x^n}{\mathrm{d}t} = \frac{\partial}{\partial \xi^\sigma} \left[\frac{\partial}{\partial t} + \sum_{\mu=1}^{D} \dot{\xi}^\mu \frac{\partial}{\partial \xi^\mu} \right] x^n = \left[\frac{\partial}{\partial t} + \sum_{\mu=1}^{D} \dot{\xi}^\mu \frac{\partial}{\partial \xi^\mu} \right] \frac{\partial x^n}{\partial \xi^\sigma}, \tag{14.41}$$

which follows since (i) partial derivative operators commute, and (ii) the partial derivative, $\partial/\partial\xi^{\sigma}$, is computed while holding the generalized velocity, $\dot{\xi}^{\mu}$, fixed. Since equations (14.40) and (14.41) are identical, we have proven the identity (14.39).

Making use of equation (14.39) for the second right hand side term in equation (14.32b) leads to

$$\sum_{n=1}^{3N}\sum_{d=1}^{D} m^{n}\dot{x}^{n}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial x^{n}}{\partial\xi^{\sigma}}\right)\delta\xi^{\sigma} = m\sum_{n=1}^{3N}\sum_{\sigma=1}^{D}\dot{x}^{n}\frac{\partial}{\partial\xi^{\sigma}}\frac{\mathrm{d}x^{n}}{\mathrm{d}t}\,\delta\xi^{\sigma} = \sum_{d=1}^{D}\frac{\partial K}{\partial\xi^{\sigma}}\,\delta\xi^{\sigma}, \tag{14.42}$$

where we again introduced the kinetic energy (14.37) of the N particle system.

14.4.4 Lagrange's equation of motion

We now combine the expression (14.29) for the work done by the generalized force against the virtual displacements, along with the identities (14.36c) and (14.42), to find

$$\sum_{d=1}^{D}\left[\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial K}{\partial\dot{\xi}^{\sigma}}\right] - \frac{\partial K}{\partial\xi^{\sigma}} - Q_{\sigma}\right]\delta\xi^{\sigma} = 0. \tag{14.43}$$

We have thus succeeded in eliminating the forces of constraint and expressing the equations in terms of just the generalized coordinates. Since each of the virtual displacements of the generalized coordinates, $\delta\xi^{\sigma}, \sigma = 1, ...D$, are linearly independent, satisfying equation (14.43) for arbitrary virtual displacements requires Lagrange's equation of motion

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial K}{\partial\dot{\xi}^{\sigma}}\right] - \frac{\partial K}{\partial\xi^{\sigma}} = Q_{\sigma} \quad \text{for } \sigma = 1,\ldots,D. \tag{14.44}$$

Lagrange's equation is a partial differential equation for the kinetic energy that is driven by generalized forces. This equation holds separately for each of the D generalized coordinates.

14.4.5 Conservative forces and the Lagrangian function

A conservative force equals to minus the spatial gradient of a potential energy scalar, P. When written using Cartesian coordinates, the potential energy for a conservative system is time independent. However, when transforming to generalized coordinates as per equation (14.15), the potential energy can pick up an explicit time dependence along with the expected dependence on generalized coordinates. We thus write the potential energy as

$$\text{potential energy} = P^{\text{cartesian}}(\boldsymbol{X}) = P^{\text{generalized}}(\xi^{1}, ..., \xi^{D}, t), \tag{14.45}$$

where $P^{\text{cartesian}}$ and $P^{\text{generalized}}$ are distinct scalar functions that both measure the potential energy. Notably, the potential energy is not a function of the Cartesian velocity nor the generalized velocity, so that

$$\frac{\partial P^{\text{cartesian}}}{\partial\dot{x}^{n}} = 0 \quad \text{and} \quad \frac{\partial P^{\text{generalized}}}{\partial\dot{\xi}^{\sigma}} = 0. \tag{14.46}$$

As per our discussion of scalar functions in Section 1.4.1, we drop the extra superscript notation on the potential energy in equation (14.45), with the understanding that the functional dependency determines which particular function is meant.[8] In this manner, the generalized

[8]We note the unfortunate duplication of P in this chapter, here for potential energy and in other places for

force takes the form

$$Q_\sigma = \sum_{n=1}^{3N} F^n \frac{\partial x^n}{\partial \xi^\sigma} = -\sum_{n=1}^{3N} \frac{\partial P(\boldsymbol{X})}{\partial x^n} \frac{\partial x^n}{\partial \xi^\sigma} = -\frac{\partial P(\xi^1, ..., \xi^D, t)}{\partial \xi^\sigma}, \tag{14.47}$$

where the second equality follows from the chain rule. Consequently, we can rewrite Lagrange's equation (14.44) as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial (K-P)}{\partial \dot{\xi}^\sigma}\right] - \frac{\partial (K-P)}{\partial \xi^\sigma} = 0 \quad \text{for } \sigma = 1, \ldots, D. \tag{14.48}$$

The difference between the kinetic energy and potential energy is referred to as the Lagrangian

$$L = K - P, \tag{14.49}$$

so that Lagrange's equation of motion (14.48) takes on the more commonly written form

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{\xi}^\sigma}\right] - \frac{\partial L}{\partial \xi^\sigma} = 0 \quad \text{for } \sigma = 1, \ldots, D. \tag{14.50}$$

The Lagrangian inherits the functional dependence of the kinetic energy as given by equation (14.38)

$$L = L(\xi^1, ..., \xi^D, \dot{\xi}^1, ..., \dot{\xi}^D, t). \tag{14.51}$$

We refer to configuration space as the D-dimensional space defined by the set of generalized coordinates. In this manner, we describe the motion of the physical system as a single point moving through a D-dimensional space. This picture contrasts to the Newtonian approach (e.g., see Section 11.6) that views N particles moving through three-dimensional space. The full state of the dynamical system is specified by the generalized coordinates plus the generalized velocities. Given the coordinates and velocities, we can then determine the Lagrangian and then, via Lagrange's partial differential equation (14.50), determine the trajectory of motion through configuration space.

In Lagrange's equation (14.50), when computing the partial derivatives, $\partial/\partial\xi^\sigma$ and $\partial/\partial\dot{\xi}^\sigma$, all other variables are held fixed. When computing the total time derivative, $\mathrm{d}/\mathrm{d}t$, we must be sure to account for the functional dependence of L according to equation (14.51), so that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{\xi}^\mu}\right] = \left[\frac{\partial}{\partial t} + \dot{\xi}^\sigma \frac{\partial}{\partial \xi^\sigma} + \ddot{\xi}^\sigma \frac{\partial}{\partial \dot{\xi}^\sigma}\right] \frac{\partial L}{\partial \dot{\xi}^\mu}. \tag{14.52}$$

14.4.6 Connection to Newton's equation of motion

As a partial differential equation for the scalar Lagrangian function, Lagrange's equation (14.50) is quite distinct from Newton's equation of motion, which is a vector ordinary differential equation for the acceleration determined by the forces. However, these two equations are directly connected since we used Newton's equation to derive Lagrange's equation.

To further reveal the equivalance, consider the unconstrained case for a single particle moving through a conservative force field, in which the Lagrangian as written in Cartesian

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the linear momentum. Notably, the linear momentum is a vector, so that it is either written using boldface $\boldsymbol{P}$ (as in equation (14.26)), or with an index, P^n (as in equation (14.24)). In contrast, the potential energy is a scalar.

coordinates as

$$L = (m/2)\,(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - P(x, y, z), \tag{14.53}$$

which leads to

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{x}}\right] = m\,\ddot{x} \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{y}}\right] = m\,\ddot{y} \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{z}}\right] = m\,\ddot{z}. \tag{14.54}$$

Lagrange's equation thus becomes Newton's equation of motion

$$\boldsymbol{F} = m\,\ddot{\boldsymbol{X}} \quad \text{with} \quad \boldsymbol{F} = -\nabla_{\boldsymbol{x}} P. \tag{14.55}$$

14.4.7 Kinetic energy in terms of generalized velocities

We wrote the kinetic energy in equation (14.37) in terms of the Cartesian velocities. Here we make use of the coordinate transformation (14.33) to expose the generalized velocities according to

$$K = \frac{m}{2}\sum_{i=1}^{N} \dot{x}^a_{(i)}\,\delta_{ab}\,\dot{x}^b_{(i)} \tag{14.56a}$$

$$= \frac{m}{2}\sum_{i=1}^{N}\left[\frac{\partial x^a_{(i)}}{\partial t} + \sum_{\sigma=1}^{D}\dot{\xi}^\sigma\,\frac{\partial x^a_{(i)}}{\partial \xi^\sigma}\right]\delta_{ab}\left[\frac{\partial x^b_{(i)}}{\partial t} + \sum_{\mu=1}^{D}\dot{\xi}^\mu\,\frac{\partial x^b_{(i)}}{\partial \xi^\mu}\right], \tag{14.56b}$$

where we made use of the notation in equation (14.11) with the implied summation over $a, b = 1, 2, 3$. Expanding this product leads to three terms, two of which arise from the time dependent coordinate transformation, whereby $\partial_t \boldsymbol{X}_{(i)} \neq 0$. The third term is given by the quadratic form

$$\frac{m}{2}\sum_{i=1}^{N}\sum_{\sigma=1}^{D}\sum_{\mu=1}^{D}\frac{\partial x^a_{(i)}}{\partial \xi^\sigma}\,\delta_{ab}\,\frac{\partial x^b_{(i)}}{\partial \xi^\mu}\,\dot{\xi}^\sigma\,\dot{\xi}^\mu = \frac{m}{2}\sum_{\sigma=1}^{D}\sum_{\mu=1}^{D} G_{\sigma\mu}(\boldsymbol{\xi})\,\dot{\xi}^\sigma\,\dot{\xi}^\mu. \tag{14.57}$$

We here introduced the symmetric tensor

$$G_{\sigma\mu} = \sum_{i=1}^{N}\frac{\partial x^a_{(i)}}{\partial \xi^\sigma}\,\delta_{ab}\,\frac{\partial x^b_{(i)}}{\partial \xi^\mu}, \tag{14.58}$$

which is the metric tensor for motion constrained to move on a manifold described by the generalized coordinates. For the special (and common) case of a time independent coordinate transformation, the metric is a function just of the generalized coordinates

$$G_{\sigma\mu} = G_{\sigma\mu}(\boldsymbol{\xi}), \tag{14.59}$$

so that the kinetic energy reduces to the quadratic form

$$K = \frac{m}{2}\sum_{\sigma=1}^{D}\sum_{\mu=1}^{D} G_{\sigma\mu}(\boldsymbol{\xi})\,\dot{\xi}^\sigma\,\dot{\xi}^\mu \quad \text{if } \partial_t \boldsymbol{X}_{(i)} = 0. \tag{14.60}$$

14.4.8 Comments

The derivation in this section closely follows sections 13, 14, and 15 of *Fetter and Walecka* (2003). In summary, the approach started from Newton's equation of motion and then introduced virtual displacements, d'Alembert's principle, and generalized coordinates. We then pursued a series of manipulations that carefully account for the functional dependencies of various quantities. In the next section we take an alternative approach that starts from Hamilton's principle of stationary action, which is a more streamlined approach that renders Lagrange's equation (referred to as the Euler-Lagrange equations) and is thus consistent with Newton's equation of motion. The more laborious approach of the current section serves to expose certain details that are useful for understanding the foundation of the theory, and that find use in different contexts within this book.

14.5 Hamilton's principle of stationary action

In this section we show that Lagrange's equations of motion (14.50) for a conservative physical system can be derived from a variational principle, known as Hamilton's principle of stationary action. Hamilton's principle states that the action, which is the time integral of the Lagrangian, is stationary if and only if Lagrange's equation of motion (now referred to as the Euler-Lagrange equation) is satisfied.

14.5.1 Notation

As for our derivation of Lagrange's equations in Section 14.4, it is useful to work with generalized coordinates in order to seamlessly account for constraints. To reduce notation clutter we use the boldface notation introduced in equation (14.12) for Cartesian coordinates. Here, the generalized coordinates and generalized velocities for a system with D degrees of freedom are organized into a list of length D according to

$$\boldsymbol{\xi} = (\xi^1, \dots, \xi^D) \quad \text{and} \quad \dot{\boldsymbol{\xi}} = (\dot{\xi}^1, \dots, \dot{\xi}^D). \tag{14.61}$$

In this manner, the kinetic energy, potential energy, and Lagrangian are written with the functional dependence

$$K = K(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t) \quad \text{and} \quad P = P(\boldsymbol{\xi}, t) \quad \text{and} \quad L = L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t). \tag{14.62}$$

We also assume the following shorthand for partial derivatives

$$\frac{\partial K}{\partial \boldsymbol{\xi}} = (\partial K/\partial \xi^1, \partial K/\partial \xi^2, \dots, \partial K/\partial \xi^D) \quad \text{and} \quad \frac{\partial K}{\partial \dot{\boldsymbol{\xi}}} = (\partial K/\partial \dot{\xi}^1, \partial K/\partial \dot{\xi}^2, \dots, \partial K/\partial \dot{\xi}^D). \tag{14.63}$$

With this notation, Lagrange's equations of motion (14.50) take the form

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{\boldsymbol{\xi}}}\right] - \frac{\partial L}{\partial \boldsymbol{\xi}} = 0. \tag{14.64}$$

Using Cartesian coordinates, we write the trajectory through space as $\boldsymbol{X}(t)$. When focused on generalized coordinates, we also refer to $\boldsymbol{\xi}(t)$ as the trajectory. This terminology follows since knowledge of each generalized coordinate is sufficient to specify the position of the particle(s) in space as a function of time.

14.5.2 The action

Consider the trajectory of an N particle system with $D \leq 3N$ degrees of freedom, and assume knowledge of the trajectory at two points in time, $t_{\mathsf{A}} < t_{\mathsf{B}}$. The action is defined as the time integral of the Lagrangian[9]

$$\mathcal{S} = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)\, \mathrm{d}t, \tag{14.65}$$

where we exposed the functional dependence (14.51) for the Lagrangian function. As defined, the action has physical dimensions of energy times time, which is the same as angular momentum

$$\mathcal{S} \;\; [\equiv] \;\; \mathrm{M}\ (\mathrm{L/T})^2\ \mathrm{T} = \mathrm{M}\ \mathrm{L}^2\ \mathrm{T}^{-1}. \tag{14.66}$$

Hamilton's principle states that the physically realized trajectory is an extremum of the action. The action integral (14.65) must have an extremum arising from the physically realized trajectory, regardless the time interval. However, the extremum is not necessarily a minimum.[10] Indeed, it is typically nontrivial to deduce the nature of the extremum given that one must compute the second variation of the action, which we did in Section 7.2.7 when studying variational calculus. Even so, we only rely on there being an extremum of the action in order to derive the equations of motion. It is for this reason that we prefer the name Hamilton's principle of stationary action rather than the commonly used principle of least action.

14.5.3 Variation of the action

To prove that the action is stationary for the physical trajectory, $\boldsymbol{\xi}(t)$, consider a perturbed trajectory via[11]

$$\boldsymbol{\xi}'(t) = \boldsymbol{\xi}(t) + \epsilon\, \boldsymbol{\mathcal{X}}(t). \tag{14.67}$$

As so defined, we have the perturbation written

$$\delta\boldsymbol{\xi}(t) = \boldsymbol{\xi}'(t) - \boldsymbol{\xi}(t) = \epsilon\, \boldsymbol{\mathcal{X}}(t), \tag{14.68}$$

and the corresponding perturbation to the generalized velocity

$$\delta\dot{\boldsymbol{\xi}}(t) = \dot{\boldsymbol{\xi}}'(t) - \dot{\boldsymbol{\xi}}(t) = \epsilon\, \dot{\boldsymbol{\mathcal{X}}}(t) = (\mathrm{d}/\mathrm{d}t)\, \delta\boldsymbol{\xi}. \tag{14.69}$$

In these equations, ϵ is a non-dimensional constant that scales the perturbation function, $\boldsymbol{\mathcal{X}}(t)$. Furthermore, the perturbation function vanishes at the initial time and final time

$$\boldsymbol{\mathcal{X}}(t_{\mathsf{A}}) = \boldsymbol{\mathcal{X}}(t_{\mathsf{B}}) = 0, \tag{14.70}$$

and $\boldsymbol{\mathcal{X}}(t)$ is consistent with the forces of constraint applied to the system. For example, if the motion is constrained to the surface of a spherical shell, then so too is $\boldsymbol{\mathcal{X}}$. Furthermore, note that equation (14.69) reveals that variations in the generalized velocities are dependent on variations of the generalized coordinates.

We refer to $\delta\boldsymbol{\xi}(t)$ as the variation of the trajectory, with Figure 14.1 providing an illustration.

[9] In some chapters in VOLUME 2, we use the symbol $\mathcal{S}$ for the entropy per mass. The distinct meanings for the same symbol will be clear from the context.

[10] This point is emphasized in a footnote in Section 2 of *Landau and Lifshitz* (1976).

[11] Be sure to distinguish the trajectory perturbation, $\boldsymbol{\mathcal{X}}(t)$, from the Cartesian expression for the physical trajectory, $\boldsymbol{X}(t)$.

Variations provide a method to probe physically unrealized paths for the purpose of characterizing and understanding what is special about the physically realized path. This approach builds from the virtual displacements considered in Section 14.3.4. Indeed, the path variations are virtual displacements from the physical path, though with the size of the variation, as measured by ϵ, not necessarily small. Like a virtual displacement, path variations are made during a single time instance so that $\mathrm{d}t = 0$.

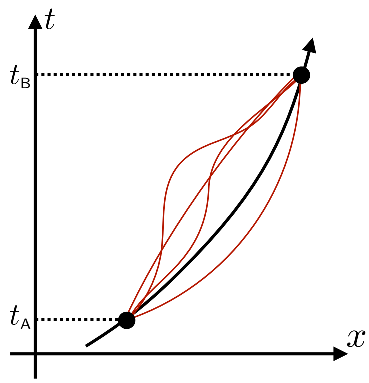

FIGURE 14.1: The physically realized trajectory, $\boldsymbol{\xi}(t)$, is depicted by the black line, with a sampling of the infinite number of possible variations, $\boldsymbol{\xi}'(t) = \boldsymbol{\xi}(t) + \delta\boldsymbol{\xi}(t)$, depicted in red. We assume knowledge of the trajectory at the start time, t_{A}, and end time, t_{B}. Hence, the physical trajectory and its variations are coincident at times t_{A} and t_{B}. Variations provide a means to probe physically unrealized paths for the purpose of characterizing and understanding what is special about the physically realized path.

The Lagrangian function takes on the following form when evaluated with the modified trajectory (14.67)

$$L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t) = L(\boldsymbol{\xi} + \epsilon\,\boldsymbol{\chi}, \dot{\boldsymbol{\xi}} + \epsilon\,\dot{\boldsymbol{\chi}}, t) \approx L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t) + \epsilon \left[\boldsymbol{\chi} \cdot \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \boldsymbol{\xi}} + \dot{\boldsymbol{\chi}} \cdot \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \dot{\boldsymbol{\xi}}} \right]. \tag{14.71}$$

We only expand the Taylor series to leading order since we are only looking for conditions for which the action has a vanishing first variation. The notation in equation (14.71) is a shorthand for the more complete sums over the degrees of freedom

$$\boldsymbol{\chi} \cdot \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \boldsymbol{\xi}} = \sum_{\sigma=1}^{D} \chi^{\sigma} \, \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \xi^{\sigma}} \quad \text{and} \quad \dot{\boldsymbol{\chi}} \cdot \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \dot{\boldsymbol{\xi}}} = \sum_{\sigma=1}^{D} \dot{\chi}^{\sigma} \, \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \dot{\xi}^{\sigma}}. \tag{14.72}$$

From equation (14.71) we deduce the variation of the Lagrangian that arises from variation of the trajectories

$$\delta L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t) = L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t) - L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t) \approx \epsilon \left[\boldsymbol{\chi} \cdot \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \boldsymbol{\xi}} + \dot{\boldsymbol{\chi}} \cdot \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \dot{\boldsymbol{\xi}}} \right], \tag{14.73}$$

along with the corresponding variation of the action

$$\delta \mathcal{S} = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} \delta L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t) \, \mathrm{d}t. \tag{14.74}$$

Integrating by parts on the final term in equation (14.73) leads to the equivalent expression for variation of the Lagrangian

$$\delta L = \epsilon\, \boldsymbol{\mathcal{X}} \cdot \left[\frac{\partial L}{\partial \boldsymbol{\xi}} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{\boldsymbol{\xi}}} \right) \right] + \frac{\mathrm{d}}{\mathrm{d}t} \left[\epsilon\, \boldsymbol{\mathcal{X}} \cdot \frac{\partial L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t)}{\partial \dot{\boldsymbol{\xi}}} \right]. \tag{14.75}$$

Since the perturbation function, $\boldsymbol{\mathcal{X}}$, vanishes at times t_{A} and t_{B} according to equation (14.70), the total time derivative plays no role in the action variation, in which case

$$\delta \mathcal{S} = \epsilon \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} \boldsymbol{\mathcal{X}} \cdot \left[\frac{\partial L}{\partial \boldsymbol{\xi}} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{\boldsymbol{\xi}}} \right) \right] \mathrm{d}t. \tag{14.76}$$

14.5.4 Stationary action and the Euler-Lagrange equations

Equation (14.76) says that variation of the action vanishes (i.e., the action is stationary) for all values of $\boldsymbol{\mathcal{X}}$ if and only if the integrand is zero for each of the generalized coordinates. We are thus led to the condition

$$\delta \mathcal{S} = 0 \Longleftrightarrow \frac{\mathrm{d}}{\mathrm{d}t} \left[\frac{\partial L}{\partial \dot{\xi}^{\sigma}} \right] - \frac{\partial L}{\partial \xi^{\sigma}} = 0 \quad \text{for } \sigma = 1, \ldots, D, \tag{14.77}$$

which is Lagrange's equation of motion (14.50) derived in Section 14.4 starting from Newton's equation of motion. In the context of Hamilton's principle, Lagrange's equations (14.77) are referred to as the Euler-Lagrange equations, given the work by Euler on variational principles.

14.5.5 Derivative of the action with respect to ϵ

The discussion in Section 14.5.3 illustrates how to vary the action, and its usage becomes second nature once doing it a few times. We here present a slightly alternative approach that, in the end, is totally equivalant to the variation approach but which offers a few more insights.

Consider the action evaluated on the perturbed trajectory

$$\mathcal{S}' = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t)\, \mathrm{d}t = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} L(\boldsymbol{\xi} + \epsilon\, \boldsymbol{\mathcal{X}}, \dot{\boldsymbol{\xi}} + \epsilon\, \dot{\boldsymbol{\mathcal{X}}}', t)\, \mathrm{d}t. \tag{14.78}$$

Now take the derivative of the perturbed action with respect to ϵ

$$\frac{\mathrm{d}\mathcal{S}'}{\mathrm{d}\epsilon} = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} \left[\frac{\partial L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t)}{\partial \boldsymbol{\xi}'} \cdot \frac{\partial \boldsymbol{\xi}'}{\partial \epsilon} + \frac{\partial L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t)}{\partial \dot{\boldsymbol{\xi}}'} \cdot \frac{\partial \dot{\boldsymbol{\xi}}'}{\partial \epsilon} \right] \mathrm{d}t. \tag{14.79}$$

Now evaluate the derivatives of the trajectory and the velocity to find

$$\frac{\mathrm{d}\mathcal{S}'}{\mathrm{d}\epsilon} = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} \left[\frac{\partial L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t)}{\partial \boldsymbol{\xi}'} \cdot \boldsymbol{\mathcal{X}} + \frac{\partial L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t)}{\partial \dot{\boldsymbol{\xi}}'} \cdot \dot{\boldsymbol{\mathcal{X}}} \right] \mathrm{d}t, \tag{14.80}$$

with integration by parts and use of the initial and final conditions yielding

$$\frac{\mathrm{d}\mathcal{S}'}{\mathrm{d}\epsilon} = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} \boldsymbol{\mathcal{X}} \cdot \left[\frac{\partial L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t)}{\partial \boldsymbol{\xi}'} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L(\boldsymbol{\xi}', \dot{\boldsymbol{\xi}}', t)}{\partial \dot{\boldsymbol{\xi}}'} \right) \right] \mathrm{d}t. \tag{14.81}$$

This expression has the same look as the variation in equation (14.76), and yet here the action is evaluated on the perturbed trajectory rather than the physically realized trajectory. Hamilton's principle says that the action is extremal for the physically realized trajectory, which means that

$$\left[\frac{\mathrm{d}\mathcal{S}'}{\mathrm{d}\epsilon}\right]_{\epsilon=0} = 0. \tag{14.82}$$

That is, Hamilton's principle says that the physically realized trajectory extremizes the action, which means that the action has a zero derivative at $\epsilon = 0$. As expected, this statement leads to the same Euler-Lagrange equations (14.77) as derived using the variational approach.

Considering the small parameter to be continuous, we can formally connect this discussion to the δ variation operator by writing

$$\left[\frac{\mathrm{d}\mathcal{S}'}{\mathrm{d}\epsilon}\right]\mathrm{d}\epsilon = \delta\mathcal{S}' \quad \text{and} \quad \frac{\mathrm{d}\boldsymbol{\xi}'}{\mathrm{d}\epsilon}\,\mathrm{d}\epsilon = \delta\boldsymbol{\xi}'. \tag{14.83}$$

14.5.6 Mechanical equivalence of Lagrangians

The Euler-Lagrange equation (14.77) remains unchanged if we multiply the Lagrangian by a constant. This property, though seemingly trivial, has important implications for scaling properties of the energy of physical systems. We deduce such properties in Section 14.6. Additionally, the Euler-Lagrange equation is unchanged by adding the total time derivative of an arbitrary function of the generalized coordinates and time

$$L^{\text{new}} = L^{\text{old}} + \frac{\mathrm{d}\Gamma(\boldsymbol{\xi}, t)}{\mathrm{d}t}. \tag{14.84}$$

To show the equivalance, note that the action transforms into

$$\mathcal{S}^{\text{new}} = \mathcal{S}^{\text{old}} + \Gamma[\boldsymbol{\xi}(t_{\mathsf{B}}), t_{\mathsf{B}}] - \Gamma[\boldsymbol{\xi}(t_{\mathsf{A}}), t_{\mathsf{A}}]. \tag{14.85}$$

The added terms are evaluated at t_{A} and t_{B}, where the trajectories are held fixed. Hence, these terms have zero variation so that

$$\delta\mathcal{S}^{\text{new}} = \delta\mathcal{S}^{\text{old}}, \tag{14.86}$$

which then means that the associated Euler-Lagrange equation is unchanged.

14.5.7 Summary of the method

As a recap, we here defined the action (14.65) as the time integral of the Lagrangian between a start time, t_{A}, and end time, $t_{\mathsf{B}} > t_{\mathsf{A}}$, with the position of the trajectory known (specified) at the start and end times, t_{A} and t_{B}. Besides knowing all details of the physical system at $t = t_{\mathsf{A}}$ and at $t = t_{\mathsf{B}}$, there is nothing special about these two times. For example, we do not imagine the physical system to have started from some rest state at t_{A}. Instead, we only imagine that information about a physical system is somehow known at at the two times, t_{A} and t_{B}. The physical question concerns the trajectory in between these two time instances. To determine this trajectory, we consider how the action is modified by considering an arbitrary variation of the trajectory for times between the end times, again with the trajectory fixed at the start and end times. We found that the Euler-Lagrange equations (14.77) are satisfied if and only if the action is stationary with respect to trajectory variations. That is, the physically realized trajectory is that trajectory that makes the action stationary (an extremum). This result constitutes Hamilton's principle of stationary action.

Since the physically realized trajectory is the one that satisfies the Euler-Lagrange equations, all other trajectories are unphysical and as such they are not constrained by dynamical principles. Instead, these virtual trajectories are only constrained by the geometry of the system and by the need to respect the prescribed conditions at times t_A and t_B. As such, Hamilton's principle tests all kinematically possible trajectories and selects the dynamically unique trajectory through insisting on stationarity of the action. Uniqueness of the dynamical trajectory constitutes a tacit assumption of Hamilton's principle. Proof of uniqueness follows from uniqueness of solutions to the Euler-Lagrange differential equations.

From a mathematical perspective, Lagrangian mechanics illustrates the notions of manifolds and tangent spaces. Namely, each point on the manifold where a physical system lives can be represented by a local generalized coordinate, and the corresponding velocity vector lives in the tangent space at this point. The accumulation of all points on the manifold then leads to the accumulation of all tangent spaces, otherwise known as the tangent bundle. The Lagrangian function, which is a functional of the trajectory and its velocity, acts to connect the tangent bundle to the manifold.

14.5.8 Some philosophical points

As already noted, Hamilton's principle provides an alternative to Newton's laws for the formulation of mechanics. Indeed, it is remarkable how complementary these two approaches are. Namely, Newton's laws are concerned with forces and accelerations acting at a point in space and time. Stating Newton's law of motion is relatively straightforward and, with the benefit of centuries of hindsight and experience, rather self-evident.[12] In contrast, Hamilton's principle says that the difference between the kinetic and potential energies, as integrated on a trajectory over a time interval, is extremized by the physically realized trajectory. Understanding the mathematical and physical statement of Hamilton's principle, and making use of it operationally, requires nontrivial formulational effort via the methods of analytical mechanics. Even so, upon mastering its formulation, the application of Hamilton's principle via the Euler-Lagrange equations can prove of great practical value for solving physical problems that are outside the reach of Newtonian methods.

The Newtonian description offers the canonical cause and effect perspective, whereby a force (the cause) creates an effect (the acceleration). In contrast, Hamilton's principle states that the physically realize motion takes place so that the action (an integral over time) is an extrema. From this view, Hamilton's principle can be viewed as offering a purpose for the motion, namely to extremize the action.[13] We do not pursue this discussion further as it goes beyond our goals, though the interested reader might find Chapter 14 of *Yourgrau and Mandelstam* (1968) of interest for its philosophical insights.

14.6 Mechanical similarity and the virial theorem

As noted in Section 14.5.6, the Euler-Lagrange equation (14.77) remains unchanged if we multiply the Lagrangian by a constant. In this section we uncover a suite of rather general scaling properties that can be inferred as a result of this seemingly trivial symmetry. Remarkably, these properties are found without solving the equations of motion.

[12] Being self-evident in retrospect is often a characteristic of a foundational physical concept.

[13] A teleological explanation considers phenomena in terms of the purpose it serves rather than of the cause by which it arises.

14.6.1 Potential energy as a homogeneous function

To derive the advertised results, we must assume that the potential energy is a homogeneous function, which means that it scales according to

$$P(\lambda\, x^1, \lambda\, x^2, \lambda\, x^3) = \lambda^\gamma\, P(x^1, x^2, x^3), \tag{14.87}$$

where λ is a non-dimensional scale factor and γ is an integer that sets the degree of homogeneity. The left hand side of equation (14.87) is the potential energy evaluated with each of its Cartesian coordinates scaled by the same number, λ. That is, the left hand side has length scales modified by λ. The right hand side is the potential energy evaluated with the unscaled coordinates (unscaled length), but it is multiplied by the scale, λ, raised to the power γ.

Note that we represented the potential energy in equation (14.87) in terms of the Cartesian coordinates, since doing so readily ensures that scaling is consistently applied to each of the distinct coordinates by the same number. We also only considered the potential energy for a single particle. Extending to an arbitrary number of particles simply means adding more coordinates to the potential energy, with the scale, λ, also applied to these additional coordinates.

We prove Euler's theorem for homogeneous functions in VOLUME 2 in our study of thermodynamics, which when applied to potential energy constitutes the identity

$$\gamma\, P(\boldsymbol{X}) = x^a\, \partial P(\boldsymbol{X})/\partial x^a. \tag{14.88}$$

This equation says that the potential energy, as multiplied by the degree of homogeneity, γ, has an equivalent expression in terms of the coordinates multiplied by the corresponding partial derivative of the potential energy. Introducing the conservative force associated with the potential energy

$$\boldsymbol{F} = -\nabla P, \tag{14.89}$$

we see that Euler's theorem for potential energy means that

$$\gamma\, P = -\boldsymbol{X} \cdot \boldsymbol{F} \tag{14.90}$$

We emphasize that the rather simple expression (14.90) for the potential energy holds only for potential energies that are homogeneous functions. Though restrictive, there are a number of physically interesting systems with homogeneous potential energies, including the following.

- $\gamma = -1$: The potential energy of particles in a gravity field or electrostatic field is inversely related to the distance between the particles, so that $\gamma = -1$.
- $\gamma = 1$: A particle moving in a uniform field, such as a uniform gravity field, has potential energy with $\gamma = 1$.
- $\gamma = 2$: Small amplitude oscillations, such as those exhibited by a pendulum (Section 15.1), simple harmonic oscillator (Section 15.7), or linear waves (VOLUME 4) have $\gamma = 2$.

Note that the trivial case of $\gamma = 0$ corresponds to a potential energy that is a constant, and thus leads to a zero force.

14.6.2 Mechanical similarity

Consider a physical system in which we scale the length by λ, just as considered for the potential energy in equation (14.87). This scaling alters the Lagrangian in a manner that alters the Euler-Lagrange equations. However, if we couple scaling of the length with a corresponding scaling of the time, then it is conceivable that we can find a combination of these two scalings that results in an overall factor multiplying the Lagrangian, in which case the Euler-Lagrange equations are unaltered. We refer to this confluence of scaling as mechanical similarity.

We are thus led to examine symmetry of the Lagrangian under the space-time scaling transformation

$$x^a \to \lambda\, x^a \Longleftrightarrow x^{a'}/x^a = \lambda \quad \text{and} \quad t \to \beta\, t \Longleftrightarrow t'/t = \beta, \tag{14.91}$$

where λ and β are two non-dimensional scales. This space-time scaling leads to a scaling of the velocity components

$$\dot{x}^a \to (\lambda/\beta)\, \dot{x}^a. \tag{14.92}$$

Since the kinetic energy is proportional to the square of the velocity, the scaling of space and time given by equation (14.91) renders the Lagrangian scaling

$$L = K - P \to (\lambda/\beta)^2\, K - \lambda^\gamma\, P. \tag{14.93}$$

In order for this scaling to not alter the Euler-Lagrange equations requires

$$\beta = \lambda^{1-\gamma/2} \Longrightarrow (t'/t) = \beta = \lambda^{1-\gamma/2} = (x'/x)^{1-\gamma/2}. \tag{14.94}$$

which we refer to as mechanical similarity. This equation means that if we scale time by β, so that $(t'/t) = \beta$, then we do not change the equations of motion if length is scaled according to $(l/l')^{1-\gamma/2}$. We can now infer the following results according to the scaling of the potential energy presented in Section 14.6.1.

- $\gamma = -1$ leads to $(t'/t) = (x'/x)^{3/2}$, or $(t'/t)^2 = (x'/x)^3$. This equality accords with Kepler's third law, which states that the square of the orbit period scales as the cube of the orbit size.
- $\gamma = 1$ leads to $(t'/t) = (x'/x)^{1/2}$, or $(t'/t)^2 = (x'/x)$. This equality accords with the quadratic time dependence for a particle freely falling through a uniform gravity field.
- $\gamma = 2$ leads to $(t'/t) = 1$, which accords with the amplitude independence for the period of small oscillations exhibited by a pendulum.

14.6.3 Virial theorem and time averaged energy

To derive the virial theorem, we assume that the physical system maintains motion within a bounded region of space and with bounded velocity. This assumption is maintained by geophysical motion. The virial theorem provides a relation between the time averaged kinetic energy and time averaged potential energy for such bounded motion.

The kinetic energy is a quadratic function of velocity. Hence, as a function of velocity, the kinetic energy is a homogeneous function of degree two. Euler's theorem for homogeneous functions thus means that

$$2\, K = m\, \delta_{ab}\, \dot{x}^a\, \dot{x}^b/2 = \dot{x}^a\, (\partial K/\partial \dot{x}^a) = \dot{x}^a\, \mathcal{P}_a, \tag{14.95}$$

where we introduced the Cartesian momentum,

$$\mathcal{P}_a \equiv \frac{\partial L}{\partial \dot{x}^a} = \frac{\partial K}{\partial \dot{x}^a}, \tag{14.96}$$

which is generalized in Section 14.9 for generalized coordinates. Here it merely represents a useful shorthand, allowing us to write

$$
\begin{aligned}
2\,K &= \dot{x}^{a}\,\mathcal{P}_{a} && (14.97a)\\
&= \frac{\mathrm{d}(x^{a}\,\mathcal{P}_{a})}{\mathrm{d}t} - x^{a}\,\dot{\mathcal{P}}_{a} && (14.97b)\\
&= \frac{\mathrm{d}(x^{a}\,\mathcal{P}_{a})}{\mathrm{d}t} + x^{a}\,\partial P/\partial x^{a} && (14.97c)\\
&= \frac{\mathrm{d}(x^{a}\,\mathcal{P}_{a})}{\mathrm{d}t} + \gamma\,P. && (14.97d)
\end{aligned}
$$

In the penultimate step (14.97c) we set

$$\dot{\mathcal{P}}_{a} = -\partial P/\partial x^{a} \tag{14.98}$$

according to Newton's law of motion, and in the final step (14.97d) we made use of Euler's theorem (14.88) for the potential energy. Taking the time average of equation (14.97d), and assuming the time average of the time derivative vanishes, then leads to the relation between the time averaged kinetic energy and time averaged potential energy

$$2\,\overline{K} = \gamma\,\overline{P}. \tag{14.99}$$

Note that the time average of the derivative vanishes if the time average is computed over a long enough time, given that the motion is assumed to be bounded. Additionally, if the motion is periodic then we can exactly remove the time derivative by taking the time average over the period of the motion.

The virial theorem (14.99) relates the time averaged kinetic energy and the time averaged potential energy in physical systems with a potential energy that is a homogeneous function of degree γ. In this book we find particular use for the case of $\gamma = 2$, which corresponds to motion of a simple harmonic oscillator, in which the virial theorem says that the potential and kinetic energies have equal time averages. This result is referred to as the equipartition of energy. In addition to harmonic oscillators and small amplitude pendula, we find VOLUME 4 that the equipartition of energy holds for linear wave motion, with the potential energy of linear waves generally a homogeneous function of degree two, and the time average is an average over the phase of the wave.

14.6.4 Further reading

This section shares much with Section 10 of *Landau and Lifshitz* (1976) as well as Section 6.13 of *Marion and Thornton* (1988).

14.7 Lagrange multipliers and forces of constraint

Derivation of Lagrange's equation of motion in Section 14.4 succeeded in eliminating the forces of constraint from the formulation. Likewise in the discussion of Hamilton's principle in Section 14.5, we assumed the $D = 3N - C$ generalized coordinates directly correspond to the D degrees of freedom available to the constrained physical system. In this section we present a method to infer the C forces of constraint through the method of Lagrange multipliers, with this inference of use when aiming to study force balances.

14.7.1 Holonomic constraints and their variation

Following equation (14.13), consider $3N$ coordinates, $\xi^{\sigma}, \sigma = 1, \ldots, 3N$, and assume they are constrained by C holonomic constraints of the form

$$\Psi_c(\xi^1, \ldots, \xi^{3N}, t) = \mathcal{C}_c, \quad \text{for} \quad c = 1, \ldots, C, \tag{14.100}$$

where $\mathcal{C}_c$ are constants. This equation is nearly the same as equation (14.13), only now we allow the coordinates to be non-Cartesian if that suits to the problem.[14] Varying the trajectory then leads to a variation of the constraint equations (14.100) that takes the form

$$\delta\Psi_c = \sum_{\sigma=1}^{3N} \frac{\partial \Psi_c}{\partial \xi^{\sigma}} \delta\xi^{\sigma} = 0, \quad \text{for} \quad c = 1, \ldots, C. \tag{14.101}$$

This equation provides C constraints that must be satisfied by the $3N$ variations, $\delta\xi^{\sigma}$. Recall that a variation is a virtual process (Section 14.3.4) and so it occurs at a time instance, which explains why there is no $\mathrm{d}t$ term in equation (14.101). Also note that variation of a constant vanishes, so that $\delta\,\mathcal{C}_c = 0$, thus rendering a zero on the right hand side of equation (14.101).

14.7.2 Lagrange multipliers and the modified Euler-Lagrange equations

Recall the variation of the action given by equation (14.76), only now write this variation assuming there are $3N$ coordinates

$$\delta\mathcal{S} = \int_{t_A}^{t_B} \sum_{\sigma=1}^{3N} \delta\xi^{\sigma} \left[\frac{\partial L}{\partial \xi^{\sigma}} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\xi}^{\sigma}}\right)\right] \mathrm{d}t = 0. \tag{14.102}$$

Contrary to what we did in Section 14.5.4, we cannot separately set $\delta\xi^{\sigma}$ to zero to then render the Euler-Lagrange equations (14.77). The reason is that the $3N$ variations, $\delta\xi^{\sigma}$, are constrained according to equation (14.103), thus making the variations dependent on one another. The method of Lagrange multipliers allows for us to incorporate the constraints while keeping all $3N$ coordinates.

To do so, introduce C Lagrange multipliers, $\Lambda^c(t)$, which are functions of time and whose form is to be specified below. Multiply each of the C constraint equations (14.101) by Λ^c and sum over all of the constraints

$$\sum_{c=1}^{C} \Lambda^c\,\delta\Psi_c = \sum_{c=1}^{C}\sum_{\sigma=1}^{3N} \Lambda^c \frac{\partial \Psi_c}{\partial \xi^{\sigma}}\,\delta\xi^{\sigma} = 0. \tag{14.103}$$

Now the action variation in equation (14.102) is unchanged by adding zero to the right hand side, and if we add zero in the form of equation (14.103) then that brings the constraints into the action variation via

$$\delta\mathcal{S} = \int_{t_A}^{t_B} \sum_{\sigma=1}^{3N} \delta\xi^{\sigma} \left[\frac{\partial L}{\partial \xi^{\sigma}} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\xi}^{\sigma}}\right) + \sum_{c=1}^{C} \Lambda^c \frac{\partial \Psi_c}{\partial \xi^{\sigma}}\right] \mathrm{d}t = 0. \tag{14.104}$$

[14] Strictly, the $3N$ coordinates, ξ^{σ}, are not "generalized coordinates" since that term refers to coordinates that directly correspond to the $D = 3N - C$ degrees of freedom. Even so, that distinction is not important for what follows.

This equation has $D = 3N - C$ independent variations and C constrained variations. Let us organize the coordinates, ξ^σ, so that the $\sigma = 1, \ldots, D$ coordinates have independent variations, whereas the remaining are constrained. Hence, coefficients of the independent variations in equation (14.104) must vanish in order to satisfy Hamilton's stationary action principle, using the same logic as for Section 14.5.4. For the remaining $\sigma = D+1, \ldots, D+C$ constrained variations, we can choose the C Lagrange multipliers so that the coefficients of these dependent variations also vanish. In this manner the coefficients multiplying all of the $3N$ variations, $\delta\xi^\sigma$, vanish, which then leads to the modified Euler-Lagrange equations along with the C holonomic constraints

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{\xi}^\sigma}\right] - \frac{\partial L}{\partial \xi^\sigma} = \sum_{c=1}^{C} \Lambda^c \frac{\partial \Psi_c}{\partial \xi^\sigma} \quad \text{for } \sigma = 1, \ldots, 3N \tag{14.105a}$$

$$\Psi_c(\xi^1, \ldots, \xi^{3N}, t) = \mathcal{C}_c \qquad \text{for} \quad c = 1, \ldots, C. \tag{14.105b}$$

14.7.3 The extended Lagrangian with multipliers

The modified Euler-Lagrange equation (14.105a) and corresponding holonomic constraints (14.105b) can be summarized by defining the extended Lagrangian as

$$L^* = L + \sum_{c=1}^{C} \Lambda^c \, \Psi_c. \tag{14.106}$$

As defined, the extended Lagrangian has the functional depedence

$$L^* = L(\xi^1, \ldots, \xi^{3N}, \dot{\xi}^1, \ldots, \dot{\xi}^{3N}, t, \Lambda^1, \ldots, \Lambda^C), \tag{14.107}$$

so that it is a function of the coordinates, ξ^σ, velocities, $\dot{\xi}^\sigma$, and time, t, as well as the Lagrange multipliers, $\Lambda^1, \ldots, \Lambda^C$. We thus find the partial derivatives

$$\frac{\partial L^*}{\partial \dot{\xi}^\sigma} = \frac{\partial L}{\partial \dot{\xi}^\sigma} \tag{14.108a}$$

$$\frac{\partial L^*}{\partial \xi^\sigma} = \frac{\partial L}{\partial \xi^\sigma} + \sum_{c=1}^{C} \Lambda^c \frac{\partial \Psi_c}{\partial \xi^\sigma} \tag{14.108b}$$

$$\frac{\partial L^*}{\partial \Lambda^c} = \Psi_c. \tag{14.108c}$$

In this manner we can write the Euler-Lagrange equation (14.105a) and corresponding holonomic constraints (14.105b) as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L^*}{\partial \dot{\xi}^\sigma}\right] = \frac{\partial L^*}{\partial \xi^\sigma} \tag{14.109a}$$

$$\frac{\partial L^*}{\partial \Lambda^c} = \Psi_c. \tag{14.109b}$$

14.7.4 Determining the forces of constraint

There are $3N$ equations in the modified Euler-Lagrange equation (14.105a), and C constraints in equation (14.105b). By adding to the burden of what is to be determined, we now have the means to infer the forces of constraint. For that purpose, re-introduce the kinetic energy and potential energy to write equation (14.105a) as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial K}{\partial \dot{\xi}^{\sigma}}\right] - \frac{\partial K}{\partial \xi^{\sigma}} = -\frac{\partial P}{\partial \xi^{\sigma}} + \sum_{c=1}^{C} \Lambda^{c} \frac{\partial \Psi_{c}}{\partial \xi^{\sigma}} \quad \text{for } \sigma = 1, \dots, 3N, \tag{14.110}$$

where the potential energy is independent of the coordinate velocities, $\partial P/\partial \dot{\xi}^{\sigma} = 0$, as follows from equation (14.46). Recalling Lagrange's equation in the form (14.44), we identify the right hand side of equation (14.110) as the generalized force

$$Q_{\sigma} = -\frac{\partial P}{\partial \xi^{\sigma}} + \sum_{c=1}^{C} \Lambda^{c} \frac{\partial \Psi_{c}}{\partial \xi^{\sigma}} \quad \text{for } \sigma = 1, \dots, 3N. \tag{14.111}$$

Now $-\partial P/\partial \xi^{\sigma}$ is the applied conservative force, so we identify the remaining term is the force of constraint

$$Q_{\sigma}^{\text{constraint}} = \sum_{c=1}^{C} \Lambda^{c} \frac{\partial \Psi_{c}}{\partial \xi^{\sigma}} \quad \text{for } \sigma = 1, \dots, 3N. \tag{14.112}$$

14.7.5 Comments

Determining the forces of constraint is most readily understood through some explicit examples, such as those presented in Chapter 15.

14.8 Symmetries and conservation laws

Part of the allure of Hamilton's principle and the corresponding Euler-Lagrange equations concerns the ability to deduce quantities that are conserved by the physically realized dynamical system. These dynamical conservation laws are associated with symmetries, with this connection between symmetries (kinematics) and conservation laws (dynamics) first made by *Noether* (1918) (see *Noether and Tavel* (2018) for an English translation). In this section we focus on space and time symmetries. In particular, we show that when space is homogeneous (i.e., there is no distinction between points in space) then linear momentum is conserved; the conservation of angular momentum arises from space isotropy (i.e., there is no distinction between directions in space); and conservation of mechanical energy arises when time is homogeneous (i.e., there is no distinction between each time instance). The present section focuses on deriving the mathematical expressions of the conservation law. In so doing, we further our understanding of Hamilton's principle and the associated manipulations needed for its practical use. We study implications of these conservation laws in Chapter 13 for particles viewed in a rotating reference frame, such as appropriate for the non-inertial terrestrial observers on a rotating planet.

To connect a symmetry to a conservation law, it is sufficient to focus on the Lagrangian since it encapsulates the mechanics. That is, we examine how the Lagrangian is affected by the chosen symmetry operation. Furthermore, since we are concerned with conservation laws realized by the physical system, the Lagrangian satisfies the Euler-Lagrange equations of motion. Additionally, we here only consider continuous space and time symmetries. Therefore, to determine implications of a symmetry it is sufficient to examine how the Lagrangian varies under an infinitesimal symmetry transformation. This treatment is convenient since infinitesimal transformations are simpler mathematically than finite transformations. In effect, we only need to go to leading order in a Taylor expansion to deduce the conservation laws. Additionally, we remove all forces of constraint, so that it is sufficient to work with Cartesian coordinates.

So in brief, we observe that if the mechanical system is to respect a particular symmetry, then its action must remain invariant under the symmetry transformation; i.e., its variation must vanish. Variation of the action vanishes if variation of the Lagrangian vanishes, with a zero variation of the Lagrangian directly leading to the differential conservation law associated with the symmetry.[15]

14.8.1 Free particle motion

Before we work through the conservation laws arising from space-time symmetries, we here study what is perhaps the simplest physical system, the free particle. Doing so provides a warm-up to the more general systems considered next. Free particle motion occurs when a particle experiences no forces at all (neither applied nor constraining) in an inertial reference frame. We infer from this statement that the particle can only move in an empty space that is homogeneous and isotropic, since otherwises it would necessarily experience a force that altered its path.[16]

For the particle to experience space as homogeneous (the particle does not care about its spatial position) and time as homogeneous (the particle does not care about the origin of time), then the Lagrangian for the particle cannot contain any explicit dependence on space or time,

$$\frac{\partial L}{\partial t} = 0 \quad \text{and} \quad \frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial z} = 0, \tag{14.113}$$

where it is convenient to use Cartesian coordinates since there are no constraints. For mechanics to be the same for all space directions (space posseses isotropy), then the Lagrangian must be independent of the velocity direction. We are led to a Lagrangian that is a function just of the velocity magnitude, or more conveniently the velocity squared. That is, the free particle Lagrangian is given by the kinetic energy

$$L = \tfrac{1}{2} m \, (\dot{x}^2 + \dot{y}^2 + \dot{z}^2), \tag{14.114}$$

where we chose Cartesian coordinates since the motion is unconstrained. We already knew this result must hold since the potential energy is zero in the absence of forces. Even so, this discussion serves to illustrate the power of arguments based on symmetry, with such arguments fundamental to how we make practical use of Hamilton's principle.

With the Lagrangian given by just the kinetic energy for unconstrained motion, then the Euler-Lagrange equation leads to constancy of the velocity

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{y}} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{z}} = 0 \Longrightarrow \dot{\boldsymbol{X}} = \text{constant}. \tag{14.115}$$

This result is Newton's first law discussed in Chapter 11, and sometimes referred to as the law

[15] In Section 14.5.6 we found that the action is unaffected the total time derivative of a function that depends only on the generalized coordinates and time. Even so, to derive the conservation laws in this section, we set the Lagrangian variation to zero.

[16] This statement carries important nuances when moving to general relativity, which is outside of our scope. So for this book, we consider space to be Euclidean, so that free particle motion is along a straight line in this space. Furthermore, it is notable that free particle motion appears to be forced motion when viewed from a non-inertial reference frame (Sections 11.4 and 11.5). As such, the symmetries of a non-inertial reference frame are generally distinct from an inertial reference frame. We here focus on inertial reference frames since it is here that free particle motion in empty space experiences no force, and velocity remains constant.

of inertia.

14.8.2 Space homogeneity and linear momentum conservation

We here examine the implications of an N particle system moving through a space that is homogeneous in an inertial reference frame. Moving beyond the free particle case of Section 14.8.1, we allow for conservative forces of interaction between the particles as measured by the potential energy function, $P(\boldsymbol{X})$. To respect spatial homogeneity requires mechanical properties of the system to be unchanged if we shift the coordinate position for each particle by an arbitrary amount. For ease in deriving the associated conservation law we examine how the Lagrangian changes when considering an infinitesimal shift in all of the particle positions

$$\boldsymbol{X}_{(i)} \rightarrow \boldsymbol{X}_{(i)} + \boldsymbol{\epsilon} \Longrightarrow \delta \boldsymbol{X}_{(i)} = \boldsymbol{\epsilon}, \tag{14.116}$$

where $\boldsymbol{\epsilon}$ is a constant displacement vector that is the same for all particles. A constant shift in particle positions does nothing to the particle velocities. Hence, upon shifting all particle positions by a constant, the Lagrangian becomes, to first order in $\boldsymbol{\epsilon}$,

$$L(\boldsymbol{X}', \dot{\boldsymbol{X}}', t) = L(\boldsymbol{X} + \boldsymbol{\epsilon}, \dot{\boldsymbol{X}}, t) \approx L(\boldsymbol{X}, \dot{\boldsymbol{X}}, t) + \epsilon^a \sum_{i=1}^{N} \frac{\partial L}{\partial x^a_{(i)}}, \tag{14.117}$$

so that variation of the Lagrangian is

$$\delta L = \epsilon^a \sum_{i=1}^{N} \frac{\partial L}{\partial x^a_{(i)}}. \tag{14.118}$$

A zero variation of the Lagrangian ensures that the physical system is unaffected by shifts in the particle positions. Since the perturbation vector, $\boldsymbol{\epsilon}$, is arbitrary, the Lagrangian has zero variation only if

$$\delta L = 0 \Longrightarrow \sum_{i=1}^{N} \frac{\partial L}{\partial x^a_{(i)}} = 0 \quad \text{for } a = 1, 2, 3. \tag{14.119}$$

Using this result in the Euler-Lagrange equation (14.77) leads to a conservation law for each of the three spatial directions

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{i=1}^{N} \frac{\partial L}{\partial \dot{x}_{(i)}} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_{i=1}^{N} \frac{\partial L}{\partial \dot{y}_{(i)}} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_{i=1}^{N} \frac{\partial L}{\partial \dot{z}_{(i)}} = 0. \tag{14.120}$$

Evidently, when space is homogeneous and an N particle system respects this symmetry, then the linear momentum of the system is a constant of the motion

$$\sum_{i=1}^{N} \frac{\partial L}{\partial \dot{x}^a_{(i)}} = m \sum_{i=1}^{N} \dot{x}^a_{(i)} = \text{constant}. \tag{14.121}$$

This result is the law of inertia for the N particle system. Note that if there is spatial symmetry in only a select number of directions, then only the corresponding momenta are conserved.

Another implication for the homogeneity condition (14.119) is that the sum of the forces acting on the N particle system vanishes

$$\sum_{i=1}^{N} \frac{\partial L}{\partial x^a_{(i)}} = -\sum_{i=1}^{N} \frac{\partial P}{\partial x^a_{(i)}} = \sum_{i=1}^{N} F^a_{(i)} = 0. \tag{14.122}$$

For the case of $N = 1$ we recover the free particle discussed in Section 14.8.1, whereby the particle experiences no forces if it moves through homogeneous space. However, an N particle system respects spatial homogeneity if the sum of all the inter-particle forces vanishes. Equation (14.122) is a statement of Newton's third law as studied in Section 11.6.2, where we also noted that a central force between particles provides a prominent example of a force that respects the third law.

14.8.3 Space isotropy and angular momentum conservation

We now consider the conservation law associated with the isotropy of space. Spatial isotropy means that the mechanical system is invariant under the same rotation of both the positions and velocities for the N particles. Let $\delta\boldsymbol{\phi}$ be an infinitesimal rotation vector and consider the following variation of the Cartesian representations of the particle positions and velocities

$$\delta \boldsymbol{X}_{(i)} = \delta\boldsymbol{\phi} \times \boldsymbol{X}_{(i)} \quad \text{and} \quad \delta \dot{\boldsymbol{X}}_{(i)} = \delta\boldsymbol{\phi} \times \dot{\boldsymbol{X}}_{(i)}, \tag{14.123}$$

which made use of Section 11.3 for how infinitesimal rigid rotations affect changes to vectors. The corresponding variation of the Lagrangian is given by

$$\delta L = \sum_{i=1}^{N} \left[\frac{\partial L}{\partial x^a_{(i)}} \, \delta x^a_{(i)} + \frac{\partial L}{\partial \dot{x}^a_{(i)}} \, \delta \dot{x}^a_{(i)} \right] = \sum_{i=1}^{N} \frac{\mathrm{d}}{\mathrm{d}t} \left[\frac{\partial L}{\partial \dot{x}^a_{(i)}} \, \delta x^a_{(i)} \right], \tag{14.124}$$

where we used the Euler-Lagrange equation (14.77) to reach the second equality. With the kinetic energy in terms of Cartesian coordinates given by equation (14.37), we can introduce the Cartesian component of linear momentum

$$P^a_{(i)} = \frac{\partial L}{\partial \dot{x}^a_{(i)}}, \tag{14.125}$$

in which case the Lagrangian has zero variation under infinitesimal rotations if

$$\sum_{i=1}^{N} \frac{\mathrm{d}}{\mathrm{d}t} \left[P^a_{(i)} \, \delta x^a_{(i)} \right] = \delta\boldsymbol{\phi} \cdot \sum_{i=1}^{N} \frac{\mathrm{d}}{\mathrm{d}t} (\boldsymbol{X}_{(i)} \times \boldsymbol{P}_{(i)}) = 0. \tag{14.126}$$

With $\delta\boldsymbol{\phi}$ arbitrary, we are left with conservation of angular momentum for the N particle system

$$\sum_{i=1}^{N} \frac{\mathrm{d}}{\mathrm{d}t} (\boldsymbol{X}_{(i)} \times \boldsymbol{P}_{(i)}) = 0. \tag{14.127}$$

For a space that possesses partial isotropy, that component of angular momentum associated with the isotropy is conserved. This case is particularly relevant for planetary physics, whereby planetary rotation reduces spatial isotropy to just that of the rotational axis and conservation of the axial angular momentum (see Section 13.5.1).

14.8.4 Time homogeneity and Hamiltonian evolution

We now consider symmetry under time translations, and thus allow for the use of generalized coordinates (and thus the possibility of forces of constraint) rather than Cartesian coordinates.

Time homogeneity

Consider how a mechanical system behaves under a shift in time, $t \to t + \epsilon$, while keeping the generalized positions and generalized velocities unchanged. In this case the Lagrangian becomes

$$L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t + \epsilon) \approx L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t) + \epsilon\, \partial_t L \Longrightarrow \delta L = \epsilon\, \partial_t L. \tag{14.128}$$

Hence, for a mechanical system to manifest time homogeneity the Lagrangian must have no explicit time dependence,

$$\left[\frac{\partial L}{\partial t}\right]_{\xi^\sigma, \dot{\xi}^\sigma} = 0 \Longleftrightarrow \text{time symmetry.} \tag{14.129}$$

Derivation of the Hamiltonian evolution

The total time derivative of the Lagrangian (following equation (14.52)) takes on the form

$$\frac{\mathrm{d}L}{\mathrm{d}t} - \left[\frac{\partial L}{\partial t}\right]_{\xi^\sigma, \dot{\xi}^\sigma} = \sum_{\sigma=1}^{D} \left[\dot{\xi}^\sigma \frac{\partial L}{\partial \xi^\sigma} + \ddot{\xi}^\sigma \frac{\partial L}{\partial \dot{\xi}^\sigma}\right] \tag{14.130a}$$

$$= \sum_{\sigma=1}^{D} \left[\dot{\xi}^\sigma \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{\xi}^\sigma} + \ddot{\xi}^\sigma \frac{\partial L}{\partial \dot{\xi}^\sigma}\right] \tag{14.130b}$$

$$= \sum_{\sigma=1}^{D} \frac{\mathrm{d}}{\mathrm{d}t} \left[\dot{\xi}^\sigma \frac{\partial L}{\partial \dot{\xi}^\sigma}\right], \tag{14.130c}$$

where the second equality used the Euler-Lagrange equation (14.77). We are thus led to

$$\frac{\mathrm{d}H}{\mathrm{d}t} = -\left[\frac{\partial L}{\partial t}\right]_{\xi^\sigma, \dot{\xi}^\sigma} \quad \text{with} \quad H = \left[\sum_{\sigma=1}^{D} \dot{\xi}^\sigma \frac{\partial L}{\partial \dot{\xi}^\sigma} - L\right], \tag{14.131}$$

where H is known as the Hamiltonian. Evidently, if there is no explicit time dependence in the Lagrangian, then the Hamiltonian is a constant of the motion

$$\left[\frac{\partial L}{\partial t}\right]_{\xi^\sigma, \dot{\xi}^\sigma} = 0 \Longrightarrow \frac{\mathrm{d}H}{\mathrm{d}t} = 0. \tag{14.132}$$

Connecting the Hamiltonian to the mechanical energy

From Section 14.4.7 we wrote the kinetic energy in terms of generalized coordinates. In the special case of a time independent transformation between Cartesian and generalized coordinates, the kinetic energy is given by equation (14.57)

$$K = \frac{m}{2} \sum_{\sigma=1}^{D} \sum_{\mu=1}^{D} G_{\sigma\mu}(\boldsymbol{\xi})\, \dot{\xi}^\sigma\, \dot{\xi}^\mu, \tag{14.133}$$

with $G_{\sigma\mu}(\boldsymbol{\xi})$ components to the metric tensor represented using generalized coordinates. We thus find

$$\frac{\partial L}{\partial \dot{\xi}^{\nu}} = \frac{\partial K}{\partial \dot{\xi}^{\nu}} = \frac{m}{2} \sum_{\sigma=1}^{D} \sum_{\mu=1}^{D} G_{\sigma\mu} \, (\delta^{\sigma}{}_{\nu} \, \dot{\xi}^{\mu} + \delta^{\mu}{}_{\nu} \, \dot{\xi}^{\sigma}) = m \sum_{\sigma=1}^{D} G_{\sigma\nu} \, \dot{\xi}^{\sigma}, \tag{14.134}$$

where we used $G_{\sigma\mu} = G_{\mu\sigma}$ for the final equality. In this case, the Hamiltonian equals to the total mechanical energy

$$H = \sum_{\nu=1}^{D} \dot{\xi}^{\nu} \, \frac{\partial L}{\partial \dot{\xi}^{\nu}} - L = m \sum_{\nu=1}^{D} \sum_{\sigma=1}^{D} G_{\sigma\nu} \, \dot{\xi}^{\nu} \, \dot{\xi}^{\sigma} - \frac{m}{2} \sum_{\nu=1}^{D} \sum_{\sigma=1}^{D} G_{\sigma\nu} \, \dot{\xi}^{\nu} \, \dot{\xi}^{\sigma} + P = K + P, \tag{14.135}$$

which is the familiar form of mechanical energy found in our study of energetics in Newtonian mechanics from Section 11.2.5.

Further distinguishing the Hamiltonian and the mechanical energy

We emphasize that the Hamiltonian is a constant of the motion if the Lagrangian has no explicit time dependence; e.g., equation (14.132). However, the Hamiltonian equals to the mechanical energy only if there is a time independent coordinate transformation between Cartesian and generalized coordinates. Typically $\partial_t L = 0$ means that $\partial_t \boldsymbol{X}_{(i)} = 0$. However, such is not always that case (e.g., see page 82 of *Fetter and Walecka* (2003)). So when $\partial_t L = 0$ but $\partial_t \boldsymbol{X}_{(i)} \neq 0$, the Hamiltonian remains a constant of the motion, but the mechanical energy is time dependent. Hence, we must qualify our earlier statement that time homogeneity leads to a constant mechanical energy. The more precise statement is that the Hamiltonian is a constant of the motion when time does not explicitly appear in the Lagrangian; however, the Hamiltonian is equal to the mechanical energy only if there is a time independent relation between Cartesian coordinates and generalized coordinates.

14.8.5 A critical operational pointer

Symmetries and corresponding conserved quantities are a crucial element in the analysis of motion. However, it is crucial not to make use of conserved quantities until the full suite of Euler-Lagrange equations have been derived. The reason is that the Lagrangian must be written as a function of the generalized coordinates, generalized velocities, and time, with these coordinates independent. If one wishes, for some reason, to rewrite the Lagrangian in terms of a conserved quantity, then doing so corrupts the functional dependencies of the Lagrangian, which is a recipe for getting the wrong partial derivatives in the Euler-Lagrange equation. We work through an example with the two-body problem of Newtonian gravity in Exercise 15.4. In that problem, we see that introducing the conserved angular momentum into the Lagrangian produces a fatal sign error to the radial equation of motion.

14.8.6 Further study

Conservation laws and symmetries in classical mechanics are lucidly discussed in Chapters 1 and 2 of *Landau and Lifshitz* (1976). See also the discussions in *Goldstein* (1980), *Fetter and Walecka* (2003), *José and Saletan* (1998), *Taylor* (2005), and *Tong* (2025). For videos discussing these topics, see this online lecture from the Space Time series and this online lecture from Physics with Elliot. This essay about Emmy Noether provides insights into this mathematician whose work, conducted under some very unfortunate circumstances, forever connected symmetry and conservation laws, with this connection providing the basis for nearly all modern theories of physics.

14.9 Hamiltonian mechanics

In this section we develop the rudiments of Hamiltonian mechanics. Mathematically, we relate Hamiltonian mechanics to Lagrangian mechanics through a Legendre transformation. Hamiltonian mechanics, and its corresponding phase space, offers a powerful framework to expose foundational properties of motion of classical systems.

14.9.1 Generalized (also canonical) momenta

We define the generalized momenta, also known as the canonical momenta, according to[17]

$$\mathcal{P}_\sigma \equiv \frac{\partial L}{\partial \dot{\xi}^\sigma} = \mathcal{P}_\sigma(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t) \quad \text{for} \quad \sigma = 1, \dots, D, \tag{14.136}$$

where the functional dependence is inherited from the Lagrangian. Also note the care in placing the σ index downstairs in the covariant position. In terms of the generalized momenta, the Euler-Lagrange equation (14.77) takes the form

$$\frac{\mathrm{d}\mathcal{P}_\sigma}{\mathrm{d}t} = \dot{\mathcal{P}}_\sigma = \frac{\partial L}{\partial \xi^\sigma} \quad \text{for} \quad \sigma = 1, \dots, D. \tag{14.137}$$

Evidently, if the Lagrangian has no explicit dependence on a particular generalized coordinate, then the corresponding generalized momenta is a constant of the motion. Such generalized coordinates are termed cyclic coordinates, and we have more to say about cyclic coordinates in Section 14.9.5.

14.9.2 Legendre transformation

In terms of the generalized momenta, the Hamiltonian function (14.132) is written

$$H = -L + \sum_{\sigma=1}^{D} \dot{\xi}^\sigma \, \mathcal{P}_\sigma. \tag{14.138}$$

This definition of the Hamiltonian represents a Legendre transformation from the Lagrangian to the Hamiltonian, whereby we swap around some of the functional dependence. Namely, we already know that the Lagrangian is a function of generalized coordinates, generalized velocities, and time. In this subsection we show that the Hamiltonian is a function of the generalized coordinates, generalized momenta, and time.

[17] Generalized momenta is synonymous with canonical momenta; they are operationally identical. Although a matter of style and not substance, one often finds generalized momenta referred to more when working in Lagrangian mechanics, whereas canonical momenta is used for Hamiltonian mechanics. Even though we are here focused on Hamiltonian mechanics, we typically use the term generalized momenta in order to emphasize its direct correspondence with generalized coordinates and generalized velocities.

Exact differential of the Hamiltonian

The Lagrangian dependence (14.51)

$$L = L(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, t), \tag{14.139}$$

leads to the exact differential of the Hamiltonian (14.138)

$$\mathrm{d}H = -\frac{\partial L}{\partial t}\,\mathrm{d}t + \sum_{\sigma=1}^{D}\left[-\frac{\partial L}{\partial \xi^\sigma}\,\mathrm{d}\xi^\sigma - \frac{\partial L}{\partial \dot{\xi}^\sigma}\,\mathrm{d}\dot{\xi}^\sigma + \mathcal{P}_\sigma\,\mathrm{d}\dot{\xi}^\sigma + \dot{\xi}^\sigma\,\mathrm{d}\mathcal{P}_\sigma\right]. \tag{14.140}$$

With the generalized momenta given by equation (14.136), the exact differential reduces to

$$\mathrm{d}H = -\frac{\partial L}{\partial t}\,\mathrm{d}t + \sum_{\sigma=1}^{D}\left[-\frac{\partial L}{\partial \xi^\sigma}\,\mathrm{d}\xi^\sigma + \dot{\xi}^\sigma\,\mathrm{d}\mathcal{P}_\sigma\right]. \tag{14.141}$$

This relation means that the Hamiltonian has the following dependence

$$H = H(\xi^1, \ldots, \xi^D, \mathcal{P}_1, \ldots, \mathcal{P}_D, t) = H(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}}, t). \tag{14.142}$$

Namely, the Hamiltonian is a function of the generalized coordinates, generalized momenta, and a (possibly) explicit function of time. Furthermore, as for the Lagrangian function, the Hamiltonian is an implicit function of time as realized through time dependence of the generalized coordinates and generalized momenta.

Relations between partial derivatives

The functional dependence (14.142) means that the exact differential of the Hamiltonian is given by

$$\mathrm{d}H = \left[\frac{\partial H}{\partial t}\right]_{\xi^\sigma, \mathcal{P}_\sigma}\mathrm{d}t + \sum_{\sigma=1}^{D}\left(\left[\frac{\partial H}{\partial \xi^\sigma}\right]_{\mathcal{P}_\sigma, t}\mathrm{d}\xi^\sigma + \left[\frac{\partial H}{\partial \mathcal{P}_\sigma}\right]_{\xi^\sigma, t}\mathrm{d}\mathcal{P}_\sigma\right), \tag{14.143}$$

which is equal also to equation (14.141), here rewritten as

$$\mathrm{d}H = -\left[\frac{\partial L}{\partial t}\right]_{\xi^\sigma, \dot{\xi}^\sigma}\mathrm{d}t + \sum_{\sigma=1}^{D}\left(-\left[\frac{\partial L}{\partial \xi^\sigma}\right]_{\dot{\xi}^\sigma, t}\mathrm{d}\xi^\sigma + \dot{\xi}^\sigma\,\mathrm{d}\mathcal{P}_\sigma\right). \tag{14.144}$$

In both equations (14.143) and (14.144) we exposed subscripts on the partial derivatives to be clear on what coordinates are held fixed when computing the derivatives. Equating terms between equations (14.143) and (14.144) leads to the identities

$$\left[\frac{\partial H}{\partial t}\right]_{\xi^\sigma, \mathcal{P}_\sigma} = -\left[\frac{\partial L}{\partial t}\right]_{\xi^\sigma, \dot{\xi}^\sigma} \tag{14.145a}$$

$$\left[\frac{\partial H}{\partial \xi^\sigma}\right]_{\mathcal{P}_\sigma, t} = -\left[\frac{\partial L}{\partial \xi^\sigma}\right]_{\dot{\xi}^\sigma, t} \tag{14.145b}$$

$$\left[\frac{\partial H}{\partial \mathcal{P}_\sigma}\right]_{\xi^\sigma, t} = \frac{\mathrm{d}\xi^\sigma}{\mathrm{d}t}. \tag{14.145c}$$

The first and second identities relate derivatives in phase space (left hand side) to derivatives in configuration space (right hand side). The third identity is one of Hamilton's equation of motion, as discussed next.

14.9.3 Hamilton's equations of motion

The manipulations in Section 14.9.2 are mathematical identities that arise from the coordinate dependencies of the Lagrangian and Hamiltonian, and the Legendre transformation (14.138) that swaps around these dependencies. We now return to physics by assuming the Lagrangian satisfies the Euler-Lagrange equations as written in the form (14.137)

$$\dot{\mathcal{P}}_\sigma = \left[\frac{\partial L}{\partial \xi^\sigma}\right]_{\dot{\xi}^\sigma, t}, \tag{14.146}$$

which brings the exact differential (14.144) to

$$\mathrm{d}H = \left[\frac{\partial H}{\partial t}\right]_{\xi^\sigma, \mathcal{P}_\sigma} \mathrm{d}t + \sum_{\sigma=1}^{D} \left(-\dot{\mathcal{P}}_\sigma \, \mathrm{d}\xi^\sigma + \dot{\xi}^\sigma \, \mathrm{d}\mathcal{P}_\sigma\right). \tag{14.147}$$

Comparing to equation (14.143) leads to Hamilton's equations of motion

$$\dot{\mathcal{P}}_\sigma = \frac{\mathrm{d}\mathcal{P}_\sigma}{\mathrm{d}t} = -\left[\frac{\partial H}{\partial \xi^\sigma}\right]_{\mathcal{P}_\sigma, t} \tag{14.148a}$$

$$\dot{\xi}^\sigma = \frac{\mathrm{d}\xi^\sigma}{\mathrm{d}t} = \left[\frac{\partial H}{\partial \mathcal{P}_\sigma}\right]_{\xi^\sigma, t}. \tag{14.148b}$$

These are $2D$ first order differential equations that relate partial derivatives of the Hamiltonian to the total time derivative of the generalized coordinates and generalized momenta. These equations are to be compared to the Euler-Lagrange equations (14.77), which are D second order differential equations for the generalized coordinates.

As a result of Hamilton's equations (14.148a) and (14.148b), we can readily compute the time derivative of the Hamiltonian

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \left[\frac{\partial H}{\partial t}\right]_{\xi^\sigma, \mathcal{P}_\sigma} + \sum_{\sigma=1}^{D} \left(\left[\frac{\partial H}{\partial \xi^\sigma}\right]_{\mathcal{P}_\sigma, t} \dot{\xi}^\sigma + \left[\frac{\partial H}{\partial \mathcal{P}_\sigma}\right]_{\xi^\sigma, t} \dot{\mathcal{P}}_\sigma\right) \tag{14.149a}$$

$$= \left[\frac{\partial H}{\partial t}\right]_{\xi^\sigma, \mathcal{P}_\sigma} + \sum_{\sigma=1}^{D} \left(-\dot{\mathcal{P}}_\sigma \, \dot{\xi}^\sigma + \dot{\mathcal{P}}_\sigma \, \dot{\xi}^\sigma\right) \tag{14.149b}$$

$$= \left[\frac{\partial H}{\partial t}\right]_{\xi^\sigma, \mathcal{P}_\sigma} \tag{14.149c}$$

$$= -\left[\frac{\partial L}{\partial t}\right]_{\xi^\sigma, \dot{\xi}^\sigma}, \tag{14.149d}$$

where the final equality follows from the mathematical identity (14.145a). Evidently, the Hamiltonian is a constant of the motion if it has no explicit time dependence. This result corresponds to Noether's theorem as discussed in Section 14.8.4.

14.9.4 Some key operational details

When deriving Hamilton's equations of motion for a particular physical problem, we typically start by writing down the Lagrangian and then using the Legendre transformation (14.138) to derive the Hamiltonian. Before performing the partial derivatives in Hamilton's equations (14.149c)-(14.148b), it is essential to express the Hamiltonian as a function of the generalized coordinates and generalized momenta. Doing so requires removing all appearances of the generalized velocity in favor of the generalized momenta.

In much of this section, we exposed subscripts on the partial derivatives to identify what variables are held fixed when taking the derivatives. Such care is important for developing a proper understanding of the equations and their manipulation. We also found such care important for developing Lagrangian mechanics earlier in this chapter. For both formulations, some exposure to the methodology readily allows one to dispense with the extra clutter of the subscripts. Even so, it can be quite valuable to expose the subscripts when debugging any particular formulation.

14.9.5 Cyclic versus ignorable coordinates

We commented in Section 14.9.1 that if the Lagrangian is independent of a coordinate, referred to as a cyclic coordinate, then its corresponding generalized momenta is a constant of the motion. Even so, the corresponding generalized velocity is not necessarily zero and so it cannot be ignored. However, in the Hamiltonian formulation we can ignore a cyclic coordinate since the generalized momenta is now a time-independent parameter of the Hamiltonian. As such, there is no corresponding equation of motion. We thus find the Hamiltonian dynamical system has two less degrees of freedom, one for the cyclic coordinate and one for the corresponding constant generalized momenta, thus motivating the name ignorable coordinate when a cyclic coordinate is encountered in the Hamiltonian formulation.

14.9.6 Modified Hamilton's principle

Hamilton's variational principle in the form of equation (14.77) leads to the Euler-Lagrange equations. We can derive a related variational principle that leads to Hamilton's equations. To do so, insert the definition of the Hamiltonian from equation (14.138) into the action to render

$$\mathcal{S} = \int_{t_\mathsf{A}}^{t_\mathsf{B}} L(\xi^\sigma, \dot{\xi}^\sigma, t)\,\mathrm{d}t = \int_{t_\mathsf{A}}^{t_\mathsf{B}} \left[-H(\xi^\sigma, \mathcal{P}_\sigma, t) + \sum_{\sigma=1}^{D} \dot{\xi}^\sigma\,\mathcal{P}_\sigma \right] \mathrm{d}t. \tag{14.150}$$

We now consider a variation of the generalized coordinates and generalized momenta, in which the action is varied according to

$$\delta\mathcal{S} = \sum_{\sigma=1}^{D} \int_{t_\mathsf{A}}^{t_\mathsf{B}} \left[-\frac{\partial H}{\partial \xi^\sigma}\,\delta\xi^\sigma - \frac{\partial H}{\partial \mathcal{P}_\sigma}\,\delta\mathcal{P}_\sigma + \delta\dot{\xi}^\sigma\,\mathcal{P}_\sigma + \dot{\xi}^\sigma\,\delta\mathcal{P}_\sigma \right] \mathrm{d}t. \tag{14.151}$$

Variations of the generalized momenta and generalized coordinates are independent. However, variation of the generalized coordinates and generalized velocities are connected just like when deriving the Euler-Lagrange equations in Section 14.5.3. Hence, an integration by parts leads to

$$\int_{t_\mathsf{A}}^{t_\mathsf{B}} \delta\dot{\xi}^\sigma\,\mathcal{P}_\sigma\,\mathrm{d}t = \int_{t_\mathsf{A}}^{t_\mathsf{B}} \frac{\mathrm{d}(\delta\xi^\sigma)}{\mathrm{d}t}\,\mathcal{P}_\sigma\,\mathrm{d}t = \int_{t_\mathsf{A}}^{t_\mathsf{B}} \left[\frac{\mathrm{d}(\delta\xi^\sigma\,\mathcal{P}_\sigma)}{\mathrm{d}t} - \delta\xi^\sigma\,\dot{\mathcal{P}}_\sigma \right] \mathrm{d}t. \tag{14.152}$$

The total time derivative in the final expression vanishes since the coordinate variation vanishes at the initial and final times. We are thus left with the action variation

$$\delta \mathcal{S} = \sum_{\sigma=1}^{D} \int_{t_\mathsf{A}}^{t_\mathsf{B}} \left[-\left(\dot{\mathcal{P}}_\sigma + \frac{\partial H}{\partial \xi^\sigma} \right) \delta \xi^\sigma + \left(\dot{\xi}^\sigma - \frac{\partial H}{\partial \mathcal{P}_\sigma} \right) \delta \mathcal{P}_\sigma \right] \mathrm{d}t. \tag{14.153}$$

Since $\delta\xi^\sigma$ and $\delta\mathcal{P}_\sigma$ are independent, we realize a stationary action if and only if Hamilton's equations are satisfied

$$\dot{\xi}^\sigma = \frac{\partial H}{\partial \mathcal{P}_\sigma} \quad \text{and} \quad \dot{\mathcal{P}}_\sigma = -\frac{\partial H}{\partial \xi^\sigma}. \tag{14.154}$$

Evidently, the modified Hamilton's principle results in Hamilton's equations, and it does so without making any assumptions about the values for $\delta\mathcal{P}_\sigma$ at the initial and final times.

14.9.7 Phase space flow and Liouville's theorem

In the Hamiltonian formulation of mechanics, we specify the physical state of a system by giving its point in phase space, as determined by the generalized positions, ξ^σ, and generalized momenta, $\mathcal{P}_\sigma$. Knowledge of the precise phase space point then, through Hamilton's equations, determines the precise evolution of the system. Such determinism also holds for Newtonian mechanics and Lagrangian mechanics, using their respective dynamical equations.

The determinism of classical physics was dethroned by two major achievements of 20th century physics: quantum mechanics and determinstic chaos. Quantum mechanics showed that we cannot specify the position and momentum simultaneously, no matter how precise the measurement apparatus. Hence, there is an intrinsic uncertainty in any physical system. In determinstic chaos, we confront the practical limitations of any measurement. Namely, as described by *Lorenz* (1963), arbitrarily small, but nonzero, uncertainty, can grow to a finite level of uncertainty in certain nonlinear dynamical systems. Determinstic chaos provides a conceptual foundation for understanding limits to the predictability of turbulent fluid motions, including geophysical fluids.

Motivated by the above notions, we introduce a level of humility in describing the motion of a physical system. In particular, for classical systems we allow for uncertainty by introducing a phase space particle density, and then study how that density evolves. As part of this analysis we establish Liouville's theorem, which is of particular importance in the study of Hamiltonian dynamical systems viewed as the flow of matter within phase space. The formulation has direct analogs to the derivation of mass continuity in fluid kinematics, as studied in VOLUME 2.

Phase space volume and phase space number density

Consider a region, $\mathcal{R}$, of classical phase space. Emulating the kinematics of fluid flow studied in VOLUME 2, we examine two kinds of regions. The first region considers $\mathcal{R}$ to be fixed in phase space, with this region corresponding to the Eulerian reference frame of fluid kinematics, with the volume of an infinitesimal version of $\mathcal{R}$ written

$$\mathrm{d}V = \mathrm{d}\xi^1 \, \mathrm{d}\mathcal{P}_1 \, \mathrm{d}\xi^2 \, \mathrm{d}\mathcal{P}_2 \ldots \mathrm{d}\xi^D \, \mathrm{d}\mathcal{P}_D = \Pi_{\sigma=1}^{D} \, \mathrm{d}\xi^\sigma \, \mathrm{d}\mathcal{P}_\sigma. \tag{14.155}$$

The second region considers every point within $\mathcal{R}$ to evolve according to the phase space flow generated by the Hamiltonian, with this region corresponding to the Lagrangian reference frame of fluid kinematics and with an infinitesimal volume written

$$\delta V = \delta\xi^1\,\delta\mathcal{P}_1\,\delta\xi^2\,\delta\mathcal{P}_2 \ldots \delta\xi^D\,\delta\mathcal{P}_D = \Pi_{\sigma=1}^{D}\,\delta\xi^\sigma\,\delta\mathcal{P}_\sigma. \tag{14.156}$$

Notably, we make use of the δ symbol for infinitesimal regions that move with the phase space flow and the exact differential, d, for regions fixed in phase space.[18]

To count the number of particles contained within a region of phase space at any instance, we introduce the particle density function, $\rho(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}}, t)$. By definition, the number of particles, $N_{\mathcal{R}}$, within the region, $\mathcal{R}$, is given by

$$N_{\mathcal{R}} = \rho(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}}, t)\,\mathrm{d}V. \tag{14.157}$$

If we know the precise positions for each particle at some time, say $t = 0$, then the density is a Dirac delta on phase space, thus reflecting the exact information about the initial position of each particle. For more realistic cases, the initial phase space position is uncertain, in which case ρ is no longer a Dirac delta.

Budget for particles within a fixed region of phase space

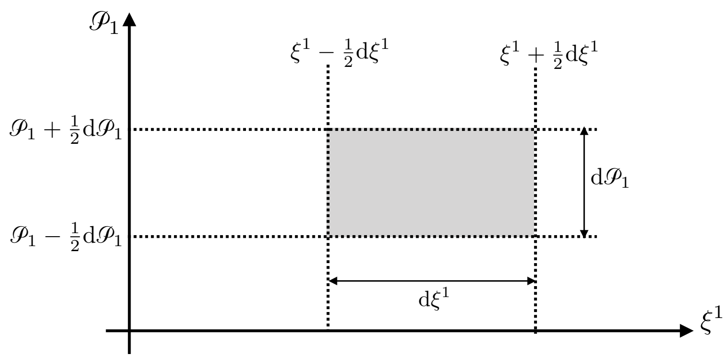

FIGURE 14.2: Illustrating an infinitesimal and region of phase space, exposing here just the generalized coordinate, ξ^1, and corresponding generalized momentum, $\mathcal{P}_1$. The remaining $2(D-1)$ coordinate directions are suppressed. Bounds to this region are given by $\xi^1 \pm \frac{1}{2}\mathrm{d}\xi^1$ and $\mathcal{P}_1 \pm \frac{1}{2}\mathrm{d}\mathcal{P}_1$.

We now develop the budget equation for the number of particles flowing through a fixed region of phase space, with Figure 14.2 illustrating a piece of that phase space volume. Namely, by taking the (Eulerian) time derivative of equation (14.157) we have

$$\frac{\partial N_{\mathcal{R}}}{\partial t} = \frac{\partial \rho(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}}, t)}{\partial t}\,\mathrm{d}V, \tag{14.158}$$

where $\partial_t(\mathrm{d}V) = 0$ since we are examining the time evolution of particles within a fixed region of phase space. Our goal is to determine the kinematic evolution equation for $\partial N_{\mathcal{R}}/\partial t$.

This goal is rather straightforward, if we assume that the number of particles in $\mathcal{R}$ evolves only through transport across the boundaries of the region. That is, we assume that particles are neither created nor destroyed, so that the physics remains locally causal. Referring to Figure 14.2 , the number of particles entering the region, $\mathcal{R}$, by crossing the boundary, $\xi^1 - \frac{1}{2}\mathrm{d}\xi^1$, is

[18]The δ in equation (14.156) is not a variation operator, nor is it a Dirac delta. Rather, it is here used to represent an infinitesimal that follows the phase space flow.

given by

$$\text{particles entering across } \xi^1 - \tfrac{1}{2}\mathrm{d}\xi^1 = (\rho\,\dot{\xi^1})_{\xi^1-\mathrm{d}\xi^1/2}\,\mathrm{d}t\,\mathrm{d}\mathcal{P}_1\left[\Pi_{\sigma=2}^{D}\mathrm{d}\xi^\sigma\,\mathrm{d}\mathcal{P}_\sigma\right]. \tag{14.159}$$

Likewise, the particles leaving $\mathcal{R}$, by passing through the right boundary are

$$\text{particles leaving across } \xi^1 + \tfrac{1}{2}\mathrm{d}\xi^1 = (\rho\,\dot{\xi^1})_{\xi^1+\mathrm{d}\xi^1/2}\,\mathrm{d}t\,\mathrm{d}\mathcal{P}_1\left[\Pi_{\sigma=2}^{D}\mathrm{d}\xi^\sigma\,\mathrm{d}\mathcal{P}_\sigma\right]. \tag{14.160}$$

We have similar expressions for motion through the generalized momentum boundaries $\mathrm{d}\mathcal{P}_1 \pm \frac{1}{2}\mathrm{d}\mathcal{P}_1$. Bringing these results together, taking the infinitesimal limit, and dividing by the phase space volume, $\mathrm{d}V$, leads to the continuity equation

$$\frac{\partial\rho}{\partial t} = -\sum_{\sigma=1}^{D}\left[\frac{\partial(\rho\,\dot{\xi}^\sigma)}{\partial\xi^\sigma} + \frac{\partial(\rho\,\dot{\mathcal{P}}_\sigma)}{\partial\mathcal{P}_\sigma}\right]. \tag{14.161}$$

Expanding the product rule and rearrangement leads to

$$\frac{\partial\rho}{\partial t} + \dot{\xi}^\sigma\,\frac{\partial\rho}{\partial\xi^\sigma} + \dot{\mathcal{P}}_\sigma\,\frac{\partial\rho}{\mathcal{P}_\sigma} = -\rho\left[\frac{\partial\dot{\xi}^\sigma}{\partial\xi^\sigma} + \frac{\partial\dot{\mathcal{P}}_\sigma}{\partial\mathcal{P}_\sigma}\right], \tag{14.162}$$

where we made use of the summation convention to reduce clutter. The right hand side is the phase space density times the convergence of the phase space flow. The left hand side is the time derivative following the phase space flow, and it corresponds to the material time derivative (or Lagrangian time derivative) of fluid kinematics. We are motivated to introduce the phase space flow time derivative

$$\frac{\mathrm{D}^{\mathsf{ph}}}{\mathrm{D}t} \equiv \frac{\partial}{\partial t} + \dot{\xi}^\sigma\,\frac{\partial}{\partial\xi^\sigma} + \dot{\mathcal{P}}_\sigma\,\frac{\partial}{\mathcal{P}_\sigma}, \tag{14.163}$$

so that the phase space continuity equation (14.162) takes the form

$$\frac{\mathrm{D}^{\mathsf{ph}}\rho}{\mathrm{D}t} = -\rho\left[\frac{\partial\dot{\xi}^\sigma}{\partial\xi^\sigma} + \frac{\partial\dot{\mathcal{P}}_\sigma}{\partial\mathcal{P}_\sigma}\right]. \tag{14.164}$$

This equation says that when following the phase space flow, the phase space density increases when moving through a region where the phase space flow converges. However, as shown next, it turns out that the phase space flow has exactly zero convergence, so that ρ remains a constant when following the flow.

Liouville's theorem: phase space flow is non-divergent

Derivation of the continuity equation (14.164) follows from kinematics. We now introduce dynamics by making use of Hamilton's equations (14.154) to express $\dot{\xi}^\sigma$ and $\dot{\mathcal{P}}_\sigma$. In particular, we find that the phase space flow has zero divergence as per

$$\frac{\partial\dot{\xi}^\sigma}{\partial\xi^\sigma} + \frac{\partial\dot{\mathcal{P}}_\sigma}{\partial\mathcal{P}_\sigma} = \frac{\partial^2 H}{\partial\mathcal{P}_\sigma\partial\xi^\sigma} - \frac{\partial^2 H}{\partial\xi^\sigma\partial\mathcal{P}_\sigma} = 0, \tag{14.165}$$

which follows from equality of the mixed partial derivatives. We thus find that the phase space density remains fixed when following the phase space flow

$$\frac{\mathrm{D}^{\mathsf{ph}}\rho}{\mathrm{D}t} = 0. \tag{14.166}$$

This result constitutes the first part of Liouville's theorem. Written in its expanded form, and using Hamilton's equations, we find

$$\frac{\partial \rho}{\partial t} = -\dot{\xi}^{\sigma}\,\frac{\partial \rho}{\partial \xi^{\sigma}} - \dot{\mathcal{P}}_{\sigma}\,\frac{\partial \rho}{\partial \mathcal{P}_{\sigma}} \tag{14.167a}$$

$$= -\frac{\partial H}{\partial \mathcal{P}_{\sigma}}\,\frac{\partial \rho}{\partial \xi^{\sigma}} + \frac{\partial H}{\partial \xi^{\sigma}}\,\frac{\partial \rho}{\partial \xi^{\sigma}} \tag{14.167b}$$

$$\equiv -\{\rho, H\}, \tag{14.167c}$$

where the final equality introduced the Poisson bracket

$$\{F, G\} = \frac{\partial F}{\partial \xi^{\sigma}}\,\frac{\partial G}{\partial \mathcal{P}_{\sigma}} - \frac{\partial F}{\partial \mathcal{P}_{\sigma}}\,\frac{\partial G}{\partial \xi^{\sigma}}. \tag{14.168}$$

Liouville's theorem: invariant phase space volume

The second part of Liouville's theorem follows by focusing on a region, $\mathcal{R}$, that follows the phase space flow, thus corresponding to a Lagrangian region in fluid kinematics. By definition, each point within this region follows a trajectory determined by Hamilton's equations. Hence, the region maintains a fixed number of particles, in which case

$$\frac{\mathrm{D}^{\mathsf{ph}} N_{\mathcal{R}}}{\mathrm{D}t} = \frac{\mathrm{D}^{\mathsf{ph}}(\rho\,\delta V)}{\mathrm{D}t} = 0. \tag{14.169}$$

Since the phase space density is constant following the flow, as per equation (14.166), we find that the phase space volume remains constant following the flow

$$\frac{\mathrm{D}^{\mathsf{ph}}(\delta V)}{\mathrm{D}t} = 0. \tag{14.170}$$

Comments

Liouville's theorem is a foundational element of statistical mechanics, with derivations of this theorem found, for example, in Section A.13 of *Reif* (1965), and Section 2.2 from *Pathria and Beale* (2021). Section 46 of *Landau and Lifshitz* (1976) and Section 10.2 of *Tong* (2025) derive Liouville's theorem using methods reminiscent of our study of Lagrangian fluid kinematics found in VOLUME 2.

14.9.8 Hamilton-Jacobi equation for the on-shell action

Thus far in this chapter, we have used the action as an intermediate quantity used to derive the equations of motion (Euler-Lagrange equations or Hamilton's equations) through invoking Hamilton's variational principle. For this purpose, the action is a functional of the trajectory whose values are prescribed at the initial and final times

$$\mathcal{S} = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} L[\boldsymbol{\xi}(\tau), \dot{\boldsymbol{\xi}}(\tau), \tau]\,\mathrm{d}\tau, \tag{14.171}$$

where L is the Lagrangian function. As a shorthand terminology, we refer to this as the off-shell action, meaning that it is evaluated using fields or trajectories that are arbitrary trial configurations, and so do not necessarily satisfy the Euler-Lagrange or Hamilton's equations of motion.[19] We make use of off-shell fields when varying the action.

Defining the on-shell action

In the present section we take a different perspective by studying properties of the on-shell version of the action, in which it is evaluated by the physically realized trajectory that satisfies the equations of motion.[20] To emphasize the distinction, we write the physically realized or on-shell action as

$$\mathcal{W} = \mathcal{S}^{\mathsf{onshell}}. \tag{14.172}$$

The on-shell action is a function of the trajectory at the prescribed initial and final times, as well as the time difference

$$\mathcal{W} = \mathcal{W}[\boldsymbol{\xi}(t_{\mathsf{A}}), \boldsymbol{\xi}(t_{\mathsf{B}}), t_{\mathsf{B}} - t_{\mathsf{A}}]. \tag{14.173}$$

Derivative of the on-shell action with respect to a generalized coordinate

Let use return to the action variation from Section 14.5.3. Upon varying the trajectories we find that the action variation is given by

$$\delta\mathcal{S} = \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} \delta\boldsymbol{\xi} \cdot \left[\frac{\partial L}{\partial \boldsymbol{\xi}} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\boldsymbol{\xi}}}\right)\right] \mathrm{d}t + [\delta\boldsymbol{\xi} \cdot \boldsymbol{\mathcal{P}}]_{t_{\mathsf{A}}}^{t_{\mathsf{B}}}, \tag{14.174}$$

in which we introduced the generalized momentum to the boundary term

$$\boldsymbol{\mathcal{P}} = \frac{\partial L}{\partial \dot{\boldsymbol{\xi}}}. \tag{14.175}$$

In deriving the Euler-Lagrange equations, we prescribe the initial and final positions for the trajectories at the boundary times, t_{A} and t_{B}. Here, we take the alternative perspective by assuming the trajectories satisfy the Euler-Lagrange equations (on-shell trajectories), yet now allow the initial and final times and positions to be variable. In this manner, variation of the on-shell action is given by the boundary terms

$$\delta\mathcal{W} = [\delta\boldsymbol{\xi} \cdot \boldsymbol{\mathcal{P}}]_{t_{\mathsf{A}}}^{t_{\mathsf{B}}}. \tag{14.176}$$

This identity means, for example, that if we take the partial derivative of the on-shell action, with respect to the generalized coordinate at the initial time or final time, then we get the value of the generalized momentum at the initial time or final time

$$\frac{\partial\mathcal{W}}{\partial\xi^{\sigma}(t_{\mathsf{A}})} = \mathcal{P}_{\sigma}(t_{\mathsf{A}}) \quad \text{and} \quad \frac{\partial\mathcal{W}}{\partial\xi^{\sigma}(t_{\mathsf{B}})} = \mathcal{P}_{\sigma}(t_{\mathsf{B}}). \tag{14.177}$$

[19] Off-shell refers to the mass-energy condition arising in quantum field theory. For our purposes, it is just a name.

[20] Elements of this section follow from Section 43 of *Landau and Lifshitz* (1976) and Section 10.7 of *Tong* (2025).

Exact differential of the on-shell action

To build on the previous results, return to the expression for the on-shell action, and let the upper time limit be variable yet the lower limit be fixed

$$\mathcal{W} = \int_{t_{\mathsf{A}}}^{t} L[\boldsymbol{\xi}(\tau), \dot{\boldsymbol{\xi}}(\tau), \tau] \, \mathrm{d}\tau. \tag{14.178}$$

Here, we assume the generalized coordinate at the initial time is fixed, yet the final time is variable. It follows that the total time derivative of the on-shell action is

$$\frac{\mathrm{d}\mathcal{W}}{\mathrm{d}t} = L[\boldsymbol{\xi}(t), \dot{\boldsymbol{\xi}}(t), t] = -H + \boldsymbol{\mathcal{P}} \cdot \dot{\boldsymbol{\xi}}, \tag{14.179}$$

where the second equality made use of the Legendre transformation (14.138). It follows that the exact differential of the on-shell action is

$$\mathrm{d}\mathcal{W} = -H \, \mathrm{d}t + \boldsymbol{\mathcal{P}} \cdot \mathrm{d}\boldsymbol{\xi}. \tag{14.180}$$

It follows that the on-shell action is a function of time and the generalized coordinates,

$$\mathcal{W} = \mathcal{W}(\boldsymbol{\xi}, t), \tag{14.181}$$

which means it is a function on configuration space. Correspondingly,

$$\frac{\mathrm{d}\mathcal{W}}{\mathrm{d}t} = \frac{\partial \mathcal{W}}{\partial t} + \frac{\partial \mathcal{W}}{\partial \xi^{\sigma}} \, \dot{\xi}^{\sigma} = -H + \boldsymbol{\mathcal{P}} \cdot \mathrm{d}\boldsymbol{\xi}, \tag{14.182}$$

where

$$\frac{\partial \mathcal{W}}{\partial t} = -H(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}}, t) \quad \text{and} \quad \frac{\partial \mathcal{W}}{\partial \xi^{\sigma}} = \mathcal{P}_{\sigma}. \tag{14.183}$$

Substituting the second identity into the first leads to the Hamilton-Jacobi equation

$$\frac{\partial \mathcal{W}}{\partial t} = -H(\boldsymbol{\xi}, \partial \mathcal{W}/\partial \boldsymbol{\xi}, t), \tag{14.184}$$

which is a first order nonlinear partial differential equation for the on-shell action, again written in configuration space (not phase space). Notably, it is a first order equation enabled by building in the initial conditions at $t = t_{\mathsf{A}}$.

Connection to Hamilton's equation

Solving the Hamilton-Jacobi equation (14.184) for the on-shell action, $\mathcal{W}(\boldsymbol{\xi}, t)$, then yields, through its partial derivatives in equation (14.183), the Hamiltonian and the generalized momenta at the arbitrary time, t. We here show that having the on-shell action provides information sufficient to solve for the full evolution. To do so, it is sufficient to show that we can recover Hamilton's equations from the on-shell action.

Taking the partial derivative of the Hamiltonian with respect to the generalized momenta yields the evolution of the generalized coordinate

$$\left. \frac{\partial H}{\partial \mathcal{P}_{\sigma}} \right|_{\mathcal{P}_{\sigma} = \partial \mathcal{W}/\partial \xi^{\sigma}} = \dot{\xi}^{\sigma}, \tag{14.185}$$

where $\mathcal{P}_\sigma = \partial \mathcal{W}/\partial \xi^\sigma$ and $-H = \partial \mathcal{W}/\partial t$. Equation (14.185) is the first of Hamilton's equations. Now from equation (14.183), the total time derivative of the generalized momentum is given by

$$\dot{\mathcal{P}}_\sigma = \frac{\mathrm{d}\mathcal{P}_\sigma}{\mathrm{d}t} = \frac{\mathrm{d}(\partial \mathcal{W}/\partial \xi^\sigma)}{\mathrm{d}t} = \frac{\partial^2 \mathcal{W}}{\partial t \partial \xi^\sigma} + \frac{\partial^2 \mathcal{W}}{\partial \xi^\sigma \partial \xi^\lambda} \frac{\mathrm{d}\xi^\lambda}{\mathrm{d}t}. \tag{14.186}$$

To compute $\partial^2 \mathcal{W}/\partial t \partial \xi^\sigma$, we make use of the Hamilton-Jacobi equation (14.184), being careful with the $\partial/\partial \xi^\sigma$ derivative, so that

$$-\frac{\partial}{\partial \xi^\sigma}\frac{\partial \mathcal{W}}{\partial t} = \frac{\partial H(\boldsymbol{\xi}, \partial \mathcal{W}/\partial \boldsymbol{\xi}, t)}{\partial \xi^\sigma} \tag{14.187a}$$

$$= \frac{\partial H}{\partial \xi^\sigma} + \frac{\partial H}{\partial \mathcal{P}_\lambda}\frac{\partial \mathcal{P}_\lambda}{\partial \xi^\sigma} \tag{14.187b}$$

$$= \frac{\partial H}{\partial \xi^\sigma} + \frac{\partial H}{\partial \mathcal{P}_\lambda}\frac{\partial^2 \mathcal{W}}{\partial \xi^\sigma \partial \xi^\lambda} \tag{14.187c}$$

$$= \frac{\partial H}{\partial \xi^\sigma} + \dot{\xi}^\lambda \frac{\partial^2 \mathcal{W}}{\partial \xi^\sigma \partial \xi^\lambda}, \tag{14.187d}$$

where in some of the intermediate steps we wrote $\partial \mathcal{W}/\partial \xi^\lambda = \mathcal{P}_\lambda$ as a shorthand. Bringing equation (14.187d) back into equation (14.186) recovers the second of Hamilton's equations

$$\dot{\mathcal{P}}_\sigma = -\frac{\partial H}{\partial \xi^\sigma}. \tag{14.188}$$

We conclude that knowledge of the on-shell action, $\mathcal{W}(\boldsymbol{\xi}, t)$, which is a function of the generalized coordinates and time, allows us to compute the full evolution of the mechanical system. We encounter the on-shell action and the Hamilton-Jacobi equation (14.184) in our study of geometrical optics in VOLUME 4. This formulation of classical mechanics also proved quite important in the early days of quantum mechanics.[21]

14.9.9 Comparing Hamiltonian mechanics and Lagrangian mechanics

We close this section by highlighting a few points of comparison between Hamiltonian mechanics and Lagrangian mechanics, and start by comparing the phase space of Hamiltonian mechanics to the configuration space of Lagrangian mechanics. Specification of a point within the $2D$ dimensional phase space provides a full specification of the state of a dynamical system, with Hamilton's equations then determining the evolution of the system through phase space. As seen in this section, particularly in our discussion of Liouville's theorem in Section 14.9.7, the geometry of flow within phase space renders insights into the nature of classical dynamical motion. Herein lies the core reason many favor Hamiltonian dynamics for their studies of classical motion.[22]

The configuration space of Lagrangian mechanics has D dimensions, as defined by the generalized coordinates. We view motion of a D dimensional physical system as the motion of a point within D dimensional configuration space, which contrasts to the viewing the motion of many particles in three-dimensional space. The geometry of configuration space is determined through constraints imposed on the system, such as the spherical geometry of

[21] See Section 10.8 of *Tong* (2025) for a discussion of how classical mechanics, as formulated via Hamilton-Jacobi theory, has an appearance very much like quantum mechanics.

[22] Mathematically, the Hamiltonian connects the position on a manifold to the cotangent space of the manifold, which is where the generalized momenta live.

particles constrained to move on the surface of a sphere. Notably, a point in configuration space does not fully specify the dynamical state of the system, which is a fundamental distinction from the phase space of Hamiltonian mechanics.

In our use of analytical mechanics in this book, we mostly use Lagrangian mechanics for practical applications since it is generally the most straightforward way to derive the equations of motion. We exemplify the method via a suite of case studies in Chapter 15. We also use Lagrangian mechanics to formulate the equations of continuum field theory and fluid mechanics using Hamilton's variational principle in VOLUME 4. We do, however, make use of the general theorems derived via Hamiltonian mechanics and its phase space formulation, with Liouville's theorem from Section 14.9.7 a primary example. We also encounter the notion of an adiabatic invariant when studying waves in VOLUME 4, with general properties of adiabatic invariants derived using phase space arguments (see Section 49 in *Landau and Lifshitz* (1976) or Section 10.6 of *Tong* (2025)). Finally, we note that the Hamilton-Jacobi formulation of Hamiltonian mechanics has direct relevance to our study of geometric optics, also encountered in VOLUME 4.

14.10 Exercises

EXERCISE 14.1: PRESSURE FROM AN IDEAL GAS

Consider an ideal gas with N identical classical particles of mass, m, contained in a closed domain of volume V. Assume the gas is in thermodynamic equilibrium with no gravity field, so that the temperature and pressure are uniform. Use a variant of the virial theorem from Section 14.6.3 to prove that the pressure within the ideal gas is given by

$$p = \tfrac{2}{3}\,\overline{\varepsilon}/V, \tag{14.189}$$

where $\overline{\varepsilon}$ is the time average of the total kinetic energy of the gas.

Hint: compute the time derivative

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{\mathsf{n}=1}^{\mathsf{N}} \boldsymbol{P}_{\mathsf{n}} \cdot \boldsymbol{X}_{\mathsf{n}}, \tag{14.190}$$

where $\boldsymbol{X}_{\mathsf{n}}(t)$ is the Cartesian position of ideal gas particle n at time t, and $\boldsymbol{P}_{\mathsf{n}} = m\,\dot{\boldsymbol{X}}_{\mathsf{n}}$ is the linear momentum. Identify the term

$$\text{virial} = -\frac{1}{2}\sum_{\mathsf{n}=1}^{\mathsf{N}} \boldsymbol{F}_{\mathsf{n}} \cdot \boldsymbol{X}_{\mathsf{n}}, \tag{14.191}$$

known as the virial, which measures how the forces act relative to the system's size. Also note that for an ideal gas there are no inter-molecular forces, so the only force arises from interactions with the domain boundary.

EXERCISE 14.2: KINETIC ENERGY AS A HOMOGENEOUS FUNCTION

Following the discussion from Section 14.4.7, assume a time independent coordinate transformation between Cartesian coordinates, x^a, and generalized coordinates, ξ^μ. In this case, prove the following identity holds

$$2\,K = \sum_{\lambda=1}^{D} \dot{\xi}^{\lambda}\,(\partial K/\partial \dot{\xi}^{\lambda}), \tag{14.192}$$

where K is the kinetic energy. This identity holds, rather trivially, for Cartesian coordinates. In this exercise we prove it also holds for generalized coordinates, where the coordinate transformation is time independent. Following from Euler's theorem for homogeneous functions, we conclude that for time independent coordinate transformations, the kinetic energy is a degree two homogeneous function of the generalized velocities (see Section 14.6.1 for a discussion of the potential energy).

EXERCISE 14.3: THERMODYNAMIC EQUILIBRIUM AND LIOUVILLE'S THEOREM

In equilibrium statistical mechanics, the phase space density from Section 14.9.7 is interpreted as the density of particles in an ensemble. As noted in Sections 2.1 and 2.2 of *Pathria and Beale* (2021), thermodynamic equilibrium of an isolated physical system corresponds to a phase space density that is constant at each point in phase space,

$$\partial\rho/\partial t = 0. \tag{14.193}$$

Show that this condition of stationarity holds if the phase space density is a function just of the Hamiltonian

$$\rho = \rho[H(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}})], \tag{14.194}$$

where the Hamiltonian has no explicit time dependence since we are concerned here with isolated physical systems. As noted in Section 2.2 of *Pathria and Beale* (2021), this functional dependence leads to the various ensemble density functions (i.e., ensemble distributions) used in statistical physics, such as the microcanonical distribution and the canonical distribution.

EXERCISE 14.4: MAUPERTUIS' PRINCIPLE

In Chapter 7 we studied geodesics from a mathematical perspective, with free particles tracing out geodesics on a manifold. Here, we extend these ideas by allowing for the particle to feel a conservative force, and so is it no longer free. Even so, we can interpret the motion as that of a geodesic according to the action

$$\mathcal{S} = m \int_{t_{\mathsf{A}}}^{t_{\mathsf{B}}} H_{\sigma\mu}\,\frac{\mathrm{d}\xi^{\sigma}}{\mathrm{d}s}\,\frac{\mathrm{d}\xi^{\mu}}{\mathrm{d}s}\,\mathrm{d}s, \tag{14.195}$$

where $H_{\sigma\mu}$ are components to an energy dependent metric tensor, and s is an arc-length parameter defined according to

$$\mathrm{d}s \equiv m^{-1}\,[E - P(\boldsymbol{\xi})]\,\mathrm{d}t, \tag{14.196}$$

where E is the constant energy of the trajectory and $P(\boldsymbol{\xi})$ is the potential energy. By extremizing this action, we find that trajectories follow geodesics for the metric, $H_{\sigma\mu}$. This geodesic interpretation of classical motion is referred to as Maupertuis principle.

EXERCISE 14.5: POISSON BRACKETS AND ANGULAR MOMENTUM

The angular momentum for a particle is given by

$$\boldsymbol{L} = \boldsymbol{X} \times \boldsymbol{P}, \tag{14.197}$$

where $\boldsymbol{X}$ is the Cartesian position vector relative to the origin, and $\boldsymbol{P}$ is the linear (mechanical) momentum. In terms of generalized coordinates and generalized momenta, the Poisson bracket is given by

$$\{F, G\} = \frac{\partial F}{\partial \xi^\sigma}\frac{\partial G}{\partial \mathcal{P}_\sigma} - \frac{\partial F}{\partial \mathcal{P}_\sigma}\frac{\partial G}{\partial \xi^\sigma}. \tag{14.198}$$

Show that the angular momentum, as a function on phase space with Cartesian coordinates, $\boldsymbol{L}(\boldsymbol{X}, \boldsymbol{P})$, satisfies

$$\{L_i, L_j\} = \epsilon_{ij}{}^k\, L_k. \tag{14.199}$$

EXERCISE 14.6: LIOUVILLE'S THEOREM FOR FALLING PARTICLES

Here we verify that Liouville's theorem from Section 14.9.7 holds for the case of freely falling particles. Assume a vertical distribution of particles with constant mass, m, that are initially contained in the vertical region, $z = z_\circ \pm h$, and that have the initial vertical momentum within the range $p_\circ \pm P$, where h and P are constants (see Figure 14.3). All the particles fall under the effects of a uniform gravitational acceleration, $\boldsymbol{h} = -g\,\hat{\boldsymbol{z}}$. Show that the phase space volume remains constant as the particles fall.

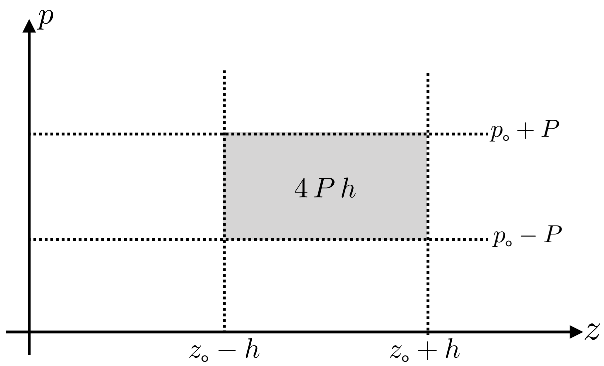

FIGURE 14.3: Initial phase space region for the particles falling under the effects of a uniform gravity field as considered in Exercise 14.6. The vertical axis is the vertical momentum and the horizontal axis is the vertical position. Particles are assumed to be initially distributed within the shaded rectangular region of phase space. Liouville's theorem says that the region volume remains constant as the particles evolve.

Chapter 15

Case studies in analytical mechanics

The discussion in Chapter 14 certainly does support the name analytical mechanics, given that there is a nontrival degree of analysis needed to reformulate the laws of mechanics beyond their Newtonian expression. In this chapter, we work through some case studies to illustrate the use of analytical mechanics for practical problems, many of which have direct relevance to geophysical motions. The case studies also further exemplify and clarify the foundations of the theory while exploring interesting dynamical systems. Most of the case studies make use of Lagrangian mechanics, as that generally leads most directly to the equations of motion. Even so, in some cases we find it useful to also discuss Hamiltonian mechanics to emphasize certain conceptual points.

CHAPTER GUIDE

This chapter relies on the formulation of analytical mechanics from Chapter 14. We also make use of material from the study of particle mechanics around a rotating planet in Chapter 12. Many of the examples considered here are fundamental to our studies of geophysical fluid motion in later volumes.

15.1 Planar simple pendulum

Consider a planar simple pendulum as illustrated in Figure 15.1, where motion is restricted to the x-z plane. This pendulum consists of a point mass, m, attached to a massless string of fixed length, ℓ, oscillating around a fixed point within a constant gravity field with acceleration $-g\,\hat{\boldsymbol{z}}$. We derive the equations of motion for this system using both Newtonian mechanics and Lagrangian mechanics.

Before diving into the specifics of the simple pendulum, we note that it exemplifies the two key properties of any oscillatory system, including linear waves studied in VOLUME 4. The first property is a restoring force that kicks in when the system is displaced a small distance from its equilibrium. For the simple pendulum the restoring force is provided by the gravitational acceleration. The second property is inertia, as provided by the mass of the pendulum, which causes the system to overshoot its equilibrium position. The exchange between the restoring force and inertia lead to oscillations.

15.1.1 Free particle motion using polar coordinates

Before addressing the simple pendulum, we find it useful to consider the equations for a free particle moving in the x-z plane without gravity and without any string. Cartesian coordinates provide the natural set of coordinates for describing this motion, in which the $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{z}}$ components to the Cartesian velocity velocity remain constant in space and time. We also consider polar coordinates, with the polar angle defined by φ as depicted in Figure 15.1, anticipating the motion of a simple pendulum using these coordinates. The coordinate

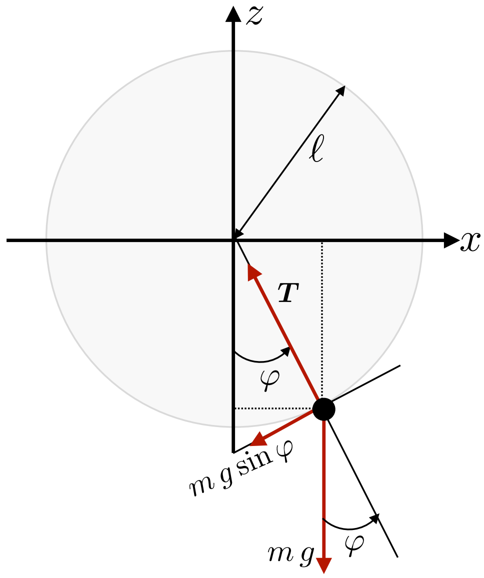

FIGURE 15.1: A simple pendulum is comprised of a point mass, m, attached to a massless string of fixed length, ℓ, with the string attached to a fixed point taken as the origin of a Cartesian coordinate system. The string makes an angle, φ, with respect to the vertical, and gravity points down with a constant acceleration, $-g\,\hat{\boldsymbol{z}}$. The force provided by the string on the particle, $\boldsymbol{T}$, points radially inward and balances the projection of the weight along the string plus the centrifugal acceleration due to motion of the particle. The point mass moves along the circumference of the circle. The string force is directed orthogonal to the motion so that it does no work on the point mass, and as such the string provides a force of constraint. The component of the weight directed along the circumference forces motion of the oscillator, with this force having magnitude $m\,g|\sin\varphi|$.

transformation between Cartesian and polar is given by

$$x = r\sin\varphi \tag{15.1a}$$
$$z = -r\cos\varphi \tag{15.1b}$$
$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{x}}\,\sin\varphi - \hat{\boldsymbol{z}}\cos\varphi \tag{15.1c}$$
$$\hat{\boldsymbol{\varphi}} = \hat{\boldsymbol{x}}\,\cos\varphi + \hat{\boldsymbol{z}}\sin\varphi. \tag{15.1d}$$

The Cartesian unit vectors are fixed in space, but the time derivatives of the polar unit vectors are

$$\dot{\hat{\boldsymbol{r}}} = \dot{\varphi}\,\hat{\boldsymbol{\varphi}} \quad \text{and} \quad \dot{\hat{\boldsymbol{\varphi}}} = -\dot{\varphi}\,\hat{\boldsymbol{r}}. \tag{15.2}$$

These results render the particle position, its velocity, and its acceleration

$$\boldsymbol{X} = r\,(\hat{\boldsymbol{x}}\,\sin\varphi - \hat{\boldsymbol{z}}\cos\varphi) = r\,\hat{\boldsymbol{r}} \tag{15.3a}$$
$$\dot{\boldsymbol{X}} = \dot{r}\,\hat{\boldsymbol{r}} + r\,\dot{\varphi}\,\hat{\boldsymbol{\varphi}} \tag{15.3b}$$
$$\ddot{\boldsymbol{X}} = (\ddot{r} - r\,\dot{\varphi}^2)\,\hat{\boldsymbol{r}} + (2\,\dot{r}\,\dot{\varphi} + r\,\ddot{\varphi})\,\hat{\boldsymbol{\varphi}}. \tag{15.3c}$$

Free particle motion has zero acceleration,

$$\ddot{x} = 0 \quad \text{and} \quad \ddot{z} = 0, \tag{15.4}$$

which means each Cartesian velocity component is a space and time constant. This trivial result has some nuance when written in polar coordinates

$$\ddot{r} - r\,\dot{\varphi}^2 = 0 \quad \text{and} \quad 2\,\dot{r}\,\dot{\varphi} + r\,\ddot{\varphi} = 0. \tag{15.5}$$

Of relevance to the pendulum, observe that forces are needed for both equations to maintain a fixed r.

Pursuing a Lagrangian mechanics approach, we compute the Euler-Lagrange equation (14.77) using (r, φ) as the two generalized coordinates. For the free particle, the Lagrangian equals to the kinetic energy

$$L = m\,(\dot{r}^2 + r^2\,\dot{\varphi}^2)/2, \tag{15.6}$$

so that the two Euler-Lagrange equations are

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{r}} = \frac{\partial L}{\partial r} \Longrightarrow \ddot{r} = r\,\dot{\varphi}^2 \tag{15.7a}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\varphi}} = \frac{\partial L}{\partial \varphi} \Longrightarrow \frac{\mathrm{d}\,(r^2\,\dot{\varphi})}{\mathrm{d}t} = 0. \tag{15.7b}$$

The radial equation says that the radial acceleration balances the centrifugal acceleration. The angular equation reflects the property of a Lagrangian that is independent of the generalized coordinate, φ, so that its corresponding generalized momentum, here the angular momentum computed relative to the origin, is a constant of the motion.

15.1.2 Newtonian mechanics of the simple pendulum

Now allow the particle to feel the constant gravitational field and the force by the string. Notably, the string force keeps the particle at a fixed distance, $r = \ell$, from the origin so that the particle motion only has a single degree of freedom. The position, velocity, and acceleration for the particle are thus given by

$$\boldsymbol{X} = \ell\,(\hat{\boldsymbol{x}} \sin\varphi - \hat{\boldsymbol{z}} \cos\varphi) = \ell\,\hat{\boldsymbol{r}} \tag{15.8a}$$

$$\dot{\boldsymbol{X}} = \ell\,\dot{\varphi}\,(\hat{\boldsymbol{x}} \cos\varphi + \hat{\boldsymbol{z}} \sin\varphi) = \ell\,\dot{\varphi}\,\hat{\boldsymbol{\varphi}} \tag{15.8b}$$

$$\ddot{\boldsymbol{X}} = \ell\,\ddot{\varphi}\,\hat{\boldsymbol{\varphi}} - \ell\,\dot{\varphi}^2\,\hat{\boldsymbol{r}}, \tag{15.8c}$$

with polar coordinates now ideally suited to the physical system. The force provided by the string points radially inward and it must be sufficient to exactly balance the projection of the particle weight along the string direction, plus the centrifugal acceleration of the particle as it oscillates along the circumference of the circle

$$\boldsymbol{T} = -m\,(g \cos\varphi + \ell\,\dot{\varphi}^2)\,\hat{\boldsymbol{r}}. \tag{15.9}$$

This string force constrains the particle to remain a fixed distance, $r = \ell$, from the center of the circle. Since the particle motion is directed along the circumference and the string force is radially directed, then the string force does no work on the particle. Evidently, the force provided by the string is a force of constraint. We reconsider this force of constraint from the Lagrangian perspective in Section 15.1.6.

Newton's equation of motion takes the form

$$m\,\ddot{\boldsymbol{X}} = -m\,g\,\hat{\boldsymbol{z}} + \boldsymbol{T}. \tag{15.10}$$

Projecting into the angular direction, $\hat{\boldsymbol{\varphi}}$, eliminates the string force and leads to the nonlinear oscillator equation

$$\ddot{\varphi} + (g/\ell) \sin\varphi = 0, \tag{15.11}$$

where we used

$$\hat{\boldsymbol{z}} \cdot \hat{\boldsymbol{\varphi}} = \sin\varphi. \tag{15.12}$$

When the angle is small we can approximate $\sin\varphi \approx \varphi$, in which case we have a linear oscillator equation

$$\ddot{\varphi} + (g/\ell)\,\varphi = 0, \tag{15.13}$$

with natural angular frequency,

$$\omega_\circ = \sqrt{g/\ell} \tag{15.14}$$

and period $2\,\pi/\omega_\circ = 2\,\pi\,\sqrt{\ell/g}$.

15.1.3 Euler-Lagrange equation for the simple pendulum

With a single degree of freedom for the simple pendulum, we choose the angle, φ, as the generalized coordinate. The kinetic energy, potential energy (referenced to $z = 0$), and Lagrangian are given by

$$K = (m/2)\,(\dot{x}^2 + \dot{z}^2) = (m/2)\,\ell^2\,\dot{\varphi}^2 \tag{15.15a}$$

$$P = -m\,g\,\ell\cos\varphi \tag{15.15b}$$

$$L = m\,(\ell^2\,\dot{\varphi}^2/2 + g\,\ell\cos\varphi), \tag{15.15c}$$

with the corresponding Euler-Lagrange equation (14.77)

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial \dot{\varphi}}\right] = \frac{\partial L}{\partial \varphi} \Longrightarrow \ddot{\varphi} + (g/\ell)\sin\varphi = 0. \tag{15.16}$$

As expected, we derived the same oscillator equation (15.11) as when using Newtonian mechanics. However, the path to this equation is somewhat more streamlined since we had no need to consider forces.

15.1.4 The Hamiltonian is a constant of the motion

The coordinate transformation between the Cartesian coordinates and generalized coordinate does not depend explicitly on time, nor does the Lagrangian. Hence, from Section 14.8.4 we know that the Hamiltonian equals to the mechanical energy, and it is a constant of the motion

$$H = -L + \dot{\varphi}\,\frac{\partial L}{\partial \dot{\varphi}} = K + P. \tag{15.17}$$

We readily confirm that $\dot{H} = 0$ by computing

$$\dot{K} + \dot{P} = m\,\ell^2\,\dot{\varphi}\,[\ddot{\varphi} + (g/\ell)\sin\varphi] = 0, \tag{15.18}$$

where the second equality follows from the Euler-Lagrange equation of motion (15.16).

15.1.5 Hamilton's equations of motion

When proving that the Hamiltonian is a constant of the motion in Section 15.1.4, we did not first write the Hamiltonian in terms of the generalized coordinates and momenta. Instead, we directly computed the time derivative and used the Euler-Lagrange equation to show that $\dot{H} = 0$. However, as mentioned in Section 14.9.4, to derive Hamilton's equations of motion we must first write the Hamiltonian in terms of the generalized coordinates and generalized momenta. The generalized momenta for the pendulum is given by

$$\mathcal{P}_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m\,\ell^2\,\dot{\varphi}, \tag{15.19}$$

which is the angular momentum relative to the origin, which means the Hamiltonian is

$$H(\varphi, \mathcal{P}_\varphi) = \frac{\mathcal{P}_\varphi^2}{2\,m\,\ell^2} - m\,g\,\ell\cos\varphi. \tag{15.20}$$

Hamilton's equations of motion (14.148a) and (14.148b) take on the general form

$$\dot{\mathcal{P}}_\sigma = -\frac{\partial H}{\partial \xi^\sigma} \quad \text{and} \quad \dot{\xi}^\sigma = \frac{\partial H}{\partial \mathcal{P}_\sigma}, \tag{15.21}$$

which for the pendulum are

$$\dot{\mathcal{P}}_\varphi = -m\,g\,\ell\sin\varphi \quad \text{and} \quad \dot{\varphi} = \frac{\mathcal{P}_\varphi}{m\,\ell^2}. \tag{15.22}$$

Reorganization of the $\dot{\mathcal{P}}_\varphi$ equation renders the Euler-Lagrange equation (15.16). In contrast, the $\dot{\varphi}$ equation is identical to the definition of the generalized momenta given by equation (15.19), so that this equation offers no new information.

15.1.6 Force of constraint provided by the string

Following the Lagrange multiplier method from Section 14.7, we here derive the force of constraint acting on the particle. As already noted, the force of constraint is provided by the string, and this force is always orthogonal to the particle motion. Indeed, we wrote this force by inspection in equation (15.9). Here we compute this force via the Lagrange multiplier. In particular, we follow the approach in Section 14.7 and identify the single constraint equation as

$$\Psi(r) = r = \ell, \tag{15.23}$$

so that $\partial\Psi/\partial r = 1$ and equation (14.112) then says that the Lagrange multiplier is the force of constraint corresponding to the radial coordinate

$$\Lambda = Q_r. \tag{15.24}$$

Following the procedure detailed in Section 14.7.3, we consider the extended Lagrangian (14.106) for the pendulum

$$L = m\,(\dot{r}^2 + r^2\,\dot{\varphi}^2)/2 + m\,g\,r\cos\varphi + \Lambda\,(r - \ell). \tag{15.25}$$

The corresponding Euler-Lagrange equations are

$$m\,\ddot{r} = m\,r\,\dot{\varphi}^2 + m\,g\cos\varphi + \Lambda \tag{15.26a}$$

$$m\,\frac{\mathrm{d}(r^2\,\dot{\varphi})}{\mathrm{d}t} = -m\,g\,r\sin\varphi \tag{15.26b}$$

$$r = \ell. \tag{15.26c}$$

The angular equation of motion (15.26a) remains unaffected by the Lagrange multiplier since there are no constraints on motion in the angular direction. So the set of three equations for the three unknowns, (r, φ, Λ), is given by

$$2\,\dot{r}\,\dot{\varphi} + r\,\ddot{\varphi} + g\sin\varphi = 0 \quad \text{and} \quad \Lambda = m\,(\ddot{r} - r\,\dot{\varphi}^2 - g\cos\varphi) \quad \text{and} \quad r = \ell. \tag{15.27}$$

Satisfying the constraint, $r = \ell$, means that $\dot{r} = 0$ and $\ddot{r} = 0$ so that the force of constraint is

$$Q_r = \Lambda = -m\,(\ell\,\dot{\varphi}^2 + g\cos\varphi) < 0, \tag{15.28}$$

where the inequality follows since $\ell\,\dot{\varphi}^2 + g\cos\varphi > 0$. This result agrees with equation (15.9) for the force provided by the string, $\boldsymbol{T}$, so that we identify

$$\boldsymbol{T} = Q_r\,\hat{\boldsymbol{r}} = -m\,(\ell\,\dot{\varphi}^2 + g\cos\varphi)\,\hat{\boldsymbol{r}}. \tag{15.29}$$

15.1.7 Comments

As anticipated, the Euler-Lagrange equation of motion is both equal to Newton's equation of motion and simple to derive. The Euler-Lagrange equation is simpler to derive largely due to the scalar nature of the Lagrangian versus the vector nature of Newton's equation of motion. Namely, it is easier to derive the kinetic energy and potential energy than the acceleration and forces. Even though we do not need forces for analytical mechanics, the calculation of forces offers great physical insights into the nature of the physical system. Conservative forces are deduced as per Newtonian mechanics via the gradient of the force potential, and forces of constraint are inferred through the method of Lagrange multipliers.

15.2 Variable length pendulum and adiabatic invariants

What if we let the length of the pendulum's string be a prescribed function of time? That is, we let the string length be a time dependent parameter of the physical system. In this case the Hamiltonian is an explicit function of time, through the string length, and so mechanical energy is not a constant of the motion. Correspondingly, the force of constraint acting along the string does work as the string changes, with this work the cause of the nonzero time derivative of the Hamiltonian. We are familiar with forces of constraint providing no work on a physical system, with this property enshrined in D'Alembert's principle studied in Section 14.2.1. Indeed, at any particular instance, the forces of constraint perform no work on a virtual displacement. However, when allowing time to evolve then the force of constraint acting along the string does work and that work results in an energy change.

Along with a time change in the pendulum's energy, there is a time change to the pendulum's angular frequency. Quite remarkably, the ratio of the energy to the angular frequency remains nearly constant for small amplitude oscillations, so long as changes to the string length are slow. The ratio of the energy to frequency is known as an adiabatic invariant. This adiabatic invariant arises when studying linear waves moving through a time dependent media.

15.2.1 Work done by the force of constraint

Recall from Section 15.1.6 that we introduced the extended Lagrangian with a Lagrange multiplier, Λ, representing the force of constraint,

$$L = m\,(\dot{r}^2 + r^2\,\dot{\varphi}^2)/2 + m\,g\,r\cos\varphi + \Lambda\,(r - \ell), \tag{15.30}$$

where (r, φ, Λ) are three generalized coordinates, and the corresponding Hamiltonian is

$$H = -L + \dot{\varphi}\,\frac{\partial L}{\partial \dot{\varphi}} + \dot{r}\,\frac{\partial L}{\partial \dot{r}} = m\,(\dot{r}^2 + r^2\,\dot{\varphi}^2)/2 - m\,g\,r\cos\varphi - \Lambda\,(r - \ell). \tag{15.31}$$

The corresponding Euler-Lagrange equations (15.27) are here repeated for completeness

$$2\,\dot{r}\,\dot{\varphi} + r\,\ddot{\varphi} + g\sin\varphi = 0 \quad \text{and} \quad \Lambda = m\,(\ddot{r} - r\,\dot{\varphi}^2 - g\cos\varphi) \quad \text{and} \quad r(t) = \ell(t). \tag{15.32}$$

Evidently, the force of constraint, Λ, picks up a new contribution from the second time derivative, $\ddot{r}$, whereas this contribution vanished in Section 15.1.6 where we assumed ℓ has a fixed length.

Both the Lagrangian (15.30) and Hamiltonian (15.31) are explicit functions of time via the time dependent string length, $\ell(t)$. Hence, the Hamiltonian has a time derivative, which can be evaluated in either of the following manners

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \left[\frac{\partial H}{\partial t}\right]_{r,\varphi,\mathcal{P}_r,\mathcal{P}_\varphi} = \frac{\partial H}{\partial \ell}\,\frac{\mathrm{d}\ell}{\mathrm{d}t} = -\left[\frac{\partial L}{\partial t}\right]_{r,\varphi,r,\dot{r},\dot{\varphi}} = \dot{\ell}\,\Lambda, \tag{15.33}$$

so that the energy of the pendulum changes due to work done by the force of constraint acting in the presence of a changing string length. We can double-check the result (15.33) by making use of the equations of motion (15.32), which now requires us to set $r = \ell$ so that

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \dot{H} = m\,\dot{l}\,(\ddot{l} + \ell\,\dot{\varphi}^2 - g\cos\varphi) + m\,\ell\,\dot{\varphi}\,(\ell\,\ddot{\varphi} + g\sin\varphi) = m\,\dot{\ell}\,(\ddot{\ell} - \ell\,\dot{\varphi}^2 - g\cos\varphi) = \dot{\ell}\,\Lambda. \tag{15.34}$$

15.2.2 Small amplitude oscillations

Assuming the angular displacements of the pendulum are small leads to the Lagrangian and Hamiltonian for a linear variable length pendulum

$$L = m\,\ell^2\,(\dot{\varphi}^2 - (g/\ell)\,\varphi^2)/2 + m\,g\,\ell + m\,\dot{\ell}^2/2, \tag{15.35a}$$
$$H = m\,\ell^2\,(\dot{\varphi}^2 + (g/\ell)\,\varphi^2)/2 - m\,g\,\ell - m\,\dot{\ell}^2/2, \tag{15.35b}$$

along with the corresponding Euler-Lagrange equation of motion

$$\frac{\mathrm{d}(\ell^2\,\dot{\varphi})}{\mathrm{d}t} + g\,\ell\,\varphi = 0. \tag{15.36}$$

Note that the term $m\,\ell^2\,(\dot{\varphi}^2 - (g/\ell)\,\varphi^2)/2$ appearing in the Lagrangian (15.35a) is the kinetic energy minus the potential energy for a simple harmonic oscillator, as studied in Section 15.7. We thus identify the angular frequency of the oscillator as

$$\omega^2(t) = g/\ell(t), \tag{15.37}$$

with $2\pi/\omega$ equal to the period of the oscillations.

15.2.3 Two time scales for the pendulum

For the linear pendulum with a fixed length, the angular displacement is given by $\varphi = A\cos(\omega\, t)$, where A is a constant amplitude and we set the initial conditions so that $\varphi(t=0) = A$. If the string length changes very slowly, then we expect the oscillations to continue yet with a time dependent frequency and amplitude. We are thus motivated to consider the ansatz

$$\varphi(t) = A(t)\cos\theta(t), \tag{15.38}$$

where the amplitude, $A(t)$, has a time dependence, as does the phase, $\theta(t)$. Furthermore, we note that the time scale for the phase derivative accords with the angular frequency

$$\dot{\theta} \approx \omega. \tag{15.39}$$

In Section 15.2.4 we find equality holds to leading order.

Let T^{-1} be the inverse time scale (considered here as an angular frequency) for the changes in the string length

$$\dot{\ell}/\ell = T^{-1}. \tag{15.40}$$

We assume that this inverse time scale is much smaller than the angular frequency for the pendulum, so that

$$\dot{\ell}/\ell = T^{-1} \ll \omega \approx \dot{\theta} \Longrightarrow \dot{\ell}^2 \ll \ell\, g, \tag{15.41}$$

where the second inequality follows since $\omega^2 = g/\ell$. The inequality $T^{-1} \ll \dot{\theta}$ means that the phase of the oscillation varies rapidly relative to the time scale for changes to the string length. Another corresponding assumption concerns the amplitude of the oscillation, which we assume also changes slowly

$$\dot{A}/A = T^{-1} \ll \omega. \tag{15.42}$$

Finally, the relation $\omega^2 = g/\ell$, along with the inequality (15.41), shows that relative changes to the angular frequency are also small

$$2\,\dot{\omega}/\omega = -\dot{\ell}/\ell \Longrightarrow |\dot{\omega}| \ll \omega^2 = g/\ell. \tag{15.43}$$

These scaling relations can be summarized by stating that the phase of the oscillation changes rapidly whereas all other quantities change slowly.

In the Lagrangian (15.35a) and Hamiltonian (15.35b), we encounter the sum, $m\,g\,\ell + m\,\dot{\ell}^2/2$. The term, $-m\,g\,\ell$, is the gravitational potential energy of a pendulum at zero angular displacement, $\varphi = 0$. Over the time scale of an oscillation, this term is roughly a constant and it can be ignored by merely readjusting the zero of the potential energy. Likewise, the term $m\,\dot{\ell}^2/2$ is the kinetic energy associated with changes to the string length. This term changes slowly and it is much smaller than the kinetic energy from angular motion, $m\,\ell^2\,\dot{\varphi}^2/2$. Hence, we also ignore this term in the following analysis.

15.2.4 Adiabatic invariant

Making use of the ansatz (15.38) yields time changes to the pendulum's angle

$$\dot{\varphi} = \dot{A}\cos\theta - A\,\dot{\theta}\sin\theta \approx -A\,\dot{\theta}\sin\theta, \tag{15.44}$$

where we made use of the assumed scales in Section 15.2.3 to reach the approximation. We are thus led to the approximate form of the Lagrangian (15.35a) (recall we are ignoring the terms $m\,g\,\ell + m\,\dot{\ell}^2/2$)

$$L \approx m\,\ell^2\,A^2\,(\dot{\theta}^2 \sin^2\theta - \omega^2 \cos^2\theta)/2. \tag{15.45}$$

As part of the time scale separation in Section 15.2.3, we assume the phase of the oscillation changes rapidly relative to changes in the angular frequency as well as changes to $\dot{\theta}$ and ℓ. Hence, we are motivated to perform a phase average,[1] in which case $\sin^2\theta$ and $\cos^2\theta$ are replaced by 1/2

$$\langle L\rangle = m\,\ell^2\,A^2\,(\dot{\theta}^2 - \omega^2)/4. \tag{15.46}$$

We treat the amplitude and phase as generalized coordinates that are connected to the angular displacement via the oscillator ansatz (15.38). Their respective phase averaged Euler-Lagrange equations are

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\langle L\rangle}{\partial\dot{A}} = \frac{\partial\langle L\rangle}{\partial A} \Longrightarrow \dot{\theta}^2 = \omega^2 \tag{15.47a}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\langle L\rangle}{\partial\dot{\theta}} = \frac{\partial\langle L\rangle}{\partial\theta} \Longrightarrow \frac{\mathrm{d}(\ell^2\,A^2\,\dot{\theta})}{\mathrm{d}t} = 0. \tag{15.47b}$$

Equation (15.47a) says that the phase has a time derivative equal to the instantaneous frequency. It also results in a zero phase averaged Lagrangian

$$\langle L\rangle = 0. \tag{15.48}$$

To interpret equation (15.47b) note that the phase averaged Hamiltonian (15.35b) is (recall we are ignoring the terms $m\,g\,\ell + m\,\dot{\ell}^2/2$)

$$\langle H\rangle = m\,\ell^2\,A^2\,(\dot{\theta}^2 + \omega^2)/4 = m\,\ell^2\,A^2\,\omega^2/2, \tag{15.49}$$

so that the Euler-Lagrange equation (15.47b) can be written

$$\frac{\mathrm{d}(H/\omega)}{\mathrm{d}t} = 0. \tag{15.50}$$

Although energy and angular frequency change due to the variable string length, their ratio is constant when the oscillations are small and the changes to the string length are slow. The ratio, H/ω, is known as the wave action, and equation (15.50) says that it is an adiabatic invariant. Conservation of the wave action results from symmetry of the phase in the phase averaged Lagrangian (15.46). That is, we can shift the phase by a constant and not alter the Lagrangian. This result exemplifies the importance of identifying any (sometimes quite innocent looking) symmetries of a dynamical system since they inevitably lead to a conservation law.

15.2.5 Further study

General discussions of adiabatic invariants are given in Section 49 of *Landau and Lifshitz* (1976), Section 11.7 of *Goldstein* (1980), Section 6.4 of *José and Saletan* (1998), and Section 10.6 of *Tong* (2025). Those discussions are accessible to those having worked through the presentation of Hamiltonian mechanics in Section 14.9. See also Section 7.1 of *Salmon* (1998) for a discussion of the adiabatic invariant for the harmonic oscillator. We encounter adiabatic invariants in

[1] We introduced phase averaging in Section 6.1.2 when studying Fourier analysis.

VOLUME 4 when studying waves in a media that gently varies in space and time, with the wave action the corresponding adiabatic invariant.

Adiabatic invariants formed an important part in the foundations of quantum mechanics. For example, the energy of a photon is given by

$$E = \hbar\omega, \tag{15.51}$$

where $\hbar = h/2\pi$ is the reduced Planck's constant, thus revealing that $E/\omega = \hbar$ is indeed a constant.

15.3 Free particle in a rotating reference frame

In Chapters 11 and 12 we derived the equations of motion for a particle as described within a rotating reference frame, which is the usual frame for studying geophysical fluid motions. In those earlier derivations we made use of a Newtonian approach that carefully worked through the vector analysis to derive the Coriolis acceleration and centrifugal acceleration. Here we make use of a Lagrangian approach. Assuming some skills with Cartesian tensors, the Lagrangian approach proves to be somewhat simpler than the Newtonian approach.

So consider the free particle moving in three dimensional space, in which the Lagrangian equals to the kinetic energy

$$L = K = \tfrac{1}{2} m\, \dot{\boldsymbol{X}}_{\mathsf{I}}^{2}, \tag{15.52}$$

where we write $\dot{\boldsymbol{X}}_{\mathsf{I}}$ for the velocity of the particle as measured by an inertial observer. Now make use of the relation derived in Section 11.5 between the velocity seen by an inertial observer and that seen by a rotating observer

$$\dot{\boldsymbol{X}}_{\mathsf{I}} = \dot{\boldsymbol{X}} + \boldsymbol{\Omega} \times \boldsymbol{X}, \tag{15.53}$$

where $\boldsymbol{X}$ is the position measured by the rotating observer, and $\boldsymbol{\Omega}$ is the angular rotation rate of the reference frame. We assume that $\dot{\boldsymbol{\Omega}} = 0$, which is a very good approximation for geophysical fluid motions of concern in this book.

The kinetic energy thus takes the form in terms of the rotating variables

$$K = \tfrac{1}{2} m\, [\dot{\boldsymbol{X}} \cdot \dot{\boldsymbol{X}} + 2\, \dot{\boldsymbol{X}} \cdot (\boldsymbol{\Omega} \times \boldsymbol{X}) + (\boldsymbol{\Omega} \times \boldsymbol{X})^{2}]. \tag{15.54}$$

To help take the partial derivatives for the Euler-Lagrange equations, we make use of some (admittedly a bit pedantic) Cartesian tensor algebra (see Chapter 1) to write

$$K = \tfrac{1}{2} m\, [\delta_{ab}\, \dot{X}^{a}\, \dot{X}^{b} + 2\, \epsilon_{abc}\, \dot{X}^{a}\, \Omega^{b}\, X^{c} + \Omega^{2}\, \delta_{ab}\, X^{a}\, X^{b} - (\delta_{ab}\, \Omega^{a}\, X^{b})\, (\delta_{cd}\, \Omega^{c}\, X^{d})]. \tag{15.55}$$

We thus find

$$\frac{2}{m}\, \frac{\partial L}{\partial X^{s}} = 2\, \epsilon_{abc}\, \dot{X}^{a}\, \Omega^{b}\, \delta^{c}{}_{s} + 2\, \Omega^{2}\, \delta_{as}\, X^{a} - 2\, (\boldsymbol{\Omega} \cdot \boldsymbol{X})\, \delta_{as}\, \Omega^{a} \tag{15.56a}$$

$$\frac{2}{m}\, \frac{\partial L}{\partial \dot{X}^{s}} = 2\, \dot{X}^{a}\, \delta_{as} + 2\, \delta^{a}{}_{s}\, (\boldsymbol{\Omega} \times \boldsymbol{X})_{a}. \tag{15.56b}$$

When the dust settles we find the Euler-Lagrange equation

$$\ddot{\boldsymbol{X}} = -2\, (\boldsymbol{\Omega} \times \boldsymbol{X}) - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{X}), \tag{15.57}$$

which has the expected Coriolis acceleration (first right hand side term) and centrifugal acceleration on the right hand side (see Chapter 12 and 13 for a detailed discussion of these accelerations).

In Section 15.6 we study motion on a geopoential in a rotating reference frame using spherical coordinates. Again, we find the methods of Lagrangian mechanics offer a straightforward approach that sidesteps many of the difficulties of the Newtonian approach.

15.4 The Foucault pendulum with an f-plane approximation

In this section we use Lagrangian mechanics to study motion of the Foucault pendulum, which is a frictionless spherical pendulum in a gravity field on a rotating planet. To illustrate the rotation of the planet, we setup the pendulum to oscillate in a fixed plane as viewed in an inertial reference frame, and as such it precesses when viewed from the planetary rotating reference frame. In this manner, the Foucault pendulum provides a remarkable illustration of the planetary Coriolis acceleration, and hence of the planetary rotation.

15.4.1 Assumptions about the Coriolis acceleration

We illustrate the Foucault pendulum in Figure 15.2, which is a spherical pendulum that feels the effects of the earth's rotation. For any point on the earth, the planetary rotation has a projection in the local vertical and local meridional directions (see Figure 12.4). However, for our study of the Foucault pendulum we only consider the local vertical component to the planetary rotation. This approximate approach accords with our study of large-scale fluid flows in this book. Even though the Foucault pendulum is tiny compared to the planetary fluid flows, the motion of the particle respects the same assumptions relevant for large-scale fluid motions. We detail these assumptions in the next paragraph.

As studied in Section 12.9.8, the meridional component to the planetary rotation leads to a Coriolis acceleration that has both a radial component and longitudinal component. The radial Coriolis acceleration is far smaller than the effective gravitational acceleration so that this contribution to the Coriolis acceleration can be ignored. The longitudinal Coriolis acceleration is determined by the vertical velocity of the particle. For the Foucault pendulum we assume the vertical velocity is much smaller than the horizontal velocity. Hence, this piece of the longitudinal Coriolis acceleration is negligible compared to the other portion that arises from horizontal motion. For these two reasons, the most significant portion of the Coriolis acceleration acting on the pendulum arises just from the local vertical component to the planetary rotation. As such, the Coriolis acceleration acts just in the local horizontal tangent plane.

Furthermore, since the horizontal length scales of the pendulum are tiny relative to the earth's radius, we assume the Coriolis parameter (equation (12.114))

$$f = 2\,\Omega \sin\phi, \tag{15.58}$$

to be constant and evaluated at the fixed point of the pendulum, which we refer to as the f-plane approximation. In equation (15.58), Ω is the angular frequency of the planetary rotation (Section 12.1), and ϕ is the planetary latitude (Figure 12.3).

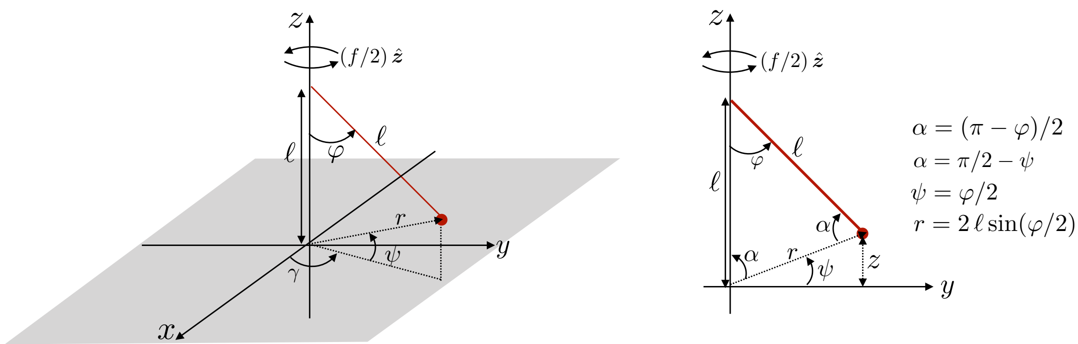

FIGURE 15.2: A Foucault pendulum is a frictionless spherical pendulum on a rotating planet. Taking the f-plane approximation, we assume rotation of the reference frame occurs around the local vertical axis with angular frequency $(f/2)\,\hat{\boldsymbol{z}}$, with $f = 2\,\Omega \sin\phi$ the Coriolis parameter and ϕ the latitude of the pendulum when at rest. To avoid confusion with the planetary spherical angles (where we use ϕ for latitude and λ for longitude as in Figure 12.3), we here use the local spherical coordinates with $\psi \geq 0$ the angle the pendulum makes with the horizontal x-y plane, and $0 \leq \gamma \leq 2\pi$ the angle it makes with the x-axis in a counter-clockwise sense. The angle made by the string relative to the vertical axis is $\varphi \geq 0$, just as for the simple pendulum in Figure 15.1. The right panel shows a projection onto the y-z plane ($\gamma = \pi/2$), which clearly illustrates the relation $\psi = \varphi/2$. Furthermore, the law of cosines expresses the radial position as a function of the angle, φ, and the string length, $r^2 = 2\,\ell^2\,(1 - \cos\varphi) = 4\,\ell^2 \sin^2(\varphi/2)$.

15.4.2 Position vector for the point mass

To avoid confusion with the planetary spherical coordinates of Figure 12.3, we write the pendulum's spherical angles as ψ (angle relative to the horizontal plane with $\psi \geq 0$ for the pendulum) and γ (angle relative to $\hat{\boldsymbol{x}}$, with $0 \leq \gamma \leq 2\pi$). In this manner, the coordinates for the point mass at the end of the string are given by

$$x = r \cos\psi \cos\gamma = 2\,\ell \sin(\varphi/2) \cos(\varphi/2) \cos\gamma = \ell \sin\varphi \cos\gamma \tag{15.59a}$$

$$y = r \cos\psi \sin\gamma = 2\,\ell \sin(\varphi/2) \cos(\varphi/2) \sin\gamma = \ell \sin\varphi \sin\gamma \tag{15.59b}$$

$$z = r \sin\psi = 2\,\ell \sin^2(\varphi/2) = \ell\,(1 - \cos\varphi). \tag{15.59c}$$

For these equations we made use of the law of cosines to write the radial distance from the origin, r, in terms of the string length, ℓ, and the angle of the string from the vertical, φ,

$$r = \ell \sqrt{2\,(1 - \cos\varphi)} = 2\,\ell \sin(\varphi/2). \tag{15.60}$$

Equations (15.59a)–(15.59c) provide an expression for the particle position

$$\boldsymbol{X}(t) = \ell\,[\hat{\boldsymbol{x}} \sin\varphi \cos\gamma + \hat{\boldsymbol{y}} \sin\varphi \sin\gamma + \hat{\boldsymbol{z}}\,(1 - \cos\varphi)], \tag{15.61}$$

which is written in terms of just the two angles, φ and γ, which serve as our generalized coordinates.

15.4.3 Lagrangian for the Foucault pendulum

We are describing motion of the particle within a rotating reference frame, with the horizontal axes rotating around the vertical as viewed in a local tangent plane on the planet. Hence, from the rigid-body rotation study in Section 11.3.3, we have (see equation (11.42)) the time change of the local horizontal Cartesian unit vectors

$$\dot{\hat{\boldsymbol{x}}} = (f/2)\,\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = (f/2)\,\hat{\boldsymbol{y}} \quad \text{and} \quad \dot{\hat{\boldsymbol{y}}} = (f/2)\,\hat{\boldsymbol{z}} \times \hat{\boldsymbol{y}} = -(f/2)\,\hat{\boldsymbol{x}}. \tag{15.62}$$

The corresponding velocity vector for the particle is

$$\begin{aligned}\dot{\boldsymbol{X}} &= \ell\,\dot{\varphi}\cos\varphi\,(\hat{\boldsymbol{x}}\cos\gamma + \hat{\boldsymbol{y}}\sin\gamma) + \ell\,\dot{\gamma}\sin\varphi\,(-\hat{\boldsymbol{x}}\sin\gamma + \hat{\boldsymbol{y}}\cos\gamma) \\ &\qquad + \ell\,\hat{\boldsymbol{z}}\,\dot{\varphi}\sin\varphi + (f/2)\,\hat{\boldsymbol{z}} \times \boldsymbol{X},\end{aligned} \tag{15.63}$$

with rearrangement leading to

$$\begin{aligned}\dot{\boldsymbol{X}} &= \ell\,\hat{\boldsymbol{x}}\,(\dot{\varphi}\cos\varphi\cos\gamma - (\dot{\gamma} + f/2)\sin\varphi\sin\gamma) \\ &\qquad + \ell\,\hat{\boldsymbol{y}}\,(\dot{\varphi}\cos\varphi\sin\gamma + (\dot{\gamma} + f/2)\sin\varphi\cos\gamma) + \ell\,\hat{\boldsymbol{z}}\,\dot{\varphi}\sin\varphi.\end{aligned} \tag{15.64}$$

Notice the appearance of $\dot{\gamma} + f/2$, which is the net angular frequency for motion around the vertical axis.

Computing the squared velocity and rearranging then leads to the particle's kinetic energy

$$K = (m\,\ell^2/2)\,[\dot{\varphi}^2 + (\dot{\gamma} + f/2)^2 \sin^2\varphi], \tag{15.65}$$

which arises from motion in the two angular directions plus the rigid-body motion of the rotating reference frame. The gravitational potential energy, referenced to the coordinate system origin, is given by

$$P = m\,g\,\ell\,(1 - \cos\varphi), \tag{15.66}$$

so that the Lagrangian is

$$L = K - P = (m\,\ell^2/2)\,[\dot{\varphi}^2 + (\dot{\gamma} + f/2)^2 \sin^2\varphi] + m\,g\,\ell\,(\cos\varphi - 1). \tag{15.67}$$

15.4.4 Euler-Lagrange equations of motion

The absence of an explicit appearance of γ in the Lagrangian (15.67) means that the γ component of the generalized momentum, $\mathcal{P}_\gamma$, is a constant of the motion

$$\frac{\mathrm{d}\mathcal{P}_\gamma}{\mathrm{d}t} = 0 \quad \text{with} \quad \mathcal{P}_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = m\,\ell^2\,(\dot{\gamma} + f/2)\sin^2\varphi. \tag{15.68}$$

We interpret $\mathcal{P}_\gamma$ as the axial angular momentum for the unconstrained rotational motion around the local vertical axis, with $\ell\sin\varphi$ the distance of the particle from the vertical axis and $\dot{\gamma} + f/2$ the net angular frequency for motion around the axis.[2] The φ equation is derived from

$$\frac{\partial L}{\partial \dot{\varphi}} = m\,\ell^2\,\dot{\varphi} \quad \text{and} \quad \frac{\partial L}{\partial \varphi} = m\,\ell^2\,(\dot{\gamma} + f/2)^2 \sin\varphi\cos\varphi - m\,g\,\ell\sin\varphi, \tag{15.69}$$

which leads to

$$\ddot{\varphi} + (g/\ell)\sin\varphi = (\dot{\gamma} + f/2)^2 \sin\varphi\cos\varphi. \tag{15.70}$$

[2] Compare this result to our study in Section 13.5 of axial angular momentum for planetary motion. In that discussion we considered motion around the planetary rotational axis. Here, the rotational axis for the Foucault pendulum is approximated by the local vertical, as per our assumptions discussed in Section 15.4.1.

This equation reduces to the planar pendulum equation (15.16) for the special case of motion within a vertical plane so that $\dot{\gamma} = 0$, and in the absence of planetary rotation so that $f = 0$. It also reduces to the planar pendulum for the case where $\dot{\gamma} = -f/2$, in which the pendulum contains zero axial angular momentum. We return to the case of zero axial angular momentum in Section 15.4.6.

15.4.5 Mechanical energy conservation

Since the Cartesian expression (15.61) for the position vector has no explicit time dependence, and neither does the Lagrangian (15.61), we know that the Hamiltonian equals to the mechanical energy and it is a constant of the motion

$$H = K + P = (m\,\ell^2/2)\,[\dot{\varphi}^2 + (\dot{\gamma} + f/2)^2 \sin^2\varphi] + m\,g\,\ell\,(1 - \cos\varphi). \tag{15.71}$$

Using the Euler-Lagrange equations of motion (15.68) and (15.70), it is straightforward to show that (see Exercise 15.3)

$$\frac{\mathrm{d}H}{\mathrm{d}t} = 0. \tag{15.72}$$

15.4.6 The case of zero axial angular momentum

Let us view the Foucault pendulum from a non-rotating inertial frame off the planet. Upon release from a position of rest, the inertial observer sees a pendulum that oscillates in a plane that is fixed with respect to the non-rotating inertial frame. In contrast, the rotating reference frame sees the pendulum's oscillation plane rotate oppositvely to the planet. The situation is directly analogous to the frictionless motion of a particle on a rotating table as illustrated in Figure 13.4. For the Foucault pendulum, the particle is constrained to oscillate on the end of a string. However, its motion couples to the rotation and so the rotating reference frame experiences a Coriolis acceleration. In picturing the motion, it is important to remember the assumption of zero friction, so that the apparatus holding the pendulum can rotate with the planet and yet not impart any frictional force to cause the pendulum to also rotate with the planet. In this manner, the pendulum maintains a fixed planar motion as viewed in the inertial reference frame.

Our thought experiment focuses on the special case of zero axial angular momentum (where, again, the axis here refers to the local vertical axis of the pendulum). In this case, the Euler-Lagrange equations of motion reduce to the relatively simple set

$$(\dot{\gamma} + f/2)\sin^2\varphi = 0 \quad \text{and} \quad \ddot{\varphi} + (g/\ell)\sin\varphi = 0. \tag{15.73}$$

The first equation is generally satisfied with $\dot{\gamma} = -f/2$, whereas the second equation is that for the planar pendulum from Section 15.1. The zero axial angular momentum motion decouples the evolution equations for γ and φ.

Evidently, with $\dot{\gamma} = -f/2$ then γ evolves as $\gamma = -(f/2)\,t$. We refer to this motion as a precession of the pendulum motion around the vertical rotational axis. The precession of γ means that the Foucault pendulum is seen by a rotating frame observer to have its oscillation plane rotate opposite to the planetary rotation (anti-cyclonically). In the northern hemisphere the pendulum's rotational plane precesses clockwise, whereas it precesses counter-clockwise in the southern hemisphere. Again, as per the motion of a frictionless ball on a rotating table seen in Figure 13.4, the precession can be interpreted as the persistent effects of the rightward deflection by the Coriolis acceleration in the northern hemisphere and leftward deflction in the southern (see Section 13.6). The period of the precession is

$$T^{\text{precession}} = \frac{2\,\pi}{|f|/2} = \frac{2\,\pi}{\Omega|\sin\phi|}. \tag{15.74}$$

Hence, a Foucault pendulum positioned at either poles (where the latitude, $\phi = \pm\pi/2$) exhibits a complete precession once per day, whereas at the equator ($\phi = 0$) there is no precession (formally an infinite period). Observe that the precession period is twice the period of inertial oscillations studied in Section 13.4.

15.4.7 Comments

Our treatment of Foucault's pendulum is distinguished from that given in Section 12 of *Fetter and Walecka* (2003) and Example 9-5 of *Marion and Thornton* (1988). First, we made use of Lagrangian mechanics whereas both of these alternative treatments use Newtonian mechanics. Consequently, we had no occasion to consider the forces acting within the string, as these forces of constraint are removed when using the angles φ and γ as generalized coordinates. Second, we made use of an approximate Coriolis acceleration as described in Section 15.4.1, where we only considered the local vertical component of the earth's rotation. The standard treatments reduce to this same approximate Coriolis acceleration, though they generally wait until later in the analysis to introduce the assumption. We found it useful to introduce the approximation earlier since it corresponds closely to how we treat the Coriolis acceleration for motion of the large-scale circulation in the ocean and atmosphere. It also simplifies the analysis.

Third, we focused on the zero axial angular momentum case, which trivially yields the precession rate $\dot{\gamma} = -f/2$. Since the axial angular momentum is a constant of the motion, we can set it to a zero value initially and it will remain so throughout the experiment (ignoring friction). It is this situation that is realized by a careful initial release of the Foucault pendulum so as to not introduce any lateral bias to the motion. In this manner, the spherical pendulum oscillates in a fixed plane when viewed by an inertial observer, while the planetary rotation occurs underneath the oscillating pendulum so that the planetary observer sees a precession of the oscillation plane.

15.5 Motion on constant potential energy surfaces

In this section we study motion constrained to move on surfaces of constant potential energy. Since the potential energy is a function of the coordinates (and not the velocity), the constraint is holonomic, which then means we can use generalized coordinates and/or Lagrange multipliers.

15.5.1 Anticipating the results

For each of the following examples, gravity provides the external conservative force and we ignore friction as well as any other external forces. The first case considers motion in the presence of a horizontally symmetric gravitational potential, which leads to a vertically directed gravitational acceleration. Motion in the horizontal directions is free (again, we ignore friction and any other forces). Hence, the two horizontal linear momentum components are constants of the motion, as is angular momentum around the vertical axis centered anywhere in the horizontal plane. Also, since the potential energy and the Lagrangian have no explicit time dependence, the Hamiltonian (mechanical energy) is a constant of the motion. Horizontal motion is supported by a force of constraint, such as provided by a frictionless flat table, with this force exactly balancing the weight of the particle.

We next study elements of the Kepler problem, which is concerned with particle motion in the presence of a spherically symmetric gravitational potential, such as generated by a spherical mass source. In this case the generalized coordinates are the two spherical angles (latitude and longitude), with the angular momentum around any axis a constant of the motion, as is the Hamiltonian. The force of constraint must counteract the gravitational force (pointing into the center) as well as the centrifugal force arising from the angular velocity (pointing away from the center). Indeed, particles with enough kinetic energy can have sufficient centrifugal acceleration to overcome the gravitational acceleration, in which case the force of constraint must keep the particle from flying outward.

15.5.2 Lagrangian using Cartesian coordinates

The unconstrained Lagrangian in Cartesian coordinates is given by

$$L = (m/2)\,(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - P(x, y, z), \tag{15.75}$$

where we assume the potential energy has no explicit time dependence, in which case the Hamiltonian is a constant of the motion (see Section 14.8.4)

$$\dot{H} = 0 \quad \text{with} \quad H = (m/2)\,(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + P. \tag{15.76}$$

Now introduce a Lagrange multiplier, Λ, to constrain the particle motion to a surface of constant potential energy, $P = \mathcal{C}$, so that the modified Lagrangian is

$$L^* = L + \Lambda\,(P - \mathcal{C}) = K - P\,(1 - \Lambda) - \Lambda\,\mathcal{C}. \tag{15.77}$$

The corresponding Euler-Lagrange equations of motion (14.109a) and (14.109b) are

$$m\,\ddot{\boldsymbol{X}} = -(1 - \Lambda)\,\nabla P \quad \text{and} \quad P = \mathcal{C}. \tag{15.78}$$

For particle motion along the $P = \mathcal{C}$ surface we know that the exact differential of the potential energy vanishes, which means that

$$\mathrm{d}P = \nabla P \cdot \mathrm{d}\boldsymbol{X} = 0 \Longrightarrow \nabla P \cdot \dot{\boldsymbol{X}} = 0. \tag{15.79}$$

This condition means that the velocity is parallel to constant potential energy surfaces, which it must be to maintain constant potential energy. Furthermore, taking the dot product of $\dot{\boldsymbol{X}}$ with the equation of motion (15.78), and using the constraint $\nabla P \cdot \dot{\boldsymbol{X}} = 0$, reveals that the kinetic energy is a constant of the motion

$$\dot{\boldsymbol{X}} \cdot \ddot{\boldsymbol{X}} = 0 \Longrightarrow \dot{K} = 0. \tag{15.80}$$

We can readily anticipate this result since with a constant potential energy and a constant Hamiltonian, then the kinetic energy is also a constant of the motion.

The Lagrange multiplier scales the force of constraint that counteracts the conservative force arising from the gradient of the potential energy. Making use of the general expression

(14.112) yields the force of constraint

$$Q_{\sigma}^{\text{constraint}} = \Lambda \frac{\partial P}{\partial x^{\sigma}} = -\Lambda F_{\sigma}, \tag{15.81}$$

where F_{σ} is the conservative force arising from the potential energy. Setting $\Lambda = 1$ ensures that the force of constraint acts to exactly oppose the conservative force, thus rendering a zero acceleration along the σ direction. The case of $\Lambda = 1$ is encountered for the particle moving along a horizontal geopotential. However, when the particle experiences any acceleration due to curved motion (e.g., motion around the sphere), then $\Lambda \neq 1$. The reason is that in addition to the conservative force from gravity, the particle encounters the centrifugal acceleration due to its angular motion around the sphere. If the sphere is itself rotating, as considered in Section 15.6, then there are additional accelerations (planetary centrifugal and planetary Coriolis) that must be accounted for when determining the force of constraint. These points will become clearer as we work through the various examples.

15.5.3 Plane symmetric potential energy

If the potential energy arises from a gravitational force field in the $-\hat{\boldsymbol{z}}$ direction, then gravitational potential energy is given by

$$P = m\, g\, z, \tag{15.82}$$

and the corresponding force of constraint is

$$Q_{z}^{\text{constraint}} = m\, g. \tag{15.83}$$

Evidently, the force of constraint acts vertically upward to exactly balance the downward weight of the particle so that the particle remains on a fixed z surface. A particle moving on a flat frictionless table provides the canonical example of this situation, where the table provides the force of constraint that balances the particle weight.

With $\Lambda = 1$ the horizontal motion has no acceleration

$$\ddot{x} = \ddot{y} = 0. \tag{15.84}$$

It is notable that all three coordinate directions have zero acceleration. However, motion in the horizontal directions is unconstrained and so the horizontal motion is free, whereas motion in the vertical direction is constrained due to the force of constraint balancing the weight.

15.5.4 Spherically symmetric potential energy

Now consider motion in the presence of a spherically symmetric potential, such as for a particle moving around a spherical earth, and describe the motion from an inertial (non-rotating) reference frame. We make use of spherical coordinates, with the transformation between planetary Cartesian coordinates and spherical coordinates given by equations (4.84a)-(4.84c)

$$x = r \cos\phi\, \cos\lambda \quad \text{and} \quad y = r\cos\phi\, \sin\lambda \quad \text{and} \quad z = r \sin\phi, \tag{15.85}$$

where $0 \leq \lambda \leq 2\pi$ is the longitude, and $-\pi/2 \leq \phi \leq \pi/2$ is the latitude (see Figure 12.3). The velocity components are given by

$$\dot{x} = \dot{r}\cos\phi\cos\lambda - r\,\dot{\phi}\sin\phi\cos\lambda - r\,\dot{\lambda}\cos\phi\sin\lambda \tag{15.86a}$$

$$\dot{y} = \dot{r}\cos\phi\sin\lambda - r\,\dot{\phi}\sin\phi\sin\lambda + r\,\dot{\lambda}\cos\phi\cos\lambda \tag{15.86b}$$

$$\dot{z} = \dot{r}\sin\phi + r\,\dot{\phi}\cos\phi, \tag{15.86c}$$

which then leads (after a bit of algebra) to the kinetic energy

$$K = (m/2)\,(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = (m/2)\,(\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,\dot{\lambda}^2\cos^2\phi). \tag{15.87}$$

We could have written this expression down by inspection if using the spherical coordinate expression (12.54d). However, the approach shown here is useful in general. Namely, we first write the kinetic energy in Cartesian coordinates, and then transform to coordinates that respect the symmetry of the system and thus provide suitable generalized coordinates.

Assuming the particle motion moves at a radius close to that of the mean earth radius allows us to approximate the gravitational potential energy according to equation (11.133)

$$P = m\,\Phi_{\mathrm{e}} = m\,g_{\mathrm{e}}\,r, \tag{15.88}$$

where $g_{\mathrm{e}} \approx 9.8$ m s^{-2} (Section 11.7.5). We are thus led to the Lagrangian

$$L = (m/2)\,(\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,\dot{\lambda}^2\cos^2\phi) - m\,g_{\mathrm{e}}\,r. \tag{15.89}$$

Since the Lagrangian and the potential energy are not explicit functions of time, then the Hamiltonian is a constant of the motion

$$\dot{H} = 0 \quad \text{with} \quad H = (m/2)\,(\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,\dot{\lambda}^2\cos^2\phi) + m\,g_{\mathrm{e}}\,r. \tag{15.90}$$

The Euler-Lagrange equations with a Lagrange multiplier

We next modify the Lagrangian by adding a Lagrange multiplier to constrain the motion to remain on a constant potential energy surface, $P = P(R)$, in which case

$$L^{*} = L + \Lambda\,[P - P(R)] = (m/2)\,(\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,\dot{\lambda}^2\cos^2\phi) - P\,(1-\Lambda) - \Lambda\,P(R). \tag{15.91}$$

The corresponding Euler-Lagrange equations are

$$\ddot{r} = r\,(\dot{\phi}^2 + \dot{\lambda}^2\cos^2\phi) - (1-\Lambda)\,g_{\mathrm{e}} \tag{15.92a}$$

$$\frac{\mathrm{d}(r^2\,\dot{\phi})}{\mathrm{d}t} = -r^2\,\dot{\lambda}^2\cos\phi\sin\phi \tag{15.92b}$$

$$\frac{\mathrm{d}(r^2\,\dot{\lambda}\cos^2\phi)}{\mathrm{d}t} = 0 \tag{15.92c}$$

$$r = R. \tag{15.92d}$$

The third equation corresponds to angular momentum conservation arising from the spherical symmetry. Satisfying the constraint, $r = R$, means that there is no radial acceleration, $\ddot{r} = 0$, and that the potential energy is fixed at the constant

$$P = m\,g_{\mathrm{e}}\,R. \tag{15.93}$$

Equation (15.92a) thus leads to

$$m\,R^2\,(\dot{\phi}^2 + \dot{\lambda}^2\cos^2\phi) = (1-\Lambda)\,P, \tag{15.94}$$

and the Hamiltonian (which is a constant of the motion) takes the form

$$H = m\, R^2\, (\dot{\phi}^2 + \dot{\lambda}^2 \cos^2\phi)/2 + P. \tag{15.95}$$

Making use of this constant Hamiltonian in equation (15.94) gives us

$$\Lambda = 3 - 2\, H/P = 1 - 2\, K/P, \tag{15.96}$$

with the kinetic energy a constant of the motion since both H and P are also constants.[3] The radial force of constraint is thus given by

$$Q_r = \Lambda\, \frac{\partial P}{\partial r} = (1 - 2\, K/P)\, m\, g_e. \tag{15.97}$$

Interpreting the force of constraint

To help interpret the force of constraint (15.97), write the kinetic energy in the form

$$2\, K = m\, R^2\, (\dot{\phi}^2 + \dot{\lambda}^2 \cos^2\phi) = m\, (v^2 + u^2) = m\, R\, A^{\text{centrifugal}}, \tag{15.98}$$

where we introduced the spherical velocity components according to equation (12.58), and defined the centrifugal acceleration arising from the angular motion of the particle around the sphere,

$$A^{\text{centrifugal}} = R\, (\dot{\phi}^2 + \dot{\lambda}^2 \cos^2\phi) = (u^2 + v^2)/R, \tag{15.99}$$

which is positive and points outward away from the axis of rotation. We can now write the force of constraint in the physically transparent manner

$$Q_r = m\, g_e\, (1 - A^{\text{centrifugal}}/g_e). \tag{15.100}$$

Evidently, if the gravitational acceleration is greater than the centrifugal acceleration, then the force of constraint is directed radially outward with a magnitude less than the weight of the particle. Here, the force of constraint supports the effective weight of the particle, with the outward directed centrifugal acceleration making the effective weight less than the resting weight. In contrast, if the centrifugal acceleration is greater than the gravitational acceleration, then the force of constraint must point radially inward to keep the particle on a constant radius, otherwise the particle will fly outward away from the planet. Finally, if the centrifugal acceleration equals to the gravitational acceleration, then the particle remains at a fixed radius with a zero force of constraint.

15.6 Motion on geopotential surfaces

We extend the results from Section 15.5 to now consider motion on geopotential surfaces that arise from central gravity acceleartion plus planetary centrifugal acceleration. For the rotating spherical planet, the geopotentials are oblate spheroidal shaped surfaces (Section 12.8.3), whereas they are parabolic for the case of a rotating plane (Section 15.6.3). The force of constraint accounts for the gravitational acceleration, the centrifugal acceleration from the particle motion, the centrifugal acceleration of the rotating reference frame, and a portion of

[3] Again, $H = K + P$ is a constant due to time symmetry whereas P is a constant due to the force of constraint that keeps the particle at a fixed radius.

the Coriolis acceleration. We greatly simplify the analysis by moving from spherical coordinates to geopotential coordinates as motivated by our work in Section 12.10.3.

15.6.1 Oblate spheroidally symmetric potential energy

Building from the analysis in Sections 15.5.3 and 15.5.4, consider a rotating spherical planet with constant angular rotation rate, $\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}$, directed through the vertical planetary Cartesian axis. We describe motion from the rotating reference frame, in which the planetary spherical coordinates and planetary Cartesian coordinates are fixed with respect to the rotating planet. This is the same convention taken in Chapter 12 and used throughout this book. As we show, the particle motion maintains a constant axial angular momentum (due to zonal symmetry), a constant Hamiltonian (due to time symmetry), and constant geopotential (due to the imposed constraint). Notably, however, the kinetic energy is not a constant of the motion.

The Lagrangian and the oblate spheroidal geopotential

The spherical coordinate expression for the particle's kinetic energy is

$$K = (m/2)\,(\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,(\dot{\lambda}+\Omega)^2\cos^2\phi). \tag{15.101}$$

The only change relative to equation (15.87) is the presence of Ω, which arises from the rigid-body rotation of the planet around the polar axis, and our choice to use spherical coordinates attached to the rotating planet. The gravitational potential energy is just like for the non-rotating case, thus leading to the Lagrangian

$$\begin{aligned}
L &= K - P && (15.102\text{a})\\
&= (m/2)\,(\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,(\dot{\lambda}+\Omega)^2\cos^2\phi) - m\,g_{\mathsf{e}}\,r && (15.102\text{b})\\
&= (m/2)\,[\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,\dot{\lambda}\,(\dot{\lambda}+2\,\Omega)\cos^2\phi] - m\,(g_{\mathsf{e}}\,r - (1/2)\,\Omega^2\,r^2\cos^2\phi) && (15.102\text{c})\\
&= K^{\mathsf{trunc}} - m\,\Phi. && (15.102\text{d})
\end{aligned}$$

The third equality introduced the truncated kinetic energy

$$K^{\mathsf{trunc}} = K - (m/2)\,\Omega^2\,r^2\cos^2\phi = (m/2)\,[\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,\dot{\lambda}\,(\dot{\lambda}+2\,\Omega)\cos^2\phi], \tag{15.103}$$

which is the kinetic energy absent the contribution from the centrifugal acceleration from the rigid-body planetary rotation. It is notable that K^{trunc} is not guaranteed to be non-negative. In equation (15.102d) we also introduced the geopotential for a rotating spherical planet

$$\Phi(r,\phi) = g_{\mathsf{e}}\,r - (1/2)\,\Omega^2\,r^2\cos^2\phi, \tag{15.104}$$

which we discussed in Section (12.8.3). In particular, surfaces of constant geopotential have larger radius at the equator than at the pole, which characterizes the earth's oblate spheroidal shape.

The Lagrange multiplier and Euler-Lagrange equations

Introducing a Lagrange multiplier to enforce the constraint of motion on a constant geopotential leads to the modified Lagrangian

$$L^{*} = L + \Lambda\,m\,(\Phi - \Phi^{\text{const}}) = K^{\mathsf{trunc}} - m\,\Phi\,(1-\Lambda) - \Lambda\,m\,\Phi^{\text{const}}, \tag{15.105}$$

where Φ^{const} is a constant. The corresponding Euler-Lagrange equations of motion are

$$\ddot{r} = r\,\dot{\phi}^2 + r\,\dot{\lambda}\,(\dot{\lambda} + 2\,\Omega)\cos^2\phi - (1-\Lambda)\,\partial\Phi/\partial r \tag{15.106a}$$

$$\frac{\mathrm{d}(r^2\,\dot{\phi})}{\mathrm{d}t} = -r^2\,\dot{\lambda}\,(\dot{\lambda} + 2\,\Omega)\cos\phi\sin\phi - (1-\Lambda)\,\partial\Phi/\partial\phi \tag{15.106b}$$

$$\frac{\mathrm{d}(r^2\,(\dot{\lambda} + \Omega)\cos^2\phi)}{\mathrm{d}t} = 0 \tag{15.106c}$$

$$\Phi = \Phi^{\text{const}}. \tag{15.106d}$$

In the absence of planetary rotation, with $\Omega = 0$, equations (15.106a)–(15.106d) reduce to the Euler-Lagrange equations (15.92a)–(15.92d) for motion around the non-rotating sphere. Furthermore, with some work, and in the absence of the geopotential constraint (i.e., $\Lambda = 0$), we can massage these equations to agree with the spherical coordinate equations (12.124)–(12.126) describing motion around a rotating sphere and derived using Newtonian methods.

Energy constants of the motion

Because the Lagrangian and the gravitational potential energy both have no explicit time dependence, the Hamiltonian, H, is a constant of the motion (see Section 14.8.4), where

$$H = K + m\,g_{\mathsf{e}}\,r = K^{\mathsf{trunc}} + (m/2)\,\Omega^2\,r^2\cos^2\phi + m\,g_{\mathsf{e}}\,r = K^{\mathsf{trunc}} + m\,\Phi + m\,\Omega^2\,r^2\cos^2\phi. \tag{15.107}$$

With the motion constrained to maintain a constant geopotential, $\dot{\Phi} = 0$ and $\dot{H} = 0$, then

$$\frac{\mathrm{d}(K + m\,g_{\mathsf{e}}\,r)}{\mathrm{d}t} = 0 \quad \text{and} \quad \frac{\mathrm{d}(K^{\mathsf{trunc}} + m\,\Omega^2\,r^2\cos^2\phi)}{\mathrm{d}t} = 0. \tag{15.108}$$

In contrast to the case of motion observed from a fixed (inertial) reference frame as studied in Section 15.5.4, the kinetic energy is here not a constant of the motion since the particle is not constrained to a constant spherical radius, $\dot{r} \neq 0$. Rather, the particle is constrained to a constant geopotential, which has an oblate spheroidal shape. This facet of the motion adds complexity to the description, which we alleviate in Section 15.6.2 by moving to geopotential coordinates. Until then, it is useful to offer some more details of the spherical coordinate formulation.

Velocity on the geopotential

Motion along the geopotential means that

$$\dot{\boldsymbol{X}} \cdot \nabla\Phi = 0, \tag{15.109}$$

so that the particle velocity is directed parallel to the geopotential surface. Writing the velocity in spherical coordinates according to equation (12.54d)

$$\dot{\boldsymbol{X}} = r\cos\phi\,(\dot{\lambda} + \Omega)\,\hat{\boldsymbol{\lambda}} + r\,\dot{\phi}\,\hat{\boldsymbol{\phi}} + \dot{r}\,\hat{\boldsymbol{r}}, \tag{15.110}$$

and using the spherical coordinate gradient operator (4.127)

$$\nabla = \hat{\boldsymbol{\lambda}}\,(r\,\cos\phi)^{-1}\,\partial_\lambda + \hat{\boldsymbol{\phi}}\,r^{-1}\,\partial_\phi + \hat{\boldsymbol{r}}\,\partial_r, \tag{15.111}$$

renders

$$\dot{\boldsymbol{X}} \cdot \nabla\Phi = \dot{\phi}\,\partial\Phi/\partial\phi + \dot{r}\,\partial\Phi/\partial r = 0, \tag{15.112}$$

which means that

$$r\,\Omega^2\cos\phi\,(r\,\dot{\phi}\sin\phi - \dot{r}\cos\phi) + \dot{r}\,g_{\mathsf{e}} = 0. \tag{15.113}$$

Evidently, any radial motion must be accompanied by meridional motion in order to remain on the geopotential. Furthermore, if there is no radial motion then there is no meridional motion, in which case the particle moves along a constant latitude circle.

15.6.2 Oblate spheroidally symmetric earth[4]

The analysis in Section 15.6.1 can be simplified by acknowledging that the earth has evolved to approximate the oblate spheroidal shape of the geopotential (15.104). Following *Veronis* (1973) and *Gill* (1982), and as detailed in Section 12.10.3 (see in particular Figure 12.5), we interpret the radial direction as aligned with the effective gravitational acceleration (central gravity plus planetary centrifugal), so that the geopotential takes on the much simpler form

$$\Phi = g\,r, \tag{15.114}$$

where g is the effective gravitational acceleration and geopotential surfaces are now surfaces of constant r. In this manner the constrained Lagrangian is

$$L = (m/2)\,(\dot{r}^2 + r^2\,\dot{\phi}^2 + r^2\,\dot{\lambda}\,(\dot{\lambda} + 2\,\Omega)\cos^2\phi) - m\,g\,r + m\,\Lambda\,(\Phi - \Phi^{\text{const}}). \tag{15.115}$$

This Lagrangian is nearly the same as for the non-rotating sphere given by equation (15.91), with the exception of the $2\,\Omega$ term. The force of constraint is required to keep the particle moving on a constant geopotential, which here means the particle maintains a fixed $r = R$.

To examine the force of constraint, we generate the Euler-Lagrange equations

$$\ddot{r} = r\,\dot{\phi}^2 + r\,\dot{\lambda}\,(\dot{\lambda} + 2\,\Omega)\cos^2\phi - (1 - \Lambda)\,g \tag{15.116a}$$

$$\frac{\mathrm{d}(r^2\,\dot{\phi})}{\mathrm{d}t} = -r^2\,\dot{\lambda}\,(\dot{\lambda} + 2\,\Omega)\cos\phi\sin\phi \tag{15.116b}$$

$$\frac{\mathrm{d}(r^2\,(\dot{\lambda} + \Omega)\cos^2\phi)}{\mathrm{d}t} = 0 \tag{15.116c}$$

$$\Phi = \Phi^{\text{const}} = g\,R. \tag{15.116d}$$

With $r = R$ for the constrained motion, the radial equation (15.116a) allows us to solve for the Lagrange multiplier

$$\Lambda = 1 - (R/g)\,(\dot{\phi}^2 + \dot{\lambda}^2\cos^2\phi + 2\,\Omega\,\dot{\lambda}\cos^2\phi) = 1 - g^{-1}\,(A^{\mathsf{centrifugal}} + A^{\mathsf{Coriolis}}), \tag{15.117}$$

where we introduced the centrifugal acceleration arising from the angular particle motion (as for the non-rotating sphere in equation (15.99))

$$A^{\mathsf{centrifugal}} = R\,(\dot{\phi}^2 + \dot{\lambda}^2\cos^2\phi), \tag{15.118}$$

[4] The analysis in this section is inspired by *Early* (2012).

as well as the Coriolis acceleration

$$A^{\text{Coriolis}} = 2\,\Omega\,R\,\dot{\lambda}\cos^2\phi, \tag{15.119}$$

which is positive for eastward motion and negative if westward. The force of constraint is thus given by

$$Q_r = \Lambda\,\frac{\partial P}{\partial r} = m\,g\,[1 - (A^{\text{centrifugal}} + A^{\text{Coriolis}})/g]. \tag{15.120}$$

Compared to the non-rotating case in equation (15.100), we here see the inclusion of the Coriolis acceleration contribution as well as the minor difference between g_{e} and g.

15.6.3 An f-plane geopotential surface

We now return to planar symmetry as in Section 15.5.3, here examining motion of a particle as viewed in a rotating reference frame with constant angular rotation rate, $(f/2)\,\hat{\boldsymbol{z}}$. Furthermore, we assume the motion is on an f-plane (VOLUME 2), which means that surfaces of constant geopotential from Section 15.6.2 are now approximated as horizontal, and it is only the vertical component of planetary rotation that is considered.

This particle has a kinetic energy

$$K = (m/2)\,(\dot{x}_{\text{I}}^2 + \dot{y}_{\text{I}}^2 + \dot{z}_{\text{I}}^2) = (m/2)\,[(\dot{x} - f\,y/2)^2 + (\dot{y} + f\,x/2)^2 + \dot{z}^2], \tag{15.121}$$

where

$$\dot{X}_{\text{I}} = \dot{\boldsymbol{X}} + (f/2)\,\hat{\boldsymbol{z}} \times \boldsymbol{X} \tag{15.122}$$

relates the velocity measured in the inertial reference frame, $\dot{\boldsymbol{X}}_{\text{I}}$, to the velocity, $\dot{\boldsymbol{X}}$, measured in the rotating reference frame (see Section 11.3.3). Expanding the kinetic energy leads to

$$K = (m/2)\,[\dot{x}\,(\dot{x} - f\,y) + \dot{y}\,(\dot{y} + f\,x) + \dot{z}^2] + (m/2)\,(f/2)^2\,(x^2 + y^2). \tag{15.123}$$

We are thus led to the Lagrangian

$$L^{\text{cartesian}} = (m/2)\,[\dot{x}\,(\dot{x} - f\,y) + \dot{y}\,(\dot{y} + f\,x) + \dot{z}^2] - m\,\Phi^{\text{parabolic}}, \tag{15.124}$$

where we introduced the geopotential

$$\Phi^{\text{parabolic}} = g_{\text{e}}\,z - (1/2)\,(f/2)^2\,(x^2 + y^2), \tag{15.125}$$

whose isosurfaces are parabolic.

Following the geopotential coordinate approach from Section 15.6.2, we reinterpret the local vertical direction as perpendicular to surfaces of constant $\Phi^{\text{parabolic}}$, in which case the Lagrangian (15.124) simplifies to

$$L = (m/2)\,[\dot{x}\,(\dot{x} - f\,y) + \dot{y}\,(\dot{y} + f\,x) + \dot{z}^2] - m\,\Phi, \tag{15.126}$$

with the geopotential

$$\Phi = g\,z. \tag{15.127}$$

To constrain the particle motion to remain on a constant geopotential, $\Phi = \Phi^{\text{const}}$, we introduce a Lagrange multiplier to produce the modified Lagrangian

$$L^{*} = (m/2)\,[\dot{x}\,(\dot{x} - f\,y) + \dot{y}\,(\dot{y} + f\,x) + \dot{z}^{2}] - m\,\Phi + \Lambda\,(\Phi - \Phi^{\text{const}}), \tag{15.128}$$

which generates the Euler-Lagrange equations

$$\frac{\mathrm{d}(\dot{x} - f\,y)}{\mathrm{d}t} = 0 \tag{15.129a}$$

$$\frac{\mathrm{d}(\dot{y} + f\,x)}{\mathrm{d}t} = 0 \tag{15.129b}$$

$$\frac{\mathrm{d}\dot{z}}{\mathrm{d}t} = -(1 - \Lambda)\,g \tag{15.129c}$$

$$\Phi = \Phi^{\text{const}}. \tag{15.129d}$$

To maintain zero vertical acceleration requires a unit Lagrange multiplier

$$\Lambda = 1, \tag{15.130}$$

which was also found in Section 15.5.3 for motion on a flat non-rotating plane. As for that case, we here find that the force of constraint must exactly balance the weight of the particle. Contrary to the full earth case in Section 15.6.2, there is no contribution to the force of constraint from the Coriolis acceleration. The reason is that we here only consider the local vertical component of the earth's rotation, whereas in Section 15.6.2 we made no such approximation.

The horizontal equations of motion (15.129a) and (15.129b) describe the f-plane inertial oscillations studied in Section 13.4. This horizontal motion is that of a free particle on an f-plane and as observed in the rotating reference frame. Consistent with the motion being free in the horizontal, it maintains a constant kinetic energy

$$K = m\,(\dot{x}^{2} + \dot{y}^{2})/2 \quad \text{and} \quad \dot{K} = m\,(\dot{x}\,\ddot{x} + \dot{y}\,\ddot{y}) = m\,f\,(\dot{x}\,\dot{y} - \dot{y}\,\dot{x}) = 0. \tag{15.131}$$

15.7 Harmonic oscillator

We here study the mechanics of harmonic oscillators, starting with a single harmonic oscillator and then considering a line of coupled oscillators. The continuum limit of the coupled oscillators provides a mechanical interpretation of longitudinal waves, such as the acoustic waves studied in VOLUME 4. The continuum limit also serves the starting point for the Lagrangian mechanics of continuous media as in VOLUME 4.

15.7.1 Physical picture

Consider a tiny piece of matter (idealized as a point) with constant mass, m. Assume this mass is constrained to move along one direction on a frictionless table. Equivalently, assume the mass moves in one direction in a vacuum without any force. Let the coordinate position, x, of the mass be measured by the trajectory, $x = X(t)$, and apply a horizontal force to the mass that is a function just of the position, $F(x)$. Newton's equation of motion for the point mass is given by

$$m\,\ddot{X} = F(x), \tag{15.132}$$

where $F(x)$ is evaluated at the horizontal position of the mass, $x = X(t)$. Now assume the point mass only moves a small distance from its equilibrium position, $x = \Delta$, which is defined by the position where the force vanishes, $F(\Delta) = 0$. We thus approximate the horizontal force by its leading order Taylor expansion

$$F(x) \approx F(\Delta) + (x - \Delta) \left[\frac{\mathrm{d}F(x)}{\mathrm{d}x} \right]_{x=\Delta} = (x - \Delta) \left[\frac{\mathrm{d}F(x)}{\mathrm{d}x} \right]_{x=\Delta}, \tag{15.133}$$

where higher order terms are dropped, and we set $F(\Delta) = 0$.

15.7.2 Oscillations from a Hooke's law restoring force

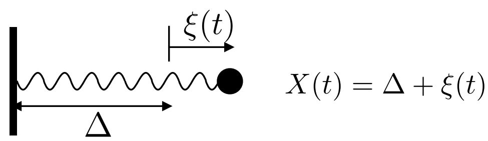

FIGURE 15.3: A simple harmonic oscillator as realized by an object of fixed mass m (approximated as a point mass) attached to a massless spring with motion in just one direction. The only force acting on the mass arises from the spring, whose spring constant, Γ, is constant. The trajectory of the point mass, $X(t) = \Delta + \xi(t)$, measures the position relative to the wall on the left, with Δ the equilibrium position (where the spring force vanishes), and $\xi(t)$ the displacement from the equilibrium position. We here depict the system where the spring is extended to the right so that $\xi(t) > 0$. In this case, the restoring force from the spring accelerates the mass to the left.

Now specialize the force to be restorative so that

$$F(x) = -\Gamma\,(x - \Delta) \quad \text{with} \quad \Gamma = \left[\frac{\mathrm{d}F(x)}{\mathrm{d}x} \right]_{x=\Delta} > 0. \tag{15.134}$$

This linear force provides the canonical Hooke's law force, such as realized by an idealized massless spring as depicted in Figure 15.3. Introducing the displacement relative to an equilibrium position (see Figure 15.3)

$$\xi(t) = X(t) - \Delta, \tag{15.135}$$

along with the Hooke's law restoring force (15.134), we find Newton's law for the particle trajectory takes the form of the linear oscillator equation

$$m\,\ddot{X} + \Gamma\,(X - \Delta) = m\,\ddot{\xi} + \Gamma\,\xi = 0 \Longrightarrow \ddot{\xi} + \omega_{\circ}^2\,\xi = 0 \quad \text{with} \quad \omega_{\circ}^2 = \Gamma/m. \tag{15.136}$$

A solution to the linear oscillator equation can be written

$$X(t) - \Delta = \xi(t) = A\cos(\omega_{\circ}\,t + \alpha), \tag{15.137}$$

where A is the oscillation amplitude and α a phase shift, with both A and α time independent and specified by initial conditions. We thus find that the Hooke's law restoring force leads to an oscillatory trajectory (15.137) centered on the equilibrium position, $x = \Delta$. The angular frequency, $\omega_{\circ}$, determines the period of oscillation according to

$$T^{\text{per}} = 2\pi/\omega_{\circ} = 2\pi\,(m/\Gamma)^{1/2}. \tag{15.138}$$

The period increases with the square root of the mass (longer period for larger mass) and decreases with the square root of the Hooke's law restoring constant (shorter period for larger

Hooke's law constant).

15.7.3 Hamiltonian as a constant of the motion

The Hooke's law force has no explicit dependence on time, $\partial F/\partial t = 0$, so that the oscillating point mass has no concern for the time origin. From our discussion of time symmetry in Section 14.8.4, we know that the simple harmonic oscillator maintains a constant Hamiltonian, which is here also equal to the mechanical energy. A constancy of the Hamiltonian via Noether's theorem then means that the oscillator exchanges mechanical energy between its kinetic energy and potential energy while holding their sum constant.

Potential energy and kinetic energy of the oscillator

During its motion, the point mass experiences work done by the spring and this work is given by

$$W = -\int_{\Delta}^{\Delta+\xi} F \,\mathrm{d}x = \Gamma \int_{\Delta}^{\Delta+\xi} (x - \Delta)\,\mathrm{d}x = \Gamma\,\xi^2/2. \tag{15.139}$$

This work to displace the mass renders a potential energy for the point mass relative to the zero potential energy it possesses when located at its equilibrium position with $\xi = 0$. Through the work-energy theorem from Section 11.2.4, we find that temporal changes in the potential energy are associated with temporal changes in kinetic energy

$$W = -m \int_{\Delta}^{\Delta+\xi} \ddot{X}\,\mathrm{d}x = -m \int_{t_0}^{t} \ddot{X}\,\dot{X}\,\mathrm{d}t = -\frac{m}{2}\int_{t_0}^{t} \frac{\mathrm{d}\dot{X}^2}{\mathrm{d}t}\,\mathrm{d}t = -K(t) + K(t_0), \tag{15.140}$$

where

$$K = m\,\dot{X}^2/2 = m\,\dot{\xi}^2/2 \tag{15.141}$$

is the kinetic energy for the oscillating particle. Hence, as the point mass oscillates, changes to its potential energy are exactly compensated by equal and opposite changes to its kinetic energy, thus reflecting the constant mechanical energy for the harmonic oscillator. Another way to see that the mechanical energy remains constant is to note that the sum of the potential plus kinetic energies has a zero time derivative

$$(\mathrm{d}/\mathrm{d}t)\,[m\,\dot{\xi}^2 + \Gamma\,\xi^2]/2 = \dot{\xi}\,[m\,\ddot{\xi} + \Gamma\,\xi] = 0, \tag{15.142}$$

where the final equality made use of the equation of motion (15.136). Indeed, making use of the trajectory (15.137) we readily see that the mechanical energy is given by the constant

$$[m\,\dot{\xi}^2 + \Gamma\,\xi^2]/2 = (A^2/2)\,[m\,\omega_\circ^2 \sin^2(\omega_\circ\,t + \alpha) + \Gamma\cos^2(\omega_\circ\,t + \alpha)] = \Gamma\,A^2/2, \tag{15.143}$$

where we set $\omega_\circ^2 = \Gamma/m$ as per equation (15.136). The total energy is thus proportional to the square of the amplitude and linearly proportional to the Hooke's law constant.

15.7.4 Equipartition of time averaged energies

When time averaged over a single oscillation period of length $2\pi/\omega_\circ$, the averaged kinetic energy and potential energy are identical

$$\frac{\omega_\circ}{2\pi}\int_0^{2\pi/\omega_\circ} (m\,\dot{\xi}^2/2)\,\mathrm{d}t = \Gamma\,A^2/4 \tag{15.144a}$$

$$\frac{\omega_\circ}{2\pi}\int_0^{2\pi/\omega_\circ}(\Gamma\,\xi^2/2)\,\mathrm{d}t = \Gamma\,A^2/4. \tag{15.144b}$$

Energy equipartition means that within a single oscillation period, there is an exact exchange between kinetic energy and potential energy, so that their time averages are identical.

The equipartition of energy is a rather generic property of linear oscillating systems, and we encounter it again for linear waves in VOLUME 4. It follows as a general result of the virial theorem proved in Section 14.6.3. In particular, it holds for systems in which the potential energy is a homogeneous function of degree two (see Section 14.6.1).

15.7.5 Lagrangian formulation

Having established the potential energy (15.139) for the oscillator, we can write the Lagrangian as

$$L = \tfrac{1}{2}(m\,\dot{\xi}^2 - \Gamma\,\xi^2) = \tfrac{m}{2}(\dot{\xi}^2 - \omega_\circ^2\,\xi^2), \tag{15.145}$$

with ξ a suitable generalized coordinate that captures the single degree of freedom for the oscillator. The Euler-Lagrange equation is

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial\dot{\xi}} = \frac{\partial L}{\partial\xi} \Longrightarrow \ddot{\xi} + \omega_\circ^2\,\xi = 0, \tag{15.146}$$

which is the same as equation (15.136) derived from Newtonian methods.

15.7.6 Further study

The study of harmonic oscillators can be found in most classical mechanics texts, with our treatment following Sections 3.1 and 3.2 of *Marion and Thornton* (1988).

15.8 Coupled harmonic oscillators

We here extend the study of a single harmonic oscillator in Section 15.7 to the case of N identical frictionless point mass particles connected by massless linear Hooke's law springs with identical spring constant, Γ. As depicted in Figure 15.4, let

$$X_{\mathsf{n}}(t) = \mathsf{n}\,\Delta + \xi_{\mathsf{n}}(t) \tag{15.147}$$

be the position of particle, n, relative to the left-most rigid wall, with $x = \mathsf{n}\,\Delta$ the equilibrium position for this particle and $\xi_{\mathsf{n}}(t)$ the displacement of the particle at time t from its equilibrium. The ξ_{n} displacements serve as generalized coordinates for this system with N degrees of freedom.[5]

To develop the equation of motion, observe that the force acting on a particle is comprised of two terms associated with the two springs, one on the left and one on the right. Consider first the end particles where one of its springs is attached to a rigid wall. For convenience, we label a point at the left wall as $\mathsf{n} = 0$ and $\mathsf{n} = N + 1$ for a point on the right wall. For the $\mathsf{n} = 1$

[5] In Section 14.3.1 that we wrote $X_{(i)}$ for the Cartesian position of particle i. This notation is important to distinguish the particle label from a tensor index. In this section, we instead use the upright n to write X_{n} to distinguish the particle label, n, from a tensor index, n. Additionally, in this section we are only concerned with motion along a single direction, so that any label is a particle label.

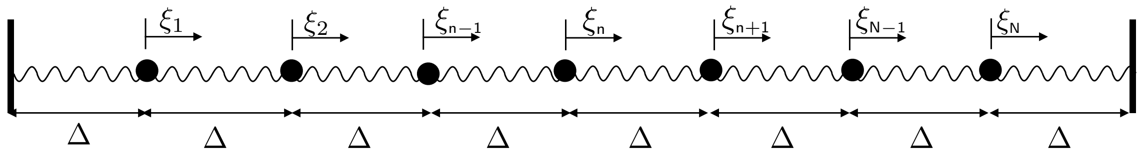

FIGURE 15.4: A line of $N = 7$ coupled simple harmonic oscillators as realized by identical point objects with fixed mass, m, each attached to two massless springs and moving in one direction. The only horizontal force acting on the masses arises from the springs, each with identical spring constants, Γ, and equilibrium length, Δ. The trajectory of any particular mass, $X_{\mathsf{n}}(t)$, measures the position relative to the equilibrium at $X_{\mathsf{n}}(t) = 0$, where the force from the springs vanishes. The system is here shown in its equilibrium where the distance between each mass is Δ, and the displacments all vanish, $\xi_{\mathsf{n}} = 0$. Each point mass is attached to two springs, with the end springs attached to a rigid wall.

mass we have the force from the left spring given by

$$F_{\mathsf{n}=0\to\mathsf{n}=1} = -\Gamma\,\xi_1, \tag{15.148}$$

just as for a single harmonic oscillator considered in Section 15.7.2. The force on particle 1 from the right spring is given by

$$F_{\mathsf{n}=2\to\mathsf{n}=1} = -\Gamma\,(\xi_1 - \xi_2). \tag{15.149}$$

We understand the nature of this force by noting that if $\xi_1 > 0$ then the spring is compressed to the right, so that the restoring force is to the left as the spring expands. Conversely, if $\xi_2 > 0$ then the spring is expanded to the right, in which case the particle is itself accelerated to the right. Hence, the equation of motion for this mass is given by

$$m\,\ddot{\xi}_1 = -\Gamma\,\xi_1 - \Gamma\,(\xi_1 - \xi_2). \tag{15.150}$$

Anticipating the force balance that acts on the interior masses, we introduce two dummy deviations, each with fixed value of zero,

$$\xi_0 \equiv 0 \quad \text{and} \quad \xi_{\mathsf{N}+1} \equiv 0. \tag{15.151}$$

We thus have the equation of motion for the $\mathsf{n} = 1$ point mass given by

$$M\,\ddot{\xi}_1 = -\Gamma\,(\xi_1 - \xi_0) - \Gamma\,(\xi_1 - \xi_2), \tag{15.152}$$

and analogous considerations hold for the $\mathsf{n} = N$ point mass

$$m\,\ddot{\xi}_{\mathsf{N}} = -\Gamma\,(\xi_{\mathsf{N}} - \xi_{\mathsf{N}-1}) - \Gamma\,(\xi_{\mathsf{N}} - \xi_{\mathsf{N}+1}). \tag{15.153}$$

The interior masses have analogous equations so that the generic equation of motion for particles $\mathsf{n} = 1, \mathsf{N}$ is given by

$$m\,\ddot{\xi}_{\mathsf{n}} = -\Gamma\,(\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1}) - \Gamma\,(\xi_{\mathsf{n}} - \xi_{\mathsf{n}+1}) \Longrightarrow \ddot{\xi}_{\mathsf{n}} = \omega_{\circ}^2\,(\xi_{\mathsf{n}+1} - 2\,\xi_{\mathsf{n}} + \xi_{\mathsf{n}-1}). \tag{15.154}$$

It is useful to write the case for $\mathsf{N} = 2$ oscillators for reference

$$\ddot{\xi}_1 = \omega_{\circ}^2\,(-2\,\xi_1 + \xi_2) \tag{15.155a}$$

$$\ddot{\xi}_2 = \omega_{\circ}^2\,(-2\,\xi_2 + \xi_1), \tag{15.155b}$$

as well as for $\mathsf{N} = 3$ oscillators

$$\ddot{\xi}_1 = \omega_\circ^2 \, (-2\,\xi_1 + \xi_2) \tag{15.156a}$$
$$\ddot{\xi}_2 = \omega_\circ^2 \, (\xi_3 - 2\,\xi_2 + \xi_1) \tag{15.156b}$$
$$\ddot{\xi}_3 = \omega_\circ^2 \, (-2\,\xi_3 + \xi_2). \tag{15.156c}$$

15.8.1 Mechanical energy for coupled oscillators

Following our discussion of mechanical energy for a single oscillator in Section 15.7.3, we multiply the equation of motion (15.154) by $\dot{\xi}_{\mathsf{n}}$ and sum over the $\mathsf{n} = 1, N$ oscillators. The acceleration term leads to the time change for the total kinetic energy of the full coupled oscillator system

$$m \sum_{\mathsf{n}=1}^{\mathsf{N}} \dot{\xi}_{\mathsf{n}} \, \ddot{\xi}_{\mathsf{n}} = \frac{m}{2} \sum_{\mathsf{n}=1}^{\mathsf{N}} \frac{\mathrm{d}\dot{\xi}_{\mathsf{n}}^2}{\mathrm{d}t}. \tag{15.157}$$

Summation over the left portion of the force in equation (15.154) can be written

$$\sum_{\mathsf{n}=1}^{\mathsf{N}} \dot{\xi}_{\mathsf{n}} \, (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1}) = \sum_{\mathsf{n}=1}^{\mathsf{N}+1} \dot{\xi}_{\mathsf{n}} \, (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1}), \tag{15.158}$$

which follows since $\xi_{\mathsf{N}+1} \equiv 0$ so that the extra term in the summation vanishes. The summation over the right portion of the force in equation (15.154) can be written

$$\sum_{\mathsf{n}=1}^{\mathsf{N}} \dot{\xi}_{\mathsf{n}} \, (\xi_{\mathsf{n}} - \xi_{\mathsf{n}+1}) = \sum_{\mathsf{n}=2}^{\mathsf{N}+1} \dot{\xi}_{\mathsf{n}-1} \, (\xi_{\mathsf{n}-1} - \xi_{\mathsf{n}}) = \sum_{\mathsf{n}=1}^{\mathsf{N}+1} \dot{\xi}_{\mathsf{n}-1} \, (\xi_{\mathsf{n}-1} - \xi_{\mathsf{n}}). \tag{15.159}$$

To reach the first equality we changed indices on the terms in the summation, and modified the summation limits accordingly. For the second equality we expanded the summation range by noting that $\xi_0 \equiv 0$, thus allowing us to bring the lower summation limit from $\mathsf{n} = 2$ to $\mathsf{n} = 1$. Combining equations (15.158) and (15.159) leads to

$$\Gamma \sum_{\mathsf{n}=1}^{\mathsf{N}} \dot{\xi}_{\mathsf{n}} \, (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1}) + \Gamma \sum_{\mathsf{n}=1}^{\mathsf{N}} \dot{\xi}_{\mathsf{n}} \, (\xi_{\mathsf{n}} - \xi_{\mathsf{n}+1}) = \frac{\Gamma}{2} \frac{\mathrm{d}}{\mathrm{d}t} \sum_{\mathsf{n}=1}^{\mathsf{N}+1} (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1})^2, \tag{15.160}$$

which is the time derivative of the potential energy, P, arising from the expansion and contraction of the Hooke's law springs, where

$$P = \frac{\Gamma}{2} \sum_{\mathsf{n}=1}^{\mathsf{N}+1} (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1})^2 = \frac{\Gamma}{2} \left[\xi_1^2 + \sum_{\mathsf{n}=2}^{\mathsf{N}} (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1})^2 + \xi_{\mathsf{N}}^2 \right], \tag{15.161}$$

where we set $\xi_0 = 0$ and $\xi_{\mathsf{N}+1} = 0$.

Bringing the kinetic energy and potential energy together leads to the conservation of mechanical energy for the coupled oscillator system

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{\mathsf{n}=1}^{\mathsf{N}+1} \left[\frac{m}{2} \, \dot{\xi}_{\mathsf{n}}^2 + \frac{\Gamma}{2} \, (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1})^2 \right] = 0. \tag{15.162}$$

Just as for the single oscillator, we see that the coupled oscillator system maintains a fixed mechanical energy in which energy is exchanged between kinetic and potential energy reservoirs.

15.8.2 Lagrangian formulation

The Lagrangian for the coupled oscillator system is given by

$$L = \sum_{\mathsf{n}=1}^{\mathsf{N}+1} \left[\frac{m}{2} (\dot{\xi}_{\mathsf{n}})^2 - \frac{\Gamma}{2} (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1})^2 \right]. \tag{15.163}$$

To derive the Euler-Lagrange equations we require the following derivatives

$$\frac{\partial L}{\partial \dot{\xi}_{\mathsf{p}}} = m \sum_{\mathsf{n}=1}^{\mathsf{N}+1} \dot{\xi}_{\mathsf{n}} \, \delta_{\mathsf{n},\mathsf{p}} = m \, \dot{\xi}_{\mathsf{p}}, \tag{15.164a}$$

$$\frac{\partial L}{\partial \xi_{\mathsf{p}}} = \Gamma \sum_{\mathsf{n}=1}^{\mathsf{N}+1} (\xi_{\mathsf{n}} - \xi_{\mathsf{n}-1}) \, (\delta_{\mathsf{n},\mathsf{p}} - \delta_{\mathsf{n}-1,\mathsf{p}}) = -\Gamma \, (\xi_{\mathsf{p}+1} - 2 \, \xi_{\mathsf{p}} + \xi_{\mathsf{p}-1}), \tag{15.164b}$$

which lead to the Euler-Lagrange equation

$$\ddot{\xi}_{\mathsf{p}} = \omega_{\circ}^2 \, (\xi_{\mathsf{p}+1} - 2 \, \xi_{\mathsf{p}} + \xi_{\mathsf{p}-1}). \tag{15.165}$$

As expected, the Euler-Lagrange equation of motion agrees with Newton's equation (15.154) derived using Newtonian methods.

15.8.3 Further study

Our presentation of coupled harmonic oscillators was inspired by Section 24 of *Fetter and Walecka* (2003), though we made use of a kinematic treatment anticipating the generalized Lagrangian mean approach used in fluid mechanics and introduced in VOLUME 4. The continuum limit of coupled oscillators serves as the starting point for our treatment of classical field theory in VOLUME 4.

15.9 Exercises

EXERCISE 15.1: ON-SHELL ACTION FOR THE FREE PARTICLE
Evaluate the on-shell action, $\mathcal{W}(\boldsymbol{\xi}, t)$, for a free particle moving in three-dimensional space (recall Section 14.9.8). That is, evaluate the action using the solution to the Euler-Lagrange equations. Express the result solely in terms of the initial position and time, $\boldsymbol{X}(t_{\mathsf{A}}), t_{\mathsf{A}}$, along with the final position and time, $\boldsymbol{X}(t_{\mathsf{B}}), t_{\mathsf{B}}$. Verify the partial derivatives (14.183)

$$\frac{\partial \mathcal{W}}{\partial t} = -H(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}}, t) \quad \text{and} \quad \frac{\partial \mathcal{W}}{\partial \xi^{\sigma}} = \mathcal{P}_{\sigma}. \tag{15.166}$$

EXERCISE 15.2: ON-SHELL ACTION FOR THE SIMPLE HARMONIC OSCILLATOR (a) Evaluate the on-shell action, $\mathcal{W}(\boldsymbol{\xi}, t)$, for a simple harmonic oscillator with mass, m, and angular frequency, $\omega_{\circ}$. That is, evaluate the action using the solution to the Euler-Lagrange equations. Write the solution in terms of the prescribed trajectory values, $\xi(t_{\mathsf{A}})$, $\xi(t_{\mathsf{B}})$, as well as the prescribed time difference, $T = t_{\mathsf{B}} - t_{\mathsf{A}}$.

(b) Verify the partial derivatives (14.183)

$$\frac{\partial \mathcal{W}}{\partial t} = -H(\boldsymbol{\xi}, \boldsymbol{\mathcal{P}}, t) \quad \text{and} \quad \frac{\partial \mathcal{W}}{\partial \xi^{\sigma}} = \mathcal{P}_{\sigma}. \tag{15.167}$$

For this part, choose time so that $t_{\mathsf{A}} = 0$ and $\xi(0) = 0$, which loses no generality since the solution is periodic.

EXERCISE 15.3: CONSERVATION OF ENERGY FOR THE FOUCAULT PENDULUM
Using the Euler-Lagrange equations of motion (15.68) and (15.70), show that $\dot{H} = 0$ for the Foucault pendulum, where the Hamiltonian is given by equation (15.71).

EXERCISE 15.4: EQUATIONS FOR THE TWO-BODY PROBLEM WITH GRAVITY
In this exercise we use Lagrangian mechanics to formulate the equations of motion for two gravitating massive spherical bodies. This problem is the foundation for Kepler's laws for orbital mechanics. We do not pursue the problem to the stage of obtaining a solution, rather we are only interested in the formulation of the equations of motion. The equations derived in this exercise form the starting point for the analysis of the Kepler problem of planetary orbits. Further analysis can be found in most classical mechanics texts.

Recall from Section 11.7 that the gravitational potential energy is given by

$$V(r) = -\frac{G\, m_1\, m_2}{r}, \tag{15.168}$$

with m_1 and m_2 the masses of the two bodies, r the distance between their centers, and G is Newton's gravitational constant.

(a) Write the Lagrangian in terms of the center of mass position, $\boldsymbol{R}$, the relative position, $\boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$, the total mass, $M = m_1 + m_2$, and the reduced mass, $\mu = m_1\, m_2/(m_1 + m_2)$.
(b) Determine the component of the angular momentum vector, $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$, that is a constant of the motion, where $\boldsymbol{p}$ is the generalized momentum conjugate to $\boldsymbol{r}$.
(c) Write the radial equation of motion in the form

$$\mu\, \ddot{r} = -\partial_r V_{\mathsf{eff}}(r), \tag{15.169}$$

and determine $V_{\mathsf{eff}}(r)$.
(d) Introduce the conserved angular momentum into the Lagrangian. Show that this premature introduction of a conserved quantity produces a radial equation of motion with the wrong sign for the contribution of the angular momentum. This example serves as an important lesson Introduce the conserved angular momentum into the Lagrangian. Show that this premature introduction of a conserved quantity produces a radial equation of motion with the wrong sign for the contribution of the angular momentum. This example serves as an important lesson (raised in Section 14.8.5) that we compute the Euler-Lagrange equations without making use of any conserved quantities.

EXERCISE 15.5: THE DOUBLE PENDULUM
Consider the double pendulum as shown in Figure 15.5.

(a) Derive the kinetic energy for this system.
(b) Derive the gravitational potential energy for this system.
(c) Derive the Euler-Lagrange equations of motion for this system.

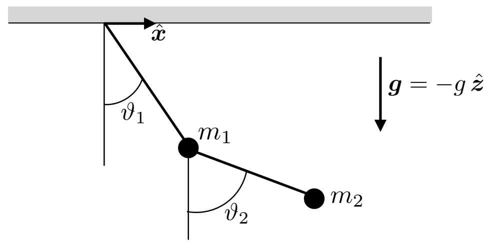

FIGURE 15.5: The double pendulum consists of two masses, m_1 and m_2, along with two massless and straight rods of length ℓ_1 and ℓ_2. The rods form an angle, ϑ_1 and ϑ_2, relative to the vertical, with positive angles as shown and negative for the opposite. The two angles serve as the generalized coordinates for this two degree of freedom dynamical system. In Exercise 15.5 we derive the equations of motion for this system using Lagrangian mechanics.

Part III

Exercises and solutions

We conclude each chapter of this book with a suite of exercises, and here we present solutions to the exercises. Working through these exercises, in full detail, is an integral part of learning and doing physics. Indeed, there is no replacement for struggle and head-scratching to support the physics problem-solving brain muscle. However, with the advent of AI tools, one can readily access AI generated solutions. It goes without saying that over-reliance on AI tools greatly compromises one's ability to develop the skills necessary to know whether the AI solution is correct. In light of this situation, we provide some worked solutions, which generally go beyond what is expected from a student learning the material for the first time. Namely, the solutions presented here aim to be instructional as well as utilitarian. For this reason, we expose many of the intermediate steps needed to derive a solution, further supporting an in-depth learning of how to independently solve physics problems. Additionally, exposing details helps to identify when an incorrect result follows from a physical/conceptual error (e.g., incorrect setup of the problem solution) or from a mathematical error (e.g., sign error).

We observe that most people are not born with *a priori* physics problem solving skills. Rather, it takes extensive practice to develop the necessary brain muscle. The student who values the ability to solve physics problems should resist the temptation to quickly flip pages to read solutions. Time pondering an exercise is time well spent learning how to do theoretical physics in a manner needed to pursue novel research. In the remainder of this section we offer specific pointers of use when diving into a physics problem.

General pointers

Here are a few general pointers that can serve to motivate problem solving, either for class work or more broadly for research.

Clear thinking leads to clear communication

Clear communication is the sign of clear thinking. Some people communicate better in writing, where one has the opportunity to carefully organize thoughts and refine the writing. Others are better at speaking, where spontaneous and interactive reflections and experiences can bolster the clarity of a presentation.

As inspiration for both the clear and obscure, pick up a textbook or lecture notes and analyze the presentation for clarity. Where is the presentation confusing? Where is the material crystal clear? Then pick up a journal article and perform the same analysis. What is appealing? What is unappealing? Then go to the internet and find a science or engineering lecture, old or new. What makes the speaker engaging and clear, or boring and obscure?

Empathy is key

Empathy is a basic facet of effective communication and teaching, where the writer, speaker, or teacher places their mind inside that of an interested and smart reader or listener. Identify with their quest to understand new ideas and to comprehend the foundations and assumptions. Are the assumptions justified based on the audience? How compelling is the scientific story? Are missing steps crucial to understanding or easily dispensed with for streamlining the presentation?

clarity evolves and honesty is essential

Although poor communication hinders our ability to digest new ideas and concepts, it is also important to appreciate that some material is tough no matter how well it is communicated.

We should aim to make a subject matter as simple as possible, but not simpler (paraphrasing Einstein). Furthermore, it sometimes takes a few generations of teaching before some scientific material can be sufficiently digested to allow for the core conceptual nuggets to be revealed. As an example, try reading Newton or Maxwell's original works as compared to a modern presentation of Classical Mechanics or Electromagnetism. So as we strive for clear communication, we cannot presume that clarity is sufficient to remove the struggles everyone experiences when learning.

Additionally, it is essential to recognize that everyone makes mistakes, either in fundamentals or practices. The toughest part of making mistakes is often the self-imposed shame and embarrassment. However, mistakes offer significant opportunities for learning and advancing, with honesty and humility critical for identifying weaknesses, both in our own work and those of others.

Specific pointers on problem solving

CHECK FOR DIMENSIONAL CONSISTENCY

The symbols we use in mathematical physics correspond to geometrical objects (e.g., points, vectors, tensors) describing a physical concept (e.g., position in space, velocity, temperature, angular momentum, stress). Hence, the symbols generally carry physical dimensions. The physical dimensions we are concerned with in this book are length (L), time (T), mass (M), and temperature. We do not consider electromagnetism or the quantum mechanical world. Physical dimensions of the equations must be self-consistent. For example, if one writes an equation $A = B$, where A and B have different physical dimensions, then the equation makes no physical sense. Something is wrong. Although not always sufficient to uncover errors, dimensional analysis is an incredibly powerful necessary step in debugging the maths.

CHECK FOR TENSORIAL CONSISTENCY

In the same way that mathematical equations in physics need to maintain dimensional consistency, they must also respect tensor rules. For example, the equation $A = B$ makes mathematical sense if A and B are both scalars. Likewise, $\mathbf{A} = \mathbf{B}$ makes sense if $\mathbf{A}$ and $\mathbf{B}$ are both vectors. However, if both $\mathbf{A}$ and $\mathbf{B}$ are vectors, then the equation $\mathbf{A} = \nabla \cdot \mathbf{B}$ does not make sense because the left hand side is a vector and the right hand side is a scalar. A more subtle example is when $\mathbf{A}$ is a vector yet $\mathbf{B}$ is an axial vector. In this case, $\mathbf{A}$ remains invariant under a change from right hand to left hand coordinates whereas $\mathbf{B}$ flips sign. Maintaining basic tensorial rules can be considered the next level of sophistication beyond dimensional analysis.

USE WORDS AND PICTURES

Words and pictures are important elements in explaining a physical concept and/or a problem in physics. Hence, it is good practice to liberally sprinkle sentences in between the key equations for the purpose of explaining what the maths means using clear English. Here are some practical payoffs to the student for this style of presentation.

- The process of explaining the maths using words and pictures requires one to dive deeper into the logic of a physics problem. Doing so often reveals weak points, incomplete or unmentioned assumptions, and errors. This process is a very important learning stage in preparing to stand in front of an audience to present results and to answer questions. It is a key facet of research and teaching.

- Physics teachers are often more forgiving of math errors if you convince the teacher that you have a sensible physical understanding of the problem. Plain English and pictures are very useful means for this purpose.

There is often more than one path to a solution

In physics, there is often more than one path to a problem solution or to the formulation of a concept. Pursuing distinct paths offers novel physical and mathematical insights, exposes otherwise hidden assumptions, and allows one to double-check the veracity of a solution. Some of the most profound advances in physics came from pursuing distinct formulations. One example concerns the distinct formulation of mechanics offered by Newton (1642-1746), and then later by Lagrange (1736-1813) and then Hamilton (1805-1865). Had Lagrange or Hamilton rested on the merits of their predecessors, we may well have had a very different intellectual evolution of 19th and 20th century physics.

Part IV

End matter

Appendix A

List of Acronyms

AI artificial intelligence xvii

GFD geophysical fluid dynamics viii

GFM geophysical fluid mechanics vii

GVC generalized vertical coordinate xx, 4

PDE partial differential equation 217

SAL self-attraction and loading 386

Appendix B

List of Symbols

Many symbols encountered in this book are defined local to their usage and are not used far outside of that location. Many other symbols appear in a variety of places and are included in the tables given below. Additionally, we generally aim to respect the following conventions.

- Many symbols are adorned with extra labels. One usage exposes tensor indices, with tensor indices written using the slanted math font, such as F^i for the component i of the vector $\mathbf{F}$. Another usage expresses part of the name for the symbol, with the label written with the upright sans serif. Examples include the "b" in η_{b} for the position of the bottom solid boundary of a fluid domain, and the "h" in ∇_{h} for the horizontal gradient operator.
- We strive for unique symbols to represent distinct mathematical and/or physical objects. Yet that goal must confront the multitude of mathematical expressions appearing in this book. We have chosen, on rare occasions, to allow some symbols to carry multiple meanings. In such cases we emphasize the particular meaning of the symbol to help avoid confusion with its alternative meaning.

NON-DIMENSIONAL NUMBERS

SYMBOL	NAME	MEANING
Bu	Burger	Bu = (deformation radius/horizontal length scale of flow)^2 = $(L_{\mathsf{d}}/L)^2$
Db	Deborah	Db = relaxation time/observation time
Ek	Ekman	Ek = vertical frictional acceleration/planetary Coriolis acceleration
Fr	Froude	Fr = fluid particle speed/fluid wave speed = U/c
Ge	Geostrophic	Ge = horizontal accelerations from Coriolis/pressure acceleration = $f\,U\,L\,\rho_{\mathsf{b}}/p$
Kn	Knudsen	Kn = molecular mean free path/macroscopic length scale
Ma	Mach	Ma = fluid particle speed/sound wave speed = U/c_{s}
Pr	Prandtl	Pr = viscosity/diffusivity = μ/κ
Pe	Peclet	Pe = advective transport/diffusive transport = $U\,L/\kappa$
Re	Reynolds	Re = inertial acceleration/frictional acceleration = $U\,L/\nu$
Ri	Richardson	Ri = squared buoyancy frequency/squared vertical shear
Ro	Rossby	Ro = horizontal inertial acceleration/planetary Coriolis acceleration = $U/(f\,L)$

LATIN SYMBOLS AND THEIR MEANING

SYMBOL	MEANING
$\mathcal{A}$	wave action
$A^{\mathrm{L}}(\boldsymbol{a}, T)$	Lagrangian representation of a fluid property as a function of material coordinates and time
$\boldsymbol{a}$	coordinate position for a fluid particle using arbitrary material/Lagrangian coordinates
$\mathbf{A}, \boldsymbol{A}$	second order skew symmetric tensor with elements satisfying $A^{mn} = -A^{nm}$
A^{v}	Avogadro's number: $A^{\mathsf{v}} = 6.0222 \times 10^{23}$ mole^{-1}
$\mathbf{B}$	baroclinicity vector: $\mathbf{B} = \nabla\rho \times (-\rho^{-1}\,\nabla p) = (\nabla\rho \times \nabla p)/\rho^2$
$\mathcal{B}$	base (or reference) manifold for describing the space of continuum matter
b	Archimidean buoyancy with $b > 0$ for relatively light fluid: $b = -g\,(\rho - \rho_{\circ})/\rho_{\circ}$
C	tracer concentration = mass of tracer per mass of fluid = tracer mass fraction
C_{d}	dimensionless bottom drag coefficient: $C_{\mathsf{d}} > 0$
$\mathcal{C}$	circulation of velocity around the boundary of a surface $\mathcal{C} \equiv \oint_{\partial\mathcal{S}} \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{r}$
c_{grav}	shallow water gravity wave speed: $c_{\mathsf{grav}} = \sqrt{g\,H}$
$\boldsymbol{c}_{\mathsf{g}}$	wave group velocity, given by wavevector gradient of dispersion relation: $\boldsymbol{c}_{\mathsf{g}} = \nabla_{\boldsymbol{k}}\varpi(\boldsymbol{k})$
$\boldsymbol{c}_{\mathsf{p}}$	wave phase velocity: $\boldsymbol{c}_{\mathsf{p}} = C_{\mathsf{p}}\,\hat{\boldsymbol{k}}$
C_{p}	wave phase speed
c_{s}	sound speed: $c_{\mathsf{s}}^{-2} = [\partial\rho/\partial p]_{\Theta,S}$
c_{p}	heat capacity at constant pressure: $c_{\mathsf{p}} = [\partial\mathcal{H}/\partial T]_{p,C}$
$\mathbf{E}, \boldsymbol{E}$	second order eddy transport tensor for tracers, and with elements E^{mn}
$\mathbb{E}^1, \mathbb{E}^2, \mathbb{E}^3$	one (line), two (plane), and three dimensional Euclidean space
$\mathcal{E}$	total energy per mass of a fluid element = sum of internal plus mechanical energies
$\boldsymbol{e}_a$	basis vectors for a chosen coordinate system, with index $a = 1, 2, 3$ for 3-dimensional space
$\boldsymbol{e}^a$	basis one-forms for a chosen coordinate system, with index $a = 1, 2, 3$ for 3-dimensional space
f	Coriolis parameter, also the planetary vorticity: $f = 2\,\Omega\,\sin\phi$
$f_{\circ}$	Coriolis parameter at a particular latitude: $f_{\circ} = 2\,\Omega\,\sin\phi_0$
$\mathbf{F}$	frictional acceleration vector
$F^{i}{}_{I}$	deformation matrix, which transforms between $\boldsymbol{x}$-space (Eulerian) and $\boldsymbol{a}$-space (Lagrangian)
G	water mass transformation, with dimensions of mass per time
$G = G^{\mathrm{grv}}$	Newton's gravitational constant: $G = 6.674 \times 10^{-11}$ N m^2 kg^{-2} $= 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}
$G(\boldsymbol{x}\vert\boldsymbol{x}_0)$	Green's function with $\boldsymbol{x}$ the observation point (or field point) and $\boldsymbol{x}_0$ the source point
$\widetilde{G}(\boldsymbol{x}\vert\boldsymbol{x}_0)$	modified Green's function for Laplace's operator with Neumann boundary conditions
$G^{\ddagger}(\boldsymbol{x}\vert\boldsymbol{x}_0)$	adjoint Green's function for non-self adjoint operators such as the diffusion operator
$\mathcal{G}(\boldsymbol{x}\vert\boldsymbol{x}_0)$	free space Green's function; i.e., the Green's function without boundaries
$\mathbf{G}$	velocity gradient tensor with elements $G^{i}{}_{j}$
$\mathcal{G}$	Gibbs potential per mass of a fluid element
g_{e}	gravitational acceleration from central gravity due to just the mass of the planet
g	effective gravitational acceleration from central gravity + planetary centrifugal as evaluated at the Earth's surface: $g \approx 9.8$ m s^{-2}

LATIN SYMBOLS AND THEIR MEANING

SYMBOL	MEANING
g^{r}	reduced gravity defined between two shallow water layers: $g^{\mathsf{r}}_{\mathsf{k}+1/2} = g\,(\rho_{\mathsf{k}+1} - \rho_{\mathsf{k}})/\rho_{\mathsf{ref}} \ll g$
$\mathbb{g}$	metric tensor (symmetric positive definite second order tensor) with components $\mathbb{g}_{ab}$
g	square root of the metric tensor determinant: $\mathsf{g} = \sqrt{\det(\mathbb{g}_{mn})}$
g^{E}	square root of the metric tensor determinant using Eulerian coordinates: $\mathsf{g}^{\mathsf{E}} = \sqrt{\det(\mathbb{g}(\boldsymbol{x})}$
g^{L}	square root of the metric tensor determinant using Lagrangian coordinates: $\mathsf{g}^{\mathsf{L}} = \sqrt{\det(\mathbb{g}(\boldsymbol{a}, T)}$
h_{k}	layer thickness for a shallow water fluid: $h_{\mathsf{k}} = \eta_{\mathsf{k}-1/2} - \eta_{\mathsf{k}+1/2} = \delta_{\mathsf{k}}\eta_{\mathsf{k}-1/2}$
h	layer thickness for a continuously stratified fluid: $h = \overline{\mathsf{h}}\,\delta\sigma$
h	specific thickness for a generalized vertical coordinate: $\mathsf{h} = \partial z/\partial\sigma = 1/(\partial\sigma/\partial z)$
$\mathcal{H}(x)$	Heaviside step function: $\mathcal{H}(x) = 0$ for $x < 0$ whereas $\mathcal{H}(x) = 1$ for $x > 0$
H	vertical length scale of the flow under consideration
H	sometimes used as depth of the ocean bottom: $z = -H(x, y) = \eta_{\mathsf{b}}(x, y)$
H	Hamiltonian energy function
$\mathcal{H}$	Hamiltonian density used in field theory; dimensions energy per volume (when in 3d space)
$\mathscr{H}$	enthalpy per mass of a fluid element
$\mathbf{I}$	unit tensor or Kronecker tensor: $\mathbf{I} = \delta^{ab}\,\boldsymbol{e}_a \otimes \boldsymbol{e}_b = \delta^{a}{}_{b}\,\boldsymbol{e}_a \otimes \boldsymbol{e}^b = \delta_{a}{}^{b}\,\boldsymbol{e}^a \otimes \boldsymbol{e}_b = \delta_{ab}\,\boldsymbol{e}^a \otimes \boldsymbol{e}^b$
$\mathscr{I}$	internal energy per mass of a fluid element
i	$\mathrm{i} = \sqrt{-1}$ used for imaginary numbers
i, j, k	tensor indices/labels for Eulerian coordinates
I, J, K	tensor indices/labels for Lagrangian coordinates
$\mathrm{Im}[\,]$	imaginary part of a complex number; e.g., $\mathrm{Im}[e^{-\mathrm{i}\,\omega\,t}] = -\sin(\omega\,t)$
$\mathbf{J}$	tracer flux; for material tracers the dimensions are mass per time per area
$\boldsymbol{k}$	wavevector (dimensions inverse length) for a wave of wavelength $\Lambda = 2\pi/\lvert\boldsymbol{k}\rvert$
$\hat{\boldsymbol{k}}$	unit vector in the direction of a wave: $\boldsymbol{k} = \hat{\boldsymbol{k}}\,\lvert\boldsymbol{k}\rvert$ (as distinct from the vertical unit vector, $\hat{\boldsymbol{z}}$)
$\lvert\boldsymbol{k}\rvert$	wavenumber: $\lvert\boldsymbol{k}\rvert = 2\pi/\Lambda$
K	kinetic energy for a particle of mass m: $K = m\,\boldsymbol{V}\cdot\boldsymbol{V}/2$
K	kinetic energy for a system of N particles, $\sum_{n=1}^{N} m^n\,\boldsymbol{V}^n \cdot \boldsymbol{V}^n$
$\mathcal{K}$	kinetic energy per mass of a fluid element arising from macroscopic motion: $\mathcal{K} = \boldsymbol{v}\cdot\boldsymbol{v}/2$
$\mathcal{K}^{\mathrm{hyd}}$	kinetic energy per mass for an approximate hydrostatic flow: $\mathcal{K}^{\mathrm{hyd}} = \boldsymbol{u}\cdot\boldsymbol{u}/2$
$\mathcal{K}^{\mathrm{sw}}$	kinetic energy per horizontal area for a shallow water layer: $\mathcal{K}^{\mathrm{sw}} = \rho\,h\,\boldsymbol{u}\cdot\boldsymbol{u}/2$
$\mathbf{K}, \boldsymbol{K}$	positive and symmetric second order tensor parameterizing diffusive mixing
k	integer index to label a layer in a shallow water model with $\mathsf{k} = 1, N$ layers ($\mathsf{k} = 1$ is top layer)
k_{B}	Boltzmann constant: $k_{\mathsf{B}} = 1.3806 \times 10^{-23}\ \mathrm{m}^2\ \mathrm{kg}\ \mathrm{s}^{-2}\ \mathrm{K}^{-1} = R^{\mathsf{g}}/A^{\mathsf{v}}$
k_{R}	Rossby height/depth: $k_{\mathsf{R}} = \lvert\boldsymbol{k}\rvert\,N/f_0$ with horizontal wavenumber $\lvert\boldsymbol{k}\rvert = \sqrt{k_x^2 + k_y^2}$
L	Lagrangian used in Lagrangian mechanics: kinetic minus potential energies: $L = K - P$
L	length scale for a particular physical feature and commonly used in scale analysis

SYMBOL	MEANING
$\mathcal{L}$	Lagrangian density used in field theory; dimensions energy per volume (when in 3d space)
L_{d}	deformation radius: (a) shallow water $L_{\mathsf{d}} = \sqrt{g\,H}/f$; (b) continuous internal $L_{\mathsf{d}} = H\,N/f$
$\mathcal{M}$	mechanical energy per mass of a fluid element arising from macroscopic motion
$\mathcal{M}^{\rm sw}$	mechanical energy per area of a shallow water fluid column: $\mathcal{M}^{\rm sw} = \mathcal{K}^{\rm sw} + \mathcal{P}^{\rm sw}$
$\mathbf{M}$	moment of inertia tensor
$\mathbf{M}$	potential momentum vector: $\mathbf{M} = \mathbf{u} + 2\,\boldsymbol{\Omega}\times\mathbf{X}$
M	Montgomergy potential for continuously stratified fluid $M = \varphi - b\,z$
M	mass, as in the mass of a fluid region, $M = \int_{\mathcal{R}} \rho\,\mathrm{d}V$
$M^{\rm dyn}_{\mathsf{k}}$	Montgomergy potential for a shallow water layer: $M^{\rm dyn}_{\mathsf{k}} = \sum_{\mathsf{j}=0}^{\mathsf{k}-1} g^{\mathsf{r}}_{\mathsf{j}+1/2}\,\eta_{\mathsf{j}+1/2}$
M^{air}	mass per mole of air: $M^{\mathsf{air}} = 28.8\times10^{-3}$ kg mole^{-1}
N	buoyancy frequency
$\mathcal{O}$	order of magnitude
P	potential energy of a physical system, with corresponding force $\boldsymbol{F} = -\nabla P$
$\mathcal{P}^{\rm sw}_{\mathsf{k}}$	potential energy per horizontal area for a shallow water fluid: $\mathcal{P}^{\rm sw}_{\mathsf{k}} = g\,\rho_{\mathsf{k}}\int_{\eta_{\mathsf{k}+1/2}}^{\eta_{\mathsf{k}-1/2}} z\,\mathrm{d}z$
$\mathcal{P}$	phase of a wave
$\mathcal{P}_\sigma$	generalized momentum for discrete particle system: $\mathcal{P}_\sigma = \partial L/\partial\dot{\xi}^\sigma$
$\mathcal{P}$	generalized momentum density for continuous media: $\mathcal{P} = \partial\mathcal{L}/\partial(\partial_t\psi)$
$\mathbf{P}$	linear momentum of a physical system
p	pressure at a point in the fluid
p_{a}	pressure applied to the ocean surface from the atmosphere or cryosphere
p_{b}	pressure at the bottom of a fluid column, at the fluid-solid earth interface
$p_{\rm slp}$	sea level pressure with an area average, $\langle p_{\rm slp}\rangle = 101.325\times10^3$ N m^{-2}
$p_{\mathsf{k}-1/2}$	hydrostatic pressure at the layer interface with vertical position $z = \eta_{\mathsf{k}-1/2}$
$p^{\rm dyn}_{\mathsf{k}}$	dynamic pressure in a shallow water layer: $p^{\rm dyn}_{\mathsf{k}} = \rho_{\mathsf{ref}}\sum_{\mathsf{j}=0}^{\mathsf{k}-1} g^{\mathsf{r}}_{\mathsf{j}+1/2}\,\eta_{\mathsf{j}+1/2}$
P_{k}	pressure integrated over a shallow water layer: $P_{\mathsf{k}} \equiv \int_{\eta_{\mathsf{k}+1/2}}^{\eta_{\mathsf{k}-1/2}} p_{\mathsf{k}}(z)\,\mathrm{d}z = h_{\mathsf{k}}\,(g\,\rho_{\mathsf{k}}\,h_{\mathsf{k}}/2 + p_{\mathsf{k}-1/2})$
Q	potential vorticity for continuously stratified (Ertel PV) or shallow water (Rossby PV)
q	quasi-geostrophic potential vorticity either for a continuous fluid or shallow water fluid
Q_{m}	mass flux (mass per horizontal area per time) across ocean surface: $Q_{\mathsf{m}} > 0$ enters ocean
$\mathcal{Q}_{\mathsf{m}}$	mass flux (mass per surface area per time) across ocean surface: $\mathcal{Q}_{\mathsf{m}}\,\mathrm{d}\mathcal{S} = Q_{\mathsf{m}}\,\mathrm{d}A$
Q_{C}	turbulent tracer flux (tracer per horiz area per time) across ocean surface: $Q_{\mathsf{C}} > 0$ enters ocean
$\mathcal{Q}_{\mathsf{C}}$	turbulent tracer flux (tracer per surface area per time) across ocean surface: $QCcal\,\mathrm{d}\mathcal{S} = Q_{\mathsf{C}}\,\mathrm{d}A$
r	radial distance of a point relative to an origin
$\mathbf{R}$	rotation rate tensor: $2\,R^m{}_n = \nabla_n v^m - \nabla^m v_n = -2\,R_n{}^m$
$\mathbb{R}^1$	real number line
$\mathbb{R}^2$	two-dimensional space of real numbers
$\mathbb{R}^3$	three-dimensional space of real numbers
R	radius of a sphere

LATIN SYMBOLS AND THEIR MEANING

SYMBOL	MEANING
R_{e}	radius of sphere whose volume approximates that of the earth: $R_{\mathsf{e}} = 6.371 \times 10^{6}\,\mathrm{m}$
R^{g}	universal gas constant: $R^{\mathsf{g}} = 8.314$ J mole^{-1} K^{-1} $= 8.314$ kg m^{2} s^{-2} mole^{-1} K^{-1}
R^{air}	specific gas constant for air: $R^{\mathsf{air}} = R^{\mathsf{g}}/M^{\mathsf{air}} = 2.938 \times 10^{2}$ m^{2} s^{-2} K^{-1}
$\mathcal{R}$	arbitrary region or manifold
$\mathcal{R}^{a}{}_{b}$	orthogonal rotation matrix
Re[ ]	real part of a complex number; e.g., $\mathrm{Re}[e^{-\mathrm{i}\,\omega\,t}] = \cos(\omega\,t)$
$\mathcal{S}$	spatial manifold
$\mathcal{S}$	entropy per mass of a fluid element
$\mathcal{S} = \mathcal{S}^{\mathsf{action}}$	action: time integral of the Lagrangian: $\mathcal{S} = \int_{t_A}^{t_B} L\,\mathrm{d}t$
$\mathbf{S}$	strain rate tensor: $2\,\mathbf{S} = \nabla\mathbf{v} + (\nabla\mathbf{v})^{\mathrm{T}}$
$\mathbf{S}^{\mathsf{dev}}$	deviatoric strain rate tensor: $\mathbf{S}^{\mathsf{dev}} = \mathbf{S} - S^{q}{}_{q}/3$
$\boldsymbol{S}$	neutral slope vector: $\boldsymbol{S} = \nabla_{\gamma} z = \hat{\boldsymbol{x}}\,S_x + \hat{\boldsymbol{y}}\,S_y$
S_{b}	slope of a base state buoyancy surface a horizontal direction; e.g., $S_{\mathsf{b}} = -\partial_y b_{\mathsf{b}}/N^{2}$
S	salt concentration = mass of salt in a fluid element per mass of seawater
S	Absolute Salinity, generically referred to as salinity: $S = 1000\,\mathsf{S}$
s	expression for a generic surface: $s = s(x, y, z, t)$.
s	arc-length along a curve $\boldsymbol{x}(s)$ with infinitesimal increment $\mathrm{d}s = \sqrt{\mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x}}$
$\hat{\boldsymbol{s}}$	unit tangent to a curve, also written as $\hat{\boldsymbol{s}} = \hat{\boldsymbol{t}}$ (see below)
sgn	sign function related to Heaviside step function via $\mathrm{sgn}(x) = 2\,\mathcal{H}(x) - 1$
T	absolute thermodynamic *in situ* temperature (Kelvin if in a thermodynamic equation)
T	time scale for a particular physical process and commonly used in scale analysis
T	time (universal Newtonian time) measured in the Lagrangian reference frame
t	time (universal Newtonian time) measured in the Eulerian reference frame
τ	general symbol for time as considered in the tensor analysis chapters
$\mathbf{T}$	stress tensor with natural elements $T^{m}{}_{n}$
$\mathbb{T}^{\mathsf{kinetic}}$	kinetic stress tensor: $\mathbb{T}^{\mathsf{kinetic}} = -\rho\,\mathbf{v}\otimes\mathbf{v}$
$\mathbb{T}^{\mathsf{sw\ kinetic}}$	kinetic stress tensor for shallow water fluid: $\mathbb{T}^{\mathsf{sw\ kinetic}} = -\rho\,\mathbf{u}\otimes\mathbf{u}$
$\hat{\boldsymbol{t}}$	unit tangent to a curve: $\hat{\boldsymbol{t}} = \mathrm{d}\boldsymbol{x}/\mathrm{d}s$, where s is the arc-length so that $\mathrm{d}s = \sqrt{\mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x}}$
$\mathbf{u}$	horizontal velocity of a fluid particle, with Cartesian representation: $\boldsymbol{u} = \hat{\boldsymbol{x}}\,u + \hat{\boldsymbol{y}}\,v$
U	horizontal velocity scale of the flow under consideration
$\mathbf{U}$	depth integrated horizontal velocity: $\mathbf{U} = \int_{\eta_{\mathsf{b}}}^{\eta}\mathbf{u}\,\mathrm{d}z$
V	volume, as in the volume of a fluid region, $V = \int_{\mathcal{R}}\mathrm{d}V$
$\mathbf{v}$	velocity of a fluid particle: $\mathbf{v} = \mathrm{D}\mathbf{x}/\mathrm{D}t$, with Cartesian components $\boldsymbol{v} = \hat{\boldsymbol{x}}\,u + \hat{\boldsymbol{y}}\,v + \hat{\boldsymbol{z}}\,w$
$\mathbf{v}^{*}$	eddy-induced velocity
$\mathbf{v}^{\dagger}$	residual velocity of a fluid particle: $\mathbf{v}^{\dagger} = \mathbf{v} + \mathbf{v}^{*}$
$\mathbf{v}^{(b)}$	velocity of a point on a region boundary
$\boldsymbol{v}^{\mathrm{L}}(\boldsymbol{a}, T)$	Lagrangian velocity of a fluid particle so that $\boldsymbol{v}^{\mathrm{L}}(\boldsymbol{a}, T) = \boldsymbol{v}[\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{a}, T), t = T]$
$\mathbf{v}_{\mathrm{I}}$	velocity of a fluid particle measured in the inertial/absolute reference frame: $\mathbf{v}_{\mathrm{I}} = \mathbf{v} + \boldsymbol{\Omega}\times\mathbf{x}$
W	vertical velocity scale of the flow under consideration
$\mathcal{W}$	on-shell action
w	vertical component to the velocity: $w = \mathrm{D}z/\mathrm{D}t$
w^{dia}	dia-surface flux = volume per surface area per time crossing a σ-surface: $w^{\mathsf{dia}} = (1/\lvert\nabla\sigma\rvert)\,\dot{\sigma}$
$w^{(\dot{\sigma})}$	dia-surface velocity = volume per *horizontal area* per time crossing σ-surface: $w^{(\dot{\sigma})} = \dot{\sigma}\,\partial z/\partial\sigma$

LATIN SYMBOLS AND THEIR MEANING

SYMBOL	MEANING
(x, y, z)	triplet of Cartesian coordinates
$\mathbf{x}$	spatial position as a line segment with an arrow pointing from an origin to the position of a particle
$\boldsymbol{x}$	spatial position represented by either general coordinates or Cartesian coordinates
$\mathring{\mathbf{x}}$	initial position for a fluid particle using arbitrary coordinates
$(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$	triplet of Cartesian unit vectors oriented in a righthand sense
$\mathbf{X}(t)$	position for a point particle defining a trajectory through space-time
$\boldsymbol{X}(\boldsymbol{a}, T)$	position of a material fluid particle expressed using material coordinates
z_σ	specific thickness for a generalized vertical coordinate: $z_\sigma = \partial z/\partial \sigma = \mathsf{h}$

GREEK SYMBOLS AND THEIR MEANING

SYMBOL	MEANING
α	thermal expansion: $\alpha = -\rho^{-1}\,\partial\rho/\partial\theta$ or $\alpha = -\rho^{-1}\,\partial\rho/\partial\Theta$ or $\alpha = -\rho^{-1}\,\partial\rho/\partial T$
α_T	thermal expansion in terms of *in situ* temp: $\alpha = -\rho^{-1}\,\partial\rho/\partial T$
$\alpha^{(\Theta)}$	thermal expansion in terms of Conservative Temperature: $\alpha^{(\Theta)} = -\rho^{-1}\,\partial\rho/\partial\Theta$
α_{aspect}	aspect ratio; ratio of vertical to horizonal scales of the flow: $\alpha_{\text{aspect}} = H/L$
$\beta, \beta^{(\text{S})}$	haline (saline) contraction coefficient: $\beta = \beta^{(\text{S})} = \rho^{-1}\,\partial\rho/\partial S$
β	meridional derivative of planetary vorticity: $\beta = \partial_y f$
$\hat{\boldsymbol{\gamma}}$	dianeutral unit direction perpendicular to the neutral tangent plane
δ_{ab}	Kronecker delta, which is the metric for Euclidean space with Cartesian coordinates
δ^a_b	components to the Kronecker tensor in arbitrary coordinates
ϵ	kinetic energy dissipation from viscosity (energy per time per mass)
ϵ_{ab}	components to the permutation symbol in two space dimensions
ϵ_{abc}	components to the permutation symbol in three space dimensions
ε_{abc}	components to the Levi-Civita symbol in three space dimensions: $\varepsilon_{abc} = \sqrt{\det(\mathbb{g}_{ab})}\,\epsilon_{abc}$
ζ	vertical component to the relative vorticity; e.g., $\zeta = \partial_x v - \partial_y u$
ζ_{a}	vertical component to the absolute vorticity; e.g., $\zeta_{\mathsf{a}} = f + \zeta$
η	vertical position of the free upper surface of a fluid domain: $z = \eta(x, y, t)$
η	vertical position of a generalized vertical coordinate surface: $z = \eta(x, y, \sigma, t)$, with σ the generalized vertical coordinate
$\eta_{\mathsf{k}-1/2}$	vertical position of the top interface of the k shallow water layer
$\eta_{\mathsf{k}+1/2}$	vertical position of the lower interface of the k shallow water layer
$\eta_{\mathsf{b}} = -H$	vertical position of static solid-earth boundary: $z = \eta_{\mathsf{b}}(x, y) = -H(x, y)$
θ	potential temperature
Θ	Conservative Temperature
κ	molecular kinematic diffusivity
κ_T	molecular diffusivity for *in situ* temperature in water: $\kappa_T = 1.4 \times 10^{-7}\ \text{m}^2\ \text{s}^{-1}$
κ_S	molecular diffusivity for salt in water: $\kappa_S = 1.5 \times 10^{-9}\ \text{m}^2\ \text{s}^{-1}$
κ_{eddy}	kinematic eddy diffusivity: $\kappa_{\text{eddy}} \gg \kappa$
Λ	wavelength of a wave: $\Lambda = 2\,\pi/\lvert\boldsymbol{k}\rvert$, where $\boldsymbol{k}$ is the wavevector and $\lvert\boldsymbol{k}\rvert$ the wavenumber.
$\lambda\!\!\!^{-}$	reduced wavelength of a wave: $\lambda\!\!\!^{-} = \Lambda/(2\pi) = 1/\lvert\boldsymbol{k}\rvert$.
λ	longitude on the sphere: $0 \leq \lambda \leq 2\pi$
μ_n	chemical potential for constituent n within a fluid (energy per mass)
$\tilde{\mu}_n$	chemical potential for constituent n within a fluid (energy per mole number)
μ	relative chemical potential for a binary fluid
μ	chemical potential for seawater: $\mu = \mu_{\text{salt}} - \mu_{\text{water}}$
μ_{vsc}	dynamic viscosity $= \rho\,\nu$
ν_{s}	specific volume: $\nu_{\mathsf{s}} = \rho^{-1}$
ν	molecular kinematic viscosity
ν_{air}	molecular kinematic viscosity of air: $\nu_{\text{air}} \approx 1.3 \times 10^{-5}\ \text{m}^2\ \text{s}^{-1}$
ν_{water}	molecular kinematic viscosity of fresh water: $\nu_{\text{water}} \approx 10^{-6}\ \text{m}^2\ \text{s}^{-1}$
ν_{eddy}	eddy viscosity: $\nu_{\text{eddy}} \gg \nu$
ξ^a	a'th component to a generalized coordinate
Π	Exner function
Π	Boussinesq dynamic enthalpy

GREEK SYMBOLS AND THEIR MEANING

SYMBOL	MEANING
ρ	Eulerian *in situ* density (mass per volume) of a fluid element: $\rho = \rho(\boldsymbol{x}, t)$
ρ^{L}	mass density following a fluid particle trajectory (Lagrangian mass density): $\rho^{\mathsf{L}} = \rho^{\mathsf{L}}(\boldsymbol{a}, T)$
$\mathring{\rho}^{\mathsf{L}}$	initial mass density in Lagrangian space-time: $\mathring{\rho}^{\mathsf{L}} = \rho^{\mathsf{L}}(\boldsymbol{a}, T = t_0)$
$\rho_{\circ}$	constant reference density used for the Boussinesq ocean
ρ_{ref}	constant reference density used for the shallow water fluid
ϱ	potential density referenced to a specified pressure
σ	generalized vertical coordinate, $\sigma = \sigma(x, y, z, t)$
$\boldsymbol{\tau}$	stress vector such as from winds or bottom stresses acting on the ocean
$\mathbb{T}$	frictional stress tensor
φ	pressure divided by the Boussinesq reference density: $\varphi = p/\rho_{\circ}$
φ	sometimes used as the variable for parameteriing a curve
ϕ	latitude on the sphere: $-\pi/2 \leq \phi \leq \phi/2$
Φ_{e}	gravitational potential from a spherical and homogeneous earth
Φ	geopotential from central gravity plus planetary centrifugal; also, potential energy per mass
$\boldsymbol{\Phi}$	inverse flow map that generates an inverse mapping of the fluid continuum: $\boldsymbol{a} = \boldsymbol{\Phi}(\boldsymbol{x}, t)$
$\boldsymbol{\varphi}$	motion field that maps the fluid continuum as time evolves: $\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{a}, T)$
ψ	streamfunction for two-dimensional non-divergent flow: $\boldsymbol{u} = \hat{\boldsymbol{z}} \times \nabla\psi$
$\boldsymbol{\Psi}$	vector streamfunction for three-dimensional non-divergent flow: $\boldsymbol{v} = \nabla \times \boldsymbol{\Psi}$
$\boldsymbol{\omega}$	relative vorticity: $\boldsymbol{\omega} = \nabla \times \boldsymbol{v}$
ω	angular frequency for a wave so that the wave period is $2\pi/\lvert\omega\rvert$
ϖ	dispersion relation for linear waves, relating angular frequency to the wavevector: $\omega = \varpi(\boldsymbol{k})$
$\boldsymbol{\Omega}$	angular velocity for a rotating reference frame
$\boldsymbol{\Omega}$	earth's angular velocity oriented through the north pole: $\lvert\boldsymbol{\Omega}\rvert = 7.2921 \times 10^{-5}\ \mathrm{s}^{-1}$

MATHEMATICAL OPERATIONS AND RELATIONS

SYMBOL	MEANING
$[\equiv]$	"has dimensions" for use in referring to the physical dimensions
$\times$	vector cross product
$\boldsymbol{\nabla}$	covariant derivative operator, which acts on a (p,q) tensor to produce a $(p,q+1)$ tensor.
∇	gradient operator
∇_{h}	horizontal gradient operator on constant z surface: $\nabla_{\mathrm{h}} = \hat{\boldsymbol{x}}\,(\partial/\partial x)_z + \hat{\boldsymbol{y}}\,(\partial/\partial y)_z = \hat{\boldsymbol{x}}\,\partial_x + \hat{\boldsymbol{y}}\,\partial_y$
$\nabla\cdot$	divergence operator that acts on a vector to produce a scalar
$\nabla\times$	curl operator
$\nabla_{\mathrm{h}\sigma}$	horizontal gradient on constant σ-surface: $\nabla_{\mathrm{h}\sigma} = \hat{\boldsymbol{x}}\,(\partial/\partial x)_\sigma + \hat{\boldsymbol{y}}\,(\partial/\partial y)_\sigma$
$\partial/\partial\sigma$	vertical partial derivative with general vertical coordinate: $\partial_\sigma = \partial/\partial\sigma = \partial/\partial\sigma = (\partial z/\partial\sigma)\,\partial/\partial z$
$\partial/\partial t$	Eulerian time derivative acting at a fixed spatial position, $\boldsymbol{x}$, also written as ∂_t
$[\partial/\partial t]_\sigma$	time derivative computed on constant σ-surface
$\mathrm{D}/\mathrm{D}t$	material, Lagrangian, or substantial time derivative following a fluid particle
$\mathrm{D_r}/\mathrm{D}t$	time derivative following a ray (integral lines of the group velocity): $\mathrm{D_r}/\mathrm{D}t = \partial/\partial t + \boldsymbol{c}_{\mathrm{g}}\cdot\nabla$
$\mathrm{D_g}/\mathrm{D}t$	time derivative following the horizontal geostrophic flow $\mathrm{D_g}/\mathrm{D}t = \partial/\partial t + \boldsymbol{u}_{\mathrm{g}}\cdot\nabla_{\mathrm{h}}$
$đ$	inexact differential operator commonly found in thermodynamics
δ	virtual displacement (also the variation) for Lagrangian mechanics and Hamilton's principle
δ	differential increment that signals an object following the fluid flow
$\delta(x)$	one-dimensional Dirac delta with dimensions of inverse length
$\delta^{(2)}(\boldsymbol{x})$	two-dimensional Dirac delta with dimensions of inverse area
$\delta(\boldsymbol{x})$	three-dimensional Dirac delta with dimensions of inverse volume
$\delta(t)$	temporal Dirac delta with dimensions of inverse time
Δ	finite difference increment in space: $\Delta_x, \Delta_y, \Delta_z, \Delta_\sigma$
$\mathrm{d}A$	infinitesimal horizontal area element: $\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y$
d^3a	infinitesimal region of material space: $\mathrm{d}^3a = \mathrm{d}a\,\mathrm{d}b\,\mathrm{d}c$
$\mathrm{d}\mathcal{S}$	infinitesimal area element on a surface
$\mathrm{d}V$	infinitesimal volume element, sometimes written $\mathrm{d}V = \mathrm{d}\boldsymbol{x}$
$\mathrm{d}\boldsymbol{x}$	infinitesimal volume element, with Cartesian expression $\mathrm{d}\boldsymbol{x} = \mathrm{d}V = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$
δV	infinitesimal volume for a region moving with the fluid (Lagrangian region)
$\int_{\mathcal{R}}\mathrm{d}V$	volume integral over an arbitrary region, $\mathcal{R}$
$\int_{\mathcal{R}(\boldsymbol{v})}\mathrm{d}V$	volume integral over a region following the fluid flow (Lagrangian integral)
$\int_{\mathcal{S}}\mathrm{d}\mathcal{S}$	surface integral over an arbitrary surface $\mathcal{S}$
$\oint_{\partial\mathcal{R}}\mathrm{d}\mathcal{S}$	surface integral over a closed surface $\partial\mathcal{R}$ that bounds the volume $\mathcal{R}$
$\oint\mathrm{d}\ell$	closed line integral over a periodic domain
$\oint_{\partial\mathcal{S}}\mathrm{d}\ell$	counter-clockwise closed line integral over the boundary of a surface, $\partial\mathcal{S}$
$\sim$	"similar to" or "scales as"
$\approx$	approximately equal to
$\dot{\Psi}$	time derivative following a trajectory; for fluid particle trajectories then, $\dot{\Psi} = \mathrm{D}\Psi/\mathrm{D}t$

Appendix C

GLOSSARY OF CONCEPTS AND TERMS

absolute simultaneity Absolute simultaneity results from the use of universal Newtonian time, in which all observers agree on past, present, and future. Namely, with universal Newtonian time, an event occuring at one point in space at time, t, also occurs at the same time everywhere else in space. 346

acceleration The second time derivative of a particle trajectory, or the first time derivative of the velocity, defines the acceleration of the particle. 352

action The action is the time integral of the Lagrangian density. Hamilton's principle states that the functional variation of the action is stationary for physically realized time evolution. xi, 191, 346, 463, 464

active tracer An active tracer affects the flow. Consequently, the tracer equation for active tracers is nonlinear. Temperature and matter concentration are two active tracers, in which these two tracers impact on the density, with density then modifying pressure that then modifies the flow. 327

active transformation We consider two versions of a symmetry transformation. Active transformations alter the physical system (e.g., move an experiment from one point in space to another), whereas passive transformations alter the mathematical description of the physics (e.g., change coordinates). 353

adiabatic invariant An adiabatic is a dynamical quantity that remains (approximately) constant when external parameters of the system change slowly compared to the system's natural timescales. The wave action is the primary adiabatic invariant that we encounter in this book. 491, 501, 504

adjoint causal Green's function The adjoint causal Green's function arises in evolutionary partial differential equations, such as the diffusion equation and wave equation. It refers to the solution to the initial-boundary value problem that solves the adjoint problem, and for which it satisfies the backward causality condition. 295

adjoint Green's function The adjoint Green's function solves the Green's function problem for the adjoint operator. For partial differential operators that are not self-adjoint, such as the diffusion operator, then the adjoint Green's function is not equal to the Green's function. 246

advection equation The advection equation, $(\partial_t + \boldsymbol{v} \cdot \nabla)\, C = 0$, is the canonical transport equation that arises from an Eulerian perspective on the evolution of a scalar field in the presence of fluid flow. We encounter the advection equation thoughout this book. 327

affine parameter An affine parameter is a parameter for a geodesic in which the tangent vector is parallel transported along the curve, giving constant-speed, force-free motion and the simplest form of the geodesic equation. Affine parameters are unique up to a constant offset and a constant rescaling. 201

analytical mechanics We refer to the combined approach of Lagrangian mechanics and Hamiltonian mechanics as analytical mechanics, with Lagrangian and Hamiltonian mechanics related through a Legrendre transformation. Both are logically equal to Newtonian mechanics, and yet they offer further conceptual insights and practical advantages beyond the Newtonian approach. In particular, rather than maintain forces at the center of the dynamical formualation (Newton), the Lagrangian and Hamiltonian approaches put energy at the forefront, and they each formulate the equations of motion through Hamilton's principle of stationary action. 346, 447, 449, 495

angular momentum Angular momentum is the moment of the linear momentum as computed around a chosen origin. For a particle of constant mass, m, the angular momentum is, $\mathbf{L} = \mathbf{X} \times \mathbf{P} = m\, \mathbf{X} \times \mathbf{V}$. 361, 428, 444

angular velocity Angular velocity measures the rotational velocity. It is oriented according to the right hand rule and it has magnitude generally measured in units of radians per second. The angular speed of the planet is mostly due to the planet's spin around its axis plus its orbit around the sun, and it has a value $\Omega = 7.2921 \times 10^{-5}\ \mathrm{s}^{-1}$. It is notable that finite rotations do not commute in three dimensional space. However, we can define the angular velocity as a vector since it lives in the tangent space (Lie algebra), not in the rotation group itself (Lie group $SO(3)$). 358, 392

anholonomic basis A set of anholonomic basis vectors (also called non-holonomic or non-coordinate) span the tangent space at each point of a manifold but are not necessarily tangent to coordinate lines. Consequently, anholonomic basis vectors cannot be transformed into a coordinate basis. 106, 116, 394, 395, 397, 399, 403

anti-symmetric tensor An anti-symmetric (also skew-symmetric) tensor equals to minus its transpose, $\boldsymbol{A} = -\boldsymbol{A}^{\mathsf{T}}$. 33

atlas An atlas is the sum charts sufficient to cover a manifold with overlapping coordinate systems. A one-to-one mapping realizes the transformation (i.e., coordinate transformation) between charts occurs in regions where the charts overlap. 83

Avogadro's number A mole of matter is defined as the mass of the material substance that contains Avogadro's number of that substance, where $A^{\mathsf{v}} = 6.022 \times 10^{23}\ \mathrm{mole}^{-1}$. Avogadro's number is the proportionality constant converting from one molar mass of a substance to the mass of a substance. Avogadro's number is conventionally specified so that one mole of the carbon isotope, ^{12}C, contains exactly 12 grams. Hence, 12 grams of ^{12}C contains 6.022×10^{23} atoms of ^{12}C. Avogadro's number provides a connection between scales active in the microscopic world of atoms and molecules to the macroscopic world. 72, 352

axial angular momentum The angular momentum relative to the planetary rotational axis is referred to as the axial angular momentum. 362, 428, 444

axial vector An axial vector (also referred to as a pseudo-vector) changes sign when changing the handedness of the coordinate system. Angular momentum and angular velocity are axial vectors. 21, 431

Beltrami flow Beltrami flow is defined by non-divergent velocity field whose vorticity is linearly proportional to the velocity: $\nabla \times \boldsymbol{v} = \lambda\, \boldsymbol{v}$. 70

biharmonic equation The biharmonic equation is the fourth order elliptic equation, $\nabla^4 \psi = \Lambda$, with Λ a source. 280

biharmonic operator The biharmonic operator is the squared Laplacian operator, $\nabla^4 = \nabla^2\, \nabla^2$. Biharmonic operators arise in elasticity theory as well as for Stokes flow in fluid mechanics. For geophysical fluid mechanics, biharmonic operators are commonly used for numerical reasons to enhance the scale-selectivity of dissipation. 317

biorthogonality relation The biorthogonality relation refers to the identity between a basis vector and basis one-form: $\mathbb{g}(\boldsymbol{e}_a, \boldsymbol{e}^b) = \delta_a{}^b = \delta^b{}_a$. It is also referred to as the duality condition. For Euclidean space, we sometimes write the shorthand $\mathbb{g}(\boldsymbol{e}_a, \boldsymbol{e}^b) = \boldsymbol{e}_a \cdot \boldsymbol{e}^b$, though one should exercise care not to confuse this expression with the inner product between two vectors. 92, 94, 124, 130, 397

body forces A body force acts throughout the extent of a fluid element, and it is synonymous with external force. Examples include the gravitational force, as well as the Coriolis and centrifugal forces arising from the rotating planetary reference frame. These forces are also known as long range forces. xix, 375

boundary Green's function The boundary Green's function is the inward normal gradient of the Green's function at the boundary. The boundary Green's function acts as mediator between information prescribed along the boundary and interior. 247, 305

boundary propagator The boundary propagator is the boundary Green's function for the diffusion and advection-diffusion problem, which incorporate time dependence along with space dependence. 250, 305, 306, 317

brachistochrone The brachistochrone is a smooth curve upon which a frictionless bead moves under the effects from gravity. The brachistochrone problem in variational calculus is concerned with finding the shape of the curve that minimizes the time for a bead to travels the shortest time between two points, A and B, where these two points are not underneath one another. 207, 208

budget equations Much of this book is concerned with deriving and understanding equations that describe the evolution of fluid properties, with such equations (differential or integral) derived from physical principles such as Newton's laws of motion, Hamilton's principle of stationary action, Noether's theorem, thermodynamic laws, mass conservation, and vorticity mechanics. These are the budget equations that form the theoretical foundation of continuum mechanics. ix

calculus of variations The calculus of variations is the mathematical method for determining the functions that make a functional (and integral), such as an action or energy, stationary under small variations. Realizing the extrema requires the integrand to satisfy the Euler-Lagrange differential equations. We make use of the calculus of variations when using Hamilton's principle for mechanics. 191

canonical momenta See generalized momenta. 480

Cartesian coordinates Cartesian coordinates, typically written (x, y, z), are the familiar coordinates from vector calculus and linear algegra. Each coordinate measures the distance from an arbitrary origin in directions defined by the orthonormal directions, $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$, each of which are each fixed in space. 7

Cartesian tensor Cartesian tensors are tensors living in Euclidean space making use of Cartesian coordinates. Cartesian tensors are sufficient for much of this book, though not for curvilinear orthogonal coordinates (e.g., spherical coordinates), Lagrangian coordinates, or generalized vertical coordinates. 7

catenary A catenary is the curve assumed by a uniform, static, flexible chain or cable hanging under its own weight when supported at two points and acted on only by gravity. 209, 210

catenoid A catenoid is a minimal surface obtained by rotating a catenary about a horizontal axis. Equivalently, it is the surface of revolution with zero mean curvature everywhere. 215

Cauchy problem The Cauchy problem is the name given to the initial value problem for the diffusion equation or the wave equation over all space, so that we are only concerned with initial conditions. 286, 323

Cauchy-Stokes decomposition theorem The Cauchy-Stokes decomposition theorem says that the arbitrary motion of a region in a continuous media can be decomposed into a uniform translation, dilation along three perpendicular axes, plus a rigid body rotation. 275, 351

causal free space Green's function The causal free space Green's function arises in evolutionary partial differential equations, such as the diffusion equation and wave equation. It refers to the solution in the absence of spatial boundary conditions (other than decay at spatial infinity) and with zero values for times before the initial time. 292

causal Green's function The causal Green's function arises in evolutionary partial differential equations, such as the diffusion equation and wave equation. It refers to the solution to the initial-boundary value problem that respects causality, so that the solution does not depend on information prior to the initial condition time. 293, 334

causality condition The causality condition means that the Green's function vanishes throughout all of space for times prior to the time, $t_\circ$, at which the Dirac source occurs. 287, 292

central force A central force is one that acts along the line between the two particles, so that $(\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)} = 0$. Newton's gravity force between two particles, as well as the Coulomb electrostatic force between two charges, are central forces. 370

centrifugal The centrifugal acceleration points towards the convex side of a turning trajectory; i.e., outward. Its opposing partner, the centripetal acceleration, points toward the center. The centrifugal acceleration arises from accelerated motion due to motion along a curved path. It is the centrifugal acceleration that pulls one away from the center of a merry-go-round whereas one's arms and hands provide the balancing centripetal acceleration. For the rotating Earth, gravity provides the centripetal acceleration to counteract the centrifugal acceleration of the rotating planet. 382, 406

centripetal The centripetal acceleration points towards the concave side of a turning trajectory, with centripetal referring to the center. Its opposing partner, the centrifugal acceleration, points away from the center, thus pointing to the convex side of a curved trajectory. The centripetal acceleration is required to keep an object moving along a curved path. It is the centrifugal acceleration that pulls one away from the center of a merry-go-round whereas one's arms and hands provide the balancing centripetal acceleration. For orbital motion, gravity provides the centripetal acceleration to counteract the centrifugal acceleration. 406

characteristic curve A characteristic curve is a curve along which a partial differential equation reduces to an ordinary differential equation. Equivalently, a characteristic curve is a trajectory in the independent-variable space along which a partial differential equation reduces to an an ordinary differential equation, so that the solution can be constructed by following these curves. 219, 220, 322, 323, 327

chart A chart on a manifold is a particular set of coordinates, ξ^a used to map points on a manifold, $\mathcal{P}$, to the real numbers (the coordinates), $\xi^a = \xi^a(\mathcal{P})$. Different charts can overlap, and on the overlaps there are smooth coordinate transformations between the charts. 83

Christoffel symbols Christoffel symbols are built from the covariant derivative of the basis vectors, thus encoding the spatial dependence of the basis vectors. They are not tensors, but they do serve a central role in continuum physics by providing the metric connection that allows the covariant derivative to compute the difference between tensors in a non-flat space and/or using non-Cartesian coordinates. They serve as the coordinate dependent expansion coefficients for the Levi-Civita connection. 30, 103, 105, 203

completeness Completeness refers to the ability of any complete basis, $P_m(x)$, to represent the Dirac delta as an infinite sum or spectral decomposition, as per $\delta(x-y) = w(y)\sum_m P_m(y)\,P_m(x)$. 157, 233, 263, 299, 335

composition property The composition property (also called the semigroup property) of the Green's function for the diffusion equation expresses how solutions at different times can be composed via convolution. This property connects the Green's function to Markov processes, in which the composition property is known as the *Chapman-Kolmogorov relation* (*Gardiner*, 1985). 298

configuration space Configuration space is defined by generalized coordinates, thus specifying where a physical system is located at a particular time instance. The dimension of configuration space is D, where D is the number of dynamical degrees of freedom. A physical state in Lagrangian mechanics is specified by the configuration space (generalized coordinates) plus the tangent to that space as specified by the generalized velocities. The Euler-Lagrange equations specify the evolution of a physical system as it moves through configuration space. 461, 489, 490

congruence A manifold filling set of integral curves forms a congruence. 87

conservation law A conservation law expresses the invariance of a dynamical property as the system evolves. Conservation laws are associated with symmetries as per Noether's theorem. 421

conservation of indices Conservation of indices refers to the need for a valid tensor equation to have tensor indices match across an equal sign. If they do not match, then there is a problem with the equation. 78

conservative force If the work done on a particle by a force is independent of the path taken between the points, then the force is said to be conservative. 355, 460

conservative vector field As conservative vector field has zero curl and so can be written as the gradient of a scalar field. 69

contact force A contact force acts on the boundary of a fluid element, with examples including stresses from pressure and from friction. Contact forces are local forces. Contact forces are sometimes referred to as internal forces, since they arise from local interactions internal to the fluid, as distinct from body forces that arise from long range external forces that act throughout the body of a fluid element. Contact forces are also called tractions in some areas of continuum mechanics. Contact forces are molecular in origin, though we are unconcerned in this book with details of the molecular dynamics leading to these forces. Contact forces act on a region of a continuous media through the area integrated stresses acting on the boundary enclosing the region. xix

continuum approximation The continuum approximation assumes that mathematical limits for fluid volumes tending to zero are reached on length and time scales very large compared to molecular space and time scales. The temporal realization of the continuum approximation is based on recognizing that macroscopic motion associated with fluid flows (e.g., advection, waves, and mixing) evolves with time scales far longer than the time scales of molecular motions. Hence, from a macroscopic perspective, the continuum approximation leads us to assume that all fluid motions are continuous in both space and time. 352

contraction Contraction refers to the summation of covariant and contravariant indices to produce a tensor of one lower order. For example, the contraction $F^a G_a$ produces a scalar known as the scalar product between two first order tensors, and the contraction $T^a{}_a$ produces the trace of a second order tensor. 34

contravariant index Contravariant index placement refers to the upstairs placement of a tensor index on the components to the representation of a tensor. For example, the contravariant representation of the velocity vector is written $\boldsymbol{v} = \boldsymbol{e}_m v^m$, with v^m the contravariant representation of the velocity using a particular coordinate system. The dual indices are the covariant indices, which live downstairs. We connect covariant and contravariant representations through a contraction with the metric tensor, g. 7, 78, 81, 397

convolution The convolution of two functions is an operation that produces a new function describing how the shape of one function is smeared or filtered by another. In the context of Green's functions, we write the solution to a linear partial differential equation in terms of the convolution of the Green's function with boundary data and source data. 235

coordinate basis A set of coordinate basis vectors has the geometric interpretation as the basis for a vector local to a point on a manifold, and they can be written as the coordinate partial derivative operators, $\boldsymbol{e}_a = \partial_a$. We do not always use coordinate basis vectors, instead sometimes making use of orthonormal basis directions,$\hat{\boldsymbol{e}}_a$, as motivated by Cartesian tensor analysis. However, coordinate basis vectors are very useful for Lagrangian coordinates. 11

coordinate representation A coordinate representation of a tensor refers to the numerical values of the expansion coefficients used to represent the tensor in terms of a chosen coordinate system. For example, the coordinate representation of the velocity vector is $\mathbf{v} = v^a \, \boldsymbol{e}_a$, where the v^a coefficients are the coordinate representations in terms of the basis, $\boldsymbol{e}_a$. The coordinate representation depends on the chosen coordinates, whereas the vector is a geometric object that remains independent of coordinates. 9

coordinates Coordinates provide a numerical representation for points in space. They provide the means to assign numbers to tensors, which are otherwise purely geometrical objects. 30

Coriolis acceleration The Coriolis acceleration, $-2\,\boldsymbol{\Omega} \times \mathbf{v}$, arises when describing motion in a rotating reference frame, with $\boldsymbol{\Omega}$ the angular velocity of the rotating frame. The velocity, $\mathbf{v}$, is the velocity relative to the rotating reference frame, so that the Coriolis acceleration vanishes for motion that is at rest in the rotating frame. 391, 401, 411, 430

Coriolis parameter The Coriolis parameter is the radial component of the Earth's planetary rotation vector: $f \equiv 2\,\boldsymbol{\Omega} \cdot \hat{\boldsymbol{r}} = 2\,\Omega \sin\phi$. It is a monotonically increasing function of latitude, moving from its minimum, $f = -2\Omega$, at the south pole to its maximum, $f = 2\Omega$, at the north pole. This monotonically increasing property is of fundamental importance to ocean and atmosphere flows. 413

Coriolis' theorem Coriolis' theorem provides the inertial acceleration as represented by coordinates in a rigidly rotating reference frame. This rotating reference frame representation introduces the following non-inertial accelerations: Coriolis acceleration, $-2\,\boldsymbol{\Omega} \times \mathbf{v}$, centrifugal acceleration, $-\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})$, and the Poincaré acceleration that arises from time dependence of the rotating reference frame, $-\mathrm{d}\boldsymbol{\Omega}/\mathrm{d}t \times \boldsymbol{X}$. In geophysical fluid mechanics, we typically assume $\mathrm{d}\boldsymbol{\Omega}/\mathrm{d}t = 0$, so that we are only concerned with non-inertial accelerations from planetary Coriolis and planetary centrifugal. 368

cotangent space The cotangent space, $T^*_{\mathcal{P}}$, is a linear vector space defined at each point, $\mathcal{P}$, of a smooth manifold. It is the space where one-forms live, and as such it is the space dual to the tangent space, $T_{\mathcal{P}}$. The accumulation of cotangent spaces for a manifold define the cotangent bundle. The generalized momenta in Hamiltonian mechanics lives in the cotangent space of the manifold on which motion occurs. 93, 490

covariance Covariance means that the physical equations are form invariant under a chosen symmetry operation. The fundamental equations of geophysical fluid mechanics respect Galilean covariance, and they do so using any arbitrary coordinate system. Additionally, the Eulerian form of the equations respect coordinate invariance whereby they take on the same form for all time independent coordinates. 76

covariant derivative The covariant derivative is the generalization of the partial derivative operator that transforms as a $(0,1)$ tensor with arbitrary coordinates. 77, 102

covariant index Covariant index placement refers to the downstairs placement of a tensor index on the components to the representation of a tensor. For example, the covariant derivative is represented as $\boldsymbol{\nabla} = \boldsymbol{e}^a \, \nabla_m$, with ∇_m the covariant representation of the derivative operator using a particular coordinate system. The dual indices are contravariant, which live upstairs. We connect covariant and contravariant representations through a contraction with the metric tensor, $\mathbb{g}$. 7, 13, 77, 78, 81, 397

cyclic coordinate If the Lagrangian has no explicit dependence on a particular generalized coordinate, then the corresponding generalized momenta is a constant of the motion. Such generalized coordinates are termed cyclic. 480, 483

cycloid A cycloid is a plane curve traced by a fixed point on the rim of a circle as the circle rolls without slipping along a straight line. The cycloid is the solution to the brachistochrone problem, which seeks the shape of a curve of fastest descent for a frictionless particle falling under uniform gravity field. 208

D'Alembert's formula D'Alembert's formula is an expression for the general solution to the wave equation, $(\partial_{tt} - c^2\nabla^2)\psi = 0$. 324, 343

D'Alembert's principle D'Alembert's principle states that at any instance along the trajectory of a physical system, the forces of constraint perform no work since the forces are applied perpendicular to the motion. It allows for the reformulation of Newton's laws so that problems in dynamics look like problems in statics. Formally, it states that for a mechanical system with constraints, the total virtual work of the applied forces plus the inertial forces vanishes for any virtual displacement consistent with the constraints. This principle forms a foundational element in the development of Lagrangian mechanics. 451, 456, 501

Deborah number The characterization of whether a material is a solid or fluid depends on the time scale of the macroscopic observation, t_{observe}, versus the time scale for the internal relaxations within the material, t_{relax}. The ratio $t_{\mathsf{relax}}/t_{\mathsf{observe}}$ is referred to as the Deborah number, Db. For the fluid mechanics considered in this book, we are concerned with tiny Deborah numbers, in which the relaxation time scales are determined by the relatively rapid molecular collisions that take place on time scales of order 10^{-10} s, whereas the observation time scales are determined by macroscropic deformations that are $\approx 10^0$ s. In turn, rigid bodies have a zero Deborah number since the particles comprising the rigid body remain rigidly fixed relative to one another, so that $t_{\mathsf{relax}} = 0$. 358

degrees of freedom For each degree of freedom in a dynamical system, there is a corresponding generalized coordinate. For example, a particle that moves in three dimensional Euclidean space has at most three independent degrees of freedom, meaning that three numbers are needed to specify the particle's spatial position at a particular time. 450

determinstic chaos Deterministic chaos is the emergence of seemingly random behavior from deterministic laws, arising from nonlinear dynamics and the associated sensitivity to initial conditions. Deterministic chaos plays a central role in the prediction of geophysical flows, with such flows highly nonlinear and turbulent. 352, 484

deviatoric tensor The deviatoric tensor is determined by removing the trace from a second order tensor, so that the deviatoric tensor has zero-trace. 34

diagnostic equation A diagnostic equation determines the value of a field at a particular time instance. An example is the non-divergence condition, $\nabla \cdot \boldsymbol{v} = 0$, satisfied by velocity in a Boussinesq ocean. There are generally no time derivatives appearing in diagnostic equations, though this property is generally a function of the chosen coordinate system. xi

differential rotation Using planetary Cartesian and spherical coordinates, earth's angular velocity vector is given by $\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}} = \Omega\,(\hat{\boldsymbol{\phi}}\,\cos\phi + \hat{\boldsymbol{r}}\,\sin\phi)$. We see that the angular velocity vector has a spherical representation that is a function of latitudinal position, with components in the local radial and local meridional directions. This spatial dependence to the local expression of planetary rotation is known as differential rotation. 366, 411, 431

diffusion equation The diffusion equation is the canonical parabolic partial differential equation. It describes the diffusion of scalar fields such as tracers. It also describes the diffusion of vorticity in the presence of Laplacian friction. 286

diffusion tensor The diffusion tensor is a second order symmetric and positive-definite tensor with dimensions of squared length per time. It is used to parameterize the downgradient eddy diffusive fluxes of tracers arising from turbulent motions. 261, 293

dilatation Dilitation refers to the change in volume per unit volume of a fluid element, and it is measured by the velocity divergence, $\nabla\cdot\boldsymbol{v}$. 351

Dirac delta The Dirac delta provides an idealization of a point source, $\delta(\boldsymbol{x})$, and it is formally infinite when evaluated at $\boldsymbol{x}=0$ whereas it vanishes at all other points. The Dirac delta plays a central role in the theory of Green's functions. When multiplied by mass, it provides the mass density for a point particle. In mathematics, the Dirac delta is known as a generalized function or a distribution. 147, 235, 328, 376, 485

Dirac delta sheet The Dirac delta sheet refers to the ability to absorb the Neumann boundary condition into a modified interior source. By doing so, the boundary value problem has a modified interior source that is proportional to a Dirac delta (hence the term Dirac delta sheet), but it now has a homogeneous boundary condition. This reformulation of the Neumann boundary condition is commonly pursued in the geophysical fluids literature, such as when studying potential vorticity and for water mass analysis. 257, 305

direction cosines In Cartesian coordinates, the matrix transforming from one coordinate system to another is a rotation matrix. Since the Cartesian basis vectors all normalized, elements of the rotation matrix are given by the cosine of the angle between the respective basis vectors. As such, the rotation matrix is sometimes referred to as the direction cosines matrix. 28

Dirichlet boundary condition The Dirichlet boundary condition prescribes the value of a function along the boundary. 225, 228, 242, 285, 287, 293, 328

dispersive waves Dispersive waves are characterized by phases whose speeds are a function of the wavenumber, so that the phase velocity is distinct from the group velocity. As a result, a localized disturbance that initially has many wavelengths will spread out or change shape as it propagates. Most waves we study in this book are dispersive, such as surface gravity waves, internal gravity waves, inertia waves, and Rossby waves. 321

divergence theorem The divergence theorem, also knows as Gauss's divergence theorem, provides a relation between the volume integral of the divergence of a vector to the boundary integral of the vector projected onto the outward normal: $\int_{\mathcal{R}} \boldsymbol{\nabla}\cdot\mathbf{F}\,\mathrm{d}V = \oint_{\partial\mathcal{R}} \mathbf{F}\cdot\hat{\mathbf{n}}\,\mathrm{d}\mathcal{S}$. There are many corollaries to this theorem that we use in this book, one that says for Cartesian tensors that $\int_{\mathcal{R}} \nabla\phi\,\mathrm{d}V = \oint_{\partial\mathcal{R}} \phi\,\hat{\boldsymbol{n}}\,\mathrm{d}\mathcal{S}$, which we use in formulating the effects from pressure in a weak formulation of the equations of motion. 291

domain of influence For a wave disturbance acting at a point in space-time, the domain of influence is the region of space-time that can be affected by the disturbance, given the propagation speed of the wave. 321, 325

duality condition The duality condition refers to the identity between a basis vector and basis one-form: $\boldsymbol{e}_a \cdot \boldsymbol{e}^b = \delta_a{}^b = \delta^b{}_a$. It is also referred to as the biorthogonality relation. For Euclidean space, we sometimes write the shorthand $\mathbb{g}(\boldsymbol{e}_a, \boldsymbol{e}^b) = \boldsymbol{e}_a \cdot \boldsymbol{e}^b$, though one should exercise care not to confuse this expression with the inner product between two vectors. 92

Duhamel's superposition integral Duhamel's superposition integral for time dependent partial differential equations allows us to move any source function from the partial differential operator to the initial conditions, and vice versa. It says that the forced solution is built by time integrating the retarded values of the unforced solution from the initial time, $t = 0$, to the current time, t. 290, 325

dynamical constraint A dynamical constraint arises from a conservation law that requires the motion to preserve a property unchanged as the system evolves. Conservation laws are associated with symmetries as per Noether's theorem. 421, 422, 450, 451

dynamics Dynamics is that part of mechanics that is concerned with the causes of motion that arise through the action of forces. Dynamics, through Newton's laws, provide the means to determine how motion is altered in the presence of forces. 349

earth gravitational acceleration The earth's gravitational accceleration, $\geq$, is the gravitational acceleration at the surface of the earth, and it is used throughout geophysical fluid mechanics: $g_{\mathsf{e}} = G\, M_{\mathsf{e}}/R_{\mathsf{e}}^2 \approx 9.8 \text{ m s}^{-2}$. 377

effective gravity The Earth's gravitational field (assumed static) as well as the planetary centrifugal acceleration (due to planetary rotation, assumed constant) both give rise to conservative forces. The combined gravitational and planetary centrifugal accelerations are encapsulated by the gradient of the geopotential, Φ, so that the effective gravitational force acting on a point particle of mass, m, is given by $\mathbf{F}_{\mathsf{geo}} = -m\, \nabla\Phi$. 356, 407, 408

eigenfunction See eigenvalue problem. 227, 260

eigenvalue See eigenvalue problem. 227

eigenvalue problem An eigenvalue problem is a linear operator equation of the form, $\mathcal{L}\,\Psi = \lambda\,\Psi$, where $\mathcal{L}$ is a linear operator, λ is a number, known as the eigenvalue, and Ψ is the corresponding eigenfunction. Evidently, the action of a linear operator onto an eigenfunction is identical to multiplying the eigenfunction by a number. Details of the eigenfunctions and eigenvalues depend on the boundary conditions and coordinates. For systems that are self-adjoint, the set of eigenfunctions forms a complete basis for the linear space on which the operator is self-adjoint. Eigenvalues can be discrete or continuous. 227, 260, 299, 335

Einstein summation convention The Einstein summation convention says that we drop the summation symbol while repeated indices (one upstairs and one downstairs) are summed over their range. The summation convention proves central to a variety of tensor manipulations, and so it is important to make friends with it. 10, 453

elements pillar The elements pillar of geophysical fluid mechanics comprises the physical and mathematical formulation of conceptual models used to garner insight into rotating and stratified fluid motion. This pillar is concerned with setting the stage by deductively and descriptively exposing how physical concepts are mathematically expressed to describe geophysical fluid flows. ix

elliptic partial differential equation An elliptic partial differential equation (PDE) is a second order PDE whose coefficients satisfy certain positivity conditions that ensure it behaves like Laplace's equation. The solutions are smooth and non-propagating, thus commonly found when studying static or equilibrium phenomena. 217, 241, 321, 376

emergent phenomena pillar The emergent phenomena pillar of geophysical fluid mechanics studies solutions to equations that describe phenomena, such as waves, instabilities, turbulence, and general circulation, all of which emerge from the fundamental equations based on first principles. These phenomena can emerge in manners that are far from simple to understand deductively, particularly when considering nonlinear behavior such as turbulence. ix

emergent scale There are two general types of dimensional scales that we use to non-dimensionalize a mathematical physics equation. One is the emergent scale, which emerges from the flow itself. Emergent scales, such as the length scale and velocity scale of the flow, are specified by the subjective interest of the theorist though these scales are not under direct control. That is, we choose to focus on flows with a particular scale for purposes of examining the corresponding equations that describe that flow regime. A key example concerns our study of planetary geostrophy and quasi-geostrophy, where we choose to focus on flows of a particular scale where the Coriolis acceleration is of leading order importance. xii

equilibrium tide The equilibrium tide is the shape of the sea surface on a spherically symmetric and non-rotating planet with no land masses. It also assumes all transients have vanished, so that it is the shape of the sea surface assuming it responds instantaneously to astronomical gravity forcing. 385, 386

equipartition of energy The virial theorem says that for physical systems with a potential energy that is a homogeneous function of degree, $\gamma = 2$, then the potential and kinetic energies have equal time averages. This result is referred to as the equipartition of energy. In addition to harmonic oscillators and small amplitude pendula, the equipartition of energy holds for linear wave motion, with the potential energy of linear waves generally a homogeneous function of degree two. 471, 522

error function The error function, $\mathrm{erf}(\xi) = \frac{2}{\sqrt{\pi}} \int_0^{\xi} \mathrm{e}^{-\lambda^2}\, \mathrm{d}\lambda$, along with the complement error function, $\mathrm{erfc}(\xi) = 1 - \mathrm{erf}(\xi) = \frac{2}{\sqrt{\pi}} \int_{\xi}^{\infty} \mathrm{e}^{-\lambda^2}\, \mathrm{d}\lambda$, appear frequently in solutions to the diffusion equation. The following properties for the error function are quite useful: $\mathrm{erf}(\xi) = -\mathrm{erf}(-\xi)$, $\mathrm{erf}(0) = 0$, $\mathrm{erf}(\infty) = 1$, as well as for the complementary error function, $\mathrm{erfc}(0) = 1$, $\mathrm{erfc}(\infty) = 0$. 318, 319

Euclidean space Euclidean space is the space of the classical physics considered in this book. It has zero curvature and can be fully described by Cartesian coordinates, for which the Euclidean metric tensor is given by the Kronecker delta. Euclidean space forms the background space on which we study fluid motion. Our main concern is geophysical fluid flows, in which the flow takes place on a rotating spherical planet and/or on stratification surfaces. We also describe fluid motion using space-time dependent Lagrangian coordinates. In each of these more general cases, the geometry and coordinates used to describe the motion are non-Euclidean and non-Cartesian. Even so, because the motion takes place with a Euclidean background geometry, the mathematical description inherits properties of the background Euclidean space, in particular the Kronecker metric. 3, 4, 7, 346, 352

Euler equation of variational calculus The Euler equation of variational calculus is the differential equation that arises from producing a stationary functional (and integral). Satisfying the Euler equation ensures that the functional is stationary. 191, 196, 197, 216

Euler identity The Euler identity is $\mathrm{e}^{-\mathrm{i}\,\omega\,t} = \cos(\omega\,t) - \mathrm{i}\,\sin(\omega\,t)$. Withing this book, the Euler identity is used throughout Fourier analysis and wave mechanics. 163

Euler's theorem Euler's theorem for homogeneous functions provides precise relation between a function's homogeneity and its partial derivatives. We make use of this theorem in our study of mechanics, through the virial theorem, and in thermodynamics. 469, 492

Euler-Lagrange equation The Euler-Lagrange equation is a differential equation that results from Hamilton's principle of stationary action. In classical mechanics, the Euler-Lagrange equation is the same as Newton's equation of motion. 196, 447, 449, 463, 466

Eulerian reference frame An Eulerian reference frame describes fluid motion relative to a fixed position ($\boldsymbol{x}$-space), commonly referred to as the laboratory frame. This kinematic description measures fluid properties as the fluid streams by a fixed observer. It is not concerned with determining trajectories. Instead, Eulerian kinematics focuses on fluid properties as continuous fields that are functions of the space position, $\boldsymbol{x}$, and time, t. 80, 81, 115, 484

evolution equation An evolution equation determines the time tendency (Eulerian evolution) of a quantity such as the temperature or velocity. Terms in the prognostic equation are referred to as time tendencies. Evolution equations are also referred to as prognostic equations. x, 312

external forces An external force acts throughout the extent of a fluid element and it arises from a force field that is external to the fluid element. It is synonymous with body force, with examples including the gravitational force, as well as the Coriolis and centrifugal forces arising from the rotating planetary reference frame. These forces are also known as long range forces. 369

external scale There are two general types of dimensional scales that we use to non-dimensionalize a mathematical physics equation. One is the external scale, with examples in this book being the gravitational acceleration, Coriolis parameter, and specified background or reference state. External scales are set by the geophysical parameter regime in which the flow occurs, and as such they are under direct control of the theorist or experimentalist. The other scale is emergent, and is a property of the flow. xii

f-plane The f-plane is a tangent plane approximation (tangent to the geoid) that makes use of Cartesian coordinates for studying geophysical fluid motion local to a point on the rotating planet and using a constant Coriolis parameter. Since motion is assumed to be on a constant geopotential, the f-plane makes use of the effective gravitational acceleration that includes both central gravity acceleration plus the planetary centrifugal acceleration. 425, 506, 518

Feynman's trick Feynman's trick (also called differentiation under the integral sign) is a method for evaluating complicated integrals by introducing a parameter and then differentiating with respect to that parameter. 240

field point In the context of Green's function theory, the field point refers to the point in space-time where the Green's function is sampled or observed. It contrasts to the source point, where the Dirac source is located. 235–237

flow map The flow map smoothly and continuously deforms the matter continuum through space as the fluid moves, and it is written as $\boldsymbol{\varphi}(\boldsymbol{a}, T)$. The flow map provides the trajectory for the fluid particle specified by the material coordinate, $\boldsymbol{a}$. So in this sense we can consider the flow map as the accumulation of all fluid particle pathlines, and with its time derivative, $\partial_T \boldsymbol{\varphi}(\boldsymbol{a}, T)$, providing the velocity of the fluid particle, $\partial_T \boldsymbol{\varphi}(\boldsymbol{a}, T) = \boldsymbol{v}[\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{a}, T), t = T] = \boldsymbol{v}^{\mathrm{L}}(\boldsymbol{a}, T)$. The motion field is the reason there is a flow map, so that the nomenclature "motion field" and "flow map" are used interchangeably in this book since they both refer to movement of the continuum. 12

forces of constraint There are any number of further constraints that arise from details of the particular motion, such as the imposition of spatial boundaries, rods or springs connecting particles, or other geometric conditions. From the perspective of classical mechanics, such constraints might be considered purely kinematical. For example, the motion of a frictionless bead threaded by a wire is constrained to move along a trajectory defined by the shape of the wire. Similarly, there are forces of constraint that impose the no-normal flow boundary condition on fluid flow. Such constraints reduce the spatial degrees of freedom of the particle motion. However, it is rarely simple, and often impossible in practice, to provide an *a priori* determination of the forces of constraint (also called reactive forces) that maintain the constraints. Since Newtonian mechanics requires all forces for determining the motion, it offers an impractical basis for studying systems where the forces of constraint are unknown *a priori*. 450, 451

Foucault pendulum A Foucault pendulum is a long, freely swinging pendulum that demonstrates Earth's rotation by exhibiting a slow precession of its plane of oscillation relative to the Earth's surface. In an inertial frame, the plane of swing of the pendulum remains fixed. Because the Earth rotates beneath it, an rotating terrestrial observer sees the swing plane rotate at a latitude-dependent rate determined by the Coriolis acceleration. 427, 506

Fourier analysis Fourier analysis is the representation of a field or signal as a superposition of sinusoidal basis functions (sines, cosines, or complex exponentials), each with a definite wavenumber (or frequency) and amplitude. We make use of Fourier analysis in our study of wave mechanics throughout this book. 161

Fourier series A Fourier series expresses a periodic function over a finite domain as a sum of sine and cosine waves (or complex exponentials) whose frequencies are integer multiples of a fundamental frequency or wavenumber. 167

Fréchet derivative The Fréchet derivative provides a generalization of an ordinary derivative. It measures how a functional (a mapping from functions to numbers) changes when the input function is varied. We make use of such derivatives when working with variational calculus. It is effectively the same as the functional derivative. 195, 196

Fredholm alternative The Fredholm alternative states that a linear differential equation of the form, $\mathcal{L}\psi = \Lambda$, has a solution only so long as the volume forcing, Λ, plus the boundary forcing (coming through the boundary conditions) is orthogonal to every function within null space of the adjoint operator. Alternatively, we say that if the volume and/or boundary forcing excite a zero eigenmode, then there is no solution. 265

free particle A free particle moves through space without feeling any forces. From Newton's law of motion, a free particle conserves its linear momentum. We also consider particle motion as that constrained to move along a constant geopotential and yet without feeling any other forces, and in this sense it is a free particle on a geopotential. 198, 443, 451, 475

free-space Green's function The free-space Green's function is the Green's function defined in unbounded space. It is sometimes referred to as the fundamental solution to the partial differential equation. 235, 236, 258, 280, 376

functional A functional is a function of functions that returns a number. For example, the action is a function of the particle trajectory, and Hamilton's principle says that the physically realized trajectory produces a stationary action with respect to variations of the trajectory. 191, 192

functional derivative The functional derivative (effectively the same as the Fréchet derivative) provides a generalization of an ordinary derivative. It measures how a functional (a mapping from functions to numbers) changes when the input function is varied. We make use of such derivatives when working with variational calculus. 195, 196

Galilean boost Galilean boost refers to a transformation in which the velocity is modified by a constant velocity. For velocity independent forces, Newtonian physics remains unchanged by a Galilean boost since since Newton's law of motion is Galilean covariant. 354

Galilean covariance Galilean covariance, also known as Galilean relativity, means that we can move between inertial reference frames without altering the physics, so long as the forces are velocity-independent and boundaries are absent. For such cases, all inertial reference frames are equally valid for studying physics, and the physical equations remain invariant under a Galilean transformation. 353

Galilean space-time Galilean space-time is the marriage of Euclidean space and Newtonian universal time. Galilean space-time is used throughout this book in our study of the non-relativistic motion of matter embedded in a background Euclidean space. 4, 76, 346, 349, 352, 451

Galilean transformation A Galilean transformation is a linear transformation from one inertial reference frame to another, operationally realized by shifting the space coordinate by $\boldsymbol{x} \rightarrow \boldsymbol{x} + \boldsymbol{U}\,t$, where $\boldsymbol{U}$ is a constant velocity. We generally expect the mechanical equations to be unchanged when undergoing a Galilean transformation, unless the forces have a dependence on the velocity. 4, 79, 363

gauge symmetry A gauge symmetry arises from a redundancy in the mathematical description of a physical system. In fluid mechanics, there is a gauge symmetry associated with the ability to add a constant to the streamfunction in a horizontally non-divergent flow, with the constant not affecting the velocity field. For three dimensional non-divergent flow, the vector streamfunction is arbitrary up to the gradient of a scalar, with the details of this scalar irrelevant to the physics of the fluid flow. We also encounter gauge symmetries when defining the flux of a scalar field. Since the convergence of the flux affects time changes to the scalar, the flux is arbitrary up to the curl of a scalar. The presence of gauge symmetries affords us some freedom to choose a particular gauge to suite our subjective needs, with the choice not altering the objective physics. A standard reference for gauge symmetry, in the context of electromagnetism, is given in Section 27-4 in Volume II of *Feynman et al.* (1963). 272

generalized coordinates A generalized coordinate is any variable that can be used to specify a degree of dynamical freedom in a dynamical system. For free particle motion in Euclidean space, we typically find it most convenient to use Cartesian coordinates, whereas other sorts of motion can find it far more convenient to use non-Cartesian coordinates (which can have arbitrary physical dimensions), thus motivating the name generalized coordinates. When formulated using Lagrangian or Hamitonian mechanics, the equations of classical mechanics take the same form regardless the details of the generalized coordinates. 450, 453, 454

generalized force A generalized force is the representation of the applied force using generalized coordinates. 457

generalized Lagrangian mean The generalized Lagrangian mean refers to a kinematic method used to decompose the flow into a Lagrangian mean plus a fluctuation relative to the mean. It is a hybrid Eulerian/Lagrangian method that introduces an Eulerian disturbance field to measure the position of a fluid particle relative to its mean position. 525

generalized Legendre equation The generalized Legendre equation is an ordinary differential equation that arises from a study of solutions to Laplace's equation on the sphere. Its solutions are associated Legendre functions. 232

generalized momenta Given a generalized coordinate, ξ^σ, generalized velocity, $\dot{\xi}^\sigma$, and Lagrangian, $L(\xi^\sigma, \dot{\xi}^\sigma, t)$, then the corresponding generalized momenta, p_σ, is given by $p_\sigma = \partial L/\partial \dot{\xi}^\sigma$. The generalized momenta, also known as the canonical momenta, plays a central role in Hamiltonian mechanics. Notably, the phase space is described by generalized coordinates and generalized momenta. 480

generalized orthogonal coordinates Generalized orthogonal (curvilinear) coordinates are defined by a diagonal metric tensor with elements defining the stretching or compression in the respective orthogonal directions. They represent generalizations of the spherical coordinates and cylindrical-polar, with the additional possibility of stretching in the vertical direction. We only consider time-independent generalized orthogonal coordinates in this book. Such coordinates are distinct from generalized vertical coordinates, which are non-orthogonal and are time dependent. 116, 117, 133

generalized vertical coordinate A generalized vertical coordinate, σ, has a one-to-one invertible relation with the geopotential vertical coordinate, z, so that $\sigma = \sigma(x, y, z, t)$, and yet this coordinate is typically not orthogonal to the horizontal Cartesian coordinates. Generalized vertical coordinates are commonly used as the basis for numerical ocean and atmosphere models, and frequently used for theoretical formulations. xx, 4, 5, 73, 82, 83, 110, 115, 135

geodesic A geodesic is a curve that extremizes the free particle action or locally minimizes (or extremizes) distance or proper time, depending on the context either in pure mathematics or mechanics. A straight line is a geodesic on a flat plane whereas a great circle is a geodesic on the surface of a sphere. 198, 203, 492

geophysical fluid dynamics Geophysical fluid dynamics is a branch of fluid mechanics concerned with natural fluid motion on a rotating and gravitating body such as a planet or star. viii

geophysical fluid mechanics Geophysical fluid mechanics is a branch of theoretical physics concerned with natural fluid motion on a rotating and gravitating body such as a planet or star, making use of concepts and methods from classical continuum mechanics and thermodynamics. vii

geopotential The geopotential for the rotating earth is the sum of the gravitational potential plus the potential from the planetary centrifugal acceleration. A resting ocean forms the shape of a geopotential, which is approximated as an oblate spheroid with slightly larger radius at the equator than the poles. 356, 408, 414, 444

geopotential height The geopotential height equals to the geopotential divided by the gravitational acceleration. In atmospheric circulation studies it is often useful to know the geopotential thickness of a column bounded by isobars. 384

geopotential vertical coordinate The geopotential vertical coordinate measures the vertical distance from the geoid defined by a resting ocean surface. It is commonly written as z, with $z > 0$ moving away from the earth center. 417

geostrophic balance The geostrophic balance is a diagnostic balance between the pressure gradient acceleration and the Coriolis acceleration. It is well maintained for the large-scale and low frequency middle to high latitude motions in the atmosphere and ocean. Geostrophic balance does not hold near the equator, since the Coriolis parameter vanishes there. xi

global instabilities Global fluid instabilities arise from the constructive interference of waves and so involve the solution of an eigenvalue problem to determine properties of unstable waves. At most, a necessary condition can be derived to determine whether a global instability exists. Global instabilities are also referred to as wave instabilities. xxi

great circle Great circles are geodesics on the surface of a sphere. They provide the shortest distance between any two points on the sphere. To create a great circle, consider any two points on the surface of the sphere with position vectors, $\boldsymbol{x}_{\mathsf{A}}$ and $\boldsymbol{x}_{\mathsf{B}}$, relative to the sphere's center. The plane defined by $\boldsymbol{x}_{\mathsf{A}}$ and $\boldsymbol{x}_{\mathsf{B}}$ passes through the sphere's center and intersects sphere's surface to form a circle. The great circle the shortest of the two segments of the arc along this circle. 205

Green's function The Green's function is the formal inverse of a linear differential operator. Knowledge of the Green's function allows one to write the solution to a linear differential equation as an integral, with the Green's function acting as a kernel. 4, 5, 147, 159, 190, 235, 242, 291, 329, 376

Green's theorem Green's theorem is the expression of Stokes' theorem on a plane. 70

Hamilton's equations of motion Hamilton's equations of motion are the dynamical equations arising in Hamiltonian mechanics. 482

Hamilton's principle Hamilton's principle of stationary action is a variational method used to derive the equations of motion for a non-dissipative dynamical system, with the corresponding differential equations of motion known as the Euler-Lagrange equations. In classical mechanics, Hamilton's principle results in the same equations as Newton's second law. However, Hamilton's principle has widespread use beyond classical mechanics, making it part of most theories of physics. xiii, 191, 192, 346, 447, 463, 464

Hamilton-Jacobi equation The Hamilton-Jacobi equation .The Hamilton–Jacobi equation is a first-order partial differential equation for the principal function (the on-shell action) whose solution completely determines the dynamics of a classical mechanical system. Provides a powerful method for solving integrable systems; forms the bridge between classical mechanics and geometrical optics; and underlies the WKB approximation and the classical limit of quantum mechanics. 489

Hamiltonian The Hamiltonian is the energy-like function whose partial derivatives generate the full time evolution of a dynamical system. In many cases, the Hamiltonian is identical to the mechanical energy of a physical system, but that identical is not general. 478

Hamiltonian mechanics Hamiltonian mechanics is that area of analytical mechanics formulated in the in the phase space defined by generalized coordinates and canonical momenta The Lagrangian mechanics of configuration space is related via a Legendre transformation to the Hamiltonian mechanics of phase space. The Euler-Lagrange equations of motion are derived by Hamilton's variational principle, which states that physical systems evolve so as to produce a stationary action. When applied to classical particle and classical continuum mechanics, Lagrangian and Hamiltonian mechanics is logically equivalent to Newtonian mechanics. Rather than maintain forces at the center of the dynamical formulation (Newton), the Lagrangian and Hamiltonian approaches put energy at the forefront, and they each formulate the equations of motion through Hamilton's variational principle of stationary action. The energy approach is generally far simpler in practice than the force approach, thus making analytical mechanics an indispensible tool for engineering mechanics. We make use of Hamiltonian mechanics when working with ray theory, and it forms the starting point in formulating quantum mechanics using linear operator theory. 447, 449, 480

harmonic function A harmonic function has zero Laplacian: $\nabla^2\psi = 0$. Harmonic functions satisfy the remarkable mean value property, whereby the value of a harmonic function at a point $\boldsymbol{x}$ within an open region of a domain, $\mathcal{R}$, equals to the average of ψ taken over the surface of a sphere within $\mathcal{R}$ that is centered at $\boldsymbol{x}$. 223, 230, 250

harmonic oscillator A harmonic oscillator is a dynamical system that experiences a linear restoring force toward an equilibrium position. Its defining feature is that the force is quadratic in the displacement from equilibrium, leading to sinusoidal motion. We sometimes refer to a simple harmonic oscillator, which is a one-dimensional harmonic oscillator with no friction. 519

Heaviside step function The Heaviside step function, $\mathcal{H}(x)$, also known as the unit step function, is a discontinuous non-dimensional mathematical function that outputs 0 for negative input values and unity for positive input values. It takes values $\mathcal{H}(\tau) = 0$ for $\tau < 0$ and $\mathcal{H}(\tau) = 1$ for $\tau > 0$. The derivative of a Heaviside step function is the Dirac delta: $\mathrm{d}\mathcal{H}/\mathrm{d}x = \delta(x)$. Note that we do not define the Heaviside at $x = 0$, though some authors give it a value of $\mathcal{H}(0) = 1/2$. For our purposes, the properties of the Heaviside step function remain unchanged whether it is defined at $x = 0$ or not. See footnote on page 20 of *Stakgold* (2000a) for more details. 150, 152, 292, 308

Heisenberg uncertainty principle The Heisenberg uncertainty principle is the quantum wave mechanical realizaton of the uncertainty relation holding in Fourier analysis. That is, the Heisenberg uncertainty principle is the Fourier-based statement that non-commuting observables cannot be sharply known at the same time. The uncertainty reflects physics, not measurement flaws. 187

Helmholtz decomposition In characterizing the kinematic properties of vector fields, such as the velocity vector for a moving fluid, *Helmholtz* (1867) introduced a method to decompose any vector into two distinct components whose properties are readily analyzed: one component vector has a zero divergence and the second vector has a zero curl. More formally, the Helmholtz decomposition says that any sufficiently smooth and rapidly decaying vector field can be decomposed uniquely into the sum of a gradient of a scalar potential and the curl of a vector potential. This Helmholtz decomposition, and its generalization to the Hodge decomposition, has extensive applications throughout fluid mechanics. 271

Helmholtz equation the Helmholtz equation arises from assuming the monochromatic time dependence, $\psi(\boldsymbol{x}, t) = e^{-\mathrm{i}\,\omega\,t}\,\Psi(\boldsymbol{x})$, for a solution to the wave equation, $(\partial_{tt} - c^2\,\nabla^2)\psi = 0$. It takes the form, $[\nabla^2 + (\omega/c)^2]\,\Psi = 0$. The Helmholtz equation arise when we are uninterested in the initial value problem for waves, but instead are interested in the steady state where the wave field is fully established. 241, 279

heuristic A heuristic technique or argument employs a practical method not guaranteed to be fully rational or deductive from all perspectives, but is sufficient for establishing a self-consistent formalism. The study of the Dirac delta is an example where physical heuristics established a formalism whose mathematical rigor followed later. 147

holonomic constraints When constraints are placed on the spatial position of the particles, then they are referred to as holonomic constraints. For example, the motion of a particle restricted to a frictionless two dimensional surface has its three Cartesian coordinates related by a function of the form, $\Psi[x(t), y(t), z(t)] = \mathcal{C}$, where $\mathcal{C}$ is a constant. For example, if the particle is confined to move on the surface of a sphere of radius, R, then the constraint takes the form $\Psi[x(t), y(t), z(t)] = x(t)^2 + y(t)^2 + z(t)^2 = R^2$. By this restriction the particle has two independent spatial degrees of freedom rather than the three available without the constraint. We refer to holonomic constraints as integrable since any differential holonomic constraint can be integrated to yield the algebraic constraint. 451

homogeneous Space is homogeneous if there is no distinction between one point and another. Likewise, a tensor possesses the homogeneity property if it is the same at each point in space. A spatially constant temperature field is an example of a homogeneous scalar tensor field. Noether's theorem says that linear momentum is conserved for physical systems possessing space homogeneity. Likewise, mechanical energy is conserved for physical systems possing homogeneity in time. 31, 32, 474

homogeneous function Consider a suite of Q independent variables, $X_1, X_2, ...X_Q$, and an arbitrary function of these variables, $F(X_1, X_2, ...X_Q)$. We say that this function is a homogeneous function of degree γ if $F(\lambda\, X_1, \lambda\, X_2, ...\lambda\, X_Q) = \lambda^{\gamma}\, F(X_1, X_2, ...X_Q)$, with λ an arbitrary scalar. The left hand side is the function evaluated with each of the independent variables scaled by the same number, λ. The right hand side is the function evaluated with the unscaled variables, but multiplied by the scale raised to the power γ. Euler's theorem for homogeneous functions provides a precise relation between a function's homogeneity and its partial derivatives. We encounter properties of homogeneous functions when establishing the virial theoram in mechanics and when studying thermodynamic functions. 469

Hooke's law Hooke's law provides a linear relationship between stress and strain in elastic media. The simplest form is for a linear harmonic oscillator, but more general forms are used in the study of elastic media. 520

hydrodynamics A branch of fluid mechanics concerned with the flow of a homogeneous (constant density) incompressible fluid. vii

hydrostatic approximation The hydrostatic balance is a diagnostic balance between the vertical pressure gradient force and the weight of fluid. The exact hydrostatic balance holds for a static fluid in a gravity field. The approximate hydrostatic balance holds quite well for moving fluids with scales of motion such that the vertical scales are far smaller than the horizontal scales. 436

hydrostatic balance The hydrostatic balance is a diagnostic balance between the vertical pressure gradient force and the weight of fluid. The exact hydrostatic balance holds for a static fluid in a gravity field. The approximate hydrostatic balance holds quite well for moving fluids with scales of motion such that the vertical scales are far smaller than the horizontal scales. xi

hydrostatic primitive equations The hydrostatic primitive equations refer to the equations of motion for a fluid on a rotating planet, with the vertical momentum equation given by the hydrostatic balance. The horizontal equations are based on the traditional approximation in which the fluid thickness is much smaller than the planetary radius. The term "primitive" means that the prognostic dynamical field is the velocity rather than vorticity and divergence. *Smagorinsky* (1963) was an early proponent of the hydrostatic primitive equations for use in studying the large-scale ocean and atmospheric circulation. These equations form the basis for many general circulation models of the atmosphere and ocean. 433

hyperbolic partial differential equation A hyperbolic partial differential equation a second order PDE whose solutions have finite propagation speed, can exhibit oscillations, and typically require initial conditions in time. Disturbances propagate with finite speed along characteristic curves (or surfaces). 222, 321, 322

ignorable coordinate If the Lagrangian is independent of a coordinate, referred to as a cyclic coordinate, then its corresponding generalized momenta is a constant of the motion. Even so, the corresponding generalized velocity is not necessarily zero and so it cannot be ignored. However, in the Hamiltonian formulation we can ignore a cyclic coordinate since the generalized momenta is now a time-independent parameter of the Hamiltonian. As such, there is no corresponding equation of motion. We thus find the Hamiltonian dynamical system has two less degrees of freedom, one for the cyclic coordinate and one for the corresponding constant generalized momenta, thus motivating the name ignorable coordinate when a cyclic coordinate is encountered in the Hamiltonian formulation. 483

impulse The impulse is the time integral of a force. If the force is further idealized to occur just at a single time instance, and it is normalized to unity, then we have the unit impulse, which is the integral of the Dirac delta. 159, 353

impulse response function The impulse response function is the response of a dynamical system to unit impulse. If the dynamical system is linear, then the impulse response function equals to the Green's function for the initial value problem. 159, 307

index gymnastics Index index gymnastics refers to the rules for manipulating the representation of tensors. 10, 25, 78

inertial oscillation An inertial oscillation is the free horizontal motion of a fluid element on a rotating Earth that occurs when the only restoring influence is the Coriolis force, with no pressure-gradient force or friction acting. Motion is clockwise in the northern hemisphere and counter-clockwise in the southern. This motion arises from a balance between the Coriolis acceleration of the rotating frame (planetary Coriolis) and the centrifugal acceleration arising from the particle's circular motion. Inertial oscillations appear in inertial waves, in which the only restoring force is the Coriolis force. Inertial waves are the limit of inertia-gravity waves as the stratification vanishes. Note that inertial oscillations are not geostrophic, as acceleration is leading order and the flow is unbalanced since pressure gradients are neglected. 426, 427

inertial reference frame An inertial frame is a reference frame in which Newton's law of motion holds in its canonical form, such as $\mathbf{F} = m\,\mathsf{A}$ for a constant mass particle. When viewed in non-intertial reference frames, Newton's law of motion picks up non-inertial forces such as the Coriolis and centrifugal forces encountered when describing motion from the rotating planetary reference frame. 353, 363

integral curve An integral curve is defined so that a given vector field is tangent to each point along the curve. For example, fluid particle trajectories are integral curves for the fluid velocity field. 87

integral surface An integral surface is a surface on which the solution to a partial differential equation is constant. The space in which the surface lives is determined by both the independent variables and the dependent variables appearing in the differential equation. 219

internal force Internal forces arise from interactions between material elements within a physical system. 369

irreversible process A physical process that results in the increase of entropy. Processes that increase the entropy of a fluid particle include the mixing of momentum such as through viscous friction; the mixing of matter such as through the diffusion of constituents in a multi-component fluid; and the mixing of enthalpy (diffusion of heat) in a fluid with variable temperature. vii

isopycnal An isopycnal is a surface of constant potential density, which serves also as a surface of constant globally referenced Archimedean buoyancy. Isopycnal models are commonly used to study the flow of stratified perfect fluids. They are the continuous expression of an adiabatic stacked shallow water model. 82, 202

isotropy Space possesses isotropy if there is no distinction between any direction. Likewise, a tensor is isotropic if it remains unchanged under any rotation. 31, 32, 474, 475, 477

Jacobian of transformation The Jacobian of transformation is the determinant of the transformation matrix used to transform coordinate representations of tensors from one set of coordinates to another. The Jacobian of transformation is one-to-one if the Jacobian does not vanish. 91

Jacobian operator The Jacobian operator in the horizontal plane is given by $J(\psi, C) = \hat{\boldsymbol{z}} \cdot (\nabla\psi \times \nabla C) = \partial_y \psi \, \partial_y C - \partial_y \psi \, \partial_x C$. It is commonly used to express advection of a scalar, C, by a horizontally non-divergent flow, $\boldsymbol{u} = \hat{\boldsymbol{z}} \times \nabla\psi$, in which case $\boldsymbol{u} \cdot \nabla\psi = \hat{\boldsymbol{z}} \cdot (\nabla\psi \times \nabla C) = J(\psi, C)$. 69

Kepler problem The Kepler problem is the classical mechanics problem of determining the motion of a particle subject to a central inverse square force, such as Newtonian gravity. 511

Kepler's third law In orbital mechanics, Kepler's third law states that the square of the orbit period scales as the cube of the orbit size. 470

kinematic boundary condition The kinematic boundary conditions arise from the kinematic constraints on fluid motion when encountering a boundary. The simplest kinematic boundary condition is the no-normal flow, in which $\boldsymbol{n} \cdot \boldsymbol{v} = 0$ for flow encountering a static and material boundary, such as the solid-earth, with $\hat{\boldsymbol{n}}$ the outward normal. When the boundary is material and yet moves, then $\boldsymbol{n} \cdot (\boldsymbol{v} - \boldsymbol{v}^{(\mathsf{b})}) = 0$, where $\boldsymbol{v}^{(\mathsf{b})}$ is the veloctiy of a point attached to the moving boundary. When the surface allows for fluid to cross, then the kinematic boundary condition is written $\rho \, (\mathbf{v} - \mathbf{v}^{(\mathsf{b})}) \cdot \hat{\mathbf{n}} = -\mathcal{Q}_{\mathsf{m}}$, where $\mathcal{Q}_{\mathsf{m}}$ is the mass per time per area crossing the surface. If the boundary is the ocean surface, and the boundary is a monotonic function of vertical, then we can write the surface ocean kinematic boundary condition as $w + \rho^{-1} \, Q_{\mathsf{m}} = (\partial_t + \boldsymbol{u} \cdot \nabla)\eta$, where $z = \eta(x, y, t)$ is the vertical position of the ocean free surface, and Q_{m} is the mass per time per horizontal area of fluid crossing the boundary. 273, 274

kinematics Kinematics is that part of mechanics that is concerned with the intrinsic properties of motion (position, velocity, and acceleration), including properties of the space and time in which motion occurs with the nature of motion. 349

Kronecker delta The Kronecker delta symbol, δ_{ab}, equals to unity if the indices $a = b$ and zero otherwise. The Kronecker delta is the Cartesian coordinate representation of the metric for Euclidean space using Cartesian coordinates. 7, 12, 14, 117, 156, 167, 397, 459

Kronecker tensor The Kronecker tensor is a symmetric $(1, 1)$ tensor with components written $\delta_a{}^b = \delta^b{}_a$, whose values equal to unity if the indices $a = b$ and zero otherwise. The Kronecker tensor has the same numerical values in any coordinate system, and is sometimes referred to as the identity tensor, $\mathcal{I} = \delta_a{}^b \, \boldsymbol{e}^a \otimes \boldsymbol{e}_b$. 11, 14

Lagrange multiplier A Lagrange multiplier is an extra dependent variable or field that multiplies the constraint and then added to the Lagrangian. When forming the Euler-Lagrange equations, the Lagrange multiplier enforces the constraint. It is thus proportional to the force of constraint. x, 450, 471

Lagrange's equation of motion In mechanics, the Lagrange equation of motion is equivalent to Newton's equation of motion, and yet it eliminates the forces of constraint and is written fully in terms of generalized coordinates. 456, 460

Lagrangian In the context of Lagrangian and Hamiltonian mechanics, the Lagrangian is the difference between the kinetic energy and potential energy. Its time integral forms the action used in Hamilton's principle of stationary action. 191, 193, 346, 461

Lagrangian coordinate A Lagrangian coordinate (also material coordinate) provides continuum of markers that distinguish fluid particles. These coordinates are used in the Lagrangian or material reference frame. There is a one-to-one relation between Lagrangian coordinates and Eulerian coordinates, thus providing an invertible mapping between the Eulerian and Lagrangian reference frames. 82

Lagrangian mechanics Lagrangian mechanics is an area of analytical mechanics formulated in the configuration space defined by generalized coordinates and their corresponding time derivatives (generalized velocities). The Lagrangian mechanics of configuration space is related via a Legendre transformation to the Hamiltonian mechanics of phase space. The Euler-Lagrange equations of motion are derived by Hamilton's variational principle, which states that physical systems evolve so as to produce a stationary action. When applied to classical particle and classical continuum mechanics, Lagrangian and Hamiltonian mechanics is logically equivalent to Newtonian mechanics. Rather than maintain forces at the center of the dynamical formulation (Newton), the Lagrangian and Hamiltonian approaches put energy at the forefront, and they each formulate the equations of motion through Hamilton's principle of stationary action. The energy approach is generally far simpler in practice than the force approach, thus making analytical mechanics an indispensible tool for engineering mechanics. Lagrangian mechanics has extensions to the study wave mechanics as well as all other areas of theoretical physics such as electrodynamics, general relativity (Einstein-Hilbert action), and quantum mechanics (Feynman path integrals). 447, 449

Lagrangian reference frame A Lagrangian reference frame is defined by motion of material fluid particles; i.e., it is a reference frame that is comoving with the continuum of fluid particles ($\boldsymbol{a}$-space). The mechanical description aims to determine the continuum of trajectories, with each trajectory delineated by a continuous material coordinate that labels each fluid particle. The Lagrangian reference frame is non-inertial since fluid particles generally experience accelerations via changes to their speed and/or direction. 5, 73, 80, 82, 484

Lagrangian time The Lagrangian time is the time measured in the Lagrangian reference frame. Lagrangian time is the same as time measured in the Eulerian reference frame, since we work with universal Newtonian time. Nonetheless, it is convenient to make the distinction when measuring how fluid properties change since these changes are subject to motion of the observer. 80

Lagrangian time derivative The Lagrangian time derivative measures the evolution of a fluid property along a trajectory defined by a fluid particle. This time derivative is also referred to as the material or substantive time derivative. 80

Lamé coefficients Lamé coefficients are the square roots of the diagonal metric tensor components in orthogonal curvilinear coordinates. 134

Laplace transform The Laplace transform is an integral transform that converts a function of time (usually defined for $t \geq 0$) into a function of a complex variable. It is widely used to solve differential equations and analyze linear systems. 240

Laplace's equation Laplace's equation, $\nabla^2\psi = 0$, is the canonical elliptic partial differential equation. Solutions are said to be harmonic functions and they satisfy a variety of integral properties. 222

Laplace-Beltrami operator The Laplace-Beltrami operator is the Laplacian written using arbitrary coordinates. It is written $\}^{-1}\,\partial_{\overline{m}}\,(\}\,\mathbb{g}^{\overline{mn}}\,\partial_{\overline{n}}C$, where $\}$ is the square root of the determinant of the metric tensor written using the $\xi^{\overline{m}}$ coordinates. 107, 145

Laplacian operator The Laplacian operator is built from the squared gradient operator, $\nabla\cdot\nabla = \nabla^2$. In Cartesian coordinates we have $\nabla^2 = \partial_{xx} + \partial_{yy} + \partial_{zz}$, with other coordinates having specific forms. 223

Legendre condition The Lengendre condition states that a necessary (though not sufficient) condition for the action to be a local minimum is for $\partial^2 L/\partial y'^2 > 0$, where $L = L[y(x), y'(x), x]]$ is the Lagrangian functional. This condition has its generalization to higher dimensional systems, in which the second derivative becomes a matrix whose eigenvalues must all be positive in order to satisfy the necessary condition. 197

Legendre transformation A Legendre transformation in thermodynamics and mechanics allows us to introduce new thermodynamic potentials that have modified natural functional dependencies. In thermodynamics, we can replace an extensive natural variable (like entropy or volume) in a thermodynamic potential with its conjugate intensive variable (temperature or pressure). Doing so is motivated if the new thermodynamic potential is more suited to experimental or theoretical conditions. The Lagrangian formulation of mechanics is related to the Hamiltonian formulation via a Legendre transformation. 480

Levi-Civita connection The Levi-Civita connection is the unique affine connection on a Riemannian manifold that is torsion-free, and is metric compatible (covariant derivative of the metric vanishes). That connection itself is a geometric object, as it provides a rule for how to take covariant derivatives of tensor fields. The expansion coefficients for the Levi-Civita connection as known as the Christoffel symbols. 103, 104

Levi-Civita tensor The Levi-Civita tensor is a third order tensor whose $(0,3)$ coordinate representation has indices that are totally anti-symmetric: $\varepsilon_{abc} = -\varepsilon_{bac} = -\varepsilon_{acb} = \varepsilon_{cba}$. In Cartesian coordinates, the Levi-Civita tensor is represented by the permutation symbol, ϵ_{abc}. 20, 69

Lie algebra A Lie algebra is a vector space, where linear combinations make sense (this is why angular velocity is a vector). A Lie group is not a vector space (e.g., rotations in three-dimensional space do not commute) but the Lie algebra is. A Lie group is a smooth space of symmetry operations. Its tangent space at the identity transformation contains the infinitesimal generators, which form the corresponding Lie algebra. 363

Lie bracket The Lie bracket between two vectors in a tangent space produces a third vector, also in the tangent space, $[\mathbf{u}, \mathbf{v}] = \mathbf{w}$. The vector $\mathbf{w}$ measures the non-commutivity of directional derivatives computed along integral curves of $\mathbf{u}$ and $\mathbf{v}$. 145

Lie group A Lie group is a smooth family of symmetry operations form a group under composition (associative, identity exists, inverse exists); depends smoothly on one or more real parameters described by a manifold (a smooth geometric space). An example Lie group is the $SO(3)$ group (special orthogonal) formed by the rotational matrices in three-dimensional space. A Lie group is a smooth space of symmetry operations; its tangent space at the identity transformation contains the infinitesimal generators, which form the corresponding Lie algebra. 362, 363

linear momentum The linear momentum of matter is given by the mass times its velocity. Time changes to the linear momentum are determined by the forces acting on the system, as per Newton's second law of motion. 353, 443

linear operator In this book, we consider a linear operator to be the combination of a linear differential operator plus linear boundary conditions. 229, 260

linear oscillator equation The linear oscillator equation is the second order linear ordinary differential equation that describes small oscillations about a stable equilibrium under a linear restoring force. We encounter linear oscillator equations in the study of Hooke's law restorative forces as well as in the study of linear waves. 520

Liouville's theorem Liouville's theorem states that the volume of an infinitesimal region of phase space remains constant when following trajectories. This result is directly analogous to the case of a fluid that evolves with a non-divergent velocity. 484, 487

local instabilities Local fluid instabilities are afforded a local necessary and sufficient condition to determine whether the fluid base state is unstable to perturbations. Gravitational instability provides the canicial example, along with centrifugal and symmetric instabilities. Local instabilities are also referred to as parcel instabilities. xxi

manifold For our purposes, a manifold is any set of points whereby any open subset of the manifold can be mapped the real numbers. In this manner, a manifold is locally Euclidean. 83, 468

mass conservation In our formulation of fluid mechanics, we assume that matter is neither created nor destroyed anywhere within the fluid continuum, and furthermore that the fluid remains in a single phase. These assumptions allow us to formulate differential and integral expressions for mass conservation. 42

material coordinate A material coordinate (also Lagrangian coordinate) provides continuum of markers that distinguish fluid particles. These coordinates are used in the Lagrangian or material reference frame. There is a one-to-one relation between material coordinates and Eulerian coordinates, thus providing an invertible mapping between the Eulerian and Lagrangian reference frames. 82

material time derivative The material time derivative measures the evolution of a fluid property along a trajectory defined by a fluid particle. This time derivative is also referred to as the Lagrangian or substantive time derivative. 80, 486

Maupertuis principle Maupertuis' principle states that, at fixed energy, the actual trajectory of a conservative mechanical system extremizes the abbreviated action $\int \mathbf{p} \cdot \mathrm{d}\mathbf{q}$, making dynamics equivalent to geodesic motion in an energy-dependent metric. 492

mean value property The mean value property states that the value of a harmonic function, $\nabla^2 \psi = 0$, at a point $\boldsymbol{x}_0$ within an open region of $\mathcal{R}$ equals to the average of ψ taken over the surface of a sphere within $\mathcal{R}$ that is centered at $\boldsymbol{x}_0$. 224, 287

mechanical energy Mechanical energy of a fluid is a dynamical property formed by adding the energy due to motion of fluid elements (kinetic energy) to the energy arising from the position of a fluid element within the geopotential field (potential energy from the effective gravitational field). 356

mechanical momentum Mechanical momentum is the linear momentum appearing in Newton's second law, and it is the momentum directly associated with motion. For a classical mechanical system, the mechanical momentum equals to the mass times the velocity. In many physical systems, the mechanical momenta is the same as the generalized (or canonical) momenta. However, for other systems, such as motion within an electromagnetic field, the mechanical momenta differs from the generalized momenta. 457

mechanical similarity Consider a physical system in which we scale the length by λ. This scaling alters the Lagrangian in a manner that alters the Euler-Lagrange equations. However, if we couple scaling of the length with a corresponding scaling of the time, then it is conceivable that we can find a combination of these two scalings that results in an overall factor multiplying the Lagrangian, in which case the Euler-Lagrange equation are unaltered. We refer to this confluence of scaling as mechanical similarity. 470

mechanical work Mechanical work refers to the work done by mechanical forces acting on matter along the time integrated trajectory. Equivalently, it is the time integral of the power. 355, 437

method of characteristics The method of characteristics transforms a partial differential equation into a family of ordinary differential equations along special curves, called characteristic curves, in the space of independent variables. By solving these ordinary differential equations, one can construct the solution of the partial differential equation. Equivalently, the method of characteristics is a formalism that determines the functional dependence of solutions to partial differential equations This method is quite useful for exposing general properties of the solutions, even for those cases where the solution is not analytically available. 217, 219

method of images The method of images (or method of mirror images) is a mathematical tool for deriving the Green's function for a boundary value problem by combining a selected set of free space Green's functions, some of which live outside the domain of interest but serve to satisfy the desired homogeneous boundary condition. Once the boundary condition is enforced by construction, the problem reduces to evaluating the Green's function from the real and image sources in unbounded space. 258, 319

metric A metric tensor is a symmetric second order tensor that provides the means to measure the distance between two points in space. The Kronecker delta is the Cartesian coordinate representation of the Euclidean space metric tensor. When working with alternative coordinates (e.g., spherical, generalized vertical coordinates), the coordinate representation becomes less trivial. 12, 14, 89, 397

metric acceleration The metric acceleration is that contribution to the acceleration arising from time dependence of the unit vectors that appears when taking the time derivative of the velocity vector. 401, 410, 411, 434

metricity Metricity (also metric compatibility of the connection), holds for manifolds with zero torsion on which the covariant derivative of the metric tensor vanishes, $\nabla_m \mathrm{g}_{ab} = 0$. We assume metricity throughout this book. When metricity holds we find that the connection preserves the metric under parallel transport, so that lengths of vectors are preserved under parallel transport, angles between vectors are preserved, and inner products are invariant along transported vectors. In this manner, the geometry is rigid with respect to the metric. The fundamental theorem of Riemannian geometry states that for a given metric, there is a unique connection that is (i) torsion-free, and (ii) metric-compatible, with this connection known as the Levi–Civita connection, which is used in standard general relativity and classical differential geometry, as well as this book. 105

minimum-maximum principle for diffusion The minimum-maximum principle for diffusion states that the solution to the diffusion equation can never evolve within a region to have extreme values larger in magnitude than values given by the initial conditions and by values given by the boundary conditions. 288

modified Green's function The modified Green's function is used for the Poisson boundary value problem with Neumann boundary conditions. It arises from arbitrariness of the solution up to a constant, which precludes existence of the regular Green's function. The modified Green's function ensures self-consistency of the source with boundary data, and it does so by introducing a constant to the source, in addition to the usual Dirac delta. 253

modulus The modulus is the amplitude of a complex number. 163

moment arm The moment arm is the distance from the axis around which the angular momentum is computed. 405

moment of inertia tensor The moment of inertial tensor, $\mathbf{I}$ (or I_{mn}), is a symmetric, second-order tensor that quantifies a rigid body's or continuous mass distribution's resistance to rotational acceleration about any arbitrary axis. It serves as the rotational analogue to scalar inertial mass, relating the angular momentum vector. The diagonal elements of the tensor measure the inertia about a chosen axis, whereas the off-diagonal elements quantify the dynamic asymmetry or tendency of the body to experience cross-axis rotational torques when spun about a non-principal axis. 361

momentum based viewpoint Determining the forces, either directly or indirectly, provides physical insight into the cause of fluid flow and its changes. This approach is referred to a momentum based viewpoint since it is based on working directly with the momentum equation (i.e., Newton's second law of motion). This viewpoint is distinct from a vorticity viewpoint whereby the primary concern is with terms contributing to the evolution of vorticity. x

motion field The motion field, $\boldsymbol{\varphi}$, is the mathematical object (or a "machine" using the language of *Misner et al.* (1973)) that generates a nonlinear time dependent and invertible flow map that continuously and smoothly reshapes the continuum as it moves through Euclidean space as time evolves. The motion field is the reason there is a flow map, so that the nomenclature "motion field" and "flow map" are used interchangeably in this book. Both refer to movement of the continuum as time progresses. 12

musical nomenclature The musical nomenclature is sometimes used for distinguishing the representation of second order tensors. The natural representation of a second order tensor occurs with one tensor index upstairs and the other downstairs. The natural representation is sometimes denoted by $(1, 1)$, to indicate the number of indices up and down. The sharp representation, or $(2, 0)$ representation, is when the tensor is represented with both indices upstairs. Finally, the flat representation or $(0, 2)$ representation is where both indices are downstairs. 19

Neumann boundary condition The Neumann boundary condition prescribes the normal derivative of a function along the boundary. For tracers, the Neumann boundary condition is often referred to as the flux boundary condition since by prescribing the normal derivative we prescribe the flux. 225, 251, 285, 287, 293, 328

Newton's first law In an inertial reference frame, every massive body remains at rest or in uniform motion unless acted on by a net force. This law is sometimes referred to as the law of inertia. 351, 475

Newton's gravitational constant $G = 6.674 \times 10^{-11}$ N m^2 kg^{-2} $= 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2} is Newton's gravitational constant, which is the fundamental constant as part of Newtonian gravity. 148, 370, 375, 526

Newton's second law In an inertial reference frame, application of a net force to a body alters its linear momentum. Mathematically, this law is written $\dot{\mathbf{P}} = \mathbf{F}$, where $\mathbf{P}$ is the linear momentum, $\mathbf{F}$ is the force, and $\dot{\mathbf{P}}$ is the time derivative of momentum. For a constant mass region of matter, Newton's law takes the form $m\,\mathbf{a} = \mathbf{F}$, and $\mathbf{a}$ is the acceleration. 351

Newton's third law To each action there is an equal and oppositely directed reaction. The third law holds for central forces, such as arise in Newtonian gravity and electrostatics. However, it does not hold for all forces, such as the Lorentz force acting on a moving charged particle. The strong form of Newton's third law is satisfied if the force of interaction between two particles is a central force (i.e., $(\mathbf{X}_{(i)} - \mathbf{X}_{(j)}) \times \mathbf{F}_{(ji)} = 0$) and it satisfies $\mathbf{F}_{(ij)} = -\mathbf{F}_{(ji)}$. The weak form of Newton's third law is satisfied if $\mathbf{F}_{(ij)} = -\mathbf{F}_{(ji)}$ whereas the force is not central. 351, 370, 477

Newtonian time Newtonian time refers to the universal time, t, that is a monotonically increasing parameter that allows for the unambiguous distinction between past, present, and future. Hence, an event occuring at one point in space at time, t, also occurs at the same time everywhere else in space. Newtonian time is the time considered in this book since all motions are assumed to be at speeds well below that of light, so that relativistic effects are irrelevant. 4, 75, 81, 346, 352

no-slip boundary condition The no-slip boundary condition means that fluid adheres to solid boundaries due to the role of viscosity, so that there is zero relative flow between the fluid and solid. The no-slip boundary condition leads to boundary layer formation for fluids flows adjacent to solid boundaries. The no-slip boundary condition cannot be imposed for perfect fluids, since such fluids have no viscosity. As discussed in the historical essay by *Anderson* (2005), it was the work of Prandtl in 1905 that first exposed the fundamental nature of the no-slip boundary condition, and its role in establishing boundary layers around solid bodies immersed in a fluid flow. Since the no-slip condition specifies the value of the velocity at the wall (fluid and wall have same velocity), it is often characterized mathematically as a kinematic boundary condition. However, it involves assumptions about stresses at the wall and so physically it arises as a result of dynamic processes, and so can be considered a dynamic boundary condition from this perspective. 273, 275

Noether's theorem Noether's theorem states that for any symmetry of a physical system, there is a corresponding dynamical conservation law. For example, a classical particle system that exhibits Galilean symmetry maintains a constant linear momentum, constant mechanical energy, and constant angular momentum. A conservation law provides a dynamical constraint on the motion. The deduction of dynamical constraints is naturally arrived at using the methods of analytical mechanics, particularly through Hamilton's principle of stationary action. Noether's theorem is one of the most important foundational elements in 20th century mathematical physics. 353, 421, 423, 437, 450

non-coordinate basis A non-coordinate set of basis vectors (also called non-holonomic or anholonomic) span the tangent space at each point of a manifold but are not necessarily tangent to coordinate lines. Consequently, non-coordinate basis vectors cannot be transformed into a coordinate basis. 116, 117

non-dimensionalization Non-dimensionalization is the process of removing all physical dimensions from an equation of motion, and in turn to identify a set of non-dimensional numbers that characterize a particular flow regime. xii

non-dispersive waves Non-dispersive waves are have wave phases whose speeds are independent of the wavenumber, so that the phase velocity is identical to the group velocity. Examples include acoustic waves and shallow water gravity waves in a non-rotating reference frame. 321

non-divergent flow Non-divergent flow is defined by $\boldsymbol{\nabla} \cdot \mathbf{v} = 0$. This flow is realized by the Boussinesq ocean, which is the most general case of non-divergent flow. It is also realized by flow within a fluid of constant density. In two-dimensions, we consider the mechanics of horizontally non-divergent barotropic flow, whereby $\boldsymbol{\nabla} \cdot \mathbf{u} = 0$. 220, 271, 274

non-holonomic constraint Non-holonomic constraints are generally non-integrable and require methods that are not fully covered by the generalized coordinates or Lagrange multipliers considered in this book. 452

null space If $\mathcal{L}$ is a differential operator, then the null space is the space consisting of all homogeneous solutions, $\mathcal{L}\psi = 0$. 230, 260, 264

oceanic oblique coordinates Oblique coordinates are non-orthogonal coordinates on the plane that are used to illustrate some of the basis tensor analysis required to handle generalized vertical coordinates. 110

on-shell The terms on-shell and off-shell, originating from quantum field theory from the relativistic mass-shell condition, serve as useful shorthand even in classical mechanics and variational calculus. They mean those fields that satisfy the equations of motion (on-shell) versus those that do not (off-shell).. 488, 525

one way wave equation The one-way wave equation is another name for the one-dimensional advection equation with a constant advection velocity. It takes the form $(\partial_t \pm U\,\partial_x)\psi = 0$, where U is a constant velocity. It represents the simplest hyperbolic partial differential equation, and can be thought of as the square root of the wave equation, $(\partial_{tt} - U^2\,\partial_{xx})\psi = 0$. 322

one-form One forms are tensors that are dual to vectors. 13, 92, 397

open region An open region, $\mathcal{R}$, is a set or domain that includes none of its boundary points. We pose boundary and initial-boundary value problems on open regions, with prescribed data given on the boundary, $\partial\mathcal{R}$. 224, 248

orbital motion Orbital motion is the motion of a body's center of mass around another body. We are mostly concerned in this book with the orbital motion of the Earth around the Sun, as well as the orbital motion of the Moon around the Earth. 382, 392

orthogonal transformation An orthogonal transformation preserves the lengths of vectors and the angles between them. As a matrix operation, an orthogonal transformation has its inverse equal to the matrix transpose. 27

orthonormal basis vectors Orthonormal basis vectors, written as $\hat{\boldsymbol{e}}_a$, are commonly used to represent tensors following the approach in Cartesian tensor analysis. Examples include the Cartesian orthonormal basis, $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$, and the spherical orthonormal basis, $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{r}})$. 11

parabolic partial differential equation A parabolic partial differential equation a second order PDE that describes diffusive processes, in which disturbances smooth out over time but do not propagate as waves. The spatial portion of the parabolic operator is typically elliptical, so that a time independent parabolic equation reduces to an elliptic equation. 222, 285, 321

parcel instabilities Parcel instabilities are afforded a local necessary and sufficient condition to determine whether the fluid base state is unstable to perturbations. Gravitational instability provides the canicial example, along with centrifugal and symmetric instabilities. Parcel instabilities are also referred to as local instabilities. xxi

parity Parity is discrete symmetry operation that swaps the sign of coordinates. Functions that have odd parity change signs under the parity transformation $(f(x) = -f(-x))$, whereas functions that keey their signs are even parity $(f(x) = f(-x))$. 168

Parseval's identity Parseval's identity states that the total energy (or variance) of a wave field is the same whether computed it in $\boldsymbol{x}$-space or $\boldsymbol{k}$-space. It is also referred to as the Bessel-Parseval identity for discrete Fourier transforms and the Parseval-Plancherel formula for continuous Fourier transforms. 171, 178

partial differential equation A partial differential equation (PDE) is simply a differential equation with more than a single independent coordinate. In this book we encounter a suite of linear PDEs, such as elliptic (e.g., Poisson's equation), parabolic (e.g., diffusion equation), and hyperbolic (e.g., non-dispersive wave equation). We also encounter nonlinear PDEs, which constitute the governing differential equations for fluid mechanics. 217

passive tracer A passive tracer is a scalar field that satisfies the tracer equation, but it has zero impact on the velocity and is thus dynamically passive. Hence, a passive tracer provides a means to probe the advective-diffusive features of the flow without modifying it. 327

passive transformation We consider two versions of a symmetry transformation. Active transformations alter the physical system (e.g., move an experiment from one point in space to another), whereas passive transformations alter the mathematical description of the physics (e.g., change coordinates). 353

perfect fluid A fluid that flows in the absence of irreversible processes so that the motion is reversible and the specific entropy remains constant following a fluid particle. A perfectfluid is a continuum of infinitesimal material fluid parcels. Some authors use the term *ideal fluid*, but we eschew that term to avoid confusion with *ideal gas*. vii

permutation symbol The permutation symbol is a three-index object that has indices that are totally anti-symmetric: $\epsilon_{abc} = -\epsilon_{bac} = -\epsilon_{acb} = \epsilon_{cba}$. It is the Cartesian coordinate representation of the Levi-Civita tensor, ε. 20

phase The phase of a wave refers to the argument of the exponential or the trignometric portion of a linear wave. A phase change by 2π radians returns to the same wave pattern. 163

phase average A phase average refers to an average of a field computed over the phase of the field. This choice for an averaging operation is particularly relevant when the fluctuating field involves quasi-linear waves. 165

phase space Phase space defined by the generalized coordinates plus the generalized momenta. A point in phase space specifies the complete state of a mechanical system (it is a solution to the equations of motion), with trajectories (flow) in phase space specified by solving Hamilton's equations. The dimension of phase space is $2D$, where D is the number of dynamical degress of freedom. 447, 480, 484, 490

Phillip's layering instability The Phillip's layer instability arises from the spatial dependence of the eddy diffusivity. It explains how a vertically sheared turbulence in a stably stratified fluid can organize into horizontal layers separated by sharp density interfaces. 289

physical dimension The physical dimension of an object refers to its dimensionality in terms of length, mass, and time. For example, the dimensions of the velocity vector are length per time, L T^{-1}, whereas the dimensions of momentum are mass times velocity, ML T^{-1}. Physical dimensions are distinguished from physical units, such as meters, kilograms, and seconds. For this book, we only make use of the dimensions of mass, lenth, and time, as we have no concern for electric charge or other physical dimensions that appear in other areas of physics. 9

physical tensor components When decomposing a tensor into its components using generalized orthogonal coordinates, it is useful to define the physical tensor components as those components that each have the same physical dimensions. We do so by attaching the stretching functions to the tensor components, thus decomposing the tensor into its components in terms of the non-coordinate unit basis vectors. 119, 135

physically formal A mathematical argument is said to be physically formal in that is provides a deductive formulation that is motivated from a physical perspective rather than serving the needs for a mathematically rigorous presentation. The prime example of a physically formal presentation is given by our study of the Dirac delta, whereby a mathematically rigorous treatment is far more detailed than provided in this book. 147

planetary Cartesian coordinates Planetary Cartesian coordinates are Cartesian coordinates with their origin at the planet center and that rotate with the planet. 128, 391, 397, 402, 425

planetary centrifugal acceleration Planetary centrifugal acceleration, $\boldsymbol{A}_{\mathsf{centrifugal}} = -\nabla\Phi_{\mathsf{centrifugal}} = \Omega^2\,(x\,\hat{\boldsymbol{x}} + y\,\hat{\boldsymbol{y}})$, arises from motion on the rotating planet, and it points outward from the polar rotation axies. 391, 401, 411, 426

planetary spherical coordinates Planetary spherical coordinates are spherical coordinates with their origin at the planet center and that rotate with the planet. 128, 391, 397, 403

planetary vorticity Planetary vorticity refers to the vorticity imparted to every fluid due to its existence in a rotating planetary reference frame. xix

plumb line The plumb line defines the local vertical direction, which is parallel to the effective gravity. The angle between the radial gravity and the effective gravity is a function of the rotation rate, Earth radius, gravitational acceleration, and latitude. 417, 419, 446

point particle A point particle has zero dimensions and so it occupies a single point in space. It is the simplest idealization of matter that forms the basis for Newtonian particle mechanics. 350

Poisson bracket The Poisson is a bilinear operation on functions of phase space that encodes the fundamental structure of Hamiltonian mechanics and determines how observables evolve in time. For any two phase space functions F, G, that are dependent on the generalized coordinates, ξ^σ, and generalized momenta, $\mathcal{P}_\sigma$, we have the Poisson bracket given by $\{F, G\} = \partial F/\partial\xi^\sigma\,\partial G/\partial\mathcal{P}_\sigma - \partial F/\partial\mathcal{P}_\sigma\,\partial G/\partial\xi^\sigma$. The Poisson bracket provides a symplectic structure to phase space, where a symplectic structure provides an anti-symmetric pairing of the $D + D$ dimensional phase space coordinates (the generalized coordinates and generalized momenta). 487, 493

Poisson's equation Poisson's equation, $\nabla^2\Psi = \sigma$, is a linear elliptic partial differential equation where the Laplacian of a scalar function, Ψ, equals to a source term, σ. This equation arises in many contexts within geophysical fluid mechanics. 223, 226, 242, 287, 291, 295, 302, 375

position space Position space refers to the space defined by the position in space, $\boldsymbol{x}$, of a field. This terminology is used for the study of waves, in which position space is dual to the wavevector space, $\boldsymbol{k}$. The Fourier transform provides the means to move between $\boldsymbol{k}$-space and $\boldsymbol{x}$-space. 161

potential energy Potential energy of a fluid is a dynamical property arising from the presence of the fluid element within the gravitational field of the planet plus the centrifugal potential of the rotating planetary reference frame. It is sometimes referred to as the potential energy from the effective gravitational acceleration, or equivalently from the geopotential. The potential energy for a fluid element of mass $\rho\,\delta V$ is given by $\Phi\,\rho\,\delta V$, where Φ is the geopotential. 355

potential momentum Potential momentum, $\boldsymbol{M} = \boldsymbol{V}_{\mathsf{cartesian}} + 2\,\boldsymbol{\Omega} \times \boldsymbol{X}$, is a constant of the motion for point particles moving along directions parallel to the geopotential (so long as the geopotential is a constant). Potential momentum is useful for the study of fronts on an f-plane, where the component of potential momentum along the front remains materially conserved for inviscid flows. 415, 424, 444

potential vorticity Potential vorticity is a strategically chosen component of the vorticity vector that melds mechanics (vorticity) to thermodynamics (stratification). Material conservation properties of potential vorticity render important constraints on fluid motion, thus promoting it as a primary field in the study of geophysical fluid mechanics. xix, 423

power Power is the scalar product of the velocity and force. The time integral of the power, as integrated along a trajectory, equals to the work done by the force on the matter. 355, 437

precession Precession is the gradual rotation of a system's axis of motion caused by a torque or anisotropy, with angular momentum changing direction but not necessarily magnitude. For example, the oscillation plane for the Foucault pendulum precesses due to the earth's rotation. 509

principle of equivalence The principle of equivalence states that the same mass, m, determines the gravitational force as well as any kinematic forces arising from acceleration. That is, in an inertial reference frame, $m\,\mathbf{g}$ is the gravitational force, and $m\,\ddot{\mathbf{X}}$ is the inertial force, with Newton's law yielding $\ddot{\mathbf{X}} = \mathbf{g}$. The principle of equivalence originates from the work of Galileo and it forms a key element to Einstein's relativity theory. 365

prognostic equation A prognostic equation determines the time tendency (Eulerian evolution) of a quantity such as the temperature or velocity. Terms in the prognostic equation are referred to as time tendencies. A prognostic equation is also referred to as an evolution equation. x, xi, 312

proper time Proper time is the time measured by an observer in the rest frame of a particle. Proper time is the natural parameter for particle trajectories. Since we use Newtonian time in this book, the proper time is the same as all other times. 80

Pythagora's theorem Pythagora's theorem provides the means to measure distance between two points in Euclidean space. It does so using the Kronecker delta, δ_{ab}. 14

quotient rule If the contraction of two objects yields a tensor, and if one of these objects is a tensor, then so too is the other. This quotient rule allows us to deduce the tensorial nature of an object based on whether that object is constructed from tensors. 77

radians Radians defines the objective means to measure the angular displacement around a circle. In particular, $\Delta\vartheta = s/R$,where s is the arc-distance along the circle's perimete, and R is the circle's radius. We make use of radians rather than degrees, since radians is an objective measure of angular displacement whereas degrees is subjective. 358

Rayleigh drag Rayleigh drag is a particular form of friction that makes use of the acceleration, $-\gamma\,\boldsymbol{v}$, with γ^{-1} having dimensions of time. Rayleigh drag adds an acceleration that drags all flow towards a state of rest. Notably, Rayleigh drag has zero scale selectivity, which contrasts to Laplacian friction suggested by kinetic theory, whereby friction acts preferentially at the small scales. 357

real fluid A fluid whose flow is affected by irreversible processes arising from momentum mixing (nonzero viscous friction); enthalpy mixing (nonzero diffusivity for temperature); matter mixing (nonzero diffusivity of matter constituents); and through sources such as radiation and chemical reactions. The specific entropy increases following a fluid particle moving in a real fluid. vii

reciprocity condition The reciprocity condition refers to properties of Green's functions according to their behavior when the source and field points are swapped. Green's functions for self adjoint operators, such as the Laplacian, have full symmetry under such swaps. For parabolic operators, such as the diffusion equation, reciprocity involves the adjoint Green's function and more care is needed. 245, 263, 292, 295

response functions Response functions measure the change in a thermodynamic property as the system is forced in some manner. The heat capacity, thermal expansion coefficient, and haline contraction coefficient, are three response functions commonly encountered in ocean and atmospheric fluid mechanics. 285

Riemannian differential geometry Riemannian differential geometry is the study of geometry for manifolds endowed with a metric tensor. 73

right hand rule The right hand rule is a convention used to specify the orientation (positive or negative) of the vector cross product. Here, we orient the right hand's out-stretched index finger with the first vector, and the middle finger with the second vector, with the thumb then defining the direction of the cross product. 20

rigid-body Rigid bodies occupy a finite region of space and are comprised of a continuum of matter particles that maintain fixed relative positions, thus constraining the media to maintain constant mass and volume. Some consider rigid bodies synonymous with solid-bodies. We make the distinction since solids are not perfectly rigid, as characterized by a nonzero Deborah number. 350, 357, 453

rigid-body rotation Consider a point fixed on a rigid body, and let the rigid body exhibit pure rotational motion with rotational velocity $\boldsymbol{\Omega}$. The velocity of that rigid-body rotation is given by $U_{\mathsf{rigid}} = \boldsymbol{\Omega} \times \mathbf{x}$, where $\mathbf{x}$ is the position of the point on the body relative to the center of the body. For example, a point on the surface of a rotating sphere exhibits rigid-body rotation with $\boldsymbol{\Omega}$ the rotational velocity around the axis of rotation. 358, 359, 368, 393, 394

Robin boundary condition The Robin or mixed boundary condition combines aspects of the Neumann and Dirichlet boundary conditions by specifying the sum of the normal derivative along the boundary (like the Neumann condition), as well as the value along the boundary (like the Dirichlet condition). 328

rotation rate tensor The rotation rate tensor is the anti-symmetric portion of the velocity gradient tensor, with elements given by twice the vorticity, $R^{mn} = -\epsilon^{mnp}\,\omega_p/2$ with $\boldsymbol{\omega} = \nabla \times \boldsymbol{v}$ the vorticity. The rotation rate tensor measures the ability of fluid flow to rotate without dilating fluid elements. 139

scalar field A scalar field provides a zeroth order tensor at each point in space-time. Temperature and mass density are two scalar fields encountered in fluid mechanics. 17

scalar product The scalar product is the contraction of a vector with a one-form that produces a number. 13, 15

Schrödinger equation The Schrödinger equation is a fundamental equation of wave mechanics, either classical or quantum. The time-independent Schrödinger equation is an eigenvalue problem whereby we seek energy eigenfunctions and eigenvalues for the self-adjoint Hamiltonian operator. 281

screened Poisson equation The screened Poisson equation is $(-\nabla^2 + \mu^2)\,\psi = \Lambda$, with $\mu^2 > 0$. This equation occurs in the Yukawa theory of mesons and the Debye-Hückel screening of ionic solutions. Although not of direct use for our studies of geophysical fluid mechanics, we see in VOLUME 4 that there is a direct connection to the Helmholtz equation (where $\mu^2 < 0$), which is important for geophysical wave theory. 279, 280

self-adjoint A self-adjoint operator is a linear operator that equals to its adjoint, where adjoint is defined relative to a chosen inner product. For real matrices, the adjoint is the transpose, so that a symmetric matrix is a self-adjoint matrix. For complex operators, self-adjoint refers to Hermitian operators (such as considered in quantum mechanics). For partial differential equations, self-adjointness is determined by properties of the differential operator, the boundary conditions, and the inner product. The eigenfunctions of self-adjoint operators form a complete set over the space of functions that satisfy the same boundary conditions as the eigenfunctions. We make particular use of of this property when performing an eigenfunction expansion of Green's functions. 227, 229, 245, 260, 295, 296

self-attraction and loading Self-attraction and loading (SAL) refers to the movement of ocean mass that modifies the Earth's gravity field, which in turn affects the ocean tides. The loading term arises from alterations in the mass felt by the solid Earth that leads the crust to expand or compress. Self-attraction arises from modifications to the gravity field due to self-gravity of both the load-deformed solid Earth and the ocean tide itself. Locally, SAL can contribute to roughly 20% of the ocean tide amplitude, so that accurate ocean tide modeling must include SAL. 386

separation of variables The separation of variables method decomposes the solution to a partial differential equation into the product of solutions to ordinary differential equations. This method is available only in certain physical systems possessing symmetries. In outline form, the method considers a partial differential equation of the form $\mathcal{L}\Psi(x, y, z) = 0$, and seeks a solution of the form, $\Psi(x, y, z) = A(x)\, B(y)\, C(z)$. If the method is available, then the differential operator, $\mathcal{L}$, and corresponding initia/boundary conditions, is said to be separable. 227

shallow fluid approximation The shallow fluid approximation is based on noting that for large-scale fluid mechanics of the atmosphere and iocean, the thickness of fluid is tiny relative to the radius of the planet. Hence, we write the radial coordinate equal to the earth's radius, $r = R_{\mathrm{e}} + z \approx R_{\mathrm{e}}$, where r appears as a multiplier (or divisor), but not as a derivative operator. The shallow fluid approximation is distinct from the shallow water approximation. 138, 436

sifting property The sifting property is a basic property of the Dirac delta, whereby the integral of the Dirac delta times another function yields the function evaluated at the point where the Dirac delta is singular. In one dimension, this property reads $\int_{-\infty}^{\infty} \delta(x)\, f(x)\, \mathrm{d}x = f(0)$. 147, 149, 263

simply connected A domain is simply connected if a closed curve can be continuously shrunk to a point while remaining in the domain. For example, the domain of the global ocean on a water covered planet is a simply connected manifold. Adding land masses in the form of islands or continents makes the ocean domain non-simply connected. 271

skew-symmetric tensor A skew-symmetric (also anti-symmetric) tensor equals to minus its transpose, $\boldsymbol{A} = -\boldsymbol{A}^{\mathsf{T}}$. 33

Snell's law Snell's law of refraction describes how a ray (of light, sound, or any wave) bends when it passes across an interface between two media in which its propagation speed differs. Mathematically, Snell's law says that $\mathsf{n}_1 \sin\theta_1 = \mathsf{n}_2 \sin\theta_2$, where n is the index of refraction, and θ is the angle the ray makes with the normal to the surface separating the two media. A media with larger index of refraction causes the ray to bend closer to the normal direction. 211, 212

source point In the context of Green's function theory, the source point refers to the point in space-time where the Dirac source is located. It contrasts to the field point, where the Green's function is sampled or observed. 235–237

spectral decomposition Completeness of any basis for a vector space allow one to represent an arbitrary member of that vector space in terms of the basis. This representation is referred to as a spectral decomposition. We make use of spectral decompositions when performing Fourier analysis, where the basis has an infinite number of elements. We also use spectral decomposition to express the Green's function for certain linear partial differential operators. 157, 161, 260, 299, 335

spherical harmonics Spherical harmonics are eigenfunctions for the Laplacian operator on a sphere of unit radius. They form a complete orthonormal basis for functions on the sphere, and they are used quite frequently to represent functions in geophysical fluid mechanics. 230, 232

spinning motion Spinning motion is rotation of a body around its own center of mass. 382, 392

statistical mechanics Statistical mechanics is an area of mathematical physics that aims to explain the physical properties of macroscopic matter using properties of the microscopic mechanics. 72

step response function The step response function is the response of a dynamical system to a force that turns on and remains on after some initial time, as per a Heaviside step function. 308

steric setup A steric setup refers to sea level rise near the coast that arises from the generally lower density near the coast. xiii

Stokes' theorem The vector calculus form of Stokes' theorem says that for a simply connected surface, $\oint_{\partial\Omega} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} = \int_{\Omega} (\nabla \times \boldsymbol{A}) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\mathcal{S}$, where $\boldsymbol{A}$ is a vector, Ω is a simply connected surface with boundary $\partial\Omega$, $\mathrm{d}\boldsymbol{x}$ is a line increment around the boundary, and $\hat{\boldsymbol{n}}$ is an outward normal to the surface and which defines the orientation. 70

strain rate tensor The strain rate tensor, $\boldsymbol{S}$, is the symmetric portion of the velocity gradient tensor. The strain rate tensor measures the ability of fluid flow to deform fluid elements through stretching, tilting, straining, and dilation. The strain rate tensor is called the deformation rate tensor in the continuum mechnics literature, such as *Tromp* (2025a). 139

streamlines A streamline is a curve in space that is tangent, at each time instance, to the fluid particle velocity. Streamlines and pathlines are identical only for steady flows, whereas they are generally distinct for unsteady flows. 220

subharmonic function A subharmonic function, ψ, satisfies the Poisson equation with a non-negative source. 226

super harmonic function A super harmonic function, ψ, satisfies the Poisson equation with a non-positive source. 226

superposition principle The superposition principle states that the sum of any two solutions to the same linear equation is itself a solution. Correspondingly, the response of a physical system to two sources is the same as the sum of the response of the system to each source applied separately. The superposition principle is the basis for the Green's function method, and for Fourier analysis. 235, 304

Sverdrup balance The Sverdrup balance is a diagnostic balance between vertically integrated meridional transport and the wind stress curl. In particular, a positive wind stress curl leads to northward vertically integrated flow. The traditional form of the Sverdrup balance arises when we assume the flow takes place over a flat bottom and the free surface undulations are negligible, so that $\beta \, v = \hat{\boldsymbol{z}} \cdot (\nabla \times \boldsymbol{F}^{\mathsf{wind}})$. This balance helps to explain the steady equatorward ocean circulation appearing in the eastern portion of middle latitude gyres. xi

symmetric tensor A symmetric tensor equals to its transpose, $\boldsymbol{S} = \boldsymbol{S}^{\mathsf{T}}$. 33

symmetry A symmetry refers to a transformation or operation that can be performed on the physical system without changing any physics. We consider two versions of a symmetry transformation. Active transformations alter the physical system (e.g., move an experiment from one point in space to another), whereas passive transformations alter the mathematical description of the physics (e.g., change coordinates). 353, 421, 423

tangent bundle The tangent bundle is the accumulation of all points and their tangent spaces over a smooth manifold. Elements of the tangent bundle are specified by fixing the point and its tangent space, so that the the dimension of a tangent bundle is twice that of the manifold. 86, 468

tangent plane The tangent plane makes use of Cartesian coordinates defined local to a point on the rotating planet. The f-plane and β-plane are the two common forms of the tangent plane approximation encounted in this book. It is important to note that the tangent plane approximation is not based on assuming a locally flat sphere. Rather, the tangent plane approximation as based on assuming a locally flat geopotential. 425

tangent plane Cartesian coordinates Tangent plane Cartesian coordinates are Cartesian coordinates defined local to a point on the rotating planet, with the approximation based on assuming a locally flat geopotential. The f-plane and β-plane are the two forms of the tangent plane approximation encounted in this book. It is important to note that the tangent plane approximation is not based on assuming a locally flat sphere. Rather, the tangent plane approximation as based on assuming a locally flat geopotential. Consequently, the approximation makes use of the effective gravitational acceleration (minus the gradient of the geopotential) that includes both central gravity acceleration plus the planetary centrifugal acceleration. 402, 425

tangent space The tangent space, $T_{\mathcal{P}}\mathcal{M}$, is the linear vector space defined at each point, $\mathcal{P}$, of a smooth manifold, $\mathcal{M}$. The geometric view of tensor fields considers them to live in the tangent space at each point of a manifold. 86, 93, 363, 468

teleological A teleological explanation considers phenomena in terms of the purpose it serves rather than of the cause by which it arises. The Newtonian description of motion offers the canonical cause and effect perspective, whereby a force (the cause) creates an effect (the acceleration). In contrast, Hamilton's principle of stationary action states that the physically realize motion takes place so that the action (an integral over time) is an extrema. From this view, Hamilton's principle can be viewed as offering a purpose for the motion, namely to extremize the action. 468

tensor Tensors are geometric objects that provide that natural home for objective physical objects such as velocity, acceleration, forces, and stresses. To perform analysis with tensors requires them to be represented using a chosen coordinate system with a particular origin, with the representation of a tensor a function of the coordinates. However, the geometric nature of a tensor means that it remains unaffected by the choice of coordinates or by the coordinate origin. 2, 7, 11

tensor fields A tensor field provides a tensor at each point in space and time. We make use of tensor fields in the study of geophysical fluid mechanics. 17

tensor order The order of a tensor is defined by the number of basis vectors and/or one-forms needed to represent the tensor. For example, a scalar is a zeroth order tensor, a vector is a first order tensor, and a diffusion tensor is a second order tensor. The order of a tensor is distinct from its rank, with the rank determined by the dimension of the space in which the tensor is represented. 13

tensor product A tensor product between two tensors produces a third tensor whose order is the sum of the terms. For example, the tensor product of two $(1,0)$ vectors produces a $(2,0)$ tensor, such as the second order kinetic stress tensor, $\mathbb{T}^{\mathsf{kinetic}} = -\rho\, \mathbf{v} \otimes \mathbf{v}$. In components, we have $(\boldsymbol{v} \otimes \boldsymbol{v})^{ab} = v^a\, v^b$. The tensor product is an expression for tensors of the outer product from linear algebra. 19, 77

tidal acceleration Tidal acceleration is the differential gravitational acceleration that acts on a finite sized object placed in a non-uniform gravitational field. The tidal acceleration falls off as the inverse cube of the distance rather than the more familiar inverse square. 378, 379

time tendency Those terms in a prognostic Eulerian equation that contribute to the time evolution are referred to as time tendencies. For prognostic equations, knowledge of the processes contributing to the net time tendency enables a prediction of flow properties. x, 312

torque Torque is the moment of the force, $\mathbf{X} \times \mathbf{F}$, as computed around a chosen origin. Torques lead to time changes in the angular momentum through $\mathrm{d}\mathbf{L}/\mathrm{d}t = \mathbf{X} \times \mathbf{F}$. 361

torsion Torsion measures the failure of covariant derivatives to commute. Torsion measures the failure of infinitesimal parallelograms to close when you move vectors around in a manifold using a given connection. In this book, as in Einstein's general relativity, we assume a Levi-Civita connection that is torsion-free. However, torsion is important for some areas of continuum mechanics (dislocations) and non-Einstein theories of gravity. 73, 106

trace The trace of a second order tensor, $\boldsymbol{T}$ is the contraction, $T^{a}{}_{a}$. 33

trajectory A trajectory is space-time particle path/curve that is parameterized by time, and it is represented by the time evolving spatial position, $\mathbf{X}(t)$, which is sometimes written with the lowercase, $\mathbf{x}(t)$. Also, the time parameter is sometimes written T or τ. The trajectory for a point mass particle in classical particle mechanics has its natural extension to the trajectories of fluid particles that form the foundation to Lagrangian kinematics. 80, 351, 352

transformation matrix The transformation matrix is the matrix of partial derivatives that provides the operational means to transform tensor components between one set of coordinates and another. For the coordinate transformation to be invertible, the Jacobian of the transformation matrix (i.e., its determinant) must never vanish. Although the transformation matrix carries two indices, it is not a tensor. Instead, it is a matrix operator used to transform from one set of basis vectors to another. 26, 90, 398

variation A variation refers to a change to a function as occurs when taking a functional derivative. In physics, a variation is a virtual displacement (yet not constrained to be small), which is not necessarily a dynamically realized change, though virtual displacements and variations must satisfy kinematic constraints. Variations are used to see how some quantity (usually an action, energy, or another functional) responds. It is a trial modification used for testing extremal conditions. 193, 455, 464

vector A vector is a first order $(1, 0)$ tensor. In the geometric interpretation of tensors, a vector is the directional derivative along a curve on a manifold, and it lives within the tangent space of a manifold. 84, 396

vector cross product Geometrically, the vector cross product provides a means to measure area associated with the parallelogram defined by two vectors, and to specify the orientation of that parallelogram. Algebraically, the vector cross product is the skew-symmetric product between two vectors. Tensorially, the vector cross product of two vectors yields a one-form that is orthogonal to either of the two vectors. 20, 69

vector field A vector field provides a vector at each point of space and time. 18, 87

velocity The time derivative of a particle trajectory defines the velocity of the particle. For a fluid, the Lagrangian time derivative of the flow map defines the velocity field. 352

velocity gradient tensor The velocity gradient tensor, $\boldsymbol{G}$, is the Eulerian means to quantify flow deformation. Its elements are the spatial derivatives of the velocity field. It is commonly decomposed into the symmetric strain rate tensor (also known as the deformation rate tensor) plus the anti-symmetric rotation tensor. 139

virial In classical mechanics, the virial characterizes the relationship between the linear momentum, forces, and position. It plays a central role in deriving the virial theorem, which connects time averaged kinetic and potential energies for bound systems. For a system of N particles, the virial as defined by **?** is given by $-\frac{1}{2}\sum_{n=1}^{N}\boldsymbol{F}_n \cdot \boldsymbol{X}_n$, where $\boldsymbol{F}_n$ is the force acting on particle n, and $\boldsymbol{X}_n$ is its position. As so defined, the virial measures how forces act relative to the system's size. An alternative definition gives the virial as $\frac{1}{2}\sum_{n=1}^{N}\boldsymbol{P}_n \cdot \boldsymbol{X}_n$, where $\boldsymbol{P}_n$ is the linear momentum. 491

virial theorem The virial theorem relates the time average of the total kinetic energy to the time average of a certain function of the potential energy. It is derived through use of Euler's theorem for homogeneous functions. One particular example of the virial theorem concerns its ability to explain why the phase averaged kinetic energy equals to the phase averaged potential energy within a linear wave. 470, 522

virtual displacement A virtual displacement is a tiny spatial displacement of a physical system that occurs at a fixed time instance. Virtual displacements respect the forces of constraint, which means they are kinematically admissable displacements. However, they are not bound by dynamical constraints and as such they can be unphysical. 455

vorticity Vorticity is the curl of the velocity, $\boldsymbol{\omega} = \nabla \times \boldsymbol{v}$. It plays a leading role in the study of geophysical fluid flows, where it is important to distinguish the relative vorticity, $\boldsymbol{\omega} = \nabla \times \boldsymbol{v}$, from the planetary vorticity, $2\,\boldsymbol{\Omega}$. xix

vorticity based viewpoint A variety of vorticity constraints offer the means to deduce flow properties without determining forces, thus prompting the vorticity based viewpoint that is distinct from the momentum-based approach, thus prompting the importance of vortex mechanics in geophysical fluid mechanics. x, 585

wave action Wave action is wave energy divided by the wave's intrinsic frequency, and it is an approximatly conserved quantity for waves in slowly varying media. Wave action is the adiabatic invariant associated with phase averaging. It directly corresponds to the case in particle mechanics, where action variables are conserved under slow parameter variation. 504, 505

wave equation The wave equation generally refers to the hyperbolic partial differential equation $(\partial_{tt} - c^2 \nabla^2)\psi = 0$, where $c^2 > 0$ is the squared wave speed. This equation describes non-dispersive acoustic and shallow water gravity waves. It is but one of many other wave equations encountered in geophysical fluid mechanics. 322

wave instabilities Wave instabilities arise from the constructive interference of waves and so involve the solution of an eigenvalue problem to determine properties of unstable waves. At most, a necessary condition can be derived to determine whether a wave instability exists. Wave instabilities are also referred to as global instabilities. xxi

wavenumber The wavenumber is the magnitude of the wavevector, $|\boldsymbol{k}|$. The wavenumber measures 2π times the number of waves per unit length, so that the wavenumber measures the spatial angular frequency of a wave. 167

wavevector space Wavevector space refers to the space defined by the wavevector, $\boldsymbol{k}$, of a field, which is a space dual to the position space, $\boldsymbol{x}$. The Fourier transform provides the means to move between $\boldsymbol{k}$-space and $\boldsymbol{x}$-space. 161

work-energy theorem The work-energy theorem of classical mechanics says that the net work done on a physical system over a time interval equals to the change in the kinetic energy. 351, 355, 356

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